Main Routine

Equipment Data Collection Gateway

v1.0.1.9 - March 26, 2020 added DN005, DN006, DN007
DN005 - add Health Alarm to Overview Server Status
DN006 - support for multiple Offline Lines
DN007 - increase Line count from 20 to 29
v1.20 - added Carrier Tracking
DN014 - v1.21 added Health to Server Interface
EDC2 - v2.00 added PdM with QualityRx, TrendRx and Server Interface
EDC2 - v2.01 added Event Configuration
EDC2 - v2.03 added Cycle Data Type
EDC3 - v3.00 added Time Sync, Web Diagnostic, Tool Scan updates, Quality RB3 and RB4, throttle Enet Utilization Msgs
v3.01 added better RIng Buffer distirbution and process N Lines/Scan of RxBuffer
v3.02 added Quality RB3 & RB4, faster RB processing toTM/SQL, reduced PdM scan time
v3.02 added Data Refresh Command to Event Mask Refresh
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion

aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

1

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE

Source	1
Dest	Prog.Routine95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD

Source A	Prog.RoutineScan[Prog.Routine]-1733165816
Source B	1
Dest	Prog.RoutineScan[Prog.Routine]-1733165816

2

Reset / Configuration / Initialization Routines

Configuration and Initialization Routine

JSR

Routine Name	_009_ConfigInit

Main Routine

Initialize Equipment Event Masks and Data Refresh for All Equipment Packets 0-7
Ignore, Maint, Fault, Alarm, Alert

3

HMI toggle to Send Event Masks for all Equipment EventMaskSend_AutoPB

HMI toggle to Send Event Masks for all Equipment EventMaskSend_AutoPB (U)

manual toggle to Send Event Masks for all Equipment EventMaskSend_ManualPB (U)

EventMaskSend (L)

EventMaskSend_Init (L)

manual toggle to Send Event Masks for all Equipment EventMaskSend_ManualPB

Any FEC First Scan FEC_EqDrvFault_FirstScan

s:FS

EventMaskSend

JSR
Routine Name _021_InitializeEventMasks

Time Sync with SNTP
Set Time Zone and DST
Update CLx Status w/ TImestamp, Scan etc.

4

EDC Gateway TimeSync from SNTP -- used by all Equipment for TimeSync

JSR
Routine Name _029_CLxTimeSyncSNTP

Set TimeZone and determine whether to apply Daylight Savings Time

JSR
Routine Name _029_CLxTimeZoneDST

Monitors CLx Status

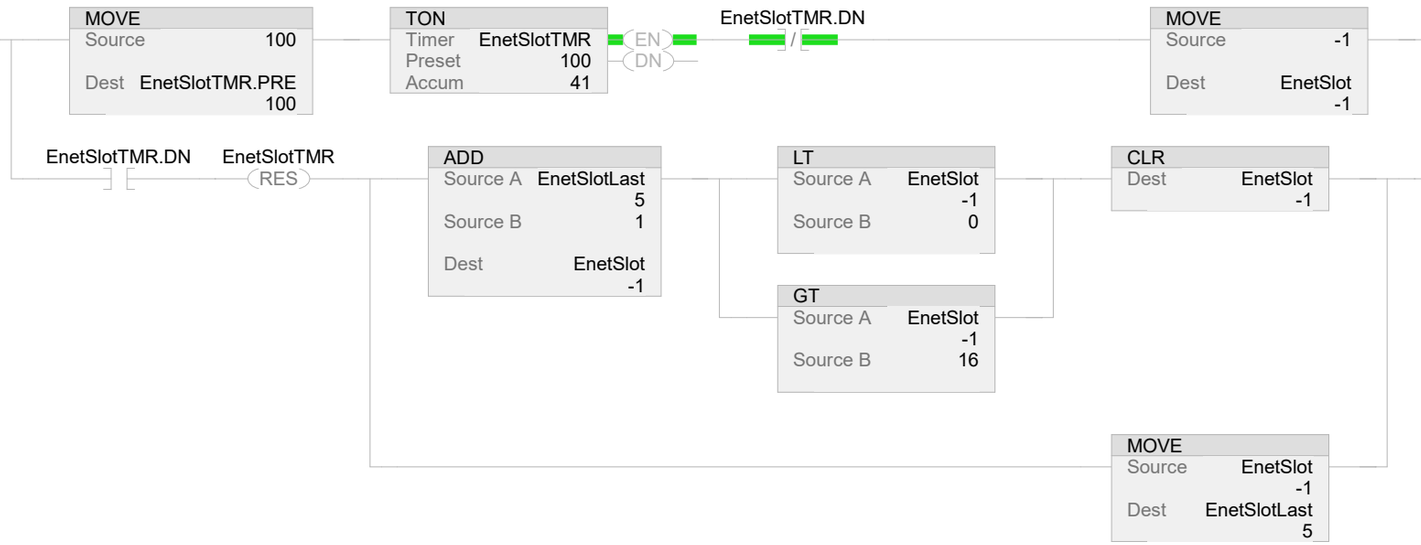
JSR
Routine Name _030_CLxStatus

Main Routine

CLx Ethernet Status and Utilization
Increment Slot Number

Service 1 Slot/Timer to reduce Communication Utilization
(Controller Message Utilization is highly impacted by Enet Utilization messages when Controller Scan Time is low)

5



Main Routine

CLx Ethernet Status and Utilization
Slots 0-3

6

EQ
Source A EnetSlot
-1
Source B 0

Enet Module Status
JSR
Routine Name _031_Enet00_NW_L8x

EQ
Source A EnetSlot
-1
Source B 1

Enet Module Status
JSR
Routine Name _031_Enet01_GW_L9x_THIS

EQ
Source A EnetSlot
-1
Source B 2

Enet Module Status
JSR
Routine Name _031_Enet02_PdM_L8x

EQ
Source A EnetSlot
-1
Source B 3

Enet Module Status
JSR
Routine Name _031_Enet03_FEC1_L8x

Main Routine

CLx Ethernet Status and Utilization
Slots 4-7

7

Enet Module Status

EQ
Source A EnetSlot -1
Source B 4

JSR
Routine Name _031_Enet04_FEC2_L8x

Enet Module Status

EQ
Source A EnetSlot -1
Source B 5

JSR
Routine Name _031_Enet05_FEC3_L8x

Enet Module Status

EQ
Source A EnetSlot -1
Source B 6

JSR
Routine Name _031_Enet06_FEC4_L8x

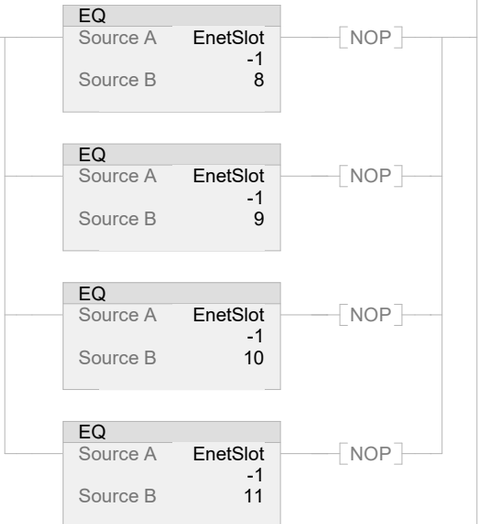
EQ
Source A EnetSlot -1
Source B 7

NOP

Main Routine

CLx Ethernet Status and Utilization
Slots 8-11

8



Main Routine

CLx Ethernet Status and Utilization
Slots 12-16

9

EQ
Source A EnetSlot -1
Source B 12
NOP

EQ
Source A EnetSlot -1
Source B 13
NOP

EQ
Source A EnetSlot -1
Source B 14
NOP

EQ
Source A EnetSlot -1
Source B 15
NOP

EQ
Source A EnetSlot -1
Source B 16
NOP

Constants
(ASCII Table Initialization, Strings, etc.)

10

Define ASCII Table
JSR
Routine Name _038_AsciiTable

Define Global Constants
JSR
Routine Name _039_Constants

Update Performance Summary

PerfUpdate limits PerformanceSummary Point Iterations to conserve Scan Time.
Default PerfUpdate=1 -- this can be raised during debug for faster updates at the expense of Driver Performance.

11

Update Performance Summary
JSR
Routine Name _090_PerfSummaryLine

Main Routine

Reset Program Scan Clear after all Programs have been reset

12

CLx Module Status
Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>
U

13

Reset Performance Summary for ALL Points
-- Automatic from Gateway Status Display Reset PB
-- Manual Reset by toggling Perf.CTL.ManualReset

EDC Comm Driver
Performance Auto
Request to Reset
Performance Values
for ALL Equipment
Perf.CTL.AutoReset
<EDC.Perf.CTL.AutoReset>

EDC Comm Driver
Performance Manual
Request to Reset
Performance Values
for ALL Equipment
Perf.CTL.ManualReset
<EDC.Perf.CTL.ManualReset>

EDC Comm Driver
Performance Auto
Request to Reset
Performance Values
for ALL Equipment
Perf.CTL.AutoReset
<EDC.Perf.CTL.AutoReset>
U

EDC Comm Driver
Performance Manual
Request to Reset
Performance Values
for ALL Equipment
Perf.CTL.ManualReset
<EDC.Perf.CTL.ManualReset>
U

Reset Performance Summary

JSR
Routine Name _096_PerfSummaryReset

Manual Reset of Gateway Status Errors
EDC_StatusErrorReset
L

CLx Module Status
Clear Scan History
CLX.TaskScan.Clear
<EDC.CLX.TaskScan.Clear>
L

CLx Module Status
Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>
L

Ethernet Module
Status Clear Scan
History
Enet00.CPU_Util.Clear
<EDC_Enet00.CPU_Util.Clear>
L

Ethernet Module
Status Clear Scan
History
Enet01.CPU_Util.Clear
<EDC_Enet01.CPU_Util.Clear>
L

Ethernet Module
Status Clear Scan
History
Enet02.CPU_Util.Clear
<EDC_Enet02.CPU_Util.Clear>
L

Ethernet Module
Status Clear Scan
History
Enet03.CPU_Util.Clear
<EDC_Enet03.CPU_Util.Clear>
L

Ethernet Module
Status Clear Scan
History
Enet04.CPU_Util.Clear
<EDC_Enet04.CPU_Util.Clear>
L

Ethernet Module
Status Clear Scan
History
Enet05.CPU_Util.Clear
<EDC_Enet05.CPU_Util.Clear>
L

Ethernet Module
Status Clear Scan
History
Enet06.CPU_Util.Clear
<EDC_Enet06.CPU_Util.Clear>
L

Main Routine

Manually Reset Performance Stats for a Single Equipment
-- Set Perf.CTL.ManualResetLNi and EQi to Equipment Index and Toggle Perf.CTL.ManualResetEquip

14

EDC Comm Driver Performance Manual Request to Reset Performance Values for One Equipment
Perf.CTL.ManualResetEquip
<EDC.Perf.CTL.ManualResetEquip>

Equipment array Line Index

MOVE	
Source	Perf.CTL.ManualResetLNi <EDC.Perf.CTL.ManualResetLNi> 0
Dest	ResetLNi 29

Equipment array Equip Index

MOVE	
Source	Perf.CTL.ManualResetEQi <EDC.Perf.CTL.ManualResetEQi> 0
Dest	ResetEQi 100

EDC Comm Driver Performance Manual Request to Reset Performance Values for One Equipment
Perf.CTL.ManualResetEquip
<EDC.Perf.CTL.ManualResetEquip>
(U)

Reset Performance Summary for LNi, EQi

JSR	
Routine Name	_098_PerfSummaryResetEquip

Transaction Log Sanpshot

Capture snapshot of Transaction Log and Error Log
(Triggered externally from TransLog Spreadsheet or internally from User-defined Trigger Logic)

15

Transaction Log Control External Snapshot Command (from TransLog spreadsheet)
Trans.SnapExt
<EDC.Trans.SnapExt>

Transaction Log Control Internal Snapshot Command Enabled (User Logic)
Trans.SnapIntEN
<EDC.Trans.SnapIntEN>

Transaction and Error Log Snapshot

JSR	
Routine Name	_080_TransLogSnapshot

Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

16

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Routine
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Fault.Type
0
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Fault.Code
0
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.FaultCount
0
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.LOGi
0
Dest Trans.Opt[5]
<EDC.Trans.Opt[5]>
-1

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 9003
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Main Routine

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

17

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

18

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
120
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan Events
CLxProgScanHigh.0
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

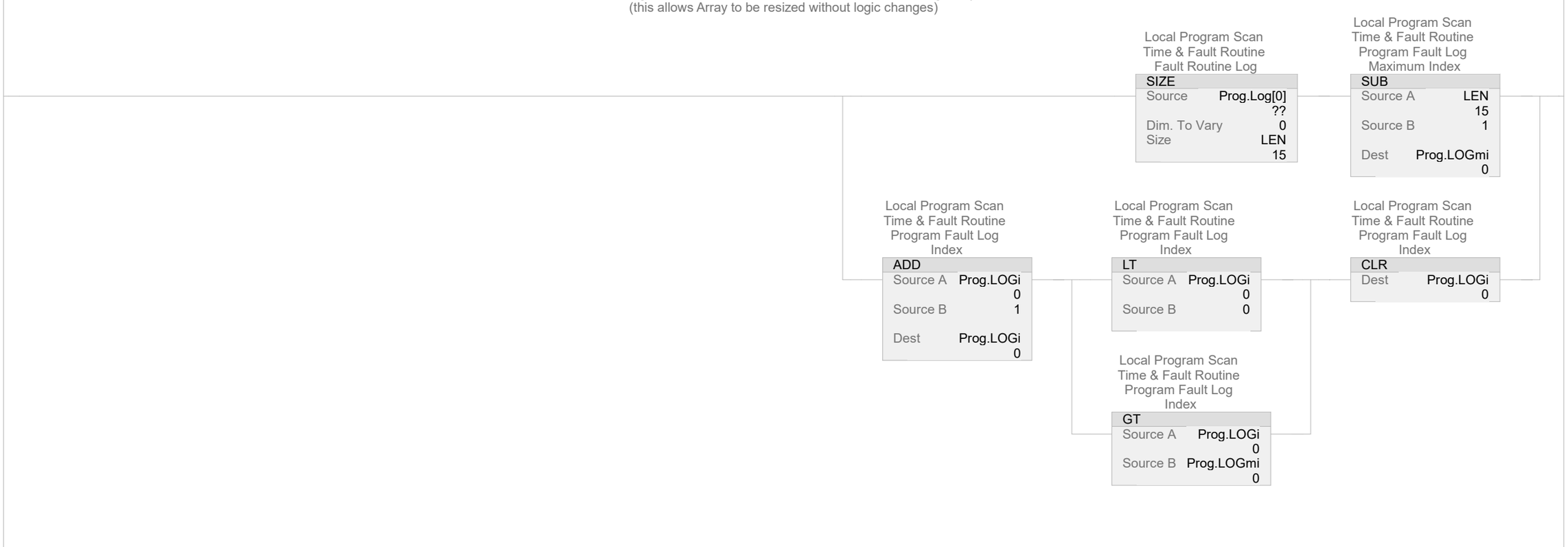
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
95
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source Perf.CTL.LNi
<EDC.Perf.CTL.LNi>
18
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source Perf.CTL.EQi
<EDC.Perf.CTL.EQi>
30
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 2
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration and Initialization Routine

EDC Gateway Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 9
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

****

Preparation for creating Master program

1. set CFG.ResetManual (on Rung 3) to reset configuration
2. change to Program Mode
3. set CFG_FullDownload (this rung)
4. save program with Tag upload

****

(there's no simple way to detect Program Download to know when to generate a Full Config Download -- best method is to manually set CFG_FullDownload before saving the Master Program)

2

AFI

Request Full Configuration Download (from Gway Reset)
CFG_FullDownload
L

Configuration and Initialization Routine

Reset ALL Structures Prior to Configuration & Initialization
In Logix Program Mode, set ResetManual then switch to Run Mode -- combines Reset with each Program s:FS Inits for clean Program
(add Config and Init on following rungs for Deployed Program)

EDC Config Manual Reset -- Clears all Data Structures (use this to create Clean file before configuration)
CFG.ResetManual
<EDC.CFG.ResetManual>

3

Reset all Structures (manually invoked to create clean Master for distribution)

JSR
Routine Name _010_Reset

EDC Config Manual Reset -- Clears all Data Structures (use this to create Clean file before configuration)
CFG.ResetManual
<EDC.CFG.ResetManual>
U

EDC Config Reset ONS (Log Transaction)
CFG.ResetONS
<EDC.CFG.ResetONS>

Configuration and Initialization Routine

Configure EDC Parameters

ConfigAuto - set by Download (also performs ConfigInit automatically)
ConfigManual - set manually to force Configuration

v3.02e Add ResetONS to force Config (this avoids RB Pointer Errors caused by missing array configuration)

4

EDC Config Auto Configuration -- Loads Driver Configuration after Download
CFG.ConfigAuto
<EDC.CFG.ConfigAuto>

EDC Config Auto Initialization -- Initializes Driver after Download
CFG.InitAuto
<EDC.CFG.InitAuto>
L

EDC Config Manual Configuration -- Loads Driver Configuration
CFG.ConfigManual
<EDC.CFG.ConfigManual>

EDC Config Manual Initialization -- Initializes Driver
CFG.InitManual
<EDC.CFG.InitManual>
L

EDC Config Reset ONS (Log Transaction)
CFG.ResetONS
<EDC.CFG.ResetONS>

S:FS

EDC Config Auto Configuration -- Loads Driver Configuration after Download
CFG.ConfigAuto
<EDC.CFG.ConfigAuto>
U

EDC Config Manual Configuration -- Loads Driver Configuration
CFG.ConfigManual
<EDC.CFG.ConfigManual>
U

EDC Config Configuration ONS (Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configure Gateway Settings
JSR
Routine Name _011_Config

Configure Gateway CLx Module Settings
JSR
Routine Name _012_ConfigCLx

Configure Gateway Line Settings
JSR
Routine Name _014_ConfigLine

Configuration and Initialization Routine

Initiaize EDC Parameters

v3.02e Add ResetONS to force Config (this avoids RB Pointer Errors caused by missing array configuration)

5

EDC Config Auto Initialization -- Initializes Driver after Download
CFG.InitAuto
<EDC.CFG.InitAuto>

EDC Config Manual Initialization -- Initializes Driver
CFG.InitManual
<EDC.CFG.InitManual>

EDC Config Reset ONS (Log Transaction)
CFG.ResetONS
<EDC.CFG.ResetONS>

S:FS

EDC Config Auto Initialization -- Initializes Driver after Download
CFG.InitAuto
<EDC.CFG.InitAuto>
U

EDC Config Manual Initialization -- Initializes Driver
CFG.InitManual
<EDC.CFG.InitManual>
U

EDC Config Initialization ONS (Log Transaction)
CFG.InitONS
<EDC.CFG.InitONS>

Initialize CLx Module

JSR
Routine Name _020_InitializeCLx

(End)

Reset all Structures
(manually invoked to create clean Master for distribution)

Reset ALL Structures Prior to Configuration & Initialization

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 10
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Clear EDC Data Structures

2

Equipment Data Buffer

FLL
Source 0
Dest EDC_EqRxBuffer
Length 1

Equipment Data Buffer

FLL
Source 0
Dest EDC_EqRxBufferAux
Length 1

Equipment Data

FLL
Source 0
Dest EDC_EqRxData
Length 1

Equipment Data

FLL
Source 0
Dest EDC_EqRxDataAux
Length 1

Callout Table (LNi, EQi) Equipment Receive Callout Live Table (for external OPC access)

SIZE
Source EDC_EqCallout[0,0] ??
Dim. To Vary 0
Size LEN1 29

Callout Table (LNi, EQi) Equipment Receive Callout Live Table (for external OPC access)

SIZE
Source EDC_EqCallout[0,0] ??
Dim. To Vary 1
Size LEN2 100

Temp Result for Array Size

MUL
Source A LEN1 29
Source B LEN2 100
Dest LEN 15

Callout Table (LNi, EQi) Equipment Receive Callout Live Table (for external OPC access)

FLL
Source 0
Dest EDC_EqCallout[0,0]
Length LEN

Reset all Structures
(manually invoked to create clean Master for distribution)

Clear EDC Fault Ring Buffers

3

Equipment Data Fault/Status Message
SIZE
Source EDC_RB_Fault01_Rec01[0]
??
Dim. To Vary 0
Size LEN
15

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault01_Read
Length 1

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault01_Rec01[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault01_Rec02[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault01_Rec03[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault02_Read
Length 1

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault02_Rec01[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault02_Rec02[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault02_Rec03[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault03_Read
Length 1

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault03_Rec01[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault03_Rec02[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault03_Rec03[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault04_Read
Length 1

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault04_Rec01[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault04_Rec02[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault04_Rec03[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault05_Read
Length 1

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault05_Rec01[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault05_Rec02[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault05_Rec03[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault06_Read
Length 1

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault06_Rec01[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault06_Rec02[0]
Length LEN

Equipment Data Fault/Status Message
FLL
Source 0
Dest EDC_RB_Fault06_Rec03[0]
Length LEN

Reset all Structures
(manually invoked to create clean Master for distribution)
Clear EDC Quality Ring Buffers

4

Equipment Data Quality/Health Message
SIZE
Source EDC_RB_Quality01_Rec01[0] ??
Dim. To Vary 0
Size LEN 15

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality01_Read
Length 1

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality01_Rec01[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality01_Rec02[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality01_Rec03[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality01_Rec04[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality01_Rec05[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality02_Read
Length 1

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality02_Rec01[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality02_Rec02[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality02_Rec03[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality02_Rec04[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality02_Rec05[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality03_Read
Length 1

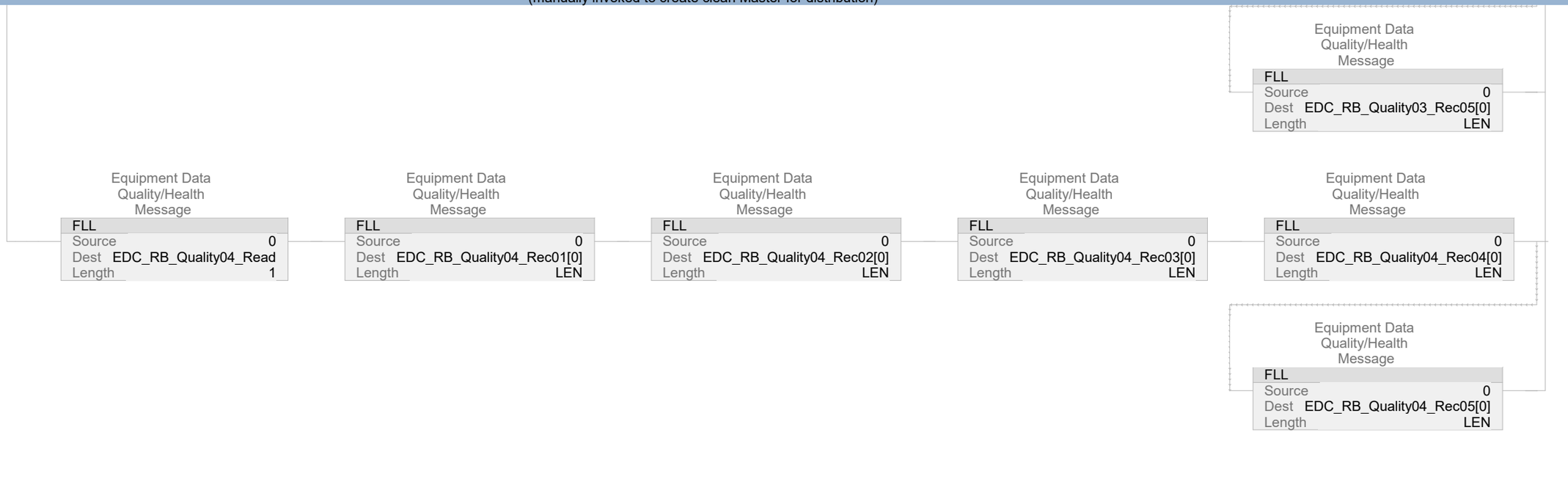
Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality03_Rec01[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality03_Rec02[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality03_Rec03[0]
Length LEN

Equipment Data Quality/Health Message
FLL
Source 0
Dest EDC_RB_Quality03_Rec04[0]
Length LEN

Reset all Structures
(manually invoked to create clean Master for distribution)



Reset all Structures
(manually invoked to create clean Master for distribution)
Clear EDC Trend Ring Buffers

5

Equipment Data Trend Message
SIZE
Source EDC_RB_Trend01_Rec01[0] ??
Dim. To Vary 0
Size LEN 15

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend01_Read
Length 1

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend01_Rec01[0]
Length LEN

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend01_Rec02[0]
Length LEN

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend01_Rec03[0]
Length LEN

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend01_Rec04[0]
Length LEN

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend01_Rec05[0]
Length LEN

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend02_Read
Length 1

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend02_Rec01[0]
Length LEN

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend02_Rec02[0]
Length LEN

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend02_Rec03[0]
Length LEN

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend02_Rec04[0]
Length LEN

Equipment Data Trend Message
FLL
Source 0
Dest EDC_RB_Trend02_Rec05[0]
Length LEN

Reset all Structures
(manually invoked to create clean Master for distribution)

Clear EDC Summary Ring Buffers

6

Equipment Data Summary Message
SIZE
Source EDC_RB_Sum01_Rec01[0] ??
Dim. To Vary 0
Size LEN 15

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum01_Read
Length 1

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum01_Rec01[0]
Length LEN

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum01_Rec02[0]
Length LEN

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum01_Rec03[0]
Length LEN

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum01_Rec04[0]
Length LEN

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum01_Rec05[0]
Length LEN

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum02_Read
Length 1

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum02_Rec01[0]
Length LEN

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum02_Rec02[0]
Length LEN

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum02_Rec03[0]
Length LEN

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum02_Rec04[0]
Length LEN

Equipment Data Summary Message
FLL
Source 0
Dest EDC_RB_Sum02_Rec05[0]
Length LEN

Reset all Structures
(manually invoked to create clean Master for distribution)
v3.02 Clear EDC Cycle Ring Buffers

7

Equipment Data Cycle Message
SIZE
Source EDC_RB_Cycle01_Rec01[0]
??
Dim. To Vary 0
Size LEN
15

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle01_Read
Length 1

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle01_Rec01[0]
Length LEN

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle01_Rec02[0]
Length LEN

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle01_Rec03[0]
Length LEN

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle01_Rec04[0]
Length LEN

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle01_Rec05[0]
Length LEN

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle02_Read
Length 1

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle02_Rec01[0]
Length LEN

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle02_Rec02[0]
Length LEN

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle02_Rec03[0]
Length LEN

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle02_Rec04[0]
Length LEN

Equipment Data Cycle Message
FLL
Source 0
Dest EDC_RB_Cycle02_Rec05[0]
Length LEN

Reset all Structures
(manually invoked to create clean Master for distribution)

8

Clear EDC Transaction Logs

Error Log Transaction Log

SIZE	
Source	EDC_TransErr[0]
	??
Dim. To Vary	0
Size	LEN
	15

Error Log Transaction Log

FLL	
Source	0
Dest	EDC_TransErr[0]
Length	LEN

Error Log Snapshot Transaction Log

FLL	
Source	0
Dest	EDC_TransErrSnap[0]
Length	LEN

Transaction Log Transaction Log

SIZE	
Source	EDC_TransLog[0]
	??
Dim. To Vary	0
Size	LEN
	15

Transaction Log Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[0]
Length	LEN

Transaction Log Snapshot Transaction Log

FLL	
Source	0
Dest	EDC_TransLogSnap[0]
Length	LEN

9

Clear Program Structure

Local Program Scan Time & Fault Routine

FLL	
Source	0
Dest	Prog
Length	1

10

Clear EDC Control Structures
DN007 Split EquipStatus Tag

Equipment Data Collection

FLL	
Source	0
Dest	EDC
Length	1

HMI

FLL	
Source	0
Dest	EDC_HMI
Length	1

Server Interface Status

FLL	
Source	0
Dest	EDC_Status01
Length	1

Event Rx/Tx Status

FLL	
Source	0
Dest	EDC_Status02
Length	1

Tool, Trend, Summary Rx/Tx Status

FLL	
Source	0
Dest	EDC_Status03
Length	1

Tool, Trend, Summary Rx/Tx Status

FLL	
Source	0
Dest	EDC_Status04
Length	1

11

Reset Equipment Event Text, Fragment Statistics and Errors
(before Download All to defrag the Event table for faster searches)

Gateway Reset CFGResetEventText (L)

Gateway Reset CFGResetAviTrack (L)

Gateway Reset CFGResetPdm (L)

Reset all Structures
(manually invoked to create clean Master for distribution)
Initalize Ring Buffer Write Pointers to avoid FULL Ring Buffers after Reset
Read Ptr = 0, Write Ptr = 1

12

Equipment Data Collection Current/Last Read Tag	Equipment Data Collection Next Write Tag	Equipment Data Collection Next Write Record
MOVE Source 1 Dest EDC.RB_Fault01.ReadTagPtr 1	MOVE Source 1 Dest EDC.RB_Fault01.WriteTagPtr 1	MOVE Source 1 Dest EDC.RB_Fault01.WriteRecPtr 1271
MOVE Source 1 Dest EDC.RB_Fault02.ReadTagPtr 1	MOVE Source 1 Dest EDC.RB_Fault02.WriteTagPtr 1	MOVE Source 1 Dest EDC.RB_Fault02.WriteRecPtr 318
MOVE Source 1 Dest EDC.RB_Fault03.ReadTagPtr 2	MOVE Source 1 Dest EDC.RB_Fault03.WriteTagPtr 2	MOVE Source 1 Dest EDC.RB_Fault03.WriteRecPtr 3240
MOVE Source 1 Dest EDC.RB_Fault04.ReadTagPtr 1	MOVE Source 1 Dest EDC.RB_Fault04.WriteTagPtr 1	MOVE Source 1 Dest EDC.RB_Fault04.WriteRecPtr 1054
MOVE Source 1 Dest EDC.RB_Fault05.ReadTagPtr 3	MOVE Source 1 Dest EDC.RB_Fault05.WriteTagPtr 3	MOVE Source 1 Dest EDC.RB_Fault05.WriteRecPtr 2698
Equipment Data Collection Current/Last Read Tag	Equipment Data Collection Next Write Tag	Equipment Data Collection Next Write Record
MOVE	MOVE	MOVE

Reset all Structures
(manually invoked to create clean Master for distribution)

MOVE
Source 1
Dest EDC.RB_Fault06.ReadTagPtr 3

MOVE
Source 1
Dest EDC.RB_Fault06.WriteTagPtr 3

MOVE
Source 1
Dest EDC.RB_Fault06.WriteRecPtr 2434

Initalize Ring Buffer Write Pointers to avoid FULL Ring Buffers after Reset
Read Ptr = 0, Write Ptr = 1

13

Equipment Data Collection Current/Last Read Tag	Equipment Data Collection Next Write Tag	Equipment Data Collection Next Write Record
MOVE Source 1 Dest EDC.RB_Quality01.ReadTagPtr 2	MOVE Source 1 Dest EDC.RB_Quality01.WriteTagPtr 2	MOVE Source 1 Dest EDC.RB_Quality01.WriteRecPtr 440
MOVE Source 1 Dest EDC.RB_Quality02.ReadTagPtr 2	MOVE Source 1 Dest EDC.RB_Quality02.WriteTagPtr 2	MOVE Source 1 Dest EDC.RB_Quality02.WriteRecPtr 984
MOVE Source 1 Dest EDC.RB_Quality03.ReadTagPtr 4	MOVE Source 1 Dest EDC.RB_Quality03.WriteTagPtr 4	MOVE Source 1 Dest EDC.RB_Quality03.WriteRecPtr 1183
MOVE Source 1 Dest EDC.RB_Quality04.ReadTagPtr 1	MOVE Source 1 Dest EDC.RB_Quality04.WriteTagPtr 1	MOVE Source 1 Dest EDC.RB_Quality04.WriteRecPtr 440

Reset all Structures
(manually invoked to create clean Master for distribution)

14

Initalize Ring Buffer Write Pointers to avoid FULL Ring Buffers after Reset
Read Ptr = 0, Write Ptr = 1

Equipment Data Collection Current/Last Read Tag	Equipment Data Collection Next Write Tag	Equipment Data Collection Next Write Record
MOVE	MOVE	MOVE
Source 1	Source 1	Source 1
Dest EDC.RB_Trend01.ReadTagPtr 4	Dest EDC.RB_Trend01.WriteTagPtr 4	Dest EDC.RB_Trend01.WriteRecPtr 326
Equipment Data Collection Current/Last Read Tag	Equipment Data Collection Next Write Tag	Equipment Data Collection Next Write Record
MOVE	MOVE	MOVE
Source 1	Source 1	Source 1
Dest EDC.RB_Trend02.ReadTagPtr 3	Dest EDC.RB_Trend02.WriteTagPtr 3	Dest EDC.RB_Trend02.WriteRecPtr 202

15

Initalize Ring Buffer Write Pointers to avoid FULL Ring Buffers after Reset
Read Ptr = 0, Write Ptr = 1

Equipment Data Collection Current/Last Read Tag	Equipment Data Collection Next Write Tag	Equipment Data Collection Next Write Record
MOVE	MOVE	MOVE
Source 1	Source 1	Source 1
Dest EDC.RB_Sum01.ReadTagPtr 1	Dest EDC.RB_Sum01.WriteTagPtr 1	Dest EDC.RB_Sum01.WriteRecPtr 88
Equipment Data Collection Current/Last Read Tag	Equipment Data Collection Next Write Tag	Equipment Data Collection Next Write Record
MOVE	MOVE	MOVE
Source 1	Source 1	Source 1
Dest EDC.RB_Sum02.ReadTagPtr 4	Dest EDC.RB_Sum02.WriteTagPtr 4	Dest EDC.RB_Sum02.WriteRecPtr 46

Reset all Structures
(manually invoked to create clean Master for distribution)
Initalize Ring Buffer Write Pointers to avoid FULL Ring Buffers after Reset
Read Ptr = 0, Write Ptr = 1

16

v3.02e Add Cycle RB Reset

Equipment Data Collection Current/Last Read Tag	Equipment Data Collection Next Write Tag	Equipment Data Collection Next Write Record
MOVE	MOVE	MOVE
Source 1	Source 1	Source 1
Dest EDC.RB_Cycle01.ReadTagPtr 2	Dest EDC.RB_Cycle01.WriteTagPtr 2	Dest EDC.RB_Cycle01.WriteRecPtr 547

Equipment Data Collection Current/Last Read Tag	Equipment Data Collection Next Write Tag	Equipment Data Collection Next Write Record
MOVE	MOVE	MOVE
Source 1	Source 1	Source 1
Dest EDC.RB_Cycle02.ReadTagPtr 4	Dest EDC.RB_Cycle02.WriteTagPtr 4	Dest EDC.RB_Cycle02.WriteRecPtr 437

(End)

Configure Gateway Settings

Gateway Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	11
Dest	Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1733165816
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1733165816

Re-enable Fault Handlers

2

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

U

Rx Message
Message Timeout Limits and Transaction Number Rollover

3

EDC Config Rx Partial Message Timeout Value (msec)

MOVE	
Source	10000
Dest	CFG.PartialMsgTimeout <EDC.CFG.PartialMsgTimeout> 10000

EDC Config Rx Live Message Timeout Value (msec)

MUL	
Source A	2.5
Source B	60000
Dest	CFG.LiveMsgTimeout <EDC.CFG.LiveMsgTimeout> 150000

EDC Config Rx Buffer Message Timeout Value (msec)

MUL	
Source A	5.5
Source B	60000
Dest	CFG.BufferMsgTimeout <EDC.CFG.BufferMsgTimeout> 330000

EDC Config Rx Buffer Message Transaction Number Rollover

MOVE	
Source	9999
Dest	CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Configure Gateway Settings

Ring Buffer Rx Msg Complete Delay Times
-- slow is used after recovery from comm loss
-- fast is normal condition (<50 msec caused TM errors in eBW testing)

0 time for fast will work, but TM log errors for
'Worker thread could not signal the Scheduler thread'
and if enough errors, TM will stop running

Mar 31, 2025 - change Fast from 50 to 100 (still seeing TM errors with 50 during backlogs)
Oct 20, 2025 - TM v14 deploy; change Slow from 200 to 100, Fast from 200 to 0

4

RingBuffer Read Complete Slow Delay Time

MOVE
Source 100
Dest CfgRbDelaySlow 100

RingBuffer Read Complete Fast Delay Time

MOVE
Source 0
Dest CfgRbDelayFast 0

Configure Gateway Settings

Transaction Log

-- TransLogAll is OFF by default -- manually toggle ON to log Keep Alive Messages during debug/deployment.
-- Initialize Size
Initialize Transaction Error Log Size

5

EDC Config Log all Transactions (includes all Keep Alives)
CFG.TransLogAll
<EDC.CFG.TransLogAll>
U

Transaction Log
Transaction Log
SIZE
Source EDC_TransLog[0]
??
Dim. To Vary 0
Size CFG.TransLOG.n
<EDC.CFG.TransLOG.n>
250

EDC Config Array Max Index
SUB
Source A CFG.TransLOG.n
<EDC.CFG.TransLOG.n>
250
Source B 1
Dest CFG.TransLOG.mi
<EDC.CFG.TransLOG.mi>
249

Error Log
Transaction Log
SIZE
Source EDC_TransErr[0]
??
Dim. To Vary 0
Size CFG.TransERR.n
<EDC.CFG.TransERR.n>
100

EDC Config Array Max Index
SUB
Source A CFG.TransERR.n
<EDC.CFG.TransERR.n>
100
Source B 1
Dest CFG.TransERR.mi
<EDC.CFG.TransERR.mi>
99

Transaction Log
Control SBR External Load -- Optional Parameters
SIZE
Source Trans.Opt[0]
<EDC.Trans.Opt[0]>
-1
Dim. To Vary 0
Size Trans.OptN
<EDC.Trans.OptN>
6

Configure Gateway Settings

Transaction Log Filters

-- Filters are OFF by default -- manually toggle EN and set filter values for Code, LNi and EQi

6

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>
U

Transaction Log Control Code Filter Low Value
MOVE
Source 0
Dest Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0

Transaction Log Control Code Filter High Value
MOVE
Source 0
Dest Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>
U

Transaction Log Control LNi Filter Low Value
MOVE
Source 0
Dest Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0

Transaction Log Control LNi Filter High Value
MOVE
Source 0
Dest Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>
U

Transaction Log Control EQi Filter Low Value
MOVE
Source 0
Dest Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0

Transaction Log Control EQi Filter High Value
MOVE
Source 0
Dest Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

(End)

Configure Gateway CLx Module Settings

Configure CLx Settings

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 12
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

********** GATEWAY-SPECIFIC CONFIGURATION REQUIRED **********

Configure CLx Processor Slot (0-16)

2

EDC Config CLx Processor Slot (0-16)

MOVE
Source 1
Dest CFG.CLx.ProcSlot <EDC.CFG.CLx.ProcSlot> 1

Configure Message Timeout
This forces a Timeout for Unconnected or Connected Timeout failure.
(On occasion, have seen Messages hang indefinitely.)

This should be set higher than Unconnected and Connected Timeouts above.

3

EDC Config CLx Message Timeout (msec)

MOVE
Source 5000
Dest CFG.CLx.MsgTimeout <EDC.CFG.CLx.MsgTimeout> 5000

Configure Gateway CLx Module Settings

Configure Unconnected & Connected Timeout for ENET Socket Messages
Unconnected: 4,000,000 = 4.0 seconds (CLx default = 30000000 = 30 seconds)
Connection Rate: 1,000,000 = 4.0 seconds (CLx default = 7500000 = 7.5 seconds)
Connection Rate Timeout Multiplier: 0 = *4 (CLx default = 0 = *4 multiplier)

4

EDC Config CLx Unconnected Message Timeout

MOVE	
Source	4000000
Dest	CFG.CLx.UC_Timeout
	<EDC.CFG.CLx.UC_Timeout>
	4000000

EDC Config CLx Connected Message Timeout Rate

DIV	
Source A	CFG.CLx.UC_Timeout
	<EDC.CFG.CLx.UC_Timeout>
	4000000
Source B	4
Dest	CFG.CLx.ConnectionRate
	<EDC.CFG.CLx.ConnectionRate>
	1000000

EDC Config CLx Connected Message Timeout Multiplier

MOVE	
Source	0
Dest	CFG.CLx.ConnectionMult
	<EDC.CFG.CLx.ConnectionMult>
	0

Configure the Maximum Point Response Messages to be enabled simultaneously
(throttle ENBT loading)

5

EDC Config Max Number of Enabled Messages allowed

MOVE	
Source	40
Dest	CFG.CLx.MsgEnLimit
	<EDC.CFG.CLx.MsgEnLimit>
	40

Configure the Performance Summary Update quantity
(number of Points to update each scan)

Default Setting is 1 -- requires 30 scans to update 60 Points
30 scans * 100msec/scan = 3 seconds worst case

Increasing this setting will increase SCAN TIME

6

EDC Config Performance Summary -- Updates per scan

MOVE	
Source	1
Dest	CFG.CLx.PerfUpdate
	<EDC.CFG.CLx.PerfUpdate>
	1

Configure Gateway CLx Module Settings

Configure Path for CLx Module Messages
Data[0] = 1 (CLx Backplane)
Data[1] = n (CLx Slot Number)

7

EDC Config CLx Module Path for Messages

FLL
Source 0
Dest CFG.CLx.Path
<EDC.CFG.CLx.Path>
Length 1

EDC Config CLx Module Path for Messages

MOVE
Source 2
Dest CFG.CLx.Path.LEN
<EDC.CFG.CLx.Path.LEN>
2

EDC Config CLx Module Path for Messages

MOVE
Source 1
Dest CFG.CLx.Path.DATA[0]
<EDC.CFG.CLx.Path.DATA[0]>
'$01'

EDC Config CLx Module Path for Messages

MOVE
Source CFG.CLx.ProcSlot
<EDC.CFG.CLx.ProcSlot>
1
Dest CFG.CLx.Path.DATA[1]
<EDC.CFG.CLx.Path.DATA[1]>
'$01'

Limit for detecting High Scan times
(in microseconds -- 100,000 = 0.1 seconds)

8

Scan Time High Limit (microseconds)

MOVE
Source 100000
Dest CLxScanHighLimit
100000

(End)

Configure Gateway Line Settings

Configure EDC Driver Line and Equip Settings

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	14
Dest	Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1733165816
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1733165816

Configure Gateway Line Settings

Line Array

-- DisableLineLoop is OFF by default -- manually toggle ON to debug a single Line (all other Lines will be disabled).
-- Use Lines Array to determine maximum values for LNi
-- Set initial values for Line Loop Start Index and End Index

2

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>
U

Equipment Data Collection Equipment Data Collection - Line
SIZE
Source EDC.Line[0] ??
Dim. To Vary 0
Size CFG.LN.n <EDC.CFG.LN.n> 29

EDC Config Array Max Index
SUB
Source A CFG.LN.n <EDC.CFG.LN.n> 29
Source B 1
Dest CFG.LN.mi <EDC.CFG.LN.mi> 28

EDC Config Array Start Index
MOVE
Source 0
Dest CFG.LN.si <EDC.CFG.LN.si> 0

EDC Config Array End Index
MOVE
Source CFG.LN.mi <EDC.CFG.LN.mi> 28
Dest CFG.LN.ei <EDC.CFG.LN.ei> 28

Configure Gateway Line Settings

Equipment Array

-- DisableEqLoop is OFF by default -- manually toggle ON to debug a single Equipment (all other Equipment will be disabled).
-- Use Equipment Array to determine maximum values for LNi

(EQsi & EQei are configurd for each Line)

3

EDC Config Disable Equipment Loop
CFG.DisableEqLoop
<EDC.CFG.DisableEqLoop>
U

Equipment Data Collection Equipment Data Collection - Line Equipment
SIZE
Source EDC.Line[0].Equip[0] ??
Dim. To Vary 0
Size CFG.EQ.n <EDC.CFG.EQ.n> 100

EDC Config Array Max Index
SUB
Source A CFG.EQ.n <EDC.CFG.EQ.n> 100
Source B 1
Dest CFG.EQ.mi <EDC.CFG.EQ.mi> 99

Offline Group Array

-- Use Array to determine maximum values for GRPi
-- Set initial values for Group Loop Start Index and End Index

4

Equipment Data Collection Equipment Data Collection - Line Equipment Data
SIZE
Source EDC.Offline.Data[0] ??
Dim. To Vary 0
Size CFG.GRP.n <EDC.CFG.GRP.n> 4

EDC Config Array Max Index
SUB
Source A CFG.GRP.n <EDC.CFG.GRP.n> 4
Source B 1
Dest CFG.GRP.mi <EDC.CFG.GRP.mi> 3

EDC Config Array Start Index
MOVE
Source 0
Dest CFG.GRP.si <EDC.CFG.GRP.si> 0

EDC Config Array End Index
MOVE
Source 3
Dest CFG.GRP.ei <EDC.CFG.GRP.ei> 3

Configure Gateway Line Settings

Fault/Status Event Array

-- Use Array to determine maximum values for EVi
-- Set initial values for Event Loop Start Index and End Index

5

Equipment Data Collection Equipment Event Table First N Fault/Status Events

SIZE
Source: EDC.Line[0].Equip[0].Event[0] ??
Dim. To Vary: 0
Size: CFG.EV.n <EDC.CFG.EV.n>: 10

EDC Config Array Max Index

SUB
Source A: CFG.EV.n <EDC.CFG.EV.n>: 10
Source B: 1
Dest: CFG.EV.mi <EDC.CFG.EV.mi>: 9

EDC Config Array Start Index

MOVE
Source: 0
Dest: CFG.EV.si <EDC.CFG.EV.si>: 0

EDC Config Array End Index

MOVE
Source: CFG.EV.mi <EDC.CFG.EV.mi>: 9
Dest: CFG.EV.ei <EDC.CFG.EV.ei>: 9

Set Default EQ Index Configuration for each Line
(Indexes will be increased as EQ communicates)

6

Configure Gateway Line Settings

FOR
Routine Name: _015_ConfigLineLoop
Index: CfgLNi: 29
Initial Value: 0
Terminal Value: CFG.LN.mi <EDC.CFG.LN.mi>
Step Size: 1

(End)

Configure Gateway Line Settings

Configure TCP/IP Driver Point Settings

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 15
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Configure Gateway Line Settings

Equipment Array
-- Set initial values for Equipment Loop Start Index and End Index
(Status logic will adjust EQei for last-configured Equipment to avoid scanning unused EQi)

(Hardcode to 5 to cover the default Equipment - AVI, Andon, CNV, Exit, Entrance - Callout Message updates EQei to include new Equipment)

2

Equipment Data Collection Lines and Equipment Array

LT
Source A EDC.Line[CfgLNi].CFG.EQ.si ??
Source B 0

Equipment Data Collection Lines and Equipment Array

MOVE
Source 0
Dest EDC.Line[CfgLNi].CFG.EQ.si ??

Equipment Data Collection Lines and Equipment Array

GT
Source A EDC.Line[CfgLNi].CFG.EQ.si ??
Source B CFG.EQ.mi <EDC.CFG.EQ.mi> 99

Equipment Data Collection Lines and Equipment Array

LE
Source A EDC.Line[CfgLNi].CFG.EQ.ei ??
Source B 0

Equipment Data Collection Lines and Equipment Array

MOVE
Source 5
Dest EDC.Line[CfgLNi].CFG.EQ.ei ??

Equipment Data Collection Lines and Equipment Array

GT
Source A EDC.Line[CfgLNi].CFG.EQ.ei ??
Source B CFG.EQ.mi <EDC.CFG.EQ.mi> 99

DN006 - support multiple Offline Lines
Initialize all Lines - Offline Lines are latched after this routine is executed

*** Gateway Configuration moved to Server Download ***

3

AFI

Equipment Data Collection Lines and Equipment Array
EDC.Line[CfgLNi].CFG.Offline

U

Configure Gateway Line Settings

(End)

Initialize CLx Module

Initialize CLx Module

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 20
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Re-enable any Disabled features

2

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>
U

Initialize and Start Module & Scan Diagnostics

3

CLx Module Status Clear Scan History
CLX.TaskScan.Clear
<EDC.CLX.TaskScan.Clear>
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>
L

Initialize FEC EqDrv Events
Each FEC Messages 1 DINT of Fault Events

4

SIZE
Source FEC_EqDrvFault[0] 16#0000_0000
Dim. To Vary 0
Size LEN 15

FLL
Source 0
Dest FEC_EqDrvFault[0]
Length LEN

Initialize CLx Module

Initialize the Entry Status Mask and Entry Status Run Check values used to determine if the Module is Running
(used with GSV MODULE EntryStatus command)
-- EntryStatusMask = 16#F000
-- StatusRunChk = 16#4000

5

Ethernet Module Status GSV Module - Entry Status Mask (hF000)

MOVE
Source 61440
Dest Enet00.EntryStatusMask
<EDC_Enet00.EntryStatusMask>
16#f000

Ethernet Module Status GSV Module - Entry Status Run Check (h4000)

MOVE
Source 16384
Dest Enet00.EntryStatusRunChk
<EDC_Enet00.EntryStatusRunChk>
16#4000

Initialize the Entry Status Mask and Entry Status Run Check values used to determine if the Module is Running
(used with GSV MODULE EntryStatus command)
-- EntryStatusMask = 16#F000
-- StatusRunChk = 16#4000

6

Ethernet Module Status GSV Module - Entry Status Mask (hF000)

MOVE
Source 61440
Dest Enet01.EntryStatusMask
<EDC_Enet01.EntryStatusMask>
16#f000

Ethernet Module Status GSV Module - Entry Status Run Check (h4000)

MOVE
Source 16384
Dest Enet01.EntryStatusRunChk
<EDC_Enet01.EntryStatusRunChk>
16#4000

Initialize the Entry Status Mask and Entry Status Run Check values used to determine if the Module is Running
(used with GSV MODULE EntryStatus command)
-- EntryStatusMask = 16#F000
-- StatusRunChk = 16#4000

7

Ethernet Module Status GSV Module - Entry Status Mask (hF000)

MOVE
Source 61440
Dest Enet02.EntryStatusMask
<EDC_Enet02.EntryStatusMask>
16#f000

Ethernet Module Status GSV Module - Entry Status Run Check (h4000)

MOVE
Source 16384
Dest Enet02.EntryStatusRunChk
<EDC_Enet02.EntryStatusRunChk>
16#4000

Initialize CLx Module

Initialize the Entry Status Mask and Entry Status Run Check values used to determine if the Module is Running
(used with GSV MODULE EntryStatus command)
-- EntryStatusMask = 16#F000
-- StatusRunChk = 16#4000

8

Ethernet Module Status GSV Module - Entry Status Mask (hF000)

MOVE
Source 61440
Dest Enet03.EntryStatusMask
<EDC_Enet03.EntryStatusMask> 16#f000

Ethernet Module Status GSV Module - Entry Status Run Check (h4000)

MOVE
Source 16384
Dest Enet03.EntryStatusRunChk
<EDC_Enet03.EntryStatusRunChk> 16#4000

Initialize the Entry Status Mask and Entry Status Run Check values used to determine if the Module is Running
(used with GSV MODULE EntryStatus command)
-- EntryStatusMask = 16#F000
-- StatusRunChk = 16#4000

9

Ethernet Module Status GSV Module - Entry Status Mask (hF000)

MOVE
Source 61440
Dest Enet04.EntryStatusMask
<EDC_Enet04.EntryStatusMask> 16#f000

Ethernet Module Status GSV Module - Entry Status Run Check (h4000)

MOVE
Source 16384
Dest Enet04.EntryStatusRunChk
<EDC_Enet04.EntryStatusRunChk> 16#4000

Initialize the Entry Status Mask and Entry Status Run Check values used to determine if the Module is Running
(used with GSV MODULE EntryStatus command)
-- EntryStatusMask = 16#F000
-- StatusRunChk = 16#4000

10

Ethernet Module Status GSV Module - Entry Status Mask (hF000)

MOVE
Source 61440
Dest Enet05.EntryStatusMask
<EDC_Enet05.EntryStatusMask> 16#f000

Ethernet Module Status GSV Module - Entry Status Run Check (h4000)

MOVE
Source 16384
Dest Enet05.EntryStatusRunChk
<EDC_Enet05.EntryStatusRunChk> 16#4000

Initialize CLx Module

Initialize the Entry Status Mask and Entry Status Run Check values used to determine if the Module is Running
(used with GSV MODULE EntryStatus command)
-- EntryStatusMask = 16#F000
-- StatusRunChk = 16#4000

11

Ethernet Module Status GSV Module - Entry Status Mask (hF000)

MOVE	
Source	61440
Dest	Enet06.EntryStatusMask <EDC_Enet06.EntryStatusMask> 16#f000

Ethernet Module Status GSV Module - Entry Status Run Check (h4000)

MOVE	
Source	16384
Dest	Enet06.EntryStatusRunChk <EDC_Enet06.EntryStatusRunChk> 16#4000

Ethernet Module Inhibit Check (h6000)

12

MOVE	
Source	16#6000
Dest	ENet_InhibChk 16#6000

(End)

Initialize Equipment Event Masks and Data Refresh for All Equipment Packets 0-7
Ignore, Maint, Fault, Alarm, Alert

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 21
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Initialize Loop Controls

2

EventMaskSend_Init

EventMaskSend_Init
U

Event Lookup Index

MOVE
Source 0
Dest EvLKi 272

Event Lookup End Index

SUB
Source A EDC_Event.CTL.EqQty 272
Source B 1
Dest EvLKei 271

Set MaskSend for all Packets of all Masks for 1 Equipment
(this will generate 40 Callout messages)

3

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].MaintMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].FaultMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].AlertMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend 2#0000_0000

v3.02 DataRefreshReq for all Packets
(this will generate 8 Fault Event messages)

4

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].DataRefreshReq 2#0000_0000

Increment Index
Clear Send command when complete

5

Event Lookup Index

ADD
Source A EvLKi 272
Source B 1
Dest EvLKi 272

Event Lookup Index

GT
Source A EvLKi 272
Source B EvLKei 271

EventMaskSend
(U)

(End)

EDC Gateway TimeSync from SNTP -- used by all Equipment for TimeSync

EDC Gateway TimeSync from SNTP Time Server
used by all Equipment for TimeSync

Coordinated System Time
TimeSync from EDC Gateway CST Master to NW, PdM and FEC

Controller | Properties | Date/Time -- Set Time Zone and Enable 'Adjust for Daylight Saving (+01:00)
Controller | Properties | Date/Time -- Enable Time Synchronization & confirm 'Is a synchronized time slave'
Controller | Properties | Date/Time | Advanced -- Priority 1 (Master Override) set to 127 & confirm Grandmaster Clock = EDC_Gway

EDC Gateway is the CST Master with Priority 1 = 127
NW, PdM, FEC are CST Slaves with Priority 1 = 128

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 29
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Coordinated System Time
Get CST Status and Value

CST_CurrentStatus.2 = System time master (should be on)
CST_CurrentStatus.3 = Synchronized (should be on)
(Note: CST_CurrentStatus.3 is NOT equivalent to the 'is a synchronized time slave' indications on Date/Time Property)

2

Coordinated System Time Status

GSV
Class Name CST
Instance Name
Attribute Name CurrentStatus
Dest CST_CurrentStatus 16#011e

Coordinated System Time Value

GSV
Class Name CST
Instance Name
Attribute Name CurrentValue
Dest CST_CurrentValue[0] 16#6e06_850d

EDC Gateway TimeSync from SNTP -- used by all Equipment for TimeSync

This AOI_SNTP_QUERY instruction retrieves the current time from a NTP (Network Time Protocol) server and returns the date to SNTP_Time_Retrieved and optionally the controller's clock as well.

Update "SNTP_SERVER_ADDRESS" tag with <IP address> or <Hostname> of the NTP server.
ntp.toyota.com = 10.83.48.1

Fill in Query_Interval_in_Min with the number of minutes to periodocally update between successful queries. 0 disables query.
Fill in Query_Error_in_Min with the number fo minutes to retry after an error.
Fill in the Sync_Controller_Clock with a 1 to automatically update controller clock, 0 otherwise.
Next_Query_in_Min will display minutes remaining before the next update.

IMPORTANT! Set the Enable instruction to a 1 to enable the instruction. Set Trigger Update Now if an immediate update is desired (Enable_Instruction must also be set).
Set Sync controller clock to have the controllers clock updated by the instruction at the time of a query.

3

SNTP Query and Clock Sync Instruction
SNTP_AOI_TAG.Enable_Instruction
L

SNTP Query and Clock Sync Instruction
SNTP_AOI_TAG.Sync_Controller_Clock
L

SNTP Query and Clock Sync Instruction

AOI_SNTP_QUERY	
AOI_SNTP_QUERY	SNTP_AOI_TAG
Enable_Instruction	1
Trigger_Update_Now	0
Sync_Controller_Clock	1
Enet_Slot_Number	1
Query_Interval_in_Min	5
Query_Error_Retry_in_Min	1
Next_Query_in_Min	4.9781833
UTC_Time	SNTP_TIME_RETRIEVED
Server_Address	EDC_CfgGateway.SntpServerIP
MSG_Data	SNTP_MSG_DATA
SKT_Create	SNTP_MSG_CREATE
SKT_Open	SNTP_MSG_OPEN
SKT_Write	SNTP_MSG_WRITE
SKT_Read	SNTP_MSG_READ
SKT_Delete	SNTP_MSG_DELETE

EN
DN
ER

SNTP Query and Clock Sync Instruction

MOVE	
Source	1
Dest	SNTP_AOI_TAG.Enet_Slot_Number 1

SNTP Query and Clock Sync Instruction
Defines how often to poll the SNT server for time. Units in minutes.

MOVE	
Source	5
Dest	SNTP_AOI_TAG.Query_Interval_in_Min 5

SNTP Query and Clock Sync Instruction

MOVE	
Source	1
Dest	SNTP_AOI_TAG.Query_Error_Retry_in_Min 1

SNTP Query and Clock Sync Instruction
SNTP_AOI_TAG.Trigger_Update_Now

s:fs

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

LT	
Source A	CLX.SysTime[0] <EDC.CLX.SysTime[0]> 2026
Source B	2024

Logix
Wall Clock
Invalid Year
(force time sync)
SNTP_InvalidYear
ONS

ADD	
Source A	SNTP_InvalidYearCount 0
Source B	1
Dest	SNTP_InvalidYearCount 0

Toggle On for
Manual Time Sync
SNTP_QueryNow

Toggle On for
Manual Time Sync
SNTP_QueryNow
U

EDC Gateway TimeSync from SNTP -- used by all Equipment for TimeSync

Diagnostic Counts and Last Sync

4

SNTP Query and Clock Sync Instruction The SNTP AOI instruction is attempting to retrieve the time.
SNTP_AOI_TAG.EN

Logix Wallclock Time Sync Enable
SNTP_EN
ONS

GSV
Class Name WallClockTime
Instance Name
Attribute Name DateTime
Dest SNTP_EnDateTime[0]
2026

MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest SNTP_EnScanCount
466539348

ADD
Source A SNTP_EnCount
21945
Source B 1
Dest SNTP_EnCount
21945

SNTP Query and Clock Sync Instruction Time was retrieved.
SNTP_AOI_TAG.DN

Logix Wallclock Time Sync Done
SNTP_DN
ONS

Date/Time or Time Interval

COP
Source SNTP_TIME_RETRIEVED
Dest SNTP_LastTimeRetrieved
Length 1

GSV
Class Name WallClockTime
Instance Name
Attribute Name DateTime
Dest SNTP_DnDateTime[0]
2026

MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest SNTP_DnScanCount
466539352

ADD
Source A SNTP_DnCount
21931
Source B 1
Dest SNTP_DnCount
21931

SNTP Query and Clock Sync Instruction There was an error retrieving the time.
SNTP_AOI_TAG.ER

Logix Wallclock Time Sync Error
SNTP_ER
ONS

ADD
Source A SNTP_ErCount
14
Source B 1
Dest SNTP_ErCount
14

EDC Gateway TimeSync from SNTP -- used by all Equipment for TimeSync

Initialize Socket Messages
Create

1 - TCP 2- UPD

5

MOVE
Source 2
Dest SNTP_MSG_DATA.Create_Source.Type
2

Address Family Must
be 2

MOVE
Source 2
Dest SNTP_MSG_DATA.Create_Source.Addr.Family
2

Port Number

MOVE
Source 0
Dest SNTP_MSG_DATA.Create_Source.Addr.Port
0

MOVE
Source 0
Dest SNTP_MSG_DATA.Create_Source.Addr.IP_4th_Octect
0

MOVE
Source 0
Dest SNTP_MSG_DATA.Create_Source.Addr.IP_3rd_Octect
0

MOVE
Source 0
Dest SNTP_MSG_DATA.Create_Source.Addr.IP_2nd_Octect
0

MOVE
Source 0
Dest SNTP_MSG_DATA.Create_Source.Addr.IP_1st_Octect
0

Initialize Socket Messages
Open

Dest_Addr is loaded programatically

6

MOVE
Source 2000
Dest SNTP_MSG_DATA.Open_Connection.Timeout
2000

EDC Gateway TimeSync from SNTP -- used by all Equipment for TimeSync

Initialize Socket Messages
Write

SNTP_MSG_DATA.Write_Source.Buffer = $0B$00$00$00

Set TimeZone and
determine whether to apply Daylight Savings Time

Determine whether to apply Daylight Savings Time

SNTP retrieves the UTC-0 time from the Toyota time server
The Time Zone configuration (Eastern, Central, Mountain, Pacific) is applied for the correct UTC offset.
The DST offset must be enabled during DST and disabled otherwise.
(The DST can be enabled/disabled manually or programatically - EDC will do this programatically.
DST is not handled automatically by the CLx because the local rules are inconsistent and have changed over time.
It's better to program this than burn it into the firmware.)

The current DST rules:
DST Starts on the 2nd Sunday of March at 2:00am
DST Ends on the 1st Sunday of November at 2:00am

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 29
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Set TimeZone and determine whether to apply Daylight Savings Time

Set Time Zone
(Controller | Properties | Date/Time settings)

Continuously Check Time Zone setting and Set if not same as Config value
Need to set Time Zone before determining DST status

Note: Properties | Date/Time | TimeZone configuration selector shows parenthesis around the UTC value - but the GSV/SSV strings do not have the parenthesis and won't work with them
TZ config selection - (UTC-06:00) Central Time (US & Canada)
GSV/SSV string - UTC-06:00 Central Time (US & Canada)

2

Wall Clock Time
Get Time Zone String

GSV	
Class Name	WallClockTime
Instance Name	
Attribute Name	TimeZoneString
Dest	WCT_GetTimeZoneString 'UTC-06:00 Central Ti'

Wall Clock Time
Get Time Zone String

EQ	
Source A	WCT_GetTimeZoneString 'UTC-06:00 Central Ti'
Source B	EDC_CfgGateway.TimeZoneString 'UTC-06:00 Central Ti'

Wall Clock Time
Time Zone
matches Config
WCT_TimeZoneOK

Wall Clock Time
Time Zone
SSV Delay Timer

MOVE	
Source	0
Dest	WCT_SetTimeZoneDelayTMR.PRE 0

Wall Clock Time
Time Zone
SSV Delay Timer

TON		
Timer	WCT_SetTimeZoneDelayTMR	EN
Preset	0	DN
Accum	29	

Wall Clock Time
Time Zone
SSV Delay Timer
WCT_SetTimeZoneDelayTMR.DN

s:fs

Wall Clock Time
Time Zone
matches Config
WCT_TimeZoneOK

Gateway Config
Download Wall Clock
Time Zone String

SSV	
Class Name	WallClockTime
Instance Name	
Attribute Name	TimeZoneString
Source	EDC_CfgGateway.TimeZoneString 'UTC-06:00 Central Ti'

Wall Clock Time
Time Zone
SSV Delay Timer
WCT_SetTimeZoneDelayTMR
RES

Wall Clock Time
Time Zone SSV
Count

ADD	
Source A	WCT_TimeZoneSetCount 254530
Source B	1
Dest	WCT_TimeZoneSetCount 254530

DST Enable/Disable Commands for SSV

3

Wall Clock Time
Set Disable DST
(0=disable
1= adjust)

MOVE	
Source	0
Dest	WCT_SetDisableDST 0

Wall Clock Time
Set Enable DST
(0=disable
1=enable)

MOVE	
Source	1
Dest	WCT_SetEnableDST 1

Set TimeZone and
determine whether to apply Daylight Savings Time

If DST not used at this location, Set DST to Disable

4

Gateway Config
Download Wall Clock
Daylight Savings
Time Enable
EDC_CfgGateway.DstEnable
/

Wall Clock Time
Enable DST
WCT_EnableDST
U

Code Rule
for DST
0=Disable
1=Disable Dec-Feb
2=Enable Apr-Oct
3=Enable Mar
4=Enable Nov
5=Disable Mar/Nov

MOVE
Source 0
Dest WCT_DSTrule
2

lbl_SetDST
JMP

If DST used at this location,

Determine DST Enable/Disable in 4 steps:
- Disable for Dec, Jan, Feb
- Enable for Apr thru Oct
- In March, Enable if Previous Sunday on or after 8th
- In Nov, Enable if Previous Sunday on or before 0

Previous Sunday is determined by the Day of Week and Day of Month

5

NOP

refresh the CLx Wallclock and Day of Week if the Time Zone was updated above

CLX.SysTime[1] = Month
CLX.SysTime[2] = Day of Month

6

Wall Clock Time
Time Zone
matches Config
WCT_TimeZoneOK
/

Monitors CLx Status

JSR
Routine Name _030_CLxStatus

Set TimeZone and
determine whether to apply Daylight Savings Time

Disable DST for Dec, Jan, Feb

7

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

EQ
Source A CLX.SysTime[1]
<EDC.CLX.SysTime[1]>
10
Source B 12

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

LE
Source A CLX.SysTime[1]
<EDC.CLX.SysTime[1]>
10
Source B 2

Wall Clock Time
Enable DST
WCT_EnableDST
U

Code Rule
for DST
0=Disable
1=Disable Dec-Feb
2=Enable Apr-Oct
3=Enable Mar
4=Enable Nov
5=Disable Mar/Nov

MOVE
Source 1
Dest WCT_DSTrule
2

lbl_SetDST
JMP

Enable for Apr thru Oct

8

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

LIMIT
Low Limit 4
Test CLX.SysTime[1]
<EDC.CLX.SysTime[1]>
10
High Limit 10

Wall Clock Time
Enable DST
WCT_EnableDST
L

Code Rule
for DST
0=Disable
1=Disable Dec-Feb
2=Enable Apr-Oct
3=Enable Mar
4=Enable Nov
5=Disable Mar/Nov

MOVE
Source 2
Dest WCT_DSTrule
2

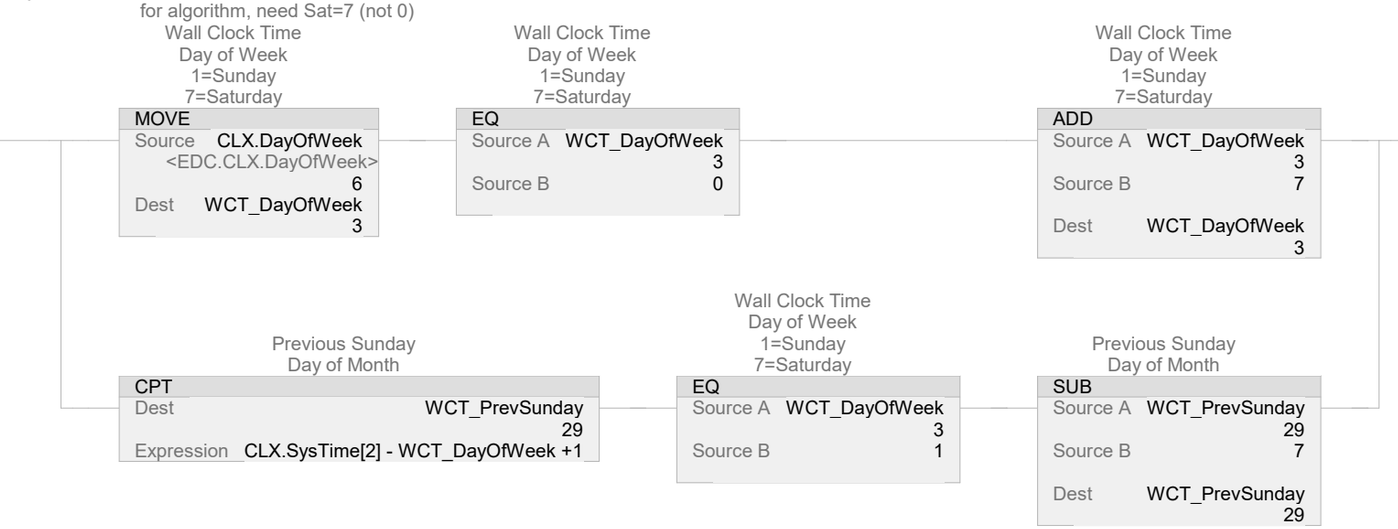
lbl_SetDST
JMP

Set TimeZone and
determine whether to apply Daylight Savings Time

Calculate the Day of Month for the Previous Sunday
using the Day of Month and Day of Week

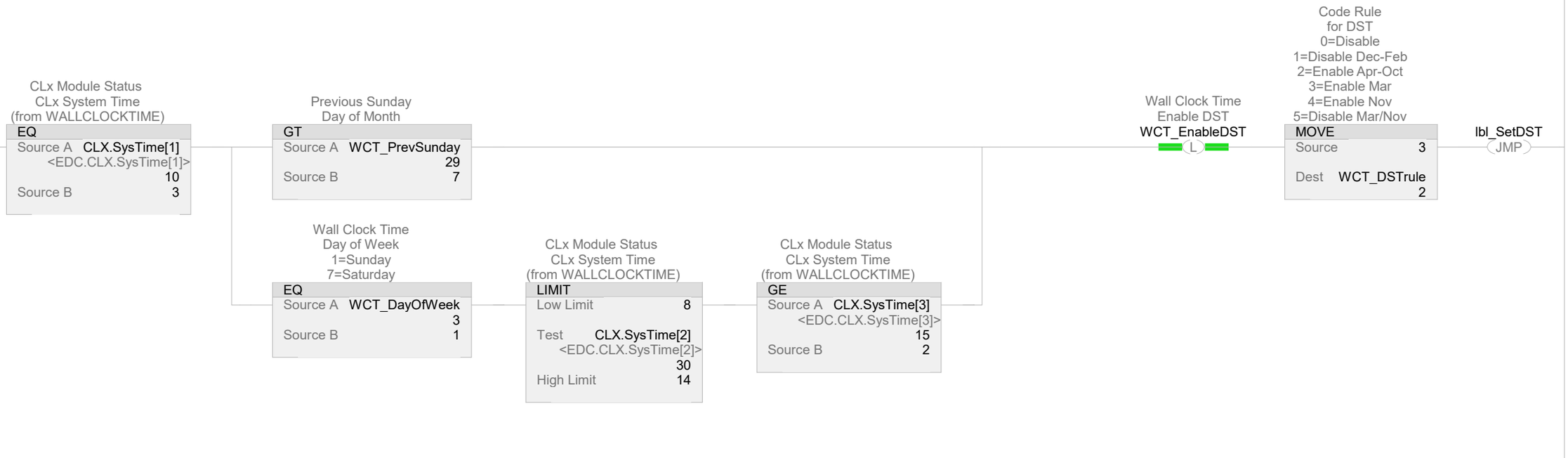
DayOfWeek - Sat=0, Sun=1, Mon=2, Tue=3, Wed=4, Th=5, Fri=6
for algorithm, need Sat=7 (not 0)

9



Set TimeZone and
determine whether to apply Daylight Savings Time
- In March, Enable if Previous Sunday is after 7th
or if this is the 2nd Sunday after 2am

10



Set TimeZone and
determine whether to apply Daylight Savings Time
- In Nov, Enable if not Sunday and Previous Sunday on or before 0
or if this is the 1st Sunday before 2am and DST is on
or if this is the 1st Sunday before 1am and DST is off
(DST check avoids toggling betwen ST/DST after the 1 hour fallback)

11

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

EQ
Source A CLX.SysTime[1]
<EDC.CLX.SysTime[1]>
10
Source B 11

Wall Clock Time
Get Apply DST
(0=disable
1=enable)

GSV
Class Name WallClockTime
Instance Name
Attribute Name ApplyDST
Dest WCT_GetApplyDST
1

Wall Clock Time
Day of Week
1=Sunday
7=Saturday

NE
Source A WCT_DayOfWeek
3
Source B 1

Previous Sunday
Day of Month

LE
Source A WCT_PrevSunday
29
Source B 0

Wall Clock Time
Enable DST
WCT_EnableDST

Code Rule
for DST
0=Disable
1=Disable Dec-Feb
2=Enable Apr-Oct
3=Enable Mar
4=Enable Nov
5=Disable Mar/Nov

MOVE
Source 4
Dest WCT_DSTrule
2

Wall Clock Time
Day of Week
1=Sunday
7=Saturday

EQ
Source A WCT_DayOfWeek
3
Source B 1

Previous Sunday
Day of Month

LE
Source A WCT_PrevSunday
29
Source B 0

Wall Clock Time
Get Apply DST
(0=disable
1=enable)

EQ
Source A WCT_GetApplyDST
1
Source B 1

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

LT
Source A CLX.SysTime[3]
<EDC.CLX.SysTime[3]>
15
Source B 2

Wall Clock Time
Get Apply DST
(0=disable
1=enable)

EQ
Source A WCT_GetApplyDST
1
Source B 0

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

LT
Source A CLX.SysTime[3]
<EDC.CLX.SysTime[3]>
15
Source B 1

lbl_SetDST

Set TimeZone and
determine whether to apply Daylight Savings Time

JMP

if reaching this point, then it's Mar before DST starts or Nov after DST ends

Code Rule
for DST
0=Disable
1=Disable Dec-Feb
2=Enable Apr-Oct
3=Enable Mar
4=Enable Nov
5=Disable Mar/Nov

Wall Clock Time
Enable DST
WCT_EnableDST

12 U

MOVE
Source 5
Dest WCT_DSTrule 2

Set DST enable/disable

lbl_SetDST
LBL

13

Wall Clock Time
Enable DST
WCT_EnableDST

Wall Clock Time Set
Apply DST (0=disable
1=enable)

MOVE
Source WCT_SetEnableDST 1
Dest WCT_SetApplyDST 1

Wall Clock Time
Enable DST
WCT_EnableDST

/

Wall Clock Time Set
Apply DST (0=disable
1=enable)

MOVE
Source WCT_SetDisableDST 0
Dest WCT_SetApplyDST 1

Set TimeZone and
determine whether to apply Daylight Savings Time

Get Daylight Saving setting
(Controller | Properties | Date/Time settings)

Continuously Check values and Set if not same as Config values

14

Wall Clock Time
Get Apply DST
(0=disable
1=enable)

GSV
Class Name WallClockTime
Instance Name
Attribute Name ApplyDST
Dest WCT_GetApplyDST
1

Wall Clock Time
Get DST Adjustment
(minutes)

GSV
Class Name WallClockTime
Instance Name
Attribute Name DSTAdjustment
Dest WCT_GetDSTAdjustment
60

Wall Clock Time
Get Apply DST
(0=disable
1=enable)

EQ
Source A WCT_GetApplyDST
1
Source B WCT_SetApplyDST
1

Wall Clock Time
Get DST Adjustment
(minutes)

EQ
Source A WCT_GetDSTAdjustment
60
Source B EDC_CfgGateway.DstAdjustment
60

Wall Clock Time
DST
matches Config
WCT_DSTok

Wall Clock Time
DST
SSV Delay Timer

MOVE
Source 0
Dest WCT_SetDstDelayTMR.PRE
0

Wall Clock Time
DST
SSV Delay Timer

TON
Timer WCT_SetDstDelayTMR
Preset 0
Accum 1
EN
DN

Wall Clock Time
DST
SSV Delay Timer
WCT_SetDstDelayTMR.DN

s:fs

Wall Clock Time
DST
matches Config
WCT_DSTok
/

Wall Clock Time Set
Apply DST (0=disable
1=enable)

SSV
Class Name WallClockTime
Instance Name
Attribute Name ApplyDST
Source WCT_SetApplyDST
1

Gateway Config
Download Wall Clock
Daylight Savings
Time Adjustment
(minutes)

SSV
Class Name WallClockTime
Instance Name
Attribute Name DSTAdjustment
Source EDC_CfgGateway.DstAdjustment
60

Wall Clock Time
DST
SSV Delay Timer
WCT_SetDstDelayTMR
RES

ADD
Source A WCT_DstSetCount
22
Source B
1
Dest WCT_DstSetCount
22

Set TimeZone and
determine whether to apply Daylight Savings Time

Send WCT DST Tool Record
every hour and on every update
(EqDriver will send whenever data <> 0 and then clear)

remove AFI for option to record every minute vs. every hour

15

AFI

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

NE
Source A CLX.SysTime[4]
<EDC.CLX.SysTime[4]> 19
Source B LastMinute 0

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

NE
Source A CLX.SysTime[3]
<EDC.CLX.SysTime[3]> 15
Source B LastHour 15

SendToolRecord

Wall Clock Time
Time Zone
matches Config
WCT_TimeZoneOK
/

SendToolRecord_ONS
ONS

Wall Clock Time
DST
matches Config
WCT_DSTok
/

SendToolRecord

MOVE
Source CLX.SysTime[3]
<EDC.CLX.SysTime[3]> 15
Dest LastHour 15

MOVE
Source CLX.SysTime[4]
<EDC.CLX.SysTime[4]> 19
Dest LastMinute 0

Year
COP
Source CLX.SysTime[0]
<EDC.CLX.SysTime[0]>
Dest WCT_DstTool[0]
Length 7

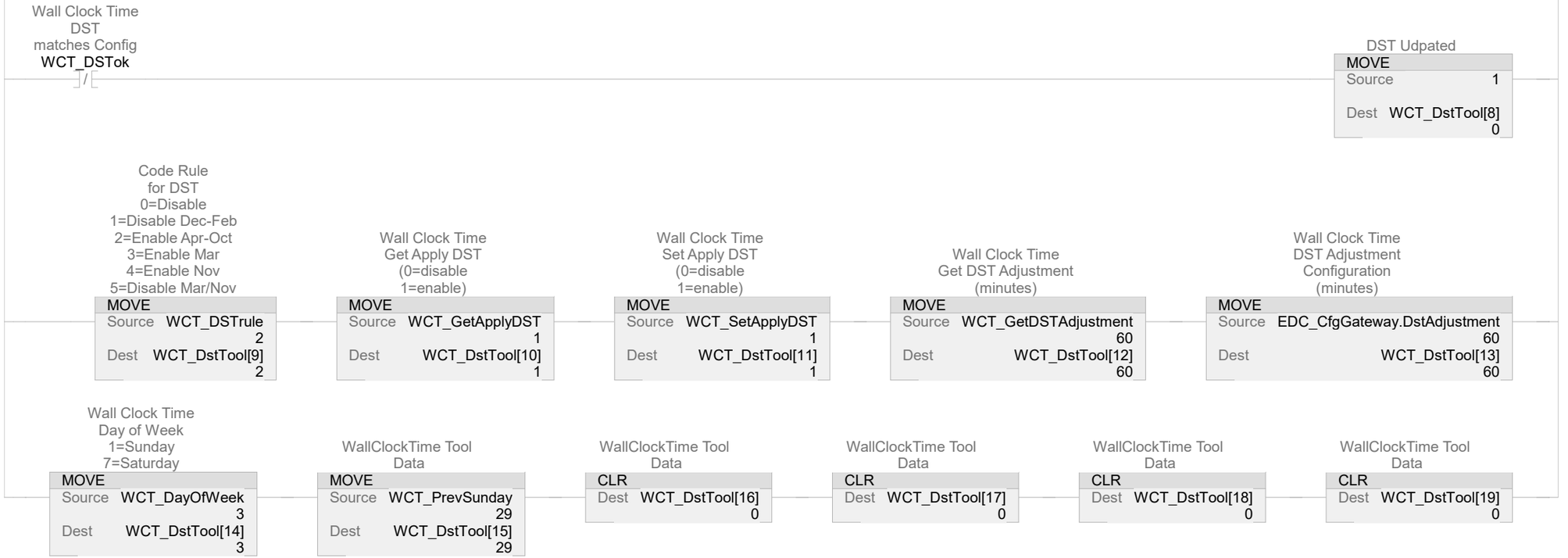
Time Zone Udpated
CLR
Dest WCT_DstTool[7] 0

DST Udpated
CLR
Dest WCT_DstTool[8] 0

Wall Clock Time
Time Zone
matches Config
WCT_TimeZoneOK
/

Time Zone Udpated
MOVE
Source 1
Dest WCT_DstTool[7] 0

Set TimeZone and
determine whether to apply Daylight Savings Time



(End)

Monitors CLx Status

Service CLx Controller Functions

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 30
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Monitors CLx Status

Enable Web Diagnostics - Enable SNMP
v36 Firmware disables by default
L8x DiagEn SINT[5] array values = 1,80,0,6,1 and Custom Message w/ ServiceCode 4c, Instance 1, Class f5, Attribute 0
L9x DiagEn USINT[5] array values = 1,161,0,17,1 and Custom Message w/ ServiceCode 4c, Instance 2, Class f5, Attribute 0

2

s:fs

CLxWebDiagEn_ManualPB

CLxWebDiagEn_ManualPB
U

ControlLogix
Web Diagnostics
Enable Message
Source Element
MOVE
Source 1
Dest CLxWebDiagEn[0]
1

ControlLogix
Web Diagnostics
Enable Message
Source Element
MOVE
Source 161
Dest CLxWebDiagEn[1]
161

ControlLogix
Web Diagnostics
Enable Message
Source Element
MOVE
Source 0
Dest CLxWebDiagEn[2]
0

ControlLogix
Web Diagnostics
Enable Message
Source Element
MOVE
Source 17
Dest CLxWebDiagEn[3]
17

ControlLogix
Web Diagnostics
Enable Message
Source Element
MOVE
Source 1
Dest CLxWebDiagEn[4]
1

ControlLogix
Web Diagnostics
Enable Message
MSG
Message Control CLxWebDiagEnMsg ...
EN
DN
ER

Monitors CLx Status

Controller Product, Revision, Serial Number, Name
(Note: Revision Major/Minor of 20/11 = v20.011)

3

S:FS

CLx Module Status Controller Type (56=L63; see GSV Help for Lookup Table)

GSV
Class Name: ControllerDevice
Instance Name:
Attribute Name: ProductCode
Dest: CLX.Version.ProductCode
<EDC.CLX.Version.ProductCode>: 309

High Byte = Minor
Low Byte = Major

GSV
Class Name: ControllerDevice
Instance Name:
Attribute Name: ProductRev
Dest: ProductRev
16#0b26

CLx Module Status Controller Serial Number (Hex)

GSV
Class Name: ControllerDevice
Instance Name:
Attribute Name: SerialNumber
Dest: CLX.Version.SerialNumber
<EDC.CLX.Version.SerialNumber>: 16#016b_5c03

CLx Module Status Controller Name

GSV
Class Name: Controller
Instance Name:
Attribute Name: Name
Dest: CLX.Version.Name
<EDC.CLX.Version.Name>: 'eAssy_EDC_Gway'

CLx Module Status Controller Major Product Revision

CLR
Dest: CLX.Version.ProductRevMajor
<EDC.CLX.Version.ProductRevMajor>: 38

CLx Module Status Controller Minor Product Revision

CLR
Dest: CLX.Version.ProductRevMinor
<EDC.CLX.Version.ProductRevMinor>: 11

CLx Module Status Controller Major Product Revision

BTD
Source: ProductRev
16#0b26
Source Bit: 0
Dest: CLX.Version.ProductRevMajor
<EDC.CLX.Version.ProductRevMajor>: 38
Dest Bit: 0
Length: 8

CLx Module Status Controller Minor Product Revision

BTD
Source: ProductRev
16#0b26
Source Bit: 8
Dest: CLX.Version.ProductRevMinor
<EDC.CLX.Version.ProductRevMinor>: 11
Dest Bit: 0
Length: 8

Count Enabled Messages

-- Clear Message Enabled Count
-- Count Active Messages in each Scan

4

CLx Module Status CLx Enabled Message Count

CLR
Dest: CLX.MsgEnCount
<EDC.CLX.MsgEnCount>: 0

Monitors CLx Status

Increment Scan Counter and Initialize Scan Timer
v3.02 Data Refresh only works with positive scan counts

5

CLx Module Status
CLx Scan Counter

ADD
Source A CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Source B 1
Dest CLX.ScanCount
<EDC.CLX.ScanCount>
466525176

CLx Module Status
CLx Scan Counter

LE
Source A CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Source B 0

CLx Module Status
CLx Scan Counter

MOVE
Source 1
Dest CLX.ScanCount
<EDC.CLX.ScanCount>
466525176

Monitors CLx Status

Create DateStamp & TimeStamp DINTs for Transaction Logs
Note: All events on same scan will have same Date/Time Stamp

[0]-[2] = Year YYYY, Month MM, Day DD
[3]-[6] = Hour HH, Minute MM, Second SS, microsec UUUUUU
[7] = YY, [8] = YYYYMMDD, [9] = HHMMSS
[10] = last SS, [11] = msec, [12] = fractional sec .ss

6

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

GSV
Class Name WallClockTime
Instance Name
Attribute Name LocalDateTime
Dest CLX.SysTime[0]
<EDC.CLX.SysTime[0]>
2026

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DIV
Source A CLX.SysTime[6]
<EDC.CLX.SysTime[6]>
904687
Source B 1000
Dest CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DIV
Source A CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Source B 10
Dest CLX.SysTime[12]
<EDC.CLX.SysTime[12]>
90

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

NE
Source A CLX.SysTime[5]
<EDC.CLX.SysTime[5]>
21
Source B CLX.SysTime[10]
<EDC.CLX.SysTime[10]>
21

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

MOVE
Source CLX.SysTime[5]
<EDC.CLX.SysTime[5]>
21
Dest CLX.SysTime[10]
<EDC.CLX.SysTime[10]>
21

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

MOD
Source A CLX.SysTime[0]
<EDC.CLX.SysTime[0]>
2026
Source B 100
Dest CLX.SysTime[7]
<EDC.CLX.SysTime[7]>
26

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

CPT
Dest CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Expression CLX.SysTime[0]*10000+(CLX.SysTime[1]*100)+CLX.SysTime[2]

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

CPT
Dest CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Expression CLX.SysTime[3]*10000+(CLX.SysTime[4]*100)+CLX.SysTime[5]

Monitors CLx Status

Create DateTime Stamp STRING from SysTime arrays
-- Clear & Initialize to 0's for auto-padding of single-digit Month, Day, etc.
-- Insert each Date/Time field based on string length to use 0-pad

- Date (yyyymmdd)
- Time (hhmmss)
- millisec (sss)

7

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

FLL
Source 0
Dest CLX.DateTime
<EDC.CLX.DateTime>
Length 1

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

MOVE
Source 17
Dest CLX.DateTime.LEN
<EDC.CLX.DateTime.LEN>
17

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

FLL
Source ascii._0
<EDC.ascii._0>
Dest CLX.DateTime.DATA[0]
<EDC.CLX.DateTime.DATA[0]>
Length CLX.DateTime.LEN
<EDC.CLX.DateTime.LEN>

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest TempString
'481'

Temp Result for
Array Size

SUB
Source A 9
Source B TempString.LEN
3
Dest LEN
15

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

DELETE
Source CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'
Qty TempString.LEN
3
Start LEN
15
Dest CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

INSERT
Source A CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'
Source B TempString
'481'
Start LEN
15
Dest CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest TempString
'481'

Temp Result for
Array Size

SUB
Source A 15
Source B TempString.LEN
3
Dest LEN
15

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

DELETE
Source CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'
Qty TempString.LEN
3
Start LEN
15
Dest CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

INSERT
Source A CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'
Source B TempString
'481'
Start LEN
15
Dest CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest TempString
'481'

Temp Result for
Array Size

SUB
Source A 18
Source B TempString.LEN
3
Dest LEN
15

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

DELETE
Source CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'
Qty TempString.LEN
3
Start LEN
15
Dest CLX.DateTime

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

INSERT
Source A CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'
Source B TempString
'481'
Start LEN
15
Dest CLX.DateTime

Monitors CLx Status

Dest CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'

Dest CLX.DateTime
<EDC.CLX.DateTime>
'20261030151921904'

DHF014 -- External Point Status Monitor

Calcluate seconds in hour for use as Heartbeat (0-3600)

8

CLx Module Status
CLx System Time
Seconds in Current
Hour (mm*60)+ss

CPT
Dest CLX.SecondsInHour
<EDC.CLX.SecondsInHour>
1161
Expression CLX.SysTime[4]*60+CLX.SysTime[5]

CLx Transaction Timestamp for Messages

9

CLx Module Status
Year (YYYY)

MOVE
Source CLX.SysTime[0]
<EDC.CLX.SysTime[0]>
2026
Dest CLX.TransTime.Year
<EDC.CLX.TransTime.Year>
2026

CLx Module Status
Month (MM)

MOVE
Source CLX.SysTime[1]
<EDC.CLX.SysTime[1]>
10
Dest CLX.TransTime.Month
<EDC.CLX.TransTime.Month>
10

CLx Module Status
Day (DD)

MOVE
Source CLX.SysTime[2]
<EDC.CLX.SysTime[2]>
30
Dest CLX.TransTime.Day
<EDC.CLX.TransTime.Day>
30

CLx Module Status
Hour (HH)

MOVE
Source CLX.SysTime[3]
<EDC.CLX.SysTime[3]>
15
Dest CLX.TransTime.Hour
<EDC.CLX.TransTime.Hour>
15

CLx Module Status
Minute (MM)

MOVE
Source CLX.SysTime[4]
<EDC.CLX.SysTime[4]>
19
Dest CLX.TransTime.Min
<EDC.CLX.TransTime.Min>
19

CLx Module Status
Second (SS)

MOVE
Source CLX.SysTime[5]
<EDC.CLX.SysTime[5]>
21
Dest CLX.TransTime.Sec
<EDC.CLX.TransTime.Sec>
21

CLx Module Status
Fractional Seconds
(.ss)

MOVE
Source CLX.SysTime[12]
<EDC.CLX.SysTime[12]>
90
Dest CLX.TransTime.FracSec
<EDC.CLX.TransTime.FracSec>
90

CLx Module Status
Year (YYYY)

LT
Source A CLX.TransTime.Year
<EDC.CLX.TransTime.Year>
2026
Source B 2017

CLx Module Status
CLx Transaction Time
Invalid
CLX.TransTimeInvalid
<EDC.CLX.TransTimeInvalid>

Monitors CLx Status

Update the Day-of-Week every minute
(could update on the hour, but this minimizes the window for incorrect DOW without updating every scan)

10

S:FS

MOVE
Source -1
Dest PrevMinute
24

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

EQ
Source A CLX.SysTime[4]
<EDC.CLX.SysTime[4]>
19
Source B PrevMinute
24

lbl_SkipDOW
JMP

Uses PLC Time to
determine Day of
Week
(0=Saturday, 6 =
Friday)

11

aoiEDC_DayOfWeek_v1_2
aoiEDC_DayOfWeek_v1_... aoi_DayofWeek_v1_2 ...
DateTime CLX.TransTime
<EDC.CLX.TransTime>
DayOfWeek CLX.DayOfWeek
<EDC.CLX.DayOfWeek>

MOVE
Source CLX.SysTime[4]
<EDC.CLX.SysTime[4]>
19
Dest PrevMinute
24

12

lbl_SkipDOW
LBL

NOP

Monitors CLx Status

Get Wallclock CurrentValue - the number of microseconds elapsed since 0000 hours 1 January 1970.
(NOTE: there could be a negative value on the scans where SNTP updates the wallclock, but the overall accuracy should be unaffected)

the CurrentValue will be more accurate than a free-running millisecond-based TON instruction
(with TON, was losing ~1/2-second in a 12-hour shift)

13

CLx Module Status
CLx WallClock
Current Value
(microseconds)

GSV
Class Name WallClockTime
Instance Name
Attribute Name CurrentValue
Dest CLX.WallclockCurrentValue
<EDC.CLX.WallclockCurrentValue>
1793391561904697

CLx Module Status
CLx WallClock
Increment
(microseconds)

SUB
Source A CLX.WallclockCurrentValue
<EDC.CLX.WallclockCurrentValue>
1793391561904697
Source B CLX.WallclockPreviousValue
<EDC.CLX.WallclockPreviousValue>
1793391561904697
Dest CLX.WallclockIncrement
<EDC.CLX.WallclockIncrement>
13612

CLx Module Status
CLx WallClock
Previous Value
(microseconds)

MOVE
Source CLX.WallclockCurrentValue
<EDC.CLX.WallclockCurrentValue>
1793391561904697
Dest CLX.WallclockPreviousValue
<EDC.CLX.WallclockPreviousValue>
1793391561904697

CLx Task Scan Time
-- Clear Values
-- Initialize Array Length

14

CLx Module Status
Clear Scan History
CLX.TaskScan.Clear
<EDC.CLX.TaskScan.Clear>

S:FS

CLx Module Status
CLx Task Scan Time

FLL
Source 0
Dest CLX.TaskScan
<EDC.CLX.TaskScan>
Length 1

Task
High Scan
Count

CLR
Dest CLxTaskScanHighCount
5

CLx Module Status
Enable Scan Monitor
CLX.TaskScan.EN
<EDC.CLX.TaskScan.EN>

L

CLx Module Status
History Array

SIZE
Source CLX.TaskScan.History[0]
<EDC.CLX.TaskScan.History[0]>
15114
Dim. To Vary 0
Size CLX.TaskScan.AvgCTL.LEN
<EDC.CLX.TaskScan.AvgCTL.LEN>
100

Monitors CLx Status

15

CLx Task Scan Time
-- Store New Value in History Array
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value

CLx Module Status
Enable Scan Monitor
CLX.TaskScan.EN
<EDC.CLX.TaskScan.EN>

CLx Module Status
Last Scan
GSV
Class Name Task
Instance Name THIS
Attribute Name LastScanTime
Dest CLX.TaskScan.Last
<EDC.CLX.TaskScan.Last>
13604

CLx Module Status
History Array
MOVE
Source CLX.TaskScan.Last
<EDC.CLX.TaskScan.Last>
13604
Dest CLX.TaskScan.History[CLX.TaskScan.i]
<EDC.CLX.TaskScan.History[CLX.TaskScan.i]>
11588

CLx Module Status
Last Scan
GT
Source A CLX.TaskScan.Last
<EDC.CLX.TaskScan.Last>
13604
Source B CLX.TaskScan.Max
<EDC.CLX.TaskScan.Max>
863980

CLx Module Status
Max Scan
MOVE
Source CLX.TaskScan.Last
<EDC.CLX.TaskScan.Last>
13604
Dest CLX.TaskScan.Max
<EDC.CLX.TaskScan.Max>
863980

CLx Module Status
Last Scan
LT
Source A CLX.TaskScan.Last
<EDC.CLX.TaskScan.Last>
13604
Source B CLX.TaskScan.Min
<EDC.CLX.TaskScan.Min>
10450

CLx Module Status
Last Scan
GT
Source A CLX.TaskScan.Last
<EDC.CLX.TaskScan.Last>
13604
Source B 0

CLx Module Status
Min Scan
MOVE
Source CLX.TaskScan.Last
<EDC.CLX.TaskScan.Last>
13604
Dest CLX.TaskScan.Min
<EDC.CLX.TaskScan.Min>
10450

CLx Module Status
Min Scan
EQ
Source A CLX.TaskScan.Min
<EDC.CLX.TaskScan.Min>
10450
Source B 0

CLx Module Status
History Array Index
ADD
Source A CLX.TaskScan.i
<EDC.CLX.TaskScan.i>
94
Source B 1
Dest CLX.TaskScan.i
<EDC.CLX.TaskScan.i>
94

CLx Module Status
History Array Index
GE
Source A CLX.TaskScan.i
<EDC.CLX.TaskScan.i>
94
Source B CLX.TaskScan.AvgCTL.LEN
<EDC.CLX.TaskScan.AvgCTL.LEN>
100

CLx Module Status
History Array Index
CLR
Dest CLX.TaskScan.i
<EDC.CLX.TaskScan.i>
94

CLx Module Status
History Array Index

CLx Module Status
Avg Scan CONTROL
CLX.TaskScan.AvgCTL
<EDC.CLX.TaskScan.AvgCTL>
RES

CLx Module Status
Avg Scan CONTROL
AVE
Array CLX.TaskScan.History[0]
EN

Monitors CLx Status

LT
Source A CLX.TaskScan.i
<EDC.CLX.TaskScan.i>
94
Source B 0

<EDC.CLX.TaskScan.History[0]>
Dim. To Vary 0
Dest CLX.TaskScan.Avg
<EDC.CLX.TaskScan.Avg>
14049
Control CLX.TaskScan.AvgCTL
<EDC.CLX.TaskScan.AvgCTL>
Length 100
Position 0
DN
ER

CLx Configuration & Initialization Program Scan Time
-- Clear Values
-- Initialize Array Length

16

CLx Module Status
Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

S:FS

CLx Module Status
CLx Total Program
Scan Time (All
Programs)

FLL
Source 0
Dest CLX.ProgScan
<EDC.CLX.ProgScan>
Length 1

CLx Module Status
Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>
L

CLx Module Status
Enable Scan Monitor
CLX.ProgScan.EN
<EDC.CLX.ProgScan.EN>
L

CLx Module Status
History Array

SIZE
Source CLX.ProgScan.History[0]
<EDC.CLX.ProgScan.History[0]>
12307
Dim. To Vary 0
Size CLX.ProgScan.AvgCTL.LEN
<EDC.CLX.ProgScan.AvgCTL.LEN>
100

Monitors CLx Status

CLx Configuration & Initialization Program Scan Time
-- Last Value & History Array are summed by logic in each Program
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value

17

CLx Module Status
Enable Scan Monitor
CLX.ProgScan.EN
S:FS <EDC.CLX.ProgScan.EN>

CLx Module Status
Last Scan
MOVE
Source CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp> 6320
Dest CLX.ProgScan.Last
<EDC.CLX.ProgScan.Last> 13391

CLx Module Status
History Array
MOVE
Source CLX.ProgScan.Last
<EDC.CLX.ProgScan.Last> 13391
Dest CLX.ProgScan.History[Prog.Scan.i]
<EDC.CLX.ProgScan.History[Prog.Scan.i]> 13038

CLx Module Status
Last Scan
GT
Source A CLX.ProgScan.Last
<EDC.CLX.ProgScan.Last> 13391
Source B CLX.ProgScan.Max
<EDC.CLX.ProgScan.Max> 860145

CLx Module Status
Max Scan
MOVE
Source CLX.ProgScan.Last
<EDC.CLX.ProgScan.Last> 13391
Dest CLX.ProgScan.Max
<EDC.CLX.ProgScan.Max> 860145

CLx Module Status
Last Scan
LT
Source A CLX.ProgScan.Last
<EDC.CLX.ProgScan.Last> 13391
Source B CLX.ProgScan.Min
<EDC.CLX.ProgScan.Min> 9678

CLx Module Status
Last Scan
GT
Source A CLX.ProgScan.Last
<EDC.CLX.ProgScan.Last> 13391
Source B 0

CLx Module Status
Min Scan
MOVE
Source CLX.ProgScan.Last
<EDC.CLX.ProgScan.Last> 13391
Dest CLX.ProgScan.Min
<EDC.CLX.ProgScan.Min> 9678

CLx Module Status
Min Scan
EQ
Source A CLX.ProgScan.Min
<EDC.CLX.ProgScan.Min> 9678
Source B 0

CLx Module Status
History Array Index
ADD
Source A CLX.ProgScan.i
<EDC.CLX.ProgScan.i> 94
Source B 1
Dest CLX.ProgScan.i
<EDC.CLX.ProgScan.i> 94

CLx Module Status
History Array Index
GE
Source A CLX.ProgScan.i
<EDC.CLX.ProgScan.i> 94
Source B CLX.ProgScan.AvgCTL.LEN
<EDC.CLX.ProgScan.AvgCTL.LEN> 100

CLx Module Status
History Array Index
CLR
Dest CLX.ProgScan.i
<EDC.CLX.ProgScan.i> 94

CLx Module Status
Avg Scan CONTROL
CLX.ProgScan.AvgCTL
<EDC.CLX.ProgScan.AvgCTL>
RES

CLx Module Status
Avg Scan CONTROL
AVE
Array CLX.ProgScan.History[0]
EN

CLx Module Status
History Array Index

Monitors CLx Status

LT
Source A CLX.ProgScan.i
<EDC.CLX.ProgScan.i>
94
Source B 0

<EDC.CLX.ProgScan.History[0]>
Dim. To Vary 0
Dest CLX.ProgScan.Avg
<EDC.CLX.ProgScan.Avg>
12474
Control CLX.ProgScan.AvgCTL
<EDC.CLX.ProgScan.AvgCTL>
Length 100
Position 0
DN
ER

Log Transactions from Configuration Program

(Clear Trans.LOGi when TransLog is initialized on Reset)

18

EDC Config Reset ONS
(Log Transaction)
CFG.ResetONS
<EDC.CFG.ResetONS>

Transaction Log Control -- Transaction Log Index
CLR
Dest Trans.LOGi
<EDC.Trans.LOGi>
211

Transaction Log Control -- Transaction Error Log Index
CLR
Dest Trans.ERRi
<EDC.Trans.ERRi>
62

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 9000
Input Par 0
Input Par -1
Input Par -1
Input Par 0

EDC Config Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 9001
Input Par 0
Input Par -1
Input Par -1
Input Par 0

EDC Config Initialization ONS
(Log Transaction)
CFG.InitONS
<EDC.CFG.InitONS>

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 9002
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Monitors CLx Status

DHF010 -- Log Transaction for Task Scan Exceeds High Limit (configured for 100 msec)

19

CLx Module Status
Last Scan

GT	
Source A	CLX.TaskScan.Last
	<EDC.CLX.TaskScan.Last>
	13604
Source B	CLxScanHighLimit
	100000

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	CLX.TaskScan.Last
	<EDC.CLX.TaskScan.Last>
	13604
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	9004
Input Par	0
Input Par	-1
Input Par	-1
Input Par	0

Task
High Scan
Count

ADD	
Source A	CLxTaskScanHighCount
	5
Source B	1
Dest	CLxTaskScanHighCount
	5

CLxTaskScanHigh

Clear Total Program Scan & History Array for next Scan values

20

CLx Module Status
CLx Total Program
Scan Time - Temp
Summation

CLR	
Dest	CLX.ProgScanTemp
	<EDC.CLX.ProgScanTemp>
	6320

Initialize Program High Scan Events

21

Program High Scan
Events

CLR	
Dest	CLxProgScanHigh
	0

(End)

Enet Module Status

Enet Module Control and Status

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	31
Dest	Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1733165816
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1733165816

Enet Module Status

Get Enet Module Status

2

** Processor can't access its own EntryStatus / FaultCode **

Ethernet Module Status GSV Module - Entry Status

GSV
Class Name Module
Instance Name EDC_NW
Attribute Name EntryStatus
Dest Enet00.EntryStatus
<EDC_Enet00.EntryStatus>
16#4000

Ethernet Module Status GSV Module - Fault Code

GSV
Class Name Module
Instance Name EDC_NW
Attribute Name FaultCode
Dest Enet00.FaultCode
<EDC_Enet00.FaultCode>
16#0000

Ethernet Module Status GSV Module - Fault Info

GSV
Class Name Module
Instance Name EDC_NW
Attribute Name FaultInfo
Dest Enet00.FaultInfo
<EDC_Enet00.FaultInfo>
16#0000_0000

Ethernet Module Status GSV Module - Mode

GSV
Class Name Module
Instance Name EDC_NW
Attribute Name Mode
Dest Enet00.Mode
<EDC_Enet00.Mode>
16#0000

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet00.EntryStatus
<EDC_Enet00.EntryStatus>
16#4000
Mask Enet00.EntryStatusMask
<EDC_Enet00.EntryStatusMask>
16#f000
Compare Enet00.EntryStatusRunChk
<EDC_Enet00.EntryStatusRunChk>
16#4000

Ethernet Module Status GSV Module - Entry Status OK
Enet00.EntryStatusOk
<EDC_Enet00.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet00.EntryStatus
<EDC_Enet00.EntryStatus>
16#4000
Mask Enet00.EntryStatusMask
<EDC_Enet00.EntryStatusMask>
16#f000
Compare ENet_InhibChk
16#6000

Ethernet Module Status GSV Module - Fault Code

EQ
Source A Enet00.FaultCode
<EDC_Enet00.FaultCode>
16#0000
Source B 0

Ethernet Module Status GSV Module - Fault Code OK
Enet00.FaultCodeOk
<EDC_Enet00.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK
Enet00.EntryStatusOk
<EDC_Enet00.EntryStatusOk>

Ethernet Module Status GSV Module - Fault Code OK
Enet00.FaultCodeOk
<EDC_Enet00.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK
Enet00.StatusOK
<EDC_Enet00.StatusOK>

Enet Module Status

Ethernet Module CPU Utilization
-- Clear Values
-- Initialize Array Length

3

Ethernet Module Status Clear Scan History
Enet00.CPU_Util.Clear
<EDC_Enet00.CPU_Util.Clear>

S:FS

Ethernet Module Status Ethernet CPU Utilization

FLL
Source 0
Dest Enet00.CPU_Util
<EDC_Enet00.CPU_Util>
Length 1

Ethernet Module Status Enable Scan Monitor
Enet00.CPU_Util.EN
<EDC_Enet00.CPU_Util.EN>

L

Ethernet Module Status History Array

SIZE
Source Enet00.CPU_Util.History[0]
<EDC_Enet00.CPU_Util.History[0]>
590
Dim. To Vary 0
Size Enet00.CPU_Util.AvgCTL.LEN
<EDC_Enet00.CPU_Util.AvgCTL.LEN>
100

Enet Module Status

Ethernet Module CPU Utilization
-- Store New Value in History Array
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value
-- Get Next Value

(EN2T v2 Module CPU_Util is shifted 1 Byte in Status Packet -- use MEQ to auto detect this and correct)
DHF005 - Add CLR Enet03.CPU_Util.Last to ensure upper bytes are 0

** SPECIAL CASE -- L85 Utilization Message replaces EN2T Status Message **
5 - Task Utilization (always 100 with continuous task)
9 - MSG Utilization
11 - I/O Utilization

4

Ethernet Module
Status Enable Scan
Monitor
Enet00.CPU_Util.EN
<EDC_Enet00.CPU_Util.EN>

Ethernet Module --
L85 Ethernet Link
Status Message
EDC_Enet00_UtilMsg.EN

Ethernet Module --
L85 Ethernet Link
Status Message
EDC_Enet00_UtilMsg.DN

Ethernet Module --
L85 Ethernet Link
Status Message

MSG
Message Control EDC_Enet00_UtilMsg ...
EN
DN
ER

Ethernet Module
Status Last Scan

MUL
Source A EDC_Enet00_Util[9]
59
Source B 10
Dest Enet00.CPU_Util.Last
<EDC_Enet00.CPU_Util.Last>
580

Ethernet Module
Status History Array

MOVE
Source Enet00.CPU_Util.Last
<EDC_Enet00.CPU_Util.Last>
580
Dest Enet00.CPU_Util.History[Enet00.CPU_Util.i]
<EDC_Enet00.CPU_Util.History[Enet00.CPU_Util.i]>
600

Ethernet Module
Status History Array
Index

ADD
Source A Enet00.CPU_Util.i
<EDC_Enet00.CPU_Util.i>

Ethernet Module
Status History Array
Index

GE
Source A Enet00.CPU_Util.i
<EDC_Enet00.CPU_Util.i>

Ethernet Module
Status History Array
Index

CLR
Dest Enet00.CPU_Util.i
<EDC_Enet00.CPU_Util.i>

Enet Module Status

78
Source B 1
Dest Enet00.CPU_Util.i
<EDC_Enet00.CPU_Util.i>
78

78
Source B Enet00.CPU_Util.AvgCTL.LEN
<EDC_Enet00.CPU_Util.AvgCTL.LEN>
100

Ethernet Module Status History Array Index

LT
Source A Enet00.CPU_Util.i
<EDC_Enet00.CPU_Util.i>
78
Source B 0

Ethernet Module Status Avg Scan CONTROL
Enet00.CPU_Util.AvgCTL
<EDC_Enet00.CPU_Util.AvgCTL>
RES

78

Ethernet Module Status Avg Scan CONTROL

AVE
Array Enet00.CPU_Util.History[0]
<EDC_Enet00.CPU_Util.History[0]>
Dim. To Vary 0
Dest Enet00.CPU_Util.Avg
<EDC_Enet00.CPU_Util.Avg>
595
Control Enet00.CPU_Util.AvgCTL
<EDC_Enet00.CPU_Util.AvgCTL>
Length 100
Position 0
EN
DN
ER

Ethernet Module Status Last Scan

GT
Source A Enet00.CPU_Util.Last
<EDC_Enet00.CPU_Util.Last>
580
Source B Enet00.CPU_Util.Max
<EDC_Enet00.CPU_Util.Max>
640

Ethernet Module Status Max Scan

MOVE
Source Enet00.CPU_Util.Last
<EDC_Enet00.CPU_Util.Last>
580
Dest Enet00.CPU_Util.Max
<EDC_Enet00.CPU_Util.Max>
640

Ethernet Module Status Last Scan

LT
Source A Enet00.CPU_Util.Last
<EDC_Enet00.CPU_Util.Last>
580
Source B Enet00.CPU_Util.Min
<EDC_Enet00.CPU_Util.Min>
450

Ethernet Module Status Last Scan

GT
Source A Enet00.CPU_Util.Last
<EDC_Enet00.CPU_Util.Last>
580
Source B 0

Ethernet Module Status Min Scan

MOVE
Source Enet00.CPU_Util.Last
<EDC_Enet00.CPU_Util.Last>
580
Dest Enet00.CPU_Util.Min
<EDC_Enet00.CPU_Util.Min>
450

Ethernet Module Status Min Scan

EQ
Source A Enet00.CPU_Util.Min
<EDC_Enet00.CPU_Util.Min>
450
Source B 0

Enet Module Status

Log Enet Utilization above 50% and 75%
(only log this once per second to avoid filling log)

{v1.0.1} -- Reset Log Event Timer
DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

5

Ethernet Module Status Last Scan
GT
Source A Enet00.CPU_Util.Last <EDC_Enet00.CPU_Util.Last> 580
Source B 500

Ethernet Module Status Last Scan
LT
Source A Enet00.CPU_Util.Last <EDC_Enet00.CPU_Util.Last> 580
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet00.UtilOver50Logged <EDC_Enet00.UtilOver50Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet00.CPU_Util.Last <EDC_Enet00.CPU_Util.Last> 580
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 50% Timer
Enet00.UtilOver50_TMR <EDC_Enet00.UtilOver50_TMR>
RES

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet00.UtilOver50Logged <EDC_Enet00.UtilOver50Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8232
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status Last Scan
GE
Source A Enet00.CPU_Util.Last <EDC_Enet00.CPU_Util.Last> 580
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet00.UtilOver75Logged <EDC_Enet00.UtilOver75Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet00.CPU_Util.Last <EDC_Enet00.CPU_Util.Last> 580
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet00.UtilOver75Logged <EDC_Enet00.UtilOver75Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8233
Input Par 0
Input Par -1
Input Par -1

Enet Module Status

Input Par 0

Log Enet Utilization above 50% and 75% -- reset 1-shot after time delay to allow another Transaction log entry
(only log this once per second to avoid filling log)

DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

6

Ethernet Module
Status Ethernet CPU
Utilization above
50% 1-shot
Enet00.UtilOver50Logged
<EDC_Enet00.UtilOver50Logged>

Ethernet Module
Status Ethernet CPU
Utilization above
75% 1-shot
Enet00.UtilOver75Logged
<EDC_Enet00.UtilOver75Logged>

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer

MOVE
Source 1000
Dest Enet00.UtilOver50_TMR.PRE
<EDC_Enet00.UtilOver50_TMR.PRE>
1000

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer

TON
Timer Enet00.UtilOver50_TMR
<EDC_Enet00.UtilOver50_TMR>
Preset 1000
Accum 2054
EN
DN

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer
Enet00.UtilOver50_TMR.DN
<EDC_Enet00.UtilOver50_TMR.DN>

Ethernet Module
Status Ethernet CPU
Utilization above
50% 1-shot
Enet00.UtilOver50Logged
<EDC_Enet00.UtilOver50Logged>
U

Ethernet Module
Status Ethernet CPU
Utilization above
75% 1-shot
Enet00.UtilOver75Logged
<EDC_Enet00.UtilOver75Logged>
U

Enet Module Status

Log Enet Module Overall Status OK & Not OK
-- if Not OK, Clear Sockets Ready

7

Ethernet Module Status GSV Module - Entry Status OK
Enet00.StatusOK
<EDC_Enet00.StatusOK>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet00.StatusOkONS
<EDC_Enet00.StatusOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8201
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet00.StatusOK
<EDC_Enet00.StatusOK>
/

Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet00.StatusNotOkONS
<EDC_Enet00.StatusNotOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8203
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Enet Module Status

Log Enet Module Entry Status OK & Not OK
-- log Status Not OK each time Entry Status changs

8

Ethernet Module Status GSV Module - Entry Status OK
Enet00.EntryStatusOk
<EDC_Enet00.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet00.EntryStatusOkONS
<EDC_Enet00.EntryStatusOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet00.EntryStatus <EDC_Enet00.EntryStatus> 16#4000
Source B Enet00.EntryStatusMask <EDC_Enet00.EntryStatusMask> 16#f000
Dest Enet00.EntryStatusLast <EDC_Enet00.EntryStatusLast> 16#4000

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet00.EntryStatusLast <EDC_Enet00.EntryStatusLast> 16#4000
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8211
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet00.EntryStatusOk
<EDC_Enet00.EntryStatusOk>
/

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet00.EntryStatus <EDC_Enet00.EntryStatus> 16#4000
Mask Enet00.EntryStatusMask <EDC_Enet00.EntryStatusMask> 16#f000
Compare Enet00.EntryStatusLast <EDC_Enet00.EntryStatusLast> 16#4000

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet00.EntryStatusSame
<EDC_Enet00.EntryStatusSame>

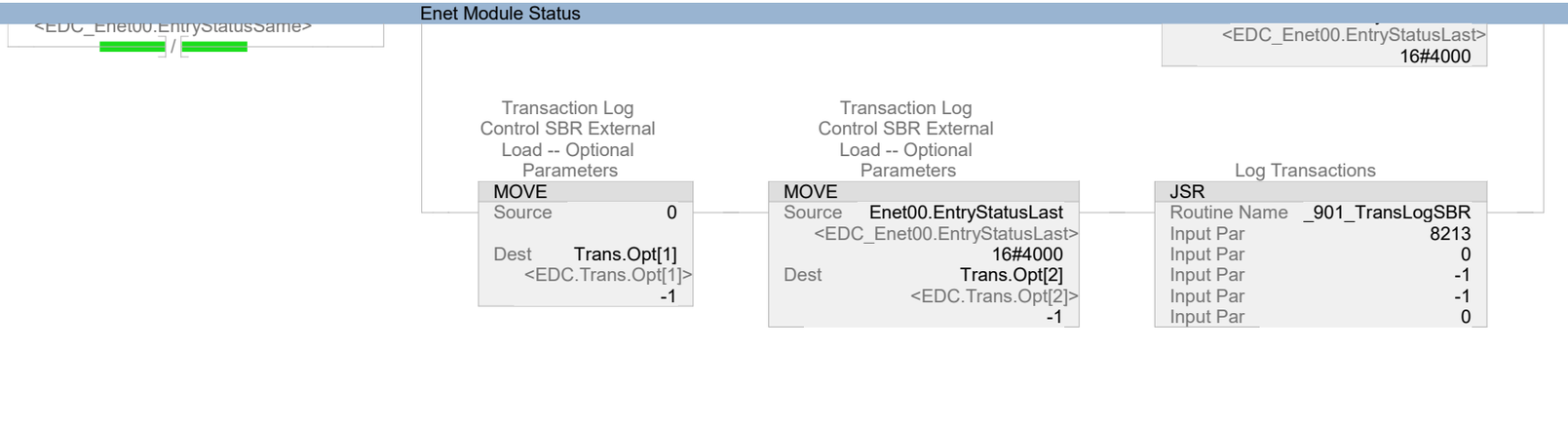
Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet00.EntryStatusNotOkONS
<EDC_Enet00.EntryStatusNotOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet00.EntryStatusSame
<EDC_Enet00.EntryStatusSame>

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet00.EntryStatus <EDC_Enet00.EntryStatus> 16#4000
Source B Enet00.EntryStatusMask <EDC_Enet00.EntryStatusMask> 16#f000
Dest Enet00.EntryStatusLast

Enet Module Status



Enet Module Status

Log Enet Module Fault Code OK & Not OK
-- log Status Not OK each time Fault Code changs

9

Ethernet Module Status GSV Module - Fault Code OK
Enet00.FaultCodeOk
<EDC_Enet00.FaultCodeOk>

Ethernet Module Status GSV Module - Fault Code OK 1-Shot
Enet00.FaultCodeOkONS
<EDC_Enet00.FaultCodeOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Enet00.FaultCode
	<EDC_Enet00.FaultCode>
	16#0000
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	8221
Input Par	0
Input Par	-1
Input Par	-1
Input Par	0

Ethernet Module Status GSV Module - Fault Code OK
Enet00.FaultCodeOk
<EDC_Enet00.FaultCodeOk>
/

Ethernet Module Status GSV Module - Fault Code Not OK 1-Shot
Enet00.FaultCodeNotOkONS
<EDC_Enet00.FaultCodeNotOkONS>
ONS

Ethernet Module Status GSV Module - Fault Code

NE	
Source A	Enet00.FaultCode
	<EDC_Enet00.FaultCode>
	16#0000
Source B	Enet00.FaultCodeLast
	<EDC_Enet00.FaultCodeLast>
	16#0005

Ethernet Module Status GSV Module - Fault Code Last Value

MOVE	
Source	Enet00.FaultCode
	<EDC_Enet00.FaultCode>
	16#0000
Dest	Enet00.FaultCodeLast
	<EDC_Enet00.FaultCodeLast>
	16#0005

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Enet00.FaultCode
	<EDC_Enet00.FaultCode>
	16#0000
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	8223
Input Par	0
Input Par	-1
Input Par	-1
Input Par	0

(End)

Enet Module Status

Enet Module Control and Status

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 31
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

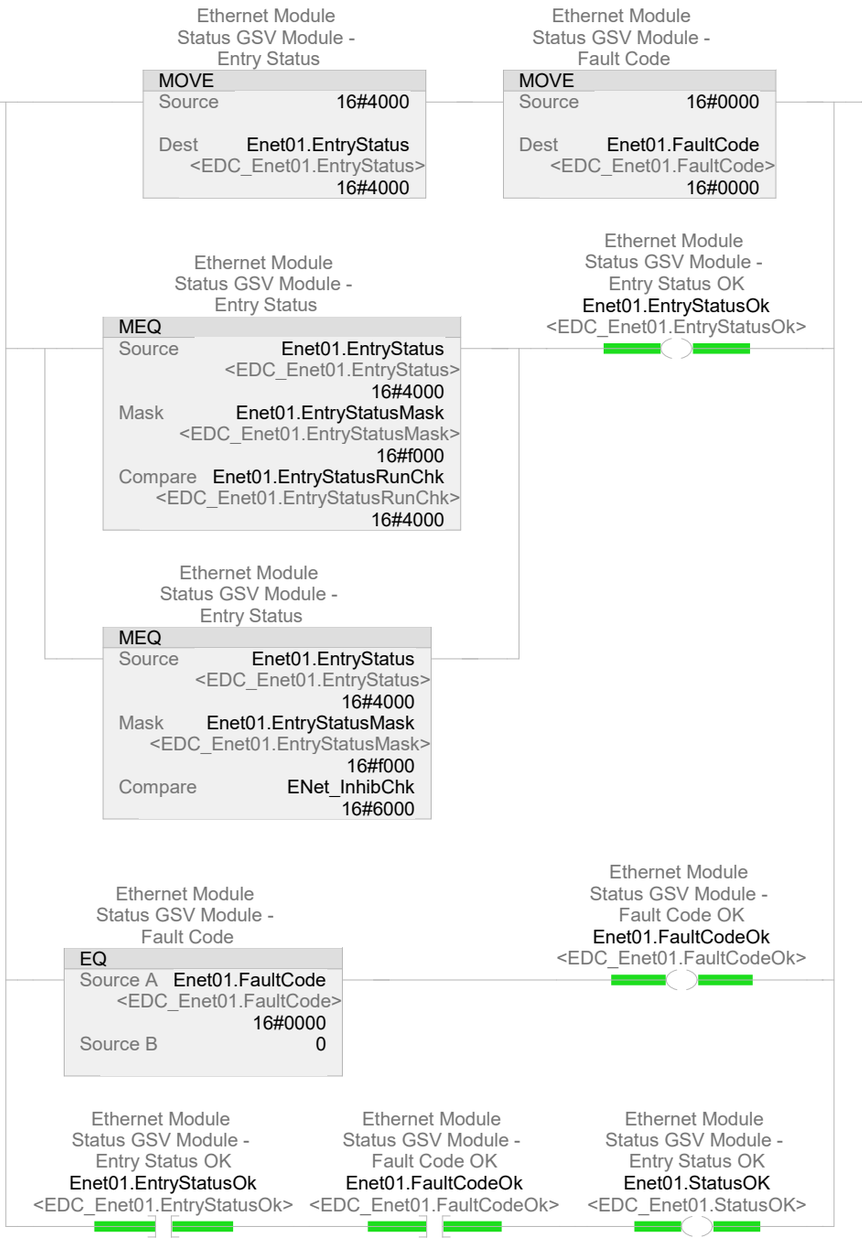
ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Enet Module Status

Get Enet Module Status

** Processor can't access its own EntryStatus / FaultCode **

2



Enet Module Status

3

Ethernet Module CPU Utilization
-- Clear Values
-- Initialize Array Length

Ethernet Module
Status Clear Scan
History
Enet01.CPU_Util.Clear
<EDC_Enet01.CPU_Util.Clear>

S:FS

Ethernet Module
Status Ethernet CPU
Utilization

FLL
Source 0
Dest Enet01.CPU_Util
<EDC_Enet01.CPU_Util>
Length 1

Ethernet Module
Status Enable Scan
Monitor
Enet01.CPU_Util.EN
<EDC_Enet01.CPU_Util.EN>

L

Ethernet Module
Status History Array

SIZE
Source Enet01.CPU_Util.History[0]
<EDC_Enet01.CPU_Util.History[0]>
390
Dim. To Vary 0
Size Enet01.CPU_Util.AvgCTL.LEN
<EDC_Enet01.CPU_Util.AvgCTL.LEN>
100

Enet Module Status

Ethernet Module CPU Utilization
-- Store New Value in History Array
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value
-- Get Next Value

(EN2T v2 Module CPU_Util is shifted 1 Byte in Status Packet -- use MEQ to auto detect this and correct)
DHF005 - Add CLR Enet03.CPU_Util.Last to ensure upper bytes are 0

** SPECIAL CASE -- L85 Utilization Message replaces EN2T Status Message **
5 - Task Utilization (always 100 with continuous task)
9 - MSG Utilization
11 - I/O Utilization

4

Ethernet Module Status Enable Scan Monitor
Enet01.CPU_Util.EN
<EDC_Enet01.CPU_Util.EN>

Ethernet Module -- L85 Ethernet Link Status Message
EDC_Enet01_UtilMsg.EN

Ethernet Module -- L85 Ethernet Link Status Message
EDC_Enet01_UtilMsg.DN

Ethernet Module -- L85 Ethernet Link Status Message

MSG
Message Control EDC_Enet01_UtilMsg ...
EN
DN
ER

Ethernet Module Status Last Scan

MUL
Source A EDC_Enet01_Util[9]
47
Source B 10
Dest Enet01.CPU_Util.Last
<EDC_Enet01.CPU_Util.Last>
440

Ethernet Module Status History Array

MOVE
Source Enet01.CPU_Util.Last
<EDC_Enet01.CPU_Util.Last>
440
Dest Enet01.CPU_Util.History[Enet01.CPU_Util.i]
<EDC_Enet01.CPU_Util.History[Enet01.CPU_Util.i]>
380

Ethernet Module Status History Array Index

ADD
Source A Enet01.CPU_Util.i
<EDC_Enet01.CPU_Util.i>

Ethernet Module Status History Array Index

GE
Source A Enet01.CPU_Util.i
<EDC_Enet01.CPU_Util.i>

Ethernet Module Status History Array Index

CLR
Dest Enet01.CPU_Util.i
<EDC_Enet01.CPU_Util.i>

Enet Module Status

78
Source B 1
Dest Enet01.CPU_Util.i
<EDC_Enet01.CPU_Util.i>
78

78
Source B Enet01.CPU_Util.AvgCTL.LEN
<EDC_Enet01.CPU_Util.AvgCTL.LEN>
100

Ethernet Module Status History Array Index
LT
Source A Enet01.CPU_Util.i
<EDC_Enet01.CPU_Util.i>
78
Source B 0

Ethernet Module Status Avg Scan CONTROL
Enet01.CPU_Util.AvgCTL
<EDC_Enet01.CPU_Util.AvgCTL>
RES

78

Ethernet Module Status Avg Scan CONTROL
AVE
Array Enet01.CPU_Util.History[0]
<EDC_Enet01.CPU_Util.History[0]>
Dim. To Vary 0
Dest Enet01.CPU_Util.Avg
<EDC_Enet01.CPU_Util.Avg>
419
Control Enet01.CPU_Util.AvgCTL
<EDC_Enet01.CPU_Util.AvgCTL>
Length 100
Position 0
EN
DN
ER

Ethernet Module Status Last Scan
GT
Source A Enet01.CPU_Util.Last
<EDC_Enet01.CPU_Util.Last>
440
Source B Enet01.CPU_Util.Max
<EDC_Enet01.CPU_Util.Max>
690

Ethernet Module Status Max Scan
MOVE
Source Enet01.CPU_Util.Last
<EDC_Enet01.CPU_Util.Last>
440
Dest Enet01.CPU_Util.Max
<EDC_Enet01.CPU_Util.Max>
690

Ethernet Module Status Last Scan
LT
Source A Enet01.CPU_Util.Last
<EDC_Enet01.CPU_Util.Last>
440
Source B Enet01.CPU_Util.Min
<EDC_Enet01.CPU_Util.Min>
240

Ethernet Module Status Min Scan
EQ
Source A Enet01.CPU_Util.Min
<EDC_Enet01.CPU_Util.Min>
240
Source B 0

Ethernet Module Status Last Scan
GT
Source A Enet01.CPU_Util.Last
<EDC_Enet01.CPU_Util.Last>
440
Source B 0

Ethernet Module Status Min Scan
MOVE
Source Enet01.CPU_Util.Last
<EDC_Enet01.CPU_Util.Last>
440
Dest Enet01.CPU_Util.Min
<EDC_Enet01.CPU_Util.Min>
240

Enet Module Status

Log Enet Utilization above 50% and 75%
(only log this once per second to avoid filling log)

{v1.0.1} -- Reset Log Event Timer
DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

5

Ethernet Module Status Last Scan
GT
Source A Enet01.CPU_Util.Last <EDC_Enet01.CPU_Util.Last> 440
Source B 500

Ethernet Module Status Last Scan
LT
Source A Enet01.CPU_Util.Last <EDC_Enet01.CPU_Util.Last> 440
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet01.UtilOver50Logged <EDC_Enet01.UtilOver50Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet01.CPU_Util.Last <EDC_Enet01.CPU_Util.Last> 440
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 50% Timer
Enet01.UtilOver50_TMR <EDC_Enet01.UtilOver50_TMR>
RES

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet01.UtilOver50Logged <EDC_Enet01.UtilOver50Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8232
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status Last Scan
GE
Source A Enet01.CPU_Util.Last <EDC_Enet01.CPU_Util.Last> 440
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet01.UtilOver75Logged <EDC_Enet01.UtilOver75Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet01.CPU_Util.Last <EDC_Enet01.CPU_Util.Last> 440
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet01.UtilOver75Logged <EDC_Enet01.UtilOver75Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8233
Input Par 0
Input Par -1
Input Par -1

Enet Module Status

Input Par 0

Log Enet Utilization above 50% and 75% -- reset 1-shot after time delay to allow another Transaction log entry
(only log this once per second to avoid filling log)

DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

6

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet01.UtilOver50Logged
<EDC_Enet01.UtilOver50Logged>

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet01.UtilOver75Logged
<EDC_Enet01.UtilOver75Logged>

Ethernet Module Status Ethernet CPU Utilization above 50% Timer

MOVE
Source 1000
Dest Enet01.UtilOver50_TMR.PRE
<EDC_Enet01.UtilOver50_TMR.PRE>
1000

Ethernet Module Status Ethernet CPU Utilization above 50% Timer

TON
Timer Enet01.UtilOver50_TMR
<EDC_Enet01.UtilOver50_TMR>
Preset 1000
Accum 0
EN
DN

Ethernet Module Status Ethernet CPU Utilization above 50% Timer
Enet01.UtilOver50_TMR.DN
<EDC_Enet01.UtilOver50_TMR.DN>

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet01.UtilOver50Logged
<EDC_Enet01.UtilOver50Logged>
U

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet01.UtilOver75Logged
<EDC_Enet01.UtilOver75Logged>
U

Enet Module Status

Log Enet Module Overall Status OK & Not OK
-- if Not OK, Clear Sockets Ready

7

Ethernet Module Status GSV Module - Entry Status OK
Enet01.StatusOK
<EDC_Enet01.StatusOK>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet01.StatusOkONS
<EDC_Enet01.StatusOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8201
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet01.StatusOK
<EDC_Enet01.StatusOK>
/

Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet01.StatusNotOkONS
<EDC_Enet01.StatusNotOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8203
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Enet Module Status

Log Enet Module Entry Status OK & Not OK
-- log Status Not OK each time Entry Status changs

8

Ethernet Module Status GSV Module - Entry Status OK
Enet01.EntryStatusOk
<EDC_Enet01.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet01.EntryStatusOkONS
<EDC_Enet01.EntryStatusOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet01.EntryStatus <EDC_Enet01.EntryStatus> 16#4000
Source B Enet01.EntryStatusMask <EDC_Enet01.EntryStatusMask> 16#f000
Dest Enet01.EntryStatusLast <EDC_Enet01.EntryStatusLast> 16#6000

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet01.EntryStatusLast <EDC_Enet01.EntryStatusLast> 16#6000
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8211
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet01.EntryStatusOk
<EDC_Enet01.EntryStatusOk>
/

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet01.EntryStatus <EDC_Enet01.EntryStatus> 16#4000
Mask Enet01.EntryStatusMask <EDC_Enet01.EntryStatusMask> 16#f000
Compare Enet01.EntryStatusLast <EDC_Enet01.EntryStatusLast> 16#6000

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet01.EntryStatusSame
<EDC_Enet01.EntryStatusSame>

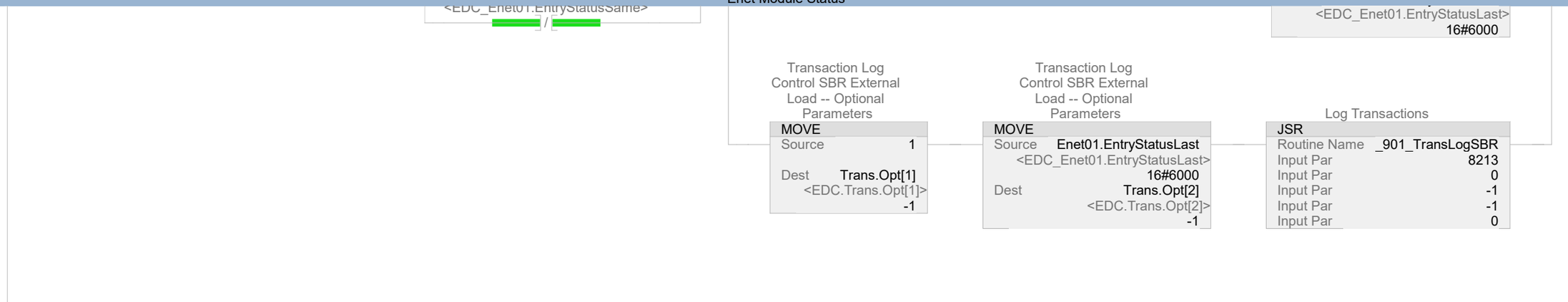
Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet01.EntryStatusNotOkONS
<EDC_Enet01.EntryStatusNotOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet01.EntryStatusSame
<EDC_Enet01.EntryStatusSame>

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet01.EntryStatus <EDC_Enet01.EntryStatus> 16#4000
Source B Enet01.EntryStatusMask <EDC_Enet01.EntryStatusMask> 16#f000
Dest Enet01.EntryStatusLast

Enet Module Status



Enet Module Status

9

Log Enet Module Fault Code OK & Not OK
-- log Status Not OK each time Fault Code changs

Ethernet Module Status GSV Module - Fault Code OK
Enet01.FaultCodeOk
<EDC_Enet01.FaultCodeOk>

Ethernet Module Status GSV Module - Fault Code OK 1-Shot
Enet01.FaultCodeOkONS
<EDC_Enet01.FaultCodeOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source Enet01.FaultCode
<EDC_Enet01.FaultCode> 16#0000
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8221
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Fault Code OK
Enet01.FaultCodeOk
<EDC_Enet01.FaultCodeOk>
/

Ethernet Module Status GSV Module - Fault Code Not OK 1-Shot
Enet01.FaultCodeNotOkONS
<EDC_Enet01.FaultCodeNotOkONS>
ONS

Ethernet Module Status GSV Module - Fault Code
NE
Source A Enet01.FaultCode
<EDC_Enet01.FaultCode> 16#0000
Source B Enet01.FaultCodeLast
<EDC_Enet01.FaultCodeLast> 16#0000

Ethernet Module Status GSV Module - Fault Code Last Value
MOVE
Source Enet01.FaultCode
<EDC_Enet01.FaultCode> 16#0000
Dest Enet01.FaultCodeLast
<EDC_Enet01.FaultCodeLast> 16#0000

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source Enet01.FaultCode
<EDC_Enet01.FaultCode> 16#0000
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8223
Input Par 0
Input Par -1
Input Par -1
Input Par 0

(End)

Enet Module Status

Enet Module Control and Status

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	31
Dest	Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1733165816
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1733165816

Enet Module Status

Get Enet Module Status

2

Ethernet Module Status GSV Module - Entry Status

GSV
Class Name Module
Instance Name PdM_Gway
Attribute Name EntryStatus
Dest Enet02.EntryStatus
<EDC_Enet02.EntryStatus>
16#4000

Ethernet Module Status GSV Module - Fault Code

GSV
Class Name Module
Instance Name PdM_Gway
Attribute Name FaultCode
Dest Enet02.FaultCode
<EDC_Enet02.FaultCode>
16#0000

Ethernet Module Status GSV Module - Fault Info

GSV
Class Name Module
Instance Name PdM_Gway
Attribute Name FaultInfo
Dest Enet02.FaultInfo
<EDC_Enet02.FaultInfo>
16#0000_0000

Ethernet Module Status GSV Module - Mode

GSV
Class Name Module
Instance Name PdM_Gway
Attribute Name Mode
Dest Enet02.Mode
<EDC_Enet02.Mode>
16#0000

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet02.EntryStatus
<EDC_Enet02.EntryStatus>
16#4000
Mask Enet02.EntryStatusMask
<EDC_Enet02.EntryStatusMask>
16#f000
Compare Enet02.EntryStatusRunChk
<EDC_Enet02.EntryStatusRunChk>
16#4000

Ethernet Module Status GSV Module - Entry Status OK

Enet02.EntryStatusOk
<EDC_Enet02.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet02.EntryStatus
<EDC_Enet02.EntryStatus>
16#4000
Mask Enet02.EntryStatusMask
<EDC_Enet02.EntryStatusMask>
16#f000
Compare ENet_InhibChk
16#6000

Ethernet Module Status GSV Module - Fault Code

EQ
Source A Enet02.FaultCode
<EDC_Enet02.FaultCode>
16#0000
Source B 0

Ethernet Module Status GSV Module - Fault Code OK

Enet02.FaultCodeOk
<EDC_Enet02.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK

Enet02.EntryStatusOk
<EDC_Enet02.EntryStatusOk>

Ethernet Module Status GSV Module - Fault Code OK

Enet02.FaultCodeOk
<EDC_Enet02.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK

Enet02.StatusOK
<EDC_Enet02.StatusOK>

Enet Module Status

3

Ethernet Module CPU Utilization
-- Clear Values
-- Initialize Array Length

Ethernet Module
Status Clear Scan
History
Enet02.CPU_Util.Clear
<EDC_Enet02.CPU_Util.Clear>

S:FS

Ethernet Module
Status Ethernet CPU
Utilization

FLL
Source 0
Dest Enet02.CPU_Util
<EDC_Enet02.CPU_Util>
Length 1

Ethernet Module
Status Enable Scan
Monitor
Enet02.CPU_Util.EN
<EDC_Enet02.CPU_Util.EN>

L

Ethernet Module
Status History Array

SIZE
Source Enet02.CPU_Util.History[0]
<EDC_Enet02.CPU_Util.History[0]>
620
Dim. To Vary 0
Size Enet02.CPU_Util.AvgCTL.LEN
<EDC_Enet02.CPU_Util.AvgCTL.LEN>
100

Enet Module Status

Ethernet Module CPU Utilization
-- Store New Value in History Array
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value
-- Get Next Value

(EN2T v2 Module CPU_Util is shifted 1 Byte in Status Packet -- use MEQ to auto detect this and correct)
DHF005 - Add CLR Enet03.CPU_Util.Last to ensure upper bytes are 0

** SPECIAL CASE -- L85 Utilization Message replaces EN2T Status Message **
5 - Task Utilization (always 100 with continuous task)
9 - MSG Utilization
11 - I/O Utilization

4

Ethernet Module
Status Enable Scan
Monitor
Enet02.CPU_Util.EN
<EDC_Enet02.CPU_Util.EN>

Ethernet Module --
L85 Ethernet Link
Status Message
EDC_Enet02_UtilMsg.EN

Ethernet Module --
L85 Ethernet Link
Status Message
EDC_Enet02_UtilMsg.DN

Ethernet Module --
L85 Ethernet Link
Status Message

MSG
Message Control EDC_Enet02_UtilMsg ...
EN
DN
ER

Ethernet Module
Status Last Scan

MUL
Source A EDC_Enet02_Util[9]
62
Source B 10
Dest Enet02.CPU_Util.Last
<EDC_Enet02.CPU_Util.Last>
620

Ethernet Module
Status History Array

MOVE
Source Enet02.CPU_Util.Last
<EDC_Enet02.CPU_Util.Last>
620
Dest Enet02.CPU_Util.History[Enet02.CPU_Util.i]
<EDC_Enet02.CPU_Util.History[Enet02.CPU_Util.i]>
620

Ethernet Module
Status History Array
Index

ADD
Source A Enet02.CPU_Util.i
<EDC_Enet02.CPU_Util.i>

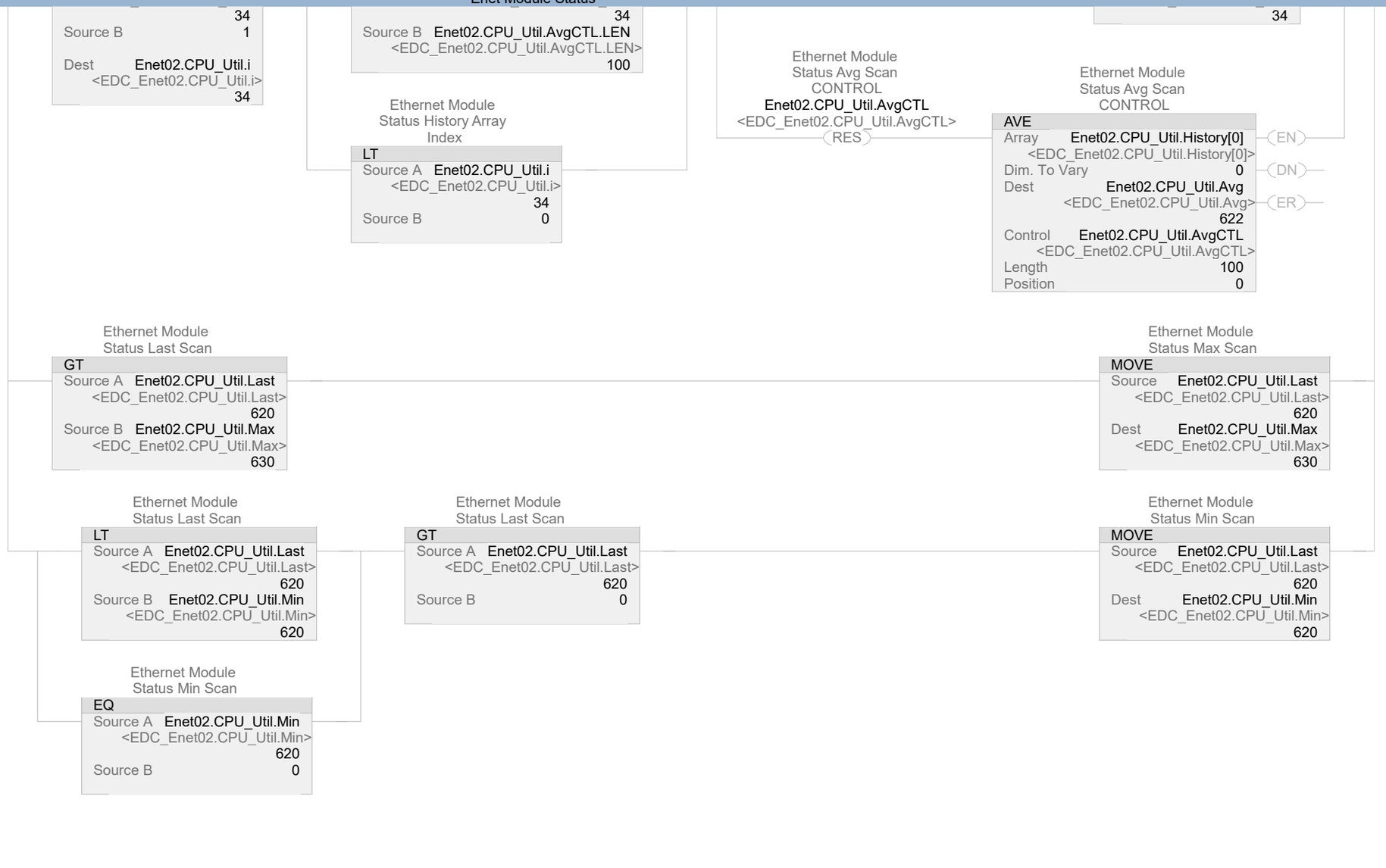
Ethernet Module
Status History Array
Index

GE
Source A Enet02.CPU_Util.i
<EDC_Enet02.CPU_Util.i>

Ethernet Module
Status History Array
Index

CLR
Dest Enet02.CPU_Util.i
<EDC_Enet02.CPU_Util.i>

Enet Module Status



Enet Module Status

Log Enet Utilization above 50% and 75%
(only log this once per second to avoid filling log)

{v1.0.1} -- Reset Log Event Timer
DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

5

Ethernet Module Status Last Scan
GT
Source A Enet02.CPU_Util.Last <EDC_Enet02.CPU_Util.Last> 620
Source B 500

Ethernet Module Status Last Scan
LT
Source A Enet02.CPU_Util.Last <EDC_Enet02.CPU_Util.Last> 620
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet02.UtilOver50Logged <EDC_Enet02.UtilOver50Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet02.CPU_Util.Last <EDC_Enet02.CPU_Util.Last> 620
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 50% Timer
Enet02.UtilOver50_TMR <EDC_Enet02.UtilOver50_TMR>
RES

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet02.UtilOver50Logged <EDC_Enet02.UtilOver50Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8232
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status Last Scan
GE
Source A Enet02.CPU_Util.Last <EDC_Enet02.CPU_Util.Last> 620
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet02.UtilOver75Logged <EDC_Enet02.UtilOver75Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet02.CPU_Util.Last <EDC_Enet02.CPU_Util.Last> 620
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet02.UtilOver75Logged <EDC_Enet02.UtilOver75Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8233
Input Par 0
Input Par -1
Input Par -1

Enet Module Status

Input Par 0

Log Enet Utilization above 50% and 75% -- reset 1-shot after time delay to allow another Transaction log entry
(only log this once per second to avoid filling log)

DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

6

Ethernet Module
Status Ethernet CPU
Utilization above
50% 1-shot
Enet02.UtilOver50Logged
<EDC_Enet02.UtilOver50Logged>

Ethernet Module
Status Ethernet CPU
Utilization above
75% 1-shot
Enet02.UtilOver75Logged
<EDC_Enet02.UtilOver75Logged>

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer

MOVE
Source 1000
Dest Enet02.UtilOver50_TMR.PRE
<EDC_Enet02.UtilOver50_TMR.PRE>
1000

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer

TON
Timer Enet02.UtilOver50_TMR
<EDC_Enet02.UtilOver50_TMR>
Preset 1000
Accum 2062
EN
DN

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer
Enet02.UtilOver50_TMR.DN
<EDC_Enet02.UtilOver50_TMR.DN>

Ethernet Module
Status Ethernet CPU
Utilization above
50% 1-shot
Enet02.UtilOver50Logged
<EDC_Enet02.UtilOver50Logged>
U

Ethernet Module
Status Ethernet CPU
Utilization above
75% 1-shot
Enet02.UtilOver75Logged
<EDC_Enet02.UtilOver75Logged>
U

Enet Module Status

Log Enet Module Overall Status OK & Not OK
-- if Not OK, Clear Sockets Ready

7

Ethernet Module Status GSV Module - Entry Status OK
Enet02.StatusOK
<EDC_Enet02.StatusOK>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet02.StatusOkONS
<EDC_Enet02.StatusOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8201
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet02.StatusOK
<EDC_Enet02.StatusOK>
/

Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet02.StatusNotOkONS
<EDC_Enet02.StatusNotOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8203
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Enet Module Status

Log Enet Module Entry Status OK & Not OK
-- log Status Not OK each time Entry Status changs

8

Ethernet Module Status GSV Module - Entry Status OK
Enet02.EntryStatusOk
<EDC_Enet02.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet02.EntryStatusOkONS
<EDC_Enet02.EntryStatusOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet02.EntryStatus <EDC_Enet02.EntryStatus> 16#4000
Source B Enet02.EntryStatusMask <EDC_Enet02.EntryStatusMask> 16#f000
Dest Enet02.EntryStatusLast <EDC_Enet02.EntryStatusLast> 16#4000

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet02.EntryStatusLast <EDC_Enet02.EntryStatusLast> 16#4000
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8211
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet02.EntryStatusOk
<EDC_Enet02.EntryStatusOk>
/

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet02.EntryStatus <EDC_Enet02.EntryStatus> 16#4000
Mask Enet02.EntryStatusMask <EDC_Enet02.EntryStatusMask> 16#f000
Compare Enet02.EntryStatusLast <EDC_Enet02.EntryStatusLast> 16#4000

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet02.EntryStatusSame
<EDC_Enet02.EntryStatusSame>

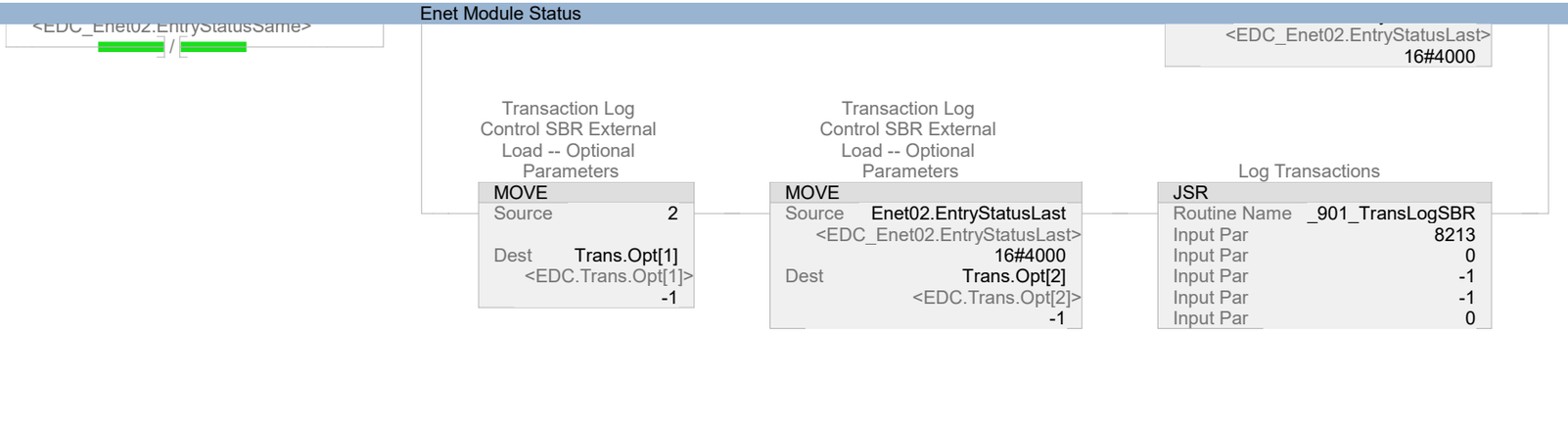
Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet02.EntryStatusNotOkONS
<EDC_Enet02.EntryStatusNotOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet02.EntryStatusSame
<EDC_Enet02.EntryStatusSame>

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet02.EntryStatus <EDC_Enet02.EntryStatus> 16#4000
Source B Enet02.EntryStatusMask <EDC_Enet02.EntryStatusMask> 16#f000
Dest Enet02.EntryStatusLast

Enet Module Status



Enet Module Status

Log Enet Module Fault Code OK & Not OK
-- log Status Not OK each time Fault Code changs

9

Ethernet Module Status GSV Module - Fault Code OK
Enet02.FaultCodeOk
<EDC_Enet02.FaultCodeOk>

Ethernet Module Status GSV Module - Fault Code OK 1-Shot
Enet02.FaultCodeOkONS
<EDC_Enet02.FaultCodeOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source Enet02.FaultCode
<EDC_Enet02.FaultCode> 16#0000
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8221
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Fault Code OK
Enet02.FaultCodeOk
<EDC_Enet02.FaultCodeOk>
/

Ethernet Module Status GSV Module - Fault Code Not OK 1-Shot
Enet02.FaultCodeNotOkONS
<EDC_Enet02.FaultCodeNotOkONS>
ONS

Ethernet Module Status GSV Module - Fault Code
NE
Source A Enet02.FaultCode
<EDC_Enet02.FaultCode> 16#0000
Source B Enet02.FaultCodeLast
<EDC_Enet02.FaultCodeLast> 16#0204

Ethernet Module Status GSV Module - Fault Code Last Value
MOVE
Source Enet02.FaultCode
<EDC_Enet02.FaultCode> 16#0000
Dest Enet02.FaultCodeLast
<EDC_Enet02.FaultCodeLast> 16#0204

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source Enet02.FaultCode
<EDC_Enet02.FaultCode> 16#0000
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8223
Input Par 0
Input Par -1
Input Par -1
Input Par 0

(End)

Enet Module Status

Enet Module Control and Status

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 31
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Enet Module Status

Get Enet Module Status

2

Ethernet Module Status GSV Module - Entry Status

GSV
Class Name Module
Instance Name EDC_FEC1
Attribute Name EntryStatus
Dest Enet03.EntryStatus
<EDC_Enet03.EntryStatus>
16#4000

Ethernet Module Status GSV Module - Fault Code

GSV
Class Name Module
Instance Name EDC_FEC1
Attribute Name FaultCode
Dest Enet03.FaultCode
<EDC_Enet03.FaultCode>
16#0000

Ethernet Module Status GSV Module - Fault Info

GSV
Class Name Module
Instance Name EDC_FEC1
Attribute Name FaultInfo
Dest Enet03.FaultInfo
<EDC_Enet03.FaultInfo>
16#0000_0000

Ethernet Module Status GSV Module - Mode

GSV
Class Name Module
Instance Name EDC_FEC1
Attribute Name Mode
Dest Enet03.Mode
<EDC_Enet03.Mode>
16#0000

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet03.EntryStatus
<EDC_Enet03.EntryStatus>
16#4000
Mask Enet03.EntryStatusMask
<EDC_Enet03.EntryStatusMask>
16#f000
Compare Enet03.EntryStatusRunChk
<EDC_Enet03.EntryStatusRunChk>
16#4000

Ethernet Module Status GSV Module - Entry Status OK

Enet03.EntryStatusOk
<EDC_Enet03.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet03.EntryStatus
<EDC_Enet03.EntryStatus>
16#4000
Mask Enet03.EntryStatusMask
<EDC_Enet03.EntryStatusMask>
16#f000
Compare ENet_InhibChk
16#6000

Ethernet Module Status GSV Module - Fault Code

EQ
Source A Enet03.FaultCode
<EDC_Enet03.FaultCode>
16#0000
Source B 0

Ethernet Module Status GSV Module - Fault Code OK

Enet03.FaultCodeOk
<EDC_Enet03.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK

Enet03.EntryStatusOk
<EDC_Enet03.EntryStatusOk>

Ethernet Module Status GSV Module - Fault Code OK

Enet03.FaultCodeOk
<EDC_Enet03.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK

Enet03.StatusOK
<EDC_Enet03.StatusOK>

Enet Module Status

3

Ethernet Module CPU Utilization
-- Clear Values
-- Initialize Array Length

Ethernet Module
Status Clear Scan
History
Enet03.CPU_Util.Clear
<EDC_Enet03.CPU_Util.Clear>

S:FS

Ethernet Module
Status Ethernet CPU
Utilization

FLL
Source 0
Dest Enet03.CPU_Util
<EDC_Enet03.CPU_Util>
Length 1

Ethernet Module
Status Enable Scan
Monitor
Enet03.CPU_Util.EN
<EDC_Enet03.CPU_Util.EN>

L

Ethernet Module
Status History Array

SIZE
Source Enet03.CPU_Util.History[0]
<EDC_Enet03.CPU_Util.History[0]>
290
Dim. To Vary 0
Size Enet03.CPU_Util.AvgCTL.LEN
<EDC_Enet03.CPU_Util.AvgCTL.LEN>
100

Enet Module Status

Ethernet Module CPU Utilization
-- Store New Value in History Array
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value
-- Get Next Value

(EN2T v2 Module CPU_Util is shifted 1 Byte in Status Packet -- use MEQ to auto detect this and correct)
DHF005 - Add CLR Enet03.CPU_Util.Last to ensure upper bytes are 0

** SPECIAL CASE -- L85 Utilization Message replaces EN2T Status Message **
5 - Task Utilization (always 100 with continuous task)
9 - MSG Utilization
11 - I/O Utilization

4

Ethernet Module Status Enable Scan Monitor
Enet03.CPU_Util.EN
<EDC_Enet03.CPU_Util.EN>

Ethernet Module -- L85 Ethernet Link Status Message
EDC_Enet03_UtilMsg.EN

Ethernet Module -- L85 Ethernet Link Status Message
EDC_Enet03_UtilMsg.DN

Ethernet Module -- L85 Ethernet Link Status Message

MSG
Message Control EDC_Enet03_UtilMsg ...
EN
DN
ER

Ethernet Module Status Last Scan

MUL
Source A EDC_Enet03_Util[9]
28
Source B 10
Dest Enet03.CPU_Util.Last
<EDC_Enet03.CPU_Util.Last>
350

Ethernet Module Status History Array

MOVE
Source Enet03.CPU_Util.Last
<EDC_Enet03.CPU_Util.Last>
350
Dest Enet03.CPU_Util.History[Enet03.CPU_Util.i]
<EDC_Enet03.CPU_Util.History[Enet03.CPU_Util.i]>
190

Ethernet Module Status History Array Index

ADD
Source A Enet03.CPU_Util.i
<EDC_Enet03.CPU_Util.i>

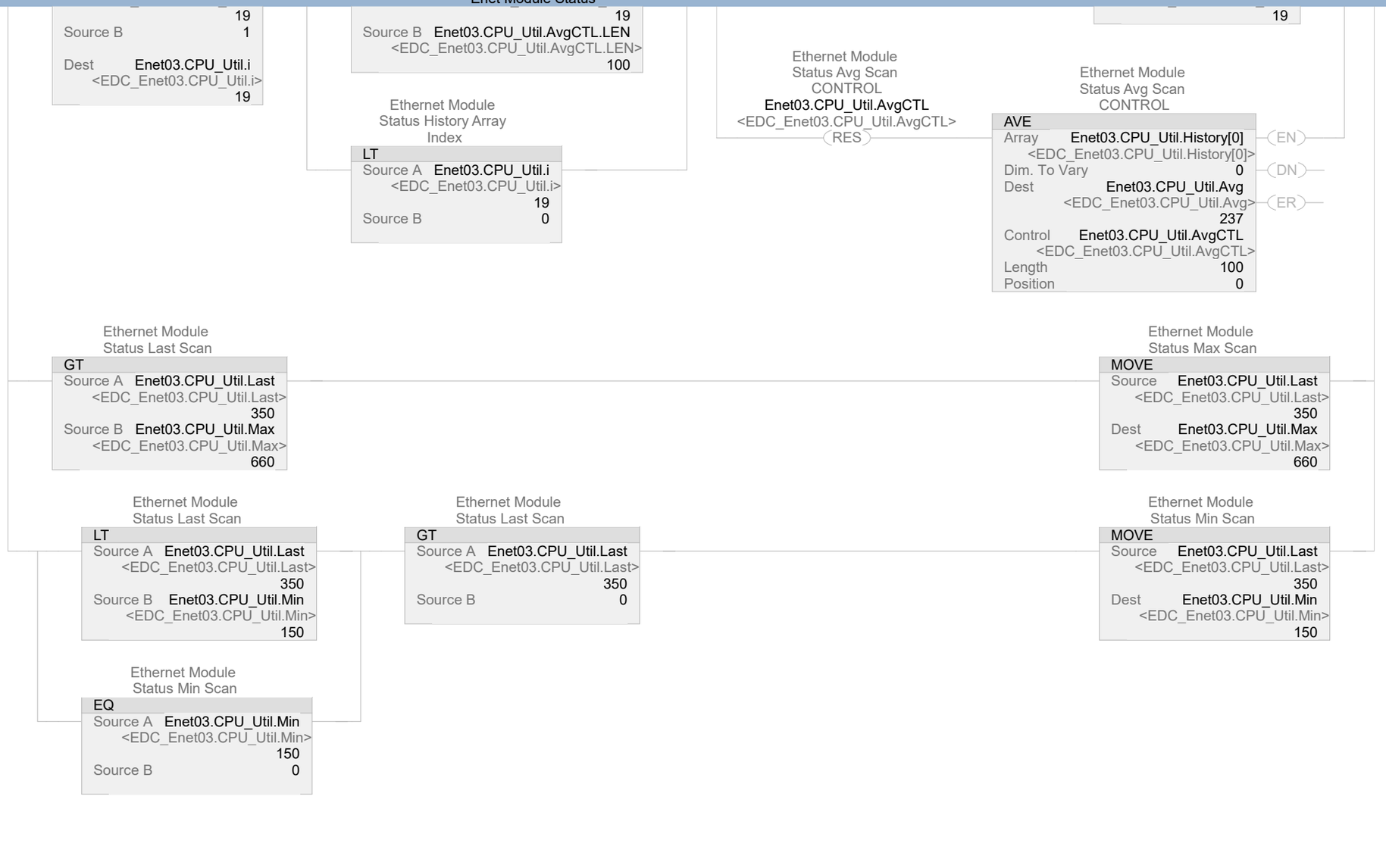
Ethernet Module Status History Array Index

GE
Source A Enet03.CPU_Util.i
<EDC_Enet03.CPU_Util.i>

Ethernet Module Status History Array Index

CLR
Dest Enet03.CPU_Util.i
<EDC_Enet03.CPU_Util.i>

Enet Module Status



Enet Module Status

Log Enet Utilization above 50% and 75%
(only log this once per second to avoid filling log)

{v1.0.1} -- Reset Log Event Timer
DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

5

Ethernet Module Status Last Scan
GT
Source A Enet03.CPU_Util.Last <EDC_Enet03.CPU_Util.Last> 350
Source B 500

Ethernet Module Status Last Scan
LT
Source A Enet03.CPU_Util.Last <EDC_Enet03.CPU_Util.Last> 350
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet03.UtilOver50Logged <EDC_Enet03.UtilOver50Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet03.CPU_Util.Last <EDC_Enet03.CPU_Util.Last> 350
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 50% Timer
Enet03.UtilOver50_TMR <EDC_Enet03.UtilOver50_TMR>
RES

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet03.UtilOver50Logged <EDC_Enet03.UtilOver50Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8232
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status Last Scan
GE
Source A Enet03.CPU_Util.Last <EDC_Enet03.CPU_Util.Last> 350
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet03.UtilOver75Logged <EDC_Enet03.UtilOver75Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet03.CPU_Util.Last <EDC_Enet03.CPU_Util.Last> 350
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet03.UtilOver75Logged <EDC_Enet03.UtilOver75Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8233
Input Par 0
Input Par -1
Input Par -1

Enet Module Status

Input Par 0

Log Enet Utilization above 50% and 75% -- reset 1-shot after time delay to allow another Transaction log entry
(only log this once per second to avoid filling log)

DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

6

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet03.UtilOver50Logged
<EDC_Enet03.UtilOver50Logged>

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet03.UtilOver75Logged
<EDC_Enet03.UtilOver75Logged>

Ethernet Module Status Ethernet CPU Utilization above 50% Timer

MOVE
Source 1000
Dest Enet03.UtilOver50_TMR.PRE
<EDC_Enet03.UtilOver50_TMR.PRE>
1000

Ethernet Module Status Ethernet CPU Utilization above 50% Timer

TON
Timer Enet03.UtilOver50_TMR
<EDC_Enet03.UtilOver50_TMR>
Preset 1000
Accum 0
EN
DN

Ethernet Module Status Ethernet CPU Utilization above 50% Timer
Enet03.UtilOver50_TMR.DN
<EDC_Enet03.UtilOver50_TMR.DN>

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet03.UtilOver50Logged
<EDC_Enet03.UtilOver50Logged>
U

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet03.UtilOver75Logged
<EDC_Enet03.UtilOver75Logged>
U

Enet Module Status

Log Enet Module Overall Status OK & Not OK
-- if Not OK, Clear Sockets Ready

7

Ethernet Module Status GSV Module - Entry Status OK
Enet03.StatusOK
<EDC_Enet03.StatusOK>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet03.StatusOkONS
<EDC_Enet03.StatusOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8201
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet03.StatusOK
<EDC_Enet03.StatusOK>
/

Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet03.StatusNotOkONS
<EDC_Enet03.StatusNotOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8203
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Enet Module Status

Log Enet Module Entry Status OK & Not OK
-- log Status Not OK each time Entry Status changs

8

Ethernet Module Status GSV Module - Entry Status OK
Enet03.EntryStatusOk
<EDC_Enet03.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet03.EntryStatusOkONS
<EDC_Enet03.EntryStatusOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Last Value

AND		
Source A	Enet03.EntryStatus	<EDC_Enet03.EntryStatus>
	16#4000	
Source B	Enet03.EntryStatusMask	<EDC_Enet03.EntryStatusMask>
	16#f000	
Dest	Enet03.EntryStatusLast	<EDC_Enet03.EntryStatusLast>
	16#4000	

Transaction Log Control SBR External Load -- Optional Parameters

MOVE		
Source	3	
Dest	Trans.Opt[1]	<EDC.Trans.Opt[1]>
	-1	

Transaction Log Control SBR External Load -- Optional Parameters

MOVE		
Source	Enet03.EntryStatusLast	<EDC_Enet03.EntryStatusLast>
	16#4000	
Dest	Trans.Opt[2]	<EDC.Trans.Opt[2]>
	-1	

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	8211
Input Par	0
Input Par	-1
Input Par	-1
Input Par	0

Ethernet Module Status GSV Module - Entry Status OK
Enet03.EntryStatusOk
<EDC_Enet03.EntryStatusOk>
/

Ethernet Module Status GSV Module - Entry Status

MEQ		
Source	Enet03.EntryStatus	<EDC_Enet03.EntryStatus>
	16#4000	
Mask	Enet03.EntryStatusMask	<EDC_Enet03.EntryStatusMask>
	16#f000	
Compare	Enet03.EntryStatusLast	<EDC_Enet03.EntryStatusLast>
	16#4000	

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet03.EntryStatusSame
<EDC_Enet03.EntryStatusSame>

Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet03.EntryStatusNotOkONS
<EDC_Enet03.EntryStatusNotOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet03.EntryStatusSame

Ethernet Module Status GSV Module - Entry Status Last Value

AND		
Source A	Enet03.EntryStatus	<EDC_Enet03.EntryStatus>
	16#4000	
Source B	Enet03.EntryStatusMask	<EDC_Enet03.EntryStatusMask>
	16#f000	
Dest	Enet03.EntryStatusLast	

Enet Module Status

<EDC_Enet03.EntryStatusSame>
/

<EDC_Enet03.EntryStatusLast>
16#4000

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet03.EntryStatusLast
<EDC_Enet03.EntryStatusLast>
16#4000
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8213
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Enet Module Status

Log Enet Module Fault Code OK & Not OK
-- log Status Not OK each time Fault Code changs

9

Ethernet Module Status GSV Module - Fault Code OK
Enet03.FaultCodeOk
<EDC_Enet03.FaultCodeOk>

Ethernet Module Status GSV Module - Fault Code OK 1-Shot
Enet03.FaultCodeOkONS
<EDC_Enet03.FaultCodeOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	3
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Enet03.FaultCode
	<EDC_Enet03.FaultCode>
	16#0000
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	8221
Input Par	0
Input Par	-1
Input Par	-1
Input Par	0

Ethernet Module Status GSV Module - Fault Code OK
Enet03.FaultCodeOk
<EDC_Enet03.FaultCodeOk>
/

Ethernet Module Status GSV Module - Fault Code Not OK 1-Shot
Enet03.FaultCodeNotOkONS
<EDC_Enet03.FaultCodeNotOkONS>
ONS

Ethernet Module Status GSV Module - Fault Code

NE	
Source A	Enet03.FaultCode
	<EDC_Enet03.FaultCode>
	16#0000
Source B	Enet03.FaultCodeLast
	<EDC_Enet03.FaultCodeLast>
	16#0005

Ethernet Module Status GSV Module - Fault Code Last Value

MOVE	
Source	Enet03.FaultCode
	<EDC_Enet03.FaultCode>
	16#0000
Dest	Enet03.FaultCodeLast
	<EDC_Enet03.FaultCodeLast>
	16#0005

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	3
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Enet03.FaultCode
	<EDC_Enet03.FaultCode>
	16#0000
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	8223
Input Par	0
Input Par	-1
Input Par	-1
Input Par	0

(End)

Enet Module Status

Enet Module Control and Status

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 31
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Enet Module Status

Get Enet Module Status

2

Ethernet Module Status GSV Module - Entry Status

GSV
Class Name Module
Instance Name EDC_FEC2
Attribute Name EntryStatus
Dest Enet04.EntryStatus
<EDC_Enet04.EntryStatus>
16#6000

Ethernet Module Status GSV Module - Fault Code

GSV
Class Name Module
Instance Name EDC_FEC2
Attribute Name FaultCode
Dest Enet04.FaultCode
<EDC_Enet04.FaultCode>
16#0000

Ethernet Module Status GSV Module - Fault Info

GSV
Class Name Module
Instance Name EDC_FEC2
Attribute Name FaultInfo
Dest Enet04.FaultInfo
<EDC_Enet04.FaultInfo>
16#0000_0000

Ethernet Module Status GSV Module - Mode

GSV
Class Name Module
Instance Name EDC_FEC2
Attribute Name Mode
Dest Enet04.Mode
<EDC_Enet04.Mode>
16#0004

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet04.EntryStatus
<EDC_Enet04.EntryStatus>
16#6000
Mask Enet04.EntryStatusMask
<EDC_Enet04.EntryStatusMask>
16#f000
Compare Enet04.EntryStatusRunChk
<EDC_Enet04.EntryStatusRunChk>
16#4000

Ethernet Module Status GSV Module - Entry Status OK

Enet04.EntryStatusOk
<EDC_Enet04.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet04.EntryStatus
<EDC_Enet04.EntryStatus>
16#6000
Mask Enet04.EntryStatusMask
<EDC_Enet04.EntryStatusMask>
16#f000
Compare ENet_InhibChk
16#6000

Ethernet Module Status GSV Module - Fault Code

EQ
Source A Enet04.FaultCode
<EDC_Enet04.FaultCode>
16#0000
Source B 0

Ethernet Module Status GSV Module - Fault Code OK

Enet04.FaultCodeOk
<EDC_Enet04.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK

Enet04.EntryStatusOk
<EDC_Enet04.EntryStatusOk>

Ethernet Module Status GSV Module - Fault Code OK

Enet04.FaultCodeOk
<EDC_Enet04.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK

Enet04.StatusOK
<EDC_Enet04.StatusOK>

Enet Module Status

3

Ethernet Module CPU Utilization
-- Clear Values
-- Initialize Array Length

Ethernet Module
Status Clear Scan
History
Enet04.CPU_Util.Clear
<EDC_Enet04.CPU_Util.Clear>

S:FS

Ethernet Module
Status Ethernet CPU
Utilization

FLL
Source 0
Dest Enet04.CPU_Util
<EDC_Enet04.CPU_Util>
Length 1

Ethernet Module
Status Enable Scan
Monitor
Enet04.CPU_Util.EN
<EDC_Enet04.CPU_Util.EN>

L

Ethernet Module
Status History Array

SIZE
Source Enet04.CPU_Util.History[0]
<EDC_Enet04.CPU_Util.History[0]>
130
Dim. To Vary 0
Size Enet04.CPU_Util.AvgCTL.LEN
<EDC_Enet04.CPU_Util.AvgCTL.LEN>
100

Enet Module Status

Ethernet Module CPU Utilization
-- Store New Value in History Array
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value
-- Get Next Value

(EN2T v2 Module CPU_Util is shifted 1 Byte in Status Packet -- use MEQ to auto detect this and correct)
DHF005 - Add CLR Enet03.CPU_Util.Last to ensure upper bytes are 0

** SPECIAL CASE -- L85 Utilization Message replaces EN2T Status Message **
5 - Task Utilization (always 100 with continuous task)
9 - MSG Utilization
11 - I/O Utilization

4

Ethernet Module Status Enable Scan Monitor
Enet04.CPU_Util.EN
<EDC_Enet04.CPU_Util.EN>

Ethernet Module -- L85 Ethernet Link Status Message
EDC_Enet04_UtilMsg.EN
/

Ethernet Module -- L85 Ethernet Link Status Message
EDC_Enet04_UtilMsg.DN

Ethernet Module -- L85 Ethernet Link Status Message
MSG
Message Control EDC_Enet04_UtilMsg ...
EN
DN
ER

Ethernet Module Status Last Scan
MUL
Source A EDC_Enet04_Util[9]
14
Source B 10
Dest Enet04.CPU_Util.Last
<EDC_Enet04.CPU_Util.Last>
130

Ethernet Module Status History Array
MOVE
Source Enet04.CPU_Util.Last
<EDC_Enet04.CPU_Util.Last>
130
Dest Enet04.CPU_Util.History[Enet04.CPU_Util.i]
<EDC_Enet04.CPU_Util.History[Enet04.CPU_Util.i]>
130

Ethernet Module Status History Array Index
ADD
Source A Enet04.CPU_Util.i
<EDC_Enet04.CPU_Util.i>

Ethernet Module Status History Array Index
GE
Source A Enet04.CPU_Util.i
<EDC_Enet04.CPU_Util.i>

Ethernet Module Status History Array Index
CLR
Dest Enet04.CPU_Util.i
<EDC_Enet04.CPU_Util.i>

Enet Module Status

75
Source B 1
Dest Enet04.CPU_Util.i
<EDC_Enet04.CPU_Util.i>
75

75
Source B Enet04.CPU_Util.AvgCTL.LEN
<EDC_Enet04.CPU_Util.AvgCTL.LEN>
100

Ethernet Module Status History Array Index

LT
Source A Enet04.CPU_Util.i
<EDC_Enet04.CPU_Util.i>
75
Source B 0

Ethernet Module Status Avg Scan CONTROL
Enet04.CPU_Util.AvgCTL
<EDC_Enet04.CPU_Util.AvgCTL>
RES

75

Ethernet Module Status Avg Scan CONTROL

AVE
Array Enet04.CPU_Util.History[0]
<EDC_Enet04.CPU_Util.History[0]>
Dim. To Vary 0
Dest Enet04.CPU_Util.Avg
<EDC_Enet04.CPU_Util.Avg>
134
Control Enet04.CPU_Util.AvgCTL
<EDC_Enet04.CPU_Util.AvgCTL>
Length 100
Position 0
EN
DN
ER

Ethernet Module Status Last Scan

GT
Source A Enet04.CPU_Util.Last
<EDC_Enet04.CPU_Util.Last>
130
Source B Enet04.CPU_Util.Max
<EDC_Enet04.CPU_Util.Max>
170

Ethernet Module Status Max Scan

MOVE
Source Enet04.CPU_Util.Last
<EDC_Enet04.CPU_Util.Last>
130
Dest Enet04.CPU_Util.Max
<EDC_Enet04.CPU_Util.Max>
170

Ethernet Module Status Last Scan

LT
Source A Enet04.CPU_Util.Last
<EDC_Enet04.CPU_Util.Last>
130
Source B Enet04.CPU_Util.Min
<EDC_Enet04.CPU_Util.Min>
120

Ethernet Module Status Last Scan

GT
Source A Enet04.CPU_Util.Last
<EDC_Enet04.CPU_Util.Last>
130
Source B 0

Ethernet Module Status Min Scan

MOVE
Source Enet04.CPU_Util.Last
<EDC_Enet04.CPU_Util.Last>
130
Dest Enet04.CPU_Util.Min
<EDC_Enet04.CPU_Util.Min>
120

Ethernet Module Status Min Scan

EQ
Source A Enet04.CPU_Util.Min
<EDC_Enet04.CPU_Util.Min>
120
Source B 0

Enet Module Status

Log Enet Utilization above 50% and 75%
(only log this once per second to avoid filling log)

{v1.0.1} -- Reset Log Event Timer
DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

5

Ethernet Module Status Last Scan
GT
Source A Enet04.CPU_Util.Last <EDC_Enet04.CPU_Util.Last> 130
Source B 500

Ethernet Module Status Last Scan
LT
Source A Enet04.CPU_Util.Last <EDC_Enet04.CPU_Util.Last> 130
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet04.UtilOver50Logged <EDC_Enet04.UtilOver50Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet04.CPU_Util.Last <EDC_Enet04.CPU_Util.Last> 130
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 50% Timer
Enet04.UtilOver50_TMR <EDC_Enet04.UtilOver50_TMR>
RES

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet04.UtilOver50Logged <EDC_Enet04.UtilOver50Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8232
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status Last Scan
GE
Source A Enet04.CPU_Util.Last <EDC_Enet04.CPU_Util.Last> 130
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet04.UtilOver75Logged <EDC_Enet04.UtilOver75Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet04.CPU_Util.Last <EDC_Enet04.CPU_Util.Last> 130
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet04.UtilOver75Logged <EDC_Enet04.UtilOver75Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8233
Input Par 0
Input Par -1
Input Par -1

Enet Module Status

Input Par 0

Log Enet Utilization above 50% and 75% -- reset 1-shot after time delay to allow another Transaction log entry
(only log this once per second to avoid filling log)

DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

6

Ethernet Module
Status Ethernet CPU
Utilization above
50% 1-shot
Enet04.UtilOver50Logged
<EDC_Enet04.UtilOver50Logged>

Ethernet Module
Status Ethernet CPU
Utilization above
75% 1-shot
Enet04.UtilOver75Logged
<EDC_Enet04.UtilOver75Logged>

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer

MOVE
Source 1000
Dest Enet04.UtilOver50_TMR.PRE
<EDC_Enet04.UtilOver50_TMR.PRE>
0

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer

TON
Timer Enet04.UtilOver50_TMR
<EDC_Enet04.UtilOver50_TMR>
Preset 0
Accum 0
EN
DN

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer
Enet04.UtilOver50_TMR.DN
<EDC_Enet04.UtilOver50_TMR.DN>

Ethernet Module
Status Ethernet CPU
Utilization above
50% 1-shot
Enet04.UtilOver50Logged
<EDC_Enet04.UtilOver50Logged>
U

Ethernet Module
Status Ethernet CPU
Utilization above
75% 1-shot
Enet04.UtilOver75Logged
<EDC_Enet04.UtilOver75Logged>
U

Enet Module Status

Log Enet Module Overall Status OK & Not OK
-- if Not OK, Clear Sockets Ready

7

Ethernet Module Status GSV Module - Entry Status OK
Enet04.StatusOK
<EDC_Enet04.StatusOK>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet04.StatusOkONS
<EDC_Enet04.StatusOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8201
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet04.StatusOK
<EDC_Enet04.StatusOK>
/

Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet04.StatusNotOkONS
<EDC_Enet04.StatusNotOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8203
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Enet Module Status

Log Enet Module Entry Status OK & Not OK
-- log Status Not OK each time Entry Status changs

8

Ethernet Module Status GSV Module - Entry Status OK
Enet04.EntryStatusOk
<EDC_Enet04.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet04.EntryStatusOkONS
<EDC_Enet04.EntryStatusOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet04.EntryStatus <EDC_Enet04.EntryStatus> 16#6000
Source B Enet04.EntryStatusMask <EDC_Enet04.EntryStatusMask> 16#f000
Dest Enet04.EntryStatusLast <EDC_Enet04.EntryStatusLast> 16#1000

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet04.EntryStatusLast <EDC_Enet04.EntryStatusLast> 16#1000
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8211
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet04.EntryStatusOk
<EDC_Enet04.EntryStatusOk>
/

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet04.EntryStatus <EDC_Enet04.EntryStatus> 16#6000
Mask Enet04.EntryStatusMask <EDC_Enet04.EntryStatusMask> 16#f000
Compare Enet04.EntryStatusLast <EDC_Enet04.EntryStatusLast> 16#1000

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet04.EntryStatusSame
<EDC_Enet04.EntryStatusSame>

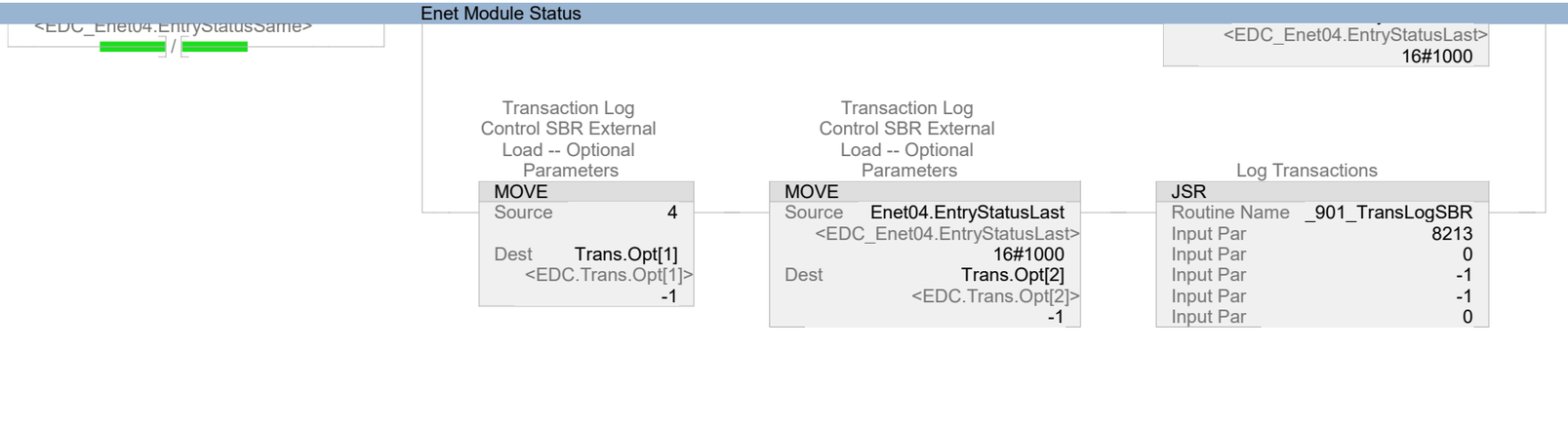
Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet04.EntryStatusNotOkONS
<EDC_Enet04.EntryStatusNotOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet04.EntryStatusSame
<EDC_Enet04.EntryStatusSame>

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet04.EntryStatus <EDC_Enet04.EntryStatus> 16#6000
Source B Enet04.EntryStatusMask <EDC_Enet04.EntryStatusMask> 16#f000
Dest Enet04.EntryStatusLast

Enet Module Status



Enet Module Status

Log Enet Module Fault Code OK & Not OK
-- log Status Not OK each time Fault Code changs

9

Ethernet Module Status GSV Module - Fault Code OK
Enet04.FaultCodeOk
<EDC_Enet04.FaultCodeOk>

Ethernet Module Status GSV Module - Fault Code OK 1-Shot
Enet04.FaultCodeOkONS
<EDC_Enet04.FaultCodeOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet04.FaultCode
<EDC_Enet04.FaultCode> 16#0000
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8221
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Fault Code OK
Enet04.FaultCodeOk
<EDC_Enet04.FaultCodeOk>
/

Ethernet Module Status GSV Module - Fault Code Not OK 1-Shot
Enet04.FaultCodeNotOkONS
<EDC_Enet04.FaultCodeNotOkONS>
ONS

Ethernet Module Status GSV Module - Fault Code

NE
Source A Enet04.FaultCode
<EDC_Enet04.FaultCode> 16#0000
Source B Enet04.FaultCodeLast
<EDC_Enet04.FaultCodeLast> 16#0315

Ethernet Module Status GSV Module - Fault Code Last Value

MOVE
Source Enet04.FaultCode
<EDC_Enet04.FaultCode> 16#0000
Dest Enet04.FaultCodeLast
<EDC_Enet04.FaultCodeLast> 16#0315

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet04.FaultCode
<EDC_Enet04.FaultCode> 16#0000
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8223
Input Par 0
Input Par -1
Input Par -1
Input Par 0

(End)

Enet Module Status

Enet Module Control and Status

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	31
Dest	Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1733165816
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1733165816

Enet Module Status

Get Enet Module Status

2

Ethernet Module Status GSV Module - Entry Status

GSV
Class Name Module
Instance Name EDC_FEC3
Attribute Name EntryStatus
Dest Enet05.EntryStatus
<EDC_Enet05.EntryStatus>
16#6000

Ethernet Module Status GSV Module - Fault Code

GSV
Class Name Module
Instance Name EDC_FEC3
Attribute Name FaultCode
Dest Enet05.FaultCode
<EDC_Enet05.FaultCode>
16#0000

Ethernet Module Status GSV Module - Fault Info

GSV
Class Name Module
Instance Name EDC_FEC3
Attribute Name FaultInfo
Dest Enet05.FaultInfo
<EDC_Enet05.FaultInfo>
16#0000_0000

Ethernet Module Status GSV Module - Mode

GSV
Class Name Module
Instance Name EDC_FEC3
Attribute Name Mode
Dest Enet05.Mode
<EDC_Enet05.Mode>
16#0004

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet05.EntryStatus
<EDC_Enet05.EntryStatus>
16#6000
Mask Enet05.EntryStatusMask
<EDC_Enet05.EntryStatusMask>
16#f000
Compare Enet05.EntryStatusRunChk
<EDC_Enet05.EntryStatusRunChk>
16#4000

Ethernet Module Status GSV Module - Entry Status OK

Enet05.EntryStatusOk
<EDC_Enet05.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet05.EntryStatus
<EDC_Enet05.EntryStatus>
16#6000
Mask Enet05.EntryStatusMask
<EDC_Enet05.EntryStatusMask>
16#f000
Compare ENet_InhibChk
16#6000

Ethernet Module Status GSV Module - Fault Code

EQ
Source A Enet05.FaultCode
<EDC_Enet05.FaultCode>
16#0000
Source B 0

Ethernet Module Status GSV Module - Fault Code OK

Enet05.FaultCodeOk
<EDC_Enet05.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK

Enet05.EntryStatusOk
<EDC_Enet05.EntryStatusOk>

Ethernet Module Status GSV Module - Fault Code OK

Enet05.FaultCodeOk
<EDC_Enet05.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK

Enet05.StatusOK
<EDC_Enet05.StatusOK>

Enet Module Status

3

Ethernet Module CPU Utilization
-- Clear Values
-- Initialize Array Length

Ethernet Module
Status Clear Scan
History
Enet05.CPU_Util.Clear
<EDC_Enet05.CPU_Util.Clear>

S:FS

Ethernet Module
Status Ethernet CPU
Utilization

FLL
Source 0
Dest Enet05.CPU_Util
<EDC_Enet05.CPU_Util>
Length 1

Ethernet Module
Status Enable Scan
Monitor
Enet05.CPU_Util.EN
<EDC_Enet05.CPU_Util.EN>

L

Ethernet Module
Status History Array

SIZE
Source Enet05.CPU_Util.History[0]
<EDC_Enet05.CPU_Util.History[0]>
0
Dim. To Vary 0
Size Enet05.CPU_Util.AvgCTL.LEN
<EDC_Enet05.CPU_Util.AvgCTL.LEN>
100

Enet Module Status

Ethernet Module CPU Utilization
-- Store New Value in History Array
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value
-- Get Next Value

(EN2T v2 Module CPU_Util is shifted 1 Byte in Status Packet -- use MEQ to auto detect this and correct)
DHF005 - Add CLR Enet03.CPU_Util.Last to ensure upper bytes are 0

** SPECIAL CASE -- L85 Utilization Message replaces EN2T Status Message **
5 - Task Utilization (always 100 with continuous task)
9 - MSG Utilization
11 - I/O Utilization

4

Ethernet Module
Status Enable Scan
Monitor
Enet05.CPU_Util.EN
<EDC_Enet05.CPU_Util.EN>

Ethernet Module --
L85 Ethernet Link
Status Message
EDC_Enet05_UtilMsg.EN

Ethernet Module --
L85 Ethernet Link
Status Message
EDC_Enet05_UtilMsg.DN

Ethernet Module --
L85 Ethernet Link
Status Message

MSG
Message Control EDC_Enet05_UtilMsg ...
EN
DN
ER

Ethernet Module
Status Last Scan

MUL
Source A EDC_Enet05_Util[9]
10
Source B 10
Dest Enet05.CPU_Util.Last
<EDC_Enet05.CPU_Util.Last>
0

Ethernet Module
Status History Array

MOVE
Source Enet05.CPU_Util.Last
<EDC_Enet05.CPU_Util.Last>
0
Dest Enet05.CPU_Util.History[Enet05.CPU_Util.i]
<EDC_Enet05.CPU_Util.History[Enet05.CPU_Util.i]>
0

Ethernet Module
Status History Array
Index

ADD
Source A Enet05.CPU_Util.i
<EDC_Enet05.CPU_Util.i>

Ethernet Module
Status History Array
Index

GE
Source A Enet05.CPU_Util.i
<EDC_Enet05.CPU_Util.i>

Ethernet Module
Status History Array
Index

CLR
Dest Enet05.CPU_Util.i
<EDC_Enet05.CPU_Util.i>

Enet Module Status

0
Source B 1
Dest Enet05.CPU_Util.i
<EDC_Enet05.CPU_Util.i>
0

0
Source B Enet05.CPU_Util.AvgCTL.LEN
<EDC_Enet05.CPU_Util.AvgCTL.LEN>
100

Ethernet Module Status History Array Index

LT
Source A Enet05.CPU_Util.i
<EDC_Enet05.CPU_Util.i>
0
Source B 0

Ethernet Module Status Avg Scan CONTROL
Enet05.CPU_Util.AvgCTL
<EDC_Enet05.CPU_Util.AvgCTL>
RES

0

Ethernet Module Status Avg Scan CONTROL

AVE
Array Enet05.CPU_Util.History[0]
<EDC_Enet05.CPU_Util.History[0]>
Dim. To Vary 0
Dest Enet05.CPU_Util.Avg
<EDC_Enet05.CPU_Util.Avg>
0
Control Enet05.CPU_Util.AvgCTL
<EDC_Enet05.CPU_Util.AvgCTL>
Length 100
Position 0
EN
DN
ER

Ethernet Module Status Last Scan

GT
Source A Enet05.CPU_Util.Last
<EDC_Enet05.CPU_Util.Last>
0
Source B Enet05.CPU_Util.Max
<EDC_Enet05.CPU_Util.Max>
0

Ethernet Module Status Max Scan

MOVE
Source Enet05.CPU_Util.Last
<EDC_Enet05.CPU_Util.Last>
0
Dest Enet05.CPU_Util.Max
<EDC_Enet05.CPU_Util.Max>
0

Ethernet Module Status Last Scan

LT
Source A Enet05.CPU_Util.Last
<EDC_Enet05.CPU_Util.Last>
0
Source B Enet05.CPU_Util.Min
<EDC_Enet05.CPU_Util.Min>
0

Ethernet Module Status Last Scan

GT
Source A Enet05.CPU_Util.Last
<EDC_Enet05.CPU_Util.Last>
0
Source B 0

Ethernet Module Status Min Scan

MOVE
Source Enet05.CPU_Util.Last
<EDC_Enet05.CPU_Util.Last>
0
Dest Enet05.CPU_Util.Min
<EDC_Enet05.CPU_Util.Min>
0

Ethernet Module Status Min Scan

EQ
Source A Enet05.CPU_Util.Min
<EDC_Enet05.CPU_Util.Min>
0
Source B 0

Enet Module Status

Log Enet Utilization above 50% and 75%
(only log this once per second to avoid filling log)

{v1.0.1} -- Reset Log Event Timer
DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

5

Ethernet Module Status Last Scan
GT
Source A Enet05.CPU_Util.Last <EDC_Enet05.CPU_Util.Last> 0
Source B 500

Ethernet Module Status Last Scan
LT
Source A Enet05.CPU_Util.Last <EDC_Enet05.CPU_Util.Last> 0
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet05.UtilOver50Logged <EDC_Enet05.UtilOver50Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 5
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet05.CPU_Util.Last <EDC_Enet05.CPU_Util.Last> 0
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 50% Timer
Enet05.UtilOver50_TMR <EDC_Enet05.UtilOver50_TMR>
RES

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet05.UtilOver50Logged <EDC_Enet05.UtilOver50Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8232
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status Last Scan
GE
Source A Enet05.CPU_Util.Last <EDC_Enet05.CPU_Util.Last> 0
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet05.UtilOver75Logged <EDC_Enet05.UtilOver75Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 5
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet05.CPU_Util.Last <EDC_Enet05.CPU_Util.Last> 0
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet05.UtilOver75Logged <EDC_Enet05.UtilOver75Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8233
Input Par 0
Input Par -1
Input Par -1

Enet Module Status

Input Par 0

Log Enet Utilization above 50% and 75% -- reset 1-shot after time delay to allow another Transaction log entry
(only log this once per second to avoid filling log)

DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

6

Ethernet Module
Status Ethernet CPU
Utilization above
50% 1-shot
Enet05.UtilOver50Logged
<EDC_Enet05.UtilOver50Logged>

Ethernet Module
Status Ethernet CPU
Utilization above
75% 1-shot
Enet05.UtilOver75Logged
<EDC_Enet05.UtilOver75Logged>

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer

MOVE
Source 1000
Dest Enet05.UtilOver50_TMR.PRE
<EDC_Enet05.UtilOver50_TMR.PRE>
0

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer

TON
Timer Enet05.UtilOver50_TMR
<EDC_Enet05.UtilOver50_TMR>
Preset 0
Accum 0
EN
DN

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer
Enet05.UtilOver50_TMR.DN
<EDC_Enet05.UtilOver50_TMR.DN>

Ethernet Module
Status Ethernet CPU
Utilization above
50% 1-shot
Enet05.UtilOver50Logged
<EDC_Enet05.UtilOver50Logged>
U

Ethernet Module
Status Ethernet CPU
Utilization above
75% 1-shot
Enet05.UtilOver75Logged
<EDC_Enet05.UtilOver75Logged>
U

Enet Module Status

Log Enet Module Overall Status OK & Not OK
-- if Not OK, Clear Sockets Ready

7

Ethernet Module Status GSV Module - Entry Status OK
Enet05.StatusOK
<EDC_Enet05.StatusOK>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet05.StatusOkONS
<EDC_Enet05.StatusOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 5
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8201
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet05.StatusOK
<EDC_Enet05.StatusOK>
/

Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet05.StatusNotOkONS
<EDC_Enet05.StatusNotOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 5
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8203
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Enet Module Status

Log Enet Module Entry Status OK & Not OK
-- log Status Not OK each time Entry Status changs

8

Ethernet Module Status GSV Module - Entry Status OK
Enet05.EntryStatusOk
<EDC_Enet05.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet05.EntryStatusOkONS
<EDC_Enet05.EntryStatusOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet05.EntryStatus <EDC_Enet05.EntryStatus> 16#6000
Source B Enet05.EntryStatusMask <EDC_Enet05.EntryStatusMask> 16#f000
Dest Enet05.EntryStatusLast <EDC_Enet05.EntryStatusLast> 16#1000

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 5
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet05.EntryStatusLast <EDC_Enet05.EntryStatusLast> 16#1000
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8211
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet05.EntryStatusOk
<EDC_Enet05.EntryStatusOk>
/

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet05.EntryStatus <EDC_Enet05.EntryStatus> 16#6000
Mask Enet05.EntryStatusMask <EDC_Enet05.EntryStatusMask> 16#f000
Compare Enet05.EntryStatusLast <EDC_Enet05.EntryStatusLast> 16#1000

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet05.EntryStatusSame
<EDC_Enet05.EntryStatusSame>

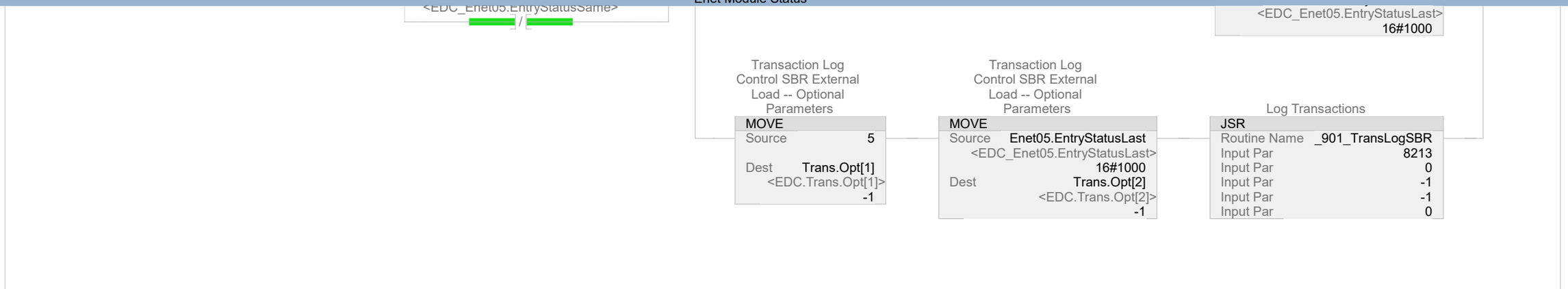
Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet05.EntryStatusNotOkONS
<EDC_Enet05.EntryStatusNotOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet05.EntryStatusSame
<EDC_Enet05.EntryStatusSame>

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet05.EntryStatus <EDC_Enet05.EntryStatus> 16#6000
Source B Enet05.EntryStatusMask <EDC_Enet05.EntryStatusMask> 16#f000
Dest Enet05.EntryStatusLast

Enet Module Status



Enet Module Status

Log Enet Module Fault Code OK & Not OK
-- log Status Not OK each time Fault Code changs

9

Ethernet Module Status GSV Module - Fault Code OK
Enet05.FaultCodeOk
<EDC_Enet05.FaultCodeOk>

Ethernet Module Status GSV Module - Fault Code OK 1-Shot
Enet05.FaultCodeOkONS
<EDC_Enet05.FaultCodeOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	5
Dest	Trans.Opt[1]
<EDC.Trans.Opt[1]>	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Enet05.FaultCode
<EDC_Enet05.FaultCode>	16#0000
Dest	Trans.Opt[2]
<EDC.Trans.Opt[2]>	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	8221
Input Par	0
Input Par	-1
Input Par	-1
Input Par	0

Ethernet Module Status GSV Module - Fault Code OK
Enet05.FaultCodeOk
<EDC_Enet05.FaultCodeOk>
/

Ethernet Module Status GSV Module - Fault Code Not OK 1-Shot
Enet05.FaultCodeNotOkONS
<EDC_Enet05.FaultCodeNotOkONS>
ONS

Ethernet Module Status GSV Module - Fault Code

NE	
Source A	Enet05.FaultCode
<EDC_Enet05.FaultCode>	16#0000
Source B	Enet05.FaultCodeLast
<EDC_Enet05.FaultCodeLast>	16#0315

Ethernet Module Status GSV Module - Fault Code Last Value

MOVE	
Source	Enet05.FaultCode
<EDC_Enet05.FaultCode>	16#0000
Dest	Enet05.FaultCodeLast
<EDC_Enet05.FaultCodeLast>	16#0315

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	5
Dest	Trans.Opt[1]
<EDC.Trans.Opt[1]>	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Enet05.FaultCode
<EDC_Enet05.FaultCode>	16#0000
Dest	Trans.Opt[2]
<EDC.Trans.Opt[2]>	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	8223
Input Par	0
Input Par	-1
Input Par	-1
Input Par	0

(End)

Enet Module Status

Enet Module Control and Status

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	31
Dest	Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1733165816
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1733165816

Enet Module Status

Get Enet Module Status

2

Ethernet Module Status GSV Module - Entry Status

GSV
Class Name Module
Instance Name EDC_FEC4
Attribute Name EntryStatus
Dest Enet06.EntryStatus
<EDC_Enet06.EntryStatus>
16#6000

Ethernet Module Status GSV Module - Fault Code

GSV
Class Name Module
Instance Name EDC_FEC4
Attribute Name FaultCode
Dest Enet06.FaultCode
<EDC_Enet06.FaultCode>
16#0000

Ethernet Module Status GSV Module - Fault Info

GSV
Class Name Module
Instance Name EDC_FEC4
Attribute Name FaultInfo
Dest Enet06.FaultInfo
<EDC_Enet06.FaultInfo>
16#0000_0000

Ethernet Module Status GSV Module - Mode

GSV
Class Name Module
Instance Name EDC_FEC4
Attribute Name Mode
Dest Enet06.Mode
<EDC_Enet06.Mode>
16#0004

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet06.EntryStatus
<EDC_Enet06.EntryStatus>
16#6000
Mask Enet06.EntryStatusMask
<EDC_Enet06.EntryStatusMask>
16#f000
Compare Enet06.EntryStatusRunChk
<EDC_Enet06.EntryStatusRunChk>
16#4000

Ethernet Module Status GSV Module - Entry Status OK

Enet06.EntryStatusOk
<EDC_Enet06.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet06.EntryStatus
<EDC_Enet06.EntryStatus>
16#6000
Mask Enet06.EntryStatusMask
<EDC_Enet06.EntryStatusMask>
16#f000
Compare ENet_InhibChk
16#6000

Ethernet Module Status GSV Module - Fault Code

EQ
Source A Enet06.FaultCode
<EDC_Enet06.FaultCode>
16#0000
Source B 0

Ethernet Module Status GSV Module - Fault Code OK

Enet06.FaultCodeOk
<EDC_Enet06.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK

Enet06.EntryStatusOk
<EDC_Enet06.EntryStatusOk>

Ethernet Module Status GSV Module - Fault Code OK

Enet06.FaultCodeOk
<EDC_Enet06.FaultCodeOk>

Ethernet Module Status GSV Module - Entry Status OK

Enet06.StatusOK
<EDC_Enet06.StatusOK>

Enet Module Status

Ethernet Module CPU Utilization
-- Clear Values
-- Initialize Array Length

3

Ethernet Module
Status Clear Scan
History
Enet06.CPU_Util.Clear
<EDC_Enet06.CPU_Util.Clear>

S:FS

Ethernet Module
Status Ethernet CPU
Utilization

FLL
Source 0
Dest Enet06.CPU_Util
<EDC_Enet06.CPU_Util>
Length 1

Ethernet Module
Status Enable Scan
Monitor
Enet06.CPU_Util.EN
<EDC_Enet06.CPU_Util.EN>

L

Ethernet Module
Status History Array

SIZE
Source Enet06.CPU_Util.History[0]
<EDC_Enet06.CPU_Util.History[0]>
0
Dim. To Vary 0
Size Enet06.CPU_Util.AvgCTL.LEN
<EDC_Enet06.CPU_Util.AvgCTL.LEN>
100

Enet Module Status

Ethernet Module CPU Utilization
-- Store New Value in History Array
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value
-- Get Next Value

(EN2T v2 Module CPU_Util is shifted 1 Byte in Status Packet -- use MEQ to auto detect this and correct)
DHF005 - Add CLR Enet03.CPU_Util.Last to ensure upper bytes are 0

** SPECIAL CASE -- L85 Utilization Message replaces EN2T Status Message **
5 - Task Utilization (always 100 with continuous task)
9 - MSG Utilization
11 - I/O Utilization

4

Ethernet Module
Status Enable Scan
Monitor
Enet06.CPU_Util.EN
<EDC_Enet06.CPU_Util.EN>

Ethernet Module --
L85 Ethernet Link
Status Message
EDC_Enet06_UtilMsg.EN

Ethernet Module --
L85 Ethernet Link
Status Message
EDC_Enet06_UtilMsg.DN

Ethernet Module --
L85 Ethernet Link
Status Message

MSG
Message Control EDC_Enet06_UtilMsg ...
EN
DN
ER

Ethernet Module
Status Last Scan

MUL
Source A EDC_Enet06_Util[9]
10
Source B 10
Dest Enet06.CPU_Util.Last
<EDC_Enet06.CPU_Util.Last>
0

Ethernet Module
Status History Array

MOVE
Source Enet06.CPU_Util.Last
<EDC_Enet06.CPU_Util.Last>
0
Dest Enet06.CPU_Util.History[Enet06.CPU_Util.i]
<EDC_Enet06.CPU_Util.History[Enet06.CPU_Util.i]>
0

Ethernet Module
Status History Array
Index

ADD
Source A Enet06.CPU_Util.i
<EDC_Enet06.CPU_Util.i>

Ethernet Module
Status History Array
Index

GE
Source A Enet06.CPU_Util.i
<EDC_Enet06.CPU_Util.i>

Ethernet Module
Status History Array
Index

CLR
Dest Enet06.CPU_Util.i
<EDC_Enet06.CPU_Util.i>

Enet Module Status

0
Source B 1
Dest Enet06.CPU_Util.i
<EDC_Enet06.CPU_Util.i>
0

0
Source B Enet06.CPU_Util.AvgCTL.LEN
<EDC_Enet06.CPU_Util.AvgCTL.LEN>
100

Ethernet Module Status History Array Index
LT
Source A Enet06.CPU_Util.i
<EDC_Enet06.CPU_Util.i>
0
Source B 0

Ethernet Module Status Avg Scan CONTROL
Enet06.CPU_Util.AvgCTL
<EDC_Enet06.CPU_Util.AvgCTL>
RES

0

Ethernet Module Status Avg Scan CONTROL
AVE
Array Enet06.CPU_Util.History[0]
<EDC_Enet06.CPU_Util.History[0]>
Dim. To Vary 0
Dest Enet06.CPU_Util.Avg
<EDC_Enet06.CPU_Util.Avg>
0
Control Enet06.CPU_Util.AvgCTL
<EDC_Enet06.CPU_Util.AvgCTL>
Length 100
Position 0
EN
DN
ER

Ethernet Module Status Last Scan
GT
Source A Enet06.CPU_Util.Last
<EDC_Enet06.CPU_Util.Last>
0
Source B Enet06.CPU_Util.Max
<EDC_Enet06.CPU_Util.Max>
0

Ethernet Module Status Max Scan
MOVE
Source Enet06.CPU_Util.Last
<EDC_Enet06.CPU_Util.Last>
0
Dest Enet06.CPU_Util.Max
<EDC_Enet06.CPU_Util.Max>
0

Ethernet Module Status Last Scan
LT
Source A Enet06.CPU_Util.Last
<EDC_Enet06.CPU_Util.Last>
0
Source B Enet06.CPU_Util.Min
<EDC_Enet06.CPU_Util.Min>
0

Ethernet Module Status Last Scan
GT
Source A Enet06.CPU_Util.Last
<EDC_Enet06.CPU_Util.Last>
0
Source B 0

Ethernet Module Status Min Scan
MOVE
Source Enet06.CPU_Util.Last
<EDC_Enet06.CPU_Util.Last>
0
Dest Enet06.CPU_Util.Min
<EDC_Enet06.CPU_Util.Min>
0

Ethernet Module Status Min Scan
EQ
Source A Enet06.CPU_Util.Min
<EDC_Enet06.CPU_Util.Min>
0
Source B 0

Enet Module Status

Log Enet Utilization above 50% and 75%
(only log this once per second to avoid filling log)

{v1.0.1} -- Reset Log Event Timer
DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

5

Ethernet Module Status Last Scan
GT
Source A Enet06.CPU_Util.Last <EDC_Enet06.CPU_Util.Last> 0
Source B 500

Ethernet Module Status Last Scan
LT
Source A Enet06.CPU_Util.Last <EDC_Enet06.CPU_Util.Last> 0
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet06.UtilOver50Logged <EDC_Enet06.UtilOver50Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 6
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet06.CPU_Util.Last <EDC_Enet06.CPU_Util.Last> 0
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 50% Timer
Enet06.UtilOver50_TMR <EDC_Enet06.UtilOver50_TMR>
RES

Ethernet Module Status Ethernet CPU Utilization above 50% 1-shot
Enet06.UtilOver50Logged <EDC_Enet06.UtilOver50Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8232
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status Last Scan
GE
Source A Enet06.CPU_Util.Last <EDC_Enet06.CPU_Util.Last> 0
Source B 750

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet06.UtilOver75Logged <EDC_Enet06.UtilOver75Logged>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 6
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
DIV
Source A Enet06.CPU_Util.Last <EDC_Enet06.CPU_Util.Last> 0
Source B 10
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Ethernet Module Status Ethernet CPU Utilization above 75% 1-shot
Enet06.UtilOver75Logged <EDC_Enet06.UtilOver75Logged>
L

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8233
Input Par 0
Input Par -1
Input Par -1

Enet Module Status

Input Par 0

Log Enet Utilization above 50% and 75% -- reset 1-shot after time delay to allow another Transaction log entry
(only log this once per second to avoid filling log)

DHF013 -- Replaced ONS/OTU with OTL/OTU for readability

6

Ethernet Module
Status Ethernet CPU
Utilization above
50% 1-shot
Enet06.UtilOver50Logged
<EDC_Enet06.UtilOver50Logged>

Ethernet Module
Status Ethernet CPU
Utilization above
75% 1-shot
Enet06.UtilOver75Logged
<EDC_Enet06.UtilOver75Logged>

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer

MOVE
Source 1000
Dest Enet06.UtilOver50_TMR.PRE
<EDC_Enet06.UtilOver50_TMR.PRE>
0

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer

TON
Timer Enet06.UtilOver50_TMR
<EDC_Enet06.UtilOver50_TMR>
Preset 0
Accum 0
EN
DN

Ethernet Module
Status Ethernet CPU
Utilization above
50% Timer
Enet06.UtilOver50_TMR.DN
<EDC_Enet06.UtilOver50_TMR.DN>

Ethernet Module
Status Ethernet CPU
Utilization above
50% 1-shot
Enet06.UtilOver50Logged
<EDC_Enet06.UtilOver50Logged>
U

Ethernet Module
Status Ethernet CPU
Utilization above
75% 1-shot
Enet06.UtilOver75Logged
<EDC_Enet06.UtilOver75Logged>
U

Enet Module Status

Log Enet Module Overall Status OK & Not OK
-- if Not OK, Clear Sockets Ready

7

Ethernet Module Status GSV Module - Entry Status OK
Enet06.StatusOK
<EDC_Enet06.StatusOK>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet06.StatusOkONS
<EDC_Enet06.StatusOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 6
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8201
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet06.StatusOK
<EDC_Enet06.StatusOK>
/

Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet06.StatusNotOkONS
<EDC_Enet06.StatusNotOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 6
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 8203
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Enet Module Status

Log Enet Module Entry Status OK & Not OK
-- log Status Not OK each time Entry Status changs

8

Ethernet Module Status GSV Module - Entry Status OK
Enet06.EntryStatusOk
<EDC_Enet06.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status OK 1-Shot
Enet06.EntryStatusOkONS
<EDC_Enet06.EntryStatusOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet06.EntryStatus <EDC_Enet06.EntryStatus> 16#6000
Source B Enet06.EntryStatusMask <EDC_Enet06.EntryStatusMask> 16#f000
Dest Enet06.EntryStatusLast <EDC_Enet06.EntryStatusLast> 16#6000

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 6
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet06.EntryStatusLast <EDC_Enet06.EntryStatusLast> 16#6000
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8211
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Entry Status OK
Enet06.EntryStatusOk
<EDC_Enet06.EntryStatusOk>

Ethernet Module Status GSV Module - Entry Status

MEQ
Source Enet06.EntryStatus <EDC_Enet06.EntryStatus> 16#6000
Mask Enet06.EntryStatusMask <EDC_Enet06.EntryStatusMask> 16#f000
Compare Enet06.EntryStatusLast <EDC_Enet06.EntryStatusLast> 16#6000

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet06.EntryStatusSame
<EDC_Enet06.EntryStatusSame>

Ethernet Module Status GSV Module - Entry Status Not OK 1-Shot
Enet06.EntryStatusNotOkONS
<EDC_Enet06.EntryStatusNotOkONS>
ONS

Ethernet Module Status GSV Module - Entry Status Same as Last Entry Status
Enet06.EntryStatusSame

Ethernet Module Status GSV Module - Entry Status Last Value

AND
Source A Enet06.EntryStatus <EDC_Enet06.EntryStatus> 16#6000
Source B Enet06.EntryStatusMask <EDC_Enet06.EntryStatusMask> 16#f000
Dest Enet06.EntryStatusLast

Enet Module Status

<EDC_Enet06.EntryStatusSame>
/

<EDC_Enet06.EntryStatusLast>
16#6000

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 6
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet06.EntryStatusLast
<EDC_Enet06.EntryStatusLast>
16#6000
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8213
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Enet Module Status

Log Enet Module Fault Code OK & Not OK
-- log Status Not OK each time Fault Code changs

9

Ethernet Module Status GSV Module - Fault Code OK
Enet06.FaultCodeOk
<EDC_Enet06.FaultCodeOk>

Ethernet Module Status GSV Module - Fault Code OK 1-Shot
Enet06.FaultCodeOkONS
<EDC_Enet06.FaultCodeOkONS>
ONS

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 6
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet06.FaultCode
<EDC_Enet06.FaultCode> 16#0000
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8221
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Ethernet Module Status GSV Module - Fault Code OK
Enet06.FaultCodeOk
<EDC_Enet06.FaultCodeOk>
/

Ethernet Module Status GSV Module - Fault Code Not OK 1-Shot
Enet06.FaultCodeNotOkONS
<EDC_Enet06.FaultCodeNotOkONS>
ONS

Ethernet Module Status GSV Module - Fault Code

NE
Source A Enet06.FaultCode
<EDC_Enet06.FaultCode> 16#0000
Source B Enet06.FaultCodeLast
<EDC_Enet06.FaultCodeLast> 16#0204

Ethernet Module Status GSV Module - Fault Code Last Value

MOVE
Source Enet06.FaultCode
<EDC_Enet06.FaultCode> 16#0000
Dest Enet06.FaultCodeLast
<EDC_Enet06.FaultCodeLast> 16#0204

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source 6
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Enet06.FaultCode
<EDC_Enet06.FaultCode> 16#0000
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 8223
Input Par 0
Input Par -1
Input Par -1
Input Par 0

(End)

Define ASCII Table

ASCII Table Enumeration
2008 Aug 22 -- Initial Release by Dan Guendelsberger (Rockwell Automation)
2009 Jun 11 -- Updated by Dan Guendelsberger
2010 Apr 08 -- Converted to Ladder Logic by Dan Guendelsberger

Creates udt-based ASCII Table for use in enumerations to avoid ASCII table lookups (and mistakes).
ascii._1 = 48
ascii._2 = 49
etc.
ascii.A = 65
ascii.B = 66
etc.
ascii._a = 97
ascii._b = 98
etc.

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 38
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Skip Value Initialize Loop if already initialized
2009Jun11 -- execute loop once to save scan time -- about 1 msec

2

ASCII Table Enumeration Control Structure SINT Array of Values 0-127 for loading ASCII udt

EQ
Source A asciiTable.Values[127] 127
Source B 127

lbl_InitDone
JMP

Define ASCII Table

Reset Index for Initialization Loop

3

ASCII Table Enumeration Control Structure Table Loop Index

CLR
Dest asciiTable.LOOPi 128

Initialize value array (0-127)

4

lbl_Init
LBL

ASCII Table Enumeration Control Structure SINT Array of Values 0-127 for loading ASCII udt

MOVE
Source asciiTable.LOOPi 128
Dest asciiTable.Values[asciiTable.LOOPi] ??

ASCII Table Enumeration Control Structure Table Loop Index

ADD
Source A asciiTable.LOOPi 128
Source B 1
Dest asciiTable.LOOPi 128

ASCII Table Enumeration Control Structure Table Loop Index

LE
Source A asciiTable.LOOPi 128
Source B 127

lbl_Init
JMP

copy value array (0-127) to the ASCII Table udt
2009Jun11 -- continuously execute to ensure ASCII values aren't mistakenly overwritten by users

5

lbl_InitDone
LBL

ASCII Table Enumeration

COP
Source asciiTable.Values[0]
Dest ascii <EDC.ascii>
Length 1

(End)

Define Global Constants

Global Constants

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 39
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

(End)

Transaction and Error Log Snapshot

Transaction Log Sanpshot

Capture snapshot of Transaction Log and Error Log
(Triggered externally from TransLog Spreadsheet or internally from User-defined Trigger Logic)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 80
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Transaction and Error Log Snapshot

****

USER LOGIC -- for triggering Snapshot of Transaction and Error Logs

****

Manually Toggle SnapIntEN to ON -- add/modify logic for the desired event for the snapshot
(when snapshot is taken, the SnapIntEN will be turned off to avoid overwriting the initial capture -- block this if overwrites are desired)

The default logic takes a Snapshot when the next Error is logged.

2

Transaction Log Control Internal Snapshot Command Enabled (User Logic) Trans.SnapIntEN <EDC.Trans.SnapIntEN>

Transaction Log Control -- Transaction Error Log Index

NE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B Trans.SnapERRi <EDC.Trans.SnapERRi> 0

Bookmark to be replaced by USER Trigger for Transaction Log Snapshot
USER_TransSnap

Transaction Log Control Internal Snapshot Command Enabled (User Logic) Trans.SnapIntEN <EDC.Trans.SnapIntEN>
U

Transaction Log Control Internal Snapshot Command Enabled (User Logic) Trans.SnapIntEN <EDC.Trans.SnapIntEN>
L

Bookmark to be replaced by USER Trigger for Transaction Log Snapshot
USER_TransSnap

Transaction Log Control Internal Snapshot Command (User Logic) Trans.SnapInt <EDC.Trans.SnapInt>
L

Default delay between trigger and snapshot is 10,000 msec to allow post-trigger data capture
(modify the Preset as desired)

3

Transaction Log Control Internal Snapshot Command (User Logic) Trans.SnapInt <EDC.Trans.SnapInt>

Transaction Log Control Internal Snapshot Delay Timer (to allow post-trigger data collection)

MOVE
Source 10000
Dest Trans.SnapIntDelay.PRE <EDC.Trans.SnapIntDelay.PRE> 0

Transaction Log Control Internal Snapshot Delay Timer (to allow post-trigger data collection)

TON
Timer Trans.SnapIntDelay <EDC.Trans.SnapIntDelay>
Preset 0
Accum 0
EN
DN

Transaction and Error Log Snapshot

Snapshot Transaction Log and Error Log

4

Transaction Log Control External Snapshot Command (from TransLog spreadsheet) **Trans.SnapExt** <EDC.Trans.SnapExt>

Transaction Log Control Internal Snapshot Command (User Logic) **Trans.SnapInt** <EDC.Trans.SnapInt>

Transaction Log Control Internal Snapshot Delay Timer (to allow post-trigger data collection) **Trans.SnapIntDelay.DN** <EDC.Trans.SnapIntDelay.DN>

Transaction Log Control External Snapshot Command (from TransLog spreadsheet) **Trans.SnapExt** <EDC.Trans.SnapExt> (U)

Transaction Log Control Internal Snapshot Command (User Logic) **Trans.SnapInt** <EDC.Trans.SnapInt> (U)

Transaction Log Control -- Transaction Log Snapshot Index

MOVE
Source Trans.LOGi <EDC.Trans.LOGi>
211
Dest Trans.SnapLOGi <EDC.Trans.SnapLOGi>
0

Transaction Log Snapshot Transaction Log

COP
Source EDC_TransLog[0]
Dest EDC_TransLogSnap[0]
Length CFG.TransLOG.n <EDC.CFG.TransLOG.n>

Transaction Log Control -- Transaction Error Snapshot Log Index

MOVE
Source Trans.ERRi <EDC.Trans.ERRi>
62
Dest Trans.SnapERRi <EDC.Trans.SnapERRi>
0

Error Log Snapshot Transaction Log

COP
Source EDC_TransErr[0]
Dest EDC_TransErrSnap[0]
Length CFG.TransERR.n <EDC.CFG.TransERR.n>

(End)

Update Performance Summary

Update Performance Summary
PerfUpdate rate determines Equipment Quantity to process each scan.
Default 1 - higher values update totals faster but increase Task Scan time

EDC Config
Performance Summary
-- Updates per scan

0

EQ
Source A CFG.CLx.PerfUpdate
<EDC.CFG.CLx.PerfUpdate>
1
Source B CFG.CLx.PerfUpdate
<EDC.CFG.CLx.PerfUpdate>
1

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan
Time & Fault Routine
Routine Number
currently Scanned

MOVE
Source 90
Dest Prog.Routine
95

Local Program Scan
Time & Fault Routine
Routine Scan
Counters (Indexed by
Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine]
-1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine]
-1733165816

Update Performance Summary

Increment Line Index
If all Lines Done, reset Line Index and update Shop totals

2

EDC Comm Driver Performance All Equipment added to Line Summary
Perf.CTL.LineDone
<EDC.Perf.CTL.LineDone>

EDC Comm Driver Performance All Equipment added to Line Summary
Perf.CTL.LineDone
<EDC.Perf.CTL.LineDone>
U

EDC Comm Driver Performance Performance Equipment Loop -- Initial Index Value
CLR
Dest Perf.CTL.EQinit
<EDC.Perf.CTL.EQinit>
30

EDC Comm Driver Performance Performance Equipment Loop -- Terminal Index Value
CLR
Dest Perf.CTL.EQterm
<EDC.Perf.CTL.EQterm>
29

EDC Comm Driver Performance Performance Line Loop -- Line Index
ADD
Source A Perf.CTL.LNi
<EDC.Perf.CTL.LNi>
18
Source B 1
Dest Perf.CTL.LNi
<EDC.Perf.CTL.LNi>
18

EDC Comm Driver Performance Performance Line Loop -- Line Index
LT
Source A Perf.CTL.LNi
<EDC.Perf.CTL.LNi>
18
Source B CFG.LN.si
<EDC.CFG.LN.si>
0

EDC Comm Driver Performance Performance Line Loop -- Line Index
GT
Source A Perf.CTL.LNi
<EDC.Perf.CTL.LNi>
18
Source B CFG.LN.ei
<EDC.CFG.LN.ei>
28

EDC Comm Driver Performance Performance Line Loop -- Line Index
MOVE
Source CFG.LN.si
<EDC.CFG.LN.si>
0
Dest Perf.CTL.LNi
<EDC.Perf.CTL.LNi>
18

Copy Line Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

3

EDC Config Array Indexes for Line-Equip
COP
Source EDC.Line[Perf.CTL.LNi].CFG.EQ
Dest LineCfgEQ
Length 1

Update Performance Summary

Update Equipment Performance Summary
-- service n+1 Equipment for LNi per scan to conserve scan time
(resume Loop from previous termination point until complete)

4

EDC Comm Driver Performance Performance Equipment Loop -- Terminal Index Value

ADD
Source A Perf.CTL.EQinit <EDC.Perf.CTL.EQinit> 30
Source B CFG.CLx.PerfUpdate <EDC.CFG.CLx.PerfUpdate> 1
Dest Perf.CTL.EQterm <EDC.Perf.CTL.EQterm> 29

EDC Comm Driver Performance Performance Equipment Loop -- Terminal Index Value

GT
Source A Perf.CTL.EQterm <EDC.Perf.CTL.EQterm> 29
Source B LineCfgEQ.ei 99

EDC Comm Driver Performance Performance Equipment Loop -- Terminal Index Value

MOVE
Source LineCfgEQ.ei 99
Dest Perf.CTL.EQterm <EDC.Perf.CTL.EQterm> 29

Update Performance Summary for LNi, EQi

FOR
Routine Name _091_PerfSummaryEquip
Index Perf.CTL.EQi <EDC.Perf.CTL.EQi> 30
Initial Value Perf.CTL.EQinit <EDC.Perf.CTL.EQinit>
Terminal Value Perf.CTL.EQterm <EDC.Perf.CTL.EQterm>
Step Size 1

EDC Comm Driver Performance Performance Equipment Loop -- Initial Index Value

ADD
Source A Perf.CTL.EQterm <EDC.Perf.CTL.EQterm> 29
Source B 1
Dest Perf.CTL.EQinit <EDC.Perf.CTL.EQinit> 30

EDC Comm Driver Performance Performance Equipment Loop -- Initial Index Value

LT
Source A Perf.CTL.EQinit <EDC.Perf.CTL.EQinit> 30
Source B 0

EDC Comm Driver Performance Performance Equipment Loop -- Initial Index Value

GT
Source A Perf.CTL.EQinit <EDC.Perf.CTL.EQinit> 30
Source B LineCfgEQ.ei 99

EDC Comm Driver Performance All Equipment added to Line Summary

Perf.CTL.LineDone <EDC.Perf.CTL.LineDone> (L)

lbl_LineDone (JMP)

Update Performance Summary

If all Line Equip updated, Calculate Averages and store Results for Line

5

EDC Comm Driver Performance Performance Equipment Loop -- Terminal Index Value

GE
Source A Perf.CTL.EQterm <EDC.Perf.CTL.EQterm> 29
Source B LineCfgEQ.ei 99

EDC Comm Driver Performance All Equipment added to Line Summary
Perf.CTL.LineDone <EDC.Perf.CTL.LineDone>
L

6

lbl_LineDone
LBL

NOP

(End)

Update Performance Summary for LNi, EQi

Update Equipment Status & Performance Summary
-- service n Equipment per scan to conserve scan time
(resume from previous termination point until complete)

Perf.CTL.LNi, Perf.CTL.EQi

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 91
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

2

Local Copy of Equip Array Tag for LNi, EQi

COP
Source EDC.Line[Perf.CTL.LNi].Equip[Perf.CTL.EQi]
Dest Equip
Length 1

Local Copy of Equip Array Status Tag for LNi, EQi

COP
Source EDC_Status01.Line[Perf.CTL.LNi].Equip[Perf.CTL.EQi]
Dest EquipStatus01
Length 1

Equipment Data Collection - Status Line Equipment

COP
Source EDC_Status02.Line[Perf.CTL.LNi].Equip[Perf.CTL.EQi]
Dest EquipStatus02
Length 1

Equipment Data Collection - Status Line Equipment

COP
Source EDC_Status03.Line[Perf.CTL.LNi].Equip[Perf.CTL.EQi]
Dest EquipStatus03
Length 1

Equipment Data Collection - Status Line Equipment

COP
Source EDC_Status04.Line[Perf.CTL.LNi].Equip[Perf.CTL.EQi]
Dest EquipStatus04
Length 1

Update Performance Summary for LNi, EQi

3

Equipment Status States

Combine all Equipment to determine Worst Case Shop and Line Status for HMI Display
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (exlude Disabled Equipment from summary)

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

COP
Source Perf.Temp.Shop.State <EDC.Perf.Temp.Shop.State>
Dest TempDINT
Length 1

OR
Source A TempDINT 638
Source B EquipStatus01.StateVal 16#0000_0202
Dest TempDINTnew 638

EDC Comm Driver Performance Status Message State
COP
Source TempDINTnew
Dest Perf.Temp.Shop.State <EDC.Perf.Temp.Shop.State>
Length 1

COP
Source Perf.Temp.Line.State <EDC.Perf.Temp.Line.State>
Dest TempDINT
Length 1

OR
Source A TempDINT 638
Source B EquipStatus01.StateVal 16#0000_0202
Dest TempDINT 638

EDC Comm Driver Performance Status Message State
COP
Source TempDINT
Dest Perf.Temp.Line.State <EDC.Perf.Temp.Line.State>
Length 1

4

Equipment Status States

Determine which Equipment has Worst Case Status for HMI Display
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (exlude Disabled Equipment from summary)

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value
GT
Source A EquipStatus01.StateVal 16#0000_0202
Source B Perf.CTL.StatusWorst <EDC.Perf.CTL.StatusWorst> 16#0000_0202

EDC Comm Driver Performance Performance Worst Status
MOVE
Source EquipStatus01.StateVal 16#0000_0202
Dest Perf.CTL.StatusWorst <EDC.Perf.CTL.StatusWorst> 16#0000_0202

EDC Comm Driver Performance Performance Worst Status - Line Index
MOVE
Source Perf.CTL.LNi <EDC.Perf.CTL.LNi> 18
Dest Perf.CTL.StatusWorstLNi <EDC.Perf.CTL.StatusWorstLNi> 0

EDC Comm Driver Performance Performance Worst Status - Equip Index
MOVE
Source Perf.CTL.EQi <EDC.Perf.CTL.EQi> 30
Dest Perf.CTL.StatusWorstEQi <EDC.Perf.CTL.StatusWorstEQi> 20

Update Performance Summary for LNi, EQi

Equipment Status - Faults
Find Next Fault to display on Gateway Status
(replaces Worst-Case Status with scroll of all Equipment with Faults)

5

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Display
Perf.CTL.FaultDisplayDwell
<EDC.Perf.CTL.FaultDisplayDwell>

EDC Comm Driver Performance Performance Line Loop -- Line Index

EQ
Source A Perf.CTL.LNi <EDC.Perf.CTL.LNi> 18
Source B Perf.CTL.FaultDisplayLNi <EDC.Perf.CTL.FaultDisplayLNi> 0

EDC Comm Driver Performance Performance Equipment Loop -- Equip Index

GT
Source A Perf.CTL.EQi <EDC.Perf.CTL.EQi> 30
Source B Perf.CTL.FaultDisplayEQi <EDC.Perf.CTL.FaultDisplayEQi> 0

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Check

AND
Source A EquipStatus01.StateVal 16#0000_0202
Source B 16#EFFFF000
Dest Perf.CTL.FaultDisplayStatusCheck <EDC.Perf.CTL.FaultDisplayStatusCheck> 16#0000_0000

EDC Comm Driver Performance Performance Line Loop -- Line Index

GT
Source A Perf.CTL.LNi <EDC.Perf.CTL.LNi> 18
Source B Perf.CTL.FaultDisplayLNi <EDC.Perf.CTL.FaultDisplayLNi> 0

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Check

NE
Source A Perf.CTL.FaultDisplayStatusCheck <EDC.Perf.CTL.FaultDisplayStatusCheck> 16#0000_0000
Source B 0

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Found
Perf.CTL.FaultDisplayFound
<EDC.Perf.CTL.FaultDisplayFound>
(L)

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Display
Perf.CTL.FaultDisplayDwell
<EDC.Perf.CTL.FaultDisplayDwell>
(L)

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Line Index

MOVE
Source Perf.CTL.LNi <EDC.Perf.CTL.LNi> 18
Dest Perf.CTL.FaultDisplayLNi <EDC.Perf.CTL.FaultDisplayLNi> 0

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Equip Index

MOVE
Source Perf.CTL.EQi <EDC.Perf.CTL.EQi> 30
Dest Perf.CTL.FaultDisplayEQi <EDC.Perf.CTL.FaultDisplayEQi> 0

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault EQID

CPT
Dest Perf.CTL.FaultDisplayEQID <EDC.Perf.CTL.FaultDisplayEQID> 964
Expression Perf.CTL.FaultDisplayLNi*100+Perf.CTL.FaultDisplayEQi

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Status

MOVE
Source EquipStatus01.StateVal 16#0000_0202
Dest Perf.CTL.FaultDisplayStatus <EDC.Perf.CTL.FaultDisplayStatus> 16#0000_0802

Update Performance Summary for LNi, EQi

Equipment Status - Faults
Display Fault for 2 Seconds before resuming search for Next Fault

6

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Found
Perf.CTL.FaultDisplayFound
<EDC.Perf.CTL.FaultDisplayFound>

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Display
Perf.CTL.FaultDisplayDwell
<EDC.Perf.CTL.FaultDisplayDwell>

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Display Timer

MOVE
Source 2000
Dest Perf.CTL.FaultDisplayDwellTMR.PRE
<EDC.Perf.CTL.FaultDisplayDwellTMR.PRE>
2000

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Display Timer

TON
Timer Perf.CTL.FaultDisplayDwellTMR
<EDC.Perf.CTL.FaultDisplayDwellTMR>
Preset 2000
Accum 0
EN
DN

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Display Timer
Perf.CTL.FaultDisplayDwellTMR.DN
<EDC.Perf.CTL.FaultDisplayDwellTMR.DN>

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Display
Perf.CTL.FaultDisplayDwell
<EDC.Perf.CTL.FaultDisplayDwell>
U

Equipment Status
Confgured Equip Count
DHF006 - changed Equip.CFG.EN to Equip.CTL.Config (include Disabled Equipment count)

7

Local Copy of Equip Array Tag for LNi, EQi Equipment Configured (by Server or Driver)
Equip.CTL.Config

EDC Comm Driver Performance Configured Equipment Count

ADD
Source A Perf.Temp.Shop.CfgCount
<EDC.Perf.Temp.Shop.CfgCount>
269
Source B 1
Dest Perf.Temp.Shop.CfgCount
<EDC.Perf.Temp.Shop.CfgCount>
269

EDC Comm Driver Performance Configured Equipment Count

ADD
Source A Perf.Temp.Line.CfgCount
<EDC.Perf.Temp.Line.CfgCount>
0
Source B 1
Dest Perf.Temp.Line.CfgCount
<EDC.Perf.Temp.Line.CfgCount>
0

Update Performance Summary for LNi, EQi

8

Callout Messages
EN, DN, ER, KA

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CalloutRx.EN
<EDC.Perf.Temp.Shop.CalloutRx.EN>
Input Par Perf.Temp.Line.CalloutRx.EN
<EDC.Perf.Temp.Line.CalloutRx.EN>
Input Par EquipStatus02.CalloutRx.EN
Return Par Perf.Temp.Shop.CalloutRx.EN
<EDC.Perf.Temp.Shop.CalloutRx.EN>
Return Par Perf.Temp.Line.CalloutRx.EN
<EDC.Perf.Temp.Line.CalloutRx.EN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CalloutRx.DN
<EDC.Perf.Temp.Shop.CalloutRx.DN>
Input Par Perf.Temp.Line.CalloutRx.DN
<EDC.Perf.Temp.Line.CalloutRx.DN>
Input Par EquipStatus02.CalloutRx.DN
Return Par Perf.Temp.Shop.CalloutRx.DN
<EDC.Perf.Temp.Shop.CalloutRx.DN>
Return Par Perf.Temp.Line.CalloutRx.DN
<EDC.Perf.Temp.Line.CalloutRx.DN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CalloutRx.ER
<EDC.Perf.Temp.Shop.CalloutRx.ER>
Input Par Perf.Temp.Line.CalloutRx.ER
<EDC.Perf.Temp.Line.CalloutRx.ER>
Input Par EquipStatus02.CalloutRx.ER
Return Par Perf.Temp.Shop.CalloutRx.ER
<EDC.Perf.Temp.Shop.CalloutRx.ER>
Return Par Perf.Temp.Line.CalloutRx.ER
<EDC.Perf.Temp.Line.CalloutRx.ER>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CalloutRx.KA
<EDC.Perf.Temp.Shop.CalloutRx.KA>
Input Par Perf.Temp.Line.CalloutRx.KA
<EDC.Perf.Temp.Line.CalloutRx.KA>
Input Par EquipStatus02.CalloutRx.KA
Return Par Perf.Temp.Shop.CalloutRx.KA
<EDC.Perf.Temp.Shop.CalloutRx.KA>
Return Par Perf.Temp.Line.CalloutRx.KA
<EDC.Perf.Temp.Line.CalloutRx.KA>

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.CalloutMsg.TransTMR.ACC
2335
Source B Perf.Temp.Shop.CalloutTransTmrAcc
<EDC.Perf.Temp.Shop.CalloutTransTmrAcc>
121605

EDC Comm Driver Performance Callout/Status Receive Message Transaction Timer

MOVE
Source Equip.CTL.CalloutMsg.TransTMR.ACC
2335
Dest Perf.Temp.Shop.CalloutTransTmrAcc
<EDC.Perf.Temp.Shop.CalloutTransTmrAcc>
121605

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.CalloutMsg.TransTMR.ACC
2335
Source B Perf.Temp.Line.CalloutTransTmrAcc
<EDC.Perf.Temp.Line.CalloutTransTmrAcc>
0

EDC Comm Driver Performance Callout/Status Receive Message Transaction Timer

MOVE
Source Equip.CTL.CalloutMsg.TransTMR.ACC
2335
Dest Perf.Temp.Line.CalloutTransTmrAcc
<EDC.Perf.Temp.Line.CalloutTransTmrAcc>
0

Update Performance Summary for LNi, EQi

Fault/Status Messages
EN, DN, ER, KA

9

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.FaultRx.EN <EDC.Perf.Temp.Shop.FaultRx.EN>
Input Par Perf.Temp.Line.FaultRx.EN <EDC.Perf.Temp.Line.FaultRx.EN>
Input Par EquipStatus02.FaultRx.EN
Return Par Perf.Temp.Shop.FaultRx.EN <EDC.Perf.Temp.Shop.FaultRx.EN>
Return Par Perf.Temp.Line.FaultRx.EN <EDC.Perf.Temp.Line.FaultRx.EN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.FaultRx.DN <EDC.Perf.Temp.Shop.FaultRx.DN>
Input Par Perf.Temp.Line.FaultRx.DN <EDC.Perf.Temp.Line.FaultRx.DN>
Input Par EquipStatus02.FaultRx.DN
Return Par Perf.Temp.Shop.FaultRx.DN <EDC.Perf.Temp.Shop.FaultRx.DN>
Return Par Perf.Temp.Line.FaultRx.DN <EDC.Perf.Temp.Line.FaultRx.DN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.FaultRx.ER <EDC.Perf.Temp.Shop.FaultRx.ER>
Input Par Perf.Temp.Line.FaultRx.ER <EDC.Perf.Temp.Line.FaultRx.ER>
Input Par EquipStatus02.FaultRx.ER
Return Par Perf.Temp.Shop.FaultRx.ER <EDC.Perf.Temp.Shop.FaultRx.ER>
Return Par Perf.Temp.Line.FaultRx.ER <EDC.Perf.Temp.Line.FaultRx.ER>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.FaultRx.KA <EDC.Perf.Temp.Shop.FaultRx.KA>
Input Par Perf.Temp.Line.FaultRx.KA <EDC.Perf.Temp.Line.FaultRx.KA>
Input Par EquipStatus02.FaultRx.KA
Return Par Perf.Temp.Shop.FaultRx.KA <EDC.Perf.Temp.Shop.FaultRx.KA>
Return Par Perf.Temp.Line.FaultRx.KA <EDC.Perf.Temp.Line.FaultRx.KA>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.FaultTx.EN <EDC.Perf.Temp.Shop.FaultTx.EN>
Input Par Perf.Temp.Line.FaultTx.EN <EDC.Perf.Temp.Line.FaultTx.EN>
Input Par EquipStatus02.FaultTx.EN
Return Par Perf.Temp.Shop.FaultTx.EN <EDC.Perf.Temp.Shop.FaultTx.EN>
Return Par Perf.Temp.Line.FaultTx.EN <EDC.Perf.Temp.Line.FaultTx.EN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.FaultTx.DN <EDC.Perf.Temp.Shop.FaultTx.DN>
Input Par Perf.Temp.Line.FaultTx.DN <EDC.Perf.Temp.Line.FaultTx.DN>
Input Par EquipStatus02.FaultTx.DN
Return Par Perf.Temp.Shop.FaultTx.DN <EDC.Perf.Temp.Shop.FaultTx.DN>
Return Par Perf.Temp.Line.FaultTx.DN <EDC.Perf.Temp.Line.FaultTx.DN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.FaultTx.ER <EDC.Perf.Temp.Shop.FaultTx.ER>
Input Par Perf.Temp.Line.FaultTx.ER <EDC.Perf.Temp.Line.FaultTx.ER>
Input Par EquipStatus02.FaultTx.ER
Return Par Perf.Temp.Shop.FaultTx.ER <EDC.Perf.Temp.Shop.FaultTx.ER>
Return Par Perf.Temp.Line.FaultTx.ER <EDC.Perf.Temp.Line.FaultTx.ER>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.FaultTx.KA <EDC.Perf.Temp.Shop.FaultTx.KA>
Input Par Perf.Temp.Line.FaultTx.KA <EDC.Perf.Temp.Line.FaultTx.KA>
Input Par EquipStatus02.FaultTx.KA
Return Par Perf.Temp.Shop.FaultTx.KA <EDC.Perf.Temp.Shop.FaultTx.KA>
Return Par Perf.Temp.Line.FaultTx.KA <EDC.Perf.Temp.Line.FaultTx.KA>

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.FaultMsg.TransTMR.ACC 3332
Source B Perf.Temp.Shop.FaultTransTmrAcc <EDC.Perf.Temp.Shop.FaultTransTmrAcc> 295161

EDC Comm Driver Performance Fault/Status Receive Message Transaction Timer

MOVE
Source Equip.CTL.FaultMsg.TransTMR.ACC 3332
Dest Perf.Temp.Shop.FaultTransTmrAcc <EDC.Perf.Temp.Shop.FaultTransTmrAcc> 295161

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.FaultMsg.TransTMR.ACC 3332
Source B Perf.Temp.Line.FaultTransTmrAcc <EDC.Perf.Temp.Line.FaultTransTmrAcc> 0

EDC Comm Driver Performance Fault/Status Receive Message Transaction Timer

MOVE
Source Equip.CTL.FaultMsg.TransTMR.ACC 3332
Dest Perf.Temp.Line.FaultTransTmrAcc <EDC.Perf.Temp.Line.FaultTransTmrAcc> 0

Update Performance Summary for LNi, EQi

Update Performance Summary for LNi, EQi

Quality/Health Messages
EN, DN, ER, KA

10

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.QualityRx.EN
<EDC.Perf.Temp.Shop.QualityRx.EN>
Input Par Perf.Temp.Line.QualityRx.EN
<EDC.Perf.Temp.Line.QualityRx.EN>
Input Par EquipStatus03.QualityRx.EN
Return Par Perf.Temp.Shop.QualityRx.EN
<EDC.Perf.Temp.Shop.QualityRx.EN>
Return Par Perf.Temp.Line.QualityRx.EN
<EDC.Perf.Temp.Line.QualityRx.EN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.QualityRx.DN
<EDC.Perf.Temp.Shop.QualityRx.DN>
Input Par Perf.Temp.Line.QualityRx.DN
<EDC.Perf.Temp.Line.QualityRx.DN>
Input Par EquipStatus03.QualityRx.DN
Return Par Perf.Temp.Shop.QualityRx.DN
<EDC.Perf.Temp.Shop.QualityRx.DN>
Return Par Perf.Temp.Line.QualityRx.DN
<EDC.Perf.Temp.Line.QualityRx.DN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.QualityRx.ER
<EDC.Perf.Temp.Shop.QualityRx.ER>
Input Par Perf.Temp.Line.QualityRx.ER
<EDC.Perf.Temp.Line.QualityRx.ER>
Input Par EquipStatus03.QualityRx.ER
Return Par Perf.Temp.Shop.QualityRx.ER
<EDC.Perf.Temp.Shop.QualityRx.ER>
Return Par Perf.Temp.Line.QualityRx.ER
<EDC.Perf.Temp.Line.QualityRx.ER>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.QualityRx.KA
<EDC.Perf.Temp.Shop.QualityRx.KA>
Input Par Perf.Temp.Line.QualityRx.KA
<EDC.Perf.Temp.Line.QualityRx.KA>
Input Par EquipStatus03.QualityRx.KA
Return Par Perf.Temp.Shop.QualityRx.KA
<EDC.Perf.Temp.Shop.QualityRx.KA>
Return Par Perf.Temp.Line.QualityRx.KA
<EDC.Perf.Temp.Line.QualityRx.KA>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.QualityTx.EN
<EDC.Perf.Temp.Shop.QualityTx.EN>
Input Par Perf.Temp.Line.QualityTx.EN
<EDC.Perf.Temp.Line.QualityTx.EN>
Input Par EquipStatus03.QualityTx.EN
Return Par Perf.Temp.Shop.QualityTx.EN
<EDC.Perf.Temp.Shop.QualityTx.EN>
Return Par Perf.Temp.Line.QualityTx.EN
<EDC.Perf.Temp.Line.QualityTx.EN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.QualityTx.DN
<EDC.Perf.Temp.Shop.QualityTx.DN>
Input Par Perf.Temp.Line.QualityTx.DN
<EDC.Perf.Temp.Line.QualityTx.DN>
Input Par EquipStatus03.QualityTx.DN
Return Par Perf.Temp.Shop.QualityTx.DN
<EDC.Perf.Temp.Shop.QualityTx.DN>
Return Par Perf.Temp.Line.QualityTx.DN
<EDC.Perf.Temp.Line.QualityTx.DN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.QualityTx.ER
<EDC.Perf.Temp.Shop.QualityTx.ER>
Input Par Perf.Temp.Line.QualityTx.ER
<EDC.Perf.Temp.Line.QualityTx.ER>
Input Par EquipStatus03.QualityTx.ER
Return Par Perf.Temp.Shop.QualityTx.ER
<EDC.Perf.Temp.Shop.QualityTx.ER>
Return Par Perf.Temp.Line.QualityTx.ER
<EDC.Perf.Temp.Line.QualityTx.ER>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.QualityTx.KA
<EDC.Perf.Temp.Shop.QualityTx.KA>
Input Par Perf.Temp.Line.QualityTx.KA
<EDC.Perf.Temp.Line.QualityTx.KA>
Input Par EquipStatus03.QualityTx.KA
Return Par Perf.Temp.Shop.QualityTx.KA
<EDC.Perf.Temp.Shop.QualityTx.KA>
Return Par Perf.Temp.Line.QualityTx.KA
<EDC.Perf.Temp.Line.QualityTx.KA>

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.QualityMsg.TransTMR.ACC
18661
Source B Perf.Temp.Shop.QualityTransTmrAcc
<EDC.Perf.Temp.Shop.QualityTransTmrAcc>
300546

EDC Comm Driver Performance Quality/Health Receive Message Transaction Timer

MOVE
Source Equip.CTL.QualityMsg.TransTMR.ACC
18661
Dest Perf.Temp.Shop.QualityTransTmrAcc
<EDC.Perf.Temp.Shop.QualityTransTmrAcc>
300546

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.QualityMsg.TransTMR.ACC
18661
Source B Perf.Temp.Line.QualityTransTmrAcc
<EDC.Perf.Temp.Line.QualityTransTmrAcc>
0

EDC Comm Driver Performance Quality/Health Receive Message Transaction Timer

MOVE
Source Equip.CTL.QualityMsg.TransTMR.ACC
18661
Dest Perf.Temp.Line.QualityTransTmrAcc
<EDC.Perf.Temp.Line.QualityTransTmrAcc>
0

Update Performance Summary for LNi, EQi

11

Trend Messages
EN, DN, ER, KA

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.TrendRx.EN <EDC.Perf.Temp.Shop.TrendRx.EN>
Input Par Perf.Temp.Line.TrendRx.EN <EDC.Perf.Temp.Line.TrendRx.EN>
Input Par EquipStatus03.TrendRx.EN
Return Par Perf.Temp.Shop.TrendRx.EN <EDC.Perf.Temp.Shop.TrendRx.EN>
Return Par Perf.Temp.Line.TrendRx.EN <EDC.Perf.Temp.Line.TrendRx.EN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.TrendRx.DN <EDC.Perf.Temp.Shop.TrendRx.DN>
Input Par Perf.Temp.Line.TrendRx.DN <EDC.Perf.Temp.Line.TrendRx.DN>
Input Par EquipStatus03.TrendRx.DN
Return Par Perf.Temp.Shop.TrendRx.DN <EDC.Perf.Temp.Shop.TrendRx.DN>
Return Par Perf.Temp.Line.TrendRx.DN <EDC.Perf.Temp.Line.TrendRx.DN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.TrendRx.ER <EDC.Perf.Temp.Shop.TrendRx.ER>
Input Par Perf.Temp.Line.TrendRx.ER <EDC.Perf.Temp.Line.TrendRx.ER>
Input Par EquipStatus03.TrendRx.ER
Return Par Perf.Temp.Shop.TrendRx.ER <EDC.Perf.Temp.Shop.TrendRx.ER>
Return Par Perf.Temp.Line.TrendRx.ER <EDC.Perf.Temp.Line.TrendRx.ER>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.TrendRx.KA <EDC.Perf.Temp.Shop.TrendRx.KA>
Input Par Perf.Temp.Line.TrendRx.KA <EDC.Perf.Temp.Line.TrendRx.KA>
Input Par EquipStatus03.TrendRx.KA
Return Par Perf.Temp.Shop.TrendRx.KA <EDC.Perf.Temp.Shop.TrendRx.KA>
Return Par Perf.Temp.Line.TrendRx.KA <EDC.Perf.Temp.Line.TrendRx.KA>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.TrendTx.EN <EDC.Perf.Temp.Shop.TrendTx.EN>
Input Par Perf.Temp.Line.TrendTx.EN <EDC.Perf.Temp.Line.TrendTx.EN>
Input Par EquipStatus03.TrendTx.EN
Return Par Perf.Temp.Shop.TrendTx.EN <EDC.Perf.Temp.Shop.TrendTx.EN>
Return Par Perf.Temp.Line.TrendTx.EN <EDC.Perf.Temp.Line.TrendTx.EN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.TrendTx.DN <EDC.Perf.Temp.Shop.TrendTx.DN>
Input Par Perf.Temp.Line.TrendTx.DN <EDC.Perf.Temp.Line.TrendTx.DN>
Input Par EquipStatus03.TrendTx.DN
Return Par Perf.Temp.Shop.TrendTx.DN <EDC.Perf.Temp.Shop.TrendTx.DN>
Return Par Perf.Temp.Line.TrendTx.DN <EDC.Perf.Temp.Line.TrendTx.DN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.TrendTx.ER <EDC.Perf.Temp.Shop.TrendTx.ER>
Input Par Perf.Temp.Line.TrendTx.ER <EDC.Perf.Temp.Line.TrendTx.ER>
Input Par EquipStatus03.TrendTx.ER
Return Par Perf.Temp.Shop.TrendTx.ER <EDC.Perf.Temp.Shop.TrendTx.ER>
Return Par Perf.Temp.Line.TrendTx.ER <EDC.Perf.Temp.Line.TrendTx.ER>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.TrendTx.KA <EDC.Perf.Temp.Shop.TrendTx.KA>
Input Par Perf.Temp.Line.TrendTx.KA <EDC.Perf.Temp.Line.TrendTx.KA>
Input Par EquipStatus03.TrendTx.KA
Return Par Perf.Temp.Shop.TrendTx.KA <EDC.Perf.Temp.Shop.TrendTx.KA>
Return Par Perf.Temp.Line.TrendTx.KA <EDC.Perf.Temp.Line.TrendTx.KA>

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.TrendMsg.TransTMR.ACC 0
Source B Perf.Temp.Shop.TrendTransTmrAcc <EDC.Perf.Temp.Shop.TrendTransTmrAcc> 276400

EDC Comm Driver Performance Trend Receive Message Transaction Timer

MOVE
Source Equip.CTL.TrendMsg.TransTMR.ACC 0
Dest Perf.Temp.Shop.TrendTransTmrAcc <EDC.Perf.Temp.Shop.TrendTransTmrAcc> 276400

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.TrendMsg.TransTMR.ACC 0
Source B Perf.Temp.Line.TrendTransTmrAcc <EDC.Perf.Temp.Line.TrendTransTmrAcc> 0

EDC Comm Driver Performance Trend Receive Message Transaction Timer

MOVE
Source Equip.CTL.TrendMsg.TransTMR.ACC 0
Dest Perf.Temp.Line.TrendTransTmrAcc <EDC.Perf.Temp.Line.TrendTransTmrAcc> 0

Update Performance Summary for LNi, EQi

12

Cycle Messages
EN, DN, ER, KA

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CycleRx.EN
<EDC.Perf.Temp.Shop.CycleRx.EN>
Input Par Perf.Temp.Line.CycleRx.EN
<EDC.Perf.Temp.Line.CycleRx.EN>
Input Par EquipStatus04.CycleRx.EN
Return Par Perf.Temp.Shop.CycleRx.EN
<EDC.Perf.Temp.Shop.CycleRx.EN>
Return Par Perf.Temp.Line.CycleRx.EN
<EDC.Perf.Temp.Line.CycleRx.EN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CycleRx.DN
<EDC.Perf.Temp.Shop.CycleRx.DN>
Input Par Perf.Temp.Line.CycleRx.DN
<EDC.Perf.Temp.Line.CycleRx.DN>
Input Par EquipStatus04.CycleRx.DN
Return Par Perf.Temp.Shop.CycleRx.DN
<EDC.Perf.Temp.Shop.CycleRx.DN>
Return Par Perf.Temp.Line.CycleRx.DN
<EDC.Perf.Temp.Line.CycleRx.DN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CycleRx.ER
<EDC.Perf.Temp.Shop.CycleRx.ER>
Input Par Perf.Temp.Line.CycleRx.ER
<EDC.Perf.Temp.Line.CycleRx.ER>
Input Par EquipStatus04.CycleRx.ER
Return Par Perf.Temp.Shop.CycleRx.ER
<EDC.Perf.Temp.Shop.CycleRx.ER>
Return Par Perf.Temp.Line.CycleRx.ER
<EDC.Perf.Temp.Line.CycleRx.ER>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CycleRx.KA
<EDC.Perf.Temp.Shop.CycleRx.KA>
Input Par Perf.Temp.Line.CycleRx.KA
<EDC.Perf.Temp.Line.CycleRx.KA>
Input Par EquipStatus04.CycleRx.KA
Return Par Perf.Temp.Shop.CycleRx.KA
<EDC.Perf.Temp.Shop.CycleRx.KA>
Return Par Perf.Temp.Line.CycleRx.KA
<EDC.Perf.Temp.Line.CycleRx.KA>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CycleTx.EN
<EDC.Perf.Temp.Shop.CycleTx.EN>
Input Par Perf.Temp.Line.CycleTx.EN
<EDC.Perf.Temp.Line.CycleTx.EN>
Input Par EquipStatus04.CycleTx.EN
Return Par Perf.Temp.Shop.CycleTx.EN
<EDC.Perf.Temp.Shop.CycleTx.EN>
Return Par Perf.Temp.Line.CycleTx.EN
<EDC.Perf.Temp.Line.CycleTx.EN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CycleTx.DN
<EDC.Perf.Temp.Shop.CycleTx.DN>
Input Par Perf.Temp.Line.CycleTx.DN
<EDC.Perf.Temp.Line.CycleTx.DN>
Input Par EquipStatus04.CycleTx.DN
Return Par Perf.Temp.Shop.CycleTx.DN
<EDC.Perf.Temp.Shop.CycleTx.DN>
Return Par Perf.Temp.Line.CycleTx.DN
<EDC.Perf.Temp.Line.CycleTx.DN>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CycleTx.ER
<EDC.Perf.Temp.Shop.CycleTx.ER>
Input Par Perf.Temp.Line.CycleTx.ER
<EDC.Perf.Temp.Line.CycleTx.ER>
Input Par EquipStatus04.CycleTx.ER
Return Par Perf.Temp.Shop.CycleTx.ER
<EDC.Perf.Temp.Shop.CycleTx.ER>
Return Par Perf.Temp.Line.CycleTx.ER
<EDC.Perf.Temp.Line.CycleTx.ER>

Add Performance Summary Counts and Times for EQi

JSR
Routine Name _095_PerfSummaryEquipSum
Input Par Perf.Temp.Shop.CycleTx.KA
<EDC.Perf.Temp.Shop.CycleTx.KA>
Input Par Perf.Temp.Line.CycleTx.KA
<EDC.Perf.Temp.Line.CycleTx.KA>
Input Par EquipStatus04.CycleTx.KA
Return Par Perf.Temp.Shop.CycleTx.KA
<EDC.Perf.Temp.Shop.CycleTx.KA>
Return Par Perf.Temp.Line.CycleTx.KA
<EDC.Perf.Temp.Line.CycleTx.KA>

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.CycleMsg.TransTMR.ACC
0
Source B Perf.Temp.Shop.CycleTransTmrAcc
<EDC.Perf.Temp.Shop.CycleTransTmrAcc>
300160

EDC Comm Driver Performance Cycle Receive Message Transaction Timer

MOVE
Source Equip.CTL.CycleMsg.TransTMR.ACC
0
Dest Perf.Temp.Shop.CycleTransTmrAcc
<EDC.Perf.Temp.Shop.CycleTransTmrAcc>
300160

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.CycleMsg.TransTMR.ACC
0
Source B Perf.Temp.Line.CycleTransTmrAcc
<EDC.Perf.Temp.Line.CycleTransTmrAcc>
0

EDC Comm Driver Performance Cycle Receive Message Transaction Timer

MOVE
Source Equip.CTL.CycleMsg.TransTMR.ACC
0
Dest Perf.Temp.Line.CycleTransTmrAcc
<EDC.Perf.Temp.Line.CycleTransTmrAcc>
0

Update Performance Summary for LNi, EQi

If all Line Equip updated, store Results for Line

13

EDC Comm Driver Performance Performance Equipment Loop -- Equip Index

GE
Source A Perf.CTL.EQi
<EDC.Perf.CTL.EQi>
30
Source B LineCfgEQ.ei
99

EDC Comm Driver Performance All Equipment added to Line Summary

Perf.CTL.LineDone
<EDC.Perf.CTL.LineDone>
L

EDC Comm Driver Performance Status Message State Value

COP
Source Perf.Temp.Line.State
<EDC.Perf.Temp.Line.State>
Dest Perf.Temp.Line.StateVal
<EDC.Perf.Temp.Line.StateVal>
Length 1

EDC Comm Driver Performance EDC Comm Driver Performance Summary for Equipment

COP
Source Perf.Temp.Line
<EDC.Perf.Temp.Line>
Dest Perf.Sum.Line[Perf.CTL.LNi]
<EDC.Perf.Sum.Line[Perf.CTL.LNi]>
Length 1

EDC Comm Driver Performance Line Performance Summary

FLL
Source 0
Dest Perf.Temp.Line
<EDC.Perf.Temp.Line>
Length 1

Update Performance Summary for LNi, EQi

If all Line Equip updated, store Results for Shop

Gateway Display - replaced Worst Status with 2-second scroll of each EQID with a Fault
If no Equipment Comm Fault to Display, show Worst Status (should be Green)
When all EQID have been searched for Faults, reset FaultDisplayLNi and FaultDisplayEQi to search again

14

EDC Comm Driver Performance All Equipment added to Line Summary
Perf.CTL.LineDone
<EDC.Perf.CTL.LineDone>

EDC Comm Driver Performance Performance Line Loop -- Line Index
GE
Source A Perf.CTL.LNi
<EDC.Perf.CTL.LNi>
18
Source B CFG.LN.ei
<EDC.CFG.LN.ei>
28

EDC Comm Driver Performance Status Message State Value
COP
Source Perf.Temp.Shop.State
<EDC.Perf.Temp.Shop.State>
Dest Perf.Temp.Shop.StateVal
<EDC.Perf.Temp.Shop.StateVal>
Length 1

EDC Comm Driver Performance Point Performance Summary
COP
Source Perf.Temp.Shop
<EDC.Perf.Temp.Shop>
Dest Perf.Sum.Shop
<EDC.Perf.Sum.Shop>
Length 1

EDC Comm Driver Performance Shop Performance Summary
FLL
Source 0
Dest Perf.Temp.Shop
<EDC.Perf.Temp.Shop>
Length 1

EDC Comm Driver Performance Performance Worst Status - Equipment ID
CPT
Dest Perf.CTL.StatusWorstEQID
<EDC.Perf.CTL.StatusWorstEQID>
964
Expression Perf.CTL.StatusWorstLNi*100+Perf.CTL.StatusWorstEQi

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Display
Perf.CTL.FaultDisplayDwell
<EDC.Perf.CTL.FaultDisplayDwell>
/

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Found
Perf.CTL.FaultDisplayFound
<EDC.Perf.CTL.FaultDisplayFound>
/

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault EQID
MOVE
Source Perf.CTL.StatusWorstEQID
<EDC.Perf.CTL.StatusWorstEQID>
964
Dest Perf.CTL.FaultDisplayEQID
<EDC.Perf.CTL.FaultDisplayEQID>
964

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Status
MOVE
Source Perf.CTL.StatusWorst
<EDC.Perf.CTL.StatusWorst>
16#0000_0202
Dest Perf.CTL.FaultDisplayStatus
<EDC.Perf.CTL.FaultDisplayStatus>
16#0000_0802

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Display
Perf.CTL.FaultDisplayDwell
<EDC.Perf.CTL.FaultDisplayDwell>
/

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Found
Perf.CTL.FaultDisplayFound
<EDC.Perf.CTL.FaultDisplayFound>
U

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Line Index
CLR
Dest Perf.CTL.FaultDisplayLNi
<EDC.Perf.CTL.FaultDisplayLNi>
0

EDC Comm Driver Performance Gateway Status - Equipment Comm Fault Equip Index
CLR
Dest Perf.CTL.FaultDisplayEQi
<EDC.Perf.CTL.FaultDisplayEQi>
0

EDC Comm Driver Performance Performance Worst Status
CLR
Dest Perf.CTL.StatusWorst
<EDC.Perf.CTL.StatusWorst>

Update Performance Summary for LNi, EQi

16#0000_0202

Copy Equip Local Tags to Arrayed Tags
Local Tag is used for speed and logic readabliity (ie. can be xref'd)

don't need to copy Equip Status back to Equip array -- no changes made

lbl_Exit

15 LBL NOP

(End)

Add Performance Summary Counts and Times for EQi

0

Update Transaction Performance Counts and TImes
(common routine for all Transactions)

Shop and Line Summary

Performance Summary Summation

SBR
Input Par PerfSumShop
Input Par PerfSumLine
Input Par PerfSumTrans

NOP

1

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 95
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Add Performance Summary Counts and Times for EQi

2

Update Transaction Performance
-- Sum Count
-- Sum Responses
-- Average Response

Equipment Transaction values *10 to restore to msec

Performance Summary Summation Transaction Counts

ADD	
Source A	PerfSumShop.Count 11
Source B	PerfSumTrans.Count 0
Dest	PerfSumShop.Count 11

Performance Summary Summation Transaction Response Time Summation

CPT	
Dest	PerfSumShop.Sum 3300030
Expression	PerfSumShop.Sum + (PerfSumTrans.Sum * 10)

Performance Summary Summation Transaction Response Time Average

DIV	
Source A	PerfSumShop.Sum 3300030
Source B	PerfSumShop.Count 11
Dest	PerfSumShop.Avg 300002

Performance Summary Summation Transaction Counts

ADD	
Source A	PerfSumLine.Count 11
Source B	PerfSumTrans.Count 0
Dest	PerfSumLine.Count 11

Performance Summary Summation Transaction Response Time Summation

CPT	
Dest	PerfSumLine.Sum 3300030
Expression	PerfSumLine.Sum + (PerfSumTrans.Sum * 10)

Performance Summary Summation Transaction Response Time Average

DIV	
Source A	PerfSumLine.Sum 3300030
Source B	PerfSumLine.Count 11
Dest	PerfSumLine.Avg 300002

Add Performance Summary Counts and Times for EQi

Update Last Transaction Performance
(Average of all Points' Last Transaction)

Equipment Transaction values *10 to restore to msec

3

Performance Summary
Equip Transaction
Transaction Response
Last Cycle has Valid
Data
PerfSumTrans.LastFlag

Performance Summary
Summation
Transaction Response
Last Point Count

ADD	
Source A	PerfSumShop.LastCount 1
Source B	1
Dest	PerfSumShop.LastCount 1

Performance Summary
Summation
Transaction Response
Last Time Summation

CPT	
Dest	PerfSumShop.LastSum 300010
Expression	PerfSumShop.LastSum + (PerfSumTrans.Last * 10)

Performance Summary
Summation
Transaction Response
Last Point Count

ADD	
Source A	PerfSumLine.LastCount 1
Source B	1
Dest	PerfSumLine.LastCount 1

Performance Summary
Summation
Transaction Response
Last Time Summation

CPT	
Dest	PerfSumLine.LastSum 300010
Expression	PerfSumLine.LastSum + (PerfSumTrans.Last * 10)

Performance Summary
Summation
Transaction Response
Last Cycle

DIV	
Source A	PerfSumShop.LastSum 300010
Source B	PerfSumShop.LastCount 1
Dest	PerfSumShop.Last 300010

Performance Summary
Summation
Transaction Response
Last Cycle

DIV	
Source A	PerfSumLine.LastSum 300010
Source B	PerfSumLine.LastCount 1
Dest	PerfSumLine.Last 300010

Add Performance Summary Counts and Times for EQi

4

Update Transaction Performance
-- Min & Max

Transaction Minimum (x.xxx)

MUL
Source A PerfSumTrans.Min 0
Source B 10
Dest PerfSumTrans_Min 0

Transaction Maximum (x.xxx)

MUL
Source A PerfSumTrans.Max 0
Source B 10
Dest PerfSumTrans_Max 0

Transaction Minimum (x.xxx)

LT
Source A PerfSumTrans_Min 0
Source B PerfSumShop.Min 0

Performance Summary Summation Transaction Response Minimum Time

MOVE
Source PerfSumTrans_Min 0
Dest PerfSumShop.Min 0

Transaction Maximum (x.xxx)

GT
Source A PerfSumTrans_Max 0
Source B PerfSumShop.Max 0

Performance Summary Summation Transaction Response Maximum Time

MOVE
Source PerfSumTrans_Max 0
Dest PerfSumShop.Max 0

Transaction Minimum (x.xxx)

LT
Source A PerfSumTrans_Min 0
Source B PerfSumLine.Min 0

Performance Summary Summation Transaction Response Minimum Time

MOVE
Source PerfSumTrans_Min 0
Dest PerfSumLine.Min 0

Transaction Maximum (x.xxx)

GT
Source A PerfSumTrans_Max 0
Source B PerfSumLine.Max 0

Performance Summary Summation Transaction Response Maximum Time

MOVE
Source PerfSumTrans_Max 0
Dest PerfSumLine.Max 0

5

Return Updated Transaction Performance

Performance Summary Summation

RET
Return Par PerfSumShop
Return Par PerfSumLine

Add Performance Summary Counts and Times for EQi

(End)

Reset Performance Summary

Reset Performance Summary
-- HMI PB
-- Configured Reset TImes

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	96
Dest	Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1733165816
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1733165816

Save last values before clearing
(TBD -- this logic could be expanded to automatically save hourly or daily counts)

2

Performance Summary Previous Values

COP	
Source	Perf.Sum <EDC.Perf.Sum>
Dest	PerfSumPrev
Length	1

EDC Comm Driver Performance Performance Summary

FLL	
Source	0
Dest	Perf.Sum <EDC.Perf.Sum>
Length	1

EDC Comm Driver Performance Performance Summary -- Temporary Values

FLL	
Source	0
Dest	Perf.Temp <EDC.Perf.Temp>
Length	1

Reset Performance Summary

Reset Equipment Transaction TImes/Counts for all Lines

3

Reset Performance Summary for LNi

FOR	
Routine Name	_097_PerfSummaryResetLine
Index	ResetLNi
	29
Initial Value	0
Terminal Value	CFG.LN.mi
	<EDC.CFG.LN.mi>
Step Size	1

Reset Performance Summary

Clear Ring Buffer Max Count
(these should be moved to a Reset routing in each Ring Buffer Program)

4

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Fault01.RecordCountMax 35

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Fault02.RecordCountMax 16

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Fault03.RecordCountMax 111

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Fault04.RecordCountMax 179

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Fault05.RecordCountMax 74

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Fault06.RecordCountMax 571

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Quality01.RecordCountMax 17

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Quality02.RecordCountMax 21

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Quality03.RecordCountMax 24

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Quality04.RecordCountMax 24

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Trend01.RecordCountMax 2

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Trend02.RecordCountMax 2

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Sum01.RecordCountMax 280

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Sum02.RecordCountMax 282

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Cycle01.RecordCountMax 3

Equipment Data Collection Ring Buffer Read Record Maximum Count
CLR
Dest EDC.RB_Cycle02.RecordCountMax 3

Reset Performance Summary

(End)

Reset Performance Summary for LNi

Reset Equipment Transaction TImes/Counts for LNi, EQi

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 97
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1733165816

Reset Equipment Transaction TImes/Counts for LNi

2

Reset Performance Summary for LNi, EQi

FOR
Routine Name _098_PerfSummaryResetEquip
Index ResetEQi 100
Initial Value 0
Terminal Value CFG.EQ.mi <EDC.CFG.EQ.mi>
Step Size 1

(End)

Reset Performance Summary for LNi, EQi

Reset Equipment Transaction TImes/Counts
for LNi, EQi

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan
Time & Fault Routine
Routine Number
currently Scanned

MOVE
Source 98
Dest Prog.Routine
95

Local Program Scan
Time & Fault Routine
Routine Scan
Counters (Indexed by
Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine]
-1733165816
Source B 1
Dest Prog.RoutineScan[Prog.Routine]
-1733165816

Reset Equipment Transaction TImes/Counts
for LNi, EQi
DN007 Split EquipStatus Tag

2

Server Interface
Status Lines and
Equipment Array

FLL
Source 0
Dest EDC_Status01.Line[ResetLNi].Equip[ResetEQi]
Length 1

Event Rx/Tx Status
Lines and Equipment
Array

FLL
Source 0
Dest EDC_Status02.Line[ResetLNi].Equip[ResetEQi]
Length 1

Tool, Trend, Summary
Rx/Tx Status Lines
and Equipment Array

FLL
Source 0
Dest EDC_Status03.Line[ResetLNi].Equip[ResetEQi]
Length 1

Tool, Trend, Summary
Rx/Tx Status Lines
and Equipment Array

FLL
Source 0
Dest EDC_Status04.Line[ResetLNi].Equip[ResetEQi]
Length 1

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 95

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 2666233
Source B 1
Dest Prog.RoutineScan[901] 2666233

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT	
Low Limit	Trans.Filter.CodeLow <EDC.Trans.Filter.CodeLow> 0
Test	Trans.Code <EDC.Trans.Code> 2021
High Limit	Trans.Filter.CodeHigh <EDC.Trans.Filter.CodeHigh> 0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT	
Low Limit	Trans.Filter.LNiLow <EDC.Trans.Filter.LNiLow> 0
Test	Trans.LNi <EDC.Trans.LNi> 4
High Limit	Trans.Filter.LNiHigh <EDC.Trans.Filter.LNiHigh> 0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT	
Low Limit	Trans.Filter.EQiLow <EDC.Trans.Filter.EQiLow> 0
Test	Trans.EQi <EDC.Trans.EQi> 21
High Limit	Trans.Filter.EQiHigh <EDC.Trans.Filter.EQiHigh> 0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

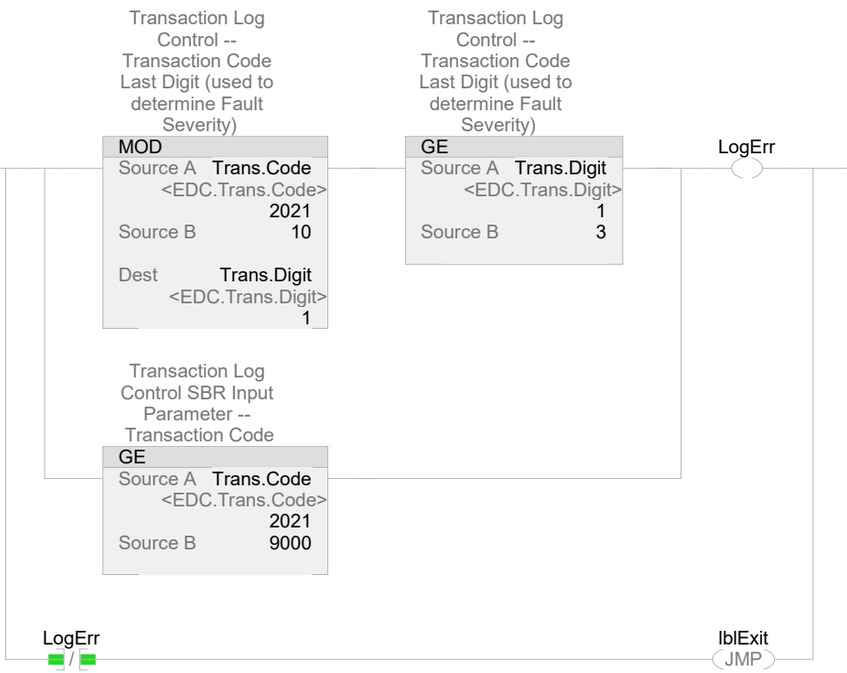
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

EDC2 - v2.01 add Event Configuration for DtOwner and DtType
v3.50 - L980 Controller; Summary KPI additions for ProdDelay, ToolChange and Configurable Option 1-5

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	101
Dest	Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1103711128
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1103711128

Configuration

2

S:FS

EDC Config Configuration ONS (Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_111_Config

Main Routine

Search ScheduleLookkup for the first HourStart where SysTime (HHMMSS) > HourStart (HHMMSS)
Store the Schedule, SchedulePrev and detect NewHour and ShiftReset

this only needs to execute on each new second since schedule can only change on the second (and typically only changes on the minute)
at end of shift, send final packet and then reset Summary data

3

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

NE
Source A CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Source B ScheduleTimePrev
151833

Schedule Lookup
Time of Prev Check

MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest ScheduleTimePrev
151833

Schedule Search FSC
ScheduleFSC
RES

Schedule Search FSC

MOVE
Source 0
Dest ScheduleFSC.POS
16

Schedule Search FSC

ADD
Source A CfgSched.ei
22
Source B 1
Dest ScheduleFSC.LEN
23

Schedule Search FSC

FSC
Control ScheduleFSC
Length 23
Position 16
Mode ALL
Expression CLX.SysTime[9]<ScheduleLookup[ScheduleFSC.POS].HourStart
EN
DN
ER

Schedule Search FSC
ScheduleFSC.FD

Schedule Lookup
Array Position

SUB
Source A ScheduleFSC.POS
16
Source B 1
Dest SchedulePOS
15

Schedule
Previous Second

COP
Source Schedule
Dest SchedulePrev
Length 1

Schedule
Current Hour

COP
Source ScheduleLookup[SchedulePOS]
Dest Schedule
Length 1

Schedule
Current Hour Hour
(0-10)

NE
Source A Schedule.Hour
9
Source B SchedulePrev.Hour
9

Schedule
New Hour
ScheduleNewHour

diagnostic

ADD
Source A SchedNewHourCount
3129
Source B 1
Dest SchedNewHourCount
3129

Schedule
Current Hour Shift
Reset (0,1)

EQ
Source A Schedule.Reset
0
Source B 1

Schedule
Current Hour Shift
Reset (0,1)

NE
Source A Schedule.Reset
0
Source B SchedulePrev.Reset
0

Schedule
Shift Reset
ScheduleShiftReset
L

diagnostic

ADD
Source A SchedShiftResetCount
285
Source B 1
Dest SchedShiftResetCount
285

Main Routine

4

Load Summary Ring Buffer Timer
periodically send EQ, Line and Group Summary if the Ring Buffer is empty
(no point in adding more Summary data if processing is backlogged - it will be old data when processed)
NewHour must be logged to close out the existing hour (bypass the buffer empty check)

Send the data on the minute change for a consistent interval

Load Summary Ring Buffer Timer
MOVE
Source 65000
Dest SumDataTMR.PRE 65000

Load Summary Ring Buffer Timer
TON
Timer SumDataTMR
Preset 65000
Accum 53558
EN
DN

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)
NE
Source A CLX.SysTime[4] <EDC.CLX.SysTime[4]> 19
Source B SysTimePrevMinute 24

MOVE
Source CLX.SysTime[4] <EDC.CLX.SysTime[4]> 19
Dest SysTimePrevMinute 24

ScheduleNewMinute
L

Load Summary Ring Buffer Timer
SumDataTMR.DN

ScheduleNewMinute

Schedule New Hour
ScheduleNewHour

EDC Ring Buffer Control Buffer Empty
RingBufferSum01.Empty <EDC.RB_Sum01.Empty>

EDC Ring Buffer Control Buffer Empty
RingBufferSum02.Empty <EDC.RB_Sum02.Empty>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RingBufferSum01.NotEmptyTMR.EN <EDC.RB_Sum01.NotEmptyTMR.EN>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RingBufferSum02.NotEmptyTMR.EN <EDC.RB_Sum02.NotEmptyTMR.EN>

Load Summary Ring Buffer by LNi
FLL
Source -1
Dest LineSumDataLoad[0]
Length 4

Load Summary Ring Buffer Timer
SumDataTMR
RES

ScheduleNewMinute
U

Main Routine

Use Schedule Shift Reset signal to reset all shift-based data
delay Reset by 1 scan to allow NewHour to send end-of-shift data before resetting values
(Reset and NewHour are enabled on same scan)

5

Schedule Shift Reset
ScheduleShiftReset

Schedule New Hour
ScheduleNewHour

Schedule Shift Reset
ScheduleShiftReset
(U)

Reset Line, Group and Equip Data by LNi

FLL
Source -1
Dest LineDataReset[0]
Length 4

ShiftResetONS
(ONS)

ShiftReset

Line Loop Configuration Index

When LNi = CfgLNi, all Equipment are checked for Callout Messages (to find new Equipment)
When LNi <> CfgLNi, only configured Equipment are checked for Callout Messages
Only 1 Line/Scan is checked for new Equipment to save scan time
(only increment if crrent CfgLNi was in Lines scanned during last Line Loop)

6

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>

Config Update Line Index

LIMIT
Low Limit LineLoopInit 15
Test CfgLNi 8
High Limit LineLoopTerm 14

Config Update Line Index

ADD
Source A CfgLNi 8
Source B 1
Dest CfgLNi 8

Config Update Line Index

LT
Source A CfgLNi 8
Source B 0

Config Update Line Index

GT
Source A CfgLNi 8
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

Config Update Line Index

MOVE
Source 0
Dest CfgLNi 8

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>

Config Update Line Index

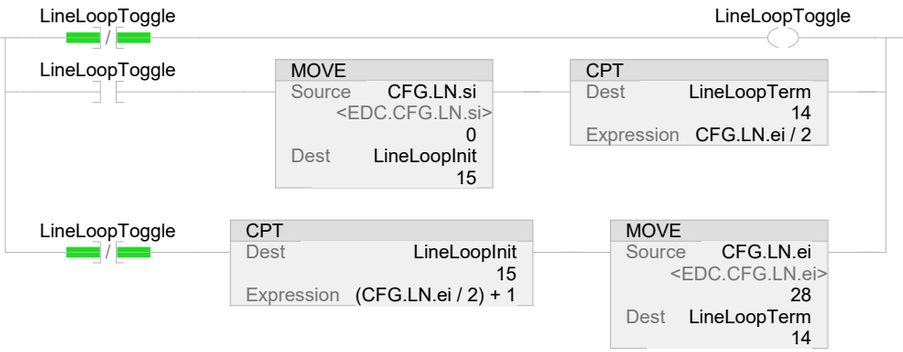
MOVE
Source _ViewLNi 9
Dest CfgLNi 8

Main Routine

Line Loop Initial/Terminal values for n Lines/Scan
reduce scan time by updating portion of lines each scan but maintain <200 msec Data Increment

Loop Increment = 0-20, 21-40
2 scans update all Lines

7

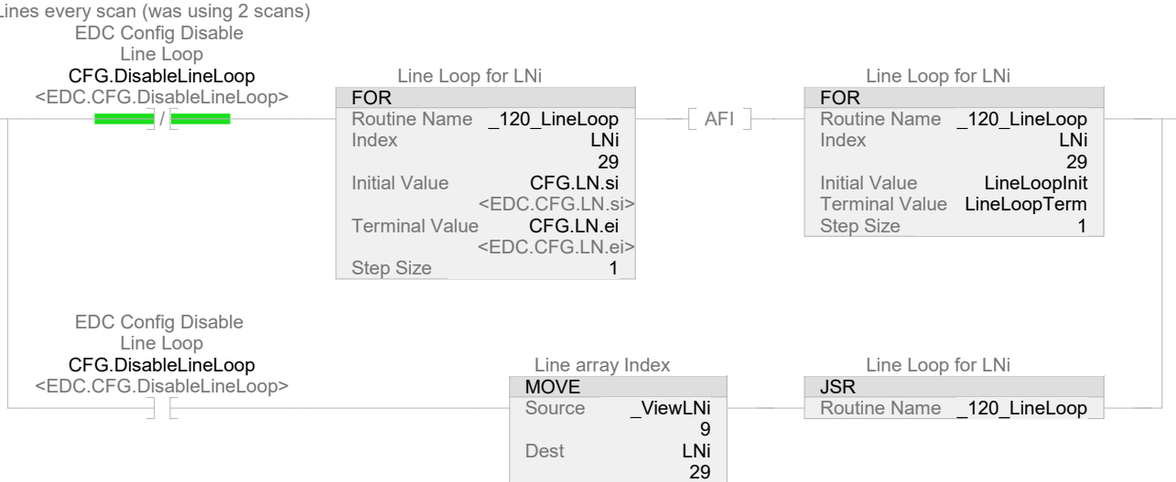


Line Loop
service all Lines each Scan for fast update of Live Callout table

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop will cause the unscanned Equipment to stop updating)

DHF008 -- service all Lines every scan (was using 2 scans)

8



Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

9

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Routine
	152
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Type
	4
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Code
	20
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.FaultCount
	0
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.LOGi
	19
Dest	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	9003
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

Main Routine

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

10

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

11

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
3696
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan Events
CLxProgScanHigh.1
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 64

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 64
Source B	1
Dest	Prog.LOGmi 49

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 19
Source B	1
Dest	Prog.LOGi 19

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 19
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 19
Source B	Prog.LOGmi 49

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 19

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
3

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
143
Dest Prog.Log[Prog.LOGi].Routine
152

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
135064652

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
20220707

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
110119

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
659

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source LNi
29
Dest Prog.Log[Prog.LOGi].Opt1
28

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source EQi
12
Dest Prog.Log[Prog.LOGi].Opt2
82

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 102
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 588226492

(End)

Configuration

Callout Message Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 111
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Configuration

Callout Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

2

Equipment Data Buffer Callout/Status Receive Message

SIZE
Source EDC_EqRxBuffer.Line[0].Equip[0].Callout[0]
1
Dim. To Vary 0
Size CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>
14

EDC Config Array Max Index

SUB
Source A CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>
14
Source B 1
Dest CFG.CalloutMSG.mi
<EDC.CFG.CalloutMSG.mi>
13

EDC Config Array End Count (udt size)

MOVE
Source 14
Dest CFG.CalloutMSG.e
<EDC.CFG.CalloutMSG.e>
14

EDC Config Array End Count (udt size)

GT
Source A CFG.CalloutMSG.e
<EDC.CFG.CalloutMSG.e>
14
Source B CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>
14

EDC Config Array End Count (udt size)

MOVE
Source CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>
14
Dest CFG.CalloutMSG.e
<EDC.CFG.CalloutMSG.e>
14

EDC Config Array End Index (used for copies)

SUB
Source A CFG.CalloutMSG.e
<EDC.CFG.CalloutMSG.e>
14
Source B 1
Dest CFG.CalloutMSG.ei
<EDC.CFG.CalloutMSG.ei>
13

Configuration

Callout Table Array
Configure Array Indexes

3

Callout Table (LNi, EQi) Equipment Receive Callout Live Table (for external OPC access)

SIZE
Source EDC_EqCallout[0,0] ??
Dim. To Vary 0
Size CFG.CalloutLN.n <EDC.CFG.CalloutLN.n> 29

EDC Config Array Max Index

SUB
Source A CFG.CalloutLN.n <EDC.CFG.CalloutLN.n> 29
Source B 1
Dest CFG.CalloutLN.mi <EDC.CFG.CalloutLN.mi> 28

Callout Table (LNi, EQi) Equipment Receive Callout Live Table (for external OPC access)

SIZE
Source EDC_EqCallout[0,0] ??
Dim. To Vary 1
Size CFG.CalloutEQ.n <EDC.CFG.CalloutEQ.n> 100

EDC Config Array Max Index

SUB
Source A CFG.CalloutEQ.n <EDC.CFG.CalloutEQ.n> 100
Source B 1
Dest CFG.CalloutEQ.mi <EDC.CFG.CalloutEQ.mi> 99

Callout Table Array Size
Predefined DINT + 40 Equipment-Defined DINTs (8 DINTs * 5 Packets)

4

Local Copy of EDC_EqCallout for this EQID

SIZE
Source TempCallout[0] 16#0000_00a3
Dim. To Vary 0
Size CFG.CalloutARR.n <EDC.CFG.CalloutARR.n> 65

EDC Config Array Max Index

SUB
Source A CFG.CalloutARR.n <EDC.CFG.CalloutARR.n> 65
Source B 1
Dest CFG.CalloutARR.mi <EDC.CFG.CalloutARR.mi> 64

Configuration

Callout Packet Count
v1.3 Fault/Callout MultiPacket support

5

EDC Config Array Size

MOVE	
Source	8
Dest	CFG.CalloutPKT.n <EDC.CFG.CalloutPKT.n> 8

EDC Config Array Max Index

SUB	
Source A	CFG.CalloutPKT.n <EDC.CFG.CalloutPKT.n> 8
Source B	1
Dest	CFG.CalloutPKT.mi <EDC.CFG.CalloutPKT.mi> 7

Callout Packet Eq-Defined Array Size (8 DINTs/Packet)
v1.3 Fault/Callout MultiPacket support

6

Equipment Data Callout/Status Message Equipment Defined Status Words

SIZE	
Source	CalloutMsg.EqDef.Word[0] 16#0000_0000
Dim. To Vary	0
Size	CFG.CalloutEqDef.n <EDC.CFG.CalloutEqDef.n> 8

EDC Config Array Max Index

SUB	
Source A	CFG.CalloutEqDef.n <EDC.CFG.CalloutEqDef.n> 8
Source B	1
Dest	CFG.CalloutEqDef.mi <EDC.CFG.CalloutEqDef.mi> 7

Zero Array used for FBC Comparison

7

SIZE	
Source	ZeroArray[0] 0
Dim. To Vary	0
Size	LEN 64

FLL	
Source	0
Dest	ZeroArray[0]
Length	LEN

Initialize Line View

8

_ViewEN
U

Configuration

Schedule Array

9

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>
U

Schedule Hours lookup array Shift Configuration from EDC Database

SIZE
Source ScheduleLookup[0] ??
Dim. To Vary 0
Size CfgSched.n 100

Schedule Array Config Array Max Index

SUB
Source A CfgSched.n 100
Source B 1
Dest CfgSched.mi 99

Schedule Array Config Array Start Index

MOVE
Source 0
Dest CfgSched.si 0

Schedule Array Config Array End Index

MOVE
Source CfgSched.mi 99
Dest CfgSched.ei 22

schedule downloaded from server
no longer need hard-coded schedule

10

lbl_Exit
JMP

Configuration

hardcode Schedule Shift Configuration
(this will be replaced by a configuration download from EDC database)

11

Schedule Hours lookup array Shift Configuration from EDC Database

FLL
Source -1
Dest ScheduleLookup[0]
Length CfgSched.n

Schedule Array Config Array Start Index

MOVE
Source 0
Dest CfgSched.si 0

Schedule Array Config Array End Index

MOVE
Source 22
Dest CfgSched.ei 22

12

MOVE
Source 0
Dest index 22

Schedule Hours lookup array

MOVE
Source 0
Dest ScheduleLookup[index].HourStart 240000

Schedule Hours lookup array

MOVE
Source 2
Dest ScheduleLookup[index].Shift 2

Schedule Hours lookup array

MOVE
Source 3
Dest ScheduleLookup[index].Qtr 2

Schedule Hours lookup array

MOVE
Source 5
Dest ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE
Source 0
Dest ScheduleLookup[index].Reset 0

Configuration

13

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	1	Source	10000	Source	2	Source	3	Source	6
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

14

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	2	Source	21200	Source	2	Source	4	Source	7
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

15

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	3	Source	31200	Source	2	Source	4	Source	8
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

16

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	4	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	40500	Source	2	Source	5	Source	9
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

17

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	5	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	50500	Source	2	Source	5	Source	10
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

18

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	6	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	60000	Source	1	Source	0	Source	0
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	1
Dest	ScheduleLookup[index].Reset 0

Configuration

19

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	7	Source	63600	Source	1	Source	1	Source	1
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

20

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	8	Source	73600	Source	1	Source	1	Source	2
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

21

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	9	Source	84200	Source	1	Source	2	Source	3
Dest	index 22	Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

22

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	10	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	94200	Source	1	Source	2	Source	4
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

23

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	11	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	111000	Source	1	Source	3	Source	5
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

24

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	12	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	121000	Source	1	Source	3	Source	6
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

25

MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE	
Source	13		Source	132700		Source	1		Source	4		Source	7
Dest	index 22		Dest	ScheduleLookup[index].HourStart 240000		Dest	ScheduleLookup[index].Shift 2		Dest	ScheduleLookup[index].Qtr 2		Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

26

MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE	
Source	14		Source	142700		Source	1		Source	4		Source	8
Dest	index 22		Dest	ScheduleLookup[index].HourStart 240000		Dest	ScheduleLookup[index].Shift 2		Dest	ScheduleLookup[index].Qtr 2		Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

27

MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE		Schedule Hours lookup array	MOVE	
Source	15		Source	152500		Source	1		Source	5		Source	9
Dest	index 22		Dest	ScheduleLookup[index].HourStart 240000		Dest	ScheduleLookup[index].Shift 2		Dest	ScheduleLookup[index].Qtr 2		Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

28

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	16	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	162500	Source	1	Source	5	Source	10
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

29

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	17	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	170000	Source	2	Source	0	Source	0
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	1
Dest	ScheduleLookup[index].Reset 0

30

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	18	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	192100	Source	2	Source	1	Source	1
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

31

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	19	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	202100	Source	2	Source	1	Source	2
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

32

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	20	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	212700	Source	2	Source	2	Source	3
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

33

MOVE		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array		Schedule Hours lookup array	
Source	21	MOVE		MOVE		MOVE		MOVE	
Dest	index 22	Source	222700	Source	2	Source	2	Source	4
		Dest	ScheduleLookup[index].HourStart 240000	Dest	ScheduleLookup[index].Shift 2	Dest	ScheduleLookup[index].Qtr 2	Dest	ScheduleLookup[index].Hour 4

Schedule Hours lookup array

MOVE	
Source	0
Dest	ScheduleLookup[index].Reset 0

Configuration

34

MOVE		Schedule Hours lookup array
Source	22	
Dest	index 22	

MOVE		Schedule Hours lookup array
Source	240000	
Dest	ScheduleLookup[index].HourStart 240000	

MOVE		Schedule Hours lookup array
Source	2	
Dest	ScheduleLookup[index].Shift 2	

MOVE		Schedule Hours lookup array
Source	2	
Dest	ScheduleLookup[index].Qtr 2	

MOVE		Schedule Hours lookup array
Source	4	
Dest	ScheduleLookup[index].Hour 4	

MOVE		Schedule Hours lookup array
Source	0	
Dest	ScheduleLookup[index].Reset 0	

35

lbl_Exit
LBL

NOP

(End)

Line Loop for LNi

Line Service Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop will cause the unscanned Equipment to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 120
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Line Loop for LNi

Copy Line Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

Reset Line bits for Accumulated EQ status
Offline Line w/ multiple Data Groups (special case)
DN006 - support multiple Offline Lines

LineDataSTS - accumulates overlapping Status of Downtime types (Tool Change, Options, Full, Short, ProdDelay)
LineDataCTL - contains exclusive Control of Downtime Type (generated after all Group-Equipment processed)
STS & CTL are needed to handle seal-in of existing Downtime Type

2

Local Copy of Line Array Tag for LNi
COP
Source EDC.Line[LNi].CFG
Dest LineCfg
Length 1

Local Copy of Line Array Tag for LNi
COP
Source EDC.Line[LNi].CTL
Dest LineCtl
Length 1

Local Copy of Line Array Tag for LNi
COP
Source EDC.Line[LNi].Data
Dest LineData
Length 1

Local Copy of Line Array Tag for LNi
FLL
Source 0
Dest LineDataSTS
Length 1

Local Copy of Line Array Tag for LNi
FLL
Source 0
Dest LineDataCTL
Length 1

Local Copy of Line Array Tag for LNi
COP
Source EDC.Line[LNi].DataCTLprev
Dest LineDataCTLprev
Length 1

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Line array Index
EQ
Source A LNi
29
Source B CFG.OfflineFirstLine
<EDC.CFG.OfflineFirstLine>
0

Offline Group Data Initialize
FOR
Routine Name _141_OfflineGroupDataInit
Index GRPi
4
Initial Value CFG.GRP.si
<EDC.CFG.GRP.si>
Terminal Value CFG.GRP.ei
<EDC.CFG.GRP.ei>
Step Size 1

Initialize Offline Groups (1st Offline Line)
OfflineFirstLine

Line array Index
EQ
Source A LNi
29
Source B CFG.OfflineLastLine
<EDC.CFG.OfflineLastLine>
28

Finalize Offline Groups (Last Offline Line)
OfflineLastLine

Nov 24, 2025 - Preserve the CycleSend signal for sending Summary Data updates when still cycling after WorkTime is off

3

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].DataCTL.CycleSend

Local Copy of Line Array Tag for LNi Cycle Count Send Summary Data
LineDataCTL.CycleSend
L

Line Loop for LNi

v3.5 Preserve the Downtime Type seal-in

4

Equipment Data Collection Lines and Equipment Array	Local Copy of Line Array Tag for LNi
EDC.Line[LNi].DataCTL.ToolChange	LineDataCTL.ToolChange
EDC.Line[LNi].DataCTL.Option01	LineDataCTL.Option01
EDC.Line[LNi].DataCTL.Option02	LineDataCTL.Option02
EDC.Line[LNi].DataCTL.Option03	LineDataCTL.Option03
EDC.Line[LNi].DataCTL.Option04	LineDataCTL.Option04
EDC.Line[LNi].DataCTL.Option05	LineDataCTL.Option05
EDC.Line[LNi].DataCTL.ProdDelay	LineDataCTL.ProdDelay
EDC.Line[LNi].DataCTL.Full	LineDataCTL.Full
EDC.Line[LNi].DataCTL.Short	LineDataCTL.Short

Line Loop for LNi

Load & Reset Summary Ring Buffer Data signals for this Line

5

Line Bit Index

MOD
Source A: LNi 29
Source B: 32
Dest: LnBTi 28

Line Word Index

CPT
Dest: LnWDi 0
Expression: (LNi - LnBTi) / 32

Load Summary Ring Buffer by LNi: LineSumDataLoad[LnWDi].[LnBTi]

Load Summary Ring Buffer for this Line: SumDataLoad

Reset Line, Group and Equip Data by LNi: LineDataReset[LnWDi].[LnBTi]

Reset Summary Data for this Line: SumDataReset

Calculate Data Increment for this Line's Scan(s) since last update.
Previous Scan(s) Time increment will be added to all Line, Group and Equipment Values that were ON.
Each Line has its own DataInc to support Line updates over multiple scans (but for performance, Callout should run all Lines on a single scan if possible)

Wallclock CurrentValue microseconds is divided by 1000 for the millisecond data increment.
The millisecond data increment is multiplied by 1000 for the microseconds to add to the Previous Value (this <1000 microseconds accumulate for a future data increment)

6

Local Copy of Line Array Tag for LNi Data Increment (microseconds)

SUB
Source A: CLX.WallclockCurrentValue <EDC.CLX.WallclockCurrentValue> 1793391561904697
Source B: LineCtl.WallclockPreviousValue 6595304822000
Dest: LineCtl.DataIncMicrosec 16234

Local Copy of Line Array Tag for LNi Data Increment (milliseconds)

DIV
Source A: LineCtl.DataIncMicrosec 16234
Source B: 1000
Dest: LineCtl.DataInc 16

Local Copy of Line Array Tag for LNi Previous Value Increment (microseconds)

MUL
Source A: LineCtl.DataInc 16
Source B: 1000
Dest: LineCtl.PrevIncMicrosec 16000

Local Copy of Line Array Tag for LNi Wallclock Previous Value (microseconds)

ADD
Source A: LineCtl.WallclockPreviousValue 6595304822000
Source B: LineCtl.PrevIncMicrosec 16000
Dest: LineCtl.WallclockPreviousValue 6595304822000

Line Loop for LNi

Equipment Loop

One set of Interface Routines is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop will cause the unscanned Equipment to stop updating)

When LNi = CfgLNi, all Equipment are checked for Callout Messages (to find new Equipment)
When LNi <> CfgLNi, only configured Equipment are checked for Callout Messages
Only 1 Line/Scan is checked for new Equipment to save scan time

7

EDC Config Disable Equipment Loop
CFG.DisableEqLoop
<EDC.CFG.DisableEqLoop>

Line array Index
NE
Source A: LNi 29
Source B: CfgLNi 8

Equipment Loop for EQi
FOR
Routine Name: _130_EquipLoop
Index: EQi 12
Initial Value: LineCfg.EQ.si
Terminal Value: LineCfg.EQ.ei
Step Size: 1

Line array Index
EQ
Source A: LNi 29
Source B: CfgLNi 8

Config Check Line LNi = CfgLNi
CfgLine

Temp Storage of LineCfg.EQ.ENei
CLR
Dest: LineCfg_EqENei_Temp 67

Equipment Loop for EQi
FOR
Routine Name: _130_EquipLoop
Index: EQi 12
Initial Value: 0
Terminal Value: CFG.EQ.mi <EDC.CFG.EQ.mi>
Step Size: 1

Local Copy of Line Array Tag for LNi Array End Index for Enabled Equipment
MOVE
Source: LineCfg_EqENei_Temp 67
Dest: LineCfg.EQ.ENei 99

EDC Config Disable Equipment Loop
CFG.DisableEqLoop
<EDC.CFG.DisableEqLoop>

Equipment array Index
MOVE
Source: _ViewEQi 18
Dest: EQi 12

Equipment Loop for EQi
JSR
Routine Name: _130_EquipLoop

Faster to JMP over AOI if not scanned
(AOI uses scan time even when Rung is scanned false)

8

Reset Summary Data for this Line
SumDataReset

lbl_SkipReset
JMP

Line Loop for LNi

Shift Reset for Line & Offline OA
Fault & Alarm Time are not reset at Shift Change

Offline Line w/ multiple Data Groups (special case)
DN006 - support multiple Offline Lines

9

Reset Summary Data for this Line
SumDataReset

Data Reset for Line, Group, Equip

aoiEDC_DataReset	
aoiEDC_DataReset	aoiDataReset ...
Data	LineData
DataCtlPrev	LineDataCTLprev

Initialize Offline Groups (1st Offline Line)
OfflineFirstLine

Offline Group Data Reset

FOR	
Routine Name	_142_OfflineGroupDataReset
Index	GRPi
	4
Initial Value	CFG.GRP.si
	<EDC.CFG.GRP.si>
Terminal Value	CFG.GRP.ei
	<EDC.CFG.GRP.ei>
Step Size	1

Line Loop for LNi

Data Increment for Line and Offline OA
Add Incremental Time to Worktime, Downtime that are ON
Add Incremental Time to to Current Alarm or Fault Duration
Calculate OA
DN006 - support multiple Offline Lines

Oct 12, 2021 - don't update Offline until all Offline Lines are updated and data is complete
(work bit was toggling between Line 0 and Line 28 update and generating many WorkEnd signals)

10

lbl_SkipReset
LBL

Prioritize STS for CTL

aoiEDC_DataCtl

aoiEDC_DataCtl	aoiDataCtl
DataSts	LineDataSTS
DataCtl	LineDataCTL

Increment Data

aoiEDC_DataInc

aoiEDC_DataInc	aoiDataInc
Data	LineData
DataCtl	LineDataCTL
DataCtlPrev	LineDataCTLprev
Increment	LineCtl.DataInc

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Finalize Offline Groups (Last Offlline Line)
OfflineLastLine

Offline Group Data Update

FOR

Routine Name	_143_OfflineGroupData
Index	GRPi
	4
Initial Value	CFG.GRP.si
	<EDC.CFG.GRP.si>
Terminal Value	CFG.GRP.ei
	<EDC.CFG.GRP.ei>
Step Size	1

Line Loop for LNi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

OA% - InDemand Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

11

Local Copy of Line Array Tag for LNi Line Enabled
LineCfg.EN

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Line Array Tag for LNi Work Time (Overall Start)
LineDataCTL.Work

Local Copy of Line Array Tag for LNi Work Time (Overall Start)
LineDataCTL.Work

Local Copy of Line Array Tag for LNi Cycle Count Send Summary Data
LineDataCTL.CycleSend

Schedule New Hour
ScheduleNewHour

Local Copy of Line Array Tag for LNi Work Time Ended
LineDataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE
Source 2
Dest SumMsg.Header.DataType 13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE
Source LNi 29
Dest SumMsg.Header.ID 3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP
Source LineData.Work
Dest SumMsg.Data.Work
Length 1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP
Source LineData.EdtOA.Maint
Dest SumMsg.Data.EdtMaint
Length 1

Equipment Data Summary Message 3 - Equip Downtime Production

COP
Source LineData.EdtOA.Prod
Dest SumMsg.Data.EdtProd
Length 1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP
Source LineData.LdtOA.Maint
Dest SumMsg.Data.LdtMaint
Length 1

Equipment Data Summary Message 5 - Line Downtime Production

COP
Source LineData.LdtOA.Prod
Dest SumMsg.Data.LdtProd
Length 1

Equipment Data Summary Message 6 -Full

COP
Source LineData.Full
Dest SumMsg.Data.Full

Equipment Data Summary Message 7 - Short

COP
Source LineData.Short
Dest SumMsg.Data.Short

Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP
Source LineData.ProdDelay
Dest SumMsg.Data.ProdDelay

Equipment Data Summary Message 10 - Tool Change (ADC)

COP
Source LineData.ToolChange
Dest SumMsg.Data.ToolChange

Line Loop for LNi

Length 1 | Length 1 | Length 1 | Length 1

Equipment Data Summary Message 11 - Config Option 1

COP
Source LineData.Option01
Dest SumMsg.Data.Opt01
Length 1

Equipment Data Summary Message 12 - Config Option 2

COP
Source LineData.Option02
Dest SumMsg.Data.Opt02
Length 1

Equipment Data Summary Message 13 - Config Option 3

COP
Source LineData.Option03
Dest SumMsg.Data.Opt03
Length 1

Equipment Data Summary Message 14 - Config Option 4

COP
Source LineData.Option04
Dest SumMsg.Data.Opt04
Length 1

Equipment Data Summary Message 15 - Config Option 5

COP
Source LineData.Option05
Dest SumMsg.Data.Opt05
Length 1

Equipment Data Summary Message Time Total

MOVE
Source LineCfg.Takt
0
Dest SumMsg.Data.Cycle.Time
0

Equipment Data Summary Message Count Total

MOVE
Source LineData.Cycle.Count
0
Dest SumMsg.Data.Cycle.Count
0

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Line Number Odd/Even
(better distribution than splitting Line Numbers into Low/High range)

OR% - All Downtime

12

SumDataLoadRB

Load Summary Data Message

JSR
Routine Name _150_SumDataMsg

RingBuffer Modulo Result

MOD
Source A LNi
29
Source B 2
Dest BufferID
1

RingBuffer Modulo Result

EQ
Source A BufferID
1
Source B 1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _151_SumLoadRB01

UseSumRB01

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _152_SumLoadRB02

Line Loop for LNi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

OR% - All Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

13

Local Copy of Line Array Tag for LNi Line Enabled
LineCfg.EN

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Line Array Tag for LNi Work Time (Overall Start)
LineDataCTL.Work

Local Copy of Line Array Tag for LNi Cycle Count Send Summary Data
LineDataCTL.CycleSend

Local Copy of Line Array Tag for LNi Cycle Count Send Summary Data
LineDataCTL.CycleSend
U

Local Copy of Line Array Tag for LNi Work Time (Overall Start)
LineDataCTL.Work
/

Schedule New Hour
ScheduleNewHour

Local Copy of Line Array Tag for LNi Work Time Ended
LineDataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE
Source 12
Dest SumMsg.Header.DataType
13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE
Source LNi
29
Dest SumMsg.Header.ID
3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP
Source LineData.Work
Dest SumMsg.Data.Work
Length 1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP
Source LineData.EdtOR.Maint
Dest SumMsg.Data.EdtMaint
Length 1

Equipment Data Summary Message 3 - Equip Downtime Production

COP
Source LineData.EdtOR.Prod
Dest SumMsg.Data.EdtProd
Length 1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP
Source LineData.LdtOR.Maint
Dest SumMsg.Data.LdtMaint
Length 1

Equipment Data Summary Message 5 - Line Downtime Production

COP
Source LineData.LdtOR.Prod
Dest SumMsg.Data.LdtProd
Length 1

Equipment Data Summary Message 6 -Full

COP
Source LineData.Full
Dest SumMsg.Data.Full

Equipment Data Summary Message 7 - Short

COP
Source LineData.Short
Dest SumMsg.Data.Short

Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP
Source LineData.ProdDelay
Dest SumMsg.Data.ProdDelay

Equipment Data Summary Message 10 - Tool Change (ADC)

COP
Source LineData.ToolChange
Dest SumMsg.Data.ToolChange

Line Loop for LNi

Length 1 | Length 1 | Length 1 | Length 1

Equipment Data Summary Message 11 - Config Option 1

COP
Source LineData.Option01
Dest SumMsg.Data.Opt01
Length 1

Equipment Data Summary Message 12 - Config Option 2

COP
Source LineData.Option02
Dest SumMsg.Data.Opt02
Length 1

Equipment Data Summary Message 13 - Config Option 3

COP
Source LineData.Option03
Dest SumMsg.Data.Opt03
Length 1

Equipment Data Summary Message 14 - Config Option 4

COP
Source LineData.Option04
Dest SumMsg.Data.Opt04
Length 1

Equipment Data Summary Message 15 - Config Option 5

COP
Source LineData.Option05
Dest SumMsg.Data.Opt05
Length 1

Equipment Data Summary Message Time Total

MOVE
Source LineCfg.Takt
0
Dest SumMsg.Data.Cycle.Time
0

Equipment Data Summary Message Count Total

MOVE
Source LineData.Cycle.Count
0
Dest SumMsg.Data.Cycle.Count
0

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Line Number Odd/Even
(better distribution than splitting Line Numbers into Low/High range)

OA% - InDemand Downtime

14

SumDataLoadRB

Load Summary Data Message

JSR
Routine Name _150_SumDataMsg

RingBuffer Modulo Result

MOD
Source A LNi
29
Source B 2
Dest BufferID
1

RingBuffer Modulo Result

EQ
Source A BufferID
1
Source B 1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _151_SumLoadRB01

UseSumRB01

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _152_SumLoadRB02

Line Loop for LNi

Copy Line working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

save updates to Line.CFG from new Equipment Callout messages

15

Equipment Data Collection Lines and Equipment Array
COP
Source LineCfg
Dest EDC.Line[LNi].CFG
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source LineCtl
Dest EDC.Line[LNi].CTL
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source LineData
Dest EDC.Line[LNi].Data
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source LineDataSTS
Dest EDC.Line[LNi].DataSTS
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source LineDataCTL
Dest EDC.Line[LNi].DataCTL
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source LineDataCTLprev
Dest EDC.Line[LNi].DataCTLprev
Length 1

Exit Routine

16

lbl_Exit
LBL

NOP

Reset Line Summary Data and Reset signals
(after Line, Group and Equipment for this LNi are updated)

17

Load Summary Ring Buffer by LNi
LineSumDataLoad[LnWDi].[LnBTi]
U

Reset Line, Group and Equip Data by LNi
LineDataReset[LnWDi].[LnBTi]
U

Check Viewing Index Limits

18

Line View Tag Index
LT
Source A _ViewLNi 9
Source B 0

Line View Tag Index
GT
Source A _ViewLNi 9
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

Line View Tag Index
CLR
Dest _ViewLNi 9

Line Loop for LNi

Copy Line Instance to Viewing Tag if selected
(_ViewLine can be used to multiplex Trends and Diagnostic Spreadsheets)

19

_ViewEN

Line array Index

EQ	
Source A	LNi 29
Source B	_ViewLNi 9

Line View Tag
(used by View
Diagnostic
Spreadsheet)

COP	
Source	EDC.Line[LNi]
Dest	_ViewLine
Length	1

(End)

Equipment Loop for EQi

Equipment Service Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop will cause the unscanned Equipment to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 130
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Equipment Loop for EQi

Skip Equipment if not Configured or Disabled
Typical Scan - only service Enabled equipment to reduce scan time
CfgLine Scan - check for new Equip Configuration - then also check if new Callout/Config message
DHF006 - Jul 17, 2017 -- changed Equip.CFG.EN to Equip.CTL.Enable and added CFG.Disabled (addition of Disabled Equipment support)
DHF014 - Dec 8, 2017 -- added Equip.CFG.Defined to CfgLine conditions to scan all Configured EQ (even if no Callout Message received)
EDC2 - Nov 15, 2022 - disable/ignore EQID 0000
EDC2 - Dec 5, 2022 - block CFG.Disable branch to allow Overview display of disabled EQID

2

Config Check Line LNi = CfgLNi: CfgLine

Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CTL.Enable

lbl_Exit (JMP)

Config Check Line LNi = CfgLNi: CfgLine

Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CTL.Enable

Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CFG.Defined

Equipment Data Buffer Lines

EQ
Source A: EDC_EqRxBuffer.Line[LNi].Equip[EQi].Callout[0] ??
Source B: 0

Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CFG.Disable

AFI

Line array Index

EQ
Source A: LNi 29
Source B: 0

Equipment array Index

EQ
Source A: EQi 12
Source B: 0

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State

FLL
Source: 0
Dest: EquipStatus01.State
Length: 1

Local Copy of Equip Array Status Tag for LNi, EQi State = Gray - Unconfigured: EquipStatus01.State.NoCFG (L)

Server Interface Status Lines and Equipment Array

COP
Source: EquipStatus01.State
Dest: EDC_Status01.Line[LNi].Equip[EQi].State
Length: 1

Equipment Loop for EQi

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

3

Local Copy of Equip Array Tag for LNi, EQi

COP	
Source	EDC.Line[LNi].Equip[EQi]
Dest	Equip
Length	1

Local Copy of Equip Array Status Tag for LNi, EQi

COP	
Source	EDC_Status01.Line[LNi].Equip[EQi]
Dest	EquipStatus01
Length	1

Equipment Data Collection - Status Line Equipment

COP	
Source	EDC_Status02.Line[LNi].Equip[EQi]
Dest	EquipStatus02
Length	1

Equipment Data Collection - Status Line Equipment

COP	
Source	EDC_Status03.Line[LNi].Equip[EQi]
Dest	EquipStatus03
Length	1

Calculate Equipment ID for Callout Table Lookup

4

Equipment ID

CPT	
Dest	EQID
	99
Expression	LNi*100+EQi

Line array Index

LIMIT	
Low Limit	0
Test	LNi
	29
High Limit	CFG.CalloutLN.mi
	<EDC.CFG.CalloutLN.mi>
	28

Equipment array Index

LIMIT	
Low Limit	0
Test	EQi
	12
High Limit	CFG.CalloutEQ.mi
	<EDC.CFG.CalloutEQ.mi>
	99

Calulated EQID Index within array limits
EQID_IndexOK

Check for good Event Looklup Index for EDC_Event EvLKi

5

Equipment-Event Index for EDC_Event Lookup

CLR	
Dest	EvLKi
	267

Local Copy of Equip Array Tag for LNi, EQi Equipment-Event Index for Lookup Tables

LIMIT	
Low Limit	0
Test	Equip.EventCTL.EvLKi
	164
High Limit	EDC_Event.CFG.EQ.mi
	2999

Equipment-Event Index for EDC_Event Lookup

MOVE	
Source	Equip.EventCTL.EvLKi
	164
Dest	EvLKi
	267

Event Lookup Index within array limits
EvLKi_IndexOK

Equipment Loop for EQi

Check for EQID match of Event Looklup Index for EDC_Event EvLKi

6

Event Lookup Index EQID Match EvLKi_EQIDmatch (U)

Calulated EQID Index within array limits EQID_IndexOK

Event Lookup Index within array limits EvLKi_IndexOK

Event Controls Equipment Fragment Statistics Array

EQ
Source A: EDC_Event.CTL.Equip[EvLKi].EQID (1916)
Source B: EQID (99)

Event Lookup Index EQID Match EvLKi_EQIDmatch (L)

If EQID mismatch, search for EQID in Event Lookup and update if found
(ServerInterface sets EvLKi during Event Download -- this should never be needed, but it provides a dynamic repair if something happens to index.)

7

Calulated EQID Index within array limits EQID_IndexOK

Event Lookup Index EQID Match EvLKi_EQIDmatch (/)

Event Configuration EQID Search

MOVE
Source: EDC_Event.CTL.EqQty (272)
Dest: EventEqFSC.LEN (262)

Event Configuration EQID Search

GT
Source A: EventEqFSC.LEN (262)
Source B: EDC_Event.CFG.EQ.n (3000)

Event Configuration EQID Search

MOVE
Source: EDC_Event.CFG.EQ.n (3000)
Dest: EventEqFSC.LEN (262)

Event Configuration EQID Search EventEqFSC (RES)

Event Configuration EQID Search

MOVE
Source: 0
Dest: EventEqFSC.POS (0)

Event Configuration EQID Search

FSC
Control: EventEqFSC
Length: 262
Position: 0
Mode: ALL
Expression: EDC_Event.CTL.Equip[EventEqFSC.POS].EQID=EQID
EN / DN / ER

Event Configuration EQID Search EventEqFSC.FD

Equipment-Event Index for EDC_Event Lookup

MOVE
Source: EventEqFSC.POS (0)
Dest: EvLKi (267)

Local Copy of Equip Array Tag for LNi, EQi Equipment-Event Index for Lookup Tables

MOVE
Source: EvLKi (267)
Dest: Equip.EventCTL.EvLKi (164)

Event Lookup Index EQID Match EvLKi_EQIDmatch (L)

Event Lookup Index Found EvLKi_Search

Equipment Loop for EQi

Callout Message
also Enables Configuration for new Equipment on the CfgLNi - one Line per scan
DHF006 - if disabled, don't service any Messages received from Equipment
EDC2 WL31 - during Full Download, block Callout until Equipment Defined or Eqiupment Configuration is complete -- prevents EqDriver from self-configuring and sending data
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

8

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable
/

Equipment Data Collection Download Started - Block Equipment Drivers until Equipment Config Complete
EDC.ServerCfg.Ctl.DownloadBlockEqDriver
/

Receive Equipment Callout Unsolicited Message
JSR
Routine Name _131_EquipCalloutRx

Local Copy of Equip Array Tag for LNi, EQi Equipment Defined (from Server Config)
Equip.CFG.Defined

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines
FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Callout[0]
Length CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>

Server Config or EqDriver Callout Message can Enable Equipment
EDC2 WL31 - during Full Config Download, Equip.CTL.Config is blocked until Equipment Configuration is complete -- along with Disable status

9

Local Copy of Equip Array Tag for LNi, EQi Equipment Defined (from Server Config)
Equip.CFG.Defined

Local Copy of Equip Array Tag for LNi, EQi Equipment Configured (by Server or Driver)
Equip.CTL.Config

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable
/

Local Copy of Equip Array Tag for LNi, EQi Driver Enabled (from Driver Callout Msg)
Equip.CFG.DriverEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Equipment Loop for EQi

Update Equipment Loop Index to include this Equipment
(should not need this, but it fixes any Loop configuration problems)

also determine the last Enabled EQi for the Line

10

Config Check Line
LNi = CfgLNi
CfgLine

Equipment array Index
GT
Source A: EQi 12
Source B: LineCfg.EQ.ei 99

Equipment array Index
LE
Source A: EQi 12
Source B: CFG.EQ.mi <EDC.CFG.EQ.mi> 99

Local Copy of Line Array Tag for LNi Array End Index
MOVE
Source: EQi 12
Dest: LineCfg.EQ.ei 99

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Equipment array Index
GT
Source A: EQi 12
Source B: LineCfg_EqENei_Temp 67

Equipment array Index
LE
Source A: EQi 12
Source B: CFG.EQ.mi <EDC.CFG.EQ.mi> 99

Temp Storage of LineCfg.EQ.ENei
MOVE
Source: EQi 12
Dest: LineCfg_EqENei_Temp 67

DHF006 - added Equip.CTL.Config and Equip.CTL.Enable
(addition of Disabled Equipment support - define CTL bits here after updating Callout status on previous rung)

11

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Calculate Equipment Data and Operational Availability (OA)
JSR
Routine Name: _140_EquipData

Equipment Loop for EQi

DHF006 - CTL.DisableEvents set by configuration to refresh Event Display to clear any existing no-Display Events

12

Local Copy of Equip Array Tag for LNi, EQi Update Events Table
Equip.EventCTL.Update

Local Copy of Equip Array Tag for LNi, EQi Equipment Events Disabled (clear existing Events in Gateway)
Equip.CTL.DisableEvents

S:FS

Update Equip Events Table for HMI Displays

JSR
Routine Name _135_EquipEvents

Local Copy of Equip Array Tag for LNi, EQi Update Events Table
Equip.EventCTL.Update
U

Local Copy of Equip Array Tag for LNi, EQi Equipment Events Disabled (clear existing Events in Gateway)
Equip.CTL.DisableEvents
U

Equipment Loop for EQi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)

OA% - InDemand Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

13

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start)
Equip.DataCTL.Work

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start)
Equip.DataCTL.Work
/
Schedule New Hour
ScheduleNewHour

Local Copy of Equip Array Tag for LNi, EQi Cycle Count Send Summary Data
Equip.DataCTL.CycleSend

Local Copy of Equip Array Tag for LNi, EQi Work Time Ended
Equip.DataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE	
Source	1
Dest	SumMsg.Header.DataType
	13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE	
Source	EQID
	99
Dest	SumMsg.Header.ID
	3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP	
Source	Equip.Data.Work
Dest	SumMsg.Data.Work
Length	1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP	
Source	Equip.Data.EdtOA.Maint
Dest	SumMsg.Data.EdtMaint
Length	1

Equipment Data Summary Message 3 - Equip Downtime Production

COP	
Source	Equip.Data.EdtOA.Prod
Dest	SumMsg.Data.EdtProd
Length	1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP	
Source	Equip.Data.LdtOA.Maint
Dest	SumMsg.Data.LdtMaint
Length	1

Equipment Data Summary Message 5 - Line Downtime Production

COP	
Source	Equip.Data.LdtOA.Prod
Dest	SumMsg.Data.LdtProd
Length	1

Equipment Data Summary Message 6 -Full

COP

Equipment Data Summary Message 7 - Short

COP

Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP

Equipment Data Summary Message 10 - Tool Change (ADC)

COP

Equipment Loop for EQi

Source	Equip.Data.Full
Dest	SumMsg.Data.Full
Length	1

Source	Equip.Data.Short
Dest	SumMsg.Data.Short
Length	1

Source	Equip.Data.ProdDelay
Dest	SumMsg.Data.ProdDelay
Length	1

Source	Equip.Data.ToolChange
Dest	SumMsg.Data.ToolChange
Length	1

Equipment Data Summary Message 11 - Config Option 1

COP

Source	Equip.Data.Option01
Dest	SumMsg.Data.Opt01
Length	1

Equipment Data Summary Message 12 - Config Option 2

COP

Source	Equip.Data.Option02
Dest	SumMsg.Data.Opt02
Length	1

Equipment Data Summary Message 13 - Config Option 3

COP

Source	Equip.Data.Option03
Dest	SumMsg.Data.Opt03
Length	1

Equipment Data Summary Message 14 - Config Option 4

COP

Source	Equip.Data.Option04
Dest	SumMsg.Data.Opt04
Length	1

Equipment Data Summary Message 15 - Config Option 5

COP

Source	Equip.Data.Option05
Dest	SumMsg.Data.Opt05
Length	1

Equipment Data Summary Message Time Total

MOVE

Source	Equip.CFG.Takt 56000
Dest	SumMsg.Data.Cycle.Time 0

Equipment Data Summary Message Count Total

MOVE

Source	Equip.Data.Cycle.Count 420
Dest	SumMsg.Data.Cycle.Count 0

Equipment Loop for EQi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Line Number Odd/Even (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi) to distribute EQID from same Line (i.e. 1000, 1001, 1002) to different buffers

OR% - All Downtime

14

SumDataLoadRB

Load Summary Data Message

JSR
Routine Name _150_SumDataMsg

ADD
Source A LNi 29
Source B EQi 12
Dest LnEqSum 35

RingBuffer Modulo Result

MOD
Source A LnEqSum 35
Source B 2
Dest BufferID 1

RingBuffer Modulo Result

EQ
Source A BufferID 1
Source B 1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _151_SumLoadRB01

UseSumRB01 /

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _152_SumLoadRB02

Equipment Loop for EQi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)

OR% - All Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

15

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start)
Equip.DataCTL.Work

Local Copy of Equip Array Tag for LNi, EQi Cycle Count Send Summary Data
Equip.DataCTL.CycleSend

Local Copy of Equip Array Tag for LNi, EQi Cycle Count Send Summary Data
Equip.DataCTL.CycleSend

U

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start)
Equip.DataCTL.Work

/

Schedule New Hour
ScheduleNewHour

Local Copy of Equip Array Tag for LNi, EQi Work Time Ended
Equip.DataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE
Source 11
Dest SumMsg.Header.DataType
13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE
Source EQID
99
Dest SumMsg.Header.ID
3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP
Source Equip.Data.Work
Dest SumMsg.Data.Work
Length 1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP
Source Equip.Data.EdtOR.Maint
Dest SumMsg.Data.EdtMaint
Length 1

Equipment Data Summary Message 3 - Equip Downtime Production

COP
Source Equip.Data.EdtOR.Prod
Dest SumMsg.Data.EdtProd
Length 1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP
Source Equip.Data.LdtOR.Maint
Dest SumMsg.Data.LdtMaint
Length 1

Equipment Data Summary Message 5 - Line Downtime Production

COP
Source Equip.Data.LdtOR.Prod
Dest SumMsg.Data.LdtProd
Length 1

Equipment Data Summary Message 6 -Full

COP

Equipment Data Summary Message 7 - Short

COP

Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP

Equipment Data Summary Message 10 - Tool Change (ADC)

COP

Equipment Loop for EQi

COP	
Source	Equip.Data.Full
Dest	SumMsg.Data.Full
Length	1

COP	
Source	Equip.Data.Short
Dest	SumMsg.Data.Short
Length	1

COP	
Source	Equip.Data.ProdDelay
Dest	SumMsg.Data.ProdDelay
Length	1

COP	
Source	Equip.Data.ToolChange
Dest	SumMsg.Data.ToolChange
Length	1

Equipment Data Summary Message 11 - Config Option 1

COP	
Source	Equip.Data.Option01
Dest	SumMsg.Data.Opt01
Length	1

Equipment Data Summary Message 12 - Config Option 2

COP	
Source	Equip.Data.Option02
Dest	SumMsg.Data.Opt02
Length	1

Equipment Data Summary Message 13 - Config Option 3

COP	
Source	Equip.Data.Option03
Dest	SumMsg.Data.Opt03
Length	1

Equipment Data Summary Message 14 - Config Option 4

COP	
Source	Equip.Data.Option04
Dest	SumMsg.Data.Opt04
Length	1

Equipment Data Summary Message 15 - Config Option 5

COP	
Source	Equip.Data.Option05
Dest	SumMsg.Data.Opt05
Length	1

Equipment Data Summary Message Time Total

MOVE	
Source	Equip.CFG.Takt 56000
Dest	SumMsg.Data.Cycle.Time 0

Equipment Data Summary Message Count Total

MOVE	
Source	Equip.Data.Cycle.Count 420
Dest	SumMsg.Data.Cycle.Count 0

Equipment Loop for EQi

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Line Number Odd/Even (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi) to distribute EQID from same Line (i.e. 1000, 1001, 1002) to different buffers

OA% - InDemand Downtime

16

SumDataLoadRB

Load Summary Data Message

JSR
Routine Name _150_SumDataMsg

ADD
Source A LNi 29
Source B EQi 12
Dest LnEqSum 35

RingBuffer Modulo Result

MOD
Source A LnEqSum 35
Source B 2
Dest BufferID 1

RingBuffer Modulo Result

EQ
Source A BufferID 1
Source B 1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _151_SumLoadRB01

UseSumRB01

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _152_SumLoadRB02

Equipment Comm Status
-- Set Default State = Configured if no State value (don't run Equipment Timer)
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)

17

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value

COP
Source EquipStatus01.State
Dest EquipStatus01.StateVal
Length 1

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value

LE
Source A EquipStatus01.StateVal 16#0000_0012
Source B 1

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State

FLL
Source 0
Dest EquipStatus01.State
Length 1

Local Copy of Equip Array Status Tag for LNi, EQi State = White - Configured (never received any Data Requests)
EquipStatus01.State.CFG
L

Equipment Loop for EQi

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

18

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value

COP	
Source	EquipStatus01.State
Dest	EquipStatus01.StateVal
Length	1

Equipment Data Collection Lines and Equipment Array

COP	
Source	Equip
Dest	EDC.Line[LNi].Equip[EQi]
Length	1

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[LNi].Equip[EQi]
Length	1

Event Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus02
Dest	EDC_Status02.Line[LNi].Equip[EQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus03
Dest	EDC_Status03.Line[LNi].Equip[EQi]
Length	1

Exit Routine

19

lbl_Exit
LBL

NOP

Check Viewing Index Limits

20

Equipment View Tag Index

LT	
Source A	_ViewEQi 18
Source B	0

Equipment View Tag Index

GT	
Source A	_ViewEQi 18
Source B	CFG.EQ.mi <EDC.CFG.EQ.mi> 99

Equipment View Tag Index

CLR	
Dest	_ViewEQi 18

Equipment Loop for EQi

Copy Equipment Instance to Viewing Tag if selected
(_ViewEquip can be used to multiplex Trends and Diagnostic Spreadsheets)

21

_ViewEN

Line array Index

EQ	
Source A	LNi
	29
Source B	_ViewLNi
	9

Equipment array Index

EQ	
Source A	EQi
	12
Source B	_ViewEQi
	18

Equipment View Tag (used by View Diagnostic Spreadsheet)

COP	
Source	EDC.Line[LNi].Equip[EQi]
Dest	_ViewEquip
Length	1

(End)

Receive Equipment Callout
Unsolicited Message

Receive Equipment Callout/Status Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 131
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)
EDC2 WL18 - Block Transaction TImer if Fault Message not enabled for PdM-only without EqDriver

2

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Local Copy of Equip Array Tag for LNi, EQi Fault Message Enabled (from Driver Callout Msg)
Equip.CFG.FaultMsgEN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.CalloutMsg.TransTMR.PRE 999999999

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.CalloutMsg.TransTMR
Preset 999999999
Accum 308
EN
DN

Receive Equipment Callout
Unsolicited Message
Transaction Timeout
Callout Message not Received
Gateway-generated Message clear stale Faults and report Callout Message Error

3

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A: Equip.CTL.CalloutMsg.TransTMR.ACC
308
Source B: CFG.LiveMsgTimeout
<EDC.CFG.LiveMsgTimeout>
150000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Callout Rx Msg ER
EquipStatus01.Update.CalloutRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans
PerfMsg: Equip.CTL.CalloutMsg
PerfTrans: EquipStatus02.CalloutRx.ER

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 1004
Input Par: Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: 0

Equipment Data Callout/Status Message

FLL
Source: 0
Dest: CalloutMsg
Length: 1

Equipment Data Callout/Status Message Transaction Time Stamp

COP
Source: CLX.TransTime
<EDC.CLX.TransTime>
Dest: CalloutMsg.TransTime
Length: 1

Equipment Data Callout/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

CPT
Dest: CalloutMsg.Header.EQID
1916
Expression: LNi*100+EQi

Equipment Data Callout/Status Message Packet ID - generates offset for 8-DINT mulitplexed data packet

MOVE
Source: 0
Dest: CalloutMsg.Header.PacketID
7

Equipment Data Callout/Status Message Gateway Message Error (from EqDriver)
CalloutMsg.PreDef._31
L

Equipment Data Callout/Status Message Alarm Status is ON (from EqDriver Alarm Mask)
CalloutMsg.PreDef._07
L

lbl_RxLoad
JMP

Receive Equipment Callout
Unsolicited Message

Transaction Timeout
Cleanup Callout Messgae ER if EQ removed from Configuration
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)

4

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable
/

Equipment Data Callout/Status Message Gateway Message Error (from EqDriver)
CalloutMsg.PreDef._31

Equipment Data Callout/Status Message
FLL
Source 0
Dest CalloutMsg
Length 1

Equipment Data Callout/Status Message Gateway Message Error (from EqDriver)
CalloutMsg.PreDef._31
U

lbl_RxLoad
JMP

No new data in Rx Buffer
Header Transaction Number =0

5

Equipment Data Buffer Lines
EQ
Source A EDC_EqRxBuffer.Line[LNi].Equip[EQi].Callout[0]
??
Source B 0

lbl_Exit
JMP

Copy Rx Buffer to local registers
DINT Array to UDT

6

Equipment Data Callout/Status Message
CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Callout[0]
Dest CalloutMsg
Length 1

Receive Equipment Callout
Unsolicited Message

No new data in Rx Buffer
Header & Footer Transaction Number =0

7

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

EQ
Source A CalloutMsg.Header.TransNum
0
Source B 0

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

EQ
Source A CalloutMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

New data in Rx Buffer
Initialize Transaction timing

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CalloutMsg.PartialTMR.EN
/

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.CalloutMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.CalloutMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CalloutMsg
PerfTrans EquipStatus02.CalloutRx.EN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 1000
Input Par Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CalloutMsg.Header.TransNum

Receive Equipment Callout
Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

9

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

EQ
Source A: CalloutMsg.Header.TransNum
0
Source B: 0

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

NE
Source A: CalloutMsg.Footer.TransNum
0
Source B: 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source: CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest: Equip.CTL.CalloutMsg.PartialTMR.PRE
10000

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer: Equip.CTL.CalloutMsg.PartialTMR
Preset: 10000
Accum: 0
EN
DN

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE
Source A: CalloutMsg.Header.TransNum
0
Source B: 0

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

EQ
Source A: CalloutMsg.Footer.TransNum
0
Source B: 0

Partial message in Rx Buffer
Wait for rest of message to arrive

10

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CalloutMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CalloutMsg.PartialTMR.DN
/

lbl_Exit
JMP

Receive Equipment Callout
Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

11

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CalloutMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CalloutMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Callout Rx Msg ER
EquipStatus01.Update.CalloutRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Footer.TransNum 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CalloutMsg
PerfTrans	EquipStatus02.CalloutRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	1005
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

lbl_NextTrans
JMP

Receive Equipment Callout
Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Callout table if mismatch

12

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Header.TransNum
	0
Source B	0

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Footer.TransNum
	0
Source B	0

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Header.TransNum
	0
Source B	CalloutMsg.Footer.TransNum
	0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.CalloutMsg.TransReset (L)

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Callout Rx Msg ER

EquipStatus01.Update.CalloutRxER (L)

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CalloutMsg
PerfTrans	EquipStatus02.CalloutRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	1006
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

lbl_NextTrans (JMP)

Receive Equipment Callout
Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

13

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Header.TransNum
	0
Source B	0

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Footer.TransNum
	0
Source B	0

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

EQ	
Source A	CalloutMsg.Header.TransNum
	0
Source B	CalloutMsg.Footer.TransNum
	0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received

Equip.CTL.CalloutMsg.TransRcvd

L

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Callout Rx DN

EquipStatus01.Update.CalloutRxOK

L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CalloutMsg
PerfTrans	EquipStatus02.CalloutRx.DN

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	1001
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

Receive Equipment Callout
Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Callout table if mismatch

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd

Expected Equipment ID

CPT	
Dest	ExpectedEQID 85
Expression	LNi*100+EQi

Equipment Data Callout/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE	
Source A	CalloutMsg.Header.EQID 1916
Source B	ExpectedEQID 85

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Callout Equipment ID Mismatch ER
EquipStatus01.Update.CalloutRxEqIdER
(L)

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Header.EQID 1916
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CalloutMsg
PerfTrans	EquipStatus02.CalloutRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	1007
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

lbl_NextTrans
(JMP)

Receive Equipment Callout
Unsolicited Message

Check new message for duplicate Transaction Number
Load Callout table but indicate issue in performance/diagnostic

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

EQ
Source A: CalloutMsg.Header.TransNum 0
Source B: Equip.CTL.CalloutMsg.TransNumLast 6451

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Callout Rx DN
EquipStatus01.Update.CalloutRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate ...
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: Equip.CTL.CalloutMsg.TransNumLast 6451
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans ...
PerfMsg: Equip.CTL.CalloutMsg
PerfTrans: EquipStatus02.CalloutRx.KA

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 1002
Input Par: Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: CalloutMsg.Header.TransNum

Receive Equipment Callout
Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load Callout table but indicate issue in performance/diagnostic

16

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Header.TransNum
	0
Source B	Equip.CTL.CalloutMsg.TransNumLast
	6451

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	CalloutMsg.Header.TransNum
	0
Source B	Equip.CTL.CalloutMsg.TransNumNext
	6452

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Callout Transaction Number Sequence ER
EquipStatus01.Update.CalloutRxSeqER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.CalloutMsg.TransNumNext
	6452
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CalloutMsg
PerfTrans	EquipStatus02.CalloutRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	1009
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

Receive Equipment Callout
Unsolicited Message
Load new message in Callout Table

17

Local Copy of Equip Array Tag for LNi, EQi Transaction Received: Equip.CTL.CalloutMsg.TransRcvd

lbl_RxLoad
LBL

Local Copy of Equip Array Tag for LNi, EQi Message Timeout: Equip.CTL.CalloutMsg.Timeout

Load Equipment Callout Message into Callout Table

JSR
Routine Name: _132_EquipCalloutRxLoad

Prepare for next message

18

Local Copy of Equip Array Tag for LNi, EQi Transaction Received: Equip.CTL.CalloutMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number

MOVE
Source: CalloutMsg.Header.TransNum 0
Dest: Equip.CTL.CalloutMsg.TransNumLast 6451

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

ADD
Source A: CalloutMsg.Header.TransNum 0
Source B: 1
Dest: Equip.CTL.CalloutMsg.TransNumNext 6452

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

LT
Source A: Equip.CTL.CalloutMsg.TransNumNext 6452
Source B: 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

GT
Source A: Equip.CTL.CalloutMsg.TransNumNext 6452
Source B: CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

MOVE
Source: 1
Dest: Equip.CTL.CalloutMsg.TransNumNext 6452

Receive Equipment Callout
Unsolicited Message
Exit path when Transaction Reset or Received

lbl_NextTrans
19 LBL NOP

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)
EDC2 - v2.01 add Event Configuration for DtOwner and DtType
(send Response with Event Masks if Server Download All complete - during Download, EqDriver will use existing Masks)

20

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CalloutMsg.TransReset

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CalloutMsg.TransReset (U)

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CalloutMsg.TransRcvd (U)

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout (U)

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CalloutMsg.PartialTMR (RES)

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.CalloutMsg.TransTMR (RES)

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.CalloutMsg.TransLast
0

Equipment Data Callout/Status Message

FLL
Source 0
Dest CalloutRspMsg
Length 1

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

Equipment Data Collection Response Download All Started
EDC.ServerCfg.Ctl.DownloadAllStarted (/)

Create Callout Response Message with Event Mask

JSR
Routine Name _134_EquipCalloutRxResponse

Equipment Data Lines

COP
Source CalloutMsg
Dest EDC_EqRxData.Line[LNi].Equip[EQi].Callout
Length 1

Equipment Data Buffer Lines

COP
Source CalloutRspMsg
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Callout[0]
Length CFG.CalloutMSG.n
<EDC.CFG.CalloutMSG.n>

Exit path when no Message, Partial Message or Corrupt Message

lbl_Exit
21 LBL NOP

Receive Equipment Callout
Unsolicited Message

(End)

Load Equipment Callout Message into Callout Table

Load Equipment Callout Message into Callout Table

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 132
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Convert Header EQID to LNi and EQi

2

MOD
Source A CalloutMsg.Header.EQID 1916
Source B 100
Dest HeaderEQi 85

CPT
Dest HeaderLNi 0
Expression (CalloutMsg.Header.EQID - HeaderEQi) / 100

Load Equipment Callout Message into Callout Table

Equipment ID outside Callout Table array limits
Log Transaction and Exit

3

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout

LT
Source A HeaderLNi 0
Source B 0

GT
Source A HeaderLNi 0
Source B CFG.CalloutLN.mi <EDC.CFG.CalloutLN.mi> 28

LT
Source A HeaderEQi 85
Source B 0

GT
Source A HeaderEQi 85
Source B CFG.CalloutEQ.mi <EDC.CFG.CalloutEQ.mi> 99

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsg.Header.EQID 1916
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsg.Header.PacketID 7
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 1013
Input Par Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CalloutMsg.Header.TransNum

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CalloutMsg
PerfTrans EquipStatus02.CalloutRx.ER

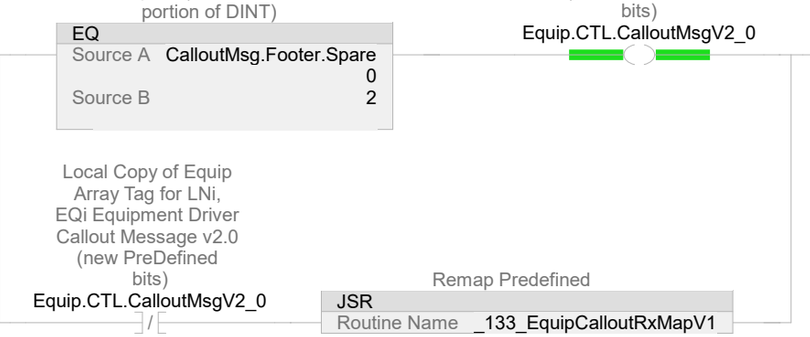
lbl_Exit
JMP

Load Equipment Callout Message into Callout Table

v2.0 - Callout Msg Version determined by Footer.Spare0=v1.6 (old PreDef layout), 2=v2.0 (new PreDef layout)
remap v1.6 to v2.0 PreDefined layout

July 25, 2022 -- blocked v1.6 remap - was clearing No Event Display (bit 15) - can't do this for v1.6 drivers until configuration is updated

4



Load Equipment Callout Message into Callout Table

Load Message into Callout Table
Set Update bits for Equip Event Table, HMI Overview Event Table and HMI Zones Event Tables
(these are used to indicate Equipment Event Table updates needed when new Callout Messages)

v1.3 Support Multiple Fault/Callout Packets
If Packet 0, Update Callout with PreDefined & EqDefined, otherwise just Load PreDefined

DN014 Health Tool Status Download from Server
Callout bits 19 (alarm) & 20 (fault) from Tool EqDriver are blocked to allow Server to control status
Callout bit 6 (EQ Fault) is turned ON when the Health Fault is ON

5

COP
Source CalloutMsg.PreDef
Dest TempDINT
Length 1

Callout Table (LNi, EQi)
MVM
Source TempDINT
16#0000_0023
Mask 16#ffe7_ffff
Dest EDC_EqCallout[HeaderLNi,HeaderEQi].PreDef
16#0000_0024

Equipment Data Callout/Status Message Packet ID - generates offset for 8-DINT mulitplexed data packet
LIMIT
Low Limit 0
Test CalloutMsg.Header.PacketID
7
High Limit CFG.CalloutPKT.mi
<EDC.CFG.CalloutPKT.mi>
7

MUL
Source A CalloutMsg.Header.PacketID
7
Source B CFG.CalloutEqDef.n
<EDC.CFG.CalloutEqDef.n>
8
Dest PktWDi
24

Callout Table (LNi, EQi)
COP
Source CalloutMsg.EqDef.Word[0]
Dest EDC_EqCallout[HeaderLNi,HeaderEQi].EqDef[PktWDi]
Length CFG.CalloutEqDef.n
<EDC.CFG.CalloutEqDef.n>

Callout Table (LNi, EQi)
EDC_EqCallout[HeaderLNi,HeaderEQi].PreDef.19
AFI
Callout Table (LNi, EQi)
EDC_EqCallout[HeaderLNi,HeaderEQi].PreDef.20

Callout Table (LNi, EQi)
EDC_EqCallout[HeaderLNi,HeaderEQi].PreDef.6
L

Local Copy of Equip Array Tag for LNi, EQi Update Events Table
Equip.EventCTL.Update
L

Local Copy of Equip Array Tag for LNi, EQi Update Events on HMI Overview Display
Equip.EventCTL.UpdateOview
L

Local Copy of Equip Array Tag for LNi, EQi Update Events on HMI Zone Displays
FLL
Source -1
Dest Equip.EventCTL.UpdateZone
Length 1

Load Equipment Callout Message into Callout Table

Set Line/Equipment Configured to enable service of Fault, Quality, Trend Messages
Update Line/Equipment scan Indexes to include this Equipment
DHF006 - changed CFG.EN to CFG.DriverEN (addition of Disabled Equipment support)
EDC2 WL08 - add Cycle Data Message

6

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout

Equipment Data Callout/Status Message Driver Enabled
CalloutMsg.Header.DriverEN

Local Copy of Line Array Tag for LNi Line Enabled
LineCfg.EN

Local Copy of Equip Array Tag for LNi, EQi Driver Enabled (from Driver Callout Msg)
Equip.CFG.DriverEN

Line array Index

GT	
Source A	LNi
	29
Source B	CFG.LN.ei
	<EDC.CFG.LN.ei>
	28

Line array Index

LE	
Source A	LNi
	29
Source B	CFG.LN.mi
	<EDC.CFG.LN.mi>
	28

EDC Config Array End Index

MOVE	
Source	LNi
	29
Dest	CFG.LN.ei
	<EDC.CFG.LN.ei>
	28

Equipment array Index

GT	
Source A	EQi
	12
Source B	LineCfg.EQ.ei
	99

Equipment array Index

LE	
Source A	EQi
	12
Source B	CFG.EQ.mi
	<EDC.CFG.EQ.mi>
	99

Local Copy of Line Array Tag for LNi Array End Index

MOVE	
Source	EQi
	12
Dest	LineCfg.EQ.ei
	99

Equipment Data Callout/Status Message Fault Message Enabled
CalloutMsg.Header.FaultMsgEN

Local Copy of Equip Array Tag for LNi, EQi Fault Message Enabled (from Driver Callout Msg)
Equip.CFG.FaultMsgEN

Equipment Data Callout/Status Message Quality Message Enabled
CalloutMsg.Header.QualityMsgEN

Local Copy of Equip Array Tag for LNi, EQi Quality Message Enabled (from Driver Callout Msg)
Equip.CFG.QualityMsgEN

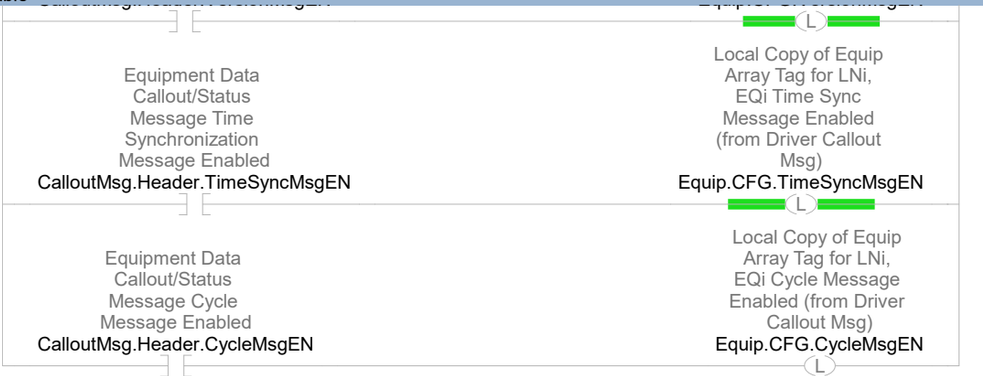
Equipment Data Callout/Status Message Trend Message Enabled
CalloutMsg.Header.TrendMsgEN

Local Copy of Equip Array Tag for LNi, EQi Trend Message Enabled (from Driver Callout Msg)
Equip.CFG.TrendMsgEN

Equipment Data Callout/Status Message Version Message Enabled
CalloutMsg.Header.VersionMsgEN

Local Copy of Equip Array Tag for LNi, EQi Version Message Enabled (from Driver Callout Msg)
Equip.CFG.VersionMsgEN

Load Equipment Callout Message into Callout Table



Load Equipment Callout Message into Callout Table

Reset Line/Equipment Configured to enable service of Fault, Quality, Trend Messages
DHF006 - changed CFG.EN to CFG.DriverEN (addition of Disabled Equipment support)
EDC2 WL08 - add Cycle Data Message

7

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CalloutMsg.Timeout

Equipment Data Callout/Status Message Driver Enabled
CalloutMsg.Header.DriverEN

Local Copy of Line Array Tag for LNi Line Enabled
LineCfg.EN

Local Copy of Equip Array Tag for LNi, EQi Driver Enabled (from Driver Callout Msg)
Equip.CFG.DriverEN

Local Copy of Equip Array Tag for LNi, EQi Fault Message Enabled (from Driver Callout Msg)
Equip.CFG.FaultMsgEN

Local Copy of Equip Array Tag for LNi, EQi Quality Message Enabled (from Driver Callout Msg)
Equip.CFG.QualityMsgEN

Local Copy of Equip Array Tag for LNi, EQi Trend Message Enabled (from Driver Callout Msg)
Equip.CFG.TrendMsgEN

Local Copy of Equip Array Tag for LNi, EQi Version Message Enabled (from Driver Callout Msg)
Equip.CFG.VersionMsgEN

Local Copy of Equip Array Tag for LNi, EQi Time Sync Message Enabled (from Driver Callout Msg)
Equip.CFG.TimeSyncMsgEN

Local Copy of Equip Array Tag for LNi, EQi Cycle Message Enabled (from Driver Callout Msg)
Equip.CFG.CycleMsgEN

Equipment Data Callout/Status Message Fault Message Enabled
CalloutMsg.Header.FaultMsgEN

Local Copy of Equip Array Tag for LNi, EQi Fault Message Enabled (from Driver Callout Msg)
Equip.CFG.FaultMsgEN

Equipment Data Callout/Status Message Quality Message Enabled
CalloutMsg.Header.QualityMsgEN

Local Copy of Equip Array Tag for LNi, EQi Quality Message Enabled (from Driver Callout Msg)
Equip.CFG.QualityMsgEN

Equipment Data Callout/Status Message Trend Message Enabled
CalloutMsg.Header.TrendMsgEN

Local Copy of Equip Array Tag for LNi, EQi Trend Message Enabled (from Driver Callout Msg)
Equip.CFG.TrendMsgEN

Equipment Data Callout/Status Message Version Message Enabled
CalloutMsg.Header.VersionMsgEN

Local Copy of Equip Array Tag for LNi, EQi Version Message Enabled (from Driver Callout Msg)
Equip.CFG.VersionMsgEN

Equipment Data Callout/Status Message Time Synchronization Message Enabled
CalloutMsg.Header.TimeSyncMsgEN

Local Copy of Equip Array Tag for LNi, EQi Time Sync Message Enabled (from Driver Callout Msg)
Equip.CFG.TimeSyncMsgEN

Load Equipment Callout Message into Callout Table

Equipment Data
Callout/Status
Message Cycle
Message Enabled
CalloutMsg.Header.CycleMsgEN

Local Copy of Equip
Array Tag for LNi,
EQi Cycle Message
Enabled (from Driver
Callout Msg)
Equip.CFG.CycleMsgEN
U

Skip Option Update if EQID Event Lookup Index is not good

Event Lookup Index
EQID Match
EvLKi_EQIDmatch

lbl_OptionsDone
JMP

8

Packet Index

MOVE
Source CalloutMsg.Header.PacketID
7
Dest PKTi
3

9

Skip Option Update if EQID Event Option 1-5 are not configured
faster to skip over AOI than solve false
(fwiw - did not see Callout scan time improvement with AOI skips)

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt01.PktConfig.[PKTi]

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt02.PktConfig.[PKTi]

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt03.PktConfig.[PKTi]

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt04.PktConfig.[PKTi]

10

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt05.PktConfig.[PKTi]

lbl_OptionsDone
JMP

Skip Option Update if EQID Event Option not configured
faster to skip over AOI than solve false

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt01.PktConfig.[PKTi]

lbl_Option2
JMP

11

Load Equipment Callout Message into Callout Table

Update Option 1

12

Use the Option Masks to determine Callout Data Packet Option Status

aoiEDC_SumOptionStatus	
aoiEDC_SumOptionStat...	aoiEDC_SumOptionStatus ...
EventData	CalloutMsg.EqDef.Word
OptMask	EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data
EventOpt	EDC_EventOpt.Equip[EvLKi].Opt01
PKTi	PKTi

KPI Option Diagnostic

ADD	
Source A	KpiOptionDiag[1] 5796936
Source B	1
Dest	KpiOptionDiag[1] 5796936

Skip Option Update if EQID Event Option not configured faster to skip over AOI than solve false

13

Event KPI Options Equipment Event KPI Options

lbl_Option2 LBL — EDC_EventOpt.Equip[EvLKi].Opt02.PktConfig.[PKTi] /

JMP lbl_Option3

Update Option 2

14

Use the Option Masks to determine Callout Data Packet Option Status

aoiEDC_SumOptionStatus	
aoiEDC_SumOptionStat...	aoiEDC_SumOptionStatus ...
EventData	CalloutMsg.EqDef.Word
OptMask	EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data
EventOpt	EDC_EventOpt.Equip[EvLKi].Opt02
PKTi	PKTi

KPI Option Diagnostic

ADD	
Source A	KpiOptionDiag[2] 1041056
Source B	1
Dest	KpiOptionDiag[2] 1041056

Skip Option Update if EQID Event Option not configured faster to skip over AOI than solve false

15

Event KPI Options Equipment Event KPI Options

lbl_Option3 LBL — EDC_EventOpt.Equip[EvLKi].Opt03.PktConfig.[PKTi] /

JMP lbl_Option4

Load Equipment Callout Message into Callout Table

Update Option 3

16

Use the Option Masks to determine Callout Data Packet Option Status

aoiEDC_SumOptionStatus	
aoiEDC_SumOptionStat...	aoiEDC_SumOptionStatus ...
EventData	CalloutMsg.EqDef.Word
OptMask	EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data
EventOpt	EDC_EventOpt.Equip[EvLKi].Opt03
PKTi	PKTi

KPI Option Diagnostic

ADD	
Source A	KpiOptionDiag[3] 1295
Source B	1
Dest	KpiOptionDiag[3] 1295

Skip Option Update if EQID Event Option not configured faster to skip over AOI than solve false

17

Event KPI Options Equipment Event KPI Options

lbl_Option4 LBL — EDC_EventOpt.Equip[EvLKi].Opt04.PktConfig.[PKTi] / — lbl_Option5 JMP

Update Option 4

18

Use the Option Masks to determine Callout Data Packet Option Status

aoiEDC_SumOptionStatus	
aoiEDC_SumOptionStat...	aoiEDC_SumOptionStatus ...
EventData	CalloutMsg.EqDef.Word
OptMask	EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data
EventOpt	EDC_EventOpt.Equip[EvLKi].Opt04
PKTi	PKTi

KPI Option Diagnostic

ADD	
Source A	KpiOptionDiag[4] 1295
Source B	1
Dest	KpiOptionDiag[4] 1295

Skip Option Update if EQID Event Option not configured faster to skip over AOI than solve false

19

Event KPI Options Equipment Event KPI Options

lbl_Option5 LBL — EDC_EventOpt.Equip[EvLKi].Opt05.PktConfig.[PKTi] / — lbl_OptionsDone JMP

Load Equipment Callout Message into Callout Table

Update Option 5

Use the Option Masks to determine Callout Data Packet Option Status

KPI Option Diagnostic

20

aoiEDC_SumOptionStatus	
aoiEDC_SumOptionStat...	aoiEDC_SumOptionStatus ...
EventData	CalloutMsg.EqDef.Word
OptMask	EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data
EventOpt	EDC_EventOpt.Equip[EvLKi].Opt05
PKTi	PKTi

ADD	
Source A	KpiOptionDiag[5] 1295
Source B	1
Dest	KpiOptionDiag[5] 1295

Option Update Done

lbl_OptionsDone

21 LBL — NOP

Callout Table Write -- Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

Transaction Log Control SBR External Load -- Optional Parameters

Log Transactions

22

MOVE	
Source	CalloutMsg.Header.EQID 1916
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

MOVE	
Source	CalloutMsg.Header.PacketID 7
Dest	Trans.Opt[2] <EDC.Trans.Opt[2]> -1

JSR	
Routine Name	_901_TransLogSBR
Input Par	1011
Input Par	Equip.CTL.CalloutMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CalloutMsg.Header.TransNum

Exit Routine

lbl_Exit

23 LBL — NOP

(End)

Remap Predefined

v2.0 - Callout Msg Version determined by Header.Cmd 0=v1.6 (old PreDef layout), 2=v2.0 (new PreDef layout)
remap v1.6 to v2.0 PreDefined layout

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 133
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

No v1 mapping needed in new EDC2 deployments
(don't want to remap accidentally if EqDriver is missing version)

2

NOP

(End)

Create Callout Response Message with Event Mask

EDC2 - v2.01 add Event Configuration for DtOwner and DtType
Create Equipment Callout Response Message with one Event Mask
Equipment Driver will use the Masks to Ignore Events, set DT Owner (Maint/Prod) and DT Type (Fault, Alarm, Alert)

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 134
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Exit if no Event Lookup Index EQID Match

2

Event Lookup Index EQID Match
EvLKi_EQIDmatch
/

lbl_Exit
JMP

Initialize Callout Response Message for PKTi
Response Message always has same PKTi as Callout Message
(no point in sending unused Packet configurations)

3

Equipment Data Callout/Status Message Packet ID - generates offset for 8-DINT mulitplexed data packet

MOVE
Source CalloutMsg.Header.PacketID 7
Dest CalloutRspMsg.Header.PacketID 3

Create Callout Response Message with Event Mask

4

Load EQID & Packet ID

Equipment Data Callout/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE	
Source	CalloutMsg.Header.EQID 1916
Dest	CalloutRspMsg.Header.EQID 1004

Equipment Data Callout/Status Message Packet ID - generates offset for 8-DINT mulitplexed data packet

MOVE	
Source	CalloutMsg.Header.PacketID 7
Dest	CalloutRspMsg.Header.PacketID 3

Packet Index

MOVE	
Source	CalloutMsg.Header.PacketID 7
Dest	PKTi 3

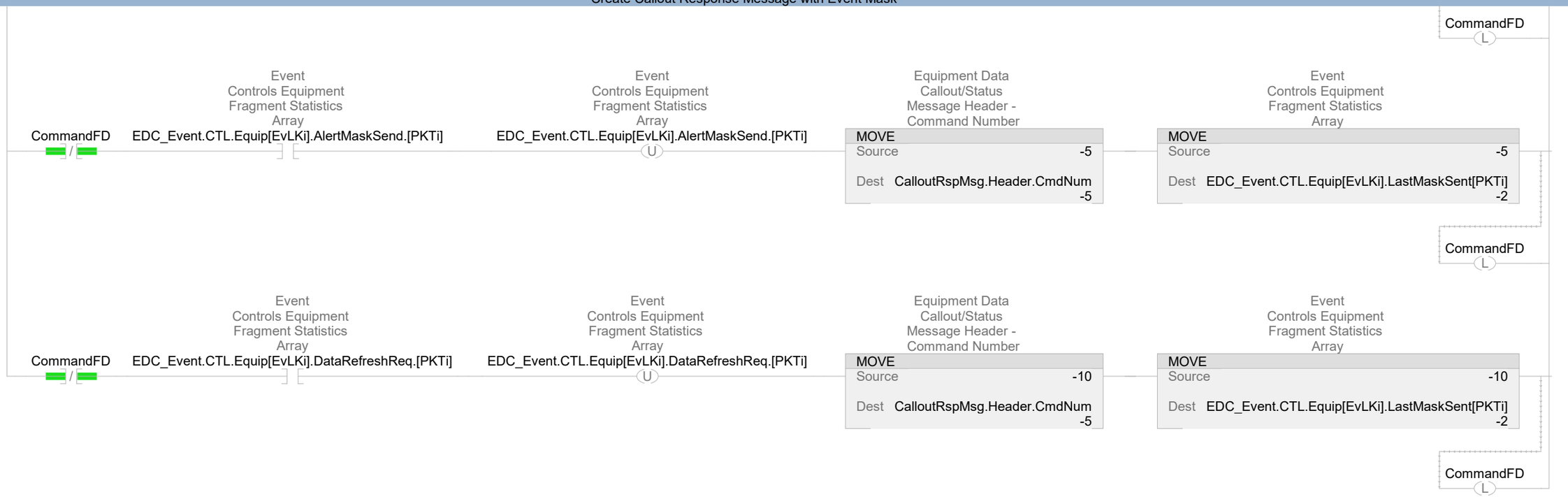
Create Callout Response Message with Event Mask

Select Packet-Command Mask for Response
Prioritize new Config items that haven't been sent
v3.0.2 Data Refresh Request Command

5

CommandFD	Event Controls Equipment Fragment Statistics Array	Event Controls Equipment Fragment Statistics Array	Equipment Data Callout/Status Message Header - Command Number	Event Controls Equipment Fragment Statistics Array
U	EDC_Event.CTL.Equip[EvLKi].MaintMaskSend.[PKTi]	EDC_Event.CTL.Equip[EvLKi].MaintMaskSend.[PKTi]	MOVE Source -1 Dest CalloutRspMsg.Header.CmdNum -5	MOVE Source -1 Dest EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
				L CommandFD
CommandFD	Event Controls Equipment Fragment Statistics Array	Event Controls Equipment Fragment Statistics Array	Equipment Data Callout/Status Message Header - Command Number	Event Controls Equipment Fragment Statistics Array
/	EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend.[PKTi]	EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend.[PKTi]	MOVE Source -2 Dest CalloutRspMsg.Header.CmdNum -5	MOVE Source -2 Dest EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
U				L CommandFD
CommandFD	Event Controls Equipment Fragment Statistics Array	Event Controls Equipment Fragment Statistics Array	Equipment Data Callout/Status Message Header - Command Number	Event Controls Equipment Fragment Statistics Array
/	EDC_Event.CTL.Equip[EvLKi].FaultMaskSend.[PKTi]	EDC_Event.CTL.Equip[EvLKi].FaultMaskSend.[PKTi]	MOVE Source -3 Dest CalloutRspMsg.Header.CmdNum -5	MOVE Source -3 Dest EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
U				L CommandFD
CommandFD	Event Controls Equipment Fragment Statistics Array	Event Controls Equipment Fragment Statistics Array	Equipment Data Callout/Status Message Header - Command Number	Event Controls Equipment Fragment Statistics Array
/	EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend.[PKTi]	EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend.[PKTi]	MOVE Source -4 Dest CalloutRspMsg.Header.CmdNum -5	MOVE Source -4 Dest EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
U				

Create Callout Response Message with Event Mask



Create Callout Response Message with Event Mask

If no undelivered Mask for this Packet, send the next Mask
Commands -1 thru -5

6

CommandFD

SUB
Source A: Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
Source B: 1
Dest: Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2

LT
Source A: Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
Source B: -5

GT
Source A: Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
Source B: -1

MOVE
Source: -1
Dest: Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2

MOVE
Source: Equipment Data Callout/Status Message Header - Command Number EDC_Event.CTL.Equip[EvLKi].LastMaskSent[PKTi] -2
Dest: CalloutRspMsg.Header.CmdNum -5

7

EQ
Source A: Equipment Data Callout/Status Message Header - Command Number CalloutRspMsg.Header.CmdNum -5
Source B: -1

COP
Source: Maintenance Event Mask Downtime Owner EDC_EventMaskPreDef.Maint
Dest: Equipment Data Callout/Status Message Data - Predefined Status CalloutRspMsg.PreDef
Length: 1

COP
Source: EDC_EventMaskMaint.Equip[EvLKi].Packet[PKTi].Data[0]
Dest: Equipment Data Callout/Status Message Equipment Defined Status Words CalloutRspMsg.EqDef.Word[0]
Length: EDC_Event.CFG.DAT.n

Create Callout Response Message with Event Mask

Ignore Event Mask
Downtime Type

8

Equipment Data Callout/Status Message Header - Command Number

EQ	
Source A	CalloutRspMsg.Header.CmdNum
	-5
Source B	-2

Equipment Data Callout/Status Message Data - Predefined Status

COP	
Source	EDC_EventMaskPreDef.Ignore
Dest	CalloutRspMsg.PreDef
Length	1

Equipment Data Callout/Status Message Equipment Defined Status Words

COP	
Source	EDC_EventMaskIgnore.Equip[EvLKi].Packet[PKTi].Data[0]
Dest	CalloutRspMsg.EqDef.Word[0]
Length	EDC_Event.CFG.DAT.n

Fault Event Mask
Downtime Type

9

Equipment Data Callout/Status Message Header - Command Number

EQ	
Source A	CalloutRspMsg.Header.CmdNum
	-5
Source B	-3

Equipment Data Callout/Status Message Data - Predefined Status

COP	
Source	EDC_EventMaskPreDef.Fault
Dest	CalloutRspMsg.PreDef
Length	1

Equipment Data Callout/Status Message Equipment Defined Status Words

COP	
Source	EDC_EventMaskFault.Equip[EvLKi].Packet[PKTi].Data[0]
Dest	CalloutRspMsg.EqDef.Word[0]
Length	EDC_Event.CFG.DAT.n

Alarm Event Mask
Downtime Type

10

Equipment Data Callout/Status Message Header - Command Number

EQ	
Source A	CalloutRspMsg.Header.CmdNum
	-5
Source B	-4

Equipment Data Callout/Status Message Data - Predefined Status

COP	
Source	EDC_EventMaskPreDef.Alarm
Dest	CalloutRspMsg.PreDef
Length	1

Equipment Data Callout/Status Message Equipment Defined Status Words

COP	
Source	EDC_EventMaskAlarm.Equip[EvLKi].Packet[PKTi].Data[0]
Dest	CalloutRspMsg.EqDef.Word[0]
Length	EDC_Event.CFG.DAT.n

Create Callout Response Message with Event Mask

Alert Event Mask
Downtime Type

11

EQ
Source A: CalloutRspMsg.Header.CmdNum
-5
Source B: -5

Equipment Data Callout/Status Message Header - Command Number

COP
Source: EDC_EventMaskPreDef.Alert
Dest: CalloutRspMsg.PreDef
Length: 1

Equipment Data Callout/Status Message Data - Predefined Status

COP
Source: EDC_EventMaskAlert.Equip[EvLKi].Packet[PKTi].Data[0]
Dest: CalloutRspMsg.EqDef.Word[0]
Length: EDC_Event.CFG.DAT.n

Equipment Data Callout/Status Message Equipment Defined Status Words

Generate the Mask Download Required to indicates which Packets have more Masks to download.
The EqDriver uses this to generate Callout Messages until all Masks are downloaded
(waiting for data events to occur in each Packet is not a reliable mechanism)
v3.02 Data Refresh Request Command

12

CPT
Dest
Expression: EDC_Event.CTL.Equip[EvLKi].MaintMaskSend OR EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend OR EDC_Event.CTL.Equip[EvLKi].FaultMaskSend OR EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend OR EDC_Event.CTL.Equip[EvLKi].AlertMaskSend

Equipment Data Callout/Status Message Footer - Spare (fills unused portion of DINT)

Create Callout Response Message with Event Mask

CalloutRspMsg.Footer.Spare
0
)R EDC_Event.CTL.Equip[EvLKi].DataRefreshReq

Exit Routine

lbl_Exit

13 LBL NOP

(End)

Update Equip Events Table for HMI Displays

Update Top-N Equipment Events
Table of Current Events and Start Time ordered by Duration (oldest first)

This method provides time-ordered list without sorting Events by duration
- Remove Existing Events from List
- Add New Events in order found until table full
- If table is full, a Remove/Add may not find Oldest New Event - it will find the Next New Event in Word, Bit order
(this is a small tradeoff to avoid sort by duration for all Events)

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 135
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

DN014 - Replace CallMsg.Header.EQID with EQID to allow Equip Event updates without Callout Msg
Calculate Equipment ID for Callout Table Lookup
Exit if EQID out of range

2

Calulated EQID Index within array limits
EQID_IndexOK
/

lbl_Exit
JMP

Update Equip Events Table for HMI Displays

Check the current Events and remove any that have turned off.
By adding/removed Events from the current Event list, the oldest events are retained, in order without sorting by Duration

Copy Callout Table to DINT array for FBC
Initialize Event Loop Pointer

DN014 - Replace CallMsg.Header.EQID with EQID to allow Equip Event updates without Callout Msg

3

Local Copy of EDC_EqCallout for this EQID

COP	
Source	EDC_EqCallout[LNi,EQi]
Dest	TempCallout[0]
Length	CFG.CalloutARR.n <EDC.CFG.CalloutARR.n>

Event Table Index

CLR	
Dest	EVi
	0

Skip Event Mask load if invalid Lookup Index default to Downtime Type of Fault

4

Event Lookup Index EQID Match
EvLKi_EQIDmatch
/

Temp Result for Array Size

MUL	
Source A	EDC_Event.CFG.PKT.n[8]
Source B	EDC_Event.CFG.DAT.n[8]
Dest	LEN 64

Equipment Event Fault Mask

FLL	
Source	-1
Dest	EquipMaskFault[0]
Length	LEN

Equipment Event Alarm Mask

FLL	
Source	0
Dest	EquipMaskAlarm[0]
Length	LEN

Equipment Event Alert Mask

FLL	
Source	0
Dest	EquipMaskAlert[0]
Length	LEN

Equipment Event No Display Mask

FLL	
Source	0
Dest	EquipMaskNoDisp[0]
Length	LEN

lbl_SkipMaskLoad
JMP

Get EqDefined Event Masks for Fault, Alarm and Alert used to determine Event Display status

5

Temp Result for Array Size

MUL	
Source A	EDC_Event.CFG.PKT.n[8]
Source B	EDC_Event.CFG.DAT.n[8]
Dest	LEN 64

Equipment Event Fault Mask

COP	
Source	EDC_EventMaskFault.Equip[EvLKi].Packet[0].Data[0]
Dest	EquipMaskFault[0]
Length	LEN

Equipment Event Alarm Mask

COP	
Source	EDC_EventMaskAlarm.Equip[EvLKi].Packet[0].Data[0]
Dest	EquipMaskAlarm[0]
Length	LEN

Equipment Event Alert Mask

COP	
Source	EDC_EventMaskAlert.Equip[EvLKi].Packet[0].Data[0]
Dest	EquipMaskAlert[0]
Length	LEN

Equipment Event No Display Mask

COP	
Source	EDC_EventMaskNoDisp.Equip[EvLKi].Packet[0].Data[0]
Dest	EquipMaskNoDisp[0]
Length	LEN

Update Equip Events Table for HMI Displays

Skip Equipment Event Mask Load

lbl_SkipMaskLoad

6

LBL

NOP

Equipment Event Display Status
- EQ InterlockOff - display only Interlock Off Event
- EQ not InterlockOff, Fault, Alarm or Alert - display no Events
- NoDisplay (Predefined.15) - display no Events (from v1.x EquDriver)
DHF006 - CTL.DisableEvents clears Events
DHF029 - Added Alert Status 5
EDC2 - v2.01 added Equip.CFG.DisableEvents to replace PreDefined.15

7

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange)
Equip.DataCTL.InterlockOff

Local Copy of Equip Array Tag for LNi, EQi Display Interlock Off Event Only
Equip.EventCTL.DispInterlockOff

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange)
Equip.DataCTL.InterlockOff

Local Copy of Equip Array Tag for LNi, EQi Fault Condition (Red)
Equip.DataCTL.Fault

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow)
Equip.DataCTL.Alarm

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue)
Equip.DataCTL.Alert

Local Copy of Equip Array Tag for LNi, EQi Display No Events
Equip.EventCTL.DispNone

EQ NoEventDisplay
TempCallout[0].15

AFI

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Local Copy of Equip Array Tag for LNi, EQi Equipment Event Display Disabled (from Server Config)
Equip.CFG.DisableEvents

Update Equip Events Table for HMI Displays

Exit Event Check Loop if Record is Empty

8

lbl_EventRemoveLoop — LBL

Local Copy of Equip Array Tag for LNi, EQi Fault/Status Event Element (Word*100) + Bit

EQ
Source A: Equip.Event[EVi].Element -1
Source B: -1

lbl_EventAdd — JMP

Check if Element Word/Bit Indexes in Range
Remove if out of Range

9

This Event local copy of Equip.Event[EVi]

COP
Source: Equip.Event[EVi]
Dest: Event
Length: 1

Display Event Removed: EventRemoved — U

This Event local copy of Equip.Event[EVi] Fault/Status Event Element Word

LIMIT
Low Limit: 0
Test: Event.Word 33
High Limit: CFG.CalloutARR.mi <EDC.CFG.CalloutARR.mi> 64

This Event local copy of Equip.Event[EVi] Fault/Status Event Element Bit

LIMIT
Low Limit: 0
Test: Event.Bit 11
High Limit: 31

Event Word Index

MOVE
Source: Event.Word 33
Dest: WDi 39

Event Bit Index

MOVE
Source: Event.Bit 11
Dest: BITi 17

EventIndexInRange

EventIndexInRange — /

lbl_EventRemove — JMP

PreDefined Event No Display
only display Fault, Alarm and Alert
v3.5 moved v2_0 inside branch; always honor NoDisp

10

Event Word Index

EQ
Source A: WDi 39
Source B: 0

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)

Equip.CTL.CalloutMsgV2_0

PreDefined Event Mask DtType Fault Mask

EDC_EventMaskPreDef.Fault.[BITi] — /

PreDefined Event Mask DtType Alarm Mask

EDC_EventMaskPreDef.Alarm.[BITi] — /

PreDefined Event Mask DtType Alert Mask

EDC_EventMaskPreDef.Alert.[BITi] — /

PreDefined Event Mask No Display Option

EDC_EventMaskPreDef.NoDisp.[BITi]

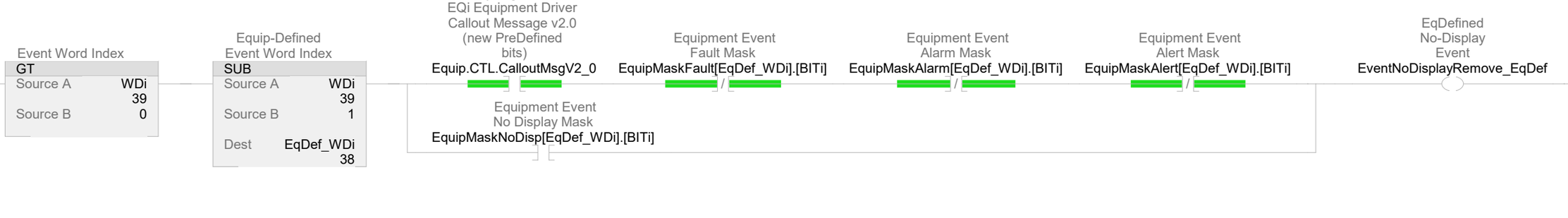
PreDefined No-Display Event

EventNoDisplayRemove_PreDef

Update Equip Events Table for HMI Displays

EqDefined Event No Display
only display Fault, Alarm and Alert
v3.5 moved v2_0 inside branch; always honor NoDisp

11



Update Equip Events Table for HMI Displays

Check if Event still On
Remove if Event is not On or Equipment/Event is No Display

PreDefined Word 0, Bits 0-7 and Bit 17 are not Displayed Events
(OverallStart, ConvRunning, FixedStopPosition, FixedStopReq, TempStopReq, ProdStopReq, Fault, Alarm, Alert)
DHF029 - Added Alert to not Displayed Events

12

Local Copy of Equip Array Tag for LNi, EQi Display No Events
Equip.EventCTL.DispNone

PreDefined No-Display Event
EventNoDisplayRemove_PreDef

EqDefined No-Display Event
EventNoDisplayRemove_EqDef

Local Copy of Equip Array Tag for LNi, EQi Display Interlock Off Event Only
Equip.EventCTL.DispInterlockOff

Remove No-Display Event from Event Viewers
EventNoDisplayRemove

Event Word Index
NE
Source A WDi 39
Source B 0

Event Bit Index
NE
Source A BITi 17
Source B 14

Event Word Index
EQ
Source A WDi 39
Source B 0

Event Bit Index
LIMIT
Low Limit 0
Test BITi 17
High Limit 7

Event Bit Index
EQ
Source A BITi 17
Source B 17

Update Equip Events Table for HMI Displays

Remove No-Display Event from Event Viewers
EventNoDisplayRemove
/

Local Copy of EDC_EqCallout for this EQID
TempCallout[WDi].[BITi]

lbl_EventNext
JMP

Remove Event
If this isn't the last Event in the Table, shift following Events up 1 position before clearing last position

13

lbl_EventRemove
LBL

Event Table Index
LT
Source A EVi
0
Source B CFG.EV.mi
<EDC.CFG.EV.mi>
9

Temp Result for Array Size
SUB
Source A CFG.EV.mi
<EDC.CFG.EV.mi>
9
Source B EVi
0
Dest LEN
64

Local Copy of Equip Array Tag for LNi, EQi Equipment Event Table First N Fault/Status Events
COP
Source Equip.Event[EVi+1]
Dest Equip.Event[EVi]
Length LEN

Display Event Removed
EventRemoved
L

Remove Event
Clear Last Position in Table

14

Local Copy of Equip Array Tag for LNi, EQi Equipment Event Table First N Fault/Status Events
FLL
Source -1
Dest Equip.Event[CFG.EV.mi]
Length 1

Update Equip Events Table for HMI Displays

Increment Index
Block Increment if Event Deleted - Event Table Shift moves next Event to current Index
If EVi exceeds Max Index, Table is full and no need to check for additional events

15

lbl_EventNext
LBL

Display Event Removed
EventRemoved
/

Event Table Index
ADD
Source A EVi 0
Source B 1
Dest EVi 0

Event Table Index
LE
Source A EVi 0
Source B CFG.EV.mi <EDC.CFG.EV.mi> 9

lbl_EventRemoveLoop
JMP

lbl_Exit
JMP

Add New Events after the Existing Events
When EventRemoveLoop ends, EVi will point to first open position in Event Table

16

lbl_EventAdd
LBL

NOP

Exit without adding Events if Equipment Display None

17

Local Copy of Equip Array Tag for LNi, EQi Display No Events
Equip.EventCTL.DispNone

lbl_Exit
JMP

Update Equip Events Table for HMI Displays

Initialize Event Search Array and Controls
Compare Inhibit bit set to enable 'One mismatch at a time' mode

18

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp
RES

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.IN
L

Search EQ Callout for Display Events FBC (Compare)
CPT
Dest EventFbcCmp.LEN 2080
Expression CFG.CalloutARR.n*32

Search EQ Callout for Display Events FBC (Compare)
MOVE
Source 0
Dest EventFbcCmp.POS 2080

Search EQ Callout for Display Events FBC (Results) EventFbcRes
RES

Search EQ Callout for Display Events FBC (Results)
MOVE
Source CFG.EV.n <EDC.CFG.EV.n> 10
Dest EventFbcRes.LEN 10

Search EQ Callout for Display Events FBC (Results)
MOVE
Source 0
Dest EventFbcRes.POS 0

Event File Bit Comparer (FBC)
Search Bit Array for On Event

19

lbl_EventAddLoop
LBL

Search EQ Callout for Display Events FBC (Results)
FBC
Source TempCallout[0]
Reference ZeroArray[0]
Result EventResArray[0]
Cmp. Control EventFbcCmp
Length 2080
Position 2080
Result Control EventFbcRes
Length 10
Position 0
EN
DN
FD
IN
ER

Update Equip Events Table for HMI Displays

If Event Found, convert Bit Number to Element Number and check Event Array and determine if exising or new
Element Number = (Word * 100) + Bit

20

Search EQ Callout for Display Events FBC (Compare)
EventFbcCmp.FD

Search EQ Callout for Display Events FBC (Results)
LIMIT
Low Limit 1
Test EventFbcRes.POS 0
High Limit CFG.EV.n <EDC.CFG.EV.n> 10

Event FBC Result Array Position
SUB
Source A EventFbcRes.POS 0
Source B 1
Dest EventResPos 0

Event FBC Result
MOVE
Source EventResArray[EventResPos] 1061
Dest EventRes 1265

Event Word Index
CPT
Dest WDi 39
Expression TRUNC(EventRes/32)

Event Bit Index
CPT
Dest BITi 17
Expression EventRes MOD 32

Event Element
CPT
Dest Element 3917
Expression WDi*100+BITi

PreDefined Event No Display if not Fault, Alarm or Alert
(Event Color is added in HMI code to save memory)
v3.5 moved v2_0 inside branch; always honor NoDisp

21

Event Word Index
EQ
Source A WDi 39
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

PreDefined Event Mask DtType Fault Mask
EDC_EventMaskPreDef.Fault.[BITi]

PreDefined Event Mask DtType Alarm Mask
EDC_EventMaskPreDef.Alarm.[BITi]

PreDefined Event Mask DtType Alert Mask
EDC_EventMaskPreDef.Alert.[BITi]

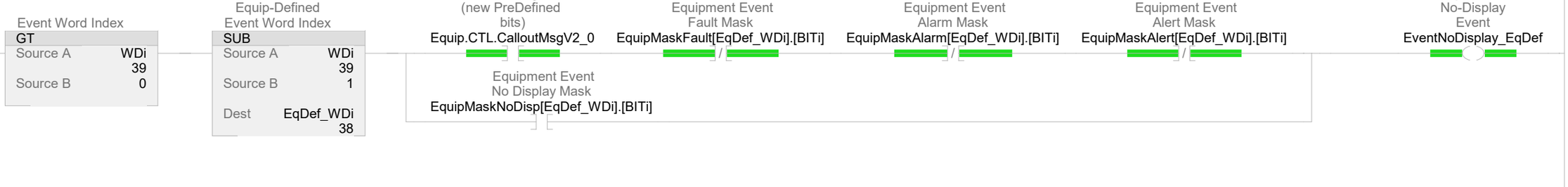
PreDefined Event Mask No Display Option
EDC_EventMaskPreDef.NoDisp.[BITi]

PreDefined No-Display Event
EventNoDisplay_PreDef

Update Equip Events Table for HMI Displays

EqDefined Event No Display if not Fault, Alarm or Alert
(Event Color is added in HMI code to save memory)
v3.5 moved v2_0 inside branch; always honor NoDisp

22



Update Equip Events Table for HMI Displays

If Event Found, convert Bit Number to Element Number and check Event Array and determine if exising or new
Element Number = (Word * 100) + Bit

PreDefined Word 0, Bits 0-7 and Bit 17 are not Displayed Events
(OverallStart, ConvRunning, FixedStopPosition, FixedStopReq, TempStopReq, ProdStopReq, Fault, Alarm, Alert)
DHF029 - Added Alert to not Displayed Events

23

Search EQ Callout for Display Events FBC (Compare)
EventFbcCmp.FD

Search EQ Callout for Display Events FBC (Results)
LIMIT
Low Limit 1
Test EventFbcRes.POS 0
High Limit CFG.EV.n <EDC.CFG.EV.n> 10

Local Copy of Equip Array Tag for LNi, EQi Display No Events
Equip.EventCTL.DispNone

PreDefined No-Display Event
EventNoDisplay_PreDef

EqDefined No-Display Event
EventNoDisplay_EqDef

Local Copy of Equip Array Tag for LNi, EQi Display Interlock Off Event Only
Equip.EventCTL.DispInterlockOff

Event Word Index
NE
Source A WDi 39
Source B 0

Event Bit Index
NE
Source A BITi 17
Source B 14

Event Word Index
EQ
Source A WDi 39
Source B 0

Event Bit Index
LIMIT
Low Limit 0
Test BITi 17
High Limit 7

Event Bit Index
EQ
Source A BITi 17
Source B 17

Block 'No Display Event' from Event Viewers
EventNoDisplay

Update Equip Events Table for HMI Displays

PreDef 19 & 20 - Health Events from Server
Use Gateway Timestamp

24

Event Word Index

EQ	
Source A	WDi 39
Source B	0

Event Bit Index

LIMIT	
Low Limit	19
Test	BITi 17
High Limit	20

Health Event from Server
EventHealth

Update Equip Events Table for HMI Displays

If Event Found and within Display Range, search existing Event Display for duplicate
if not found, add Event to Display
if found - existing Event, resume search without adding

25

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.FD

Block 'No Display Event' from Event Viewers EventNoDisplay

Search Event Display for existing (already displayed, don't need to add) EventExistFSC (RES)

Event Table Index

GT	
Source A	EVi
	0
Source B	0

Search Event Display for existing (already displayed, don't need to add)

MOVE	
Source	EVi
	0
Dest	EventExistFSC.LEN
	1

Search Event Display for existing (already displayed, don't need to add)

MOVE	
Source	0
Dest	EventExistFSC.POS
	0

Search Event Display for existing (already displayed, don't need to add)

FSC		
Control	EventExistFSC	EN
Length	1	DN
Position	0	ER
Mode	ALL	
Expression	Element=Equip.Event[EventExistFSC.POS].Element	

Search Event Display for existing (already displayed, don't need to add) EventExistFSC.FD

Local Copy of Equip Array Tag for LNi, EQi Fault/Status Event Element (Word*100) + Bit

MOVE	
Source	Element
	3917
Dest	Equip.Event[EVi].Element
	-1

Local Copy of Equip Array Tag for LNi, EQi Fault/Status Event Element Word

MOVE	
Source	WDi
	39
Dest	Equip.Event[EVi].Word
	-1

Local Copy of Equip Array Tag for LNi, EQi Fault/Status Event Element Bit

MOVE	
Source	BITi
	17
Dest	Equip.Event[EVi].Bit
	-1

Health Event from Server EventHealth

Local Copy of Equip Array Tag for LNi, EQi Transaction Time

COP	
Source	CalloutMsg.TransTime
Dest	Equip.Event[EVi].TransTime
Length	1

Event Table Index

ADD	
Source A	EVi
	0
Source B	1
Dest	EVi
	0

Health Event from Server EventHealth

Local Copy of Equip Array Tag for LNi, EQi Transaction Time

COP	
Source	CLX.TransTime
	<EDC.CLX.TransTime>
Dest	Equip.Event[EVi].TransTime
Length	1

Update Equip Events Table for HMI Displays

If Event Found and not within Display Range, decrement Result Array to make room for a different result
(otherwise Result Array fills up with Events that aren't displayed)

26

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.FD

Block 'No Display Event' from Event Viewers EventNoDisplay

Search EQ Callout for Display Events FBC (Results)

SUB
Source A EventFbcRes.POS[0]
Source B 1
Dest EventFbcRes.POS[0]

Search EQ Callout for Display Events FBC (Results)

LT
Source A EventFbcRes.POS[0]
Source B 0

Search EQ Callout for Display Events FBC (Results)

MOVE
Source 0
Dest EventFbcRes.POS[0]

If FBC Compare and Result Array not Done, AND Event Table not full -- resume the FBC Compare to look for another Event

27

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.DN

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.ER

Search EQ Callout for Display Events FBC (Results) EventFbcRes.DN

Search EQ Callout for Display Events FBC (Results) EventFbcRes.ER

Event Table Index

LE
Source A EVi[0]
Source B CFG.EV.mi <EDC.CFG.EV.mi> 9

Search EQ Callout for Display Events FBC (Compare) EventFbcCmp.EN

U

lbl_EventAddLoop

JMP

Exit Routine

28

lbl_Exit

LBL

NOP

(End)

Calculate Equipment Data and Operational Availability (OA)

Calculate Equipment Operational Availability (OA)

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

v1.6 Separate Line and Equipment Downtime

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 140
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Faster to JMP over AOI if not scanned
(AOI uses scan time even when Rung is scanned false)

2

Reset Summary Data for this Line
SumDataReset
/

lbl_SkipReset
JMP

Shift Reset for Equipment OA
Fault & Alarm Time are not reset at Shift Change

3

Reset Summary Data for this Line
SumDataReset

External Data Job Count

CLR
Dest EDC_CycleCount[LNi,EQi] ??

Data Reset for Line, Group, Equip

aoiEDC_DataReset
aoiEDC_DataReset aoiDataReset ...
Data Equip.Data
DataCtlPrev Equip.DataCTLprev

Exit if EQID out of range

4

lbl_SkipReset
LBL

Calulated EQID Index within array limits
EQID_IndexOK
/

lbl_Exit
JMP

Calculate Equipment Data and Operational Availability (OA)

If Offline, load Group & Line Indexes
Offline Equipment can be configured to use any Line for OverallStart/Running signals
Offline Equipment can be configured to be in any Group for HMI Display
DN006 - support multiple Offline Lines

5

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Offline Equipment Group Index
MOVE
Source Equip.CFGoffline.Group 0
Dest EquipGRPi 2

Offline Equipment Group Index
LT
Source A EquipGRPi 2
Source B 0

Offline Equipment Group Index
GT
Source A EquipGRPi 2
Source B CFG.GRP.ei <EDC.CFG.GRP.ei> 3

Offline Equipment Group Index
CLR
Dest EquipGRPi 2

Copy EDC_EQCallout Predefined to a local tag for named references
DN006 - support multiple Offline Lines

GroupDataSTS - accumulates overlapping Status of Downtime types (Tool Change, Options, Full, Short, ProdDelay)
GroupDataCTL - contains exclusive Control of Downtime Type (generated after all Group-Equipment processed)
STS & CTL are needed to handle seal-in of existing Downtime Type

6

Local copy of Callout Table Predefined for EQID
COP
Source EDC_EqCallout[LNi,EQi].PreDef
Dest PreDef
Length 1

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Group Array Tag for GRPi
COP
Source EDC.Offline.DataSTS[EquipGRPi]
Dest GrpDataSTS
Length 1

Local Copy of Group Array Tag for GRPi
COP
Source EDC.Offline.DataCTL[EquipGRPi]
Dest GrpDataCTL
Length 1

Calculate Equipment Data and Operational Availability (OA)

External Data Trigger
Cycle Counts

used by Telit to trigger Robot data reads
and for Summary Data

7

Local copy of Callout Table Predefined for EQID Overload is ON
PreDef._12

Local Copy of Equip Array Tag for LNi, EQi Cycle Count External Data 1-Shot
Equip.DataCTL.CycleCountONS
ONS

External Data Job Count

ADD
Source A EDC_CycleCount[LNi,EQi] ??
Source B 1
Dest EDC_CycleCount[LNi,EQi] ??

Calculate Equipment Data and Operational Availability (OA)

Equipment Data - EDT/LDT

Determine Worktime and Stop status
EDT - Equip Downtime - EQ Not Ready
LDT - Line Downtime - Line Not Running

Equipment Producing (OR%)
Equipment Available (OA%)

8

Local copy of Callout Table Predefined for EQID Line Production Work Time
PreDef._00

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start)
Equip.DataCTL.Work

Local copy of Callout Table Predefined for EQID Temp Stop Request
PreDef._04

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OR
Equip.DataCTL.EdtOR

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA)
PreDef._27

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OA
Equip.DataCTL.EdtOA

Local copy of Callout Table Predefined for EQID EQ Fixed Stop Request
PreDef._03

Local copy of Callout Table Predefined for EQID Line Conveyor Stopped at Fixed Stop Position (L10)
PreDef._02

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

Local copy of Callout Table Predefined for EQID Fault Status is ON (from EqDriver Fault Mask)
PreDef._06

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed
PreDef._14

Local copy of Callout Table Predefined for EQID Line Conveyor Running
PreDef._01

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed
PreDef._14

Local copy of Callout Table Predefined for EQID Temp Stop Request
PreDef._04

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OR
Equip.DataCTL.LdtOR

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA)
PreDef._27

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OA
Equip.DataCTL.LdtOA

Local copy of Callout Table Predefined for EQID

Local copy of Callout Table Predefined for EQID Line Conveyor

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0

Calculate Equipment Data and Operational Availability (OA)

EQ Fixed Stop Request PreDef._03

Stopped at Fixed Stop Position (L10) PreDef._02

(new PreDefined bits) Equip.CTL.CalloutMsgV2_0

Equipment Data - EDT Maint/Prod

Determine Maintenance/Production Stop status
EDT - Equip Downtime - EQ Not Ready

Equipment Producing (OR%)
Equipment Available (OA%)

Sep 2023 - seal in initial Maint/Prod for duration of Downtime

9

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OR Equip.DataCTL.EdtOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Production OR Equip.DataCTL.EdtProdOR

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed PreDef._14

Local copy of Callout Table Predefined for EQID Production Stop Request (from EqDriver Mask Prod/Maint Mask) PreDef._05

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OR Equip.DataCTL.EdtMaintOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OR Equip.DataCTL.EdtMaintOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OA Equip.DataCTL.EdtOA

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OA Equip.DataCTL.EdtMaintOA

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OR Equip.DataCTL.EdtMaintOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Production OR Equip.DataCTL.EdtProdOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OA Equip.DataCTL.EdtOA

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Production OA Equip.DataCTL.EdtProdOA

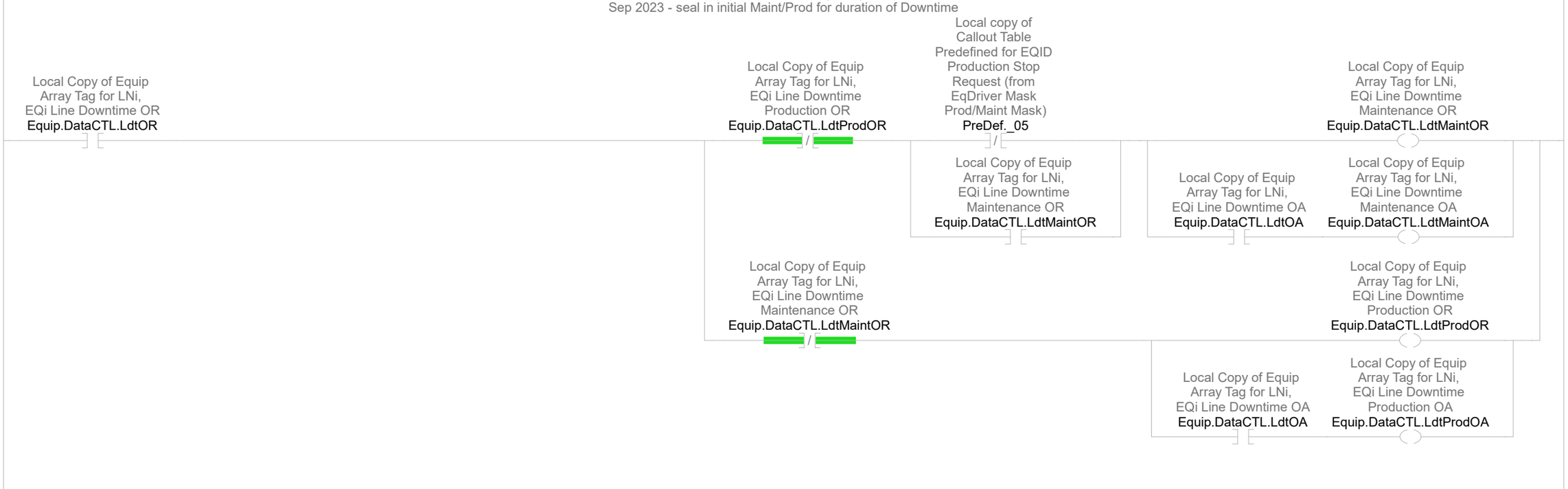
Calculate Equipment Data and Operational Availability (OA)

Equipment Data - LDT Maint/Prod

Determine Maintenance/Production Stop status
LDT - Line Downtime - Line Not Running

Equipment Producing (OR%)
Equipment Available (OA%)
Sep 2023 - seal in initial Maint/Prod for duration of Downtime

10

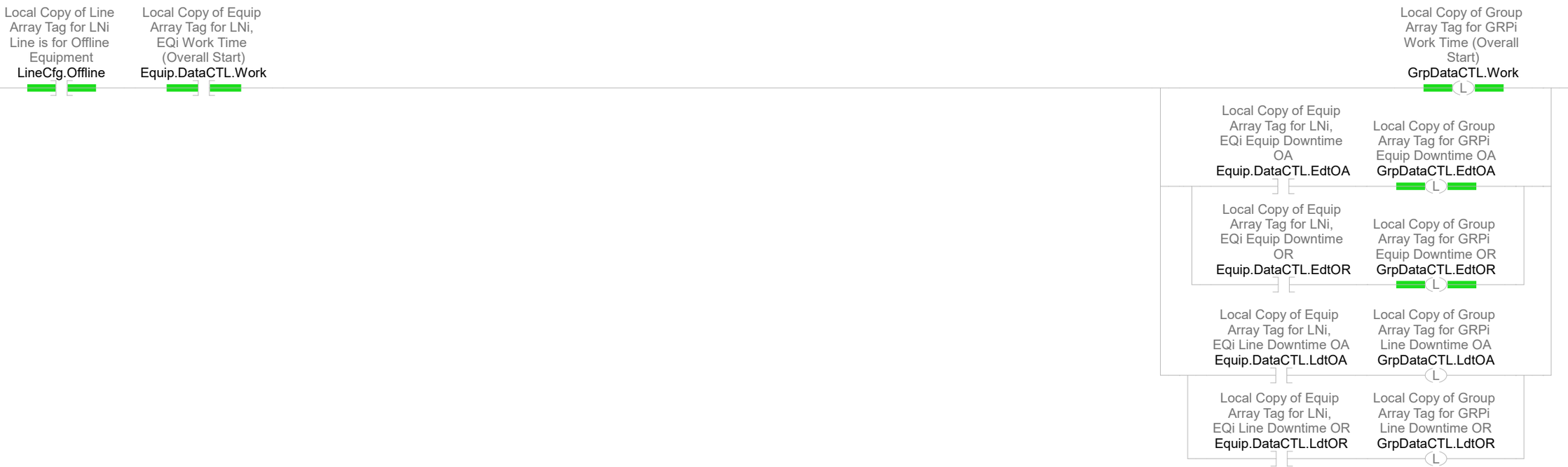


Calculate Equipment Data and Operational Availability (OA)

Group Data - EDT/LDT

Determine Worktime and Stop statusfor all EQID in Group
If Any Equip has Work/Stop, Group has Work/Stop (worst-case rollup)

11



Calculate Equipment Data and Operational Availability (OA)

Group Data - EDT Maint/Prod

Determine Maintenance/Production Downtime status for all EQID on Group
Feb 8, 2017 - changed Line Rollup to charge Maintenance if any EQ Maintenance

12

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OA
Equip.DataCTL.EdtMaintOA

Local Copy of Group Array Tag for GRPi Equip Downtime Maintenance OA
GrpDataCTL.EdtMaintOA

Local Copy of Group Array Tag for GRPi Equip Downtime Production OA
GrpDataCTL.EdtProdOA

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Maintenance OR
Equip.DataCTL.EdtMaintOR

Local Copy of Group Array Tag for GRPi Equip Downtime Maintenance OR
GrpDataCTL.EdtMaintOR

Local Copy of Group Array Tag for GRPi Equip Downtime Production OR
GrpDataCTL.EdtProdOR

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Production OA
Equip.DataCTL.EdtProdOA

Local Copy of Group Array Tag for GRPi Equip Downtime Maintenance OA
GrpDataCTL.EdtMaintOA

Local Copy of Group Array Tag for GRPi Equip Downtime Production OA
GrpDataCTL.EdtProdOA

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime Production OR
Equip.DataCTL.EdtProdOR

Local Copy of Group Array Tag for GRPi Equip Downtime Maintenance OR
GrpDataCTL.EdtMaintOR

Local Copy of Group Array Tag for GRPi Equip Downtime Production OR
GrpDataCTL.EdtProdOR

Calculate Equipment Data and Operational Availability (OA)

Group Data - LDT Maint/Prod

Determine Maintenance/Production Stop status for all EQID on Group
Feb 8, 2017 - changed Line Rollup to charge Maintenance if any EQ Maintenance

13

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Maintenance OA
Equip.DataCTL.LdtMaintOA

Local Copy of Group Array Tag for GRPi Line Downtime Maintenance OA
GrpDataCTL.LdtMaintOA

Local Copy of Group Array Tag for GRPi Line Downtime Production OA
GrpDataCTL.LdtProdOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Maintenance OR
Equip.DataCTL.LdtMaintOR

Local Copy of Group Array Tag for GRPi Line Downtime Maintenance OR
GrpDataCTL.LdtMaintOR

Local Copy of Group Array Tag for GRPi Line Downtime Production OR
GrpDataCTL.LdtProdOR

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Production OA
Equip.DataCTL.LdtProdOA

Local Copy of Group Array Tag for GRPi Line Downtime Maintenance OA
GrpDataCTL.LdtMaintOA

Local Copy of Group Array Tag for GRPi Line Downtime Production OA
GrpDataCTL.LdtProdOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Production OR
Equip.DataCTL.LdtProdOR

Local Copy of Group Array Tag for GRPi Line Downtime Maintenance OR
GrpDataCTL.LdtMaintOR

Local Copy of Group Array Tag for GRPi Line Downtime Production OR
GrpDataCTL.LdtProdOR

Add Equipment Data to Line Data
Worktime and LineDT use Rollup of all Equiment or a Single EQi designated for Line Data

14

Equipment array Index

EQ
Source A EQi
12
Source B EDC_CfgGateway.LdtEQi
0

Gateway Config Download Line Downtime Rollup Enable
EDC_CfgGateway.LdtRollupEnable

Add Equip Data to Line Data
AddToLDT

Calculate Equipment Data and Operational Availability (OA)

This Line's Last Enabled Equip Loop scan
(update Line Data rollups)

15

Equipment array Index

EQ	
Source A	EQi 12
Source B	LineCfg.EQ.ENei 99

This Line's Last Equipment
LineCfg_LastEQi

16

Equipment Maintenance
Line Maintenance rollup of all Equipment

June 2025 -- add Line Rollup rules to Maint Mode selection
if rollup disabled, include Equip Bypass in Line Maint Mode

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed
PreDef._14

Local copy of Callout Table Predefined for EQID Production Stop Request (from EqDriver Mask Prod/Maint Mask)
PreDef._05
/

Local Copy of Equip Array Tag for LNi, EQi Maintenance Mode
Equip.DataCTL.MaintMode

Add Equip Data to Line Data
AddToLDT

Gateway Config Download Line Downtime Rollup Enable
EDC_CfgGateway.LdtRollupEnable
/

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed
PreDef._14

Local Copy of Line Array Tag for LNi Maintenance Mode
LineDataCTL.MaintMode
L

Local copy of Callout Table Predefined for EQID Production Stop Request (from EqDriver Mask Prod/Maint Mask)
PreDef._05
/

Calculate Equipment Data and Operational Availability (OA)

Line Data - EDT/LDT

Determine Line Worktime and LineDT statusfor all EQID on Line
Worktime and LineDT use Rollup of all Equiment or a Single EQi designated for Line Data
EquipDT uses Rollup of all Equipment

17

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start) Equip.DataCTL.Work

Add Equip Data to Line Data AddToLDT

Local Copy of Line Array Tag for LNi Work Time (Overall Start) LineDataCTL.Work (L)

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OA Equip.DataCTL.EdtOA

Local Copy of Line Array Tag for LNi Equip Downtime OA LineDataCTL.EdtOA (L)

Local Copy of Equip Array Tag for LNi, EQi Equip Downtime OR Equip.DataCTL.EdtOR

Local Copy of Line Array Tag for LNi Equip Downtime OR LineDataCTL.EdtOR (L)

Add Equip Data to Line Data AddToLDT

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OA Equip.DataCTL.LdtOA

Local Copy of Line Array Tag for LNi Line Downtime OA LineDataCTL.LdtOA (L)

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OR Equip.DataCTL.LdtOR

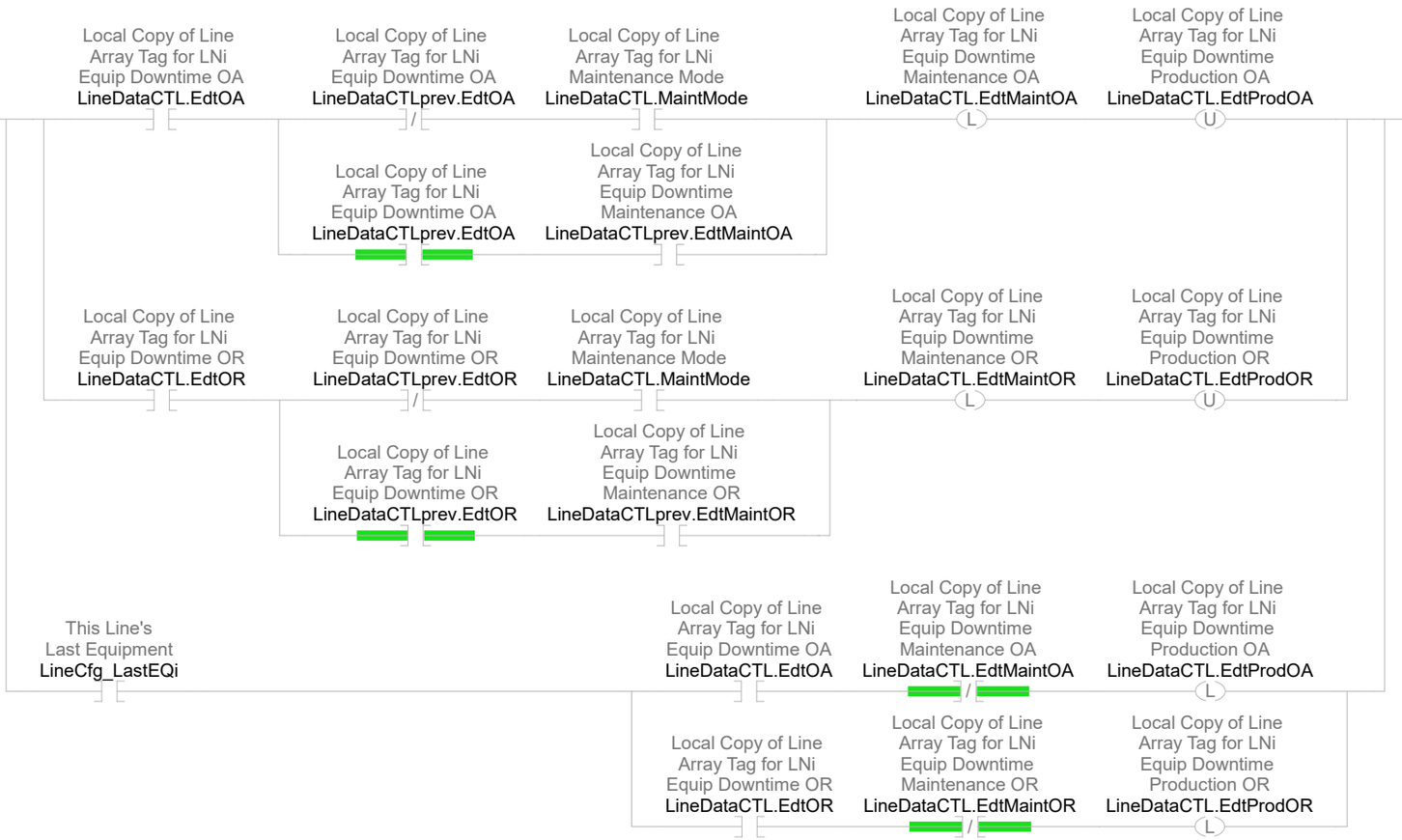
Local Copy of Line Array Tag for LNi Line Downtime OR LineDataCTL.LdtOR (L)

Calculate Equipment Data and Operational Availability (OA)

Line Data - EDT Maint/Prod

Determine Line Maintenance/Production Downtime status for all EQID on Line
Seal-in Maint status at start of Downtime if any Equip has Maint status - otherwise charge Production

18

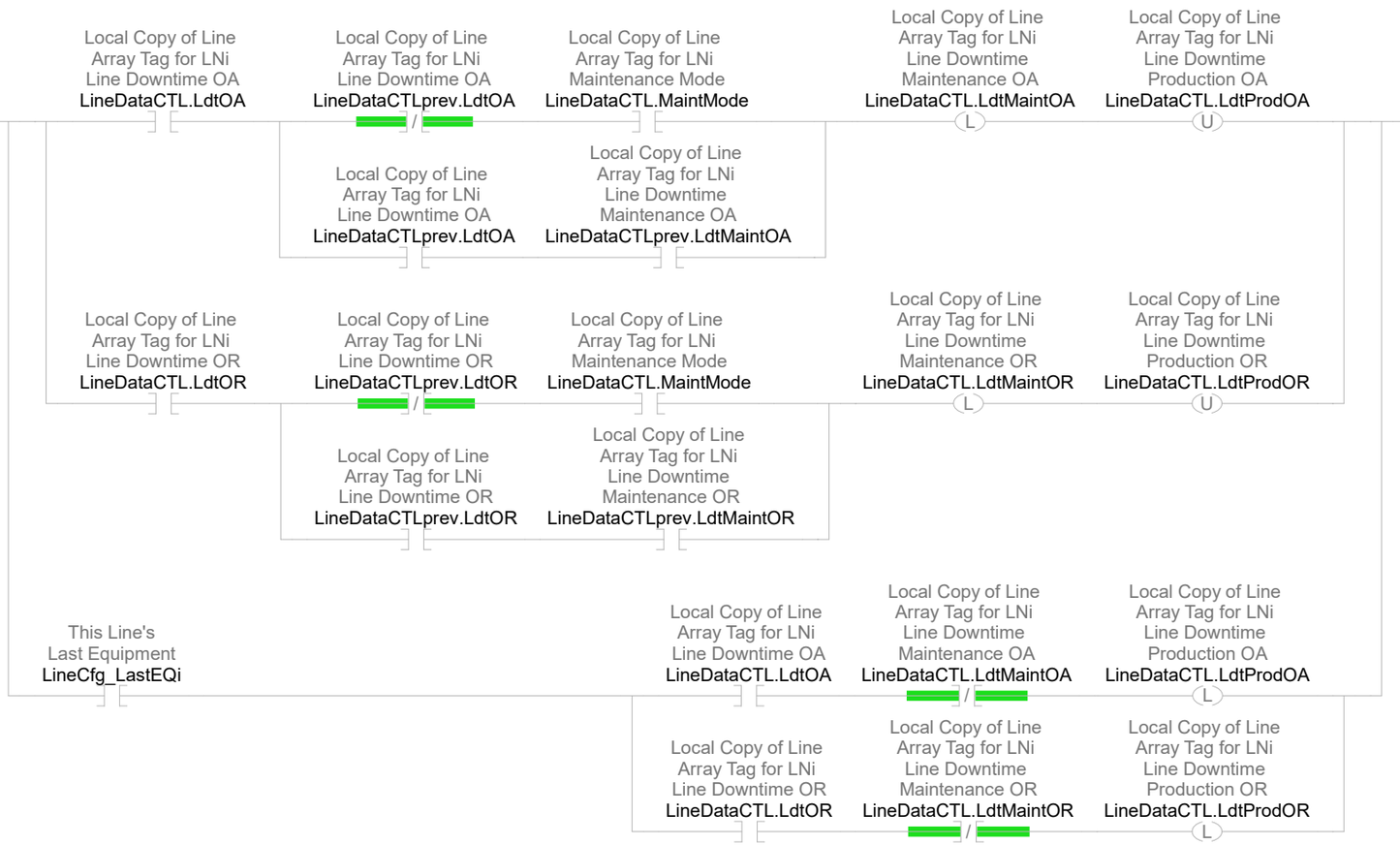


Calculate Equipment Data and Operational Availability (OA)

Line Data - LDT Maint/Prod

Determine Line Maintenance/Production Downtime status using Rollup of all Equiment or a Single EQi designated for Line Data
Seal-in Maint status at start of Downtime if any Equip has Maint status - otherwise charge Production

19



Calculate Equipment Data and Operational Availability (OA)

Determine Interlock status (Orange)
DHF029 Add Alert status
DN006 - support multiple Offline Lines

20

Local copy of Callout Table Predefined for EQID Interlock Off - Equipment Bypassed
PreDef._14

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange)
Equip.DataCTL.InterlockOff

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Group Array Tag for GRPi Interlock Off Condition (Orange)
GrpDataCTL.InterlockOff
L

Local Copy of Group Array Tag for GRPi Fault Condition (Red)
GrpDataCTL.Fault
U

Local Copy of Group Array Tag for GRPi Alarm Condition (Yellow)
GrpDataCTL.Alarm
U

Local Copy of Group Array Tag for GRPi Alert Condition (Lt Blue)
GrpDataCTL.Alert
U

Add Equip Data to Line Data
AddToLDT

Local Copy of Line Array Tag for LNi Interlock Off Condition (Orange)
LineDataCTL.InterlockOff
L

Local Copy of Line Array Tag for LNi Fault Condition (Red)
LineDataCTL.Fault
U

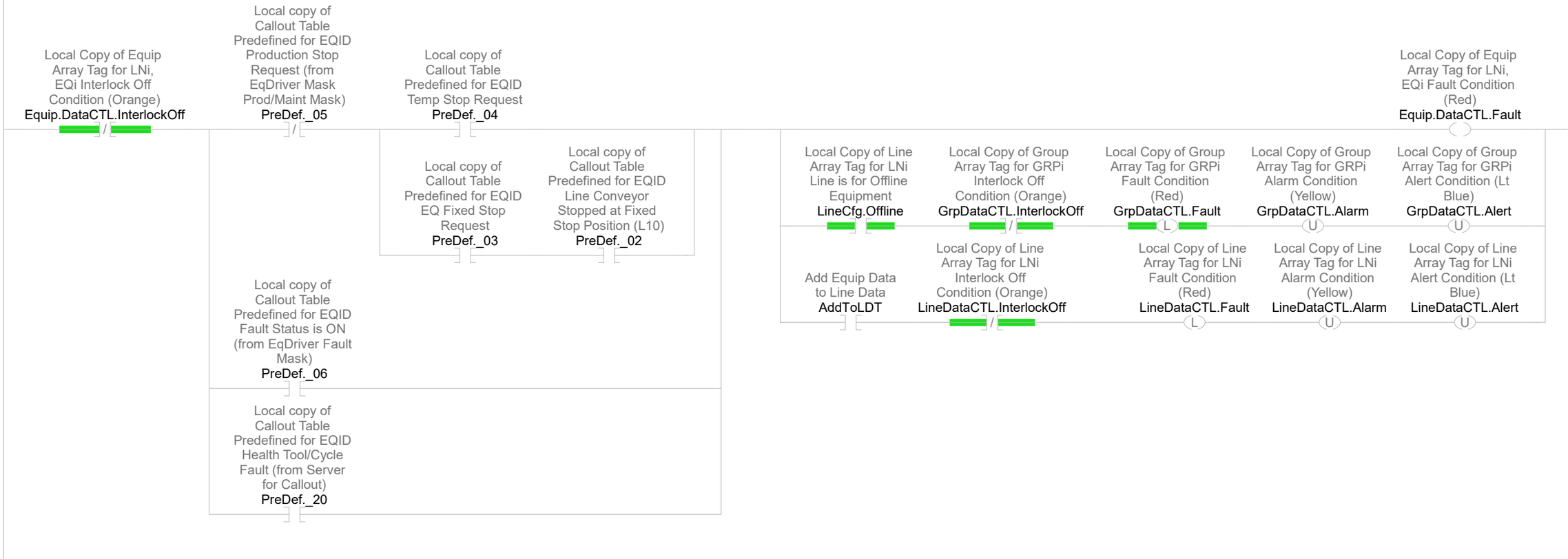
Local Copy of Line Array Tag for LNi Alarm Condition (Yellow)
LineDataCTL.Alarm
U

Local Copy of Line Array Tag for LNi Alert Condition (Lt Blue)
LineDataCTL.Alert
U

Calculate Equipment Data and Operational Availability (OA)

Determine Fault status (Red)
Offline if LNi = 0; Online if LNi <>0
DHF029 Add Alert status
DN006 - support multiple Offline Lines
DN014 - add EQ Tool Health Fault (PreDef.Spare20)

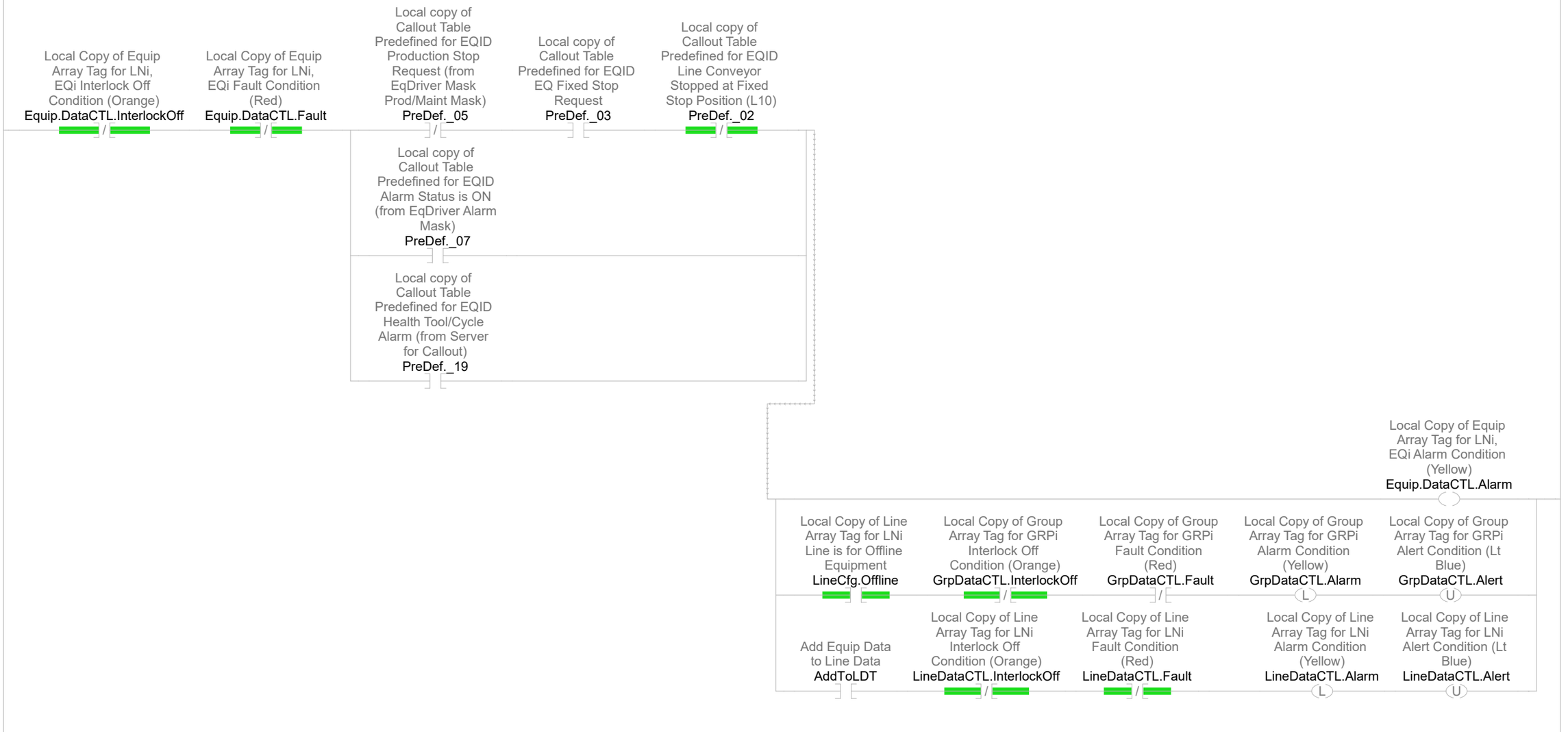
21



Calculate Equipment Data and Operational Availability (OA)

Determine Alarm status (Yellow)
DHF029 Add Alert status
DN006 - support multiple Offline Lines
DN014 - add EQ Tool Health Alarm (PreDef.Spare19)

22



Calculate Equipment Data and Operational Availability (OA)

DHF029 Determine Alert status (Lt Blue)
DN006 - support multiple Offline Lines

23

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange) Equip.DataCTL.InterlockOff

Local Copy of Equip Array Tag for LNi, EQi Fault Condition (Red) Equip.DataCTL.Fault

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow) Equip.DataCTL.Alarm

Local copy of Callout Table Predefined for EQID Alert Status is ON (from EqDriver Alert Mask) PreDef._17

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue) Equip.DataCTL.Alert

Local Copy of Line Array Tag for LNi Line is for Offline Equipment LineCfg.Offline

Local Copy of Group Array Tag for GRPi Interlock Off Condition (Orange) GrpDataCTL.InterlockOff

Local Copy of Group Array Tag for GRPi Fault Condition (Red) GrpDataCTL.Fault

Local Copy of Group Array Tag for GRPi Alarm Condition (Yellow) GrpDataCTL.Alarm

Local Copy of Group Array Tag for GRPi Alert Condition (Lt Blue) GrpDataCTL.Alert

Add Equip Data to Line Data AddToLDT

Local Copy of Line Array Tag for LNi Interlock Off Condition (Orange) LineDataCTL.InterlockOff

Local Copy of Line Array Tag for LNi Fault Condition (Red) LineDataCTL.Fault

Local Copy of Line Array Tag for LNi Alarm Condition (Yellow) LineDataCTL.Alarm

Local Copy of Line Array Tag for LNi Alert Condition (Lt Blue) LineDataCTL.Alert

Cycle

24

Local copy of Callout Table Predefined for EQID Overload is ON PreDef._12

Local Copy of Equip Array Tag for LNi, EQi Cycle Equip.DataCTL.Cycle

Local Copy of Equip Array Tag for LNi, EQi Cycle Count Send Summary Data 1-shot Equip.DataCTL.CycleSendONS

Local Copy of Equip Array Tag for LNi, EQi Cycle Count Send Summary Data Equip.DataCTL.CycleSend

Local Copy of Line Array Tag for LNi Line is for Offline Equipment LineCfg.Offline

Local Copy of Group Array Tag for GRPi Cycle GrpDataCTL.Cycle

Local Copy of Group Array Tag for GRPi Cycle Count Send Summary Data GrpDataCTL.CycleSend

Add Equip Data to Line Data AddToLDT

Local Copy of Line Array Tag for LNi Cycle LineDataCTL.Cycle

Local Copy of Line Array Tag for LNi Cycle Count Send Summary Data LineDataCTL.CycleSend

Calculate Equipment Data and Operational Availability (OA)

Assign Downtime Status if Worktime and Line Stopped w/ Request

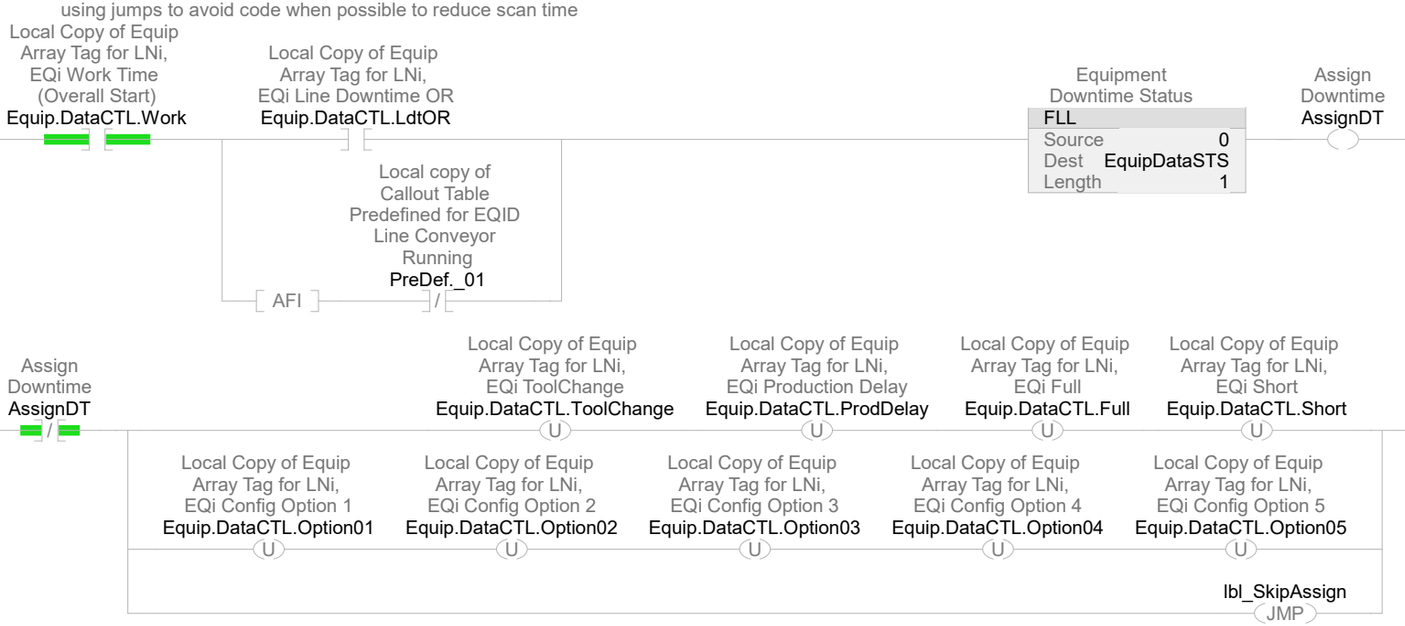
collect STS for Equip, Group and Line - more than 1 signal can be on in STS
the STS will be used to update CTL where only one signal can be on and existing DT items must be sealed-in while on

if Equip Running, no need to check for Downtime Assignment
clear the CTL bits and jump over the assignment code

Line must accumulate STS across multiple Equip
Group must accumulate STS across multiple Line/Equip

this code executes once for every configured Equip on every scan
using jumps to avoid code when possible to reduce scan time

25



Calculate Equipment Data and Operational Availability (OA)

Tool Change (ADC)
HIghest Priority - overrides all other DT

if Equip in Tool Change, no need to assign any other STS for this Equip, Group, Line

26

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA) PreDef._27

Local copy of Callout Table Predefined for EQID Tool Change (ADC) PreDef._28

Equipment Downtime Status ToolChange EquipDataSTS.ToolChange (L)

Local Copy of Line Array Tag for LNi Line is for Offline Equipment LineCfg.Offline

Local Copy of Group Array Tag for GRPi ToolChange GrpDataSTS.ToolChange (L)

Add Equip Data to Line Data AddToLDT

Local Copy of Line Array Tag for LNi ToolChange LineDataSTS.ToolChange (L)

lbl_AssignStsDone (JMP)

Production Delay

27

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA) PreDef._27

Local copy of Callout Table Predefined for EQID Delay (Prodution Hold, SubShort) PreDef._24

Equipment Downtime Status Production Delay EquipDataSTS.ProdDelay (L)

Local Copy of Line Array Tag for LNi Line is for Offline Equipment LineCfg.Offline

Local Copy of Group Array Tag for GRPi Production Delay GrpDataSTS.ProdDelay (L)

Add Equip Data to Line Data AddToLDT

Local Copy of Line Array Tag for LNi Production Delay LineDataSTS.ProdDelay (L)

Calculate Equipment Data and Operational Availability (OA)

Full

28

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA)
PreDef._27

Local copy of Callout Table Predefined for EQID Full
PreDef._26

Equipment Downtime Status Full
EquipDataSTS.Full
L

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Group Array Tag for GRPi Full
GrpDataSTS.Full
L

Add Equip Data to Line Data
AddToLDT

Local Copy of Line Array Tag for LNi Full
LineDataSTS.Full
L

Short

29

Local copy of Callout Table Predefined for EQID In Demand Off (don't charge OA)
PreDef._27

Local copy of Callout Table Predefined for EQID Short
PreDef._25

Equipment Downtime Status Short
EquipDataSTS.Short
L

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Group Array Tag for GRPi Short
GrpDataSTS.Short
L

Add Equip Data to Line Data
AddToLDT

Local Copy of Line Array Tag for LNi Short
LineDataSTS.Short
L

Skip Option Check if EQID Event Lookup Index is not good
Skip Option Check if not InDemand

30

Event Lookup Index EQID Match
EvLKi_EQIDmatch
/

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OA
Equip.DataCTL.LdtOA
/

lbl_SkipOptions
JMP

Calculate Equipment Data and Operational Availability (OA)

Option 1
Rollup to Line from all Equip (ignoare AddToLDT)

31

Event KPI Options Equipment Event KPI Options
EDC_EventOpt.Equip[EvLKi].Opt01.Status

Equipment Downtime Status Config Option 1
EquipDataSTS.Option01 (L)

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Group Array Tag for GRPi Config Option 1
GrpDataSTS.Option01 (L)

Local Copy of Line Array Tag for LNi Config Option 1
LineDataSTS.Option01 (L)

Option 2
Rollup to Line from all Equip (ignoare AddToLDT)

32

Event KPI Options Equipment Event KPI Options
EDC_EventOpt.Equip[EvLKi].Opt02.Status

Equipment Downtime Status Config Option 2
EquipDataSTS.Option02 (L)

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Group Array Tag for GRPi Config Option 2
GrpDataSTS.Option02 (L)

Local Copy of Line Array Tag for LNi Config Option 2
LineDataSTS.Option02 (L)

Calculate Equipment Data and Operational Availability (OA)

Option 3
Rollup to Line from all Equip (ignoare AddToLDT)

33

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt03.Status

Equipment
Downtime Status
Config Option 3
EquipDataSTS.Option03
L

Local Copy of Line
Array Tag for LNi
Line is for Offline
Equipment
LineCfg.Offline

Local Copy of Group
Array Tag for GRPi
Config Option 3
GrpDataSTS.Option03
L

Local Copy of Line
Array Tag for LNi
Config Option 3
LineDataSTS.Option03
L

Option 4
Rollup to Line from all Equip (ignoare AddToLDT)

34

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt04.Status

Equipment
Downtime Status
Config Option 4
EquipDataSTS.Option04
L

Local Copy of Line
Array Tag for LNi
Line is for Offline
Equipment
LineCfg.Offline

Local Copy of Group
Array Tag for GRPi
Config Option 4
GrpDataSTS.Option04
L

Local Copy of Line
Array Tag for LNi
Config Option 4
LineDataSTS.Option04
L

Calculate Equipment Data and Operational Availability (OA)

Option 5
Rollup to Line from all Equip (ignoare AddToLDT)

35

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt05.Status

Equipment
Downtime Status
Config Option 5
EquipDataSTS.Option05
L

Local Copy of Line
Array Tag for LNi
Line is for Offline
Equipment
LineCfg.Offline

Local Copy of Group
Array Tag for GRPi
Config Option 5
GrpDataSTS.Option05
L

Local Copy of Line
Array Tag for LNi
Config Option 5
LineDataSTS.Option05
L

Skip Option Check if EQID Event Lookup Index is not good

36

lbl_SkipOptions
LBL

NOP

Downtime Assign is Done
get Equip Priority Downtime (CTL) from Status (STS)

37

lbl_AssignStsDone
LBL

Prioritize STS
for CTL
aoiEDC_DataCtl
aoiEDC_DataCtl aoiDataCtl ...
DataSts EquipDataSTS
DataCtl Equip.DataCTL

skip Assign if Equip Running

38

lbl_SkipAssign
LBL

NOP

Calculate Equipment Data and Operational Availability (OA)

Equip Data Increment and OA
Add Incremental Time to Worktime, Downtime that are ON
Add Incremental Time to to Current Alarm or Fault Duration
Calculate OA
DN006 - support multiple Offline Lines

39

Increment Data

aoiEDC_DataInc	
aoiEDC_DataInc	aoiDataInc
Data	Equip.Data
DataCtl	Equip.DataCTL
DataCtlPrev	Equip.DataCTLprev
Increment	LineCtl.DataInc

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Equipment Data Collection Offline Group Data Status

COP	
Source	GrpDataSTS
Dest	EDC.Offline.DataSTS[EquipGRPi]
Length	1

Equipment Data Collection Offline Group Data Control

COP	
Source	GrpDataCTL
Dest	EDC.Offline.DataCTL[EquipGRPi]
Length	1

Exit Routine

40

lbl_Exit
LBL

NOP

(End)

Offline Group Data Initialize

Initialize, Reset or Update Offline Group Operational Availability (OA) Controls

One set of Interfaces is shared by all Group
GRPi = Offline Group Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 141
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Initialize Group Data Control Word

2

Local Copy of Offline Array Tag for LNi

FLL
Source 0
Dest OfflineDataCTL
Length 1

Offline Group Data Initialize

v3.5 Preserve the Downtime Type seal-in

3

Equipment Data Collection Offline Group Data Control	Local Copy of Offline Array Tag for LNi
EDC.Offline.DataCTL[GRPi].ToolChange	OfflineDataCTL.ToolChange (L)
EDC.Offline.DataCTL[GRPi].Option01	OfflineDataCTL.Option01 (L)
EDC.Offline.DataCTL[GRPi].Option02	OfflineDataCTL.Option02 (L)
EDC.Offline.DataCTL[GRPi].Option03	OfflineDataCTL.Option03 (L)
EDC.Offline.DataCTL[GRPi].Option04	OfflineDataCTL.Option04 (L)
EDC.Offline.DataCTL[GRPi].Option05	OfflineDataCTL.Option05 (L)
EDC.Offline.DataCTL[GRPi].ProdDelay	OfflineDataCTL.ProdDelay (L)
EDC.Offline.DataCTL[GRPi].Full	OfflineDataCTL.Full (L)
EDC.Offline.DataCTL[GRPi].Short	OfflineDataCTL.Short (L)

Offline Group Data Initialize

Nov 24, 2025 - Preserve the CycleSend signal for sending Summary Data updates when still cycling after WorkTime is off

4

Equipment Data Collection Offline Group Data Control
EDC.Offline.DataCTL[GRPi].CycleSend

Local Copy of Offline Array Tag for LNi Cycle Count Send Summary Data
OfflineDataCTL.CycleSend
L

Copy Offline working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

OfflineDataSTS is reloaded each scan by the Line/Equip Loop for comparison to OfflineDataCtlprev
OfflineDataCTL is a mixed bag -- some bits are cleared and some are retained for seal-in

5

Equipment Data Collection Offline Group Data Status

FLL	
Source	0
Dest	EDC.Offline.DataSTS[GRPi]
Length	1

Equipment Data Collection Offline Group Data Control

COP	
Source	OfflineDataCTL
Dest	EDC.Offline.DataCTL[GRPi]
Length	1

(End)

Offline Group Data Reset

Initialize, Reset or Update Offline Group Operational Availability (OA) Controls

One set of Interfaces is shared by all Group
GRPi = Offline Group Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	142
Dest	Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1103711128
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1103711128

Copy Offline Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

2

Local Copy of Offline Array Tag for LNi

COP	
Source	EDC.Offline.Data[GRPi]
Dest	OfflineData
Length	1

Shift Reset of Group Data

3

Data Reset for Line, Group, Equip

aoiEDC_DataReset	
aoiEDC_DataReset	aoiDataReset ...
Data	OfflineData
DataCtlPrev	OfflineDataCTLprev

Offline Group Data Reset

Copy Offline working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

Equipment Data Collection Offline Group Data (1-Trim, 2-Chassis, 3-Final)

COP	
Source	OfflineData
Dest	EDC.Offline.Data[GRPi]
Length	1

4

(End)

Offline Group Data Update

Initialize, Reset or Update Offline Group Operational Availability (OA) Controls

One set of Interfaces is shared by all Group
GRPi = Offline Group Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 143
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Copy Offline Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

2

Local Copy of Offline Array Tag for LNi

COP
Source EDC.Offline.DataSTS[GRPi]
Dest OfflineDataSTS
Length 1

Local Copy of Offline Array Tag for LNi

COP
Source EDC.Offline.DataCTL[GRPi]
Dest OfflineDataCTL
Length 1

Local Copy of Offline Array Tag for LNi

COP
Source EDC.Offline.DataCTLprev[GRPi]
Dest OfflineDataCTLprev
Length 1

Local Copy of Offline Array Tag for LNi

COP
Source EDC.Offline.Data[GRPi]
Dest OfflineData
Length 1

Offline Group Data Increment and OA
Add Incremental Time to Worktime, Downtime that are ON
Add Incremental Time to to Current Alarm or Fault Duration
Calculate OA

3

Prioritize STS for CTL

aoiEDC_DataCtl
aoiEDC_DataCtl aoiDataCtl
DataSts OfflineDataSTS
DataCtl OfflineDataCTL

Increment Data

aoiEDC_DataInc
aoiEDC_DataInc aoiDataInc
Data OfflineData
DataCtl OfflineDataCTL
DataCtlPrev OfflineDataCTLprev
Increment LineCtl.DataInc

Offline Group Data Update

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

OA% - InDemand Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

4

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Offline Array Tag for LNi Work Time (Overall Start)
OfflineDataCTL.Work

Local Copy of Offline Array Tag for LNi Work Time (Overall Start)
OfflineDataCTL.Work
/

Local Copy of Group Array Tag for GRPi Cycle Count Send Summary Data
GrpDataCTL.CycleSend

Schedule New Hour
ScheduleNewHour

Local Copy of Offline Array Tag for LNi Work Time Ended
OfflineDataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE
Source 3
Dest SumMsg.Header.DataType 13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE
Source GRPi 4
Dest SumMsg.Header.ID 3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP
Source OfflineData.Work
Dest SumMsg.Data.Work
Length 1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP
Source OfflineData.EdtOA.Maint
Dest SumMsg.Data.EdtMaint
Length 1

Equipment Data Summary Message 3 - Equip Downtime Production

COP
Source OfflineData.EdtOA.Prod
Dest SumMsg.Data.EdtProd
Length 1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP
Source OfflineData.LdtOA.Maint
Dest SumMsg.Data.LdtMaint
Length 1

Equipment Data Summary Message 5 - Line Downtime Production

COP
Source OfflineData.LdtOA.Prod
Dest SumMsg.Data.LdtProd
Length 1

Equipment Data Summary Message 6 -Full

COP
Source OfflineData.Full
Dest SumMsg.Data.Full
Length 1

Equipment Data Summary Message 7 - Short

COP
Source OfflineData.Short
Dest SumMsg.Data.Short
Length 1

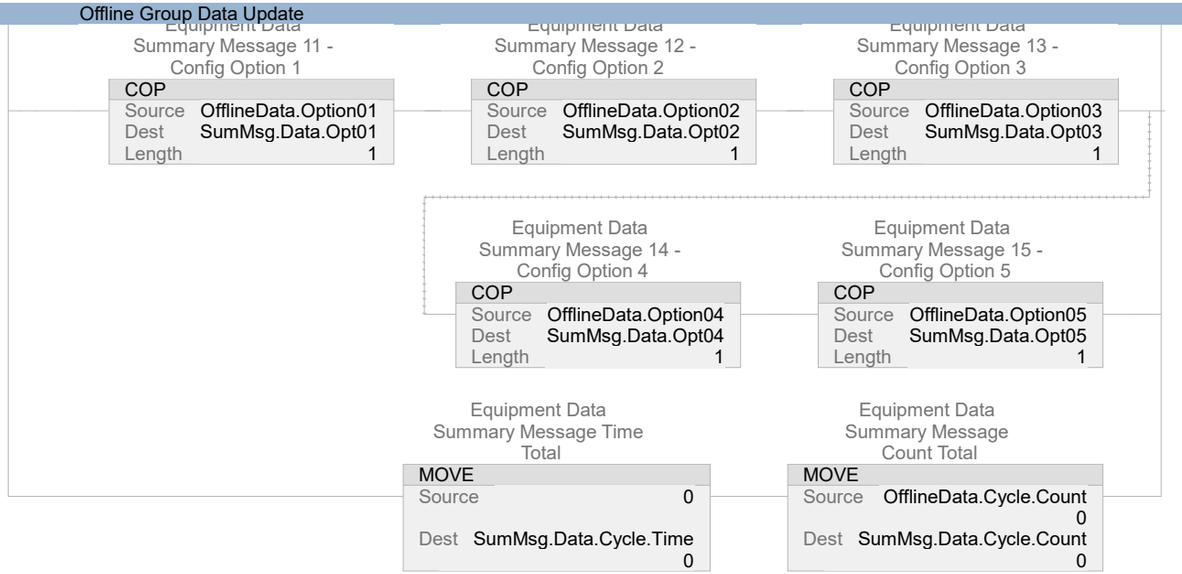
Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP
Source OfflineData.ProdDelay
Dest SumMsg.Data.ProdDelay
Length 1

Equipment Data Summary Message 10 - Tool Change (ADC)

COP
Source OfflineData.ToolChange
Dest SumMsg.Data.ToolChange
Length 1

Offline Group Data Update



Offline Group Data Update

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Group Number Odd/Even
(better distribution than splitting Line Numbers into Low/High range)

OR% - All Downtime

5

SumDataLoadRB

Load Summary Data Message

JSR
Routine Name _150_SumDataMsg

RingBuffer Modulo Result

MOD
Source A GRPi
4
Source B 2
Dest BufferID
1

RingBuffer Modulo Result

EQ
Source A BufferID
1
Source B 1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _151_SumLoadRB01

UseSumRB01

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR
Routine Name _152_SumLoadRB02

Offline Group Data Update

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

OR% - All Downtime
Send Time in INTEGER msec - SQL RB Store will convert to FLOAT seconds
Allow Cycle Count updates when not in Worktime
at New Hour, send a final packet for the Previous Hour ID to accurately close out the hour

6

Load Summary Ring Buffer for this Line
SumDataLoad

Local Copy of Offline Array Tag for LNi Work Time (Overall Start)
OfflineDataCTL.Work

Local Copy of Group Array Tag for GRPi Cycle Count Send Summary Data
GrpDataCTL.CycleSend

Local Copy of Group Array Tag for GRPi Cycle Count Send Summary Data
GrpDataCTL.CycleSend
U

Local Copy of Offline Array Tag for LNi Work Time (Overall Start)
OfflineDataCTL.Work
/

Schedule New Hour
ScheduleNewHour

Local Copy of Offline Array Tag for LNi Work Time Ended
OfflineDataCTL.WorkEnd

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

MOVE	
Source	13
Dest	SumMsg.Header.DataType
	13

Equipment Data Summary Message Header - Equipment ID, LineID or GroupID

MOVE	
Source	GRPi
	4
Dest	SumMsg.Header.ID
	3

SumDataLoadRB

Equipment Data Summary Message 1 - Work Data

COP	
Source	OfflineData.Work
Dest	SumMsg.Data.Work
Length	1

Equipment Data Summary Message 2 - Equip Downtime Maintenance

COP	
Source	OfflineData.EdtOR.Maint
Dest	SumMsg.Data.EdtMaint
Length	1

Equipment Data Summary Message 3 - Equip Downtime Production

COP	
Source	OfflineData.EdtOR.Prod
Dest	SumMsg.Data.EdtProd
Length	1

Equipment Data Summary Message 4 - Line Downtime Maintenance

COP	
Source	OfflineData.LdtOR.Maint
Dest	SumMsg.Data.LdtMaint
Length	1

Equipment Data Summary Message 5 - Line Downtime Production

COP	
Source	OfflineData.LdtOR.Prod
Dest	SumMsg.Data.LdtProd
Length	1

Equipment Data Summary Message 6 -Full

COP	
Source	OfflineData.Full

Equipment Data Summary Message 7 - Short

COP	
Source	OfflineData.Short

Equipment Data Summary Message 9 - Production Delay (Sub Short)

COP	
Source	OfflineData.ProdDelay

Equipment Data Summary Message 10 - Tool Change (ADC)

COP	
Source	OfflineData.ToolChange

Offline Group Data Update

Dest SumMsg.Data.Full
Length 1

Dest SumMsg.Data.Short
Length 1

Dest SumMsg.Data.ProdDelay
Length 1

Dest SumMsg.Data.ToolChange
Length 1

Equipment Data Summary Message 11 - Config Option 1

COP
Source OfflineData.Option01
Dest SumMsg.Data.Opt01
Length 1

Equipment Data Summary Message 12 - Config Option 2

COP
Source OfflineData.Option02
Dest SumMsg.Data.Opt02
Length 1

Equipment Data Summary Message 13 - Config Option 3

COP
Source OfflineData.Option03
Dest SumMsg.Data.Opt03
Length 1

Equipment Data Summary Message 14 - Config Option 4

COP
Source OfflineData.Option04
Dest SumMsg.Data.Opt04
Length 1

Equipment Data Summary Message 15 - Config Option 5

COP
Source OfflineData.Option05
Dest SumMsg.Data.Opt05
Length 1

Equipment Data Summary Message Time Total

MOVE
Source 0
Dest SumMsg.Data.Cycle.Time
0

Equipment Data Summary Message Count Total

MOVE
Source OfflineData.Cycle.Count
0
Dest SumMsg.Data.Cycle.Count
0

Offline Group Data Update

Summary Data to Server via Summary Ring Buffer
Periodically during Work and whenever Work Ends

Sep 2023 - change RB channel allocation to Group Number Odd/Even
(better distribution than splitting Line Numbers into Low/High range)

OA% - InDemand Downtime

7

SumDataLoadRB

Load Summary Data Message

JSR	
Routine Name	_150_SumDataMsg

RingBuffer Modulo Result

MOD	
Source A	GRPi 4
Source B	2
Dest	BufferID 1

RingBuffer Modulo Result

EQ	
Source A	BufferID 1
Source B	1

UseSumRB01

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR	
Routine Name	_151_SumLoadRB01

UseSumRB01 (/)

UseSumRB02

Load Equipment,Line, Group Summary Data into Ring Buffer

JSR	
Routine Name	_152_SumLoadRB02

Copy Offline working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

8

Equipment Data Collection Offline Group Data Control

COP	
Source	OfflineDataCTL
Dest	EDC.Offline.DataCTL[GRPi]
Length	1

Equipment Data Collection Offline Group Data Control previous values

COP	
Source	OfflineDataCTLprev
Dest	EDC.Offline.DataCTLprev[GRPi]
Length	1

Equipment Data Collection Offline Group Data (1-Trim, 2-Chassis, 3-Final)

COP	
Source	OfflineData
Dest	EDC.Offline.Data[GRPi]
Length	1

(End)

Load Summary Data Message

Load Equipment, Line, Group Summary Data into Message

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 150
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Increment Transaction Number

2

ADD
Source A SumTransactionNumber 7244
Source B 1
Dest SumTransactionNumber 7244

LT
Source A SumTransactionNumber 7244
Source B 1

GT
Source A SumTransactionNumber 7244
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

MOVE
Source 1
Dest SumTransactionNumber 7244

Load Summary Data Message

Load Message Header, Timestamp and Footer Data
SumMsg Header.ID, Header.DataType and Data are loaded by calling routine
Header.Spare is used for Schedule Hour 1-10
for Schedule Hour, load the SchedulePrev.Hour to properly handle hour changes; accum data is for the previous minute; data for the current minute hasn't been collected yet
(i.e. final data packet for hour 0 is sent at the start of hour 1, and the Schedule Hour must be 0 for this packet)

*** SPECIAL CASE ***
for end of shift closeout record - clear the fractional seconds to avoid assigning record to the next shift
(SQL is using Transaction Timestamp to determine ShiftDate, ShiftID - Gateway must send a shift closeout record to land exactly on the SQL ShiftWindowEnd datetime)

3

Equipment Data Summary Message Header - Transaction Number (must match Footer)

MOVE
Source SumTransactionNumber 7244
Dest SumMsg.Header.TransNum 7244

Equipment Data Summary Message Header - Command Number

MOVE
Source 0
Dest SumMsg.Header.CmdNum 0

Equipment Data Summary Message Header - Schedule Hour 0-10

MOVE
Source SchedulePrev.Hour 9
Dest SumMsg.Header.Hour 9

Equipment Data Summary Message Transaction Time Stamp

COP
Source CLX.TransTime <EDC.CLX.TransTime>
Dest SumMsg.TransTime
Length 1

Schedule Shift Reset

ScheduleShiftReset

Equipment Data Summary Message Fractional Seconds (.ss)

CLR
Dest SumMsg.TransTime.FracSec 1

Equipment Data Summary Message Footer - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest SumMsg.Footer.Spare 0

Equipment Data Summary Message Footer - Transaction Number (must match Footer)

MOVE
Source SumTransactionNumber 7244
Dest SumMsg.Footer.TransNum 7244

(End)

Load Equipment,Line, Group Summary Data into Ring Buffer

Load Equipment, Line, Group Summary Data Message into Ring Buffer

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 151
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Load Equipment,Line, Group Summary Data into Ring Buffer

2

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

EDC Ring Buffer Control Next Write Tag

LT
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B RingBufferSum01.MaxTagPtr <EDC.RB_Sum01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

LT
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B RingBufferSum01.MaxRecPtr <EDC.RB_Sum01.MaxRecPtr> 199

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RingBufferSum01.WritePtrErr <EDC.RB_Sum01.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Dest RingBufferSum01.WritePtrErrTag <EDC.RB_Sum01.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Dest RingBufferSum01.WritePtrErrRec <EDC.RB_Sum01.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2413
Input Par 0
Input Par LNi
Input Par EQi
Input Par -1

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B RingBufferSum01.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1

EDC Ring Buffer Control Next Write Record

EQ
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B RingBufferSum01.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87

EDC Ring Buffer Control Buffer Full - Block Writes
RingBufferSum01.FullBlockWrite <EDC.RB_Sum01.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RingBufferSum01.WriteBlockErr <EDC.RB_Sum01.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RingBufferSum01.WriteBlockErrCnt <EDC.RB_Sum01.WriteBlockErrCnt> 0
Source B 1
Dest RingBufferSum01.WriteBlockErrCnt <EDC.RB_Sum01.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum01.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2414
Input Par 0
Input Par LNi
Input Par EQi
Input Par -1

lbl_Exit
JMP

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 1

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum01_Rec01[RingBufferSum01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 2

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum01_Rec02[RingBufferSum01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 3

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum01_Rec03[RingBufferSum01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 4

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum01_Rec04[RingBufferSum01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 5

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum01_Rec05[RingBufferSum01.WriteRecPtr]
Length 1

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum01.WriteTagPtr
	<EDC.RB_Sum01.WriteTagPtr>
	1
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum01.WriteRecPtr
	<EDC.RB_Sum01.WriteRecPtr>
	88
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum01.ReadTagPtr
	<EDC.RB_Sum01.ReadTagPtr>
	1
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum01.ReadRecPtr
	<EDC.RB_Sum01.ReadRecPtr>
	87
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2411
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	-1

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B 1
Dest RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

EDC Ring Buffer Control Next Write Record

LT
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B RingBufferSum01.MaxRecPtr <EDC.RB_Sum01.MaxRecPtr> 199

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RingBufferSum01.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

EDC Ring Buffer Control Next Write Tag

ADD
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 1
Dest RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B RingBufferSum01.MaxTagPtr <EDC.RB_Sum01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RingBufferSum01.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

Load Equipment,Line, Group Summary Data into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD

Source A	RingBufferSum01.ShiftCnt <EDC.RB_Sum01.ShiftCnt> 113402
Source B	1
Dest	RingBufferSum01.ShiftCnt <EDC.RB_Sum01.ShiftCnt> 113402

8

lbl_Exit

LBL

NOP

(End)

Load Equipment,Line, Group Summary Data into Ring Buffer

Load Equipment, Line, Group Summary Data Message into Ring Buffer

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 152
Dest Prog.Routine 143

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1103711128
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1103711128

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B RingBufferSum02.MaxTagPtr <EDC.RB_Sum02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

LT
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B RingBufferSum02.MaxRecPtr <EDC.RB_Sum02.MaxRecPtr> 199

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RingBufferSum02.WritePtrErr <EDC.RB_Sum02.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Dest RingBufferSum02.WritePtrErrTag <EDC.RB_Sum02.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Dest RingBufferSum02.WritePtrErrRec <EDC.RB_Sum02.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2413
Input Par 0
Input Par LNi
Input Par EQi
Input Par -1

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B RingBufferSum02.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4

EDC Ring Buffer Control Next Write Record

EQ
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B RingBufferSum02.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45

EDC Ring Buffer Control Buffer Full - Block Writes
RingBufferSum02.FullBlockWrite <EDC.RB_Sum02.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RingBufferSum02.WriteBlockErr <EDC.RB_Sum02.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RingBufferSum02.WriteBlockErrCnt <EDC.RB_Sum02.WriteBlockErrCnt> 0
Source B 1
Dest RingBufferSum02.WriteBlockErrCnt <EDC.RB_Sum02.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RingBufferSum02.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2414
Input Par 0
Input Par LNi
Input Par EQi
Input Par -1

lbl_Exit
JMP

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr
<EDC.RB_Sum02.WriteTagPtr>
4
Source B 1

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum02_Rec01[RingBufferSum02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr
<EDC.RB_Sum02.WriteTagPtr>
4
Source B 2

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum02_Rec02[RingBufferSum02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr
<EDC.RB_Sum02.WriteTagPtr>
4
Source B 3

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum02_Rec03[RingBufferSum02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr
<EDC.RB_Sum02.WriteTagPtr>
4
Source B 4

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum02_Rec04[RingBufferSum02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RingBufferSum02.WriteTagPtr
<EDC.RB_Sum02.WriteTagPtr>
4
Source B 5

Equipment Data Summary Message

COP
Source SumMsg
Dest EDC_RB_Sum02_Rec05[RingBufferSum02.WriteRecPtr]
Length 1

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum02.WriteTagPtr
	<EDC.RB_Sum02.WriteTagPtr>
	4
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum02.WriteRecPtr
	<EDC.RB_Sum02.WriteRecPtr>
	46
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum02.ReadTagPtr
	<EDC.RB_Sum02.ReadTagPtr>
	4
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RingBufferSum02.ReadRecPtr
	<EDC.RB_Sum02.ReadRecPtr>
	45
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2411
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	-1

Load Equipment,Line, Group Summary Data into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B 1
Dest RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

EDC Ring Buffer Control Next Write Record

LT
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B RingBufferSum02.MaxRecPtr <EDC.RB_Sum02.MaxRecPtr> 199

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RingBufferSum02.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

EDC Ring Buffer Control Next Write Tag

ADD
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 1
Dest RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B RingBufferSum02.MaxTagPtr <EDC.RB_Sum02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RingBufferSum02.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

Load Equipment,Line, Group Summary Data into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD

Source A	RingBufferSum02.ShiftCnt <EDC.RB_Sum02.ShiftCnt> 113130
Source B	1
Dest	RingBufferSum02.ShiftCnt <EDC.RB_Sum02.ShiftCnt> 113130

8

lbl_Exit

LBL

NOP

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

Transaction Log
Control SBR Input
Parameter --
Transaction Code

0

SBR	
Input Par	Trans.Code
	<EDC.Trans.Code>
Input Par	Trans.Time
	<EDC.Trans.Time>
Input Par	Trans.LNi
	<EDC.Trans.LNi>
Input Par	Trans.EQi
	<EDC.Trans.EQi>
Input Par	Trans.Num
	<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan
Time & Fault Routine
Routine Number
currently Scanned

MOVE	
Source	901
Dest	Prog.Routine
	143

Local Program Scan
Time & Fault Routine
Routine Scan
Counters (Indexed by
Routine Number)

ADD	
Source A	Prog.RoutineScan[901]
	-1502204510
Source B	1
Dest	Prog.RoutineScan[901]
	-1502204510

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

ADD
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: 1
Dest: Trans.LOGi <EDC.Trans.LOGi> 211

LT
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: 0

GE
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

CLR
Dest: Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

ADD
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: 1
Dest: Trans.LOGx <EDC.Trans.LOGx> 212

LT
Source A: Trans.LOGx <EDC.Trans.LOGx> 212
Source B: 0

GE
Source A: Trans.LOGx <EDC.Trans.LOGx> 212
Source B: CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

CLR
Dest: Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

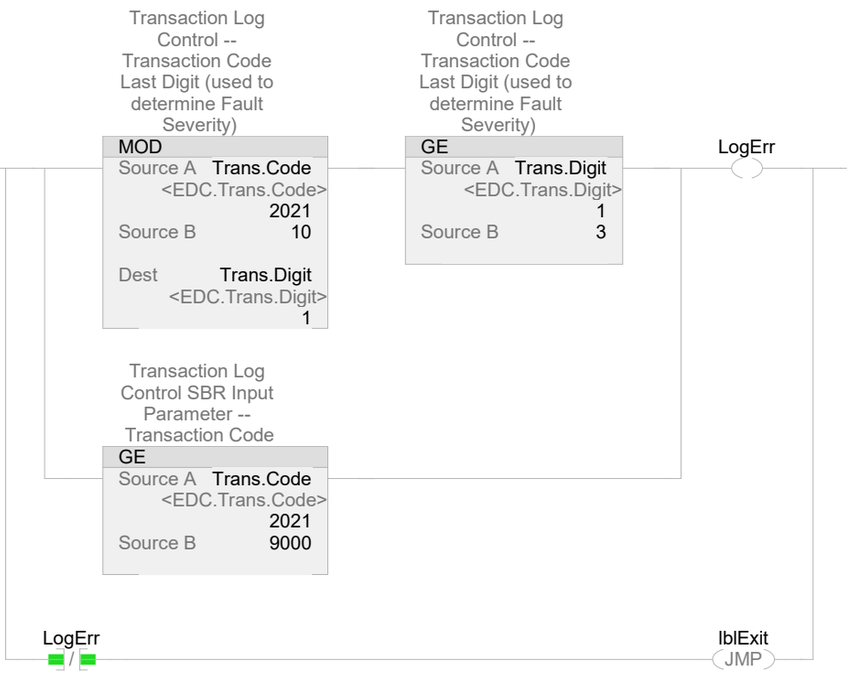
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

EDC2 - v2.00 added PdM with QualityRx, TrendRx and Server Interface
EDC2 - v2.03 added Cycle Data Type
EDC3 - v3.00 added Quality RB03 and RB04, Scanner Tool updates
v3.50 - L980 Controller; add BulkFaultMsg Processing

0

S:FS

Program Version

aoiEDC_ProgVersion

aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE

Source	201
Dest	Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD

Source A	Prog.RoutineScan[Prog.Routine] -1348352226
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1348352226

Configuration

2

S:FS

Toggle ON to update Configuration ManualConfig

Toggle ON to update Configuration ManualConfig

U

Configuration

JSR

Routine Name	_211_Config

Initialize Line Loop Count to Service n Lines / Scan

3

CLR

Dest	LineLoopCount 9

Main Routine

Increment Line Loop Index

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop may cause the unscanned Equipment to lose data if Equipment Buffers overflow)

4

lbl_LineLoop
LBL

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>

Line array Index
ADD
Source A LNi 27
Source B 1
Dest LNi 27

Line array Index
LT
Source A LNi 27
Source B CFG.LN.si <EDC.CFG.LN.si> 0

Line array Index
MOVE
Source CFG.LN.si <EDC.CFG.LN.si> 0
Dest LNi 27

EDC Config Disable Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>

Line array Index
MOVE
Source _ViewLNi 0
Dest LNi 27

Line array Index
GT
Source A LNi 27
Source B CFG.LN.ei <EDC.CFG.LN.ei> 28

Line Loop

One set of Interface Routines is shared by all Equipment
LNi, EQi = Line, Equipment Index

5

Line Loop for LNi
JSR
Routine Name _220_LineLoop

Increment Line Loop Count
Loop until all Max Lines serviced

service n Lines per Scan
set n to optimize EqDriver RxBuffer processing speed vs Gateway Scan & Class 3 Utilization

6

ADD
Source A LineLoopCount 9
Source B 1
Dest LineLoopCount 9

LT
Source A LineLoopCount 9
Source B CfgLineLoopMax 9

lbl_LineLoop
JMP

Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

7

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Routine
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Fault.Type
0
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Fault.Code
0
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.FaultCount
0
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.LOGi
0
Dest Trans.Opt[5]
<EDC.Trans.Opt[5]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 9003
Input Par 0
Input Par LNi
Input Par EQi
Input Par 0

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Main Routine

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

8

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

9

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
62
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan Events
CLxProgScanHigh.2
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

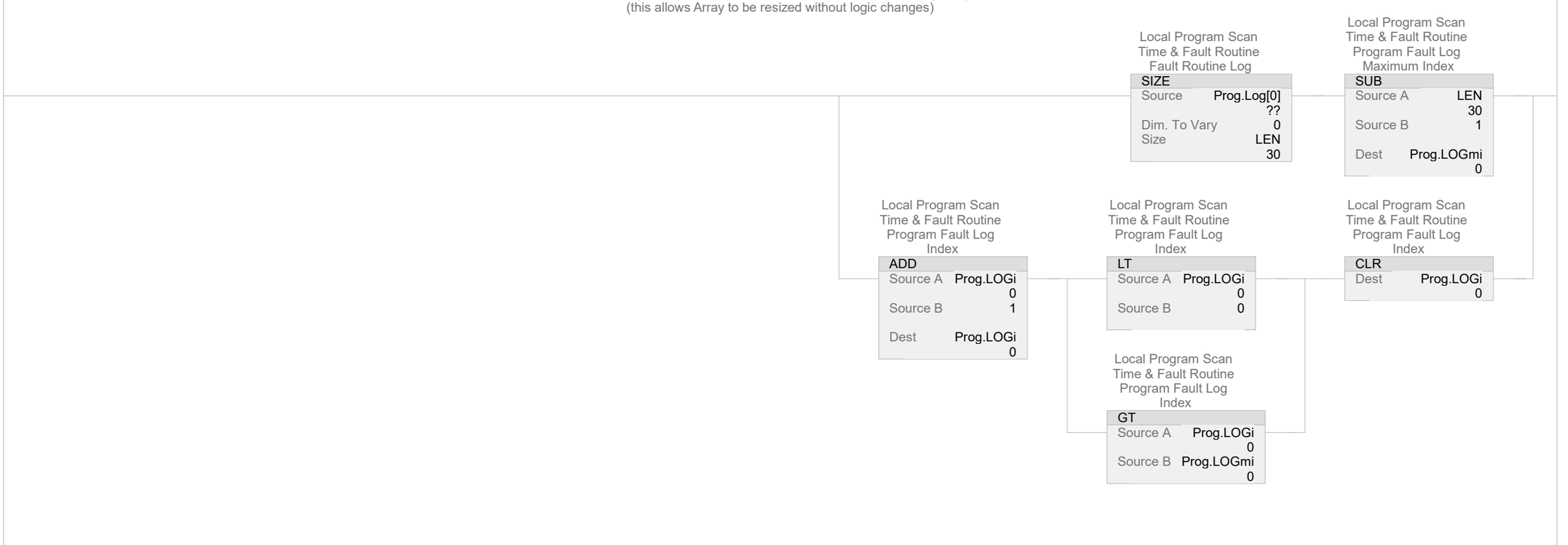
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

v1.19 Added CNVi, CARi, DATi

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
225
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source LNi
27
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source EQi
26
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source CNVi
25
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source CARi
2
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source DATi
1
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 202
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>
/

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Gateway Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 211
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

EquipRxBufferData Program - Initialize Line Loop Max to Service n Lines / Scan

Servicing more Lines/Scan allows EqDrivers to send data faster
The trade-off is an increase in Gateway scan time and Class 3 Message utiliation
(if you increase the Lines/Scan -- verify Gateway has the capacity!)

v3.5 use the Line array config to service ~750-1000 of 3000 Equipment/scan (target 3-4 scans for all Lines)

2

RxBufferData Quantity of Lines to service in one scan

DIV
Source A CFG.LN.n <EDC.CFG.LN.n> 29
Source B 3
Dest CfgLineLoopMax 9

Configuration

3

Bulk Fault Buffer Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

Equipment Data
Buffer Fault/Status
Receive Messages
(Header DINT, 10
Msgs 14 DINT/each,
Footer DINT

SIZE
Source EDC_EqRxBufferFault.Line[0].Equip[0].Fault[0]
0
Dim. To Vary 0
Size CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>
143

EDC Config Array Max
Index

SUB
Source A CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>
143
Source B 1
Dest CFG.BulkFaultBufferMSG.mi
<EDC.CFG.BulkFaultBufferMSG.mi>
142

EDC Config Array End
Count (udt size)

MOVE
Source 143
Dest CFG.BulkFaultBufferMSG.e
<EDC.CFG.BulkFaultBufferMSG.e>
143

EDC Config Array End
Count (udt size)

GT
Source A CFG.BulkFaultBufferMSG.e
<EDC.CFG.BulkFaultBufferMSG.e>
143
Source B CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>
143

EDC Config Array End
Count (udt size)

MOVE
Source CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>
143
Dest CFG.BulkFaultBufferMSG.e
<EDC.CFG.BulkFaultBufferMSG.e>
143

EDC Config Array End
Index (used for
copies)

SUB
Source A CFG.BulkFaultBufferMSG.e
<EDC.CFG.BulkFaultBufferMSG.e>
143
Source B 1
Dest CFG.BulkFaultBufferMSG.ei
<EDC.CFG.BulkFaultBufferMSG.ei>
142

Configuration

Bulk Fault Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

4

Equipment Data
Bulk Fault/Status
Message Equipment
Data Fault/Status
Message

SIZE	
Source	BulkFaultMsg.Msg[0] ??
Dim. To Vary	0
Size	CFG.BulkFaultMSG.n <EDC.CFG.BulkFaultMSG.n> 10

EDC Config Array Max
Index

SUB	
Source A	CFG.BulkFaultMSG.n <EDC.CFG.BulkFaultMSG.n> 10
Source B	1
Dest	CFG.BulkFaultMSG.mi <EDC.CFG.BulkFaultMSG.mi> 9

Configuration

Fault Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

5

Equipment Data Buffer Fault/Status Receive Message

SIZE
Source EDC_EqRxBuffer.Line[0].Equip[0].Fault[0]
1
Dim. To Vary 0
Size CFG.FaultMSG.n <EDC.CFG.FaultMSG.n> 14

EDC Config Array Max Index

SUB
Source A CFG.FaultMSG.n <EDC.CFG.FaultMSG.n> 14
Source B 1
Dest CFG.FaultMSG.mi <EDC.CFG.FaultMSG.mi> 13

EDC Config Array End Count (udt size)

MOVE
Source 14
Dest CFG.FaultMSG.e <EDC.CFG.FaultMSG.e> 14

EDC Config Array End Count (udt size)

GT
Source A CFG.FaultMSG.e <EDC.CFG.FaultMSG.e> 14
Source B CFG.FaultMSG.n <EDC.CFG.FaultMSG.n> 14

EDC Config Array End Count (udt size)

MOVE
Source CFG.FaultMSG.n <EDC.CFG.FaultMSG.n> 14
Dest CFG.FaultMSG.e <EDC.CFG.FaultMSG.e> 14

EDC Config Array End Index (used for copies)

SUB
Source A CFG.FaultMSG.e <EDC.CFG.FaultMSG.e> 14
Source B 1
Dest CFG.FaultMSG.ei <EDC.CFG.FaultMSG.ei> 13

Configuration

Quality Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

6

Equipment Data Buffer Quality/Health Receive Message

SIZE
Source EDC_EqRxBuffer.Line[0].Equip[0].Quality[0]
0
Dim. To Vary 0
Size CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>
39

EDC Config Array Max Index

SUB
Source A CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>
39
Source B 1
Dest CFG.QualityMSG.mi
<EDC.CFG.QualityMSG.mi>
38

EDC Config Array End Count (udt size)

MOVE
Source 39
Dest CFG.QualityMSG.e
<EDC.CFG.QualityMSG.e>
39

EDC Config Array End Count (udt size)

GT
Source A CFG.QualityMSG.e
<EDC.CFG.QualityMSG.e>
39
Source B CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>
39

EDC Config Array End Count (udt size)

MOVE
Source CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>
39
Dest CFG.QualityMSG.e
<EDC.CFG.QualityMSG.e>
39

EDC Config Array End Index (used for copies)

SUB
Source A CFG.QualityMSG.e
<EDC.CFG.QualityMSG.e>
39
Source B 1
Dest CFG.QualityMSG.ei
<EDC.CFG.QualityMSG.ei>
38

Configuration

Trend Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

7

Equipment Data Buffer Trend Receive Message

SIZE
Source EDC_EqRxBuffer.Line[0].Equip[0].Trend[0] 0
Dim. To Vary 0
Size CFG.TrendMSG.n <EDC.CFG.TrendMSG.n> 70

EDC Config Array Max Index

SUB
Source A CFG.TrendMSG.n <EDC.CFG.TrendMSG.n> 70
Source B 1
Dest CFG.TrendMSG.mi <EDC.CFG.TrendMSG.mi> 69

EDC Config Array End Count (udt size)

MOVE
Source 70
Dest CFG.TrendMSG.e <EDC.CFG.TrendMSG.e> 70

EDC Config Array End Count (udt size)

GT
Source A CFG.TrendMSG.e <EDC.CFG.TrendMSG.e> 70
Source B CFG.TrendMSG.n <EDC.CFG.TrendMSG.n> 70

EDC Config Array End Count (udt size)

MOVE
Source CFG.TrendMSG.n <EDC.CFG.TrendMSG.n> 70
Dest CFG.TrendMSG.e <EDC.CFG.TrendMSG.e> 70

EDC Config Array End Index (used for copies)

SUB
Source A CFG.TrendMSG.e <EDC.CFG.TrendMSG.e> 70
Source B 1
Dest CFG.TrendMSG.ei <EDC.CFG.TrendMSG.ei> 69

Configuration

8

Version Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

Equipment Data Buffer Version Receive Message

SIZE
Source EDC_EqRxBuffer.Line[0].Equip[0].Version[0]
8
Dim. To Vary 0
Size CFG.VersionMSG.n <EDC.CFG.VersionMSG.n>
30

EDC Config Array Max Index

SUB
Source A CFG.VersionMSG.n <EDC.CFG.VersionMSG.n>
30
Source B 1
Dest CFG.VersionMSG.mi <EDC.CFG.VersionMSG.mi>
29

EDC Config Array End Count (udt size)

MOVE
Source 30
Dest CFG.VersionMSG.e <EDC.CFG.VersionMSG.e>
30

EDC Config Array End Count (udt size)

GT
Source A CFG.VersionMSG.e <EDC.CFG.VersionMSG.e>
30
Source B CFG.VersionMSG.n <EDC.CFG.VersionMSG.n>
30

EDC Config Array End Count (udt size)

MOVE
Source CFG.VersionMSG.n <EDC.CFG.VersionMSG.n>
30
Dest CFG.VersionMSG.e <EDC.CFG.VersionMSG.e>
30

EDC Config Array End Index (used for copies)

SUB
Source A CFG.VersionMSG.e <EDC.CFG.VersionMSG.e>
30
Source B 1
Dest CFG.VersionMSG.ei <EDC.CFG.VersionMSG.ei>
29

Configuration

Cycle Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

9

Equipment Data Buffer Cycle Receive Message

SIZE
Source EDC_EqRxBufferAux.Line[0].Equip[0].Cycle[0] 0
Dim. To Vary 0
Size CFG.CycleMSG.n <EDC.CFG.CycleMSG.n> 134

EDC Config Array Max Index

SUB
Source A CFG.CycleMSG.n <EDC.CFG.CycleMSG.n> 134
Source B 1
Dest CFG.CycleMSG.mi <EDC.CFG.CycleMSG.mi> 133

EDC Config Array End Count (udt size)

MOVE
Source 134
Dest CFG.CycleMSG.e <EDC.CFG.CycleMSG.e> 134

EDC Config Array End Count (udt size)

GT
Source A CFG.CycleMSG.e <EDC.CFG.CycleMSG.e> 134
Source B CFG.CycleMSG.n <EDC.CFG.CycleMSG.n> 134

EDC Config Array End Count (udt size)

MOVE
Source CFG.CycleMSG.n <EDC.CFG.CycleMSG.n> 134
Dest CFG.CycleMSG.e <EDC.CFG.CycleMSG.e> 134

EDC Config Array End Index (used for copies)

SUB
Source A CFG.CycleMSG.e <EDC.CFG.CycleMSG.e> 134
Source B 1
Dest CFG.CycleMSG.ei <EDC.CFG.CycleMSG.ei> 133

(End)

Line Loop for LNi

Line Service Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop may cause the unscanned Equipment to lose data if Equipment Buffers overflow)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	220
Dest	Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1348352226
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1348352226

Copy Line Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

2

EDC Config Array Indexes for Line-Equip

COP	
Source	EDC.Line[LNi].CFG.EQ
Dest	LineCfgEQ
Length	1

Line Loop for LNi

Equipment Loop

One set of Interface Routines is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop may cause the unscanned Equipment to lose data if Equipment Buffers overflow)

3



Copy Line working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

only Line.CFG used - no changes to save

4



Exit Routine

5



Line Loop for LNi

Check Viewing Index Limits

6

Line View Tag Index
LT
Source A _ViewLNi 0
Source B 0

Line View Tag Index
GT
Source A _ViewLNi 0
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

Line View Tag Index
CLR
Dest _ViewLNi 0

Copy Line Instance to Viewing Tag if selected
(_ViewLine can be used to multiplex Trends and Diagnostic Spreadsheets)

7

_ViewEN

Line array Index
EQ
Source A LNi 27
Source B _ViewLNi 0

Line View Tag (used by View Diagnostic Spreadsheet)
COP
Source EDC.Line[LNi]
Dest _ViewLine
Length 1

(End)

Equipment Loop for EQi

Equipment Service Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop may cause the unscanned Equipment to lose data if Equipment Buffers overflow)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 225
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Exit if Equipment is not Configured
(reduce scan time by skipping logic for unused Equipment)
DHF006 - changed Equip.CFG.EN to Equip.CTL.Config (addition of Disabled Equipment support)
EDC2 - Nov 15, 2022 - disable/ignore EQID 0000

2

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.Config

Line array Index

EQ
Source A LNi 27
Source B 0

Equipment array Index

EQ
Source A EQi 26
Source B 0

lbl_Exit
JMP

Equipment Loop for EQi

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

3

Local Copy of Equip Array Tag for LNi, EQi

COP	
Source	EDC.Line[LNi].Equip[EQi]
Dest	Equip
Length	1

Local Copy of Equip Array Tag for LNi, EQi

COP	
Source	EDCAux.Line[LNi].Equip[EQi]
Dest	EquipAux
Length	1

Local Copy of Equip Array Status Tag for LNi, EQi

COP	
Source	EDC_Status01.Line[LNi].Equip[EQi]
Dest	EquipStatus01
Length	1

Equipment Data Collection - Status Line Equipment

COP	
Source	EDC_Status02.Line[LNi].Equip[EQi]
Dest	EquipStatus02
Length	1

Equipment Data Collection - Status Line Equipment

COP	
Source	EDC_Status03.Line[LNi].Equip[EQi]
Dest	EquipStatus03
Length	1

Equipment Data Collection - Status Line Equipment

COP	
Source	EDC_Status04.Line[LNi].Equip[EQi]
Dest	EquipStatus04
Length	1

Equipment Comm Status
-- Set Default State = Configured if no State value
(don't run Equipment Timer)

4

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value

COP	
Source	EquipStatus01.State
Dest	EquipStatus01.StateVal
Length	1

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State Value

LE	
Source A	EquipStatus01.StateVal
	16#0000_0022
Source B	1

Local Copy of Equip Array Status Tag for LNi, EQi Current Communication State

FLL	
Source	0
Dest	EquipStatus01.State
Length	1

Local Copy of Equip Array Status Tag for LNi, EQi State = White - Configured (never received any Data Requests)
EquipStatus01.State.CFG
L

Equipment Loop for EQi

Bulk Fault Event Message
DHF006 - if disabled, block FaultMsg
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

Fault messages for each Equipment should only arrive on EDC_EqRxBuffer (single) or EDC_EqRxBufferFault (bulk)
normally, one method will timeout while the other operates
if both or neither are timed-out, need to service both methods until the desired method is determined

5

Local Copy of Equip Array Tag for LNi, EQi Fault Message Enabled (from Driver Callout Msg)
Equip.CFG.FaultMsgEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Receive Equipment Bulk Fault/Status Unsolicited Message

JSR
Routine Name _227_EquipFaultRxBulk

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBufferFault.Line[LNi].Equip[EQi].Fault[0]
Length CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>

Equipment Loop for EQi

Fault Event Message
DHF006 - if disabled, block FaultMsg after sending PKT 0-7 Messages to reset all Fault bits (addition of Disabled Equipment support)
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

Fault messages for each Equipment should only arrive on EDC_EqRxBuffer (single) or EDC_EqRxBufferFault (bulk)
normally, one method will timeout while the other operates
if both or neither are timed-out, need to service both methods until the desired method is determined

6

Local Copy of Equip Array Tag for LNi, EQi Fault Message Enabled (from Driver Callout Msg)
Equip.CFG.FaultMsgEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Receive Equipment Fault/Status Unsolicited Message

JSR
Routine Name _230_EquipFaultRx

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Fault[0]
Length CFG.FaultMSG.n
<EDC.CFG.FaultMSG.n>

Equipment Loop for EQi

Quality/Health Message
DHF006 - if disabled, don't service any Messages received from Equipment
EDC2 WL18 - allow PdM Sample for Equipment without Quality Message enabled
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

7

Local Copy of Equip Array Tag for LNi, EQi Quality Message Enabled (from Driver Callout Msg)
Equip.CFG.QualityMsgEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Receive Equipment Quality/Health Unsolicited Message

JSR
Routine Name _240_EquipQualityRx

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.QualityMsg.PdmSampleSend

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Quality[0]
Length CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>

Equipment Loop for EQi

Trend Message
DHF006 - if disabled, don't service any Messages received from Equipment
EDC2 WL18 - allow PdM Sample for Equipment without Trend Message enabled
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

8

Local Copy of Equip Array Tag for LNi, EQi Trend Message Enabled (from Driver Callout Msg)
Equip.CFG.TrendMsgEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Receive Equipment Trend Unsolicited Message

JSR
Routine Name _250_EquipTrendRx

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.TrendMsg.PdmSampleSend

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Trend[0]
Length CFG.TrendMSG.n
<EDC.CFG.TrendMSG.n>

Version Message
DHF006 - if disabled, don't service any Messages received from Equipment
EDC2 WL31 - when Disabled, discard Messages from RxBuffer

9

Local Copy of Equip Array Tag for LNi, EQi Version Message Enabled (from Driver Callout Msg)
Equip.CFG.VersionMsgEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Receive Equipment Version Unsolicited Message

JSR
Routine Name _260_EquipVersionRx

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Version[0]
Length CFG.VersionMSG.n
<EDC.CFG.VersionMSG.n>

Equipment Loop for EQi

Cycle Message
DHF006 - if disabled, don't service any Messages received from Equipment
EDC2 WL31 - when Disabled, discard Messages from RxBuffer
EDC2 WL08 - add Cycle Data Message

10

Local Copy of Equip Array Tag for LNi, EQi Cycle Message Enabled (from Driver Callout Msg)
Equip.CFG.CycleMsgEN

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Receive Equipment Cycle Unsolicited Message
JSR
Routine Name _270_EquipCycleRx

Local Copy of Equip Array Tag for LNi, EQi Equipment Disabled (from Server Config)
Equip.CFG.Disable

Equipment Data Buffer Lines
FLL
Source 0
Dest EDC_EqRxBufferAux.Line[LNi].Equip[EQi].Cycle[0]
Length CFG.CycleMSG.n
<EDC.CFG.CycleMSG.n>

Save & Exit Routine

11

lbl_SaveExit
LBL

NOP

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

Equipment Data Collection Lines and Equipment Array
COP
Source Equip
Dest EDC.Line[LNi].Equip[EQi]
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source EquipAux
Dest EDCAux.Line[LNi].Equip[EQi]
Length 1

Server Interface Status Lines and Equipment Array
COP
Source EquipStatus01
Dest EDC_Status01.Line[LNi].Equip[EQi]
Length 1

Event Rx/Tx Status Lines and Equipment Array
COP
Source EquipStatus02
Dest EDC_Status02.Line[LNi].Equip[EQi]
Length 1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array
COP
Source EquipStatus03
Dest EDC_Status03.Line[LNi].Equip[EQi]
Length 1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array
COP
Source EquipStatus04
Dest EDC_Status04.Line[LNi].Equip[EQi]
Length 1

Equipment Loop for EQi

Exit Routine

13

lbl_Exit
LBL

NOP

Check Viewing Index Limits

14

Equipment View Tag Index
LT
Source A _ViewEQi 6
Source B 0

Equipment View Tag Index
GT
Source A _ViewEQi 6
Source B CFG.EQ.mi <EDC.CFG.EQ.mi> 99

Equipment View Tag Index
CLR
Dest _ViewEQi 6

Copy Equipment Instance to Viewing Tag if selected
(_ViewEquip can be used to multiplex Trends and Diagnostic Spreadsheets)

15

_ViewEN

Line array Index
EQ
Source A LNi 27
Source B _ViewLNi 0

Equipment array Index
EQ
Source A EQi 26
Source B _ViewEQi 6

Equipment View Tag (used by View Diagnostic Spreadsheet)
COP
Source EDC.Line[LNi].Equip[EQi]
Dest _ViewEquip
Length 1

(End)

Receive Equipment Bulk Fault/Status Unsolicited Message

Receive Equipment Bulk Fault/Status Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 227
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Receive Equipment Bulk Fault/Status Unsolicited Message

2

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

Transaction Timeout if both Rx & RxBulk timeout
DHF006 - Block Reset for Send Disable Message

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest EquipAux.CTL.BulkFaultMsg.TransTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer EquipAux.CTL.BulkFaultMsg.TransTMR
Preset 0
Accum 0
EN
DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
0
Source B CFG.BufferMsgTimeout
<EDC.CFG.BufferMsgTimeout>
330000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout

EquipAux.CTL.BulkFaultMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER

EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2004
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par 0

Diagnostic

ADD
Source A BulkFaultTimeoutER
13539
Source B 1
Dest BulkFaultTimeoutER
13539

Equipment Data Bulk Fault/Status Message

FLL
Source 0
Dest BulkFaultMsg
Length 1

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Copy Rx Buffer to local registers
DINT Array to UDT

Bulk Fault Message contains 1-10 Fault Messages

3

Equipment Data
Bulk Fault/Status
Message

CPS
Source EDC_EqRxBufferFault.Line[LNi].Equip[EQi].Fault[0]
Dest BulkFaultMsg
Length 1

the message has a variable length depending on the number of Records
use the Record Count to update the Footer.TransNum
14 DINT/record + 2 DINT Header

4

Equipment Data
Bulk Fault/Status
Message Header -
Record Count

LIMIT
Low Limit 1
Test BulkFaultMsg.Header.RecordCount
0
High Limit 10

CPT
Dest FLTi
16
Expression (BulkFaultMsg.Header.RecordCount * CFG.FaultMSG.e) + 2

Equipment Data
Bulk Fault/Status
Message Footer

COP
Source EDC_EqRxBufferFault.Line[LNi].Equip[EQi].Fault[FLTi]
Dest BulkFaultMsg.Footer
Length 1

No new data in Rx Buffer
Header & Footer Transaction Number =0

5

Equipment Data
Bulk Fault/Status
Message Header -
Transaction Number
(must match Footer)

EQ
Source A BulkFaultMsg.Header.TransNum
0
Source B 0

Equipment Data
Bulk Fault/Status
Message Footer -
Transaction Number
(must match Footer)

EQ
Source A BulkFaultMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

New data in Rx Buffer
Initialize Transaction timing for Bulk & Legacy path

6

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
EquipAux.CTL.BulkFaultMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
EquipAux.CTL.BulkFaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.FaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest EquipAux.CTL.BulkFaultMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source BulkFaultMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.EN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2000
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par BulkFaultMsg.Header.TransNum

Receive Equipment Bulk Fault/Status Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

7

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A BulkFaultMsg.Header.TransNum
0
Source B 0

Equipment Data Bulk Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest EquipAux.CTL.BulkFaultMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer EquipAux.CTL.BulkFaultMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Header.TransNum
0
Source B 0

Equipment Data Bulk Fault/Status Message Footer - Transaction Number (must match Footer)

EQ
Source A BulkFaultMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
EquipAux.CTL.BulkFaultMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
EquipAux.CTL.BulkFaultMsg.PartialTMR.DN

lbl_Exit
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

9

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
EquipAux.CTL.BulkFaultMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
EquipAux.CTL.BulkFaultMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER
EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: BulkFaultMsg.Footer.TransNum
0
Dest: Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans
PerfMsg: EquipAux.CTL.BulkFaultMsg
PerfTrans: EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2005
Input Par: EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: BulkFaultMsg.Header.TransNum

Diagnostic

ADD
Source A: BulkFaultPartialMsgER
272
Source B: 1
Dest: BulkFaultPartialMsgER
272

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

10

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Header.TransNum
0
Source B 0

Equipment Data Bulk Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Footer.TransNum
0
Source B 0

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Header.TransNum
0
Source B BulkFaultMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

EquipAux.CTL.BulkFaultMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER

EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source BulkFaultMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2006
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par BulkFaultMsg.Header.TransNum

Diagnostic

ADD
Source A FaultTransNumER
0
Source B 1
Dest FaultTransNumER
0

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

11

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Header.TransNum 0
Source B 0

Equipment Data Bulk Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A BulkFaultMsg.Footer.TransNum 0
Source B 0

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A BulkFaultMsg.Header.TransNum 0
Source B BulkFaultMsg.Footer.TransNum 0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Fault Rx DN
EquipStatus01.Update.FaultRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source BulkFaultMsg.Footer.TransNum 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.DN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2001
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

skip over the Message Processing Loop if a Transaction was not received

12

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd
/

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

BulkFaultMsg.Header.EQID should match each FaultMsg.Header.EQID
(not going to check FaultMsg.Header.EQID for each Message - assume they are the same)

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Expected Equipment ID
CPT
Dest ExpectedEQID 1006
Expression LNi*100+EQi

Equipment Data Bulk Fault/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)
NE
Source A BulkFaultMsg.Header.EQID 0
Source B ExpectedEQID 1006

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Equipment ID Mismatch ER
EquipStatus01.Update.FaultRxEqIdER
L

Update Status State for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source BulkFaultMsg.Header.EQID 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2007
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par BulkFaultMsg.Header.TransNum

Diagnostic
ADD
Source A BulkFaultEqidER 0
Source B 1
Dest BulkFaultEqidER 0

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

BulkFaultMsg.Header.TransNum should be incremented by the previous Message Record Count
(not going to check FaultMsg.Header.TransNum for each Message - assume they are in order)

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A BulkFaultMsg.Header.TransNum 0
Source B EquipAux.CTL.BulkFaultMsg.TransNumLast 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Fault Rx DN
EquipStatus01.Update.FaultRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EquipAux.CTL.BulkFaultMsg.TransNumLast 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg EquipAux.CTL.BulkFaultMsg
PerfTrans EquipStatus02.FaultRx.KA

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2002
Input Par EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par BulkFaultMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

BulkFaultMsg.Header.TransNum should match BulkFaultMsg.Msg[0].Header.TransNum (same as first Record)
(not going to check FaultMsg.Header.TransNum for each Message - assume they are in order)

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	BulkFaultMsg.Header.TransNum 0
Source B	EquipAux.CTL.BulkFaultMsg.TransNumLast 0

Equipment Data Bulk Fault/Status Message Header - Transaction Number (must match Footer)

NE	
Source A	BulkFaultMsg.Header.TransNum 0
Source B	EquipAux.CTL.BulkFaultMsg.TransNumNext 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Transaction Number Sequence ER
EquipStatus01.Update.FaultRxSeqER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	EquipAux.CTL.BulkFaultMsg.TransNumNext 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	EquipAux.CTL.BulkFaultMsg
PerfTrans	EquipStatus02.FaultRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2009
Input Par	EquipAux.CTL.BulkFaultMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	BulkFaultMsg.Header.TransNum

Diagnostic

ADD	
Source A	BulkFaultSeqER 37
Source B	1
Dest	BulkFaultSeqER 37

Diagnostic

MOVE	
Source	LNi 27
Dest	BulkFaultSeqER_LNi 10

Diagnostic

MOVE	
Source	EQi 26
Dest	BulkFaultSeqER_EQi 4

Diagnostic

MOVE	
Source	BulkFaultMsg.Header.TransNum 0
Dest	BulkFaultSeqER_TransNum 1

Diagnostic

MOVE	
Source	EquipAux.CTL.BulkFaultMsg.TransNumNext 0
Dest	BulkFaultSeqER_TransNumNext 42

Receive Equipment Bulk Fault/Status Unsolicited Message

Initialize Record Loop Indexes
(if invalid, assume max count and process the blanks - hopefully the sequence errors will draw attention)

16

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Record Index
CLR
Dest RECi 2

Record Max Index
SUB
Source A BulkFaultMsg.Header.RecordCount 0
Source B 1
Dest RECmi 1

Record Max Index
LT
Source A RECmi 1
Source B 0

Record Max Index
GT
Source A RECmi 1
Source B CFG.BulkFaultMSG.mi <EDC.CFG.BulkFaultMSG.mi> 9

Record Max Index
MOVE
Source CFG.BulkFaultMSG.mi <EDC.CFG.BulkFaultMSG.mi> 9
Dest RECmi 1

Message Loop
Load Next FaultMsg

17

lbl_MsgLoop
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Equipment Data Fault/Status Message
COP
Source BulkFaultMsg.Msg[RECi]
Dest FaultMsg
Length 1

TBD -- should each FaultMsg be validated for EQID and TransNum?
it's a reasonable assumption the standard EqDrv properly packages multiple FaultMsg for the BulkMsg

EQID Check
Valid TransNum Check
Duplicate TransNum Check
TransNum Sequence Check

18

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

NOP

Receive Equipment Bulk Fault/Status Unsolicited Message

Load new message in Ring Buffer
Feb 2017 - 4 RB Transactions
v1.6 - 6 RB Transactions
Mar 5, 2018 - rebalance Ring Buffers to avoid backlogs in RB4 and RB6
Sep 2023 - change RB channel allocation to modulo of Line Number (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi) to distribute EQID from same Line (i.e. 1000, 1001, 1002) to different buffers
Dec 4, 2025 - change RB channel allocation back to modulo of LNi -- all EQID on same Line must go thru same RB to avoid incorrect Candidate/Event generation for Maximo/PCR

19

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

RingBuffer Modulo Result
MOD
Source A LNi 27
Source B 6
Dest BufferID 3

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 1

UseFaultRB01

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _231_EquipFaultRxLoadRB01

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 2

UseFaultRB02

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _232_EquipFaultRxLoadRB02

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 3

UseFaultRB03

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _233_EquipFaultRxLoadRB03

RingBuffer Modulo Result
EQ
Source A BufferID 3
Source B 4

UseFaultRB04

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _234_EquipFaultRxLoadRB04

RingBuffer Modulo Result
EQ

UseFaultRB05

Load Equipment Fault/Status Message into Ring Buffer
JSR

Receive Equipment Bulk Fault/Status Unsolicited Message

EQ
Source A BufferID
3
Source B 5

UseFaultRB05

JSR
Routine Name _235_EquipFaultRxLoadRB05

Load Equipment
Fault/Status Message
into Ring Buffer

UseFaultRB01 UseFaultRB02 UseFaultRB03 UseFaultRB04 UseFaultRB05 UseFaultRB06

JSR
Routine Name _236_EquipFaultRxLoadRB06

Increment Record Index and jump to beginning of Record Loop

20

Local Copy of Equip
Array Tag for LNi,
EQi Transaction
Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Record
Index
ADD
Source A RECi
2
Source B 1
Dest RECi
2

Record
Index
LE
Source A RECi
2
Source B RECmi
1

lbl_MsgLoop
JMP

Receive Equipment Bulk Fault/Status Unsolicited Message

Prepare for next message
TransNumNext is incremented by the Record Count

21

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number

MOVE
Source BulkFaultMsg.Header.TransNum 0
Dest EquipAux.CTL.BulkFaultMsg.TransNumLast 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

ADD
Source A BulkFaultMsg.Header.TransNum 0
Source B BulkFaultMsg.Header.RecordCount 0
Dest EquipAux.CTL.BulkFaultMsg.TransNumNext 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

LT
Source A EquipAux.CTL.BulkFaultMsg.TransNumNext 0
Source B 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

MOVE
Source 1
Dest EquipAux.CTL.BulkFaultMsg.TransNumNext 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

GT
Source A EquipAux.CTL.BulkFaultMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

SUB
Source A EquipAux.CTL.BulkFaultMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999
Dest EquipAux.CTL.BulkFaultMsg.TransNumNext 0

Receive Equipment Bulk Fault/Status Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)
DHF006 - add DisableFaultMsg

BulkFaultMsg - store last FaultMsg in EDC_EqRxData
(only one of Fault and BulkFault RxBuffer will be receiving data, so no problem sharing result area)

don't reset BulkFaultMsg Timeout if FaultMsg is in use
(timeout is used to suspend scan of the unused Fault method)

22

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
EquipAux.CTL.BulkFaultMsg.TransReset

Diagnostic
ADD
Source A BulkTransReset 272
Source B 1
Dest BulkTransReset 272

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd

Diagnostic
ADD
Source A BulkTransRcvd 27916223
Source B 1
Dest BulkTransRcvd 27916223

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
EquipAux.CTL.BulkFaultMsg.Timeout

Diagnostic
ADD
Source A BulkTransTimeout 20308
Source B 1
Dest BulkTransTimeout 20308

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
EquipAux.CTL.BulkFaultMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
EquipAux.CTL.BulkFaultMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
EquipAux.CTL.BulkFaultMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
EquipAux.CTL.BulkFaultMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
EquipAux.CTL.BulkFaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)
CLR
Dest EquipAux.CTL.BulkFaultMsg.TransLast 0

Equipment Data Lines
COP
Source FaultMsg
Dest EDC_EqRxData.Line[LNi].Equip[EQi].Fault
Length 1

Equipment Data Buffer Lines
FLL
Source 0
Dest EDC_EqRxBufferFault.Line[LNi].Equip[EQi].Fault[0]
Length CFG.BulkFaultBufferMSG.n
<EDC.CFG.BulkFaultBufferMSG.n>

Exit Routine

23

lbl_Exit
LBL

NOP

(End)

Receive Equipment Fault/Status Unsolicited Message

Receive Equipment Fault/Status Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 230
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Receive Equipment Fault/Status Unsolicited Message

Send Fault Event message for PKTi 0-7 with all Events off to reset any open Events.

2

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg

Equipment Data Fault/Status Message
FLL
Source 0
Dest FaultMsg
Length 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number
EQ
Source A Equip.CTL.FaultMsg.TransNumNext 17
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number
MOVE
Source 1
Dest Equip.CTL.FaultMsg.TransNumNext 17

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)
MOVE
Source Equip.CTL.FaultMsg.TransNumNext 17
Dest FaultMsg.Header.TransNum 0

Equipment Data Fault/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)
CPT
Dest FaultMsg.Header.EQID 0
Expression (LNi * 100) + EQi

Equipment Data Fault/Status Message Header - Data Count - Total Qty of Words used in all Packets
MOVE
Source 8
Dest FaultMsg.Header.DataCount 0

Equipment Data Fault/Status Message Header - Packet ID - generates offset for 8-DINT mulitplexed data packet
MOVE
Source Equip.CTL.DisablePKTi 0
Dest FaultMsg.Header.PacketID 0

Equipment Data Fault/Status Message Transaction Time Stamp
COP
Source CLX.TransTime <EDC.CLX.TransTime>
Dest FaultMsg.TransTime
Length 1

Equipment Data Fault/Status Message Data - Predefined Status
FLL
Source 0
Dest FaultMsg.PreDef
Length 1

Equipment Data Fault/Status Message Data - Equipment Defined Status
FLL
Source 0
Dest FaultMsg.EqDef
Length 1

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)
MOVE
Source FaultMsg.Header.TransNum 0
Dest FaultMsg.Footer.TransNum 0

Receive Equipment Fault/Status Unsolicited Message

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

Transaction Timeout if both Rx & RxBulk timeout
DHF006 - Block Reset for Send Disable Message

3

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg
[/]

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.FaultMsg.TransTMR.PRE 999999999

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.FaultMsg.TransTMR
Preset 999999999
Accum 44481
EN
DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.FaultMsg.TransTMR.ACC 44481
Source B CFG.BufferMsgTimeout <EDC.CFG.BufferMsgTimeout> 330000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.FaultMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER
EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate
Status EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2004
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par 0

Diagnostic

ADD
Source A FaultTimeoutER 20453
Source B 1
Dest FaultTimeoutER 20453

Equipment Data Fault/Status Message

FLL
Source 0
Dest FaultMsg
Length 1

lbl_MsgReset
JMP

Receive Equipment Fault/Status Unsolicited Message

Copy Rx Buffer to local registers
DINT Array to UDT
DHF006 - Block Buffer Data copy for Send Disable Message

4

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg
/

Equipment Data Fault/Status Message

CPS	
Source	EDC_EqRxBuffer.Line[LNi].Equip[EQi].Fault[0]
Dest	FaultMsg
Length	1

No new data in Rx Buffer
Header & Footer Transaction Number =0

5

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

EQ	
Source A	FaultMsg.Header.TransNum
	0
Source B	0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

EQ	
Source A	FaultMsg.Footer.TransNum
	0
Source B	0

lbl_Exit
JMP

Receive Equipment Fault/Status Unsolicited Message

New data in Rx Buffer
Initialize Transaction timing for Legacy & Bulk path

6

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.FaultMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.FaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
EquipAux.CTL.BulkFaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.FaultMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.EN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2000
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Receive Equipment Fault/Status Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

7

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A FaultMsg.Header.TransNum
0
Source B 0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A FaultMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.FaultMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.FaultMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A FaultMsg.Header.TransNum
0
Source B 0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

EQ
Source A FaultMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.FaultMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.FaultMsg.PartialTMR.DN
/

lbl_Exit
JMP

Receive Equipment Fault/Status Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

9

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.FaultMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.FaultMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER
EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	FaultMsg.Footer.TransNum 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.FaultMsg
PerfTrans	EquipStatus02.FaultRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2005
Input Par	Equip.CTL.FaultMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	FaultMsg.Header.TransNum

Diagnostic

ADD	
Source A	FaultPartialMsgER 0
Source B	1
Dest	FaultPartialMsgER 0

lbl_MsgReset
JMP

Receive Equipment Fault/Status Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

10

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A FaultMsg.Header.TransNum
0
Source B 0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A FaultMsg.Footer.TransNum
0
Source B 0

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A FaultMsg.Header.TransNum
0
Source B FaultMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.FaultMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Msg ER

EquipStatus01.Update.FaultRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2006
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Diagnostic

ADD
Source A FaultTransNumER
0
Source B 1
Dest FaultTransNumER
0

lbl_MsgReset
JMP

Receive Equipment Fault/Status Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

11

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A FaultMsg.Header.TransNum
0
Source B 0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

NE
Source A FaultMsg.Footer.TransNum
0
Source B 0

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A FaultMsg.Header.TransNum
0
Source B FaultMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Fault Rx DN
EquipStatus01.Update.FaultRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.DN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2001
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Receive Equipment Fault/Status Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

12

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd

Expected Equipment ID

CPT	
Dest	ExpectedEQID 1006
Expression	LNi*100+EQi

Equipment Data Fault/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE	
Source A	FaultMsg.Header.EQID 0
Source B	ExpectedEQID 1006

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Equipment ID Mismatch ER
EquipStatus01.Update.FaultRxEqIdER (L)

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	FaultMsg.Header.EQID 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.FaultMsg
PerfTrans	EquipStatus02.FaultRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2007
Input Par	Equip.CTL.FaultMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	FaultMsg.Header.TransNum

Diagnostic

ADD	
Source A	FaultEqidER 56723
Source B	1
Dest	FaultEqidER 56723

lbl_MsgReset (JMP)

Receive Equipment Fault/Status Unsolicited Message

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

EQ
Source A FaultMsg.Header.TransNum
0
Source B Equip.CTL.FaultMsg.TransNumLast
16

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Fault Rx DN
EquipStatus01.Update.FaultRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Equip.CTL.FaultMsg.TransNumLast
16
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.KA

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2002
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Fault/Status Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A FaultMsg.Header.TransNum 0
Source B Equip.CTL.FaultMsg.TransNumLast 16

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

NE
Source A FaultMsg.Header.TransNum 0
Source B Equip.CTL.FaultMsg.TransNumNext 17

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Fault Rx Transaction Number Sequence ER
EquipStatus01.Update.FaultRxSeqER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Equip.CTL.FaultMsg.TransNumNext 17
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2009
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Diagnostic

ADD
Source A FaultSeqER 1817812
Source B 1
Dest FaultSeqER 1817812

Diagnostic

MOVE
Source LNi 27
Dest FaultSeqER_LNi 1

Diagnostic

MOVE
Source EQi 26
Dest FaultSeqER_EQi 19

Diagnostic

MOVE
Source FaultMsg.Header.TransNum 0
Dest FaultSeqER_TransNum 6136

Diagnostic

MOVE
Source Equip.CTL.FaultMsg.TransNumNext 17
Dest FaultSeqER_TransNumNext 6108

Receive Equipment Fault/Status Unsolicited Message

Receive Equipment Fault/Status Unsolicited Message

Load new message in Ring Buffer
Feb 2017 - 4 RB Transactions
v1.6 - 6 RB Transactions
Mar 5, 2018 - rebalance Ring Buffers to avoid backlogs in RB4 and RB6
Sep 2023 - change RB channel allocation to modulo of Line Number (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi) to distribute EQID from same Line (i.e. 1000, 1001, 1002) to different buffers
Dec 4, 2025 - change RB channel allocation back to modulo of LNi -- all EQID on same Line must go thru same RB to avoid incorrect Candidate/Event generation for Maximo/PCR

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd

RingBuffer Modulo Result
MOD
Source A LNi
27
Source B 6
Dest BufferID
3

RingBuffer Modulo Result
EQ
Source A BufferID
3
Source B 1

UseFaultRB01

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _231_EquipFaultRxLoadRB01

RingBuffer Modulo Result
EQ
Source A BufferID
3
Source B 2

UseFaultRB02

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _232_EquipFaultRxLoadRB02

RingBuffer Modulo Result
EQ
Source A BufferID
3
Source B 3

UseFaultRB03

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _233_EquipFaultRxLoadRB03

RingBuffer Modulo Result
EQ
Source A BufferID
3
Source B 4

UseFaultRB04

Load Equipment Fault/Status Message into Ring Buffer
JSR
Routine Name _234_EquipFaultRxLoadRB04

RingBuffer Modulo Result
EQ

UseFaultRB05

Load Equipment Fault/Status Message into Ring Buffer
JSR

Receive Equipment Fault/Status Unsolicited Message

EQ
Source A BufferID
3
Source B 5

UseFaultRB05

JSR
Routine Name _235_EquipFaultRxLoadRB05

Load Equipment
Fault/Status Message
into Ring Buffer

UseFaultRB01 UseFaultRB02 UseFaultRB03 UseFaultRB04 UseFaultRB05 UseFaultRB06

JSR
Routine Name _236_EquipFaultRxLoadRB06

Prepare for next message

16

Local Copy of Equip
Array Tag for LNi,
EQi Transaction
Received
Equip.CTL.FaultMsg.TransRcvd

Local Copy of Equip
Array Tag for LNi,
EQi Receive Message
Last Transaction
Number

MOVE
Source FaultMsg.Header.TransNum
0
Dest Equip.CTL.FaultMsg.TransNumLast
16

Local Copy of Equip
Array Tag for LNi,
EQi Receive Message
Next Transaction
Number

ADD
Source A FaultMsg.Header.TransNum
0
Source B 1
Dest Equip.CTL.FaultMsg.TransNumNext
17

Local Copy of Equip
Array Tag for LNi,
EQi Receive Message
Next Transaction
Number

LT
Source A Equip.CTL.FaultMsg.TransNumNext
17
Source B 1

Local Copy of Equip
Array Tag for LNi,
EQi Receive Message
Next Transaction
Number

GT
Source A Equip.CTL.FaultMsg.TransNumNext
17
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Local Copy of Equip
Array Tag for LNi,
EQi Receive Message
Next Transaction
Number

MOVE
Source 1
Dest Equip.CTL.FaultMsg.TransNumNext
17

Receive Equipment Fault/Status Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)
DHF006 - add DisableFaultMsg

don't reset FaultMsg Timeout if BulkFaultMessage is in use
(timeout is used to suspend scan of the unused Fault method)

17

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.FaultMsg.TransReset

Diagnostic
ADD
Source A TransReset 0
Source B 1
Dest TransReset 0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd

Diagnostic
ADD
Source A TransRcvd 59530117
Source B 1
Dest TransRcvd 59530117

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.FaultMsg.Timeout

Diagnostic
ADD
Source A TransTimeout 13563
Source B 1
Dest TransTimeout 13563

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg

Diagnostic
ADD
Source A DisableFaultMsg 192
Source B 1
Dest DisableFaultMsg 192

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.FaultMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.FaultMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.FaultMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.FaultMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.FaultMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)
CLR
Dest Equip.CTL.FaultMsg.TransLast 0

Equipment Data Lines
COP
Source FaultMsg
Dest EDC_EqRxData.Line[LNi].Equip[EQi].Fault
Length 1

Equipment Data Buffer Lines
FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Fault[0]
Length CFG.FaultMSG.n <EDC.CFG.FaultMSG.n>

Receive Equipment Fault/Status Unsolicited Message

For newly Disabled Equipment,
reset Disable Fault Msg after all Packets 0-7 have been sent to reset any open DB Events

18

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg

Local Copy of Equip Array Tag for LNi, EQi Equipment Disable Msg PKTi

ADD
Source A Equip.CTL.DisablePKTi 0
Source B 1
Dest Equip.CTL.DisablePKTi 0

Local Copy of Equip Array Tag for LNi, EQi Equipment Disable Msg PKTi

GT
Source A Equip.CTL.DisablePKTi 0
Source B 7

Local Copy of Equip Array Tag for LNi, EQi Equipment Fault Msg Disabled (clear existing Events in database)
Equip.CTL.DisableFaultMsg
U

Exit Routine

19

lbl_Exit
LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 231
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B RB_Fault01.MaxTagPtr <EDC.RB_Fault01.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B RB_Fault01.MaxRecPtr <EDC.RB_Fault01.MaxRecPtr> 3999

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Fault01.WritePtrErr <EDC.RB_Fault01.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Dest RB_Fault01.WritePtrErrTag <EDC.RB_Fault01.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Dest RB_Fault01.WritePtrErrRec <EDC.RB_Fault01.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault01.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault01.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B RB_Fault01.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B RB_Fault01.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Fault01.FullBlockWrite <EDC.RB_Fault01.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault01.WriteBlockErr <EDC.RB_Fault01.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault01.WriteBlockErrCnt <EDC.RB_Fault01.WriteBlockErrCnt> 0
Source B 1
Dest RB_Fault01.WriteBlockErrCnt <EDC.RB_Fault01.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault01.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault01.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr> 1
Source B 1

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault01_Rec01[RB_Fault01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault01.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr> 1
Source B 2

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault01_Rec02[RB_Fault01.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault01.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr> 1
Source B 3

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault01_Rec03[RB_Fault01.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault01.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr> 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault01.WriteRecPtr
<EDC.RB_Fault01.WriteRecPtr> 1271
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault01.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr> 1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault01.ReadRecPtr
<EDC.RB_Fault01.ReadRecPtr> 1270
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B 1
Dest RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B RB_Fault01.MaxRecPtr <EDC.RB_Fault01.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault01.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B 1
Dest RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B RB_Fault01.MaxTagPtr <EDC.RB_Fault01.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault01.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault01.ShiftCnt
<EDC.RB_Fault01.ShiftCnt>
69240
Source B 1
Dest RB_Fault01.ShiftCnt
<EDC.RB_Fault01.ShiftCnt>
69240

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 232
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B RB_Fault02.MaxTagPtr <EDC.RB_Fault02.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B RB_Fault02.MaxRecPtr <EDC.RB_Fault02.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Fault02.WritePtrErr <EDC.RB_Fault02.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Dest RB_Fault02.WritePtrErrTag <EDC.RB_Fault02.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Dest RB_Fault02.WritePtrErrRec <EDC.RB_Fault02.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault02.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault02.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B RB_Fault02.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B RB_Fault02.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Fault02.FullBlockWrite <EDC.RB_Fault02.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault02.WriteBlockErr <EDC.RB_Fault02.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault02.WriteBlockErrCnt <EDC.RB_Fault02.WriteBlockErrCnt> 0
Source B 1
Dest RB_Fault02.WriteBlockErrCnt <EDC.RB_Fault02.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault02.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault02.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr> 1
Source B 1

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault02_Rec01[RB_Fault02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault02.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr> 1
Source B 2

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault02_Rec02[RB_Fault02.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault02.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr> 1
Source B 3

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault02_Rec03[RB_Fault02.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault02.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr> 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault02.WriteRecPtr
<EDC.RB_Fault02.WriteRecPtr> 318
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault02.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr> 1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault02.ReadRecPtr
<EDC.RB_Fault02.ReadRecPtr> 317
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B 1
Dest RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B RB_Fault02.MaxRecPtr <EDC.RB_Fault02.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault02.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B 1
Dest RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B RB_Fault02.MaxTagPtr <EDC.RB_Fault02.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault02.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault02.ShiftCnt
<EDC.RB_Fault02.ShiftCnt>
24334
Source B 1
Dest RB_Fault02.ShiftCnt
<EDC.RB_Fault02.ShiftCnt>
24334

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 233
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B RB_Fault03.MaxTagPtr <EDC.RB_Fault03.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B RB_Fault03.MaxRecPtr <EDC.RB_Fault03.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault03.WritePtrErr <EDC.RB_Fault03.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Dest RB_Fault03.WritePtrErrTag <EDC.RB_Fault03.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Dest RB_Fault03.WritePtrErrRec <EDC.RB_Fault03.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B RB_Fault03.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B RB_Fault03.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Fault03.FullBlockWrite <EDC.RB_Fault03.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault03.WriteBlockErr <EDC.RB_Fault03.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault03.WriteBlockErrCnt <EDC.RB_Fault03.WriteBlockErrCnt> 0
Source B 1
Dest RB_Fault03.WriteBlockErrCnt <EDC.RB_Fault03.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault03.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit

JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault03.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Source B 1

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault03_Rec01[RB_Fault03.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault03.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Source B 2

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault03_Rec02[RB_Fault03.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault03.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Source B 3

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault03_Rec03[RB_Fault03.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault03.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault03.WriteRecPtr
<EDC.RB_Fault03.WriteRecPtr> 3240
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault03.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr> 2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault03.ReadRecPtr
<EDC.RB_Fault03.ReadRecPtr> 3239
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B 1
Dest RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B RB_Fault03.MaxRecPtr <EDC.RB_Fault03.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault03.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B 1
Dest RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B RB_Fault03.MaxTagPtr <EDC.RB_Fault03.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault03.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault03.ShiftCnt
<EDC.RB_Fault03.ShiftCnt>
147189
Source B 1
Dest RB_Fault03.ShiftCnt
<EDC.RB_Fault03.ShiftCnt>
147189

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 234
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B RB_Fault04.MaxTagPtr <EDC.RB_Fault04.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B RB_Fault04.MaxRecPtr <EDC.RB_Fault04.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Fault04.WritePtrErr <EDC.RB_Fault04.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Dest RB_Fault04.WritePtrErrTag <EDC.RB_Fault04.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Dest RB_Fault04.WritePtrErrRec <EDC.RB_Fault04.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault04.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault04.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B RB_Fault04.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B RB_Fault04.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053

EDC Ring Buffer Control Buffer Full - Block Writes
RB_Fault04.FullBlockWrite <EDC.RB_Fault04.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Fault04.WriteBlockErr <EDC.RB_Fault04.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault04.WriteBlockErrCnt <EDC.RB_Fault04.WriteBlockErrCnt> 0
Source B 1
Dest RB_Fault04.WriteBlockErrCnt <EDC.RB_Fault04.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault04.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr> 1
Source B 1

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault04_Rec01[RB_Fault04.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault04.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr> 1
Source B 2

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault04_Rec02[RB_Fault04.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault04.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr> 1
Source B 3

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault04_Rec03[RB_Fault04.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr> 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.WriteRecPtr
<EDC.RB_Fault04.WriteRecPtr> 1054
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr> 1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault04.ReadRecPtr
<EDC.RB_Fault04.ReadRecPtr> 1053
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B 1
Dest RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B RB_Fault04.MaxRecPtr <EDC.RB_Fault04.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault04.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B 1
Dest RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B RB_Fault04.MaxTagPtr <EDC.RB_Fault04.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault04.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault04.ShiftCnt
<EDC.RB_Fault04.ShiftCnt>
318575
Source B 1
Dest RB_Fault04.ShiftCnt
<EDC.RB_Fault04.ShiftCnt>
318575

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 235
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B RB_Fault05.MaxTagPtr <EDC.RB_Fault05.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B RB_Fault05.MaxRecPtr <EDC.RB_Fault05.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault05.WritePtrErr <EDC.RB_Fault05.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Dest RB_Fault05.WritePtrErrTag <EDC.RB_Fault05.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Dest RB_Fault05.WritePtrErrRec <EDC.RB_Fault05.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B RB_Fault05.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B RB_Fault05.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697

EDC Ring Buffer Control Buffer Full - Block Writes
RB_Fault05.FullBlockWrite <EDC.RB_Fault05.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Fault05.WriteBlockErr <EDC.RB_Fault05.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault05.WriteBlockErrCnt <EDC.RB_Fault05.WriteBlockErrCnt> 0
Source B 1
Dest RB_Fault05.WriteBlockErrCnt <EDC.RB_Fault05.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault05.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault05.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Source B 1

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault05_Rec01[RB_Fault05.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault05.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Source B 2

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault05_Rec02[RB_Fault05.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag
EQ
Source A RB_Fault05.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Source B 3

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest EDC_RB_Fault05_Rec03[RB_Fault05.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault05.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault05.WriteRecPtr
<EDC.RB_Fault05.WriteRecPtr> 2698
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault05.ReadTagPtr
<EDC.RB_Fault05.ReadTagPtr> 3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Fault05.ReadRecPtr
<EDC.RB_Fault05.ReadRecPtr> 2697
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B 1
Dest RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B RB_Fault05.MaxRecPtr <EDC.RB_Fault05.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault05.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B 1
Dest RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B RB_Fault05.MaxTagPtr <EDC.RB_Fault05.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault05.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault05.ShiftCnt
<EDC.RB_Fault05.ShiftCnt>
135824
Source B 1
Dest RB_Fault05.ShiftCnt
<EDC.RB_Fault05.ShiftCnt>
135824

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Fault/Status Message into Ring Buffer

Load Equipment Fault/Status Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 236
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B RB_Fault06.MaxTagPtr <EDC.RB_Fault06.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B RB_Fault06.MaxRecPtr <EDC.RB_Fault06.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Fault06.WritePtrErr <EDC.RB_Fault06.WritePtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Dest RB_Fault06.WritePtrErrTag <EDC.RB_Fault06.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Dest RB_Fault06.WritePtrErrRec <EDC.RB_Fault06.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2013
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

Load Equipment Fault/Status Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B RB_Fault06.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B RB_Fault06.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433

EDC Ring Buffer Control Buffer Full - Block Writes
RB_Fault06.FullBlockWrite <EDC.RB_Fault06.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Fault06.WriteBlockErr <EDC.RB_Fault06.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Fault06.WriteBlockErrCnt <EDC.RB_Fault06.WriteBlockErrCnt> 8543
Source B 1
Dest RB_Fault06.WriteBlockErrCnt <EDC.RB_Fault06.WriteBlockErrCnt> 8543

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.FaultMsg
PerfTrans EquipStatus02.FaultRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2014
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault06.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Source B 1

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault06_Rec01[RB_Fault06.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault06.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Source B 2

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault06_Rec02[RB_Fault06.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Fault06.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Source B 3

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest EDC_RB_Fault06_Rec03[RB_Fault06.WriteRecPtr]
Length 1

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.WriteRecPtr
<EDC.RB_Fault06.WriteRecPtr> 2434
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr> 3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Fault06.ReadRecPtr
<EDC.RB_Fault06.ReadRecPtr> 2433
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2011
Input Par Equip.CTL.FaultMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par FaultMsg.Header.TransNum

Load Equipment Fault/Status Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B 1
Dest RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B RB_Fault06.MaxRecPtr <EDC.RB_Fault06.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Fault06.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B 1
Dest RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B RB_Fault06.MaxTagPtr <EDC.RB_Fault06.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Fault06.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

Load Equipment Fault/Status Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Fault06.ShiftCnt
<EDC.RB_Fault06.ShiftCnt>
344695
Source B 1
Dest RB_Fault06.ShiftCnt
<EDC.RB_Fault06.ShiftCnt>
344695

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Receive Equipment Quality/Health Unsolicited Message

Receive Equipment Quality/Health Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 240
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Receive Equipment Quality/Health Unsolicited Message

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

EDC2 WL18 - allow PdM Sample for Equipment without Quality Message enabled
Block Transaction TImer if Quality Message not enabled -- PdM Sample uses the Quality RB but doesn't use the EqDriver Quality Message

2

Local Copy of Equip Array Tag for LNi, EQi Quality Message Enabled (from Driver Callout Msg)
Equip.CFG.QualityMsgEN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.QualityMsg.TransTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.QualityMsg.TransTMR
Preset 0
Accum 0
EN
DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.QualityMsg.TransTMR.ACC
0
Source B CFG.BufferMsgTimeout
<EDC.CFG.BufferMsgTimeout>
330000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.QualityMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Quality Rx Msg ER
EquipStatus01.Update.QualityRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2104
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par 0

Diagnostic

ADD
Source A QualityTimeoutER
39960
Source B 1
Dest QualityTimeoutER
39960

MOVE
Source LNi
27
Dest xxQualityLNi
10

MOVE
Source EQi
26
Dest xxQualityEQi
15

Equipment Data Quality/Health Message

FLL
Source 0
Dest QualityMsg
Length 1

lbl_MsgReset
JMP

Receive Equipment Quality/Health Unsolicited Message

PdM - Bypass Quality Tool RxBuffer processing when PdM Sample is ready and valid Tool Config/Msg generated
EqDriver and PdmDriver are sharing the Quality Tool RingBuffer interface

Nov 19, 2022 - When EQ has multple tools with mix of Sample and Trigger, a Triggered Tool can incorrectly use another Tool's Sample signal.
Triggered Tools should not scan the Sample routine. (Moved JSR 249 here and then jumping over RxBuffer/Trigger)

3

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver	Pdm Send Data Collection Sample	Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver	
Equip.CTL.QualityMsg.PdmSampleSend	JSR Routine Name _249_EquipQualityRxPdmSampleSend	Equip.CTL.QualityMsg.PdmSampleSend	lbl_LoadMsg JMP

Receive Equipment Quality/Health Unsolicited Message

Copy Rx Buffer to local registers
DINT Array to UDT

v1.6 - Quality Msg Version determined by Header.PacketType 0=v1.5 (5 Data), 1=v1.6 (10 Data)

4

Equipment Data Quality/Health Message Header

CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Quality[0]
Dest QualityMsg.Header
Length 1

Equipment Data Quality/Health Message Header - PacketType (0=5 Data, 1=10 Data)

NE
Source A QualityMsg.Header.PacketType
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6
L

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Header - PacketType (0=5 Data, 1=10 Data)

EQ
Source A QualityMsg.Header.PacketType
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6
U

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6

Equipment Data Quality/Health Message

CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Quality[0]
Dest QualityMsg
Length 1

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6
/

Equipment Data Quality/Health Message

CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Quality[0]
Dest QualityMsgv1_5
Length 1

Receive Equipment Quality/Health Unsolicited Message

No new data in Rx Buffer
Header & Footer Transaction Number =0

v1.6 - Quality Msg Version determined by Header.PacketSizer 0=v1.5 (5 Data), 1=v1.6 (10 Data)

5

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

EQ
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

EQ
Source A QualityMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

EQ
Source A QualityMsgv1_5.Header.TransNum
7709
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

EQ
Source A QualityMsgv1_5.Footer.TransNum
0
Source B 0

v1.6 - Map v1.5 Quality Message to Message Structure for compatibility

6

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Quality Message v1.6 (10-data packet)
Equip.CTL.QualtyMsgV1_6

Map v1.5 Quality Message to v1.6 data structure

JSR
Routine Name _246_EquipQualityRxMapV1_5

Receive Equipment Quality/Health Unsolicited Message

New data in Rx Buffer
Initialize Transaction timing

7

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.QualityMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.QualityMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)
CLR
Dest Equip.CTL.QualityMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source QualityMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.EN

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2100
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Receive Equipment Quality/Health Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

8

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

EQ
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

NE
Source A QualityMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.QualityMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.QualityMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

EQ
Source A QualityMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

9

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.QualityMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.QualityMsg.PartialTMR.DN

lbl_Exit
JMP

Receive Equipment Quality/Health Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

10

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.QualityMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.QualityMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Quality Rx Msg ER
EquipStatus01.Update.QualityRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2105
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Diagnostic

ADD
Source A QualityPartialMsgER
125
Source B 1
Dest QualityPartialMsgER
125

lbl_MsgReset
JMP

Receive Equipment Quality/Health Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

11

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

NE
Source A QualityMsg.Footer.TransNum
0
Source B 0

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B QualityMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.QualityMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Quality Rx Msg ER

EquipStatus01.Update.QualityRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2106
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Diagnostic

ADD
Source A QualityTransNumER
4
Source B 1
Dest QualityTransNumER
4

lbl_MsgReset
JMP

Receive Equipment Quality/Health Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

12

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum 0
Source B 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

NE
Source A QualityMsg.Footer.TransNum 0
Source B 0

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

EQ
Source A QualityMsg.Header.TransNum 0
Source B QualityMsg.Footer.TransNum 0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received

Equip.CTL.QualityMsg.TransRcvd
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Quality Rx DN

EquipStatus01.Update.QualityRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityMsg.Footer.TransNum 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.DN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2101
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Receive Equipment Quality/Health Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Expected Equipment ID

CPT	
Dest	ExpectedEQID 1006
Expression	LNi*100+EQi

Equipment Data Quality/Health Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE	
Source A	QualityMsg.Header.EQID 0
Source B	ExpectedEQID 1006

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Quality Rx Equipment ID Mismatch ER
EquipStatus01.Update.QualityRxEqIdER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	QualityMsg.Header.EQID 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.QualityMsg
PerfTrans	EquipStatus03.QualityRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2107
Input Par	Equip.CTL.QualityMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	QualityMsg.Header.TransNum

Diagnostic

ADD	
Source A	QualityEqidER 2112
Source B	1
Dest	QualityEqidER 2112

lbl_MsgReset
JMP

Receive Equipment Quality/Health Unsolicited Message

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

EQ
Source A: QualityMsg.Header.TransNum 0
Source B: Equip.CTL.QualityMsg.TransNumLast 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Quality Rx DN
EquipStatus01.Update.QualityRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate ...
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: Equip.CTL.QualityMsg.TransNumLast 0
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans ...
PerfMsg: Equip.CTL.QualityMsg
PerfTrans: EquipStatus03.QualityRx.KA

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2102
Input Par: Equip.CTL.QualityMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: QualityMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Quality/Health Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B Equip.CTL.QualityMsg.TransNumLast
0

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

NE
Source A QualityMsg.Header.TransNum
0
Source B Equip.CTL.QualityMsg.TransNumNext
0

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Quality Rx Transaction Number Sequence ER
EquipStatus01.Update.QualityRxSeqER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Equip.CTL.QualityMsg.TransNumNext
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2109
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Diagnostic

ADD
Source A QualitySeqER
267938
Source B 1
Dest QualitySeqER
267938

Diagnostic

MOVE
Source LNi
27
Dest QualitySeqER_LNi
9

Diagnostic

MOVE
Source EQi
26
Dest QualitySeqER_EQi
36

Diagnostic

MOVE
Source QualityMsg.Header.TransNum
0
Dest QualitySeqER_TransNum
7168

Diagnostic

MOVE
Source Equip.CTL.QualityMsg.TransNumNext
0
Dest QualitySeqER_TransNumNext
8597

Receive Equipment Quality/Health Unsolicited Message

16

Prepare for next message

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number

MOVE
Source QualityMsg.Header.TransNum 0
Dest Equip.CTL.QualityMsg.TransNumLast 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

ADD
Source A QualityMsg.Header.TransNum 0
Source B 1
Dest Equip.CTL.QualityMsg.TransNumNext 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

LT
Source A Equip.CTL.QualityMsg.TransNumNext 0
Source B 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

GT
Source A Equip.CTL.QualityMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

MOVE
Source 1
Dest Equip.CTL.QualityMsg.TransNumNext 0

Receive Equipment Quality/Health Unsolicited Message

BTD signed-SINT ToolNum to DINT to support Tools 0-255
used in PdmTrigger Start/Send and Carrier Track

17

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

ToolNum converted from signed-SINT to unsigned

CLR
Dest QualityMsg_Data_ToolNum 26

ToolNum converted from signed-SINT to unsigned

BTD
Source QualityMsg.Data.ToolNum 0
Source Bit 0
Dest QualityMsg_Data_ToolNum 26
Dest Bit 0
Length 8

v2.00 Pdm - Start Trigger
Command -11 = PdM Start Trigger for this EQID-ToolID
Command -12 = PdM Stop/Send Trigger for this EQID-ToolID
Command -13 = PdM Start & Stop/Send Trigger for thies EQID-ToolID (send current data value)

18

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Equipment Data Quality/Health Message Header - Command Number

EQ
Source A QualityMsg.Header.CmdNum 0
Source B -11

Equipment Data Quality/Health Message Header - Command Number

EQ
Source A QualityMsg.Header.CmdNum 0
Source B -13

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerStart
(L)

Pdm Start Data Collection Trigger

JSR
Routine Name _247_EquipQualityRxPdmTriggerStart

Receive Equipment Quality/Health Unsolicited Message

v2.00 Pdm - Send Trigger
Command -11 = PdM Start Trigger for this EQID-ToolID
Command -12 = PdM Stop/Send Trigger for this EQID-ToolID
Command -13 = PdM Start & Stop/Send Trigger for thies EQID-ToolID (send current data value)

19

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Equipment Data Quality/Health Message Header - Command Number
EQ
Source A QualityMsg.Header.CmdNum
0
Source B -12

Equipment Data Quality/Health Message Header - Command Number
EQ
Source A QualityMsg.Header.CmdNum
0
Source B -13

Don't send Trigger Commands to RingBuffer
Equipment Data Quality/Health Message Header - Command Number
CLR
Dest QualityMsg.Header.CmdNum
0

Local Copy of Equip Array Tag for LNi, EQi PdM Send Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerSend
L

Pdm Send Data Collection Trigger
JSR
Routine Name _248_EquipQualityRxPdmTriggerSend

PdM - Bypass Quality Tool RxBuffer processing when PdM Sample is ready
EqDriver and PdmDriver are sharing the Quality Tool RingBuffer interface

20

lbl_LoadMsg
LBL

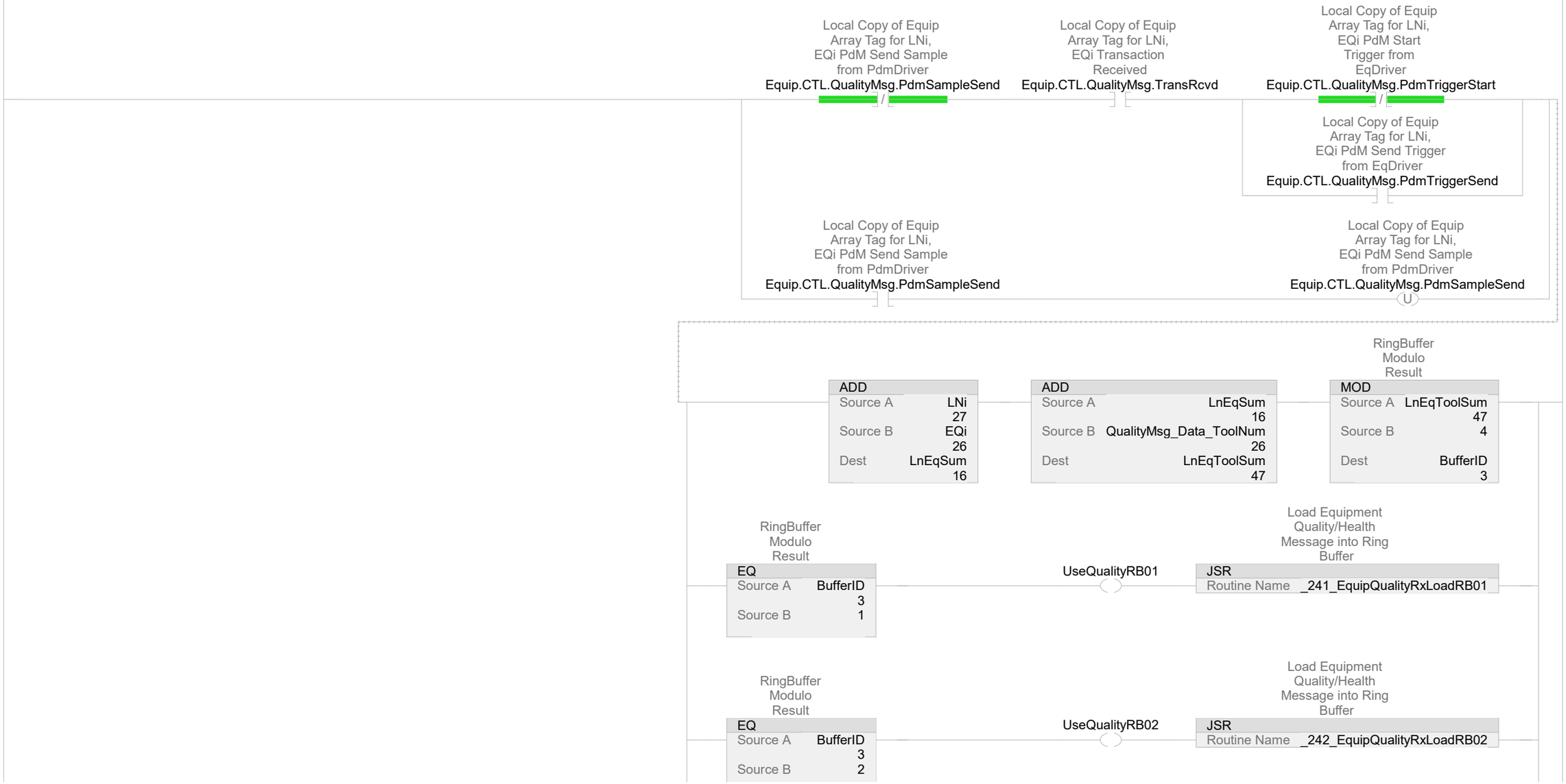
NOP

Receive Equipment Quality/Health Unsolicited Message

Load new message in Ring Buffer
PdM - Ignore PdM Start Trigger Messages (unless Send Trigger also on)
PdM - service PdM Sample Send

Sep 2023 - change RB channel allocation to Line Number Odd/Even (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi + ToolNum) to distribute Tools from same Equipment to different buffers

21



Receive Equipment Quality/Health Unsolicited Message

RingBuffer Modulo Result

EQ
Source A BufferID
Source B 3

UseQualityRB03

Load Equipment Quality/Health Message into Ring Buffer

JSR
Routine Name _243_EquipQualityRxLoadRB03

UseQualityRB01 UseQualityRB02 UseQualityRB03 UseQualityRB04

Load Equipment Quality/Health Message into Ring Buffer

JSR
Routine Name _244_EquipQualityRxLoadRB04

v1.19 -- Load new message in Tracking Conveyor-Carrier arrays
if Tool is configured for Carrier Data
PdM - Ignore Pdm Sample Send Commands and Start Trigger Messages (unless Send Trigger also on)

22

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Quality Message Enabled for Carrier Tracking Data (from Server Config)
Equip.CFG.QualityMsgTrackEN

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerStart

Local Copy of Equip Array Tag for LNi, EQi PdM Send Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerSend

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.QualityMsg.PdmSampleSend

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

JSR
Routine Name _245_EquipQualityRxCarrierTrack

Receive Equipment Quality/Health Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)

23

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.QualityMsg.TransReset

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.QualityMsg.Timeout

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.QualityMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.QualityMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.QualityMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerStart
U

Local Copy of Equip Array Tag for LNi, EQi PdM Send Trigger from EqDriver
Equip.CTL.QualityMsg.PdmTriggerSend
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.QualityMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.QualityMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.QualityMsg.TransLast
0

Equipment Data Lines

COP
Source QualityMsg
Dest EDC_EqRxData.Line[LNi].Equip[EQi].Quality
Length 1

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Quality[0]
Length CFG.QualityMSG.n
<EDC.CFG.QualityMSG.n>

Exit Routine

24

lbl_Exit
LBL

NOP

(End)

Load Equipment Quality/Health Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 241
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Quality/Health Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B RB_Quality01.MaxTagPtr <EDC.RB_Quality01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B RB_Quality01.MaxRecPtr <EDC.RB_Quality01.MaxRecPtr> 1299

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Quality01.WritePtrErr <EDC.RB_Quality01.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Dest RB_Quality01.WritePtrErrTag <EDC.RB_Quality01.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Dest RB_Quality01.WritePtrErrRec <EDC.RB_Quality01.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality01.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality01.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2113
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

Load Equipment Quality/Health Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B RB_Quality01.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B RB_Quality01.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Quality01.FullBlockWrite <EDC.RB_Quality01.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Quality01.WriteBlockErr <EDC.RB_Quality01.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Quality01.WriteBlockErrCnt <EDC.RB_Quality01.WriteBlockErrCnt> 0
Source B 1
Dest RB_Quality01.WriteBlockErrCnt <EDC.RB_Quality01.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2114
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
EQ Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 1	COP Source QualityMsg Dest EDC_RB_Quality01_Rec01[RB_Quality01.WriteRecPtr] Length 1
EQ Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 2	COP Source QualityMsg Dest EDC_RB_Quality01_Rec02[RB_Quality01.WriteRecPtr] Length 1
EQ Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 3	COP Source QualityMsg Dest EDC_RB_Quality01_Rec03[RB_Quality01.WriteRecPtr] Length 1
EQ Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 4	COP Source QualityMsg Dest EDC_RB_Quality01_Rec04[RB_Quality01.WriteRecPtr] Length 1
EQ Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 5	COP Source QualityMsg Dest EDC_RB_Quality01_Rec05[RB_Quality01.WriteRecPtr] Length 1

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.WriteTagPtr
<EDC.RB_Quality01.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.WriteRecPtr
<EDC.RB_Quality01.WriteRecPtr>
440
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality01.ReadRecPtr
<EDC.RB_Quality01.ReadRecPtr>
439
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2111
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B 1
Dest RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B RB_Quality01.MaxRecPtr <EDC.RB_Quality01.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality01.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B 1
Dest RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B RB_Quality01.MaxTagPtr <EDC.RB_Quality01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality01.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

Load Equipment Quality/Health Message into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD
Source A RB_Quality01.ShiftCnt
<EDC.RB_Quality01.ShiftCnt>
86102
Source B 1
Dest RB_Quality01.ShiftCnt
<EDC.RB_Quality01.ShiftCnt>
86102

Exit Routine

lbl_Exit

8 LBL NOP

(End)

Load Equipment Quality/Health Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 242
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Quality/Health Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B RB_Quality02.MaxTagPtr <EDC.RB_Quality02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B RB_Quality02.MaxRecPtr <EDC.RB_Quality02.MaxRecPtr> 1299

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Quality02.WritePtrErr <EDC.RB_Quality02.WritePtrErr>
(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Dest RB_Quality02.WritePtrErrTag <EDC.RB_Quality02.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Dest RB_Quality02.WritePtrErrRec <EDC.RB_Quality02.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality02.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality02.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2113
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984

Load Equipment Quality/Health Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B RB_Quality02.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B RB_Quality02.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Quality02.FullBlockWrite <EDC.RB_Quality02.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Quality02.WriteBlockErr <EDC.RB_Quality02.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Quality02.WriteBlockErrCnt <EDC.RB_Quality02.WriteBlockErrCnt> 0
Source B 1
Dest RB_Quality02.WriteBlockErrCnt <EDC.RB_Quality02.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2114
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
EQ Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2 Source B 1	COP Source QualityMsg Dest EDC_RB_Quality02_Rec01[RB_Quality02.WriteRecPtr] Length 1
EQ Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2 Source B 2	COP Source QualityMsg Dest EDC_RB_Quality02_Rec02[RB_Quality02.WriteRecPtr] Length 1
EQ Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2 Source B 3	COP Source QualityMsg Dest EDC_RB_Quality02_Rec03[RB_Quality02.WriteRecPtr] Length 1
EQ Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2 Source B 4	COP Source QualityMsg Dest EDC_RB_Quality02_Rec04[RB_Quality02.WriteRecPtr] Length 1
EQ Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2 Source B 5	COP Source QualityMsg Dest EDC_RB_Quality02_Rec05[RB_Quality02.WriteRecPtr] Length 1

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.WriteTagPtr
<EDC.RB_Quality02.WriteTagPtr> 2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.WriteRecPtr
<EDC.RB_Quality02.WriteRecPtr> 984
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr> 2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality02.ReadRecPtr
<EDC.RB_Quality02.ReadRecPtr> 983
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2111
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B 1
Dest RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B RB_Quality02.MaxRecPtr <EDC.RB_Quality02.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality02.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B 1
Dest RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B RB_Quality02.MaxTagPtr <EDC.RB_Quality02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality02.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2

Load Equipment Quality/Health Message into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD
Source A RB_Quality02.ShiftCnt
<EDC.RB_Quality02.ShiftCnt>
87645
Source B 1
Dest RB_Quality02.ShiftCnt
<EDC.RB_Quality02.ShiftCnt>
87645

Exit Routine

lbl_Exit

8 LBL NOP

(End)

Load Equipment Quality/Health Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 243
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Quality/Health Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B RB_Quality03.MaxTagPtr <EDC.RB_Quality03.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B RB_Quality03.MaxRecPtr <EDC.RB_Quality03.MaxRecPtr> 1299

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Quality03.WritePtrErr <EDC.RB_Quality03.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Dest RB_Quality03.WritePtrErrTag <EDC.RB_Quality03.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Dest RB_Quality03.WritePtrErrRec <EDC.RB_Quality03.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality03.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Quality03.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2113
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

Load Equipment Quality/Health Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B RB_Quality03.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B RB_Quality03.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Quality03.FullBlockWrite <EDC.RB_Quality03.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Quality03.WriteBlockErr <EDC.RB_Quality03.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Quality03.WriteBlockErrCnt <EDC.RB_Quality03.WriteBlockErrCnt> 0
Source B 1
Dest RB_Quality03.WriteBlockErrCnt <EDC.RB_Quality03.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2114
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
EQ Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4 Source B 1	COP Source QualityMsg Dest EDC_RB_Quality03_Rec01[RB_Quality03.WriteRecPtr] Length 1
EQ Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4 Source B 2	COP Source QualityMsg Dest EDC_RB_Quality03_Rec02[RB_Quality03.WriteRecPtr] Length 1
EQ Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4 Source B 3	COP Source QualityMsg Dest EDC_RB_Quality03_Rec03[RB_Quality03.WriteRecPtr] Length 1
EQ Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4 Source B 4	COP Source QualityMsg Dest EDC_RB_Quality03_Rec04[RB_Quality03.WriteRecPtr] Length 1
EQ Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4 Source B 5	COP Source QualityMsg Dest EDC_RB_Quality03_Rec05[RB_Quality03.WriteRecPtr] Length 1

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr>
4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.WriteRecPtr
<EDC.RB_Quality03.WriteRecPtr>
1183
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr>
4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality03.ReadRecPtr
<EDC.RB_Quality03.ReadRecPtr>
1182
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2111
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B 1
Dest RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B RB_Quality03.MaxRecPtr <EDC.RB_Quality03.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality03.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B 1
Dest RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B RB_Quality03.MaxTagPtr <EDC.RB_Quality03.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality03.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

Load Equipment Quality/Health Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Quality03.ShiftCnt
<EDC.RB_Quality03.ShiftCnt>
102565
Source B 1
Dest RB_Quality03.ShiftCnt
<EDC.RB_Quality03.ShiftCnt>
102565

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Quality/Health Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 244
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Quality/Health Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B RB_Quality04.MaxTagPtr <EDC.RB_Quality04.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B RB_Quality04.MaxRecPtr <EDC.RB_Quality04.MaxRecPtr> 1299

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Quality04.WritePtrErr <EDC.RB_Quality04.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Dest RB_Quality04.WritePtrErrTag <EDC.RB_Quality04.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Dest RB_Quality04.WritePtrErrRec <EDC.RB_Quality04.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2113
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

Load Equipment Quality/Health Message into Ring Buffer

3

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B RB_Quality04.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B RB_Quality04.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Quality04.FullBlockWrite <EDC.RB_Quality04.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Quality04.WriteBlockErr <EDC.RB_Quality04.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Quality04.WriteBlockErrCnt <EDC.RB_Quality04.WriteBlockErrCnt> 0
Source B 1
Dest RB_Quality04.WriteBlockErrCnt <EDC.RB_Quality04.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.QualityMsg
PerfTrans EquipStatus03.QualityRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2114
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
EQ Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1 Source B 1	COP Source QualityMsg Dest EDC_RB_Quality04_Rec01[RB_Quality04.WriteRecPtr] Length 1
EQ Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1 Source B 2	COP Source QualityMsg Dest EDC_RB_Quality04_Rec02[RB_Quality04.WriteRecPtr] Length 1
EQ Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1 Source B 3	COP Source QualityMsg Dest EDC_RB_Quality04_Rec03[RB_Quality04.WriteRecPtr] Length 1
EQ Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1 Source B 4	COP Source QualityMsg Dest EDC_RB_Quality04_Rec04[RB_Quality04.WriteRecPtr] Length 1
EQ Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1 Source B 5	COP Source QualityMsg Dest EDC_RB_Quality04_Rec05[RB_Quality04.WriteRecPtr] Length 1

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteTagPtr
<EDC.RB_Quality04.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.WriteRecPtr
<EDC.RB_Quality04.WriteRecPtr>
440
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadTagPtr
<EDC.RB_Quality04.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Quality04.ReadRecPtr
<EDC.RB_Quality04.ReadRecPtr>
439
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2111
Input Par Equip.CTL.QualityMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par QualityMsg.Header.TransNum

Load Equipment Quality/Health Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B 1
Dest RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B RB_Quality04.MaxRecPtr <EDC.RB_Quality04.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Quality04.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B 1
Dest RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B RB_Quality04.MaxTagPtr <EDC.RB_Quality04.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Quality04.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

Load Equipment Quality/Health Message into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD
Source A RB_Quality04.ShiftCnt
<EDC.RB_Quality04.ShiftCnt>
99178
Source B 1
Dest RB_Quality04.ShiftCnt
<EDC.RB_Quality04.ShiftCnt>
99178

Exit Routine

lbl_Exit

8 LBL NOP

(End)

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

v1.19 - Store Quality Message Carrier data to Tracking Conveyor-Carrier arrays

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 245
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Scanner Config for matching EQID-ToolID
(combined field allows a quick search)

2

QualityTool Scanner Search (EQID-ToolID)

CPT
Dest ScanSrch 11324
Expression (QualityMsg.Header.EQID *100) + QualityMsg_Data_ToolNum

QualityTool Scanner Search (EQID-ToolID)

NE
Source A ScanSrch 11324
Source B 0

ScanSrchCTRL
RES

MOVE
Source EDC_Track.Cfg.SCN.n 300
Dest ScanSrchCTRL.LEN 300

MOVE
Source 0
Dest ScanSrchCTRL.POS 299

FSC
Control ScanSrchCTRL
Length 300
Position 299
Mode ALL
Expression ScanSrch = EDC_Track.ScanCfg[ScanSrchCTRL.POS].Srch
EN
DN
ER

Exit Routine if Tracking Scanner not Found

3

ScanSrchCTRL.FD
/

lbl_Exit
JMP

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Copy Scanner Configuration to local tags

4

EDC Tracking Scanner Configuration

COP
Source EDC_Track.ScanCfg[ScanSrchCTRL.POS]
Dest ScanCfg
Length 1

Tracking Conveyor Index

MOVE
Source ScanCfg.ConvID
25
Dest CNVi
25

Tracking Equipment Tool Data Index

MOVE
Source ScanCfg.DataID
1
Dest DATi
1

CPT
Dest EQIDi
0
Expression ScanCfg.EQID - (ScanCfg.ConvID * 100)

Exit Routine if Tracking Scanner not Enabled

5

EDC Tracking Scanner Configuration Enabled
ScanCfg.Enable
/

lbl_Exit
JMP

Exit Routine if Invalid Tracking Conveyor Array Index

6

Tracking Conveyor Index

LT
Source A CNVi
25
Source B 0

Tracking Conveyor Index

GT
Source A CNVi
25
Source B EDC_Track.Cfg.CNV.mi
28

Carrier Tracking from EQ Quality Tools Tracking Conveyor Error Count

ADD
Source A EDC_Track.ConvIdERn
0
Source B 1
Dest EDC_Track.ConvIdERn
0

Carrier Tracking from EQ Quality Tools Tracking Conveyor Error EQID-ToolID

MOVE
Source ScanCfg.Srch
11201
Dest EDC_Track.ConvIdERsrch
0

Carrier Tracking from EQ Quality Tools Tracking Conveyor Error Index

MOVE
Source CNVi
25
Dest EDC_Track.ConvIdERi
0

lbl_Exit
JMP

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Exit Routine if Invalid Quality Message Tool Data Index
(Configuration indicates which Data word contains the Carrier Number)

7

Tracking Equipment Tool Data Index

LT
Source A DATi 1
Source B 1

Tracking Equipment Tool Data Index

GT
Source A DATi 1
Source B 10

Carrier Tracking from EQ Quality Tools Quality Message Data Error Count

ADD
Source A EDC_Track.DataIdERn 0
Source B 1
Dest EDC_Track.DataIdERn 0

Carrier Tracking from EQ Quality Tools Quality Message Data Error EQID-ToolID

MOVE
Source ScanCfg.Srch 11201
Dest EDC_Track.DataIdERsrch 0

Carrier Tracking from EQ Quality Tools Quality Message Data Error Index

MOVE
Source DATi 1
Dest EDC_Track.DataIdERi 0

lbl_Exit
JMP

Carrier Tracking from EQ Quality Tools Tracking Conveyor

ADD
Source A EDC_Track.Conv[CNVi].DataIdERn 0
Source B 1
Dest EDC_Track.Conv[CNVi].DataIdERn 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source ScanCfg.Srch 11201
Dest EDC_Track.Conv[CNVi].DataIdERsrch 0

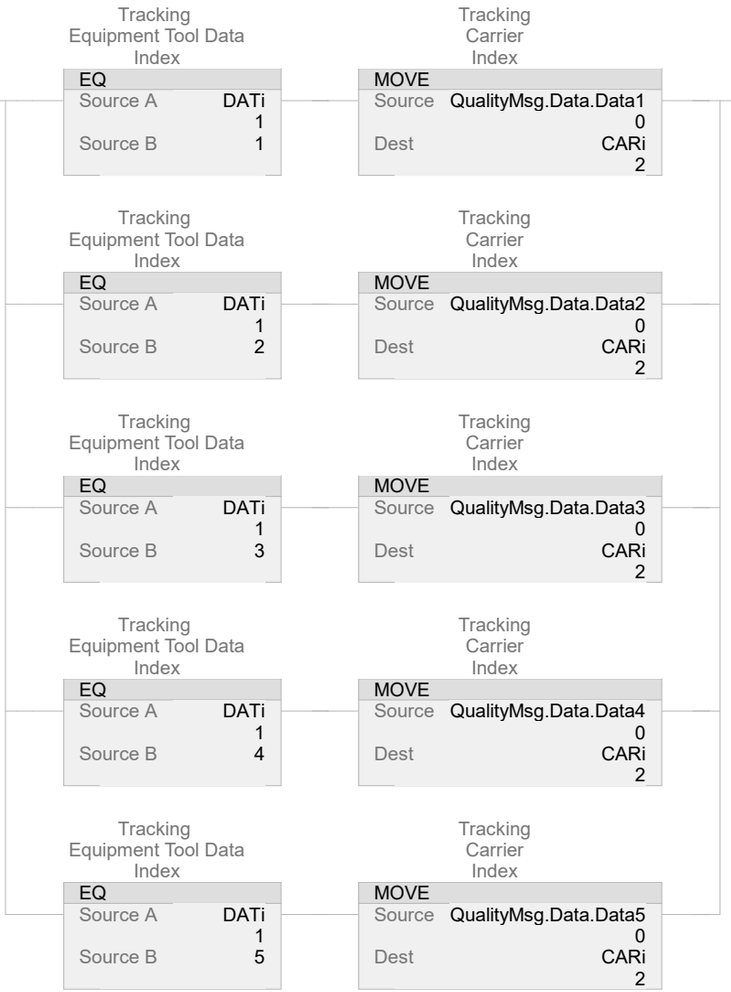
Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source DATi 1
Dest EDC_Track.Conv[CNVi].DataIdERi 0

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Get Carrier Number from Quality Data 1-5

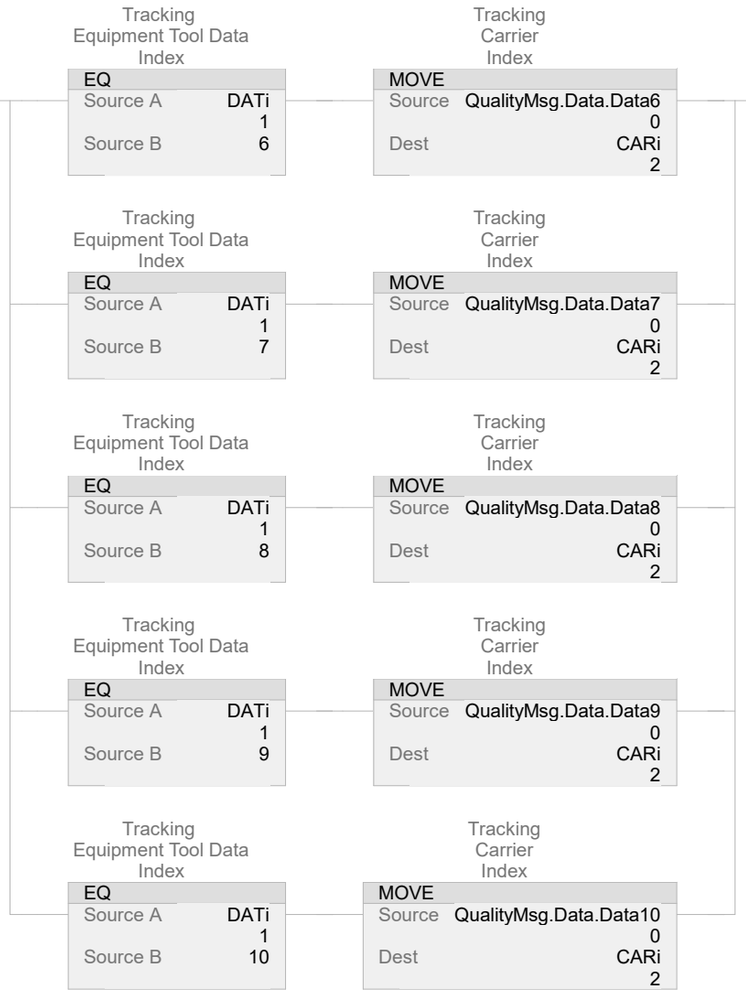
8



Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Get Carrier Number from Quality Data 6-10

9



Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

** Temporary Code to fix Rear Carrier Number **

10

xxFixRearCarrNumEN

Equipment Data Quality/Health Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

EQ
Source A QualityMsg.Header.EQID[0]
Source B 113

Equipment Data Quality/Health Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

EQ
Source A QualityMsg.Header.EQID[0]
Source B 114

Tracking Carrier Index

LIMIT
Low Limit 1
Test CARi[2]
High Limit 300

Tracking Carrier Index

ADD
Source A CARi[2]
Source B 300
Dest CARi[2]

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Check if valid Carrier Index

11

Tracking Carrier Index

LIMIT	
Low Limit	0
Test	CARi 2
High Limit	EDC_Track.Cfg.CAR.mi 999

ValidCARi

ValidCARi

Tracking Carrier Index

CLR	
Dest	CARi 2

Carrier Tracking from EQ Quality Tools Tracking Conveyor Carrier Error Count

ADD	
Source A	EDC_Track.CarrERn 0
Source B	1
Dest	EDC_Track.CarrERn 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor Carrier Error EQID-ToolID

MOVE	
Source	ScanCfg.Srch 11201
Dest	EDC_Track.CarrERsrch 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor Carrier Error Index

MOVE	
Source	CARi 2
Dest	EDC_Track.CarrERi 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

ADD	
Source A	EDC_Track.Conv[CNVi].CarrERn 0
Source B	1
Dest	EDC_Track.Conv[CNVi].CarrERn 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE	
Source	ScanCfg.Srch 11201
Dest	EDC_Track.Conv[CNVi].CarrERsrch 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE	
Source	CARi 2
Dest	EDC_Track.Conv[CNVi].CarrERi 0

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Check if valid Equipment Index

12

LIMIT
Low Limit 0
Test EQIDi 0
High Limit CFG.EQ.mi <EDC.CFG.EQ.mi> 99

ValidEQIDi

ValidEQIDi /

CLR
Dest EQIDi 0

Diagnostic
ADD
Source A ConvCarr_InvalidEQiCnt 189098
Source B 1
Dest ConvCarr_InvalidEQiCnt 189098

Check if valid Pitch Index

13

ValidEQIDi

MOVE
Source EDC.Line[CNVi].Equip[EQIDi].CFG.Pitch 0
Dest EQPi 0

LIMIT
Low Limit 0
Test EQPi 0
High Limit EDC_Track.Cfg.POS.mi 59

ValidEQPi

ValidEQPi /

CLR
Dest EQPi 0

Diagnostic
ADD
Source A ConvCarr_InvalidEQPiCnt 0
Source B 1
Dest ConvCarr_InvalidEQPiCnt 0

Combine Carrier Type (0-4) with Ano & Bno
This allows one Tracking Image to contain multiple Ano/Bno for different Door Types

14

Carrier Type & Ano
for combined search
CPT
Dest TypeAno 230986
Expression (ScanCfg.Type*100000) + QualityMsg.Data.AssyNum

Carrier Type & Bno
for combined search
CPT
Dest TypeBno 275504
Expression (ScanCfg.Type*100000) + QualityMsg.Data.BodyNum

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Search for duplicate Ano
(no need to search for multiple duplicates since duplicate is deleted every time Ano/Bno added)

15

Equipment Data Quality/Health Message Assembly Sequence Number

NE
Source A QualityMsg.Data.AssyNum
0
Source B 0

Duplicate Ano Search CONTROL
FindAno
RES

Duplicate Ano Search CONTROL

MOVE
Source EDC_Track.Cfg.CAR.n
1000
Dest FindAno.LEN
1000

Duplicate Ano Search CONTROL

FSC
Control FindAno
Length 1000
Position 999
Mode ALL
Expression TypeAno=(EDC_Track.Conv[CNVi].Carr[FindAno.POS].Type*100000+EDC_Track.Conv[CNVi].Carr[FindAno.POS].Ano)
EN
DN
ER

Duplicate Ano Search CONTROL
FindAno.FD

Carrier Tracking from EQ Quality Tools Tracking Conveyor

FLL
Source 0
Dest EDC_Track.Conv[CNVi].Carr[FindAno.POS]
Length 1

Carrier Tracking from EQ Quality Tools Duplicate Ano Deleted Count

ADD
Source A EDC_Track.DupAnoCnt
185
Source B 1
Dest EDC_Track.DupAnoCnt
185

Carrier Tracking from EQ Quality Tools Duplicate Ano Deleted EQID-ToolID

MOVE
Source QualityMsg.Header.EQID
0
Dest EDC_Track.DupAnoSrch
114

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Search for duplicate Bno
(no need to search for multiple duplicates since duplicate is deleted every time Ano/Bno added)

16

Equipment Data Quality/Health Message Assembly Body Number

NE
Source A QualityMsg.Data.BodyNum
0
Source B 0

Duplicate Bno Search CONTROL FindBno
RES

Duplicate Bno Search CONTROL

MOVE
Source EDC_Track.Cfg.CAR.n
1000
Dest FindBno.LEN
1000

Duplicate Bno Search CONTROL

FSC
Control FindBno
Length 1000
Position 999
Mode ALL
Expression TypeBno=(EDC_Track.Conv[CNVi].Carr[FindBno.POS].Type*100000+EDC_Track.Conv[CNVi].Carr[FindBno.POS].Bno)
EN
DN
ER

Duplicate Bno Search CONTROL FindBno.FD

Carrier Tracking from EQ Quality Tools Tracking Conveyor

FLL
Source 0
Dest EDC_Track.Conv[CNVi].Carr[FindBno.POS]
Length 1

Carrier Tracking from EQ Quality Tools Duplicate Bno Deleted Count

ADD
Source A EDC_Track.DupBnoCnt
3
Source B 1
Dest EDC_Track.DupBnoCnt
3

Carrier Tracking from EQ Quality Tools Duplicate Bno Deleted EQID-ToolID

MOVE
Source QualityMsg.Header.EQID
0
Dest EDC_Track.DupBnoSrch
114

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Update HMI animation for Model Type
(this provides initial animation for the Door Carriers until they reach the PDC queue - other carriers are typically in PDC or Line tracking before they are scanned)
(0=White, 1=Yellow, 3=Cyan)

17

Shop-specific code
for animation

JSR	
Routine Name	_245_EquipQualityRxCarrierTrack_CfgAnim

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Store Tracking Conveyor-Carrier Data

18

ValidCARi

Equipment Data Quality/Health Message Program Number Executed

NE
Source A QualityMsg.Data.ActPrgNum
0
Source B 0
0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.ActPrgNum
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Model
73

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.AssyNum
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Ano
30737

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.BodyNum
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Bno
75212

Equipment Data Quality/Health Message Program Number Executed

EQ
Source A QualityMsg.Data.ActPrgNum
0
Source B 0
0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.AviPrgNum
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Model
73

Carrier Tracking from EQ Quality Tools Tracking Conveyor

COP
Source QualityMsg.TransTime
Dest EDC_Track.Conv[CNVi].Carr[CARi].TransTime
Length 1

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.CycleCount
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].CycleCount
588517

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source CNVi
25
Dest EDC_Track.Conv[CNVi].Carr[CARi].Loc
12

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source EQPi
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Pos
57

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source Animation
0
Dest EDC_Track.Conv[CNVi].Carr[CARi].Anim
4

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source ScanCfg.Type
2
Dest EDC_Track.Conv[CNVi].Carr[CARi].Type
2

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Store Previous Message Diagnostic

19

Carrier Tracking from EQ Quality Tools Tracking Conveyor

COP
Source EDC_Track.Conv[CNVi].CarrCurr
Dest EDC_Track.Conv[CNVi].CarrPrev
Length 1

Store Current Message Diagnostic

20

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source CARi
2
Dest EDC_Track.Conv[CNVi].CarrCurr.Carrier
159

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.AssyNum
0
Dest EDC_Track.Conv[CNVi].CarrCurr.Ano
30983

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.BodyNum
0
Dest EDC_Track.Conv[CNVi].CarrCurr.Bno
75438

Carrier Tracking from EQ Quality Tools Tracking Conveyor

COP
Source QualityMsg.TransTime
Dest EDC_Track.Conv[CNVi].CarrCurr.TransTime
Length 1

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
Source QualityMsg.Data.CycleCount
0
Dest EDC_Track.Conv[CNVi].CarrCurr.CycleCount
588763

New Data Count

21

Diagnostic

ADD
Source A ConvCarr_NewDataCnt
4289327
Source B 1
Dest ConvCarr_NewDataCnt
4289327

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Copy EDC_Track (Conv-Carrier) to EDC_TrackAVI (Conv-Position) for AVI Overview Display

22

NOP

Remove duplicate Tracking Conveyor-Position Data for Bno
(no need to search for multiple duplicates since duplicate is deleted every time Ano/Bno added)

23

Gateway Config
Download Tracking
Allow Duplicate
Carriers on Line
Enable
EDC_CfgGateway.TrackingDupEnable

Equipment Data
Quality/Health
Message Assembly
Sequence Number

NE
Source A QualityMsg.Data.AssyNum
0
Source B 0

FindAnoPitch
RES

MOVE
Source EDC_Track.Cfg.POS.n
60
Dest FindAnoPitch.LEN
60

FSC
Control FindAnoPitch
Length 60
Position 59
Mode ALL
Expression QualityMsg.Data.AssyNum=EDC_TrackAVI.Line[CNVi].Pitch[FindAnoPitch.POS].Ano
EN
DN
ER

FindAnoPitch.FD

AVI Line Tracking
Line Queue

FLL
Source 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[FindAnoPitch.POS]
Length 1

AVI Line Tracking
Line Queue

CLR
Dest EDC_TrackAVI.Line[CNVi].Anim[FindAnoPitch.POS]
4

Diagnostic

ADD
Source A DupAnoPitchCnt
43
Source B 1
Dest DupAnoPitchCnt
43

Diagnostic

MOVE
Source QualityMsg.Header.EQID
0
Dest DupAnoPitchSrch
830

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Remove duplicate Tracking Conveyor-Position Data for Bno
(no need to search for multiple duplicates since duplicate is deleted every time Ano/Bno added)

24

Gateway Config
Download Tracking
Allow Duplicate
Carriers on Line
Enable
EDC_CfgGateway.TrackingDupEnable

Equipment Data
Quality/Health
Message Assembly
Body Number

NE
Source A QualityMsg.Data.BodyNum
0
Source B 0

FindBnoPitch
RES

MOVE
Source EDC_Track.Cfg.POS.n
60
Dest FindBnoPitch.LEN
60

FSC
Control FindBnoPitch
Length 60
Position 59
Mode ALL
Expression QualityMsg.Data.BodyNum=EDC_TrackAVI.Line[CNVi].Pitch[FindBnoPitch.POS].Bno
EN
DN
ER

FindBnoPitch.FD

AVI Line Tracking
Line Queue

FLL
Source 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[FindBnoPitch.POS]
Length 1

AVI Line Tracking
Line Queue

CLR
Dest EDC_TrackAVI.Line[CNVi].Anim[FindBnoPitch.POS]
4

Diagnostic

ADD
Source A DupBnoPitchCnt
13385
Source B 1
Dest DupBnoPitchCnt
13385

Diagnostic

MOVE
Source QualityMsg.Header.EQID
0
Dest DupBnoPitchSrch
213

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Remove duplicate Tracking Conveyor-Position Data for Cno
(no need to search for multiple duplicates since duplicate is deleted every time Ano/Bno added)

25

Gateway Config Download Tracking Allow Duplicate Carriers on Line Enable
EDC_CfgGateway.TrackingDupEnable

ValidCARi

Tracking Carrier Index
NE
Source A CARi 2
Source B 0

FindCnoPitch
RES

MOVE
Source EDC_Track.Cfg.POS.n 60
Dest FindCnoPitch.LEN 60

FSC
Control FindCnoPitch
Length 60
Position 59
Mode ALL
Expression CARi=EDC_TrackAVI.Line[CNVi].Pitch[FindCnoPitch.POS].Carrier
EN
DN
ER

FindCnoPitch.FD

AVI Line Tracking Line Queue
FLL
Source 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[FindCnoPitch.POS]
Length 1

AVI Line Tracking Line Queue
CLR
Dest EDC_TrackAVI.Line[CNVi].Anim[FindCnoPitch.POS] 4

Diagnostic
ADD
Source A DupCnoPitchCnt 415163
Source B 1
Dest DupCnoPitchCnt 415163

Diagnostic
MOVE
Source QualityMsg.Header.EQID 0
Dest DupCnoPitchSrch 114

Store Carrier data to Carrier Tracking Conveyor-Carrier arrays

Store Tracking Conveyor-Position Data
If AviPrgNum = 0, load ActPrgNum

26

ValidEQPi

AVI Line Tracking Line Queue
MOVE
Source CARi 2
Dest EDC_TrackAVI.Line[CNVi].Pitch[EQPi].Carrier 0

AVI Line Tracking Line Queue
MOVE
Source QualityMsg.Data.AssyNum 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[EQPi].Ano 0

AVI Line Tracking Line Queue
MOVE
Source QualityMsg.Data.BodyNum 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[EQPi].Bno 0

AVI Line Tracking Line Queue
MOVE
Source Animation 0
Dest EDC_TrackAVI.Line[CNVi].Anim[EQPi] 1

Equipment Data Quality/Health Message Program Number Executed
NE
Source A QualityMsg.Data.ActPrgNum 0
Source B 0

AVI Line Tracking Line Queue
MOVE
Source QualityMsg.Data.ActPrgNum 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[EQPi].Model 46

Equipment Data Quality/Health Message Program Number Executed
EQ
Source A QualityMsg.Data.ActPrgNum 0
Source B 0

AVI Line Tracking Line Queue
MOVE
Source QualityMsg.Data.AviPrgNum 0
Dest EDC_TrackAVI.Line[CNVi].Pitch[EQPi].Model 46

Exit Routine Cycle Count

27

lbl_Exit
LBL

Diagnostic
ADD
Source A ConvCarr_CycleCnt 18361303
Source B 1
Dest ConvCarr_CycleCnt 18361303

(End)

Shop-specific code for animation

Update HMI animation for Model Type
(this provides initial animation for the Door Carriers until they reach the PDC queue - other carriers are typically in PDC or Line tracking before they are scanned)
(0=White, 1=Yellow, 3=Cyan)

0

Equipment Data Quality/Health Message Assembly Sequence Number
EQ
Source A QualityMsg.Data.AssyNum 0
Source B 0

Equipment Data Quality/Health Message Assembly Body Number
EQ
Source A QualityMsg.Data.BodyNum 0
Source B 0

Empty

Animation Value
MOVE
Source 1
Dest Animation 0

Empty

Equipment Data Quality/Health Message Program Number Executed
EQ
Source A QualityMsg.Data.ActPrgNum 0
Source B 72

AnimateGH

Animation Value
MOVE
Source 3
Dest Animation 0

Equipment Data Quality/Health Message Program Number from SLCP
EQ
Source A QualityMsg.Data.AviPrgNum 0
Source B 72

Equipment Data Quality/Health Message Program Number Executed
EQ
Source A QualityMsg.Data.ActPrgNum 0
Source B 73

AnimateL

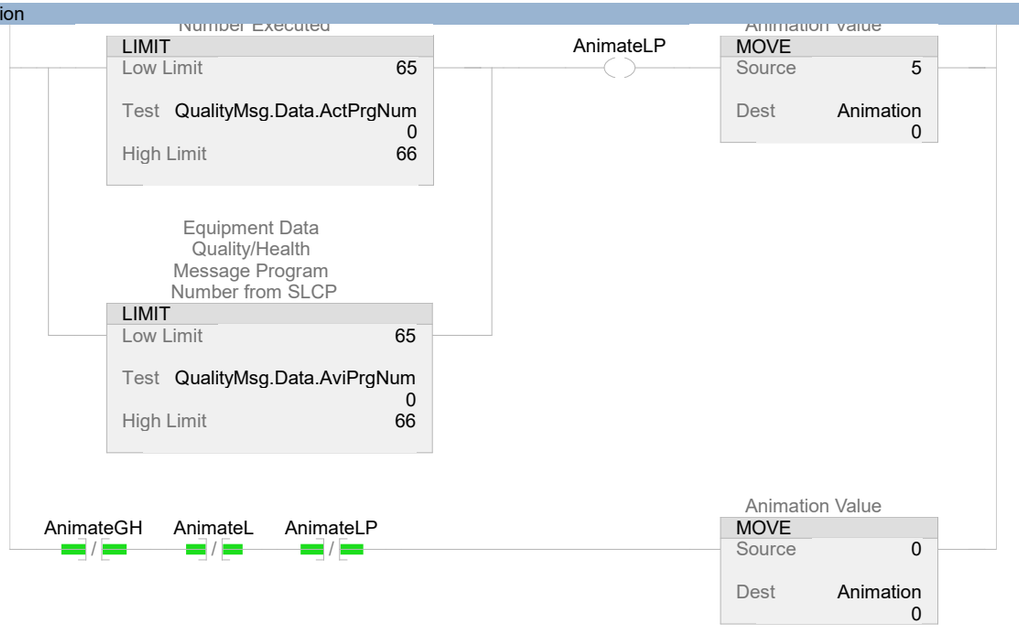
Animation Value
MOVE
Source 4
Dest Animation 0

Equipment Data Quality/Health Message Program Number from SLCP
EQ
Source A QualityMsg.Data.AviPrgNum 0
Source B 73

Equipment Data Quality/Health Message Program Number Executed

Animation Value

Shop-specific code for animation



(End)

Map v1.5 Quality Message to v1.6 data structure

v1.6 - Map v1.5 Quality Message to v1.6 data structure

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 246
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Header, TransTime and Footer

2

Equipment Data Quality/Health Message Header

COP
Source QualityMsgv1_5.Header
Dest QualityMsg.Header
Length 1

Equipment Data Quality/Health Message Transaction Time Stamp

COP
Source QualityMsgv1_5.TransTime
Dest QualityMsg.TransTime
Length 1

Equipment Data Quality/Health Message Footer

COP
Source QualityMsgv1_5.Footer
Dest QualityMsg.Footer
Length 1

Map v1.5 Quality Message to v1.6 data structure

3

Data
Move individual elements to resolve data differences

Equipment Data Quality/Health Message Tool Number

MOVE	
Source	QualityMsgv1_5.Data.ToolNum 38
Dest	QualityMsg.Data.ToolNum 0

Equipment Data Quality/Health Message Program Number from SLCP

MOVE	
Source	QualityMsgv1_5.Data.AviPrgNum 1
Dest	QualityMsg.Data.AviPrgNum 0

Equipment Data Quality/Health Message Program Number Executed

MOVE	
Source	QualityMsgv1_5.Data.ActPrgNum 1
Dest	QualityMsg.Data.ActPrgNum 0

Equipment Data Quality/Health Message Job Status

MOVE	
Source	QualityMsgv1_5.Data.JobStatus 1
Dest	QualityMsg.Data.JobStatus 0

Equipment Data Quality/Health Message Fault Code

MOVE	
Source	QualityMsgv1_5.Data.FaultCode 0
Dest	QualityMsg.Data.FaultCode 0

Equipment Data Quality/Health Message Spare

MOVE	
Source	QualityMsgv1_5.Data.Spare1 2
Dest	QualityMsg.Data.Spare1 0

Equipment Data Quality/Health Message Spare

MOVE	
Source	QualityMsgv1_5.Data.Spare2 0
Dest	QualityMsg.Data.Spare2 0

Equipment Data Quality/Health Message Data Values Used (1-5)

MOVE	
Source	QualityMsgv1_5.Data.DataCount 10
Dest	QualityMsg.Data.DataCount 0

Equipment Data Quality/Health Message Assembly Body Number

MOVE	
Source	QualityMsgv1_5.Data.BodyNum 74759
Dest	QualityMsg.Data.BodyNum 0

Equipment Data Quality/Health Message Assembly Sequence Number

MOVE	
Source	QualityMsgv1_5.Data.AssyNum 29259
Dest	QualityMsg.Data.AssyNum 0

Equipment Data Quality/Health Message VIN Number

COP	
Source	QualityMsgv1_5.Data.VinNum[0]
Dest	QualityMsg.Data.VinNum[0]
Length	20

Equipment Data Quality/Health Message Employee Badge Number for Job Release

COP	
Source	QualityMsgv1_5.Data.BadgeNum[0]
Dest	QualityMsg.Data.BadgeNum[0]
Length	12

Equipment Data Quality/Health Message Cycle Time

MOVE	
Source	QualityMsgv1_5.Data.CycleTime 0
Dest	QualityMsg.Data.CycleTime 0

Equipment Data Quality/Health Message Cycle Count

MOVE	
Source	QualityMsgv1_5.Data.CycleCount 0
Dest	QualityMsg.Data.CycleCount 0

Map v1.5 Quality Message to v1.6 data structure

4

Data
Move individual elements to resolve data differences

Equipment Data Quality/Health Message Quality Data 1	Equipment Data Quality/Health Message Quality Data 2	Equipment Data Quality/Health Message Quality Data 3	Equipment Data Quality/Health Message Quality Data 4	Equipment Data Quality/Health Message Quality Data 5
MOVE	MOVE	MOVE	MOVE	MOVE
Source QualityMsgv1_5.Data.Data1 0	Source QualityMsgv1_5.Data.Data2 0	Source QualityMsgv1_5.Data.Data3 0	Source QualityMsgv1_5.Data.Data4 0	Source QualityMsgv1_5.Data.Data5 0
Dest QualityMsg.Data.Data1 0	Dest QualityMsg.Data.Data2 0	Dest QualityMsg.Data.Data3 0	Dest QualityMsg.Data.Data4 0	Dest QualityMsg.Data.Data5 0

5

Data
Move individual elements to resolve data differences

Equipment Data Quality/Health Message Quality Data 6	Equipment Data Quality/Health Message Quality Data 7	Equipment Data Quality/Health Message Quality Data 8	Equipment Data Quality/Health Message Quality Data 9	Equipment Data Quality/Health Message Quality Data 10
MOVE	MOVE	MOVE	MOVE	MOVE
Source 0	Source 0	Source 0	Source 0	Source 0
Dest QualityMsg.Data.Data6 0	Dest QualityMsg.Data.Data7 0	Dest QualityMsg.Data.Data8 0	Dest QualityMsg.Data.Data9 0	Dest QualityMsg.Data.Data10 0

6

Data
Move individual elements to resolve data differences

Equipment Data Quality/Health Message Quality Data 1 Scaling Factor	Equipment Data Quality/Health Message Quality Data 2 Scaling Factor	Equipment Data Quality/Health Message Quality Data 3 Scaling Factor	Equipment Data Quality/Health Message Quality Data 4 Scaling Factor	Equipment Data Quality/Health Message Quality Data 5 Scaling Factor
MOVE	MOVE	MOVE	MOVE	MOVE
Source QualityMsgv1_5.Data.Scale1 0	Source QualityMsgv1_5.Data.Scale2 0	Source QualityMsgv1_5.Data.Scale3 0	Source QualityMsgv1_5.Data.Scale4 0	Source QualityMsgv1_5.Data.Scale5 0
Dest QualityMsg.Data.Scale1 0	Dest QualityMsg.Data.Scale2 0	Dest QualityMsg.Data.Scale3 0	Dest QualityMsg.Data.Scale4 0	Dest QualityMsg.Data.Scale5 0

Map v1.5 Quality Message to v1.6 data structure

Data

Move individual elements to resolve data differences

7

Equipment Data Quality/Health Message Quality Data 5 Scaling Factor	Equipment Data Quality/Health Message Quality Data 5 Scaling Factor	Equipment Data Quality/Health Message Quality Data 5 Scaling Factor	Equipment Data Quality/Health Message Quality Data 5 Scaling Factor	Equipment Data Quality/Health Message Quality Data 5 Scaling Factor
MOVE	MOVE	MOVE	MOVE	MOVE
Source 0	Source 0	Source 0	Source 0	Source 0
Dest QualityMsg.Data.Scale6 0	Dest QualityMsg.Data.Scale7 0	Dest QualityMsg.Data.Scale8 0	Dest QualityMsg.Data.Scale9 0	Dest QualityMsg.Data.Scale10 0

(End)

Pdm Start Data Collection Trigger

v2.00 Pdm - Start Trigger
Command -11 = Start Trigger for this EQID-ToolID

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 247
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

2

Pdm Start Trigger EQ-Tool Search (EQID-ToolID)

CPT
Dest ToolStartTrigSrch 944074
Expression (QualityMsg.Header.EQID*1000)+QualityMsg_Data_ToolNum

Pdm Start Trigger EQ-Tool Search (EQID-ToolID)

NE
Source A ToolStartTrigSrch 944074
Source B 0

Tool Search for Start Trigger ToolStartTrigFSC

RES

Tool Search for Start Trigger

ADD
Source A PdM_Cfg.TL.ei 914
Source B 1
Dest ToolStartTrigFSC.LEN 915

Tool Search for Start Trigger

MOVE
Source 0
Dest ToolStartTrigFSC.POS 496

Tool Search for Start Trigger

FSC
Control ToolStartTrigFSC
Length 915
Position 496
Mode ALL
Expression ToolStartTrigSrch=PdM_Tool[ToolStartTrigFSC.POS].Cfg.EqToolID

EN
DN
ER

Pdm Start Data Collection Trigger

Exit Routine if Pdm Tool not Found

3

Tool Search for Start Trigger
ToolStartTrigFSC.FD
/

lbl_Exit
JMP

Exit Routine if
Pdm Tool not Enabled
Pdm Tool Trigger not Enabled

(keep this separate from previous rung to avoid bad index when Tool not found)

4

PdM Tool Array
PdM_Tool[ToolStartTrigFSC.POS].Cfg.Enable
/

PdM Tool Array
PdM_Tool[ToolStartTrigFSC.POS].Cfg.TriggerEnable
/

lbl_Exit
JMP

Set EQID-Tool Start Trigger

5

PdM Tool Array
PdM_Tool[ToolStartTrigFSC.POS].Ctl.TriggerStart
L

Exit Routine

6

lbl_Exit
LBL

NOP

(End)

Pdm Send Data Collection Trigger

v2.00 Pdm - Send Trigger
Command -12 = Send Trigger for this EQID-ToolID

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 248
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

2

Pdm Send Trigger EQ-Tool Search (EQID-ToolID)

CPT
Dest ToolSendTrigSrch 1010021
Expression (QualityMsg.Header.EQID*1000)+QualityMsg_Data_ToolNum

Pdm Send Trigger EQ-Tool Search (EQID-ToolID)

NE
Source A ToolSendTrigSrch 1010021
Source B 0

Tool Search for Send Trigger

ToolSendTrigFSC
RES

Tool Search for Send Trigger

ADD
Source A PdM_Cfg.TL.ei 914
Source B 1
Dest ToolSendTrigFSC.LEN 915

Tool Search for Send Trigger

MOVE
Source 0
Dest ToolSendTrigFSC.POS 827

Tool Search for Send Trigger

FSC
Control ToolSendTrigFSC
Length 915
Position 827
Mode ALL
Expression ToolSendTrigSrch=PdM_Tool[ToolSendTrigFSC.POS].Cfg.EqToolID
EN
DN
ER

Pdm Send Data Collection Trigger

Exit Routine if Pdm Tool not Found

3

Tool Search for Send Trigger
ToolSendTrigFSC.FD
/
lbl_Exit
JMP

Exit Routine if
Pdm Tool not Enabled
Pdm Tool Trigger not Enabled

(keep this separate from previous rung to avoid bad index when Tool not found)

4

PdM Tool Array
PdM_Tool[ToolSendTrigFSC.POS].Cfg.Enable
/
lbl_Exit
JMP
PdM Tool Array
PdM_Tool[ToolSendTrigFSC.POS].Cfg.TriggerEnable
/

Set EQID-Tool Send Trigger

5

PdM Tool Array
PdM_Tool[ToolSendTrigFSC.POS].Ctl.TriggerSend
L

Load Pdm Data 1-10 to Quality Message

6

PdM Data Loop Index
CLR
Dest
LOOPi
10

Load Pdm Data 1-10 to Quality Message
Skip to next Data if this Data not Enabled

7

lbl_DataLoop
LBL
PdM Tool Array
PdM_Tool[ToolSendTrigFSC.POS].Cfg.Data[LOOPi].Enable
/
lbl_NextData
JMP

Pdm Send Data Collection Trigger

Load Data Value

8

MOVE
Source PdM_Tool[ToolSendTrigFSC.POS].Array[LOOPi].Data
??
Dest PdmData
64

MOVE
Source PdM_Tool[ToolSendTrigFSC.POS].Array[LOOPi].Scale
??
Dest PdmScale
1000

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data1
0

Equipment Data Quality/Health Message Quality Data 1 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale1
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 1

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data2
0

Equipment Data Quality/Health Message Quality Data 2 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale2
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 2

Equipment Data Quality/Health Message Quality Data 3
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data3
0

Equipment Data Quality/Health Message Quality Data 3 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale3
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 3

Equipment Data Quality/Health Message Quality Data 4
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data4
0

Equipment Data Quality/Health Message Quality Data 4 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale4
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 4

Equipment Data Quality/Health Message Quality Data 5
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 5 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale5
0

Pdm Send Data Collection Trigger



Pdm Send Data Collection Trigger

Increment Loop Counter

9

lbl_NextData
LBL

PdM Data Loop Index
ADD
Source A LOOPi 10
Source B 1
Dest LOOPi 10

PdM Data Loop Index
LT
Source A LOOPi 10
Source B 10

lbl_DataLoop
JMP

Exit Routine

10

lbl_Exit
LBL

NOP

(End)

Pdm Send Data Collection Sample

v2.00 Pdm - Load Sample to Quality Message
Pdm Driver sets SampleSend and ToolID for Quality Tool to Send

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 249
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

PdM Tool sets SampleSend and ToolID

2

Pdm Send Sample EQ-Tool Search (EQID-ToolID)

CPT
Dest ToolSendSmplSrch 79032
Expression (((LNi*100)+EQi)*1000)+Equip.CTL.QualityMsg.PdmSampleToolID

Pdm Send Sample EQ-Tool Search (EQID-ToolID)

NE
Source A ToolSendSmplSrch 79032
Source B 0

Tool Search for Send Sample

ToolSendSmplFSC
RES

Tool Search for Send Sample

ADD
Source A PdM_Cfg.TL.ei 914
Source B 1
Dest ToolSendSmplFSC.LEN 915

Tool Search for Send Sample

MOVE
Source 0
Dest ToolSendSmplFSC.POS 532

Tool Search for Send Sample

FSC
Control ToolSendSmplFSC
Length 915
Position 532
Mode ALL
Expression ToolSendSmplSrch=PdM_Tool[ToolSendSmplFSC.POS].Cfg.EqToolID
EN
DN
ER

Pdm Send Data Collection Sample

Exit Routine if Pdm Tool not Found

3

Tool Search for Send Sample
ToolSendSmplFSC.FD
/

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.QualityMsg.PdmSampleSend
U

lbl_Exit
JMP

Exit Routine if Pdm Tool not Enabled

4

PdM Tool Array
PdM_Tool[ToolSendSmplFSC.POS].Cfg.Enable
/

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.QualityMsg.PdmSampleSend
U

lbl_Exit
JMP

Initialize QualityMsg
PdM Sample generates its own Header, etc.

5

Equipment Data Quality/Health Message

FLL	
Source	0
Dest	QualityMsg
Length	1

Pdm Send Data Collection Sample

Increment Transaction Number
Pdm uses Transaction Numbers 10,000-10,999 to avoid conflicts with EqDriver

6

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

ADD
Source A: Equip.CTL.QualityMsg.PdmSampleTransNum 0
Source B: 1
Dest: Equip.CTL.QualityMsg.PdmSampleTransNum 0

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

LT
Source A: Equip.CTL.QualityMsg.PdmSampleTransNum 0
Source B: 10001

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

GT
Source A: Equip.CTL.QualityMsg.PdmSampleTransNum 0
Source B: 10999

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

MOVE
Source: 10001
Dest: Equip.CTL.QualityMsg.PdmSampleTransNum 0

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

MOVE
Source: Equip.CTL.QualityMsg.PdmSampleTransNum 0
Dest: QualityMsg.Header.TransNum 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

MOVE
Source: Equip.CTL.QualityMsg.PdmSampleTransNum 0
Dest: QualityMsg.Footer.TransNum 0

Transaction Time

7

Equipment Data Quality/Health Message Transaction Time Stamp

COP
Source: CLX.TransTime <EDC.CLX.TransTime>
Dest: QualityMsg.TransTime
Length: 1

Pdm Send Data Collection Sample

8

Equipment ID & Tool Number

Equipment Data Quality/Health Message Header - PacketType (0=5 Data, 1=10 Data)

MOVE
Source 1
Dest QualityMsg.Header.PacketType
0

Equipment Data Quality/Health Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source PdM_Tool[ToolSendSmplFSC.POS].Cfg.EQID
79
Dest QualityMsg.Header.EQID
0

Equipment Data Quality/Health Message Tool Number

BTD
Source PdM_Tool[ToolSendSmplFSC.POS].Cfg.ToolID
32
Source Bit 0
Dest QualityMsg.Data.ToolNum
0
Dest Bit 0
Length 8

9

Job Status

Equipment Data Quality/Health Message Job Status

MOVE
Source 1
Dest QualityMsg.Data.JobStatus
0

10

Load Pdm Data 1-10 to Quality Message

PdM Data Loop Index

CLR
Dest LOOPi
10

11

Load Pdm Data 1-10 to Quality Message
Skip to next Data if this Data not Enabled

lbl_DataLoop
LBL

PdM Tool Array
PdM_Tool[ToolSendSmplFSC.POS].Cfg.Data[LOOPi].Enable
/

lbl_NextData
JMP

Pdm Send Data Collection Sample

12

Load Data Value
(default to Average if illegal Type)

MOVE
Source PdM_Tool[ToolSendSmplFSC.POS].Array[LOOPi].Data
??
Dest PdmData
64

MOVE
Source PdM_Tool[ToolSendSmplFSC.POS].Array[LOOPi].Scale
??
Dest PdmScale
1000

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data1
0

Equipment Data Quality/Health Message Quality Data 1 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale1
0

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 1
Dest QualityMsg.Data.DataCount
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 1

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data2
0

Equipment Data Quality/Health Message Quality Data 2 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale2
0

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 2
Dest QualityMsg.Data.DataCount
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 2

Equipment Data Quality/Health Message Quality Data 3
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data3
0

Equipment Data Quality/Health Message Quality Data 3 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale3
0

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 3
Dest QualityMsg.Data.DataCount
0

PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 3

Equipment Data Quality/Health Message Quality Data 4
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data4
0

Equipment Data Quality/Health Message Quality Data 4 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale4
0

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 4
Dest QualityMsg.Data.DataCount
0

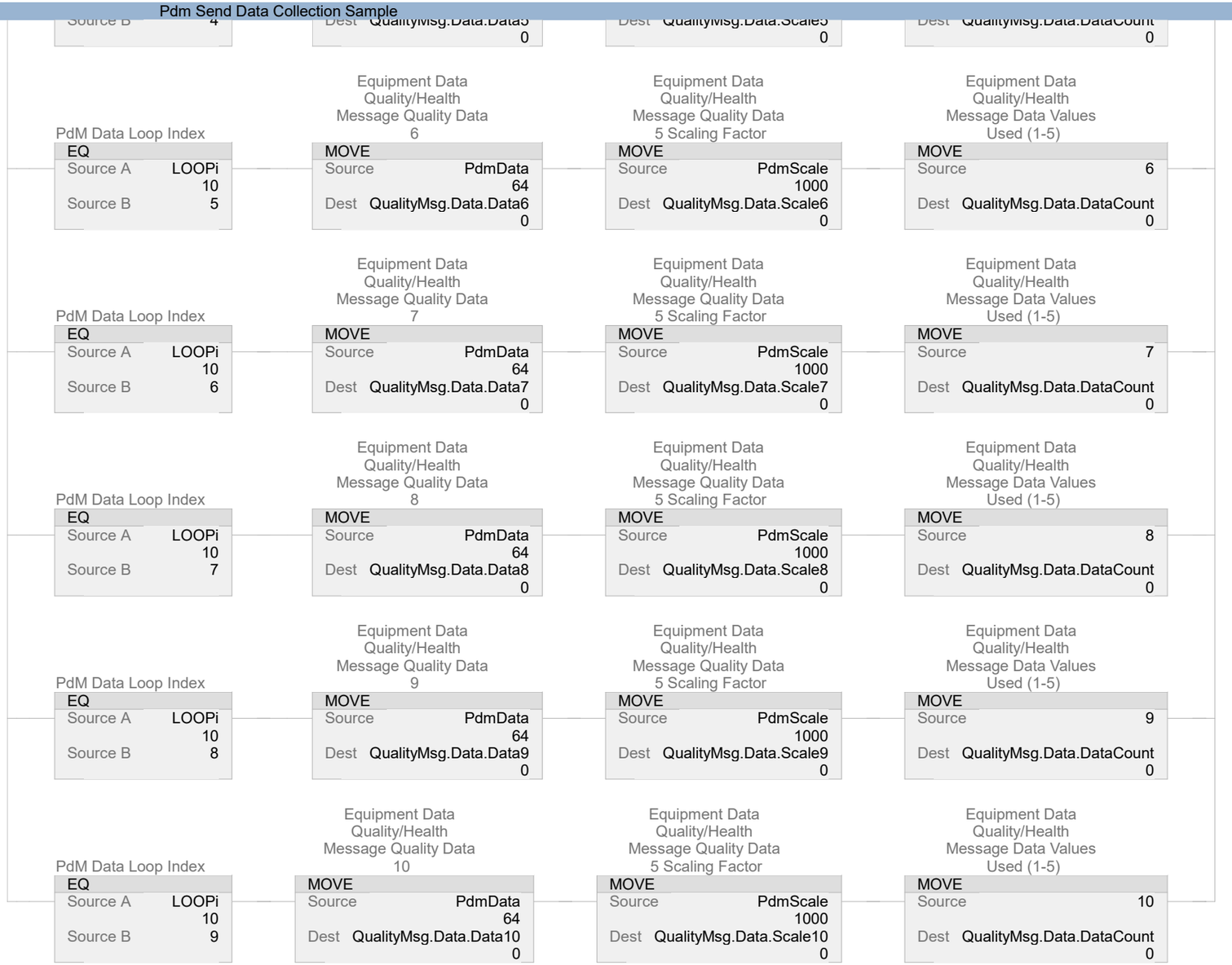
PdM Data Loop Index
EQ
Source A LOOPi
10
Source B 4

Equipment Data Quality/Health Message Quality Data 5
MOVE
Source PdmData
64
Dest QualityMsg.Data.Data5

Equipment Data Quality/Health Message Quality Data 5 Scaling Factor
MOVE
Source PdmScale
1000
Dest QualityMsg.Data.Scale5

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 5
Dest QualityMsg.Data.DataCount

Pdm Send Data Collection Sample



Pdm Send Data Collection Sample

Increment Loop Counter

13

lbl_NextData
LBL

PdM Data Loop Index
ADD
Source A LOOPi 10
Source B 1
Dest LOOPi 10

PdM Data Loop Index
LT
Source A LOOPi 10
Source B 10

lbl_DataLoop
JMP

Exit Routine

14

lbl_Exit
LBL

NOP

(End)

Receive Equipment Trend Unsolicited Message

Receive Equipment Trend Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 250
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Receive Equipment Trend Unsolicited Message

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

EDC2 WL18 - allow PdM Sample for Equipment without Quality Message enabled
Block the Transaction TImer if Trend Message not enabled -- PdM Sample uses the Trend RB but doesn't use the EqDriver Trend Message

2

Local Copy of Equip Array Tag for LNi, EQi Trend Message Enabled (from Driver Callout Msg)
Equip.CFG.TrendMsgEN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.TrendMsg.TransTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.TrendMsg.TransTMR
Preset 0
Accum 0
EN
DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.TrendMsg.TransTMR.ACC
0
Source B CFG.BufferMsgTimeout
<EDC.CFG.BufferMsgTimeout>
330000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.TrendMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Trend Rx Msg ER
EquipStatus01.Update.TrendRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2204
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par 0

MOVE
Source LNi
27
Dest xxTrendLNi
0

MOVE
Source EQi
26
Dest xxTrendEQi
53

Equipment Data Trend Message

FLL
Source 0
Dest TrendMsg
Length 1

lbl_MsgReset
JMP

PdM - Bypass Trend Tool RxBuffer processing when PdM Sample is ready
EqDriver and PdmDriver are sharing the Trend Tool RingBuffer interface

3

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.TrendMsg.PdmSampleSend

lbl_PdmSample
JMP

Receive Equipment Trend Unsolicited Message

Copy Rx Buffer to local registers
DINT Array to UDT

4

Equipment Data
Trend Message

CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Trend[0]
Dest TrendMsg
Length 1

No new data in Rx Buffer
Header & Footer Transaction Number =0

5

Equipment Data
Trend Message
Header - Transaction
Number (must match
Footer)

EQ
Source A TrendMsg.Header.TransNum
0
Source B 0

Equipment Data
Trend Message
Footer - Transaction
Number (must match
Footer)

EQ
Source A TrendMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

Receive Equipment Trend Unsolicited Message

New data in Rx Buffer
Initialize Transaction timing

6

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.TrendMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.TrendMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)
CLR
Dest Equip.CTL.TrendMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source TrendMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.EN

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2200
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

Receive Equipment Trend Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

7

Equipment Data Trend Message Header - Transaction Number (must match Footer)

EQ
Source A TrendMsg.Header.TransNum
0
Source B 0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

NE
Source A TrendMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.TrendMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.TrendMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE
Source A TrendMsg.Header.TransNum
0
Source B 0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

EQ
Source A TrendMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.TrendMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.TrendMsg.PartialTMR.DN
/

lbl_Exit
JMP

Receive Equipment Trend Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

9

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.TrendMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.TrendMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Trend Rx Msg ER
EquipStatus01.Update.TrendRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	TrendMsg.Footer.TransNum 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.TrendMsg
PerfTrans	EquipStatus03.TrendRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2205
Input Par	Equip.CTL.TrendMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	TrendMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Trend Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

10

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE
Source A TrendMsg.Header.TransNum
0
Source B 0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

NE
Source A TrendMsg.Footer.TransNum
0
Source B 0

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE
Source A TrendMsg.Header.TransNum
0
Source B TrendMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.TrendMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Trend Rx Msg ER

EquipStatus01.Update.TrendRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2206
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Trend Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

11

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE	
Source A	TrendMsg.Header.TransNum
	0
Source B	0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

NE	
Source A	TrendMsg.Footer.TransNum
	0
Source B	0

Equipment Data Trend Message Header - Transaction Number (must match Footer)

EQ	
Source A	TrendMsg.Header.TransNum
	0
Source B	TrendMsg.Footer.TransNum
	0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received

Equip.CTL.TrendMsg.TransRcvd

(L)

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Trend Rx DN

EquipStatus01.Update.TrendRxOK

(L)

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	TrendMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.TrendMsg
PerfTrans	EquipStatus03.TrendRx.DN

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2201
Input Par	Equip.CTL.TrendMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	TrendMsg.Header.TransNum

Receive Equipment Trend Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

12

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Expected Equipment ID

CPT
Dest ExpectedEQID 1006
Expression LNi*100+EQi

Equipment Data Trend Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE
Source A TrendMsg.Header.EQID 0
Source B ExpectedEQID 1006

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Trend Rx Equipment ID Mismatch ER
EquipStatus01.Update.TrendRxEqIdER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendMsg.Header.EQID 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2207
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Trend Unsolicited Message

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Equipment Data Trend Message Header - Transaction Number (must match Footer)

EQ
Source A TrendMsg.Header.TransNum
0
Source B Equip.CTL.TrendMsg.TransNumLast
0

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Trend Rx DN
EquipStatus01.Update.TrendRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Equip.CTL.TrendMsg.TransNumLast
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.KA

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2202
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Trend Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE	
Source A	TrendMsg.Header.TransNum 0
Source B	Equip.CTL.TrendMsg.TransNumLast 0

Equipment Data Trend Message Header - Transaction Number (must match Footer)

NE	
Source A	TrendMsg.Header.TransNum 0
Source B	Equip.CTL.TrendMsg.TransNumNext 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Trend Rx Transaction Number Sequence ER
EquipStatus01.Update.TrendRxSeqER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.TrendMsg.TransNumNext 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.TrendMsg
PerfTrans	EquipStatus03.TrendRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2209
Input Par	Equip.CTL.TrendMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	TrendMsg.Header.TransNum

Diagnostic

ADD	
Source A	TrendSeqER 43570
Source B	1
Dest	TrendSeqER 43570

Diagnostic

MOVE	
Source	LNi 27
Dest	TrendSeqER_LNi 5

Diagnostic

MOVE	
Source	EQi 26
Dest	TrendSeqER_EQi 25

Diagnostic

MOVE	
Source	TrendMsg.Header.TransNum 0
Dest	TrendSeqER_TransNum 5639

Diagnostic

MOVE	
Source	Equip.CTL.TrendMsg.TransNumNext 0
Dest	TrendSeqER_TransNumNext 20

Receive Equipment Trend Unsolicited Message

v2.00 Pdm - Start Trigger
Command -11 = PdM Start Trigger for this EQID-ToolID
Command -12 = PdM Stop/Send Trigger for this EQID-ToolID

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Equipment Data Trend Message Spare
EQ
Source A TrendMsg.Data.Spare1[0]
Source B -11

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.TrendMsg.PdmTriggerStart
L

Pdm Start Data Collection Trigger
JSR
Routine Name _257_EquipTrendRxPdmTriggerStart

v2.00 Pdm - Send Trigger
Command -11 = PdM Start Trigger for this EQID-ToolID
Command -12 = PdM Stop/Send Trigger for this EQID-ToolID

Don't send Trigger Commands to RingBuffer

16

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Equipment Data Trend Message Spare
EQ
Source A TrendMsg.Data.Spare1[0]
Source B -12

Local Copy of Equip Array Tag for LNi, EQi PdM Send Trigger from EqDriver
Equip.CTL.TrendMsg.PdmTriggerSend
L

Pdm Send Data Collection Trigger
JSR
Routine Name _258_EquipTrendRxPdmTriggerSend

PdM - Bypass Trend Tool RxBuffer processing when PdM Sample is ready
EqDriver and PdmDriver are sharing the Trend Tool RingBuffer interface

17

lbl_PdmSample
LBL

NOP

v2.00 Pdm - Load Sample to Trend Message
Pdm Driver sets SampleSend and ToolID for Trend Tool to Send

18

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.TrendMsg.PdmSampleSend

Pdm Send Data Collection Sample
JSR
Routine Name _259_EquipTrendRxPdmSampleSend

Receive Equipment Trend Unsolicited Message

Load new message in Ring Buffer
PdM - Ignore PdM Start Trigger Messages
PdM - service PdM Sample Send

Sep 2023 - change RB channel allocation to Line Number Odd/Even (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi + ToolNum) to distribute Tools from same Equipment to different buffers

19

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.TrendMsg.PdmTriggerStart

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.TrendMsg.PdmSampleSend

Local Copy of Equip Array Tag for LNi, EQi PdM Send Sample from PdmDriver
Equip.CTL.TrendMsg.PdmSampleSend

U

ADD
Source A: LNi 27
Source B: EQi 26
Dest: LnEqSum 16

ADD
Source A: LnEqSum 16
Source B: TrendMsg.Data.ToolNum 0
Dest: LnEqToolSum 47

RingBuffer Modulo Result
MOD
Source A: LnEqToolSum 47
Source B: 2
Dest: BufferID 3

RingBuffer Modulo Result
EQ
Source A: BufferID 3
Source B: 1

UseTrendRB01

Load Equipment Trend Message into Ring Buffer
JSR
Routine Name: _251_EquipTrendRxLoadRB01

UseTrendRB01

UseTrendRB02

Load Equipment Trend Message into Ring Buffer
JSR
Routine Name: _252_EquipTrendRxLoadRB02

Receive Equipment Trend Unsolicited Message

Prepare for next message

20

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number

MOVE
Source TrendMsg.Header.TransNum 0
Dest Equip.CTL.TrendMsg.TransNumLast 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

ADD
Source A TrendMsg.Header.TransNum 0
Source B 1
Dest Equip.CTL.TrendMsg.TransNumNext 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

LT
Source A Equip.CTL.TrendMsg.TransNumNext 0
Source B 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

GT
Source A Equip.CTL.TrendMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

MOVE
Source 1
Dest Equip.CTL.TrendMsg.TransNumNext 0

Receive Equipment Trend Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)

21

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.TrendMsg.TransReset

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.TrendMsg.Timeout

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.TrendMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.TrendMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.TrendMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi PdM Start Trigger from EqDriver
Equip.CTL.TrendMsg.PdmTriggerStart
U

Local Copy of Equip Array Tag for LNi, EQi PdM Send Trigger from EqDriver
Equip.CTL.TrendMsg.PdmTriggerSend
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.TrendMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.TrendMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.TrendMsg.TransLast
0

Equipment Data Lines

COP
Source TrendMsg
Dest EDC_EqRxData.Line[LNi].Equip[EQi].Trend
Length 1

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Trend[0]
Length CFG.TrendMSG.n
<EDC.CFG.TrendMSG.n>

Exit Routine

22

lbl_Exit
LBL

NOP

(End)

Load Equipment Trend Message into Ring Buffer

Load Equipment Trend Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 251
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Trend Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B RB_Trend01.MaxTagPtr <EDC.RB_Trend01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B RB_Trend01.MaxRecPtr <EDC.RB_Trend01.MaxRecPtr> 499

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Trend01.WritePtrErr <EDC.RB_Trend01.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Dest RB_Trend01.WritePtrErrTag <EDC.RB_Trend01.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Dest RB_Trend01.WritePtrErrRec <EDC.RB_Trend01.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend01.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend01.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2213
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

Load Equipment Trend Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B RB_Trend01.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B RB_Trend01.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Trend01.FullBlockWrite <EDC.RB_Trend01.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Trend01.WriteBlockErr <EDC.RB_Trend01.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Trend01.WriteBlockErrCnt <EDC.RB_Trend01.WriteBlockErrCnt> 0
Source B 1
Dest RB_Trend01.WriteBlockErrCnt <EDC.RB_Trend01.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2214
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Trend Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

	EDC Ring Buffer Control Next Write Tag	Equipment Data Trend Message
	EQ Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4 Source B 1	COP Source TrendMsg Dest EDC_RB_Trend01_Rec01[RB_Trend01.WriteRecPtr] Length 1
	EQ Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4 Source B 2	COP Source TrendMsg Dest EDC_RB_Trend01_Rec02[RB_Trend01.WriteRecPtr] Length 1
	EQ Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4 Source B 3	COP Source TrendMsg Dest EDC_RB_Trend01_Rec03[RB_Trend01.WriteRecPtr] Length 1
	EQ Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4 Source B 4	COP Source TrendMsg Dest EDC_RB_Trend01_Rec04[RB_Trend01.WriteRecPtr] Length 1
	EQ Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4 Source B 5	COP Source TrendMsg Dest EDC_RB_Trend01_Rec05[RB_Trend01.WriteRecPtr] Length 1

Load Equipment Trend Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.WriteTagPtr
<EDC.RB_Trend01.WriteTagPtr>
4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.WriteRecPtr
<EDC.RB_Trend01.WriteRecPtr>
326
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr>
4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend01.ReadRecPtr
<EDC.RB_Trend01.ReadRecPtr>
325
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2211
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

Load Equipment Trend Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B 1
Dest RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B RB_Trend01.MaxRecPtr <EDC.RB_Trend01.MaxRecPtr> 499

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Trend01.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B 1
Dest RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B RB_Trend01.MaxTagPtr <EDC.RB_Trend01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Trend01.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

Load Equipment Trend Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Trend01.ShiftCnt
<EDC.RB_Trend01.ShiftCnt>
2413
Source B 1
Dest RB_Trend01.ShiftCnt
<EDC.RB_Trend01.ShiftCnt>
2413

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Trend Message into Ring Buffer

Load Equipment Trend Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 252
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Trend Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B RB_Trend02.MaxTagPtr <EDC.RB_Trend02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B RB_Trend02.MaxRecPtr <EDC.RB_Trend02.MaxRecPtr> 499

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Trend02.WritePtrErr <EDC.RB_Trend02.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Dest RB_Trend02.WritePtrErrTag <EDC.RB_Trend02.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Dest RB_Trend02.WritePtrErrRec <EDC.RB_Trend02.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend02.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Trend02.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2213
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

Load Equipment Trend Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B RB_Trend02.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B RB_Trend02.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Trend02.FullBlockWrite <EDC.RB_Trend02.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Trend02.WriteBlockErr <EDC.RB_Trend02.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Trend02.WriteBlockErrCnt <EDC.RB_Trend02.WriteBlockErrCnt> 0
Source B 1
Dest RB_Trend02.WriteBlockErrCnt <EDC.RB_Trend02.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.TrendMsg
PerfTrans EquipStatus03.TrendRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2214
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Trend Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

	EDC Ring Buffer Control Next Write Tag	Equipment Data Trend Message
	EQ Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3 Source B 1	COP Source TrendMsg Dest EDC_RB_Trend02_Rec01[RB_Trend02.WriteRecPtr] Length 1
	EQ Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3 Source B 2	COP Source TrendMsg Dest EDC_RB_Trend02_Rec02[RB_Trend02.WriteRecPtr] Length 1
	EQ Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3 Source B 3	COP Source TrendMsg Dest EDC_RB_Trend02_Rec03[RB_Trend02.WriteRecPtr] Length 1
	EQ Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3 Source B 4	COP Source TrendMsg Dest EDC_RB_Trend02_Rec04[RB_Trend02.WriteRecPtr] Length 1
	EQ Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3 Source B 5	COP Source TrendMsg Dest EDC_RB_Trend02_Rec05[RB_Trend02.WriteRecPtr] Length 1

Load Equipment Trend Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.WriteTagPtr
<EDC.RB_Trend02.WriteTagPtr>
3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.WriteRecPtr
<EDC.RB_Trend02.WriteRecPtr>
202
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr>
3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Trend02.ReadRecPtr
<EDC.RB_Trend02.ReadRecPtr>
201
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2211
Input Par Equip.CTL.TrendMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par TrendMsg.Header.TransNum

Load Equipment Trend Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B 1
Dest RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B RB_Trend02.MaxRecPtr <EDC.RB_Trend02.MaxRecPtr> 499

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Trend02.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B 1
Dest RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B RB_Trend02.MaxTagPtr <EDC.RB_Trend02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Trend02.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

Load Equipment Trend Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Trend02.ShiftCnt
<EDC.RB_Trend02.ShiftCnt>
2674
Source B 1
Dest RB_Trend02.ShiftCnt
<EDC.RB_Trend02.ShiftCnt>
2674

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Pdm Start Data Collection Trigger

v2.00 Pdm - Start Trigger
Command -11 = Start Trigger for this EQID-ToolID

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 257
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

2

Pdm Start Trigger EQ-Tool Search (EQID-ToolID)

CPT
Dest TrendStartTrigSrch 53002
Expression TrendMsg.Header.EQID*1000+TrendMsg.Data.ToolNum

Pdm Start Trigger EQ-Tool Search (EQID-ToolID)

NE
Source A TrendStartTrigSrch 53002
Source B 0

Trend Search for Start Trigger TrendStartTrigFSC

RES

Trend Search for Start Trigger

ADD
Source A PdM_Cfg.TRN.ei 3
Source B 1
Dest TrendStartTrigFSC.LEN 4

Trend Search for Start Trigger

MOVE
Source 0
Dest TrendStartTrigFSC.POS 1

Trend Search for Start Trigger

FSC
Control TrendStartTrigFSC
Length 4
Position 1
Mode ALL
Expression TrendStartTrigSrch=PdM_Trend[TrendStartTrigFSC.POS].Cfg.EqToolID

EN
DN
ER

Pdm Start Data Collection Trigger

Exit Routine if Pdm Trend not Found

3

Trend Search for Start Trigger
TrendStartTrigFSC.FD
/

lbl_Exit
JMP

Exit Routine if Pdm Trend not Enabled

4

PdM Trend Array
PdM_Trend[TrendStartTrigFSC.POS].Cfg.Enable
/

lbl_Exit
JMP

Set EQID-Tool Start Trigger

5

PdM Trend Array
PdM_Trend[TrendStartTrigFSC.POS].Ctl.TriggerStart
L

Exit Routine

6

lbl_Exit
LBL

NOP

(End)

Pdm Send Data Collection Trigger

v2.00 Pdm - Send Trigger
Command -12 = Send Trigger for this EQID-ToolID

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 258
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

2

Pdm Send Trigger EQ-Tool Search (EQID-ToolID)

CPT
Dest TrendSendTrigSrch 53002
Expression TrendMsg.Header.EQID*1000+TrendMsg.Data.ToolNum

Pdm Send Trigger EQ-Tool Search (EQID-ToolID)

NE
Source A TrendSendTrigSrch 53002
Source B 0

Trend Search for Send Trigger

TrendSendTrigFSC
RES

Trend Search for Send Trigger

ADD
Source A PdM_Cfg.TRN.ei 3
Source B 1
Dest TrendSendTrigFSC.LEN 4

Trend Search for Send Trigger

MOVE
Source 0
Dest TrendSendTrigFSC.POS 1

Trend Search for Send Trigger

FSC
Control TrendSendTrigFSC
Length 4
Position 1
Mode ALL
Expression TrendSendTrigSrch=PdM_Trend[TrendSendTrigFSC.POS].Cfg.EqToolID
EN
DN
ER

Pdm Send Data Collection Trigger

Exit Routine if Pdm Trend not Found

Trend Search for Send Trigger
TrendSendTrigFSC.FD
3 [/] lbl_Exit JMP

Exit Routine if Pdm Trend not Enabled

PdM Trend Array
PdM_Trend[TrendSendTrigFSC.POS].Cfg.Enable
4 [/] lbl_Exit JMP

Exit Routine if Pdm Data not Enabled

PdM Trend Array
PdM_Trend[TrendSendTrigFSC.POS].Cfg.Data.Enable
5 [/] lbl_Exit JMP

Set EQID-Tool Send Trigger

PdM Trend Array
PdM_Trend[TrendSendTrigFSC.POS].Ctl.TriggerSend
6 L

Load Data Values

Equipment Data Trend Message Trend Data
Equipment Data Trend Message Trend Scale
Equipment Data Trend Message Data Values Used (1-120)

7
COP
Source PdM_Trend[TrendSendTrigFSC.POS].Array.Data[0]
Dest TrendMsg.Data.Trend[0]
Length 120

MOVE
Source PdM_Trend[TrendSendTrigFSC.POS].Array.Scale
1000
Dest TrendMsg.Data.Scale
0

MOVE
Source PdM_Trend[TrendSendTrigFSC.POS].Ctl.DataCount
15
Dest TrendMsg.Data.DataCount
0

Exit Routine

lbl_Exit
8 LBL NOP

(End)

Pdm Send Data Collection Sample

v2.00 Pdm - Load Sample to Trend Message
Pdm Driver sets SampleSend and ToolID for Trend Tool to Send

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 259
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

PdM Tool sets SampleSend and ToolID

2

Pdm Send Sample EQ-Tool Search (EQID-ToolID)

CPT
Dest TrendSendSmplSrch 100005
Expression (((LNi*100)+EQi)*1000)+Equip.CTL.TrendMsg.PdmSampleToolID

Pdm Send Sample EQ-Tool Search (EQID-ToolID)

NE
Source A TrendSendSmplSrch 100005
Source B 0

Trend Search for Send Sample TrendSendSmplFSC

RES

Trend Search for Send Sample

ADD
Source A PdM_Cfg.TRN.ei 3
Source B 1
Dest TrendSendSmplFSC.LEN 12

Trend Search for Send Sample

MOVE
Source 0
Dest TrendSendSmplFSC.POS 6

Trend Search for Send Sample

FSC
Control TrendSendSmplFSC
Length 12
Position 6
Mode ALL
Expression TrendSendSmplSrch=PdM_Trend[TrendSendSmplFSC.POS].Cfg.EqToolID

EN
DN
ER

Pdm Send Data Collection Sample

Exit Routine if Pdm Trend not Found

3

Trend Search for Send Sample
TrendSendSmplFSC.FD
/

lbl_Exit
JMP

Exit Routine if Pdm Trend not Enabled

4

PdM Trend Array
PdM_Trend[TrendSendSmplFSC.POS].Cfg.Enable
/

lbl_Exit
JMP

Exit Routine if Pdm Data not Enabled

5

PdM Trend Array
PdM_Trend[TrendSendSmplFSC.POS].Cfg.Data.Enable
/

lbl_Exit
JMP

Initialize TrendMsg
PdM Sample generates its own Header, etc.

6

Equipment Data Trend Message

FLL	
Source	0
Dest	TrendMsg
Length	1

Pdm Send Data Collection Sample

Increment Transaction Number
Pdm uses Transaction Numbers 10,000-10,999 to avoid conflicts with EqDriver

7

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

ADD
Source A: Equip.CTL.TrendMsg.PdmSampleTransNum 0
Source B: 1
Dest: Equip.CTL.TrendMsg.PdmSampleTransNum 0

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

LT
Source A: Equip.CTL.TrendMsg.PdmSampleTransNum 0
Source B: 10001

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

GT
Source A: Equip.CTL.TrendMsg.PdmSampleTransNum 0
Source B: 10999

Local Copy of Equip Array Tag for LNi, EQi PdM Sample EquipToolID from PdmDriver

MOVE
Source: 10001
Dest: Equip.CTL.TrendMsg.PdmSampleTransNum 0

Equipment Data Trend Message Header - Transaction Number (must match Footer)

MOVE
Source: Equip.CTL.TrendMsg.PdmSampleTransNum 0
Dest: TrendMsg.Header.TransNum 0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

MOVE
Source: Equip.CTL.TrendMsg.PdmSampleTransNum 0
Dest: TrendMsg.Footer.TransNum 0

Transaction Time

8

Equipment Data Trend Message Transaction Time Stamp

COP
Source: CLX.TransTime <EDC.CLX.TransTime>
Dest: TrendMsg.TransTime
Length: 1

Pdm Send Data Collection Sample

Equipment ID & Tool Number

9

Equipment Data Trend Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source PdM_Trend[TrendSendSmplFSC.POS].Cfg.EQID 0
Dest TrendMsg.Header.EQID 0

Equipment Data Trend Message Tool Number

MOVE
Source PdM_Trend[TrendSendSmplFSC.POS].Cfg.ToolID 0
Dest TrendMsg.Data.ToolNum 0

Job Status

10

Equipment Data Trend Message Job Status

MOVE
Source 1
Dest TrendMsg.Data.JobStatus 0

Load Data Values

Note: Rate is an INTEGER and doesn't support sample rates >1/second
(1 is loaded regardless of the Rate setting)

11

Equipment Data Trend Message Trend Data

COP
Source PdM_Trend[TrendSendSmplFSC.POS].Array.Data[0]
Dest TrendMsg.Data.Trend[0]
Length 120

Equipment Data Trend Message Trend Scale

MOVE
Source PdM_Trend[TrendSendSmplFSC.POS].Array.Scale 0
Dest TrendMsg.Data.Scale 0

Equipment Data Trend Message Data Values Used (1-120)

MOVE
Source PdM_Trend[TrendSendSmplFSC.POS].Ctl.DataCount 0
Dest TrendMsg.Data.DataCount 0

Equipment Data Trend Message Trend Rate - samples per second (1-10)

MOVE
Source 1
Dest TrendMsg.Data.Rate 0

Exit Routine

12

lbl_Exit
LBL

NOP

(End)

Receive Equipment Version Unsolicited Message

Receive Equipment Version Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 260
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

2

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.VersionMsg.TransTMR.PRE 999999999

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.VersionMsg.TransTMR
Preset 999999999
Accum 948907
EN
DN

Copy Rx Buffer to local registers
DINT Array to UDT

3

Equipment Data Version Message Format

CPS
Source EDC_EqRxBuffer.Line[LNi].Equip[EQi].Version[0]
Dest VersionMsg
Length 1

Receive Equipment Version Unsolicited Message

No new data in Rx Buffer
Header & Footer Transaction Number =0

4

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

EQ
Source A VersionMsg.Header.TransNum
0
Source B 0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

EQ
Source A VersionMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

New data in Rx Buffer
Initialize Transaction timing

5

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.VersionMsg.PartialTMR.EN
/

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.VersionMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.VersionMsg.TransLast
0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source VersionMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2300
Input Par Equip.CTL.VersionMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par VersionMsg.Header.TransNum

Receive Equipment Version Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

6

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

EQ
Source A VersionMsg.Header.TransNum
0
Source B 0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

NE
Source A VersionMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.VersionMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.VersionMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

NE
Source A VersionMsg.Header.TransNum
0
Source B 0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

EQ
Source A VersionMsg.Footer.TransNum
0
Source B 0

Partial message in Rx Buffer
Wait for rest of message to arrive

7

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.VersionMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.VersionMsg.PartialTMR.DN

lbl_Exit
JMP

Receive Equipment Version Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.VersionMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.VersionMsg.TransReset
L

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	VersionMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2305
Input Par	Equip.CTL.VersionMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	VersionMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Version Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

9

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

NE	
Source A	VersionMsg.Header.TransNum
	0
Source B	0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

NE	
Source A	VersionMsg.Footer.TransNum
	0
Source B	0

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

NE	
Source A	VersionMsg.Header.TransNum
	0
Source B	VersionMsg.Footer.TransNum
	0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.VersionMsg.TransReset

(L)

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	VersionMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2306
Input Par	Equip.CTL.VersionMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	VersionMsg.Header.TransNum

lbl_MsgReset

(JMP)

Receive Equipment Version Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

10

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

NE	
Source A	VersionMsg.Header.TransNum
	0
Source B	0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

NE	
Source A	VersionMsg.Footer.TransNum
	0
Source B	0

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

EQ	
Source A	VersionMsg.Header.TransNum
	0
Source B	VersionMsg.Footer.TransNum
	0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received

Equip.CTL.VersionMsg.TransRcvd

L

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	VersionMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2301
Input Par	Equip.CTL.VersionMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	VersionMsg.Header.TransNum

Receive Equipment Version Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

11

Local Copy of Equip Array Tag for LNi, EQi Transaction Received: Equip.CTL.VersionMsg.TransRcvd

Expected Equipment ID

CPT
Dest: ExpectedEQID 1006
Expression: LNi*100+EQi

Equipment Data Version Message Format Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE
Source A: VersionMsg.Header.EQID 0
Source B: ExpectedEQID 1006

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: VersionMsg.Header.EQID 0
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2307
Input Par: Equip.CTL.VersionMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: VersionMsg.Header.TransNum

lbl_MsgReset
JMP

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

12

Local Copy of Equip Array Tag for LNi, EQi Transaction Received: Equip.CTL.VersionMsg.TransRcvd

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

EQ
Source A: VersionMsg.Header.TransNum 0
Source B: Equip.CTL.VersionMsg.TransNumLast 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: Equip.CTL.VersionMsg.TransNumLast 0
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2302
Input Par: Equip.CTL.VersionMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: VersionMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Version Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.VersionMsg.TransRcvd

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

NE	
Source A	VersionMsg.Header.TransNum 0
Source B	Equip.CTL.VersionMsg.TransNumNext 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.VersionMsg.TransNumNext 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2309
Input Par	Equip.CTL.VersionMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	VersionMsg.Header.TransNum

Load new message in Version Table

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.VersionMsg.TransRcvd

Equipment Data Lines

COP	
Source	VersionMsg
Dest	EDC_EqRxData.Line[LNi].Equip[EQi].Version
Length	1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2311
Input Par	Equip.CTL.VersionMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	VersionMsg.Header.TransNum

Receive Equipment Version Unsolicited Message

Prepare for next message

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.VersionMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number

MOVE
Source VersionMsg.Header.TransNum 0
Dest Equip.CTL.VersionMsg.TransNumLast 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

ADD
Source A VersionMsg.Header.TransNum 0
Source B 1
Dest Equip.CTL.VersionMsg.TransNumNext 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

LT
Source A Equip.CTL.VersionMsg.TransNumNext 0
Source B 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

GT
Source A Equip.CTL.VersionMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

MOVE
Source 1
Dest Equip.CTL.VersionMsg.TransNumNext 0

Receive Equipment Version Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)

16

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.VersionMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.VersionMsg.TransReset

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.VersionMsg.Timeout

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.VersionMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.VersionMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.VersionMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.VersionMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.VersionMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.VersionMsg.TransLast
0

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBuffer.Line[LNi].Equip[EQi].Version[0]
Length CFG.VersionMSG.n
<EDC.CFG.VersionMSG.n>

Exit Routine

17

lbl_Exit
LBL

NOP

(End)

Receive Equipment Cycle Unsolicited Message

Receive Equipment Quality/Health Unsolicited Message

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	270
Dest	Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1348352226
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1348352226

Receive Equipment Cycle Unsolicited Message

Transaction Timer
Free-running Timer for Performance Timing and Transaction Log

2

Local Copy of Equip Array Tag for LNi, EQi Cycle Message Enabled (from Driver Callout Msg)
Equip.CFG.CycleMsgEN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

MOVE
Source 999999999
Dest Equip.CTL.CycleMsg.TransTMR.PRE 0

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

TON
Timer Equip.CTL.CycleMsg.TransTMR
Preset 0
Accum 0
EN
DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer

GT
Source A Equip.CTL.CycleMsg.TransTMR.ACC 0
Source B CFG.BufferMsgTimeout <EDC.CFG.BufferMsgTimeout> 330000

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CycleMsg.Timeout
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Cycle Rx Msg ER or Seq/EQID ER
EquipStatus01.Update.CycleRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate
Status EquipStatus01

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2504
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par 0

Diagnostic

MOVE
Source LNi 27
Dest CycleLNi 10

Diagnostic

MOVE
Source EQi 26
Dest CycleEQi 14

Equipment Data Cycle Message

FLL
Source 0
Dest CycleMsg
Length 1

lbl_MsgReset
JMP

Copy Rx Buffer to local registers
DINT Array to UDT

3

Equipment Data Cycle Message

CPS
Source EDC_EqRxBufferAux.Line[LNi].Equip[EQi].Cycle[0]
Dest CycleMsg
Length 1

Receive Equipment Cycle Unsolicited Message

No new data in Rx Buffer
Header & Footer Transaction Number =0

4

Equipment Data
Cycle Message Header
- Transaction Number
(must match Footer)

EQ
Source A CycleMsg.Header.TransNum
0
Source B 0

Equipment Data
Cycle Message Footer
- Transaction Number
(must match Footer)

EQ
Source A CycleMsg.Footer.TransNum
0
Source B 0

lbl_Exit
JMP

New data in Rx Buffer
Initialize Transaction timing

5

Local Copy of Equip
Array Tag for LNi,
EQi Partial
MessageTimer
Equip.CTL.CycleMsg.PartialTMR.EN
/

Local Copy of Equip
Array Tag for LNi,
EQi Transaction
Timer
Equip.CTL.CycleMsg.TransTMR
RES

Local Copy of Equip
Array Tag for LNi,
EQi Transaction
Timer Accum Last
(used to calculate
differential)

CLR
Dest Equip.CTL.CycleMsg.TransLast
0

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source CycleMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance
Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.EN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2500
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

Receive Equipment Cycle Unsolicited Message

Check for partial message in Rx Buffer
Header or Footer Transaction Number <>0

6

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

EQ
Source A CycleMsg.Header.TransNum
0
Source B 0

Equipment Data Cycle Message Footer - Transaction Number (must match Footer)

NE
Source A CycleMsg.Footer.TransNum
0
Source B 0

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE
Source A CycleMsg.Header.TransNum
0
Source B 0

Equipment Data Cycle Message Footer - Transaction Number (must match Footer)

EQ
Source A CycleMsg.Footer.TransNum
0
Source B 0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

MOVE
Source CFG.PartialMsgTimeout
<EDC.CFG.PartialMsgTimeout>
10000
Dest Equip.CTL.CycleMsg.PartialTMR.PRE
0

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer

TON
Timer Equip.CTL.CycleMsg.PartialTMR
Preset 0
Accum 0
EN
DN

Partial message in Rx Buffer
Wait for rest of message to arrive

7

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CycleMsg.PartialTMR.EN

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CycleMsg.PartialTMR.DN

lbl_Exit
JMP

Receive Equipment Cycle Unsolicited Message

Partial message in Rx Buffer timeout
Clear Rx Buffer so Equipment can send another message
Discard Data

8

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CycleMsg.PartialTMR.DN

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CycleMsg.TransReset
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Cycle Rx Msg ER or Seq/EQID ER
EquipStatus01.Update.CycleRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate

aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	CycleMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Performance Transaction

aoiEDC_PerfTrans

aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CycleMsg
PerfTrans	EquipStatus04.CycleRx.ER

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	2505
Input Par	Equip.CTL.CycleMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CycleMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Cycle Unsolicited Message

Check for new message in Rx Buffer w/ Header/Footer mismatch
Header & Footer Transaction Number <>0 and <>
Exit without loading Ring Buffer if mismatch

9

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE	
Source A	CycleMsg.Header.TransNum
	0
Source B	0

Equipment Data Cycle Message Footer - Transaction Number (must match Footer)

NE	
Source A	CycleMsg.Footer.TransNum
	0
Source B	0

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE	
Source A	CycleMsg.Header.TransNum
	0
Source B	CycleMsg.Footer.TransNum
	0

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset

Equip.CTL.CycleMsg.TransReset (L)

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Cycle Rx Msg ER or Seq/EQID ER

EquipStatus01.Update.CycleRxER (L)

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CycleMsg.Footer.TransNum
	0
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CycleMsg
PerfTrans	EquipStatus04.CycleRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2506
Input Par	Equip.CTL.CycleMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CycleMsg.Header.TransNum

lbl_MsgReset (JMP)

Receive Equipment Cycle Unsolicited Message

Check for new message in Rx Buffer
Header & Footer Transaction Number <>0 and match

10

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE
Source A CycleMsg.Header.TransNum
0
Source B 0
0

Equipment Data Cycle Message Footer - Transaction Number (must match Footer)

NE
Source A CycleMsg.Footer.TransNum
0
Source B 0
0

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

EQ
Source A CycleMsg.Header.TransNum
0
Source B CycleMsg.Footer.TransNum
0

Local Copy of Equip Array Tag for LNi, EQi Transaction Received

Equip.CTL.CycleMsg.TransRcvd
L

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Cycle Rx DN

EquipStatus01.Update.CycleRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleMsg.Footer.TransNum
0
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.DN

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2501
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

Receive Equipment Cycle Unsolicited Message

Check for Equipment ID msimatch with LNi, EQi
Equipment configuration determines which RxBuffer LNi, EQi is loaded
Exit without loading Ring Buffer if mismatch

11

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

Expected Equipment ID

CPT
Dest ExpectedEQID 1006
Expression LNi*100+EQi

Equipment Data Cycle Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE
Source A CycleMsg.Header.EQID 0
Source B ExpectedEQID 1006

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Cycle Rx Msg ER or Seq/EQID ER
EquipStatus01.Update.CycleRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleMsg.Header.EQID 0
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2507
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Cycle Unsolicited Message

Check new message for duplicate Transaction Number
Exit without loading Ring Buffer if duplicate
(probably Keep Alive from EQ - no need to pass these to server)

12

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

EQ
Source A: CycleMsg.Header.TransNum 0
Source B: Equip.CTL.CycleMsg.TransNumLast 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Green - Cycle Rx DN
EquipStatus01.Update.CycleRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate ...
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: Equip.CTL.CycleMsg.TransNumLast 0
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans ...
PerfMsg: Equip.CTL.CycleMsg
PerfTrans: EquipStatus04.CycleRx.KA

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2502
Input Par: Equip.CTL.CycleMsg.TransTMR.ACC
Input Par: LNi
Input Par: EQi
Input Par: CycleMsg.Header.TransNum

lbl_MsgReset
JMP

Receive Equipment Cycle Unsolicited Message

Check new message for Transaction Number Sequence Error
expecting increment
Load date in Ring Buffer in either case
(otherwise this will never recover from a sequence error)

13

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE	
Source A	CycleMsg.Header.TransNum 0
Source B	Equip.CTL.CycleMsg.TransNumLast 0

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

NE	
Source A	CycleMsg.Header.TransNum 0
Source B	Equip.CTL.CycleMsg.TransNumNext 0

Local Copy of Equip Array Status Tag for LNi, EQi State = Red - Cycle Rx Msg ER or Seq/EQID ER
EquipStatus01.Update.CycleRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Equip.CTL.CycleMsg.TransNumNext 0
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	Equip.CTL.CycleMsg
PerfTrans	EquipStatus04.CycleRx.ER

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2509
Input Par	Equip.CTL.CycleMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CycleMsg.Header.TransNum

Diagnostic

ADD	
Source A	CycleSeqER 1953
Source B	1
Dest	CycleSeqER 1953

Diagnostic

MOVE	
Source	LNi 27
Dest	CycleSeqER_LNi 11

Diagnostic

MOVE	
Source	EQi 26
Dest	CycleSeqER_EQi 32

Diagnostic

MOVE	
Source	CycleMsg.Header.TransNum 0
Dest	CycleSeqER_TransNum 2163

Diagnostic

MOVE	
Source	Equip.CTL.CycleMsg.TransNumNext 0
Dest	CycleSeqER_TransNumNext 2152

Receive Equipment Cycle Unsolicited Message

Load new message in Ring Buffer

Sep 2023 - change RB channel allocation to Line Number Odd/Even (better distribution than splitting Line Numbers into Low/High range)
Nov 2024 v3.01 - change RB channel allocation to modulo (LNi + EQi + ToolNum) to distribute Tools from same Equipment to different buffers

14

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

ADD
Source A: LNi 27
Source B: EQi 26
Dest: LnEqSum 16

ADD
Source A: LnEqSum 16
Source B: CycleMsg.Data.SeqNum 0
Dest: LnEqSeqSum 27

RingBuffer Modulo Result
MOD
Source A: LnEqSeqSum 27
Source B: 2
Dest: BufferID 3

RingBuffer Modulo Result
EQ
Source A: BufferID 3
Source B: 1

UseCycleRB01

Load Equipment Cycle Message into Ring Buffer
JSR
Routine Name: _271_EquipCycleRxLoadRB01

UseCycleRB01 (/)

UseCycleRB02

Load Equipment Cycle Message into Ring Buffer
JSR
Routine Name: _272_EquipCycleRxLoadRB02

Receive Equipment Cycle Unsolicited Message

Prepare for next message

15

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Receive Message Last Transaction Number

MOVE
Source CycleMsg.Header.TransNum 0
Dest Equip.CTL.CycleMsg.TransNumLast 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

ADD
Source A CycleMsg.Header.TransNum 0
Source B 1
Dest Equip.CTL.CycleMsg.TransNumNext 0

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

LT
Source A Equip.CTL.CycleMsg.TransNumNext 0
Source B 1

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

GT
Source A Equip.CTL.CycleMsg.TransNumNext 0
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Local Copy of Equip Array Tag for LNi, EQi Receive Message Next Transaction Number

MOVE
Source 1
Dest Equip.CTL.CycleMsg.TransNumNext 0

Receive Equipment Cycle Unsolicited Message

Transaction Reset or Received
Clear Rx Buffer so Equipment can send next Message
(Equipment checks Header Transaction Number = 0 for clear to send)

16

lbl_MsgReset
LBL

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CycleMsg.TransReset

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CycleMsg.Timeout

Local Copy of Equip Array Tag for LNi, EQi Transaction Reset
Equip.CTL.CycleMsg.TransReset
U

Local Copy of Equip Array Tag for LNi, EQi Transaction Received
Equip.CTL.CycleMsg.TransRcvd
U

Local Copy of Equip Array Tag for LNi, EQi Message Timeout
Equip.CTL.CycleMsg.Timeout
U

Local Copy of Equip Array Tag for LNi, EQi Partial MessageTimer
Equip.CTL.CycleMsg.PartialTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer
Equip.CTL.CycleMsg.TransTMR
RES

Local Copy of Equip Array Tag for LNi, EQi Transaction Timer Accum Last (used to calculate differential)

CLR
Dest Equip.CTL.CycleMsg.TransLast
0

Equipment Data Lines

COP
Source CycleMsg
Dest EDC_EqRxDataAux.Line[LNi].Equip[EQi].Cycle
Length 1

Equipment Data Buffer Lines

FLL
Source 0
Dest EDC_EqRxBufferAux.Line[LNi].Equip[EQi].Cycle[0]
Length CFG.CycleMSG.n
<EDC.CFG.CycleMSG.n>

Exit Routine

17

lbl_Exit
LBL

NOP

(End)

Load Equipment Cycle Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 271
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Cycle Message into Ring Buffer

2

EDC Ring Buffer Control Next Write Tag
LT
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag
GT
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B RB_Cycle01.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record
LT
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B RB_Cycle01.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB_Cycle01.WritePtrErr <EDC.RB_Cycle01.WritePtrErr>
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Dest RB_Cycle01.WritePtrErrTag <EDC.RB_Cycle01.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Dest RB_Cycle01.WritePtrErrRec <EDC.RB_Cycle01.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle01.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB_Cycle01.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2513
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

Load Equipment Cycle Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B RB_Cycle01.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B RB_Cycle01.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Cycle01.FullBlockWrite <EDC.RB_Cycle01.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Cycle01.WriteBlockErr <EDC.RB_Cycle01.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Cycle01.WriteBlockErrCnt <EDC.RB_Cycle01.WriteBlockErrCnt> 0
Source B 1
Dest RB_Cycle01.WriteBlockErrCnt <EDC.RB_Cycle01.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle01.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle01.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2514
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Cycle Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Cycle Message
EQ Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2 Source B 1	COP Source CycleMsg Dest EDC_RB_Cycle01_Rec01[RB_Cycle01.WriteRecPtr] Length 1
EQ Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2 Source B 2	COP Source CycleMsg Dest EDC_RB_Cycle01_Rec02[RB_Cycle01.WriteRecPtr] Length 1
EQ Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2 Source B 3	COP Source CycleMsg Dest EDC_RB_Cycle01_Rec03[RB_Cycle01.WriteRecPtr] Length 1
EQ Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2 Source B 4	COP Source CycleMsg Dest EDC_RB_Cycle01_Rec04[RB_Cycle01.WriteRecPtr] Length 1
EQ Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2 Source B 5	COP Source CycleMsg Dest EDC_RB_Cycle01_Rec05[RB_Cycle01.WriteRecPtr] Length 1

Load Equipment Cycle Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB_Cycle01.WriteTagPtr
	<EDC.RB_Cycle01.WriteTagPtr>
	2
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB_Cycle01.WriteRecPtr
	<EDC.RB_Cycle01.WriteRecPtr>
	547
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB_Cycle01.ReadTagPtr
	<EDC.RB_Cycle01.ReadTagPtr>
	2
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB_Cycle01.ReadRecPtr
	<EDC.RB_Cycle01.ReadRecPtr>
	546
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2511
Input Par	Equip.CTL.CycleMsg.TransTMR.ACC
Input Par	LNi
Input Par	EQi
Input Par	CycleMsg.Header.TransNum

Load Equipment Cycle Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B 1
Dest RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B RB_Cycle01.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Cycle01.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1
Dest RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B RB_Cycle01.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Cycle01.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

Load Equipment Cycle Message into Ring Buffer

Increment Buffer Count for Current Shift

7

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

ADD
Source A RB_Cycle01.ShiftCnt <EDC.RB_Cycle01.ShiftCnt> 7564
Source B 1
Dest RB_Cycle01.ShiftCnt <EDC.RB_Cycle01.ShiftCnt> 7564

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Load Equipment Cycle Message into Ring Buffer

Load Equipment Quality/Health Message into Ring Buffer

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 272
Dest Prog.Routine 225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1348352226
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1348352226

Load Equipment Cycle Message into Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B RB_Cycle02.MaxTagPtr <EDC.RB_Cycle02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B RB_Cycle02.MaxRecPtr <EDC.RB_Cycle02.MaxRecPtr> 999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Cycle02.WritePtrErr <EDC.RB_Cycle02.WritePtrErr>

(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Dest RB_Cycle02.WritePtrErrTag <EDC.RB_Cycle02.WritePtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Dest RB_Cycle02.WritePtrErrRec <EDC.RB_Cycle02.WritePtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2513
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437

Load Equipment Cycle Message into Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

3

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B RB_Cycle02.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4

EDC Ring Buffer Control Next Write Record

EQ
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B RB_Cycle02.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436

EDC Ring Buffer Control Buffer Full - Block Writes

RB_Cycle02.FullBlockWrite <EDC.RB_Cycle02.FullBlockWrite>

EDC Ring Buffer Control Diagnostic - Manual Reset

RB_Cycle02.WriteBlockErr <EDC.RB_Cycle02.WriteBlockErr>

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A RB_Cycle02.WriteBlockErrCnt <EDC.RB_Cycle02.WriteBlockErrCnt> 0
Source B 1
Dest RB_Cycle02.WriteBlockErrCnt <EDC.RB_Cycle02.WriteBlockErrCnt> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg Equip.CTL.CycleMsg
PerfTrans EquipStatus04.CycleRx.ER

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2514
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

lbl_Exit
JMP

Load Equipment Cycle Message into Ring Buffer

RB Write -- Load Message into Ring Buffer
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag	Equipment Data Cycle Message
EQ Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4 Source B 1	COP Source CycleMsg Dest EDC_RB_Cycle02_Rec01[RB_Cycle02.WriteRecPtr] 1 Length 1
EQ Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4 Source B 2	COP Source CycleMsg Dest EDC_RB_Cycle02_Rec02[RB_Cycle02.WriteRecPtr] 1 Length 1
EQ Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4 Source B 3	COP Source CycleMsg Dest EDC_RB_Cycle02_Rec03[RB_Cycle02.WriteRecPtr] 1 Length 1
EQ Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4 Source B 4	COP Source CycleMsg Dest EDC_RB_Cycle02_Rec04[RB_Cycle02.WriteRecPtr] 1 Length 1
EQ Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4 Source B 5	COP Source CycleMsg Dest EDC_RB_Cycle02_Rec05[RB_Cycle02.WriteRecPtr] 1 Length 1

Load Equipment Cycle Message into Ring Buffer

RB Write -- Log Transaction

5

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr>
4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.WriteRecPtr
<EDC.RB_Cycle02.WriteRecPtr>
437
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.ReadTagPtr
<EDC.RB_Cycle02.ReadTagPtr>
4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB_Cycle02.ReadRecPtr
<EDC.RB_Cycle02.ReadRecPtr>
436
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2511
Input Par Equip.CTL.CycleMsg.TransTMR.ACC
Input Par LNi
Input Par EQi
Input Par CycleMsg.Header.TransNum

Load Equipment Cycle Message into Ring Buffer

RB Write -- Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B 1
Dest RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437

EDC Ring Buffer Control Next Write Record

LT
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B RB_Cycle02.MaxRecPtr <EDC.RB_Cycle02.MaxRecPtr> 999

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB_Cycle02.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B 1
Dest RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B RB_Cycle02.MaxTagPtr <EDC.RB_Cycle02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB_Cycle02.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4

Load Equipment Cycle Message into Ring Buffer

Increment Buffer Count for Current Shift

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

7

ADD
Source A RB_Cycle02.ShiftCnt
<EDC.RB_Cycle02.ShiftCnt>
7368
Source B 1
Dest RB_Cycle02.ShiftCnt
<EDC.RB_Cycle02.ShiftCnt>
7368

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine
225

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901]
-1329175072
Source B 1
Dest Prog.RoutineScan[901]
-1329175072

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

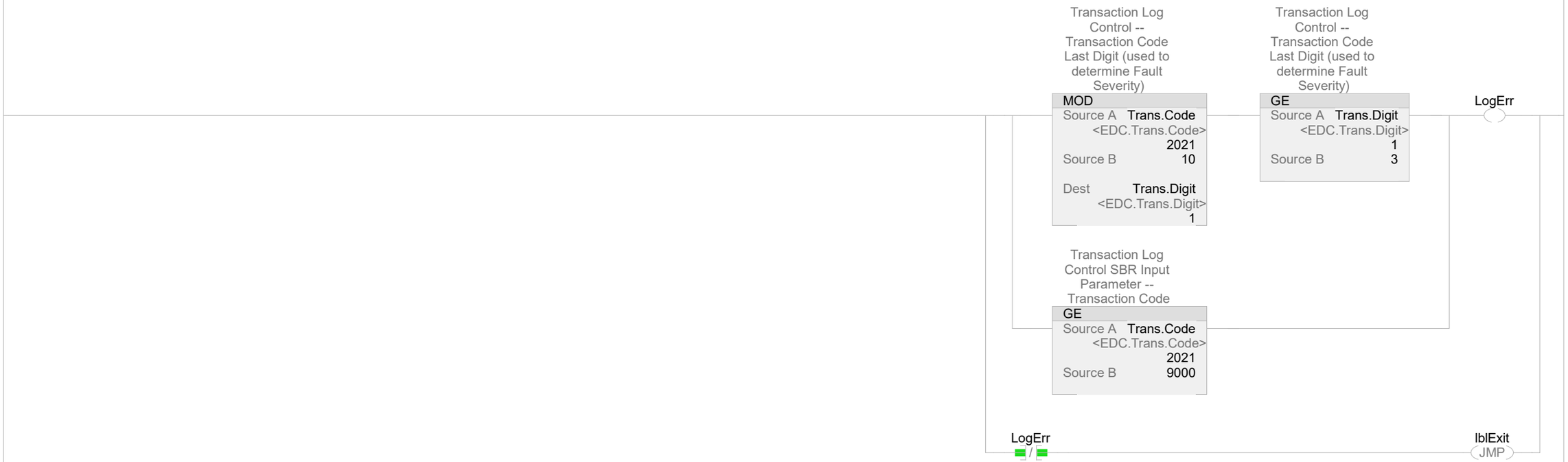
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Fault Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	301
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562848711
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562848711

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_311_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _320_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Fault01.ShiftCnt>
69240
Dest RB.PrevShiftCnt
<EDC.RB_Fault01.PrevShiftCnt>
85089

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Fault01.ShiftCnt>
69240

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 15
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.3
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 4000

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 4000
Source B	1
Dest	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 0
Source B	1
Dest	Prog.LOGi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 0
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 0
Source B	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 0

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
320
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr>
1
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault01.WriteRecPtr>
1271
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr>
1
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault01.ReadRecPtr>
1270
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 302
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562848711
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562848711

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 311
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562848711
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562848711

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Fault/Status Message

SIZE
Source EDC_RB_Fault01_Rec01[0] ??
Dim. To Vary 0
Size LEN 4000

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 4000
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Fault01.MaxRecPtr> 3999

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 3
Dest RB.MaxTagPtr <EDC.RB_Fault01.MaxTagPtr> 3

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A: RB.TransactionNumber <EDC.RB_Fault01.TransactionNumber> 284
Source B: 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source: 1
Dest: RB.TransactionNumber <EDC.RB_Fault01.TransactionNumber> 284

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Fault01.WritePtrErr>
U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Fault01.ReadPtrErr>
U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 320
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562848711
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562848711

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault01.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Source B RB.MaxRecPtr <EDC.RB_Fault01.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Fault01.ReadPtrErr>
(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Dest RB.ReadPtrErrTag <EDC.RB_Fault01.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Dest RB.ReadPtrErrRec <EDC.RB_Fault01.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2023
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Fault01.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Fault01.MaxTagPtr> 3
Dest RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Fault01.MaxRecPtr> 3999
Dest RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Fault01.ReadNextRecPtr> 1271
Source B RB.MaxRecPtr <EDC.RB_Fault01.MaxRecPtr> 3999

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault01.ReadNextTagPtr> 1

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Fault01.ReadNextRecPtr> 1271

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Fault01.ReadNextTagPtr> 1
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault01.ReadNextTagPtr> 1

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Fault01.ReadNextTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault01.MaxTagPtr> 3

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Fault01.ReadNextRecPtr> 1271

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault01.ReadNextTagPtr> 1

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Source B RB.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Source B RB.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Fault01.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault01.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Fault01.ReadNextTagPtr> 1
Source B RB.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Fault01.ReadNextRecPtr> 1271
Source B RB.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault01.Empty>

Buffer Pointers
Combine Tag & Record Pointers

5

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Fault01.WritePtr> 1271
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Fault01.ReadPtr> 1270
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Fault01.ReadNextPtr> 1271
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Fault01.WritePtr> 1271
Source B RB.ReadPtr <EDC.RB_Fault01.ReadPtr> 1270

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault01.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Fault01.WritePtr> 1271
Source B RB.ReadPtr <EDC.RB_Fault01.ReadPtr> 1270

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Fault01.WritePtr> 1271
Source B RB.ReadNextPtr <EDC.RB_Fault01.ReadNextPtr> 1271

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault01.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Fault01.WritePtr> 1271
Source B RB.ReadNextPtr <EDC.RB_Fault01.ReadNextPtr> 1271

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault01.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Fault01.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

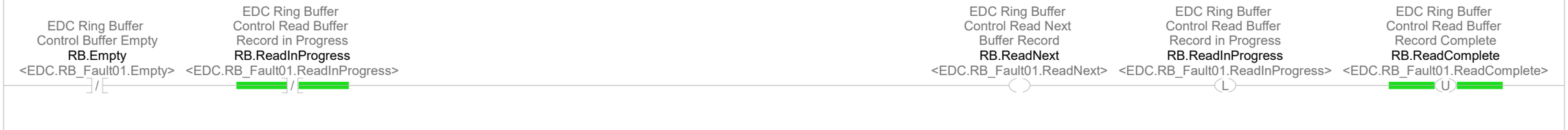
MOVE
Source RB.RecordCount <EDC.RB_Fault01.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Fault01.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

35 35

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault01.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Source B RB.MaxRecPtr <EDC.RB_Fault01.MaxRecPtr> 3999

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault01.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1

Process RingBuffer Reads from Transaction Manager

9

Read Next
Load Read Record (Tags 01-05)

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault01.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr>
Source B 1
1

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault01_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr>
Source B 2
1

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault01_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr>
Source B 3
1

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault01_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault01_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault01.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Fault01.TransactionNumber>
284
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Fault01.TransactionNumber>
284

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Fault01.TransactionNumber>
284
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Fault01.TransactionNumber>
284
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Fault01.TransactionNumber>
284

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault01.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Fault01_Read.Header.EQID>
119
Source B 100
Dest EQi
16

CPT
Dest LNi
19
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
19

CLR
Dest PerfEQi
16

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault01.ReadValidEQID>
U

LIMIT
Low Limit 0
Test LNi
19
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
16
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
19
Dest PerfLNi
19

MOVE
Source EQi
16
Dest PerfEQi
16

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault01.ReadValidEQID>
L

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault01.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus02
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Log Transaction & Performance

13

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault01.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault01.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault01.ReadRecPtr> 1270
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2020
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Fault01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.EN

Process RingBuffer Reads from Transaction Manager

14

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Fault01.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault01.TransactionNumber>
284
Source B RB.Interface
<EDC.RB_Fault01.Interface>
284

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault01.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Fault01.Interface>
284
Dest RB.InterfaceNEG
<EDC.RB_Fault01.InterfaceNEG>
-284

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault01.TransactionNumber>
284
Source B RB.InterfaceNEG
<EDC.RB_Fault01.InterfaceNEG>
-284

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Fault01.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

15

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault01.ReadComplete>

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN
/

EDC Ring Buffer
Control Read
Complete Delay Timer
MOVE
Source CfgRbDelaySlow 100
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Fault01.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer
Control Read
Complete Delay Timer
TON
Timer RB.ReadCompleteDelayTMR <EDC.RB_Fault01.ReadCompleteDelayTMR>
Preset 0
Accum 14
EN
DN

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN

EDC Ring Buffer
Control Read
Complete Delay Timer
MOVE
Source CfgRbDelayFast 0
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Fault01.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer
Control Read
Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Fault01.ReadCompleteDelayTMR.DN>

EDC Ring Buffer
Control Read Buffer
Record Complete
Delay
RB.ReadCompleteDelay
<EDC.RB_Fault01.ReadCompleteDelay>

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Fault01.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Fault01.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Complete 1-shot
RB.ReadCompleteONS
<EDC.RB_Fault01.ReadCompleteONS>
ONS

Read Complete Log Transaction & Performance

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Green - Fault Ring Buffer Tx DN
EquipStatus01.Update.FaultTxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault01.WriteRecPtr>
1271
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault01.ReadRecPtr>
1270
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2021
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Fault01.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Fault01.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Fault01.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault01.ReadResend>

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault01.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault01.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault01.ReadResend>

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault01.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer
TON
Timer RB.ReadInProgressTMR <EDC.RB_Fault01.ReadInProgressTMR>
Preset 5000
Accum 0
EN
DN

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Fault01.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Fault01.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault01.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Fault Tx Msg ER
EquipStatus01.Update.FaultTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault01.WriteRecPtr>
1271
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault01.ReadRecPtr>
1270
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2024
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Fault01.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Fault01.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Fault01.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault01.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Fault01.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Fault01.KeepAliveTMR>
Preset 300000
Accum 17509
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Fault01.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Fault01.KeepAlive>

JSR
Routine Name _321_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Fault01.KeepAliveTMR>
RES

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
Feb2017 - Changed to: Alarm if not below 1000 for 15 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Fault01.Empty>

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault01.RecordCount> 0
Source B 100

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MUL
Source A 60000
Source B 5
Dest RB.NotEmptyTMR.PRE <EDC.RB_Fault01.NotEmptyTMR.PRE> 300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR <EDC.RB_Fault01.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC <EDC.RB_Fault01.NotEmptyTMR.ACC> 0
Source B RB.NotEmptyMax <EDC.RB_Fault01.NotEmptyMax> 300007

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC <EDC.RB_Fault01.NotEmptyTMR.ACC> 0
Dest RB.NotEmptyMax <EDC.RB_Fault01.NotEmptyMax> 300007

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full RB.Full <EDC.RB_Fault01.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer RB.NotEmptyTMR.DN <EDC.RB_Fault01.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm RB.BufferAlarm <EDC.RB_Fault01.BufferAlarm>

Process RingBuffer Reads from Transaction Manager

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault01.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Event Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus02
Dest	EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	321
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562848711
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562848711

2

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr>
1
Source B 1

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault01_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr>
1
Source B 2

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault01_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr>
1
Source B 3

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault01_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Fault/Status Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum 7
Source B 1
Dest KeepAliveMsg.Header.CmdNum 7

Equipment Data Fault/Status Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum 7
Source B 1

Equipment Data Fault/Status Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum 7
Source B 9999

Equipment Data Fault/Status Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum 7

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr>
1
Source B 1

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault01_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr>
1
Source B 2

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault01_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr>
1
Source B 3

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault01_Rec03[RB.WriteRecPtr]
Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault01.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault01.WriteRecPtr>
1271
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault01.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault01.ReadRecPtr>
1270
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2022
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Fault01.KeepAliveTMR.ACC>
17509
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271
Source B RB.MaxRecPtr <EDC.RB_Fault01.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault01.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault01.WriteRecPtr> 1271

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault01.WriteTagPtr> 1

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 188439186
Source B 1
Dest Prog.RoutineScan[901] 188439186

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT	
Low Limit	Trans.Filter.CodeLow <EDC.Trans.Filter.CodeLow> 0
Test	Trans.Code <EDC.Trans.Code> 2021
High Limit	Trans.Filter.CodeHigh <EDC.Trans.Filter.CodeHigh> 0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT	
Low Limit	Trans.Filter.LNiLow <EDC.Trans.Filter.LNiLow> 0
Test	Trans.LNi <EDC.Trans.LNi> 4
High Limit	Trans.Filter.LNiHigh <EDC.Trans.Filter.LNiHigh> 0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT	
Low Limit	Trans.Filter.EQiLow <EDC.Trans.Filter.EQiLow> 0
Test	Trans.EQi <EDC.Trans.EQi> 21
High Limit	Trans.Filter.EQiHigh <EDC.Trans.Filter.EQiHigh> 0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

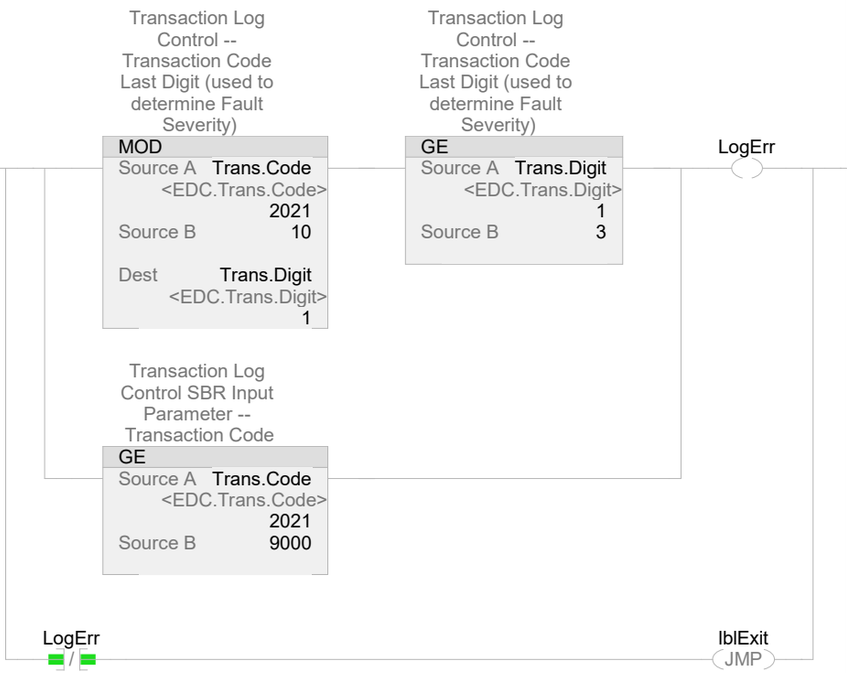
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Fault Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	301
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562847992
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562847992

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_311_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _320_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Fault02.ShiftCnt>
24334
Dest RB.PrevShiftCnt
<EDC.RB_Fault02.PrevShiftCnt>
25487

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Fault02.ShiftCnt>
24334

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 16
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.3
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

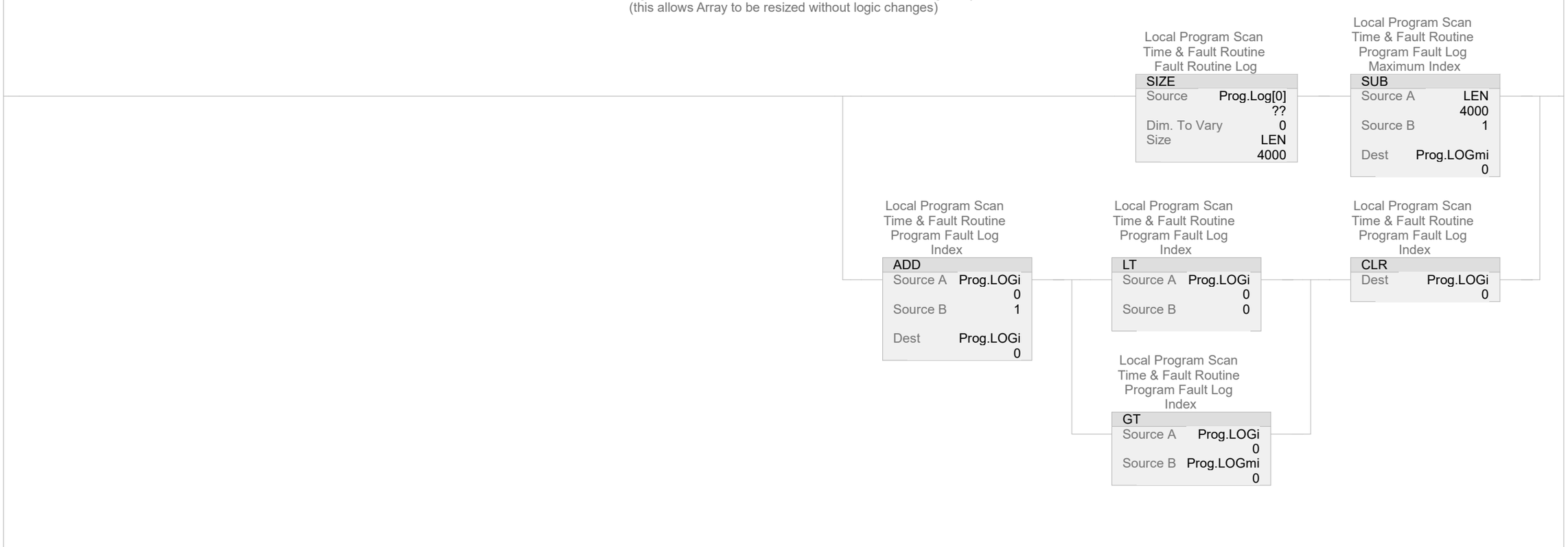
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
320
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr>
1
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault02.WriteRecPtr>
318
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr>
1
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault02.ReadRecPtr>
317
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 302
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562847992
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562847992

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 311
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562847992
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562847992

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Fault/Status Message

SIZE
Source EDC_RB_Fault02_Rec01[0] ??
Dim. To Vary 0
Size LEN 4000

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 4000
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Fault02.MaxRecPtr> 3999

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 3
Dest RB.MaxTagPtr <EDC.RB_Fault02.MaxTagPtr> 3

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A: RB.TransactionNumber <EDC.RB_Fault02.TransactionNumber> 2785
Source B: 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source: 1
Dest: RB.TransactionNumber <EDC.RB_Fault02.TransactionNumber> 2785

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Fault02.WritePtrErr>
U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Fault02.ReadPtrErr>
U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 320
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562847992
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562847992

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault02.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Source B RB.MaxRecPtr <EDC.RB_Fault02.MaxRecPtr> 3999

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Fault02.ReadPtrErr>

(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Dest RB.ReadPtrErrTag <EDC.RB_Fault02.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Dest RB.ReadPtrErrRec <EDC.RB_Fault02.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2023
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Fault02.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Fault02.MaxTagPtr> 3
Dest RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Fault02.MaxRecPtr> 3999
Dest RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault02.ReadNextTagPtr> 1

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Fault02.ReadNextRecPtr> 318

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Fault02.ReadNextRecPtr> 318
Source B RB.MaxRecPtr <EDC.RB_Fault02.MaxRecPtr> 3999

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Fault02.ReadNextRecPtr> 318

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Fault02.ReadNextTagPtr> 1
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault02.ReadNextTagPtr> 1

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Fault02.ReadNextTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault02.MaxTagPtr> 3

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault02.ReadNextTagPtr> 1

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Source B RB.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Source B RB.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Fault02.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault02.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Fault02.ReadNextTagPtr> 1
Source B RB.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Fault02.ReadNextRecPtr> 318
Source B RB.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault02.Empty>

Buffer Pointers
Combine Tag & Record Pointers

5

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Fault02.WritePtr> 318
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Fault02.ReadPtr> 317
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Fault02.ReadNextPtr> 318
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault02.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Fault02.WritePtr> 318
Source B RB.ReadPtr <EDC.RB_Fault02.ReadPtr> 317

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault02.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Fault02.WritePtr> 318
Source B RB.ReadPtr <EDC.RB_Fault02.ReadPtr> 317

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault02.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault02.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Fault02.WritePtr> 318
Source B RB.ReadNextPtr <EDC.RB_Fault02.ReadNextPtr> 318

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault02.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Fault02.WritePtr> 318
Source B RB.ReadNextPtr <EDC.RB_Fault02.ReadNextPtr> 318

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault02.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault02.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Fault02.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

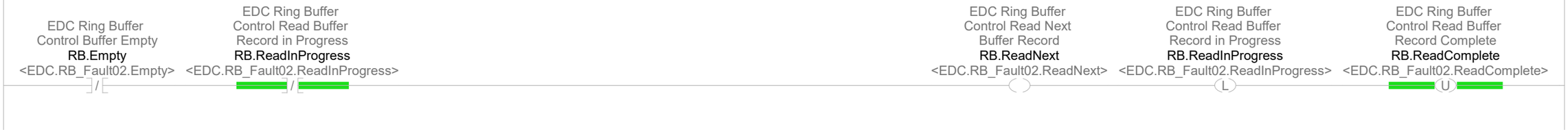
MOVE
Source RB.RecordCount <EDC.RB_Fault02.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Fault02.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

16 16

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault02.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Source B RB.MaxRecPtr <EDC.RB_Fault02.MaxRecPtr> 3999

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault02.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1

Process RingBuffer Reads from Transaction Manager

9

Read Next
Load Read Record (Tags 01-05)

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault02.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr>
Source B 1
1

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault02_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr>
Source B 2
1

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault02_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr>
Source B 3
1

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault02_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault02_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault02.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault02.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Fault02.TransactionNumber>
2785
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Fault02.TransactionNumber>
2785

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Fault02.TransactionNumber>
2785
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Fault02.TransactionNumber>
2785
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Fault02.TransactionNumber>
2785

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault02.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault02.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Fault02_Read.Header.EQID>
220
Source B 100
Dest EQi
31

CPT
Dest LNi
2
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
2

CLR
Dest PerfEQi
31

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault02.ReadValidEQID>
U

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault02.ReadValidEQID>
L

LIMIT
Low Limit 0
Test LNi
2
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
31
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
2
Dest PerfLNi
2

MOVE
Source EQi
31
Dest PerfEQi
31

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault02.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus02
Length 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault02.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault02.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault02.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault02.ReadRecPtr> 317
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2020
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Fault02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.EN

Process RingBuffer Reads from Transaction Manager

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

14

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Fault02.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault02.TransactionNumber>
2785
Source B RB.Interface
<EDC.RB_Fault02.Interface>
2785

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault02.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Fault02.Interface>
2785
Dest RB.InterfaceNEG
<EDC.RB_Fault02.InterfaceNEG>
-2785

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault02.TransactionNumber>
2785
Source B RB.InterfaceNEG
<EDC.RB_Fault02.InterfaceNEG>
-2785

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Fault02.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

15

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Fault02.ReadComplete>

Server Status TM Transaction OK Enable High Speed RB ServerStat_TMokTMR.DN

MOVE
Source CfgRbDelaySlow 100
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Fault02.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR <EDC.RB_Fault02.ReadCompleteDelayTMR>
Preset 0
Accum 16
EN
DN

Server Status TM Transaction OK Enable High Speed RB ServerStat_TMokTMR.DN

MOVE
Source CfgRbDelayFast 0
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Fault02.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer RB.ReadCompleteDelayTMR.DN <EDC.RB_Fault02.ReadCompleteDelayTMR.DN>

EDC Ring Buffer Control Read Buffer Record Complete Delay RB.ReadCompleteDelay <EDC.RB_Fault02.ReadCompleteDelay>

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Fault02.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault02.ReadComplete>

EDC Ring Buffer
Control Read Buffer
Record Complete
1-shot
RB.ReadCompleteONS
<EDC.RB_Fault02.ReadCompleteONS>
ONS

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault02.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi State = Green -
Fault Ring Buffer Tx
DN
EquipStatus01.Update.FaultTxOK
L

Update Status State
for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault02.WriteRecPtr>
318
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault02.ReadRecPtr>
317
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2021
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault02.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault02.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Fault02.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction
aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Fault02.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Fault02.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault02.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault02.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault02.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault02.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault02.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer

TON
Timer RB.ReadInProgressTMR <EDC.RB_Fault02.ReadInProgressTMR> EN
Preset 5000 DN
Accum 0

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Fault02.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Fault02.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault02.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault02.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Fault Tx Msg ER
EquipStatus01.Update.FaultTxER
L

Update Status State for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault02.WriteRecPtr>
318
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault02.ReadRecPtr>
317
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2024
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Fault02.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Fault02.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Fault02.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault02.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Fault02.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Fault02.KeepAliveTMR>
Preset 300000
Accum 6691
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Fault02.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Fault02.KeepAlive>

JSR
Routine Name _321_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Fault02.KeepAliveTMR>
RES

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
Feb2017 - Changed to: Alarm if not below 1000 for 15 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Fault02.Empty>

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault02.RecordCount> 0
Source B 100

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MUL
Source A 60000
Source B 5
Dest RB.NotEmptyTMR.PRE <EDC.RB_Fault02.NotEmptyTMR.PRE> 0

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR <EDC.RB_Fault02.NotEmptyTMR>
Preset 0
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC <EDC.RB_Fault02.NotEmptyTMR.ACC> 0
Source B RB.NotEmptyMax <EDC.RB_Fault02.NotEmptyMax> 0

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC <EDC.RB_Fault02.NotEmptyTMR.ACC> 0
Dest RB.NotEmptyMax <EDC.RB_Fault02.NotEmptyMax> 0

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full RB.Full <EDC.RB_Fault02.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer RB.NotEmptyTMR.DN <EDC.RB_Fault02.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm RB.BufferAlarm <EDC.RB_Fault02.BufferAlarm>

Process RingBuffer Reads from Transaction Manager

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault02.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP
Source EquipStatus01
Dest EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length 1

Event Rx/Tx Status Lines and Equipment Array

COP
Source EquipStatus02
Dest EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Length 1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	321
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562847992
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562847992

2

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr>
1
Source B 1

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault02_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr>
1
Source B 2

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault02_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr>
1
Source B 3

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault02_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Fault/Status Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1
Dest KeepAliveMsg.Header.CmdNum 1

Equipment Data Fault/Status Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1

Equipment Data Fault/Status Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 9999

Equipment Data Fault/Status Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum 1

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr> 1
Source B 1

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault02_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr> 1
Source B 2

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault02_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr> 1
Source B 3

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault02_Rec03[RB.WriteRecPtr]
Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault02.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault02.WriteRecPtr>
318
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault02.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault02.ReadRecPtr>
317
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2022
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

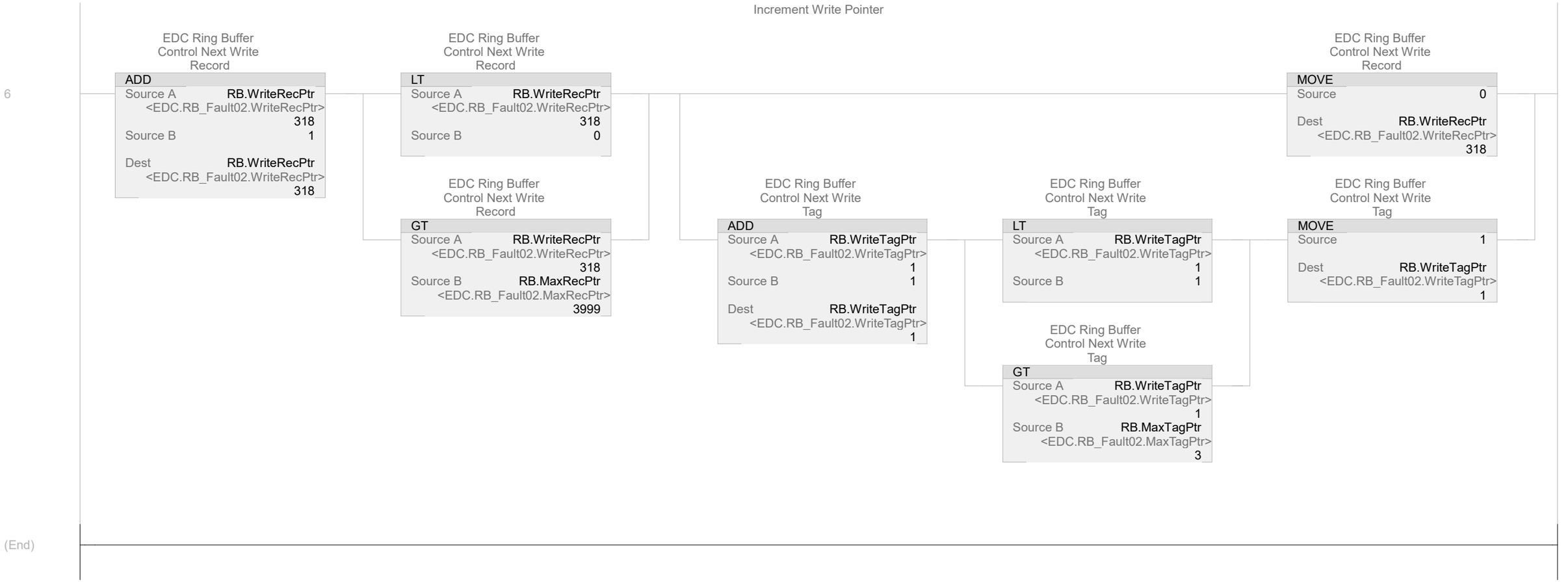
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Fault02.KeepAliveTMR.ACC>
6691
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.KA


Increment Write Pointer
6
EDC Ring Buffer Control Next Write Record
ADD
Source A RB.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
EDC Ring Buffer Control Next Write Record
LT
Source A RB.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B 0
EDC Ring Buffer Control Next Write Record
GT
Source A RB.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
Source B RB.MaxRecPtr <EDC.RB_Fault02.MaxRecPtr> 3999
EDC Ring Buffer Control Next Write Tag
ADD
Source A RB.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
EDC Ring Buffer Control Next Write Tag
LT
Source A RB.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B 1
EDC Ring Buffer Control Next Write Tag
GT
Source A RB.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault02.MaxTagPtr> 3
EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault02.WriteRecPtr> 318
EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault02.WriteTagPtr> 1
(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 175303045
Source B 1
Dest Prog.RoutineScan[901] 175303045

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

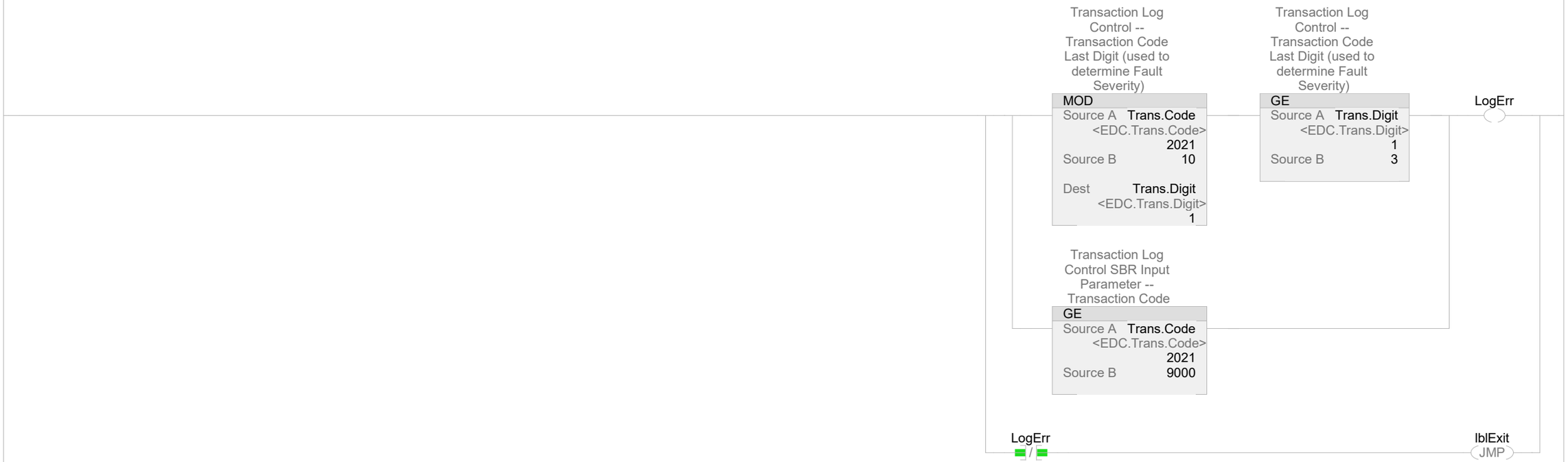
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Fault Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	301
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562847274
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562847274

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

Configuration

JSR	
Routine Name	_311_Config

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _320_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Fault03.ShiftCnt>
147189
Dest RB.PrevShiftCnt
<EDC.RB_Fault03.PrevShiftCnt>
172357

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Fault03.ShiftCnt>
147189

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 14
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.3
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

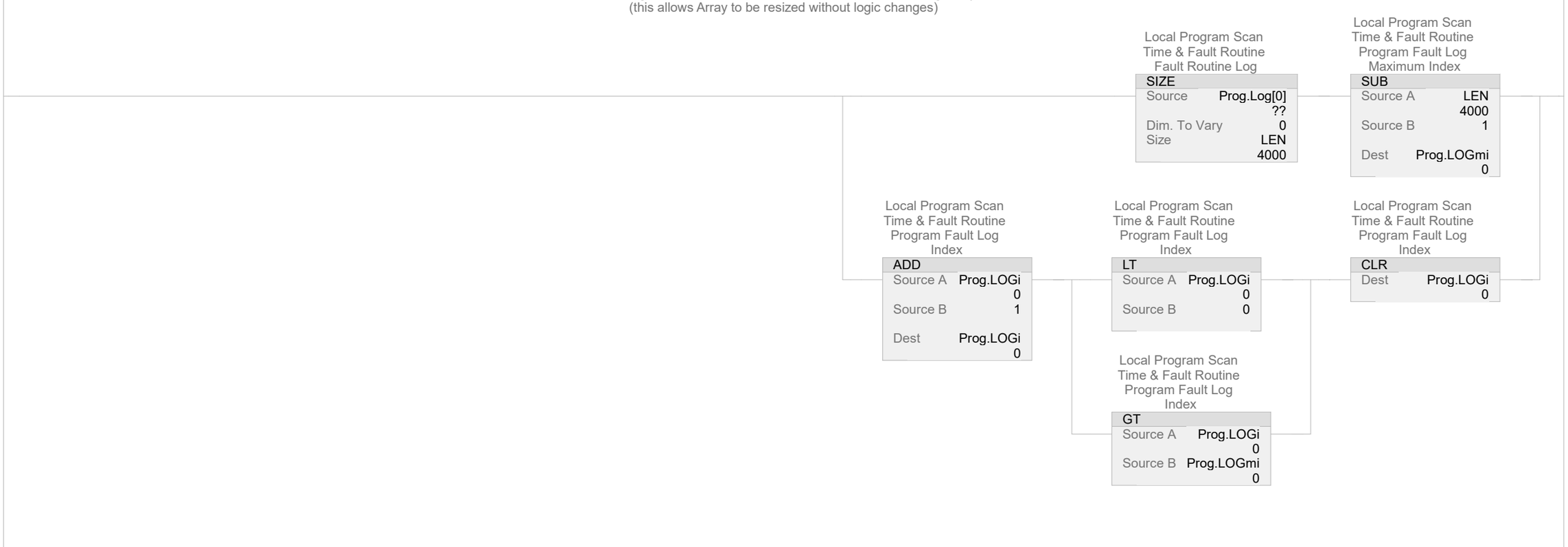
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
320
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr>
2
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault03.WriteRecPtr>
3240
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr>
2
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault03.ReadRecPtr>
3239
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 302
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562847274
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562847274

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>
/

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 311
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562847274
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562847274

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Fault/Status Message

SIZE
Source EDC_RB_Fault03_Rec01[0] ??
Dim. To Vary 0
Size LEN 4000

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 4000
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Fault03.MaxRecPtr> 3999

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 3
Dest RB.MaxTagPtr <EDC.RB_Fault03.MaxTagPtr> 3

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A RB.TransactionNumber <EDC.RB_Fault03.TransactionNumber> 5071
Source B 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber <EDC.RB_Fault03.TransactionNumber> 5071

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Fault03.WritePtrErr>

U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Fault03.ReadPtrErr>

U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 320
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562847274
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562847274

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Fault03.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Source B RB.MaxRecPtr <EDC.RB_Fault03.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags
EDC Ring Buffer Control Diagnostic - Manual Reset
RB.ReadPtrErr <EDC.RB_Fault03.ReadPtrErr>
(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Dest RB.ReadPtrErrTag <EDC.RB_Fault03.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value
MOVE
Source RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Dest RB.ReadPtrErrRec <EDC.RB_Fault03.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2023
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Fault03.TransactionNumber>

EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source RB.MaxTagPtr <EDC.RB_Fault03.MaxTagPtr> 3
Dest RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source RB.MaxRecPtr <EDC.RB_Fault03.MaxRecPtr> 3999
Dest RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Fault03.ReadNextRecPtr> 3240
Source B RB.MaxRecPtr <EDC.RB_Fault03.MaxRecPtr> 3999

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Dest RB.ReadNextTagPtr <EDC.RB_Fault03.ReadNextTagPtr> 2

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Fault03.ReadNextRecPtr> 3240

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Fault03.ReadNextTagPtr> 2
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault03.ReadNextTagPtr> 2

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Fault03.ReadNextTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Fault03.MaxTagPtr> 3

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Fault03.ReadNextRecPtr> 3240

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault03.ReadNextTagPtr> 2

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Source B RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Source B RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Fault03.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault03.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Fault03.ReadNextTagPtr> 2
Source B RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Fault03.ReadNextRecPtr> 3240
Source B RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault03.Empty>

Buffer Pointers
Combine Tag & Record Pointers

5

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Fault03.WritePtr> 7240
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Fault03.ReadPtr> 7239
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Fault03.ReadNextPtr> 7240
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault03.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Fault03.WritePtr> 7240
Source B RB.ReadPtr <EDC.RB_Fault03.ReadPtr> 7239

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault03.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Fault03.WritePtr> 7240
Source B RB.ReadPtr <EDC.RB_Fault03.ReadPtr> 7239

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault03.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault03.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Fault03.WritePtr> 7240
Source B RB.ReadNextPtr <EDC.RB_Fault03.ReadNextPtr> 7240

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault03.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Fault03.WritePtr> 7240
Source B RB.ReadNextPtr <EDC.RB_Fault03.ReadNextPtr> 7240

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault03.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault03.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Fault03.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

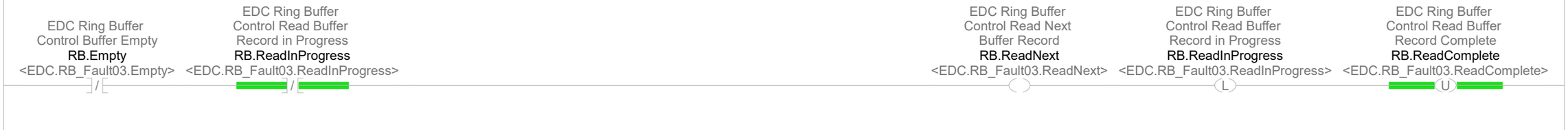
MOVE
Source RB.RecordCount <EDC.RB_Fault03.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Fault03.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

111 111

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault03.ReadNext>

EDC Ring Buffer Control Current/Last Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239
Source B RB.MaxRecPtr <EDC.RB_Fault03.MaxRecPtr> 3999

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr> 3239

EDC Ring Buffer Control Current/Last Read Tag

ADD
Source A RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Fault03.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr> 2

Process RingBuffer Reads from Transaction Manager

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault03.ReadNext>

Read Next
Load Read Record (Tags 01-05)

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr>
Source B 2
1

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault03_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault03_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr>
Source B 2
2

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault03_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault03_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr>
Source B 2
3

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault03_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault03_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault03.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault03.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Fault03.TransactionNumber>
5071
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Fault03.TransactionNumber>
5071

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Fault03.TransactionNumber>
5071
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Fault03.TransactionNumber>
5071
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Fault03.TransactionNumber>
5071

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault03.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault03.ReadInProgressTimeout>

MOD
Source A: RB_Read.Header.EQID <EDC_RB_Fault03_Read.Header.EQID> 951
Source B: 100
Dest: EQi 35

CPT
Dest: LNi 9
Expression: (RB_Read.Header.EQID-EQi)/100

CLR
Dest: PerfLNi 9

CLR
Dest: PerfEQi 35

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault03.ReadValidEQID>
U

LIMIT
Low Limit: 0
Test: LNi 9
High Limit: CFG.LN.mi <EDC.CFG.LN.mi> 28

LIMIT
Low Limit: 0
Test: EQi 35
High Limit: CFG.EQ.mi <EDC.CFG.EQ.mi> 99

MOVE
Source: LNi 9
Dest: PerfLNi 9

MOVE
Source: EQi 35
Dest: PerfEQi 35

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault03.ReadValidEQID>
L

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault03.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi

COP
Source: EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest: EquipStatus01
Length: 1

Equipment Data Collection - Status Line Equipment

COP
Source: EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Dest: EquipStatus02
Length: 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault03.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault03.ReadInProgressTimeout>

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault03.WriteRecPtr>
3240
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault03.ReadRecPtr>
3239
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2020
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault03.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault03.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.EN

Process RingBuffer Reads from Transaction Manager

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

14

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Fault03.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault03.TransactionNumber>
5071
Source B RB.Interface
<EDC.RB_Fault03.Interface>
5071

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault03.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Fault03.Interface>
5071
Dest RB.InterfaceNEG
<EDC.RB_Fault03.InterfaceNEG>
-5071

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault03.TransactionNumber>
5071
Source B RB.InterfaceNEG
<EDC.RB_Fault03.InterfaceNEG>
-5071

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Fault03.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

15

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault03.ReadComplete>

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN
/

EDC Ring Buffer
Control Read
Complete Delay Timer
MOVE
Source CfgRbDelaySlow 100
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Fault03.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer
Control Read
Complete Delay Timer
TON
Timer RB.ReadCompleteDelayTMR <EDC.RB_Fault03.ReadCompleteDelayTMR>
Preset 0
Accum 15
EN
DN

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN

EDC Ring Buffer
Control Read
Complete Delay Timer
MOVE
Source CfgRbDelayFast 0
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Fault03.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer
Control Read
Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Fault03.ReadCompleteDelayTMR.DN>

EDC Ring Buffer
Control Read Buffer
Record Complete
Delay
RB.ReadCompleteDelay
<EDC.RB_Fault03.ReadCompleteDelay>

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Fault03.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Fault03.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Complete 1-shot
RB.ReadCompleteONS
<EDC.RB_Fault03.ReadCompleteONS>
ONS

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault03.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Green - Fault Ring Buffer Tx DN
EquipStatus01.Update.FaultTxOK
L

Update Status State for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Read Complete
Log Transaction & Performance

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr>: 2
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]>: -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr>: 3240
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]>: -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr>: 2
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]>: -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr>: 3239
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]>: -1

Log Transactions
JSR
Routine Name: _901_TransLogSBR
Input Par: 2021
Input Par: 0
Input Par: LNi
Input Par: EQi
Input Par: RB.TransactionNumber <EDC.RB_Fault03.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault03.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest: TempPerfMsg.TransLast: 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source: RB.ReadInProgressTMR.ACC <EDC.RB_Fault03.ReadInProgressTMR.ACC>: 0
Dest: TempPerfMsg.TransTMR.ACC: 110

Performance Transaction
aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Fault03.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Fault03.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault03.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault03.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault03.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault03.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault03.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer

TON
Timer RB.ReadInProgressTMR <EDC.RB_Fault03.ReadInProgressTMR> EN
Preset 5000 DN
Accum 0

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Fault03.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Fault03.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault03.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault03.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Fault Tx Msg ER
EquipStatus01.Update.FaultTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr>: 2
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr>: 3240
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadTagPtr <EDC.RB_Fault03.ReadTagPtr>: 2
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadRecPtr <EDC.RB_Fault03.ReadRecPtr>: 3239
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]>: -1

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2024
Input Par: 0
Input Par: LNi
Input Par: EQi
Input Par: RB.TransactionNumber <EDC.RB_Fault03.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault03.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest: TempPerfMsg.TransLast: 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source: RB.ReadInProgressTMR.ACC <EDC.RB_Fault03.ReadInProgressTMR.ACC>: 0
Dest: TempPerfMsg.TransTMR.ACC: 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans
PerfMsg: TempPerfMsg
PerfTrans: EquipStatus02.FaultTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Fault03.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Fault03.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault03.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Fault03.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Fault03.KeepAliveTMR>
Preset 300000
Accum 6379
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Fault03.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Fault03.KeepAlive>

JSR
Routine Name _321_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Fault03.KeepAliveTMR>
RES

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
Feb2017 - Changed to: Alarm if not below 1000 for 15 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Fault03.Empty>

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault03.RecordCount> 0
Source B 100

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MUL
Source A 60000
Source B 5
Dest RB.NotEmptyTMR.PRE <EDC.RB_Fault03.NotEmptyTMR.PRE> 300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR <EDC.RB_Fault03.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC <EDC.RB_Fault03.NotEmptyTMR.ACC> 0
Source B RB.NotEmptyMax <EDC.RB_Fault03.NotEmptyMax> 300014

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC <EDC.RB_Fault03.NotEmptyTMR.ACC> 0
Dest RB.NotEmptyMax <EDC.RB_Fault03.NotEmptyMax> 300014

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full RB.Full <EDC.RB_Fault03.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer RB.NotEmptyTMR.DN <EDC.RB_Fault03.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm RB.BufferAlarm <EDC.RB_Fault03.BufferAlarm>

Process RingBuffer Reads from Transaction Manager

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault03.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP
Source EquipStatus01
Dest EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length 1

Event Rx/Tx Status Lines and Equipment Array

COP
Source EquipStatus02
Dest EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Length 1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	321
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562847274
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562847274

2

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr> 2
Source B 1

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault03_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr> 2
Source B 2

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault03_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr> 2
Source B 3

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault03_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Fault/Status Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum
2
Source B 1
Dest KeepAliveMsg.Header.CmdNum
2

Equipment Data Fault/Status Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum
2
Source B 1

Equipment Data Fault/Status Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum
2
Source B 9999

Equipment Data Fault/Status Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum
2

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Source B 1

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault03_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Source B 2

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault03_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Source B 3

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault03_Rec03[RB.WriteRecPtr]
Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault03.WriteTagPtr> 2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault03.WriteRecPtr> 3240
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault03.ReadTagPtr> 2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault03.ReadRecPtr> 3239
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2022
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault03.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault03.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

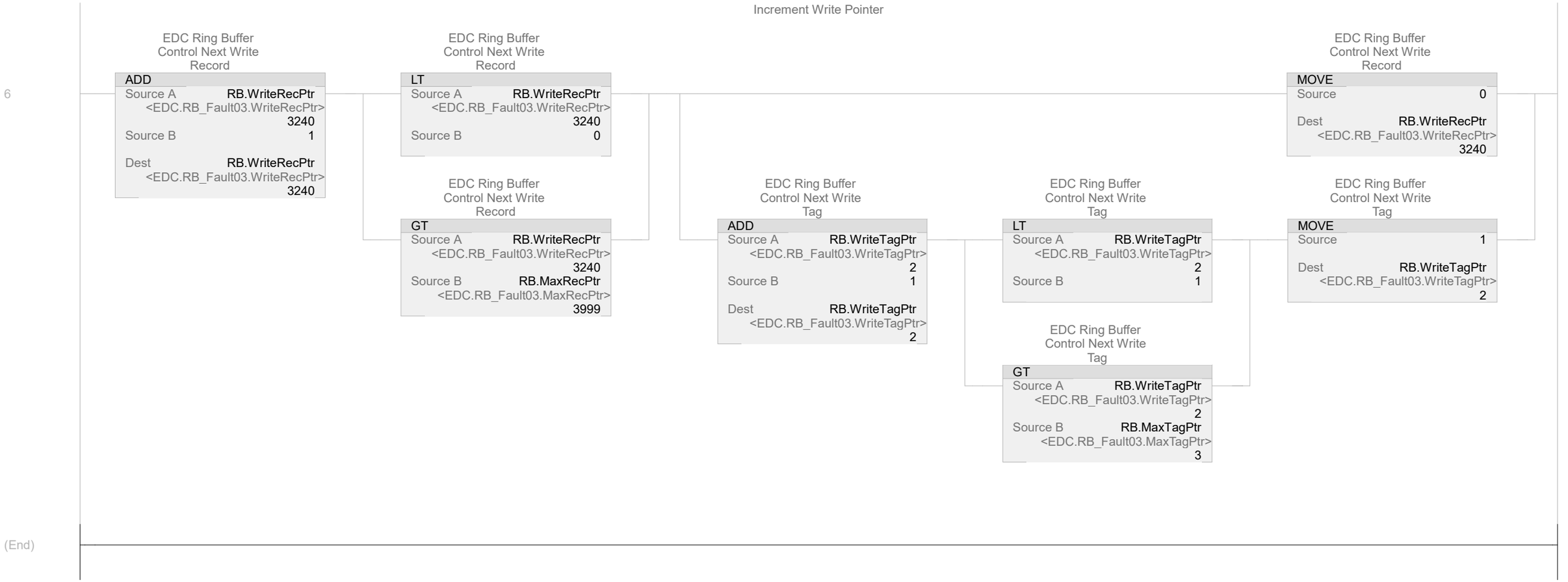
CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Fault03.KeepAliveTMR.ACC> 6379
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.KA


Increment Write Pointer
6
EDC Ring Buffer Control Next Write Record
ADD
Source A RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
EDC Ring Buffer Control Next Write Record
LT
Source A RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B 0
EDC Ring Buffer Control Next Write Record
GT
Source A RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
Source B RB.MaxRecPtr <EDC.RB_Fault03.MaxRecPtr> 3999
EDC Ring Buffer Control Next Write Tag
ADD
Source A RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
EDC Ring Buffer Control Next Write Tag
LT
Source A RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B 1
EDC Ring Buffer Control Next Write Tag
GT
Source A RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Fault03.MaxTagPtr> 3
EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault03.WriteRecPtr> 3240
EDC Ring Buffer Control Next Write Tag
MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault03.WriteTagPtr> 2
(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR	
Input Par	Trans.Code
	<EDC.Trans.Code>
Input Par	Trans.Time
	<EDC.Trans.Time>
Input Par	Trans.LNi
	<EDC.Trans.LNi>
Input Par	Trans.EQi
	<EDC.Trans.EQi>
Input Par	Trans.Num
	<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	901
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[901] 196673847
Source B	1
Dest	Prog.RoutineScan[901] 196673847

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

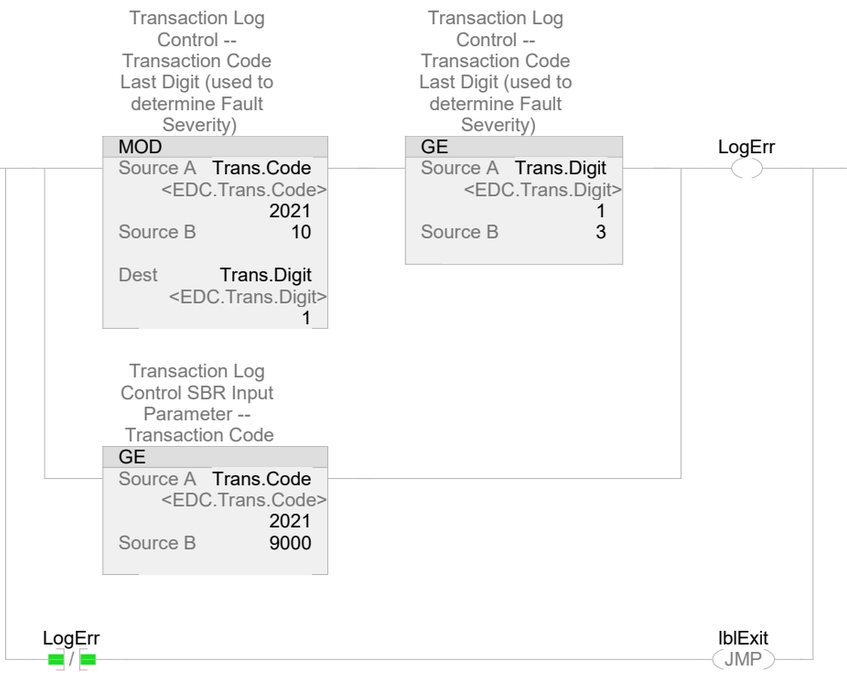
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP	
Source	EDC_TransLog[Trans.LOGi]
Dest	EDC_TransErr[Trans.ERRi]
Length	1

Error Log

FLL	
Source	0
Dest	EDC_TransErr[Trans.ERRx]
Length	1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL	
Source	-1
Dest	Trans.Opt[0] <EDC.Trans.Opt[0]>
Length	Trans.OptN <EDC.Trans.OptN>

(End)

Main Routine

Fault Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	301
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562846556
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562846556

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

Configuration

JSR	
Routine Name	_311_Config

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _320_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Fault04.ShiftCnt>
318575
Dest RB.PrevShiftCnt
<EDC.RB_Fault04.PrevShiftCnt>
357752

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Fault04.ShiftCnt>
318575

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A: Prog.Scan.Last 22
Source B: CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A: ProgScanHighCount 0
Source B: 1
Dest: ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.3
(L)

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest: ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 4000

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 4000
Source B	1
Dest	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 0
Source B	1
Dest	Prog.LOGi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 0
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 0
Source B	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 0

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
320
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr>
1
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault04.WriteRecPtr>
1054
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr>
1
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault04.ReadRecPtr>
1053
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 302
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562846556
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562846556

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 311
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562846556
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562846556

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Fault/Status Message

SIZE
Source EDC_RB_Fault04_Rec01[0] ??
Dim. To Vary 0
Size LEN 4000

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 4000
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Fault04.MaxRecPtr> 3999

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 3
Dest RB.MaxTagPtr <EDC.RB_Fault04.MaxTagPtr> 3

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A: RB.TransactionNumber <EDC.RB_Fault04.TransactionNumber> 4971
Source B: 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source: 1
Dest: RB.TransactionNumber <EDC.RB_Fault04.TransactionNumber> 4971

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Fault04.WritePtrErr>
U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Fault04.ReadPtrErr>
U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 320
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562846556
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562846556

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault04.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Source B RB.MaxRecPtr <EDC.RB_Fault04.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Fault04.ReadPtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Dest RB.ReadPtrErrTag <EDC.RB_Fault04.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Dest RB.ReadPtrErrRec <EDC.RB_Fault04.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2023
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Fault04.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Fault04.MaxTagPtr> 3
Dest RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Fault04.MaxRecPtr> 3999
Dest RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Fault04.ReadNextRecPtr> 1054
Source B RB.MaxRecPtr <EDC.RB_Fault04.MaxRecPtr> 3999

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault04.ReadNextTagPtr> 1

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Fault04.ReadNextRecPtr> 1054

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Fault04.ReadNextTagPtr> 1
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault04.ReadNextTagPtr> 1

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Fault04.ReadNextTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault04.MaxTagPtr> 3

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Fault04.ReadNextRecPtr> 1054

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault04.ReadNextTagPtr> 1

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Source B RB.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Source B RB.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Fault04.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault04.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Fault04.ReadNextTagPtr> 1
Source B RB.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Fault04.ReadNextRecPtr> 1054
Source B RB.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault04.Empty>

5

Buffer Pointers
Combine Tag & Record Pointers

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Fault04.WritePtr> 1054
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Fault04.ReadPtr> 1053
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Fault04.ReadNextPtr> 1054
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault04.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Fault04.WritePtr> 1054
Source B RB.ReadPtr <EDC.RB_Fault04.ReadPtr> 1053

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault04.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Fault04.WritePtr> 1054
Source B RB.ReadPtr <EDC.RB_Fault04.ReadPtr> 1053

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault04.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault04.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Fault04.WritePtr> 1054
Source B RB.ReadNextPtr <EDC.RB_Fault04.ReadNextPtr> 1054

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault04.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Fault04.WritePtr> 1054
Source B RB.ReadNextPtr <EDC.RB_Fault04.ReadNextPtr> 1054

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault04.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault04.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Fault04.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

MOVE
Source RB.RecordCount <EDC.RB_Fault04.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Fault04.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

179

179

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault04.ReadNext>

EDC Ring Buffer Control Current/Last Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Source B RB.MaxRecPtr <EDC.RB_Fault04.MaxRecPtr> 3999

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053

EDC Ring Buffer Control Current/Last Read Tag

ADD
Source A RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault04.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1

Process RingBuffer Reads from Transaction Manager

9

Read Next
Load Read Record (Tags 01-05)

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault04.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr>
Source B 1

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault04_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault04_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr>
Source B 2

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault04_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault04_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr>
Source B 3

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault04_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault04_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault04.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault04.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Fault04.TransactionNumber>
4971
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Fault04.TransactionNumber>
4971

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Fault04.TransactionNumber>
4971
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Fault04.TransactionNumber>
4971
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Fault04.TransactionNumber>
4971

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault04.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault04.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Fault04_Read.Header.EQID>
1009
Source B 100
Dest EQi
3

CPT
Dest LNi
10
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
10

CLR
Dest PerfEQi
3

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault04.ReadValidEQID>
U

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault04.ReadValidEQID>
L

LIMIT
Low Limit 0
Test LNi
10
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
3
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
10
Dest PerfLNi
10

MOVE
Source EQi
3
Dest PerfEQi
3

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault04.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus02
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Log Transaction & Performance

13

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault04.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault04.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault04.ReadRecPtr> 1053
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2020
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Fault04.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault04.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.EN

Process RingBuffer Reads from Transaction Manager

14

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Fault04.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault04.TransactionNumber>
4971
Source B RB.Interface
<EDC.RB_Fault04.Interface>
4971

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault04.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Fault04.Interface>
4971
Dest RB.InterfaceNEG
<EDC.RB_Fault04.InterfaceNEG>
-4971

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault04.TransactionNumber>
4971
Source B RB.InterfaceNEG
<EDC.RB_Fault04.InterfaceNEG>
-4971

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Fault04.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

15

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault04.ReadComplete>

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN
/

EDC Ring Buffer
Control Read
Complete Delay Timer
MOVE
Source CfgRbDelaySlow 100
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Fault04.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer
Control Read
Complete Delay Timer
TON
Timer RB.ReadCompleteDelayTMR <EDC.RB_Fault04.ReadCompleteDelayTMR>
Preset 0
Accum 15
EN
DN

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN

EDC Ring Buffer
Control Read
Complete Delay Timer
MOVE
Source CfgRbDelayFast 0
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Fault04.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer
Control Read
Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Fault04.ReadCompleteDelayTMR.DN>

EDC Ring Buffer
Control Read Buffer
Record Complete
Delay
RB.ReadCompleteDelay
<EDC.RB_Fault04.ReadCompleteDelay>

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Fault04.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault04.ReadComplete>

EDC Ring Buffer
Control Read Buffer
Record Complete
1-shot
RB.ReadCompleteONS
<EDC.RB_Fault04.ReadCompleteONS>
ONS

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault04.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi State = Green -
Fault Ring Buffer Tx
DN
EquipStatus01.Update.FaultTxOK
L

Update Status State
for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.WriteTagPtr
	<EDC.RB_Fault04.WriteTagPtr>
	1
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.WriteRecPtr
	<EDC.RB_Fault04.WriteRecPtr>
	1054
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.ReadTagPtr
	<EDC.RB_Fault04.ReadTagPtr>
	1
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.ReadRecPtr
	<EDC.RB_Fault04.ReadRecPtr>
	1053
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2021
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber
	<EDC.RB_Fault04.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault04.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC
	<EDC.RB_Fault04.ReadInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

Performance
Transaction

aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Fault04.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Fault04.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault04.ReadResend>

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault04.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault04.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault04.ReadResend>

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault04.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer
TON
Timer RB.ReadInProgressTMR <EDC.RB_Fault04.ReadInProgressTMR>
Preset 5000
Accum 0
EN
DN

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Fault04.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Fault04.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault04.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault04.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Fault Tx Msg ER
EquipStatus01.Update.FaultTxER
L

Update Status State for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault04.WriteRecPtr>
1054
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault04.ReadRecPtr>
1053
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2024
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault04.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault04.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Fault04.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Fault04.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Fault04.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault04.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Fault04.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Fault04.KeepAliveTMR>
Preset 300000
Accum 195
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Fault04.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Fault04.KeepAlive>

JSR
Routine Name _321_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Fault04.KeepAliveTMR>
RES

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
Feb2017 - Changed to: Alarm if not below 1000 for 15 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Fault04.Empty>

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault04.RecordCount> 0
Source B 100

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MUL
Source A 60000
Source B 5
Dest RB.NotEmptyTMR.PRE <EDC.RB_Fault04.NotEmptyTMR.PRE> 300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR <EDC.RB_Fault04.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC <EDC.RB_Fault04.NotEmptyTMR.ACC> 0
Source B RB.NotEmptyMax <EDC.RB_Fault04.NotEmptyMax> 300017

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC <EDC.RB_Fault04.NotEmptyTMR.ACC> 0
Dest RB.NotEmptyMax <EDC.RB_Fault04.NotEmptyMax> 300017

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full RB.Full <EDC.RB_Fault04.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer RB.NotEmptyTMR.DN <EDC.RB_Fault04.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm RB.BufferAlarm <EDC.RB_Fault04.BufferAlarm>

Process RingBuffer Reads from Transaction Manager

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault04.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Event Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus02
Dest	EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	321
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562846556
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562846556

2

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr> 1
Source B 1

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault04_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr> 1
Source B 2

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault04_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr> 1
Source B 3

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault04_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Fault/Status Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum
2
Source B
1
Dest KeepAliveMsg.Header.CmdNum
2

Equipment Data Fault/Status Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum
2
Source B
1

Equipment Data Fault/Status Message Header - Command Number

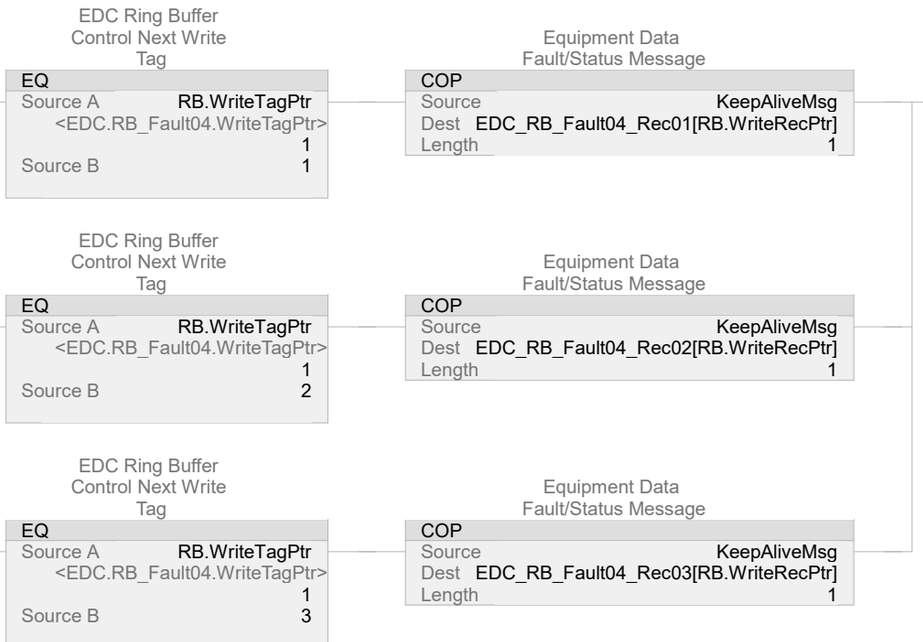
GT
Source A KeepAliveMsg.Header.CmdNum
2
Source B
9999

Equipment Data Fault/Status Message Header - Command Number

MOVE
Source
1
Dest KeepAliveMsg.Header.CmdNum
2

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05



5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault04.WriteTagPtr> 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault04.WriteRecPtr> 1054
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault04.ReadTagPtr> 1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault04.ReadRecPtr> 1053
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2022
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault04.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault04.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Fault04.KeepAliveTMR.ACC> 195
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.KA

6

Increment Write Pointer

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054
Source B RB.MaxRecPtr <EDC.RB_Fault04.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Fault04.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault04.WriteRecPtr> 1054

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault04.WriteTagPtr> 1

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 269832112
Source B 1
Dest Prog.RoutineScan[901] 269832112

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

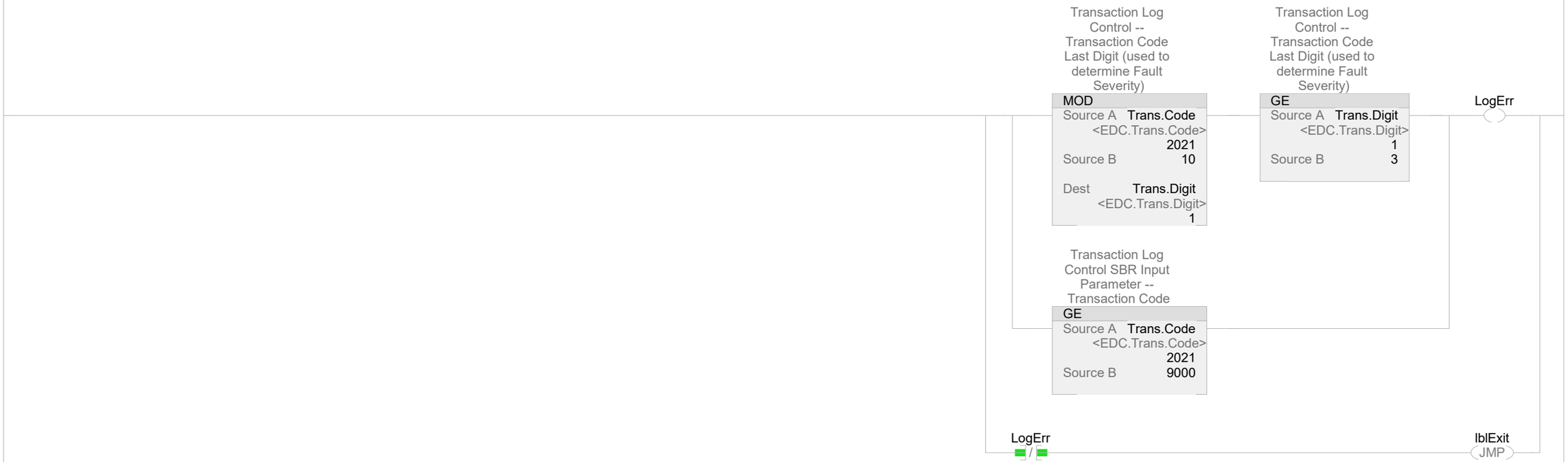
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Fault Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion ...
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	301
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562845909
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562845909

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_311_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _320_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Fault05.ShiftCnt>
135824
Dest RB.PrevShiftCnt
<EDC.RB_Fault05.PrevShiftCnt>
161029

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Fault05.ShiftCnt>
135824

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 16
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.3
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 4000

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 4000
Source B	1
Dest	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 0
Source B	1
Dest	Prog.LOGi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 0
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 0
Source B	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 0

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
320
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr>
3
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault05.WriteRecPtr>
2698
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault05.ReadTagPtr>
3
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault05.ReadRecPtr>
2697
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 302
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562845909
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562845909

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 311
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562845909
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562845909

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Fault/Status Message

SIZE
Source EDC_RB_Fault05_Rec01[0] ??
Dim. To Vary 0
Size LEN 4000

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 4000
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Fault05.MaxRecPtr> 3999

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 3
Dest RB.MaxTagPtr <EDC.RB_Fault05.MaxTagPtr> 3

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A: RB.TransactionNumber <EDC.RB_Fault05.TransactionNumber> 7887
Source B: 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source: 1
Dest: RB.TransactionNumber <EDC.RB_Fault05.TransactionNumber> 7887

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Fault05.WritePtrErr>
U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Fault05.ReadPtrErr>
U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 320
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562845909
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562845909

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Fault05.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Source B RB.MaxRecPtr <EDC.RB_Fault05.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Fault05.ReadPtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Dest RB.ReadPtrErrTag <EDC.RB_Fault05.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Dest RB.ReadPtrErrRec <EDC.RB_Fault05.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2023
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Fault05.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Fault05.MaxTagPtr> 3
Dest RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Fault05.MaxRecPtr> 3999
Dest RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Dest RB.ReadNextTagPtr <EDC.RB_Fault05.ReadNextTagPtr> 3

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Fault05.ReadNextRecPtr> 2698

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Fault05.ReadNextRecPtr> 2698
Source B RB.MaxRecPtr <EDC.RB_Fault05.MaxRecPtr> 3999

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Fault05.ReadNextRecPtr> 2698

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Fault05.ReadNextTagPtr> 3
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault05.ReadNextTagPtr> 3

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Fault05.ReadNextTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Fault05.MaxTagPtr> 3

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault05.ReadNextTagPtr> 3

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Source B RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Source B RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Fault05.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault05.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Fault05.ReadNextTagPtr> 3
Source B RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Fault05.ReadNextRecPtr> 2698
Source B RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault05.Empty>

5

Buffer Pointers
Combine Tag & Record Pointers

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Fault05.WritePtr> 10698
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Fault05.ReadPtr> 10697
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Fault05.ReadNextPtr> 10698
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault05.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Fault05.WritePtr> 10698
Source B RB.ReadPtr <EDC.RB_Fault05.ReadPtr> 10697

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault05.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Fault05.WritePtr> 10698
Source B RB.ReadPtr <EDC.RB_Fault05.ReadPtr> 10697

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault05.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault05.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Fault05.WritePtr> 10698
Source B RB.ReadNextPtr <EDC.RB_Fault05.ReadNextPtr> 10698

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault05.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Fault05.WritePtr> 10698
Source B RB.ReadNextPtr <EDC.RB_Fault05.ReadNextPtr> 10698

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault05.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault05.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Fault05.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

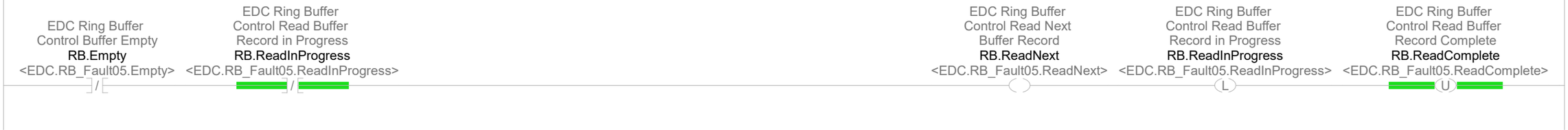
MOVE
Source RB.RecordCount <EDC.RB_Fault05.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Fault05.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

74 74

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault05.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Source B RB.MaxRecPtr <EDC.RB_Fault05.MaxRecPtr> 3999

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Fault05.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3

Process RingBuffer Reads from Transaction Manager

9

Read Next
Load Read Record (Tags 01-05)

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault05.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault05.ReadTagPtr>
Source B 3
1

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault05_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault05_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault05.ReadTagPtr>
Source B 3
2

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault05_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault05_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault05.ReadTagPtr>
Source B 3
3

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault05_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault05_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault05.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault05.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Fault05.TransactionNumber>
7887
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Fault05.TransactionNumber>
7887

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Fault05.TransactionNumber>
7887
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Fault05.TransactionNumber>
7887
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Fault05.TransactionNumber>
7887

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault05.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault05.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Fault05_Read.Header.EQID>
531
Source B 100
Dest EQi
31

CPT
Dest LNi
5
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
5

CLR
Dest PerfEQi
31

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault05.ReadValidEQID>
U

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault05.ReadValidEQID>
L

LIMIT
Low Limit 0
Test LNi
5
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
31
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
5
Dest PerfLNi
5

MOVE
Source EQi
31
Dest PerfEQi
31

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault05.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus02
Length 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault05.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault05.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr> 2697
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2020
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Fault05.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault05.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.EN

Process RingBuffer Reads from Transaction Manager

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

14

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Fault05.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault05.TransactionNumber>
7887
Source B RB.Interface
<EDC.RB_Fault05.Interface>
7887

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault05.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Fault05.Interface>
7887
Dest RB.InterfaceNEG
<EDC.RB_Fault05.InterfaceNEG>
-7887

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault05.TransactionNumber>
7887
Source B RB.InterfaceNEG
<EDC.RB_Fault05.InterfaceNEG>
-7887

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Fault05.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

15

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Fault05.ReadComplete>

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN
/

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source CfgRbDelaySlow 100
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Fault05.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR <EDC.RB_Fault05.ReadCompleteDelayTMR>
Preset 0
Accum 12
EN
DN

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source CfgRbDelayFast 0
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Fault05.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Fault05.ReadCompleteDelayTMR.DN>

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Fault05.ReadCompleteDelay>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault05.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Fault05.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Complete 1-shot
RB.ReadCompleteONS
<EDC.RB_Fault05.ReadCompleteONS>
ONS

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault05.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Green - Fault Ring Buffer Tx DN
EquipStatus01.Update.FaultTxOK
L

Update Status State for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr>: 3
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]>: -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr>: 2698
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]>: -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr>: 3
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]>: -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr>: 2697
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]>: -1

Log Transactions
JSR
Routine Name: _901_TransLogSBR
Input Par: 2021
Input Par: 0
Input Par: LNi
Input Par: EQi
Input Par: RB.TransactionNumber <EDC.RB_Fault05.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault05.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest: TempPerfMsg.TransLast: 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source: RB.ReadInProgressTMR.ACC <EDC.RB_Fault05.ReadInProgressTMR.ACC>: 0
Dest: TempPerfMsg.TransTMR.ACC: 110

Performance Transaction
aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Fault05.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Fault05.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault05.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault05.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault05.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault05.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault05.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer

TON
Timer RB.ReadInProgressTMR <EDC.RB_Fault05.ReadInProgressTMR> EN
Preset 5000 DN
Accum 0

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Fault05.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Fault05.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault05.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault05.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Fault Tx Msg ER
EquipStatus01.Update.FaultTxER
L

Update Status State for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr>: 3
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]>: -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr>: 2698
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]>: -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.ReadTagPtr <EDC.RB_Fault05.ReadTagPtr>: 3
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]>: -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.ReadRecPtr <EDC.RB_Fault05.ReadRecPtr>: 2697
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]>: -1

Log Transactions
JSR
Routine Name: _901_TransLogSBR
Input Par: 2024
Input Par: 0
Input Par: LNi
Input Par: EQi
Input Par: RB.TransactionNumber <EDC.RB_Fault05.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault05.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest: TempPerfMsg.TransLast: 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source: RB.ReadInProgressTMR.ACC <EDC.RB_Fault05.ReadInProgressTMR.ACC>: 0
Dest: TempPerfMsg.TransTMR.ACC: 110

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans
PerfMsg: TempPerfMsg
PerfTrans: EquipStatus02.FaultTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Fault05.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Fault05.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault05.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Fault05.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Fault05.KeepAliveTMR>
Preset 300000
Accum 18401
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Fault05.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Fault05.KeepAlive>

JSR
Routine Name _321_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Fault05.KeepAliveTMR>
RES

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
Feb2017 - Changed to: Alarm if not below 1000 for 15 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Fault05.Empty>

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault05.RecordCount> 0
Source B 100

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MUL
Source A 60000
Source B 5
Dest RB.NotEmptyTMR.PRE <EDC.RB_Fault05.NotEmptyTMR.PRE> 300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR <EDC.RB_Fault05.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC <EDC.RB_Fault05.NotEmptyTMR.ACC> 0
Source B RB.NotEmptyMax <EDC.RB_Fault05.NotEmptyMax> 300008

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC <EDC.RB_Fault05.NotEmptyTMR.ACC> 0
Dest RB.NotEmptyMax <EDC.RB_Fault05.NotEmptyMax> 300008

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full RB.Full <EDC.RB_Fault05.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer RB.NotEmptyTMR.DN <EDC.RB_Fault05.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm RB.BufferAlarm <EDC.RB_Fault05.BufferAlarm>

Process RingBuffer Reads from Transaction Manager

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault05.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP
Source EquipStatus01
Dest EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length 1

Event Rx/Tx Status Lines and Equipment Array

COP
Source EquipStatus02
Dest EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Length 1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	321
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562845909
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562845909

2

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault05.ReadTagPtr> 3
Source B 1

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault05_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault05.ReadTagPtr> 3
Source B 2

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault05_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault05.ReadTagPtr> 3
Source B 3

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault05_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Fault/Status Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum
2
Source B
1
Dest KeepAliveMsg.Header.CmdNum
2

Equipment Data Fault/Status Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum
2
Source B
1

Equipment Data Fault/Status Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum
2
Source B
9999

Equipment Data Fault/Status Message Header - Command Number

MOVE
Source
1
Dest KeepAliveMsg.Header.CmdNum
2

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Source B 1

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault05_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Source B 2

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault05_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr> 3
Source B 3

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault05_Rec03[RB.WriteRecPtr]
Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault05.WriteTagPtr>
3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault05.WriteRecPtr>
2698
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault05.ReadTagPtr>
3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault05.ReadRecPtr>
2697
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2022
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault05.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault05.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Fault05.KeepAliveTMR.ACC>
18401
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698
Source B RB.MaxRecPtr <EDC.RB_Fault05.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Fault05.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault05.WriteRecPtr> 2698

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault05.WriteTagPtr> 3

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 174019291
Source B 1
Dest Prog.RoutineScan[901] 174019291

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

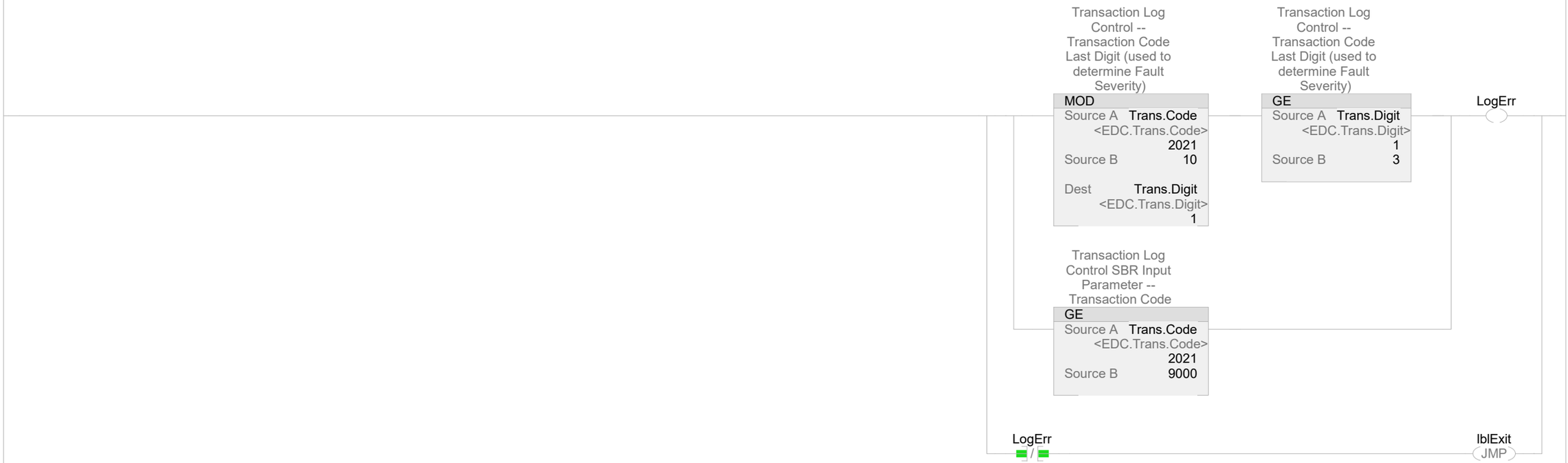
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Fault Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	301
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562845105
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562845105

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_311_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _320_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Fault06.ShiftCnt>
344695
Dest RB.PrevShiftCnt
<EDC.RB_Fault06.PrevShiftCnt>
401978

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Fault06.ShiftCnt>
344695

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 14
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.3
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 4000

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 4000
Source B	1
Dest	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 0
Source B	1
Dest	Prog.LOGi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 0
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 0
Source B	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 0

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
320
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr>
3
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault06.WriteRecPtr>
2434
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr>
3
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault06.ReadRecPtr>
2433
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 302
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562845105
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562845105

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>
/

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 311
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562845105
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562845105

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Fault/Status Message

SIZE
Source EDC_RB_Fault06_Rec01[0] ??
Dim. To Vary 0
Size LEN 4000

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 4000
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Fault06.MaxRecPtr> 3999

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 3
Dest RB.MaxTagPtr <EDC.RB_Fault06.MaxTagPtr> 3

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A: RB.TransactionNumber <EDC.RB_Fault06.TransactionNumber>
1105
Source B: 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source: 1
Dest: RB.TransactionNumber <EDC.RB_Fault06.TransactionNumber>
1105

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset
RB.WritePtrErr <EDC.RB_Fault06.WritePtrErr>
U

EDC Ring Buffer Control Diagnostic - Manual Reset
RB.ReadPtrErr <EDC.RB_Fault06.ReadPtrErr>
U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 320
Dest Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562845105
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562845105

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Fault06.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Source B RB.MaxRecPtr <EDC.RB_Fault06.MaxRecPtr> 3999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Fault06.ReadPtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Dest RB.ReadPtrErrTag <EDC.RB_Fault06.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Dest RB.ReadPtrErrRec <EDC.RB_Fault06.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2023
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Fault06.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Fault06.MaxTagPtr> 3
Dest RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Fault06.MaxRecPtr> 3999
Dest RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Dest RB.ReadNextTagPtr <EDC.RB_Fault06.ReadNextTagPtr> 3

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Fault06.ReadNextRecPtr> 2434

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Fault06.ReadNextRecPtr> 2434
Source B RB.MaxRecPtr <EDC.RB_Fault06.MaxRecPtr> 3999

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Fault06.ReadNextRecPtr> 2434

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Fault06.ReadNextTagPtr> 3
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault06.ReadNextTagPtr> 3

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Fault06.ReadNextTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Fault06.MaxTagPtr> 3

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Fault06.ReadNextTagPtr> 3

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Source B RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Source B RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Fault06.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault06.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Fault06.ReadNextTagPtr> 3
Source B RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Fault06.ReadNextRecPtr> 2434
Source B RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault06.Empty>

Buffer Pointers
Combine Tag & Record Pointers

5

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Fault06.WritePtr> 10434
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Fault06.ReadPtr> 10433
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Fault06.ReadNextPtr> 10434
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault06.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Fault06.WritePtr> 10434
Source B RB.ReadPtr <EDC.RB_Fault06.ReadPtr> 10433

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault06.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Fault06.WritePtr> 10434
Source B RB.ReadPtr <EDC.RB_Fault06.ReadPtr> 10433

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault06.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Fault06.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Fault06.WritePtr> 10434
Source B RB.ReadNextPtr <EDC.RB_Fault06.ReadNextPtr> 10434

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault06.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Fault06.WritePtr> 10434
Source B RB.ReadNextPtr <EDC.RB_Fault06.ReadNextPtr> 10434

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Fault06.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault06.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Fault06.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

MOVE
Source RB.RecordCount <EDC.RB_Fault06.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Fault06.RecordCountMax>

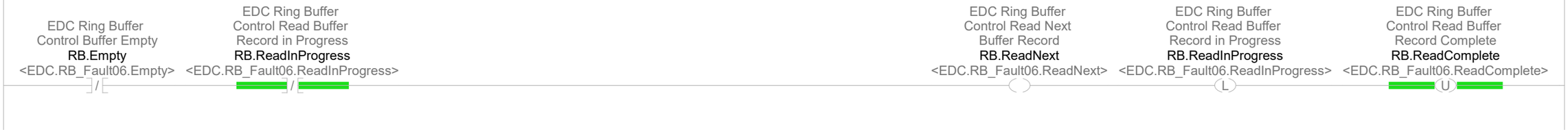
Process RingBuffer Reads from Transaction Manager

571

571

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault06.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Source B RB.MaxRecPtr <EDC.RB_Fault06.MaxRecPtr> 3999

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Fault06.MaxTagPtr> 3

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3

Process RingBuffer Reads from Transaction Manager

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault06.ReadNext>

Read Next
Load Read Record (Tags 01-05)

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr>
Source B 1

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault06_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault06_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr>
Source B 2

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault06_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault06_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr>
Source B 3

Equipment Data
Fault/Status Message

COP
Source EDC_RB_Fault06_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Fault06_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault06.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault06.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Fault06.TransactionNumber>
1105
Source B 1

Dest RB.TransactionNumber
<EDC.RB_Fault06.TransactionNumber>
1105

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Fault06.TransactionNumber>
1105
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Fault06.TransactionNumber>
1105
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1

Dest RB.TransactionNumber
<EDC.RB_Fault06.TransactionNumber>
1105

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Fault06.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault06.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Fault06_Read.Header.EQID>
83
Source B 100
Dest EQi
76

CPT
Dest LNi
0
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
0

CLR
Dest PerfEQi
76

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault06.ReadValidEQID>
U

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault06.ReadValidEQID>
L

LIMIT
Low Limit 0
Test LNi
0
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
76
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
0
Dest PerfLNi
0

MOVE
Source EQi
76
Dest PerfEQi
76

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Fault06.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus02
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Log Transaction & Performance

13

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Fault06.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault06.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2020
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Fault06.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault06.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.EN

Process RingBuffer Reads from Transaction Manager

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

14

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Fault06.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault06.TransactionNumber>
1105
Source B RB.Interface
<EDC.RB_Fault06.Interface>
1105

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault06.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Fault06.Interface>
1105
Dest RB.InterfaceNEG
<EDC.RB_Fault06.InterfaceNEG>
-1105

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Fault06.TransactionNumber>
1105
Source B RB.InterfaceNEG
<EDC.RB_Fault06.InterfaceNEG>
-1105

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Fault06.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

15

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Fault06.ReadComplete>

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN
/

EDC Ring Buffer
Control Read
Complete Delay Timer
MOVE
Source CfgRbDelaySlow
100
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Fault06.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer
Control Read
Complete Delay Timer
TON
Timer RB.ReadCompleteDelayTMR
<EDC.RB_Fault06.ReadCompleteDelayTMR>
Preset 0
Accum 16
EN
DN

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN

EDC Ring Buffer
Control Read
Complete Delay Timer
MOVE
Source CfgRbDelayFast
0
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Fault06.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer
Control Read
Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Fault06.ReadCompleteDelayTMR.DN>

EDC Ring Buffer
Control Read Buffer
Record Complete
Delay
RB.ReadCompleteDelay
<EDC.RB_Fault06.ReadCompleteDelay>

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Fault06.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Fault06.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Complete 1-shot
RB.ReadCompleteONS
<EDC.RB_Fault06.ReadCompleteONS>
ONS

Read Complete Log Transaction & Performance

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault06.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Green - Fault Ring Buffer Tx DN
EquipStatus01.Update.FaultTxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Dest	Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadTagPtr <EDC.RB_Fault06.ReadTagPtr> 3
Dest	Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadRecPtr <EDC.RB_Fault06.ReadRecPtr> 2433
Dest	Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2021
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber <EDC.RB_Fault06.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault06.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC <EDC.RB_Fault06.ReadInProgressTMR.ACC> 0
Dest	TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Fault06.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Fault06.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault06.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault06.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault06.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Fault06.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Fault06.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer

TON
Timer RB.ReadInProgressTMR <EDC.RB_Fault06.ReadInProgressTMR> EN
Preset 5000 DN
Accum 0

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Fault06.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Fault06.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Fault06.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault06.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Fault Tx Msg ER
EquipStatus01.Update.FaultTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr>
3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault06.WriteRecPtr>
2434
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr>
3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault06.ReadRecPtr>
2433
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2024
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault06.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault06.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Fault06.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Fault06.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Fault06.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Fault06.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Fault06.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Fault06.KeepAliveTMR>
Preset 300000
Accum 527
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Fault06.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Fault06.KeepAlive>

JSR
Routine Name _321_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Fault06.KeepAliveTMR>
RES

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
Feb2017 - Changed to: Alarm if not below 1000 for 15 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Fault06.Empty>

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Fault06.RecordCount> 0
Source B 100

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MUL
Source A 60000
Source B 5
Dest RB.NotEmptyTMR.PRE <EDC.RB_Fault06.NotEmptyTMR.PRE> 300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR <EDC.RB_Fault06.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC <EDC.RB_Fault06.NotEmptyTMR.ACC> 0
Source B RB.NotEmptyMax <EDC.RB_Fault06.NotEmptyMax> 300017

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC <EDC.RB_Fault06.NotEmptyTMR.ACC> 0
Dest RB.NotEmptyMax <EDC.RB_Fault06.NotEmptyMax> 300017

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full RB.Full <EDC.RB_Fault06.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer RB.NotEmptyTMR.DN <EDC.RB_Fault06.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm RB.BufferAlarm <EDC.RB_Fault06.BufferAlarm>

Process RingBuffer Reads from Transaction Manager

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault06.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Event Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus02
Dest	EDC_Status02.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	321
Dest	Prog.Routine 320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562845105
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562845105

2

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr> 3
Source B 1

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault06_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr> 3
Source B 2

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault06_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr> 3
Source B 3

Equipment Data Fault/Status Message

COP
Source EDC_RB_Fault06_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Fault/Status Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1
Dest KeepAliveMsg.Header.CmdNum 1

Equipment Data Fault/Status Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1

Equipment Data Fault/Status Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 9999

Equipment Data Fault/Status Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum 1

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Source B 1

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault06_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Source B 2

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault06_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr> 3
Source B 3

Equipment Data Fault/Status Message

COP
Source KeepAliveMsg
Dest EDC_RB_Fault06_Rec03[RB.WriteRecPtr]
Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Fault06.WriteTagPtr>
3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Fault06.WriteRecPtr>
2434
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Fault06.ReadTagPtr>
3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Fault06.ReadRecPtr>
2433
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2022
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Fault06.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Fault06.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Fault06.KeepAliveTMR.ACC>
527
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus02.FaultTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434
Source B RB.MaxRecPtr <EDC.RB_Fault06.MaxRecPtr> 3999

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Fault06.MaxTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Fault06.WriteRecPtr> 2434

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Fault06.WriteTagPtr> 3

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine
320

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901]
325718362
Source B 1
Dest Prog.RoutineScan[901]
325718362

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT	
Low Limit	Trans.Filter.CodeLow <EDC.Trans.Filter.CodeLow> 0
Test	Trans.Code <EDC.Trans.Code> 2021
High Limit	Trans.Filter.CodeHigh <EDC.Trans.Filter.CodeHigh> 0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT	
Low Limit	Trans.Filter.LNiLow <EDC.Trans.Filter.LNiLow> 0
Test	Trans.LNi <EDC.Trans.LNi> 4
High Limit	Trans.Filter.LNiHigh <EDC.Trans.Filter.LNiHigh> 0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT	
Low Limit	Trans.Filter.EQiLow <EDC.Trans.Filter.EQiLow> 0
Test	Trans.EQi <EDC.Trans.EQi> 21
High Limit	Trans.Filter.EQiHigh <EDC.Trans.Filter.EQiHigh> 0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

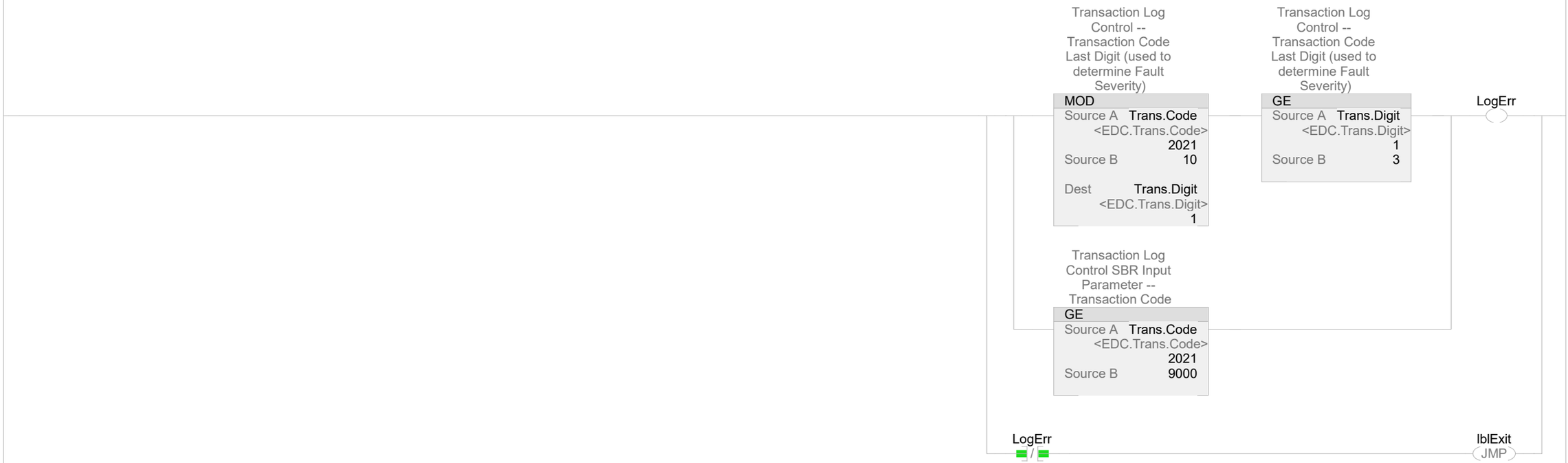
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Quality Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	401
Dest	Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562889430
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562889430

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_411_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _420_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Quality01.ShiftCnt>
86102
Dest RB.PrevShiftCnt
<EDC.RB_Quality01.PrevShiftCnt>
101079

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Quality01.ShiftCnt>
86102

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A: Prog.Scan.Last 12
Source B: CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A: ProgScanHighCount 0
Source B: 1
Dest: ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.4
(L)

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest: ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

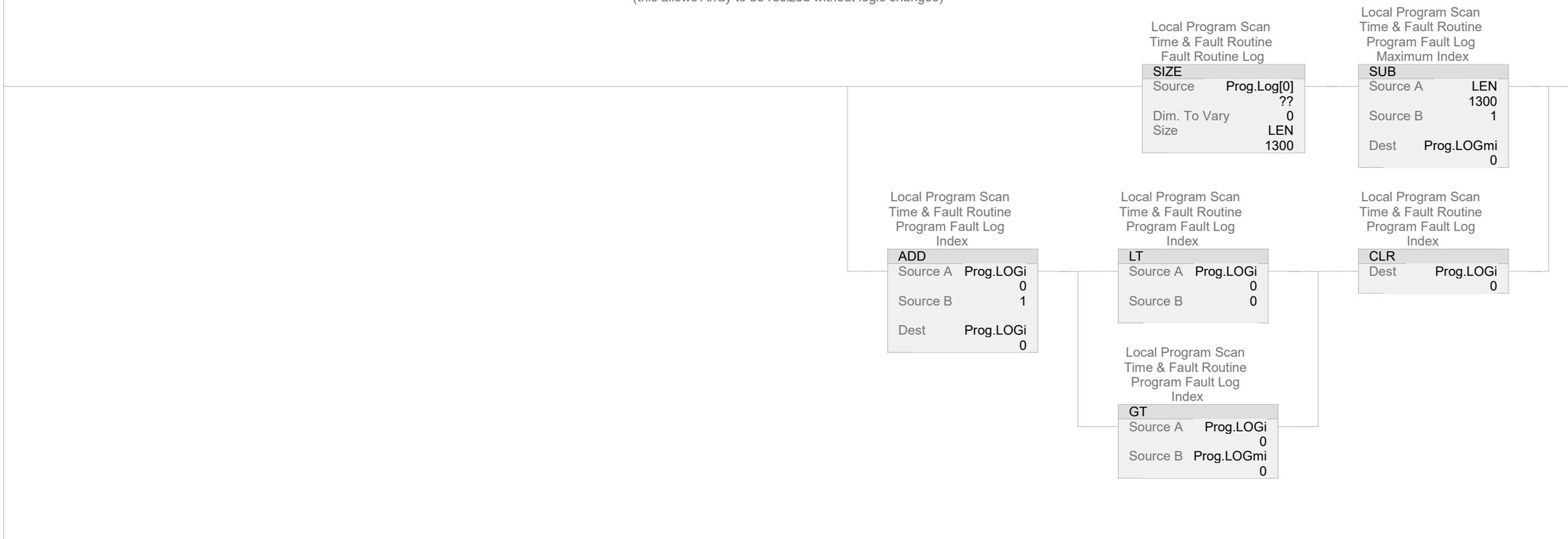
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
420
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality01.WriteTagPtr>
2
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality01.WriteRecPtr>
440
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr>
2
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality01.ReadRecPtr>
439
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 402
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562889430
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562889430

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>
/

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 411
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562889430
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562889430

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Quality/Health Message

SIZE
Source EDC_RB_Quality01_Rec01[0] ??
Dim. To Vary 0
Size LEN 1300

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 1300
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Quality01.MaxRecPtr> 1299

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Quality01.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A RB.TransactionNumber <EDC.RB_Quality01.TransactionNumber> 8080
Source B 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber <EDC.RB_Quality01.TransactionNumber> 8080

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Quality01.WritePtrErr>

U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Quality01.ReadPtrErr>

U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 420
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562889430
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562889430

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Quality01.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Source B RB.MaxRecPtr <EDC.RB_Quality01.MaxRecPtr> 1299

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Quality01.ReadPtrErr>

(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Dest RB.ReadPtrErrTag <EDC.RB_Quality01.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Dest RB.ReadPtrErrRec <EDC.RB_Quality01.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2123
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Quality01.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Quality01.MaxTagPtr> 5
Dest RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Quality01.MaxRecPtr> 1299
Dest RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Quality01.ReadNextRecPtr> 440
Source B RB.MaxRecPtr <EDC.RB_Quality01.MaxRecPtr> 1299

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Dest RB.ReadNextTagPtr <EDC.RB_Quality01.ReadNextTagPtr> 2

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Quality01.ReadNextRecPtr> 440

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Quality01.ReadNextTagPtr> 2
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Quality01.ReadNextTagPtr> 2

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Quality01.ReadNextTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Quality01.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Quality01.ReadNextRecPtr> 440

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Quality01.ReadNextTagPtr> 2

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Source B RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Source B RB.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Quality01.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality01.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Quality01.ReadNextTagPtr> 2
Source B RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Quality01.ReadNextRecPtr> 440
Source B RB.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality01.Empty>

Buffer Pointers
Combine Tag & Record Pointers

5

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Quality01.WritePtr> 1740
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Quality01.ReadPtr> 1739
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Quality01.ReadNextPtr> 1740
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Quality01.WritePtr> 1740
Source B RB.ReadPtr <EDC.RB_Quality01.ReadPtr> 1739

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality01.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Quality01.WritePtr> 1740
Source B RB.ReadPtr <EDC.RB_Quality01.ReadPtr> 1739

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Quality01.WritePtr> 1740
Source B RB.ReadNextPtr <EDC.RB_Quality01.ReadNextPtr> 1740

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality01.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Quality01.WritePtr> 1740
Source B RB.ReadNextPtr <EDC.RB_Quality01.ReadNextPtr> 1740

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Quality01.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Quality01.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

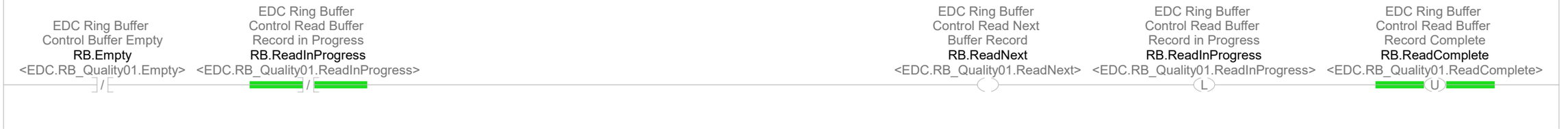
MOVE
Source RB.RecordCount <EDC.RB_Quality01.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Quality01.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

17 17

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Quality01.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439
Source B RB.MaxRecPtr <EDC.RB_Quality01.MaxRecPtr> 1299

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Quality01.ReadRecPtr> 439

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Quality01.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Quality01.ReadTagPtr> 2

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality01.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr>
Source B 1

Equipment Data
Quality/Health
Message

COP
Source EDC_RB_Quality01_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Quality01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr>
Source B 2

Equipment Data
Quality/Health
Message

COP
Source EDC_RB_Quality01_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Quality01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr>
Source B 3

Equipment Data
Quality/Health
Message

COP
Source EDC_RB_Quality01_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Quality01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr>
Source B 4

Equipment Data
Quality/Health
Message

COP
Source EDC_RB_Quality01_Rec04[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Quality01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr>
Source B 5

Equipment Data
Quality/Health
Message

COP
Source EDC_RB_Quality01_Rec05[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Quality01_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality01.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Quality01.TransactionNumber>
8080
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Quality01.TransactionNumber>
8080

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Quality01.TransactionNumber>
8080
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Quality01.TransactionNumber>
8080
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Quality01.TransactionNumber>
8080

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality01.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Quality01_Read.Header.EQID>
1003
Source B 100
Dest EQi
20

CPT
Dest LNi
7
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
7

CLR
Dest PerfEQi
20

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality01.ReadValidEQID>
U

LIMIT
Low Limit 0
Test LNi
7
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
20
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
7
Dest PerfLNi
7

MOVE
Source EQi
20
Dest PerfEQi
20

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality01.ReadValidEQID>
L

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality01.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus03
Length 1

Process RingBuffer Reads from Transaction Manager

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Read Next
Log Transaction & Performance

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality01.ReadInProgressTimeout>

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality01.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality01.WriteRecPtr>
440
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality01.ReadRecPtr>
439
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2120
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Quality01.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality01.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.EN

Process RingBuffer Reads from Transaction Manager

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

14

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Quality01.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Quality01.TransactionNumber>
8080
Source B RB.Interface
<EDC.RB_Quality01.Interface>
8080

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Quality01.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Quality01.Interface>
8080
Dest RB.InterfaceNEG
<EDC.RB_Quality01.InterfaceNEG>
-8080

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Quality01.TransactionNumber>
8080
Source B RB.InterfaceNEG
<EDC.RB_Quality01.InterfaceNEG>
-8080

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Quality01.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

15

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Quality01.ReadComplete>

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN
/

MOVE
Source CfgRbDelaySlow
100
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Quality01.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR
<EDC.RB_Quality01.ReadCompleteDelayTMR>
Preset 0
Accum 14
EN
DN

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN

MOVE
Source CfgRbDelayFast
0
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Quality01.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer Control Read Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Quality01.ReadCompleteDelayTMR.DN>

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Quality01.ReadCompleteDelay>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality01.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Quality01.ReadComplete>

EDC Ring Buffer
Control Read Buffer
Record Complete
1-shot
RB.ReadCompleteONS
<EDC.RB_Quality01.ReadCompleteONS>
ONS

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality01.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi State = Green -
Quality Ring Buffer
Tx DN
EquipStatus01.Update.QualityTxOK
L

Update Status State
for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.WriteTagPtr
	<EDC.RB_Quality01.WriteTagPtr>
	2
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.WriteRecPtr
	<EDC.RB_Quality01.WriteRecPtr>
	440
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.ReadTagPtr
	<EDC.RB_Quality01.ReadTagPtr>
	2
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.ReadRecPtr
	<EDC.RB_Quality01.ReadRecPtr>
	439
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2121
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber
	<EDC.RB_Quality01.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality01.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC
	<EDC.RB_Quality01.ReadInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

Performance
Transaction

aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Quality01.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Quality01.ReadComplete> /

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Quality01.ReadResend> /

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Quality01.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Quality01.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Quality01.ReadResend> U

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Quality01.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer
TON
Timer RB.ReadInProgressTMR <EDC.RB_Quality01.ReadInProgressTMR> EN
Preset 5000 DN
Accum 0

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Quality01.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Quality01.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality01.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Quality Tx Msg ER
EquipStatus01.Update.QualityTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteTagPtr
	<EDC.RB_Quality01.WriteTagPtr>
	2
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteRecPtr
	<EDC.RB_Quality01.WriteRecPtr>
	440
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadTagPtr
	<EDC.RB_Quality01.ReadTagPtr>
	2
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadRecPtr
	<EDC.RB_Quality01.ReadRecPtr>
	439
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2124
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber
	<EDC.RB_Quality01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC
	<EDC.RB_Quality01.ReadInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EquipStatus03.QualityTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Quality01.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Quality01.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality01.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Quality01.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Quality01.KeepAliveTMR>
Preset 300000
Accum 801
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Quality01.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Quality01.KeepAlive>

JSR
Routine Name _421_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Quality01.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality01.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Quality01.NotEmptyTMR.PRE>
300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Quality01.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Quality01.NotEmptyTMR.ACC>
0
Source B RB.NotEmptyMax
<EDC.RB_Quality01.NotEmptyMax>
300000

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Quality01.NotEmptyTMR.ACC>
0
Dest RB.NotEmptyMax
<EDC.RB_Quality01.NotEmptyMax>
300000

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Quality01.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Quality01.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Quality01.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality01.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus03
Dest	EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	421
Dest	Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562889430
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562889430

2

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr> 2
Source B 1

Equipment Data Quality/Health Message

COP
Source EDC_RB_Quality01_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr> 2
Source B 2

Equipment Data Quality/Health Message

COP
Source EDC_RB_Quality01_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr> 2
Source B 3

Equipment Data Quality/Health Message

COP
Source EDC_RB_Quality01_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr> 2
Source B 4

Equipment Data Quality/Health Message

COP
Source EDC_RB_Quality01_Rec04[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr> 2
Source B 5

Equipment Data Quality/Health Message

COP
Source EDC_RB_Quality01_Rec05[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Quality/Health Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1
Dest KeepAliveMsg.Header.CmdNum 1

Equipment Data Quality/Health Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1

Equipment Data Quality/Health Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 9999

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum 1

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

	EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
	EQ Source A RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 1	COP Source KeepAliveMsg Dest EDC_RB_Quality01_Rec01[RB.WriteRecPtr] Length 1
	EQ Source A RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 2	COP Source KeepAliveMsg Dest EDC_RB_Quality01_Rec02[RB.WriteRecPtr] Length 1
	EQ Source A RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 3	COP Source KeepAliveMsg Dest EDC_RB_Quality01_Rec03[RB.WriteRecPtr] Length 1
	EQ Source A RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 4	COP Source KeepAliveMsg Dest EDC_RB_Quality01_Rec04[RB.WriteRecPtr] Length 1
	EQ Source A RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2 Source B 5	COP Source KeepAliveMsg Dest EDC_RB_Quality01_Rec05[RB.WriteRecPtr] Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality01.WriteTagPtr> 2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality01.WriteRecPtr> 440
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality01.ReadTagPtr> 2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality01.ReadRecPtr> 439
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2122
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Quality01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Quality01.KeepAliveTMR.ACC> 801
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440
Source B RB.MaxRecPtr <EDC.RB_Quality01.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Quality01.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Quality01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Quality01.WriteTagPtr> 2

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 155951549
Source B 1
Dest Prog.RoutineScan[901] 155951549

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi
<EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi
<EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi
<EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi
<EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n
<EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi
<EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi
<EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx
<EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx
<EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx
<EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n
<EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx
<EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

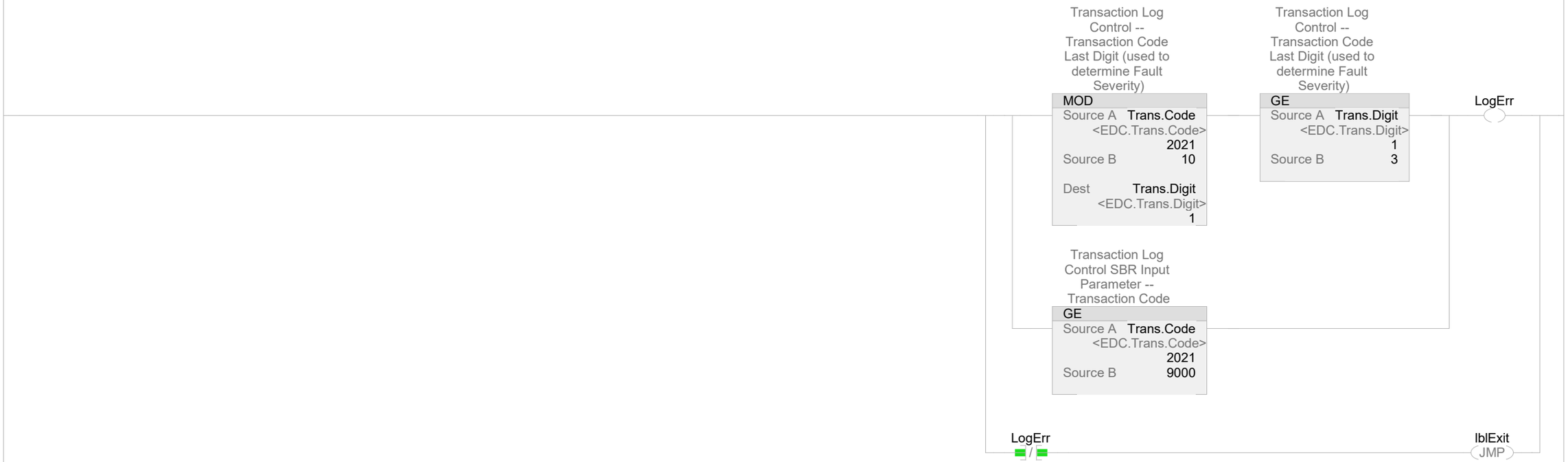
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Quality Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	401
Dest	Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562861518
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562861518

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_411_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _420_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Quality02.ShiftCnt>
87645
Dest RB.PrevShiftCnt
<EDC.RB_Quality02.PrevShiftCnt>
102048

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Quality02.ShiftCnt>
87645

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 13
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.4
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

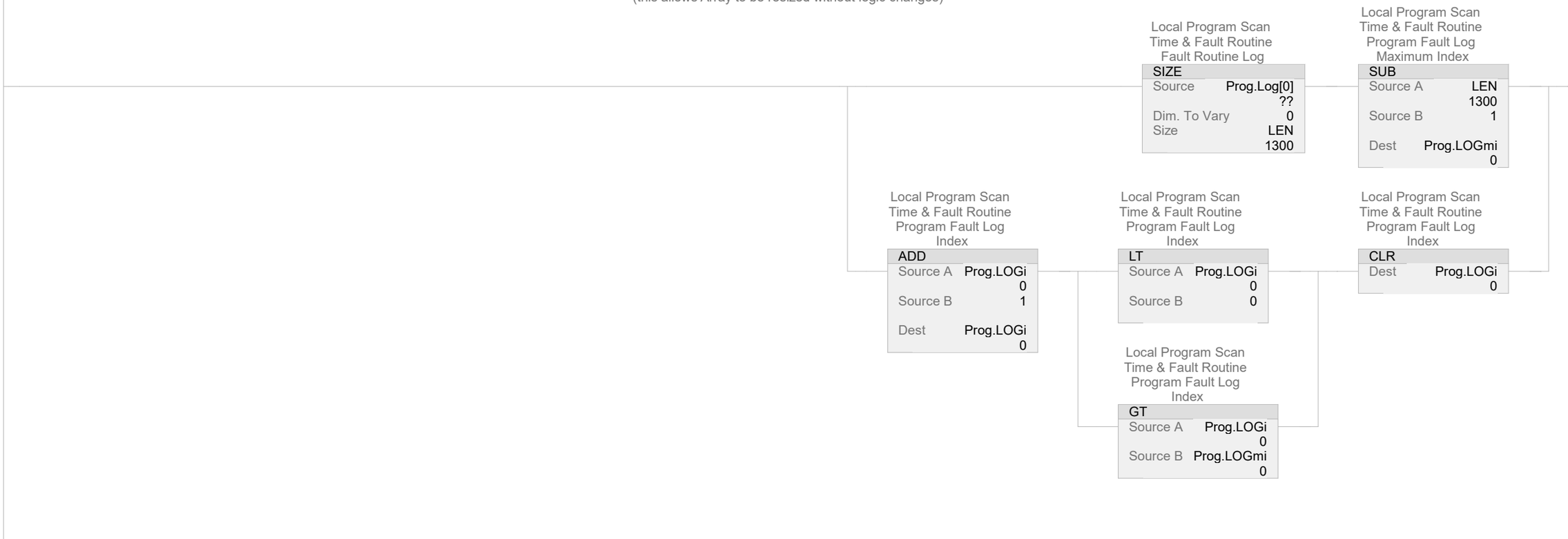
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
420
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality02.WriteTagPtr>
2
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality02.WriteRecPtr>
984
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr>
2
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality02.ReadRecPtr>
983
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 402
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562861518
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562861518

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 411
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562861518
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562861518

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Quality/Health Message

SIZE
Source EDC_RB_Quality02_Rec01[0] ??
Dim. To Vary 0
Size LEN 1300

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 1300
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Quality02.MaxRecPtr> 1299

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Quality02.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A RB.TransactionNumber <EDC.RB_Quality02.TransactionNumber> 5548
Source B 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber <EDC.RB_Quality02.TransactionNumber> 5548

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Quality02.WritePtrErr>

U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Quality02.ReadPtrErr>

U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 420
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562861518
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562861518

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Quality02.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Source B RB.MaxRecPtr <EDC.RB_Quality02.MaxRecPtr> 1299

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Quality02.ReadPtrErr>

(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Dest RB.ReadPtrErrTag <EDC.RB_Quality02.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Dest RB.ReadPtrErrRec <EDC.RB_Quality02.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2123
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Quality02.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Quality02.MaxTagPtr> 5
Dest RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Quality02.MaxRecPtr> 1299
Dest RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Quality02.ReadNextRecPtr> 984
Source B RB.MaxRecPtr <EDC.RB_Quality02.MaxRecPtr> 1299

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Dest RB.ReadNextTagPtr <EDC.RB_Quality02.ReadNextTagPtr> 2

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Quality02.ReadNextRecPtr> 984

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Quality02.ReadNextTagPtr> 2
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Quality02.ReadNextTagPtr> 2

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Quality02.ReadNextTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Quality02.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Quality02.ReadNextRecPtr> 984

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Quality02.ReadNextTagPtr> 2

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr>
2
Source B RB.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr>
2

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr>
983
Source B RB.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr>
984

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Quality02.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality02.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Quality02.ReadNextTagPtr>
2
Source B RB.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr>
2

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Quality02.ReadNextRecPtr>
984
Source B RB.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr>
984

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality02.Empty>

Buffer Pointers
Combine Tag & Record Pointers

5

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Quality02.WritePtr>
2284
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Quality02.ReadPtr>
2283
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Quality02.ReadNextPtr>
2284
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality02.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Quality02.WritePtr> 2284
Source B RB.ReadPtr <EDC.RB_Quality02.ReadPtr> 2283

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality02.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Quality02.WritePtr> 2284
Source B RB.ReadPtr <EDC.RB_Quality02.ReadPtr> 2283

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality02.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality02.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Quality02.WritePtr> 2284
Source B RB.ReadNextPtr <EDC.RB_Quality02.ReadNextPtr> 2284

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality02.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Quality02.WritePtr> 2284
Source B RB.ReadNextPtr <EDC.RB_Quality02.ReadNextPtr> 2284

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality02.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Quality02.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Quality02.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

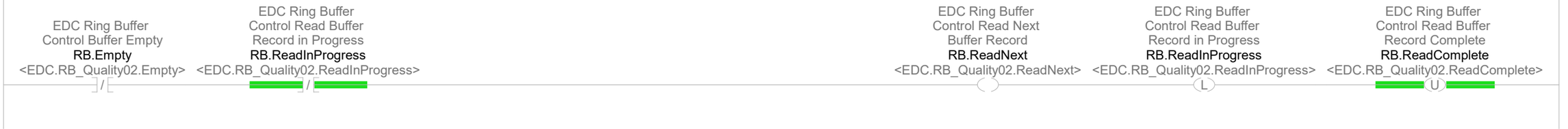
MOVE
Source RB.RecordCount <EDC.RB_Quality02.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Quality02.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

21 21

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Quality02.ReadNext>

EDC Ring Buffer Control Current/Last Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983
Source B RB.MaxRecPtr <EDC.RB_Quality02.MaxRecPtr> 1299

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr> 983

EDC Ring Buffer Control Current/Last Read Tag

ADD
Source A RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Quality02.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr> 2

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality02.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr>
Source B 1

Equipment Data
Quality/Health
Message

COP
Source EDC_RB_Quality02_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Quality02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr>
Source B 2

Equipment Data
Quality/Health
Message

COP
Source EDC_RB_Quality02_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Quality02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr>
Source B 3

Equipment Data
Quality/Health
Message

COP
Source EDC_RB_Quality02_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Quality02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr>
Source B 4

Equipment Data
Quality/Health
Message

COP
Source EDC_RB_Quality02_Rec04[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Quality02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr>
Source B 5

Equipment Data
Quality/Health
Message

COP
Source EDC_RB_Quality02_Rec05[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Quality02_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

10

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality02.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality02.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Quality02.TransactionNumber>
5548
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Quality02.TransactionNumber>
5548

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Quality02.TransactionNumber>
5548
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Quality02.TransactionNumber>
5548
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Quality02.TransactionNumber>
5548

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality02.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality02.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Quality02_Read.Header.EQID>
1103
Source B 100
Dest EQi
31

CPT
Dest LNi
9
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
9

CLR
Dest PerfEQi
31

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality02.ReadValidEQID>
U

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality02.ReadValidEQID>
L

LIMIT
Low Limit 0
Test LNi
9
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
31
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
9
Dest PerfLNi
9

MOVE
Source EQi
31
Dest PerfEQi
31

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality02.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus03
Length 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality02.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality02.ReadInProgressTimeout>

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality02.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality02.WriteRecPtr>
984
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality02.ReadRecPtr>
983
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2120
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Quality02.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality02.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.EN

Process RingBuffer Reads from Transaction Manager

14

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Quality02.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Quality02.TransactionNumber>
5548
Source B RB.Interface
<EDC.RB_Quality02.Interface>
5548

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Quality02.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Quality02.Interface>
5548
Dest RB.InterfaceNEG
<EDC.RB_Quality02.InterfaceNEG>
-5548

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Quality02.TransactionNumber>
5548
Source B RB.InterfaceNEG
<EDC.RB_Quality02.InterfaceNEG>
-5548

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Quality02.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

15

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Quality02.ReadComplete>

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN

MOVE
Source CfgRbDelaySlow
100
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Quality02.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR
<EDC.RB_Quality02.ReadCompleteDelayTMR>
Preset 0
Accum 12
EN
DN

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN

MOVE
Source CfgRbDelayFast
0
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Quality02.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer Control Read Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Quality02.ReadCompleteDelayTMR.DN>

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Quality02.ReadCompleteDelay>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality02.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Quality02.ReadComplete>

EDC Ring Buffer
Control Read Buffer
Record Complete
1-shot
RB.ReadCompleteONS
<EDC.RB_Quality02.ReadCompleteONS>
ONS

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality02.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi State = Green -
Quality Ring Buffer
Tx DN
EquipStatus01.Update.QualityTxOK
L

Update Status State
for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality02.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality02.WriteRecPtr>
984
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality02.ReadRecPtr>
983
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2121
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Quality02.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality02.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Quality02.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction
aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Quality02.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Quality02.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Quality02.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Quality02.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Quality02.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Quality02.ReadResend>

U

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Quality02.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer

TON
Timer RB.ReadInProgressTMR <EDC.RB_Quality02.ReadInProgressTMR>
Preset 5000
Accum 0
EN
DN

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Quality02.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Quality02.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality02.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality02.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Quality Tx Msg ER
EquipStatus01.Update.QualityTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr>: 2
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr>: 984
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadTagPtr <EDC.RB_Quality02.ReadTagPtr>: 2
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadRecPtr <EDC.RB_Quality02.ReadRecPtr>: 983
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]>: -1

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2124
Input Par: 0
Input Par: LNi
Input Par: EQi
Input Par: RB.TransactionNumber <EDC.RB_Quality02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest: TempPerfMsg.TransLast: 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source: RB.ReadInProgressTMR.ACC <EDC.RB_Quality02.ReadInProgressTMR.ACC>: 0
Dest: TempPerfMsg.TransTMR.ACC: 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans
PerfMsg: TempPerfMsg
PerfTrans: EquipStatus03.QualityTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Quality02.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Quality02.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality02.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Quality02.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Quality02.KeepAliveTMR>
Preset 300000
Accum 1134
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Quality02.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Quality02.KeepAlive>

JSR
Routine Name _421_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Quality02.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality02.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Quality02.NotEmptyTMR.PRE>
300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Quality02.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Quality02.NotEmptyTMR.ACC>
0
Source B RB.NotEmptyMax
<EDC.RB_Quality02.NotEmptyMax>
300001

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Quality02.NotEmptyTMR.ACC>
0
Dest RB.NotEmptyMax
<EDC.RB_Quality02.NotEmptyMax>
300001

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Quality02.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Quality02.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Quality02.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality02.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus03
Dest	EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	421
Dest	Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562861518
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562861518

2

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr
	<EDC.RB_Quality02.ReadTagPtr>
	2
Source B	1

Equipment Data Quality/Health Message

COP	
Source	EDC_RB_Quality02_Rec01[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr
	<EDC.RB_Quality02.ReadTagPtr>
	2
Source B	2

Equipment Data Quality/Health Message

COP	
Source	EDC_RB_Quality02_Rec02[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr
	<EDC.RB_Quality02.ReadTagPtr>
	2
Source B	3

Equipment Data Quality/Health Message

COP	
Source	EDC_RB_Quality02_Rec03[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr
	<EDC.RB_Quality02.ReadTagPtr>
	2
Source B	4

Equipment Data Quality/Health Message

COP	
Source	EDC_RB_Quality02_Rec04[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr
	<EDC.RB_Quality02.ReadTagPtr>
	2
Source B	5

Equipment Data Quality/Health Message

COP	
Source	EDC_RB_Quality02_Rec05[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Quality/Health Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1
Dest KeepAliveMsg.Header.CmdNum 1

Equipment Data Quality/Health Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1

Equipment Data Quality/Health Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 9999

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum 1

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

	EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
	EQ	COP
	Source A RB.WriteTagPtr	Source KeepAliveMsg
	<EDC.RB_Quality02.WriteTagPtr> 2	Dest EDC_RB_Quality02_Rec01[RB.WriteRecPtr]
	Source B 1	Length 1
	EQ	COP
	Source A RB.WriteTagPtr	Source KeepAliveMsg
	<EDC.RB_Quality02.WriteTagPtr> 2	Dest EDC_RB_Quality02_Rec02[RB.WriteRecPtr]
	Source B 2	Length 1
	EQ	COP
	Source A RB.WriteTagPtr	Source KeepAliveMsg
	<EDC.RB_Quality02.WriteTagPtr> 2	Dest EDC_RB_Quality02_Rec03[RB.WriteRecPtr]
	Source B 3	Length 1
	EQ	COP
	Source A RB.WriteTagPtr	Source KeepAliveMsg
	<EDC.RB_Quality02.WriteTagPtr> 2	Dest EDC_RB_Quality02_Rec04[RB.WriteRecPtr]
	Source B 4	Length 1
	EQ	COP
	Source A RB.WriteTagPtr	Source KeepAliveMsg
	<EDC.RB_Quality02.WriteTagPtr> 2	Dest EDC_RB_Quality02_Rec05[RB.WriteRecPtr]
	Source B 5	Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality02.WriteTagPtr> 2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality02.WriteRecPtr> 984
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality02.ReadTagPtr> 2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality02.ReadRecPtr> 983
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2122
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Quality02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Quality02.KeepAliveTMR.ACC> 1134
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984
Source B RB.MaxRecPtr <EDC.RB_Quality02.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Quality02.WriteRecPtr> 984

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Quality02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Quality02.WriteTagPtr> 2

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 250515315
Source B 1
Dest Prog.RoutineScan[901] 250515315

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

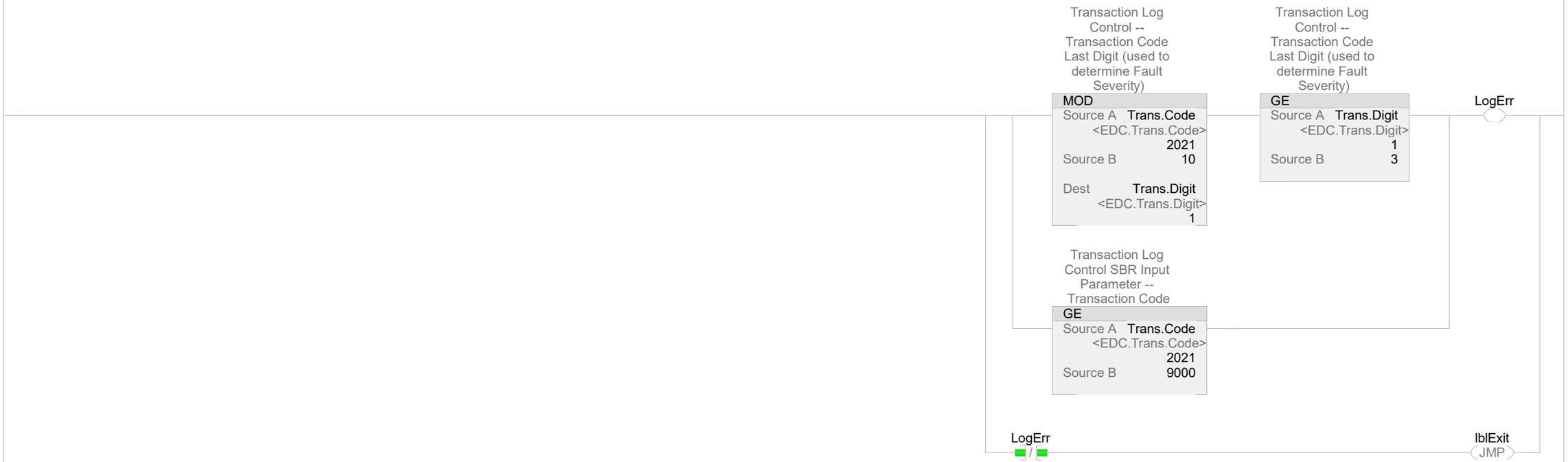
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9



Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Quality Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion ...
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	401
Dest	Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -418249704
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -418249704

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_411_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _420_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Quality03.ShiftCnt>
102565
Dest RB.PrevShiftCnt
<EDC.RB_Quality03.PrevShiftCnt>
125974

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Quality03.ShiftCnt>
102565

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 13
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.4
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 1300

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 1300
Source B	1
Dest	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 0
Source B	1
Dest	Prog.LOGi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 0
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 0
Source B	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 0

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
420
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr>
4
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality03.WriteRecPtr>
1183
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr>
4
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality03.ReadRecPtr>
1182
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(L)

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 402
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -418249704
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -418249704

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>
/

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 411
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -418249704
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -418249704

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Quality/Health Message

SIZE
Source EDC_RB_Quality03_Rec01[0] ??
Dim. To Vary 0
Size LEN 1300

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 1300
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Quality03.MaxRecPtr> 1299

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Quality03.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A RB.TransactionNumber <EDC.RB_Quality03.TransactionNumber> 316
Source B 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber <EDC.RB_Quality03.TransactionNumber> 316

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Quality03.WritePtrErr>

U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Quality03.ReadPtrErr>

U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 420
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -418249704
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -418249704

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Quality03.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Source B RB.MaxRecPtr <EDC.RB_Quality03.MaxRecPtr> 1299

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Quality03.ReadPtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Dest RB.ReadPtrErrTag <EDC.RB_Quality03.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Dest RB.ReadPtrErrRec <EDC.RB_Quality03.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2123
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Quality03.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Quality03.MaxTagPtr> 5
Dest RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Quality03.MaxRecPtr> 1299
Dest RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Quality03.ReadNextRecPtr> 1183
Source B RB.MaxRecPtr <EDC.RB_Quality03.MaxRecPtr> 1299

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Dest RB.ReadNextTagPtr <EDC.RB_Quality03.ReadNextTagPtr> 4

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Quality03.ReadNextRecPtr> 1183

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Quality03.ReadNextTagPtr> 4
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Quality03.ReadNextTagPtr> 4

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Quality03.ReadNextTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Quality03.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Quality03.ReadNextRecPtr> 1183

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Quality03.ReadNextTagPtr> 4

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Source B RB.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Source B RB.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Quality03.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality03.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Quality03.ReadNextTagPtr> 4
Source B RB.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Quality03.ReadNextRecPtr> 1183
Source B RB.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality03.Empty>

Buffer Pointers
Combine Tag & Record Pointers

5

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Quality03.WritePtr> 5083
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Quality03.ReadPtr> 5082
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Quality03.ReadNextPtr> 5083
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality03.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Quality03.WritePtr> 5083
Source B RB.ReadPtr <EDC.RB_Quality03.ReadPtr> 5082

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality03.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Quality03.WritePtr> 5083
Source B RB.ReadPtr <EDC.RB_Quality03.ReadPtr> 5082

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality03.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality03.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Quality03.WritePtr> 5083
Source B RB.ReadNextPtr <EDC.RB_Quality03.ReadNextPtr> 5083

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality03.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Quality03.WritePtr> 5083
Source B RB.ReadNextPtr <EDC.RB_Quality03.ReadNextPtr> 5083

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality03.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Quality03.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Quality03.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

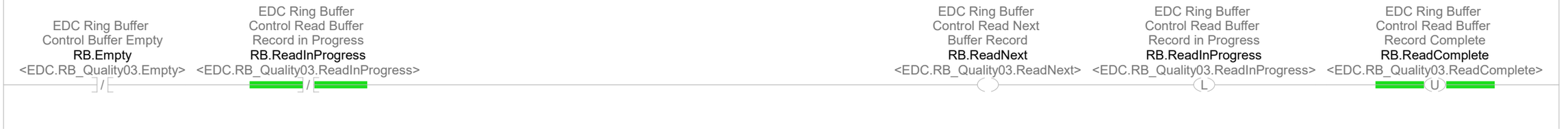
MOVE
Source RB.RecordCount <EDC.RB_Quality03.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Quality03.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

24 24

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Quality03.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Source B RB.MaxRecPtr <EDC.RB_Quality03.MaxRecPtr> 1299

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Quality03.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality03.ReadNext>

EDC Ring Buffer Control Current/Last Read Tag	Equipment Data Quality/Health Message
EQ Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> Source B 4 1	COP Source EDC_RB_Quality03_Rec01[RB.ReadRecPtr] Dest RB_Read <EDC_RB_Quality03_Read> Length 1
EQ Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> Source B 4 2	COP Source EDC_RB_Quality03_Rec02[RB.ReadRecPtr] Dest RB_Read <EDC_RB_Quality03_Read> Length 1
EQ Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> Source B 4 3	COP Source EDC_RB_Quality03_Rec03[RB.ReadRecPtr] Dest RB_Read <EDC_RB_Quality03_Read> Length 1
EQ Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> Source B 4 4	COP Source EDC_RB_Quality03_Rec04[RB.ReadRecPtr] Dest RB_Read <EDC_RB_Quality03_Read> Length 1
EQ Source A RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> Source B 4 5	COP Source EDC_RB_Quality03_Rec05[RB.ReadRecPtr] Dest RB_Read <EDC_RB_Quality03_Read> Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality03.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality03.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Quality03.TransactionNumber>
316
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Quality03.TransactionNumber>
316

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Quality03.TransactionNumber>
316
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Quality03.TransactionNumber>
316
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Quality03.TransactionNumber>
316

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality03.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality03.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Quality03_Read.Header.EQID>
61
Source B 100
Dest EQi
30

CPT
Dest LNi
9
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
9

CLR
Dest PerfEQi
30

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality03.ReadValidEQID>
U

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality03.ReadValidEQID>
L

LIMIT
Low Limit 0
Test LNi
9
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
30
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
9
Dest PerfLNi
9

MOVE
Source EQi
30
Dest PerfEQi
30

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality03.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus03
Length 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Quality03.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality03.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality03.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Quality03.ReadRecPtr> 1182
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2120
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Quality03.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality03.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.EN

Process RingBuffer Reads from Transaction Manager

14

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Quality03.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Quality03.TransactionNumber>
316
Source B RB.Interface
<EDC.RB_Quality03.Interface>
316

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Quality03.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Quality03.Interface>
316
Dest RB.InterfaceNEG
<EDC.RB_Quality03.InterfaceNEG>
-316

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Quality03.TransactionNumber>
316
Source B RB.InterfaceNEG
<EDC.RB_Quality03.InterfaceNEG>
-316

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Quality03.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

15

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Quality03.ReadComplete>

Server Status TM Transaction OK Enable High Speed RB ServerStat_TMokTMR.DN

MOVE
Source CfgRbDelaySlow 100
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Quality03.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR <EDC.RB_Quality03.ReadCompleteDelayTMR>
Preset 0
Accum 15
EN
DN

Server Status TM Transaction OK Enable High Speed RB ServerStat_TMokTMR.DN

MOVE
Source CfgRbDelayFast 0
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Quality03.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer RB.ReadCompleteDelayTMR.DN <EDC.RB_Quality03.ReadCompleteDelayTMR.DN>

EDC Ring Buffer Control Read Buffer Record Complete Delay RB.ReadCompleteDelay <EDC.RB_Quality03.ReadCompleteDelay>

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Quality03.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Quality03.ReadComplete>

EDC Ring Buffer
Control Read Buffer
Record Complete
1-shot
RB.ReadCompleteONS
<EDC.RB_Quality03.ReadCompleteONS>
ONS

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality03.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi State = Green -
Quality Ring Buffer
Tx DN
EquipStatus01.Update.QualityTxOK
L

Update Status State
for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr>
4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality03.WriteRecPtr>
1183
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr>
4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality03.ReadRecPtr>
1182
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2121
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Quality03.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality03.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Quality03.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction
aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Quality03.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Quality03.ReadComplete> /

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Quality03.ReadResend> /

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Quality03.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Quality03.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Quality03.ReadResend> U

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Quality03.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer
TON
Timer RB.ReadInProgressTMR <EDC.RB_Quality03.ReadInProgressTMR> EN
Preset 5000 DN
Accum 0

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Quality03.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Quality03.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality03.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality03.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Quality Tx Msg ER
EquipStatus01.Update.QualityTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr>: 4
Dest: Trans.Opt[1]
<EDC.Trans.Opt[1]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteRecPtr
<EDC.RB_Quality03.WriteRecPtr>: 1183
Dest: Trans.Opt[2]
<EDC.Trans.Opt[2]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr>: 4
Dest: Trans.Opt[3]
<EDC.Trans.Opt[3]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadRecPtr
<EDC.RB_Quality03.ReadRecPtr>: 1182
Dest: Trans.Opt[4]
<EDC.Trans.Opt[4]>: -1

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2124
Input Par: 0
Input Par: LNi
Input Par: EQi
Input Par: RB.TransactionNumber
<EDC.RB_Quality03.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality03.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest: TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source: RB.ReadInProgressTMR.ACC
<EDC.RB_Quality03.ReadInProgressTMR.ACC>: 0
Dest: TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans
PerfMsg: TempPerfMsg
PerfTrans: EquipStatus03.QualityTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Quality03.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Quality03.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality03.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Quality03.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Quality03.KeepAliveTMR>
Preset 300000
Accum 484
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Quality03.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Quality03.KeepAlive>

JSR
Routine Name _421_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Quality03.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality03.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Quality03.NotEmptyTMR.PRE>
300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Quality03.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Quality03.NotEmptyTMR.ACC>
0
Source B RB.NotEmptyMax
<EDC.RB_Quality03.NotEmptyMax>
300000

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Quality03.NotEmptyTMR.ACC>
0
Dest RB.NotEmptyMax
<EDC.RB_Quality03.NotEmptyMax>
300000

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Quality03.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Quality03.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Quality03.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality03.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus03
Dest	EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	421
Dest	Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -418249704
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -418249704

2

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr> 4
Source B 1

Equipment Data Quality/Health Message

COP
Source EDC_RB_Quality03_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr> 4
Source B 2

Equipment Data Quality/Health Message

COP
Source EDC_RB_Quality03_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr> 4
Source B 3

Equipment Data Quality/Health Message

COP
Source EDC_RB_Quality03_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr> 4
Source B 4

Equipment Data Quality/Health Message

COP
Source EDC_RB_Quality03_Rec04[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr> 4
Source B 5

Equipment Data Quality/Health Message

COP
Source EDC_RB_Quality03_Rec05[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Quality/Health Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1
Dest KeepAliveMsg.Header.CmdNum 1

Equipment Data Quality/Health Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1

Equipment Data Quality/Health Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 9999

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum 1

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr> 4
Source B 1

Equipment Data Quality/Health Message

COP
Source KeepAliveMsg
Dest EDC_RB_Quality03_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr> 4
Source B 2

Equipment Data Quality/Health Message

COP
Source KeepAliveMsg
Dest EDC_RB_Quality03_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr> 4
Source B 3

Equipment Data Quality/Health Message

COP
Source KeepAliveMsg
Dest EDC_RB_Quality03_Rec03[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr> 4
Source B 4

Equipment Data Quality/Health Message

COP
Source KeepAliveMsg
Dest EDC_RB_Quality03_Rec04[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr> 4
Source B 5

Equipment Data Quality/Health Message

COP
Source KeepAliveMsg
Dest EDC_RB_Quality03_Rec05[RB.WriteRecPtr]
Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality03.WriteTagPtr> 4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality03.WriteRecPtr> 1183
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality03.ReadTagPtr> 4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality03.ReadRecPtr> 1182
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2122
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Quality03.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality03.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Quality03.KeepAliveTMR.ACC> 484
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183
Source B RB.MaxRecPtr <EDC.RB_Quality03.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Quality03.WriteRecPtr> 1183

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Quality03.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Quality03.WriteTagPtr> 4

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 324617884
Source B 1
Dest Prog.RoutineScan[901] 324617884

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

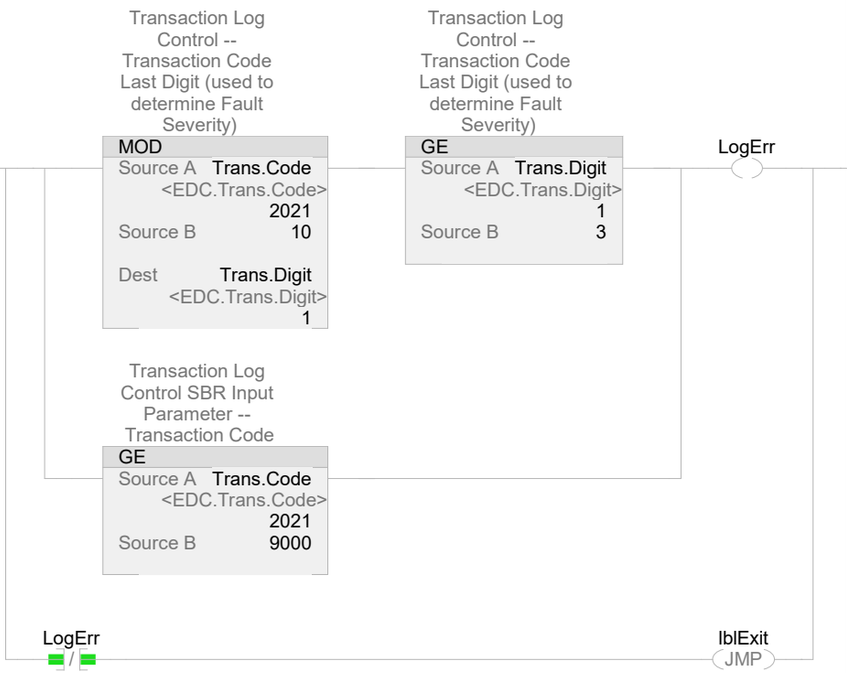
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Quality Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	401
Dest	Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -307561562
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -307561562

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

Configuration

JSR	
Routine Name	_411_Config

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _420_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Quality04.ShiftCnt>
99178
Dest RB.PrevShiftCnt
<EDC.RB_Quality04.PrevShiftCnt>
121316

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Quality04.ShiftCnt>
99178

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 14
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.4
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 1300

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 1300
Source B	1
Dest	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 0
Source B	1
Dest	Prog.LOGi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 0
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 0
Source B	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 0

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
420
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality04.WriteTagPtr>
1
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality04.WriteRecPtr>
440
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality04.ReadTagPtr>
1
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality04.ReadRecPtr>
439
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(L)

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 402
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -307561562
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -307561562

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 411
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -307561562
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -307561562

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Quality/Health Message

SIZE
Source EDC_RB_Quality04_Rec01[0] ??
Dim. To Vary 0
Size LEN 1300

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 1300
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Quality04.MaxRecPtr> 1299

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Quality04.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A RB.TransactionNumber <EDC.RB_Quality04.TransactionNumber> 4763
Source B 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber <EDC.RB_Quality04.TransactionNumber> 4763

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Quality04.WritePtrErr>

U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Quality04.ReadPtrErr>

U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 420
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -307561562
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -307561562

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Quality04.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439
Source B RB.MaxRecPtr <EDC.RB_Quality04.MaxRecPtr> 1299

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Quality04.ReadPtrErr>

(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1
Dest RB.ReadPtrErrTag <EDC.RB_Quality04.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439
Dest RB.ReadPtrErrRec <EDC.RB_Quality04.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2123
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Quality04.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Quality04.MaxTagPtr> 5
Dest RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Quality04.MaxRecPtr> 1299
Dest RB.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Quality04.ReadNextRecPtr> 440
Source B RB.MaxRecPtr <EDC.RB_Quality04.MaxRecPtr> 1299

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1
Dest RB.ReadNextTagPtr <EDC.RB_Quality04.ReadNextTagPtr> 1

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Quality04.ReadNextRecPtr> 440

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Quality04.ReadNextTagPtr> 1
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Quality04.ReadNextTagPtr> 1

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Quality04.ReadNextTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Quality04.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Quality04.ReadNextRecPtr> 440

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Quality04.ReadNextTagPtr> 1

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1
Source B RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Quality04.ReadRecPtr> 439
Source B RB.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Quality04.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality04.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Quality04.ReadNextTagPtr> 1
Source B RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Quality04.ReadNextRecPtr> 440
Source B RB.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality04.Empty>

5

Buffer Pointers
Combine Tag & Record Pointers

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Quality04.WritePtr> 440
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Quality04.ReadPtr> 439
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Quality04.ReadNextPtr> 440
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality04.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Quality04.WritePtr> 440
Source B RB.ReadPtr <EDC.RB_Quality04.ReadPtr> 439

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality04.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Quality04.WritePtr> 440
Source B RB.ReadPtr <EDC.RB_Quality04.ReadPtr> 439

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality04.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality04.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Quality04.WritePtr> 440
Source B RB.ReadNextPtr <EDC.RB_Quality04.ReadNextPtr> 440

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality04.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Quality04.WritePtr> 440
Source B RB.ReadNextPtr <EDC.RB_Quality04.ReadNextPtr> 440

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Quality04.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Quality04.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Quality04.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

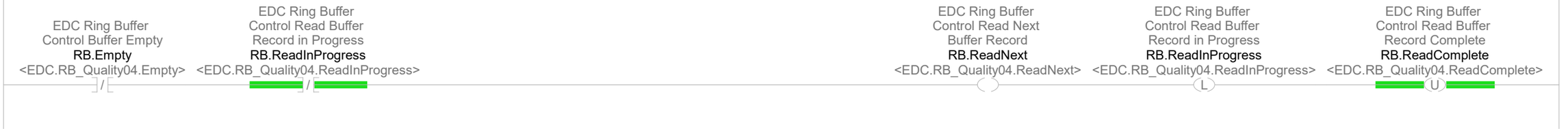
MOVE
Source RB.RecordCount <EDC.RB_Quality04.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Quality04.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

24 24

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality04.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Record

ADD
Source A RB.ReadRecPtr
<EDC.RB_Quality04.ReadRecPtr>
439
Source B
1
Dest RB.ReadRecPtr
<EDC.RB_Quality04.ReadRecPtr>
439

EDC Ring Buffer
Control Current/Last
Read Record

LT
Source A RB.ReadRecPtr
<EDC.RB_Quality04.ReadRecPtr>
439
Source B
0

EDC Ring Buffer
Control Current/Last
Read Record

GT
Source A RB.ReadRecPtr
<EDC.RB_Quality04.ReadRecPtr>
439
Source B RB.MaxRecPtr
<EDC.RB_Quality04.MaxRecPtr>
1299

EDC Ring Buffer
Control Current/Last
Read Record

MOVE
Source
0
Dest RB.ReadRecPtr
<EDC.RB_Quality04.ReadRecPtr>
439

EDC Ring Buffer
Control Current/Last
Read Tag

ADD
Source A RB.ReadTagPtr
<EDC.RB_Quality04.ReadTagPtr>
1
Source B
1
Dest RB.ReadTagPtr
<EDC.RB_Quality04.ReadTagPtr>
1

EDC Ring Buffer
Control Current/Last
Read Tag

LT
Source A RB.ReadTagPtr
<EDC.RB_Quality04.ReadTagPtr>
1
Source B
1

EDC Ring Buffer
Control Current/Last
Read Tag

GT
Source A RB.ReadTagPtr
<EDC.RB_Quality04.ReadTagPtr>
1
Source B RB.MaxTagPtr
<EDC.RB_Quality04.MaxTagPtr>
5

EDC Ring Buffer
Control Current/Last
Read Tag

MOVE
Source
1
Dest RB.ReadTagPtr
<EDC.RB_Quality04.ReadTagPtr>
1

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality04.ReadNext>

EDC Ring Buffer Control Current/Last Read Tag	Equipment Data Quality/Health Message
EQ Source A: RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1 Source B: 1	COP Source: EDC_RB_Quality04_Rec01[RB.ReadRecPtr] Dest: RB_Read <EDC_RB_Quality04_Read> Length: 1
EQ Source A: RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1 Source B: 2	COP Source: EDC_RB_Quality04_Rec02[RB.ReadRecPtr] Dest: RB_Read <EDC_RB_Quality04_Read> Length: 1
EQ Source A: RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1 Source B: 3	COP Source: EDC_RB_Quality04_Rec03[RB.ReadRecPtr] Dest: RB_Read <EDC_RB_Quality04_Read> Length: 1
EQ Source A: RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1 Source B: 4	COP Source: EDC_RB_Quality04_Rec04[RB.ReadRecPtr] Dest: RB_Read <EDC_RB_Quality04_Read> Length: 1
EQ Source A: RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr> 1 Source B: 5	COP Source: EDC_RB_Quality04_Rec05[RB.ReadRecPtr] Dest: RB_Read <EDC_RB_Quality04_Read> Length: 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality04.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality04.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Quality04.TransactionNumber>
4763
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Quality04.TransactionNumber>
4763

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Quality04.TransactionNumber>
4763
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Quality04.TransactionNumber>
4763
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Quality04.TransactionNumber>
4763

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality04.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality04.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Quality04_Read.Header.EQID>
1006
Source B 100
Dest EQi
6

CPT
Dest LNi
10
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
10

CLR
Dest PerfEQi
6

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality04.ReadValidEQID>
U

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality04.ReadValidEQID>
L

LIMIT
Low Limit 0
Test LNi
10
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
6
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
10
Dest PerfLNi
10

MOVE
Source EQi
6
Dest PerfEQi
6

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality04.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus03
Length 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Quality04.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality04.ReadInProgressTimeout>

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality04.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality04.WriteRecPtr>
440
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality04.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality04.ReadRecPtr>
439
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2120
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Quality04.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality04.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.EN

Process RingBuffer Reads from Transaction Manager

14

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Quality04.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Quality04.TransactionNumber>
4763
Source B RB.Interface
<EDC.RB_Quality04.Interface>
4763

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Quality04.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Quality04.Interface>
4763
Dest RB.InterfaceNEG
<EDC.RB_Quality04.InterfaceNEG>
-4763

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Quality04.TransactionNumber>
4763
Source B RB.InterfaceNEG
<EDC.RB_Quality04.InterfaceNEG>
-4763

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Quality04.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

15

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Quality04.ReadComplete>

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN

MOVE
Source CfgRbDelaySlow
100
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Quality04.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR
<EDC.RB_Quality04.ReadCompleteDelayTMR>
Preset 0
Accum 12
EN
DN

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN

MOVE
Source CfgRbDelayFast
0
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Quality04.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer Control Read Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Quality04.ReadCompleteDelayTMR.DN>

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Quality04.ReadCompleteDelay>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality04.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Quality04.ReadComplete>

EDC Ring Buffer
Control Read Buffer
Record Complete
1-shot
RB.ReadCompleteONS
<EDC.RB_Quality04.ReadCompleteONS>
ONS

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality04.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi State = Green -
Quality Ring Buffer
Tx DN
EquipStatus01.Update.QualityTxOK
L

Update Status State
for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.WriteTagPtr
	<EDC.RB_Quality04.WriteTagPtr>
	1
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.WriteRecPtr
	<EDC.RB_Quality04.WriteRecPtr>
	440
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.ReadTagPtr
	<EDC.RB_Quality04.ReadTagPtr>
	1
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	RB.ReadRecPtr
	<EDC.RB_Quality04.ReadRecPtr>
	439
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2121
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber
	<EDC.RB_Quality04.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Quality04.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC
	<EDC.RB_Quality04.ReadInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

Performance
Transaction

aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Quality04.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Quality04.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend
RB.ReadResend
<EDC.RB_Quality04.ReadResend>

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE
<EDC.RB_Quality04.ReadInProgressTMR.PRE>
5000

EDC Ring Buffer Control Read Buffer Record Resend
RB.ReadResend
<EDC.RB_Quality04.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend
RB.ReadResend
<EDC.RB_Quality04.ReadResend>
U

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE
<EDC.RB_Quality04.ReadInProgressTMR.PRE>
5000

EDC Ring Buffer Control Read in Progress Timer
TON
Timer RB.ReadInProgressTMR
<EDC.RB_Quality04.ReadInProgressTMR>
Preset 5000
Accum 0
EN
DN

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR.DN
<EDC.RB_Quality04.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality04.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Quality04.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality04.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Quality Tx Msg ER
EquipStatus01.Update.QualityTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteTagPtr
	<EDC.RB_Quality04.WriteTagPtr>
	1
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteRecPtr
	<EDC.RB_Quality04.WriteRecPtr>
	440
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadTagPtr
	<EDC.RB_Quality04.ReadTagPtr>
	1
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadRecPtr
	<EDC.RB_Quality04.ReadRecPtr>
	439
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2124
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber
	<EDC.RB_Quality04.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality04.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC
	<EDC.RB_Quality04.ReadInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EquipStatus03.QualityTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Quality04.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Quality04.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality04.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Quality04.KeepAliveTMR.PRE> 300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Quality04.KeepAliveTMR>
Preset 300000
Accum 319
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Quality04.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Quality04.KeepAlive>

JSR
Routine Name _421_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Quality04.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Quality04.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Quality04.NotEmptyTMR.PRE> 300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Quality04.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Quality04.NotEmptyTMR.ACC> 0
Source B RB.NotEmptyMax
<EDC.RB_Quality04.NotEmptyMax> 300001

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Quality04.NotEmptyTMR.ACC> 0
Dest RB.NotEmptyMax
<EDC.RB_Quality04.NotEmptyMax> 300001

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Quality04.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Quality04.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Quality04.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality04.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus03
Dest	EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	421
Dest	Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -307561562
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -307561562

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

2

EDC Ring Buffer Control Current/Last Read Tag		Equipment Data Quality/Health Message	
EQ		COP	
Source A	RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr>	Source	EDC_RB_Quality04_Rec01[RB.ReadRecPtr]
	1	Dest	KeepAliveMsg
Source B	1	Length	1

EDC Ring Buffer Control Current/Last Read Tag		Equipment Data Quality/Health Message	
EQ		COP	
Source A	RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr>	Source	EDC_RB_Quality04_Rec02[RB.ReadRecPtr]
	1	Dest	KeepAliveMsg
Source B	2	Length	1

EDC Ring Buffer Control Current/Last Read Tag		Equipment Data Quality/Health Message	
EQ		COP	
Source A	RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr>	Source	EDC_RB_Quality04_Rec03[RB.ReadRecPtr]
	1	Dest	KeepAliveMsg
Source B	3	Length	1

EDC Ring Buffer Control Current/Last Read Tag		Equipment Data Quality/Health Message	
EQ		COP	
Source A	RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr>	Source	EDC_RB_Quality04_Rec04[RB.ReadRecPtr]
	1	Dest	KeepAliveMsg
Source B	4	Length	1

EDC Ring Buffer Control Current/Last Read Tag		Equipment Data Quality/Health Message	
EQ		COP	
Source A	RB.ReadTagPtr <EDC.RB_Quality04.ReadTagPtr>	Source	EDC_RB_Quality04_Rec05[RB.ReadRecPtr]
	1	Dest	KeepAliveMsg
Source B	5	Length	1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Quality/Health Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1
Dest KeepAliveMsg.Header.CmdNum 1

Equipment Data Quality/Health Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1

Equipment Data Quality/Health Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 9999

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum 1

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

4

	EDC Ring Buffer Control Next Write Tag	Equipment Data Quality/Health Message
	EQ Source A: RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr>: 1 Source B: 1	COP Source: KeepAliveMsg Dest: EDC_RB_Quality04_Rec01[RB.WriteRecPtr] Length: 1
	EQ Source A: RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr>: 1 Source B: 2	COP Source: KeepAliveMsg Dest: EDC_RB_Quality04_Rec02[RB.WriteRecPtr] Length: 1
	EQ Source A: RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr>: 1 Source B: 3	COP Source: KeepAliveMsg Dest: EDC_RB_Quality04_Rec03[RB.WriteRecPtr] Length: 1
	EQ Source A: RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr>: 1 Source B: 4	COP Source: KeepAliveMsg Dest: EDC_RB_Quality04_Rec04[RB.WriteRecPtr] Length: 1
	EQ Source A: RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr>: 1 Source B: 5	COP Source: KeepAliveMsg Dest: EDC_RB_Quality04_Rec05[RB.WriteRecPtr] Length: 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Quality04.WriteTagPtr> 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Quality04.WriteRecPtr> 440
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Quality04.ReadTagPtr> 1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Quality04.ReadRecPtr> 439
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2122
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Quality04.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Quality04.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Quality04.KeepAliveTMR.ACC> 319
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.QualityTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440
Source B RB.MaxRecPtr <EDC.RB_Quality04.MaxRecPtr> 1299

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Quality04.WriteRecPtr> 440

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Quality04.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Quality04.WriteTagPtr> 1

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 420

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 339379987
Source B 1
Dest Prog.RoutineScan[901] 339379987

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

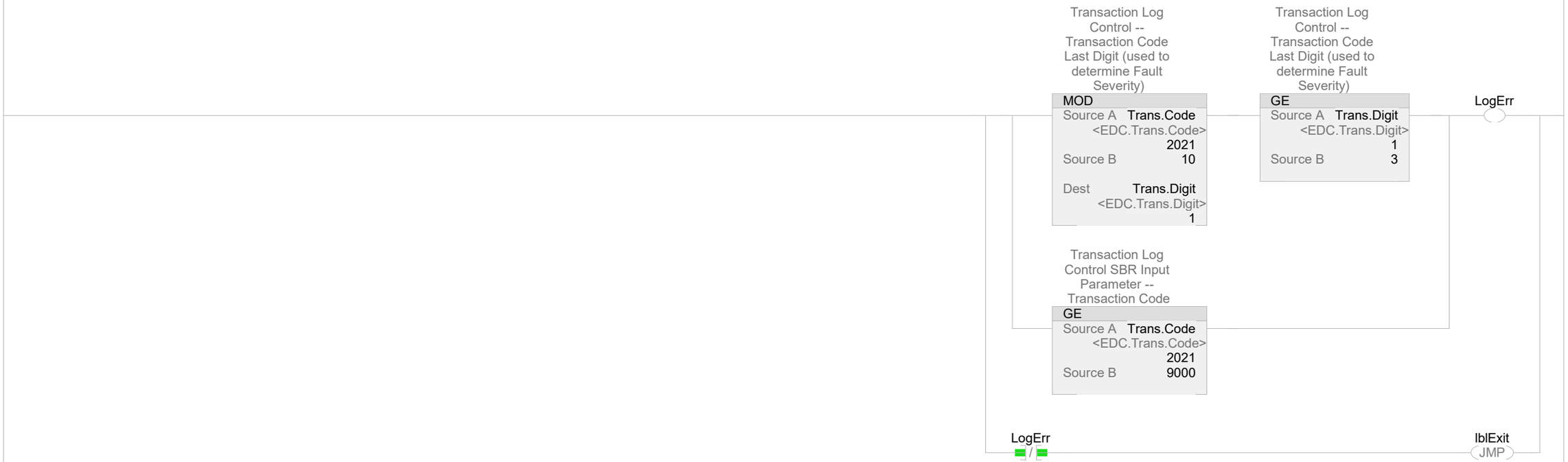
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Trend Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	501
Dest	Prog.Routine 520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562840552
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562840552

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

Configuration

JSR	
Routine Name	_511_Config

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _520_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Trend01.ShiftCnt>
2413
Dest RB.PrevShiftCnt
<EDC.RB_Trend01.PrevShiftCnt>
2553

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Trend01.ShiftCnt>
2413

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 13
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.5
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

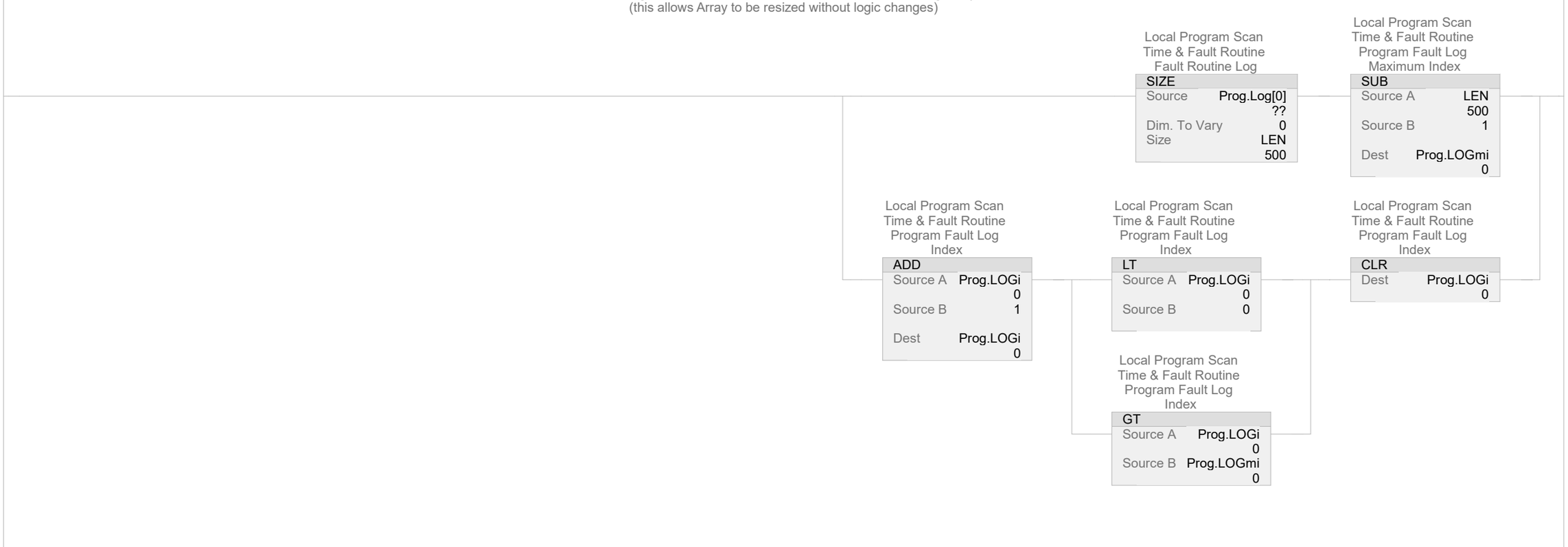
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
520
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Trend01.WriteTagPtr>
4
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Trend01.WriteRecPtr>
326
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr>
4
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Trend01.ReadRecPtr>
325
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 502
Dest Prog.Routine 520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562840552
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562840552

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>
/

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 511
Dest Prog.Routine 520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562840552
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562840552

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Trend Message

SIZE
Source EDC_RB_Trend01_Rec01[0] ??
Dim. To Vary 0
Size LEN 500

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 500
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Trend01.MaxRecPtr> 499

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Trend01.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A RB.TransactionNumber <EDC.RB_Trend01.TransactionNumber> 2037
Source B 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber <EDC.RB_Trend01.TransactionNumber> 2037

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Trend01.WritePtrErr>

U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Trend01.ReadPtrErr>

U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 520
Dest Prog.Routine 520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562840552
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562840552

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A: RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Source B: 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A: RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Source B: RB.MaxTagPtr <EDC.RB_Trend01.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A: RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Source B: 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A: RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Source B: RB.MaxRecPtr <EDC.RB_Trend01.MaxRecPtr> 499

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Trend01.ReadPtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source: RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Dest: RB.ReadPtrErrTag <EDC.RB_Trend01.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source: RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Dest: RB.ReadPtrErrRec <EDC.RB_Trend01.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2223
Input Par: 0
Input Par: -1
Input Par: -1
Input Par: RB.TransactionNumber <EDC.RB_Trend01.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source: 1
Dest: RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record

MOVE
Source: 0
Dest: RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source: RB.MaxTagPtr <EDC.RB_Trend01.MaxTagPtr> 5
Dest: RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source: RB.MaxRecPtr <EDC.RB_Trend01.MaxRecPtr> 499
Dest: RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Trend01.ReadNextRecPtr> 326
Source B RB.MaxRecPtr <EDC.RB_Trend01.MaxRecPtr> 499

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Dest RB.ReadNextTagPtr <EDC.RB_Trend01.ReadNextTagPtr> 4

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Trend01.ReadNextRecPtr> 326

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Trend01.ReadNextTagPtr> 4
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Trend01.ReadNextTagPtr> 4

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Trend01.ReadNextTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Trend01.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Trend01.ReadNextRecPtr> 326

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Trend01.ReadNextTagPtr> 4

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Source B RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Source B RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Trend01.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Trend01.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Trend01.ReadNextTagPtr> 4
Source B RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Trend01.ReadNextRecPtr> 326
Source B RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Trend01.Empty>

Buffer Pointers
Combine Tag & Record Pointers

5

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Trend01.WritePtr> 1826
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Trend01.ReadPtr> 1825
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Trend01.ReadNextPtr> 1826
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Trend01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Trend01.WritePtr> 1826
Source B RB.ReadPtr <EDC.RB_Trend01.ReadPtr> 1825

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Trend01.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Trend01.WritePtr> 1826
Source B RB.ReadPtr <EDC.RB_Trend01.ReadPtr> 1825

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Trend01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Trend01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Trend01.WritePtr> 1826
Source B RB.ReadNextPtr <EDC.RB_Trend01.ReadNextPtr> 1826

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Trend01.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Trend01.WritePtr> 1826
Source B RB.ReadNextPtr <EDC.RB_Trend01.ReadNextPtr> 1826

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Trend01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Trend01.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Trend01.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

MOVE
Source RB.RecordCount <EDC.RB_Trend01.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Trend01.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

2 2

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Trend01.Empty>	EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Trend01.ReadInProgress>	EDC Ring Buffer Control Read Next Buffer Record RB.ReadNext <EDC.RB_Trend01.ReadNext>	EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Trend01.ReadInProgress>	EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Trend01.ReadComplete>
7 /	/	( )	L	U

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Trend01.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Source B RB.MaxRecPtr <EDC.RB_Trend01.MaxRecPtr> 499

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Trend01.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Trend01.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr>
4
Source B 1

Equipment Data
Trend Message

COP
Source EDC_RB_Trend01_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Trend01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr>
4
Source B 2

Equipment Data
Trend Message

COP
Source EDC_RB_Trend01_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Trend01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr>
4
Source B 3

Equipment Data
Trend Message

COP
Source EDC_RB_Trend01_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Trend01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr>
4
Source B 4

Equipment Data
Trend Message

COP
Source EDC_RB_Trend01_Rec04[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Trend01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr>
4
Source B 5

Equipment Data
Trend Message

COP
Source EDC_RB_Trend01_Rec05[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Trend01_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Trend01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Trend01.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Trend01.TransactionNumber>
2037
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Trend01.TransactionNumber>
2037

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Trend01.TransactionNumber>
2037
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Trend01.TransactionNumber>
2037
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Trend01.TransactionNumber>
2037

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Trend01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Trend01.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Trend01_Read.Header.EQID>
528
Source B 100
Dest EQi
25

CPT
Dest LNi
5
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
5

CLR
Dest PerfEQi
25

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Trend01.ReadValidEQID>
U

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Trend01.ReadValidEQID>
L

LIMIT
Low Limit 0
Test LNi
5
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
25
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
5
Dest PerfLNi
5

MOVE
Source EQi
25
Dest PerfEQi
25

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Trend01.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus03
Length 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Trend01.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Trend01.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2220
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Trend01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.TrendTx.EN

Process RingBuffer Reads from Transaction Manager

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

14

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Trend01.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Trend01.TransactionNumber>
2037
Source B RB.Interface
<EDC.RB_Trend01.Interface>
2037

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Trend01.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Trend01.Interface>
2037
Dest RB.InterfaceNEG
<EDC.RB_Trend01.InterfaceNEG>
-2037

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Trend01.TransactionNumber>
2037
Source B RB.InterfaceNEG
<EDC.RB_Trend01.InterfaceNEG>
-2037

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Trend01.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

15

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Trend01.ReadComplete>

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN
/

EDC Ring Buffer
Control Read
Complete Delay Timer

MOVE
Source CfgRbDelaySlow
100
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Trend01.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer
Control Read
Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR
<EDC.RB_Trend01.ReadCompleteDelayTMR>
Preset 0
Accum 15
EN
DN

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN

EDC Ring Buffer
Control Read
Complete Delay Timer

MOVE
Source CfgRbDelayFast
0
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Trend01.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer
Control Read
Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Trend01.ReadCompleteDelayTMR.DN>

EDC Ring Buffer
Control Read Buffer
Record Complete
Delay
RB.ReadCompleteDelay
<EDC.RB_Trend01.ReadCompleteDelay>

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Trend01.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Trend01.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Complete 1-shot
RB.ReadCompleteONS
<EDC.RB_Trend01.ReadCompleteONS>
ONS

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Green - Trend Ring Buffer Tx DN
EquipStatus01.Update.TrendTxOK
L

Update Status State for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr>
4
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr>
326
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr>
4
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source: RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr>
325
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name: _901_TransLogSBR
Input Par: 2221
Input Par: 0
Input Par: LNi
Input Par: EQi
Input Par: RB.TransactionNumber <EDC.RB_Trend01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest: TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source: RB.ReadInProgressTMR.ACC <EDC.RB_Trend01.ReadInProgressTMR.ACC>
0
Dest: TempPerfMsg.TransTMR.ACC
110

Performance Transaction
aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.TrendTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Trend01.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Trend01.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend
RB.ReadResend
<EDC.RB_Trend01.ReadResend>

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE
<EDC.RB_Trend01.ReadInProgressTMR.PRE>
5000

EDC Ring Buffer Control Read Buffer Record Resend
RB.ReadResend
<EDC.RB_Trend01.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend
RB.ReadResend
<EDC.RB_Trend01.ReadResend>
U

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE
<EDC.RB_Trend01.ReadInProgressTMR.PRE>
5000

EDC Ring Buffer Control Read in Progress Timer
TON
Timer RB.ReadInProgressTMR
<EDC.RB_Trend01.ReadInProgressTMR>
Preset 5000
Accum 0
EN
DN

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR.DN
<EDC.RB_Trend01.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Trend01.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Trend01.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Trend Tx Msg ER
EquipStatus01.Update.TrendTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteTagPtr
	<EDC.RB_Trend01.WriteTagPtr>
	4
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteRecPtr
	<EDC.RB_Trend01.WriteRecPtr>
	326
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadTagPtr
	<EDC.RB_Trend01.ReadTagPtr>
	4
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadRecPtr
	<EDC.RB_Trend01.ReadRecPtr>
	325
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2224
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber
	<EDC.RB_Trend01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC
	<EDC.RB_Trend01.ReadInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EquipStatus03.TrendTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Trend01.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Trend01.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Trend01.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Trend01.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Trend01.KeepAliveTMR>
Preset 300000
Accum 206916
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Trend01.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Trend01.KeepAlive>

JSR
Routine Name _521_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Trend01.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Trend01.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Trend01.NotEmptyTMR.PRE>
300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Trend01.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Trend01.NotEmptyTMR.ACC>
0
Source B RB.NotEmptyMax
<EDC.RB_Trend01.NotEmptyMax>
300001

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Trend01.NotEmptyTMR.ACC>
0
Dest RB.NotEmptyMax
<EDC.RB_Trend01.NotEmptyMax>
300001

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Trend01.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Trend01.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Trend01.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend01.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus03
Dest	EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 521
Dest Prog.Routine 520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562840552
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562840552

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

2

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr> 4
Source B 1

Equipment Data Trend Message

COP
Source EDC_RB_Trend01_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr> 4
Source B 2

Equipment Data Trend Message

COP
Source EDC_RB_Trend01_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr> 4
Source B 3

Equipment Data Trend Message

COP
Source EDC_RB_Trend01_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr> 4
Source B 4

Equipment Data Trend Message

COP
Source EDC_RB_Trend01_Rec04[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend01.ReadTagPtr> 4
Source B 5

Equipment Data Trend Message

COP
Source EDC_RB_Trend01_Rec05[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Trend Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum
2
Source B 1
Dest KeepAliveMsg.Header.CmdNum
2

Equipment Data Trend Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum
2
Source B 1

Equipment Data Trend Message Header - Command Number

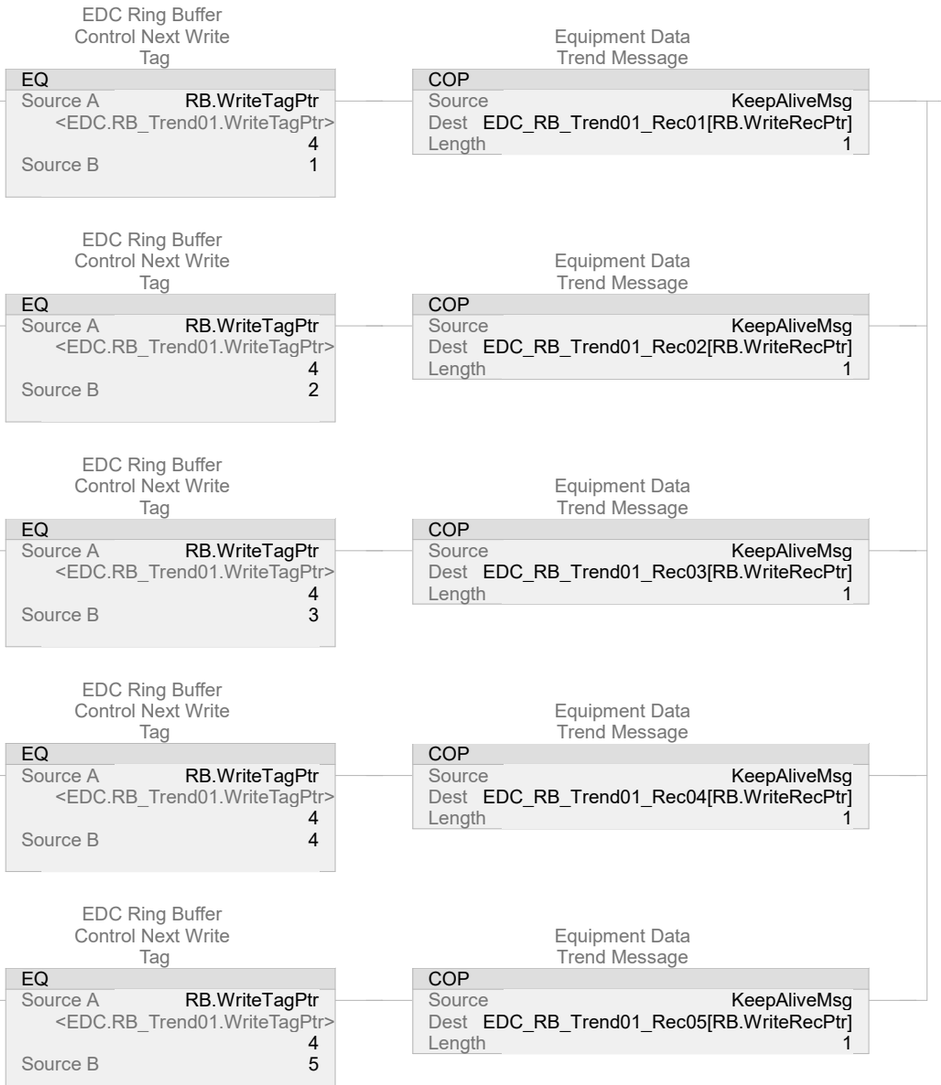
GT
Source A KeepAliveMsg.Header.CmdNum
2
Source B 9999

Equipment Data Trend Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum
2

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05



5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Trend01.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Trend01.ReadRecPtr> 325
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2222
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Trend01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID <EDC.RB_Trend01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC <EDC.RB_Trend01.KeepAliveTMR.ACC> 206916
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.TrendTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326
Source B RB.MaxRecPtr <EDC.RB_Trend01.MaxRecPtr> 499

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Trend01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Trend01.WriteRecPtr> 326

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Trend01.WriteTagPtr> 4

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine
520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901]
22328558
Source B 1
Dest Prog.RoutineScan[901]
22328558

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

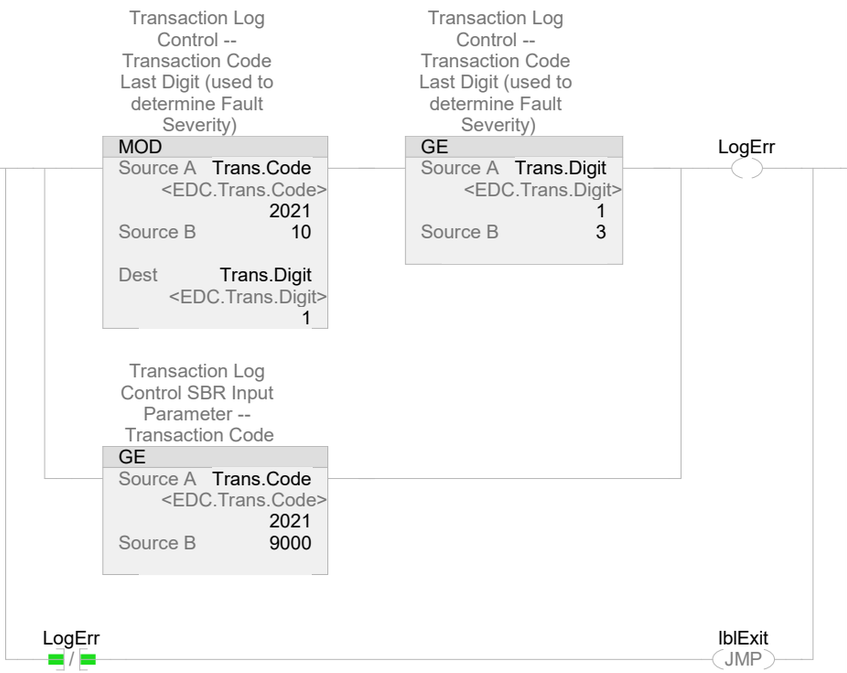
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Trend Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	501
Dest	Prog.Routine 520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -562839822
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -562839822

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_511_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _520_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Trend02.ShiftCnt>
2674
Dest RB.PrevShiftCnt
<EDC.RB_Trend02.PrevShiftCnt>
3106

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Trend02.ShiftCnt>
2674

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 15
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.5
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 500

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 500
Source B	1
Dest	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 0
Source B	1
Dest	Prog.LOGi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 0
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 0
Source B	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 0

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
520
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Trend02.WriteTagPtr>
3
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Trend02.WriteRecPtr>
202
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr>
3
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Trend02.ReadRecPtr>
201
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(L)

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 502
Dest Prog.Routine 520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562839822
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562839822

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 511
Dest Prog.Routine 520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562839822
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562839822

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Trend Message

SIZE
Source EDC_RB_Trend02_Rec01[0] ??
Dim. To Vary 0
Size LEN 500

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 500
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Trend02.MaxRecPtr> 499

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Trend02.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A RB.TransactionNumber <EDC.RB_Trend02.TransactionNumber> 6415
Source B 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber <EDC.RB_Trend02.TransactionNumber> 6415

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Trend02.WritePtrErr>

U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Trend02.ReadPtrErr>

U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 520
Dest Prog.Routine 520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562839822
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562839822

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Trend02.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Source B RB.MaxRecPtr <EDC.RB_Trend02.MaxRecPtr> 499

EDC Ring Buffer Control Diagnostic - Manual Reset
RB.ReadPtrErr <EDC.RB_Trend02.ReadPtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Dest RB.ReadPtrErrTag <EDC.RB_Trend02.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Dest RB.ReadPtrErrRec <EDC.RB_Trend02.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2223
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Trend02.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Trend02.MaxTagPtr> 5
Dest RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Trend02.MaxRecPtr> 499
Dest RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Dest RB.ReadNextTagPtr <EDC.RB_Trend02.ReadNextTagPtr> 3

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Trend02.ReadNextRecPtr> 202

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Trend02.ReadNextRecPtr> 202
Source B RB.MaxRecPtr <EDC.RB_Trend02.MaxRecPtr> 499

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Trend02.ReadNextRecPtr> 202

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Trend02.ReadNextTagPtr> 3
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Trend02.ReadNextTagPtr> 3

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Trend02.ReadNextTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Trend02.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Trend02.ReadNextTagPtr> 3

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Source B RB.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Source B RB.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Trend02.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Trend02.ReadInProgress>
/

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Trend02.ReadNextTagPtr> 3
Source B RB.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Trend02.ReadNextRecPtr> 202
Source B RB.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Trend02.Empty>

Buffer Pointers
Combine Tag & Record Pointers

5

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Trend02.WritePtr> 1202
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Trend02.ReadPtr> 1201
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Trend02.ReadNextPtr> 1202
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Trend02.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr
<EDC.RB_Trend02.WritePtr>
1202
Source B RB.ReadPtr
<EDC.RB_Trend02.ReadPtr>
1201

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount
<EDC.RB_Trend02.RecordCount>
0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr
<EDC.RB_Trend02.WritePtr>
1202
Source B RB.ReadPtr
<EDC.RB_Trend02.ReadPtr>
1201

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount
<EDC.RB_Trend02.RecordCount>
0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Trend02.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr
<EDC.RB_Trend02.WritePtr>
1202
Source B RB.ReadNextPtr
<EDC.RB_Trend02.ReadNextPtr>
1202

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount
<EDC.RB_Trend02.RecordCount>
0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr
<EDC.RB_Trend02.WritePtr>
1202
Source B RB.ReadNextPtr
<EDC.RB_Trend02.ReadNextPtr>
1202

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount
<EDC.RB_Trend02.RecordCount>
0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount
<EDC.RB_Trend02.RecordCount>
0
Source B RB.RecordCountMax
<EDC.RB_Trend02.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

MOVE
Source RB.RecordCount
<EDC.RB_Trend02.RecordCount>
0
Dest RB.RecordCountMax
<EDC.RB_Trend02.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

2 2

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Trend02.Empty>	EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Trend02.ReadInProgress>	EDC Ring Buffer Control Read Next Buffer Record RB.ReadNext <EDC.RB_Trend02.ReadNext>	EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Trend02.ReadInProgress>	EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Trend02.ReadComplete>
7 /	/		L	U

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Trend02.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Source B RB.MaxRecPtr <EDC.RB_Trend02.MaxRecPtr> 499

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Trend02.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Trend02.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr>
Source B 3 1

Equipment Data
Trend Message

COP
Source EDC_RB_Trend02_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Trend02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr>
Source B 3 2

Equipment Data
Trend Message

COP
Source EDC_RB_Trend02_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Trend02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr>
Source B 3 3

Equipment Data
Trend Message

COP
Source EDC_RB_Trend02_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Trend02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr>
Source B 3 4

Equipment Data
Trend Message

COP
Source EDC_RB_Trend02_Rec04[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Trend02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr>
Source B 3 5

Equipment Data
Trend Message

COP
Source EDC_RB_Trend02_Rec05[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Trend02_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Trend02.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Trend02.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Trend02.TransactionNumber>
6415
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Trend02.TransactionNumber>
6415

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Trend02.TransactionNumber>
6415
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Trend02.TransactionNumber>
6415
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Trend02.TransactionNumber>
6415

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Trend02.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Trend02.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Trend02_Read.Header.EQID>
531
Source B 100
Dest EQi
4

CPT
Dest LNi
15
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
15

CLR
Dest PerfEQi
4

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Trend02.ReadValidEQID>
U

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Trend02.ReadValidEQID>
L

LIMIT
Low Limit 0
Test LNi
15
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
4
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
15
Dest PerfLNi
15

MOVE
Source EQi
4
Dest PerfEQi
4

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Trend02.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus03
Length 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Trend02.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Trend02.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Trend02.ReadTagPtr> 3
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Trend02.ReadRecPtr> 201
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2220
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Trend02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.TrendTx.EN

Process RingBuffer Reads from Transaction Manager

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

14

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Trend02.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Trend02.TransactionNumber>
6415
Source B RB.Interface
<EDC.RB_Trend02.Interface>
6415

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Trend02.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Trend02.Interface>
6415
Dest RB.InterfaceNEG
<EDC.RB_Trend02.InterfaceNEG>
-6415

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Trend02.TransactionNumber>
6415
Source B RB.InterfaceNEG
<EDC.RB_Trend02.InterfaceNEG>
-6415

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Trend02.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

15

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Trend02.ReadComplete>

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN
/

MOVE
Source CfgRbDelaySlow
100
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Trend02.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer
Control Read
Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR
<EDC.RB_Trend02.ReadCompleteDelayTMR>
Preset 0
Accum 14
EN
DN

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN

MOVE
Source CfgRbDelayFast
0
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Trend02.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer
Control Read
Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Trend02.ReadCompleteDelayTMR.DN>

EDC Ring Buffer
Control Read Buffer
Record Complete
Delay
RB.ReadCompleteDelay
<EDC.RB_Trend02.ReadCompleteDelay>

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Trend02.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Trend02.ReadComplete>

EDC Ring Buffer
Control Read Buffer
Record Complete
1-shot
RB.ReadCompleteONS
<EDC.RB_Trend02.ReadCompleteONS>
ONS

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Trend02.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi State = Green -
Trend Ring Buffer Tx
DN
EquipStatus01.Update.TrendTxOK
L

Update Status State
for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Trend02.WriteTagPtr>
3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Trend02.WriteRecPtr>
202
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr>
3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Trend02.ReadRecPtr>
201
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2221
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Trend02.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Trend02.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Trend02.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction
aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.TrendTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Trend02.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Trend02.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Trend02.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Trend02.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Trend02.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Trend02.ReadResend>

U

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Trend02.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer

TON
Timer RB.ReadInProgressTMR <EDC.RB_Trend02.ReadInProgressTMR>
Preset 5000
Accum 0
EN
DN

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Trend02.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Trend02.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Trend02.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend02.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Trend Tx Msg ER
EquipStatus01.Update.TrendTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteTagPtr
	<EDC.RB_Trend02.WriteTagPtr>
	3
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteRecPtr
	<EDC.RB_Trend02.WriteRecPtr>
	202
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadTagPtr
	<EDC.RB_Trend02.ReadTagPtr>
	3
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadRecPtr
	<EDC.RB_Trend02.ReadRecPtr>
	201
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2224
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber
	<EDC.RB_Trend02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC
	<EDC.RB_Trend02.ReadInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EquipStatus03.TrendTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Trend02.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Trend02.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Trend02.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Trend02.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Trend02.KeepAliveTMR>
Preset 300000
Accum 226874
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Trend02.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Trend02.KeepAlive>

JSR
Routine Name _521_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Trend02.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Trend02.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Trend02.NotEmptyTMR.PRE>
300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Trend02.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Trend02.NotEmptyTMR.ACC>
0
Source B RB.NotEmptyMax
<EDC.RB_Trend02.NotEmptyMax>
300004

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Trend02.NotEmptyTMR.ACC>
0
Dest RB.NotEmptyMax
<EDC.RB_Trend02.NotEmptyMax>
300004

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Trend02.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Trend02.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Trend02.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend02.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus03
Dest	EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 521
Dest Prog.Routine 520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -562839822
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -562839822

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

2

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr> 3
Source B 1

Equipment Data Trend Message

COP
Source EDC_RB_Trend02_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr> 3
Source B 2

Equipment Data Trend Message

COP
Source EDC_RB_Trend02_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr> 3
Source B 3

Equipment Data Trend Message

COP
Source EDC_RB_Trend02_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr> 3
Source B 4

Equipment Data Trend Message

COP
Source EDC_RB_Trend02_Rec04[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr> 3
Source B 5

Equipment Data Trend Message

COP
Source EDC_RB_Trend02_Rec05[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Trend Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum
2
Source B 1
Dest KeepAliveMsg.Header.CmdNum
2

Equipment Data Trend Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum
2
Source B 1

Equipment Data Trend Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum
2
Source B 9999

Equipment Data Trend Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum
2

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Trend02.WriteTagPtr> 3
Source B 1

Equipment Data Trend Message

COP
Source KeepAliveMsg
Dest EDC_RB_Trend02_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Trend02.WriteTagPtr> 3
Source B 2

Equipment Data Trend Message

COP
Source KeepAliveMsg
Dest EDC_RB_Trend02_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Trend02.WriteTagPtr> 3
Source B 3

Equipment Data Trend Message

COP
Source KeepAliveMsg
Dest EDC_RB_Trend02_Rec03[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Trend02.WriteTagPtr> 3
Source B 4

Equipment Data Trend Message

COP
Source KeepAliveMsg
Dest EDC_RB_Trend02_Rec04[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Trend02.WriteTagPtr> 3
Source B 5

Equipment Data Trend Message

COP
Source KeepAliveMsg
Dest EDC_RB_Trend02_Rec05[RB.WriteRecPtr]
Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Trend02.WriteTagPtr>
3
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Trend02.WriteRecPtr>
202
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Trend02.ReadTagPtr>
3
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Trend02.ReadRecPtr>
201
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2222
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Trend02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Trend02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Trend02.KeepAliveTMR.ACC>
226874
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.TrendTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202
Source B RB.MaxRecPtr <EDC.RB_Trend02.MaxRecPtr> 499

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3
Source B RB.MaxTagPtr <EDC.RB_Trend02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Trend02.WriteRecPtr> 202

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Trend02.WriteTagPtr> 3

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine
520

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901]
21824920
Source B 1
Dest Prog.RoutineScan[901]
21824920

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

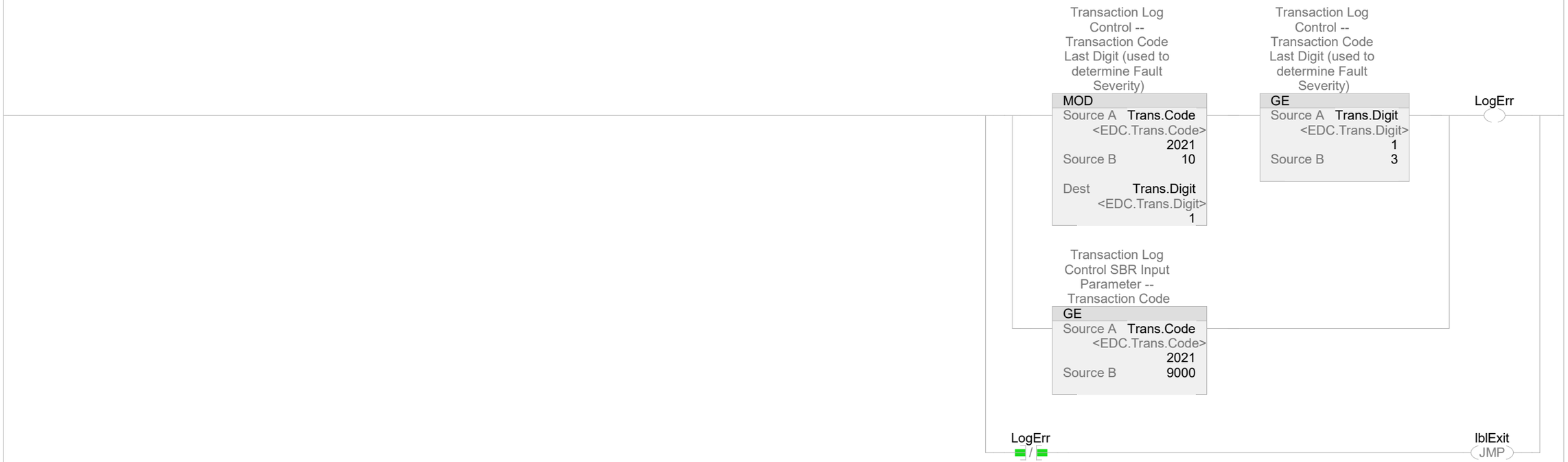
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9



Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Summary Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	601
Dest	Prog.Routine 620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -850801397
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -850801397

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_611_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _620_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Sum01.ShiftCnt>
113402
Dest RB.PrevShiftCnt
<EDC.RB_Sum01.PrevShiftCnt>
132552

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Sum01.ShiftCnt>
113402

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A Prog.Scan.Last 14
Source B CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A ProgScanHighCount 0
Source B 1
Dest ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.6
L

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
620
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Sum01.WriteTagPtr>
1
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Sum01.WriteRecPtr>
88
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr>
1
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Sum01.ReadRecPtr>
87
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 602
Dest Prog.Routine 620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -850801397
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -850801397

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>
/

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 611
Dest Prog.Routine 620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -850801397
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -850801397

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Summary Message

SIZE
Source EDC_RB_Sum01_Rec01[0] ??
Dim. To Vary 0
Size LEN 200

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 200
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Sum01.MaxRecPtr> 199

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Sum01.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A RB.TransactionNumber <EDC.RB_Sum01.TransactionNumber> 4705
Source B 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber <EDC.RB_Sum01.TransactionNumber> 4705

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Sum01.WritePtrErr>
U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Sum01.ReadPtrErr>
U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 620
Dest Prog.Routine 620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -850801397
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -850801397

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Sum01.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Source B RB.MaxRecPtr <EDC.RB_Sum01.MaxRecPtr> 199

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Sum01.ReadPtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Dest RB.ReadPtrErrTag <EDC.RB_Sum01.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Dest RB.ReadPtrErrRec <EDC.RB_Sum01.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2423
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Sum01.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Sum01.MaxTagPtr> 5
Dest RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Sum01.MaxRecPtr> 199
Dest RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Sum01.ReadNextRecPtr> 88
Source B RB.MaxRecPtr <EDC.RB_Sum01.MaxRecPtr> 199

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Dest RB.ReadNextTagPtr <EDC.RB_Sum01.ReadNextTagPtr> 1

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Sum01.ReadNextRecPtr> 88

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Sum01.ReadNextTagPtr> 1
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Sum01.ReadNextTagPtr> 1

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Sum01.ReadNextTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Sum01.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Sum01.ReadNextRecPtr> 88

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Sum01.ReadNextTagPtr> 1

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Source B RB.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Source B RB.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

EDC Ring Buffer Control Buffer Full RB.Full <EDC.RB_Sum01.Full>

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Sum01.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Sum01.ReadNextTagPtr> 1
Source B RB.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Sum01.ReadNextRecPtr> 88
Source B RB.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Sum01.Empty>

5

Buffer Pointers
Combine Tag & Record Pointers

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Sum01.WritePtr> 88
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Sum01.ReadPtr> 87
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Sum01.ReadNextPtr> 88
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Sum01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Sum01.WritePtr> 88
Source B RB.ReadPtr <EDC.RB_Sum01.ReadPtr> 87

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Sum01.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Sum01.WritePtr> 88
Source B RB.ReadPtr <EDC.RB_Sum01.ReadPtr> 87

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Sum01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Sum01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Sum01.WritePtr> 88
Source B RB.ReadNextPtr <EDC.RB_Sum01.ReadNextPtr> 88

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Sum01.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Sum01.WritePtr> 88
Source B RB.ReadNextPtr <EDC.RB_Sum01.ReadNextPtr> 88

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Sum01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Sum01.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Sum01.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

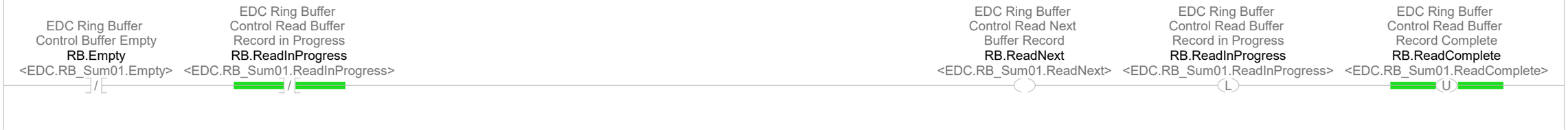
MOVE
Source RB.RecordCount <EDC.RB_Sum01.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Sum01.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

280 280

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Sum01.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Source B RB.MaxRecPtr <EDC.RB_Sum01.MaxRecPtr> 199

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Sum01.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Sum01.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr>
Source B 1
1

Equipment Data
Summary Message

COP
Source EDC_RB_Sum01_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Sum01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr>
Source B 2
1

Equipment Data
Summary Message

COP
Source EDC_RB_Sum01_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Sum01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr>
Source B 3
1

Equipment Data
Summary Message

COP
Source EDC_RB_Sum01_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Sum01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr>
Source B 4
1

Equipment Data
Summary Message

COP
Source EDC_RB_Sum01_Rec04[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Sum01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr>
Source B 5
1

Equipment Data
Summary Message

COP
Source EDC_RB_Sum01_Rec05[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Sum01_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Sum01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Sum01.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Sum01.TransactionNumber>
4705
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Sum01.TransactionNumber>
4705

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Sum01.TransactionNumber>
4705
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Sum01.TransactionNumber>
4705
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Sum01.TransactionNumber>
4705

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

DataType - 1=Equip, 2=Line, 3=OfflineGroup

11

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Sum01.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Sum01.ReadInProgressTimeout>

MOD
Source A: RB_Read.Header.ID <EDC_RB_Sum01_Read.Header.ID>
3
Source B: 100
Dest: EQi
3

CPT
Dest: LNi
0
Expression: (RB_Read.Header.ID-EQi)/100

CLR
Dest: PerfLNi
0

CLR
Dest: PerfEQi
0

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum01.ReadValidEQID>
(U)

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

EQ
Source A: RB_Read.Header.DataType <EDC_RB_Sum01_Read.Header.DataType>
13
Source B: 1

LIMIT
Low Limit: 0
Test: LNi
0
High Limit: CFG.LN.mi <EDC.CFG.LN.mi>
28

LIMIT
Low Limit: 0
Test: EQi
3
High Limit: CFG.EQ.mi <EDC.CFG.EQ.mi>
99

MOVE
Source: LNi
0
Dest: PerfLNi
0

MOVE
Source: EQi
3
Dest: PerfEQi
0

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum01.ReadValidEQID>
(L)

Process RingBuffer Reads from Transaction Manager

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data Collection - Status Line Equipment

COP
Source EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus03
Length 1

Read Next
Log Transaction & Performance

13

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Sum01.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Sum01.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Sum01.ReadTagPtr> 1
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Sum01.ReadRecPtr> 87
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2420
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Sum01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.SumTx.EN

Process RingBuffer Reads from Transaction Manager

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

14

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Sum01.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Sum01.TransactionNumber>
4705
Source B RB.Interface
<EDC.RB_Sum01.Interface>
4705

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Sum01.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Sum01.Interface>
4705
Dest RB.InterfaceNEG
<EDC.RB_Sum01.InterfaceNEG>
-4705

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Sum01.TransactionNumber>
4705
Source B RB.InterfaceNEG
<EDC.RB_Sum01.InterfaceNEG>
-4705

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Sum01.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

15

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Sum01.ReadComplete>

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN
/

EDC Ring Buffer
Control Read
Complete Delay Timer
MOVE
Source CfgRbDelaySlow
100
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Sum01.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer
Control Read
Complete Delay Timer
TON
Timer RB.ReadCompleteDelayTMR
<EDC.RB_Sum01.ReadCompleteDelayTMR>
Preset 0
Accum 14
EN
DN

Server Status
TM Transaction OK
Enable High Speed RB
ServerStat_TMokTMR.DN

EDC Ring Buffer
Control Read
Complete Delay Timer
MOVE
Source CfgRbDelayFast
0
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Sum01.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer
Control Read
Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Sum01.ReadCompleteDelayTMR.DN>

EDC Ring Buffer
Control Read Buffer
Record Complete
Delay
RB.ReadCompleteDelay
<EDC.RB_Sum01.ReadCompleteDelay>

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Sum01.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Sum01.ReadComplete>

EDC Ring Buffer
Control Read Buffer
Record Complete
1-shot
RB.ReadCompleteONS
<EDC.RB_Sum01.ReadCompleteONS>
ONS

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Sum01.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi State = Green -
Sum Ring Buffer Tx
DN
EquipStatus01.Update.SumTxOK
L

Update Status State
for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Sum01.WriteTagPtr>
1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Sum01.WriteRecPtr>
88
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr>
1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Sum01.ReadRecPtr>
87
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2421
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Sum01.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Sum01.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Sum01.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction
aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.SumTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Sum01.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Sum01.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Sum01.ReadResend>

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Sum01.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Sum01.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Sum01.ReadResend>

U

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Sum01.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer

TON
Timer RB.ReadInProgressTMR <EDC.RB_Sum01.ReadInProgressTMR> EN
Preset 5000 DN
Accum 0

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Sum01.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Sum01.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Sum01.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Sum Tx Msg ER
EquipStatus01.Update.SumTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteTagPtr
	<EDC.RB_Sum01.WriteTagPtr>
	1
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteRecPtr
	<EDC.RB_Sum01.WriteRecPtr>
	88
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadTagPtr
	<EDC.RB_Sum01.ReadTagPtr>
	1
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadRecPtr
	<EDC.RB_Sum01.ReadRecPtr>
	87
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2424
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber
	<EDC.RB_Sum01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC
	<EDC.RB_Sum01.ReadInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EquipStatus03.SumTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Sum01.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Sum01.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Sum01.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Sum01.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Sum01.KeepAliveTMR>
Preset 300000
Accum 35253
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Sum01.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Sum01.KeepAlive>

JSR
Routine Name _621_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Sum01.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Sum01.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Sum01.NotEmptyTMR.PRE>
300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Sum01.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Sum01.NotEmptyTMR.ACC>
0
Source B RB.NotEmptyMax
<EDC.RB_Sum01.NotEmptyMax>
300001

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Sum01.NotEmptyTMR.ACC>
0
Dest RB.NotEmptyMax
<EDC.RB_Sum01.NotEmptyMax>
300001

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Sum01.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Sum01.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Sum01.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum01.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus03
Dest	EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	621
Dest	Prog.Routine 620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -850801397
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -850801397

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

2

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr> 1
Source B 1

Equipment Data Summary Message

COP
Source EDC_RB_Sum01_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr> 1
Source B 2

Equipment Data Summary Message

COP
Source EDC_RB_Sum01_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr> 1
Source B 3

Equipment Data Summary Message

COP
Source EDC_RB_Sum01_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr> 1
Source B 4

Equipment Data Summary Message

COP
Source EDC_RB_Sum01_Rec04[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr> 1
Source B 5

Equipment Data Summary Message

COP
Source EDC_RB_Sum01_Rec05[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Summary Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1
Dest KeepAliveMsg.Header.CmdNum 1

Equipment Data Summary Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1

Equipment Data Summary Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 9999

Equipment Data Summary Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum 1

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Sum01.WriteTagPtr> 1
Source B 1

Equipment Data Summary Message

COP
Source KeepAliveMsg
Dest EDC_RB_Sum01_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Sum01.WriteTagPtr> 1
Source B 2

Equipment Data Summary Message

COP
Source KeepAliveMsg
Dest EDC_RB_Sum01_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Sum01.WriteTagPtr> 1
Source B 3

Equipment Data Summary Message

COP
Source KeepAliveMsg
Dest EDC_RB_Sum01_Rec03[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Sum01.WriteTagPtr> 1
Source B 4

Equipment Data Summary Message

COP
Source KeepAliveMsg
Dest EDC_RB_Sum01_Rec04[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Sum01.WriteTagPtr> 1
Source B 5

Equipment Data Summary Message

COP
Source KeepAliveMsg
Dest EDC_RB_Sum01_Rec05[RB.WriteRecPtr]
Length 1

5

Log Transaction
Only update Keep Alive Performance if Equipment (DataType = 1)

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Sum01.WriteTagPtr> 1
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Sum01.WriteRecPtr> 88
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Sum01.ReadTagPtr> 1
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Sum01.ReadRecPtr> 87
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2422
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Sum01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum01.ReadValidEQID>

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

EQ
Source A KeepAliveMsg.Header.DataType 11
Source B 1

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Sum01.KeepAliveTMR.ACC> 35253
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans (...)
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.SumTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88
Source B RB.MaxRecPtr <EDC.RB_Sum01.MaxRecPtr> 199

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1
Source B RB.MaxTagPtr <EDC.RB_Sum01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Sum01.WriteRecPtr> 88

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Sum01.WriteTagPtr> 1

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine
620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901]
276864929
Source B 1
Dest Prog.RoutineScan[901]
276864929

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log

FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log

MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log

MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log

MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log

MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log

MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log

MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log

MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log

MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log

MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log

MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log

MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log

MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

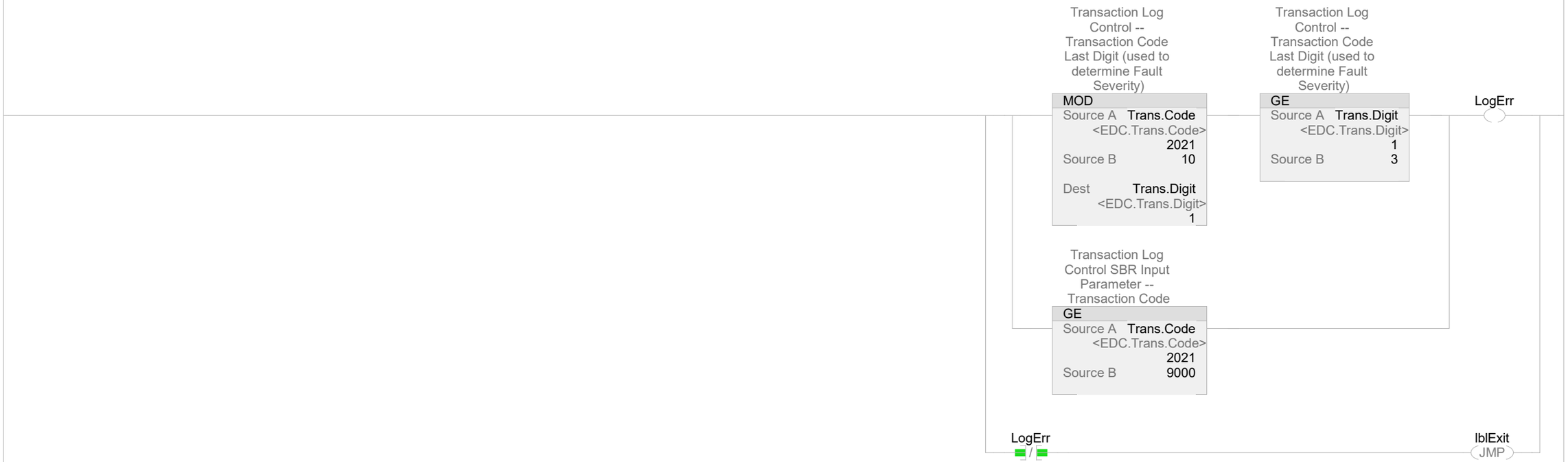
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Summary Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	601
Dest	Prog.Routine 620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -850800771
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -850800771

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_611_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _620_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Sum02.ShiftCnt>
113130
Dest RB.PrevShiftCnt
<EDC.RB_Sum02.PrevShiftCnt>
132844

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Sum02.ShiftCnt>
113130

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A: Prog.Scan.Last 13
Source B: CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A: ProgScanHighCount 0
Source B: 1
Dest: ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.6 (L)

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest: ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 200

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 200
Source B	1
Dest	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 0
Source B	1
Dest	Prog.LOGi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 0
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 0
Source B	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 0

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
620
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Sum02.WriteTagPtr>
4
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Sum02.WriteRecPtr>
46
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr>
4
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Sum02.ReadRecPtr>
45
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 602
Dest Prog.Routine 620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -850800771
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -850800771

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 611
Dest Prog.Routine 620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -850800771
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -850800771

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Summary Message

SIZE
Source EDC_RB_Sum02_Rec01[0] ??
Dim. To Vary 0
Size LEN 200

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 200
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Sum02.MaxRecPtr> 199

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Sum02.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A: RB.TransactionNumber <EDC.RB_Sum02.TransactionNumber> 1264
Source B: 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source: 1
Dest: RB.TransactionNumber <EDC.RB_Sum02.TransactionNumber> 1264

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Sum02.WritePtrErr>
U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Sum02.ReadPtrErr>
U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _101_RingBufferInit Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	620
Dest	Prog.Routine 620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -850800771
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -850800771

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Source B 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Sum02.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Source B 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Source B RB.MaxRecPtr <EDC.RB_Sum02.MaxRecPtr> 199

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Sum02.ReadPtrErr>

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Dest RB.ReadPtrErrTag <EDC.RB_Sum02.ReadPtrErrTag> 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Dest RB.ReadPtrErrRec <EDC.RB_Sum02.ReadPtrErrRec> 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2423
Input Par 0
Input Par -1
Input Par -1
Input Par RB.TransactionNumber <EDC.RB_Sum02.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source RB.MaxTagPtr <EDC.RB_Sum02.MaxTagPtr> 5
Dest RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source RB.MaxRecPtr <EDC.RB_Sum02.MaxRecPtr> 199
Dest RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Dest RB.ReadNextTagPtr <EDC.RB_Sum02.ReadNextTagPtr> 4

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Sum02.ReadNextRecPtr> 46

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Sum02.ReadNextRecPtr> 46
Source B RB.MaxRecPtr <EDC.RB_Sum02.MaxRecPtr> 199

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Sum02.ReadNextRecPtr> 46

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Sum02.ReadNextTagPtr> 4
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Sum02.ReadNextTagPtr> 4

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Sum02.ReadNextTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Sum02.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Sum02.ReadNextTagPtr> 4

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Source B RB.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Source B RB.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

EDC Ring Buffer Control Buffer Full RB.Full <EDC.RB_Sum02.Full>

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Sum02.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Sum02.ReadNextTagPtr> 4
Source B RB.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Sum02.ReadNextRecPtr> 46
Source B RB.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Sum02.Empty>

5

Buffer Pointers
Combine Tag & Record Pointers

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Sum02.WritePtr> 646
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Sum02.ReadPtr> 645
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Sum02.ReadNextPtr> 646
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Sum02.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Sum02.WritePtr> 646
Source B RB.ReadPtr <EDC.RB_Sum02.ReadPtr> 645

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Sum02.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Sum02.WritePtr> 646
Source B RB.ReadPtr <EDC.RB_Sum02.ReadPtr> 645

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Sum02.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Sum02.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Sum02.WritePtr> 646
Source B RB.ReadNextPtr <EDC.RB_Sum02.ReadNextPtr> 646

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Sum02.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Sum02.WritePtr> 646
Source B RB.ReadNextPtr <EDC.RB_Sum02.ReadNextPtr> 646

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Sum02.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Sum02.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Sum02.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

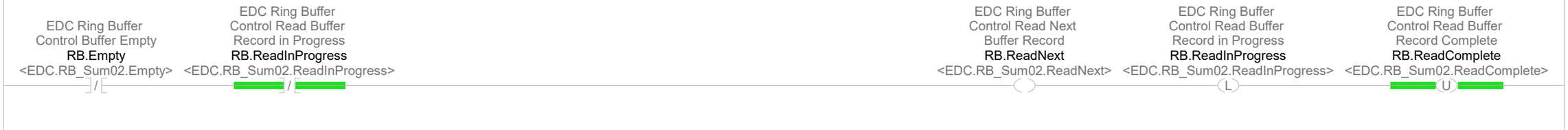
MOVE
Source RB.RecordCount <EDC.RB_Sum02.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Sum02.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

282 282

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Sum02.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Source B RB.MaxRecPtr <EDC.RB_Sum02.MaxRecPtr> 199

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Sum02.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Sum02.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr>
Source B 1

Equipment Data
Summary Message

COP
Source EDC_RB_Sum02_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Sum02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr>
Source B 2

Equipment Data
Summary Message

COP
Source EDC_RB_Sum02_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Sum02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr>
Source B 3

Equipment Data
Summary Message

COP
Source EDC_RB_Sum02_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Sum02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr>
Source B 4

Equipment Data
Summary Message

COP
Source EDC_RB_Sum02_Rec04[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Sum02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr>
Source B 5

Equipment Data
Summary Message

COP
Source EDC_RB_Sum02_Rec05[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Sum02_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Sum02.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Sum02.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Sum02.TransactionNumber>
1264
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Sum02.TransactionNumber>
1264

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Sum02.TransactionNumber>
1264
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Sum02.TransactionNumber>
1264
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Sum02.TransactionNumber>
1264

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

DataType - 1=Equip, 2=Line, 3=OfflineGroup

11

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Sum02.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Sum02.ReadInProgressTimeout>

MOD
Source A: RB_Read.Header.ID <EDC_RB_Sum02_Read.Header.ID>
Source B: 2 100
Dest: EQi 2

CPT
Dest: LNi 0
Expression: (RB_Read.Header.ID-EQi)/100

CLR
Dest: PerfLNi 0

CLR
Dest: PerfEQi 0

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum02.ReadValidEQID>
(U)

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

EQ
Source A: RB_Read.Header.DataType <EDC_RB_Sum02_Read.Header.DataType>: 13
Source B: 1

LIMIT
Low Limit: 0
Test: LNi 0
High Limit: CFG.LN.mi <EDC.CFG.LN.mi>: 28

LIMIT
Low Limit: 0
Test: EQi 2
High Limit: CFG.EQ.mi <EDC.CFG.EQ.mi>: 99

MOVE
Source: LNi 0
Dest: PerfLNi 0

MOVE
Source: EQi 2
Dest: PerfEQi 0

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum02.ReadValidEQID>
(L)

Process RingBuffer Reads from Transaction Manager

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum02.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data Collection - Status Line Equipment

COP
Source EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus03
Length 1

Read Next
Log Transaction & Performance

13

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Sum02.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Sum02.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Dest Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Sum02.ReadTagPtr> 4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Sum02.ReadRecPtr> 45
Dest Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2420
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Sum02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.SumTx.EN

Process RingBuffer Reads from Transaction Manager

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

14

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Sum02.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Sum02.TransactionNumber>
1264
Source B RB.Interface
<EDC.RB_Sum02.Interface>
1264

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Sum02.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Sum02.Interface>
1264
Dest RB.InterfaceNEG
<EDC.RB_Sum02.InterfaceNEG>
-1264

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Sum02.TransactionNumber>
1264
Source B RB.InterfaceNEG
<EDC.RB_Sum02.InterfaceNEG>
-1264

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Sum02.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

15

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Sum02.ReadComplete>

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN
/

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source CfgRbDelaySlow
100
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Sum02.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR
<EDC.RB_Sum02.ReadCompleteDelayTMR>
Preset 0
Accum 13
EN
DN

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source CfgRbDelayFast
0
Dest RB.ReadCompleteDelayTMR.PRE
<EDC.RB_Sum02.ReadCompleteDelayTMR.PRE>
0

EDC Ring Buffer Control Read Complete Delay Timer
RB.ReadCompleteDelayTMR.DN
<EDC.RB_Sum02.ReadCompleteDelayTMR.DN>

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Sum02.ReadCompleteDelay>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Sum02.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Sum02.ReadComplete>

EDC Ring Buffer
Control Read Buffer
Record Complete
1-shot
RB.ReadCompleteONS
<EDC.RB_Sum02.ReadCompleteONS>
ONS

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Sum02.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi State = Green -
Sum Ring Buffer Tx
DN
EquipStatus01.Update.SumTxOK
L

Update Status State
for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Sum02.WriteTagPtr>
4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Sum02.WriteRecPtr>
46
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr>
4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Sum02.ReadRecPtr>
45
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2421
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Sum02.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Sum02.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Sum02.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction
aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.SumTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Sum02.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Sum02.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Resend
RB.ReadResend
<EDC.RB_Sum02.ReadResend>

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE
<EDC.RB_Sum02.ReadInProgressTMR.PRE>
5000

EDC Ring Buffer Control Read Buffer Record Resend
RB.ReadResend
<EDC.RB_Sum02.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend
RB.ReadResend
<EDC.RB_Sum02.ReadResend>
U

EDC Ring Buffer Control Read in Progress Timer
MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE
<EDC.RB_Sum02.ReadInProgressTMR.PRE>
5000

EDC Ring Buffer Control Read in Progress Timer
TON
Timer RB.ReadInProgressTMR
<EDC.RB_Sum02.ReadInProgressTMR>
Preset 5000
Accum 0
EN
DN

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR.DN
<EDC.RB_Sum02.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Sum02.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Sum02.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum02.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Sum Tx Msg ER
EquipStatus01.Update.SumTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteTagPtr
	<EDC.RB_Sum02.WriteTagPtr>
	4
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteRecPtr
	<EDC.RB_Sum02.WriteRecPtr>
	46
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadTagPtr
	<EDC.RB_Sum02.ReadTagPtr>
	4
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadRecPtr
	<EDC.RB_Sum02.ReadRecPtr>
	45
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2424
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber
	<EDC.RB_Sum02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC
	<EDC.RB_Sum02.ReadInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EquipStatus03.SumTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Sum02.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Sum02.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Sum02.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Sum02.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Sum02.KeepAliveTMR>
Preset 300000
Accum 35046
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Sum02.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Sum02.KeepAlive>

JSR
Routine Name _621_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Sum02.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Sum02.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Sum02.NotEmptyTMR.PRE>
300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Sum02.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Sum02.NotEmptyTMR.ACC>
0
Source B RB.NotEmptyMax
<EDC.RB_Sum02.NotEmptyMax>
300001

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Sum02.NotEmptyTMR.ACC>
0
Dest RB.NotEmptyMax
<EDC.RB_Sum02.NotEmptyMax>
300001

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Sum02.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Sum02.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Sum02.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum02.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus03
Dest	EDC_Status03.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	621
Dest	Prog.Routine 620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -850800771
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -850800771

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

2

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr> 4
Source B 1

Equipment Data Summary Message

COP
Source EDC_RB_Sum02_Rec01[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr> 4
Source B 2

Equipment Data Summary Message

COP
Source EDC_RB_Sum02_Rec02[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr> 4
Source B 3

Equipment Data Summary Message

COP
Source EDC_RB_Sum02_Rec03[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr> 4
Source B 4

Equipment Data Summary Message

COP
Source EDC_RB_Sum02_Rec04[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr> 4
Source B 5

Equipment Data Summary Message

COP
Source EDC_RB_Sum02_Rec05[RB.ReadRecPtr]
Dest KeepAliveMsg
Length 1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data Summary Message Header - Command Number

ADD
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1
Dest KeepAliveMsg.Header.CmdNum 1

Equipment Data Summary Message Header - Command Number

LT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 1

Equipment Data Summary Message Header - Command Number

GT
Source A KeepAliveMsg.Header.CmdNum 1
Source B 9999

Equipment Data Summary Message Header - Command Number

MOVE
Source 1
Dest KeepAliveMsg.Header.CmdNum 1

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Next Write Tag

EQ	
Source A	RB.WriteTagPtr
	<EDC.RB_Sum02.WriteTagPtr>
	4
Source B	1

Equipment Data Summary Message

COP	
Source	KeepAliveMsg
Dest	EDC_RB_Sum02_Rec01[RB.WriteRecPtr]
Length	1

EDC Ring Buffer Control Next Write Tag

EQ	
Source A	RB.WriteTagPtr
	<EDC.RB_Sum02.WriteTagPtr>
	4
Source B	2

Equipment Data Summary Message

COP	
Source	KeepAliveMsg
Dest	EDC_RB_Sum02_Rec02[RB.WriteRecPtr]
Length	1

EDC Ring Buffer Control Next Write Tag

EQ	
Source A	RB.WriteTagPtr
	<EDC.RB_Sum02.WriteTagPtr>
	4
Source B	3

Equipment Data Summary Message

COP	
Source	KeepAliveMsg
Dest	EDC_RB_Sum02_Rec03[RB.WriteRecPtr]
Length	1

EDC Ring Buffer Control Next Write Tag

EQ	
Source A	RB.WriteTagPtr
	<EDC.RB_Sum02.WriteTagPtr>
	4
Source B	4

Equipment Data Summary Message

COP	
Source	KeepAliveMsg
Dest	EDC_RB_Sum02_Rec04[RB.WriteRecPtr]
Length	1

EDC Ring Buffer Control Next Write Tag

EQ	
Source A	RB.WriteTagPtr
	<EDC.RB_Sum02.WriteTagPtr>
	4
Source B	5

Equipment Data Summary Message

COP	
Source	KeepAliveMsg
Dest	EDC_RB_Sum02_Rec05[RB.WriteRecPtr]
Length	1

5

Log Transaction
Only update Keep Alive Performance if Equipment (DataType = 1)

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Sum02.WriteTagPtr> 4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Sum02.WriteRecPtr> 46
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Sum02.ReadTagPtr> 4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Sum02.ReadRecPtr> 45
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]> -1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2422
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Sum02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Sum02.ReadValidEQID>

Equipment Data Summary Message Header - Data Type - 1=Equip, 2=Line, 3=Group

EQ
Source A KeepAliveMsg.Header.DataType 11
Source B 1

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Sum02.KeepAliveTMR.ACC> 35046
Dest TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus03.SumTx.KA

6

Increment Write Pointer

EDC Ring Buffer Control Next Write Record

ADD
Source A: RB.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B: 1
Dest: RB.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

EDC Ring Buffer Control Next Write Record

LT
Source A: RB.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B: 0

EDC Ring Buffer Control Next Write Record

GT
Source A: RB.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46
Source B: RB.MaxRecPtr <EDC.RB_Sum02.MaxRecPtr> 199

EDC Ring Buffer Control Next Write Tag

ADD
Source A: RB.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B: 1
Dest: RB.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A: RB.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B: 1

EDC Ring Buffer Control Next Write Tag

GT
Source A: RB.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4
Source B: RB.MaxTagPtr <EDC.RB_Sum02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

MOVE
Source: 0
Dest: RB.WriteRecPtr <EDC.RB_Sum02.WriteRecPtr> 46

EDC Ring Buffer Control Next Write Tag

MOVE
Source: 1
Dest: RB.WriteTagPtr <EDC.RB_Sum02.WriteTagPtr> 4

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine
620

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901]
413552736
Source B 1
Dest Prog.RoutineScan[901]
413552736

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

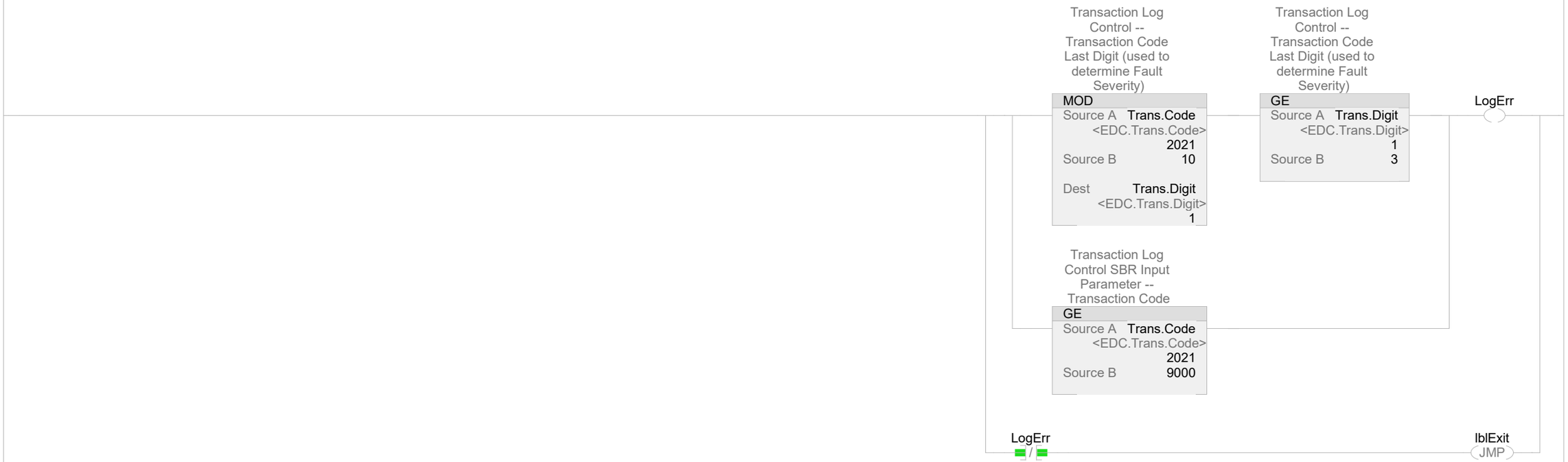
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP	
Source	EDC_TransLog[Trans.LOGi]
Dest	EDC_TransErr[Trans.ERRi]
Length	1

Error Log

FLL	
Source	0
Dest	EDC_TransErr[Trans.ERRx]
Length	1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL	
Source	-1
Dest	Trans.Opt[0] <EDC.Trans.Opt[0]>
Length	Trans.OptN <EDC.Trans.OptN>

(End)

Main Routine

Cycle Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1101
Dest	Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806558467
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806558467

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Configuration

JSR	
Routine Name	_1111_Config

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _1120_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Cycle01.ShiftCnt>
7564
Dest RB.PrevShiftCnt
<EDC.RB_Cycle01.PrevShiftCnt>
8603

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Cycle01.ShiftCnt>
7564

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A: Prog.Scan.Last 13
Source B: CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A: ProgScanHighCount 0
Source B: 1
Dest: ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.11 (L)

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest: ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

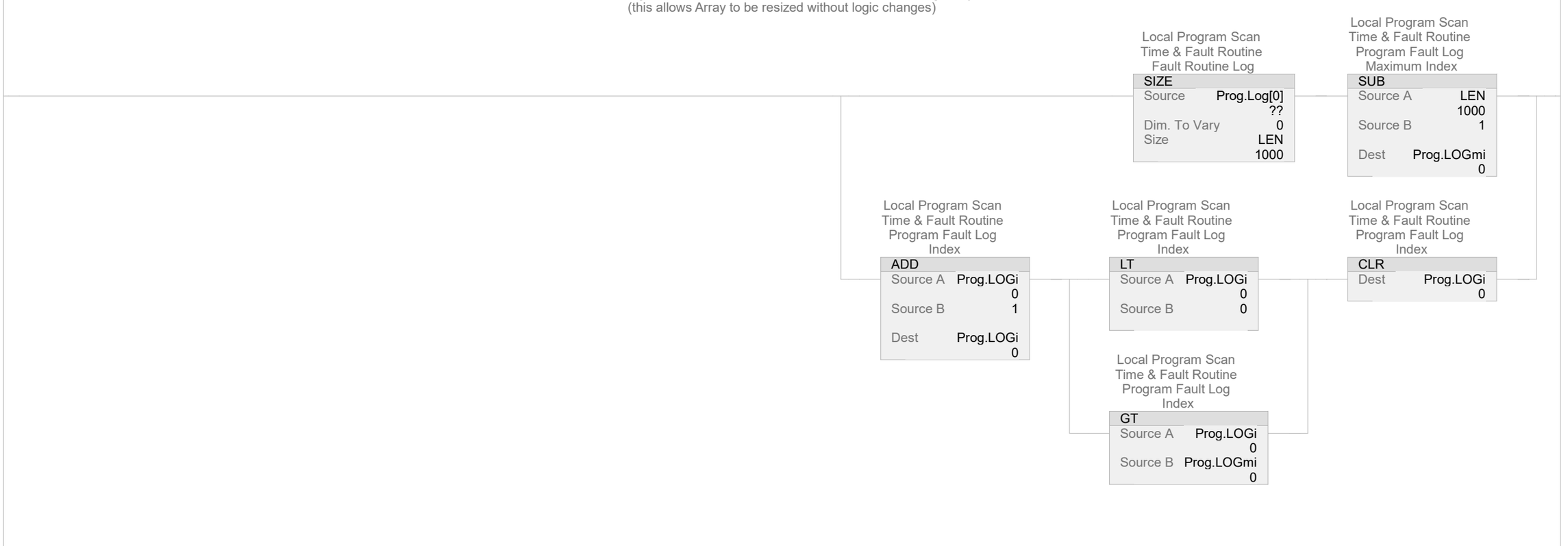
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
1120
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr>
2
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Cycle01.WriteRecPtr>
547
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
2
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Cycle01.ReadRecPtr>
546
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(L)

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1102
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806558467
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806558467

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1111
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806558467
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806558467

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Cycle Message

SIZE
Source EDC_RB_Cycle01_Rec01[0] ??
Dim. To Vary 0
Size LEN 1000

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 1000
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A: RB.TransactionNumber <EDC.RB_Cycle01.TransactionNumber> 6814
Source B: 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source: 1
Dest: RB.TransactionNumber <EDC.RB_Cycle01.TransactionNumber> 6814

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Cycle01.WritePtrErr>
U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Cycle01.ReadPtrErr>
U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _1111_Config Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1120
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806558467
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806558467

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A: RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr>: 2
Source B: 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A: RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr>: 2
Source B: RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr>: 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A: RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr>: 546
Source B: 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A: RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr>: 546
Source B: RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr>: 999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Cycle01.ReadPtrErr>

(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source: RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr>: 2
Dest: RB.ReadPtrErrTag <EDC.RB_Cycle01.ReadPtrErrTag>: 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source: RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr>: 546
Dest: RB.ReadPtrErrRec <EDC.RB_Cycle01.ReadPtrErrRec>: 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr>: 2
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr>: 547
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr>: 2
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr>: 546
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]>: -1

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2523
Input Par: 0
Input Par: -1
Input Par: -1
Input Par: RB.TransactionNumber <EDC.RB_Cycle01.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source: 1
Dest: RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr>: 2

EDC Ring Buffer Control Next Write Record

MOVE
Source: 0
Dest: RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr>: 547

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source: RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr>: 5
Dest: RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr>: 2

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source: RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr>: 999
Dest: RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr>: 546

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Cycle01.ReadNextRecPtr> 547
Source B RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Dest RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Cycle01.ReadNextRecPtr> 547

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Cycle01.ReadNextRecPtr> 547

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Buffer Full RB.Full <EDC.RB_Cycle01.Full>

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle01.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Cycle01.ReadNextTagPtr> 2
Source B RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Cycle01.ReadNextRecPtr> 547
Source B RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Cycle01.Empty>

5

Buffer Pointers
Combine Tag & Record Pointers

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Cycle01.WritePtr> 1547
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Cycle01.ReadPtr> 1546
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Cycle01.ReadNextPtr> 1547
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Cycle01.WritePtr> 1547
Source B RB.ReadPtr <EDC.RB_Cycle01.ReadPtr> 1546

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Cycle01.WritePtr> 1547
Source B RB.ReadPtr <EDC.RB_Cycle01.ReadPtr> 1546

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle01.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Cycle01.WritePtr> 1547
Source B RB.ReadNextPtr <EDC.RB_Cycle01.ReadNextPtr> 1547

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Cycle01.WritePtr> 1547
Source B RB.ReadNextPtr <EDC.RB_Cycle01.ReadNextPtr> 1547

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Cycle01.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

MOVE
Source RB.RecordCount <EDC.RB_Cycle01.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Cycle01.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

3 3

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7

EDC Ring Buffer Control Buffer Empty RB.Empty <EDC.RB_Cycle01.Empty>	EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle01.ReadInProgress>	EDC Ring Buffer Control Read Next Buffer Record RB.ReadNext <EDC.RB_Cycle01.ReadNext>	EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle01.ReadInProgress>	EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Cycle01.ReadComplete>
/	/		L	U

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Cycle01.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546
Source B RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr> 546

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Cycle01.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
Source B 1

Equipment Data
Cycle Message

COP
Source EDC_RB_Cycle01_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Cycle01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
Source B 2

Equipment Data
Cycle Message

COP
Source EDC_RB_Cycle01_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Cycle01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
Source B 3

Equipment Data
Cycle Message

COP
Source EDC_RB_Cycle01_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Cycle01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
Source B 4

Equipment Data
Cycle Message

COP
Source EDC_RB_Cycle01_Rec04[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Cycle01_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
Source B 5

Equipment Data
Cycle Message

COP
Source EDC_RB_Cycle01_Rec05[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Cycle01_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Cycle01.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle01.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Cycle01.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle01.ReadInProgressTimeout>

MOD
Source A: RB_Read.Header.EQID <EDC_RB_Cycle01_Read.Header.EQID> 965
Source B: 100
Dest: EQi 19

CPT
Dest: LNi 4
Expression: (RB_Read.Header.EQID-EQi)/100

CLR
Dest: PerfLNi 4

CLR
Dest: PerfEQi 19

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>
(U)

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>
(L)

LIMIT
Low Limit: 0
Test: LNi 4
High Limit: CFG.LN.mi <EDC.CFG.LN.mi> 28

LIMIT
Low Limit: 0
Test: EQi 19
High Limit: CFG.EQ.mi <EDC.CFG.EQ.mi> 99

MOVE
Source: LNi 4
Dest: PerfLNi 4

MOVE
Source: EQi 19
Dest: PerfEQi 19

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi

COP
Source: EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest: EquipStatus01
Length: 1

Equipment Data Collection - Status Line Equipment

COP
Source: EDC_Status04.Line[PerfLNi].Equip[PerfEQi]
Dest: EquipStatus04
Length: 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Cycle01.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle01.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr>
2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr>
547
Dest Trans.Opt[2] <EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr>
2
Dest Trans.Opt[3] <EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Cycle01.ReadRecPtr>
546
Dest Trans.Opt[4] <EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2520
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Cycle01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg TempPerfMsg
PerfTrans EquipStatus04.CycleTx.EN

Process RingBuffer Reads from Transaction Manager

14

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Cycle01.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814
Source B RB.Interface
<EDC.RB_Cycle01.Interface>
6814

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Cycle01.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Cycle01.Interface>
6814
Dest RB.InterfaceNEG
<EDC.RB_Cycle01.InterfaceNEG>
-6814

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>
6814
Source B RB.InterfaceNEG
<EDC.RB_Cycle01.InterfaceNEG>
-6814

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Cycle01.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

15

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Cycle01.ReadComplete>

Server Status TM Transaction OK Enable High Speed RB ServerStat_TMokTMR.DN /

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source CfgRbDelaySlow 100
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Cycle01.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR <EDC.RB_Cycle01.ReadCompleteDelayTMR>
Preset 0
Accum 13
EN
DN

Server Status TM Transaction OK Enable High Speed RB ServerStat_TMokTMR.DN

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source CfgRbDelayFast 0
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Cycle01.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer RB.ReadCompleteDelayTMR.DN <EDC.RB_Cycle01.ReadCompleteDelayTMR.DN>

EDC Ring Buffer Control Read Buffer Record Complete Delay RB.ReadCompleteDelay <EDC.RB_Cycle01.ReadCompleteDelay>

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle01.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

Read Complete
Log Transaction & Performance

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Cycle01.ReadComplete>

EDC Ring Buffer
Control Read Buffer
Record Complete
1-shot
RB.ReadCompleteONS
<EDC.RB_Cycle01.ReadCompleteONS>
ONS

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi State = Green -
Cycle Tx DN
EquipStatus01.Update.CycleTxOK
L

Update Status State
for Transactions
aoiEDC_StatusUpdate
aoiEDC_StatusUpdate aoiEDC_StatusUpdate ...
Status EquipStatus01

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Cycle01.WriteRecPtr>
547
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Cycle01.ReadRecPtr>
546
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2521
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
Accum Last (used to
calculate
differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data
Collection - Line
Equipment Control
for Message
Transaction Timer
MOVE
Source RB.ReadInProgressTMR.ACC
<EDC.RB_Cycle01.ReadInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance
Transaction
aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus04.CycleTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle01.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Cycle01.ReadComplete> /

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Cycle01.ReadResend> /

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Cycle01.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Cycle01.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Cycle01.ReadResend> U

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Cycle01.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer

TON
Timer RB.ReadInProgressTMR <EDC.RB_Cycle01.ReadInProgressTMR> EN
Preset 5000 DN
Accum 0

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Cycle01.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Cycle01.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle01.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Cycle Tx Msg ER
EquipStatus01.Update.CycleTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate	
aoiEDC_StatusUpdate	aoiEDC_StatusUpdate ...
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteTagPtr
	<EDC.RB_Cycle01.WriteTagPtr>
	2
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.WriteRecPtr
	<EDC.RB_Cycle01.WriteRecPtr>
	547
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadTagPtr
	<EDC.RB_Cycle01.ReadTagPtr>
	2
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	RB.ReadRecPtr
	<EDC.RB_Cycle01.ReadRecPtr>
	546
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	2524
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber
	<EDC.RB_Cycle01.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE	
Source	RB.ReadInProgressTMR.ACC
	<EDC.RB_Cycle01.ReadInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EquipStatus04.CycleTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Cycle01.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Cycle01.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Cycle01.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Cycle01.KeepAliveTMR.PRE>
300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Cycle01.KeepAliveTMR>
Preset 300000
Accum 34823
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Cycle01.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Cycle01.KeepAlive>

JSR
Routine Name _1121_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Cycle01.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Cycle01.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Cycle01.NotEmptyTMR.PRE>
300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Cycle01.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Cycle01.NotEmptyTMR.ACC>
0
Source B RB.NotEmptyMax
<EDC.RB_Cycle01.NotEmptyMax>
300003

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Cycle01.NotEmptyTMR.ACC>
0
Dest RB.NotEmptyMax
<EDC.RB_Cycle01.NotEmptyMax>
300003

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Cycle01.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Cycle01.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Cycle01.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus04
Dest	EDC_Status04.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1121
Dest	Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806558467
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806558467

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

2

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B	1

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle01_Rec01[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B	2

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle01_Rec02[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B	3

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle01_Rec03[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B	4

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle01_Rec04[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr <EDC.RB_Cycle01.ReadTagPtr> 2
Source B	5

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle01_Rec05[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data
Cycle Message Header
- Command Number

ADD	
Source A	KeepAliveMsg.Header.CmdNum 1
Source B	1
Dest	KeepAliveMsg.Header.CmdNum 1

Equipment Data
Cycle Message Header
- Command Number

LT	
Source A	KeepAliveMsg.Header.CmdNum 1
Source B	1

Equipment Data
Cycle Message Header
- Command Number

GT	
Source A	KeepAliveMsg.Header.CmdNum 1
Source B	9999

Equipment Data
Cycle Message Header
- Command Number

MOVE	
Source	1
Dest	KeepAliveMsg.Header.CmdNum 1

4

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle01_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr> 2
Source B 2

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle01_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr> 2
Source B 3

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle01_Rec03[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr> 2
Source B 4

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle01_Rec04[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr> 2
Source B 5

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle01_Rec05[RB.WriteRecPtr]
Length 1

5

Log Transaction

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle01.ReadValidEQID>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.WriteTagPtr
<EDC.RB_Cycle01.WriteTagPtr>
2
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.WriteRecPtr
<EDC.RB_Cycle01.WriteRecPtr>
547
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.ReadTagPtr
<EDC.RB_Cycle01.ReadTagPtr>
2
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source RB.ReadRecPtr
<EDC.RB_Cycle01.ReadRecPtr>
546
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 2522
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Cycle01.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Cycle01.KeepAliveTMR.ACC>
34823
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus04.CycleTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547
Source B RB.MaxRecPtr <EDC.RB_Cycle01.MaxRecPtr> 999

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2
Source B RB.MaxTagPtr <EDC.RB_Cycle01.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Cycle01.WriteRecPtr> 547

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Cycle01.WriteTagPtr> 2

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 91892768
Source B 1
Dest Prog.RoutineScan[901] 91892768

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

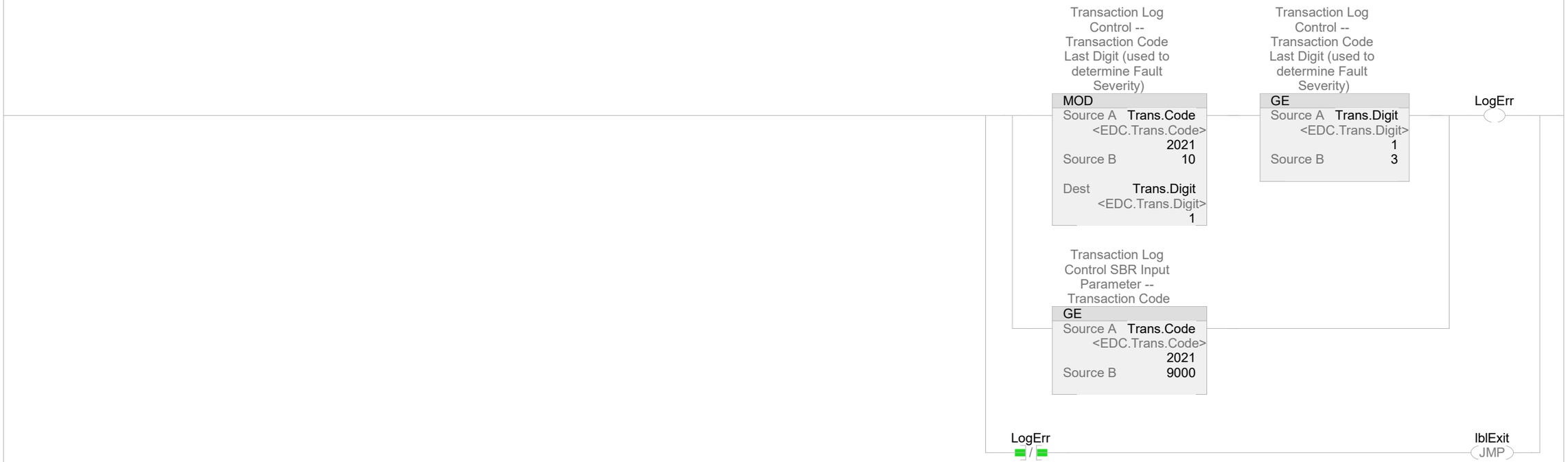
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Cycle Ring Buffer
v3.02 Dec2024 - replace handshake bit with TransNum (complete) or negative TransNum (resend)
integer TransNum eliminates the Read Complete Delay Time since the Gateway does not need to reset the handshake bit
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1101
Dest	Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806557869
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806557869

Confgure and Initialize Ring Buffer Pointers and Transacations
v3.02e - add ConfigONS

2

S:FS

Configuration

JSR	
Routine Name	_1111_Config

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Main Routine

Process RingBuffer Reads by Transaction Manager to SQL

3

Process RingBuffer Reads from Transaction Manager

JSR
Routine Name _1120_RingBuffer

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

4

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount
0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Shift Reset
Store Prev Shift and Clear Current Shift

5

ShiftReset

EDC Ring Buffer Control Diagnostic - Auto Update every shift

MOVE
Source RB.ShiftCnt
<EDC.RB_Cycle02.ShiftCnt>
7368
Dest RB.PrevShiftCnt
<EDC.RB_Cycle02.PrevShiftCnt>
8544

EDC Ring Buffer Control Diagnostic - Auto Reset every shift

CLR
Dest RB.ShiftCnt
<EDC.RB_Cycle02.ShiftCnt>
7368

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

6

Local Program Scan Time & Fault Routine Last Scan

GT
Source A: Prog.Scan.Last 16
Source B: CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A: ProgScanHighCount 0
Source B: 1
Dest: ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.11
(L)

CLx Module Status Clear Scan History

CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count

CLR
Dest: ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

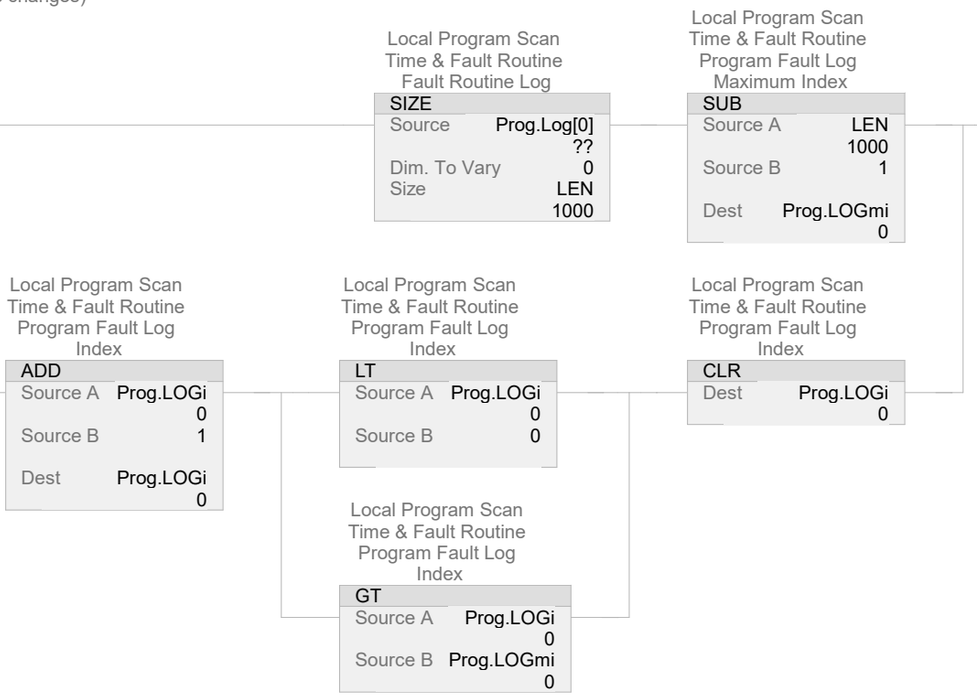
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
1120
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source RB.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr>
4
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source RB.WriteRecPtr
<EDC.RB_Cycle02.WriteRecPtr>
437
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source RB.ReadTagPtr
<EDC.RB_Cycle02.ReadTagPtr>
4
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source RB.ReadRecPtr
<EDC.RB_Cycle02.ReadRecPtr>
436
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(L)

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1102
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806557869
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806557869

5

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

6

(End)

Configuration

Ring Buffer Compare

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1111
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806557869
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806557869

Configure Ring Buffer Array size and Tag Count
(Ring Buffers with multiple tags MUST use same array size on all Tags.)

2

Equipment Data Cycle Message

SIZE
Source EDC_RB_Cycle02_Rec01[0] ??
Dim. To Vary 0
Size LEN 1000

EDC Ring Buffer Control Maximum Record Pointer

SUB
Source A LEN 1000
Source B 1
Dest RB.MaxRecPtr <EDC.RB_Cycle02.MaxRecPtr> 999

EDC Ring Buffer Control Maximum Tag Pointer

MOVE
Source 5
Dest RB.MaxTagPtr <EDC.RB_Cycle02.MaxTagPtr> 5

Configuration

Check for uninitialized Transaction Number

3

EDC Ring Buffer Control TM Transaction Trigger

EQ
Source A: RB.TransactionNumber <EDC.RB_Cycle02.TransactionNumber> 8702
Source B: 0

EDC Ring Buffer Control TM Transaction Trigger

MOVE
Source: 1
Dest: RB.TransactionNumber <EDC.RB_Cycle02.TransactionNumber> 8702

Reset Ring Buffer Faults caused by missing Config

4

EDC Ring Buffer Control Diagnostic - Manual Reset RB.WritePtrErr <EDC.RB_Cycle02.WritePtrErr>
U

EDC Ring Buffer Control Diagnostic - Manual Reset RB.ReadPtrErr <EDC.RB_Cycle02.ReadPtrErr>
U

(End)

Process RingBuffer Reads from Transaction Manager

Extract Read Records from RingBuffer for Transaction Manager
(supports multiple Tags for Ring Buffers with >2M data)

NOTE: Configure Tag Count in _1111_Config Rung 2

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1120
Dest	Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806557869
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806557869

Process RingBuffer Reads from Transaction Manager

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

2

EDC Ring Buffer Control Current/Last Read Tag

LT
Source A: RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr>: 4
Source B: 1

EDC Ring Buffer Control Current/Last Read Tag

GT
Source A: RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr>: 4
Source B: RB.MaxTagPtr <EDC.RB_Cycle02.MaxTagPtr>: 5

EDC Ring Buffer Control Current/Last Read Record

LT
Source A: RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr>: 436
Source B: 0

EDC Ring Buffer Control Current/Last Read Record

GT
Source A: RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr>: 436
Source B: RB.MaxRecPtr <EDC.RB_Cycle02.MaxRecPtr>: 999

Manually Reset Error Indicator & Tags

EDC Ring Buffer Control Diagnostic - Manual Reset

RB.ReadPtrErr <EDC.RB_Cycle02.ReadPtrErr>

(L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source: RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr>: 4
Dest: RB.ReadPtrErrTag <EDC.RB_Cycle02.ReadPtrErrTag>: 0

EDC Ring Buffer Control Diagnostic - Manual Reset Value

MOVE
Source: RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr>: 436
Dest: RB.ReadPtrErrRec <EDC.RB_Cycle02.ReadPtrErrRec>: 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr>: 4
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr>: 437
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr>: 4
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]>: -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr>: 436
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]>: -1

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2523
Input Par: 0
Input Par: -1
Input Par: -1
Input Par: RB.TransactionNumber <EDC.RB_Cycle02.TransactionNumber>

EDC Ring Buffer Control Next Write Tag

MOVE
Source: 1
Dest: RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr>: 4

EDC Ring Buffer Control Next Write Record

MOVE
Source: 0
Dest: RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr>: 437

EDC Ring Buffer Control Current/Last Read Tag

MOVE
Source: RB.MaxTagPtr <EDC.RB_Cycle02.MaxTagPtr>: 5
Dest: RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr>: 4

EDC Ring Buffer Control Current/Last Read Record

MOVE
Source: RB.MaxRecPtr <EDC.RB_Cycle02.MaxRecPtr>: 999
Dest: RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr>: 436

Process RingBuffer Reads from Transaction Manager

Next Read Location
ReadTagPointer & ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextTagPointer & ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

GT
Source A RB.ReadNextRecPtr <EDC.RB_Cycle02.ReadNextRecPtr> 437
Source B RB.MaxRecPtr <EDC.RB_Cycle02.MaxRecPtr> 999

EDC Ring Buffer Control Next Read Tag

MOVE
Source RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4
Dest RB.ReadNextTagPtr <EDC.RB_Cycle02.ReadNextTagPtr> 4

EDC Ring Buffer Control Next Read Record

ADD
Source A RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436
Source B 1
Dest RB.ReadNextRecPtr <EDC.RB_Cycle02.ReadNextRecPtr> 437

EDC Ring Buffer Control Next Read Tag

ADD
Source A RB.ReadNextTagPtr <EDC.RB_Cycle02.ReadNextTagPtr> 4
Source B 1
Dest RB.ReadNextTagPtr <EDC.RB_Cycle02.ReadNextTagPtr> 4

EDC Ring Buffer Control Next Read Tag

GT
Source A RB.ReadNextTagPtr <EDC.RB_Cycle02.ReadNextTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Cycle02.MaxTagPtr> 5

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest RB.ReadNextRecPtr <EDC.RB_Cycle02.ReadNextRecPtr> 437

EDC Ring Buffer Control Next Read Tag

MOVE
Source 1
Dest RB.ReadNextTagPtr <EDC.RB_Cycle02.ReadNextTagPtr> 4

Process RingBuffer Reads from Transaction Manager

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Current/Last Read Tag

EQ
Source A RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4
Source B RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4

EDC Ring Buffer Control Current/Last Read Record

EQ
Source A RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436
Source B RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Cycle02.Full>

EDC Ring Buffer Control Read Buffer Record in Progress
RB.ReadInProgress
<EDC.RB_Cycle02.ReadInProgress>

EDC Ring Buffer Control Next Read Tag

EQ
Source A RB.ReadNextTagPtr <EDC.RB_Cycle02.ReadNextTagPtr> 4
Source B RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4

EDC Ring Buffer Control Next Read Record

EQ
Source A RB.ReadNextRecPtr <EDC.RB_Cycle02.ReadNextRecPtr> 437
Source B RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Cycle02.Empty>

5

Buffer Pointers
Combine Tag & Record Pointers

EDC Ring Buffer Control Next Write Tag/Record

CPT
Dest RB.WritePtr <EDC.RB_Cycle02.WritePtr> 3437
Expression (RB.WriteTagPtr-1)*(RB.MaxRecPtr+1)+RB.WriteRecPtr

EDC Ring Buffer Control Current/Last Read Tag/Record

CPT
Dest RB.ReadPtr <EDC.RB_Cycle02.ReadPtr> 3436
Expression (RB.ReadTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadRecPtr

EDC Ring Buffer Control Next Read Tag/Record

CPT
Dest RB.ReadNextPtr <EDC.RB_Cycle02.ReadNextPtr> 3437
Expression (RB.ReadNextTagPtr-1)*(RB.MaxRecPtr+1)+RB.ReadNextRecPtr

Process RingBuffer Reads from Transaction Manager

Buffer Count

6

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle02.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GT
Source A RB.WritePtr <EDC.RB_Cycle02.WritePtr> 3437
Source B RB.ReadPtr <EDC.RB_Cycle02.ReadPtr> 3436

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle02.RecordCount> 0
Expression RB.WritePtr-RB.ReadPtr

EDC Ring Buffer Control Next Write Tag/Record

LE
Source A RB.WritePtr <EDC.RB_Cycle02.WritePtr> 3437
Source B RB.ReadPtr <EDC.RB_Cycle02.ReadPtr> 3436

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle02.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadPtr-RB.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle02.ReadInProgress>

EDC Ring Buffer Control Next Write Tag/Record

GE
Source A RB.WritePtr <EDC.RB_Cycle02.WritePtr> 3437
Source B RB.ReadNextPtr <EDC.RB_Cycle02.ReadNextPtr> 3437

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle02.RecordCount> 0
Expression RB.WritePtr-RB.ReadNextPtr

EDC Ring Buffer Control Next Write Tag/Record

LT
Source A RB.WritePtr <EDC.RB_Cycle02.WritePtr> 3437
Source B RB.ReadNextPtr <EDC.RB_Cycle02.ReadNextPtr> 3437

EDC Ring Buffer Control Ring Buffer Read Record Count

CPT
Dest RB.RecordCount <EDC.RB_Cycle02.RecordCount> 0
Expression (RB.MaxRecPtr+1)*RB.MaxTagPtr-(RB.ReadNextPtr-RB.WritePtr)

EDC Ring Buffer Control Ring Buffer Read Record Count

GT
Source A RB.RecordCount <EDC.RB_Cycle02.RecordCount> 0
Source B RB.RecordCountMax <EDC.RB_Cycle02.RecordCountMax>

EDC Ring Buffer Control Ring Buffer Read Record Maximum Count

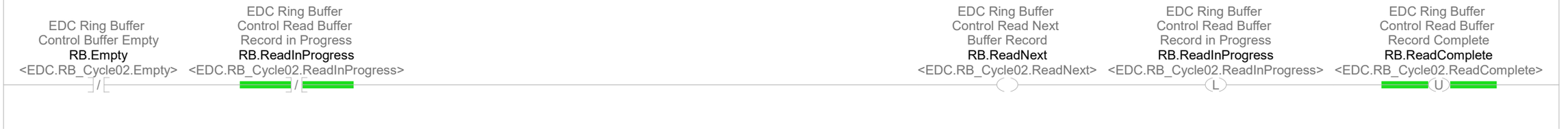
MOVE
Source RB.RecordCount <EDC.RB_Cycle02.RecordCount> 0
Dest RB.RecordCountMax <EDC.RB_Cycle02.RecordCountMax>

Process RingBuffer Reads from Transaction Manager

3 3

Read Next
Ready for next Transaction Manager Read
DHF019 - Remove ONS - not needed and allowed ReadNext to hang on power up

7



Process RingBuffer Reads from Transaction Manager

Read Next
Increment Read Pointer

8

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Cycle02.ReadNext>

EDC Ring Buffer Control Current/Last Read Record
ADD
Source A RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436
Source B 1
Dest RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436

EDC Ring Buffer Control Current/Last Read Record
LT
Source A RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436
Source B 0

EDC Ring Buffer Control Current/Last Read Record
GT
Source A RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436
Source B RB.MaxRecPtr <EDC.RB_Cycle02.MaxRecPtr> 999

EDC Ring Buffer Control Current/Last Read Record
MOVE
Source 0
Dest RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436

EDC Ring Buffer Control Current/Last Read Tag
ADD
Source A RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4
Source B 1
Dest RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4

EDC Ring Buffer Control Current/Last Read Tag
LT
Source A RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4
Source B 1

EDC Ring Buffer Control Current/Last Read Tag
GT
Source A RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Cycle02.MaxTagPtr> 5

EDC Ring Buffer Control Current/Last Read Tag
MOVE
Source 1
Dest RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4

Process RingBuffer Reads from Transaction Manager

Process RingBuffer Reads from Transaction Manager

Read Next
Load Read Record (Tags 01-05)

9

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Cycle02.ReadNext>

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Cycle02.ReadTagPtr>
Source B 1

Equipment Data
Cycle Message

COP
Source EDC_RB_Cycle02_Rec01[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Cycle02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Cycle02.ReadTagPtr>
Source B 2

Equipment Data
Cycle Message

COP
Source EDC_RB_Cycle02_Rec02[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Cycle02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Cycle02.ReadTagPtr>
Source B 3

Equipment Data
Cycle Message

COP
Source EDC_RB_Cycle02_Rec03[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Cycle02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Cycle02.ReadTagPtr>
Source B 4

Equipment Data
Cycle Message

COP
Source EDC_RB_Cycle02_Rec04[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Cycle02_Read>
Length 1

EDC Ring Buffer
Control Current/Last
Read Tag

EQ
Source A RB.ReadTagPtr
<EDC.RB_Cycle02.ReadTagPtr>
Source B 5

Equipment Data
Cycle Message

COP
Source EDC_RB_Cycle02_Rec05[RB.ReadRecPtr]
Dest RB_Read
<EDC_RB_Cycle02_Read>
Length 1

Process RingBuffer Reads from Transaction Manager

Read Next
Increment Transaction Number
(Triggers TM Transaction on change)

10

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Cycle02.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle02.ReadInProgressTimeout>

EDC Ring Buffer
Control TM
Transaction Trigger

ADD
Source A RB.TransactionNumber
<EDC.RB_Cycle02.TransactionNumber>
8702
Source B 1
Dest RB.TransactionNumber
<EDC.RB_Cycle02.TransactionNumber>
8702

EDC Ring Buffer
Control TM
Transaction Trigger

LT
Source A RB.TransactionNumber
<EDC.RB_Cycle02.TransactionNumber>
8702
Source B 1

EDC Ring Buffer
Control TM
Transaction Trigger

GT
Source A RB.TransactionNumber
<EDC.RB_Cycle02.TransactionNumber>
8702
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

EDC Ring Buffer
Control TM
Transaction Trigger

MOVE
Source 1
Dest RB.TransactionNumber
<EDC.RB_Cycle02.TransactionNumber>
8702

Process RingBuffer Reads from Transaction Manager

Read Next
Calculate LNi, EQi for Transaction Logs & Performance

11

EDC Ring Buffer
Control Read Next
Buffer Record
RB.ReadNext
<EDC.RB_Cycle02.ReadNext>

EDC Ring Buffer
Control Read Buffer
Record in Progress
Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle02.ReadInProgressTimeout>

MOD
Source A RB_Read.Header.EQID
<EDC_RB_Cycle02_Read.Header.EQID>
1011
Source B 100
Dest EQi
11

CPT
Dest LNi
10
Expression (RB_Read.Header.EQID-EQi)/100

CLR
Dest PerfLNi
10

CLR
Dest PerfEQi
11

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Cycle02.ReadValidEQID>
U

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Cycle02.ReadValidEQID>
L

LIMIT
Low Limit 0
Test LNi
10
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
11
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MOVE
Source LNi
10
Dest PerfLNi
10

MOVE
Source EQi
11
Dest PerfEQi
11

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

12

EDC Ring Buffer
Control Read Data
has Valid Equipment
ID
RB.ReadValidEQID
<EDC.RB_Cycle02.ReadValidEQID>

Local Copy of Equip
Array Tag for LNi,
EQi

COP
Source EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus01
Length 1

Equipment Data
Collection - Status
Line Equipment

COP
Source EDC_Status04.Line[PerfLNi].Equip[PerfEQi]
Dest EquipStatus04
Length 1

Process RingBuffer Reads from Transaction Manager

13

Read Next
Log Transaction & Performance

EDC Ring Buffer Control Read Next Buffer Record
RB.ReadNext
<EDC.RB_Cycle02.ReadNext>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle02.ReadInProgressTimeout>

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr>
4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr>
437
Dest Trans.Opt[2] <EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr>
4
Dest Trans.Opt[3] <EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr>
436
Dest Trans.Opt[4] <EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2520
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber <EDC.RB_Cycle02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans
PerfMsg TempPerfMsg
PerfTrans EquipStatus04.CycleTx.EN

Process RingBuffer Reads from Transaction Manager

14

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

EDC Ring Buffer
Control Read Buffer
Record in Progress
RB.ReadInProgress
<EDC.RB_Cycle02.ReadInProgress>

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Cycle02.TransactionNumber>
8702
Source B RB.Interface
<EDC.RB_Cycle02.Interface>
8702

EDC Ring Buffer
Control Read Buffer
Record Complete
RB.ReadComplete
<EDC.RB_Cycle02.ReadComplete>
L

EDC Ring Buffer
Control TM Handshake
negative value

NEG
Source RB.Interface
<EDC.RB_Cycle02.Interface>
8702
Dest RB.InterfaceNEG
<EDC.RB_Cycle02.InterfaceNEG>
-8702

EDC Ring Buffer
Control TM
Transaction Trigger

EQ
Source A RB.TransactionNumber
<EDC.RB_Cycle02.TransactionNumber>
8702
Source B RB.InterfaceNEG
<EDC.RB_Cycle02.InterfaceNEG>
-8702

EDC Ring Buffer
Control Read Buffer
Record Resend
RB.ReadResend
<EDC.RB_Cycle02.ReadResend>
L

Process RingBuffer Reads from Transaction Manager

Read Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
No longer need delaly -- replaced Handshake bit with echo of TransNum
Slow/Fast delay added - no delay caused TM thread errors

15

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Cycle02.ReadComplete>

Server Status TM Transaction OK Enable High Speed RB ServerStat_TMokTMR.DN /

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source CfgRbDelaySlow 100
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Cycle02.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer RB.ReadCompleteDelayTMR <EDC.RB_Cycle02.ReadCompleteDelayTMR>
Preset 0
Accum 14
EN
DN

Server Status TM Transaction OK Enable High Speed RB ServerStat_TMokTMR.DN

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source CfgRbDelayFast 0
Dest RB.ReadCompleteDelayTMR.PRE <EDC.RB_Cycle02.ReadCompleteDelayTMR.PRE> 0

EDC Ring Buffer Control Read Complete Delay Timer RB.ReadCompleteDelayTMR.DN <EDC.RB_Cycle02.ReadCompleteDelayTMR.DN>

EDC Ring Buffer Control Read Buffer Record Complete Delay RB.ReadCompleteDelay <EDC.RB_Cycle02.ReadCompleteDelay>

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle02.ReadInProgress>
U

Process RingBuffer Reads from Transaction Manager

16

EDC Ring Buffer Control Read Buffer Record Complete
RB.ReadComplete
<EDC.RB_Cycle02.ReadComplete>

EDC Ring Buffer Control Read Buffer Record Complete 1-shot
RB.ReadCompleteONS
<EDC.RB_Cycle02.ReadCompleteONS>
ONS

Read Complete
Log Transaction & Performance

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle02.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Green - Cycle Tx DN
EquipStatus01.Update.CycleTxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdate

aoiEDC_StatusUpdate	aoiEDC_StatusUpdate
Status	EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Dest	Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Dest	Trans.Opt[2] <EDC.Trans.Opt[2]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr> 4
Dest	Trans.Opt[3] <EDC.Trans.Opt[3]> -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr> 436
Dest	Trans.Opt[4] <EDC.Trans.Opt[4]> -1

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	2521
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	RB.TransactionNumber <EDC.RB_Cycle02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR

Dest	TempPerfMsg.TransLast 110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE

Source	RB.ReadInProgressTMR.ACC <EDC.RB_Cycle02.ReadInProgressTMR.ACC> 0
Dest	TempPerfMsg.TransTMR.ACC 110

Performance Transaction

aoiEDC_PerfTrans

Process RingBuffer Reads from Transaction Manager

aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus04.CycleTx.DN

Read in Progress - No Response from TM Timer
Increment Transaction Number to Retry TM Message Trigger

reduced retry from 62 to 5 seconds

17

EDC Ring Buffer Control Read Buffer Record in Progress RB.ReadInProgress <EDC.RB_Cycle02.ReadInProgress>

EDC Ring Buffer Control Read Buffer Record Complete RB.ReadComplete <EDC.RB_Cycle02.ReadComplete> /

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Cycle02.ReadResend> /

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 5000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Cycle02.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Cycle02.ReadResend>

EDC Ring Buffer Control Read Buffer Record Resend RB.ReadResend <EDC.RB_Cycle02.ReadResend> U

EDC Ring Buffer Control Read in Progress Timer

MOVE
Source 2000
Dest RB.ReadInProgressTMR.PRE <EDC.RB_Cycle02.ReadInProgressTMR.PRE> 5000

EDC Ring Buffer Control Read in Progress Timer

TON
Timer RB.ReadInProgressTMR <EDC.RB_Cycle02.ReadInProgressTMR> EN
Preset 5000 DN
Accum 0

EDC Ring Buffer Control Read in Progress Timer RB.ReadInProgressTMR.DN <EDC.RB_Cycle02.ReadInProgressTMR.DN>

EDC Ring Buffer Control Read Buffer Record in Progress Timeout RB.ReadInProgressTimeout <EDC.RB_Cycle02.ReadInProgressTimeout>

Process RingBuffer Reads from Transaction Manager

Read in Progress - No Response from TM Timer
Log Transaction & Performance

18

EDC Ring Buffer Control Read Buffer Record in Progress Timeout
RB.ReadInProgressTimeout
<EDC.RB_Cycle02.ReadInProgressTimeout>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle02.ReadValidEQID>

Local Copy of Equip Array Tag for LNi, EQi State = Red - Cycle Tx Msg ER
EquipStatus01.Update.CycleTxER
L

Update Status State for Transactions

aoiEDC_StatusUpdate
aoiEDC_StatusUpdate: aoiEDC_StatusUpdate
Status: EquipStatus01

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr>
4
Dest: Trans.Opt[1] <EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr>
437
Dest: Trans.Opt[2] <EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadTagPtr <EDC.RB_Cycle02.ReadTagPtr>
4
Dest: Trans.Opt[3] <EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: RB.ReadRecPtr <EDC.RB_Cycle02.ReadRecPtr>
436
Dest: Trans.Opt[4] <EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 2524
Input Par: 0
Input Par: LNi
Input Par: EQi
Input Par: RB.TransactionNumber <EDC.RB_Cycle02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest: TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source: RB.ReadInProgressTMR.ACC <EDC.RB_Cycle02.ReadInProgressTMR.ACC>
0
Dest: TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans
PerfMsg: TempPerfMsg
PerfTrans: EquipStatus04.CycleTx.ER

EDC Ring Buffer Control Read in Progress Timer
RB.ReadInProgressTMR
<EDC.RB_Cycle02.ReadInProgressTMR>
RES

Process RingBuffer Reads from Transaction Manager

RB Write -- Keep Alive Periodic Resend Last Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

19

EDC Ring Buffer Control Read Buffer Record Complete Delay
RB.ReadCompleteDelay
<EDC.RB_Cycle02.ReadCompleteDelay>

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Cycle02.Empty>

EDC Ring Buffer Control Keep Alive Timer

MOVE
Source 300000
Dest RB.KeepAliveTMR.PRE
<EDC.RB_Cycle02.KeepAliveTMR.PRE> 300000

EDC Ring Buffer Control Keep Alive Timer

TON
Timer RB.KeepAliveTMR
<EDC.RB_Cycle02.KeepAliveTMR>
Preset 300000
Accum 34122
EN
DN

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR.DN
<EDC.RB_Cycle02.KeepAliveTMR.DN>

EDC Ring Buffer Control Read Buffer Keep Alive Resend Last
RB.KeepAlive
<EDC.RB_Cycle02.KeepAlive>

JSR
Routine Name _1121_RingBufferKeepAlive

EDC Ring Buffer Control Keep Alive Timer
RB.KeepAliveTMR
<EDC.RB_Cycle02.KeepAliveTMR>
RES

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

20

EDC Ring Buffer Control Buffer Empty
RB.Empty
<EDC.RB_Cycle02.Empty>
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest RB.NotEmptyTMR.PRE
<EDC.RB_Cycle02.NotEmptyTMR.PRE> 300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer RB.NotEmptyTMR
<EDC.RB_Cycle02.NotEmptyTMR>
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A RB.NotEmptyTMR.ACC
<EDC.RB_Cycle02.NotEmptyTMR.ACC> 0
Source B RB.NotEmptyMax
<EDC.RB_Cycle02.NotEmptyMax> 300001

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source RB.NotEmptyTMR.ACC
<EDC.RB_Cycle02.NotEmptyTMR.ACC> 0
Dest RB.NotEmptyMax
<EDC.RB_Cycle02.NotEmptyMax> 300001

Process RingBuffer Reads from Transaction Manager

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

21

EDC Ring Buffer Control Buffer Full
RB.Full
<EDC.RB_Cycle02.Full>

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
RB.NotEmptyTMR.DN
<EDC.RB_Cycle02.NotEmptyTMR.DN>

EDC Ring Buffer Control Read Buffer Alarm
RB.BufferAlarm
<EDC.RB_Cycle02.BufferAlarm>

Copy Equipment working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)
DN007 Split EquipStatus Tag

22

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle02.ReadValidEQID>

Server Interface Status Lines and Equipment Array

COP	
Source	EquipStatus01
Dest	EDC_Status01.Line[PerfLNi].Equip[PerfEQi]
Length	1

Tool, Trend, Summary Rx/Tx Status Lines and Equipment Array

COP	
Source	EquipStatus04
Dest	EDC_Status04.Line[PerfLNi].Equip[PerfEQi]
Length	1

(End)

RB Write -- Periodic Resend Transaction
Resend last message if RingBuffer is EMPTY for Keep Alive to TM/SQL
(Server will discard duplicate Transaction Numbers)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1121
Dest	Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806557869
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806557869

Use Read Tag Pointer to store Read Record in Temp
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

2

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr
	<EDC.RB_Cycle02.ReadTagPtr>
	4
Source B	1

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle02_Rec01[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr
	<EDC.RB_Cycle02.ReadTagPtr>
	4
Source B	2

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle02_Rec02[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr
	<EDC.RB_Cycle02.ReadTagPtr>
	4
Source B	3

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle02_Rec03[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr
	<EDC.RB_Cycle02.ReadTagPtr>
	4
Source B	4

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle02_Rec04[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

EDC Ring Buffer Control Current/Last Read Tag

EQ	
Source A	RB.ReadTagPtr
	<EDC.RB_Cycle02.ReadTagPtr>
	4
Source B	5

Equipment Data Cycle Message

COP	
Source	EDC_RB_Cycle02_Rec05[RB.ReadRecPtr]
Dest	KeepAliveMsg
Length	1

3

Increment Command Number to indicate Keep Alive Count
(used by TM Stored Procedure to detect difference between Keep Alive and Duplicate)

DHF024 Mar05, 2019 - Keep Alive CmdNum limited to 1-9999

Equipment Data
Cycle Message Header
- Command Number

ADD	
Source A	KeepAliveMsg.Header.CmdNum 1
Source B	1
Dest	KeepAliveMsg.Header.CmdNum 1

Equipment Data
Cycle Message Header
- Command Number

LT	
Source A	KeepAliveMsg.Header.CmdNum 1
Source B	1

Equipment Data
Cycle Message Header
- Command Number

GT	
Source A	KeepAliveMsg.Header.CmdNum 1
Source B	9999

Equipment Data
Cycle Message Header
- Command Number

MOVE	
Source	1
Dest	KeepAliveMsg.Header.CmdNum 1

Use Write Tag Pointer to store Temp Record
(Read and Write records may be in different Tags at Rollovers)
Tags 01-05

4

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr> 4
Source B 1

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle02_Rec01[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr> 4
Source B 2

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle02_Rec02[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr> 4
Source B 3

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle02_Rec03[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr> 4
Source B 4

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle02_Rec04[RB.WriteRecPtr]
Length 1

EDC Ring Buffer Control Next Write Tag

EQ
Source A RB.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr> 4
Source B 5

Equipment Data Cycle Message

COP
Source KeepAliveMsg
Dest EDC_RB_Cycle02_Rec05[RB.WriteRecPtr]
Length 1

5

Log Transaction

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteTagPtr
<EDC.RB_Cycle02.WriteTagPtr>
4
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.WriteRecPtr
<EDC.RB_Cycle02.WriteRecPtr>
437
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadTagPtr
<EDC.RB_Cycle02.ReadTagPtr>
4
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source RB.ReadRecPtr
<EDC.RB_Cycle02.ReadRecPtr>
436
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 2522
Input Par 0
Input Par LNi
Input Par EQi
Input Par RB.TransactionNumber
<EDC.RB_Cycle02.TransactionNumber>

EDC Ring Buffer Control Read Data has Valid Equipment ID
RB.ReadValidEQID
<EDC.RB_Cycle02.ReadValidEQID>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source RB.KeepAliveTMR.ACC
<EDC.RB_Cycle02.KeepAliveTMR.ACC>
34122
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EquipStatus04.CycleTx.KA

Increment Write Pointer

6

EDC Ring Buffer Control Next Write Record

ADD
Source A RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B 1
Dest RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437

EDC Ring Buffer Control Next Write Record

LT
Source A RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437
Source B RB.MaxRecPtr <EDC.RB_Cycle02.MaxRecPtr> 999

EDC Ring Buffer Control Next Write Tag

ADD
Source A RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B 1
Dest RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4

EDC Ring Buffer Control Next Write Tag

LT
Source A RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B 1

EDC Ring Buffer Control Next Write Tag

GT
Source A RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4
Source B RB.MaxTagPtr <EDC.RB_Cycle02.MaxTagPtr> 5

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest RB.WriteRecPtr <EDC.RB_Cycle02.WriteRecPtr> 437

EDC Ring Buffer Control Next Write Tag

MOVE
Source 1
Dest RB.WriteTagPtr <EDC.RB_Cycle02.WriteTagPtr> 4

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 1120

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 83719673
Source B 1
Dest Prog.RoutineScan[901] 83719673

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

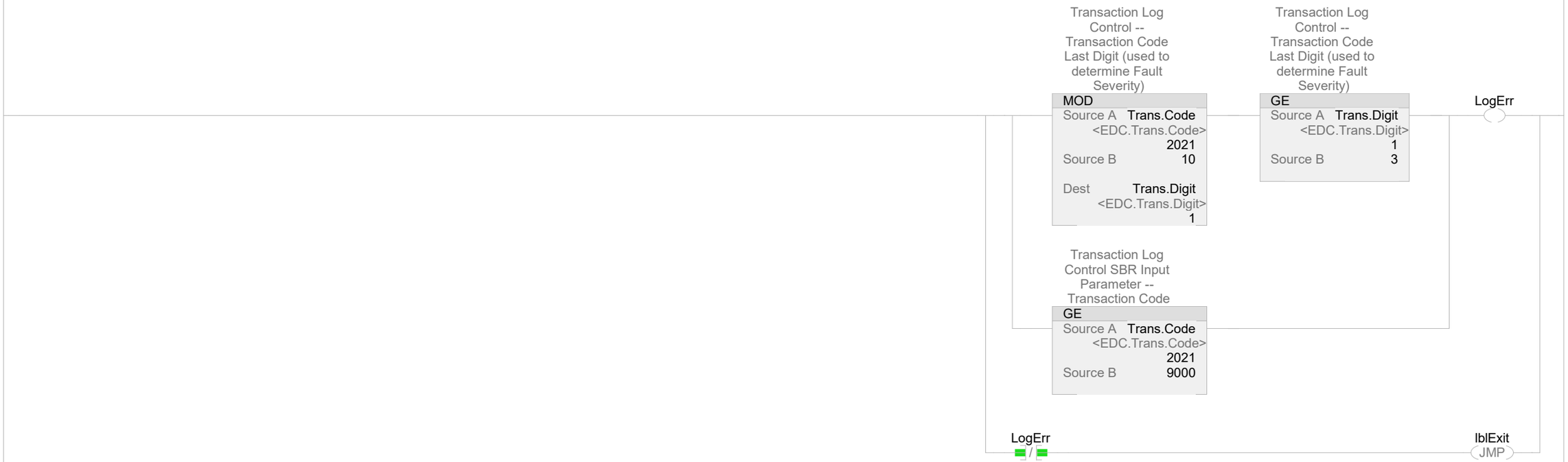
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

HMI
v3.50 - L980 Controller; new HMI Position Array in parallel with Left, Right, Middle

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion ...
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	701
Dest	Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 163891661
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 163891661

Configuration
v3.02e - add ConfigONS

2

s:FS

Configuration

JSR	
Routine Name	_711_Config

EDC Config Configuration ONS (Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Update HMI Config (from Server Interface)
CfgHmi_Update

Update HMI Config (from Server Interface)
CfgHmi_Update
U

Main Routine

Periodically Initialize Event Tables during Configuration Download
(this forces an update of the Event Descriptions from 'Undefined' as they are loaded
and it's faster/easier to wipe the Events clear and rebuild than selectiviely update the Descriptions)

3

Line Loop Complete - Update HMI Displays
LineLoopComplete

HMI Refresh HMI Events after Config All
EDC_HMI.CTL.DownloadAllRefresh

HMI Refresh HMI Events after Config All
EDC_HMI.CTL.DownloadAllRefresh
(U)

Initialize Overview and Zone Event Tables
JSR
Routine Name _712_EventsInit

Update and Initialize HMI Display Zones
Copy Temp Zones to Display and Clear after processing alll Line/Equip Loops
Line Loop updates n-Lines/Scan - only update/initialize Zones when all Lines updated

4

Line Loop Complete - Update HMI Displays
LineLoopComplete

Zone Init Loop for ZNi
FOR
Routine Name _714_ZoneInit
Index ZNi
21
Initial Value EDC_HMI.CFG.ZN.si
Terminal Value EDC_HMI.CFG.ZN.ei
Step Size 1

Main Routine

DHF028 Clear any unused Status Tables
Store any updates to Current Status Table
Initialize for next round of Line/Equip Loop processing

5

Line Loop Complete - Update HMI Displays
LineLoopComplete

Status Table Index
LT
Source A STi 1
Source B 4

HMI HMI Status Tables Display
FLL
Source 0
Dest EDC_HMI.Status[4]
Length 1

DTOS
Source 5
Dest EDC_HMI.Status[4].Name '5'

HMI First EQID on Display
COP
Source strHyphen
Dest EDC_HMI.Status[4].EQIDfirst
Length 1

HMI Last EQID on Display
COP
Source strHyphen
Dest EDC_HMI.Status[4].EQIDlast
Length 1

Status Table Index
LT
Source A STi 1
Source B 3

HMI HMI Status Tables Display
FLL
Source 0
Dest EDC_HMI.Status[3]
Length 1

DTOS
Source 4
Dest EDC_HMI.Status[3].Name '4'

HMI First EQID on Display
COP
Source strHyphen
Dest EDC_HMI.Status[3].EQIDfirst
Length 1

HMI Last EQID on Display
COP
Source strHyphen
Dest EDC_HMI.Status[3].EQIDlast
Length 1

Status Table Index
LT
Source A STi 1
Source B 2

HMI HMI Status Tables Display
FLL
Source 0
Dest EDC_HMI.Status[2]
Length 1

DTOS
Source 3
Dest EDC_HMI.Status[2].Name '3'

HMI First EQID on Display
COP
Source strHyphen
Dest EDC_HMI.Status[2].EQIDfirst
Length 1

HMI Last EQID on Display
COP
Source strHyphen
Dest EDC_HMI.Status[2].EQIDlast
Length 1

Status Table Index
LT
Source A STi 1
Source B 1

HMI HMI Status Tables Display
FLL
Source 0
Dest EDC_HMI.Status[1]
Length 1

DTOS
Source 2
Dest EDC_HMI.Status[1].Name '2'

HMI First EQID on Display
COP
Source strHyphen
Dest EDC_HMI.Status[1].EQIDfirst
Length 1

HMI Last EQID on Display
COP
Source strHyphen
Dest EDC_HMI.Status[1].EQIDlast
Length 1

Status Table Row Index
GT
Source A StRi 15
Source B 0

Status Table Name Index
DTOS
Source StNi 2
Dest HmiStatus.Name ''

HMI Equip Status Displays
COP
Source HmiStatus
Dest EDC_HMI.Status[STi]
Length 1

HMI Status Tables Display
FLL
Source 0
Dest HmiStatus
Length 1

Status Table Index
CLR
Dest STi 1

Status Table Row Index
CLR
Dest StRi 15

Status Table Name Index
MOVE
Source 1
Dest StNi 2

Main Routine

Update Overview Fault/Status Events
Table of Current Events, Start Time, Duration

Update all Equipment Periodically to remove old Events from Configuration changes to Display
Increased from 1 to 6 seconds to reduce Scan Time
Typically, changes in EQ Callout bits will trigger Event updates - periodic update cleans up any non-Event driven changes.

6

HMI Overview Event Update Timer

MOVE
Source 6000
Dest EDC_HMI.CTL.OviewEventTMR.PRE 6000

HMI Overview Event Update Timer

TON
Timer EDC_HMI.CTL.OviewEventTMR
Preset 6000
Accum 4105
EN
DN

Rotate Overview Line/Offline Popup Banners
Update all Banners together
Line Loop updates n-Lines/Scan - set Popup Rotate once for each Line
v3.5 Disabled Popup Rotate (no longer used on Overview)

7

AFI

HMI Overview Line/Group Popup Rotate Timer

MOVE
Source 5000
Dest EDC_HMI.CTL.OviewPopupTMR.PRE 0

HMI Overview Line/Group Popup Rotate Timer

TON
Timer EDC_HMI.CTL.OviewPopupTMR
Preset 0
Accum 0
EN
DN

HMI Overview Line/Group Popup Rotate Timer
EDC_HMI.CTL.OviewPopupTMR.DN

HMI Overview Line/Group Popup Rotate Timer
EDC_HMI.CTL.OviewPopupTMR
RES

Rotate all Line Popup Banners Setup
LinePopupSetup
L

Rotate all Offline Group Popup Banners
OfflinePopupRotate

Line Loop Complete - Update HMI Displays
LineLoopComplete

Rotate all Line Popup Banners
LinePopupRotate

Rotate all Line Popup Banners Setup
LinePopupSetup
U

Rotate all Line Popup Banners
LinePopupRotate
U

Rotate all Line Popup Banners Setup
LinePopupSetup

Rotate all Line Popup Banners
LinePopupRotate
L

Offline Groups
Name & OA Update
Popup Display Rotate Timer

8

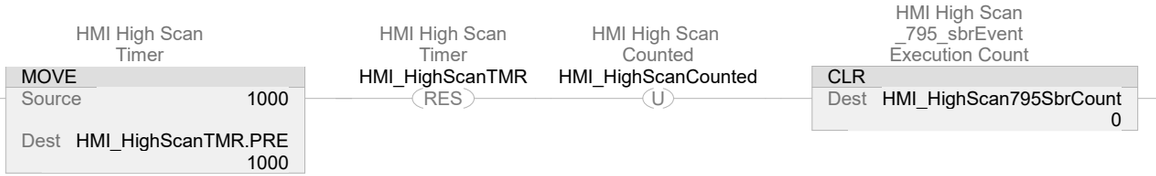
Offline Group Loop for GRPi

FOR
Routine Name _715_OviewOfflineInit
Index GRPi 4
Initial Value EDC_HMI.CFG.GRP.si
Terminal Value EDC_HMI.CFG.GRP.ei
Step Size 1

Main Routine

Initialize HMI High Scan Timer
This timer prevents Overview and Zone 1-20 Event Display Updates from exceeding 1 second/scan.
During Full Configuration downloads, clearing and setting of Event Masks can cause many event changes.

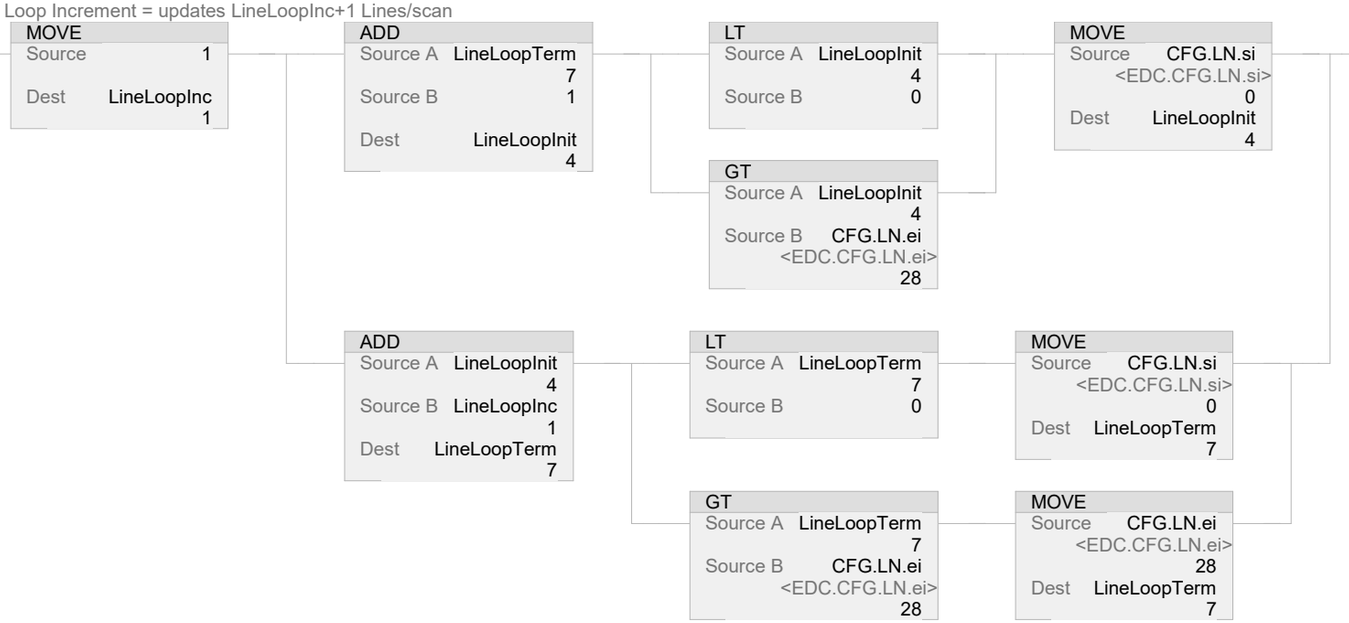
9



Line Loop Initial/Terminal values for n Lines/Scan
reduce scan time by updating portion of lines each scan but maintain <500 msec update of HMI data

Loop Increment = updates LineLoopInc+1 Lines/scan

10



Main Routine

Line Loop
service n Lines each Scan

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop will cause the unscanned Equipment to stop updating)

11

EDC Config Disable
Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>

Line Loop for LNi

FOR	
Routine Name	_720_LineLoop
Index	LNi
	4
Initial Value	LineLoopInit
Terminal Value	LineLoopTerm
Step Size	1

EDC Config Disable
Line Loop
CFG.DisableLineLoop
<EDC.CFG.DisableLineLoop>

Line array Index

MOVE	
Source	_ViewLNi
	0
Dest	LNi
	4

Line Loop for LNi

JSR	
Routine Name	_720_LineLoop

Line Loop Complete
used to update HMI Displays
(takes multiple scans to update all Lines/Equipment - this is moment for updating Zones, Events, Popups w/ complete data)

12

EDC Config Array End
Index

EQ	
Source A	CFG.LN.ei
	<EDC.CFG.LN.ei>
	28
Source B	LineLoopTerm
	7

Line Loop Complete -
Update HMI Displays
LineLoopComplete

Update Overview Fault/Status Events
Table of Current Events, Start Time, Duration

Update all Equipment Periodically for Configuration changes to Display

13

Line Loop Complete -
Update HMI Displays
LineLoopComplete

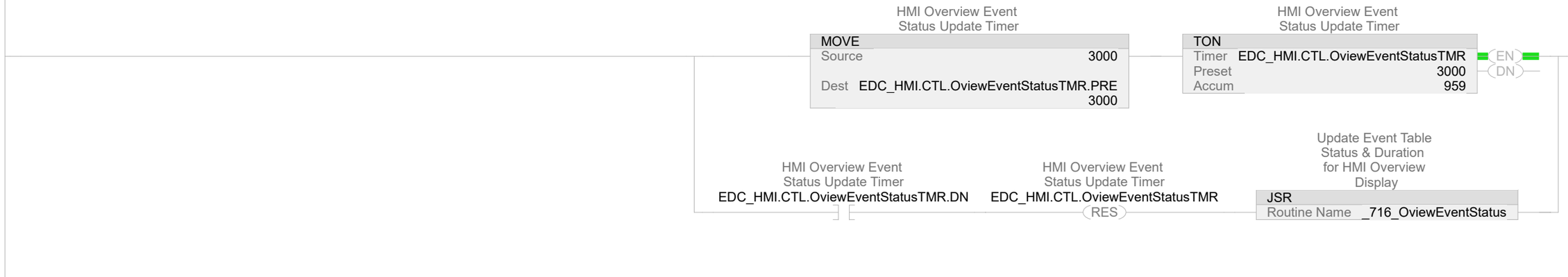
HMI Overview Event
Update Timer
EDC_HMI.CTL.OviewEventTMR.DN

HMI Overview Event
Update Timer
EDC_HMI.CTL.OviewEventTMR
RES

Main Routine

14

Update Overview Event Status and Duration
Increased from 500msec to 3 second to reduce Scan Time
(display is in 0.1 minute increments)



Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

15

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Routine
	702
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Type
	4
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Code
	21
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.FaultCount
	0
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.LOGi
	23
Dest	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	9003
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	0

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Main Routine

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

16

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

17

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
39
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
3
Source B
1
Dest ProgScanHighCount
3

Program High Scan Events
CLxProgScanHigh.7
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
3

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

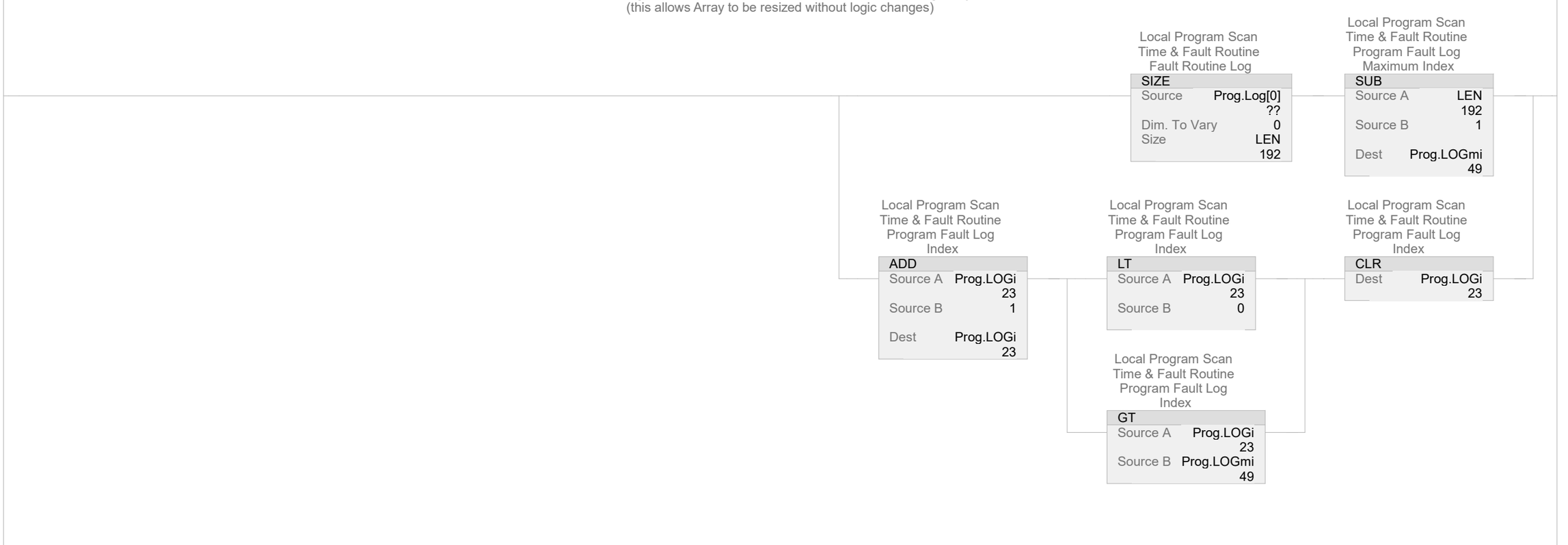
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
4428

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
750
Dest Prog.Log[Prog.LOGi].Routine
702

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
49925311

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
20220628

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
143141

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
866

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source LNi
4
Dest Prog.Log[Prog.LOGi].Opt1
12

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source EQi
26
Dest Prog.Log[Prog.LOGi].Opt2
2

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source GRPi
4
Dest Prog.Log[Prog.LOGi].Opt3
4

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source ZNi
21
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source ZnRi
99
Dest Prog.Log[Prog.LOGi].Opt5
99

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 702
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

5

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 369168272

6

(End)

Configuration

HMI Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 711
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Disable Common Line Equipment on Overview Dispaly
(i.e. Assembly Andon, AVI, Conveyor, Ent Lifter and Exit Lifter assignments for EQID xx00-04)

*** Gateway Configuration moved to Server Download ***

2

NOP

Configuration

3

Overview Pitch Array
Configure Array Indexes

HMI HMI Overview
Pitch Data

SIZE	
Source	EDC_HMI.Oview.Line[0].Pitch[0] ??
Dim. To Vary	0
Size	EDC_HMI.CFG.PIT.n 41

HMI Array Max Index

SUB	
Source A	EDC_HMI.CFG.PIT.n 41
Source B	1
Dest	EDC_HMI.CFG.PIT.mi 40

Gateway Config
Download HMI Common
Line Equipment
Enable
EDC_CfgGateway.HmiComEquipEnable

HMI Array Start
Index

MOVE	
Source	1
Dest	EDC_HMI.CFG.PIT.si 1

HMI Array End Index

MOVE	
Source	EDC_HMI.CFG.PIT.mi 40
Dest	EDC_HMI.CFG.PIT.ei 40

Gateway Config
Download HMI Common
Line Equipment
Enable
EDC_CfgGateway.HmiComEquipEnable

HMI Array Start
Index

MOVE	
Source	0
Dest	EDC_HMI.CFG.PIT.si 1

4

Zone Table Array
Configure Array Indexes

HMI HMI Zone Tables
Display

SIZE	
Source	EDC_HMI.Zone[0] ??
Dim. To Vary	0
Size	EDC_HMI.CFG.ZN.n 21

HMI Array Max Index

SUB	
Source A	EDC_HMI.CFG.ZN.n 21
Source B	1
Dest	EDC_HMI.CFG.ZN.mi 20

HMI Array Start
Index

MOVE	
Source	0
Dest	EDC_HMI.CFG.ZN.si 0

HMI Array End Index

MOVE	
Source	EDC_HMI.CFG.ZN.mi 20
Dest	EDC_HMI.CFG.ZN.ei 20

Configuration

5

Zone Table Row Array
Configure Array Indexes

HMI HMI Zone Table Row

SIZE	
Source	EDC_HMI.Zone[0].Row[0] ??
Dim. To Vary	0
Size	EDC_HMI.CFG.ZnR.n 100

HMI Array Max Index

SUB	
Source A	EDC_HMI.CFG.ZnR.n 100
Source B	1
Dest	EDC_HMI.CFG.ZnR.mi 99

HMI Array Start Index

MOVE	
Source	0
Dest	EDC_HMI.CFG.ZnR.si 0

HMI Array End Index

MOVE	
Source	EDC_HMI.CFG.ZnR.mi 99
Dest	EDC_HMI.CFG.ZnR.ei 99

6

DHF028 Status Table Array
Configure Array Indexes

HMI HMI Status Tables Display

SIZE	
Source	EDC_HMI.Status[0] ??
Dim. To Vary	0
Size	EDC_HMI.CFG.ST.n 5

HMI Array Max Index

SUB	
Source A	EDC_HMI.CFG.ST.n 5
Source B	1
Dest	EDC_HMI.CFG.ST.mi 4

HMI Array Start Index

MOVE	
Source	0
Dest	EDC_HMI.CFG.ST.si 0

HMI Array End Index

MOVE	
Source	EDC_HMI.CFG.ST.mi 4
Dest	EDC_HMI.CFG.ST.ei 4

Configuration

7

DHF028 Status Table Row Array
Configure Array Indexes

HMI HMI StatusTable Row

SIZE
Source EDC_HMI.Status[0].Row[0] ??
Dim. To Vary 0
Size EDC_HMI.CFG.StR.n 100

HMI Array Max Index

SUB
Source A EDC_HMI.CFG.StR.n 100
Source B 1
Dest EDC_HMI.CFG.StR.mi 99

HMI Array Start Index

MOVE
Source 0
Dest EDC_HMI.CFG.StR.si 0

HMI Array End Index

MOVE
Source EDC_HMI.CFG.StR.mi 99
Dest EDC_HMI.CFG.StR.ei 99

8

Overview Offline Array
Configure Array Indexes

HMI HMI Overview OffLine Data

SIZE
Source EDC_HMI.Oview.Offline[0] ??
Dim. To Vary 0
Size EDC_HMI.CFG.GRP.n 4

HMI Array Max Index

SUB
Source A EDC_HMI.CFG.GRP.n 4
Source B 1
Dest EDC_HMI.CFG.GRP.mi 3

HMI Array Start Index

MOVE
Source 0
Dest EDC_HMI.CFG.GRP.si 0

HMI Array End Index

MOVE
Source EDC_HMI.CFG.GRP.mi 3
Dest EDC_HMI.CFG.GRP.ei 3

Configuration

9

Overview Offline Equipment Array
Configure Array Indexes

HMI HMI Overview Equip Data

SIZE	
Source	EDC_HMI.Oview.Offline[0].Equip[0] ??
Dim. To Vary	0
Size	EDC_HMI.CFG.GrpR.n 100

HMI Array Max Index

SUB	
Source A	EDC_HMI.CFG.GrpR.n 100
Source B	1
Dest	EDC_HMI.CFG.GrpR.mi 99

HMI Array Start Index

MOVE	
Source	0
Dest	EDC_HMI.CFG.GrpR.si 0

HMI Array End Index

MOVE	
Source	EDC_HMI.CFG.GrpR.mi 99
Dest	EDC_HMI.CFG.GrpR.ei 99

10

Event Table Array
Configure Array Indexes

common array size for all Overview & Zone Event Displays
(actual rows displayed will be fewer than calculated in table -- larger table provides more accurate Date/Time when many Events turning on/off)

HMI Event Table Row

SIZE	
Source	HmiEvent[0] ??
Dim. To Vary	0
Size	EDC_HMI.CFG.EV.n 200

HMI Array Max Index

SUB	
Source A	EDC_HMI.CFG.EV.n 200
Source B	1
Dest	EDC_HMI.CFG.EV.mi 199

HMI Array Start Index

MOVE	
Source	0
Dest	EDC_HMI.CFG.EV.si 0

HMI Array End Index

MOVE	
Source	EDC_HMI.CFG.EV.mi 199
Dest	EDC_HMI.CFG.EV.ei 199

Configuration

11

Event Text Table Array
Configure Array Indexes

Lookup Table for all EQ/Event Text Descriptions - Tag01

Event Text Lookup
Event Text
Configuration

SIZE
Source EDC_EventTxt01[0]
??
Dim. To Vary 0
Size EDC_HMI.CFG.EvTxt01.n
32500

HMI Array Max Index

SUB
Source A EDC_HMI.CFG.EvTxt01.n
32500
Source B 1
Dest EDC_HMI.CFG.EvTxt01.mi
32499

HMI Array Start
Index

MOVE
Source 0
Dest EDC_HMI.CFG.EvTxt01.si
0

HMI Array End Index

ADD
Source A EDC_HMI.CFG.EvTxt01.si
0
Source B EDC_HMI.CFG.EvTxt01.mi
32499
Dest EDC_HMI.CFG.EvTxt01.ei
32499

12

Event Text Table Array
Configure Array Indexes

Lookup Table for all EQ/Event Text Descriptions - Tag02
si, se continue from Tag01 array

Event Text Lookup
Event Text
Configuration

SIZE
Source EDC_EventTxt02[0]
??
Dim. To Vary 0
Size EDC_HMI.CFG.EvTxt02.n
32500

HMI Array Max Index

SUB
Source A EDC_HMI.CFG.EvTxt02.n
32500
Source B 1
Dest EDC_HMI.CFG.EvTxt02.mi
32499

HMI Array Start
Index

ADD
Source A EDC_HMI.CFG.EvTxt01.ei
32499
Source B 1
Dest EDC_HMI.CFG.EvTxt02.si
32500

HMI Array End Index

ADD
Source A EDC_HMI.CFG.EvTxt02.si
32500
Source B EDC_HMI.CFG.EvTxt02.mi
32499
Dest EDC_HMI.CFG.EvTxt02.ei
64999

Configuration

13

Event Text Table Array
Configure Array Indexes

Lookup Table for all EQ/Event Text Descriptions - Tag03
si, se continue from Tag02 array

Event Text Lookup
Event Text
Configuration

SIZE	
Source	EDC_EventTxt03[0] ??
Dim. To Vary	0
Size	EDC_HMI.CFG.EvTxt03.n 32500

HMI Array Max Index

SUB	
Source A	EDC_HMI.CFG.EvTxt03.n 32500
Source B	1
Dest	EDC_HMI.CFG.EvTxt03.mi 32499

HMI Array Start
Index

ADD	
Source A	EDC_HMI.CFG.EvTxt02.ei 64999
Source B	1
Dest	EDC_HMI.CFG.EvTxt03.si 65000

HMI Array End Index

ADD	
Source A	EDC_HMI.CFG.EvTxt03.si 65000
Source B	EDC_HMI.CFG.EvTxt03.mi 32499
Dest	EDC_HMI.CFG.EvTxt03.ei 97499

14

Event Text Table Array
Configure Array Indexes

Lookup Table for all EQ/Event Text Descriptions - Tag04
si, se continue from Tag03 array

Event Text Lookup
Event Text
Configuration

SIZE	
Source	EDC_EventTxt04[0] ??
Dim. To Vary	0
Size	EDC_HMI.CFG.EvTxt04.n 32500

HMI Array Max Index

SUB	
Source A	EDC_HMI.CFG.EvTxt04.n 32500
Source B	1
Dest	EDC_HMI.CFG.EvTxt04.mi 32499

HMI Array Start
Index

ADD	
Source A	EDC_HMI.CFG.EvTxt03.ei 97499
Source B	1
Dest	EDC_HMI.CFG.EvTxt04.si 97500

HMI Array End Index

ADD	
Source A	EDC_HMI.CFG.EvTxt04.si 97500
Source B	EDC_HMI.CFG.EvTxt04.mi 32499
Dest	EDC_HMI.CFG.EvTxt04.ei 129999

Configuration

Event Text Table Array
Configure Array Indexes

Lookup Table for all EQ/Event Text Descriptions - Tag05
si, se continue from Tag04 array

15

Event Text Lookup
Event Text
Configuration

SIZE	
Source	EDC_EventTxt05[0] ??
Dim. To Vary	0
Size	EDC_HMI.CFG.EvTxt05.n 32500

HMI Array Max Index

SUB	
Source A	EDC_HMI.CFG.EvTxt05.n 32500
Source B	1
Dest	EDC_HMI.CFG.EvTxt05.mi 32499

HMI Array Start
Index

ADD	
Source A	EDC_HMI.CFG.EvTxt04.ei 129999
Source B	1
Dest	EDC_HMI.CFG.EvTxt05.si 130000

HMI Array End Index

ADD	
Source A	EDC_HMI.CFG.EvTxt05.si 130000
Source B	EDC_HMI.CFG.EvTxt05.mi 32499
Dest	EDC_HMI.CFG.EvTxt05.ei 162499

Configuration

Event Text Table Array
Configure Array Indexes

Lookup Table for all EQ/Event Text Descriptions - Tag06
si, se continue from Tag05 array

Tag06 will be used for Config downloads but can also store Events
some portion of Tag06 must be reserved for Config download

16

Event Text Lookup
Event Text
Configuration

SIZE
Source EDC_EventTxt06[0] ??
Dim. To Vary 0
Size EDC_HMI.CFG.EvTxt06.n 32500

HMI Array Max Index

SUB
Source A EDC_HMI.CFG.EvTxt06.n 32500
Source B 1
Dest EDC_HMI.CFG.EvTxt06.mi 32499

HMI Array Start
Index

ADD
Source A EDC_HMI.CFG.EvTxt05.ei 162499
Source B 1
Dest EDC_HMI.CFG.EvTxt06.si 162500

HMI Array End Index

ADD
Source A EDC_HMI.CFG.EvTxt06.si 162500
Source B EDC_HMI.CFG.EvTxt06.mi 32499
Dest EDC_HMI.CFG.EvTxt06.ei 194999

Server Status Table Array
Configure Array Indexes

17

HMI HMI Overview
Server Status Block

SIZE
Source EDC_HMI.Oview.Status[0] ??
Dim. To Vary 0
Size EDC_HMI.CFG.Stat.n 15

HMI Array Max Index

SUB
Source A EDC_HMI.CFG.Stat.n 15
Source B 1
Dest EDC_HMI.CFG.Stat.mi 14

HMI Array Start
Index

MOVE
Source 1
Dest EDC_HMI.CFG.Stat.si 1

HMI Array End Index

MOVE
Source 12
Dest EDC_HMI.CFG.Stat.ei 12

Configuration

18

Load String Constants for Space & Dash
Used for concatenation with Line & Equipment Names

MOVE
Source 1
Dest strSpace.LEN
1

MOVE
Source ascii.Space
<EDC.ascii.Space>
' '
Dest strSpace.DATA[0]
' '

MOVE
Source 1
Dest strHyphen.LEN
1

MOVE
Source ascii.Hyphen
<EDC.ascii.Hyphen>
'-'
Dest strHyphen.DATA[0]
'-'

MOVE
Source 1
Dest strPeriod.LEN
1

MOVE
Source ascii.Period
<EDC.ascii.Period>
'.'
Dest strPeriod.DATA[0]
'.'

19

Default Alarm Text
"wb "
(Callout Element number - Word and BIt to be Inserted after w & b)

HMI Event
Description
'Element'

MOVE
Source 3
Dest EDC_HMI.CFG.EventElement.LEN
3

HMI Event
Description
'Element'

MOVE
Source ascii.P
<EDC.ascii.P>
'P'
Dest EDC_HMI.CFG.EventElement.DATA[0]
'P'

HMI Event
Description
'Element'

MOVE
Source ascii._r
<EDC.ascii._r>
'r'
Dest EDC_HMI.CFG.EventElement.DATA[1]
'r'

HMI Event
Description
'Element'

MOVE
Source ascii.Period
<EDC.ascii.Period>
'.'
Dest EDC_HMI.CFG.EventElement.DATA[2]
'.'

Configuration

Default Alarm Text
" Undefined"
(appended to 'WnnBnn ' when Alarm Text not found)

20

HMI Event Description 'Undefined'

MOVE
Source 9
Dest EDC_HMI.CFG.EventUndefined.LEN 9

HMI Event Description 'Undefined'

MOVE
Source ascii.U <EDC.ascii.U> 'U'
Dest EDC_HMI.CFG.EventUndefined.DATA[0] 'U'

HMI Event Description 'Undefined'

MOVE
Source ascii._n <EDC.ascii._n> 'n'
Dest EDC_HMI.CFG.EventUndefined.DATA[1] 'n'

HMI Event Description 'Undefined'

MOVE
Source ascii._d <EDC.ascii._d> 'd'
Dest EDC_HMI.CFG.EventUndefined.DATA[2] 'd'

HMI Event Description 'Undefined'

MOVE
Source ascii._e <EDC.ascii._e> 'e'
Dest EDC_HMI.CFG.EventUndefined.DATA[3] 'e'

HMI Event Description 'Undefined'

MOVE
Source ascii._f <EDC.ascii._f> 'f'
Dest EDC_HMI.CFG.EventUndefined.DATA[4] 'f'

HMI Event Description 'Undefined'

MOVE
Source ascii._i <EDC.ascii._i> 'i'
Dest EDC_HMI.CFG.EventUndefined.DATA[5] 'i'

HMI Event Description 'Undefined'

MOVE
Source ascii._n <EDC.ascii._n> 'n'
Dest EDC_HMI.CFG.EventUndefined.DATA[6] 'n'

HMI Event Description 'Undefined'

MOVE
Source ascii._e <EDC.ascii._e> 'e'
Dest EDC_HMI.CFG.EventUndefined.DATA[7] 'e'

HMI Event Description 'Undefined'

MOVE
Source ascii._d <EDC.ascii._d> 'd'
Dest EDC_HMI.CFG.EventUndefined.DATA[8] 'd'

Default Alarm Text
"0"
(used to 0-pad Word & Bit numbers)

21

MOVE
Source 1
Dest str0.LEN 1

MOVE
Source ascii._0 <EDC.ascii._0> '0'
Dest str0.DATA[0] '0'

Configuration

Default Equipment Names
"E"

22

MOVE
Source 1
Dest strE.LEN
1

MOVE
Source ascii.E
<EDC.ascii.E>
'E'
Dest strE.DATA[0]
'E'

(End)

Initialize Overview and Zone Event Tables

HMI Event Initialization

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 712
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Initialize Overview Display Events

2

HMI HMI Event Table Row

FLL
Source 0
Dest EDC_HMI.Oview.Event[0]
Length EDC_HMI.CFG.EV.n

Initialize Zone Index

3

CLR
Dest EventZNi 21

Initialize Zone Display Events

4

lbl_ZoneLoop
LBL

HMI Maintenance Zone Displays

FLL
Source 0
Dest EDC_HMI.Zone[EventZNi].Event[0]
Length EDC_HMI.CFG.EV.n

Initialize Overview and Zone Event Tables

Increment Zone Index

5

ADD	
Source A	EventZNi 21
Source B	1
Dest	EventZNi 21

LE	
Source A	EventZNi 21
Source B	EDC_HMI.CFG.ZN.ei 20

lbl_ZoneLoop
JMP

(End)

Zone Init Loop for ZNi

Update and Initialize HMI Display Zones
Copy Temp Zones to Display and Clear after processing alll Line/Equip Loops
Line Loop updates n-Lines/Scan - only update/initialize Zones when all Lines updated

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 714
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Copy Temporary Work to Display
Initialize Temporary Work

2

HMI Maintenance Zone Displays

COP
Source EDC_HMI.Zone[ZNi].RowTemp[0]
Dest EDC_HMI.Zone[ZNi].Row[0]
Length EDC_HMI.CFG.ZnR.n

HMI Maintenance Zone Displays

FLL
Source 0
Dest EDC_HMI.Zone[ZNi].RowTemp[0]
Length EDC_HMI.CFG.ZnR.n

HMI Maintenance Zone Displays

CLR
Dest EDC_HMI.Zone[ZNi].ZnRi ??

Zone Init Loop for ZNi

Update Zone Event Status and Duration
Increased from 500msec to 3 second to reduce Scan Time
(display is in 0.1 minute increments)

small offset for ZNi to avoid 20 JSRs on same scan

3

HMI Zone Event
Status Update Timer

CPT
Dest EDC_HMI.CTL.ZoneEventStatusTMR[ZNi].PRE
??
Expression 3000 + (ZNi * 10)

HMI Zone Event
Status Update Timer

TON
Timer EDC_HMI.CTL.ZoneEventStatusTMR[ZNi]
Preset ??
Accum ??
EN
DN

HMI Zone Event
Status Update Timer
EDC_HMI.CTL.ZoneEventStatusTMR[ZNi].DN

HMI Zone Event
Status Update Timer
EDC_HMI.CTL.ZoneEventStatusTMR[ZNi]
RES

Update Event Table
Status & Duration
for HMI Zone Display

JSR
Routine Name _717_ZoneEventStatus

Reset Zone Refresh when all Equipment Events have been updated
(only reset once on first zone in Loop)

4

Zone Table Index

EQ
Source A ZNi
21
Source B 0

Refresh
Zone Events

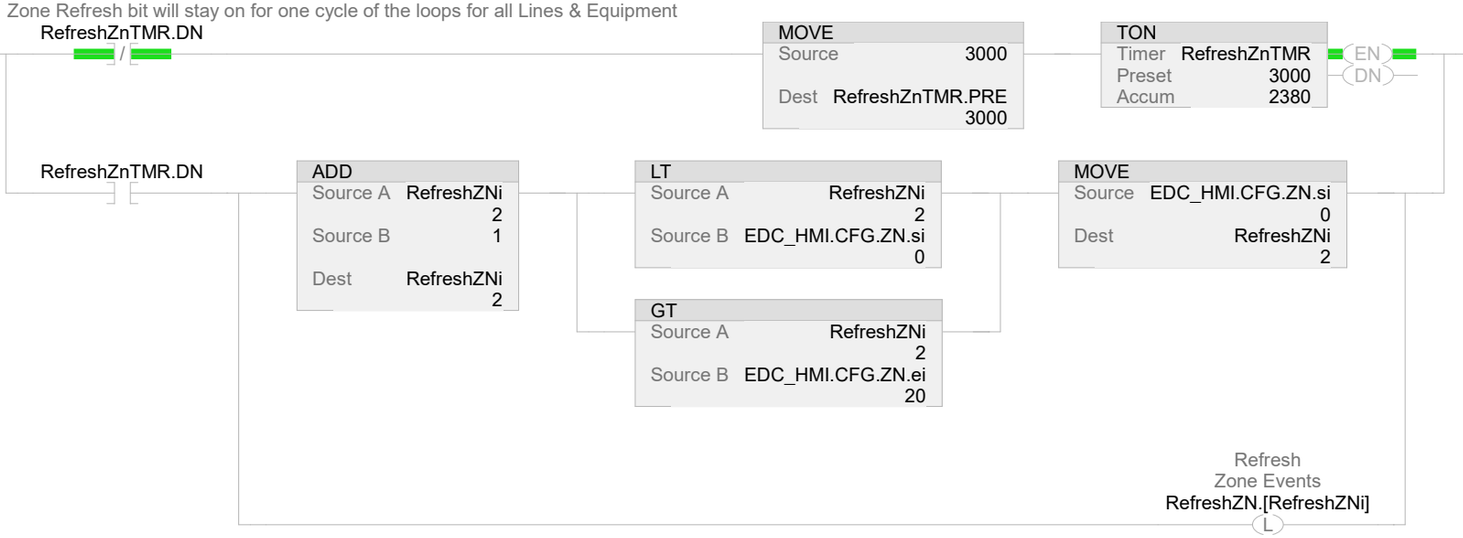
CLR
Dest RefreshZN
16#0000_0000

Zone Init Loop for ZNi

Periodically Update all Equipment Events for 1 Zone to remove any old Events
(i.e. Configuration change removed Event from a Zone)

Zone Refresh bit will stay on for one cycle of the loops for all Lines & Equipment

5



(End)

Offline Group Loop for GRPi

Update Offline Group Header

One set of Interfaces is shared by all Group
LNi=0, GRPi = Offline Group Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 715
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Copy Offline Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

2

HMI Control Line

COP
Source EDC_HMI.CTL.Offline[GRPi]
Dest HmiCtlOffline
Length 1

Offline Group Loop for GRPi

Initialize Group Data when all Lines are Complete
DN006 - support multiple Offline Lines

3

Line Loop Complete - Update HMI Displays
LineLoopComplete

HMI Overview OffLine Data
FLL
Source 0
Dest HmiOviewOfliine[GRPi]
Length 1

HMI Overview OffLine Data Equipment Fault/Alarm Popup Display
COP
Source EDC_HMI.Oview.Offline[GRPi].Popup
Dest HmiOviewOfliine[GRPi].Popup
Length 1

HMI Overview OffLine Data OffLine Name
COP
Source EDC.Offline.CFG[GRPi].Name
Dest HmiOviewOfliine[GRPi].Name
Length 1

HMI Overview OffLine Data Offline OR%
MOVE
Source EDC.Offline.Data[GRPi].EdtOR.Avail ??
Dest HmiOviewOfliine[GRPi].OR ??

HMI Overview OffLine Data Offline OA%
MOVE
Source EDC.Offline.Data[GRPi].EdtOA.Avail ??
Dest HmiOviewOfliine[GRPi].OA ??

Popup Display Rotate Timer

4

Rotate all Offline Group Popup Banners
OfflinePopupRotate

HMI Control Line Overview Line Popup - Rotate Time
HmiCtlOffline.PopupRotate
L

Copy Offline working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

5

HMI HMI Offline Display Control
COP
Source HmiCtlOffline
Dest EDC_HMI.CTL.Offline[GRPi]
Length 1

Exit Routine

6

lbl_Exit
LBL

NOP

Offline Group Loop for GRPi

(End)

Update Event Table Status & Duration for HMI Overview Display

Update Overview Display Fault/Status Event Status and Duration
Table of Current Events, Start Time, Duration

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 716
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Update the Event Status and Duration

Copy Overview Event Table to/from Subroutine Tag

2

HMI Event Table Row

COP
Source EDC_HMI.Oview.Event[0]
Dest HmiEvent[0]
Length EDC_HMI.CFG.EV.n

Update Event Table Status & Duration for HMI Display

JSR
Routine Name _796_sbrEventStatus

HMI HMI Event Table Row

COP
Source HmiEvent[0]
Dest EDC_HMI.Oview.Event[0]
Length EDC_HMI.CFG.EV.n

(End)

Update Event Table Status & Duration for HMI Zone Display

Update Zone Display Fault/Status Event Status and Duration
Table of Current Events, Start Time, Duration

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index
ZNi = Zone Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 717
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Update the Event Status and Duration

Copy Overview Event Table to/from Subroutine Tag

2

HMI Event Table Row

COP
Source EDC_HMI.Zone[ZNi].Event[0]
Dest HmiEvent[0]
Length EDC_HMI.CFG.EV.n

Update Event Table Status & Duration for HMI Display

JSR
Routine Name _796_sbrEventStatus

HMI Maintenance Zone Displays

COP
Source HmiEvent[0]
Dest EDC_HMI.Zone[ZNi].Event[0]
Length EDC_HMI.CFG.EV.n

(End)

Line Loop for LNi

Line Service Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop will cause the unscanned Equipment to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	720
Dest	Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 163891661
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 163891661

Line Loop for LNi

Copy Line Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

Reset Line bits for Accumulated EQ status
DN006 - support multiple Offline Lines

2

Local Copy of Line Array Tag for LNi

COP
Source EDC.Line[LNi].CFG
Dest LineCfg
Length 1

Local Copy of Line Array Tag for LNi

COP
Source EDC.Line[LNi].Data
Dest LineData
Length 1

HMI Control Line

COP
Source EDC_HMI.CTL.Line[LNi]
Dest HmiCtlLine
Length 1

HMI Overview Line Local Tag

FLL
Source 0
Dest HmiOviewLine
Length 1

AFI

HMI Overview Line Local Tag Equipment Fault/Alarm Popup Display

COP
Source EDC_HMI.Oview.Line[LNi].Popup
Dest HmiOviewLine.Popup
Length 1

Local Copy of Line Array Tag for LNi Line is for Offline Equipment

LineCfg.Offline

Line array Index

EQ
Source A LNi
4
Source B CFG.OfflineFirstLine
<EDC.CFG.OfflineFirstLine>
0

Initialize Offline Groups (1st Offline Line)

OfflineFirstLine

Line array Index

EQ
Source A LNi
4
Source B CFG.OfflineLastLine
<EDC.CFG.OfflineLastLine>
28

Finalize Offline Groups (Last Offline Line)

OfflineLastLine

Line Loop for LNi

Line Andon Status

v1.6 - removed Andon Status Produced/Consumed for Line Status (ViewSE gets directly from Andon)

3

HMI Overview Line Local Tag Line Short Name

COP
Source LineCfg.ShortName
Dest HmiOviewLine.Name
Length 1

HMI Overview Line Local Tag Line OA%

MOVE
Source LineData.LdtOA.Avail 93.248924
Dest HmiOviewLine.OA 93.0506

HMI Overview Line Local Tag Line OR%

MOVE
Source LineData.LdtOR.Avail 83.600845
Dest HmiOviewLine.OR 86.394806

HMI Overview Line Local Tag Line Status (Andon Status Colors)

MOVE
Source 0
Dest HmiOviewLine.Status 0

AFI

HMI Overview Line Local Tag Line Status (Andon Status Colors)

MOVE
Source 1
Dest HmiOviewLine.Status 0

AFI

HMI Overview Line Local Tag Line Status (Andon Status Colors)

MOVE
Source 2
Dest HmiOviewLine.Status 0

AFI

HMI Overview Line Local Tag Line Status (Andon Status Colors)

MOVE
Source 3
Dest HmiOviewLine.Status 0

Popup Display Rotate Timer

4

Rotate all Line Popup Banners
LinePopupRotate

HMI Control Line Overview Line Popup - Rotate Time
HmiCtlLine.PopupRotate
(L)

Line Loop for LNi

Equipment Loop

One set of Interface Routines is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop will cause the unscanned Equipment to stop updating)

5

EDC Config Disable Equipment Loop
CFG.DisableEqLoop
<EDC.CFG.DisableEqLoop>

Equipment Loop for EQi

FOR
Routine Name _730_EquipLoop
Index EQi
26
Initial Value LineCfg.EQ.si
Terminal Value LineCfg.EQ.ei
Step Size 1

EDC Config Disable Equipment Loop
CFG.DisableEqLoop
<EDC.CFG.DisableEqLoop>

Equipment array Index

MOVE
Source _ViewEQi
0
Dest EQi
26

Equipment Loop for EQi

JSR
Routine Name _730_EquipLoop

Copy Line working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

save updates to Line.CFG from new Equipment Callout messages
DN006 - support multiple Offline Lines

6

HMI HMI Line Display Control

COP
Source HmiCtlLine
Dest EDC_HMI.CTL.Line[LNi]
Length 1

HMI Line Data Array by Conv Index

COP
Source HmiOviewLine
Dest EDC_HMI.Oview.Line[LNi]
Length 1

Finalize Offline Groups (Last Offlline Line)
OfflineLastLine

HMI HMI Overview OffLine Data

COP
Source HmiOviewOfliine[0]
Dest EDC_HMI.Oview.Offline[0]
Length EDC_HMI.CFG.GRP.n

Line Loop for LNi

Exit Routine

7

lbl_Exit
LBL

NOP

Check Viewing Index Limits

8

Line View Tag Index
LT
Source A _ViewLNi 0
Source B 0

Line View Tag Index
GT
Source A _ViewLNi 0
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

Line View Tag Index
CLR
Dest _ViewLNi 0

Copy Line Instance to Viewing Tag if selected
(_ViewLine can be used to multiplex Trends and Diagnostic Spreadsheets)

9

_ViewEN

Line array Index
EQ
Source A LNi 4
Source B _ViewLNi 0

Line View Tag (used by View Diagnostic Spreadsheet)
COP
Source EDC.Line[LNi]
Dest _ViewLine
Length 1

(End)

Equipment Loop for EQi

Equipment Service Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop will cause the unscanned Equipment to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	730
Dest	Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 163891661
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 163891661

Equipment Loop for EQi

Monitor HMI High Scan Timer
This timer prevents Overview and Zone 1-20 Event Display Updates from exceeding 1 second/scan.
During Full Configuration downloads, clearing and setting of Event Masks can cause many event changes.

2

HMI High Scan Timer
TON
Timer HMI_HighScanTMR
Preset 1000
Accum 0
EN
DN

HMI High Scan Timer
HMI_HighScanTMR.DN

HMI High Scan Counted
HMI_HighScanCounted

HMI High Scan Counted
HMI_HighScanCounted
L

HMI High Scan Count
ADD
Source A HMI_HighScanCount
904
Source B 1
Dest HMI_HighScanCount
904

HMI High Scan LNi
MOVE
Source LNi
4
Dest HMI_HighScanLastLNi
11

HMI High Scan EQi
MOVE
Source EQi
26
Dest HMI_HighScanLastEQi
33

HMI High Scan Routine Count
MOVE
Source HMI_HighScan795SbrCount
0
Dest HMI_HighScan795SbrCountLast
31

Program High Scan Events
CLxProgScanHigh.29
L
BRK

Service Equip Loop for EQi
Equip Configured or Line/Group PopupRotate needs reset
DHF006 - changed Equip.CFG.EN to Equip.CTL.Config (addition of Disabled Equipment support)

3

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.Config

UpdateEQi

HMI Control Line Overview Line Popup - Rotate Time
HmiCtlLine.PopupRotate

Equipment array Index
EQ
Source A EQi
26
Source B LineCfg.EQ.ei
21

Equipment Loop for EQi

Equipment is not Configured - Exit
Special Case - must update Line/Offline Popup Display if EQi = ei and any PopupUpdates required
(only happens if EQei if not configure - abnormal, but possible)

4

UpdateEQi
[/]

lbl_Exit
JMP

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

5

Local Copy of Equip Array Tag for LNi, EQi

COP	
Source	EDC.Line[LNi].Equip[EQi]
Dest	Equip
Length	1

Equipment Loop for EQi

Common Equipment Pitch
EQi 0-4 are shown in standard Line locations on Overview Display
(AVI, Andon, Conv, Ent, Exit)
DN006 - support multiple Offline Lines

Feb 9, 2021 - bypass Pitch=0; assign Andon, AVI, Coneyor to Pitch 1 for AVI Tracking data
v3.5 replaced hard-coded common equipment EQID with configuration

6

Gateway Config Download HMI Common Line Equipment Enable
EDC_CfgGateway.HmiComEquipEnable

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Local Copy of Equip Array Tag for LNi, EQi Pitch ID (0-350)

EQ
Source A Equip.CFG.Pitch[15]
Source B 0

Equipment array Index

EQ
Source A EQi[26]
Source B EDC_CfgGateway.HmiComAviEQi[9]

Equipment array Index

EQ
Source A EQi[26]
Source B EDC_CfgGateway.HmiComAndonEQi[1]

Equipment array Index

EQ
Source A EQi[26]
Source B EDC_CfgGateway.HmiComConvEQi[2]

Equipment array Index

EQ
Source A EQi[26]
Source B EDC_CfgGateway.HmiComEntEQi[3]

Equipment array Index

EQ
Source A EQi[26]
Source B EDC_CfgGateway.HmiComExitEQi[4]

EQi = 0-4
AVI, Andon, Conv, Ent, Exit shown in Common Line loccations
CommonLineEquip

Equipment Loop for EQi

Special Case
Show New, Unconfigured Equipment in Pitch 1, Middle Position
DHF006 - changed Equip.CFG.EN to Equip.CTL.Config (addition of Disabled Equipment support)
DN006 - support multiple Offline Lines

7

Gateway Config Download HMI Common Line Equipment Enable: EDC_CfgGateway.HmiComEquipEnable

Local Copy of Equip Array Tag for LNi, EQi Equipment Configured (by Server or Driver): Equip.CTL.Config

Local Copy of Line Array Tag for LNi Line is for Offline Equipment: LineCfg.Offline

EQi = 0-4 AVI, Andon, Conv, Ent, Exit shown in Common Line loccations: CommonLineEquip

Local Copy of Equip Array Tag for LNi, EQi Pitch ID (0-350)

EQ
Source A: Equip.CFG.Pitch
15
Source B: 0

Equipment Data Collection Lines and Equipment Array

MOVE
Source: 1
Dest: EDC.Line[LNi].Equip[EQi].CFG.Pitch
0

Equipment Data Collection Lines and Equipment Array

MOVE
Source: 3
Dest: EDC.Line[LNi].Equip[EQi].CFG.Position
0

CPT
Dest: EQID
1811
Expression: LNi*100+EQi

Local Copy of Equip Array Tag for LNi, EQi Equipment Name

EQ
Source A: Equip.CFG.Name.LEN
10
Source B: 0

DTOS
Source: EQID
1811
Dest: EDC.Line[LNi].Equip[EQi].CFG.Name
??

CONCAT
Source A: strE
'E'
Source B: EDC.Line[LNi].Equip[EQi].CFG.Name
??
Dest: EDC.Line[LNi].Equip[EQi].CFG.Name
??

Local Copy of Equip Array Tag for LNi, EQi Equipment Short Name

EQ
Source A: Equip.CFG.ShortName.LEN
5
Source B: 0

DTOS
Source: EQID
1811
Dest: EDC.Line[LNi].Equip[EQi].CFG.ShortName
??

CONCAT
Source A: strE
'E'
Source B: EDC.Line[LNi].Equip[EQi].CFG.ShortName
??
Dest: EDC.Line[LNi].Equip[EQi].CFG.ShortName
??

Equipment Loop for EQi

Update Overview Line/Pitch/Equipment DIsplay
Update Overview Offline/Group/Equipment DIsplay
LNi=0 is Offline
DHF006 - changed Equip.CFG.EN to Equip.CTL.Config (addition of Disabled Equipment support)
DN006 - support multiple Offline Lines

8

Local Copy of Equip Array Tag for LNi, EQi Equipment Configured (by Server or Driver)
Equip.CTL.Config

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Overview Line/Pitch/Equipment Display
JSR
Routine Name _731_Oview

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Overview Offline Group/Equipment Display
JSR
Routine Name _732_OviewOffline

Equipment Loop for EQi

Overview Line Popup DIsplay for Equipment Fault/Alarm
Overview Offline Group Popup DIsplay for Equipment Fault/Alarm
(All Offline Groups use Offline PopupRotate for enable - reset on last EQi.)

Rotate between each active Fault/Alarm and Line Display
DN006 - support multiple Offline Lines

9

HMI Control Line Overview Line Popup - Rotate Time
HmiCtlLine.PopupRotate

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline
/

Overview Line Popup Display for Equipment Fault/Alarm

JSR
Routine Name _733_OviewPopup

Local Copy of Line Array Tag for LNi Line is for Offline Equipment
LineCfg.Offline

Overview Offline Popup Display for Equipment Fault/Alarm

FOR
Routine Name _734_OviewOfflinePopup
Index GRPi
4
Initial Value EDC_HMI.CFG.GRP.si
Terminal Value EDC_HMI.CFG.GRP.ei
Step Size 1

Equipment array Index

EQ
Source A EQi
26
Source B LineCfg.EQ.ei
21

HMI Control Line Overview Line Popup - Rotate Time
HmiCtlLine.PopupRotate
U

Equipment Loop for EQi

Update Overview Fault/Status Events
Table of Current Events, Start Time, Duration

Update if Equipment Events have changed in new Callout Message
and Periodically for Configuration changes to Display
DHF006 - changed Equip.CFG.EN to Equip.CTL.Config (addition of Disabled Equipment support)

10

Local Copy of Equip Array Tag for LNi, EQi Equipment Configured (by Server or Driver)
Equip.CTL.Config

Local Copy of Equip Array Tag for LNi, EQi Update Events on HMI Overview Display
Equip.EventCTL.UpdateOview

HMI Overview Event Update Timer
EDC_HMI.CTL.OviewEventTMR.DN

Update Event Table for HMI Overview Display

JSR	
Routine Name	_735_OviewEvent

Local Copy of Equip Array Tag for LNi, EQi Update Events on HMI Overview Display
Equip.EventCTL.UpdateOview
U

Overview Status
(no code required - reads directly from TM-written tags)

11

AFI

Update Status Block for HMI Overview Display

JSR	
Routine Name	_736_OviewStatus

Zone Loop

One set of Interface Routines is shared by all Zones
LNi, EQi, Zni = Line, Equipment, Zone Index

The Zone Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Zone to be scanned continuously.
(WARNING: Disabling the Zone FOR Loop will cause the unscanned Zones to stop updating)

12

Local Copy of Equip Array Tag for LNi, EQi Equipment Configured (by Server or Driver)
Equip.CTL.Config

Zone Loop for ZNi

FOR	
Routine Name	_740_ZoneLoop
Index	ZNi
	21
Initial Value	EDC_HMI.CFG.ZN.si
Terminal Value	EDC_HMI.CFG.ZN.ei
Step Size	1

Equipment Loop for EQi

Copy Working Tags to Equipment Arrayed Data
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

13

Equipment Data Collection Lines and Equipment Array

COP
Source Equip.EventCTL
Dest EDC.Line[LNi].Equip[EQi].EventCTL
Length 1

DHF028 Update Status Table DIsplays
Add Equpment to Status Table if configured

Local Copy of Equip Array Tag for LNi, EQi Equipment Configured (by Server or Driver)
Equip.CTL.Config

14

StatusTable Displays

JSR
Routine Name _750_StatusTable

Exit Routine

lbl_Exit
LBL

15

NOP

Check Viewing Index Limits

16

Equipment View Tag Index

LT
Source A _ViewEQi
0
Source B 0

Equipment View Tag Index

GT
Source A _ViewEQi
0
Source B CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

Equipment View Tag Index

CLR
Dest _ViewEQi
0

Equipment Loop for EQi

Copy Equipment Instance to Viewing Tag if selected
(_ViewEquip can be used to multiplex Trends and Diagnostic Spreadsheets)

17

_ViewEN

Line array Index

EQ	
Source A	LNi
	4
Source B	_ViewLNi
	0

Equipment array Index

EQ	
Source A	EQi
	26
Source B	_ViewEQi
	0

Equipment View Tag (used by View Diagnostic Spreadsheet)

COP	
Source	EDC.Line[LNi].Equip[EQi]
Dest	_ViewEquip
Length	1

(End)

Overview Line/Pitch/Equipment Display

Overview Line/Pitch/Equipment Display

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 731
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Overview Line/Pitch/Equipment Display

If Online, Verify Pitch and Position Index
or Hardcoded EQ Index (AVI, Andon, Conv, Ent, Exit)

Offline if LNi = 0; Online if LNi <>0

2

Local Copy of Equip Array Tag for LNi, EQi Pitch ID (0-350)

LIMIT
Low Limit: EDC_HMI.CFG.PIT.si 1
Test: Equip.CFG.Pitch 15
High Limit: EDC_HMI.CFG.PIT.ei 40

Local Copy of Equip Array Tag for LNi, EQi Position ID (0-5)

LIMIT
Low Limit: 0
Test: Equip.CFG.Position 2
High Limit: 5

Pitch Index

MOVE
Source: Equip.CFG.Pitch 15
Dest: PITi 23

MOVE
Source: Equip.CFG.Position 2
Dest: POSi 1

IndexOk

EQi = 0-4 AVI, Andon, Conv, Ent, Exit shown in Common Line loccations

CommonLineEquip

Pitch Index

CLR
Dest: PITi 23

CLR
Dest: POSi 1

IndexOk
/

Diagnostic

MOVE
Source: Equip.CFG.Pitch 15
Dest: BadPITi 0

Diagnostic

MOVE
Source: Equip.CFG.Position 2
Dest: BadPOSi 0

lbl_Exit
JMP

Overview Line/Pitch/Equipment Display

Load Equipment Display to Temp
DHF006 - add Equip.CTL.Enable for State 0 (addition of Disabled Equipment support)
DHF029 - Added Alert Status 5

3

HMI Overview Equip Data Equipment Configured TempOviewPos.Visible — L

HMI Overview Equip Data Equipment Short Name
COP
Source Equip.CFG.ShortName
Dest TempOviewPos.Name
Length 1

HMI Overview Equip Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)
CLR
Dest TempOviewPos.Status 1

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled) Equip.CTL.Enable — /

HMI Overview Equip Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)
MOVE
Source 0
Dest TempOviewPos.Status 1

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled) Equip.CTL.Enable

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue) Equip.DataCTL.Alert — /

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow) Equip.DataCTL.Alarm — /

Local Copy of Equip Array Tag for LNi, EQi Fault Condition (Red) Equip.DataCTL.Fault — /

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange) Equip.DataCTL.InterlockOff — /

HMI Overview Equip Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)
MOVE
Source 1
Dest TempOviewPos.Status 1

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue) Equip.DataCTL.Alert

HMI Overview Equip Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)
MOVE
Source 5
Dest TempOviewPos.Status 1

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow) Equip.DataCTL.Alarm

HMI Overview Equip Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)
MOVE
Source 2
Dest TempOviewPos.Status 1

Local Copy of Equip Array Tag for LNi

HMI Overview Equip Data Equipment Status (0-Gray

Overview Line/Pitch/Equipment Display

Array Tag for LNi,
EQi Fault Condition
(Red)
Equip.DataCTL.Fault

Status (0-Gray,
1-Green, 2-Yellow,
3-Red, 4-Orange)

MOVE
Source 3
Dest TempOviewPos.Status[1]

Local Copy of Equip
Array Tag for LNi,
EQi Interlock Off
Condition (Orange)
Equip.DataCTL.InterlockOff

HMI Overview
Equip Data Equipment
Status (0-Gray,
1-Green, 2-Yellow,
3-Red, 4-Orange)

MOVE
Source 4
Dest TempOviewPos.Status[1]

Overview Line/Pitch/Equipment Display

If Online, Load Temp to pre-assigned Positions
AVI, Andon, Conv, Ent, Exit

Offline if LNi = 0; Online if LNi <>0
v3.5 replaced hard-coded common equipment EQID with configuration

4

EQi = 0-4
AVI, Andon, Conv, Ent, Exit shown in Common Line loccations
CommonLineEquip

Equipment array Index
EQ
Source A EQi 26
Source B EDC_CfgGateway.HmiComAviEQi 9

HMI Overview Line Local Tag AVI Equipment
COP
Source TempOviewPos
Dest HmiOviewLine.AVI
Length 1

lbl_Exit
JMP

Equipment array Index
EQ
Source A EQi 26
Source B EDC_CfgGateway.HmiComAndonEQi 1

HMI Overview Line Local Tag Andon Equipment
COP
Source TempOviewPos
Dest HmiOviewLine.Andon
Length 1

Equipment array Index
EQ
Source A EQi 26
Source B EDC_CfgGateway.HmiComConvEQi 2

HMI Overview Line Local Tag Conveyor Equipment
COP
Source TempOviewPos
Dest HmiOviewLine.Conv
Length 1

Equipment array Index
EQ
Source A EQi 26
Source B EDC_CfgGateway.HmiComEntEQi 3

HMI Overview Line Local Tag Entrance Equipment
COP
Source TempOviewPos
Dest HmiOviewLine.Ent
Length 1

Equipment array Index
EQ
Source A EQi 26
Source B EDC_CfgGateway.HmiComExitEQi 4

HMI Overview Line Local Tag Exit Equipment
COP
Source TempOviewPos
Dest HmiOviewLine.Exit
Length 1

Overview Line/Pitch/Equipment Display

v3.5 add Position 0-5
(from SQL CfgPosition.PositionID)

Support multiple Equip in one Position by loading worst-case status with AOI (Fault-Alarm-Alert-Bypass-Running-Disable)
Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

5

Determines
worst-case Pitch
status for multiple
Equipment

aoiEDC_HmiPitchStatus

aoiEDC_HmiPitchStatu...	aoiEDC_HmiPitchStatus ...
TempPos	TempOviewPos
HmiPos	HmiOviewLine.Pitch[PITi].Pos[POSi]

Overview Line/Pitch/Equipment Display

If Online, Load Temp to Pitch Positions
East, West, Middle

Offline if LNi = 0; Online if LNi <>0

Support multiple Equip in one block by loading worst-case status with AOI (Fault-Alarm-Alert-Bypass-Running-Disable)
Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

JMP to desired Position Update
faster to skip over AOI than solve false
(fwiw - did not see HMI scan time improvement with AOI skips)

6

Local Copy of Equip Array Tag for LNi, EQi Position ID (0-5)

LE
Source A Equip.CFG.Position 2
Source B 1

lbl_East JMP

Local Copy of Equip Array Tag for LNi, EQi Position ID (0-5)

EQ
Source A Equip.CFG.Position 2
Source B 2

lbl_West JMP

Local Copy of Equip Array Tag for LNi, EQi Position ID (0-5)

GE
Source A Equip.CFG.Position 2
Source B 3

lbl_Mid JMP

Legacy Position - East
aka Position 1

7

lbl_East LBL

Determines worst-case Pitch status for multiple Equipment

aoiEDC_HmiPitchStatus
aoiEDC_HmiPitchStatu... aoiEDC_HmiPitchStatus
TempPos TempOviewPos
HmiPos HmiOviewLine.Pitch[PITi].East

lbl_Exit JMP

Overview Line/Pitch/Equipment Display

Legacy Position - West
aka Position 2

Determines
worst-case Pitch
status for multiple
Equipment

8

lbl_West
LBL

aoiEDC_HmiPitchStatus
aoiEDC_HmiPitchStatu... aoiEDC_HmiPitchStatus ...
TempPos TempOviewPos
HmiPos HmiOviewLine.Pitch[PITi].West

lbl_Exit
JMP

Legacy Position - Middle
combined Positions 3-5

Determines
worst-case Pitch
status for multiple
Equipment

9

lbl_MId
LBL

aoiEDC_HmiPitchStatus
aoiEDC_HmiPitchStatu... aoiEDC_HmiPitchStatus ...
TempPos TempOviewPos
HmiPos HmiOviewLine.Pitch[PITi].Middle

Exit Routine

10

lbl_Exit
LBL

NOP

(End)

Overview Offline Group/Equipment Display

Overview Offline Group/Equipment Display

One set of Interfaces is shared by all Equipment
LNi=0, EQi = Offline, Equipment Index
GRPi = Group Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 732
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Overview Offline Group/Equipment Display

Load Equipment Display to Temp
DHF006 - add Equip.CTL.Enable for State 0 (addition of Disabled Equipment support)
DHF029 - Added Alert Status 5

2

HMI Overview Equip Data Equipment Configured TempOviewPos.Visible

L

HMI Overview Equip Data Equipment Short Name

COP
Source Equip.CFG.ShortName
Dest TempOviewPos.Name
Length 1

HMI Overview Equip Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

CLR
Dest TempOviewPos.Status
1

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled) Equip.CTL.Enable

/

HMI Overview Equip Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 0
Dest TempOviewPos.Status
1

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled) Equip.CTL.Enable

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue) Equip.DataCTL.Alert

/

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow) Equip.DataCTL.Alarm

/

Local Copy of Equip Array Tag for LNi, EQi Fault Condition (Red) Equip.DataCTL.Fault

/

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange) Equip.DataCTL.InterlockOff

/

HMI Overview Equip Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 1
Dest TempOviewPos.Status
1

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue) Equip.DataCTL.Alert

HMI Overview Equip Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 5
Dest TempOviewPos.Status
1

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow) Equip.DataCTL.Alarm

HMI Overview Equip Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 2
Dest TempOviewPos.Status
1

Local Copy of Equip Array Tag for LNi

HMI Overview Equip Data Equipment Status (0-Gray

Overview Offline Group/Equipment Display

Array Tag for LNi,
EQi Fault Condition
(Red)
Equip.DataCTL.Fault

Status (0-Gray,
1-Green, 2-Yellow,
3-Red, 4-Orange)

MOVE	
Source	3
Dest	TempOviewPos.Status[1]

Local Copy of Equip
Array Tag for LNi,
EQi Interlock Off
Condition (Orange)
Equip.DataCTL.InterlockOff

HMI Overview
Equip Data Equipment
Status (0-Gray,
1-Green, 2-Yellow,
3-Red, 4-Orange)

MOVE	
Source	4
Dest	TempOviewPos.Status[1]

If Offline, load Group Index
Offline Equipment can be configured to be in any Group for HMI Display
DN006 - support multiple Offline Lines

3

Offline Equipment
Group Index

MOVE	
Source	Equip.CFGoffline.Group[-1]
Dest	EquipGRPi[2]

Offline Equipment
Group Index

LT	
Source A	EquipGRPi[2]
Source B	0

Offline Equipment
Group Index

GT	
Source A	EquipGRPi[2]
Source B	CFG.GRP.ei <EDC.CFG.GRP.ei>[3]

Offline Equipment
Group Index

CLR	
Dest	EquipGRPi[2]

Overview Offline Group/Equipment Display

If Offline Load Temp to Group Row
Group Row determined by Equipment Row, Col configuration
DHF007 - Increased Row count from 6 to 10
DN006 - support multiple Offline Lines, Row count from 10 to 12

4

Offline Group Display Index

CPT
Dest GrpRi 34
Expression (Equip.CFGoffline.DisplayCol-1)*12+(Equip.CFGoffline.DisplayRow-1)

Offline Group Display Index

LT
Source A GrpRi 34
Source B 0

Offline Group Display Index

MOVE
Source 0
Dest GrpRi 34

Offline Group Display Index

GT
Source A GrpRi 34
Source B EDC_HMI.CFG.GrpR.mi 99

Offline Group Display Index

MOVE
Source EDC_HMI.CFG.GrpR.mi 99
Dest GrpRi 34

HMI Overview OffLine Data HMI Overview Equip Data

COP
Source TempOviewPos
Dest HmiOviewOfliine[EquipGRPi].Equip[GrpRi]
Length 1

(End)

Overview Line Popup Display for Equipment Fault/Alarm

Overview Line Popup Display for Equipment Fault/Alarm

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 733
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Check for Next Equipment to Display
v1.4 Block Popup if No Display is on
DHF006 - block popup if Disabled (addition of Disabled Equipment support)

2

Local Copy of Equip Array Tag for LNi, EQi Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Local Copy of Equip Array Tag for LNi, EQi Display No Events
Equip.EventCTL.DispNone

Local Copy of Equip Array Tag for LNi, EQi Display Interlock Off Event Only
Equip.EventCTL.DispInterlockOff

Equipment array Index

GT
Source A EQi 26
Source B HmiCtlLine.PopupCurrEQi 0

Rotate to Next Popup
LinePopupRotateNext

Overview Line Popup Display for Equipment Fault/Alarm

3

Load Next Equipment to Display
convert Duration from msec to minutes
DHF029 - Added Alert Status 5 (remove AFI to enable)

Rotate to Next Popup
LinePopupRotateNext

AFI

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue)
Equip.DataCTL.Alert

HMI Overview Line Local Tag Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 5
Dest HmiOviewLine.Popup.Status 0

HMI Overview Line Local Tag Equipment Status Duration (minutes)

DIV
Source A Equip.Data.Alert.Time 0
Source B 60000
Dest HmiOviewLine.Popup.Duration 0.0

Rotate to Next Popup Load
LinePopupRotateLoad

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow)
Equip.DataCTL.Alarm

HMI Overview Line Local Tag Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 2
Dest HmiOviewLine.Popup.Status 0

HMI Overview Line Local Tag Equipment Status Duration (minutes)

DIV
Source A Equip.Data.Alarm.Time 0
Source B 60000
Dest HmiOviewLine.Popup.Duration 0.0

Local Copy of Equip Array Tag for LNi, EQi Fault Condition (Red)
Equip.DataCTL.Fault

HMI Overview Line Local Tag Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 3
Dest HmiOviewLine.Popup.Status 0

HMI Overview Line Local Tag Equipment Status Duration (minutes)

DIV
Source A Equip.Data.Fault.Time 0
Source B 60000
Dest HmiOviewLine.Popup.Duration 0.0

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange)
Equip.DataCTL.InterlockOff

HMI Overview Line Local Tag Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 4
Dest HmiOviewLine.Popup.Status 0

HMI Overview Line Local Tag Equipment Status Duration (minutes)

DIV
Source A Equip.Data.InterlockOff.Time 0
Source B 60000
Dest HmiOviewLine.Popup.Duration 0.0

Overview Line Popup Display for Equipment Fault/Alarm

Load Next Equipment to Display
DHF029 - Added Alert Status 5 (remove AFI to enable)

4

Rotate to Next Popup Load
LinePopupRotateLoad

HMI Overview Line Local Tag Equipment Data Loaded
HmiOviewLine.Popup.Visible
L

HMI Overview Line Local Tag Equipment Name
COP
Source Equip.CFG.Name
Dest HmiOviewLine.Popup.Name
Length 1

HMI Overview Line Local Tag Equipment ID (LNi/EQi)
CPT
Dest HmiOviewLine.Popup.EQID
0
Expression (LNi * 100) + EQi

HMI Control Line Overview Line Popup - Current Equipment Display
MOVE
Source EQi
26
Dest HmiCtlLine.PopupCurrEQi
0

HMI Control Line Overview Line Popup - Rotate Time
HmiCtlLine.PopupRotate
U

Display Line if no more Equipment to Display
DHF015 Dec 14, 2017 -- changed No-Popup state from 0 to -1 so EQID xx00 can display on Popup

5

HMI Control Line Overview Line Popup - Rotate Time
HmiCtlLine.PopupRotate

Equipment array Index
GE
Source A EQi
26
Source B LineCfg.EQ.ei
21

HMI Overview Line Local Tag Equipment Fault/Alarm Popup Display
FLL
Source 0
Dest HmiOviewLine.Popup
Length 1

HMI Control Line Overview Line Popup - Current Equipment Display
MOVE
Source -1
Dest HmiCtlLine.PopupCurrEQi
0

HMI Control Line Overview Line Popup - Rotate Time
HmiCtlLine.PopupRotate
U

(End)

Overview Offline Popup Display for Equipment Fault/Alarm

Initialize, Reset or Update Offline Group Operational Controls

One set of Interfaces is shared by all Group
LNi=0, EQi = Offline, Equipment Index
GRPi = Offline Group Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 734
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Exit Routine if not ready for next Popup Rotate

2

HMI HMI Offline Display Control
EDC_HMI.CTL.Offline[GRPi].PopupRotate
/

lbl_Exit
JMP

Copy Offline Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

3

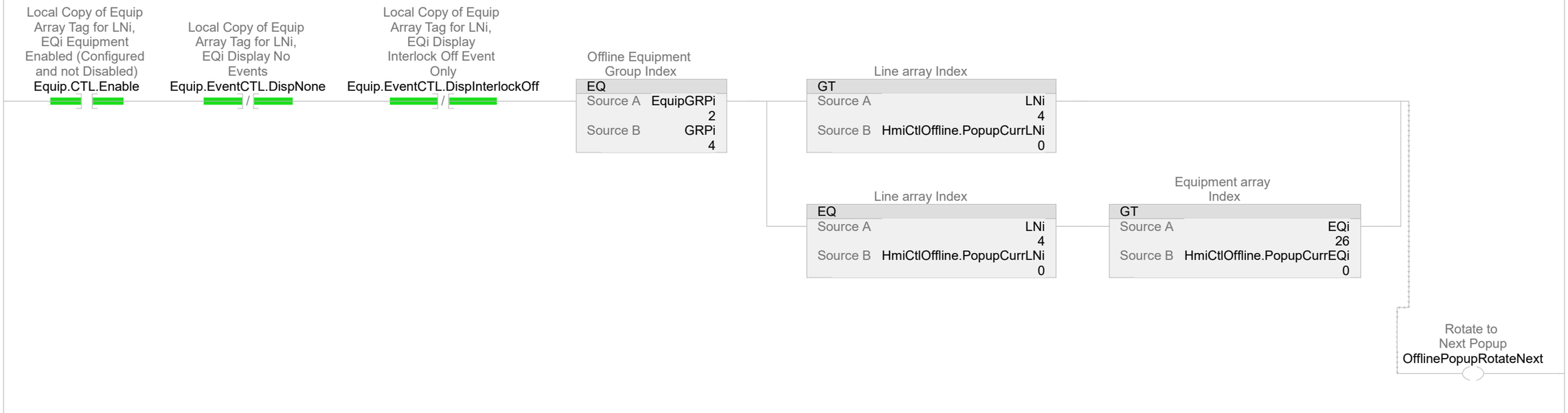
HMI Control Line

COP
Source EDC_HMI.CTL.Offline[GRPi]
Dest HmiCtlOffline
Length 1

Overview Offline Popup Display for Equipment Fault/Alarm

Check for Next Equipment to Display
v1.4 Block Popup if No Display is on
DHF006 - block popup if Disabled (addition of Disabled Equipment support)
DN006 - support multiple Offline Lines

4



Overview Offline Popup Display for Equipment Fault/Alarm

Load Next Equipment to Display
convert Duration from msec to minutes
DHF029 - Added Alert Status 5 (remove AFI to enable)
DN006 - support multiple Offline Lines

5

Rotate to Next Popup
OfflinePopupRotateNext

AFI

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue)
Equip.DataCTL.Alert

HMI Overview OffLine Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 5
Dest HmiOviewOfliine[GRPi].Popup.Status
??

HMI Overview OffLine Data Equipment Status Duration (minutes)

DIV
Source A Equip.Data.Alert.Time
0
Source B 60000
Dest HmiOviewOfliine[GRPi].Popup.Duration
??

Rotate to Next Popup Load
OfflinePopupRotateLoad

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow)
Equip.DataCTL.Alarm

HMI Overview OffLine Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 2
Dest HmiOviewOfliine[GRPi].Popup.Status
??

HMI Overview OffLine Data Equipment Status Duration (minutes)

DIV
Source A Equip.Data.Alarm.Time
0
Source B 60000
Dest HmiOviewOfliine[GRPi].Popup.Duration
??

Local Copy of Equip Array Tag for LNi, EQi Fault Condition (Red)
Equip.DataCTL.Fault

HMI Overview OffLine Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 3
Dest HmiOviewOfliine[GRPi].Popup.Status
??

HMI Overview OffLine Data Equipment Status Duration (minutes)

DIV
Source A Equip.Data.Fault.Time
0
Source B 60000
Dest HmiOviewOfliine[GRPi].Popup.Duration
??

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange)
Equip.DataCTL.InterlockOff

HMI Overview OffLine Data Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source 4
Dest HmiOviewOfliine[GRPi].Popup.Status
??

HMI Overview OffLine Data Equipment Status Duration (minutes)

DIV
Source A Equip.Data.InterlockOff.Time
0
Source B 60000
Dest HmiOviewOfliine[GRPi].Popup.Duration
??

Overview Offline Popup Display for Equipment Fault/Alarm

6

Load Next Equipment to Display
DN006 - support multiple Offline Lines

Rotate to Next Popup Load: OfflinePopupRotateLoad

HMI Overview OffLine Data Equipment Data Loaded: HmiOviewOfliine[GRPi].Popup.Visible (L)

COP
Source: Equip.CFG.Name
Dest: HmiOviewOfliine[GRPi].Popup.Name
Length: 1

HMI Overview OffLine Data Equipment Name

CPT
Dest: HmiOviewOfliine[GRPi].Popup.EQID
??
Expression: (LNi * 100) + EQi

HMI Overview OffLine Data Equipment ID (LNi/EQi)

MOVE
Source: LNi
4
Dest: HmiCtlOffline.PopupCurrLNi
0

HMI Control Line Overview Line Popup - Current Line Display

MOVE
Source: EQi
26
Dest: HmiCtlOffline.PopupCurrEQi
0

HMI Control Line Overview Line Popup - Current Equipment Display

HMI Control Line Overview Line Popup - Rotate Time: HmiCtlOffline.PopupRotate (U)

7

Hide Popup Banner and Display Group if no more Equipment to Display
DHF015 Dec 14, 2017 -- changed No-Popup state from 0 to -1 so EQID xx00 can display on Popup

HMI Control Line Overview Line Popup - Rotate Time: HmiCtlOffline.PopupRotate

Finalize Offline Groups (Last Offline Line): OfflineLastLine

Equipment array Index

GE
Source A: EQi
26
Source B: LineCfg.EQ.ei
21

FLL
Source: 0
Dest: HmiOviewOfliine[GRPi].Popup
Length: 1

HMI Overview OffLine Data Equipment Fault/Alarm Popup Display

MOVE
Source: -1
Dest: HmiCtlOffline.PopupCurrLNi
0

HMI Control Line Overview Line Popup - Current Line Display

MOVE
Source: -1
Dest: HmiCtlOffline.PopupCurrEQi
0

HMI Control Line Overview Line Popup - Current Equipment Display

HMI Control Line Overview Line Popup - Rotate Time: HmiCtlOffline.PopupRotate (U)

Overview Offline Popup Display for Equipment Fault/Alarm

Copy Offline working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

8

HMI HMI Offline
Display Control

COP	
Source	HmiCtlOffline
Dest	EDC_HMI.CTL.Offline[GRPi]
Length	1

Exit Routine

9

lbl_Exit
LBL

NOP

(End)

Update Event Table for HMI Overview Display

Update Overview Display Fault/Status Events
Table of Current Events, Start Time, Duration

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 735
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

2

Local Copy of Equip Array Tag for LNi, EQi Maintenance Rank Assignments (A-D = Values 1-4)

LT
Source A Equip.CFG.RankChar 2
Source B EDC_HMI.CFG.Oview.RankChar 4

Local Copy of Equip Array Tag for LNi, EQi Maintenance Rank Assignments (A-D = Values 1-4)

EQ
Source A Equip.CFG.RankChar 2
Source B EDC_HMI.CFG.Oview.RankChar 4

Local Copy of Equip Array Tag for LNi, EQi Maintenance Risk Rank Assignments (4-1)

GE
Source A Equip.CFG.RankNum 4
Source B EDC_HMI.CFG.Oview.RankNum 1

Equipment Rank >= ZNi Rank Filter
RankMatch

Update Event Table for HMI Overview Display

Check Equipment Rank for match with Event Table Rank Configuration
Overview Event Table can be filtered by Equipment Rank

3

Equipment Rank >=
ZNi Rank Filter
RankMatch

Equipment Zone/Rank
in Display Config
DisplayMatch

Add/Remove Events for this Equipment
Copy Overview Event Table to/rfrom Subroutine Tag

4

HMI Event Table Row

COP	
Source	EDC_HMI.Oview.Event[0]
Dest	HmiEvent[0]
Length	EDC_HMI.CFG.EV.n

Update Event Table
for HMI Display

JSR	
Routine Name	_795_sbrEvent

HMI HMI Event Table
Row

COP	
Source	HmiEvent[0]
Dest	EDC_HMI.Oview.Event[0]
Length	EDC_HMI.CFG.EV.n

(End)

Update Status Block for HMI Overview Display

Update Status Block for HMI Overview Display
Table of Last Process Times, Backlog Times/Counts, Total Records and Errors.
- Fault Status
- Fault Ring Buffer
- Quality Ring Buffer
- Trend Ring Buffer
- Summary Ring Buffer
- CIS Messages

ServerInterface Program loads data values from TM Transaction to EDC_HMI tag and updates Status colors
(more efficient to upate colors once when data loaded)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 736
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

(End)

Zone Loop for ZNi

Zone Loop

One set of Interface Routines is shared by all Zones
LNi, EQi, Zni = Line, Equipment, Zone Index

The Zone Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Zone to be scanned continuously.
(WARNING: Disabling the Zone FOR Loop will cause the unscanned Zones to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 740
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Check Equipment Zone Assignments for match with Table Zone Assignments
Exit if not matched and Zone Update not required

2

Check if Equipment Zone is one of the Zones configured for this Display Item

aoiEDC_HmiZoneCheck
aoiEDC_HmiZoneCheck aoiEDC_HmiZoneCheck ...
EquipZone Equip.CFG.Zone
DispZone EDC_HMI.CFG.Zone[ZNi].ZoneMask
ZoneMatch ZoneMatch
0

Equipment in ZNi Zone Filter
ZoneMatch
/

Refresh Zone Events
RefreshZN.[ZNi]
/

lbl_Exit
JMP

Zone Loop for ZNi

Check Equipment Rank Assignments for match with Zone Table Rank Assignments
Each Zone Table can display any combination of Zone and Rank
Exit if not matched and Zone Update not required

3

Local Copy of Equip Array Tag for LNi, EQi Maintenance Rank Assignments (A-D = Values 1-4)

LT
Source A Equip.CFG.RankChar[2]
Source B EDC_HMI.CFG.Zone[ZNi].RankChar[??]

Equipment Rank >= ZNi Rank Filter
RankMatch

Local Copy of Equip Array Tag for LNi, EQi Maintenance Rank Assignments (A-D = Values 1-4)

EQ
Source A Equip.CFG.RankChar[2]
Source B EDC_HMI.CFG.Zone[ZNi].RankChar[??]

Local Copy of Equip Array Tag for LNi, EQi Maintenance Risk Rank Assignments (4-1)

GE
Source A Equip.CFG.RankNum[4]
Source B EDC_HMI.CFG.Zone[ZNi].RankNum[??]

Equipment Rank >= ZNi Rank Filter
RankMatch
/

Refresh Zone Events
RefreshZN.[ZNi]
/

lbl_Exit
JMP

Update Zone Table DIsplay for Selected Zone
Add Equpment to Zone Table if configured for this Zone

4

Equipment in ZNi Zone Filter
ZoneMatch

Equipment Rank >= ZNi Rank Filter
RankMatch

Zone Table Display

JSR
Routine Name _741_ZoneTable

Zone Loop for ZNi

Update Zone Display Fault/Status Events
Table of Current Events, Start Time, Duration

Update if Equipment Events have changed in new Callout Message
and Period Refresh to cleanup after Configuration changes to Display

5

Local Copy of Equip Array Tag for LNi, EQi Update Events on HMI Zone Displays
Equip.EventCTL.UpdateZone.[ZNi]

Refresh Zone Events
RefreshZN.[ZNi]

Update Equip Event Tables for HMI Zone Display
JSR
Routine Name _745_ZoneEvent

Local Copy of Equip Array Tag for LNi, EQi Update Events on HMI Zone Displays
Equip.EventCTL.UpdateZone.[ZNi]
U

Exit Routine

6

lbl_Exit
LBL

NOP

(End)

Zone Table Display

Zone Table Display
Equipment are shown in each Zone Table based on Equipment and Table configuration.

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index
ZNi = Zone Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 741
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Copy Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

2

Zone Table Row Index

MOVE
Source EDC_HMI.Zone[ZNi].ZnRi ??
Dest ZnRi 99

Zone Table Display

Load Zone Table Row with Equipment Data

3

HMI Zone Table Row Equipment Data Loaded
HmiZoneRow.Visible

L

HMI Zone Table Row Equipment ID (LNi/EQi)

CPT
Dest HmiZoneRow.EQID 1709
Expression LNi*100+EQi

HMI Zone Table Row Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source TempOviewPos.Status 1
Dest HmiZoneRow.Status 1

Local Copy of Line Array Tag for LNi Line Short Name

FLL
Source 0
Dest TempString
Length 1

CONCAT
Source A LineCfg.ShortName 'Eng'
Source B strHyphen '-'
Dest TempString 'IP-'

CONCAT
Source A TempString 'IP-'
Source B Equip.CFG.Name 'TA/TF 2026'
Dest HmiZoneRow.Name 'IP-IP AVI'

Load Zone Table Row with Equipment Data
convert Duration from msec to minutes

4

HMI Zone Table Row Equipment Status Maintenance Duration (minutes)

DIV
Source A Equip.Data.EdtOA.Maint.Time 0
Source B 60000
Dest HmiZoneRow.MaintDur 0.1441

HMI Zone Table Row Equipment Status Production Duration (minutes)

DIV
Source A Equip.Data.EdtOA.Prod.Time 607366
Source B 60000
Dest HmiZoneRow.ProdDur 12.245684

DHF021 Load Zone Table Row with Equipment PreDefined Faults for Status Display

5

HMI Zone Table Row Equipment Pre-Defined Faults

COP
Source EDC_EqRxData.Line[LNi].Equip[EQi].Callout.PreDef
Dest HmiZoneRow.PreDef
Length 1

Zone Table Display

Copy Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

6

HMI Maintenance Zone Displays

COP	
Source	HmiZoneRow
Dest	EDC_HMI.Zone[ZNi].RowTemp[ZnRi]
Length	1

Increment Zone Table Row Index

7

Zone Table Row Index

ADD	
Source A	ZnRi 99
Source B	1
Dest	ZnRi 99

Zone Table Row Index

LT	
Source A	ZnRi 99
Source B	EDC_HMI.CFG.ZnR.si 0

Zone Table Row Index

MOVE	
Source	EDC_HMI.CFG.ZnR.si 0
Dest	ZnRi 99

Zone Table Row Index

GT	
Source A	ZnRi 99
Source B	EDC_HMI.CFG.ZnR.ei 99

Zone Table Row Index

MOVE	
Source	EDC_HMI.CFG.ZnR.ei 99
Dest	ZnRi 99

Copy Working Tags to Equipment Arrayed Data
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

8

HMI Maintenance Zone Displays

MOVE	
Source	ZnRi 99
Dest	EDC_HMI.Zone[ZNi].ZnRi ??

(End)

Update Equip Event Tables for HMI Zone Display

Update Zone Display Fault/Status Events
Table of Current Events, Start Time, Duration

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index
ZNi = Zone Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 745
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Check Equipment Zone and Rank Assignments for match with Event Table Zone & Rank Assignments
Each Zone Event Table can be filtered by Equipment Zone and Equipment Rank

2

Equipment in ZNi Zone Filter
ZoneMatch

Equipment Rank >= ZNi Rank Filter
RankMatch

Equipment Zone/Rank in Display Config
DisplayMatch

Add/Remove Events for this Equipment
Copy Overview Event Table to/rfrom Subroutine Tag

3

HMI Event Table Row

COP
Source EDC_HMI.Zone[ZNi].Event[0]
Dest HmiEvent[0]
Length EDC_HMI.CFG.EV.n

Update Event Table for HMI Display

JSR
Routine Name _795_sbrEvent

HMI Maintenance Zone Displays

COP
Source HmiEvent[0]
Dest EDC_HMI.Zone[ZNi].Event[0]
Length EDC_HMI.CFG.EV.n

(End)

StatusTable Displays

DHF028 Status Table Display
Equipment are shown in each Status Table based on Equipment configuration.

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 750
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Load Status Table Row with Equipment Data

2

HMI StatusTable Row Equipment Data Loaded
HmiStatusRow.Visible

HMI StatusTable Row Equipment ID (LNi/EQi)

CPT
Dest HmiStatusRow.EQID 1709
Expression (LNi * 100) + EQi

HMI StatusTable Row Equipment Status (0-Gray, 1-Green, 2-Yellow, 3-Red, 4-Orange)

MOVE
Source TempOviewPos.Status 1
Dest HmiStatusRow.Status 1

FLL
Source 0
Dest TempString
Length 1

Local Copy of Line Array Tag for LNi Line Short Name

CONCAT
Source A LineCfg.ShortName 'Eng'
Source B strHyphen '-'
Dest TempString 'IP-'

CONCAT
Source A TempString 'IP-'
Source B Equip.CFG.Name 'TA/TF 2026'
Dest HmiStatusRow.Name 'IP-IP AVI'

StatusTable Displays

Load Status Table Row with Equipment Data
convert Duration from msec to minutes

3

HMI StatusTable Row Equipment Status Maintenance Duration (minutes)

DIV	
Source A	Equip.Data.EdtOA.Maint.Time 0
Source B	60000
Dest	HmiStatusRow.MaintDur 0.1441

HMI StatusTable Row Equipment Status Production Duration (minutes)

DIV	
Source A	Equip.Data.EdtOA.Prod.Time 607366
Source B	60000
Dest	HmiStatusRow.ProdDur 12.245684

DHF021 Load Status Table Row with Equipment PreDefined Faults for Status Display

4

HMI StatusTable Row Equipment Pre-Defined Faults

COP	
Source	EDC_EqRxData.Line[LNi].Equip[EQi].Callout.PreDef
Dest	HmiStatusRow.PreDef
Length	1

Copy Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

5

HMI Status Tables Display HMI StatusTable Row

COP	
Source	HmiStatusRow
Dest	HmiStatus.Row[StRi]
Length	1

Update First and Last EQID on this Display

6

MOVE	
Source	HmiStatusRow.EQID 1709
Dest	TempDINT 318

DTOS	
Source	TempDINT 318
Dest	HmiStatus.EQIDlast '1521'

Status Table Row Index

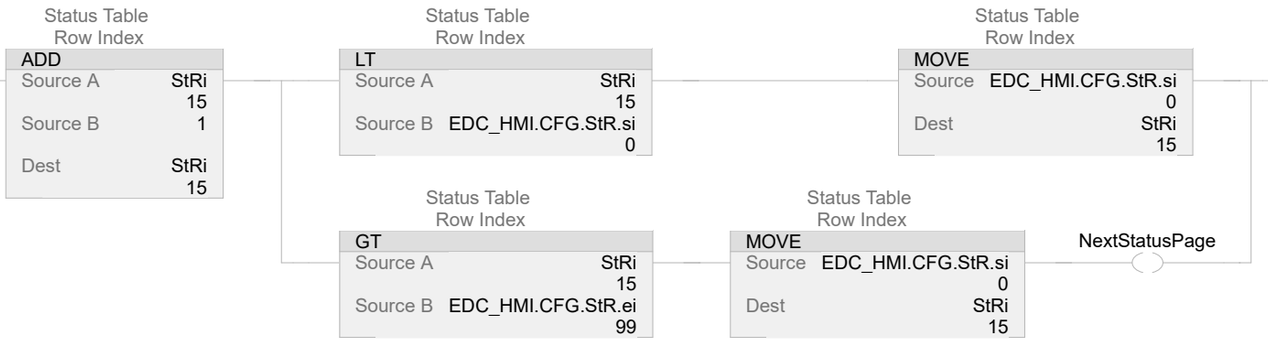
EQ	
Source A	StRi 15
Source B	0

DTOS	
Source	TempDINT 318
Dest	HmiStatus.EQIDfirst '1007'

StatusTable Displays

Increment Status Table Row Index
(100 rows / Table)

7



StatusTable Displays

Update Status Table and initialize for next Status Table
Increment Status Table Index
(initially 3 Tables)

8

NextStatusPage

Status Table Name Index	HMI Equip Status Displays	HMI Status Tables Display	Status Table Row Index
DTOS	COP	FLL	CLR
Source StNi 2	Source HmiStatus	Source 0	Dest StRi 15
Dest HmiStatus.Name "	Dest EDC_HMI.Status[STi]	Dest HmiStatus	
	Length 1	Length 1	

Status Table Index	Status Table Index	Status Table Index
ADD	LT	MOVE
Source A STi 1	Source A STi 1	Source EDC_HMI.CFG.ST.si 0
Source B 1	Source B EDC_HMI.CFG.ST.si 0	Dest STi 1
Dest STi 1		

Status Table Index	Status Table Index
GT	MOVE
Source A STi 1	Source EDC_HMI.CFG.StR.ei 99
Source B EDC_HMI.CFG.ST.ei 4	Dest STi 1

Status Table Name Index
ADD
Source A STi 1
Source B 1
Dest StNi 2

Exit Routine

9

lbl_Exit

LBL

NOP

(End)

Update Event Table for HMI Display

Update Display Fault/Status Events
Table of Current Events, Start Time, Duration

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 795
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

HMI HIgh Scan Timer
Count _795_sbrEvent executions

2

HMI High Scan _795_sbrEvent Execution Count

ADD
Source A HMI_HighScan795SbrCount 0
Source B 1
Dest HMI_HighScan795SbrCount 0

Check the current Events and remove any that have turned off.
By adding/removed Events from the current Event list, the oldest events are retained, in order without sorting by Duration

Initialize Event Loop Pointer

3

Event Display Table Index

CLR
Dest EVi 7

Update Event Table for HMI Display

Exit Event Check Loop if Record is Empty

4

lbl_EventRemoveLoop
LBL

HMI Event Table Row Equipment Data Loaded
HmiEvent[EVi].Visible
/

lbl_EventAdd
JMP

Check if Line/Equip Indexes in Range
Remove if out of Range

5

HMI Event Table Row Line Index
LIMIT
Low Limit 0
Test HmiEvent[EVi].LNi 0
High Limit CFG.LN.mi <EDC.CFG.LN.mi> 28

HMI Event Table Row Equipment Index
LIMIT
Low Limit 0
Test HmiEvent[EVi].EQi 0
High Limit CFG.EQ.mi <EDC.CFG.EQ.mi> 99

LNi, EQi Indexes OK
LnEq_IndexOk

LNi, EQi Indexes OK
LnEq_IndexOk
/

lbl_EventRemove
JMP

Check if Equip is Configured -- Remove if no Longer Configured
(need to do this here since Unconfigured EQ will not update its own Events)
DHF006 - changed Equip.CFG.EN to Equip.CTL.Enable (addition of Disabled Equipment support)

6

Event Line Index (0-28)
MOVE
Source HmiEvent[EVi].LNi 0
Dest EvLNi 10

Event Equipment Index (0-99)
MOVE
Source HmiEvent[EVi].EQi 0
Dest EvEQi 4

Equipment Data Collection Lines and Equipment Array
EDC.Line[EvLNi].Equip[EvEQi].CTL.Enable
/

lbl_EventRemove
JMP

Update Event Table for HMI Display

Check if Event is from other Equipment
Increment Loop Index if not this Equipment

7

HMI Event Table Row Line Index

NE
Source A HmiEvent[EVi].LNi 0
Source B LNi 4

HMI Event Table Row Equipment Index

NE
Source A HmiEvent[EVi].EQi 0
Source B EQi 26

IncrementEVi L
lbl_EventNext JMP

Event is from this Equipment
Remove if Equipment not in Rank Filter

8

Equipment Zone/Rank in Display Config
DisplayMatch /

lbl_EventRemove JMP

Event is from this Equipment - check if Event still On
Remove if Event not found in Equipment Event Table
Increment Loop Index if Event is found

9

Equipment Event Search
EquipFSC RES

Equipment Event Search

MOVE
Source CFG.EV.n <EDC.CFG.EV.n> 10
Dest EquipFSC.LEN 10

Equipment Event Search

MOVE
Source 0
Dest EquipFSC.POS 0

Equipment Event Search

FSC
Control EquipFSC
Length 10
Position 0
Mode ALL
Expression Equip.Event[EquipFSC.POS].Element=HmiEvent[EVi].Element
EN DN ER

Equipment Event Search
EquipFSC.FD /

lbl_EventRemove JMP

Equipment Event Search
EquipFSC.FD

IncrementEVi L
lbl_EventNext JMP

Update Event Table for HMI Display

Remove Event
If this isn't the last Event in the Table, shift following Events up 1 position before clearing last position

10

lbl_EventRemove — LBL — IncrementEVi — L

Event Display Table Index

LT
Source A: EVi, 7
Source B: EDC_HMI.CFG.EV.mi, 199

MOVE
Source: EVi, 7
Dest: i, 7

Temp Result for Array Size

SUB
Source A: EDC_HMI.CFG.EV.mi, 199
Source B: i, 7
Dest: LEN, 192

HMI Event Table Row

COP
Source: HmiEvent[i+1]
Dest: HmiEvent[i]
Length: LEN

IncrementEVi — U

Remove Event
Clear Last Position in Table
Set cleared Element to -1 to differentiate from valid Element 0 in searches

11

HMI Event Table Row

FLL
Source: 0
Dest: HmiEvent[EDC_HMI.CFG.EV.mi]
Length: 1

HMI Event Table Row Fault/Status Element (Word*100 + Bit)

MOVE
Source: -1
Dest: HmiEvent[EDC_HMI.CFG.EV.mi].Element, -1

Increment Index
Block Increment if Event Deleted - Event Table Shift moves next Event to current Index
If EVi exceeds Max Index, Table is full and no need to check for additional events

12

lbl_EventNext — LBL

IncrementEVi — U

Event Display Table Index

ADD
Source A: EVi, 7
Source B: 1
Dest: EVi, 7

Event Display Table Index

LE
Source A: EVi, 7
Source B: EDC_HMI.CFG.EV.mi, 199

lbl_EventRemoveLoop — JMP

lbl_Exit — JMP

Update Event Table for HMI Display

Add New Events after the Existing Events
When EventRemoveLoop ends, EVi will point to first open position in Event Table

Exit without Adidng Events if Equipment not in Display Filter

13

lbl_EventAdd
LBL

Equipment Zone/Rank
in Display Config
DisplayMatch
/

lbl_Exit
JMP

Initialize Equipment Event Table Index

14

Equipment Event
Index
CLR
Dest EqEVi
0

Equipment Event Table
Search for Events - Exit if no Event

15

lbl_EventAddLoop
LBL

Local Copy of Equip
Array Tag for LNi,
EQi Fault/Status
Event Element
(Word*100) + Bit
EQ
Source A Equip.Event[EqEVi].Element
-1
Source B -1

lbl_Exit
JMP

If Equipment has Event, search Event table for Element match

Initialize Search

16

Search HMI Event
Table for this Event
(already there?)
HmiEventFSC
RES

Search HMI Event
Table for this Event
(already there?)
MOVE
Source EDC_HMI.CFG.EV.n
200
Dest HmiEventFSC.LEN
200

Search HMI Event
Table for this Event
(already there?)
MOVE
Source 0
Dest HmiEventFSC.POS
6

Update Event Table for HMI Display

If Equipment has Event, search Event table for Existing Event (Element match)

17

lbl_EventAddExistCheckLoop
LBL

Search HMI Event Table for this Event (already there?)

FSC
Control HmiEventFSC
Length 200
Position 6
Mode ALL
Expression Equip.Event[EqEVi].Element=HmiEvent[HmiEventFSC.POS].Element
EN
DN
ER

If Element match found, check for LNi and EQi match
if no Match and search not done (i.e. same Element but different EQID) - resume Element search
if Match - existing, move to next EQ Element without saving

18

Search HMI Event Table for this Event (already there?)
HmiEventFSC.FD

MOVE
Source HmiEventFSC.POS
6
Dest FdEVi
6

HMI Event Table Row Equipment Data Loaded
HmiEvent[FdEVi].Visible

HMI Event Table Row Line Index

EQ
Source A HmiEvent[FdEVi].LNi
10
Source B LNi
4

HMI Event Table Row Equipment Index

EQ
Source A HmiEvent[FdEVi].EQi
4
Source B EQi
26

Eq Event Found in HMI Event Table (don't need to add)
HmiEventMatchFD

Eq Event Found in HMI Event Table (don't need to add)
HmiEventMatchFD
/

Search HMI Event Table for this Event (already there?)
HmiEventFSC.DN
/

Search HMI Event Table for this Event (already there?)
HmiEventFSC.IN
U

lbl_EventAddExistCheckLoop
JMP

Existing Event Match Found
Jump to Next Equipment Event

19

Eq Event Found in HMI Event Table (don't need to add)
HmiEventMatchFD

lbl_NextEvent
JMP

Update Event Table for HMI Display

New Event - Save Event Details
Set default Status/Duration
Status/Duration update routine will quickly overwrite - no point in doing it here

20

HMI Event Table Row Equipment Data Loaded
HmiEvent[EVi].Visible
(L)

HMI Event Table Row Line Index
MOVE
Source LNi
4
Dest HmiEvent[EVi].LNi
0

HMI Event Table Row Equipment Index
MOVE
Source EQi
26
Dest HmiEvent[EVi].EQi
0

HMI Event Table Row Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)
MOVE
Source 1
Dest HmiEvent[EVi].Status
0

HMI Event Table Row Line Short Name
COP
Source LineCfg.ShortName
Dest HmiEvent[EVi].LineName
Length 1

HMI Event Table Row Equipment Short Name
COP
Source Equip.CFG.ShortName
Dest HmiEvent[EVi].EquipName
Length 1

HMI Event Table Row Fault/Status Element (Word*100 + Bit)
MOVE
Source Equip.Event[EqEVi].Element
-1
Dest HmiEvent[EVi].Element
-1

HMI Event Table Row Transaction Time
COP
Source Equip.Event[EqEVi].TransTime
Dest HmiEvent[EVi].TransTime
Length 1

HMI Event Table Row Equipment Status Duration (minutes)
MOVE
Source 0
Dest HmiEvent[EVi].Duration
0.0

Update Event Table for HMI Display

New Event - Save Event Details
Determine Event Lookup Location for Predefined or Shared
(Otherwise use EQID Search Window)

DHF030 - Added WDi limit to EventTextShared to force search for non-shared Andon/AVI Events
DN006 - support multiple Offline Lines
EDC2 WL09 Event Configuration - remove Shared Text and add Tag04

21

Packet Data
Bit Index

MOD
Source A HmiEvent[EVi].Element
-1
Source B 100
Dest BITi
22

CPT
Dest WDi
0
Expression (HmiEvent[EVi].Element-BITi)/100 - 1

PreDefined
Event
EventPreDef

LT
Source A WDi
0
Source B 0

Update Event Table for HMI Display

New Event - Save Event Details

Load 0-padded Element WWBBinto Alarm Word/Bit template as "WW.BB" or "Pr.BB"

22

PreDefined Event
EventPreDef

HMI Event Table Row Alarm Text
COP
Source EDC_HMI.CFG.EventElement
Dest HmiEvent[EVi].AlarmText
Length 1

PreDefined Event
EventPreDef

DTOS
Source WDi 0
Dest strWD '0'

HMI Event Table Row Alarm Text
COP
Source strWD
Dest HmiEvent[EVi].AlarmText
Length 1

HMI Event Table Row Alarm Text
CONCAT
Source A HmiEvent[EVi].AlarmText ??
Source B strPeriod '.'
Dest HmiEvent[EVi].AlarmText ??

HMI Event Table Row Alarm Text
LT
Source A HmiEvent[EVi].AlarmText.LEN 0
Source B 2

HMI Event Table Row Alarm Text
INSERT
Source A HmiEvent[EVi].AlarmText ??
Source B str0 '0'
Start 1
Dest HmiEvent[EVi].AlarmText ??

Packet Data Bit Index
DTOS
Source BITi 22
Dest strBIT '29'

HMI Event Table Row Alarm Text
CONCAT
Source A HmiEvent[EVi].AlarmText ??
Source B strBIT '29'
Dest HmiEvent[EVi].AlarmText ??

HMI Event Table Row Alarm Text
CONCAT
Source A HmiEvent[EVi].AlarmText ??
Source B strSpace ' '
Dest HmiEvent[EVi].AlarmText ??

HMI Event Table Row Alarm Text
LT
Source A HmiEvent[EVi].AlarmText.LEN 0
Source B 5

HMI Event Table Row Alarm Text
INSERT
Source A HmiEvent[EVi].AlarmText ??
Source B str0 '0'
Start 4
Dest HmiEvent[EVi].AlarmText ??

Update Event Table for HMI Display

New Event - Save Event Details
Determine Event Element ID and Table Lookup Range
Event ElementID combines EQID and Event ElementID into unique 8-digit value
(PreDefined do not add EQID; EqDefined subtract 100 from Event ElementID to shift to 0-based word match of Event Configuration)

Tag01 Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment

EQID EventText Start Index (si) and End Index (ei) are used to reduce Table search area for EqDefined
(si, ei are maintained by configuration download logic)

EDC2 WL09 Event Configuration - remove Shared Text and add Tag04

23

PreDefined Event: EventPreDef

Event Equip-Element ID:
MOVE
Source HmiEvent[EVi].Element -1
Dest EventEqElemID 9360029

EqDefined Event Text Tag01 Search:
MOVE
Source EventEqElemID 9360029
Dest EventTextFSC01.POS 29262

Search Text Tag01 EventTextTag01 (L)
Search Text Tag02 EventTextTag02 (U)
Search Text Tag03 EventTextTag03 (U)
Search Text Tag04 EventTextTag04 (U)
Search Text Tag05 EventTextTag05 (U)
Search Text Tag06 EventTextTag06 (U)

PreDefined Event: EventPreDef (/)

Event Equip-Element ID:
CPT
Dest EventEqElemID 9360029
Expression (LNi*100+EQi)*10000+HmiEvent[EVi].Element-100

Equipment Defined Event: EventTextEqDef

Search Text Tag01 EventTextTag01 (U)
Search Text Tag02 EventTextTag02 (U)
Search Text Tag03 EventTextTag03 (U)
Search Text Tag04 EventTextTag04 (U)
Search Text Tag05 EventTextTag05 (U)
Search Text Tag06 EventTextTag06 (U)

EqDefined Event Text Tag01 Search EventTextFSC01 (RES)
EqDefined Event Text Tag02 Search EventTextFSC02 (RES)
EqDefined Event Text Tag03 Search EventTextFSC03 (RES)
EqDefined Event Text Tag04 Search EventTextFSC04 (RES)
EqDefined Event Text Tag05 Search EventTextFSC05 (RES)
EqDefined Event Text Tag06 Search EventTextFSC06 (RES)

Update Event Table for HMI Display

Update Event Table for HMI Display

Tag01
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

24

Equipment Defined Event
EventTextEqDef

Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup Start Index

LIMIT
Low Limit: EDC_HMI.CFG.EvTxt01.si 0
Test: Equip.EventCTL.TXTsi 40769
High Limit: EDC_HMI.CFG.EvTxt01.ei 32499

EqDefined Event Text Tag01 Search

MOVE
Source: Equip.EventCTL.TXTsi 40769
Dest: EventTextFSC01.POS 29262

EqDefined Event Text Tag01 Search

MOVE
Source: EDC_HMI.CFG.EvTxt01.n 32500
Dest: EventTextFSC01.LEN 29265

Search Text Tag01
EventTextTag01
L

Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup End Index

LIMIT
Low Limit: EDC_HMI.CFG.EvTxt01.si 0
Test: Equip.EventCTL.TXTei 41330
High Limit: EDC_HMI.CFG.EvTxt01.ei 32499

EqDefined Event Text Tag01 Search

ADD
Source A: Equip.EventCTL.TXTei 41330
Source B: 1
Dest: EventTextFSC01.LEN 29265

Tag01 Event Text Search

25

Equipment Defined Event
EventTextEqDef

Search Text Tag01
EventTextTag01

EqDefined Event Text Tag01 Search

FSC
Control: EventTextFSC01
Length: 29265
Position: 29262
Mode: ALL
Expression: EventEqElemID=EDC_EventTxt01[EventTextFSC01.POS].EqEventID
EN
DN
ER

Update Event Table for HMI Display

Tag02
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

26

Equipment Defined Event: EventTextEqDef
EqDefined Event Text Tag01 Search: EventTextFSC01.FD
Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup Start Index

LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt01.si 0
- Test: Equip.EventCTL.TXTsi 40769
- High Limit: EDC_HMI.CFG.EvTxt02.ei 64999

Search Text Tag02: EventTextTag02

EqDefined Event Text Tag02 Search

SUB
- Source A: Equip.EventCTL.TXTsi 40769
- Source B: EDC_HMI.CFG.EvTxt02.si 32500
- Dest: EventTextFSC02.POS 0

EqDefined Event Text Tag02 Search

LT
- Source A: EventTextFSC02.POS 0
- Source B: 0

EqDefined Event Text Tag02 Search

MOVE
- Source: 0
- Dest: EventTextFSC02.POS 0

EqDefined Event Text Tag02 Search

MOVE
- Source: EDC_HMI.CFG.EvTxt02.n 32500
- Dest: EventTextFSC02.LEN 1010

Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup End Index

LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt02.si 32500
- Test: Equip.EventCTL.TXTei 41330
- High Limit: EDC_HMI.CFG.EvTxt02.ei 64999

EqDefined Event Text Tag02 Search

CPT
- Dest: EventTextFSC02.LEN 1010
- Expression: Equip.EventCTL.TXTei-EDC_HMI.CFG.EvTxt02.si+1

Tag02 Event Text Search

27

Equipment Defined Event: EventTextEqDef
Search Text Tag02: EventTextTag02

EqDefined Event Text Tag02 Search

FSC
- Control: EventTextFSC02
- Length: 1010
- Position: 0
- Mode: ALL
- Expression: EventEqElemID=EDC_EventTxt02[EventTextFSC02.POS].EqEventID

EN
DN
ER

Update Event Table for HMI Display

Tag03
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

28

Equipment Defined Event: EventTextEqDef
EqDefined Event Text Tag01 Search: EventTextFSC01.FD
EqDefined Event Text Tag02 Search: EventTextFSC02.FD
Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup Start Index

LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt01.si 0
- Test: Equip.EventCTL.TXTsi 40769
- High Limit: EDC_HMI.CFG.EvTxt03.ei 97499

Search Text Tag03: EventTextTag03 (L)

EqDefined Event Text Tag03 Search

SUB
- Source A: Equip.EventCTL.TXTsi 40769
- Source B: EDC_HMI.CFG.EvTxt03.si 65000
- Dest: EventTextFSC03.POS 0

EqDefined Event Text Tag03 Search

LT
- Source A: EventTextFSC03.POS 0
- Source B: 0

EqDefined Event Text Tag03 Search

MOVE
- Source: 0
- Dest: EventTextFSC03.POS 0

EqDefined Event Text Tag03 Search

MOVE
- Source: EDC_HMI.CFG.EvTxt03.n 32500
- Dest: EventTextFSC03.LEN 32500

Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup End Index

LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt03.si 65000
- Test: Equip.EventCTL.TXTei 41330
- High Limit: EDC_HMI.CFG.EvTxt03.ei 97499

EqDefined Event Text Tag03 Search

CPT
- Dest: EventTextFSC03.LEN 32500
- Expression: Equip.EventCTL.TXTei-EDC_HMI.CFG.EvTxt03.si+1

Tag03 Event Text Search

29

Equipment Defined Event: EventTextEqDef
Search Text Tag03: EventTextTag03

EqDefined Event Text Tag03 Search

FSC
- Control: EventTextFSC03
- Length: 32500
- Position: 0
- Mode: ALL
- Expression: EventEqElemID=EDC_EventTxt03[EventTextFSC03.POS].EqEventID

(EN) (DN) (ER)

Update Event Table for HMI Display

Tag04
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

30

Equipment Defined Event: EventTextEqDef
EqDefined Event Text Tag01 Search: EventTextFSC01.FD
EqDefined Event Text Tag02 Search: EventTextFSC02.FD
EqDefined Event Text Tag03 Search: EventTextFSC03.FD

Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup Start Index

LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt01.si 0
- Test: Equip.EventCTL.TXTsi 40769
- High Limit: EDC_HMI.CFG.EvTxt04.ei 129999

Search Text Tag04: EventTextTag04 (L)

EqDefined Event Text Tag04 Search

SUB
- Source A: Equip.EventCTL.TXTsi 40769
- Source B: EDC_HMI.CFG.EvTxt04.si 97500
- Dest: EventTextFSC04.POS 0

EqDefined Event Text Tag04 Search

LT
- Source A: EventTextFSC04.POS 0
- Source B: 0

EqDefined Event Text Tag04 Search

MOVE
- Source: 0
- Dest: EventTextFSC04.POS 0

EqDefined Event Text Tag04 Search

MOVE
- Source: EDC_HMI.CFG.EvTxt04.n 32500
- Dest: EventTextFSC04.LEN 32500

Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup End Index

LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt04.si 97500
- Test: Equip.EventCTL.TXTei 41330
- High Limit: EDC_HMI.CFG.EvTxt04.ei 129999

EqDefined Event Text Tag04 Search

CPT
- Dest: EventTextFSC04.LEN 32500
- Expression: Equip.EventCTL.TXTei-EDC_HMI.CFG.EvTxt04.si+1

Tag04 Event Text Search

31

Equipment Defined Event: EventTextEqDef
Search Text Tag04: EventTextTag04

EqDefined Event Text Tag04 Search

FSC
- Control: EventTextFSC04
- Length: 32500
- Position: 0
- Mode: ALL
- Expression: EventEqElemID=EDC_EventTxt04[EventTextFSC04.POS].EqEventID

(EN) (DN) (ER)

Update Event Table for HMI Display

Tag05
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

32

Equipment Defined Event: EventTextEqDef
EqDefined Event Text Tag01 Search: EventTextFSC01.FD
EqDefined Event Text Tag02 Search: EventTextFSC02.FD
EqDefined Event Text Tag03 Search: EventTextFSC03.FD
EqDefined Event Text Tag04 Search: EventTextFSC04.FD

Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup Start Index

LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt01.si 0
- Test: Equip.EventCTL.TXTsi 40769
- High Limit: EDC_HMI.CFG.EvTxt05.ei 162499

Search Text Tag05: EventTextTag05 (L)

EqDefined Event Text Tag05 Search

SUB
- Source A: Equip.EventCTL.TXTsi 40769
- Source B: EDC_HMI.CFG.EvTxt05.si 130000
- Dest: EventTextFSC05.POS 0

EqDefined Event Text Tag05 Search

LT
- Source A: EventTextFSC05.POS 0
- Source B: 0

EqDefined Event Text Tag05 Search

MOVE
- Source: 0
- Dest: EventTextFSC05.POS 0

EqDefined Event Text Tag05 Search

MOVE
- Source: EDC_HMI.CFG.EvTxt05.n 32500
- Dest: EventTextFSC05.LEN 32500

Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup End Index

LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt05.si 130000
- Test: Equip.EventCTL.TXTei 41330
- High Limit: EDC_HMI.CFG.EvTxt05.ei 162499

EqDefined Event Text Tag05 Search

CPT
- Dest: EventTextFSC05.LEN 32500
- Expression: Equip.EventCTL.TXTei-EDC_HMI.CFG.EvTxt05.si+1

Tag05 Event Text Search

33

Equipment Defined Event: EventTextEqDef
Search Text Tag05: EventTextTag05

EqDefined Event Text Tag05 Search

FSC
- Control: EventTextFSC05
- Length: 32500
- Position: 0
- Mode: ALL
- Expression: EventEqElemID=EDC_EventTxt05[EventTextFSC05.POS].EqEventID

(EN) (DN) (ER)

Update Event Table for HMI Display

34

Tag06
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

Equipment Defined Event: EventTextEqDef
EqDefined Event Text Tag01 Search: EventTextFSC01.FD
EqDefined Event Text Tag02 Search: EventTextFSC02.FD
EqDefined Event Text Tag03 Search: EventTextFSC03.FD
EqDefined Event Text Tag04 Search: EventTextFSC04.FD
EqDefined Event Text Tag05 Search: EventTextFSC05.FD

Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup Start Index

LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt01.si 0
- Test: Equip.EventCTL.TXTsi 40769
- High Limit: EDC_HMI.CFG.EvTxt06.ei 194999

Search Text Tag06: EventTextTag06 (L)

EqDefined Event Text Tag06 Search

SUB
- Source A: Equip.EventCTL.TXTsi 40769
- Source B: EDC_HMI.CFG.EvTxt06.si 162500
- Dest: EventTextFSC06.POS 0

EqDefined Event Text Tag06 Search

LT
- Source A: EventTextFSC06.POS 0
- Source B: 0

EqDefined Event Text Tag06 Search

MOVE
- Source: 0
- Dest: EventTextFSC06.POS 0

EqDefined Event Text Tag06 Search

MOVE
- Source: EDC_HMI.CFG.EvTxt06.n 32500
- Dest: EventTextFSC06.LEN 32500

Local Copy of Equip Array Tag for LNi, EQi Alarm Text Lookup End Index

LIMIT
- Low Limit: EDC_HMI.CFG.EvTxt06.si 162500
- Test: Equip.EventCTL.TXTei 41330
- High Limit: EDC_HMI.CFG.EvTxt06.ei 194999

EqDefined Event Text Tag06 Search

CPT
- Dest: EventTextFSC06.LEN 32500
- Expression: Equip.EventCTL.TXTei-EDC_HMI.CFG.EvTxt06.si+1

35

Tag06 Event Text Search

Equipment Defined Event: EventTextEqDef
Search Text Tag06: EventTextTag06

EqDefined Event Text Tag06 Search

FSC
- Control: EventTextFSC06
- Length: 32500
- Position: 0
- Mode: ALL
- Expression: EventEqElemID=EDC_EventTxt06[EventTextFSC06.POS].EqEventID
- EN
- DN
- ER

Update Event Table for HMI Display

New Event - Save Event Details
Lookup Alarm Text by EqEvent ElemID - if not found, add 'Undefined'

36

PreDefined Event
EventPreDef

EqDefined Event Text Tag01 Search

MOVE	
Source	EventEqElemID 9360029
Dest	EventTextFSC01.POS 29262

HMI Event Table Row Alarm Text

CONCAT	
Source A	HmiEvent[EVi].AlarmText ??
Source B	EDC_EventTxt01[EventTextFSC01.POS].Text ??
Dest	HmiEvent[EVi].AlarmText ??

EventTextFD

Equipment Defined Event
EventTextEqDef

Search Text Tag01
EventTextTag01

EqDefined Event Text Tag01 Search
EventTextFSC01.FD

Equipment Defined Event
EventTextEqDef

Search Text Tag02
EventTextTag02

EqDefined Event Text Tag02 Search
EventTextFSC02.FD

HMI Event Table Row Alarm Text

CONCAT	
Source A	HmiEvent[EVi].AlarmText ??
Source B	EDC_EventTxt02[EventTextFSC02.POS].Text ??
Dest	HmiEvent[EVi].AlarmText ??

Search Text Tag03
EventTextTag03

EqDefined Event Text Tag03 Search
EventTextFSC03.FD

HMI Event Table Row Alarm Text

CONCAT	
Source A	HmiEvent[EVi].AlarmText ??
Source B	EDC_EventTxt03[EventTextFSC03.POS].Text ??
Dest	HmiEvent[EVi].AlarmText ??

Search Text Tag04
EventTextTag04

EqDefined Event Text Tag04 Search
EventTextFSC04.FD

HMI Event Table Row Alarm Text

CONCAT	
Source A	HmiEvent[EVi].AlarmText ??
Source B	EDC_EventTxt04[EventTextFSC04.POS].Text ??
Dest	HmiEvent[EVi].AlarmText ??

Search Text Tag05
EventTextTag05

EqDefined Event Text Tag05 Search
EventTextFSC05.FD

HMI Event Table Row Alarm Text

CONCAT	
Source A	HmiEvent[EVi].AlarmText ??
Source B	EDC_EventTxt05[EventTextFSC05.POS].Text ??
Dest	HmiEvent[EVi].AlarmText ??

Search Text Tag06

EqDefined Event Text Tag06 Search

HMI Event Table Row Alarm Text

Update Event Table for HMI Display

EventTextTag06 EventTextFSC06.FD

CONCAT
Source A HmiEvent[EVi].AlarmText ??
Source B EDC_EventTxt06[EventTextFSC06.POS].Text ??
Dest HmiEvent[EVi].AlarmText ??

HMI Event Table Row Alarm Text

EventTextFD
/

CONCAT
Source A HmiEvent[EVi].AlarmText ??
Source B EDC_HMI.CFG.EventUndefined 'Undefined'
Dest HmiEvent[EVi].AlarmText ??

New Event - Save Event Details
Increment Event Index

Event Display Table Index

37

ADD
Source A EVi 7
Source B 1
Dest EVi 7

Next Equipment Event

lbl_NextEvent
38 LBL

NOP

Increment Equipment Event Index
If Equipment Event Index is not Done AND Event Table is not full -- resume search for another new Event

Equipment Event Index

39

ADD
Source A EqEVi 0
Source B 1
Dest EqEVi 0

Equipment Event Index

LE
Source A EqEVi 0
Source B CFG.EV.mi <EDC.CFG.EV.mi> 9

Event Display Table Index

LE
Source A EVi 7
Source B EDC_HMI.CFG.EV.mi 199

lbl_EventAddLoop
JMP

Update Event Table for HMI Display

Exit Routine

40

lbl_Exit
LBL

NOP

(End)

Update Event Table Status & Duration for HMI Display

Update Display Fault/Status Event Status and Duration
Table of Current Events, Start Time, Duration

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 796
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 163891661
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 163891661

Update the Event Status and Duration

Initialize Event Loop Pointer

2

Event Display Table Index

CLR
Dest EVi 7

Exit Loop if no more Visible DIsplay Records

3

lbl_EventLoop
LBL

HMI Event Table Row Equipment Data Loaded
HmiEvent[EVi].Visible
/

lbl_Exit
JMP

Update Event Table Status & Duration for HMI Display

Validate Line and Equipment Array Indexes
Skip Record if Indexes out of Range
DFH006 - Bug Fix - replaced LNi, EQi with EvLNi, EvEQi (was overwriting Loop Pointers)

4

Event Line Index (0-28)

MOVE
Source HmiEvent[EVi].LNi 0
Dest EvLNi 10

Event Equipment Index (0-99)

MOVE
Source HmiEvent[EVi].EQi 0
Dest EvEQi 4

Event Line Index (0-28)

LIMIT
Low Limit CFG.LN.si <EDC.CFG.LN.si> 0
Test EvLNi 10
High Limit CFG.LN.ei <EDC.CFG.LN.ei> 28

MOVE
Source EvLNi 10
Dest ValidLNi 18

Event Equipment Index (0-99)

LIMIT
Low Limit EDC.Line[ValidLNi].CFG.EQ.si 0
Test EvEQi 4
High Limit EDC.Line[ValidLNi].CFG.EQ.ei 35

LNi, EQi Indexes OK
LnEq_IndexOk

LNi, EQi Indexes OK
LnEq_IndexOk
/

lbl_NextRow
JMP

Check for good Event Looklup Index for EDC_Event EvLKi

5

Equipment-Event Index for EDC_Event Lookup

CLR
Dest EvLKi 259

Equipment Data Collection Lines and Equipment Array

LIMIT
Low Limit 0
Test EDC.Line[EvLNi].Equip[EvEQi].EventCTL.EvLKi 198
High Limit EDC_Event.CFG.EQ.mi 2999

Equipment-Event Index for EDC_Event Lookup

MOVE
Source EDC.Line[EvLNi].Equip[EvEQi].EventCTL.EvLKi 198
Dest EvLKi 259

Event Lookup Index within array limits
EvLKi_IndexOK

Event Lookup Index for EDC_Event matches

6

Event Lookup Index within array limits
EvLKi_IndexOK

CPT
Dest EQID 1811
Expression EvLNi*100+EvEQi

Event Controls Equipment Fragment Statistics Array

EQ
Source A EDC_Event.CTL.Equip[EvLKi].EQID 1811
Source B EQID 1811

Event Lookup Index EQID Match
EvLKi_EQIDmatch

Update Event Table Status & Duration for HMI Display

Convert Event Element into Packet, Data, Bit Indexes for Mask Loolup

7

MOVE
Source HmiEvent[EVi].Element -1
Dest Element 5622

LE
Source A Element 5622
Source B 31

PreDefined Event
EventPreDef

CLR
Dest WRDi 55

Packet Index
CLR
Dest PKTi 6

Packet Data Index
CLR
Dest DATi 7

Packet Data Bit Index
MOVE
Source Element 5622
Dest BITi 22

PreDefined Event
EventPreDef
/

Packet Data Bit Index
MOD
Source A Element 5622
Source B 100
Dest BITi 22

EqDefined Event Word Index
CPT
Dest WRDi 55
Expression (Element-BITi)/100 - 1

Packet Data Index
MOD
Source A WRDi 55
Source B 8
Dest DATi 7

Packet Index
CPT
Dest PKTi 6
Expression (WRDi - DATi) / 8

Exit if no Event Mask Indexes not OK
(default to Gray Status)

8

Packet Index
LIMIT
Low Limit 0
Test PKTi 6
High Limit EDC_Event.CFG.PKT.mi 7

Packet Data Index
LIMIT
Low Limit 0
Test DATi 7
High Limit EDC_Event.CFG.DAT.mi 7

Packet Data Bit Index
LIMIT
Low Limit 0
Test BITi 22
High Limit EDC_Event.CFG.BIT.mi 31

MaskIndexesOK

9

Local Copy of Equip Array Tag for LNi, EQi
COP
Source EDC.Line[EvLNi].Equip[EvEQi]
Dest Equip
Length 1

Update Event Table Status & Duration for HMI Display

Event Status for EqDriver v1 -- uses Equipment Status
DFH006 - Bug Fix - replaced LNi, EQi with EvLNi, EvEQi (was overwriting Loop Pointers)
DHF029 - Added Alert Status 5

10

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0
/

MOVE
Source 1
Dest TempStatus 2

Local Copy of Equip Array Tag for LNi, EQi Alert Condition (Lt Blue)
Equip.DataCTL.Alert

MOVE
Source 5
Dest TempStatus 2

Local Copy of Equip Array Tag for LNi, EQi Alarm Condition (Yellow)
Equip.DataCTL.Alarm

MOVE
Source 2
Dest TempStatus 2

Local Copy of Equip Array Tag for LNi, EQi Fault Condition (Red)
Equip.DataCTL.Fault

MOVE
Source 3
Dest TempStatus 2

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange)
Equip.DataCTL.InterlockOff

MOVE
Source 4
Dest TempStatus 2

Update Event Table Status & Duration for HMI Display

Exit if no Equipment Event Configuration found
(default to Gray Status)

11

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

Event Lookup Index EQID Match
EvLKi_EQIDmatch
/

Diagnostic
ADD
Source A NoMatchCount 63002
Source B 1
Dest NoMatchCount 63002

MaskIndexesOK
/

Diagnostic
ADD
Source A BadIndexCount 0
Source B 1
Dest BadIndexCount 0

MOVE
Source 0
Dest TempStatus 2

lbl_SkipStatus
JMP

Update Event Table Status & Duration for HMI Display

Event Status for EqDriver v2 -- uses Event Mask Configuration
Start with lowest prioirty and overwrite with higher priorities
(only one bit should be on, but this ensures highest Priority is selected)

Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orangr, 5-Cyan)

12

Local Copy of Equip Array Tag for LNi, EQi Equipment Driver Callout Message v2.0 (new PreDefined bits)
Equip.CTL.CalloutMsgV2_0

MOVE
Source 1
Dest TempStatus 2

PreDefined Event
EventPreDef

DT Type = Alert
Event Mask Equipment Mask Equip Array
EDC_EventMaskAlert.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]

EventAlertMask

MOVE
Source 5
Dest TempStatus 2

PreDefined Event
EventPreDef

PreDefined Event Mask DtType Alert Mask
EDC_EventMaskPreDef.Alert.[BITi]

PreDefined Event
EventPreDef

DT Type = Alarm
Event Mask Equipment Mask Equip Array
EDC_EventMaskAlarm.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]

EventAlarmMask

MOVE
Source 2
Dest TempStatus 2

PreDefined Event
EventPreDef

PreDefined Event Mask DtType Alarm Mask
EDC_EventMaskPreDef.Alarm.[BITi]

PreDefined Event
EventPreDef

DT Type = Fault
Event Mask Equipment Mask Equip Array
EDC_EventMaskFault.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]

EventFaultMask

MOVE
Source 3
Dest TempStatus 2

PreDefined Event
EventPreDef

PreDefined Event Mask DtType Fault Mask
EDC_EventMaskPreDef.Fault.[BITi]

Local Copy of Equip Array Tag for LNi, EQi Interlock Off Condition (Orange)
Equip.DataCTL.InterlockOff

MOVE
Source 4
Dest TempStatus 2

Update Event Table Status & Duration for HMI Display

Skip Status if Event Configuration not found

13

lbl_SkipStatus
LBL

NOP

Store Event Status

HMI Event Table Row Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

14

MOVE
Source TempStatus 2
Dest HmiEvent[EVi].Status 0

Calculate Event Accum Time
Current Date/Time - Event Start Date/Time

Event Duration

15

aoiEDC_EventDur
aoiEDC_EventDur aoiEDC_EventDur ...
TransTime HmiEvent[EVi].TransTime
CurrTime CLX.TransTime <EDC.CLX.TransTime>
Duration HmiEvent[EVi].Duration

Increment Event Pointer

Event Display Table Index

Event Display Table Index

16

lbl_NextRow
LBL

ADD
Source A EVi 7
Source B 1
Dest EVi 7

LIMIT
Low Limit EDC_HMI.CFG.EV.si 0
Test EVi 7
High Limit EDC_HMI.CFG.EV.ei 199

lbl_EventLoop
JMP

Exit Routine

17

lbl_Exit
LBL

NOP

Update Event Table Status & Duration for HMI Display

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
LNi, EQi = Line, Equipment Index
Num = Transaction Number

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code <EDC.Trans.Code>
Input Par Trans.Time <EDC.Trans.Time>
Input Par Trans.LNi <EDC.Trans.LNi>
Input Par Trans.EQi <EDC.Trans.EQi>
Input Par Trans.Num <EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine 750

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901] 5014
Source B 1
Dest Prog.RoutineScan[901] 5014

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT	
Low Limit	Trans.Filter.CodeLow <EDC.Trans.Filter.CodeLow> 0
Test	Trans.Code <EDC.Trans.Code> 2021
High Limit	Trans.Filter.CodeHigh <EDC.Trans.Filter.CodeHigh> 0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT	
Low Limit	Trans.Filter.LNiLow <EDC.Trans.Filter.LNiLow> 0
Test	Trans.LNi <EDC.Trans.LNi> 4
High Limit	Trans.Filter.LNiHigh <EDC.Trans.Filter.LNiHigh> 0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT	
Low Limit	Trans.Filter.EQiLow <EDC.Trans.Filter.EQiLow> 0
Test	Trans.EQi <EDC.Trans.EQi> 21
High Limit	Trans.Filter.EQiHigh <EDC.Trans.Filter.EQiHigh> 0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

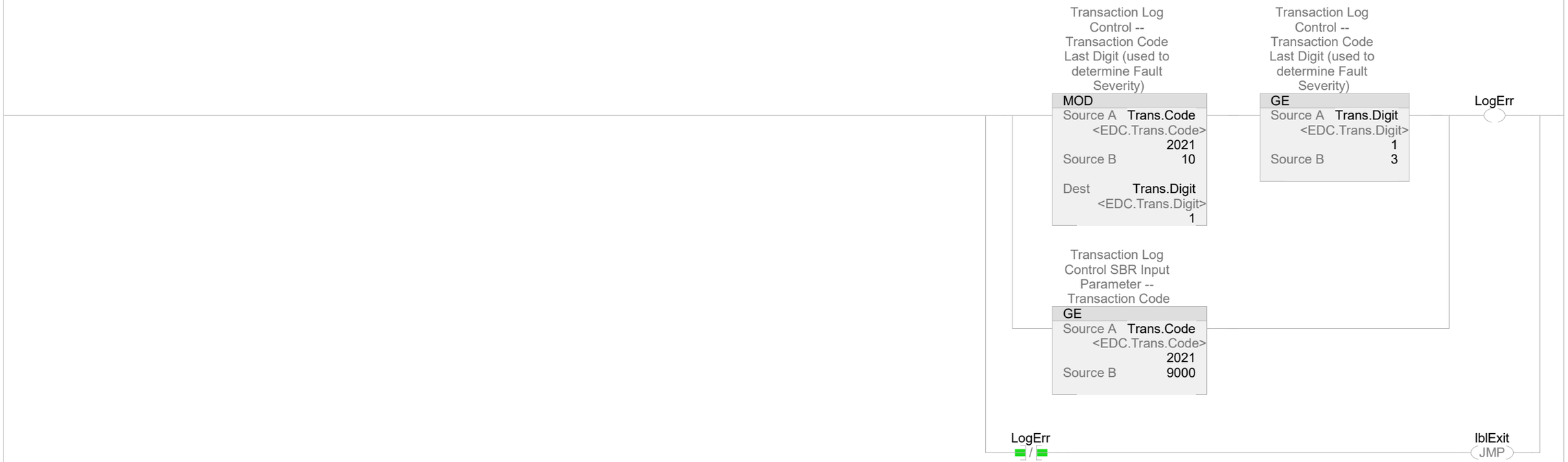
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9



Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP	
Source	EDC_TransLog[Trans.LOGi]
Dest	EDC_TransErr[Trans.ERRi]
Length	1

Error Log

FLL	
Source	0
Dest	EDC_TransErr[Trans.ERRx]
Length	1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL	
Source	-1
Dest	Trans.Opt[0] <EDC.Trans.Opt[0]>
Length	Trans.OptN <EDC.Trans.OptN>

(End)

Main Routine

EDC2 - v2.00 added PdM with QualityRx, TrendRx and Server Interface
EDC2 - v2.01 added Event DtOwner and DtType Mask Configuration; removed Shared Andon/AVI Events
EDC3 - v3.00 added Stuck Download Detection for bad configuration, expanded Server Status block
v3.50 - L980 Controller; Gateway Configuration Download; TransNum Handshake

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

1

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	801
Dest	Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806549717
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806549717

2

Reset

Reset Event Configuration
(use before Download All to defrag the Event table for faster searches)

Gateway Reset CFGResetEventText

Toggle to Reset Configuration ManualReset

Gateway Reset CFGResetEventText (U)

Toggle to Reset Configuration ManualReset (U)

AutoConfig (L)

JSR	
Routine Name	_810_Reset

Main Routine

Configuration

3

AutoConfig

Toggle to Load User Configuration ManualConfig

s:FS

AutoConfig U

Toggle to Load User Configuration ManualConfig U

Configuration

JSR

Routine Name _811_Config

Reset for Configuration and Status Data Errors

4

Manual Reset for Configuration Status Errors ManualResetER

Manual Reset for Configuration Status Errors ManualResetER U

Manual Reset of Gateway Status Errors EDC_StatusErrorReset

Manual Reset of Gateway Status Errors EDC_StatusErrorReset U

Server Interface Status Reset Errors EDC_Status01.ServerCfg.Update.ResetER L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...

Status EDC_Status01.ServerCfg

Server Interface Status Reset Errors EDC_Status01.ServerPdm.Update.ResetER L

Update PdM State for Transactions

aoiEDC_StatusUpdateServerPdm

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm ...

Status EDC_Status01.ServerPdm

Server Interface Status Reset Errors EDC_Status01.ServerStat.Update.ResetER L

Update Status State for Transactions

aoiEDC_StatusUpdateServerStatus

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerStatus ...

Status EDC_Status01.ServerStat

Server Interface Status Reset Errors EDC_Status01.ServerHealth.Update.ResetER L

Update Health State for Transactions

aoiEDC_StatusUpdateServerHealth

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerHealth ...

Status EDC_Status01.ServerHealth

Server Interface Status Reset Errors EDC_Status01.ServerMask.Update.ResetER L

Update Mask State for Transactions

aoiEDC_StatusUpdateServerMask

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerMask ...

Status EDC_Status01.ServerMask

Main Routine

Line Configuration Request

Configuration Request to Server via Transaction Manager

JSR
Routine Name _820_CfgRequest

5

PdM Configuration Request

Configuration Request to Server via Transaction Manager

JSR
Routine Name _845_PdmRequest

6

Status Request
DN005 Health Status Alarm Callout

Status Request to Server via Transaction Manager

JSR
Routine Name _860_StatRequest

Detect and Callout CIS Message Errors

JSR
Routine Name _870_StatCisErrCallout

Detect and Callout Health Status Errors

JSR
Routine Name _871_StatHealthErrCallout

7

EQ Health Request
DN014 Health Displays

EQ Health Request to Server via Transaction Manager

JSR
Routine Name _880_HealthRequest

8

Event Mask Configuration Request

Configuration Request to Server via Transaction Manager

JSR
Routine Name _890_MaskRequest

9

Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

10

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Routine
802
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Fault.Type
4
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Fault.Code
20
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.FaultCount
0
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.LOGi
30
Dest Trans.Opt[5]
<EDC.Trans.Opt[5]>
-1

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 9003
Input Par 0
Input Par -1
Input Par -1
Input Par 0

Main Routine

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

11

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

12

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
59
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan Events
CLxProgScanHigh.8
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV	
Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	-20378558

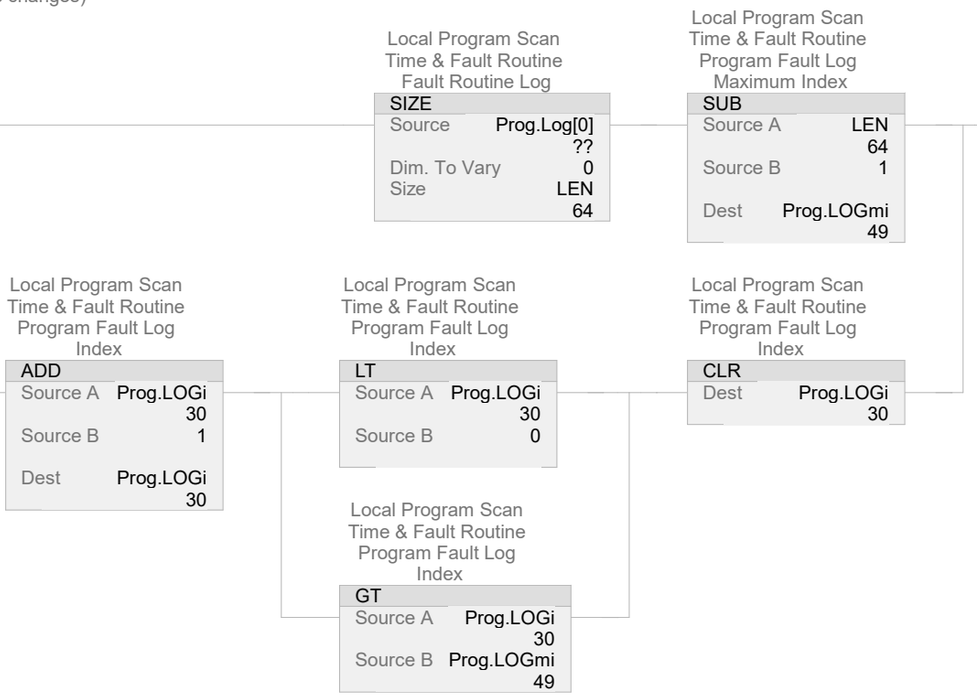
Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD	
Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

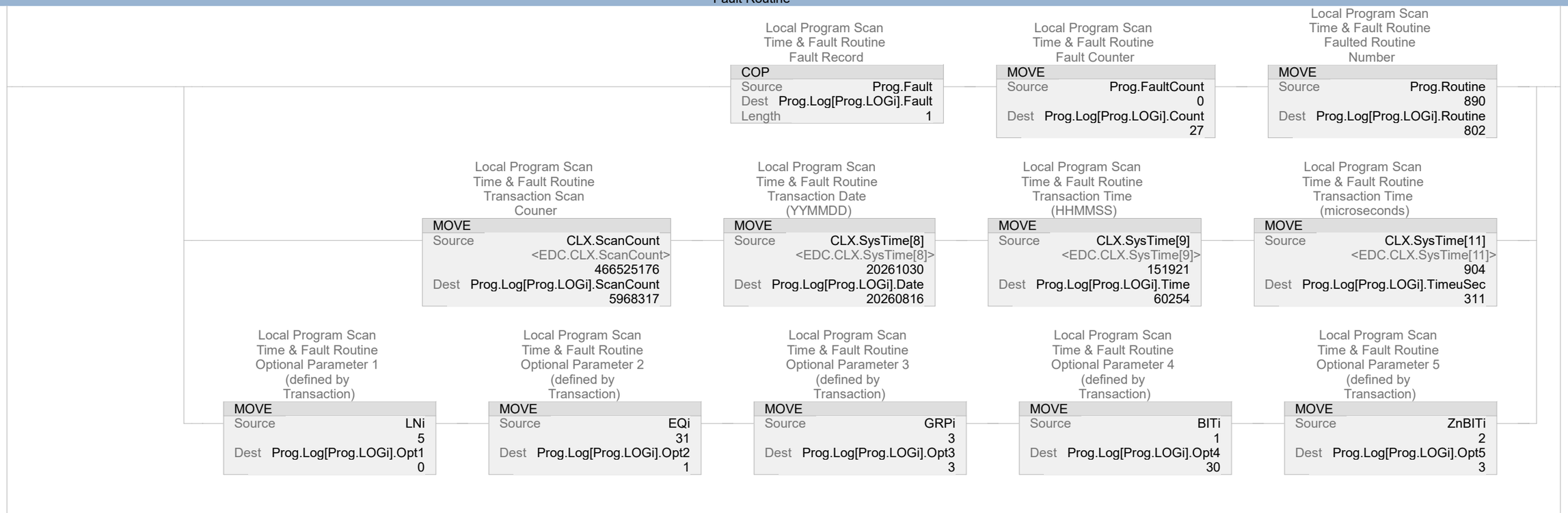
Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

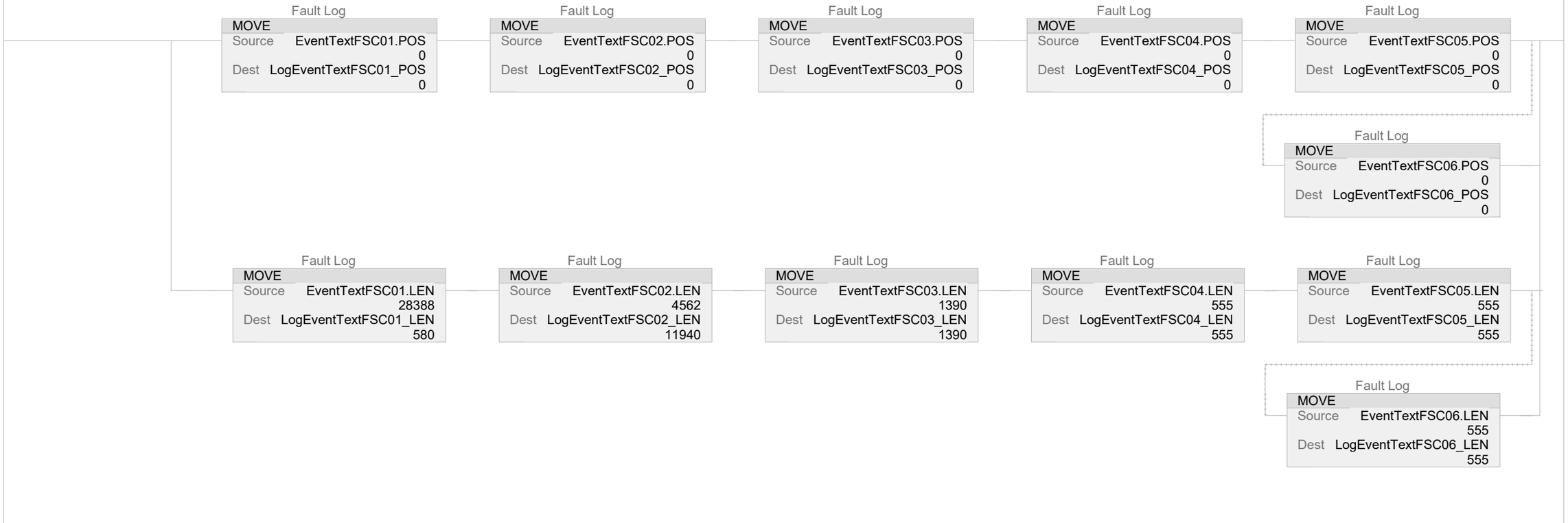
3



Fault Routine

Trap FSC Parameters

4



Fault Routine

Trap Mask Download Error

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

EQ
Source A Prog.Routine 890
Source B 896

Fault Log

ADD
Source A LogMaskFaultCount 4
Source B 1
Dest LogMaskFaultCount 4

Fault Log

MOVE
Source MaskEQi 26
Dest LogMaskEQi 13

Fault Log

MOVE
Source MaskLNi 9
Dest LogMaskLNi 1

Fault Log

MOVE
Source MaskEvLKi 164
Dest LogMaskEvLKi 91

Fault Log

MOVE
Source LOOPi 21
Dest LogLoopi 21

Fault Log

MOVE
Source TOOLi 915
Dest LogTOOLi 915

Fault Log

MOVE
Source TRNDi 4
Dest LogTRNDi 4

Fault Log

MOVE
Source MSKi 8
Dest LogMSKi 8

Fault Log

COP
Source ServerMask.DataRspLast <EDC.ServerMask.DataRspLast>
Dest LogServerMask_DataRspLast
Length 1

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

6

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log

Prog.LogFault

(L)

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

7

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 802
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Fault Routine

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

8

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type
0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code
0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow
-20378558

(End)

Server Interface Reset
(called before Full Configuration Download)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 810
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Clear Event Config Faults

2

Server Interface Status State = Red - Equip Event Array Full ER
EDC_Status01.ServerCfg.Update.EquipEventArrayFull
(U)

Reset Equipment Event Text, Fragment Statistics and Errors
(before Download All to defrag the Event table for faster searches)

3

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt01[0]
Length EDC_HMI.CFG.EvTxt01.n

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt02[0]
Length EDC_HMI.CFG.EvTxt02.n

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt03[0]
Length EDC_HMI.CFG.EvTxt03.n

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt04[0]
Length EDC_HMI.CFG.EvTxt04.n

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt05[0]
Length EDC_HMI.CFG.EvTxt05.n

Event Text Lookup Event Text Configuration

FLL
Source 0
Dest EDC_EventTxt06[0]
Length EDC_HMI.CFG.EvTxt06.n

Event Controls Event Control

FLL
Source 0
Dest EDC_Event.CTL
Length 1

CLR
Dest EventEmptyLastFD 45063

Manual Reset for Configuration Status Errors
ManualResetER
(L)

v2.0 Clear Event Masks

4

DT Type = Alarm Event Mask
FLL
Source 0
Dest EDC_EventMaskAlarm
Length 1

DT Type = Alert Event Mask
FLL
Source 0
Dest EDC_EventMaskAlert
Length 1

DT Type = Fault Event Mask
FLL
Source -1
Dest EDC_EventMaskFault
Length 1

DT Type = Ignore Event Mask
FLL
Source -1
Dest EDC_EventMaskIgnore
Length 1

DT Owner = Maintenance Event Mask
FLL
Source -1
Dest EDC_EventMaskMaint
Length 1

No Display Event Mask
FLL
Source 0
Dest EDC_EventMaskNoDisp
Length 1

May 2026 - clear Event KPI Option Masks
and Equipment KPI Option Packet Config/Status

5

KPI Option 1 Event Mask
FLL
Source 0
Dest EDC_EventMaskOpt01
Length 1

KPI Option 2 Event Mask
FLL
Source 0
Dest EDC_EventMaskOpt02
Length 1

KPI Option 3 Event Mask
FLL
Source 0
Dest EDC_EventMaskOpt03
Length 1

KPI Option 4 Event Mask
FLL
Source 0
Dest EDC_EventMaskOpt04
Length 1

KPI Option 5 Event Mask
FLL
Source -1
Dest EDC_EventMaskOpt05
Length 1

Event KPI Options
FLL
Source 0
Dest EDC_EventOpt
Length 1

Clear Carrier Scanners

6

Carrier Tracking from EQ Quality Tools EDC Tracking Scanner Configuration
FLL
Source 0
Dest EDC_Track.ScanCfg[0]
Length EDC_Track.Cfg.SCN.n

Reset Line, Group, Equip
Clear Line Loop Index

7

Line Index
CLR
Dest LNi
5

Line Loop

8

lbl_LineLoop
LBL

Equipment Index
CLR
Dest EQi
31

Reset Line Config
Reset Carrier Tracking Line Config

9

Equipment Data Collection Lines and Equipment Array
FLL
Source 0
Dest EDC.Line[LNi].CFG
Length 1

Carrier Tracking from EQ Quality Tools Tracking Conveyor
FLL
Source 0
Dest EDC_Track.Conv[LNi].Cfg
Length 1

Reset Group Config

10

Line Index
LIMIT
Low Limit 0
Test LNi
5
High Limit CFG.GRP.mi
<EDC.CFG.GRP.mi>
3

MOVE
Source LNi
5
Dest GRPi
3

Equipment Data Collection Offline Group Configuration
FLL
Source 0
Dest EDC.Offline.CFG[GRPi]
Length 1

11

Equipment Loop
Clear Equipment Event Text Search range
DN006 - support multiple Offline Lines
DN009 -add Equip.CTL to reset (otherwise deleted Equipment auto-configures as new Equipment)
v2.0 Clear Event Lookup Index
EDC2 WL09 - save CTL.*Msg to avoid Duplicate Message Errors during full download

lbl_EquipLoop
LBL

Equipment Data Collection - Line Equipment Control
COP
Source EDC.Line[LNi].Equip[EQi].CTL
Dest TempCTL
Length 1

Equipment Data Collection Lines and Equipment Array
FLL
Source 0
Dest EDC.Line[LNi].Equip[EQi].CFG
Length 1

Equipment Data Collection Lines and Equipment Array
FLL
Source 0
Dest EDC.Line[LNi].Equip[EQi].CFGoffline
Length 1

Equipment Data Collection Lines and Equipment Array
FLL
Source 0
Dest EDC.Line[LNi].Equip[EQi].CTL
Length 1

Equipment Data Collection Lines and Equipment Array
CLR
Dest EDC.Line[LNi].Equip[EQi].EventCTL.TXTsi
23544

Equipment Data Collection Lines and Equipment Array
CLR
Dest EDC.Line[LNi].Equip[EQi].EventCTL.TXTei
23779

Equipment Data Collection Lines and Equipment Array
CLR
Dest EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi
144

12

EDC2 WL09 - restore CTL.*Msg to avoid Duplicate Message Errors during full download

Equipment Data Collection Lines and Equipment Array
COP
Source TempCTL.CalloutMsg
Dest EDC.Line[LNi].Equip[EQi].CTL.CalloutMsg
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source TempCTL.FaultMsg
Dest EDC.Line[LNi].Equip[EQi].CTL.FaultMsg
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source TempCTL.QualityMsg
Dest EDC.Line[LNi].Equip[EQi].CTL.QualityMsg
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source TempCTL.TrendMsg
Dest EDC.Line[LNi].Equip[EQi].CTL.TrendMsg
Length 1

Equipment Data Collection Lines and Equipment Array
COP
Source TempCTL.VersionMsg
Dest EDC.Line[LNi].Equip[EQi].CTL.VersionMsg
Length 1

Increment Equipment Index

13

Equipment Index
ADD
Source A EQi 31
Source B 1
Dest EQi 31

Equipment Index
LE
Source A EQi 31
Source B CFG.EQ.mi <EDC.CFG.EQ.mi> 99

lbl_EquipLoop
JMP

Increment Line Index

14

Line Index
ADD
Source A LNi 5
Source B 1
Dest LNi 5

Line Index
LE
Source A LNi 5
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

lbl_LineLoop
JMP

Reset Diagnostic Counts

15

Server Response Command diagnostic count
SIZE
Source RspCmd[0] 44710
Dim. To Vary 0
Size SIZE 100

Server Response Command diagnostic count
FLL
Source 0
Dest RspCmd[0]
Length SIZE

(End)

Configuration

HMI Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 811
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Fast and Slow Request Times
Fast is used when Last Transaction Response had data

2

Server Configuration from Transaction Manager Fast Transacton Request Time Delay

MOVE
Source 0
Dest ServerCfg.Ctl.ReqFastTime <EDC.ServerCfg.Ctl.ReqFastTime> 0

Server Configuration from Transaction Manager Slow Transacton Request Time Delay

MOVE
Source 10000
Dest ServerCfg.Ctl.ReqSlowTime <EDC.ServerCfg.Ctl.ReqSlowTime> 10000

Fast and Slow Request Times
Fast is used when Last Transaction Response had data

3

Server PdM Configuration from Transaction Manager Fast Transacton Request Time Delay

MOVE
Source 0
Dest ServerPdm.Ctl.ReqFastTime <EDC.ServerPdm.Ctl.ReqFastTime> 0

Server PdM Configuration from Transaction Manager Slow Transacton Request Time Delay

MOVE
Source 10000
Dest ServerPdm.Ctl.ReqSlowTime <EDC.ServerPdm.Ctl.ReqSlowTime> 10000

Configuration

4

Fast and Slow Status Request Times
Fast is used when Last Transaction Response not last row of table

Server Status from Transaction Manager Fast Transacton Request Time Delay

MOVE
Source 0
Dest ServerStat.Ctl.ReqFastTime
<EDC.ServerStat.Ctl.ReqFastTime> 0

Server Status from Transaction Manager Slow Transacton Request Time Delay

MOVE
Source 10000
Dest ServerStat.Ctl.ReqSlowTime
<EDC.ServerStat.Ctl.ReqSlowTime> 10000

5

Fast and Slow Health Request Times
Fast is used when Last Transaction Response has new data

(Slow time is relatively fast -- with 200+ Equipment, guaranteed a refresh in ~3 minutes even if data changes aren't detected)

Server EQ Health from Transaction Manager Fast Transacton Request Time Delay

MOVE
Source 0
Dest ServerHealth.Ctl.ReqFastTime
<EDC.ServerHealth.Ctl.ReqFastTime> 0

Server EQ Health from Transaction Manager Slow Transacton Request Time Delay

MOVE
Source 1000
Dest ServerHealth.Ctl.ReqSlowTime
<EDC.ServerHealth.Ctl.ReqSlowTime> 1000

6

Fast and Slow Request Times
Fast is used when Last Transaction Response had data

Server Ignore Mask Configuration from Transaction Manager Fast Transacton Request Time Delay

MOVE
Source 0
Dest ServerMask.Ctl.ReqFastTime
<EDC.ServerMask.Ctl.ReqFastTime> 0

Server Ignore Mask Configuration from Transaction Manager Slow Transacton Request Time Delay

MOVE
Source 10000
Dest ServerMask.Ctl.ReqSlowTime
<EDC.ServerMask.Ctl.ReqSlowTime> 10000

Configuration

Reset Empty Event Last Found

7

CLR
Dest EventEmptyLastFD
45063

Configuration

Event Mask Equipment, Packet, Data and Bit Arrays
Configure Array Indexes

*** NOTE: Event CTL and EventMast Equipment Arrays should be same SIZE --- use the smaller array to set array limits ***

8

Event
Controls Event
Control Equipment
Array

SIZE
Source EDC_Event.CTL.Equip[0] ??
Dim. To Vary 0
Size EDC_Event.CFG.EQ.n 3000

DT Type = Alarm
Event Mask Event
Mask Equipment
Blocks

SIZE
Source EDC_EventMaskAlarm.Equip[0] ??
Dim. To Vary 0
Size LEN 64

Temp Result for
Array Size

LT
Source A LEN 64
Source B EDC_Event.CFG.EQ.n 3000

Event
Controls Array Size

MOVE
Source LEN 64
Dest EDC_Event.CFG.EQ.n 3000

Event
Controls Array Max
Index

SUB
Source A EDC_Event.CFG.EQ.n 3000
Source B 1
Dest EDC_Event.CFG.EQ.mi 2999

DT Type = Alarm
Event Mask Event
Mask Equipment
Packet of 8 DINTs

SIZE
Source EDC_EventMaskAlarm.Equip[0].Packet[0] ??
Dim. To Vary 0
Size EDC_Event.CFG.PKT.n 8

Event
Controls Array Max
Index

SUB
Source A EDC_Event.CFG.PKT.n 8
Source B 1
Dest EDC_Event.CFG.PKT.mi 7

DT Type = Alarm
Event Mask Event
Packet Data

SIZE
Source EDC_EventMaskAlarm.Equip[0].Packet[0].Data[0] 16#0000_0000
Dim. To Vary 0
Size EDC_Event.CFG.DAT.n 8

Event
Controls Array Max
Index

SUB
Source A EDC_Event.CFG.DAT.n 8
Source B 1
Dest EDC_Event.CFG.DAT.mi 7

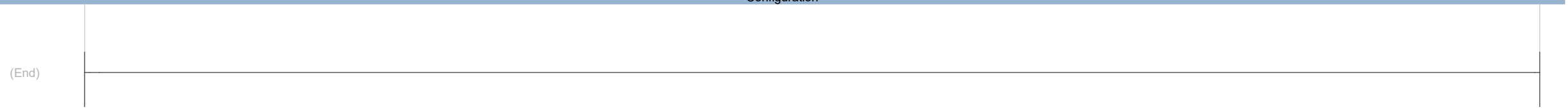
Event
Controls Array Size

MOVE
Source 32
Dest EDC_Event.CFG.BIT.n 32

Event
Controls Array Max
Index

SUB
Source A EDC_Event.CFG.BIT.n 32
Source B 1
Dest EDC_Event.CFG.BIT.mi 31

Configuration



Oview Display Zone/Mask Initialization

Overview Display Text
Create Zone/Mask text string for Overview Display
("A, B, C") for display on Event Table Zoom

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	818
Dest	Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806549717
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806549717

Overview Configuration for Rank
(Overview Mask defaults to all Zones)

2

MOVE	
Source	EDC_HMI.CFG.Oview.RankNum 1
Dest	RankNum 1

MOVE	
Source	EDC_HMI.CFG.Oview.RankChar 4
Dest	RankChar 4

Oview Display Zone/Mask Initialization

Overview shows all Zones - create ZoneList showing range of first and last Zones
(i.e. "A-T" for 20 Zones)

3

ASCII
Zone Letter

FLL	
Source	0
Dest	ZoneList
Length	1

ADD	
Source A	ascii.A <EDC.ascii.A> 'A'
Source B	EDC_HMI.CFG.ZN.si 0
Dest	ZoneLetter 'H'

MOVE	
Source	ZoneLetter 'H'
Dest	ZoneList.DATA[ZoneList.LEN] '$00'

ADD	
Source A	ZoneList.LEN 3
Source B	1
Dest	ZoneList.LEN 3

MOVE	
Source	ascii.Hyphen <EDC.ascii.Hyphen> '-'
Dest	ZoneList.DATA[ZoneList.LEN] '$00'

ADD	
Source A	ZoneList.LEN 3
Source B	1
Dest	ZoneList.LEN 3

ASCII
Zone Letter

ADD	
Source A	ascii.A <EDC.ascii.A> 'A'
Source B	EDC_HMI.CFG.ZN.mi 20
Dest	ZoneLetter 'H'

MOVE	
Source	ZoneLetter 'H'
Dest	ZoneList.DATA[ZoneList.LEN] '$00'

ADD	
Source A	ZoneList.LEN 3
Source B	1
Dest	ZoneList.LEN 3

Convert RankNum and RankChar to RankList
(i.e. "4C")

4

FLL	
Source	0
Dest	RankList
Length	1

MOVE	
Source	2
Dest	RankList.LEN 2

ADD	
Source A	RankNum 1
Source B	48
Dest	RankList.DATA[0] '1'

ADD	
Source A	RankChar 4
Source B	64
Dest	RankList.DATA[1] 'D'

Oview Display Zone/Mask Initialization

Save ZoneList and RankList to HMI Display

5

HMI Zone List ("A,B,C")

COP	
Source	ZoneList
Dest	EDC_HMI.Oview.ZoneList
Length	1

HMI Rank List ("A1,A2, B1,C1")

COP	
Source	RankList
Dest	EDC_HMI.Oview.RankList
Length	1

(End)

Zone Display Zone/Mask Initialization

Zone Display Text
Create Zone/Mask text string for Zone Display
("A, B, C") for display on n Zone Tables

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 819
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Zone Configuration for Mask and Rank

2

MOVE
Source EDC_HMI.CFG.Zone[ZNi].ZoneMask 16#0000_0000
Dest ZoneMask 16#0000_0000

MOVE
Source EDC_HMI.CFG.Zone[ZNi].RankNum 1
Dest RankNum 1

MOVE
Source EDC_HMI.CFG.Zone[ZNi].RankChar 4
Dest RankChar 4

Intialize Zone Loop

3

AddComma
U

FLL
Source 0
Dest ZoneList
Length 1

CLR
Dest LOOPi 21

Zone Display Zone/Mask Initialization

Convert ZoneMask to ZoneList
(i.e. 00-0000-0000-0000-0011 = "AB")

add each Zone letter to Zone List
eliminate commas since there are 20 Zones

4

lbl_ZoneLoop
LBL

ZoneMask.[LOOPi]

AddComma
AFI

MOVE
Source ascii.Comma
<EDC.ascii.Comma>
','
Dest ZoneList.DATA[ZoneList.LEN]
'$00'

ADD
Source A ZoneList.LEN
3
Source B 1
Dest ZoneList.LEN
3

ASCII
Zone Letter

ADD
Source A ascii.A
<EDC.ascii.A>
'A'
Source B LOOPi
21
Dest ZoneLetter
'H'

MOVE
Source ZoneLetter
'H'
Dest ZoneList.DATA[ZoneList.LEN]
'$00'

ADD
Source A ZoneList.LEN
3
Source B 1
Dest ZoneList.LEN
3

AddComma
L

Increment Loop

5

ADD
Source A LOOPi
21
Source B 1
Dest LOOPi
21

LE
Source A LOOPi
21
Source B EDC_HMI.CFG.ZN.mi
20

lbl_ZoneLoop
JMP

If No Zones assigned, set list to N/A

6

EQ
Source A ZoneList.LEN
3
Source B 0

COP
Source strNA
Dest ZoneList
Length 1

Zone Display Zone/Mask Initialization

Convert RankNum and RankChar to RankList
(i.e. "4C")

7

FLL
Source 0
Dest RankList
Length 1

MOVE
Source 2
Dest RankList.LEN
2

ADD
Source A RankNum
1
Source B 48
Dest RankList.DATA[0]
'1'

ADD
Source A RankChar
4
Source B 64
Dest RankList.DATA[1]
'D'

Save ZoneList and RankList to HMI Display

8

HMI Maintenance Zone Displays

COP
Source ZoneList
Dest EDC_HMI.Zone[ZNi].ZoneList
Length 1

HMI Maintenance Zone Displays

COP
Source RankList
Dest EDC_HMI.Zone[ZNi].RankList
Length 1

(End)

Configuration Request to Server via Transaction Manager

Configuration Data Request to SQL Server via Transaction Manager

- Logix Requests Data (-1 Command)
- Server Responds with Data (1-18 Command), No Data (0 Command) or No Response (-1)

- if Data, Logix processes Data and Sends Ack (Echo Server Command)
- Server uses Ack to set Download Timestamp for Config Record
- Server Responds with next Data or NoData

- if No Data, Logix waits and then Requests Data (-1 Command)

0 NOP

Request Commands to TM/SQL

-1 = New Data Request (delay between requests)
1-18 = Ack Data Response and New Data Request (no delay between requests)
99 - Full Download Request

1 NOP

Response Commands from Server

-1 = Data Error
0 = No Data to Download
1 = Line Configuration
2 = Offline Groups Configuration
3 = HMI Overview Display Configuration
4 = HMI Zone Display Configuration
5 = Equipment Configuration
6 = Schedule Hours Configuration
7 = Gateway Configuration
9 = Carrier Scanner Configuration
16 = Equipment Event Configuration - PreDefined
17 = Equipment Event Configuration - EqDefined
18 = Shop Configuration (Reset and Download All)

2 NOP

Configuration Request to Server via Transaction Manager

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

3

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	820
Dest	Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806549717
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806549717

Configuration Request to Server via Transaction Manager

Transaction Request Timer
Fast Interval used when last TM Response had data

4

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Response Duplicate Error
ServerCfg.Ctl.RspDupER
<EDC.ServerCfg.Ctl.RspDupER>

Server Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerCfg.Ctl.RspCmd0
<EDC.ServerCfg.Ctl.RspCmd0>

Server Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerCfg.Ctl.ReqFastTime
<EDC.ServerCfg.Ctl.ReqFastTime>
0
Dest ServerCfg.Ctl.ReqNextTMR.PRE
<EDC.ServerCfg.Ctl.ReqNextTMR.PRE>
10000

Server Configuration from Transaction Manager Request Next Record Timer

TON
Timer ServerCfg.Ctl.ReqNextTMR
<EDC.ServerCfg.Ctl.ReqNextTMR>
Preset 10000
Accum 7766
EN
DN

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Response Duplicate Error
ServerCfg.Ctl.RspDupER
<EDC.ServerCfg.Ctl.RspDupER>

Server Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerCfg.Ctl.RspCmd0
<EDC.ServerCfg.Ctl.RspCmd0>

Server Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerCfg.Ctl.ReqSlowTime
<EDC.ServerCfg.Ctl.ReqSlowTime>
10000
Dest ServerCfg.Ctl.ReqNextTMR.PRE
<EDC.ServerCfg.Ctl.ReqNextTMR.PRE>
10000

Request Next
Ready for next Transaction Manager Trigger

5

Server Configuration from Transaction Manager Request Next Record Timer
ServerCfg.Ctl.ReqNextTMR.DN
<EDC.ServerCfg.Ctl.ReqNextTMR.DN>

Server Configuration from Transaction Manager Request Next Record
ServerCfg.Ctl.ReqNext
<EDC.ServerCfg.Ctl.ReqNext>

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>
L

Server Configuration from Transaction Manager Request Record Complete
ServerCfg.Ctl.ReqComplete
<EDC.ServerCfg.Ctl.ReqComplete>
U

Configuration Request to Server via Transaction Manager

Request Next Initialize
if Response Good Data, echo Response in next Request to Ack
if not Response Good Data, send new Data Request after wait interval

Mar 24, 2024 -- added ReqInProgressTimeout
instead of resending last Request, Init a new request
(was locking up on repeated Gateway Download requests because SQL response >1 minute)

6

Server Configuration from Transaction Manager Request Next Record
ServerCfg.Ctl.ReqNext
<EDC.ServerCfg.Ctl.ReqNext>

Server Configuration from Transaction Manager Request Record in Progress Timeout
ServerCfg.Ctl.ReqInProgressTimeout
<EDC.ServerCfg.Ctl.ReqInProgressTimeout>

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Configuration Request to Server via Transaction Manager - Data Init

JSR
Routine Name _821_CfgRequestInit

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Configuration Request to Server via Transaction Manager

Request Next
Increment Transaction Number
(Triggers TM Transaction on change)

7

Server Configuration from Transaction Manager Request Next Record
ServerCfg.Ctl.ReqNext
<EDC.ServerCfg.Ctl.ReqNext>

Server Configuration from Transaction Manager Request Record in Progress Timeout
ServerCfg.Ctl.ReqInProgressTimeout
<EDC.ServerCfg.Ctl.ReqInProgressTimeout>

Server Configuration from Transaction Manager Transaction Number (TM Request Trigger)

ADD
Source A ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721
Source B 1
Dest ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721

Server Configuration from Transaction Manager Transaction Number (TM Request Trigger)

LT
Source A ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721
Source B 1

Server Configuration from Transaction Manager Transaction Number (TM Request Trigger)

GT
Source A ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Server Configuration from Transaction Manager Transaction Number (TM Request Trigger)

MOVE
Source 1
Dest ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721

Configuration Request to Server via Transaction Manager

8

Request Next
Log Transaction & Performance

Server Configuration from Transaction Manager Request Next Record
ServerCfg.Ctl.ReqNext
<EDC.ServerCfg.Ctl.ReqNext>

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	4000
Input Par	ServerCfg.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerCfg.DataReq.Command
	<EDC.ServerCfg.DataReq.Command>
Input Par	ServerCfg.Ctl.CommandID
	<EDC.ServerCfg.Ctl.CommandID>
Input Par	ServerCfg.Ctl.TransactionNumber
	<EDC.ServerCfg.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR	
Dest	TempPerfMsg.TransTMR.ACC
	110

Server Configuration from Transaction Manager Request Record in Progress Timeout
ServerCfg.Ctl.ReqInProgressTimeout
<EDC.ServerCfg.Ctl.ReqInProgressTimeout>

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	4002
Input Par	ServerCfg.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerCfg.DataReq.Command
	<EDC.ServerCfg.DataReq.Command>
Input Par	ServerCfg.Ctl.CommandID
	<EDC.ServerCfg.Ctl.CommandID>
Input Par	ServerCfg.Ctl.TransactionNumber
	<EDC.ServerCfg.Ctl.TransactionNumber>

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans
PerfMsg	TempPerfMsg
PerfTrans	EDC_Status01.ServerCfg.Msg.EN

Configuration Request to Server via Transaction Manager

Request Complete
Wait for TM Handshake to Acknowledge Request Complete
v3.5 - replace bit-handshiake with TransNum echo

9

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>

Server Configuration from Transaction Manager Transaction Number (TM Request Trigger)

EQ
Source A ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>
8721
Source B ServerCfg.Ctl.Interface
<EDC.ServerCfg.Ctl.Interface>
8721

Server Configuration from Transaction Manager Request Record Complete
ServerCfg.Ctl.ReqComplete
<EDC.ServerCfg.Ctl.ReqComplete>
L

Configuration Request to Server via Transaction Manager

Request Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
Also problems with asynchronous data from TM - handshake can update before data values.
2-Check method is fast and avoids errors for occasional slow update of data.
Wait 1 scan before 1st check to eliminate 99% of missing data issues.
2nd check after 500msec if Check 1 fails.
(anything >2 seconds will generate a TM Transaction error based on Tag Retrieval rate)

10

Server Configuration from Transaction Manager Request Record Complete
ServerCfg.Ctl.ReqComplete
<EDC.ServerCfg.Ctl.ReqComplete>

Server Configuration from Transaction Manager Request Record First Check Timer

MOVE
Source 10
Dest ServerCfg.Ctl.ReqCompleteCheckTMR.PRE
<EDC.ServerCfg.Ctl.ReqCompleteCheckTMR.PRE>
10

Server Configuration from Transaction Manager Request Record First Check Timer

TON
Timer ServerCfg.Ctl.ReqCompleteCheckTMR
<EDC.ServerCfg.Ctl.ReqCompleteCheckTMR>
Preset 10
Accum 16
EN
DN

Server Configuration from Transaction Manager Request Record First Check Timer
ServerCfg.Ctl.ReqCompleteCheckTMR.DN
<EDC.ServerCfg.Ctl.ReqCompleteCheckTMR.DN>

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Request Complete Delay Timer

MOVE
Source 500
Dest ServerCfg.Ctl.ReqCompleteDelayTMR.PRE
<EDC.ServerCfg.Ctl.ReqCompleteDelayTMR.PRE>
500

Server Configuration from Transaction Manager Request Complete Delay Timer

TON
Timer ServerCfg.Ctl.ReqCompleteDelayTMR
<EDC.ServerCfg.Ctl.ReqCompleteDelayTMR>
Preset 500
Accum 507
EN
DN

Server Configuration from Transaction Manager Request Complete Delay Timer
ServerCfg.Ctl.ReqCompleteDelayTMR.DN
<EDC.ServerCfg.Ctl.ReqCompleteDelayTMR.DN>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Configuration Request to Server via Transaction Manager

Request Complete - Response Received
Log Transaction & Performance
(Delay needed to allow all TM Data values to arrive - was receiving Handshake before Data - order of TM variables not enforced in data update.)

11

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Request Record First Check 1-shot
ServerCfg.Ctl.ReqCompleteCheckONS
<EDC.ServerCfg.Ctl.ReqCompleteCheckONS>
ONS

Server Configuration from Transaction Manager Request Complete 1st Check Count

ADD
Source A ServerCfg.Ctl.ReqComp1stChkCnt
<EDC.ServerCfg.Ctl.ReqComp1stChkCnt>
1663573
Source B 1
Dest ServerCfg.Ctl.ReqComp1stChkCnt
<EDC.ServerCfg.Ctl.ReqComp1stChkCnt>
1663573

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Server Configuration from Transaction Manager Request Record Complete 1-shot
ServerCfg.Ctl.ReqCompleteONS
<EDC.ServerCfg.Ctl.ReqCompleteONS>
ONS

Server Configuration from Transaction Manager Request Complete 2nd Check Count

ADD
Source A ServerCfg.Ctl.ReqComp2ndChkCnt
<EDC.ServerCfg.Ctl.ReqComp2ndChkCnt>
2506
Source B 1
Dest ServerCfg.Ctl.ReqComp2ndChkCnt
<EDC.ServerCfg.Ctl.ReqComp2ndChkCnt>
2506

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4001
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataReq.Command
<EDC.ServerCfg.DataReq.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Green - Msg Rx DN
EDC_Status01.ServerCfg.Update.MsgRxOK
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

Equipment Data

Configuration Request to Server via Transaction Manager

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerCfg.Msg.DN

Configuration Response from Server via Transaction Manager

JSR
Routine Name _830_CfgResponse

Request Complete Delay Reset Request

12

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>
U

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Send new Request

13

Server Configuration from Transaction Manager Request Record in Progress
ServerCfg.Ctl.ReqInProgress
<EDC.ServerCfg.Ctl.ReqInProgress>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Server Configuration from Transaction Manager Request in Progress Timer

MOVE
Source 12000
Dest ServerCfg.Ctl.ReqInProgressTMR.PRE
<EDC.ServerCfg.Ctl.ReqInProgressTMR.PRE>
12000

Server Configuration from Transaction Manager Request in Progress Timer

TON
Timer ServerCfg.Ctl.ReqInProgressTMR
<EDC.ServerCfg.Ctl.ReqInProgressTMR>
Preset 12000
Accum 0
EN
DN

Server Configuration from Transaction Manager Request in Progress Timer
ServerCfg.Ctl.ReqInProgressTMR.DN
<EDC.ServerCfg.Ctl.ReqInProgressTMR.DN>

Server Configuration from Transaction Manager Request Record in Progress Timeout
ServerCfg.Ctl.ReqInProgressTimeout
<EDC.ServerCfg.Ctl.ReqInProgressTimeout>

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Log Transaction & Performance

14

Server Configuration from Transaction Manager Request Record in Progress Timeout
ServerCfg.Ctl.ReqInProgressTimeout
<EDC.ServerCfg.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4004
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataReq.Command
<EDC.ServerCfg.DataReq.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- Msg Rx ER
EDC_Status01.ServerCfg.Update.MsgRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerCfg.Msg.ER

Server Configuration from Transaction Manager Request in Progress Timer
ServerCfg.Ctl.ReqInProgressTMR
<EDC.ServerCfg.Ctl.ReqInProgressTMR>
RES

(End)

Configuration Request to Server via Transaction Manager - Data Init

Request Next Initialize
if Response Good Data, echo Response in next Request to Server to Ack
if not Response Good Data, send new Data Request after wait interval

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 821
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Request to Server via Transaction Manager - Data Init

DN011 Separate Req & Rsp Tags
Create Data Init Packet
(FLL -1 corrupts FLOAT values and causes TM Transaction Failure; overwrite with MOV)

2

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager - Data
FLL
Source -1
Dest ServerCfgInit
Length 1

Server Configuration from Transaction Manager - Data Name
FLL
Source 0
Dest ServerCfgInit.Name
Length 1

Server Configuration from Transaction Manager - Data Short Name
FLL
Source 0
Dest ServerCfgInit.ShortName
Length 1

Server Configuration from Transaction Manager - Data Equipment Event Text
FLL
Source 0
Dest ServerCfgInit.EqEventText
Length 1

Server Configuration from Transaction Manager - Data Name
MOVE
Source 1
Dest ServerCfgInit.Name.DATA[0] '$01'

Server Configuration from Transaction Manager - Data Short Name
MOVE
Source 1
Dest ServerCfgInit.ShortName.DATA[0] '$01'

Server Configuration from Transaction Manager - Data Equipment Event Text
MOVE
Source 1
Dest ServerCfgInit.EqEventText.DATA[0] '$01'

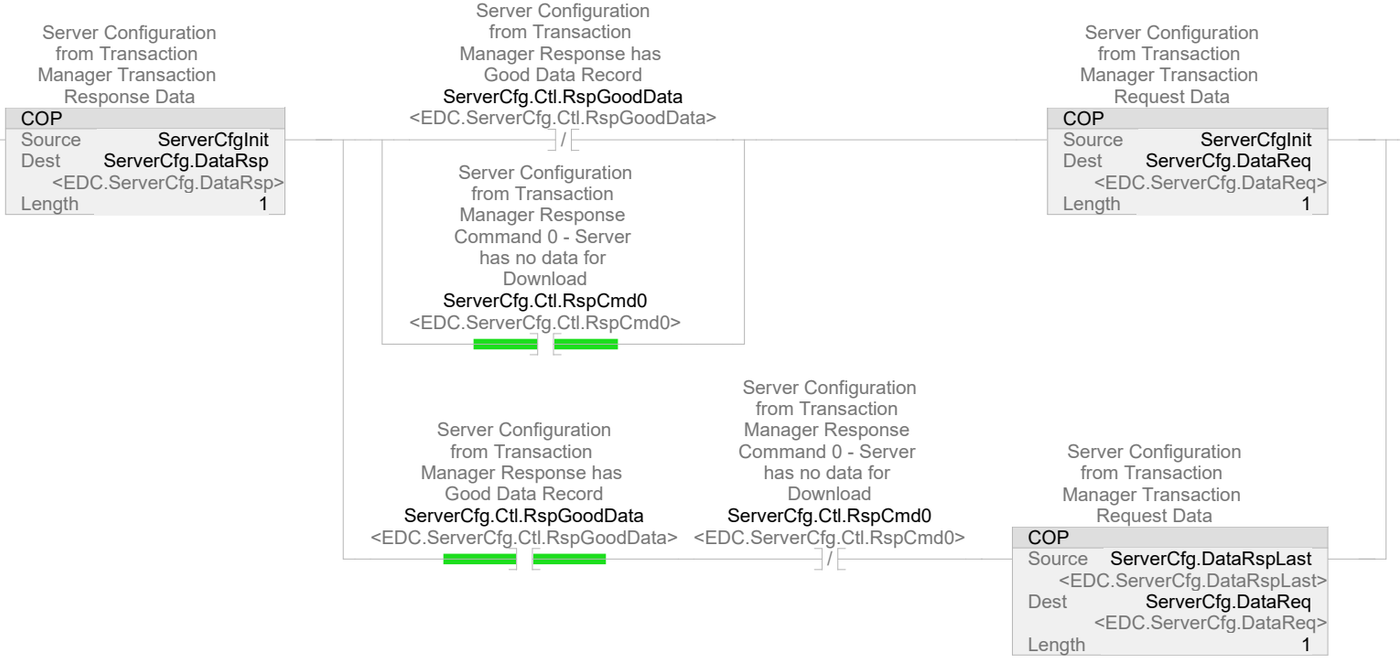
Server Configuration from Transaction Manager - Data Target OA - Green Status
MOVE
Source -1
Dest ServerCfgInit.TargetOAGrn -1.0

Server Configuration from Transaction Manager - Data Target OA - Yellow Status
MOVE
Source -1
Dest ServerCfgInit.TargetOAYel -1.0

Configuration Request to Server via Transaction Manager - Data Init

DN011 Separate Req & Rsp Tags
Clear Response to ensure new data on next DB Response
DHF017 Jan 30, 2018 -- Clear Server Request if no Ack for Good Message Response
Copy Response to Request for Good Data Ack to DB

3



Configuration Request to Server via Transaction Manager - Data Init

Config Download ALL Request - Command 99
Automatic Sunday morning Defragment of Event Text

DHF012 - Fixed Nov/Dec bug in aoiEDC_DayofWeek_v1_2
DN009 - perform Sunday Config Download ALL regardless of Status Event Defrag count
(this also removes any newly disabled Equipment from the Overview)

4

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>

CLx Module Status Hour (HH)

EQ
Source A CLX.TransTime.Hour
<EDC.CLX.TransTime.Hour>
15
Source B 6

CLx Module Status Minute (MM)

EQ
Source A CLX.TransTime.Min
<EDC.CLX.TransTime.Min>
19
Source B 0

CLx Module Status CLx Wallclock Day of Week (0=Saturday, 6=Friday)

EQ
Source A CLX.DayOfWeek
<EDC.CLX.DayOfWeek>
6
Source B 1

Server Configuration from Transaction Manager Request/Response Command

MOVE
Source 99
Dest ServerCfg.DataReq.Command
<EDC.ServerCfg.DataReq.Command>
-1

Initiate Full Download (manual set)
ManualFullDownload

Request Full Configuration Download (from Gway Reset)
CFG_FullDownload

Initiate Full Download (manual set)
ManualFullDownload
U

Request Full Configuration Download (from Gway Reset)
CFG_FullDownload
U

Store Last Request

5

Server Configuration from Transaction Manager Transaction Last Request Data

COP
Source ServerCfg.DataReq
<EDC.ServerCfg.DataReq>
Dest ServerCfg.DataReqLast
<EDC.ServerCfg.DataReqLast>
Length 1

(End)

Configuration Response from Server via Transaction Manager

Configuration Response from Server via Transaction Manager
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 830
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Detect Duplicate Config Data Response
(this indicates the database is stuck on a bad config item)

2

Server Configuration from Transaction Manager Request/Response Command

NE
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 0

Server Configuration from Transaction Manager Request/Response Command

NE
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B -1

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B ServerCfg.DataRsp.Command
<EDC.ServerCfg.DataRsp.Command>
0

Server Configuration from Transaction Manager Shop Key - for DB Ack only

EQ
Source A ServerCfg.DataRspLast.ShopKey
<EDC.ServerCfg.DataRspLast.ShopKey>
-1
Source B ServerCfg.DataRsp.ShopKey
<EDC.ServerCfg.DataRsp.ShopKey>
-1

Server Configuration from Transaction Manager Line Key - for DB Ack only

EQ
Source A ServerCfg.DataRspLast.LineKey
<EDC.ServerCfg.DataRspLast.LineKey>
-1
Source B ServerCfg.DataRsp.LineKey
<EDC.ServerCfg.DataRsp.LineKey>
-1

Server Configuration from Transaction Manager Equip Key - for DB Ack only

EQ
Source A ServerCfg.DataRspLast.EqKey
<EDC.ServerCfg.DataRspLast.EqKey>
-1
Source B ServerCfg.DataRsp.EqKey
<EDC.ServerCfg.DataRsp.EqKey>
-1

Server Configuration from Transaction Manager Equipment ID

EQ
Source A ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Source B ServerCfg.DataRsp.EqID
<EDC.ServerCfg.DataRsp.EqID>
-1

Server Configuration from Transaction Manager Line ID (0=Offline)

EQ
Source A ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Source B ServerCfg.DataRsp.LineID
<EDC.ServerCfg.DataRsp.LineID>
-1

Server Configuration from Transaction Manager Pitch ID (0-350)

EQ
Source A ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID>
-1
Source B ServerCfg.DataRsp.PitchID
<EDC.ServerCfg.DataRsp.PitchID>
-1

Server Configuration from Transaction Manager Position ID (0-5)

EQ
Source A ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID>
-1
Source B ServerCfg.DataRsp.PositionID
<EDC.ServerCfg.DataRsp.PositionID>
-1

Server Configuration from Transaction Manager Maintenance Zone Display ID (1-8)

EQ
Source A ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Source B ServerCfg.DataRsp.ZoneDispID
<EDC.ServerCfg.DataRsp.ZoneDispID>
-1

Server Configuration from Transaction Manager Offline Group ID (1-4)

EQ
Source A ServerCfg.DataRspLast.OfflineGroupID
<EDC.ServerCfg.DataRspLast.OfflineGroupID>
-1
Source B ServerCfg.DataRsp.OfflineGroupID
<EDC.ServerCfg.DataRsp.OfflineGroupID>
-1

Server Configuration from Transaction Manager Equipment Event ID - EQID & Element ID

EQ
Source A ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Source B ServerCfg.DataRsp.EqEventID
<EDC.ServerCfg.DataRsp.EqEventID>
-1

Server Configuration from Transaction Manager Equipment Event ID - EQID & Element ID

EQ
Source A ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Source B ServerCfg.DataRsp.EqEventID
<EDC.ServerCfg.DataRsp.EqEventID>
-1

Server Configuration from Transaction Manager Response Duplicate
ServerCfg.Ctl.RspDup
<EDC.ServerCfg.Ctl.RspDup>

Configuration Response from Server via Transaction Manager

3

Duplicate Message Count and Detection
(this indicates the database is stuck on a bad config item)

Server Configuration from Transaction Manager Response Duplicate
ServerCfg.Ctl.RspDup
<EDC.ServerCfg.Ctl.RspDup>

ADD
Source A RspDuplicateCount 0
Source B 1
Dest RspDuplicateCount 0

GT
Source A RspDuplicateCount 0
Source B RspDuplicateMax 53

MOVE
Source RspDuplicateCount 0
Dest RspDuplicateMax 53

GT
Source A RspDuplicateCount 0
Source B 5

Server Configuration from Transaction Manager Response Duplicate Error
ServerCfg.Ctl.RspDupER
<EDC.ServerCfg.Ctl.RspDupER>

Server Configuration from Transaction Manager Response Duplicate
ServerCfg.Ctl.RspDup
<EDC.ServerCfg.Ctl.RspDup>

CLR
Dest RspDuplicateCount 0

4

Move Response to working tag and Initialize Response Good Data
DN011 Separate Req & Rsp Tags

Moving DataRsp to DataLastRsp prevents SQL/TM from changing data while Logix is processing.

Server Configuration from Transaction Manager Transaction Last Response Data

COP
Source ServerCfg.DataRsp <EDC.ServerCfg.DataRsp>
Dest ServerCfg.DataRspLast <EDC.ServerCfg.DataRspLast>
Length 1

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Configuration Response from Server via Transaction Manager

Check Response Command for Bad Data
Valid Commands 0, 1-7, 9, 16-18

5

Server Configuration from Transaction Manager Request/Response Command

LT
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 0

Server Configuration from Transaction Manager Request/Response Command

LIMIT
Low Limit 8
Test ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
High Limit 8

Server Configuration from Transaction Manager Request/Response Command

LIMIT
Low Limit 10
Test ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
High Limit 15

Server Configuration from Transaction Manager Request/Response Command

GT
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 18

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4012
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerCfg.Msg.KA

lbl_RspComp
JMP

Configuration Response from Server via Transaction Manager

If Response Command received, process Response

6

RspCommandRcvd

Server has no new Configuration data to download

7

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 0

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Server Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerCfg.Ctl.RspCmd0
<EDC.ServerCfg.Ctl.RspCmd0>

Server Response Command diagnostic count

ADD
Source A RspCmd[0]
44710
Source B 1
Dest RspCmd[0]
44710

Process Response Command

8

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 1

Configuration Response from Server via Transaction Manager

JSR
Routine Name _831_CfgResponseLine

Server Response Command diagnostic count

ADD
Source A RspCmd[1]
29
Source B 1
Dest RspCmd[1]
29

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 2

Configuration Response from Server via Transaction Manager

JSR
Routine Name _832_CfgResponseOfflineGroup

Server Response Command diagnostic count

ADD
Source A RspCmd[2]
4
Source B 1
Dest RspCmd[2]
4

Configuration Response from Server via Transaction Manager

Process Response Command

9

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 3

Configuration Response from Server via Transaction Manager

JSR
Routine Name _833_CfgResponseOviewDisplay

Server Response Command diagnostic count

ADD
Source A RspCmd[3]
1
Source B 1
Dest RspCmd[3]
1

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 4

Configuration Response from Server via Transaction Manager

JSR
Routine Name _834_CfgResponseZoneDisplay

Server Response Command diagnostic count

ADD
Source A RspCmd[4]
21
Source B 1
Dest RspCmd[4]
21

Configuration Response from Server via Transaction Manager

Process Response Command

10

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
Source B 5

Configuration Response from Server via Transaction Manager

JSR
Routine Name _835_CfgResponseEquipment

Server Response Command diagnostic count

ADD
Source A RspCmd[5] 300
Source B 1
Dest RspCmd[5] 300

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
Source B 6

Configuration Response from Server via Transaction Manager

JSR
Routine Name _840_CfgResponseSchedule

Server Response Command diagnostic count

ADD
Source A RspCmd[6] 23
Source B 1
Dest RspCmd[6] 23

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
Source B 7

Configuration Response from Server via Transaction Manager

JSR
Routine Name _841_CfgResponseGateway

Server Response Command diagnostic count

ADD
Source A RspCmd[7] 1
Source B 1
Dest RspCmd[7] 1

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command> 0
Source B 9

Configuration Response from Server via Transaction Manager

JSR
Routine Name _839_CfgResponseCarrierScanner

Server Response Command diagnostic count

ADD
Source A RspCmd[9] 5
Source B 1
Dest RspCmd[9] 5

Configuration Response from Server via Transaction Manager

Process Response Command

11

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 16

Server Response Command diagnostic count

ADD
Source A RspCmd[16]
32
Source B 1
Dest RspCmd[16]
32

Server Configuration from Transaction Manager Download Started - Event Received
ServerCfg.Ctl.DownloadEventRcvd
<EDC.ServerCfg.Ctl.DownloadEventRcvd>

Configuration Response from Server via Transaction Manager

JSR
Routine Name _836_CfgResponseEquipmentEventTxt

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 17

Server Response Command diagnostic count

ADD
Source A RspCmd[17]
96886
Source B 1
Dest RspCmd[17]
96886

Process Response Command

12

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 18

Configuration Response from Server via Transaction Manager

JSR
Routine Name _838_CfgResponseShop

Server Response Command diagnostic count

ADD
Source A RspCmd[18]
2
Source B 1
Dest RspCmd[18]
2

Command Response Complete

13

lbl_RspComp
LBL

NOP

Configuration Response from Server via Transaction Manager

Store Last Good, Bad or Incomplete Response w/ Diagnostic Counts
Incomplete - Command Num received with incomplete data packet on on 1st Check
Bad - Incomplete data packet on 2nd Check

14

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerCfg.Ctl.RspCmd0
<EDC.ServerCfg.Ctl.RspCmd0>

Server Configuration from Transaction Manager Transaction Last Good Response Data

COP
Source ServerCfg.DataRspLast
<EDC.ServerCfg.DataRspLast>
Dest ServerCfg.DataRspLastGood
<EDC.ServerCfg.DataRspLastGood>
Length 1

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Server Configuration from Transaction Manager Response Good on 1st Check

ADD
Source A ServerCfg.Ctl.RspGood1stChkCnt
<EDC.ServerCfg.Ctl.RspGood1stChkCnt>
1715275
Source B 1
Dest ServerCfg.Ctl.RspGood1stChkCnt
<EDC.ServerCfg.Ctl.RspGood1stChkCnt>
1715275

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Server Configuration from Transaction Manager Response Good on 2nd Check

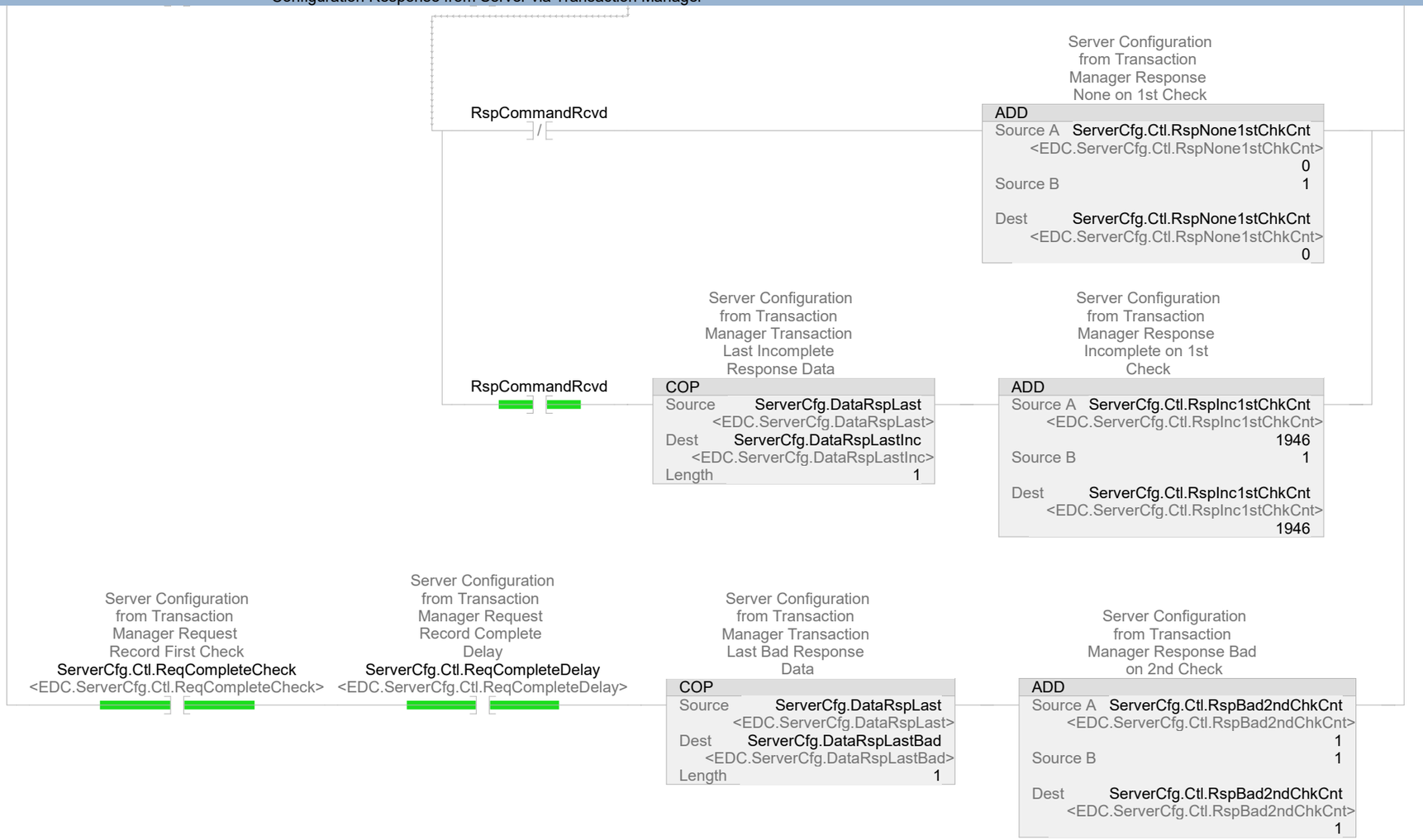
ADD
Source A ServerCfg.Ctl.RspGood2ndChkCnt
<EDC.ServerCfg.Ctl.RspGood2ndChkCnt>
3211
Source B 1
Dest ServerCfg.Ctl.RspGood2ndChkCnt
<EDC.ServerCfg.Ctl.RspGood2ndChkCnt>
3211

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Request Record First Check
ServerCfg.Ctl.ReqCompleteCheck
<EDC.ServerCfg.Ctl.ReqCompleteCheck>

Server Configuration from Transaction Manager Request Record Complete Delay
ServerCfg.Ctl.ReqCompleteDelay
<EDC.ServerCfg.Ctl.ReqCompleteDelay>

Configuration Response from Server via Transaction Manager



Configuration Response from Server via Transaction Manager

Download ALL configuration is complete

15

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>

Server Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerCfg.Ctl.RspCmd0
<EDC.ServerCfg.Ctl.RspCmd0>

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>
U

Server Configuration from Transaction Manager Response Download All Complete
ServerCfg.Ctl.DownloadAllComplete
<EDC.ServerCfg.Ctl.DownloadAllComplete>
L

Server Configuration from Transaction Manager Response Download All Complete 1-shot
ServerCfg.Ctl.DownloadAllCompleteONS
<EDC.ServerCfg.Ctl.DownloadAllCompleteONS>

HMI Refresh HMI Events after Config All
EDC_HMI.CTL.DownloadAllRefresh
L

EDC2 WL31 -- ensure BlockEqDriver doesn't get stuck on in a failed download
(better to collect data from Disabled than not collect from Enabled)

EDC2 WL31 Equipment Complete -- once download reaches the Events, all Equipment should have a good configuration
EDC2 WL31 Block EqDriver -- don't service EqDriver Messages until Equipment Configuration is complete and EQ Disable status can be re-established

16

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>
/

Server Configuration from Transaction Manager Download Started - Event Received
ServerCfg.Ctl.DownloadEventRcvd
<EDC.ServerCfg.Ctl.DownloadEventRcvd>

Server Configuration from Transaction Manager Download Started - Equipment Config Complete
ServerCfg.Ctl.DownloadEquipComplete
<EDC.ServerCfg.Ctl.DownloadEquipComplete>
L

Server Configuration from Transaction Manager Download Started - Block Equipment Drivers until Equipment Config Complete
ServerCfg.Ctl.DownloadBlockEqDriver
<EDC.ServerCfg.Ctl.DownloadBlockEqDriver>
U

Configuration Response from Server via Transaction Manager

DHF013 - Nov 27, 2017 -- Periodic Refresh of Event Text during Download ALL configuration

17

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>

Download All Periodic Refresh of Event Text

MUL
Source A 3
Source B 60000
Dest DownloadAllPeriodicRefreshTMR.PRE 180000

Download All Periodic Refresh of Event Text

TON
Timer DownloadAllPeriodicRefreshTMR
Preset 180000
Accum 0
EN
DN

Download All Periodic Refresh of Event Text
DownloadAllPeriodicRefreshTMR.DN

Download All Periodic Refresh of Event Text
DownloadAllPeriodicRefreshTMR
RES

HMI Refresh HMI Events after Config All
EDC_HMI.CTL.DownloadAllRefresh
L

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 1 - Line
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 831
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data
v3.5 only check LineID lower limit to determine Good Data (upper limit is only used to determine if LineID can be saved in array)

2

Server Configuration from Transaction Manager Line ID (0=Offline)

GE
Source A ServerCfg.DataRspLast.LineID <EDC.ServerCfg.DataRspLast.LineID> -1
Source B 0

Server Configuration from Transaction Manager Name

NE
Source A ServerCfg.DataRspLast.Name.DATA[0] <EDC.ServerCfg.DataRspLast.Name.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Short Name

NE
Source A ServerCfg.DataRspLast.ShortName.DATA[0] <EDC.ServerCfg.DataRspLast.ShortName.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Target OA - Green Status

GE
Source A ServerCfg.DataRspLast.TargetOAGrn <EDC.ServerCfg.DataRspLast.TargetOAGrn> -1.0
Source B 0

Server Configuration from Transaction Manager Target OA - Yellow Status

GE
Source A ServerCfg.DataRspLast.TargetOAYel <EDC.ServerCfg.DataRspLast.TargetOAYel> -1.0
Source B 0

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData <EDC.ServerCfg.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4013
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red-Line Rx Data ER
EDC_Status01.ServerCfg.Update.LineRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

v3.5 Check LIneID for valid Index
if SQL is configured for more Lines than Gateway, Gateway will ack the invalid LineID without saving
(the alternative is to continuously retry the invalid LineID index)

4

Server Configuration from Transaction Manager Line ID (0=Offline)

GT
Source A ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Source B CFG.LN.mi
<EDC.CFG.LN.mi>
28

lbl_ExitGoodStatus
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store

5

Line Index

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID> -1
Dest LNi
5

Equipment Data Collection Lines and Equipment Array

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC.Line[LNi].CFG.Name
Length 1

Equipment Data Collection Lines and Equipment Array

COP
Source ServerCfg.DataRspLast.ShortName
<EDC.ServerCfg.DataRspLast.ShortName>
Dest EDC.Line[LNi].CFG.ShortName
Length 1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.TargetOAGrn
<EDC.ServerCfg.DataRspLast.TargetOAGrn> -1.0
Dest EDC.Line[LNi].CFG.TargetOAGrn 96.0

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.TargetOAYel
<EDC.ServerCfg.DataRspLast.TargetOAYel> -1.0
Dest EDC.Line[LNi].CFG.TargetOAYel 94.0

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID> -1
Dest EDC.Line[LNi].CFG.Takt 56300

Check if Line is already configured for Offline

6

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].CFG.Offline

Line was configured for Offline
OffLineOn

Configuration Response from Server via Transaction Manager

DN006 - support multiple Offline Lines
If Line Name contains 'Offline' latch the Offline configuration

July 31, 2023 - added 'Other' to Offline search for eBW
Feb 12, 2024 - added UPPER for case-insensitive search of OFFLINE & OTHER

7

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].CFG.Offline
U

Server Configuration from Transaction Manager Name
UPPER
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
''
Dest strNameUPPER
'OFFLINE2'

Line Name contains 'Offline'
CLR
Dest OfflineFD
1

FIND
Source strNameUPPER
'OFFLINE2'
Search strOFFLINE
'OFFLINE'
Start 1
Result OfflineFD
1

Line Name contains 'Offline'
LIMIT
Low Limit 1
Test OfflineFD
1
High Limit strNameUPPER.LEN
8

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].CFG.Offline
L

Line Name contains 'Other'
CLR
Dest OtherFD
0

FIND
Source strNameUPPER
'OFFLINE2'
Search strOTHER
'OTHER'
Start 1
Result OtherFD
0

Line Name contains 'Other'
LIMIT
Low Limit 1
Test OtherFD
0
High Limit strNameUPPER.LEN
8

Line Index
LT
Source A LNi
5
Source B EDC.CFG.OfflineFirstLine
0

Equipment Data Collection First Offline Line (for multiline support during Processing)
MOVE
Source LNi
5
Dest EDC.CFG.OfflineFirstLine
0

Line Index
GT
Source A LNi
5
Source B EDC.CFG.OfflineLastLine
28

Equipment Data Collection Last Offline Line (for multiline support during Processing)
MOVE
Source LNi
5
Dest EDC.CFG.OfflineLastLine
28

Configuration Response from Server via Transaction Manager

when the Line's Offline status is On or stayed Off,
the OfflineFirstLine and OfflineLastLine are configured in previous rung
and can skip the update logic

8

Line was configured for Offline
OffLineOn
/

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].CFG.Offline

lbl_SkipOffline
JMP

when a Line's Offline status was On and turned Off,
must search for the new OfflineFirstLine and OfflineLastLine values

initialize Line index and OfflineFirstLine/OfflineLastLine values

9

CLR
Dest index
22

Equipment Data Collection First Offline Line (for multiline support during Processing)
CLR
Dest EDC.CFG.OfflineFirstLine
0

Equipment Data Collection Last Offline Line (for multiline support during Processing)
CLR
Dest EDC.CFG.OfflineLastLine
28

set OfflineFirstLine and OfflineLastLine values for Offline Lines

10

lbl_Offline
LBL

Equipment Data Collection Lines and Equipment Array
EDC.Line[index].CFG.Offline

LT
Source A index
22
Source B EDC.CFG.OfflineFirstLine
0

Equipment Data Collection First Offline Line (for multiline support during Processing)
MOVE
Source index
22
Dest EDC.CFG.OfflineFirstLine
0

GT
Source A index
22
Source B EDC.CFG.OfflineLastLine
28

Equipment Data Collection Last Offline Line (for multiline support during Processing)
MOVE
Source index
22
Dest EDC.CFG.OfflineLastLine
28

Configuration Response from Server via Transaction Manager

Increment Loop Index

11

ADD
Source A index 22
Source B 1
Dest index 22

LE
Source A index 22
Source B CFG.LN.mi <EDC.CFG.LN.mi> 28

lbl_Offline
JMP

12

lbl_SkipOffline
LBL

NOP

Data OK

Server Interface Status State = Red- Line Rx Data OK

Update Status State for Transactions

13

lbl_ExitGoodStatus
LBL

EDC_Status01.ServerCfg.Update.LineRxDataOK
L

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Exit Routine

14

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 2 - Offline Group
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 832
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data

2

Server Configuration from Transaction Manager Offline Group ID (1-4)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.OfflineGroupID <EDC.ServerCfg.DataRspLast.OfflineGroupID> -1
High Limit CFG.GRP.mi <EDC.CFG.GRP.mi> 3

Server Configuration from Transaction Manager Name

NE
Source A ServerCfg.DataRspLast.Name.DATA[0] <EDC.ServerCfg.DataRspLast.Name.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Short Name

NE
Source A ServerCfg.DataRspLast.ShortName.DATA[0] <EDC.ServerCfg.DataRspLast.ShortName.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Target OA - Green Status

GE
Source A ServerCfg.DataRspLast.TargetOAGrn <EDC.ServerCfg.DataRspLast.TargetOAGrn> -1.0
Source B 0

Server Configuration from Transaction Manager Target OA - Yellow Status

GE
Source A ServerCfg.DataRspLast.TargetOAYel <EDC.ServerCfg.DataRspLast.TargetOAYel> -1.0
Source B 0

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData <EDC.ServerCfg.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.OfflineGroupID
<EDC.ServerCfg.DataRspLast.OfflineGroupID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4014
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red-Offline Group Rx Data ER
EDC_Status01.ServerCfg.Update.OfflineRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store

4

MOVE
Source ServerCfg.DataRspLast.OfflineGroupID
<EDC.ServerCfg.DataRspLast.OfflineGroupID> -1
Dest GRPi
3

Equipment Data Collection Offline Group Configuration

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC.Offline.CFG[GRPi].Name
Length 1

HMI Offline Data Array by Group Index (1-Trim, 2-Chassis, 3-Final)

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC_HMI.Oview.Offline[GRPi].Name
Length 1

Equipment Data Collection Offline Group Configuration

COP
Source ServerCfg.DataRspLast.ShortName
<EDC.ServerCfg.DataRspLast.ShortName>
Dest EDC.Offline.CFG[GRPi].ShortName
Length 1

Equipment Data Collection Offline Group Configuration

MOVE
Source ServerCfg.DataRspLast.TargetOAGrn
<EDC.ServerCfg.DataRspLast.TargetOAGrn> -1.0
Dest EDC.Offline.CFG[GRPi].TargetOAGrn
96.0

Equipment Data Collection Offline Group Configuration

MOVE
Source ServerCfg.DataRspLast.TargetOAYel
<EDC.ServerCfg.DataRspLast.TargetOAYel> -1.0
Dest EDC.Offline.CFG[GRPi].TargetOAYel
94.0

Data OK

Server Interface Status State = Red-Offline Group Rx Data OK
EDC_Status01.ServerCfg.Update.OfflineRxDataOK
L

5

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

Exit Routine

lbl_Exit
6
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 3 - Overview Display
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 833
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data

2

Server Configuration from Transaction Manager Maintenance Risk Rank Assignments (4-1)

GE
Source A ServerCfg.DataRspLast.RankNum <EDC.ServerCfg.DataRspLast.RankNum> -1
Source B 0

Server Configuration from Transaction Manager Maintenance Rank Assignments (A-D = Values 1-4)

GE
Source A ServerCfg.DataRspLast.RankChar <EDC.ServerCfg.DataRspLast.RankChar> -1
Source B 0

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4015
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- HMI Oview Display Rx Data ER
EDC_Status01.ServerCfg.Update.HmiOviewRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Process Data and Store

4

HMI Maintenance Risk Rank Assignments (4-1)

MOVE
Source ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum>
-1
Dest EDC_HMI.CFG.Oview.RankNum
1

HMI Maintenance Rank Assignments (A-D = Values 1-4)

MOVE
Source ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar>
-1
Dest EDC_HMI.CFG.Oview.RankChar
4

Initialize Zone/Rank Text for Overview

5

Oview Display Zone/Mask Initialization

JSR
Routine Name _818_InitOview

Configuration Response from Server via Transaction Manager

Data OK

Server Interface
Status State = Red-
HMI Oview Display Rx
Data OK
EDC_Status01.ServerCfg.Update.HmiOviewRxDataOK

Update Status State
for Transactions
aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

6 L

Exit Routine

lbl_Exit
7 LBL NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 4 - Zone Display
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 834
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data
DHF034 June 2020 -- Remove Zone check, all values 0000-0000 thru 1111-1111 are legal
(when Zone H is checked, the value appears as a negative in the SINT signed integer)
Feb 2, 2023 - add Group Name

2

Server Configuration from Transaction Manager Maintenance Zone Display ID (1-8)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID> -1
High Limit EDC_HMI.CFG.ZN.mi 20

Server Configuration from Transaction Manager Name

NE
Source A ServerCfg.DataRspLast.Name.DATA[0]
<EDC.ServerCfg.DataRspLast.Name.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Maintenance Zone Mask (A-T)

GE
Source A ServerCfg.DataRspLast.Zone
<EDC.ServerCfg.DataRspLast.Zone> -1
Source B 0

Server Configuration from Transaction Manager Maintenance Risk Rank Assignments (4-1)

GE
Source A ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum> -1
Source B 0

Server Configuration from Transaction Manager Maintenance Rank Assignments (A-D = Values 1-4)

GE
Source A ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar> -1
Source B 0

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4016
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red-HMI Zone Display Rx Data ER
EDC_Status01.ServerCfg.Update.HmiZoneRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store
Feb 2023 - added Group Name

4

HMI Zone Display Config

MOVE
Source ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID> -1
Dest ZNi 20

HMI Zone Display Config

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC_HMI.CFG.Zone[ZNi].Name
Length 1

HMI Zone Display Config

MOVE
Source ServerCfg.DataRspLast.Zone
<EDC.ServerCfg.DataRspLast.Zone> -1
Dest EDC_HMI.CFG.Zone[ZNi].ZoneMask 16#0000_0000

HMI Zone Display Config

MOVE
Source ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum> -1
Dest EDC_HMI.CFG.Zone[ZNi].RankNum 1

HMI Zone Display Config

MOVE
Source ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar> -1
Dest EDC_HMI.CFG.Zone[ZNi].RankChar 4

Data OK

5

Server Interface Status State = Red- HMI Zone Display Rx Data OK

EDC_Status01.ServerCfg.Update.HmiZoneRxDataOK
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Initialize Zone/Rank Text for Zone w/ ZNi

6

Zone Display Zone/Mask Initialization

JSR
Routine Name _819_InitZone

Exit Routine

7

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 5 - Equipment
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 835
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Convert EQID to LNi and EQi

2

Equipment Index

MOD
Source A ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1
Source B 100
Dest EQi 31

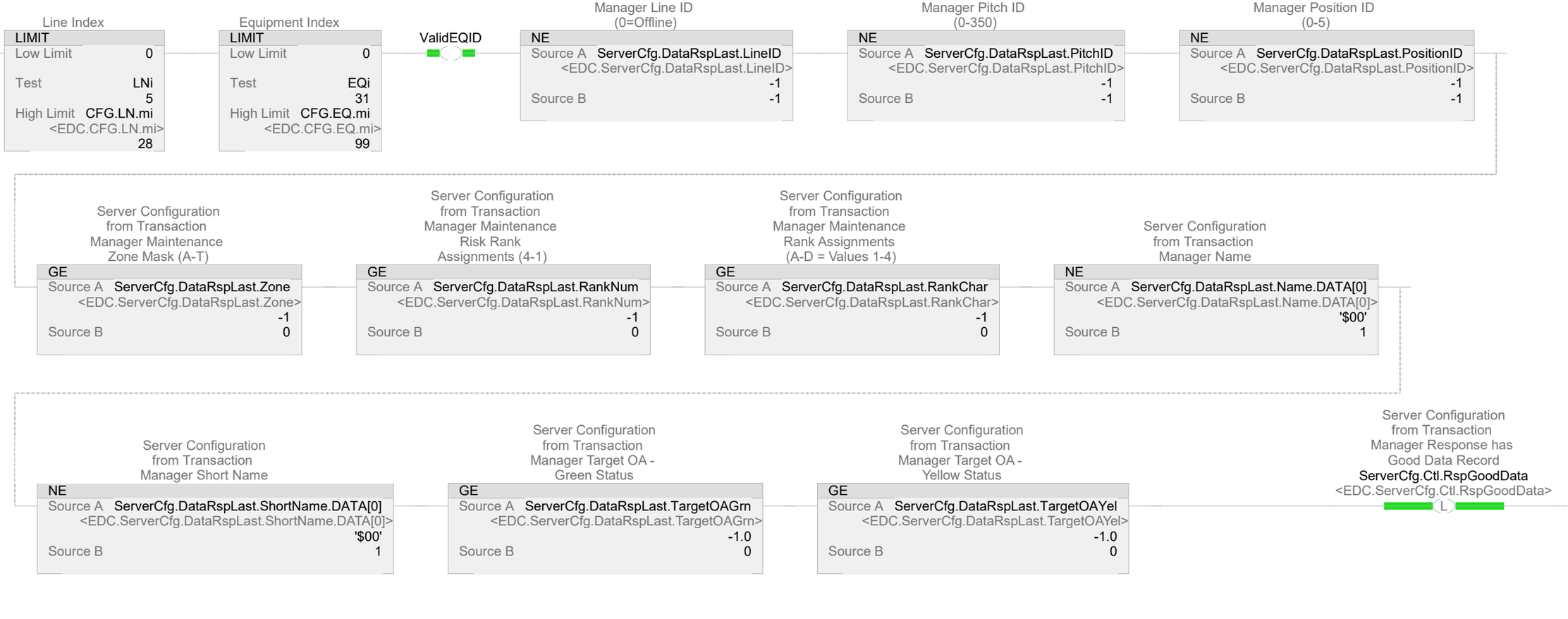
Line Index

CPT
Dest LNi 5
Expression (ServerCfg.DataRspLast.EqID - EQi) / 100

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

3



Configuration Response from Server via Transaction Manager

DN009 May 5, 2020 -- Acknowlege LNi and EQi for EQID 2900-9999
(Configuration allows EQID 1-9999 - values outside array limits are for decommissioned Equipment)

EDC2 - ack any EQID outside configured LNi, EQi
(unused EQID can be assigned to unconfigured LNi, EQi)

4

ValidEQID
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

L

lbl_DataOk
JMP

Data Error - Indexes out of Range
Log Transaction & Performance

5

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4017
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red - Equip Rx Data ER
EDC_Status01.ServerCfg.Update.EquipRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store

6

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.Line 5

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.Pitch 23

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.Position 1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.RankNum 1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.RankChar 3

Equipment Data Collection Lines and Equipment Array

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC.Line[LNi].Equip[EQi].CFG.Name
Length 1

Equipment Data Collection Lines and Equipment Array

COP
Source ServerCfg.DataRspLast.ShortName
<EDC.ServerCfg.DataRspLast.ShortName>
Dest EDC.Line[LNi].Equip[EQi].CFG.ShortName
Length 1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.TargetOAGrn
<EDC.ServerCfg.DataRspLast.TargetOAGrn> -1.0
Dest EDC.Line[LNi].Equip[EQi].CFG.TargetOAGrn 98.0

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.TargetOAYel
<EDC.ServerCfg.DataRspLast.TargetOAYel> -1.0
Dest EDC.Line[LNi].Equip[EQi].CFG.TargetOAYel 96.0

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID> -1
Dest EDC.Line[LNi].Equip[EQi].CFG.Takt 56300

Convert Zone Number to BIt
Zone 1-20 to Bits 0-19
(allows easy compare to Zone Masks)

7

Equipment Data Collection Lines and Equipment Array

CLR
Dest EDC.Line[LNi].Equip[EQi].CFG.Zone
2#0000_0000_0000_0000_0000_0000_0100_0000

Server Configuration from Transaction Manager Maintenance Zone Mask (A-T)

LIMIT
Low Limit 1
Test ServerCfg.DataRspLast.Zone
<EDC.ServerCfg.DataRspLast.Zone> -1
High Limit 20

SUB
Source A ServerCfg.DataRspLast.Zone
<EDC.ServerCfg.DataRspLast.Zone> -1
Source B 1
Dest ZnBITi 2

Equipment Data Collection Lines and Equipment Array

EDC.Line[LNi].Equip[EQi].CFG.Zone.[ZnBITi]
(L)

Configuration Response from Server via Transaction Manager

DN006 - support multiple Offline Lines

8

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.OfflineGroupID
<EDC.ServerCfg.DataRspLast.OfflineGroupID>
-1
Dest EDC.Line[LNi].Equip[EQi].CFGoffline.Group
0

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.OfflineGroupDisplayRow
<EDC.ServerCfg.DataRspLast.OfflineGroupDisplayRow>
-1
Dest EDC.Line[LNi].Equip[EQi].CFGoffline.DisplayRow
1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.OfflineGroupDisplayCol
<EDC.ServerCfg.DataRspLast.OfflineGroupDisplayCol>
-1
Dest EDC.Line[LNi].Equip[EQi].CFGoffline.DisplayCol
1

Equipment Data Collection Lines and Equipment Array

MOVE
Source ServerCfg.DataRspLast.OfflineGroupLine
<EDC.ServerCfg.DataRspLast.OfflineGroupLine>
-1
Dest EDC.Line[LNi].Equip[EQi].CFGoffline.Line
-1

Configuration Response from Server via Transaction Manager

v3.5 refresh Events if any config changes to Equipment
(don't wait for next Callout/Health update)

9

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.Update

L

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.UpdateOview

L

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.1
<EDC.ServerCfg.DataRspLast.ConfigStatus.1>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.DisableEvents

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.1
<EDC.ServerCfg.DataRspLast.ConfigStatus.1>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.DisableEvents

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.2
<EDC.ServerCfg.DataRspLast.ConfigStatus.2>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Decom

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.2
<EDC.ServerCfg.DataRspLast.ConfigStatus.2>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Decom

Equipment Data Collection Lines and Equipment Array

FLL

Source	-1
Dest	EDC.Line[LNi].Equip[EQi].EventCTL.UpdateZone
Length	1

Configuration Response from Server via Transaction Manager

Configuration Response from Server via Transaction Manager

DHF006 - New Configuration to support Disabled Equipment
Defined - exists in Configuration database
Disabled - temporarily or permanently out of commission - clear Callout table (this will clear Events)
CTL.DisableFaultMsg & CTL.DisableFaultMsgBulk sends Fault Messages for all Packets once to clear all Events in DB
CTL.DisableEvents refreshes Event Display once to clear any NoDisplay Events
v3.5 Add Data Refresh Request for re-Enabled Equipment

Bit 0 - Disable
Bit 1 - Disable Event Display
Bit 2 - Decommission

10

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Defined
L

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable
/

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.DisableFaultMsg
L

Equipment Data Collection Lines and Equipment Array
CLR
Dest EDC.Line[LNi].Equip[EQi].CTL.DisablePKTi
0

Server Configuration from Transaction Manager Command-specific Record ID for TransLog
MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 4033
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable
L

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.DisableEvents
L

Callout Table (LNi, EQi)
FLL
Source 0
Dest EDC_EqCallout[LNi,EQi]
Length 1

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>
/

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.DataRefreshReq

Configuration Response from Server via Transaction Manager

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4032
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Disable
U

EDC2 - v2.01 added Equip.CFG.DisableEvents to replace PreDefined.15 from EqDriver v1.x
Bit 0 - Disable
Bit 1 - Disable Event Display
Bit 2 - Decommission

11

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.1
<EDC.ServerCfg.DataRspLast.ConfigStatus.1>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.DisableEvents
L

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.DisableEvents
L

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.1
<EDC.ServerCfg.DataRspLast.ConfigStatus.1>
/

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.DisableEvents
U

Configuration Response from Server via Transaction Manager

v3.5 added Equip.CFG.Decom
Bit 0 - Disable
Bit 1 - Disable Event Display
Bit 2 - Decommission

12

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.2
<EDC.ServerCfg.DataRspLast.ConfigStatus.2>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Decom
L

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.2
<EDC.ServerCfg.DataRspLast.ConfigStatus.2>
/

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.Decom
U

Check for good Event Looklup Index for EDC_Event EvLKi

13

Equipment-Event Index for EDC_Event Lookup
CLR
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array
LIMIT
Low Limit 0
Test EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144
High Limit EDC_Event.CFG.EQ.mi 2999

Equipment-Event Index for EDC_Event Lookup
MOVE
Source EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144
Dest EvLKi 164

Event Lookup Index within array limits
EvLKi_IndexOK

Equipment-Event Index for EDC_Event matches

14

Event Lookup Index EQID Match
EvLKi_EQIDMatch
U

Event Lookup Index within array limits
EvLKi_IndexOK

Event Controls Equipment Fragment Statistics Array
EQ
Source A EDC_Event.CTL.Equip[EvLKi].EQID 926
Source B ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1

Event Lookup Index EQID Match
EvLKi_EQIDMatch
L

Configuration Response from Server via Transaction Manager

Search for EQID Index if not found
Store result to avoid future searches

15

Event Lookup Index EQID Match: EvLKi_EQIDMatch [/]

Fragment Statistics Equipment Search: EventEqFSC (RES)

Fragment Statistics Equipment Search — MOVE
Source 0
Dest EventEqFSC.POS 0

Fragment Statistics Equipment Search — MOVE
Source EDC_Event.CFG.EQ.n 3000
Dest EventEqFSC.LEN 3000

Fragment Statistics Equipment Search — FSC
Control EventEqFSC
Length 3000
Position 0
Mode ALL
Expression EDC_Event.CTL.Equip[EventEqFSC.POS].EQID=ServerCfg.DataRspLast.EqID
EN / DN / ER

Fragment Statistics Equipment Search: EventEqFSC.FD

Equipment-Event Index for EDC_Event Lookup — MOVE
Source EventEqFSC.POS 0
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array — MOVE
Source EvLKi 164
Dest EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144

Event Lookup Index EQID Match: EvLKi_EQIDMatch (L)

Event Lookup Index Found: EvLKi_Search

Search for Empty Index if new Equipment
Store result to avoid future searches

16

Event Lookup Index EQID Match: EvLKi_EQIDMatch [/]

Fragment Statistics Empty Search: EventEmptyFSC (RES)

Fragment Statistics Empty Search — MOVE
Source 0
Dest EventEmptyFSC.POS 271

Fragment Statistics Empty Search — MOVE
Source EDC_Event.CFG.EQ.n 3000
Dest EventEmptyFSC.LEN 3000

Fragment Statistics Empty Search — FSC
Control EventEmptyFSC
Length 3000
Position 271
Mode ALL
Expression EDC_Event.CTL.Equip[EventEmptyFSC.POS].EQID = 0
EN / DN / ER

Fragment Statistics Empty Search: EventEmptyFSC.FD

Equipment-Event Index for EDC_Event Lookup — MOVE
Source EventEmptyFSC.POS 271
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array — MOVE
Source EvLKi 164
Dest EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144

Event Controls Equipment Fragment Statistics Array — MOVE
Source ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1
Dest EDC_Event.CTL.Equip[EvLKi].EQID 926

Event Controls Equipment with Event Text Qty — ADD
Source A EDC_Event.CTL.EqQty 272
Source B 1
Dest EDC_Event.CTL.EqQty 272

Event Lookup Index EQID Match: EvLKi_EQIDMatch (L)

Event Lookup Index New: EvLKi_New

Configuration Response from Server via Transaction Manager

New or Newly-Enabled EQID Data Refresh - Initialize Event Masks to DtOwner = Maintenance, DtType = Fault and Ignore = pass data
Initialize Mask Config words (new data for EqDriver)

*** Ignore Mask is inverted from Configuration -- when OFF, the Mask bit is 1 to pass data thru MEQ ***
v3.5 Add Data Refresh Request for re-Enabled Equipment, but only clear masks for New Equipment
(Disabled Equipment Events & Masks are updated so Events work correctly if re-Enabled)

17

Event Lookup Index New
EvLKi_New

Event Lookup Index EQID Match
EvLKi_EQIDMatch

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.DataRefreshReq

Temp Result for Array Size

MUL
Source A EDC_Event.CFG.PKT.n 8
Source B EDC_Event.CFG.DAT.n 8
Dest LEN 64

Event Lookup Index New
EvLKi_New

DT Owner = Maintenance Event Mask Equipment Mask Equip Array

FLL
Source -1
Dest EDC_EventMaskMaint.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Fault Event Mask Equipment Mask Equip Array

FLL
Source -1
Dest EDC_EventMaskFault.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Alarm Event Mask Equipment Mask Equip Array

FLL
Source 0
Dest EDC_EventMaskAlarm.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Alert Event Mask Equipment Mask Equip Array

FLL
Source 0
Dest EDC_EventMaskAlert.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Ignore Event Mask Equipment Mask Equip Array

FLL
Source -1
Dest EDC_EventMaskIgnore.Equip[EvLKi].Packet[0]
Length LEN

No Display Event Mask Equipment Mask Equip Array

FLL
Source 0
Dest EDC_EventMaskNoDisp.Equip[EvLKi].Packet[0]
Length LEN

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].MaintMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].FaultMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].AlertMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend 2#0000_0000

Event Controls Equipment Fragment Statistics Array

MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].DataRefreshReq 2#0000_0000

Configuration Response from Server via Transaction Manager

Equipment Not Found and Statistics Array is Full
(shouldn't happen - requires Manual Intervention and Manual Reset of Error)

Event Lookup Index
EQID Match
EvLKi_EQIDMatch

18

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4025
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red - Equip Event Fragmentation Statistics Array Full ER

EDC_Status01.ServerCfg.Update.EventStatsArrayFull
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

Data OK
DN009 May 5, 2020 -- Ack data for any EQID 1-9999

lbl_DataOk
LBL

19

Server Interface Status State = Red - Equip Rx Data OK

EDC_Status01.ServerCfg.Update.EquipRxDataOK
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

Exit Routine

lbl_Exit
LBL

20

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 16 - Equipment Event
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing
EDC2 - v2.01 added Event DtOwner and DtType Mask Configuration; removed Shared Andon/AVI Events

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 836
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Convert EventID into sub-indexes
EqEventID -- WRDi & BITi (Word and Bit Indexes for EqEvent)
WRDi -- PKTi & DATI (8 Packets with 8 Data for Masks)

(Jan 16, 2017 -- EQID removed from ServerCfg.Data.EqEventID and sent in separate ServerCfg.Data.EQID)
Combine EQID and EqEventID into EventID
EQID = -1 for PreDefined Faults

2

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest EQID
531

Event Mask Bit Index (0-31)

MOD
Source A ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Source B 100
Dest BITi
1

Event Word Index (0-64)

CPT
Dest WRDi
59
Expression (ServerCfg.DataRspLast.EqEventID - BITi) / 100

Event Mask Data Index (0-7)

MOD
Source A WRDi
59
Source B 8
Dest DATi
3

Event Mask Packet Index (0-7)

CPT
Dest PKTi
7
Expression (WRDi - DATi) / 8

Pre-Defined Event

3

Server Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
0
Source B 16

Server Configuration from Transaction Manager Equipment Event ID - EQID & Element ID

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
High Limit 31

Line Index

MOVE
Source 0
Dest LNi
5

Equipment Index

MOVE
Source 0
Dest EQi
31

EQ EventID for Event Table storage (EEEEeeee)

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Dest EqEventID
9265901

PreDefined Event
EventTextPreDef

Configuration Response from Server via Transaction Manager

Convert EQID into sub-indexes
EQID = LNi & EQi

DN009 May 5, 2020 -- Acknowlege LNi and EQi for EQID 2900-9999
(Configuration allows EQID 1-9999 - values outside array limits are for decommissioned Equipment)

4

PreDefined Event
EventTextPreDef
/

Equipment Index
MOD
Source A EQID
531
Source B
100
Dest EQi
31

Line Index
CPT
Dest LNi
5
Expression (EQID - EQi) / 100

Line Index
LIMIT
Low Limit CFG.LN.si
<EDC.CFG.LN.si>
0
Test LNi
5
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

Valid Line Index
ValidLNi

Valid Line Index
ValidLNi
/

Line Index
LIMIT
Low Limit CFG.LN.n
<EDC.CFG.LN.n>
29
Test LNi
5
High Limit
99

Decommissioned Line
DecomLNi

Equipment Index
LIMIT
Low Limit
0
Test EQi
31
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

Valid Equipment Index
ValidEQi

Valid Line Index
ValidLNi
/

Line Index
MOVE
Source
0
Dest LNi
5

Configuration Response from Server via Transaction Manager

5

Equipment-Defined Event

PreDefined Event EventTextPreDef

Valid Line Index ValidLNi

Valid Equipment Index ValidEQi

EqDefined Event EventTextEqDef

Search Text Tag01 EventTextTag01 (U)

Search Text Tag02 EventTextTag02 (U)

Search Text Tag03 EventTextTag03 (U)

Search Text Tag04 EventTextTag04 (U)

Search Text Tag05 EventTextTag05 (U)

Search Text Tag06 EventTextTag06 (U)

Event Text Tag01 Search EventTextFSC01 (RES)

Event Text Tag02 Search EventTextFSC02 (RES)

Event Text Tag03 Search EventTextFSC03 (RES)

Event Text Tag04 Search EventTextFSC04 (RES)

Event Text Tag05 Search EventTextFSC05 (RES)

Event Text Tag06 Search EventTextFSC06 (RES)

EQ EventID for Event Table storage (EEEEeeee)

CPT

Dest	EqEventID 9265901
Expression	(EQID * 10000) + ServerCfg.DataRspLast.EqEventID

Configuration Response from Server via Transaction Manager

Good Data Packet
EDC2 - v2.01 added DtOwner (RankNum), DtType (RankChar), NoDisplay (Cor
May 2026 - added Summary Data KPI Option ID for EqDefined Events

6

PreDefined Event
EventTextPreDef

EqDefined Event
EventTextEqDef

EQ EventID for Event Table storage (EEEEeeee)

LIMIT
Low Limit 0
Test EqEventID
9265901
High Limit 99999999

Server Configuration from Transaction Manager Equipment Event Text

NE
Source A ServerCfg.DataRspLast.EqEventText.DATA[0]
<EDC.ServerCfg.DataRspLast.EqEventText.DATA[0]>
'$00'
Source B 1

Server Configuration from Transaction Manager Maintenance Risk Rank Assignments (4-1)

NE
Source A ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum>
-1
Source B -1

Server Configuration from Transaction Manager Maintenance Rank Assignments (A-D = Values 1-4)

NE
Source A ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar>
-1
Source B -1

Server Configuration from Transaction Manager Config Status Bits

NE
Source A ServerCfg.DataRspLast.ConfigStatus
<EDC.ServerCfg.DataRspLast.ConfigStatus>
-1
Source B -1

Server Configuration from Transaction Manager Offline Group Display Row

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.OfflineGroupDisplayRow
<EDC.ServerCfg.DataRspLast.OfflineGroupDisplayRow>
-1
High Limit 5

PreDefined Event
EventTextPreDef

0=Production
1=Maintenance

MOVE
Source ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum>
-1
Dest DtOwner
1

Server Configuration from Transaction Manager Equipment Event Text

EQ
Source A ServerCfg.DataRspLast.EqEventText.DATA[0]
<EDC.ServerCfg.DataRspLast.EqEventText.DATA[0]>
'$00'
Source B 1

Configuration Response from Server via Transaction Manager

gStatus)

Server Configuration
from Transaction
Manager Response has
Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

L

0=Not Used
1=Fault
2=Alarm
3=Alert
4=Info
5=Ignore

MOVE
Source ServerCfg.DataRspLast.RankChar
<EDC.ServerCfg.DataRspLast.RankChar>
-1
Dest DtType
4

0 = not defined
1-5 = Option 1-5

MOVE
Source ServerCfg.DataRspLast.OfflineGroupDisplayRow
<EDC.ServerCfg.DataRspLast.OfflineGroupDisplayRow>
-1
Dest KpiOptionID
0

MOVE
Source ServerCfg.DataRspLast.ConfigStatus
<EDC.ServerCfg.DataRspLast.ConfigStatus>
-1
Dest NoDisplay
0

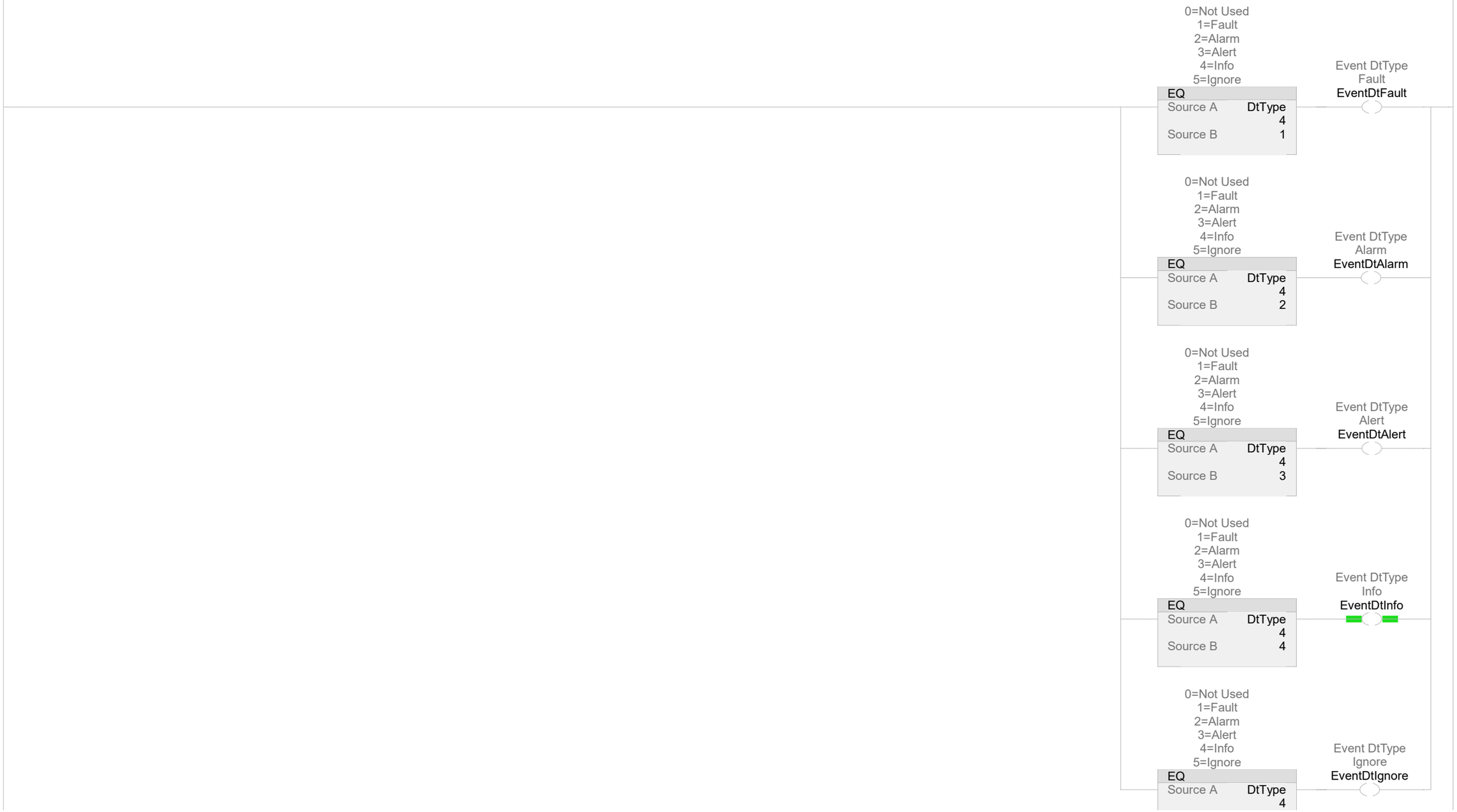
Fault Log

COP
Source ServerCfg.DataRspLast
<EDC.ServerCfg.DataRspLast>
Dest LogServerCfg_DataRespLast
Length 1

Configuration Response from Server via Transaction Manager

EDC2 - v2.01 Downtime Types

7



Configuration Response from Server via Transaction Manager

Source B 5

EDC2 - v2.01 Downtime Owner and No Display Option

8

0=Production
1=Maintenance
EQ
Source A DtOwner
1
Source B 1

Event DtOwner
Maintenance
EventDtMaint

EQ
Source A NoDisplay
0
Source B 1

Event NoDisplay
EventNoDisplay

If DtType = Info, Ignore or set for NoDisplay, don't save the Event Text - just update the Masks

9

PreDefined
Event
EventTextPreDef
/

Event DtType
Info
EventDtInfo

Event DtType
Ignore
EventDtIgnore

Event NoDisplay
EventNoDisplay

lbl_DataOk
JMP

Configuration Response from Server via Transaction Manager

DN009 May 5, 2020 -- Acknowlege LNi and EQi for EQID 2900-9999
(Configuration allows EQID 1-9999 - values outside array limits are for decommissioned Equipment)

(this shouldn't happen - Server should timestamp these complete without sending)

10

Decommissioned Line
DecomLNi

Valid Line Index
ValidLNi
/

Valid Equipment Index
ValidEQi

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Decommissioned Equipment
DecomEQi

lbl_DataOk
JMP

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

11

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	LNi
	5
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	EQi
	31
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	EqEventID
	9265901
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE	
Source	ServerCfg.DataRspLast.EqEventID
	<EDC.ServerCfg.DataRspLast.EqEventID>
	-1
Dest	ServerCfg.Ctl.CommandID
	<EDC.ServerCfg.Ctl.CommandID>
	2317

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	4023
Input Par	ServerCfg.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerCfg.DataRspLast.Command
	<EDC.ServerCfg.DataRspLast.Command>
Input Par	ServerCfg.Ctl.CommandID
	<EDC.ServerCfg.Ctl.CommandID>
Input Par	ServerCfg.Ctl.TransactionNumber
	<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red - Equip Event Rx Data ER
EDC_Status01.ServerCfg.Update.EquipEventRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig	
aoiEDC_StatusUpdateS...	aoiEDC_StatusUpdateServerConfig ...
Status	EDC_Status01.ServerCfg

lbl_Exit
JMP

Get Event Text Start/End Index for this Equipment

12

Event Text Start Index

MOVE	
Source	EDC.Line[LNi].Equip[EQi].EventCTL.TXTsi
	23544
Dest	TXTsi
	28256

Event Text End Index

MOVE	
Source	EDC.Line[LNi].Equip[EQi].EventCTL.TXTei
	23779
Dest	TXTei
	28387

Configuration Response from Server via Transaction Manager

13

Tag01
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

EqDefined Event: EventTextEqDef

Event Text Start Index

LIMIT	
Low Limit	EDC_HMI.CFG.EvTxt01.si 0
Test	TXTsi 28256
High Limit	EDC_HMI.CFG.EvTxt01.ei 32499

Search Text Tag01: EventTextTag01 (L)

Event Text Tag01 Search

MOVE	
Source	TXTsi 28256
Dest	EventTextFSC01.POS 0

Event Text Tag01 Search

MOVE	
Source	EDC_HMI.CFG.EvTxt01.n 32500
Dest	EventTextFSC01.LEN 28388

Event Text End Index

LIMIT	
Low Limit	EDC_HMI.CFG.EvTxt01.si 0
Test	TXTei 28387
High Limit	EDC_HMI.CFG.EvTxt01.ei 32499

Event Text Tag01 Search

ADD	
Source A	TXTei 28387
Source B	1
Dest	EventTextFSC01.LEN 28388

14

Tag01 Event Text Search

EqDefined Event: EventTextEqDef

Search Text Tag01: EventTextTag01

Event Text Tag01 Search

FSC	
Control	EventTextFSC01
Length	28388
Position	0
Mode	ALL
Expression	EDC_EventTxt01[EventTextFSC01.POS].EqEventID=EqEventID

EN DN ER

Configuration Response from Server via Transaction Manager

Tag02
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

15

EqDefined Event: EventTextEqDef
Event Text Tag01 Search: EventTextFSC01.FD

Event Text Start Index
LIMIT
Low Limit: EDC_HMI.CFG.EvTxt01.si 0
Test: TXTsi 28256
High Limit: EDC_HMI.CFG.EvTxt02.ei 64999

Event Text End Index
GT
Source A: TXTei 28387
Source B: EDC_HMI.CFG.EvTxt02.si 32500

Search Text Tag02: EventTextTag02 (L)

Event Text Tag02 Search
SUB
Source A: TXTsi 28256
Source B: EDC_HMI.CFG.EvTxt02.si 32500
Dest: EventTextFSC02.POS 0

Event Text Tag02 Search
LT
Source A: EventTextFSC02.POS 0
Source B: 0

Event Text Tag02 Search
MOVE
Source: 0
Dest: EventTextFSC02.POS 0

Event Text Tag02 Search
MOVE
Source: EDC_HMI.CFG.EvTxt02.n 32500
Dest: EventTextFSC02.LEN 4562

Event Text End Index
LIMIT
Low Limit: EDC_HMI.CFG.EvTxt02.si 32500
Test: TXTei 28387
High Limit: EDC_HMI.CFG.EvTxt02.ei 64999

Event Text Tag02 Search
CPT
Dest: EventTextFSC02.LEN 4562
Expression: TXTei-EDC_HMI.CFG.EvTxt02.si+1

Tag02 Event Text Search

16

EqDefined Event: EventTextEqDef
Search Text Tag02: EventTextTag02

Event Text Tag02 Search
FSC
Control: EventTextFSC02
Length: 4562
Position: 0
Mode: ALL
Expression: EDC_EventTxt02[EventTextFSC02.POS].EqEventID=EqEventID
EN
DN
ER

Configuration Response from Server via Transaction Manager

Tag03
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

17

EqDefined Event: EventTextEqDef
Event Text Tag01 Search: EventTextFSC01.FD
Event Text Tag02 Search: EventTextFSC02.FD

Event Text Start Index
LIMIT
Low Limit: EDC_HMI.CFG.EvTxt01.si 0
Test: TXTsi 28256
High Limit: EDC_HMI.CFG.EvTxt03.ei 97499

Event Text End Index
GT
Source A: TXTei 28387
Source B: EDC_HMI.CFG.EvTxt03.si 65000

Search Text Tag03: EventTextTag03 (L)

Event Text Tag03 Search
SUB
Source A: TXTsi 28256
Source B: EDC_HMI.CFG.EvTxt03.si 65000
Dest: EventTextFSC03.POS 0

Event Text Tag03 Search
LT
Source A: EventTextFSC03.POS 0
Source B: 0

Event Text Tag03 Search
MOVE
Source: 0
Dest: EventTextFSC03.POS 0

Event Text Tag03 Search
MOVE
Source: EDC_HMI.CFG.EvTxt03.n 32500
Dest: EventTextFSC03.LEN 1390

Event Text End Index
LIMIT
Low Limit: EDC_HMI.CFG.EvTxt03.si 65000
Test: TXTei 28387
High Limit: EDC_HMI.CFG.EvTxt03.ei 97499

Event Text Tag03 Search
CPT
Dest: EventTextFSC03.LEN 1390
Expression: TXTei-EDC_HMI.CFG.EvTxt03.si+1

Tag03 Event Text Search

18

EqDefined Event: EventTextEqDef
Search Text Tag03: EventTextTag03

Event Text Tag03 Search
FSC
Control: EventTextFSC03
Length: 1390
Position: 0
Mode: ALL
Expression: EDC_EventTxt03[EventTextFSC03.POS].EqEventID=EqEventID
EN
DN
ER

Configuration Response from Server via Transaction Manager

Tag04
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

19

EqDefined Event: EventTextEqDef
Event Text Tag01 Search: EventTextFSC01.FD
Event Text Tag02 Search: EventTextFSC02.FD
Event Text Tag03 Search: EventTextFSC03.FD

Event Text Start Index
LIMIT
Low Limit: EDC_HMI.CFG.EvTxt01.si 0
Test: TXTsi 28256
High Limit: EDC_HMI.CFG.EvTxt04.ei 129999

Event Text End Index
GT
Source A: TXTei 28387
Source B: EDC_HMI.CFG.EvTxt04.si 97500

Search Text Tag04: EventTextTag04 (L)

Event Text Tag04 Search
SUB
Source A: TXTsi 28256
Source B: EDC_HMI.CFG.EvTxt04.si 97500
Dest: EventTextFSC04.POS 0

Event Text Tag04 Search
LT
Source A: EventTextFSC04.POS 0
Source B: 0

Event Text Tag04 Search
MOVE
Source: 0
Dest: EventTextFSC04.POS 0

Event Text Tag04 Search
MOVE
Source: EDC_HMI.CFG.EvTxt04.n 32500
Dest: EventTextFSC04.LEN 555

Event Text End Index
LIMIT
Low Limit: EDC_HMI.CFG.EvTxt04.si 97500
Test: TXTei 28387
High Limit: EDC_HMI.CFG.EvTxt04.ei 129999

Event Text Tag04 Search
CPT
Dest: EventTextFSC04.LEN 555
Expression: TXTei-EDC_HMI.CFG.EvTxt04.si+1

Tag04 Event Text Search

20

EqDefined Event: EventTextEqDef
Search Text Tag04: EventTextTag04

Event Text Tag04 Search
FSC
Control: EventTextFSC04
Length: 555
Position: 0
Mode: ALL
Expression: EDC_EventTxt04[EventTextFSC04.POS].EqEventID=EqEventID
EN
DN
ER

Configuration Response from Server via Transaction Manager

Tag05
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

21

EqDefined Event: EventTextEqDef
Event Text Tag01 Search: EventTextFSC01.FD
Event Text Tag02 Search: EventTextFSC02.FD
Event Text Tag03 Search: EventTextFSC03.FD
Event Text Tag04 Search: EventTextFSC04.FD

Event Text Start Index — LIMIT
Low Limit	EDC_HMI.CFG.EvTxt01.si 0
Test	TXTsi 28256
High Limit	EDC_HMI.CFG.EvTxt05.ei 162499

Event Text End Index — GT
Source A	TXTei 28387
Source B	EDC_HMI.CFG.EvTxt05.si 130000

Search Text Tag05: EventTextTag05 (L)

Event Text Tag05 Search — SUB
Source A	TXTsi 28256
Source B	EDC_HMI.CFG.EvTxt05.si 130000
Dest	EventTextFSC05.POS 0

Event Text Tag05 Search — LT
Source A	EventTextFSC05.POS 0
Source B	0

Event Text Tag05 Search — MOVE
Source	0
Dest	EventTextFSC05.POS 0

Event Text Tag05 Search — MOVE
Source	EDC_HMI.CFG.EvTxt05.n 32500
Dest	EventTextFSC05.LEN 555

Event Text End Index — LIMIT
Low Limit	EDC_HMI.CFG.EvTxt05.si 130000
Test	TXTei 28387
High Limit	EDC_HMI.CFG.EvTxt05.ei 162499

Event Text Tag05 Search — CPT
Dest	EventTextFSC05.LEN 555
Expression	TXTei-EDC_HMI.CFG.EvTxt05.si+1

Tag05 Event Text Search

22

EqDefined Event: EventTextEqDef
Search Text Tag05: EventTextTag05

Event Text Tag05 Search — FSC
Control	EventTextFSC05
Length	555
Position	0
Mode	ALL
Expression	EDC_EventTxt05[EventTextFSC05.POS].EqEventID=EqEventID

(EN) (DN) (ER)

Configuration Response from Server via Transaction Manager

Tag06
Determine if Event Text could be in Tag and set Search Limits
Start search at the TXTsi
End search at the TXTei or the end of the Tag Array

23

EqDefined Event: EventTextEqDef
Event Text Tag01 Search: EventTextFSC01.FD
Event Text Tag02 Search: EventTextFSC02.FD
Event Text Tag03 Search: EventTextFSC03.FD
Event Text Tag04 Search: EventTextFSC04.FD
Event Text Tag05 Search: EventTextFSC05.FD

Event Text Start Index — LIMIT
Low Limit	EDC_HMI.CFG.EvTxt01.si 0
Test	TXTsi 28256
High Limit	EDC_HMI.CFG.EvTxt06.ei 194999

Event Text End Index — GT
Source A	TXTei 28387
Source B	EDC_HMI.CFG.EvTxt06.si 162500

Search Text Tag06 EventTextTag06 (L)

Event Text Tag06 Search — SUB
Source A	TXTsi 28256
Source B	EDC_HMI.CFG.EvTxt06.si 162500
Dest	EventTextFSC06.POS 0

Event Text Tag06 Search — LT
Source A	EventTextFSC06.POS 0
Source B	0

Event Text Tag06 Search — MOVE
Source	0
Dest	EventTextFSC06.POS 0

Event Text Tag06 Search — MOVE
Source	EDC_HMI.CFG.EvTxt06.n 32500
Dest	EventTextFSC06.LEN 555

Event Text End Index — LIMIT
Low Limit	EDC_HMI.CFG.EvTxt06.si 162500
Test	TXTei 28387
High Limit	EDC_HMI.CFG.EvTxt06.ei 194999

Event Text Tag06 Search — CPT
Dest	EventTextFSC06.LEN 555
Expression	TXTei-EDC_HMI.CFG.EvTxt06.si+1

Tag06 Event Text Search

24

EqDefined Event: EventTextEqDef
Search Text Tag06: EventTextTag06

Event Text Tag06 Search — FSC
Control	EventTextFSC06
Length	555
Position	0
Mode	ALL
Expression	EDC_EventTxt06[EventTextFSC06.POS].EqEventID=EqEventID

EN DN ER

Configuration Response from Server via Transaction Manager

Tag01
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

25

EqDefined Event EventTextEqDef
Event Text Tag01 Empty Search EventTextEmptyFSC01 (RES)
Event Text Tag01 Search EventTextFSC01.FD
Event Text Tag02 Search EventTextFSC02.FD
Event Text Tag03 Search EventTextFSC03.FD
Event Text Tag04 Search EventTextFSC04.FD
Event Text Tag05 Search EventTextFSC05.FD
Event Text Tag06 Search EventTextFSC06.FD

LT
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt01.ei 32499

Event Text Tag01 Empty Search
MOVE
Source EventEmptyLastFD 45063
Dest EventTextEmptyFSC01.POS 0

Event Text Tag01 Empty Search
MOVE
Source EDC_HMI.CFG.EvTxt01.n 32500
Dest EventTextEmptyFSC01.LEN 32500

Event Text Tag01 Empty Search
LT
Source A EventTextEmptyFSC01.POS 0
Source B 32

Event Text Tag01 Empty Search
MOVE
Source 32
Dest EventTextEmptyFSC01.POS 0

Event Text Tag01 Empty Search
FSC
Control EventTextEmptyFSC01
Length 32500
Position 0
Mode ALL
Expression EDC_EventTxt01[EventTextEmptyFSC01.POS].EqEventID = 0
EN
DN
ER

Event Text Tag01 Empty Search EventTextEmptyFSC01.FD

Event Text Tag01 Empty Search
MOVE
Source EventTextEmptyFSC01.POS 0
Dest EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

Tag02
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

26

EqDefined Event EventTextEqDef
Event Text Tag02 Empty Search EventTextEmptyFSC02 (RES)
Event Text Tag01 Search EventTextFSC01.FD (/)
Event Text Tag02 Search EventTextFSC02.FD (/)
Event Text Tag03 Search EventTextFSC03.FD (/)
Event Text Tag04 Search EventTextFSC04.FD (/)
Event Text Tag05 Search EventTextFSC05.FD (/)
Event Text Tag06 Search EventTextFSC06.FD (/)
Event Text Tag01 Empty Search EventTextEmptyFSC01.FD (/)

LT
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt02.ei 64999

Event Text Tag02 Empty Search
SUB
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt02.si 32500
Dest EventTextEmptyFSC02.POS 0

Event Text Tag02 Empty Search
MOVE
Source EDC_HMI.CFG.EvTxt02.n 32500
Dest EventTextEmptyFSC02.LEN 32500

Event Text Tag02 Empty Search
LT
Source A EventTextEmptyFSC02.POS 0
Source B 0

Event Text Tag02 Empty Search
MOVE
Source 0
Dest EventTextEmptyFSC02.POS 0

Event Text Tag02 Empty Search
FSC
Control EventTextEmptyFSC02
Length 32500
Position 0
Mode ALL
Expression EDC_EventTxt02[EventTextEmptyFSC02.POS].EqEventID = 0
(EN) (DN) (ER)

Event Text Tag02 Empty Search EventTextEmptyFSC02.FD

ADD
Source A EventTextEmptyFSC02.POS 0
Source B EDC_HMI.CFG.EvTxt02.si 32500
Dest EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

27

Tag03
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

EqDefined Event EventTextEqDef
Event Text Tag03 Empty Search EventTextEmptyFSC03 (RES)
Event Text Tag01 Search EventTextFSC01.FD
Event Text Tag02 Search EventTextFSC02.FD
Event Text Tag03 Search EventTextFSC03.FD
Event Text Tag04 Search EventTextFSC04.FD
Event Text Tag05 Search EventTextFSC05.FD
Event Text Tag06 Search EventTextFSC06.FD
Event Text Tag01 Empty Search EventTextEmptyFSC01.FD
Event Text Tag02 Empty Search EventTextEmptyFSC02.FD

LT
Source A: EventEmptyLastFD 45063
Source B: EDC_HMI.CFG.EvTxt03.ei 97499

Event Text Tag03 Empty Search
SUB
Source A: EventEmptyLastFD 45063
Source B: EDC_HMI.CFG.EvTxt03.si 65000
Dest: EventTextEmptyFSC03.POS 0

Event Text Tag03 Empty Search
MOVE
Source: EDC_HMI.CFG.EvTxt03.n 32500
Dest: EventTextEmptyFSC03.LEN 25000

Event Text Tag03 Empty Search
LT
Source A: EventTextEmptyFSC03.POS 0
Source B: 0

Event Text Tag03 Empty Search
MOVE
Source: 0
Dest: EventTextEmptyFSC03.POS 0

Event Text Tag03 Empty Search
FSC
Control: EventTextEmptyFSC03
Length: 25000
Position: 0
Mode: ALL
Expression: EDC_EventTxt03[EventTextEmptyFSC03.POS].EqEventID = 0
EN DN ER

Event Text Tag03 Empty Search EventTextEmptyFSC03.FD

ADD
Source A: EventTextEmptyFSC03.POS 0
Source B: EDC_HMI.CFG.EvTxt03.si 65000
Dest: EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

Tag04
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

28

EqDefined Event EventTextEqDef
Event Text Tag04 Empty Search EventTextEmptyFSC04 (RES)
Event Text Tag01 Search EventTextFSC01.FD
Event Text Tag02 Search EventTextFSC02.FD
Event Text Tag03 Search EventTextFSC03.FD
Event Text Tag04 Search EventTextFSC04.FD
Event Text Tag05 Search EventTextFSC05.FD
Event Text Tag06 Search EventTextFSC06.FD
Event Text Tag01 Empty Search EventTextEmptyFSC01.FD
Event Text Tag02 Empty Search EventTextEmptyFSC02.FD

Event Text Tag03 Empty Search EventTextEmptyFSC03.FD

LT
Source A: EventEmptyLastFD 45063
Source B: EDC_HMI.CFG.EvTxt04.ei 129999

Event Text Tag04 Empty Search

SUB
Source A: EventEmptyLastFD 45063
Source B: EDC_HMI.CFG.EvTxt04.si 97500
Dest: EventTextEmptyFSC04.POS 0

Event Text Tag04 Empty Search

MOVE
Source: EDC_HMI.CFG.EvTxt04.n 32500
Dest: EventTextEmptyFSC04.LEN 25000

Event Text Tag04 Empty Search

LT
Source A: EventTextEmptyFSC04.POS 0
Source B: 0

Event Text Tag04 Empty Search

MOVE
Source: 0
Dest: EventTextEmptyFSC04.POS 0

Event Text Tag04 Empty Search

FSC
Control: EventTextEmptyFSC04
Length: 25000
Position: 0
Mode: ALL
Expression: EDC_EventTxt04[EventTextEmptyFSC04.POS].EqEventID=0
(EN) (DN) (ER)

Event Text Tag04 Empty Search EventTextEmptyFSC04.FD

ADD
Source A: EventTextEmptyFSC04.POS 0
Source B: EDC_HMI.CFG.EvTxt04.si 97500
Dest: EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

29

Tag05
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

EqDefined Event EventTextEqDef
Event Text Tag05 Empty Search EventTextEmptyFSC05 (RES)
Event Text Tag01 Search EventTextFSC01.FD
Event Text Tag02 Search EventTextFSC02.FD
Event Text Tag03 Search EventTextFSC03.FD
Event Text Tag04 Search EventTextFSC04.FD
Event Text Tag05 Search EventTextFSC05.FD
Event Text Tag06 Search EventTextFSC06.FD
Event Text Tag01 Empty Search EventTextEmptyFSC01.FD
Event Text Tag02 Empty Search EventTextEmptyFSC02.FD
Event Text Tag03 Empty Search EventTextEmptyFSC03.FD
Event Text Tag04 Empty Search EventTextEmptyFSC04.FD

LT
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt05.ei 162499

Event Text Tag05 Empty Search
SUB
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt05.si 130000
Dest EventTextEmptyFSC05.POS 0

Event Text Tag05 Empty Search
MOVE
Source EDC_HMI.CFG.EvTxt05.n 32500
Dest EventTextEmptyFSC05.LEN 25000

Event Text Tag05 Empty Search
LT
Source A EventTextEmptyFSC05.POS 0
Source B 0

Event Text Tag05 Empty Search
MOVE
Source 0
Dest EventTextEmptyFSC05.POS 0

Event Text Tag05 Empty Search
FSC
Control EventTextEmptyFSC05
Length 25000
Position 0
Mode ALL
Expression EDC_EventTxt05[EventTextEmptyFSC05.POS].EqEventID=0
EN
DN
ER

Event Text Tag05 Empty Search EventTextEmptyFSC05.FD

ADD
Source A EventTextEmptyFSC05.POS 0
Source B EDC_HMI.CFG.EvTxt05.si 130000
Dest EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

30

Tag06
If Element not found, Find next empty record in Event table
Array 0-31 Reserved for Pre-Defined Events 0-31 Common to all Equipment
(Start Empty search from last Empty FD position to reduce search)

EqDefined Event EventTextEqDef
Event Text Tag06 Empty Search EventTextEmptyFSC06 (RES)
Event Text Tag01 Search EventTextFSC01.FD
Event Text Tag02 Search EventTextFSC02.FD
Event Text Tag03 Search EventTextFSC03.FD
Event Text Tag04 Search EventTextFSC04.FD
Event Text Tag05 Search EventTextFSC05.FD
Event Text Tag06 Search EventTextFSC06.FD
Event Text Tag01 Empty Search EventTextEmptyFSC01.FD
Event Text Tag02 Empty Search EventTextEmptyFSC02.FD
Event Text Tag03 Empty Search EventTextEmptyFSC03.FD
Event Text Tag04 Empty Search EventTextEmptyFSC04.FD
Event Text Tag05 Empty Search EventTextEmptyFSC05.FD

LT
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt06.ei 194999

Event Text Tag06 Empty Search
SUB
Source A EventEmptyLastFD 45063
Source B EDC_HMI.CFG.EvTxt06.si 162500
Dest EventTextEmptyFSC06.POS 0

Event Text Tag06 Empty Search
MOVE
Source EDC_HMI.CFG.EvTxt06.n 32500
Dest EventTextEmptyFSC06.LEN 25000

Event Text Tag06 Empty Search
LT
Source A EventTextEmptyFSC06.POS 0
Source B 0

Event Text Tag06 Empty Search
MOVE
Source 0
Dest EventTextEmptyFSC06.POS 0

Event Text Tag06 Empty Search
FSC
Control EventTextEmptyFSC06
Length 25000
Position 0
Mode ALL
Expression EDC_EventTxt06[EventTextEmptyFSC06.POS].EqEventID=0
EN
DN
ER

Event Text Tag06 Empty Search EventTextEmptyFSC06.FD

ADD
Source A EventTextEmptyFSC06.POS 0
Source B EDC_HMI.CFG.EvTxt06.si 162500
Dest EventEmptyLastFD 45063

Configuration Response from Server via Transaction Manager

Load Event Text array Index

31

PreDefined Event EventTextPreDef

Event Text Tag01, Tag02 or Tag03 Array Index
MOVE
Source EqEventID 9265901
Dest ARRi 28387

Overwrite Existing or Insert New EventPosFoundTag01

EventPosFound

EqDefined Event EventTextEqDef

Event Text Tag01 Search EventTextFSC01.FD

Event Text Tag01, Tag02 or Tag03 Array Index
MOVE
Source EventTextFSC01.POS 0
Dest ARRi 28387

Event Text Tag01 Empty Search EventTextEmptyFSC01.FD

Event Text Tag01, Tag02 or Tag03 Array Index
MOVE
Source EventTextEmptyFSC01.POS 0
Dest ARRi 28387

EqDefined Event EventTextEqDef

Event Text Tag02 Search EventTextFSC02.FD

Event Text Tag01, Tag02 or Tag03 Array Index
MOVE
Source EventTextFSC02.POS 0
Dest ARRi 28387

Overwrite Existing or Insert New EventPosFoundTag02

Event Text Tag02 Empty Search EventTextEmptyFSC02.FD

Event Text Tag01, Tag02 or Tag03 Array Index
MOVE
Source EventTextEmptyFSC02.POS 0
Dest ARRi 28387

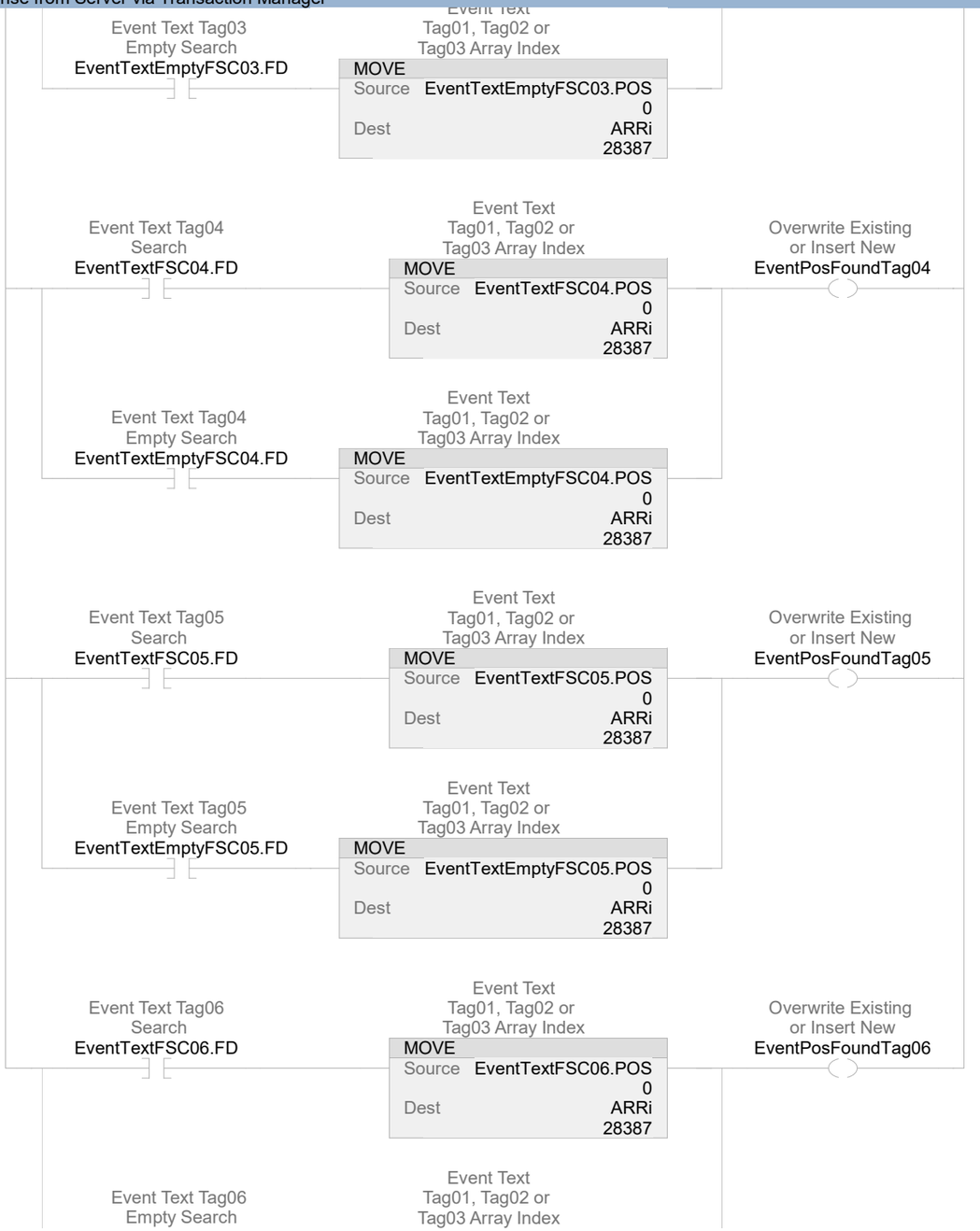
Event Text Tag03 Search EventTextFSC03.FD

Event Text Tag01, Tag02 or Tag03 Array Index
MOVE
Source EventTextFSC03.POS 0
Dest ARRi 28387

Overwrite Existing or Insert New EventPosFoundTag03

Event Text

Configuration Response from Server via Transaction Manager



Configuration Response from Server via Transaction Manager

EventTextEmptyFSC06.FD

MOVE
Source EventTextEmptyFSC06.POS
0
Dest ARRi
28387

Equipment Event Text Array Full
(shouldn't happen - requires Manual Intervention and Manual Reset of Error)

32

Overwrite Existing or Insert New EventPosFoundTag01 /
Overwrite Existing or Insert New EventPosFoundTag02 /
Overwrite Existing or Insert New EventPosFoundTag03 /
Overwrite Existing or Insert New EventPosFoundTag04 /
Overwrite Existing or Insert New EventPosFoundTag05 /
Overwrite Existing or Insert New EventPosFoundTag06 /

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4024
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red - Equip Event Array Full ER

EDC_Status01.ServerCfg.Update.EquipEventArrayFull
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store
Log Transaction

33

Overwrite Existing or Insert New EventPosFoundTag01

Event Text Lookup
MOVE
Source EqEventID 9265901
Dest EDC_EventTxt01[ARRi].EqEventID 9262317

Event Text Lookup
COP
Source ServerCfg.DataRspLast.EqEventText <EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_EventTxt01[ARRi].Text
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 1
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Overwrite Existing or Insert New EventPosFoundTag02

Event Text Lookup
MOVE
Source EqEventID 9265901
Dest EDC_EventTxt02[ARRi].EqEventID 0

Event Text Lookup
COP
Source ServerCfg.DataRspLast.EqEventText <EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_EventTxt02[ARRi].Text
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 2
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Overwrite Existing or Insert New EventPosFoundTag03

Event Text Lookup
MOVE
Source EqEventID 9265901
Dest EDC_EventTxt03[ARRi].EqEventID 0

Event Text Lookup
COP
Source ServerCfg.DataRspLast.EqEventText <EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_EventTxt03[ARRi].Text
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 3
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Overwrite Existing or Insert New EventPosFoundTag04

Event Text Lookup
MOVE
Source EqEventID 9265901
Dest EDC_EventTxt04[ARRi].EqEventID 0

Event Text Lookup
COP
Source ServerCfg.DataRspLast.EqEventText <EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_EventTxt04[ARRi].Text
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 4
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

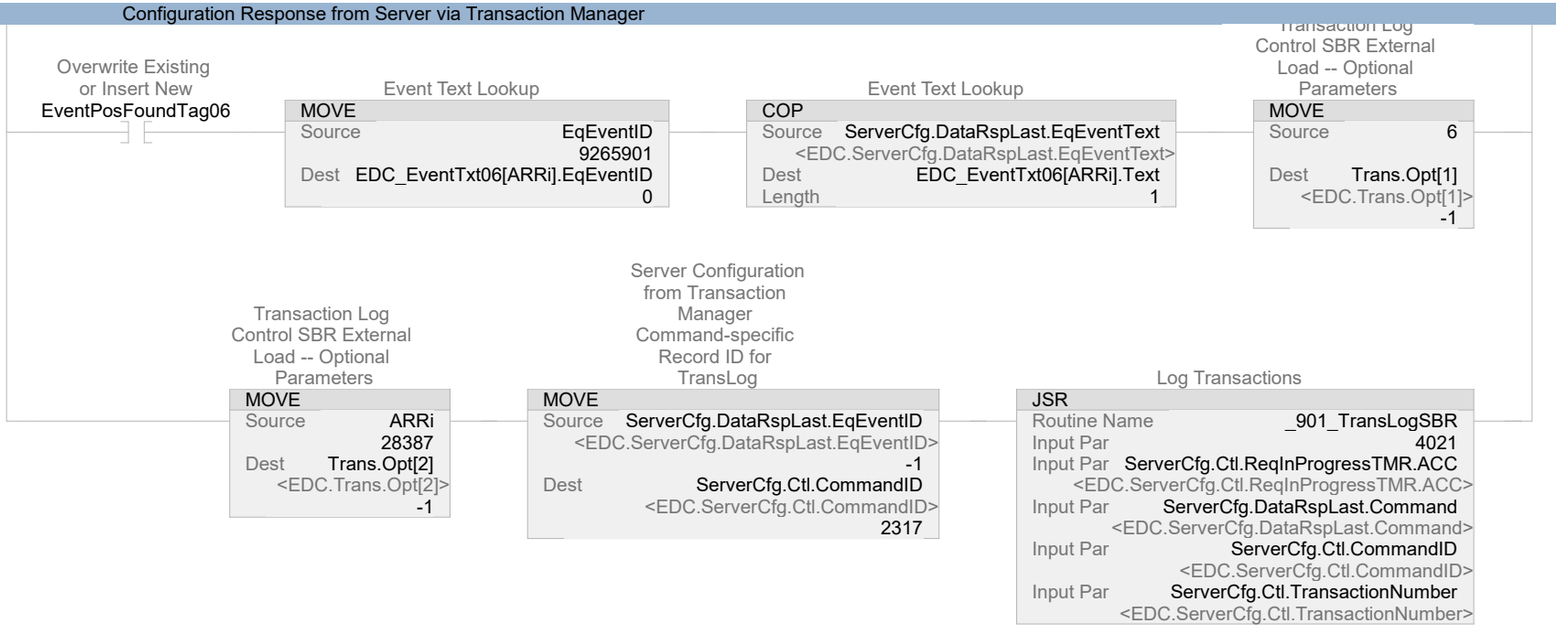
Overwrite Existing or Insert New EventPosFoundTag05

Event Text Lookup
MOVE
Source EqEventID 9265901
Dest EDC_EventTxt05[ARRi].EqEventID 0

Event Text Lookup
COP
Source ServerCfg.DataRspLast.EqEventText <EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_EventTxt05[ARRi].Text
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source 5
Dest Trans.Opt[1] <EDC.Trans.Opt[1]> -1

Configuration Response from Server via Transaction Manager



Configuration Response from Server via Transaction Manager

Update Equipment-Defined array search limits for EQID existing or new Event

34

EqDefined Event EventTextEqDef

Event Text Tag01 Search EventTextFSC01.FD

Event Text Tag01 Empty Search EventTextEmptyFSC01.FD

MOVE
Source ARRi 28387
Dest TXTi 28387

Event Text Tag02 Search EventTextFSC02.FD

Event Text Tag02 Empty Search EventTextEmptyFSC02.FD

ADD
Source A ARRi 28387
Source B EDC_HMI.CFG.EvTxt02.si 32500
Dest TXTi 28387

Event Text Tag03 Search EventTextFSC03.FD

Event Text Tag03 Empty Search EventTextEmptyFSC03.FD

ADD
Source A ARRi 28387
Source B EDC_HMI.CFG.EvTxt03.si 65000
Dest TXTi 28387

Event Text Tag04 Search EventTextFSC04.FD

Event Text Tag04 Empty Search EventTextEmptyFSC04.FD

ADD
Source A ARRi 28387
Source B EDC_HMI.CFG.EvTxt04.si 97500
Dest TXTi 28387

Event Text Tag05 Search EventTextFSC05.FD

Event Text Tag05 Empty Search EventTextEmptyFSC05.FD

ADD
Source A ARRi 28387
Source B EDC_HMI.CFG.EvTxt05.si 130000
Dest TXTi 28387

Event Text Tag06 Search EventTextFSC06.FD

Event Text Tag06 Empty Search

ADD
Source A ARRi 28387
Source B EDC_HMI.CFG.EvTxt06.si

LT
Source A TXTi 28387
Source B TXTsi 28256

Event Text Start Index

EQ
Source A TXTsi 28256
Source B 0

Equipment Data Collection Lines and Equipment Array

MOVE
Source TXTi 28387
Dest EDC.Line[LNi].Equip[EQi].EventCTL.TXTsi 23544

GT
Source A TXTi 28387
Source B TXTei 28387

Equipment Data Collection Lines and Equipment Array

MOVE
Source TXTi 28387
Dest EDC.Line[LNi].Equip[EQi].EventCTL.TXTei 23779

Configuration Response from Server via Transaction Manager

EventTextEmptyFSC06.FD

Dest 162500 TXTi 28387

Data OK
DN009 May 5, 2020 -- Ack data for any EQID 1-9999

35

lbl_DataOk
LBL

Server Interface Status State = Red - Equip Event Rx Data OK
EDC_Status01.ServerCfg.Update.EquipEventRxDataOK
L

Update Status State for Transactions
aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Update Event Text Fragment Statistics
(for New Event Text only)

36

EqDefined Event
EventTextEqDef

EventPosFound

Event Text Tag01 Empty Search
EventTextEmptyFSC01.FD

Event DtType Ignore
EventDtIgnore

Event DtType Info
EventDtInfo

Event NoDisplay
EventNoDisplay

Update Event Text Statistics
UpdateEventTextStat

Event Text Tag02 Empty Search
EventTextEmptyFSC02.FD

Event Text Tag03 Empty Search
EventTextEmptyFSC03.FD

Event Text Tag04 Empty Search
EventTextEmptyFSC04.FD

Event Text Tag05 Empty Search
EventTextEmptyFSC05.FD

Event Text Tag06 Empty Search
EventTextEmptyFSC06.FD

Configuration Response from Server via Transaction Manager

Update Event Controls
Event Statistics and Masks

37

Decommissioned Equipment
DecomEQi
EqDefined Event
EventTextEqDef
Configuration Response from Server via Transaction Manager
JSR
Routine Name _837_CfgResponseEquipmentEventCtlEq

EqDefined Event
EventTextEqDef
Configuration Response from Server via Transaction Manager
JSR
Routine Name _837_CfgResponseEquipmentEventCtlPre

Exit Routine

38

lbl_Exit
LBL
NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 17 - Equipment-Defined Event Text
Update Event Text Fragment Statistics and Masks
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 837
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check for good Event Looklup Index for EDC_Event EvLKi

2

Equipment-Event Index for EDC_Event Lookup

CLR
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array

LIMIT
Low Limit 0
Test EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144
High Limit EDC_Event.CFG.EQ.mi 2999

Equipment-Event Index for EDC_Event Lookup

MOVE
Source EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144
Dest EvLKi 164

Event Lookup Index within array limits
EvLKi_IndexOK

Equipment-Event Index for EDC_Event matches

3

Event Lookup Index EQID Match
EvLKi_EQIDMatch
U

Event Lookup Index within array limits
EvLKi_IndexOK

Event Controls Equipment Fragment Statistics Array

EQ
Source A EDC_Event.CTL.Equip[EvLKi].EQID 926
Source B EQID 531

Event Lookup Index EQID Match
EvLKi_EQIDMatch
L

Configuration Response from Server via Transaction Manager

Search for EQID Index if not found
Store result to avoid future searches

(redundant to CfgResponseEquipment - should always exist from Equipment Download)

4

Event Lookup Index EQID Match
EvLKi_EQIDMatch
/

Diagnostic
ADD
Source A EventEqFscCount 16
Source B 1
Dest EventEqFscCount 16

Fragment Statistics Equipment Search
EventEqFSC
RES

Fragment Statistics Equipment Search
MOVE
Source 0
Dest EventEqFSC.POS 0

Fragment Statistics Equipment Search
MOVE
Source EDC_Event.CFG.EQ.n 3000
Dest EventEqFSC.LEN 3000

Fragment Statistics Equipment Search
FSC
Control EventEqFSC
Length 3000
Position 0
Mode ALL
Expression EDC_Event.CTL.Equip[EventEqFSC.POS].EQID=EQID
EN
DN
ER

Fragment Statistics Equipment Search
EventEqFSC.FD

Equipment-Event Index for EDC_Event Lookup
MOVE
Source EventEqFSC.POS 0
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array
MOVE
Source EvLKi 164
Dest EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144

Event Lookup Index EQID Match
EvLKi_EQIDMatch
L

Event Lookup Index Found
EvLKi_Search

Configuration Response from Server via Transaction Manager

Search for Empty Index if new Equipment
Store result to avoid future searches

(redundant to CfgResponseEquipment - should always exist from Equipment Download)

5

Event Lookup Index EQID Match
EvLKi_EQIDMatch
/

Diagnostic
ADD
Source A EventEmptyFscCount 1
Source B 1
Dest EventEmptyFscCount 1

Fragment Statistics Empty Search
EventEmptyFSC
RES

Fragment Statistics Empty Search
MOVE
Source 0
Dest EventEmptyFSC.POS 271

Fragment Statistics Empty Search
MOVE
Source EDC_Event.CFG.EQ.n 3000
Dest EventEmptyFSC.LEN 3000

Fragment Statistics Empty Search
FSC
Control EventEmptyFSC
Length 3000
Position 271
Mode ALL
Expression EDC_Event.CTL.Equip[EventEmptyFSC.POS].EQID = 0
EN
DN
ER

Fragment Statistics Empty Search
EventEmptyFSC.FD

Equipment-Event Index for EDC_Event Lookup
MOVE
Source EventEmptyFSC.POS 271
Dest EvLKi 164

Equipment Data Collection Lines and Equipment Array
MOVE
Source EvLKi 164
Dest EDC.Line[LNi].Equip[EQi].EventCTL.EvLKi 144

Event Controls Equipment Fragment Statistics Array
MOVE
Source EQID 531
Dest EDC_Event.CTL.Equip[EvLKi].EQID 926

Event Controls Equipment with Event Text Qty
ADD
Source A EDC_Event.CTL.EqQty 272
Source B 1
Dest EDC_Event.CTL.EqQty 272

Event Lookup Index EQID Match
EvLKi_EQIDMatch
L

Event Lookup Index New
EvLKi_New

Configuration Response from Server via Transaction Manager

Equipment Not Found and Statistics Array is Full
(shouldn't happen - requires Manual Intervention and Manual Reset of Error)

Event Lookup Index
EQID Match
EvLKi_EQIDMatch

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4025
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red - Equip Event Fragmentation Statistics Array Full ER
EDC_Status01.ServerCfg.Update.EventStatsArrayFull
L

Update Status State for Transactions
aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Skip Event Text Statistics Update if new Event was not added

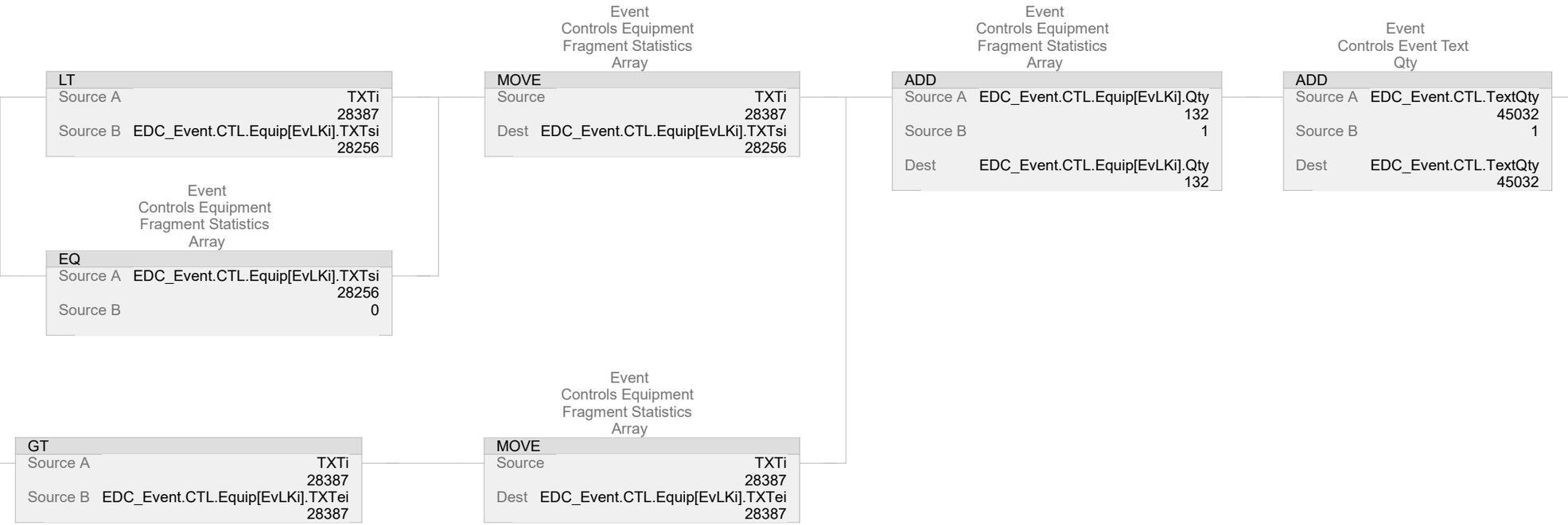
Update Event Text Statistics
UpdateEventTextStat

lbl_UpdateMasks
JMP

Configuration Response from Server via Transaction Manager

If new Event Text, Update Index Range and Increment EQ, Total Counts

8



Configuration Response from Server via Transaction Manager

Update Fragmentation for Previous Equipment
Only allow positive Fragment counts
Don't allow Previous Frag if Previous Event quantity is 0

Fragmentation = This ei - Prev si - 1
Change Fragmentation Statistics from Event% to Equip% (more meaningful - Event% overstated fragmentation)

9

Equipment-Event Index for EDC_Event Lookup

NE
Source A EvLKi 164
Source B 0

SUB
Source A EvLKi 164
Source B 1
Dest LKprv 233

MOVE
Source EDC_Event.CTL.Equip[LKprv].Frag 0
Dest PrevFrag 0

Event Controls Equipment Fragment Statistics Array

CPT
Dest EDC_Event.CTL.Equip[LKprv].Frag 0
Expression EDC_Event.CTL.Equip[EvLKi].TXTsi - EDC_Event.CTL.Equip[LKprv].TXTei - 1

Event Controls Equipment Fragment Statistics Array

LT
Source A EDC_Event.CTL.Equip[LKprv].Frag 0
Source B 0

Event Controls Equipment Fragment Statistics Array

CLR
Dest EDC_Event.CTL.Equip[LKprv].Frag 0

Event Controls Equipment Fragment Statistics Array

EQ
Source A EDC_Event.CTL.Equip[LKprv].Qty 192
Source B 0

AFI

Event Controls Event Fragment Qty

CPT
Dest EDC_Event.CTL.FragQty 4
Expression EDC_Event.CTL.FragQty + (EDC_Event.CTL.Equip[LKprv].Frag - PrevFrag)

EQ
Source A PrevFrag

Event Controls Equipment Fragment Statistics Array

NE
Source A EDC_Event.CTL.Equip[LKprv].Frag

Event Controls Event Fragment Qty

ADD
Source A EDC_Event.CTL.FragQty

Configuration Response from Server via Transaction Manager

	0
Source B	0

	0
Source B	0

	4
Source B	1
Dest	EDC_Event.CTL.FragQty 4

Configuration Response from Server via Transaction Manager

Update Fragmentation for This Equipment
Only allow positive Fragment counts

Fragmentation = This ei - Next si + 1
Change Fragmentation Statistics from Event% to Equip% (more meaningful - Event% overstated fragmentation)

10

Equipment-Event Index for EDC_Event Lookup

NE
Source A EvLKi 164
Source B EDC_Event.CFG.EQ.mi 2999

ADD
Source A EvLKi 164
Source B 1
Dest LKnxt 235

Event Controls Equipment Fragment Statistics Array

NE
Source A EDC_Event.CTL.Equip[LKnxt].TXTsi 37062
Source B 0

MOVE
Source EDC_Event.CTL.Equip[EvLKi].Frag 0
Dest PrevFrag 0

Event Controls Equipment Fragment Statistics Array

CPT
Dest EDC_Event.CTL.Equip[EvLKi].Frag 0
Expression EDC_Event.CTL.Equip[EvLKi].TXTei - EDC_Event.CTL.Equip[LKnxt].TXTsi + 1

Event Controls Equipment Fragment Statistics Array

LT
Source A EDC_Event.CTL.Equip[EvLKi].Frag 0
Source B 0

Event Controls Equipment Fragment Statistics Array

CLR
Dest EDC_Event.CTL.Equip[EvLKi].Frag 0

Event Controls Event Fragment Qty

CPT
Dest EDC_Event.CTL.FragQty 4
Expression EDC_Event.CTL.FragQty + (EDC_Event.CTL.Equip[EvLKi].Frag - PrevFrag)

AFI

Event Controls Equipment Fragment Statistics Array

EQ
Source A PrevFrag 0

Event Controls Equipment Fragment Statistics Array

NE
Source A EDC_Event.CTL.Equip[EvLKi].Frag 0

Event Controls Event Fragment Qty

ADD
Source A EDC_Event.CTL.FragQty 4

Configuration Response from Server via Transaction Manager

Source B	0

Source B	0

Source B	1
Dest	EDC_Event.CTL.FragQty 4

Update Fragmentation Percentage and Last EQ
Change Fragmentation Statistics from Event% to Equip% (more meaningful - Event% overstated fragmentation)

11

AFI

Event
Controls Event
Fragment Percentage

CPT

Dest	EDC_Event.CTL.FragPerc 1.4705882
Expression	EDC_Event.CTL.FragQty / (EDC_Event.CTL.FragQty + EDC_Event.CTL.TextQty) *100

Event
Controls Event
Fragment Percentage

CPT

Dest	EDC_Event.CTL.FragPerc 1.4705882
Expression	(EDC_Event.CTL.FragQty * 100.0) / EDC_Event.CTL.EqQty

Update Event Masks for DtOwner and DtType

12

lbl_UpdateMasks
LBL

NOP

Configuration Response from Server via Transaction Manager

New EQID - Initialize Event Masks to DtOwner = Maintenance, DtType = Fault and Ignore = pass data
Initialize Mask Config words (new data for EqDriver)

*** Ignore Mask is inverted from Configuration -- when OFF, the Mask bit is 1 to pass data thru MEQ ***
v3.02 Data Refresh Request Command

13

Event Lookup Index New
EvLKi_New

Temp Result for Array Size
MUL
Source A EDC_Event.CFG.PKT.n[8]
Source B EDC_Event.CFG.DAT.n[8]
Dest LEN
64

DT Owner = Maintenance Event Mask Equipment Mask Equip Array
FLL
Source -1
Dest EDC_EventMaskMaint.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Fault Event Mask Equipment Mask Equip Array
FLL
Source -1
Dest EDC_EventMaskFault.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Alarm Event Mask Equipment Mask Equip Array
FLL
Source 0
Dest EDC_EventMaskAlarm.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Alert Event Mask Equipment Mask Equip Array
FLL
Source 0
Dest EDC_EventMaskAlert.Equip[EvLKi].Packet[0]
Length LEN

DT Type = Ignore Event Mask Equipment Mask Equip Array
FLL
Source -1
Dest EDC_EventMaskIgnore.Equip[EvLKi].Packet[0]
Length LEN

No Display Event Mask Equipment Mask Equip Array
FLL
Source 0
Dest EDC_EventMaskNoDisp.Equip[EvLKi].Packet[0]
Length LEN

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].MaintMaskSend
2#0000_0000

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].FaultMaskSend
2#0000_0000

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend
2#0000_0000

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].AlertMaskSend
2#0000_0000

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].IgnoreMaskSend
2#0000_0000

Event Controls Equipment Fragment Statistics Array
MOVE
Source -1
Dest EDC_Event.CTL.Equip[EvLKi].DataRefreshReq
2#0000_0000

Configuration Response from Server via Transaction Manager

Validate Packet Indexes
PKTi, DATi and BITi

14

Event Mask
Packet Index
(0-7)

LIMIT
Low Limit 0
Test PKTi
7
High Limit EDC_Event.CFG.PKT.mi
7

Event Mask
Data Index
(0-7)

LIMIT
Low Limit 0
Test DATi
3
High Limit EDC_Event.CFG.DAT.mi
7

Event Mask
Bit Index
(0-31)

LIMIT
Low Limit 0
Test BITi
1
High Limit EDC_Event.CFG.BIT.mi
31

PacketIndexOK

PacketIndexOK
/

lbl_Exit
JMP

Update Event Mask for Downtime Owner = Maintenance
Update Mask Cfg for Download to EqDriver

May 2, 2023 - change Maint Mask to Maint-Fault Mask
(only charge Maintenance when DtOwner=Maint and DtType = Fault)

15

Event DtOwner
Maintenance
EventDtMaint

Event DtType
Fault
EventDtFault

DT Owner =
Maintenance
Event Mask Equipment
Mask Equip Array
EDC_EventMaskMaint.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Event
Controls Equipment
Fragment Statistics
Array
EDC_Event.CTL.Equip[EvLKi].MaintMaskSend.[PKTi]
L

Event DtOwner
Maintenance
EventDtMaint
/

DT Owner =
Maintenance
Event Mask Equipment
Mask Equip Array
EDC_EventMaskMaint.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

Event DtType
Fault
EventDtFault
/

Configuration Response from Server via Transaction Manager

Update Event Mask for Downtime Type = Fault
Update Mask Cfg for Download to EqDriver

16

Event DtType Fault EventDtFault

DT Type = Fault Event Mask Equipment Mask Equip Array EDC_EventMaskFault.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] L

Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].FaultMaskSend.[PKTi] L

Event DtType Fault EventDtFault /

DT Type = Fault Event Mask Equipment Mask Equip Array EDC_EventMaskFault.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] U

Update Event Mask for Downtime Type = Alarm
Update Mask Cfg for Download to EqDriver

17

Event DtType Alarm EventDtAlarm

DT Type = Alarm Event Mask Equipment Mask Equip Array EDC_EventMaskAlarm.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] L

Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].AlarmMaskSend.[PKTi] L

Event DtType Alarm EventDtAlarm /

DT Type = Alarm Event Mask Equipment Mask Equip Array EDC_EventMaskAlarm.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] U

Update Event Mask for Downtime Type = Alert
Update Mask Cfg for Download to EqDriver

18

Event DtType Alert EventDtAlert

DT Type = Alert Event Mask Equipment Mask Equip Array EDC_EventMaskAlert.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] L

Event Controls Equipment Fragment Statistics Array EDC_Event.CTL.Equip[EvLKi].AlertMaskSend.[PKTi] L

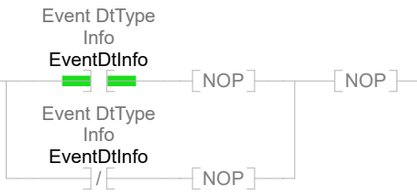
Event DtType Alert EventDtAlert /

DT Type = Alert Event Mask Equipment Mask Equip Array EDC_EventMaskAlert.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi] U

Configuration Response from Server via Transaction Manager

Downtime Type = Info
(no Mask for Info -- it's the default status if not Fault, Alarm, Alert or Ignore)

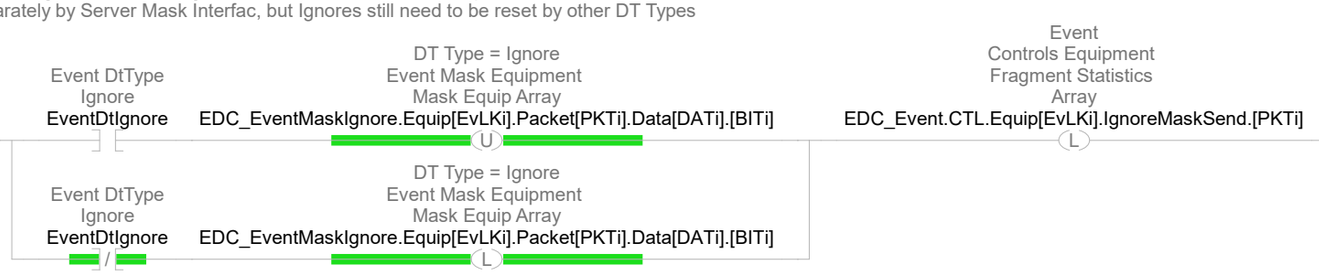
19



Update Event Mask for Downtime Type = Ignore
Update Mask Cfg for Download to EqDriver

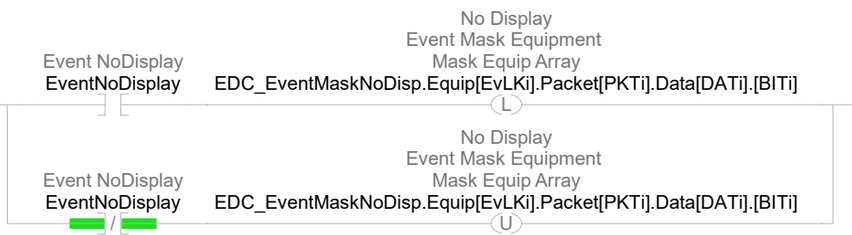
*** Ignore Mask is inverted from Configuration -- when Ignore is OFF, the Mask bit is 1 to pass data thru MEQ ***
Ignore Events are downloaded separately by Server Mask Interfac, but Ignores still need to be reset by other DT Types

20



Update Event Mask for NoDisplay

21



v3.02 Data Refresh Request Command

22



Configuration Response from Server via Transaction Manager

Set Summary Data KPI Option ID 1-5 Event Mask
and Equipment Packet Config State

23

KPI Option 1
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

KPI Option 2
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

KPI Option 3
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

KPI Option 4
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

KPI Option 5
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
U

0 = not defined
1-5 = Option 1-5
EQ
Source A KpiOptionID
0
Source B 1

KPI Option 1
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt01.PktConfig.[PKTi]
L

0 = not defined
1-5 = Option 1-5
EQ
Source A KpiOptionID
0
Source B 2

KPI Option 2
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt02.PktConfig.[PKTi]
L

0 = not defined
1-5 = Option 1-5
EQ
Source A KpiOptionID
0
Source B 3

KPI Option 3
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt03.PktConfig.[PKTi]
L

0 = not defined
1-5 = Option 1-5
EQ
Source A KpiOptionID
0
Source B 4

KPI Option 4
Event Mask Equipment
Mask Equip Array
EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt04.PktConfig.[PKTi]
L

0 = not defined

KPI Option 5
Event Mask Equipment

Event
KPI Options
Equipment Event KPI

Configuration Response from Server via Transaction Manager

1-5 = Option 1-5

EQ
Source A KpiOptionID
0
Source B 5

Mask Equip Array
EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[DATi].[BITi]
L

Options
EDC_EventOpt.Equip[EvLKi].Opt05.PktConfig.[PKTi]
L

Reset Equipment Packet Config State for KPI Option 1
if no Events Configured

24

0 = not defined
1-5 = Option 1-5

NE
Source A KpiOptionID
0
Source B 1

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[0]
16#0000_0000
Source B 0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[1]
16#0000_0000
Source B 0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[2]
16#0000_0000
Source B 0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[3]
16#0000_0000
Source B 0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[4]
16#0000_0000
Source B 0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[5]
16#0000_0000
Source B 0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[6]
16#0000_0000
Source B 0

KPI Option 1
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt01.Equip[EvLKi].Packet[PKTi].Data[7]
16#0000_0000
Source B 0

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt01.PktConfig.[PKTi]
U

Configuration Response from Server via Transaction Manager

Reset Equipment Packet Config State for KPI Option 2
if no Events Configured

25

0 = not defined
1-5 = Option 1-5

NE
Source A KpiOptionID
0
Source B 2

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[0]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[1]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[2]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[3]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[4]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[5]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[6]
16#0000_0000
Source B 0

KPI Option 2
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt02.Equip[EvLKi].Packet[PKTi].Data[7]
16#0000_0000
Source B 0

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt02.PktConfig.[PKTi]
U

Configuration Response from Server via Transaction Manager

Reset Equipment Packet Config State for KPI Option 3
if no Events Configured

26

0 = not defined
1-5 = Option 1-5

NE
Source A KpiOptionID
0
Source B 3

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[0]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[1]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[2]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[3]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[4]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[5]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[6]
16#0000_0000
Source B 0

KPI Option 3
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt03.Equip[EvLKi].Packet[PKTi].Data[7]
16#0000_0000
Source B 0

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt03.PktConfig.[PKTi]
U

Configuration Response from Server via Transaction Manager

Reset Equipment Packet Config State for KPI Option 4
if no Events Configured

27

0 = not defined
1-5 = Option 1-5

NE
Source A KpiOptionID
0
Source B 4

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[0]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[1]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[2]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[3]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[4]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[5]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[6]
16#0000_0000
Source B 0

KPI Option 4
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt04.Equip[EvLKi].Packet[PKTi].Data[7]
16#0000_0000
Source B 0

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt04.PktConfig.[PKTi]
U

Configuration Response from Server via Transaction Manager

Reset Equipment Packet Config State for KPI Option 5
if no Events Configured

28

0 = not defined
1-5 = Option 1-5

NE
Source A KpiOptionID
0
Source B 5

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[0]
16#0000_0000
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[1]
16#0000_0000
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[2]
16#0000_0000
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[3]
16#ffff_fffc
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[4]
16#ffff_ffff
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[5]
16#ffff_ffff
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[6]
16#ffff_ffff
Source B 0

KPI Option 5
Event Mask Equipment
Mask Equip Array

EQ
Source A EDC_EventMaskOpt05.Equip[EvLKi].Packet[PKTi].Data[7]
16#ffff_ffff
Source B 0

Event
KPI Options
Equipment Event KPI
Options
EDC_EventOpt.Equip[EvLKi].Opt05.PktConfig.[PKTi]
U

Configuration Response from Server via Transaction Manager

v3.5 Refresh Equipment Events if not Full Download
(full Download already has a periodic refresh)

29

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.Update
L

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.UpdateOview
L

Equipment Data Collection Lines and Equipment Array
FLL
Source -1
Dest EDC.Line[LNi].Equip[EQi].EventCTL.UpdateZone
Length 1

Exit Routine

30

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 16 - PreDefined Event Text
Update Event Text Masks

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 837
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Update Event Mask for Downtime Owner = Maintenance

May 2, 2023 - change Maint Mask to Maint-Fault Mask
(only charge Maintenance when DtOwner=Maint and DtType = Fault)

2

Event DtOwner Maintenance EventDtMaint

Event DtType Fault EventDtFault

PreDefined Event Mask DtOwner Maintenance Mask EDC_EventMaskPreDef.Maint.[BITi]
(L)

Event DtOwner Maintenance EventDtMaint
(/)

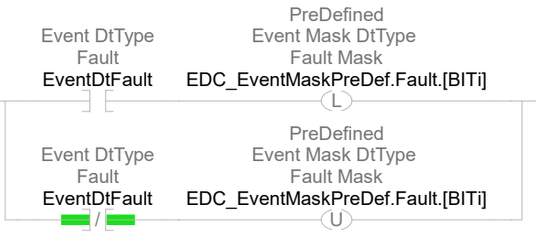
Event DtType Fault EventDtFault
(/)

PreDefined Event Mask DtOwner Maintenance Mask EDC_EventMaskPreDef.Maint.[BITi]
(U)

Configuration Response from Server via Transaction Manager

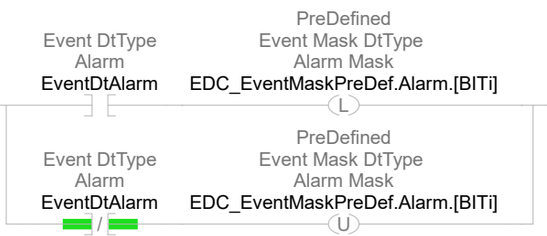
Update Event Mask for Downtime Type = Fault

3



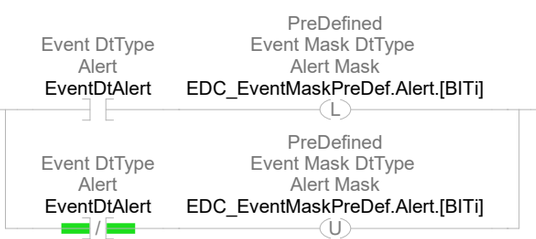
Update Event Mask for Downtime Type = Alarm

4



Update Event Mask for Downtime Type = Alert

5



Configuration Response from Server via Transaction Manager

Downtime Type = Info
(no Mask for Info -- it's the default status if not Fault, Alarm, Alert or Ignore)

6

Event DtType Info EventDtInfo — NOP — NOP
Event DtType Info EventDtInfo — / — NOP

Update Event Mask for Downtime Type = Ignore

*** Ignore Mask is inverted from Configuration -- when Ignore is OFF, the Mask bit is 1 to pass data thru MEQ ***
Ignore Events are downloaded separately by Server Mask Interfac, but Ignores still need to be reset by other DT Types

7

Event DtType Ignore EventDtIgnore — PreDefined Event Mask DtType Ignore Mask EDC_EventMaskPreDef.Ignore.[BITi] — U
Event DtType Ignore EventDtIgnore — / — PreDefined Event Mask DtType Ignore Mask EDC_EventMaskPreDef.Ignore.[BITi] — L

Update Event Mask for NoDisplay

8

Event NoDisplay EventNoDisplay — PreDefined Event Mask No Display Option EDC_EventMaskPreDef.NoDisp.[BITi] — L
Event NoDisplay EventNoDisplay — / — PreDefined Event Mask No Display Option EDC_EventMaskPreDef.NoDisp.[BITi] — U

v3.5 Refresh All Events if not Full Download
(full Download already has a periodic refresh)

9

Server Configuration from Transaction Manager Response Download All Started ServerCfg.Ctl.DownloadAllStarted <EDC.ServerCfg.Ctl.DownloadAllStarted> — /

HMI Refresh HMI Events after Config All EDC_HMI.CTL.DownloadAllRefresh — L

Configuration Response from Server via Transaction Manager

Exit Routine

10

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 8 - Shop
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 838
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data

2

Server Configuration from Transaction Manager Name

NE
Source A ServerCfg.DataRspLast.Name.DATA[0] <EDC.ServerCfg.DataRspLast.Name.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Short Name

NE
Source A ServerCfg.DataRspLast.ShortName.DATA[0] <EDC.ServerCfg.DataRspLast.ShortName.DATA[0]> '$00'
Source B 1

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4018
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- Shop Rx Data ER
EDC_Status01.ServerCfg.Update.ShopRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Full Download Request -- Reset ALL existing Configuration
(Shop Configuration command forces Reset and Download ALL configuration)

EDC2 WL31 Equipment Complete -- once download reaches the Events, all Equipment should have a good configuration
EDC2 WL31 Block EqDriver -- don't service EqDriver Messages until Equipment Configuration is complete and EQ Disable status can be re-established

4

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Server Configuration from Transaction Manager Response Download All Started
ServerCfg.Ctl.DownloadAllStarted
<EDC.ServerCfg.Ctl.DownloadAllStarted>
L

Server Configuration from Transaction Manager Response Download All Complete
ServerCfg.Ctl.DownloadAllComplete
<EDC.ServerCfg.Ctl.DownloadAllComplete>
U

Server Configuration from Transaction Manager Download Started - Equipment Config Complete
ServerCfg.Ctl.DownloadEquipComplete
<EDC.ServerCfg.Ctl.DownloadEquipComplete>
U

Server Configuration from Transaction Manager Download Started - Block Equipment Drivers until Equipment Config Complete
ServerCfg.Ctl.DownloadBlockEqDriver
<EDC.ServerCfg.Ctl.DownloadBlockEqDriver>
L

JSR
Routine Name _810_Reset

Configuration Response from Server via Transaction Manager

Process Data and Store

Equipment Data Collection Gateway Name

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC.CFG.Name
Length 1

Equipment Data Collection Shop Site ID

COP
Source ServerCfg.DataRspLast.ShortName
<EDC.ServerCfg.DataRspLast.ShortName>
Dest EDC.CFG.SiteID
Length 1

HMI Shop Name

COP
Source EDC.CFG.Name
Dest EDC_HMI.Name
Length 1

5

PdM Full Download Initialize

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Server PdM Configuration from Transaction Manager Response Download All Started
ServerPdm.Ctl.DownloadAllStarted
<EDC.ServerPdm.Ctl.DownloadAllStarted>
L

Server PdM Configuration from Transaction Manager Response Download All Complete
ServerPdm.Ctl.DownloadAllComplete
<EDC.ServerPdm.Ctl.DownloadAllComplete>
U

Configuration Initialization before full download

JSR
Routine Name _858_PdmDownloadInit

6

Mask Full Download Initialize

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Server Ignore Mask Configuration from Transaction Manager Response Download All Started
ServerMask.Ctl.DownloadAllStarted
<EDC.ServerMask.Ctl.DownloadAllStarted>
L

Server Ignore Mask Configuration from Transaction Manager Response Download All Complete
ServerMask.Ctl.DownloadAllComplete
<EDC.ServerMask.Ctl.DownloadAllComplete>
U

7

Exit Routine

lbl_Exit
LBL

NOP

8

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 9 - Quality Tool Carrier Scanner
LineID - used for Conv ID
PitchID- used for Quality Tool ID
PosID - used for Quality Tool Data ID
ZoneDispID- used for Carrier Scanner Type
ConfigStatus - used for Enable, Emtpy bits

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 839
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Convert EQID to LNi and EQi

2

Equipment Index

MOD
Source A ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1
Source B 100
Dest EQi 31

Line Index

CPT
Dest LNi 5
Expression (ServerCfg.DataRspLast.EqID - EQi) / 100

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

3

Line Index
LIMIT
Low Limit 0
Test LNi
5
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

Equipment Index
LIMIT
Low Limit 0
Test EQi
31
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

Server Configuration from Transaction Manager Line ID (0=Offline)
NE
Source A ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Source B -1

Server Configuration from Transaction Manager Pitch ID (0-350)
NE
Source A ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID>
-1
Source B -1

Server Configuration from Transaction Manager Position ID (0-5)
NE
Source A ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID>
-1
Source B -1

Server Configuration from Transaction Manager Maintenance Zone Display ID (1-8)
NE
Source A ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Source B -1

Server Configuration from Transaction Manager Name
NE
Source A ServerCfg.DataRspLast.Name.DATA[0]
<EDC.ServerCfg.DataRspLast.Name.DATA[0]>
'$00'
Source B 1

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>

4

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4019
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red-ER
Carrier Scanner Data Read/Write
EDC_Status01.ServerCfg.Update.CarrScanDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Initialize Carrier Scanner Index

5

CLR
Dest i
300

Carrier Scanner Empty Array Position Found
ScanEmptyFD
U

Carrier Scanner Matching Array Position Found
ScanMatchFD
U

Configuration Response from Server via Transaction Manager

Search for matching Scanner Record to update
(and search for first unused array position)

6

lbl_CheckPos
LBL

EQ
Source A: EDC_Track.ScanCfg[i].EQID ??
Source B: 0

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

Carrier Scanner Empty Array Position Found
ScanEmptyFD
/

MOVE
Source: i 300
Dest: ScanPOS 4

Carrier Scanner Empty Array Position Found
ScanEmptyFD
L

lbl_NextPos
JMP

Check Carrier Scanner Position for match of EQID, ToolID, DataID

7

EQ
Source A: EDC_Track.ScanCfg[i].EQID ??
Source B: ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

EQ
Source A: EDC_Track.ScanCfg[i].ToolID ??
Source B: ServerCfg.DataRspLast.PitchID <EDC.ServerCfg.DataRspLast.PitchID> -1

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

EQ
Source A: EDC_Track.ScanCfg[i].DataID ??
Source B: ServerCfg.DataRspLast.PositionID <EDC.ServerCfg.DataRspLast.PositionID> -1

Carrier Scanner Empty Array Position

MOVE
Source: i 300
Dest: ScanPOS 4

Carrier Scanner Matching Array Position Found
ScanMatchFD
L

Increment Carrier Scanner Index

8

lbl_NextPos
LBL

Carrier Scanner Matching Array Position Found
ScanMatchFD
/

ADD
Source A: i 300
Source B: 1
Dest: i 300

LE
Source A: i 300
Source B: EDC_Track.Cfg.SCN.mi 299

lbl_CheckPos
JMP

Configuration Response from Server via Transaction Manager

Carrier Scanner Array Full

9

Carrier Scanner Empty Array Position Found
ScanEmptyFD

Carrier Scanner Matching Array Position Found
ScanMatchFD

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4026
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red-Carrier Scanner Array Full ER Read/Write
EDC_Status01.ServerCfg.Update.CarrScanArrayFull
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store

LineID - used for Conv ID
PitchID- used for Quality Tool ID
PosID - used for Quality Tool Data ID
ZoneDispID- used for Carrier Scanner Type
ConfigStatus - used for Enable, Emtpy bits

10

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CFG.QualityMsgTrackEN
(L)

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration
EDC_Track.ScanCfg[ScanPOS].Enable
(U)

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration
EDC_Track.ScanCfg[ScanPOS].Empty
(U)

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.0
<EDC.ServerCfg.DataRspLast.ConfigStatus.0>

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration
EDC_Track.ScanCfg[ScanPOS].Enable
(L)

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.1
<EDC.ServerCfg.DataRspLast.ConfigStatus.1>

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration
EDC_Track.ScanCfg[ScanPOS].Empty
(L)

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID> -1
Dest EDC_Track.ScanCfg[ScanPOS].EQID
114

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

MOVE
Source ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID> -1
Dest EDC_Track.ScanCfg[ScanPOS].ToolID
1

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

MOVE
Source ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID> -1
Dest EDC_Track.ScanCfg[ScanPOS].DataID
1

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID> -1
Dest EDC_Track.ScanCfg[ScanPOS].ConvID
25

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

MOVE
Source ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID> -1
Dest EDC_Track.ScanCfg[ScanPOS].Type
4

Carrier Tracking from EQ Quality Tools Tracking Scanner Configuration

CPT
Dest EDC_Track.ScanCfg[ScanPOS].Srch
11401
Expression (ServerCfg.DataRspLast.EqID *100) + ServerCfg.DataRspLast.PitchID

Carrier Tracking from EQ Quality Tools Tracking Conveyor

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC_Track.Conv[ServerCfg.DataRspLast.LineID].Cfg.Name
Length 1

Configuration Response from Server via Transaction Manager

Data OK

Server Interface Status State = Green- Carrier Scanner Data OK

EDC_Status01.ServerCfg.Update.CarrScanDataOK

L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig

aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig

Status EDC_Status01.ServerCfg

11

Exit Routine

lbl_Exit

LBL

12

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 6 - Schedule
LineID - used for Array Index
EQID - used for ShiftID
PitchID- used for Quarter ID
PosID - used for Hour ID
ZoneDispID- used for Hour Start
EqEventID - used for Shift Reset

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	840
Dest	Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -806549717
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

2

Server Configuration from Transaction Manager Shop Key - for DB Ack only

EQ
Source A ServerCfg.DataRspLast.ShopKey <EDC.ServerCfg.DataRspLast.ShopKey> -1
Source B 1

Server Configuration from Transaction Manager Line ID (0=Offline)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.LineID <EDC.ServerCfg.DataRspLast.LineID> -1
High Limit CfgSched.mi 99

Server Configuration from Transaction Manager Equipment ID

NE
Source A ServerCfg.DataRspLast.EqID <EDC.ServerCfg.DataRspLast.EqID> -1
Source B -1

Server Configuration from Transaction Manager Pitch ID (0-350)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.PitchID <EDC.ServerCfg.DataRspLast.PitchID> -1
High Limit 5

Server Configuration from Transaction Manager Position ID (0-5)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.PositionID <EDC.ServerCfg.DataRspLast.PositionID> -1
High Limit 10

Server Configuration from Transaction Manager Maintenance Zone Display ID (1-8)

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.ZoneDispID <EDC.ServerCfg.DataRspLast.ZoneDispID> -1
High Limit 240000

Server Configuration from Transaction Manager Equipment Event ID - EQID & Element ID

LIMIT
Low Limit 0
Test ServerCfg.DataRspLast.EqEventID <EDC.ServerCfg.DataRspLast.EqEventID> -1
High Limit 1

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData <EDC.ServerCfg.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance
(share Shop status - need to expand Status to support new downloads)

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
/

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4019
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- Shop Rx Data ER
EDC_Status01.ServerCfg.Update.ShopRxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig ...
Status EDC_Status01.ServerCfg

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store
if HourStart = 240000, this is the last record -- initialize remaining array positions to -1

LineID - used for array Index
EqID - used for Shift
PitchID- used for Quarter 0-5
PosID - used for Hour 0-10
ZoneDispID- used for HourStart (0-240000)
EqEventID - used for Shift Reset (0-1)

4

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Dest index
22

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest ScheduleLookup[index].Shift
2

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID>
-1
Dest ScheduleLookup[index].Qtr
2

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID>
-1
Dest ScheduleLookup[index].Hour
4

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Dest ScheduleLookup[index].HourStart
240000

Schedule Hours lookup array

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Dest ScheduleLookup[index].Reset
0

Server Configuration from Transaction Manager Maintenance Zone Display ID (1-8)

EQ
Source A ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Source B 240000

LT
Source A index
22
Source B CfgSched.mi
99

Temp Result for Array Size

SUB
Source A CfgSched.mi
99
Source B index
22
Dest LEN
64

Schedule Hours lookup array

FLL
Source -1
Dest ScheduleLookup[index+1]
Length LEN

Configuration Response from Server via Transaction Manager

adjust array end index
used for FSC search length

Schedule Array
Config Array End
Index

5

GT
Source A index
22
Source B CfgSched.ei
22

LE
Source A index
22
Source B CfgSched.mi
99

MOVE
Source index
22
Dest CfgSched.ei
22

Server Configuration
from Transaction
Manager Maintenance
Zone Display ID
(1-8)

EQ
Source A ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Source B 240000

Data OK
(share Shop status - need to expand Status to support new downloads)

Server Interface
Status State = Red-
Shop Rx Data OK
EDC_Status01.ServerCfg.Update.ShopRxDataOK

6

L

Update Status State
for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Exit Routine

lbl_Exit

7

LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 7 - Gateway
Zone - used for TimeZoneID
EqEvent
RankNum - used for DstAdjust
Name - used for SntpServerAddress
EQID - used for LdtEQi
ConfigStatus - used for 3 config bits

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 841
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

Logix Wallclock allows Time Zone selection from -12 to +14

2

Server Configuration from Transaction Manager Shop Key - for DB Ack only

EQ
Source A ServerCfg.DataRspLast.ShopKey
<EDC.ServerCfg.DataRspLast.ShopKey>
-1
Source B 1

Server Configuration from Transaction Manager Equipment Event Text

NE
Source A ServerCfg.DataRspLast.EqEventText.DATA[0]
<EDC.ServerCfg.DataRspLast.EqEventText.DATA[0]>
'$00'
Source B 1

Server Configuration from Transaction Manager Maintenance Risk Rank Assignments (4-1)

NE
Source A ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum>
-1
Source B -1

Server Configuration from Transaction Manager Name

NE
Source A ServerCfg.DataRspLast.Name.DATA[0]
<EDC.ServerCfg.DataRspLast.Name.DATA[0]>
'$00'
Source B 1

Server Configuration from Transaction Manager Equipment ID

NE
Source A ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Source B -1

Server Configuration from Transaction Manager Response has Good Data Record

ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance
(share Shop status - need to expand Status to support new downloads)

3

Server Configuration from Transaction Manager Response has Good Data Record
ServerCfg.Ctl.RspGoodData
<EDC.ServerCfg.Ctl.RspGoodData>
[/]

Server Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerCfg.DataRspLast.ShopKey
<EDC.ServerCfg.DataRspLast.ShopKey>
-1
Dest ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
2317

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4019
Input Par ServerCfg.Ctl.ReqInProgressTMR.ACC
<EDC.ServerCfg.Ctl.ReqInProgressTMR.ACC>
Input Par ServerCfg.DataRspLast.Command
<EDC.ServerCfg.DataRspLast.Command>
Input Par ServerCfg.Ctl.CommandID
<EDC.ServerCfg.Ctl.CommandID>
Input Par ServerCfg.Ctl.TransactionNumber
<EDC.ServerCfg.Ctl.TransactionNumber>

Server Interface Status State = Red- Shop Rx Data ER
EDC_Status01.ServerCfg.Update.ShopRxDataER
(L)

Update Status State for Transactions

aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig (...)
Status EDC_Status01.ServerCfg

lbl_Exit
(JMP)

Configuration Response from Server via Transaction Manager

EqEventTxt- used for TimeZone (59-char max - expanded EqEventTxt from 40 to 82)
RankNum - used for DstAdjust
Name - used for SntpServerAddress
EQID - used for LdtEQi
ConfigStatus - used for 3 config bits

4

Gateway Config Download Wall Clock Time Zone String

COP
Source ServerCfg.DataRspLast.EqEventText
<EDC.ServerCfg.DataRspLast.EqEventText>
Dest EDC_CfgGateway.TimeZoneString
Length 1

Gateway Config Download Wall Clock Daylight Savings Time Adjustment (minutes)

MOVE
Source ServerCfg.DataRspLast.RankNum
<EDC.ServerCfg.DataRspLast.RankNum>
-1
Dest EDC_CfgGateway.DstAdjustment
60

Gateway Config Download Wall Clock Time Sync SNTP Server IP Address

COP
Source ServerCfg.DataRspLast.Name
<EDC.ServerCfg.DataRspLast.Name>
Dest EDC_CfgGateway.SntpServerIP
Length 1

Gateway Config Download Line Downtime (KPI) Equip Index when Rollup not enabled (default 0)

MOVE
Source ServerCfg.DataRspLast.EqID
<EDC.ServerCfg.DataRspLast.EqID>
-1
Dest EDC_CfgGateway.LdtEQi
0

Gateway Config Download HMI Common Line Equipment - AVI Equip Index (default 0)

MOVE
Source ServerCfg.DataRspLast.LineID
<EDC.ServerCfg.DataRspLast.LineID>
-1
Dest EDC_CfgGateway.HmiComAviEQi
9

Gateway Config Download HMI Common Line Equipment - Andon Equip Index (default 0)

MOVE
Source ServerCfg.DataRspLast.PitchID
<EDC.ServerCfg.DataRspLast.PitchID>
-1
Dest EDC_CfgGateway.HmiComAndonEQi
1

Gateway Config Download HMI Common Line Equipment - Conveyor Equip Index (default 2)

MOVE
Source ServerCfg.DataRspLast.PositionID
<EDC.ServerCfg.DataRspLast.PositionID>
-1
Dest EDC_CfgGateway.HmiComConvEQi
2

Gateway Config Download HMI Common Line Equipment - Entrance Equip Index (default 3)

MOVE
Source ServerCfg.DataRspLast.ZoneDispID
<EDC.ServerCfg.DataRspLast.ZoneDispID>
-1
Dest EDC_CfgGateway.HmiComEntEQi
3

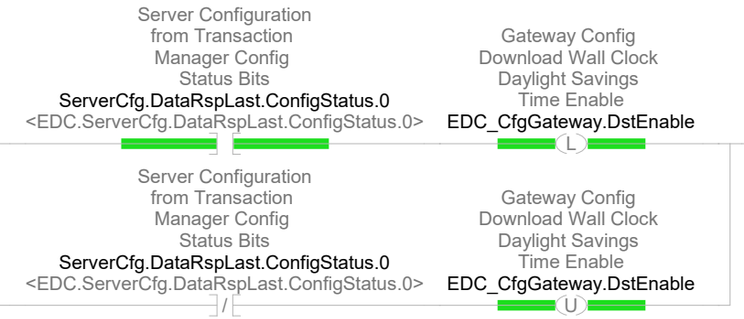
Gateway Config Download HMI Common Line Equipment - Exit Equip Index (default 4)

MOVE
Source ServerCfg.DataRspLast.EqEventID
<EDC.ServerCfg.DataRspLast.EqEventID>
-1
Dest EDC_CfgGateway.HmiComExitEQi
4

Configuration Response from Server via Transaction Manager

Bit 0 - Enable Daylight Savings Time (DST)
if enabled, applies the DST offset during DST

5



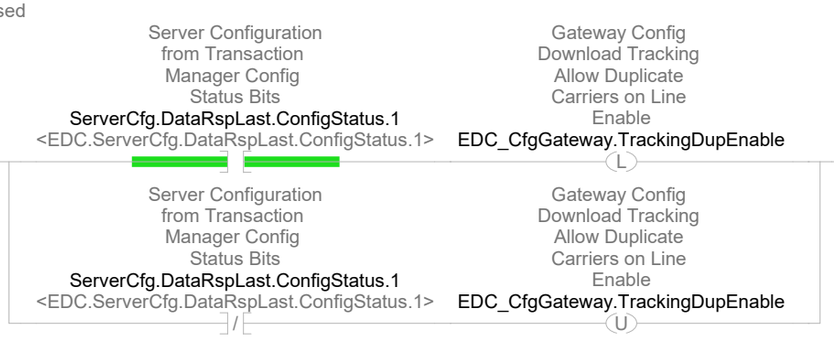
Bit 1 - Enable Tracking Duplicates

Tracking - allow duplicate Die/Mold on Line

Assembly, Paint and BodyWeld do not allow duplicate Carriers on a Line
Plastics and Stamping do allow duplicate Die/Mold on a Line

this may need to be applied at the Line level, but for now it is Shop-based

6

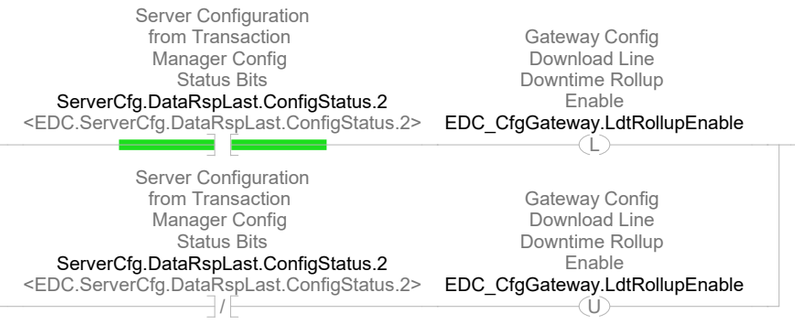


Configuration Response from Server via Transaction Manager

Bit 2 - Enable Downtime Rollup
Rollup of All Equipment worst case or one Equipment (i.e. Andon)
(rarely used - typically the Line DT = Andon Equipment DT)

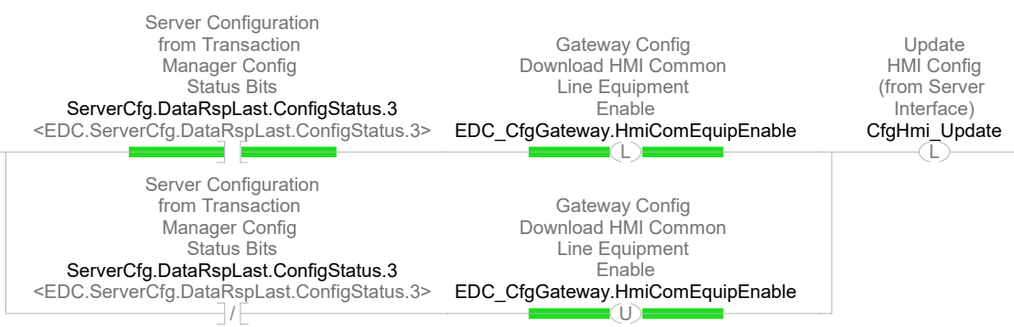
Line Downtime Configuration

7



Bit 3 - Disable HMI Overview Common Line Equipment
(used in shops with EQID xx00-xx05 assigned for a common display of AVI, Andon, Conveyor, Entrance LIfter and Exit Lifter)

8



Configuration Response from Server via Transaction Manager

Bit 4 - Enable HMI Overview for Health Alarm/Fault Event Display

9

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.4
<EDC.ServerCfg.DataRspLast.ConfigStatus.4>

Gateway Config Download HMI Display Health Events Enable
EDC_CfgGateway.HmiHealthEventEnable
L

Server Configuration from Transaction Manager Config Status Bits
ServerCfg.DataRspLast.ConfigStatus.4
<EDC.ServerCfg.DataRspLast.ConfigStatus.4>
/

Gateway Config Download HMI Display Health Events Enable
EDC_CfgGateway.HmiHealthEventEnable
U

Data OK
(share Shop status - need to expand Status to support new downloads)

10

Server Interface Status State = Red- Shop Rx Data OK
EDC_Status01.ServerCfg.Update.ShopRxDataOK
L

Update Status State for Transactions
aoiEDC_StatusUpdateServerConfig
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerConfig
Status EDC_Status01.ServerCfg

Exit Routine

11

lbl_Exit
LBL

NOP

(End)

Configuration Request to Server via Transaction Manager

Configuration Data Request to Server via Transaction Manager

- Logix Requests Data (-1 Command)
- Server Responds with Data (101-103 Command), No Data (0 Command) or No Response (-1)

- if Data, Logix processes Data and Sends Ack (Echo Server Command)
- Server uses Ack to set Download Timestamp for Config Record
- Server Responds with next Data/NoData

- if NoData, Logix waits and then Requests Data (-1 Command)

0

NOP

Request Commands to TM/SQL

-1 = New Data Request (delay between requests)
101-103 = Ack Data Response and New Data Request (no delay between requests)

1

NOP

Response Commands from Server

-1 = No Data Error
0 = No Data to Download
101 = Quality Tool Configuration
102 = Trend Tool Configuration
103 = Network Configuration

2

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

3

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 845
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Request to Server via Transaction Manager

Transaction Request Timer
Fast Interval used when last TM Response had data

4

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Response Duplicate Error
ServerPdm.Ctl.RspDupER
<EDC.ServerPdm.Ctl.RspDupER>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerPdm.Ctl.ReqFastTime
<EDC.ServerPdm.Ctl.ReqFastTime>
0
Dest ServerPdm.Ctl.ReqNextTMR.PRE
<EDC.ServerPdm.Ctl.ReqNextTMR.PRE>
10000

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Response Duplicate Error
ServerPdm.Ctl.RspDupER
<EDC.ServerPdm.Ctl.RspDupER>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerPdm.Ctl.ReqSlowTime
<EDC.ServerPdm.Ctl.ReqSlowTime>
10000
Dest ServerPdm.Ctl.ReqNextTMR.PRE
<EDC.ServerPdm.Ctl.ReqNextTMR.PRE>
10000

Server PdM Configuration from Transaction Manager Request Next Record Timer

TON
Timer ServerPdm.Ctl.ReqNextTMR
<EDC.ServerPdm.Ctl.ReqNextTMR>
Preset 10000
Accum 9505
EN
DN

Configuration Request to Server via Transaction Manager

Request Next
Ready for next Transaction Manager Trigger

Server PdM Configuration from Transaction Manager Request Next Record Timer
ServerPdm.Ctl.ReqNextTMR.DN
<EDC.ServerPdm.Ctl.ReqNextTMR.DN>

Server PdM Configuration from Transaction Manager Request Next Record
ServerPdm.Ctl.ReqNext
<EDC.ServerPdm.Ctl.ReqNext>

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>

Server PdM Configuration from Transaction Manager Request Record Complete
ServerPdm.Ctl.ReqComplete
<EDC.ServerPdm.Ctl.ReqComplete>

5

Request Next Initialize
if Response Good Data, echo Response in next Request to Ack
if not Response Good Data, send new Data Request after wait interval

Server PdM Configuration from Transaction Manager Request Next Record
ServerPdm.Ctl.ReqNext
<EDC.ServerPdm.Ctl.ReqNext>

Server PdM Configuration from Transaction Manager Request Record in Progress Timeout
ServerPdm.Ctl.ReqInProgressTimeout
<EDC.ServerPdm.Ctl.ReqInProgressTimeout>

Configuration Request to Server via Transaction Manager - Data Init

JSR
Routine Name _846_PdmRequestInit

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
298

6

Configuration Request to Server via Transaction Manager

Request Next
Increment Transaction Number
(Triggers TM Transaction on change)

7

Server PdM Configuration from Transaction Manager Request Next Record
ServerPdm.Ctl.ReqNext
<EDC.ServerPdm.Ctl.ReqNext>

Server PdM Configuration from Transaction Manager Request Record in Progress Timeout
ServerPdm.Ctl.ReqInProgressTimeout
<EDC.ServerPdm.Ctl.ReqInProgressTimeout>

Server PdM Configuration from Transaction Manager Transaction Number (TM Request Trigger)

ADD
Source A ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210
Source B 1
Dest ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210

Server PdM Configuration from Transaction Manager Transaction Number (TM Request Trigger)

LT
Source A ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210
Source B 1

Server PdM Configuration from Transaction Manager Transaction Number (TM Request Trigger)

GT
Source A ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Server PdM Configuration from Transaction Manager Transaction Number (TM Request Trigger)

MOVE
Source 1
Dest ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210

Configuration Request to Server via Transaction Manager

Request Next
Log Transaction & Performance

8

Server PdM Configuration from Transaction Manager Request Next Record
ServerPdm.Ctl.ReqNext
<EDC.ServerPdm.Ctl.ReqNext>

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	4300
Input Par	ServerPdm.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerPdm.DataReq.Command
	<EDC.ServerPdm.DataReq.Command>
Input Par	ServerPdm.Ctl.CommandID
	<EDC.ServerPdm.Ctl.CommandID>
Input Par	ServerPdm.Ctl.TransactionNumber
	<EDC.ServerPdm.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR	
Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR	
Dest	TempPerfMsg.TransTMR.ACC
	110

Server PdM Configuration from Transaction Manager Request Record in Progress Timeout
ServerPdm.Ctl.ReqInProgressTimeout
<EDC.ServerPdm.Ctl.ReqInProgressTimeout>

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	4302
Input Par	ServerPdm.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerPdm.DataReq.Command
	<EDC.ServerPdm.DataReq.Command>
Input Par	ServerPdm.Ctl.CommandID
	<EDC.ServerPdm.Ctl.CommandID>
Input Par	ServerPdm.Ctl.TransactionNumber
	<EDC.ServerPdm.Ctl.TransactionNumber>

Performance Transaction

aoiEDC_PerfTrans	
aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EDC_Status01.ServerPdm.Msg.EN

Configuration Request to Server via Transaction Manager

Request Complete
Wait for TM Handshake to Acknowledge Request Complete
v3.5 - replace bit-handshiake with TransNum echo

9

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>

Server PdM Configuration from Transaction Manager Transaction Number (TM Request Trigger)

EQ
Source A ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>
3210
Source B ServerPdm.Ctl.Interface
<EDC.ServerPdm.Ctl.Interface>
3210

Server PdM Configuration from Transaction Manager Request Record Complete
ServerPdm.Ctl.ReqComplete
<EDC.ServerPdm.Ctl.ReqComplete>
L

Configuration Request to Server via Transaction Manager

Request Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
Also problems with asynchronous data from TM - handshake can update before data values.
2-Check method is fast and avoids errors for occasional slow update of data.
Wait 1 scan before 1st check to eliminate 99% of missing data issues.
2nd check after 2000msec if Check 1 fails.
(anything >2 seconds will generate a TM Transaction error based on Tag Retrieval rate)

10

Server PdM Configuration from Transaction Manager Request Record Complete
ServerPdm.Ctl.ReqComplete
<EDC.ServerPdm.Ctl.ReqComplete>

Server PdM Configuration from Transaction Manager Request Record First Check Timer

MOVE
Source 10
Dest ServerPdm.Ctl.ReqCompleteCheckTMR.PRE
<EDC.ServerPdm.Ctl.ReqCompleteCheckTMR.PRE>
10

Server PdM Configuration from Transaction Manager Request Record First Check Timer

TON
Timer ServerPdm.Ctl.ReqCompleteCheckTMR
<EDC.ServerPdm.Ctl.ReqCompleteCheckTMR>
Preset 10
Accum 14
EN
DN

Server PdM Configuration from Transaction Manager Request Record First Check Timer
ServerPdm.Ctl.ReqCompleteCheckTMR.DN
<EDC.ServerPdm.Ctl.ReqCompleteCheckTMR.DN>

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Request Complete Delay Timer

MOVE
Source 2000
Dest ServerPdm.Ctl.ReqCompleteDelayTMR.PRE
<EDC.ServerPdm.Ctl.ReqCompleteDelayTMR.PRE>
2000

Server PdM Configuration from Transaction Manager Request Complete Delay Timer

TON
Timer ServerPdm.Ctl.ReqCompleteDelayTMR
<EDC.ServerPdm.Ctl.ReqCompleteDelayTMR>
Preset 2000
Accum 2006
EN
DN

Server PdM Configuration from Transaction Manager Request Complete Delay Timer
ServerPdm.Ctl.ReqCompleteDelayTMR.DN
<EDC.ServerPdm.Ctl.ReqCompleteDelayTMR.DN>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Configuration Request to Server via Transaction Manager

Request Complete - Response Received
Log Transaction & Performance
(Delay needed to allow all TM Data values to arrive - was receiving Handshake before Data - order of TM variables not enforced in data update.)

11

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Request Record First Check 1-shot
ServerPdm.Ctl.ReqCompleteCheckONS
<EDC.ServerPdm.Ctl.ReqCompleteCheckONS>
ONS

Server PdM Configuration from Transaction Manager Request Complete 1st Check Count
ADD
Source A ServerPdm.Ctl.ReqComp1stChkCnt
<EDC.ServerPdm.Ctl.ReqComp1stChkCnt>
663124
Source B 1
Dest ServerPdm.Ctl.ReqComp1stChkCnt
<EDC.ServerPdm.Ctl.ReqComp1stChkCnt>
663124

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Request Record Complete 1-shot
ServerPdm.Ctl.ReqCompleteONS
<EDC.ServerPdm.Ctl.ReqCompleteONS>
ONS

Server PdM Configuration from Transaction Manager Request Complete 2nd Check Count
ADD
Source A ServerPdm.Ctl.ReqComp2ndChkCnt
<EDC.ServerPdm.Ctl.ReqComp2ndChkCnt>
9982
Source B 1
Dest ServerPdm.Ctl.ReqComp2ndChkCnt
<EDC.ServerPdm.Ctl.ReqComp2ndChkCnt>
9982

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 4301
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataReq.Command
<EDC.ServerPdm.DataReq.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Server Interface Status State = Green - Msg Rx DN
EDC_Status01.ServerPdm.Update.MsgRxOK
L

Update PdM State for Transactions
aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm ...
Status EDC_Status01.ServerPdm

Configuration Request to Server via Transaction Manager

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerPdm.Msg.DN

Configuration Response from Server via Transaction Manager

JSR
Routine Name _850_PdmResponse

Request Complete Delay Reset Request

12

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>
U

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Send new Request

13

Server PdM Configuration from Transaction Manager Request Record in Progress
ServerPdm.Ctl.ReqInProgress
<EDC.ServerPdm.Ctl.ReqInProgress>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Request in Progress Timer

MOVE
Source 62000
Dest ServerPdm.Ctl.ReqInProgressTMR.PRE
<EDC.ServerPdm.Ctl.ReqInProgressTMR.PRE>
62000

Server PdM Configuration from Transaction Manager Request in Progress Timer

TON
Timer ServerPdm.Ctl.ReqInProgressTMR
<EDC.ServerPdm.Ctl.ReqInProgressTMR>
Preset 62000
Accum 0
EN
DN

Server PdM Configuration from Transaction Manager Request in Progress Timer
ServerPdm.Ctl.ReqInProgressTMR.DN
<EDC.ServerPdm.Ctl.ReqInProgressTMR.DN>

Server PdM Configuration from Transaction Manager Request Record in Progress Timeout
ServerPdm.Ctl.ReqInProgressTimeout
<EDC.ServerPdm.Ctl.ReqInProgressTimeout>

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Log Transaction & Performance

14

Server PdM Configuration from Transaction Manager Request Record in Progress Timeout
ServerPdm.Ctl.ReqInProgressTimeout
<EDC.ServerPdm.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4304
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataReq.Command
<EDC.ServerPdm.DataReq.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Server Interface Status State = Red-Msg Rx ER
EDC_Status01.ServerPdm.Update.MsgRxER
L

Update PdM State for Transactions

aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm ...
Status EDC_Status01.ServerPdm

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerPdm.Msg.ER

Server PdM Configuration from Transaction Manager Request in Progress Timer
ServerPdm.Ctl.ReqInProgressTMR
<EDC.ServerPdm.Ctl.ReqInProgressTMR>
RES

(End)

Configuration Request to Server via Transaction Manager - Data Init

Request Next Initialize
if Response Good Data, echo Response in next Request to Server to Ack
if not Response Good Data, send new Data Request after wait interval

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 846
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

DN011 Separate Req & Rsp Tags
Create Data Init Packet

2

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

/

Server PdM Configuration from Transaction Manager - Data

FLL
Source -1
Dest ServerPdmInit
Length 1

Configuration Request to Server via Transaction Manager - Data Init

DN011 Separate Req & Rsp Tags
Clear Response to ensure new data on next DB Response
DHF017 Jan 30, 2018 -- Clear Server Request if no Ack for Good Message Response
Copy Response to Request for Good Data Ack to DB

3

Server PdM Configuration from Transaction Manager Transaction Response Data

COP	
Source	ServerPdmInit
Dest	ServerPdm.DataRsp <EDC.ServerPdm.DataRsp>
Length	1

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Transaction Request Data

COP	
Source	ServerPdmInit
Dest	ServerPdm.DataReq <EDC.ServerPdm.DataReq>
Length	1

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Transaction Request Data

COP	
Source	ServerPdm.DataRspLast <EDC.ServerPdm.DataRspLast>
Dest	ServerPdm.DataReq <EDC.ServerPdm.DataReq>
Length	1

Store Last Request

4

Server PdM Configuration from Transaction Manager Transaction Last Request Data

COP	
Source	ServerPdm.DataReq <EDC.ServerPdm.DataReq>
Dest	ServerPdm.DataReqLast <EDC.ServerPdm.DataReqLast>
Length	1

Configuration Request to Server via Transaction Manager - Data Init

(End)

Configuration Response from Server via Transaction Manager

Configuration Response from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 850
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Detect Duplicate Config Data Response
(this indicates the database is stuck on a bad config item)

2

Server PdM Configuration from Transaction Manager Request/Response Command

NE
Source A ServerPdm.DataRspLast.Command <EDC.ServerPdm.DataRspLast.Command> 0
Source B 0

Server PdM Configuration from Transaction Manager Request/Response Command

NE
Source A ServerPdm.DataRspLast.Command <EDC.ServerPdm.DataRspLast.Command> 0
Source B -1

Server PdM Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerPdm.DataRspLast.Command <EDC.ServerPdm.DataRspLast.Command> 0
Source B ServerPdm.DataRsp.Command <EDC.ServerPdm.DataRsp.Command> 0

Server PdM Configuration from Transaction Manager Shop Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.ShopKey <EDC.ServerPdm.DataRspLast.ShopKey> -1
Source B ServerPdm.DataRsp.ShopKey <EDC.ServerPdm.DataRsp.ShopKey> -1

Server PdM Configuration from Transaction Manager Line Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.LineKey <EDC.ServerPdm.DataRspLast.LineKey> -1
Source B ServerPdm.DataRsp.LineKey <EDC.ServerPdm.DataRsp.LineKey> -1

Server PdM Configuration from Transaction Manager Equip Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.EqKey <EDC.ServerPdm.DataRspLast.EqKey> -1
Source B ServerPdm.DataRsp.EqKey <EDC.ServerPdm.DataRsp.EqKey> -1

Server PdM Configuration from Transaction Manager Tool Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.ToolKey <EDC.ServerPdm.DataRspLast.ToolKey> -1
Source B ServerPdm.DataRsp.ToolKey <EDC.ServerPdm.DataRsp.ToolKey> -1

Server PdM Configuration from Transaction Manager PdM Tool Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.PdmToolKey <EDC.ServerPdm.DataRspLast.PdmToolKey> -1
Source B ServerPdm.DataRsp.PdmToolKey <EDC.ServerPdm.DataRsp.PdmToolKey> -1

Server PdM Configuration from Transaction Manager PdM Trend Key - for DB Ack only

EQ
Source A ServerPdm.DataRspLast.PdmTrendKey <EDC.ServerPdm.DataRspLast.PdmTrendKey> -1
Source B ServerPdm.DataRsp.PdmTrendKey <EDC.ServerPdm.DataRsp.PdmTrendKey> -1

Server PdM Configuration from Transaction Manager EquipmentID (0001-9999) & ToolID (001-999) combined for fast searches

EQ
Source A ServerPdm.DataRspLast.EqToolID <EDC.ServerPdm.DataRspLast.EqToolID> -1
Source B ServerPdm.DataRsp.EqToolID <EDC.ServerPdm.DataRsp.EqToolID> -1

Server PdM Configuration from Transaction Manager Equipment ID (0001-9999)

EQ
Source A ServerPdm.DataRspLast.EqID <EDC.ServerPdm.DataRspLast.EqID> -1
Source B ServerPdm.DataRsp.EqID <EDC.ServerPdm.DataRsp.EqID> -1

Server PdM Configuration from Transaction Manager Tool ID (001-999)

EQ
Source A ServerPdm.DataRspLast.ToolID <EDC.ServerPdm.DataRspLast.ToolID> -1
Source B ServerPdm.DataRsp.ToolID <EDC.ServerPdm.DataRsp.ToolID> -1

Server PdM Configuration from Transaction Manager Network Index

EQ
Source A ServerPdm.DataRspLast.Data[0].NETi

Server PdM Configuration from Transaction Manager Response Duplicate
ServerPdm.Ctl.RspDup
<EDC.ServerPdm.Ctl.RspDup>

Configuration Response from Server via Transaction Manager

<EDC.ServerPdm.DataRspLast.Data[0].NETi>
-1
Source B ServerPdm.DataRsp.Data[0].NETi
<EDC.ServerPdm.DataRsp.Data[0].NETi>
-1

Duplicate Message Count and Detection
(this indicates the database is stuck on a bad config item)

3

Server PdM Configuration from Transaction Manager Response Duplicate
ServerPdm.Ctl.RspDup
<EDC.ServerPdm.Ctl.RspDup>

ADD
Source A PdmRspDuplicateCount
0
Source B
1
Dest PdmRspDuplicateCount
0

GT
Source A PdmRspDuplicateCount
0
Source B PdmRspDuplicateMax
1

MOVE
Source PdmRspDuplicateCount
0
Dest PdmRspDuplicateMax
1

GT
Source A PdmRspDuplicateCount
0
Source B
5

Server PdM Configuration from Transaction Manager Response Duplicate Error
ServerPdm.Ctl.RspDupER
<EDC.ServerPdm.Ctl.RspDupER>

Server PdM Configuration from Transaction Manager Response Duplicate
ServerPdm.Ctl.RspDup
<EDC.ServerPdm.Ctl.RspDup>

CLR
Dest PdmRspDuplicateCount
0

Move Response to working tag and Initialize Response Good Data
DN011 Separate Req & Rsp Tags

Moving DataRsp to DataLastRsp prevents SQL/TM from changing data while Logix is processing.

4

Server PdM Configuration from Transaction Manager Transaction Last Response Data

COP
Source ServerPdm.DataRsp
<EDC.ServerPdm.DataRsp>
Dest ServerPdm.DataRspLast
<EDC.ServerPdm.DataRspLast>
Length 1

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Configuration Response from Server via Transaction Manager

Check Response Command for Bad Data
Valid Commands 0, 101-103

5

Server PdM Configuration from Transaction Manager Request/Response Command

NE
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 0

Server PdM Configuration from Transaction Manager Request/Response Command

LT
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 101

Server PdM Configuration from Transaction Manager Request/Response Command

GT
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 103

Server PdM Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
298

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4312
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE

Performance Transaction

aoiEDC_PerfTrans

Configuration Response from Server via Transaction Manager

Dest	TempPerfMsg.TransLast
	110

Source	ServerPdm.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
	0
Dest	TempPerfMsg.TransTMR.ACC
	110

aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EDC_Status01.ServerPdm.Msg.KA

lbl_RspComp
JMP

If Response Command received, process Response

6

PdmRspCmdRcvd

Server has no new Configuration data to download

7

Server PdM Configuration from Transaction Manager Request/Response Command

EQ

Source A	ServerPdm.DataRspLast.Command
	<EDC.ServerPdm.DataRspLast.Command>
	0
Source B	0

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
L

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

ADD

Source A	PdmRspCmd[0]
	698142
Source B	1
Dest	PdmRspCmd[0]
	698142

Configuration Response from Server via Transaction Manager

Process Response Command

8

Server PdM Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 101

Configuration Response from Server via Transaction Manager

JSR
Routine Name _851_PdmResponseTool

ADD
Source A PdmRspCmd[1]
18929
Source B 1
Dest PdmRspCmd[1]
18929

Server PdM Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 102

Configuration Response from Server via Transaction Manager

JSR
Routine Name _852_PdmResponseTrend

ADD
Source A PdmRspCmd[2]
133
Source B 1
Dest PdmRspCmd[2]
133

Server PdM Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
0
Source B 103

Configuration Response from Server via Transaction Manager

JSR
Routine Name _853_PdmResponseNetwork

ADD
Source A PdmRspCmd[3]
7146
Source B 1
Dest PdmRspCmd[3]
7146

Command Response Complete

9

lbl_RspComp
LBL

NOP

Configuration Response from Server via Transaction Manager

Store Last Good, Bad or Incomplete Response w/ Diagnostic Counts
Incomplete - Command Num received with incomplete data packet on on 1st Check
Bad - Incomplete data packet on 2nd Check

10

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Transaction Last Good Response Data

COP
Source ServerPdm.DataRspLast
<EDC.ServerPdm.DataRspLast>
Dest ServerPdm.DataRspLastGood
<EDC.ServerPdm.DataRspLastGood>
Length 1

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Response Good on 1st Check

ADD
Source A ServerPdm.Ctl.RspGood1stChkCnt
<EDC.ServerPdm.Ctl.RspGood1stChkCnt>
653142
Source B 1
Dest ServerPdm.Ctl.RspGood1stChkCnt
<EDC.ServerPdm.Ctl.RspGood1stChkCnt>
653142

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Response Good on 2nd Check

ADD
Source A ServerPdm.Ctl.RspGood2ndChkCnt
<EDC.ServerPdm.Ctl.RspGood2ndChkCnt>
9982
Source B 1
Dest ServerPdm.Ctl.RspGood2ndChkCnt
<EDC.ServerPdm.Ctl.RspGood2ndChkCnt>
9982

Server PdM Configuration from Transaction Manager Response has

Server PdM Configuration from Transaction Manager Request

Server PdM Configuration from Transaction Manager Request Record Complete

Configuration Response from Server via Transaction Manager

Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

PdmRspCmdRcvd

Server PdM Configuration from Transaction Manager Response None on 1st Check

ADD
Source A ServerPdm.Ctl.RspNone1stChkCnt
<EDC.ServerPdm.Ctl.RspNone1stChkCnt>
0
Source B 1
Dest ServerPdm.Ctl.RspNone1stChkCnt
<EDC.ServerPdm.Ctl.RspNone1stChkCnt>
0

PdmRspCmdRcvd

Server PdM Configuration from Transaction Manager Transaction Last Incomplete Response Data

COP
Source ServerPdm.DataRspLast
<EDC.ServerPdm.DataRspLast>
Dest ServerPdm.DataRspLastInc
<EDC.ServerPdm.DataRspLastInc>
Length 1

Server PdM Configuration from Transaction Manager Response Incomplete on 1st Check

ADD
Source A ServerPdm.Ctl.RspInc1stChkCnt
<EDC.ServerPdm.Ctl.RspInc1stChkCnt>
9982
Source B 1
Dest ServerPdm.Ctl.RspInc1stChkCnt
<EDC.ServerPdm.Ctl.RspInc1stChkCnt>
9982

Server PdM Configuration from Transaction Manager Request Record First Check
ServerPdm.Ctl.ReqCompleteCheck
<EDC.ServerPdm.Ctl.ReqCompleteCheck>

Server PdM Configuration from Transaction Manager Request Record Complete Delay
ServerPdm.Ctl.ReqCompleteDelay
<EDC.ServerPdm.Ctl.ReqCompleteDelay>

Server PdM Configuration from Transaction Manager Transaction Last Bad Response Data

COP
Source ServerPdm.DataRspLast
<EDC.ServerPdm.DataRspLast>
Dest ServerPdm.DataRspLastBad
<EDC.ServerPdm.DataRspLastBad>
Length 1

Server PdM Configuration from Transaction Manager Response Bad on 2nd Check

ADD
Source A ServerPdm.Ctl.RspBad2ndChkCnt
<EDC.ServerPdm.Ctl.RspBad2ndChkCnt>
0
Source B 1
Dest ServerPdm.Ctl.RspBad2ndChkCnt
<EDC.ServerPdm.Ctl.RspBad2ndChkCnt>
0

Configuration Response from Server via Transaction Manager

Download ALL configuration is complete

11

Server PdM Configuration from Transaction Manager Response Download All Started
ServerPdm.Ctl.DownloadAllStarted
<EDC.ServerPdm.Ctl.DownloadAllStarted>

Server PdM Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerPdm.Ctl.RspCmd0
<EDC.ServerPdm.Ctl.RspCmd0>

Server PdM Configuration from Transaction Manager Response Download All Started
ServerPdm.Ctl.DownloadAllStarted
<EDC.ServerPdm.Ctl.DownloadAllStarted>
U

Server PdM Configuration from Transaction Manager Response Download All Complete
ServerPdm.Ctl.DownloadAllComplete
<EDC.ServerPdm.Ctl.DownloadAllComplete>
L

Server PdM Configuration from Transaction Manager Response Download All Complete 1-shot
ServerPdm.Ctl.DownloadAllCompleteONS
<EDC.ServerPdm.Ctl.DownloadAllCompleteONS>

Configuration Udpate after full download

JSR
Routine Name _859_PdmDownloadComplete

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 101 - Tool
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 851
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data
-- allow Tool w/out Data Configuration (Data is typically created after Tool is created and will generate another download to update config)

2

Server PdM Configuration from Transaction Manager EquipmentID (0001-9999) & ToolID (001-999) combined for fast searches

GT
Source A ServerPdm.DataRspLast.EqToolID
<EDC.ServerPdm.DataRspLast.EqToolID>
-1
Source B 0

Server PdM Configuration from Transaction Manager Equipment ID (0001-9999)

NE
Source A ServerPdm.DataRspLast.EqID
<EDC.ServerPdm.DataRspLast.EqID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Tool ID (001-999)

GT
Source A ServerPdm.DataRspLast.ToolID
<EDC.ServerPdm.DataRspLast.ToolID>
-1
Source B 0

Server PdM Configuration from Transaction Manager Tool Enable/Disable (1=Enable, 0=Disable)

NE
Source A ServerPdm.DataRspLast.ToolEnable
<EDC.ServerPdm.DataRspLast.ToolEnable>
0
Source B -1

Server PdM Configuration from Transaction Manager Trigger Enable (1=Trigger, 0=Sample)

NE
Source A ServerPdm.DataRspLast.TriggerEnable
<EDC.ServerPdm.DataRspLast.TriggerEnable>
0
Source B -1

Server PdM Configuration from Transaction Manager Sample Period ID

NE
Source A ServerPdm.DataRspLast.SamplePeriodID
<EDC.ServerPdm.DataRspLast.SamplePeriodID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Start Index

NE
Source A ServerPdm.DataRspLast.DATsi
<EDC.ServerPdm.DataRspLast.DATsi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data End Index

NE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B -1

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
L

Configuration Response from Server via Transaction Manager

Validate Data 0 config

3

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 0

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[0].NETi
<EDC.ServerPdm.DataRspLast.Data[0].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[0].DATi
<EDC.ServerPdm.DataRspLast.Data[0].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[0].TypeID
<EDC.ServerPdm.DataRspLast.Data[0].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[0].Enable
<EDC.ServerPdm.DataRspLast.Data[0].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 1 config

4

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 1

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[1].NETi
<EDC.ServerPdm.DataRspLast.Data[1].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[1].DATi
<EDC.ServerPdm.DataRspLast.Data[1].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[1].TypeID
<EDC.ServerPdm.DataRspLast.Data[1].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[1].Enable
<EDC.ServerPdm.DataRspLast.Data[1].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 2 config

5

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 2

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[2].NETi
<EDC.ServerPdm.DataRspLast.Data[2].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[2].DATi
<EDC.ServerPdm.DataRspLast.Data[2].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[2].TypeID
<EDC.ServerPdm.DataRspLast.Data[2].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[2].Enable
<EDC.ServerPdm.DataRspLast.Data[2].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 3 config

6

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 3

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[3].NETi
<EDC.ServerPdm.DataRspLast.Data[3].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[3].DATi
<EDC.ServerPdm.DataRspLast.Data[3].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[3].TypeID
<EDC.ServerPdm.DataRspLast.Data[3].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[3].Enable
<EDC.ServerPdm.DataRspLast.Data[3].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 4 config

7

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 4

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[4].NETi
<EDC.ServerPdm.DataRspLast.Data[4].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[4].DATi
<EDC.ServerPdm.DataRspLast.Data[4].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[4].TypeID
<EDC.ServerPdm.DataRspLast.Data[4].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[4].Enable
<EDC.ServerPdm.DataRspLast.Data[4].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 5 config

8

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 5

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[5].NETi
<EDC.ServerPdm.DataRspLast.Data[5].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[5].DATi
<EDC.ServerPdm.DataRspLast.Data[5].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[5].TypeID
<EDC.ServerPdm.DataRspLast.Data[5].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[5].Enable
<EDC.ServerPdm.DataRspLast.Data[5].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 6 config

9

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 6

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[6].NETi
<EDC.ServerPdm.DataRspLast.Data[6].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[6].DATi
<EDC.ServerPdm.DataRspLast.Data[6].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[6].TypeID
<EDC.ServerPdm.DataRspLast.Data[6].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[6].Enable
<EDC.ServerPdm.DataRspLast.Data[6].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 7 config

10

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 7

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[7].NETi
<EDC.ServerPdm.DataRspLast.Data[7].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[7].DATi
<EDC.ServerPdm.DataRspLast.Data[7].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[7].TypeID
<EDC.ServerPdm.DataRspLast.Data[7].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[7].Enable
<EDC.ServerPdm.DataRspLast.Data[7].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 8 config

11

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 8

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[8].NETi
<EDC.ServerPdm.DataRspLast.Data[8].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[8].DATi
<EDC.ServerPdm.DataRspLast.Data[8].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[8].TypeID
<EDC.ServerPdm.DataRspLast.Data[8].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[8].Enable
<EDC.ServerPdm.DataRspLast.Data[8].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate Data 9 config

12

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

GE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 9

Server PdM Configuration from Transaction Manager Network Index

NE
Source A ServerPdm.DataRspLast.Data[9].NETi
<EDC.ServerPdm.DataRspLast.Data[9].NETi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Index

NE
Source A ServerPdm.DataRspLast.Data[9].DATi
<EDC.ServerPdm.DataRspLast.Data[9].DATi>
-1
Source B -1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[9].TypeID
<EDC.ServerPdm.DataRspLast.Data[9].TypeID>
-1
Source B -1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[9].Enable
<EDC.ServerPdm.DataRspLast.Data[9].Enable>
/

PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
U

PdmGoodData
/

Configuration Response from Server via Transaction Manager

Validate unused Data 0 not Enabled

13

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 0

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[0].Enable
<EDC.ServerPdm.DataRspLast.Data[0].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Validate unused Data 1 not Enabled

14

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 1

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[1].Enable
<EDC.ServerPdm.DataRspLast.Data[1].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Configuration Response from Server via Transaction Manager

Validate unused Data 2 not Enabled

15

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 2

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[2].Enable
<EDC.ServerPdm.DataRspLast.Data[2].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Validate unused Data 3 not Enabled

16

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 3

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[3].Enable
<EDC.ServerPdm.DataRspLast.Data[3].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Configuration Response from Server via Transaction Manager

Validate unused Data 4 not Enabled

17

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 4

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[4].Enable
<EDC.ServerPdm.DataRspLast.Data[4].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Validate unused Data 5 not Enabled

18

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 5

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[5].Enable
<EDC.ServerPdm.DataRspLast.Data[5].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Configuration Response from Server via Transaction Manager

Validate unused Data 6 not Enabled

19

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 6

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[6].Enable
<EDC.ServerPdm.DataRspLast.Data[6].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Validate unused Data 7 not Enabled

20

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 7

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[7].Enable
<EDC.ServerPdm.DataRspLast.Data[7].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Configuration Response from Server via Transaction Manager

Validate unused Data 8 not Enabled

21

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 8

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[8].Enable
<EDC.ServerPdm.DataRspLast.Data[8].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Validate unused Data 9 not Enabled

22

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Data End Index

LT
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei>
-1
Source B 9

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[9].Enable
<EDC.ServerPdm.DataRspLast.Data[9].Enable>
PdmGoodData

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
PdmGoodData

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

23

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
/

Server PdM Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerPdm.DataRspLast.EqToolID
<EDC.ServerPdm.DataRspLast.EqToolID>
-1
Dest ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
298

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4313
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Server Interface Status State = Red-Tool Rx Data ER
EDC_Status01.ServerPdm.Update.ToolRxDataER
L

Update PdM State for Transactions

aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm
Status EDC_Status01.ServerPdm

lbl_Exit
JMP

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

24

Pdm Tool Array Search
PdmToolFSC
RES

PdM Config Array End Index

GT
Source A PdM_Cfg.TL.ei 914
Source B 0

Pdm Tool Array Search

ADD
Source A PdM_Cfg.TL.ei 914
Source B 1
Dest PdmToolFSC.LEN 915

Pdm Tool Array Search

MOVE
Source 0
Dest PdmToolFSC.POS 839

Pdm Tool Array Search

FSC
Control PdmToolFSC
Length 915
Position 839
Mode ALL
Expression PdM_Tool[PdmToolFSC.POS].Cfg.EqToolID=ServerPdm.DataRspLast.EqToolID
EN
DN
ER

Configuration Response from Server via Transaction Manager

Search Pdm Config for Empty Position

25

Pdm Tool Array Search PdmToolFSC.FD /

Pdm Tool Array Empty Search PdmToolEmptyFSC RES

Pdm Tool Array Empty Search
MOVE
Source PdM_Cfg.TL.n 3000
Dest PdmToolEmptyFSC.LEN 3000

Pdm Tool Array Empty Search
MOVE
Source 0
Dest PdmToolEmptyFSC.POS 1

Pdm Tool Array Empty Search
FSC
Control PdmToolEmptyFSC
Length 3000
Position 1
Mode ALL
Expression PdM_Tool[PdmToolEmptyFSC.POS].Cfg.EqToolID = 0
EN DN ER

Store Array Position for Update/Insert

26

Pdm Tool Array Search PdmToolFSC.FD

MOVE
Source PdmToolFSC.POS 839
Dest TLi 839

Pdm Tool Array Search PdmToolFSC.FD /

Pdm Tool Array Empty Search PdmToolEmptyFSC.FD

MOVE
Source PdmToolEmptyFSC.POS 1
Dest TLi 839

ADD
Source A Pdm_ToolCount 756
Source B 1
Dest Pdm_ToolCount 756

GT
Source A TLi 839
Source B PdM_Cfg.TL.ei 914

PdM Config Array End Index
MOVE
Source TLi 839
Dest PdM_Cfg.TL.ei 914

Tool Array is Full

27

Pdm Tool Array Search PdmToolFSC.FD /

Pdm Tool Array Empty Search PdmToolEmptyFSC.FD /

PdM Config Tool Configuration Array is Full PdM_Cfg.ToolArrayFull L

lbl_Exit JMP

Configuration Response from Server via Transaction Manager

28

Process Data and Store

Save but don't enable Tool if no Data values have been configured

PdM Tool Array
PdM_Tool[TLi].Cfg.DownloadPend
U

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.EqToolID
<EDC.ServerPdm.DataRspLast.EqToolID> -1
Dest PdM_Tool[TLi].Cfg.EqToolID 1004032

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.EqID
<EDC.ServerPdm.DataRspLast.EqID> -1
Dest PdM_Tool[TLi].Cfg.EQID 1004

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.ToolID
<EDC.ServerPdm.DataRspLast.ToolID> -1
Dest PdM_Tool[TLi].Cfg.ToolID 32

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.SamplePeriodID
<EDC.ServerPdm.DataRspLast.SamplePeriodID> -1
Dest PdM_Tool[TLi].Cfg.SamplePeriodID 5

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.DATsi
<EDC.ServerPdm.DataRspLast.DATsi> -1
Dest PdM_Tool[TLi].Cfg.DATsi 0

PdM Tool Array
MOVE
Source ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei> -1
Dest PdM_Tool[TLi].Cfg.DATei 3

PdM Tool Array
COP
Source ServerPdm.DataRspLast.Data[0]
<EDC.ServerPdm.DataRspLast.Data[0]>
Dest PdM_Tool[TLi].Cfg.Data[0]
Length 10

PdM Tool Array
PdM_Tool[TLi].Cfg.Enable
U

Server PdM Configuration from Transaction Manager Tool Enable/Disable (1=Enable, 0=Disable)
EQ
Source A ServerPdm.DataRspLast.ToolEnable
<EDC.ServerPdm.DataRspLast.ToolEnable> 0
Source B 1

Server PdM Configuration from Transaction Manager Data Start Index
NE
Source A ServerPdm.DataRspLast.DATsi
<EDC.ServerPdm.DataRspLast.DATsi> -1
Source B -1

Server PdM Configuration from Transaction Manager Data End Index
NE
Source A ServerPdm.DataRspLast.DATei
<EDC.ServerPdm.DataRspLast.DATei> -1
Source B -1

PdM Tool Array
PdM_Tool[TLi].Cfg.Enable
L

PdM Tool Array
PdM_Tool[TLi].Cfg.TriggerEnable
U

Server PdM Configuration from Transaction Manager Trigger Enable (1=Trigger, 0=Sample)
EQ
Source A ServerPdm.DataRspLast.TriggerEnable
<EDC.ServerPdm.DataRspLast.TriggerEnable> 0
Source B 1

PdM Tool Array
PdM_Tool[TLi].Cfg.TriggerEnable
L

Configuration Response from Server via Transaction Manager

Initialize Tool's Last Sample value to -1 to prevent a sample when the rate is changed
(PdM code will synch newly configured tool to the current Sample Period without sampling.)

Initialize Tool's Start signal to off to force Init/Reset of PdM Data/Array
(Config change to Data NETi/DATi wasn't clearing previously collected data)

Sunday Configuration Download will reset all Sampled tools with some data loss for the current sample.

29

PdM Tool Array
PdM_Tool[TLi].Cfg.TriggerEnable
/

PdM Tool Array
MOVE
Source -1
Dest PdM_Tool[TLi].Ctl.PeriodLast
185

PdM Tool Array
PdM_Tool[TLi].Ctl.Start
U

Data OK

30

Server Interface
Status State = Green
- Tool Rx Data OK
EDC_Status01.ServerPdm.Update.ToolRxDataOK
L

Update PdM State for
Transactions
aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm
Status EDC_Status01.ServerPdm

Exit Routine

31

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 102 - Trend

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 852
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

2

Server PdM Configuration from Transaction Manager EquipmentID (0001-9999) & ToolID (001-999) combined for fast searches

GT	
Source A	ServerPdm.DataRspLast.EqToolID
	<EDC.ServerPdm.DataRspLast.EqToolID>
	-1
Source B	0

Server PdM Configuration from Transaction Manager Equipment ID (0001-9999)

NE	
Source A	ServerPdm.DataRspLast.EqID
	<EDC.ServerPdm.DataRspLast.EqID>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Tool ID (001-999)

GT	
Source A	ServerPdm.DataRspLast.ToolID
	<EDC.ServerPdm.DataRspLast.ToolID>
	-1
Source B	0

Server PdM Configuration from Transaction Manager Tool Enable/Disable (1=Enable, 0=Disable)

NE	
Source A	ServerPdm.DataRspLast.ToolEnable
	<EDC.ServerPdm.DataRspLast.ToolEnable>
	0
Source B	-1

Server PdM Configuration from Transaction Manager Trigger Enable (1=Trigger, 0=Sample)

NE	
Source A	ServerPdm.DataRspLast.TriggerEnable
	<EDC.ServerPdm.DataRspLast.TriggerEnable>
	0
Source B	-1

Server PdM Configuration from Transaction Manager Sample Period ID

NE	
Source A	ServerPdm.DataRspLast.SamplePeriodID
	<EDC.ServerPdm.DataRspLast.SamplePeriodID>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Sample Rate ID

NE	
Source A	ServerPdm.DataRspLast.SampleRateID
	<EDC.ServerPdm.DataRspLast.SampleRateID>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Data Start Index

NE	
Source A	ServerPdm.DataRspLast.DATsi
	<EDC.ServerPdm.DataRspLast.DATsi>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Data End Index

NE	
Source A	ServerPdm.DataRspLast.DATei
	<EDC.ServerPdm.DataRspLast.DATei>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Network Index

NE	
Source A	ServerPdm.DataRspLast.Data[0].NETi
	<EDC.ServerPdm.DataRspLast.Data[0].NETi>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Data Index

NE	
Source A	ServerPdm.DataRspLast.Data[0].DATi
	<EDC.ServerPdm.DataRspLast.Data[0].DATi>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE	
Source A	ServerPdm.DataRspLast.Data[0].TypeID
	<EDC.ServerPdm.DataRspLast.Data[0].TypeID>
	-1
Source B	-1

Server PdM Configuration from Transaction Manager Response has Good Data Record

ServerPdm.Ctl.RspGoodData

<EDC.ServerPdm.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>

Server PdM Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerPdm.DataRspLast.EqToolID
<EDC.ServerPdm.DataRspLast.EqToolID>
-1
Dest ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
298

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4314
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC
<EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataRspLast.Command
<EDC.ServerPdm.DataRspLast.Command>
Input Par ServerPdm.Ctl.CommandID
<EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber
<EDC.ServerPdm.Ctl.TransactionNumber>

Server Interface Status State = Red-Trend Rx Data ER
EDC_Status01.ServerPdm.Update.TrendRxDataER
L

Update PdM State for Transactions

aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm
Status EDC_Status01.ServerPdm

lbl_Exit
JMP

Search Pdm Config for matching EQID-ToolID
(combined field allows a quick search)

4

Pdm Trend Array Search
PdmTrendFSC
RES

PdM Config Array End Index

GT
Source A PdM_Cfg.TRN.ei 3
Source B 0

Pdm Trend Array Search

ADD
Source A PdM_Cfg.TRN.ei 3
Source B 1
Dest PdmTrendFSC.LEN 4

Pdm Trend Array Search

MOVE
Source 0
Dest PdmTrendFSC.POS 3

Pdm Trend Array Search

FSC
Control PdmTrendFSC
Length 4
Position 3
Mode ALL
Expression PdM_Trend[PdmTrendFSC.POS].Cfg.EqToolID=ServerPdm.DataRspLast.EqToolID
EN
DN
ER

Configuration Response from Server via Transaction Manager

Search Pdm Config for Empty Position

5

Pdm Trend Array Search PdmTrendFSC.FD /

PdmTrendEmptyFSC (RES)

MOVE
Source PdM_Cfg.TRN.n 1000
Dest PdmTrendEmptyFSC.LEN 1000

MOVE
Source 0
Dest PdmTrendEmptyFSC.POS 3

FSC
Control PdmTrendEmptyFSC
Length 1000
Position 3
Mode ALL
Expression PdM_Trend[PdmTrendEmptyFSC.POS].Cfg.EqToolID=0
EN
DN
ER

Store Array Position for Update/Insert

6

Pdm Trend Array Search PdmTrendFSC.FD

MOVE
Source PdmTrendFSC.POS 3
Dest TRNi 3

Pdm Trend Array Search PdmTrendFSC.FD /

PdmTrendEmptyFSC.FD

MOVE
Source PdmTrendEmptyFSC.POS 3
Dest TRNi 3

ADD
Source A Pdm_TrendCount 3
Source B 1
Dest Pdm_TrendCount 3

GT
Source A TRNi 3
Source B PdM_Cfg.TRN.ei 3

PdM Config Array End Index

MOVE
Source TRNi 3
Dest PdM_Cfg.TRN.ei 3

Trend Array is Full

7

Pdm Trend Array Search PdmTrendFSC.FD /

PdmTrendEmptyFSC.FD /

PdM Config Trend Configuration Array is Full

PdM_Cfg.TrendArrayFull (L)

lbl_Exit (JMP)

Configuration Response from Server via Transaction Manager

Process Data and Store

8

PdM Trend Array
PdM_Trend[TRNi].Cfg.DownloadPend
U

PdM Trend Array
MOVE
Source ServerPdm.DataRspLast.EqToolID
<EDC.ServerPdm.DataRspLast.EqToolID> -1
Dest PdM_Trend[TRNi].Cfg.EqToolID
967001

PdM Trend Array
MOVE
Source ServerPdm.DataRspLast.EqID
<EDC.ServerPdm.DataRspLast.EqID> -1
Dest PdM_Trend[TRNi].Cfg.EQID
967

PdM Trend Array
MOVE
Source ServerPdm.DataRspLast.ToolID
<EDC.ServerPdm.DataRspLast.ToolID> -1
Dest PdM_Trend[TRNi].Cfg.ToolID
1

PdM Trend Array
MOVE
Source ServerPdm.DataRspLast.SamplePeriodID
<EDC.ServerPdm.DataRspLast.SamplePeriodID> -1
Dest PdM_Trend[TRNi].Cfg.SamplePeriodID
0

PdM Trend Array
MOVE
Source ServerPdm.DataRspLast.SampleRateID
<EDC.ServerPdm.DataRspLast.SampleRateID> -1
Dest PdM_Trend[TRNi].Cfg.SampleRateID
14

PdM Trend Array
COP
Source ServerPdm.DataRspLast.Data[0]
<EDC.ServerPdm.DataRspLast.Data[0]>
Dest PdM_Trend[TRNi].Cfg.Data
Length 1

Server PdM Configuration from Transaction Manager Tool Enable/Disable (1=Enable, 0=Disable)

PdM Trend Array
PdM_Trend[TRNi].Cfg.Enable
U

EQ
Source A ServerPdm.DataRspLast.ToolEnable
<EDC.ServerPdm.DataRspLast.ToolEnable> 0
Source B 1

PdM Trend Array
PdM_Trend[TRNi].Cfg.Enable
L

Server PdM Configuration from Transaction Manager Trigger Enable (1=Trigger, 0=Sample)

PdM Trend Array
PdM_Trend[TRNi].Cfg.TriggerEnable
U

EQ
Source A ServerPdm.DataRspLast.TriggerEnable
<EDC.ServerPdm.DataRspLast.TriggerEnable> 0
Source B 1

PdM Trend Array
PdM_Trend[TRNi].Cfg.TriggerEnable
L

Configuration Response from Server via Transaction Manager

Initialize Trend's Last Sample value to -1
to prevent a sample when the rate is changed
(PdM code will sync newly configured trend to the current Sample Period without sampling.)

9

PdM Trend Array
PdM_Trend[TRNi].Cfg.TriggerEnable
/

PdM Trend Array
MOVE
Source -1
Dest PdM_Trend[TRNi].Ctl.PeriodLast
0

PdM Trend Array
PdM_Trend[TRNi].Ctl.Start
U

Data OK

10

Server Interface
Status State = Green
- Trend Rx Data OK
EDC_Status01.ServerPdm.Update.TrendRxDataOK
L

Update PdM State for
Transactions
aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm
Status EDC_Status01.ServerPdm

Exit Routine

11

lbl_Exit
LBL

NOP

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 103 - Network

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 853
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data

2

Server PdM Configuration from Transaction Manager Shop Key - for DB Ack only

NE
Source A ServerPdm.DataRspLast.ShopKey <EDC.ServerPdm.DataRspLast.ShopKey> -1
Source B -1

Server PdM Configuration from Transaction Manager Network Index

LIMIT
Low Limit PdM_Cfg.NET.si 0
Test ServerPdm.DataRspLast.Data[0].NETi <EDC.ServerPdm.DataRspLast.Data[0].NETi> -1
High Limit PdM_Cfg.NET.mi 299

Server PdM Configuration from Transaction Manager Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

NE
Source A ServerPdm.DataRspLast.Data[0].TypeID <EDC.ServerPdm.DataRspLast.Data[0].TypeID> -1
Source B -1

Diagnostic

ADD
Source A NetDataGoodCount 74724
Source B 1
Dest NetDataGoodCount 74724

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData <EDC.ServerPdm.Ctl.RspGoodData>

L

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server PdM Configuration from Transaction Manager Response has Good Data Record
ServerPdm.Ctl.RspGoodData
<EDC.ServerPdm.Ctl.RspGoodData>
/

Diagnostic

ADD
Source A NetDataBadCount 3293
Source B 1
Dest NetDataBadCount 3293

Server PdM Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerPdm.DataRspLast.Data[0].NETi <EDC.ServerPdm.DataRspLast.Data[0].NETi> -1
Dest ServerPdm.Ctl.CommandID <EDC.ServerPdm.Ctl.CommandID> 298

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4315
Input Par ServerPdm.Ctl.ReqInProgressTMR.ACC <EDC.ServerPdm.Ctl.ReqInProgressTMR.ACC>
Input Par ServerPdm.DataRspLast.Command <EDC.ServerPdm.DataRspLast.Command>
Input Par ServerPdm.Ctl.CommandID <EDC.ServerPdm.Ctl.CommandID>
Input Par ServerPdm.Ctl.TransactionNumber <EDC.ServerPdm.Ctl.TransactionNumber>

Server Interface Status State = Red-Network Rx Data ER
EDC_Status01.ServerPdm.Update.NetworkRxDataER
L

Update PdM State for Transactions

aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm ...
Status EDC_Status01.ServerPdm

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Process Data and Store

4

MOVE
Source ServerPdm.DataRspLast.Data[0].NETi
<EDC.ServerPdm.DataRspLast.Data[0].NETi> -1
Dest NETi 299

Network Modules

MOVE
Source ServerPdm.DataRspLast.Data[0].TypeID
<EDC.ServerPdm.DataRspLast.Data[0].TypeID> -1
Dest PdM_NetMod[NETi].Cfg.TypeID 2

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[0].Enable
<EDC.ServerPdm.DataRspLast.Data[0].Enable>

Network Modules
PdM_NetMod[NETi].Cfg.Enable
L

Server PdM Configuration from Transaction Manager Enable data collection
ServerPdm.DataRspLast.Data[0].Enable
<EDC.ServerPdm.DataRspLast.Data[0].Enable>

Network Modules
PdM_NetMod[NETi].Cfg.Enable
U

Data OK

5

Server Interface Status State = Green - Reset Rx Data OK
EDC_Status01.ServerPdm.Update.ResetRxDataOK
L

Update PdM State for Transactions
aoiEDC_StatusUpdateServerPdm
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerPdm ...
Status EDC_Status01.ServerPdm

Exit Routine

6

lbl_Exit
LBL

NOP

(End)

Configuration Initialization before full download

PdM Interface Initialization
(called before Full Configuration Download)

Tools, Trends and Network are not actually reset before full download because this would cause dropouts in the Sampled PdM data
Existing Networks are updated by new configurations
Existing Tools and Trends are marked for Download Pending -- and unmarked when updated by new configuration.
Pending when the Full Configuration Download is complete, any Tools and Trends still marked Download Pending are disabled.

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 858
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Initialize Tool Loop

2

CLR
Dest TOOLi 915

Tool Loop
mark for Download Pending

3

lbl_ToolLoop
LBL

PdM Tool Array
PdM_Tool[TOOLi].Cfg.DownloadPend
L

ADD
Source A TOOLi 915
Source B 1
Dest TOOLi 915

LE
Source A TOOLi 915
Source B PdM_Cfg.TL.ei 914

lbl_ToolLoop
JMP

Configuration Initialization before full download

Initialize Trend Loop

4

CLR
Dest TRNDi 4

Trend Loop
mark for Download Pending

5

lbl_TrendLoop
LBL

PdM Trend Array
PdM_Trend[TRNDi].Cfg.DownloadPend
L

ADD
Source A TRNDi 4
Source B 1
Dest TRNDi 4

LE
Source A TRNDi 4
Source B PdM_Cfg.TRN.ei 3

lbl_TrendLoop
JMP

Reset Pdm Config Errors

6

PdM Config Tool Configuration Array is Full
PdM_Cfg.ToolArrayFull
U

PdM Config Trend Configuration Array is Full
PdM_Cfg.TrendArrayFull
U

PdM Config Tool Processing Error - Invalid EQID
PdM_Cfg.ToolInvalidEQID
U

PdM Config Tool Processing Error - Invalid Data Index
PdM_Cfg.ToolInvalidDataIndex
U

PdM Config Tool Processing Error - Invalid Sample Period
PdM_Cfg.ToolInvalidSamplePeriod
U

PdM Config Tool Processing Error - Invalid Network Data
PdM_Cfg.ToolInvalidNetworkData
U

PdM Config Trend Processing Error - Invalid EQID
PdM_Cfg.TrendInvalidEQID
U

PdM Config Trend Processing Error - Invalid Sample Period
PdM_Cfg.TrendInvalidSamplePeriod
U

PdM Config Trend Processing Error - Invalid Sample Rate
PdM_Cfg.TrendInvalidSampleRate
U

PdM Config Trend Processing Error - Invalid Network Data
PdM_Cfg.TrendInvalidNetworkData
U

Exit Routine

7

lbl_Exit
LBL

NOP

(End)

Configuration Udpate after full download

PdM Interface Initialization
(called after Full Configuration Download)

Tools, Trends and Network are not actually reset before full download because this would cause dropouts in the Sampled PdM data
Existing Networks are updated by new configurations
Existing Tools and Trends are marked for Download Pending -- and unmarked when updated by new configuration.
Pending when the Full Configuration Download is complete, any Tools and Trends still marked Download Pending are disabled.

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 859
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Initialize Tool Loop

2

CLR
Dest Pdm_ToolCount 756

CLR
Dest TOOLi 915

Configuration Udpate after full download

Tool Loop
Clear Tool Config, Control and Data if still marked for Download Pending

3

lbl_ToolLoop
LBL

PdM Tool Array
PdM_Tool[TOOLi].Cfg.DownloadPend

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TOOLi].Cfg
Length 1

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TOOLi].Ctl
Length 1

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TOOLi].Data[0]
Length PdM_Cfg.DAT.n

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TOOLi].Array[0]
Length PdM_Cfg.DAT.n

PdM Tool Array
PdM_Tool[TOOLi].Cfg.Enable

ADD
Source A Pdm_ToolCount 756
Source B 1
Dest Pdm_ToolCount 756

ADD
Source A TOOLi 915
Source B 1
Dest TOOLi 915

LE
Source A TOOLi 915
Source B PdM_Cfg.TL.ei 914

lbl_ToolLoop
JMP

Initialize Trend Loop

4

CLR
Dest Pdm_TrendCount 3

CLR
Dest TRNDi 4

Configuration Udpate after full download

Trend Loop
Clear Tool Config, Control and Data if still marked for Download Pending

5

lbl_TrendLoop
LBL

PdM Trend Array
PdM_Trend[TRNDi].Cfg.DownloadPend

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNDi].Cfg
Length 1

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNDi].Ctl
Length 1

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNDi].Data
Length 1

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNDi].Array
Length 1

PdM Trend Array
PdM_Trend[TRNDi].Cfg.Enable

ADD
Source A Pdm_TrendCount 3
Source B 1
Dest Pdm_TrendCount 3

ADD
Source A TRNDi 4
Source B 1
Dest TRNDi 4

LE
Source A TRNDi 4
Source B PdM_Cfg.TRN.ei 3

lbl_TrendLoop
JMP

Exit Routine

6

lbl_Exit
LBL

NOP

(End)

Status Request to Server via Transaction Manager

Status Data Request to Server via Transaction Manager

- Logix Requests Data (1-13 Command for each row of Status table)
- Server Responds with Data (1-13 Command) or NoData (-1 Command)

- if Data, Logix stores Data for HMI Overview display
- if NoData, Logix waits and then Requests Data

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 860
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Status Request to Server via Transaction Manager

Transaction Request Timer
Fast Interval used when last TM Response had data & not Last Table Row

SQL Job runs every 20 seconds to update the Status data
Gateway refreshes Status data every 10 seconds (2x update rate)

2

Server Status from Transaction Manager Request Record in Progress
ServerStat.Ctl.ReqInProgress
<EDC.ServerStat.Ctl.ReqInProgress>

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>

Server Status from Transaction Manager Response has Last Row Record
ServerStat.Ctl.RspLastRow
<EDC.ServerStat.Ctl.RspLastRow>

Server Status from Transaction Manager Request Next Record Timer

MOVE
Source ServerStat.Ctl.ReqFastTime
<EDC.ServerStat.Ctl.ReqFastTime>
0
Dest ServerStat.Ctl.ReqNextTMR.PRE
<EDC.ServerStat.Ctl.ReqNextTMR.PRE>
10000

Server Status from Transaction Manager Request Next Record Timer

TON
Timer ServerStat.Ctl.ReqNextTMR
<EDC.ServerStat.Ctl.ReqNextTMR>
Preset 10000
Accum 1945
EN
DN

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>

Server Status from Transaction Manager Response has Last Row Record
ServerStat.Ctl.RspLastRow
<EDC.ServerStat.Ctl.RspLastRow>

Server Status from Transaction Manager Request Next Record Timer

MOVE
Source ServerStat.Ctl.ReqSlowTime
<EDC.ServerStat.Ctl.ReqSlowTime>
10000
Dest ServerStat.Ctl.ReqNextTMR.PRE
<EDC.ServerStat.Ctl.ReqNextTMR.PRE>
10000

Request Next
Ready for next Transaction Manager Trigger

3

Server Status from Transaction Manager Request Next Record Timer
ServerStat.Ctl.ReqNextTMR.DN
<EDC.ServerStat.Ctl.ReqNextTMR.DN>

Server Status from Transaction Manager Request Next Record
ServerStat.Ctl.ReqNext
<EDC.ServerStat.Ctl.ReqNext>

Server Status from Transaction Manager Request Record in Progress
ServerStat.Ctl.ReqInProgress
<EDC.ServerStat.Ctl.ReqInProgress>
L

Server Status from Transaction Manager Request Record Complete
ServerStat.Ctl.ReqComplete
<EDC.ServerStat.Ctl.ReqComplete>
U

Status Request to Server via Transaction Manager

Request Next Initialize
if Response Good Data, request next Row of Data in Response to Server
if not Response Good Data, send new Data Request after wait interval

4

Server Status from Transaction Manager Request Next Record
ServerStat.Ctl.ReqNext
<EDC.ServerStat.Ctl.ReqNext>

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Status Request to Server via Transaction Manager - Data Init

JSR
Routine Name _861_StatRequestInit

Request Next
Increment Transaction Number
(Triggers TM Transaction on change)

5

Server Status from Transaction Manager Request Next Record
ServerStat.Ctl.ReqNext
<EDC.ServerStat.Ctl.ReqNext>

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Server Status from Transaction Manager Transaction Number (TM Request Trigger)

ADD
Source A ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319
Source B 1
Dest ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319

Server Status from Transaction Manager Transaction Number (TM Request Trigger)

LT
Source A ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319
Source B 1

Server Status from Transaction Manager Transaction Number (TM Request Trigger)

GT
Source A ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Server Status from Transaction Manager Transaction Number (TM Request Trigger)

MOVE
Source 1
Dest ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319

Status Request to Server via Transaction Manager

6

Request Next
Log Transaction & Performance

Server Status from Transaction Manager Request Next Record
ServerStat.Ctl.ReqNext
<EDC.ServerStat.Ctl.ReqNext>

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 4100
Input Par: ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par: ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
Input Par: ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
Input Par: ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest: TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR
Dest: TempPerfMsg.TransTMR.ACC
110

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 4102
Input Par: ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par: ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
Input Par: ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
Input Par: ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans ...
PerfMsg: TempPerfMsg
PerfTrans: EDC_Status01.ServerStat.Msg.EN

Status Request to Server via Transaction Manager

Request Complete
Wait for TM Handshake to Acknowledge Request Complete
v3.5 - replace bit-handshiake with TransNum echo

7

Server Status from Transaction Manager Request Record in Progress
ServerStat.Ctl.ReqInProgress
<EDC.ServerStat.Ctl.ReqInProgress>

Server Status from Transaction Manager Transaction Number (TM Request Trigger)

EQ
Source A ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>
7319
Source B ServerStat.Ctl.Interface
<EDC.ServerStat.Ctl.Interface>
7319

Server Status from Transaction Manager Request Record Complete
ServerStat.Ctl.ReqComplete
<EDC.ServerStat.Ctl.ReqComplete>
L

Request Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify

8

Server Status from Transaction Manager Request Record Complete
ServerStat.Ctl.ReqComplete
<EDC.ServerStat.Ctl.ReqComplete>

Server Status from Transaction Manager Request Complete Delay Timer

MOVE
Source 50
Dest ServerStat.Ctl.ReqCompleteDelayTMR.PRE
<EDC.ServerStat.Ctl.ReqCompleteDelayTMR.PRE>
50

Server Status from Transaction Manager Request Complete Delay Timer

TON
Timer ServerStat.Ctl.ReqCompleteDelayTMR
<EDC.ServerStat.Ctl.ReqCompleteDelayTMR>
Preset 50
Accum 55
EN
DN

Server Status from Transaction Manager Request Complete Delay Timer
ServerStat.Ctl.ReqCompleteDelayTMR.DN
<EDC.ServerStat.Ctl.ReqCompleteDelayTMR.DN>

Server Status from Transaction Manager Request Record Complete Delay
ServerStat.Ctl.ReqCompleteDelay
<EDC.ServerStat.Ctl.ReqCompleteDelay>

Server Status from Transaction Manager Request Record in Progress
ServerStat.Ctl.ReqInProgress
<EDC.ServerStat.Ctl.ReqInProgress>
U

Status Request to Server via Transaction Manager

Request Complete - Response Received
Log Transaction & Performance
(Delay needed to allow all TM Data values to arrive - was receiving Handshake before Data - order of TM variables not enforeced in data update.)

9

Server Status from Transaction Manager Request Record Complete Delay
ServerStat.Ctl.ReqCompleteDelay
<EDC.ServerStat.Ctl.ReqCompleteDelay>

Server Status from Transaction Manager Request Record Complete 1-shot
ServerStat.Ctl.ReqCompleteONS
<EDC.ServerStat.Ctl.ReqCompleteONS>
ONS

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 4101
Input Par ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
Input Par ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
Input Par ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>

Server Interface Status State = Green - Msg Rx DN
EDC_Status01.ServerStat.Update.MsgRxOK
L

Update Status State for Transactions
aoiEDC_StatusUpdateServerStatus
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerStatus ...
Status EDC_Status01.ServerStat

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)
CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer
MOVE
Source ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction
aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerStat.Msg.DN

Status Response from Server via Transaction Manager
JSR
Routine Name _865_StatResponse

Status Request to Server via Transaction Manager

Request in Progress Timeout
Send new Request

Dec 31, 2025 - reduced from 62 to 12 seconds to capture failure before next TM service restart

10

Server Status from Transaction Manager Request Record in Progress
ServerStat.Ctl.ReqInProgress
<EDC.ServerStat.Ctl.ReqInProgress>

Server Status from Transaction Manager Request Record Complete Delay
ServerStat.Ctl.ReqCompleteDelay
<EDC.ServerStat.Ctl.ReqCompleteDelay>
/

Server Status from Transaction Manager Request in Progress Timer

MOVE
Source 12000
Dest ServerStat.Ctl.ReqInProgressTMR.PRE
<EDC.ServerStat.Ctl.ReqInProgressTMR.PRE>
12000

Server Status from Transaction Manager Request in Progress Timer

TON
Timer ServerStat.Ctl.ReqInProgressTMR
<EDC.ServerStat.Ctl.ReqInProgressTMR>
Preset 12000
Accum 0
EN
DN

Server Status from Transaction Manager Request in Progress Timer
ServerStat.Ctl.ReqInProgressTMR.DN
<EDC.ServerStat.Ctl.ReqInProgressTMR.DN>

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Status Request to Server via Transaction Manager

Request in Progress Timeout
Log Transaction & Performance

11

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4104
Input Par ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
Input Par ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
Input Par ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>

Server Interface Status State = Red-Msg Rx ER
EDC_Status01.ServerStat.Update.MsgRxER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerStatus
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerStatus ...
Status EDC_Status01.ServerStat

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerStat.Msg.ER

Server Status from Transaction Manager Request in Progress Timer
ServerStat.Ctl.ReqInProgressTMR
<EDC.ServerStat.Ctl.ReqInProgressTMR>
RES

Status Request to Server via Transaction Manager

Enable RB high-speed data transfer
use the Server Status Message to monitor TM status

If TM communications stop, throttle RB speed until it has recovered for 1 minute.
This provides time for TM to restart properly if the applicaiton was restarted.
(was seeing TM Log Error = The Worker thread could not signal the Scheduler thread)

12

s:fs

Server Status from Transaction Manager Request Record Complete Delay
ServerStat.Ctl.ReqCompleteDelay
<EDC.ServerStat.Ctl.ReqCompleteDelay>

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.TT

Server Status TM Transaction OK Enable High Speed RB
ServerStat_TMokTMR.DN

Server Status TM Transaction OK Enable High Speed RB

MOVE
Source 60000
Dest ServerStat_TMokTMR.PRE 60000

Server Status TM Transaction OK Enable High Speed RB

TON
Timer ServerStat_TMokTMR
Preset 60000
Accum 60008
EN
DN

Server Status from Transaction Manager Request Record Complete Delay
ServerStat.Ctl.ReqCompleteDelay
<EDC.ServerStat.Ctl.ReqCompleteDelay>

StatCompleteONS
ONS

ADD
Source A ServerStat_CompleteCount 6917818
Source B 1
Dest ServerStat_CompleteCount 6917818

Server Status from Transaction Manager Request Record in Progress Timeout
ServerStat.Ctl.ReqInProgressTimeout
<EDC.ServerStat.Ctl.ReqInProgressTimeout>

StatTimeoutONS
ONS

ADD
Source A ServerStat_TimeoutCount 135
Source B 1
Dest ServerStat_TimeoutCount 135

Status Request to Server via Transaction Manager

(End)

Status Request to Server via Transaction Manager - Data Init

Request Next Initialize
if Response Good Data, request next Row of Data in Response to Server
if not Response Good Data, send new Data Request after wait interval

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 861
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Status Request to Server via Transaction Manager - Data Init

Initialize Data Values
(FLL -1 corrupts FLOAT values and causes TM Transaction Failure - overwrite with MOV -1)

2

Server Status from Transaction Manager Status Transaction Data

FLL
Source -1
Dest ServerStat.Data
<EDC.ServerStat.Data>
Length 1

Server Status from Transaction Manager Status Description

FLL
Source 0
Dest ServerStat.Data.Desc
<EDC.ServerStat.Data.Desc>
Length 1

Server Status from Transaction Manager Last Process Time

FLL
Source 0
Dest ServerStat.Data.LastProcTime
<EDC.ServerStat.Data.LastProcTime>
Length 1

Server Status from Transaction Manager Status Description

MOVE
Source 1
Dest ServerStat.Data.Desc.DATA[0]
<EDC.ServerStat.Data.Desc.DATA[0]>
'T'

Server Status from Transaction Manager Last Process Time

MOVE
Source 1
Dest ServerStat.Data.LastProcTime.DATA[0]
<EDC.ServerStat.Data.LastProcTime.DATA[0]>
'2'

Server Status from Transaction Manager Backlog Time (minutes)

MOVE
Source -1
Dest ServerStat.Data.BacklogTime
<EDC.ServerStat.Data.BacklogTime>
0.3

Status Request to Server via Transaction Manager - Data Init

Increment Command
(Table Row Index)
DN005 Health Status Alarm Callout

3

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>

Server Status from Transaction Manager Command-specific Record ID for TransLog

ADD
Source A ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13
Source B 1

Dest ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13

Server Status from Transaction Manager Command-specific Record ID for TransLog

LT
Source A ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13
Source B 1

Server Status from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source 1

Dest ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13

Server Status from Transaction Manager Command-specific Record ID for TransLog

GT
Source A ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13
Source B 13

Status Request to Server via Transaction Manager - Data Init

Load Command

Server Status from Transaction Manager Request/Response Command

MOVE
Source ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13
Dest ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
13

4

Store Last Request

Server Status from Transaction Manager Transaction Data Last Request

COP
Source ServerStat.Data
<EDC.ServerStat.Data>
Dest ServerStat.DataLastReq
<EDC.ServerStat.DataLastReq>
Length 1

5

(End)

Status Response from Server via Transaction Manager

Status Response from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 865
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Initialize Response Good Data

2

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>

U

Status Response from Server via Transaction Manager

Check Response Command for No Data
DN005 Health Status Alarm Callout

3

Server Status from Transaction Manager Request/Response Command

LT
Source A ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
13
Source B 1

Server Status from Transaction Manager Request/Response Command

GT
Source A ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
13
Source B 15

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4113
Input Par ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
Input Par ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
Input Par ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>

lbl_Exit
JMP

Process Response Command

4

Status Response from Server via Transaction Manager

JSR
Routine Name _866_StatResponseRow

Exit Routine

5

lbl_Exit
LBL

NOP

Status Response from Server via Transaction Manager

Store Last Response and Last Good Response

6

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>

Server Status from Transaction Manager Transaction Data Last Response

COP
Source ServerStat.Data
<EDC.ServerStat.Data>
Dest ServerStat.DataLastResp
<EDC.ServerStat.DataLastResp>
Length 1

Server Status from Transaction Manager Transaction Data Last Good Response

COP
Source ServerStat.Data
<EDC.ServerStat.Data>
Dest ServerStat.DataLastRespGood
<EDC.ServerStat.DataLastRespGood>
Length 1

(End)

Status Response from Server via Transaction Manager

Process Status Row from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 866
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Status Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data
DN005 Health Status Alarm Callout

2

Server Status from Transaction Manager Request/Response Command

EQ
Source A ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
13
Source B ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
13

Server Status from Transaction Manager Request/Response Command

EQ
Source A ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
13
Source B ServerStat.Data.StatusID
<EDC.ServerStat.Data.StatusID>
13

Server Status from Transaction Manager Status ID

LIMIT
Low Limit 1
Test ServerStat.Data.StatusID
<EDC.ServerStat.Data.StatusID>
13
High Limit EDC_HMI.CFG.Stat.n
15

Server Status from Transaction Manager Status Description

NE
Source A ServerStat.Data.Desc.DATA[0]
<EDC.ServerStat.Data.Desc.DATA[0]>
'T'
Source B 1

Server Status from Transaction Manager Last Process Time

NE
Source A ServerStat.Data.LastProcTime.DATA[0]
<EDC.ServerStat.Data.LastProcTime.DATA[0]>
'2'
Source B 1

Server Status from Transaction Manager Backlog Time (minutes)

NE
Source A ServerStat.Data.BacklogTime
<EDC.ServerStat.Data.BacklogTime>
0.3
Source B -1

Server Status from Transaction Manager Backlog Count (events)

NE
Source A ServerStat.Data.BacklogCount
<EDC.ServerStat.Data.BacklogCount>
3
Source B -1

Server Status from Transaction Manager Shift Record Count (events)

NE
Source A ServerStat.Data.ShiftTotal
<EDC.ServerStat.Data.ShiftTotal>
96603
Source B -1

Server Status from Transaction Manager Shift Error Count (events)

GE
Source A ServerStat.Data.ErrorCount
<EDC.ServerStat.Data.ErrorCount>
0
Source B 0

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>
L

Status Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

Server Status from Transaction Manager Response has Good Data Record
ServerStat.Ctl.RspGoodData
<EDC.ServerStat.Ctl.RspGoodData>

3

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 4114
Input Par: ServerStat.Ctl.ReqInProgressTMR.ACC
<EDC.ServerStat.Ctl.ReqInProgressTMR.ACC>
Input Par: ServerStat.Data.Command
<EDC.ServerStat.Data.Command>
Input Par: ServerStat.Ctl.CommandID
<EDC.ServerStat.Ctl.CommandID>
Input Par: ServerStat.Ctl.TransactionNumber
<EDC.ServerStat.Ctl.TransactionNumber>

Server Interface Status State = Red-Line Rx Data ER
EDC_Status01.ServerStat.Update.RxDataER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerStatus
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerStatus
Status EDC_Status01.ServerStat

lbl_Exit
JMP

Data Good - Reset Data ER

Server Interface Status State = Red-Line Rx Data ER
EDC_Status01.ServerStat.State.RxDataER

4

Server Interface Status Reset Errors
EDC_Status01.ServerStat.Update.ResetER
L

Update Status State for Transactions

aoiEDC_StatusUpdateServerStatus
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerStatus
Status EDC_Status01.ServerStat

Status Response from Server via Transaction Manager

5

Store Data
(Server Rows 1-15 stored in HMI Array 0-14)

Row Index

SUB
Source A ServerStat.Data.StatusID
<EDC.ServerStat.Data.StatusID>
13
Source B 1
Dest Ri
12

HMI Server Status Block

COP
Source ServerStat.Data.Desc
<EDC.ServerStat.Data.Desc>
Dest EDC_HMI.Oview.Status[Ri].Desc
Length 1

HMI Server Status Block

COP
Source ServerStat.Data.LastProcTime
<EDC.ServerStat.Data.LastProcTime>
Dest EDC_HMI.Oview.Status[Ri].LastProcTime
Length 1

HMI Server Status Block

MOVE
Source ServerStat.Data.BacklogTime
<EDC.ServerStat.Data.BacklogTime>
0.3
Dest EDC_HMI.Oview.Status[Ri].BacklogTime
0.3

HMI Server Status Block

MOVE
Source ServerStat.Data.BacklogCount
<EDC.ServerStat.Data.BacklogCount>
3
Dest EDC_HMI.Oview.Status[Ri].BacklogCount
3

HMI Server Status Block

MOVE
Source ServerStat.Data.ShiftTotal
<EDC.ServerStat.Data.ShiftTotal>
96603
Dest EDC_HMI.Oview.Status[Ri].ShiftTotal
96603

HMI Server Status Block

MOVE
Source ServerStat.Data.ErrorCount
<EDC.ServerStat.Data.ErrorCount>
0
Dest EDC_HMI.Oview.Status[Ri].ErrCount
0

Status Response from Server via Transaction Manager

Update HMI Overview Status Color

6

Server Status from Transaction Manager Backlog Time (minutes)

LT
Source A ServerStat.Data.BacklogTime <EDC.ServerStat.Data.BacklogTime> 0.3
Source B 5

HMI Server Status Block

MOVE
Source 1
Dest EDC_HMI.Oview.Status[Ri].BackColor 1

Server Status from Transaction Manager Backlog Time (minutes)

LIMIT
Low Limit 5
Test ServerStat.Data.BacklogTime <EDC.ServerStat.Data.BacklogTime> 0.3
High Limit 10

HMI Server Status Block

MOVE
Source 2
Dest EDC_HMI.Oview.Status[Ri].BackColor 1

Server Status from Transaction Manager Backlog Time (minutes)

GE
Source A ServerStat.Data.BacklogTime <EDC.ServerStat.Data.BacklogTime> 0.3
Source B 10

HMI Server Status Block

MOVE
Source 3
Dest EDC_HMI.Oview.Status[Ri].BackColor 1

Last Row of Table received (delay before next table update)

7

Row Index

GE
Source A Ri 12
Source B EDC_HMI.CFG.Stat.ei 12

Server Status from Transaction Manager Response has Last Row Record ServerStat.Ctl.RspLastRow <EDC.ServerStat.Ctl.RspLastRow>

Status Response from Server via Transaction Manager

Exit Routine

8

lbl_Exit
LBL

NOP

(End)

Detect and Callout CIS Message Errors

Detect and Callout CIS Message Errors
- Each Error increment latches SendCallout
- When SendCallout and 1-minute dwell since last – set Callout bit for 30 seconds
(this provides 30 seconds on/off for each event)
(also guarantees first & last events will be annunciated regardless of quantity during a 'burst' of events)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 870
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Server Status Table
Load CIS Row Index
(SQL Status ID 6 = Gateway Status Row 5)

2

MOVE
Source 5
Dest CisServerStatRow 5

Detect new CIS Error

3

HMI Server Status Block

NE
Source A EDC_HMI.Oview.Status[CisServerStatRow].ErrCount 0
Source B CisLastErrCount 0

MOVE
Source EDC_HMI.Oview.Status[CisServerStatRow].ErrCount 0
Dest CisLastErrCount 0

NE
Source A CisLastErrCount 0
Source B 0

CisNewError
(L)

Detect and Callout CIS Message Errors

Send CIS Error to Callout for 30 seconds
Block new Errors until this Error sent and Delay is done
(might not send Callout for each Error, but at minimum will indicate First and Last in a burst without bugging the heck out of the recipient)

4

CisNewError CisErrorSendDelay CisNewError CisSendErrorTMR.DN

CisErrorSend

MOVE
Source 30000
Dest CisSendErrorTMR.PRE 30000

TON
Timer CisSendErrorTMR
Preset 30000
Accum 0
EN
DN

CisErrorSend CisErrorSendDelay

Server Status from Transaction Manager Response has CIS Error - Send Callout Bit
ServerStat.Ctl.CisErrorSendCallout
<EDC.ServerStat.Ctl.CisErrorSendCallout>

Delay 30 seconds before sending any new Errors

5

CisErrorSend CisErrorSendDelay

MOVE
Source 30000
Dest CisSendDelayTMR.PRE 30000

TON
Timer CisSendDelayTMR
Preset 30000
Accum 0
EN
DN

CisSendDelayTMR.DN

CisErrorSendDelay

(End)

Detect and Callout Health Status Errors

DN005 Health Status Alarm Callout
Detect and Callout Health Status Errors
- Each Error increment latches SendCallout
- When SendCallout and 1-minute dwell since last – set Callout bit for 30 seconds
(this provides 30 seconds on/off for each event)
(also guarantees first & last events will be annunciated regardless of quantity during a 'burst' of events)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 871
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Server Status Table
Load Health Status Row Index
(SQL Status ID 7 = Gateway Status Row 6)

2

MOVE
Source 6
Dest HealthServStatRow 6

Detect new Health Status Error

3

HMI Server Status Block

NE
Source A EDC_HMI.Oview.Status[HealthServStatRow].ErrCount 0
Source B HealthLastErrCount 0

MOVE
Source EDC_HMI.Oview.Status[HealthServStatRow].ErrCount 0
Dest HealthLastErrCount 0

NE
Source A HealthLastErrCount 0
Source B 0

HealthNewError
L

Detect and Callout Health Status Errors

Send Health Status Error to Callout for 30 seconds
Block new Errors until this Error sent and Delay is done
(might not send Callout for each Error, but at minimum will indicate First and Last in a burst without bugging the heck out of the recipient)

4

HealthNewError HealthErrorSendDelay HealthNewError HealthSendErrorTMR.DN

HealthErrorSend

MOVE
Source 30000
Dest HealthSendErrorTMR.PRE 30000

TON
Timer HealthSendErrorTMR
Preset 30000
Accum 0
EN
DN

HealthErrorSend HealthErrorSendDelay

Equipment Data Collection Response has Health Status Error - Send Callout Bit
EDC.ServerStat.Ctl.HealthStatusSendCallout

Delay 30 seconds before sending any new Errors

5

HealthErrorSend HealthErrorSendDelay

MOVE
Source 30000
Dest HealthSendDelayTMR.PRE 30000

TON
Timer HealthSendDelayTMR
Preset 30000
Accum 0
EN
DN

HealthSendDelayTMR.DN HealthErrorSendDelay

(End)

EQ Health Request to Server via Transaction Manager

EQ Health Data Request to Server via Transaction Manager
DN014 Health Displays

- Logix Requests Data
- Server Responds with Data (EQID and Status) or NoData (0 Command)

- if Data, Logix stores Data for Callout and Acks
- if NoData, Logix waits and then Requests Data

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 880
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

EQ Health Request to Server via Transaction Manager

Transaction Request Timer
Fast Interval used when last TM Response had good response with new data

2

Server EQ Health from Transaction Manager Request Record in Progress
ServerHealth.Ctl.ReqInProgress
<EDC.ServerHealth.Ctl.ReqInProgress>

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>

Server EQ Health from Transaction Manager Request Next Record Timer

MOVE
Source ServerHealth.Ctl.ReqFastTime
<EDC.ServerHealth.Ctl.ReqFastTime>
0
Dest ServerHealth.Ctl.ReqNextTMR.PRE
<EDC.ServerHealth.Ctl.ReqNextTMR.PRE>
1000

Server EQ Health from Transaction Manager Request Next Record Timer

TON
Timer ServerHealth.Ctl.ReqNextTMR
<EDC.ServerHealth.Ctl.ReqNextTMR>
Preset 1000
Accum 678
EN
DN

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>

Server EQ Health from Transaction Manager Request Next Record Timer

MOVE
Source ServerHealth.Ctl.ReqSlowTime
<EDC.ServerHealth.Ctl.ReqSlowTime>
1000
Dest ServerHealth.Ctl.ReqNextTMR.PRE
<EDC.ServerHealth.Ctl.ReqNextTMR.PRE>
1000

Request Next
Ready for next Transaction Manager Trigger

3

Server EQ Health from Transaction Manager Request Next Record Timer
ServerHealth.Ctl.ReqNextTMR.DN
<EDC.ServerHealth.Ctl.ReqNextTMR.DN>

Server EQ Health from Transaction Manager Request Next Record
ServerHealth.Ctl.ReqNext
<EDC.ServerHealth.Ctl.ReqNext>

Server EQ Health from Transaction Manager Request Record in Progress
ServerHealth.Ctl.ReqInProgress
<EDC.ServerHealth.Ctl.ReqInProgress>
L

Server EQ Health from Transaction Manager Request Record Complete
ServerHealth.Ctl.ReqComplete
<EDC.ServerHealth.Ctl.ReqComplete>
U

EQ Health Request to Server via Transaction Manager

Request Next Initialize
if Response Good Data, echo Response in next Request to Ack
if not Response Good Data, send new Data Request after wait interval

4

Server EQ Health from Transaction Manager Request Next Record
ServerHealth.Ctl.ReqNext
<EDC.ServerHealth.Ctl.ReqNext>

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

EQ Helath Request to Server via Transaction Manager - Data Init

JSR
Routine Name _881_HealthRequestInit

Request Next
Increment Transaction Number
(Triggers TM Transaction on change)

5

Server EQ Health from Transaction Manager Request Next Record
ServerHealth.Ctl.ReqNext
<EDC.ServerHealth.Ctl.ReqNext>

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

Server EQ Health from Transaction Manager Transaction Number (TM Request Trigger)

ADD
Source A ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750
Source B 1
Dest ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750

Server EQ Health from Transaction Manager Transaction Number (TM Request Trigger)

LT
Source A ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750
Source B 1

Server EQ Health from Transaction Manager Transaction Number (TM Request Trigger)

GT
Source A ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750
Source B CFG.MsgTransNumRollover
<EDC.CFG.MsgTransNumRollover>
9999

Server EQ Health from Transaction Manager Transaction Number (TM Request Trigger)

MOVE
Source 1
Dest ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750

EQ Health Request to Server via Transaction Manager

6

Request Next
Log Transaction & Performance

Server EQ Health from Transaction Manager Request Next Record
ServerHealth.Ctl.ReqNext
<EDC.ServerHealth.Ctl.ReqNext>

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	4200
Input Par	ServerHealth.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par	0
Input Par	ServerHealth.Data.EQID
	<EDC.ServerHealth.Data.EQID>
Input Par	ServerHealth.Ctl.TransactionNumber
	<EDC.ServerHealth.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR

Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR

Dest	TempPerfMsg.TransTMR.ACC
	110

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	4202
Input Par	ServerHealth.Ctl.ReqInProgressTMR.ACC
	<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par	0
Input Par	ServerHealth.Data.EQID
	<EDC.ServerHealth.Data.EQID>
Input Par	ServerHealth.Ctl.TransactionNumber
	<EDC.ServerHealth.Ctl.TransactionNumber>

Performance Transaction

aoiEDC_PerfTrans

aoiEDC_PerfTrans	EDC_aoiPerfTrans
PerfMsg	TempPerfMsg
PerfTrans	EDC_Status01.ServerHealth.Msg.EN

EQ Health Request to Server via Transaction Manager

Request Complete
Wait for TM Handshake to Acknowledge Request Complete
v3.5 - replace bit-handshiake with TransNum echo

7

Server EQ Health from Transaction Manager Request Record in Progress
ServerHealth.Ctl.ReqInProgress
<EDC.ServerHealth.Ctl.ReqInProgress>

Server EQ Health from Transaction Manager Transaction Number (TM Request Trigger)

EQ
Source A ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>
3750
Source B ServerHealth.Ctl.Interface
<EDC.ServerHealth.Ctl.Interface>
3750

Server EQ Health from Transaction Manager Request Record Complete
ServerHealth.Ctl.ReqComplete
<EDC.ServerHealth.Ctl.ReqComplete>
L

Request Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify

8

Server EQ Health from Transaction Manager Request Record Complete
ServerHealth.Ctl.ReqComplete
<EDC.ServerHealth.Ctl.ReqComplete>

Server EQ Health from Transaction Manager Request Complete Delay Timer

MOVE
Source 50
Dest ServerHealth.Ctl.ReqCompleteDelayTMR.PRE
<EDC.ServerHealth.Ctl.ReqCompleteDelayTMR.PRE>
50

Server EQ Health from Transaction Manager Request Complete Delay Timer

TON
Timer ServerHealth.Ctl.ReqCompleteDelayTMR
<EDC.ServerHealth.Ctl.ReqCompleteDelayTMR>
Preset 50
Accum 62
EN
DN

Server EQ Health from Transaction Manager Request Complete Delay Timer
ServerHealth.Ctl.ReqCompleteDelayTMR.DN
<EDC.ServerHealth.Ctl.ReqCompleteDelayTMR.DN>

Server EQ Health from Transaction Manager Request Record Complete Delay
ServerHealth.Ctl.ReqCompleteDelay
<EDC.ServerHealth.Ctl.ReqCompleteDelay>

Server EQ Health from Transaction Manager Request Record in Progress
ServerHealth.Ctl.ReqInProgress
<EDC.ServerHealth.Ctl.ReqInProgress>
U

EQ Health Request to Server via Transaction Manager

Request Complete - Response Received
Log Transaction & Performance
(Delay needed to allow all TM Data values to arrive - was receiving Handshake before Data - order of TM variables not enforeced in data update.)

9

Server EQ Health from Transaction Manager Request Record Complete Delay
ServerHealth.Ctl.ReqCompleteDelay
<EDC.ServerHealth.Ctl.ReqCompleteDelay>

Server EQ Health from Transaction Manager Request Record Complete 1-shot
ServerHealth.Ctl.ReqCompleteONS
<EDC.ServerHealth.Ctl.ReqCompleteONS>
ONS

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 4201
Input Par: ServerHealth.Ctl.ReqInProgressTMR.ACC
<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par: 0
Input Par: ServerHealth.Data.EQID
<EDC.ServerHealth.Data.EQID>
Input Par: ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>

Server Interface Status State = Green - Msg Rx DN
EDC_Status01.ServerHealth.Update.MsgRxOK
L

Update Health State for Transactions

aoiEDC_StatusUpdateServerHealth
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerHealth ...
Status EDC_Status01.ServerHealth

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerHealth.Ctl.ReqInProgressTMR.ACC
<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerHealth.Msg.DN

EQ Health Response from Server via Transaction Manager

JSR
Routine Name _885_HealthResponse

EQ Health Request to Server via Transaction Manager

Request in Progress Timeout
Send new Request

Dec 31, 2025 - reduced from 62 to 12 seconds to capture failure before next TM service restart

10

Server EQ Health from Transaction Manager Request Record in Progress
ServerHealth.Ctl.ReqInProgress
<EDC.ServerHealth.Ctl.ReqInProgress>

Server EQ Health from Transaction Manager Request Record Complete Delay
ServerHealth.Ctl.ReqCompleteDelay
<EDC.ServerHealth.Ctl.ReqCompleteDelay>

Server EQ Health from Transaction Manager Request in Progress Timer

MOVE
Source 12000
Dest ServerHealth.Ctl.ReqInProgressTMR.PRE
<EDC.ServerHealth.Ctl.ReqInProgressTMR.PRE>
12000

Server EQ Health from Transaction Manager Request in Progress Timer

TON
Timer ServerHealth.Ctl.ReqInProgressTMR
<EDC.ServerHealth.Ctl.ReqInProgressTMR>
Preset 12000
Accum 0
EN
DN

Server EQ Health from Transaction Manager Request in Progress Timer
ServerHealth.Ctl.ReqInProgressTMR.DN
<EDC.ServerHealth.Ctl.ReqInProgressTMR.DN>

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

EQ Health Request to Server via Transaction Manager

Request in Progress Timeout
Log Transaction & Performance

11

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 4204
Input Par: ServerHealth.Ctl.ReqInProgressTMR.ACC
<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par: 0
Input Par: ServerHealth.Data.EQID
<EDC.ServerHealth.Data.EQID>
Input Par: ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>

Server Interface Status State = Red- Msg Rx ER
EDC_Status01.ServerHealth.Update.MsgRxER
L

Update Health State for Transactions

aoiEDC_StatusUpdateServerHealth
aoiEDC_StatusUpdateS...: aoiEDC_StatusUpdateServerHealth
Status: EDC_Status01.ServerHealth

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest: TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source: ServerHealth.Ctl.ReqInProgressTMR.ACC
<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
0
Dest: TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans: EDC_aoiPerfTrans
PerfMsg: TempPerfMsg
PerfTrans: EDC_Status01.ServerHealth.Msg.ER

Server EQ Health from Transaction Manager Request in Progress Timer
ServerHealth.Ctl.ReqInProgressTMR
<EDC.ServerHealth.Ctl.ReqInProgressTMR>
RES

EQ Health Request to Server via Transaction Manager

use the Server Health Message to monitor TM status

12

S:FS

Server EQ Health from Transaction Manager Request Record Complete Delay
ServerHealth.Ctl.ReqCompleteDelay
<EDC.ServerHealth.Ctl.ReqCompleteDelay>

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

Server Health TM Transaction OK
ServerHealth_TMokTMR.TT

Server Health TM Transaction OK
ServerHealth_TMokTMR.DN

Server Health TM Transaction OK

MOVE
Source 60000
Dest ServerHealth_TMokTMR.PRE
60000

Server Health TM Transaction OK

TON
Timer ServerHealth_TMokTMR
Preset 60000
Accum 60014
EN
DN

Server EQ Health from Transaction Manager Request Record Complete Delay
ServerHealth.Ctl.ReqCompleteDelay
<EDC.ServerHealth.Ctl.ReqCompleteDelay>

HealthCompleteONS
ONS

ADD
Source A ServerHealth_CompleteCount
5602672
Source B 1
Dest ServerHealth_CompleteCount
5602672

Server EQ Health from Transaction Manager Request Record in Progress Timeout
ServerHealth.Ctl.ReqInProgressTimeout
<EDC.ServerHealth.Ctl.ReqInProgressTimeout>

HealthTimeoutONS
ONS

ADD
Source A ServerHealth_TimeoutCount
133
Source B 1
Dest ServerHealth_TimeoutCount
133

(End)

EQ Helath Request to Server via Transaction Manager - Data Init

Request Next Initialize
if Response Good Data, echo Response in next Request to Ack
if not Response Good Data, send new Data Request after wait interval

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 881
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Initialize Data Values
(FLL -1 corrupts FLOAT values and causes TM Transaction Failure - overwrite with MOV -1)

2

Server EQ Health from Transaction Manager Status Transaction Data

FLL
Source -1
Dest ServerHealth.Data <EDC.ServerHealth.Data>
Length 1

Server EQ Health from Transaction Manager Last Process Time

FLL
Source 0
Dest ServerHealth.Data.LastProcTime <EDC.ServerHealth.Data.LastProcTime>
Length 1

Server EQ Health from Transaction Manager Last Process Time

MOVE
Source 1
Dest ServerHealth.Data.LastProcTime.DATA[0] <EDC.ServerHealth.Data.LastProcTime.DATA[0]> 'O'

EQ Helath Request to Server via Transaction Manager - Data Init

Ack Command if Good

3

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

Server EQ Health from Transaction Manager Request/Response EQID

MOVE
Source ServerHealth.DataLastResp.EQID
<EDC.ServerHealth.DataLastResp.EQID>
1010
Dest ServerHealth.Data.EQID
<EDC.ServerHealth.Data.EQID>
1010

Server EQ Health from Transaction Manager Status

MOVE
Source ServerHealth.DataLastResp.Status
<EDC.ServerHealth.DataLastResp.Status>
0
Dest ServerHealth.Data.Status
<EDC.ServerHealth.Data.Status>
0

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

NOP

Store Last Request

4

Server EQ Health from Transaction Manager Transaction Data Last Request

COP
Source ServerHealth.Data
<EDC.ServerHealth.Data>
Dest ServerHealth.DataLastReq
<EDC.ServerHealth.DataLastReq>
Length 1

(End)

EQ Health Response from Server via Transaction Manager

Health Response from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 885
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Initialize Response Good Data

2

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

U

Move Command to EQID
(Callout Table Index)

3

MOVE
Source ServerHealth.Data.EQID <EDC.ServerHealth.Data.EQID> 1010
Dest EQID 531

Equipment Index

MOD
Source A EQID 531
Source B 100
Dest EQi 31

Line Index

CPT
Dest LNi 5
Expression (EQID - EQi) / 100

EQ Health Response from Server via Transaction Manager

Check Response Command for Legal Callout Table Index (i.e. EQID)

4

Line Index

LT
Source A LNi
5
Source B 0

Line Index

GT
Source A LNi
5
Source B CFG.CalloutLN.mi
<EDC.CFG.CalloutLN.mi>
28

Equipment Index

LT
Source A EQi
31
Source B 0

Equipment Index

GT
Source A EQi
31
Source B CFG.CalloutEQ.mi
<EDC.CFG.CalloutEQ.mi>
99

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4213
Input Par ServerHealth.Ctl.ReqInProgressTMR.ACC
<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par 0
Input Par ServerHealth.Data.EQID
<EDC.ServerHealth.Data.EQID>
Input Par ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>

lbl_Exit
JMP

Process Response Command

Allowable Data Values enforced by Data Base - no Data checking other than array indexes

5

EQ Health Response from Server via Transaction Manager

JSR
Routine Name _886_HealthResponseEQID

Exit Routine

6

lbl_Exit
LBL

NOP

EQ Health Response from Server via Transaction Manager

Store Last Response and Last Good Response

7

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

Server EQ Health from Transaction Manager Transaction Data Last Response

COP
Source ServerHealth.Data
<EDC.ServerHealth.Data>
Dest ServerHealth.DataLastResp
<EDC.ServerHealth.DataLastResp>
Length 1

Server EQ Health from Transaction Manager Transaction Data Last Good Response

COP
Source ServerHealth.Data
<EDC.ServerHealth.Data>
Dest ServerHealth.DataLastRespGood
<EDC.ServerHealth.DataLastRespGood>
Length 1

(End)

EQ Health Response from Server via Transaction Manager

Process EQID Callout from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 886
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Check Indexes for Valid Configuration Data

2

Line Index

LIMIT
Low Limit 0
Test LNi 5
High Limit CFG.CalloutLN.mi <EDC.CFG.CalloutLN.mi> 28

Equipment Index

LIMIT
Low Limit 0
Test EQi 31
High Limit CFG.CalloutEQ.mi <EDC.CFG.CalloutEQ.mi> 99

Server EQ Health from Transaction Manager Status

LIMIT
Low Limit 0
Test ServerHealth.Data.Status <EDC.ServerHealth.Data.Status> 0
High Limit 4

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData <EDC.ServerHealth.Ctl.RspGoodData>

L

EQ Health Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

3

Server EQ Health from Transaction Manager Response has Good Data Record
ServerHealth.Ctl.RspGoodData
<EDC.ServerHealth.Ctl.RspGoodData>

Log Transactions

JSR
Routine Name: _901_TransLogSBR
Input Par: 4214
Input Par: ServerHealth.Ctl.ReqInProgressTMR.ACC
<EDC.ServerHealth.Ctl.ReqInProgressTMR.ACC>
Input Par: 0
Input Par: ServerHealth.Data.EQID
<EDC.ServerHealth.Data.EQID>
Input Par: ServerHealth.Ctl.TransactionNumber
<EDC.ServerHealth.Ctl.TransactionNumber>

Server Interface Status State = Red-Line Rx Data ER
EDC_Status01.ServerHealth.Update.RxDataER
L

Update Health State for Transactions

aoiEDC_StatusUpdateServerHealth
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerHealth ...
Status EDC_Status01.ServerHealth

lbl_Exit
JMP

EQ Health Response from Server via Transaction Manager

Store Data
Update EQID Callout for Health Alarm (19) and Health Fault (20)

States 1 & 2 - Low & High Alarms (Yellow)
States 3 & 4 - Low & High Faults (Red)

New Data if Last Process Time is NULL or if PreDef bits Change

4

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>
(U)

Server EQ Health from Transaction Manager Last Process Time

EQ
Source A ServerHealth.Data.LastProcTime.LEN
<EDC.ServerHealth.Data.LastProcTime.LEN>
19
Source B 0

Gateway Config Download HMI Display Health Events Enable
EDC_CfgGateway.HmiHealthEventEnable

Server EQ Health from Transaction Manager Status

EQ
Source A ServerHealth.Data.Status
<EDC.ServerHealth.Data.Status>
0
Source B 0

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>
(L)

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19
(U)

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20
(U)

Server EQ Health from Transaction Manager Status

LIMIT
Low Limit 1
Test ServerHealth.Data.Status
<EDC.ServerHealth.Data.Status>
0
High Limit 2

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19
/

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>
(L)

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19
(L)

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20
(U)

Server EQ Health from Transaction Manager Status

LIMIT

Callout Table (LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19

Server EQ Health from Transaction Manager Response has New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>

EQ Health Response from Server via Transaction Manager

LIMIT
Low Limit 3
Test ServerHealth.Data.Status
<EDC.ServerHealth.Data.Status>
0
High Limit 4

Callout Table
(LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20
/

L

Callout Table
(LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19
U

Callout Table
(LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20
L

Oct 4, 2021 - Disabled Health Alarm/Fault bits from EDC Server

5

Gateway Config
Download HMI Display
Health Events Enable
EDC_CfgGateway.HmiHealthEventEnable
/

Callout Table
(LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.19
U

Callout Table
(LNi, EQi)
EDC_EqCallout[LNi,EQi].PreDef.20
U

If new Data, Verify LNi, EQi Limits are OK - if not exit
(next Callout Message will update the status if this fails, but that could take minutes)

v3.5 Update Equipment Events with/without NewData to force update after configuration changes

6

Server EQ Health
from Transaction
Manager Response has
New Data
ServerHealth.Ctl.RspNewData
<EDC.ServerHealth.Ctl.RspNewData>

Line Index
LIMIT
Low Limit 0
Test LNi
5
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

Equipment Index
LIMIT
Low Limit 0
Test EQi
31
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

LimitsOK

LimitsOK
/

lbl_Exit
JMP

EQ Health Response from Server via Transaction Manager

Force an update of the Overview & Zone Events
(same actions taken when a new Callout Message is received)

7

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.Update
L

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].EventCTL.UpdateOview
L

Equipment Data Collection Lines and Equipment Array

FLL
Source -1
Dest EDC.Line[LNi].Equip[EQi].EventCTL.UpdateZone
Length 1

Exit Routine

8

lbl_Exit
LBL

NOP

(End)

Configuration Request to Server via Transaction Manager

Configuration Data Request to Server via Transaction Manager

- Logix Requests Data (-1 Command)
- Server Responds with Data (201 Command) or NoData (0 Command)

- if Data, Logix processes Data and Sends Ack (Echo Server Command)
- Server uses Ack to set Download Timestamp for Config Record
- Server Responds with next Data/NoData

- if NoData, Logix waits and then Requests Data (-1 Command)

0

NOP

Request Commands to TM/SQL

-1 = New Data Request (delay between requests)
201 = Ack Data Response and New Data Request (no delay between requests)

1

NOP

Response Commands from Server

-1 = Data Error
0 = No Data to Download
201 = Event Ignore Mask

2

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

3

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 890
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Request to Server via Transaction Manager

Transaction Request Timer
Fast Interval used when last TM Response had data

4

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Response Duplicate Error
ServerMask.Ctl.RspDupER
<EDC.ServerMask.Ctl.RspDupER>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerMask.Ctl.ReqFastTime
<EDC.ServerMask.Ctl.ReqFastTime>
0
Dest ServerMask.Ctl.ReqNextTMR.PRE
<EDC.ServerMask.Ctl.ReqNextTMR.PRE>
10000

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Response Duplicate Error
ServerMask.Ctl.RspDupER
<EDC.ServerMask.Ctl.RspDupER>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Request Next Record Timer

MOVE
Source ServerMask.Ctl.ReqSlowTime
<EDC.ServerMask.Ctl.ReqSlowTime>
10000
Dest ServerMask.Ctl.ReqNextTMR.PRE
<EDC.ServerMask.Ctl.ReqNextTMR.PRE>
10000

Server Ignore Mask Configuration from Transaction Manager Request Next Record Timer

TON
Timer ServerMask.Ctl.ReqNextTMR
<EDC.ServerMask.Ctl.ReqNextTMR>
Preset 10000
Accum 7220
EN
DN

Configuration Request to Server via Transaction Manager

Request Next
Ready for next Transaction Manager Trigger

5

Server Ignore Mask Configuration from Transaction Manager Request Next Record Timer
ServerMask.Ctl.ReqNextTMR.DN
<EDC.ServerMask.Ctl.ReqNextTMR.DN>

Server Ignore Mask Configuration from Transaction Manager Request Next Record
ServerMask.Ctl.ReqNext
<EDC.ServerMask.Ctl.ReqNext>

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>
L

Server Ignore Mask Configuration from Transaction Manager Request Record Complete
ServerMask.Ctl.ReqComplete
<EDC.ServerMask.Ctl.ReqComplete>
U

Request Next Initialize
if Response Good Data, echo Response in next Request to Ack
if not Response Good Data, send new Data Request after wait interval

6

Server Ignore Mask Configuration from Transaction Manager Request Next Record
ServerMask.Ctl.ReqNext
<EDC.ServerMask.Ctl.ReqNext>

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress Timeout
ServerMask.Ctl.ReqInProgressTimeout
<EDC.ServerMask.Ctl.ReqInProgressTimeout>

Configuration Request to Server via Transaction Manager - Data Init

JSR
Routine Name _891_MaskRequestInit

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>
/

Server Ignore Mask Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
-1

Configuration Request to Server via Transaction Manager

Request Next
Increment Transaction Number
(Triggers TM Transaction on change)

7

Server Ignore Mask Configuration from Transaction Manager Request Next Record
ServerMask.Ctl.ReqNext
<EDC.ServerMask.Ctl.ReqNext>

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress Timeout
ServerMask.Ctl.ReqInProgressTimeout
<EDC.ServerMask.Ctl.ReqInProgressTimeout>

Server Ignore Mask Configuration from Transaction Manager Transaction Number (TM Request Trigger)

ADD
Source A ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber> 1372
Source B 1
Dest ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber> 1372

Server Ignore Mask Configuration from Transaction Manager Transaction Number (TM Request Trigger)

LT
Source A ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber> 1372
Source B 1

Server Ignore Mask Configuration from Transaction Manager Transaction Number (TM Request Trigger)

GT
Source A ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber> 1372
Source B CFG.MsgTransNumRollover <EDC.CFG.MsgTransNumRollover> 9999

Server Ignore Mask Configuration from Transaction Manager Transaction Number (TM Request Trigger)

MOVE
Source 1
Dest ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber> 1372

Configuration Request to Server via Transaction Manager

8

Request Next
Log Transaction & Performance

Server Ignore Mask Configuration from Transaction Manager Request Next Record
ServerMask.Ctl.ReqNext
<EDC.ServerMask.Ctl.ReqNext>

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	4400
Input Par	ServerMask.Ctl.ReqInProgressTMR.ACC <EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerMask.DataReq.Command <EDC.ServerMask.DataReq.Command>
Input Par	ServerMask.Ctl.CommandID <EDC.ServerMask.Ctl.CommandID>
Input Par	ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR

Dest	TempPerfMsg.TransLast
	110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

CLR

Dest	TempPerfMsg.TransTMR.ACC
	110

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress Timeout
ServerMask.Ctl.ReqInProgressTimeout
<EDC.ServerMask.Ctl.ReqInProgressTimeout>

Log Transactions

JSR

Routine Name	_901_TransLogSBR
Input Par	4402
Input Par	ServerMask.Ctl.ReqInProgressTMR.ACC <EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par	ServerMask.DataReq.Command <EDC.ServerMask.DataReq.Command>
Input Par	ServerMask.Ctl.CommandID <EDC.ServerMask.Ctl.CommandID>
Input Par	ServerMask.Ctl.TransactionNumber <EDC.ServerMask.Ctl.TransactionNumber>

Performance Transaction

aoiEDC_PerfTrans

aoiEDC_PerfTrans	EDC_aoiPerfTrans ...
PerfMsg	TempPerfMsg
PerfTrans	EDC_Status01.ServerMask.Msg.EN

Configuration Request to Server via Transaction Manager

Request Complete
Wait for TM Handshake to Acknowledge Request Complete
v3.5 - replace bit-handshiake with TransNum echo

9

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>

Server Ignore Mask Configuration from Transaction Manager Transaction Number (TM Request Trigger)

EQ
Source A ServerMask.Ctl.TransactionNumber
<EDC.ServerMask.Ctl.TransactionNumber>
1372
Source B ServerMask.Ctl.Interface
<EDC.ServerMask.Ctl.Interface>
1372

Server Ignore Mask Configuration from Transaction Manager Request Record Complete
ServerMask.Ctl.ReqComplete
<EDC.ServerMask.Ctl.ReqComplete>
L

Configuration Request to Server via Transaction Manager

Request Complete Delay
Don't reset TM Handshake too soon or it will cause a TM Error on Read Verify
Also problems with asynchronous data from TM - handshake can update before data values.
2-Check method is fast and avoids errors for occasional slow update of data.
Wait 1 scan before 1st check to eliminate 99% of missing data issues.
2nd check after 2000msec if Check 1 fails.
(anything >2 seconds will generate a TM Transaction error based on Tag Retrieval rate)

10

Server Ignore Mask Configuration from Transaction Manager Request Record Complete
ServerMask.Ctl.ReqComplete
<EDC.ServerMask.Ctl.ReqComplete>

Server Ignore Mask Configuration from Transaction Manager Request Record First Check Timer

MOVE
Source 10
Dest ServerMask.Ctl.ReqCompleteCheckTMR.PRE
<EDC.ServerMask.Ctl.ReqCompleteCheckTMR.PRE>
10

Server Ignore Mask Configuration from Transaction Manager Request Record First Check Timer

TON
Timer ServerMask.Ctl.ReqCompleteCheckTMR
<EDC.ServerMask.Ctl.ReqCompleteCheckTMR>
Preset 10
Accum 15
EN
DN

Server Ignore Mask Configuration from Transaction Manager Request Record First Check Timer
ServerMask.Ctl.ReqCompleteCheckTMR.DN
<EDC.ServerMask.Ctl.ReqCompleteCheckTMR.DN>

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Request Complete Delay Timer

MOVE
Source 2000
Dest ServerMask.Ctl.ReqCompleteDelayTMR.PRE
<EDC.ServerMask.Ctl.ReqCompleteDelayTMR.PRE>
2000

Server Ignore Mask Configuration from Transaction Manager Request Complete Delay Timer

TON
Timer ServerMask.Ctl.ReqCompleteDelayTMR
<EDC.ServerMask.Ctl.ReqCompleteDelayTMR>
Preset 2000
Accum 2009
EN
DN

Server Ignore Mask Configuration from Transaction Manager Request Complete Delay Timer
ServerMask.Ctl.ReqCompleteDelayTMR.DN
<EDC.ServerMask.Ctl.ReqCompleteDelayTMR.DN>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Configuration Request to Server via Transaction Manager

Request Complete - Response Received
Log Transaction & Performance
(Delay needed to allow all TM Data values to arrive - was receiving Handshake before Data - order of TM variables not enforced in data update.)

11

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Request Record First Check 1-shot
ServerMask.Ctl.ReqCompleteCheckONS
<EDC.ServerMask.Ctl.ReqCompleteCheckONS>
ONS

Server Ignore Mask Configuration from Transaction Manager Request Complete 1st Check Count
ADD
Source A ServerMask.Ctl.ReqComp1stChkCnt
<EDC.ServerMask.Ctl.ReqComp1stChkCnt>
551295
Source B 1
Dest ServerMask.Ctl.ReqComp1stChkCnt
<EDC.ServerMask.Ctl.ReqComp1stChkCnt>
551295

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete 1-shot
ServerMask.Ctl.ReqCompleteONS
<EDC.ServerMask.Ctl.ReqCompleteONS>
ONS

Server Ignore Mask Configuration from Transaction Manager Request Complete 2nd Check Count
ADD
Source A ServerMask.Ctl.ReqComp2ndChkCnt
<EDC.ServerMask.Ctl.ReqComp2ndChkCnt>
537946
Source B 1
Dest ServerMask.Ctl.ReqComp2ndChkCnt
<EDC.ServerMask.Ctl.ReqComp2ndChkCnt>
537946

Log Transactions
JSR
Routine Name _901_TransLogSBR
Input Par 4401
Input Par ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par ServerMask.DataReq.Command
<EDC.ServerMask.DataReq.Command>
Input Par ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
Input Par ServerMask.Ctl.TransactionNumber
<EDC.ServerMask.Ctl.TransactionNumber>

Server Interface Status State = Green - Msg Rx DN
EDC_Status01.ServerMask.Update.MsgRxOK
L

Update Mask State for Transactions
aoiEDC_StatusUpdateServerMask
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerMask ...
Status EDC_Status01.ServerMask

Configuration Request to Server via Transaction Manager

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerMask.Msg.DN

Configuration Response from Server via Transaction Manager

JSR
Routine Name _895_MaskResponse

Request Complete Delay Reset Request

12

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>
U

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Send new Request

13

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress
ServerMask.Ctl.ReqInProgress
<EDC.ServerMask.Ctl.ReqInProgress>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Request in Progress Timer

MOVE
Source 62000
Dest ServerMask.Ctl.ReqInProgressTMR.PRE
<EDC.ServerMask.Ctl.ReqInProgressTMR.PRE>
62000

Server Ignore Mask Configuration from Transaction Manager Request in Progress Timer

TON
Timer ServerMask.Ctl.ReqInProgressTMR
<EDC.ServerMask.Ctl.ReqInProgressTMR>
Preset 62000
Accum 0
EN
DN

Server Ignore Mask Configuration from Transaction Manager Request in Progress Timer
ServerMask.Ctl.ReqInProgressTMR.DN
<EDC.ServerMask.Ctl.ReqInProgressTMR.DN>

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress Timeout
ServerMask.Ctl.ReqInProgressTimeout
<EDC.ServerMask.Ctl.ReqInProgressTimeout>

Configuration Request to Server via Transaction Manager

Request in Progress Timeout
Log Transaction & Performance

14

Server Ignore Mask Configuration from Transaction Manager Request Record in Progress Timeout
ServerMask.Ctl.ReqInProgressTimeout
<EDC.ServerMask.Ctl.ReqInProgressTimeout>

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4404
Input Par ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par ServerMask.DataReq.Command
<EDC.ServerMask.DataReq.Command>
Input Par ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
Input Par ServerMask.Ctl.TransactionNumber
<EDC.ServerMask.Ctl.TransactionNumber>

Server Interface Status State = Red- Msg Rx ER
EDC_Status01.ServerMask.Update.MsgRxER
L

Update Mask State for Transactions

aoiEDC_StatusUpdateServerMask
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerMask (...)
Status EDC_Status01.ServerMask

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans (...)
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerMask.Msg.ER

Server Ignore Mask Configuration from Transaction Manager Request in Progress Timer
ServerMask.Ctl.ReqInProgressTMR
<EDC.ServerMask.Ctl.ReqInProgressTMR>
RES

(End)

Configuration Request to Server via Transaction Manager - Data Init

Request Next Initialize
if Response Good Data, echo Response in next Request to Server to Ack
if not Response Good Data, send new Data Request after wait interval

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 891
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

DN011 Separate Req & Rsp Tags
Create Data Init Packet

2

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager - Data

FLL
Source -1
Dest ServerMaskInit
Length 1

Configuration Request to Server via Transaction Manager - Data Init

DN011 Separate Req & Rsp Tags
Clear Response to ensure new data on next DB Response
DHF017 Jan 30, 2018 -- Clear Server Request if no Ack for Good Message Response
Copy Response to Request for Good Data Ack to DB

3

Server Ignore Mask Configuration from Transaction Manager Transaction Response Data

COP
Source ServerMaskInit
Dest ServerMask.DataRsp <EDC.ServerMask.DataRsp>
Length 1

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Transaction Request Data

COP
Source ServerMaskInit
Dest ServerMask.DataReq <EDC.ServerMask.DataReq>
Length 1

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Transaction Request Data

COP
Source ServerMask.DataRspLast <EDC.ServerMask.DataRspLast>
Dest ServerMask.DataReq <EDC.ServerMask.DataReq>
Length 1

Store Last Request

4

Server Ignore Mask Configuration from Transaction Manager Transaction Last Request Data

COP
Source ServerMask.DataReq <EDC.ServerMask.DataReq>
Dest ServerMask.DataReqLast <EDC.ServerMask.DataReqLast>
Length 1

Configuration Request to Server via Transaction Manager - Data Init

(End)

Configuration Response from Server via Transaction Manager

Configuration Response from Server via Transaction Manager

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 895
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Detect Duplicate Config Data Response
(this indicates the database is stuck on a bad config item)

2

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

NE
Source A ServerMask.DataRspLast.Command <EDC.ServerMask.DataRspLast.Command> 0
Source B 0

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

NE
Source A ServerMask.DataRspLast.Command <EDC.ServerMask.DataRspLast.Command> 0
Source B -1

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerMask.DataRspLast.Command <EDC.ServerMask.DataRspLast.Command> 0
Source B ServerMask.DataRsp.Command <EDC.ServerMask.DataRsp.Command> 0

Server Ignore Mask Configuration from Transaction Manager Shop Key - for DB Ack only

EQ
Source A ServerMask.DataRspLast.ShopKey <EDC.ServerMask.DataRspLast.ShopKey> -1
Source B ServerMask.DataRsp.ShopKey <EDC.ServerMask.DataRsp.ShopKey> -1

Server Ignore Mask Configuration from Transaction Manager Line Key - for DB Ack only

EQ
Source A ServerMask.DataRspLast.LineKey <EDC.ServerMask.DataRspLast.LineKey> -1
Source B ServerMask.DataRsp.LineKey <EDC.ServerMask.DataRsp.LineKey> -1

Server Ignore Mask Configuration from Transaction Manager Equip Key - for DB Ack only

EQ
Source A ServerMask.DataRspLast.EqKey <EDC.ServerMask.DataRspLast.EqKey> -1
Source B ServerMask.DataRsp.EqKey <EDC.ServerMask.DataRsp.EqKey> -1

Server Ignore Mask Configuration from Transaction Manager LineID (00-99)

EQ
Source A ServerMask.DataRspLast.LineID <EDC.ServerMask.DataRspLast.LineID> -1
Source B ServerMask.DataRsp.LineID <EDC.ServerMask.DataRsp.LineID> -1

Server Ignore Mask Configuration from Transaction Manager Equipment ID (0001-9999)

EQ
Source A ServerMask.DataRspLast.EqID <EDC.ServerMask.DataRspLast.EqID> -1
Source B ServerMask.DataRsp.EqID <EDC.ServerMask.DataRsp.EqID> -1

Server Ignore Mask Configuration from Transaction Manager Event Packet ID (0-7)

EQ
Source A ServerMask.DataRspLast.PacketID <EDC.ServerMask.DataRspLast.PacketID> -1
Source B ServerMask.DataRsp.PacketID <EDC.ServerMask.DataRsp.PacketID> -1

Server Ignore Mask Configuration from Transaction Manager Response Duplicate
ServerMask.Ctl.RspDup <EDC.ServerMask.Ctl.RspDup>

Configuration Response from Server via Transaction Manager

Duplicate Message Count and Detection
(this indicates the database is stuck on a bad config item)

3

Server Ignore Mask Configuration from Transaction Manager Response Duplicate
ServerMask.Ctl.RspDup
<EDC.ServerMask.Ctl.RspDup>

ADD
Source A MaskRspDuplicateCount 0
Source B 1
Dest MaskRspDuplicateCount 0

GT
Source A MaskRspDuplicateCount 0
Source B MaskRspDuplicateMax 50

MOVE
Source MaskRspDuplicateCount 0
Dest MaskRspDuplicateMax 50

GT
Source A MaskRspDuplicateCount 0
Source B 5

Server Ignore Mask Configuration from Transaction Manager Response Duplicate
ServerMask.Ctl.RspDup
<EDC.ServerMask.Ctl.RspDup>

CLR
Dest MaskRspDuplicateCount 0

Server Ignore Mask Configuration from Transaction Manager Response Duplicate Error
ServerMask.Ctl.RspDupER
<EDC.ServerMask.Ctl.RspDupER>

Configuration Response from Server via Transaction Manager

Move Response to working tag and Initialize Response Good Data
DN011 Separate Req & Rsp Tags

Moving DataRsp to DataLastRsp prevents SQL/TM from changing data while Logix is processing.

4

Server Ignore Mask Configuration from Transaction Manager Transaction Last Response Data

COP
Source ServerMask.DataRsp <EDC.ServerMask.DataRsp>
Dest ServerMask.DataRspLast <EDC.ServerMask.DataRspLast>
Length 1

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData <EDC.ServerMask.Ctl.RspGoodData>

U

Configuration Response from Server via Transaction Manager

Check Response Command for Bad Data
Commands 201

5

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

LT
Source A ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
0
Source B 201

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

GT
Source A ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
0
Source B 201

Server Ignore Mask Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source -1
Dest ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4412
Input Par ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
Input Par ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
Input Par ServerMask.Ctl.TransactionNumber
<EDC.ServerMask.Ctl.TransactionNumber>

Equipment Data Collection - Line Equipment Control for Message Transaction Timer Accum Last (used to calculate differential)

CLR
Dest TempPerfMsg.TransLast
110

Equipment Data Collection - Line Equipment Control for Message Transaction Timer

MOVE
Source ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
0
Dest TempPerfMsg.TransTMR.ACC
110

Performance Transaction

aoiEDC_PerfTrans
aoiEDC_PerfTrans EDC_aoiPerfTrans ...
PerfMsg TempPerfMsg
PerfTrans EDC_Status01.ServerMask.Msg.KA

lbl_RspComp
JMP

If Response Command received, process Response

6

MaskRspCmdRcvd

Configuration Response from Server via Transaction Manager

Server has no new Configuration data to download

7

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
0
Source B 0

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>
L

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

ADD
Source A MaskRspCmd[0]
0
Source B 1
Dest MaskRspCmd[0]
0

Process Response Command

8

Server Ignore Mask Configuration from Transaction Manager Request/Response Command

EQ
Source A ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
0
Source B 201

Configuration Response from Server via Transaction Manager

JSR
Routine Name _896_MaskResponseIgnMask

ADD
Source A MaskRspCmd[1]
13555
Source B 1
Dest MaskRspCmd[1]
13555

Command Response Complete

9

lbl_RspComp
LBL

NOP

Configuration Response from Server via Transaction Manager

Store Last Good, Bad or Incomplete Response w/ Diagnostic Counts
Incomplete - Command Num received with incomplete data packet on on 1st Check
Bad - Incomplete data packet on 2nd Check

10

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Transaction Last Good Response Data

COP
Source ServerMask.DataRspLast
<EDC.ServerMask.DataRspLast>
Dest ServerMask.DataRspLastGood
<EDC.ServerMask.DataRspLastGood>
Length 1

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Response Good on 1st Check

ADD
Source A ServerMask.Ctl.RspGood1stChkCnt
<EDC.ServerMask.Ctl.RspGood1stChkCnt>
13349
Source B 1
Dest ServerMask.Ctl.RspGood1stChkCnt
<EDC.ServerMask.Ctl.RspGood1stChkCnt>
13349

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Response Good on 2nd Check

ADD
Source A ServerMask.Ctl.RspGood2ndChkCnt
<EDC.ServerMask.Ctl.RspGood2ndChkCnt>
0
Source B 1
Dest ServerMask.Ctl.RspGood2ndChkCnt
<EDC.ServerMask.Ctl.RspGood2ndChkCnt>
0

Server Ignore Mask Configuration from Transaction Manager Response has

Server Ignore Mask Configuration from Transaction Manager Request

Server Ignore Mask Configuration from Transaction Manager Request Record Complete

Configuration Response from Server via Transaction Manager

Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

MaskRspCmdRcvd

Server Ignore Mask Configuration from Transaction Manager Response None on 1st Check

ADD
Source A ServerMask.Ctl.RspNone1stChkCnt <EDC.ServerMask.Ctl.RspNone1stChkCnt> 47
Source B 1
Dest ServerMask.Ctl.RspNone1stChkCnt <EDC.ServerMask.Ctl.RspNone1stChkCnt> 47

MaskRspCmdRcvd

Server Ignore Mask Configuration from Transaction Manager Transaction Last Incomplete Response Data

COP
Source ServerMask.DataRspLast <EDC.ServerMask.DataRspLast>
Dest ServerMask.DataRspLastInc <EDC.ServerMask.DataRspLastInc>
Length 1

Server Ignore Mask Configuration from Transaction Manager Response Incomplete on 1st Check

ADD
Source A ServerMask.Ctl.RspInc1stChkCnt <EDC.ServerMask.Ctl.RspInc1stChkCnt> 537899
Source B 1
Dest ServerMask.Ctl.RspInc1stChkCnt <EDC.ServerMask.Ctl.RspInc1stChkCnt> 537899

Server Ignore Mask Configuration from Transaction Manager Request Record First Check
ServerMask.Ctl.ReqCompleteCheck
<EDC.ServerMask.Ctl.ReqCompleteCheck>

Server Ignore Mask Configuration from Transaction Manager Request Record Complete Delay
ServerMask.Ctl.ReqCompleteDelay
<EDC.ServerMask.Ctl.ReqCompleteDelay>

Server Ignore Mask Configuration from Transaction Manager Transaction Last Bad Response Data

COP
Source ServerMask.DataRspLast <EDC.ServerMask.DataRspLast>
Dest ServerMask.DataRspLastBad <EDC.ServerMask.DataRspLastBad>
Length 1

Server Ignore Mask Configuration from Transaction Manager Response Bad on 2nd Check

ADD
Source A ServerMask.Ctl.RspBad2ndChkCnt <EDC.ServerMask.Ctl.RspBad2ndChkCnt> 537946
Source B 1
Dest ServerMask.Ctl.RspBad2ndChkCnt <EDC.ServerMask.Ctl.RspBad2ndChkCnt> 537946

Configuration Response from Server via Transaction Manager

Download ALL configuration is complete

11

Server Ignore Mask Configuration from Transaction Manager Response Download All Started
ServerMask.Ctl.DownloadAllStarted
<EDC.ServerMask.Ctl.DownloadAllStarted>

Server Ignore Mask Configuration from Transaction Manager Response Command 0 - Server has no data for Download
ServerMask.Ctl.RspCmd0
<EDC.ServerMask.Ctl.RspCmd0>

Server Ignore Mask Configuration from Transaction Manager Response Download All Started
ServerMask.Ctl.DownloadAllStarted
<EDC.ServerMask.Ctl.DownloadAllStarted>
U

Server Ignore Mask Configuration from Transaction Manager Response Download All Complete
ServerMask.Ctl.DownloadAllComplete
<EDC.ServerMask.Ctl.DownloadAllComplete>
L

Server Ignore Mask Configuration from Transaction Manager Response Download All Complete 1-shot
ServerMask.Ctl.DownloadAllCompleteONS
<EDC.ServerMask.Ctl.DownloadAllCompleteONS>

(End)

Configuration Response from Server via Transaction Manager

Process Configuration from Server via Transaction Manager

Command 201 - Event Ignore Mask
DHF022 - ServCfg.DataLastResp replaces SerfCfg.Data for all Store/Response to ensure no changes during processing

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 896
Dest Prog.Routine 890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -806549717
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -806549717

Configuration Response from Server via Transaction Manager

Convert EQID to LNi and EQi

EDC2 - ack any EQID outside configured LNi, EQi without storing config data
(unused EQID can be assigned to unconfigured LNi, EQi)

2

MOD
Source A ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Source B 100
Dest MaskEQi
26

CPT
Dest MaskLNi
9
Expression (ServerMask.DataRspLast.EqID - MaskEQi) / 100

LIMIT
Low Limit 0
Test MaskLNi
9
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test MaskEQi
26
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

MaskValidEQID

Server Ignore Mask Configuration from Transaction Manager Command-specific Record ID for TransLog

MaskValidEQID
/

MOVE
Source ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Dest ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
-1

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>
L

lbl_DataOk
JMP

Configuration Response from Server via Transaction Manager

Validate Packet Index

v3.5 - ack any Packet outside array limits without storing config data
(Events be created outside limits)

3

Server Ignore Mask Configuration from Transaction Manager Event Packet ID (0-7)

LIMIT
Low Limit 0
Test ServerMask.DataRspLast.PacketID
<EDC.ServerMask.DataRspLast.PacketID>
-1
High Limit EDC_Event.CFG.PKT.mi
7

MaskValidPacketID

Server Ignore Mask Configuration from Transaction Manager Command-specific Record ID for TransLog

MaskValidPacketID
/

MOVE
Source ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Dest ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
-1

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>
L

lbl_DataOk
JMP

Get Eq-Event Lookup Index

4

MOVE
Source EDC.Line[MaskLNi].Equip[MaskEQi].EventCTL.EvLKi
164
Dest MaskEvLKi
164

LIMIT
Low Limit 0
Test MaskEvLKi
164
High Limit EDC_Event.CFG.EQ.mi
2999

MaskGoodLImits

MaskGoodLImits
/

lbl_SkipDataCheck
JMP

Configuration Response from Server via Transaction Manager

Check Indexes for Valid Configuration Data

Mask values can be 0, -1, etc. - no way to validate and ensure full packet received
FooterID is last element and must match EqID

Verify EQID has a valid Event Lookup index

5

Server Ignore Mask Configuration from Transaction Manager Equipment ID (0001-9999)

NE
Source A ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Source B -1

Server Ignore Mask Configuration from Transaction Manager Event Packet ID (0-7)

NE
Source A ServerMask.DataRspLast.PacketID
<EDC.ServerMask.DataRspLast.PacketID>
-1
Source B -1

Server Ignore Mask Configuration from Transaction Manager Equipment ID (0001-9999)

EQ
Source A ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Source B ServerMask.DataRspLast.FooterID
<EDC.ServerMask.DataRspLast.FooterID>
-1

Server Ignore Mask Configuration from Transaction Manager Equipment ID (0001-9999)

EQ
Source A ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Source B EDC_Event.CTL.Equip[MaskEvLKi].EQID
926

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>
L

Skip Data Check if Indexes invalid

6

lbl_SkipDataCheck
LBL

NOP

Configuration Response from Server via Transaction Manager

Data Error - Indexes out of Range
Log Transaction & Performance

7

Server Ignore Mask Configuration from Transaction Manager Response has Good Data Record
ServerMask.Ctl.RspGoodData
<EDC.ServerMask.Ctl.RspGoodData>

Server Ignore Mask Configuration from Transaction Manager Command-specific Record ID for TransLog

MOVE
Source ServerMask.DataRspLast.EqID
<EDC.ServerMask.DataRspLast.EqID>
-1
Dest ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
-1

Log Transactions

JSR
Routine Name _901_TransLogSBR
Input Par 4413
Input Par ServerMask.Ctl.ReqInProgressTMR.ACC
<EDC.ServerMask.Ctl.ReqInProgressTMR.ACC>
Input Par ServerMask.DataRspLast.Command
<EDC.ServerMask.DataRspLast.Command>
Input Par ServerMask.Ctl.CommandID
<EDC.ServerMask.Ctl.CommandID>
Input Par ServerMask.Ctl.TransactionNumber
<EDC.ServerMask.Ctl.TransactionNumber>

Server Interface Status State = Red-Ignore Mask Rx Data ER
EDC_Status01.ServerMask.Update.IgMaskRxDataER
L

Update Mask State for Transactions

aoiEDC_StatusUpdateServerMask
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerMask ...
Status EDC_Status01.ServerMask

lbl_Exit
JMP

Configuration Response from Server via Transaction Manager

Store the Ignore Mask for the EQ-Event Packet
Set the Mask Send bit for updating EqDriver
v3.02 Data Refresh Request Command

8

Server Ignore Mask Configuration from Transaction Manager Packet Data Mask

SIZE
Source ServerMask.DataRspLast.Data[0]
<EDC.ServerMask.DataRspLast.Data[0]>
0
Dim. To Vary 0
Size DataLEN
8

DT Type = Ignore Event Mask Equipment Mask Equip Array

COP
Source ServerMask.DataRspLast.Data[0]
<EDC.ServerMask.DataRspLast.Data[0]>
Dest EDC_EventMaskIgnore.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[0]
Length DataLEN

Event Controls Equipment Fragment Statistics Array
EDC_Event.CTL.Equip[MaskEvLKi].IgnoreMaskSend.[ServerMask.DataRspLast.PacketID]
L

Event Controls Equipment Fragment Statistics Array
EDC_Event.CTL.Equip[MaskEvLKi].DataRefreshReq.[ServerMask.DataRspLast.PacketID]
L

Initialize Loop

9

CLR
Dest MSKi
8

Configuration Response from Server via Transaction Manager

update Fault, Alarm, Alert Masks to clear Ignore items
(when set to Ignore, the Event will no longer be downloaded to update Fault, Alarm, Alert)

10

lbl_MaskLoop
LBL

DT Type = Fault
Event Mask Equipment
Mask Equip Array

AND
Source A EDC_EventMaskFault.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Source B EDC_EventMaskIgnore.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Dest EDC_EventMaskFault.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??

DT Type = Alarm
Event Mask Equipment
Mask Equip Array

AND
Source A EDC_EventMaskAlarm.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Source B EDC_EventMaskIgnore.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Dest EDC_EventMaskAlarm.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??

DT Type = Alert
Event Mask Equipment
Mask Equip Array

AND
Source A EDC_EventMaskAlert.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Source B EDC_EventMaskIgnore.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??
Dest EDC_EventMaskAlert.Equip[MaskEvLKi].Packet[ServerMask.DataRspLast.PacketID].Data[MSKi] ??

ADD
Source A MSKi 8
Source B 1
Dest MSKi 8

LT
Source A MSKi 8
Source B DataLEN 8

lbl_MaskLoop
JMP

Configuration Response from Server via Transaction Manager

Data OK

Server Interface
Status State = Green
- Ignore Mask Rx
Data OK

Update Mask State
for Transactions

11

lbl_DataOk
LBL

EDC_Status01.ServerMask.Update.IgMaskDataOK
L

aoiEDC_StatusUpdateServerMask
aoiEDC_StatusUpdateS... aoiEDC_StatusUpdateServerMask ...
Status EDC_Status01.ServerMask

Exit Routine

12

lbl_Exit
LBL

NOP

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 901
Dest Prog.Routine
890

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[901]
447423535
Source B 1
Dest Prog.RoutineScan[901]
447423535

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

Transaction Log Control -- Transaction Log Index

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.LOGi <EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx <EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log

FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log

MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log

MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log

MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log

MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log

MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log

MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log

MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log

MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log

MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log

MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log

MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log

MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

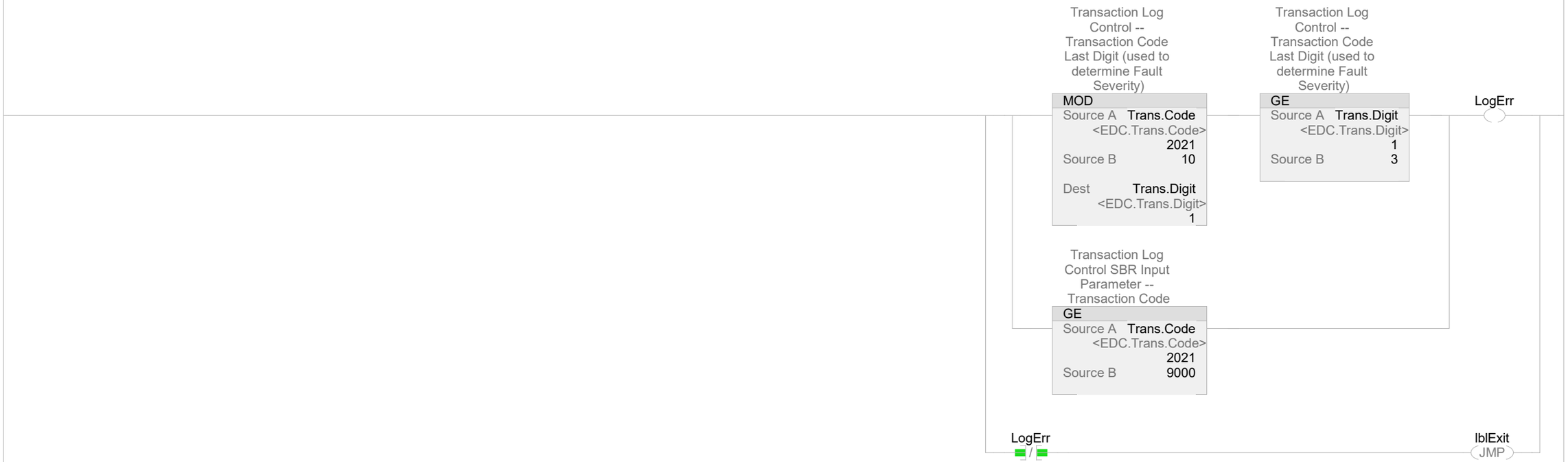
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

v1.19 - added Carrier Tracking
v3.00 Tracking updates for Carrier
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion ...
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

1

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	901
Dest	Prog.Routine 920

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -638139455
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -638139455

2

Reset

Reset Event Configuration
(use before Download All to defrag the Event table for faster searches)

Gateway Reset
CFGResetAviTrack

Toggle to Reset Configuration
ManualReset

Gateway Reset
CFGResetAviTrack
(U)

Toggle to Reset Configuration
ManualReset
(U)

AutoConfig
(L)

Reset

JSR	
Routine Name	_910_Reset

Main Routine

Configuration

3

AutoConfig

Toggle to Load Configuration ManualConfig

HMI Refresh HMI Events after Config All EDC_HMI.CTL.DownloadAllRefresh

s:FS

AutoConfig U

Toggle to Load Configuration ManualConfig U

Configuration

JSR

Routine Name _911_Config

Check for Carrier Requests from MC
Generate Carrier Responses for MC

(MC has the Request & Response Messages)

4

Carrier Requests from MC

JSR

Routine Name _920_CarrierRequest

Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

5

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Routine
	902
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Type
	4
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Code
	20
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.FaultCount
	0
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.LOGi
	34
Dest	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1

Log Transactions

JSR	
Routine Name	_991_TransLogSBR
Input Par	9003
Input Par	0
Input Par	-1
Input Par	-1
Input Par	0

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Main Routine

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

6

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

7

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
7
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan Events
CLxProgScanHigh.9
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV	
Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	2090626792

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD	
Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 50

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 50
Source B	1
Dest	Prog.LOGmi 49

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 34
Source B	1
Dest	Prog.LOGi 34

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 34
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 34
Source B	Prog.LOGmi 49

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 34

Fault Routine

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
912

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
920
Dest Prog.Log[Prog.LOGi].Routine
902

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
-269636260

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
20200928

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
104033

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
424

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source PITi
60
Dest Prog.Log[Prog.LOGi].Opt1
42

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source ConvNum
7
Dest Prog.Log[Prog.LOGi].Opt2
3

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 902
Dest Prog.Routine 920

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -638139455
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -638139455

5

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 2090626792

6

(End)

Reset

HMI Reset

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 910
Dest Prog.Routine 920

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -638139455
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -638139455

Reset EDC Tracking Arrays

2

Carrier Tracking from EQ Quality Tools

FLL
Source 0
Dest EDC_Track
Length 1

AVI Line Tracking

FLL
Source 0
Dest EDC_TrackAVI
Length 1

(End)

Configuration

HMI Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 911
Dest Prog.Routine 920

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -638139455
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -638139455

Carrier Tracking Conveyor Array

2

Carrier Tracking from EQ Quality Tools EDC Tracking Conveyor

SIZE
Source EDC_Track.Conv[0] ??
Dim. To Vary 0
Size EDC_Track.Cfg.CNV.n 29

Carrier Tracking from EQ Quality Tools Array Max Index

SUB
Source A EDC_Track.Cfg.CNV.n 29
Source B 1
Dest EDC_Track.Cfg.CNV.mi 28

Carrier Tracking from EQ Quality Tools Array Start Index

MOVE
Source 0
Dest EDC_Track.Cfg.CNV.si 0

Carrier Tracking from EQ Quality Tools Array End Index

MOVE
Source EDC_Track.Cfg.CNV.mi 28
Dest EDC_Track.Cfg.CNV.ei 28

Configuration

Carrier Tracking Carrier Array

3

Carrier Tracking from EQ Quality Tools EDC Tracking Conveyor Carrier

SIZE
Source EDC_Track.Conv[0].Carr[0] ??
Dim. To Vary 0
Size EDC_Track.Cfg.CAR.n 1000

Carrier Tracking from EQ Quality Tools Array Max Index

SUB
Source A EDC_Track.Cfg.CAR.n 1000
Source B 1
Dest EDC_Track.Cfg.CAR.mi 999

Carrier Tracking from EQ Quality Tools Array Start Index

MOVE
Source 0
Dest EDC_Track.Cfg.CAR.si 0

Carrier Tracking from EQ Quality Tools Array End Index

MOVE
Source EDC_Track.Cfg.CAR.mi 999
Dest EDC_Track.Cfg.CAR.ei 999

Reqest & Response Message Carrier Position Array

4

AVI Line Tracking Carrier Tracking for AVI Tracking Image

SIZE
Source EDC_TrackAVI.Line[0].Pitch[0] ??
Dim. To Vary 0
Size EDC_Track.Cfg.POS.n 60

Carrier Tracking from EQ Quality Tools Array Max Index

SUB
Source A EDC_Track.Cfg.POS.n 60
Source B 1
Dest EDC_Track.Cfg.POS.mi 59

Carrier Tracking from EQ Quality Tools Array Start Index

MOVE
Source 0
Dest EDC_Track.Cfg.POS.si 0

Carrier Tracking from EQ Quality Tools Array End Index

MOVE
Source EDC_Track.Cfg.POS.mi 59
Dest EDC_Track.Cfg.POS.ei 59

Configuration

Carrier Scanner Configuration Array

5

Carrier Tracking from EQ Quality Tools EDC Tracking Scanner Configuration

SIZE
Source EDC_Track.ScanCfg[0] ??
Dim. To Vary 0
Size EDC_Track.Cfg.SCN.n 300

Carrier Tracking from EQ Quality Tools Array Max Index

SUB
Source A EDC_Track.Cfg.SCN.n 300
Source B 1
Dest EDC_Track.Cfg.SCN.mi 299

Carrier Tracking from EQ Quality Tools Array Start Index

MOVE
Source 0
Dest EDC_Track.Cfg.SCN.si 0

Carrier Tracking from EQ Quality Tools Array End Index

MOVE
Source EDC_Track.Cfg.SCN.mi 299
Dest EDC_Track.Cfg.SCN.ei 299

Animation Model Configuraiton

East Assembly
72 = H = Highlander (Tool PrgNum =1)
73 = I = Sienna (Tool PrgNum = 2)

6

Carrier Tracking from EQ Quality Tools Animation Model Value from AVI Tracking

MOVE
Source 72
Dest EDC_Track.Cfg.AnimationModel 72

Carrier Tracking from EQ Quality Tools Animation Model Value from Tool PrgAct

MOVE
Source 1
Dest EDC_Track.Cfg.AnimationPrgNum 1

(End)

Carrier Requests from MC

Carrier Request from MC

0 - TransNum (Header)
1 - ConvNum (AVI Conveyor Number)
2 - SearchCmd (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)
3 - SearchCnv (Carrier Tracking Array)
4 - Conv Queue (1=Line, 2=Rear, 3=PDC, 4=PDCL)
5 - Carrier Type (0=None, 1=DoorFR, 2=DoorFL, 3=DoorRR, 4=DoorRL)
10-189 - Data (60 x Carrier, Ano, Bno)
199 - TransNum (Footer)

ConvQueue = 4 is special case for Door L PDC - need to update Carrier Tracking but not AVI Tracking

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 920
Dest Prog.Routine 920

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -638139455
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -638139455

Check for new, non-zero, matching TransNum in Header and Footer

2

Carrier Request Data Packet from Master Controller (MC)

NE
Source A EDC_CarrierReqData[0] 3831
Source B TransNum 5649

Carrier Request Data Packet from Master Controller (MC)

NE
Source A EDC_CarrierReqData[0] 3831
Source B 0

Carrier Request Data Packet from Master Controller (MC)

EQ
Source A EDC_CarrierReqData[0] 3831
Source B EDC_CarrierReqData[199] 3831

NewTransaction
L

ADD
Source A NewReqCnt 192372266
Source B 1
Dest NewReqCnt 192372266

NewTransaction
/

lbl_Exit
JMP

Carrier Requests from MC

Copy Carrier Request to working registers

3

Transaction Number

MOVE
Source EDC_CarrierReqData[0] 3831
Dest TransNum 5649

AVI Conveyor Number (T1, T2, T2, etc)

MOVE
Source EDC_CarrierReqData[1] 10
Dest ConvNum 7

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

MOVE
Source EDC_CarrierReqData[2] 3
Dest SearchCmd 3

Carrier Tracking Conveyor Number (Trim, Door, etc)

MOVE
Source EDC_CarrierReqData[3] 10
Dest SearchCnv 7

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

MOVE
Source EDC_CarrierReqData[4] 1
Dest ConvQueue 1

Carrier Type (0=None, 1=DoorFR, 2=DoorFL, 3=DoorRR, 4=DoorRL)

MOVE
Source EDC_CarrierReqData[5] 0
Dest CarrType 0

Carrier Tracking for AVI Tracking Image

COP
Source EDC_CarrierReqData[10]
Dest MsgReqQueue[0]
Length EDC_Track.Cfg.POS.n

Validate Search Conveyor
Abandon Transaction Processing if outside array limits

4

Carrier Tracking Conveyor Number (Trim, Door, etc)

LT
Source A SearchCnv 7
Source B 0

Carrier Tracking Conveyor Number (Trim, Door, etc)

GT
Source A SearchCnv 7
Source B EDC_Track.Cfg.CNV.mi 28

NewTransaction
U

ADD
Source A SearchCnvERcnt 0
Source B 1
Dest SearchCnvERcnt 0

lbl_Exit
JMP

Carrier Requests from MC

Validate Search Command
Abandon Transaction Processing if outside array limits

5

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

LT
Source A SearchCmd 3
Source B 1

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

GT
Source A SearchCmd 3
Source B 4

NewTransaction
U

ADD
Source A SearchCmdERcnt 130
Source B 1
Dest SearchCmdERcnt 130

lbl_Exit
JMP

Generate Carrier Response

6

Carrier Requests to MC

JSR
Routine Name _930_CarrierResponse

Exit Routine

7

lbl_Exit
LBL

NOP

(End)

Carrier Requests to MC

Carrier Request from MC

0 - TransNum (Header)
1 - ConvNum (AVI Conveyor Number)
2 - SearchCmd (1=Ano, 2=Bno, 3=Ano & Bno)
3 - SearchCnv (Carrier Tracking Array)
4 - Conv Queue (1=Line, 2=Rear, 3=PDC, 4=PDCL)
5 - Carrier Type (0=None, 1=DoorFR, 2=DoorFL, 3=DoorRR, 4=DoorRL)
10-189 - Data (60 x Carrier, Ano, Bno)
199 - TransNum (Footer)

ConvQueue = 4 is special case for Door L PDC - need to update Carrier Tracking but not AVI Tracking

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 930
Dest Prog.Routine 920

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -638139455
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -638139455

Copy Carrier Request to Carrier Response working registers

2

Carrier Tracking for AVI Tracking Image

COP
Source MsgReqQueue[0]
Dest MsgRspQueue[0]
Length EDC_Track.Cfg.POS.n

Initialize the Pitch Position Index

3

Pitch Index

CLR
Dest PITi 60

Carrier Requests to MC

Initialize the Carrier Response

4

lbl_LinePos
LBL

Carrier Found
FD
U

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

NE
Source A SearchCmd
3
Source B 4

Carrier Tracking for AVI Tracking Image Carrier Number

CLR
Dest MsgRspQueue[PlTi].Carrier
??

Validate Conveyor Number
Abandon Transaction Processing if outside array limits

5

AVI Conveyor Number (T1, T2, T2, etc)

LT
Source A ConvNum
7
Source B 0

AVI Conveyor Number (T1, T2, T2, etc)

GT
Source A ConvNum
7
Source B EDC_Track.Cfg.CNV.mi
28

lbl_SkipAnimation
JMP

Update HMI animation for Model Type

May 2023 - custom code for 4 Models
'4' or '5' = 52 or 53 -- Highlander (0-white)
'6', '7' or '8' = 54, 55 or 56 -- Grand Highlander (3-cyan)
'A' or 'B' = 65 or 66 -- Lexus (4-dodger blue)
'C' = 67 -- Lexus PHEV (5-peach orange)

6

Shop-specific code for animation

JSR
Routine Name _930_CarrierResponse_CfgAnim

Carrier Requests to MC

Update HMI animation for Model Type
72 = H = Highlander
73 = I = Sienna

7

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

EQ	
Source A	ConvQueue
	1
Source B	1

AVI Line Tracking Line Queue

MOVE	
Source	Animation
	0
Dest	EDC_TrackAVI.Line[ConvNum].Anim[PITi]
	??

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

EQ	
Source A	ConvQueue
	1
Source B	2

AVI Line Tracking Rear Queue

MOVE	
Source	Animation
	0
Dest	EDC_TrackAVI.Rear[ConvNum].Anim[PITi]
	??

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

EQ	
Source A	ConvQueue
	1
Source B	3

AVI Line Tracking PDC Queue

MOVE	
Source	Animation
	0
Dest	EDC_TrackAVI.Pdc[ConvNum].Anim[PITi]
	??

8

lbl_SkipAnimation

LBL

NOP

Combine Carrier Type (0-4) with Ano & Bno
This allows one Tracking Image to contain multiple Ano/Bno for different Door Types

9

Carrier Type & Ano for combined search

CPT	
Dest	TypeAno
	30841
Expression	CarrType*100000+MsgRspQueue[PITi].Ano

Carrier Type & Bno for combined search

CPT	
Dest	TypeBno
	75307
Expression	CarrType*100000+MsgRspQueue[PITi].Bno

Carrier Requests to MC

Search for matching Ano (Command 1)

10

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

EQ
Source A SearchCmd 3
Source B 1

FindAno
RES

CLR
Dest FindAno.POS 0

MOVE
Source EDC_Track.Cfg.CAR.n 1000
Dest FindAno.LEN 60

FSC
Control FindAno
Length 60
Position 0
Mode ALL
Expression TypeAno=(EDC_Track.Conv[SearchCnv].Carr[FindAno.POS].Type*100000+EDC_Track.Conv[SearchCnv].Carr[FindAno.POS].Ano)
EN
DN
ER

FindAno.FD

Carrier Found
FD
L

ADD
Source A FindAnoCnt 0
Source B 1
Dest FindAnoCnt 0

Carrier Index
MOVE
Source FindAno.POS 0
Dest CARi 0

ADD
Source A SearchCmd1Cnt 0
Source B 1
Dest SearchCmd1Cnt 0

Carrier Requests to MC

Search for matching Bno (Command 2 or 3)
(Command 3 requires additional step below)

11

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

EQ
Source A SearchCmd
3
Source B 2

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

EQ
Source A SearchCmd
3
Source B 3

FindBno
RES

CLR
Dest FindBno.POS
999

MOVE
Source EDC_Track.Cfg.CAR.n
1000
Dest FindBno.LEN
1000

FSC
Control FindBno
Length 1000
Position 999
Mode ALL
Expression TypeBno=(EDC_Track.Conv[SearchCnv].Carr[FindBno.POS].Type*100000+EDC_Track.Conv[SearchCnv].Carr[FindBno.POS].Bno)
EN
DN
ER

FindBno.FD

Carrier Index

MOVE
Source FindBno.POS
999
Dest CARi
0

ADD
Source A FindBnoCnt
-716242866
Source B 1
Dest FindBnoCnt
-716242866

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

EQ
Source A SearchCmd
3
Source B 2

Carrier Found

FD
L

ADD
Source A SearchCmd2Cnt
1
Source B 1
Dest SearchCmd2Cnt
1

Carrier Requests to MC

Search for matching Ano (Command 3)

12

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

EQ
Source A SearchCmd
3
Source B 3

FindBno.FD

Carrier Tracking for AVI Tracking Image Assembly Number

NE
Source A MsgRspQueue[PITi].Ano
??
Source B 0

Carrier Tracking for AVI Tracking Image Assembly Number

EQ
Source A MsgRspQueue[PITi].Ano
??
Source B EDC_Track.Conv[SearchCnv].Carr[FindBno.POS].Ano
0

Carrier Type (0=None, 1=DoorFR, 2=DoorFL, 3=DoorRR, 4=DoorRL)

EQ
Source A CarrType
0
Source B EDC_Track.Conv[SearchCnv].Carr[FindBno.POS].Type
0

Carrier Found FD
(L)

ADD
Source A SearchCmd3Cnt
-2112789932
Source B 1
Dest SearchCmd3Cnt
-2112789932

Skip Search

13

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

EQ
Source A SearchCmd
3
Source B 4

Carrier Found FD
(L)

ADD
Source A SearchCmd4Cnt
15442200
Source B 1
Dest SearchCmd4Cnt
15442200

Carrier Requests to MC

Determine if Carrier Data should be updated in ConvLoc & ConvPos
(carrier can existing on multple lines since Queues have 60 positions - don't want to overwrite C1 Pos with old T3 Pos)

Branch 1 -- If Conveyor Number = current Conveyor Location, update Pitch Position (updates as carrier moves on line)
Branch 2 -- If Conveyor Number <> current Conveyor Location and Pitch Position < current Pitch, update (carrier on new line); blocked for PDC Queues
Branch 3 -- Special Case for Trim 1 PDC Queue - need because Trim Carrier Scanner is initializing these Jobs after they are in the Queue

14

Carrier Found
FD

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

EQ	
Source A	SearchCmd
	3
Source B	4

AVI Conveyor Number (T1, T2, T2, etc)

EQ	
Source A	ConvNum
	7
Source B	EDC_Track.Conv[SearchCnv].Carr[CARi].Loc
	0

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

NE	
Source A	ConvQueue
	1
Source B	3

AVI Conveyor Number (T1, T2, T2, etc)

NE	
Source A	ConvNum
	7
Source B	EDC_Track.Conv[SearchCnv].Carr[CARi].Loc
	0

Pitch Index

LE	
Source A	PITi
	60
Source B	EDC_Track.Conv[SearchCnv].Carr[CARi].Pos
	0

UpdateLocPos

Carrier Requests to MC

If the Carrier search was successful,
Store Carrier Number to Response and store Model to Carrier Tracking (add to the Ano/Bno from EqDriver)
Update Conveyor Number and Pitch Position to Carrier Tracking

15

Search Command (1=Ano, 2=Bno, 3=Ano & Bno, 4=NoSearch)

NE
- Source A: SearchCmd = 3
- Source B: 4

Carrier Found

FD

Carrier Tracking for AVI Tracking Image Carrier Number

MOVE
- Source: CARi = 0
- Dest: MsgRspQueue[PITi].Carrier = ??

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
- Source: MsgReqQueue[PITi].Model = ??
- Dest: EDC_Track.Conv[SearchCnv].Carr[CARi].Model = 0

UpdateLocPos

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
- Source: ConvNum = 7
- Dest: EDC_Track.Conv[SearchCnv].Carr[CARi].Loc = 0

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

LIMIT
- Low Limit: 1
- Test: ConvQueue = 1
- High Limit: 2

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
- Source: PITi = 60
- Dest: EDC_Track.Conv[SearchCnv].Carr[CARi].Pos = 0

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

LIMIT
- Low Limit: 3
- Test: ConvQueue = 1
- High Limit: 4

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MUL
- Source A: PITi = 60
- Source B: -1
- Dest: EDC_Track.Conv[SearchCnv].Carr[CARi].Pos = 0

Carrier Tracking from EQ Quality Tools Tracking Conveyor

MOVE
- Source: Animation = 0
- Dest: EDC_Track.Conv[SearchCnv].Carr[CARi].Anim = 0

Carrier Requests to MC

Increment Pitch Position

16

lbl_NextPos
LBL

Pitch Index
ADD
Source A PITi 60
Source B 1
Dest PITi 60

Pitch Index
LE
Source A PITi 60
Source B EDC_Track.Cfg.POS.mi 59

lbl_LinePos
JMP

Validate Conveyor Number
Abandon Transaction Processing if outside array limits

17

AVI Conveyor Number (T1, T2, T2, etc)
LT
Source A ConvNum 7
Source B 0

AVI Conveyor Number (T1, T2, T2, etc)
GT
Source A ConvNum 7
Source B EDC_Track.Cfg.CNV.mi 28

ADD
Source A ConvNumERcnt 0
Source B 1
Dest ConvNumERcnt 0

lbl_SaveRsp
JMP

Carrier Requests to MC

Validate Tracking Queue Number
Abandon Transaction Processing if outside array limits

18

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

LT
Source A ConvQueue 1
Source B 1

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

GT
Source A ConvQueue 1
Source B 4

ADD
Source A ConvQueueERcnt 0
Source B 1
Dest ConvQueueERcnt 0

lbl_SaveRsp
JMP

Copy Response Carrier Array to Queue
Conveyor and Queue (Line, Rear, PDC) determined by Request Packet

19

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

EQ
Source A ConvQueue 1
Source B 1

AVI Line Tracking Line Queue

COP
Source MsgRspQueue[0]
Dest EDC_TrackAVI.Line[ConvNum].Pitch[0]
Length EDC_Track.Cfg.POS.n

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

EQ
Source A ConvQueue 1
Source B 2

AVI Line Tracking Rear Queue

COP
Source MsgRspQueue[0]
Dest EDC_TrackAVI.Rear[ConvNum].Pitch[0]
Length EDC_Track.Cfg.POS.n

AVI Conveyor Queue (1=Line, 2=Rear, 3=PDC)

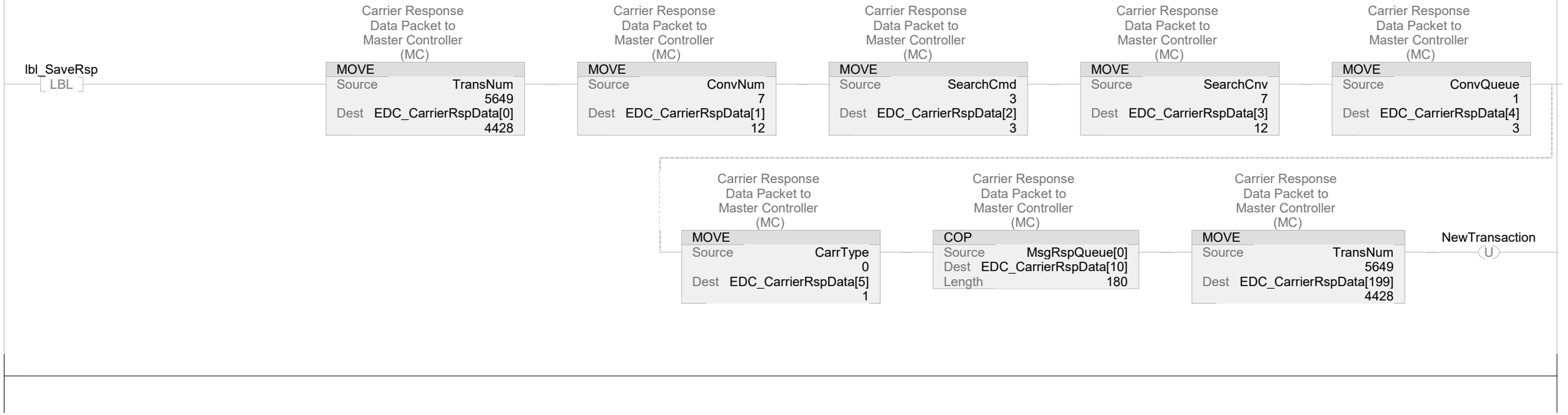
EQ
Source A ConvQueue 1
Source B 3

AVI Line Tracking PDC Queue

COP
Source MsgRspQueue[0]
Dest EDC_TrackAVI.Pdc[ConvNum].Pitch[0]
Length EDC_Track.Cfg.POS.n

Carrier Requests to MC

Copy working registers to Carrier Response



Shop-specific code for animation

Update HMI animation for Model Type

May 2023 - custom code for 4 Models
'4' or '5' = 52 or 53 -- Highlander (0-white)
'6', '7' or '8' = 54, 55 or 56 -- Grand Highlander (3-cyan)
'A' or 'B' = 65 or 66 -- Lexus (4-dodger blue)
'C' = 67 -- Lexus PHEV (5-peach orange)

0

Carrier Tracking for AVI Tracking Image Assembly Number

EQ
Source A MsgReqQueue[PITi].Ano ??
Source B 0

Empty

Animation Value
MOVE
Source 1
Dest Animation 0

Empty

EQ
Source A 72
Source B MsgReqQueue[PITi].Model ??

AnimateGH

Animation Value
MOVE
Source 3
Dest Animation 0

Carrier Tracking for AVI Tracking Image Vehicle Model

EQ
Source A MsgReqQueue[PITi].Model ??
Source B 73

AnimateL

Animation Value
MOVE
Source 4
Dest Animation 0

Carrier Tracking for AVI Tracking Image Vehicle Model

LIMIT
Low Limit 65
Test MsgReqQueue[PITi].Model ??
High Limit 66

AnimateLP

Animation Value
MOVE
Source 5
Dest Animation 0

AnimateGH AnimateL AnimateLP

AnimateH

Animation Value
MOVE
Source 0
Dest Animation 0

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
PNTi = Transaction Point Index (-1 = n/a)

Transaction Log Control SBR Input Parameter -- Transaction Code

0

SBR
Input Par Trans.Code
<EDC.Trans.Code>
Input Par Trans.Time
<EDC.Trans.Time>
Input Par Trans.LNi
<EDC.Trans.LNi>
Input Par Trans.EQi
<EDC.Trans.EQi>
Input Par Trans.Num
<EDC.Trans.Num>

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 991
Dest Prog.Routine
920

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[991]
17
Source B 1
Dest Prog.RoutineScan[991]
17

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

ADD
Source A Trans.LOGi
<EDC.Trans.LOGi>
211
Source B 1
Dest Trans.LOGi
<EDC.Trans.LOGi>
211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi
<EDC.Trans.LOGi>
211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi
<EDC.Trans.LOGi>
211
Source B CFG.TransLOG.n
<EDC.CFG.TransLOG.n>
250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi
<EDC.Trans.LOGi>
211

Generate Log Index for clearing Next Transaction

4

ADD
Source A Trans.LOGi
<EDC.Trans.LOGi>
211
Source B 1
Dest Trans.LOGx
<EDC.Trans.LOGx>
212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx
<EDC.Trans.LOGx>
212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx
<EDC.Trans.LOGx>
212
Source B CFG.TransLOG.n
<EDC.CFG.TransLOG.n>
250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx
<EDC.Trans.LOGx>
212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

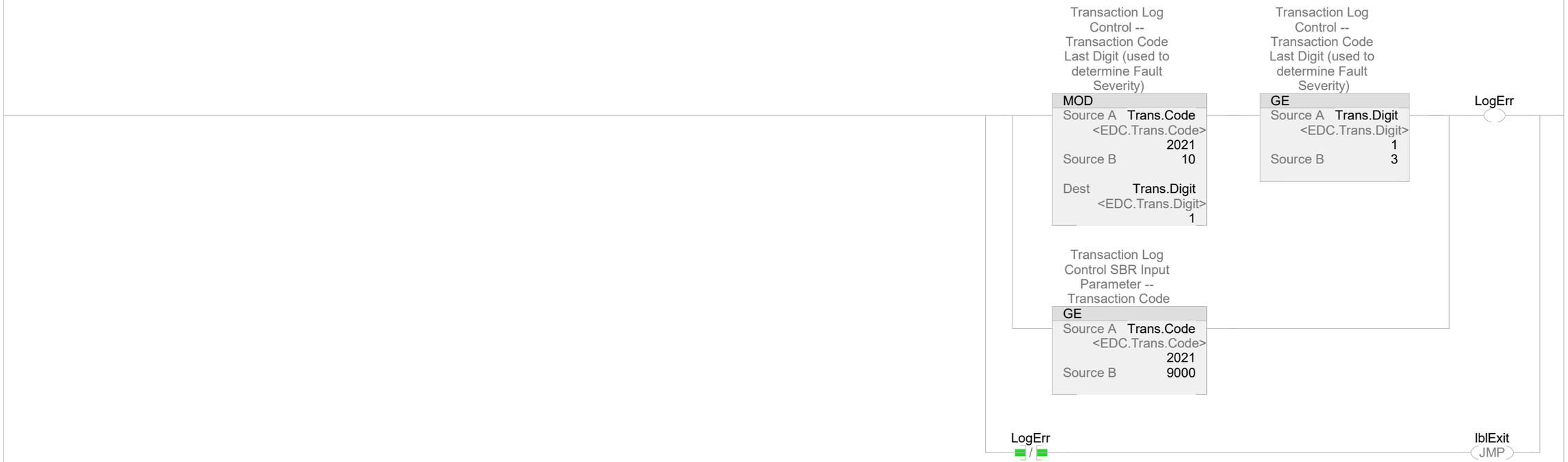
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

Smartwatch Zone Display

Convert Zone Event Display to Line-Equip list for Andon Smart Watch Popups

0

S:FS

Program Version

aoiEDC_ProgVersion

aoiEDC_ProgVersion	EDC_aoiProgVersion ...
Major	1
Minor	0
Build	1
Patch	8
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE

Source	101
Dest	Prog.Routine 130

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD

Source A	Prog.RoutineScan[Prog.Routine] -1861622393
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1861622393

Initialize Zone Event Max Index
Configure 'Red Only' Events for Display

2

Local Copy of SW Zone Array Tag for ZNi HMI Event Table Row

SIZE

Source	SwZone.Event[0] ??
Dim. To Vary	0
Size	LEN 10

Smartwatch Zone Event Array Max Index

SUB

Source A	LEN 10
Source B	1
Dest	ZoneEVmi 9

Only Display Equipment with Red Status

CfgRedEventsOnly

L

Main Routine

Zone Loop Initial/Terminal values for n Zones/Scan
reduce scan time by updating portion of lines each scan but maintain <500 msec update of HMI data

3

Loop Increment = updates ZoneLoopInc+1 Zones/scan

MOVE
Source 1
Dest ZoneLoopInc 1

ADD
Source A ZoneLoopTerm 16
Source B 1
Dest ZoneLoopInit 15

LT
Source A ZoneLoopInit 15
Source B 1

GT
Source A ZoneLoopInit 15
Source B EDC_HMI.CFG.ZN.ei 20

MOVE
Source 1
Dest ZoneLoopInit 15

ADD
Source A ZoneLoopInit 15
Source B ZoneLoopInc 1
Dest ZoneLoopTerm 16

LT
Source A ZoneLoopTerm 16
Source B 1

GT
Source A ZoneLoopTerm 16
Source B EDC_HMI.CFG.ZN.ei 20

MOVE
Source 1
Dest ZoneLoopTerm 16

MOVE
Source EDC_HMI.CFG.ZN.ei 20
Dest ZoneLoopTerm 16

4

G/L AREA

Zone_PlannedStop_Status.1 Zone_PlannedStop_Status.2 Zone_PlannedStop_Status.3 Zone_PlannedStop_Status.4

SHOP AREA

Zone_PlannedStop_Status.0

Equipment Receive Callout Live Table (for external OPC access) Data - Predefined Status

EDC_EqCallout[18,0].PreDef.0

AFI

xxxTest_Status.0

5

Zone_PlannedStop_Status.1

AFI

xxxTest_Status.1

Main Routine

6
AFI
xxxTest_Status.2
Zone_PlannedStop_Status.2

7
AFI
xxxTest_Status.3
Zone_PlannedStop_Status.3

8
AFI
xxxTest_Status.4
G/L AREA
Zone_PlannedStop_Status.4

Zone Loop
service n Zones each Scan

The Zone Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Zone to be scanned continuously.
(WARNING: Disabling Zone FOR Loop will cause the unscanned Zones to stop updating)

9
Disable Zone Loop
CFG_DisableZoneLoop
Zone Loop for ZNi
FOR
Routine Name _120_ZoneLoop
Index ZNi
19
Initial Value ZoneLoopInit
Terminal Value ZoneLoopTerm
Step Size 1
Disable Zone Loop
CFG_DisableZoneLoop
Zone Table Index
MOVE
Source _ViewZNi
0
Dest ZNi
19
Zone Loop for ZNi
JSR
Routine Name _120_ZoneLoop

Andon SmartWatch Main Routine

10
Smart Watch Andon Display
JSR
Routine Name _a1_Main

Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

11

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Routine
702
Dest Trans.Opt[1]
<EDC.Trans.Opt[1]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Fault.Type
4
Dest Trans.Opt[2]
<EDC.Trans.Opt[2]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.Log[Prog.LOGi].Fault.Code
20
Dest Trans.Opt[3]
<EDC.Trans.Opt[3]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.FaultCount
0
Dest Trans.Opt[4]
<EDC.Trans.Opt[4]>
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source Prog.LOGi
45
Dest Trans.Opt[5]
<EDC.Trans.Opt[5]>
-1

Log Transactions

JSR
Routine Name _z901_TransLogSBR
Input Par 9003
Input Par 0
Input Par LNi
Input Par EQi
Input Par 0

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Main Routine

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

12

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

13

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
111
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan Events
CLxProgScanHigh.30
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
123940

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
130
Dest Prog.Log[Prog.LOGi].Routine
702

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
315549464

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
20170319

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
93552

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
874

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source LNi
9
Dest Prog.Log[Prog.LOGi].Opt1
10

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source EQi
17
Dest Prog.Log[Prog.LOGi].Opt2
10

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source GRPi
4
Dest Prog.Log[Prog.LOGi].Opt3
4

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source ZNi
19
Dest Prog.Log[Prog.LOGi].Opt4
6

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source ZnRi
99
Dest Prog.Log[Prog.LOGi].Opt5
17

Store Index Values for debug

4

Fault Routine Diagnostic Index
MOVE
Source GRPi
4
Dest FaultGRPI
0

Fault Routine Diagnostic Index
MOVE
Source sbrGRPi
15
Dest FaultSbrGRPI
0

Fault Routine Diagnostic Index
MOVE
Source LOOPi
4
Dest FaultILOOPi
0

Fault Routine Diagnostic Index
MOVE
Source BITi
104
Dest FaultIBITi
0

Fault Routine Diagnostic Index
MOVE
Source DATi
2
Dest FaultDATI
0

Fault Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(L)

5

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	102
Dest	Prog.Routine 130

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] -1861622393
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] -1861622393

6

Fault Routine

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

7

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type
0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code
0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow
1417957687

(End)

Zone Loop for ZNi

Zone Loop
service n Zones each Scan

The Zone Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Zone to be scanned continuously.
(WARNING: Disabling Zone FOR Loop will cause the unscanned Zones to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 120
Dest Prog.Routine 130

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1861622393
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1861622393

Copy Zone Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

2

Local Copy of Zone Array Tag for ZNi

COP
Source EDC_HMI.Zone[ZNi]
Dest HmiZone
Length 1

Local Copy of SW Zone Array Tag for ZNi

COP
Source EDC_HMI_ZoneSW[ZNi]
Dest SwZone
Length 1

SW Zone Array Tag New Display Values for ZNi

FLL
Source 0
Dest SwZoneNew
Length 1

SwZoneNewFull
U

Zone Table Index

EQ
Source A ZNi 19
Source B 1

FirstZone

Local Copy of SW Zone Array Tag for All Zones

COP
Source EDC_HMI_ZoneSW[0]
Dest SwZoneAll
Length 1

SW Zone Array Tag New Display Values for All Zones

FLL
Source 0
Dest SwZoneAllNew
Length 1

SwZoneAllNewFull
U

Zone Table Index

EQ
Source A ZNi 19
Source B EDC_HMI.CFG.ZN.ei 20

LastZone

Zone Loop for ZNi

Event Loop

The Event Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Event to be scanned continuously.
(WARNING: Disabling the Event FOR Loop will cause the unscanned Events to stop updating)

3

Disable Event Loop
CFG_DisableEventLoop
/

Event Loop for ZNi, EVi
FOR
Routine Name _130_EventLoop
Index EVi 0
Initial Value EDC_HMI.CFG.EV.si
Terminal Value EDC_HMI.CFG.EV.ei
Step Size 1

Disable Event Loop
CFG_DisableEventLoop

MOVE
Source _ViewEVi 0
Dest EVi 0

Event Loop for ZNi, EVi
JSR
Routine Name _130_EventLoop

Determine Alarm value
if new Equipment added to table, New Alarm must be >= Existing Alarm value to generate Smartwatch Alarm

4

SW Zone Array Tag New Display Values for ZNi Event Table Alarm Count
MOVE
Source SwZoneNew.EVi 0
Dest SwZoneNew.Alarm 0

SW Zone Array Tag New Display Values for ZNi Event Table New Alarm Flag
SwZoneNew.NewAlarm

SW Zone Array Tag New Display Values for ZNi Event Table Row Index
LE
Source A SwZoneNew.EVi 0
Source B SwZone.Alarm 0

SW Zone Array Tag New Display Values for ZNi Event Table Alarm Count
ADD
Source A SwZone.Alarm 0
Source B 1
Dest SwZoneNew.Alarm 0

LastZone

SW Zone Array Tag New Display Values for All Zones Event Table Alarm Count
MOVE
Source SwZoneAllNew.EVi 1
Dest SwZoneAllNew.Alarm 0

SW Zone Array Tag New Display Values for All Zones Event Table New Alarm Flag
SwZoneAllNew.NewAlarm

SW Zone Array Tag New Display Values for All Zones Event Table Row Index
LE
Source A SwZoneAllNew.EVi 1
Source B SwZoneAll.Alarm 1

SW Zone Array Tag New Display Values for All Zones Event Table Alarm Count
ADD
Source A SwZoneAll.Alarm 1
Source B 1
Dest SwZoneAllNew.Alarm 0

Zone Loop for ZNi

Copy Zone working tags to Arrayed Tags
Working Tag is used for speed and logic readabliity (ie. can be xref'd)

5

HMI Zone Tables
Smart Watch
Display

COP	
Source	SwZoneNew
Dest	EDC_HMI_ZoneSW[ZNi]
Length	1

HMI Zone Tables
Smart Watch
Display

LastZone

COP	
Source	SwZoneAllNew
Dest	EDC_HMI_ZoneSW[0]
Length	1

Exit Routine

6

lbl_Exit
LBL

NOP

Check Viewing Index Limits

7

LT	
Source A	_ViewZNi 0
Source B	0

GT	
Source A	_ViewZNi 0
Source B	EDC_HMI.CFG.ZN.mi 20

CLR	
Dest	_ViewZNi 0

Zone Loop for ZNi

Copy Line Instance to Viewing Tag if selected
(_ViewZone can be used to multiplex Trends and Diagnostic Spreadsheets)

8

_ViewEN

Zone Table Index

EQ	
Source A	ZNi 19
Source B	_ViewZNi 0

Zone View Tag
(used by View
Diagnostic
Spreadsheet)

COP	
Source	EDC_HMI.Zone[ZNi]
Dest	_ViewZone
Length	1

(End)

Event Loop for ZNi, EVi

Event Loop

The Event Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Event to be scanned continuously.
(WARNING: Disabling the Event FOR Loop will cause the unscanned Events to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 130
Dest Prog.Routine 130

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] -1861622393
Source B 1
Dest Prog.RoutineScan[Prog.Routine] -1861622393

Break from Loop if Event is not Visible
(all Displayed Events have been serviced)

2

Local Copy of Zone Array Tag for ZNi Equipment Data Loaded
HmiZone.Event[EVi].Visible

Zone_PlannedStop_Status.[ZNi]

SHOP AREA
Zone_PlannedStop_Status.0

BRK

Event Loop for ZNi, EVi

Exit from Loop if Event is not Red
(based on Configuration to display only Red events or Events Greater than X Time)

3

Only Display Equipment with Red Status
CfgRedEventsOnly

Local Copy of Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

NE
Source A HmiZone.Event[EVi].Status
5
Source B
3

Local Copy of Zone Array Tag for ZNi Equipment Status Duration (minutes)

LT
Source A HmiZone.Event[EVi].Duration
0.0
Source B Zone_Min_Event_Times[0]
0.001

Local Copy of Zone Array Tag for ZNi Equipment Status Duration (minutes)

LT
Source A HmiZone.Event[EVi].Duration
0.0
Source B Zone_Min_Event_Times[ZNi]
0.001

Zone_PlannedStop_Status.[ZNi]

SHOP AREA
Zone_PlannedStop_Status.0

lbl_Exit
JMP

Copy Event Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

4

Zone_PlannedStop_Status.[ZNi]

SHOP AREA
Zone_PlannedStop_Status.0

Local Copy of Event Array Tag for ZNi, EVi

COP
Source HmiZone.Event[EVi]
Dest HmiZoneEvent
Length 1

Event Loop for ZNi, EVi

Copy Event Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

Local Copy of Event Array Tag for ZNi, EVi

5

Zone_PlannedStop_Status.[ZNi]

SHOP AREA
Zone_PlannedStop_Status.0

COP
Source Zone_PlannedStop_Event
Dest HmiZoneEvent
Length 1

If Zone Event List is Full, skip Zone Event logic

6

SwZoneNewFull

lbl_SkipNewEvent
JMP

Event Loop for ZNi, EVi

Search Smartwatch Zone New Event Table for duplicate Equipment
(only add 1 Equipment entry for multiple Events)

If not duplicate, add to Smartwatch Zone Event Table
and update worst-case Status

7

SW Zone New Event Duplicate Search NewEventDupFSC — RES

SW Zone Array Tag New Display Values for ZNi Event Table Row Index

GT
Source A: SwZoneNew.EVi = 0
Source B: 0

SW Zone New Event Duplicate Search

MOVE
Source: 0
Dest: NewEventDupFSC.POS = 0

SW Zone New Event Duplicate Search

MOVE
Source: SwZoneNew.EVi = 0
Dest: NewEventDupFSC.LEN = 1

SW Zone New Event Duplicate Search

FSC
Control: NewEventDupFSC
Length: 1
Position: 0
Mode: ALL
Expression: HmiZoneEvent.LNi*100+HmiZoneEvent.EQi=(SwZoneNew.Event[NewEventDupFSC.POS].LNi*100+SwZoneNew.Event[NewEventDupFSC.POS].EQi)
EN / DN / ER

SW Zone New Event Duplicate Search NewEventDupFSC.FD — XIC / XIO

SW Zone Array Tag New Display Values for ZNi HMI Event Table Row

COP
Source: HmiZoneEvent
Dest: SwZoneNew.Event[SwZoneNew.EVi]
Length: 1

SW Zone Array Tag New Display Values for ZNi Event Table Row Index

ADD
Source A: SwZoneNew.EVi = 0
Source B: 1
Dest: SwZoneNew.EVi = 0

Local Copy of Event Array Tag for ZNi, EVi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

GT
Source A: HmiZoneEvent.Status = 3
Source B: SwZoneNew.Status = 0

SW Zone Array Tag New Display Values for ZNi Event Table Equip Status

MOVE
Source: HmiZoneEvent.Status = 3
Dest: SwZoneNew.Status = 0

Event Loop for ZNi, EVi

Verify Zone EVi within range
(keep searching Events when table full to detect new Eq Alarm)

8

SW Zone Array Tag
New Display Values
for ZNi Event Table
Row Index

LT
Source A SwZoneNew.EVi 0
Source B 0

SW Zone Array Tag
New Display Values
for ZNi Event Table
Row Index

MOVE
Source 0
Dest SwZoneNew.EVi 0

SW Zone Array Tag
New Display Values
for ZNi Event Table
Row Index

GT
Source A SwZoneNew.EVi 0
Source B ZoneEVmi 9

SW Zone Array Tag
New Display Values
for ZNi Event Table
Row Index

MOVE
Source ZoneEVmi 9
Dest SwZoneNew.EVi 0

SwZoneNewFull
(L)

Event Loop for ZNi, EVi

If Event added to SW Zone Event table,
Search Smartwatch Zone Existing Event Table to determine if New Equipment to be Alarmed
(no need to search if NewAlarm flag is already set)

9

SW Zone Array Tag New Display Values for ZNi Event Table New Alarm Flag: SwZoneNew.NewAlarm [/]

SW Zone New Event Duplicate Search: NewEventDupFSC.FD [/]

Local Copy of SW Zone Array Tag for ZNi Event Table Row Index

GT
- Source A: SwZone.EVi
- Source A: 0
- Source B: 0

SW Zone Existing Events Duplicate Search: ExistEventDupFSC (RES)

SW Zone Existing Events Duplicate Search

MOVE
- Source: 0
- Dest: ExistEventDupFSC.POS
- Dest: 0

SW Zone Existing Events Duplicate Search

ADD
- Source A: ZoneEVmi
- Source A: 9
- Source B: 1
- Dest: ExistEventDupFSC.LEN
- Dest: 10

SW Zone Existing Events Duplicate Search

FSC
- Control: ExistEventDupFSC (EN, DN, ER)
- Length: 10
- Position: 0
- Mode: ALL
- Expression: (HmiZoneEvent.LNi *100) + HmiZoneEvent.EQi = (SwZone.Event[ExistEventDupFSC.POS].LNi * 100) + SwZone.Event[ExistEventDupFSC.POS].EQi

SW Zone Existing Events Duplicate Search: ExistEventDupFSC.FD [/]

SW Zone Array Tag New Display Values for ZNi Event Table New Alarm Flag: SwZoneNew.NewAlarm (L)

10

lbl_SkipNewEvent (LBL)

NOP

If All Zones Event List is Full, skip All Zones Event logic

11

SwZoneAllNewFull

lbl_SkipAllNewEvent (JMP)

Event Loop for ZNi, EVi

Search Smartwatch ALL Zone New Event Table for duplicate Equipment
(only add 1 Equipment entry for multiple Events)

If not duplicate, add to Smartwatch ALL Zone Event Table

12

SW ALL Zone New Event Duplicate Search AllNewEventDupFSC
RES

SW Zone Array Tag New Display Values for All Zones Event Table Row Index
GT
Source A: SwZoneAllNew.EVi 1
Source B: 0

SW ALL Zone New Event Duplicate Search
MOVE
Source: 0
Dest: AllNewEventDupFSC.POS 0

SW ALL Zone New Event Duplicate Search
MOVE
Source: SwZoneAllNew.EVi 1
Dest: AllNewEventDupFSC.LEN 1

SW ALL Zone New Event Duplicate Search
FSC
Control: AllNewEventDupFSC
Length: 1
Position: 0
Mode: ALL
Expression: (HmiZoneEvent.LNi *100) + HmiZoneEvent.EQi = (SwZoneAllNew.Event[AllNewEventDupFSC.POS].LNi * 100) + SwZoneAllNew.Event[AllNewEventDupFSC.POS].EQi
EN
DN
ER

SW ALL Zone New Event Duplicate Search AllNewEventDupFSC.FD
/

SW Zone Array Tag New Display Values for All Zones HMI Event Table Row
COP
Source: HmiZoneEvent
Dest: SwZoneAllNew.Event[SwZoneAllNew.EVi]
Length: 1

SW Zone Array Tag New Display Values for All Zones Event Table Row Index
ADD
Source A: SwZoneAllNew.EVi 1
Source B: 1
Dest: SwZoneAllNew.EVi 1

Local Copy of Event Array Tag for ZNi, EVi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)
GT
Source A: HmiZoneEvent.Status 3
Source B: SwZoneAllNew.Status 3

SW Zone Array Tag New Display Values for All Zones Event Table Equip Status
MOVE
Source: HmiZoneEvent.Status 3
Dest: SwZoneAllNew.Status 3

Event Loop for ZNi, EVi

13

Verify Zone EVi within range
(keep searching Events when table full to detect new Eq Alarm)

SW Zone Array Tag
New Display Values
for All Zones Event
Table Row Index

LT
Source A SwZoneAllNew.EVi
1
Source B 0

SW Zone Array Tag
New Display Values
for All Zones Event
Table Row Index

MOVE
Source 0
Dest SwZoneAllNew.EVi
1

SW Zone Array Tag
New Display Values
for All Zones Event
Table Row Index

GT
Source A SwZoneAllNew.EVi
1
Source B ZoneEVmi
9

SW Zone Array Tag
New Display Values
for All Zones Event
Table Row Index

MOVE
Source ZoneEVmi
9
Dest SwZoneAllNew.EVi
1

SwZoneAllNewFull
(L)

Event Loop for ZNi, EVi

If Event added to SW ALL Zone Event table,
Search Smartwatch ALL Zone Existing Event Table to determine if New Equipment to be Alarmed
(no need to search if NewAlarm flag is already set)

14

SW Zone Array Tag New Display Values for All Zones Event Table New Alarm Flag: SwZoneAllNew.NewAlarm [/]

SW ALL Zone New Event Duplicate Search: AllNewEventDupFSC.FD [/]

Local Copy of SW Zone Array Tag for All Zones Event Table Row Index

GT
- Source A: SwZoneAll.EVi
- Source B: 1
- 0

SW ALL Zone Existing Events Duplicate Search: AllExistEventDupFSC (RES)

SW ALL Zone Existing Events Duplicate Search

MOVE
- Source: 0
- Dest: AllExistEventDupFSC.POS
- 0

SW ALL Zone Existing Events Duplicate Search

ADD
- Source A: ZoneEVmi
- 9
- Source B: 1
- Dest: AllExistEventDupFSC.LEN
- 10

SW ALL Zone Existing Events Duplicate Search

FSC
- Control: AllExistEventDupFSC (EN)
- Length: 10 (DN)
- Position: 0 (ER)
- Mode: ALL
- Expression: (HmiZoneEvent.LNi *100) + HmiZoneEvent.EQi = (SwZoneAll.Event[AllExistEventDupFSC.POS].LNi * 100) + SwZoneAll.Event[AllExistEventDupFSC.POS].EQi

SW ALL Zone Existing Events Duplicate Search: AllExistEventDupFSC.FD [/]

SW Zone Array Tag New Display Values for All Zones Event Table New Alarm Flag: SwZoneAllNew.NewAlarm (L)

15

lbl_SkipAllNewEvent (LBL)

NOP

Stop Looking for New Events if Zone and All Zone Event tables are full

16

SwZoneNewFull SwZoneAllNewFull

BRK

Event Loop for ZNi, EVi

Copy Working Tags to Equipment Arrayed Data
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

17

NOP

Exit Routine

lbl_Exit

18

LBL

NOP

Check Viewing Index Limits

19

LT
Source A _ViewEVi 0
Source B 0

GT
Source A _ViewEVi 0
Source B EDC_HMI.CFG.EV.ei 199

CLR
Dest _ViewEVi 0

Copy Equipment Instance to Viewing Tag if selected
(_ViewEquip can be used to multiplex Trends and Diagnostic Spreadsheets)

20

_ViewEN

Zone Table Index

EQ
Source A ZNi 19
Source B _ViewZNi 0

EQ
Source A EVi 0
Source B _ViewEVi 0

HMI Event Table Row

COP
Source EDC_HMI.Zone[ZNi].Event[EVi]
Dest _ViewEvent
Length 1

(End)

Smart Watch Andon Display

Smart Watch Andon Display

Displays Group Icons, Line Status/KPI and Shop KPI
v2.2 Mar2018 - Split Group Channel into Heartbeat, Static and Dynamic Channels to avoid synchronized updates of all data every second
v2.3 Mar2018 - changed CGP (Common Gateway Platform) FT_Mobility controller tag to generic MDI (Mobile Data Interface)
v2.4 Apr2018 - add JSON Left/Right Braces and String Data Double Quotes to all Watch Strings
v2.5 Apr2018 - remove JSON Left/Right Braces; remove Test Logic
v2.6 Apr2018 - replaced y_RTOS_Truncate with y_RTOS_2Decimal for Real to String n.nn conversion
v2.7 Apr2018 - removed extra leading spaces from Icon Text to avoid text wrap on watch display
v2.8 Apr2018 - implemented GRPmi configuration in b_GRP0_AllErrors

0

NOP

Configure
v3.02e - add ConfigONS

Toggle On to update
Smart Watch
Configuration
a_ManualConfig
U

Configuration
JSR
Routine Name _a2_Config

1

s:FS

EDC Config
Configuration ONS
(Log Transaction)
CFG.ConfigONS
<EDC.CFG.ConfigONS>

Toggle On to update
Smart Watch
Configuration
a_ManualConfig

Initialize
v3.02e - add InitONS

Toggle On to update
Smart Watch
Initialization
a_ManualInit
U

Initialization
JSR
Routine Name _a3_Init

2

s:FS

EDC Config
Initialization ONS
(Log Transaction)
CFG.InitONS
<EDC.CFG.InitONS>

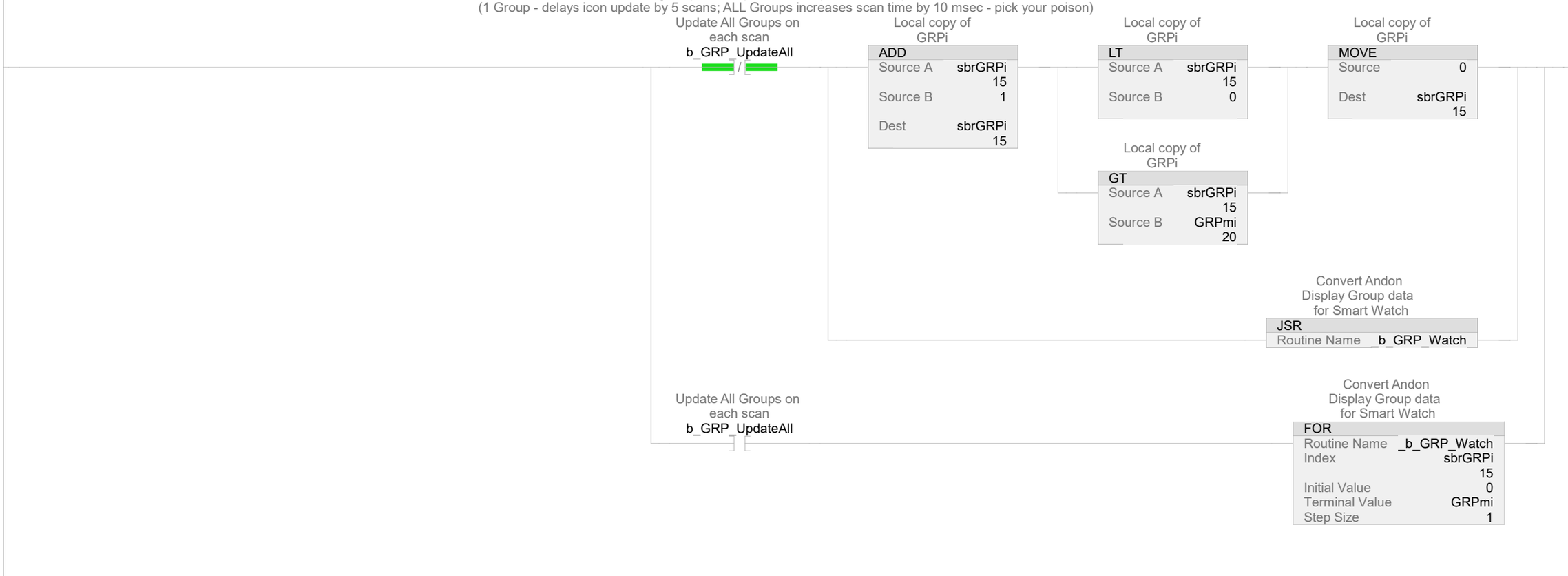
Toggle On to update
Smart Watch
Initialization
a_ManualInit

Smart Watch Andon Display

Convert Groups 0-4 Icons to Watch Data
Populate Watch Display Tags to support backward compatibility with original Watch Mobile Data Interface (MDI)

Next 2 Rungs: Option to update 1 Group or ALL Groups each scan
(1 Group - delays icon update by 5 scans; ALL Groups increases scan time by 10 msec - pick your poison)

3



Smart Watch Andon Display

Convert Groups 0-4 Watch Data to Mobile Data Interface (MDI) String
Group 0 = All Groups Combined

4

Update All Groups on each scan
b_GRP_UpdateAll

Convert Andon Display Group Smart Watch to Mobility String
JSR
Routine Name _b_GRP_WatchString

Update All Groups on each scan
b_GRP_UpdateAll

Convert Andon Display Group Smart Watch to Mobility String
FOR
Routine Name _b_GRP_WatchString
Index sbrGRPi
15
Initial Value 0
Terminal Value GRPmi
Step Size 1

Create Line Status and KPI Watch Data and convert to Mobile Data Interface (MDI) String

5

Line KPI data
JSR
Routine Name _c_Line_KPI

Line Status
JSR
Routine Name _c_Line_Status

Line CGP String
JSR
Routine Name _c_Line_WatchString

Create Shop KPI Watch Data and convert to Mobile Data Interface (MDI) String

6

Shop KPI data
JSR
Routine Name _d_Shop_KPI

Shop CGP String
JSR
Routine Name _d_Shop_WatchString

Create Heartbeat and convert to Mobile Data Interface (MDI) String

7

Heartbeat CGP String (Date/Time)
JSR
Routine Name _e_Heartbeat_WatchString

Create KPI Popup Display and convert to Mobile Data Interface (MDI) String

8

xxxTest_KpiDisplayEN

KPI Popup Display
JSR
Routine Name _f_KpiDisplay

Smart Watch Andon Display

Create Scrolling Message Popup Display and convert to Mobile Data Interface (MDI) String

9

xxxTest_ScrollingMsgEN

Andon Scrolling Message Popup Display

JSR

Routine Name _g_ScrollMsgDisplay

(End)

Configuration

Smart Watch Configuration

0

NOP

Set Maximum Group Used
Feb 2023 - increased from 4 to 20

1

Group Max Index

MOVE	
Source	20
Dest	GRPmi 20

Update 1 Group or ALL Groups each scan
1 Group - delays icon update by 5 scans; ALL Groups increases scan time by 10 msec - pick your poison

2

Update All Groups on each scan
b_GRP_UpdateAll
U

Configuration

Initialize Mobile Data Interface (MDI) Communication Channels
udt Tx/Rx array sizes must match
Latch TxEn and RxEn for each Comm Channel in use

3

Smartwatch Interface
FLL
Source 0
Dest MDI
Length 1

Smartwatch Interface Transmit Channels
SIZE
Source MDI.Tx[0]
'"hour": 15, "min": 2'
Dim. To Vary 0
Size MDI.Size
64

Smartwatch Interface Transmit Channel Config (1=Enabled)
FLL
Source -1
Dest MDI.TxEn
Length 1

Smartwatch Interface Receive Channel Config (1=Enabled)
FLL
Source 0
Dest MDI.RxEn
Length 1

Smartwatch Interface Config Change (Logix Set, Mobility Reset)
MOVE
Source 1
Dest MDI.Config
1

KPI Display
Display KPI Popups when no Event Popups

4

Andon Display KPI Item Display Quantity
MOVE
Source 4
Dest KPI.Cfg.ItemQTY
4

Andon Display KPI Enable KPI Display when no Events
KPI.Cfg.DisplayEN
L

Andon Display KPI Enable Display of 2 KPI Events in Popups (if off, displays 1 KPI Event in Popups)
KPI.Cfg.Display2EN
L

Configuration

Scrolling Message Display

5

Andon Display Scrolling Message Scrolling Message Character Display Length

MOVE
Source 9
Dest ScrollMsg.Cfg.DisplayLEN
9

Andon Display Scrolling Message Enable Scrolling Message Display

ScrollMsg.Cfg.DisplayEN

L

(End)

Initialization

Smart Watch Initialize

0

NOP

Color Code constants

1

Constant - Color Code
MOVE
Source 16#000000
Dest conBlack 16#0000_0000

Constant - Color Code
MOVE
Source 16#0000FF
Dest conBlue 16#0000_00ff

Constant - Color Code
MOVE
Source 16#996633
Dest conBrown 16#0099_6633

Constant - Color Code
MOVE
Source 16#00FFFF
Dest conCyan 16#0000_ffff

Constant - Color Code
MOVE
Source 16#555555
Dest conDarkGray 16#0055_5555

Constant - Color Code
MOVE
Source 16#808080
Dest conGray 16#0080_8080

Constant - Color Code
MOVE
Source 16#00FF00
Dest conGreen 16#0000_ff00

Constant - Color Code
MOVE
Source 16#AAAAAA
Dest conLightGray 16#00aa_aaaa

Constant - Color Code
MOVE
Source 16#FF00FF
Dest conMagenta 16#00ff_00ff

Constant - Color Code
MOVE
Source 16#FF8000
Dest conOrange 16#00ff_8000

Constant - Color Code
MOVE
Source 16#800080
Dest conPurple 16#0080_0080

Constant - Color Code
MOVE
Source 16#FF0000
Dest conRed 16#00ff_0000

Constant - Color Code
MOVE
Source 16#FFFF00
Dest conYellow 16#00ff_ff00

Constant - Color Code
MOVE
Source 16#FFFFFF
Dest conWhite 16#00ff_ffff

ASCII and String constants

2

ASCII Space
MOVE
Source 16#20
Dest ascSpace ' '

ASCII Period
MOVE
Source 16#2E
Dest ascPeriod '.'

ASCII Zero
MOVE
Source 16#30
Dest asc0 48

ASCII Quote
MOVE
Source 16#22
Dest ascDblQuote '"'

MOVE
Source 1
Dest strPeriod.LEN 1

MOVE
Source 1
Dest strDblQuote.LEN 1

MOVE
Source ascPeriod '.'
Dest strPeriod.DATA[0] '.'

MOVE
Source ascDblQuote '"'
Dest strDblQuote.DATA[0] '"'

Initialization

Group Icon Text Array Size

3

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

SIZE	
Source	sbrGRP_WatchCtl.Watch.Popup[0].Text.DATA[0] 'O'
Dim. To Vary	0
Size	GroupIconTextLEN 12

Clear Internal controls

4

Local copy of GRPn tag

FLL	
Source	0
Dest	sbrGRP
Length	1

Local copy of GRP_WatchCtl[n] .Watch

FLL	
Source	0
Dest	sbrGRP_Watch
Length	1

Local copy of GRP_WatchCtl[n]

FLL	
Source	0
Dest	sbrGRP_WatchCtl
Length	1

Local copy of GRP_WatchCtl[n] .WatchPrev

FLL	
Source	0
Dest	sbrGRP_WatchPrev
Length	1

Andon Display WatchOS Control

FLL	
Source	0
Dest	sbrGRP_WatchStringCtl
Length	1

Local Copy of Group Watch String

FLL	
Source	0
Dest	sbrGRP_WatchStringDyn
Length	1

Local Copy of Group Watch String

FLL	
Source	0
Dest	sbrGRP_WatchStringStat
Length	1

Clear Previous to force initial String load

5

Line Status and KPI Data Previous

FLL	
Source	0
Dest	LinePrev
Length	1

Shop KPI Data Previous

FLL	
Source	0
Dest	ShopPrev
Length	1

Heartbeat Data Previous

FLL	
Source	0
Dest	HeartbeatPrev[0]
Length	13

Andon Display WatchOS Control Setup Registers for Watch Display - Previous Values

FLL	
Source	0
Dest	GRP_WatchCtl[0].WatchPrev
Length	1

Andon Display WatchOS Control Setup Registers for Watch Display - Previous Values

FLL	
Source	0
Dest	GRP_WatchCtl[1].WatchPrev
Length	1

Andon Display WatchOS Control Setup Registers for Watch Display - Previous Values

FLL	
Source	0
Dest	GRP_WatchCtl[2].WatchPrev
Length	1

Andon Display WatchOS Control Setup Registers for Watch Display - Previous Values

FLL	
Source	0
Dest	GRP_WatchCtl[3].WatchPrev
Length	1

Andon Display WatchOS Control Setup Registers for Watch Display - Previous Values

FLL	
Source	0
Dest	GRP_WatchCtl[4].WatchPrev
Length	1

Initialization

(End)

Convert Andon Display Group data for Smart Watch

Update Group Watch Strings
Copy Array tags to Local Working tags

0

Local copy of GRP_WatchCtl[n]

COP	
Source	GRP_WatchCtl[sbrGRPi]
Dest	sbrGRP_WatchCtl
Length	1

Copy Zone Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

1

Local Copy of SW Zone Array Tag for ZNi

COP	
Source	EDC_HMI_ZoneSW[sbrGRPi]
Dest	SwZone
Length	1

Convert Andon Display Group data for Smart Watch

Popup Display Registers
Group Description

2

AFI

Local copy of GRPi

DTOS
Source sbrGRPi 15
Dest sbrGRP_WatchCtl.Group '8'

Group Zone Text

CONCAT
Source A strGroup_Zone 'EDC Group '
Source B sbrGRP_WatchCtl.Group '8'
Dest sbrGRP_WatchCtl.Watch.Desc 'Gr8-Gr08'

Local copy of GRPi

EQ
Source A sbrGRPi 15
Source B 0

Local copy of GRP_WatchCtl[n] Group Number String

COP
Source strGroup_NumberALL
Dest sbrGRP_WatchCtl.Group
Length 1

Local copy of GRPi

DTOS
Source sbrGRPi 15
Dest sbrGRP_WatchCtl.Group '8'

CONCAT
Source A strGroup 'Gr'
Source B sbrGRP_WatchCtl.Group '8'
Dest sbrGRP_WatchCtl.Watch.Desc 'Gr8-Gr08'

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

CONCAT
Source A sbrGRP_WatchCtl.Watch.Desc 'Gr8-Gr08'
Source B strDash '-'
Dest sbrGRP_WatchCtl.Watch.Desc 'Gr8-Gr08'

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

CONCAT
Source A sbrGRP_WatchCtl.Watch.Desc 'Gr8-Gr08'
Source B EDC_HMI.CFG.Zone[sbrGRPi].Name ??
Dest sbrGRP_WatchCtl.Watch.Desc 'Gr8-Gr08'

Popup Display Registers
Hearbeat (HH:MM:SS)

3

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source CLX.SysTime[3] <EDC.CLX.SysTime[3]> 15
Dest sbrGRP_WatchCtl.Watch.HB.Hour 15

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source CLX.SysTime[4] <EDC.CLX.SysTime[4]> 19
Dest sbrGRP_WatchCtl.Watch.HB.Minute 25

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source CLX.SysTime[5] <EDC.CLX.SysTime[5]> 21
Dest sbrGRP_WatchCtl.Watch.HB.Second 0

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source 0
Dest sbrGRP_WatchCtl.Watch.HB.Spare 0

Convert Andon Display Group data for Smart Watch

Popup Display Registers
Alarm

Alarm On Delay - minimum time between Alarms

4

Local copy of
GRP_WatchCtl[n]
Alarm On Delay Timer

MOVE
Source 0
Dest sbrGRP_WatchCtl.AlarmOnDelayTMR.PRE
0

Local copy of
GRP_WatchCtl[n]
Alarm On Delay Timer

TON
Timer sbrGRP_WatchCtl.AlarmOnDelayTMR
Preset 0
Accum 0
EN
DN

Popup Display Registers
Alarm

5

Local copy of
GRP_WatchCtl[n]
Alarm On Delay Timer
sbrGRP_WatchCtl.AlarmOnDelayTMR.DN

Local copy of
GRP_WatchCtl[n]
Setup Registers for
Watch Display

MOVE
Source SwZone.Alarm
0
Dest sbrGRP_WatchCtl.Watch.Alarm
1

Local copy of
GRP_WatchCtl[n]
Alarm On Delay Timer
sbrGRP_WatchCtl.AlarmOnDelayTMR
RES

Popup 1
Initialize Color/Text for Blank Display
Color = 0 for Black Background
Text = SPACES for Blank Display

6

Local copy of
GRP_WatchCtl[n]
Setup Registers for
Watch Display

MOVE
Source conBlack
16#0000_0000
Dest sbrGRP_WatchCtl.Watch.Popup[0].Color
16#00ff_0000

Local copy of
GRP_WatchCtl[n]
Setup Registers for
Watch Display

MOVE
Source 1
Dest sbrGRP_WatchCtl.Watch.Popup[0].Text.LEN
9

Local copy of
GRP_WatchCtl[n]
Setup Registers for
Watch Display

FLL
Source ascSpace
Dest sbrGRP_WatchCtl.Watch.Popup[0].Text.DATA[0]
Length 1

Convert Andon Display Group data for Smart Watch

Popup 2
Initialize Color/Text for No Display
Color = 0 for Black Background
Text = SPACES for Blank Display

7

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source conBlack
16#0000_0000
Dest sbrGRP_WatchCtl.Watch.Popup[1].Color
16#0000_0000

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source 1
Dest sbrGRP_WatchCtl.Watch.Popup[1].Text.LEN
1

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

FLL
Source ascSpace
Dest sbrGRP_WatchCtl.Watch.Popup[1].Text.DATA[0]
Length 1

Convert Andon Display Group data for Smart Watch

Display Rotation Timer

8

Local copy of GRP_WatchCtl[n] Popup Rotation Timer

MOVE
Source 2500
Dest sbrGRP_WatchCtl.PopupRotateTMR.PRE 2500

Local copy of GRP_WatchCtl[n] Popup Rotation Timer

TON
Timer sbrGRP_WatchCtl.PopupRotateTMR
Preset 2500
Accum 905
EN
DN

Local copy of GRP_WatchCtl[n] Popup Rotation Timer sbrGRP_WatchCtl.PopupRotateTMR.DN

Local copy of GRP_WatchCtl[n] Popup Rotation Index

ADD
Source A sbrGRP_WatchCtl.POPi 0
Source B 2
Dest sbrGRP_WatchCtl.POPi 0

Local copy of GRP_WatchCtl[n] Popup Rotation Timer sbrGRP_WatchCtl.PopupRotateTMR

RES

Local copy of GRP_WatchCtl[n] Popup Rotation Index

LT
Source A sbrGRP_WatchCtl.POPi 0
Source B 0

Local copy of GRP_WatchCtl[n] Popup Rotation Index

MOVE
Source 0
Dest sbrGRP_WatchCtl.POPi 0

Local copy of GRP_WatchCtl[n] Popup Rotation Index

GE
Source A sbrGRP_WatchCtl.POPi 0
Source B SwZone.EVi 0

Exit if no Line-Equip to display in Popup 1

9

MOVE
Source sbrGRP_WatchCtl.POPi 0
Dest POPi 0

GE
Source A POPi 0
Source B SwZone.EVi 0

lbl_Exit

JMP

Convert Andon Display Group data for Smart Watch

Popup 1
Load Line Name
and initialize Normalization loop

Local copy of
GRP_WatchCtl[n]
Setup Registers for
Watch Display

10

COP
Source SwZone.Event[POPi].LineName
Dest sbrGRP_WatchCtl.Watch.Popup[0].Text
Length 1

MOVE
Source 1
Dest PRPi 4

MOVE
Source sbrGRP_WatchCtl.Watch.Popup[0].Text.LEN 9
Dest PRPmi 3

Popup 1
Convert Line Name to Normal
(i.e. FSMP to Fsmp)

Local copy of
GRP_WatchCtl[n]
Setup Registers for
Watch Display

Local copy of
GRP_WatchCtl[n]
Setup Registers for
Watch Display

11

lbl_Popup1
LBL

LE
Source A PRPi 4
Source B PRPmi 3

LIMIT
Low Limit 65
Test sbrGRP_WatchCtl.Watch.Popup[0].Text.DATA[PRPi] 'S'
High Limit 90

ADD
Source A sbrGRP_WatchCtl.Watch.Popup[0].Text.DATA[PRPi] 'S'
Source B 32
Dest sbrGRP_WatchCtl.Watch.Popup[0].Text.DATA[PRPi] 'S'

ADD
Source A PRPi 4
Source B 1
Dest PRPi 4

LE
Source A PRPi 4
Source B PRPmi 3

lbl_Popup1
JMP

Convert Andon Display Group data for Smart Watch

12

Popup 1
Load Event to Watch Color & Alarm Text

Dec17, 2018 - replaced LineName with Normalized LineName (FSMP is Fsmp) and change '-' to '.' to avoid line-wrap

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

CONCAT
Source A sbrGRP_WatchCtl.Watch.Popup[0].Text 'Off.StatW'
Source B strPeriod '.'
Dest sbrGRP_WatchCtl.Watch.Popup[0].Text 'Off.StatW'

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

CONCAT
Source A sbrGRP_WatchCtl.Watch.Popup[0].Text 'Off.StatW'
Source B SwZone.Event[POPi].EquipName ??
Dest sbrGRP_WatchCtl.Watch.Popup[0].Text 'Off.StatW'

Local Copy of SW Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

EQ
Source A SwZone.Event[POPi].Status 0
Source B 0

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source conGray 16#0080_8080
Dest sbrGRP_WatchCtl.Watch.Popup[0].Color 16#00ff_0000

Local Copy of SW Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

EQ
Source A SwZone.Event[POPi].Status 0
Source B 1

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source conWhite 16#00ff_ffff
Dest sbrGRP_WatchCtl.Watch.Popup[0].Color 16#00ff_0000

Local Copy of SW Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

EQ
Source A SwZone.Event[POPi].Status 0
Source B 2

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source conYellow 16#00ff_ff00
Dest sbrGRP_WatchCtl.Watch.Popup[0].Color 16#00ff_0000

Local Copy of SW Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

EQ

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE

Convert Andon Display Group data for Smart Watch

Source A	SwZone.Event[POPi].Status 0
Source B	3

Source	conRed 16#00ff_0000
Dest	sbrGRP_WatchCtl.Watch.Popup[0].Color 16#00ff_0000

Local Copy of SW Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

EQ

Source A	SwZone.Event[POPi].Status 0
Source B	4

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE

Source	conOrange 16#00ff_8000
Dest	sbrGRP_WatchCtl.Watch.Popup[0].Color 16#00ff_0000

Exit if no Line-Equip to display in Popup 2

13

ADD

Source A	sbrGRP_WatchCtl.POPi 0
Source B	1
Dest	POPi 0

GE

Source A	POPi 0
Source B	SwZone.EVi 0

lbl_Exit
JMP

Popup 2
Load Line Name
and initialize Normalization loop

14

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

COP

Source	SwZone.Event[POPi].LineName
Dest	sbrGRP_WatchCtl.Watch.Popup[1].Text
Length	1

MOVE

Source	1
Dest	PRPi 4

MOVE

Source	sbrGRP_WatchCtl.Watch.Popup[1].Text.LEN 1
Dest	PRPmi 3

Convert Andon Display Group data for Smart Watch

15

lbl_Popup2
LBL

LE
Source A PRPi 4
Source B PRPmi 3

Popup 2
Convert Line Name to Normal
(i.e. FSMP to Fsmp)
Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

LIMIT
Low Limit 65
Test sbrGRP_WatchCtl.Watch.Popup[1].Text.DATA[PRPi] 'S'
High Limit 90

ADD
Source A PRPi 4
Source B 1
Dest PRPi 4

LE
Source A PRPi 4
Source B PRPmi 3

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

ADD
Source A sbrGRP_WatchCtl.Watch.Popup[1].Text.DATA[PRPi] 'S'
Source B 32
Dest sbrGRP_WatchCtl.Watch.Popup[1].Text.DATA[PRPi] 'S'

lbl_Popup2
JMP

Convert Andon Display Group data for Smart Watch

Popup 2
Load Event to Watch Color & Alarm Text

Dec17, 2018 - replaced LineName with Normalized LineName (FSMP is Fsmp) and change '-' to '.' to avoid line-wrap

16

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

CONCAT
Source A sbrGRP_WatchCtl.Watch.Popup[1].Text ''
Source B strPeriod '.'
Dest sbrGRP_WatchCtl.Watch.Popup[1].Text ''

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

CONCAT
Source A sbrGRP_WatchCtl.Watch.Popup[1].Text ''
Source B SwZone.Event[POPi].EquipName ??
Dest sbrGRP_WatchCtl.Watch.Popup[1].Text ''

Local Copy of SW Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

EQ
Source A SwZone.Event[POPi].Status 0
Source B 0

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source conGray 16#0080_8080
Dest sbrGRP_WatchCtl.Watch.Popup[1].Color 16#0000_0000

Local Copy of SW Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

EQ
Source A SwZone.Event[POPi].Status 0
Source B 1

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source conWhite 16#00ff_ffff
Dest sbrGRP_WatchCtl.Watch.Popup[1].Color 16#0000_0000

Local Copy of SW Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

EQ
Source A SwZone.Event[POPi].Status 0
Source B 2

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE
Source conYellow 16#00ff_ff00
Dest sbrGRP_WatchCtl.Watch.Popup[1].Color 16#0000_0000

Local Copy of SW Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

EQ

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

MOVE

Convert Andon Display Group data for Smart Watch

Local Copy of SW Zone Array Tag for ZNi Equipment Status (0-Gray, 1-White, 2-Yellow, 3-Red, 4-Orange, 5-Cyan)

EQ
Source A SwZone.Event[POPi].Status
0
Source B 3

MOVE
Source conRed
16#00ff_0000
Dest sbrGRP_WatchCtl.Watch.Popup[1].Color
16#0000_0000

Local copy of GRP_WatchCtl[n] Setup Registers for Watch Display

EQ
Source A SwZone.Event[POPi].Status
0
Source B 4

MOVE
Source conOrange
16#00ff_8000
Dest sbrGRP_WatchCtl.Watch.Popup[1].Color
16#0000_0000

Copy Local Tags to Arrays

Andon Display WatchOS Control

17

lbl_Exit
LBL

COP
Source sbrGRP_WatchCtl
Dest GRP_WatchCtl[sbrGRPi]
Length 1

(End)

Convert Andon Display Group Smart Watch to Mobility String

0

Update Group Watch Strings
Copy Array tags to Local Working tags

Local copy of GRP_WatchCtl[n].Watch

COP	
Source	GRP_WatchCtl[sbrGRPi].Watch
Dest	sbrGRP_Watch
Length	1

Local copy of GRP_WatchCtl[n].WatchPrev

COP	
Source	GRP_WatchCtl[sbrGRPi].WatchPrev
Dest	sbrGRP_WatchPrev
Length	1

Andon Display WatchOS Control

COP	
Source	GRP_WatchStringCtl[sbrGRPi]
Dest	sbrGRP_WatchStringCtl
Length	1

1

Update Group Static String Timer
(forces periodic update of String when data is not changing)
Select Presets that prevent synchronization of the Channel updates during production breaks

Andon Display WatchOS Control Watch Static String Update Timer

CPT	
Dest	sbrGRP_WatchStringCtl.UpdateStatTMR.PRE 24000
Expression	21000 + (sbrGRPi * 1000)

Andon Display WatchOS Control Watch Static String Update Timer

TON		
Timer	sbrGRP_WatchStringCtl.UpdateStatTMR	EN
Preset	24000	DN
Accum	22521	

2

If Watch Group Static Data hasn't changed, don't update Watch Group String
(text manipulation is SLOW)

Andon Display WatchOS Control Watch Static String Update Timer
sbrGRP_WatchStringCtl.UpdateStatTMR.DN

Local copy of GRP_WatchCtl[n].Watch

EQ	
Source A	sbrGRP_Watch.Desc 'Gr17-Gr17'
Source B	sbrGRP_WatchPrev.Desc 'Gr9-Gr09'

lbl_SkipStaticUpdate
JMP

Convert Andon Display Group Smart Watch to Mobility String

3

Increment Heartbeat to force Watch Data Update
Reset Group Static String Timer

Andon Display WatchOS Control Watch Static String Heartbeat

ADD
Source A: sbrGRP_WatchStringCtl.UpdateStatHB 1331
Source B: 1
Dest: sbrGRP_WatchStringCtl.UpdateStatHB 1331

Andon Display WatchOS Control Watch Static String Heartbeat

LT
Source A: sbrGRP_WatchStringCtl.UpdateStatHB 1331
Source B: 1

Andon Display WatchOS Control Watch Static String Heartbeat

GT
Source A: sbrGRP_WatchStringCtl.UpdateStatHB 1331
Source B: 9999

Andon Display WatchOS Control Watch Static String Heartbeat

MOVE
Source: 1
Dest: sbrGRP_WatchStringCtl.UpdateStatHB 1331

Andon Display WatchOS Control Watch Static String Update Timer
sbrGRP_WatchStringCtl.UpdateStatTMR
RES

4

Update Watch Group Static String
Update Prev values

Convert Andon Display Group Smart Watch to Mobility String

JSR
Routine Name: _b_GRP_WatchStringStat

AFI

Andon Display WatchOS Control Setup Registers for Watch Display - Previous Values

COP
Source: sbrGRP_Watch
Dest: GRP_WatchCtl[sbrGRPi].WatchPrev
Length: 1

Convert Andon Display Group Smart Watch to Mobility String

Copy Watch Group String Data to MDI Channels

5

Local copy of GRPi

EQ	
Source A	sbrGRPi 15
Source B	0

MOVE	
Source	9
Dest	TXi 37

Smartwatch Interface Transmit Channels

COP	
Source	sbrGRP_WatchStringStat
Dest	MDI.Tx[TXi]
Length	1

Local copy of GRPi

LIMIT	
Low Limit	1
Test	sbrGRPi 15
High Limit	4

CPT	
Dest	TXi 37
Expression	sbrGRPi *2 - 1

Local copy of GRPi

LIMIT	
Low Limit	5
Test	sbrGRPi 15
High Limit	20

CPT	
Dest	TXi 37
Expression	sbrGRPi *2 + 6

6

lbl_SkipStaticUpdate

LBL

NOP

Update Group Dynamic String Timer
(forces periodic update of String when data is not changing)
Select Presets that prevent synchronization of the Channel updates during production breaks

7

Andon Display WatchOS Control Watch Dynamic String Update Timer

CPT	
Dest	sbrGRP_WatchStringCtl.UpdateDynTMR.PRE 5300
Expression	5000 + (sbrGRPi * 100)

Andon Display WatchOS Control Watch Dynamic String Update Timer

TON	
Timer	sbrGRP_WatchStringCtl.UpdateDynTMR
Preset	5300
Accum	0

EN
DN

Convert Andon Display Group Smart Watch to Mobility String

If Watch Group Data hasn't changed, don't update Watch Group String
(text manipulation is SLOW)

8

Andon Display WatchOS Control Watch Dynamic String Update Timer
sbrGRP_WatchStringCtl.UpdateDynTMR.DN

Local copy of GRP_WatchCtl[n].Watch
EQ
Source A sbrGRP_Watch.Alarm 0
Source B sbrGRP_WatchPrev.Alarm 0

Local copy of GRP_WatchCtl[n].Watch
EQ
Source A sbrGRP_Watch.Popup[0].Color 16#0000_0000
Source B sbrGRP_WatchPrev.Popup[0].Color 16#0000_0000

Local copy of GRP_WatchCtl[n].Watch
EQ
Source A sbrGRP_Watch.Popup[1].Color 16#0000_0000
Source B sbrGRP_WatchPrev.Popup[1].Color 16#0000_0000

Local copy of GRP_WatchCtl[n].Watch
EQ
Source A sbrGRP_Watch.Popup[0].Text ''
Source B sbrGRP_WatchPrev.Popup[0].Text ''

Local copy of GRP_WatchCtl[n].Watch
EQ
Source A sbrGRP_Watch.Popup[1].Text ''
Source B sbrGRP_WatchPrev.Popup[1].Text ''

lblExit
JMP

Reset Group Dynamic String Timer

9

Andon Display WatchOS Control Watch Dynamic String Heartbeat
ADD
Source A sbrGRP_WatchStringCtl.UpdateDynHB 8765
Source B 1
Dest sbrGRP_WatchStringCtl.UpdateDynHB 8765

Andon Display WatchOS Control Watch Dynamic String Heartbeat
LT
Source A sbrGRP_WatchStringCtl.UpdateDynHB 8765
Source B 1

Andon Display WatchOS Control Watch Dynamic String Heartbeat
GT
Source A sbrGRP_WatchStringCtl.UpdateDynHB 8765
Source B 9999

Andon Display WatchOS Control Watch Dynamic String Heartbeat
MOVE
Source 1
Dest sbrGRP_WatchStringCtl.UpdateDynHB 8765

Andon Display WatchOS Control Watch Dynamic String Update Timer
sbrGRP_WatchStringCtl.UpdateDynTMR
RES

Convert Andon Display Group Smart Watch to Mobility String

Update Watch Group Dynamic String
Update Prev values

Convert Andon Display Group Smart Watch to Mobility String

10

JSR	
Routine Name	_b_GRP_WatchStringDyn

Copy Watch Group String Data to MDI Channels

11

Local copy of GRPi

EQ	
Source A	sbrGRPi 15
Source B	0

MOVE	
Source	10
Dest	TXi 37

Smartwatch Interface Transmit Channels

COP	
Source	sbrGRP_WatchStringDyn
Dest	MDI.Tx[TXi]
Length	1

Local copy of GRPi

LIMIT	
Low Limit	1
Test	sbrGRPi 15
High Limit	4

CPT	
Dest	TXi 37
Expression	sbrGRPi *2

Local copy of GRPi

LIMIT	
Low Limit	5
Test	sbrGRPi 15
High Limit	20

CPT	
Dest	TXi 37
Expression	sbrGRPi *2 + 7

Convert Andon Display Group Smart Watch to Mobility String

Exit from Routine

Copy Local Working tags to Array tags

12

lblExit
LBL

Andon Display WatchOS Control Setup Registers for Watch Display - Previous Values

COP
Source sbrGRP_Watch
Dest GRP_WatchCtl[sbrGRPi].WatchPrev
Length 1

Andon Display WatchOS Control

COP
Source sbrGRP_WatchStringCtl
Dest GRP_WatchStringCtl[sbrGRPi]
Length 1

13

xxxTest_KpiDisplayEN
SAW_AndonKpiMsg.EN

TON
Timer tINS_Andon_Read
Preset 5000
Accum 2013
EN
DN

tINS_Andon_Read.DN

MSG
Message Control SAW_AndonKpiMsg ...
EN
DN
ER

14

xxxTest_ScrollingMsgEN
SAW_msgINS_Andon_Read2.EN

TON
Timer tINS_Andon_Read2
Preset 2000
Accum 797
EN
DN

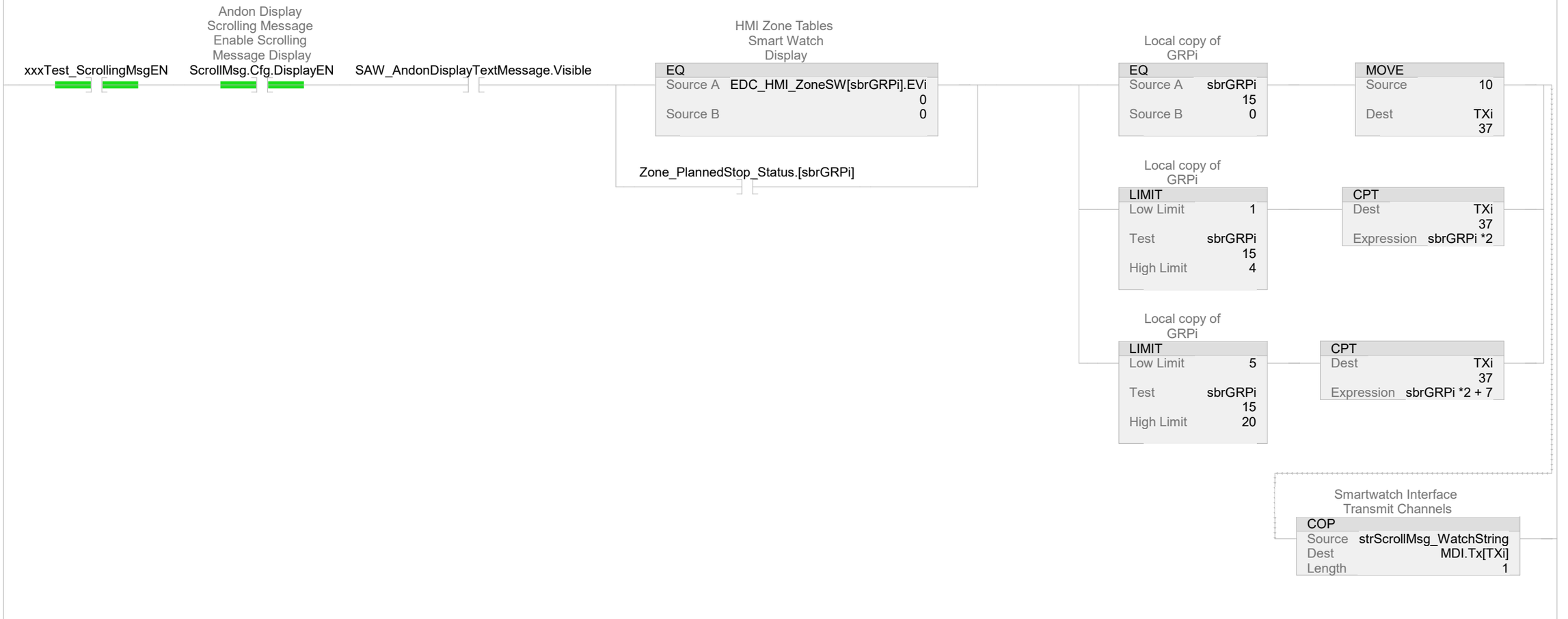
tINS_Andon_Read2.DN

MSG
Message Control SAW_msgINS_Andon_Read2 ...
EN
DN
ER

Convert Andon Display Group Smart Watch to Mobility String

If Scrolling Message Display Enabled,
Overwrite Watch Group String Data with Scrolling Message Popups when no Events to Display

15



Convert Andon Display Group Smart Watch to Mobility String

If KPI Display Enabled,
Overwrite Watch Group String Data with KPI Popups when no Events to Display

16

xxxTest_KpiDisplayEN

Andon Display
KPI Enable KPI
Display
when no Events
KPI.Cfg.DisplayEN

Andon Display
Scrolling Message
Enable Scrolling
Message Display
ScrollMsg.Cfg.DisplayEN

SAW_AndonDisplayTextMessage.Visible

HMI Zone Tables
Smart Watch
Display

EQ
Source A EDC_HMI_ZoneSW[sbrGRPi].EVi 0
Source B 0

Zone_PlannedStop_Status.[sbrGRPi]

Local copy of
GRPi

EQ
Source A sbrGRPi 15
Source B 0

MOVE
Source 10
Dest TXi 37

Local copy of
GRPi

LIMIT
Low Limit 1
Test sbrGRPi 15
High Limit 4

CPT
Dest TXi 37
Expression sbrGRPi *2

Local copy of
GRPi

LIMIT
Low Limit 5
Test sbrGRPi 15
High Limit 20

CPT
Dest TXi 37
Expression sbrGRPi *2 + 7

Smartwatch Interface
Transmit Channels

COP
Source strKPI_WatchString
Dest MDI.Tx[TXi]
Length 1

(End)

Convert Andon Display Group Smart Watch to Mobility String

Convert AndonDisplayWatch scalar data to Mobile Data Interface (MDI) string
Example JSON object data

"alarm": 2
, "topColor": 16711680
, "topText": "9F EQ"
, "btmColor": 16776960
, "btmText": "12F EQ"
, "HB": 59

0

NOP

Alarm Count

1

Local Copy of Group Watch String

COP
Source strGroupDynAlarm
Dest sbrGRP_WatchStringDyn
Length 1

Local copy of GRP_WatchCtl[n].Watch

DTOS
Source sbrGRP_Watch.Alarm
0
Dest TempString
'5393'

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringDyn
'"alarm": 0, "topColo'
Source B TempString
'5393'
Dest sbrGRP_WatchStringDyn
'"alarm": 0, "topColo'

Top Half Color
Popup 0 Color Code

2

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringDyn
'"alarm": 0, "topColo'
Source B strGroupDynTopColor
', "topColor": '
Dest sbrGRP_WatchStringDyn
'"alarm": 0, "topColo'

Local copy of GRP_WatchCtl[n].Watch

DTOS
Source sbrGRP_Watch.Popup[0].Color
16#0000_0000
Dest TempString
'5393'

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringDyn
'"alarm": 0, "topColo'
Source B TempString
'5393'
Dest sbrGRP_WatchStringDyn
'"alarm": 0, "topColo'

Convert Andon Display Group Smart Watch to Mobility String

3

Top Half Text
Popup 0 Text

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	strGroupDynTopText ', "topText": '
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	strDblQuote '"'
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	sbrGRP_Watch.Popup[0].Text ' '
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	strDblQuote '"'
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

4

Bottom Half Color
Popup 1 Color Code

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	strGroupDynBtmColor ', "btmColor": '
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

Local copy of GRP_WatchCtl[n] .Watch

DTOS	
Source	sbrGRP_Watch.Popup[1].Color 16#0000_0000
Dest	TempString '5393'

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	TempString '5393'
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

5

Bottom Half Text
Popup 1 Text

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	strGroupDynBtmText ', "btmText": '
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	strDblQuote '"'
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	sbrGRP_Watch.Popup[1].Text ' '
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	strDblQuote '"'
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

Convert Andon Display Group Smart Watch to Mobility String

6

Heartbeat

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	strHeartbeat ', "HB": '
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

Andon Display WatchOS Control Watch Dynamic String Heartbeat

DTOS	
Source	sbrGRP_WatchStringCtl.UpdateDynHB 8765
Dest	TempString '5393'

Local Copy of Group Watch String

CONCAT	
Source A	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'
Source B	TempString '5393'
Dest	sbrGRP_WatchStringDyn '"alarm": 0, "topColo'

Exit from Routine

7

lblExit
LBL

NOP

(End)

Convert Andon Display Group Smart Watch to Mobility String

Convert AndonDisplayWatch scalar data to Mobile Data Interface (MDI) string
Example JSON object data

"group": "1"
, "desc": "F1 Group 1"
, "HB": 59

0

NOP

Group Number

1

Local Copy of Group Watch String

COP
Source strGroupStatNumber
Dest sbrGRP_WatchStringStat
Length 1

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringStat '"group": "2", "desc"'
Source B strDblQuote '"'
Dest sbrGRP_WatchStringStat '"group": "2", "desc"'

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringStat '"group": "2", "desc"'
Source B sbrGRP_WatchCtl.Group '8'
Dest sbrGRP_WatchStringStat '"group": "2", "desc"'

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringStat '"group": "2", "desc"'
Source B strDblQuote '"'
Dest sbrGRP_WatchStringStat '"group": "2", "desc"'

Description
Group Name

2

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringStat '"group": "2", "desc"'
Source B strGroupStatDesc ', "desc": '
Dest sbrGRP_WatchStringStat '"group": "2", "desc"'

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringStat '"group": "2", "desc"'
Source B strDblQuote '"'
Dest sbrGRP_WatchStringStat '"group": "2", "desc"'

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringStat '"group": "2", "desc"'
Source B sbrGRP_Watch.Desc 'Gr17-Gr17'
Dest sbrGRP_WatchStringStat '"group": "2", "desc"'

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringStat '"group": "2", "desc"'
Source B strDblQuote '"'
Dest sbrGRP_WatchStringStat '"group": "2", "desc"'

Heartbeat

3

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringStat '"group": "2", "desc"'
Source B strHeartbeat ', "HB": '
Dest sbrGRP_WatchStringStat '"group": "2", "desc"'

Andon Display WatchOS Control Watch Static String Heartbeat

DTOS
Source sbrGRP_WatchStringCtl.UpdateStatHB 1331
Dest TempString '5393'

Local Copy of Group Watch String

CONCAT
Source A sbrGRP_WatchStringStat '"group": "2", "desc"'
Source B TempString '5393'
Dest sbrGRP_WatchStringStat '"group": "2", "desc"'

Convert Andon Display Group Smart Watch to Mobility String

Exit from Routine

4

lblExit
LBL

NOP

(End)

Line KPI data

Line KPI Data
and Foreground/Background colors

0 NOP

Line KPI data

Line OA%
set fore/back colors to match Line Status Border

1

Line Status and KPI Data Line OA%

MOVE
Source EDC.Line[18].Data.LdtOA.Avail 98.64098
Dest Line.OA 98.65577

AFI

Line Status and KPI Data Line OA%

GT
Source A Line.OA 98.65577
Source B 98

Line Status and KPI Data Line OA Foreground Color

MOVE
Source conWhite 16#00ff_ffff
Dest Line.OAfore 16#0000_0000

Line Status and KPI Data Line OA Background Color

MOVE
Source conGreen 16#0000_ff00
Dest Line.OAback 16#0000_ff00

Line Status and KPI Data Line OA%

LIMIT
Low Limit 96
Test Line.OA 98.65577
High Limit 98

Line Status and KPI Data Line OA Foreground Color

MOVE
Source conBlack 16#0000_0000
Dest Line.OAfore 16#0000_0000

Line Status and KPI Data Line OA Background Color

MOVE
Source conYellow 16#00ff_ff00
Dest Line.OAback 16#0000_ff00

Line Status and KPI Data Line OA%

LT
Source A Line.OA 98.65577
Source B 96

Line Status and KPI Data Line OA Foreground Color

MOVE
Source conWhite 16#00ff_ffff
Dest Line.OAfore 16#0000_0000

Line Status and KPI Data Line OA Background Color

MOVE
Source conRed 16#00ff_0000
Dest Line.OAback 16#0000_ff00

AFI

Line Status and KPI Data Line OA Foreground Color

MOVE
Source Line.StatusFore 16#00ff_ffff
Dest Line.OAfore 16#0000_0000

Line Status and KPI Data Line OA Background Color

MOVE
Source Line.StatusBack 16#00ff_0000
Dest Line.OAback 16#0000_ff00

Zone_PlannedStop_Status.[0]

Line Status and KPI Data Line OA Foreground Color

MOVE
Source conBlack 16#0000_0000
Dest Line.OAfore 16#0000_0000

Line Status and KPI Data Line OA Background Color

MOVE
Source conGreen 16#0000_ff00
Dest Line.OAback 16#0000_ff00

Zone_PlannedStop_Status.[0]

Line Status and KPI Data Line OA Foreground Color

MOVE
Source conBlack 16#0000_0000
Dest Line.OAfore 16#0000_0000

Line Status and KPI Data Line OA Background Color

MOVE
Source conWhite 16#00ff_ffff
Dest Line.OAback 16#0000_ff00

Line KPI data

Line KPI data

Line OR%
set fore/back colors to match Line Status Border

2

Line Status and KPI Data Line OR%

MOVE	
Source	EDC.Line[18].Data.LdtOR.Avail 87.86823
Dest	Line.OR 87.99378

AFI

Line Status and KPI Data Line OR%

GT	
Source A	Line.OR 87.99378
Source B	98

Line Status and KPI Data Line OR Foreground Color

MOVE	
Source	conWhite 16#00ff_ffff
Dest	Line.ORfore 16#0000_0000

Line Status and KPI Data Line OR Background Color

MOVE	
Source	conGreen 16#0000_ff00
Dest	Line.ORback 16#0000_ff00

Line Status and KPI Data Line OR%

LIMIT	
Low Limit	96
Test	Line.OR 87.99378
High Limit	98

Line Status and KPI Data Line OR Foreground Color

MOVE	
Source	conBlack 16#0000_0000
Dest	Line.ORfore 16#0000_0000

Line Status and KPI Data Line OR Background Color

MOVE	
Source	conYellow 16#00ff_ff00
Dest	Line.ORback 16#0000_ff00

Line Status and KPI Data Line OR%

LT	
Source A	Line.OR 87.99378
Source B	96

Line Status and KPI Data Line OR Foreground Color

MOVE	
Source	conWhite 16#00ff_ffff
Dest	Line.ORfore 16#0000_0000

Line Status and KPI Data Line OR Background Color

MOVE	
Source	conRed 16#00ff_0000
Dest	Line.ORback 16#0000_ff00

AFI

Line Status and KPI Data Line OR Foreground Color

MOVE	
Source	Line.StatusFore 16#00ff_ffff
Dest	Line.ORfore 16#0000_0000

Line Status and KPI Data Line OR Background Color

MOVE	
Source	Line.StatusBack 16#00ff_0000
Dest	Line.ORback 16#0000_ff00

Zone_PlannedStop_Status.[0] /

Line Status and KPI Data Line OR Foreground Color

MOVE	
Source	conBlack 16#0000_0000
Dest	Line.ORfore 16#0000_0000

Line Status and KPI Data Line OR Background Color

MOVE	
Source	conGreen 16#0000_ff00
Dest	Line.ORback 16#0000_ff00

Zone_PlannedStop_Status.[0]

Line Status and KPI Data Line OR Foreground Color

MOVE	
Source	conBlack 16#0000_0000
Dest	Line.ORfore 16#0000_0000

Line Status and KPI Data Line OR Background Color

MOVE	
Source	conWhite 16#00ff_ffff
Dest	Line.ORback 16#0000_ff00

Line KPI data

3

Line Off Target
set fore/back colors to match Line Status Border

Line Status and KPI Data Line Off Target
MOVE
Source 999
Dest Line.OffTarg 999

AFI

Line Status and KPI Data Line Off Target Foreground Color
MOVE
Source conWhite 16#00ff_ffff
Dest Line.OffTargFore 16#00ff_ffff

Line Status and KPI Data Line Off Target Background Color
MOVE
Source conGreen 16#0000_ff00
Dest Line.OffTargBack 16#00ff_0000

Line Status and KPI Data Line Off Target Foreground Color
MOVE
Source Line.StatusFore 16#00ff_ffff
Dest Line.OffTargFore 16#00ff_ffff

Line Status and KPI Data Line Off Target Background Color
MOVE
Source Line.StatusBack 16#00ff_0000
Dest Line.OffTargBack 16#00ff_0000

Line KPI data

Line Off
set fore/back colors to match Line Status Border

4

Line Status and KPI Data Line Off
MOVE
Source EDC_CycleCount[18,0] 405
Dest Line.Off 406

AFI

Line Status and KPI Data Line Off Foreground Color
MOVE
Source conWhite 16#00ff_ffff
Dest Line.OffFore 16#0000_0000

Line Status and KPI Data Line Off Background Color
MOVE
Source conGreen 16#0000_ff00
Dest Line.OffBack 16#0000_ff00

AFI

Line Status and KPI Data Line Off Foreground Color
MOVE
Source Line.StatusFore 16#00ff_ffff
Dest Line.OffFore 16#0000_0000

Line Status and KPI Data Line Off Background Color
MOVE
Source Line.StatusBack 16#00ff_0000
Dest Line.OffBack 16#0000_ff00

Zone_PlannedStop_Status.[0] /

Line Status and KPI Data Line Off Foreground Color
MOVE
Source conBlack 16#0000_0000
Dest Line.OffFore 16#0000_0000

Line Status and KPI Data Line Off Background Color
MOVE
Source conGreen 16#0000_ff00
Dest Line.OffBack 16#0000_ff00

Zone_PlannedStop_Status.[0]

Line Status and KPI Data Line Off Foreground Color
MOVE
Source conBlack 16#0000_0000
Dest Line.OffFore 16#0000_0000

Line Status and KPI Data Line Off Background Color
MOVE
Source conWhite 16#00ff_ffff
Dest Line.OffBack 16#0000_ff00

(End)

Line Status

Line Status
and Foreground/Background colors

0

NOP

Interlock Off
Black on Orange

1

HMI Zone Tables
Smart Watch
Display Event Table
Equip Status

EQ
Source A EDC_HMI_ZoneSW[0].Status 3
Source B 4

Line Status and KPI
Data Line Status

COP
Source strLineStatus_Other
Dest Line.Status
Length 1

Line Status and KPI
Data Line Status
Foreground Color

MOVE
Source conBlack 16#0000_0000
Dest Line.StatusFore 16#00ff_ffff

Line Status and KPI
Data Line Status
Background Color

MOVE
Source conOrange 16#00ff_8000
Dest Line.StatusBack 16#00ff_0000

Fault
White on Red

2

HMI Zone Tables
Smart Watch
Display Event Table
Equip Status

EQ
Source A EDC_HMI_ZoneSW[0].Status 3
Source B 3

Line Status and KPI
Data Line Status

COP
Source strLineStatus_Work
Dest Line.Status
Length 1

Line Status and KPI
Data Line Status
Foreground Color

MOVE
Source conWhite 16#00ff_ffff
Dest Line.StatusFore 16#00ff_ffff

Line Status and KPI
Data Line Status
Background Color

MOVE
Source conRed 16#00ff_0000
Dest Line.StatusBack 16#00ff_0000

Warning
Black on Yellow

3

HMI Zone Tables
Smart Watch
Display Event Table
Equip Status

EQ
Source A EDC_HMI_ZoneSW[0].Status 3
Source B 2

Line Status and KPI
Data Line Status

COP
Source strLineStatus_Short
Dest Line.Status
Length 1

Line Status and KPI
Data Line Status
Foreground Color

MOVE
Source conBlack 16#0000_0000
Dest Line.StatusFore 16#00ff_ffff

Line Status and KPI
Data Line Status
Background Color

MOVE
Source conYellow 16#00ff_ff00
Dest Line.StatusBack 16#00ff_0000

Line Status

Plan Stop
Black on White

4

HMI Zone Tables
Smart Watch
Display Event Table
Equip Status

EQ	
Source A	EDC_HMI_ZoneSW[0].Status
	3
Source B	1

Line Status and KPI
Data Line Status

COP	
Source	strLineStatus_PlanStop
Dest	Line.Status
Length	1

Line Status and KPI
Data Line Status
Foreground Color

MOVE	
Source	conBlack
	16#0000_0000
Dest	Line.StatusFore
	16#00ff_ffff

Line Status and KPI
Data Line Status
Background Color

MOVE	
Source	conWhite
	16#00ff_ffff
Dest	Line.StatusBack
	16#00ff_0000

Run
White on Green

5

HMI Zone Tables
Smart Watch
Display Event Table
Equip Status

EQ	
Source A	EDC_HMI_ZoneSW[0].Status
	3
Source B	0

Line Status and KPI
Data Line Status

COP	
Source	strLineStatus_Running
Dest	Line.Status
Length	1

Line Status and KPI
Data Line Status
Foreground Color

MOVE	
Source	conWhite
	16#00ff_ffff
Dest	Line.StatusFore
	16#00ff_ffff

Line Status and KPI
Data Line Status
Background Color

MOVE	
Source	conGreen
	16#0000_ff00
Dest	Line.StatusBack
	16#00ff_0000

(End)

Line CGP String

Convert Line scalar data to Mobile Data Interface (MDI) string
Example JSON object data

"stat": "AVI"
, "statFore": 0
, "statBack": 16744448
, "OA": 98.0
, "OAfore": 0
, "OAback": 16776960
, "OR": 100.0
, "ORfore": 16777215
, "ORback": 65280
, "LOtarg": 259
, "LOtargFore": 16777215
, "LOtargBack": 0
, "LO": 243
, "LOfore": 16777215
, "LOback": 0
, "HB": 59

0

NOP

Update Line String Timer
(forces periodic update of Line String when data is not changing)
Select Presets that prevent synchronization of the Channel updates during production breaks

1

Line Watch String Update Timer

MOVE
Source 6000
Dest LineStringUpdateTMR.PRE 6000

Line Watch String Update Timer

TON
Timer LineStringUpdateTMR
Preset 6000
Accum 4042
EN
DN

If Watch Line Data hasn't changed, don't update Watch Line String
(text manipulation is SLOW)

2

Line Watch String Update Timer
LineStringUpdateTMR.DN
/

Line Status and KPI Data Line OA%

EQ
Source A Line.OA 98.65577
Source B LinePrev.OA 98.65553

Line Status and KPI Data Line OR%

EQ
Source A Line.OR 87.99378
Source B LinePrev.OR 87.99749

Line Status and KPI Data Line Off Target

EQ
Source A Line.OffTarg 999
Source B LinePrev.OffTarg 999

Line Status and KPI Data Line Off

EQ
Source A Line.Off 406
Source B LinePrev.Off 406

lblExit
JMP

Line CGP String

Reset Line String Timer

3

Line Watch String Update Heartbeat

ADD
Source A LineStringUpdateHB 5963
Source B 1
Dest LineStringUpdateHB 5963

Line Watch String Update Heartbeat

LT
Source A LineStringUpdateHB 5963
Source B 1

Line Watch String Update Heartbeat

GT
Source A LineStringUpdateHB 5963
Source B 9999

Line Watch String Update Heartbeat

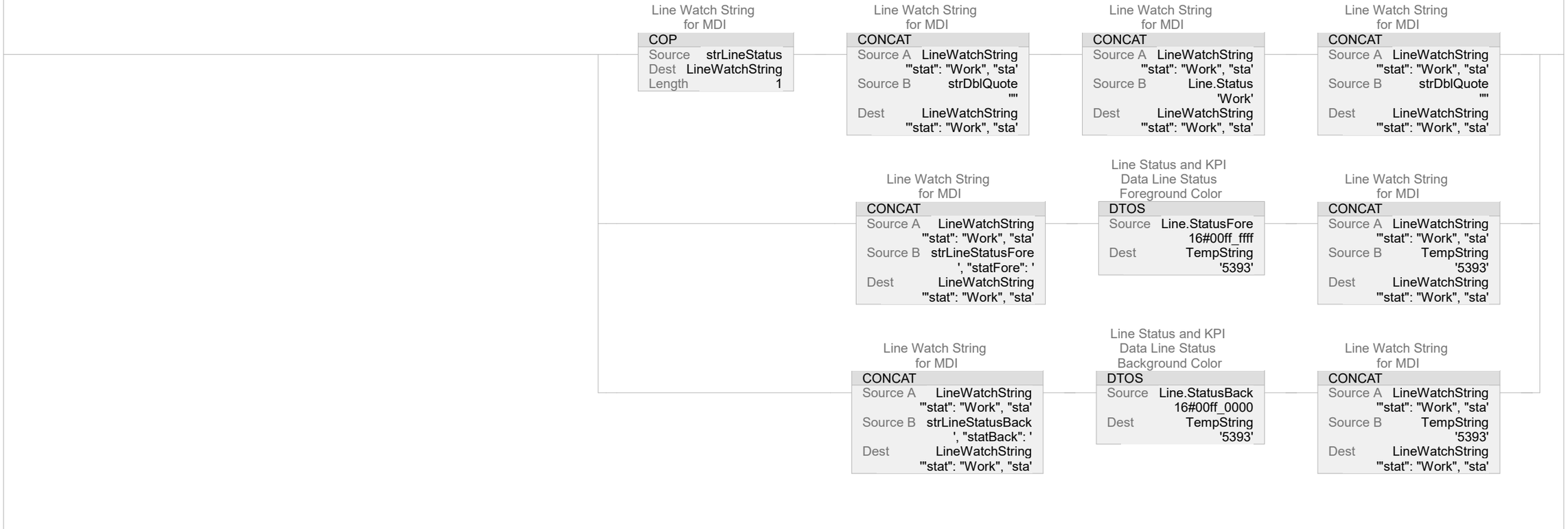
MOVE
Source 1
Dest LineStringUpdateHB 5963

Line Watch String Update Timer
LineStringUpdateTMR
RES

Line CGP String

Line Status

4



Line CGP String

Line OA%

v2.6 - REAL values change constantly - causing continuous update to CGP channel -- convert REAL to DINT values for conversion to STRING
(this also reduces the number of decimal positions generated by RTOS)

5

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B strLineOA ', "OA": '
Dest LineWatchString '"stat": "Work", "sta'

Line Status and KPI Data Line OA%

DTOS
Source Line.OA 98.65577
Dest TempString '5393'

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B TempString '5393'
Dest LineWatchString '"stat": "Work", "sta'

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B strLineOAfore ', "OAfore": '
Dest LineWatchString '"stat": "Work", "sta'

Line Status and KPI Data Line OA Foreground Color

DTOS
Source Line.OAfore 16#0000_0000
Dest TempString '5393'

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B TempString '5393'
Dest LineWatchString '"stat": "Work", "sta'

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B strLineOAback ', "OAback": '
Dest LineWatchString '"stat": "Work", "sta'

Line Status and KPI Data Line OA Background Color

DTOS
Source Line.OAback 16#0000_ff00
Dest TempString '5393'

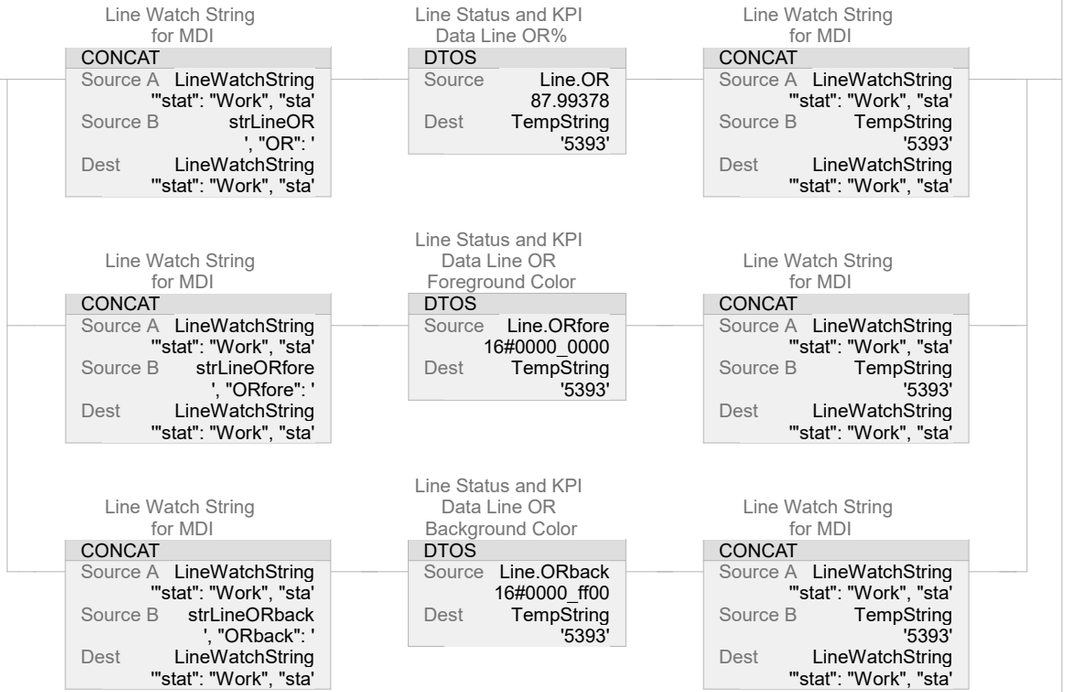
Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B TempString '5393'
Dest LineWatchString '"stat": "Work", "sta'

Line CGP String

Line OR%

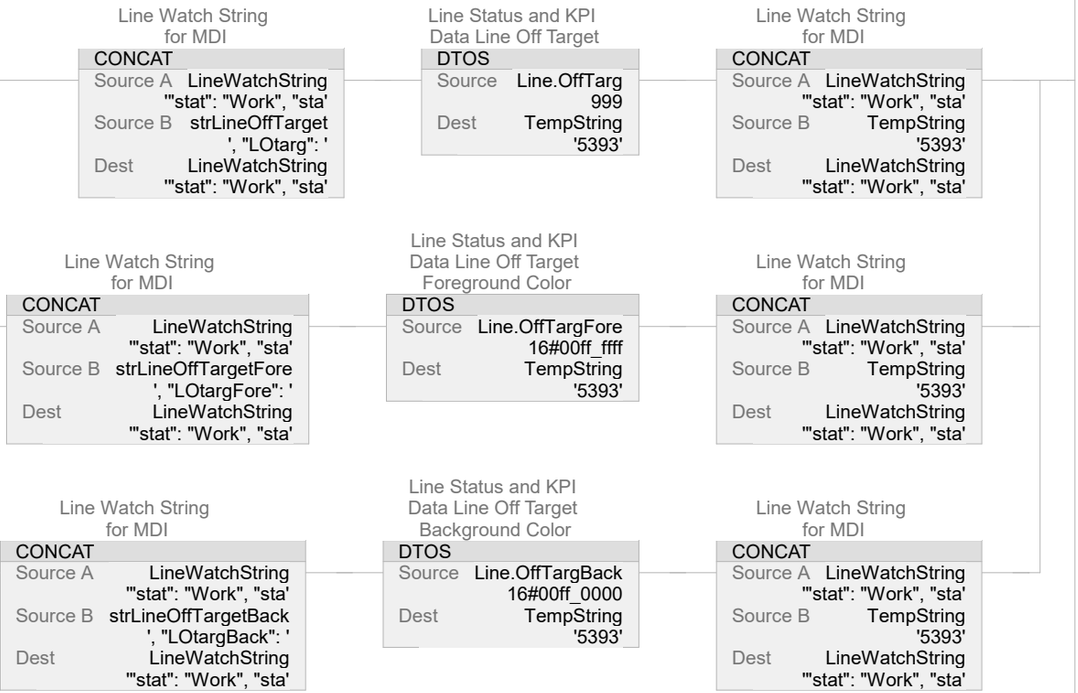
6



Line CGP String

Line Off Target

7



Line CGP String

Line Off

8

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B strLineOff ', "LO": '
Dest LineWatchString '"stat": "Work", "sta'

Line Status and KPI Data Line Off

DTOS
Source Line.Off 406
Dest TempString '5393'

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B TempString '5393'
Dest LineWatchString '"stat": "Work", "sta'

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B strLineOffFore ', "LOfore": '
Dest LineWatchString '"stat": "Work", "sta'

Line Status and KPI Data Line Off Foreground Color

DTOS
Source Line.OffFore 16#0000_0000
Dest TempString '5393'

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B TempString '5393'
Dest LineWatchString '"stat": "Work", "sta'

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B strLineOffBack ', "LOback": '
Dest LineWatchString '"stat": "Work", "sta'

Line Status and KPI Data Line Off Background Color

DTOS
Source Line.OffBack 16#0000_ff00
Dest TempString '5393'

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B TempString '5393'
Dest LineWatchString '"stat": "Work", "sta'

Heartbeat Count (1-9999)

9

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B strHeartbeat ', "HB": '
Dest LineWatchString '"stat": "Work", "sta'

Line Watch String Update Heartbeat

DTOS
Source LineStringUpdateHB 5963
Dest TempString '5393'

Line Watch String for MDI

CONCAT
Source A LineWatchString '"stat": "Work", "sta'
Source B TempString '5393'
Dest LineWatchString '"stat": "Work", "sta'

Line CGP String

Copy Line String to MDI Channel

10

Smartwatch Interface Transmit Channels

COP	
Source	LineWatchString
Dest	MDI.Tx[11]
Length	1

Store Previous Values

11

Line Status and KPI Data Previous

COP	
Source	Line
Dest	LinePrev
Length	1

Exit from Routine

12

lblExit
LBL

NOP

(End)

Shop KPI data

Shop KPI Data
and Foreground/Background colors

0

NOP

Shop DRR%

1

Shop KPI Data Shop Direct Run Rate %
MOVE
Source 0
Dest Shop.DRR 0.0

Shop KPI Data Shop Direct Run Rate %
GT
Source A Shop.DRR 0.0
Source B 98

Shop KPI Data Shop DRR Foreground Color
MOVE
Source conWhite 16#00ff_ffff
Dest Shop.DRRfore 16#00ff_ffff

Shop KPI Data Shop DRR Background Color
MOVE
Source conGreen 16#0000_ff00
Dest Shop.DRRback 16#00ff_0000

Shop KPI Data Shop Direct Run Rate %
LIMIT
Low Limit 96
Test Shop.DRR 0.0
High Limit 98

Shop KPI Data Shop DRR Foreground Color
MOVE
Source conBlack 16#0000_0000
Dest Shop.DRRfore 16#00ff_ffff

Shop KPI Data Shop DRR Background Color
MOVE
Source conYellow 16#00ff_ff00
Dest Shop.DRRback 16#00ff_0000

Shop KPI Data Shop Direct Run Rate %
LT
Source A Shop.DRR 0.0
Source B 96

Shop KPI Data Shop DRR Foreground Color
MOVE
Source conWhite 16#00ff_ffff
Dest Shop.DRRfore 16#00ff_ffff

Shop KPI Data Shop DRR Background Color
MOVE
Source conRed 16#00ff_0000
Dest Shop.DRRback 16#00ff_0000

Shop KPI data

Shop OR%

2

Shop KPI Data Shop OR%

MOVE
Source 0
Dest Shop.OR 0.0

Shop KPI Data Shop OR%

GT
Source A Shop.OR 0.0
Source B 98

Shop KPI Data Shop OR Foreground Color

MOVE
Source conWhite 16#00ff_ffff
Dest Shop.ORfore 16#00ff_ffff

Shop KPI Data Shop OR Background Color

MOVE
Source conGreen 16#0000_ff00
Dest Shop.ORback 16#00ff_0000

Shop KPI Data Shop OR%

LIMIT
Low Limit 96
Test Shop.OR 0.0
High Limit 98

Shop KPI Data Shop OR Foreground Color

MOVE
Source conBlack 16#0000_0000
Dest Shop.ORfore 16#00ff_ffff

Shop KPI Data Shop OR Background Color

MOVE
Source conYellow 16#00ff_ff00
Dest Shop.ORback 16#00ff_0000

Shop KPI Data Shop OR%

LT
Source A Shop.OR 0.0
Source B 96

Shop KPI Data Shop OR Foreground Color

MOVE
Source conWhite 16#00ff_ffff
Dest Shop.ORfore 16#00ff_ffff

Shop KPI Data Shop OR Background Color

MOVE
Source conRed 16#00ff_0000
Dest Shop.ORback 16#00ff_0000

Shop KPI data

Shop Plan
set fore/back colors to match Line Status Border

3

Shop KPI Data Shop Plan

MOVE	
Source	999
Dest	Shop.Plan 999

Shop KPI Data Shop Plan Foreground Color

MOVE	
Source	conWhite 16#00ff_ffff
Dest	Shop.PlanFore 16#0000_0000

Shop KPI Data Shop Plan Background Color

MOVE	
Source	conBlack 16#0000_0000
Dest	Shop.PlanBack 16#0000_ff00

AFI

Shop KPI Data Shop Plan Foreground Color

MOVE	
Source	Line.StatusFore 16#00ff_ffff
Dest	Shop.PlanFore 16#0000_0000

Shop KPI Data Shop Plan Background Color

MOVE	
Source	Line.StatusBack 16#00ff_0000
Dest	Shop.PlanBack 16#0000_ff00

Zone_PlannedStop_Status.[0] /

Shop KPI Data Shop Plan Foreground Color

MOVE	
Source	conBlack 16#0000_0000
Dest	Shop.PlanFore 16#0000_0000

Shop KPI Data Shop Plan Background Color

MOVE	
Source	conGreen 16#0000_ff00
Dest	Shop.PlanBack 16#0000_ff00

Zone_PlannedStop_Status.[0]

Shop KPI Data Shop Plan Foreground Color

MOVE	
Source	conBlack 16#0000_0000
Dest	Shop.PlanFore 16#0000_0000

Shop KPI Data Shop Plan Background Color

MOVE	
Source	conWhite 16#00ff_ffff
Dest	Shop.PlanBack 16#0000_ff00

Shop Line Off

4

Shop KPI Data Shop Line Off

MOVE	
Source	0
Dest	Shop.Off 0

Shop KPI Data Shohp Line Off Foreground Color

MOVE	
Source	conWhite 16#00ff_ffff
Dest	Shop.OffFore 16#00ff_ffff

Shop KPI Data Shop Line Off Background Color

MOVE	
Source	conBlack 16#0000_0000
Dest	Shop.OffBack 16#0000_0000

Shop KPI data

Shop OT

5

Shop KPI Data Shop Overtime		Shop KPI Data Shop Overtime Foreground Color		Shop KPI Data Shop Overtime Background Color	
MOVE		MOVE		MOVE	
Source	0	Source	conWhite 16#00ff_ffff	Source	conBlack 16#0000_0000
Dest	Shop.OT 0.0	Dest	Shop.OTfore 16#00ff_ffff	Dest	Shop.OTback 16#0000_0000

(End)

Shop CGP String

Convert Shop scalar data to Mobile Data Interface (MDI) string
Example JSON object data

"DRR": 79.0
, "DRRfore": 16777215
, "DRRback": 16711680
, "OR": 92.5
, "ORfore": 16777215
, "ORback": 16711680
, "plan": 351
, "planFore": 16777215
, "planBack": 0
, "LO": 249
, "LOfore": 16777215
, "LOback": 0
, "OT": 0.0
, "OTfore": 16777215
, "OTback": 0
, "HB": 59

0

NOP

Update Shop String Timer
(forces periodic update of Shop String when data is not changing)
Select Presets that prevent synchronization of the Channel updates during production breaks

1

Shop Watch String Update Timer

MOVE
Source 31000
Dest ShopStringUpdateTMR.PRE 31000

Shop Watch String Update Timer

TON
Timer ShopStringUpdateTMR
Preset 31000
Accum 23157
EN
DN

If Watch Shop Data hasn't changed, don't update Watch Shop String
(text manipulation is SLOW)

2

Shop Watch String Update Timer
ShopStringUpdateTMR.DN
/

Shop KPI Data Shop Direct Run Rate %

EQ
Source A Shop.DRR 0.0
Source B ShopPrev.DRR 0.0

Shop KPI Data Shop OR%

EQ
Source A Shop.OR 0.0
Source B ShopPrev.OR 0.0

Shop KPI Data Shop Plan

EQ
Source A Shop.Plan 999
Source B ShopPrev.Plan 999

Shop KPI Data Shop Line Off

EQ
Source A Shop.Off 0
Source B ShopPrev.Off 0

Shop KPI Data Shop Overtime

EQ
Source A Shop.OT 0.0
Source B ShopPrev.OT 0.0

lblExit
JMP

Shop CGP String

Reset Shop String Timer

3

Shop Watch String Update Heartbeat

ADD
Source A ShopStringUpdateHB 4340
Source B 1
Dest ShopStringUpdateHB 4340

Shop Watch String Update Heartbeat

LT
Source A ShopStringUpdateHB 4340
Source B 1

Shop Watch String Update Heartbeat

GT
Source A ShopStringUpdateHB 4340
Source B 9999

Shop Watch String Update Heartbeat

MOVE
Source 1
Dest ShopStringUpdateHB 4340

Shop Watch String Update Timer
ShopStringUpdateTMR
RES

Shop CGP String

Shop DRR%

4

Shop Watch String for MDI

COP
Source strShopDRR
Dest ShopWatchString
Length 1

Shop KPI Data Shop Direct Run Rate %

DTOS
Source Shop.DRR
0.0
Dest TempString
'5393'

Shop Watch String for MDI

CONCAT
Source A ShopWatchString
'"DRR": 0, "DRRfore":'
Source B TempString
'5393'
Dest ShopWatchString
'"DRR": 0, "DRRfore":'

Shop Watch String for MDI

CONCAT
Source A ShopWatchString
'"DRR": 0, "DRRfore":'
Source B strShopDRRfore
', "DRRfore": '
Dest ShopWatchString
'"DRR": 0, "DRRfore":'

Shop KPI Data Shop DRR Foreground Color

DTOS
Source Shop.DRRfore
16#00ff_ffff
Dest TempString
'5393'

Shop Watch String for MDI

CONCAT
Source A ShopWatchString
'"DRR": 0, "DRRfore":'
Source B TempString
'5393'
Dest ShopWatchString
'"DRR": 0, "DRRfore":'

Shop Watch String for MDI

CONCAT
Source A ShopWatchString
'"DRR": 0, "DRRfore":'
Source B strShopDRRback
', "DRRback": '
Dest ShopWatchString
'"DRR": 0, "DRRfore":'

Shop KPI Data Shop DRR Background Color

DTOS
Source Shop.DRRback
16#00ff_0000
Dest TempString
'5393'

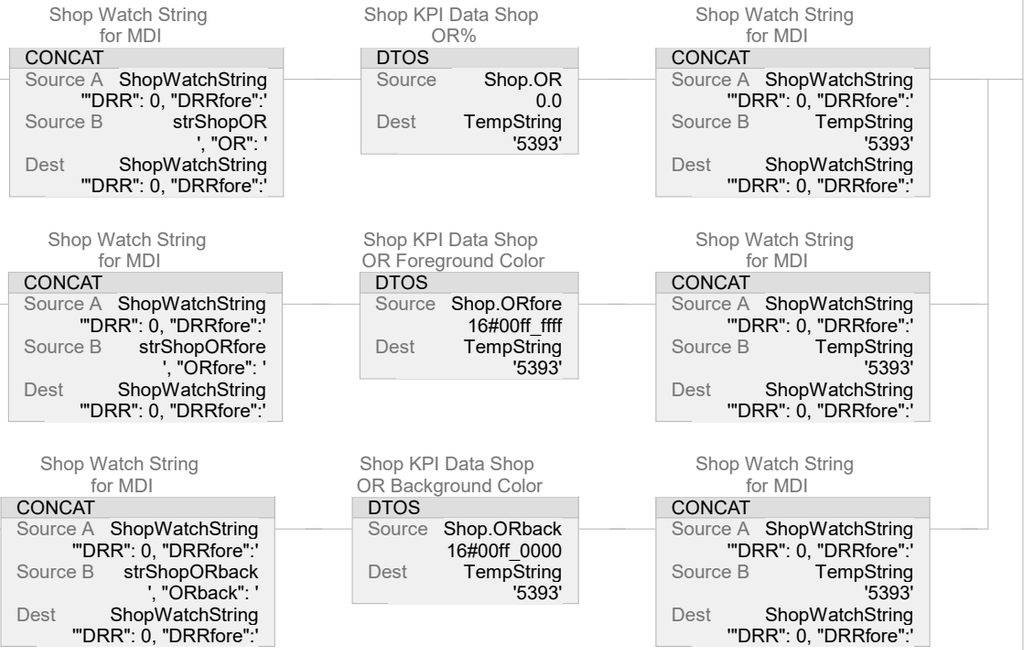
Shop Watch String for MDI

CONCAT
Source A ShopWatchString
'"DRR": 0, "DRRfore":'
Source B TempString
'5393'
Dest ShopWatchString
'"DRR": 0, "DRRfore":'

Shop CGP String

Shop OR%

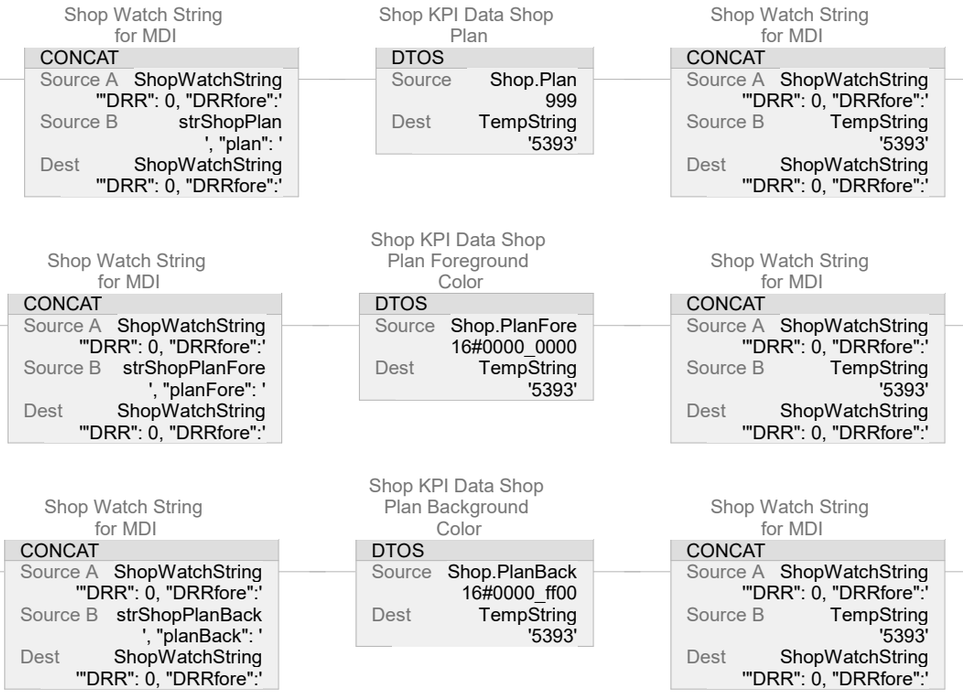
5



Shop CGP String

Shop Plan

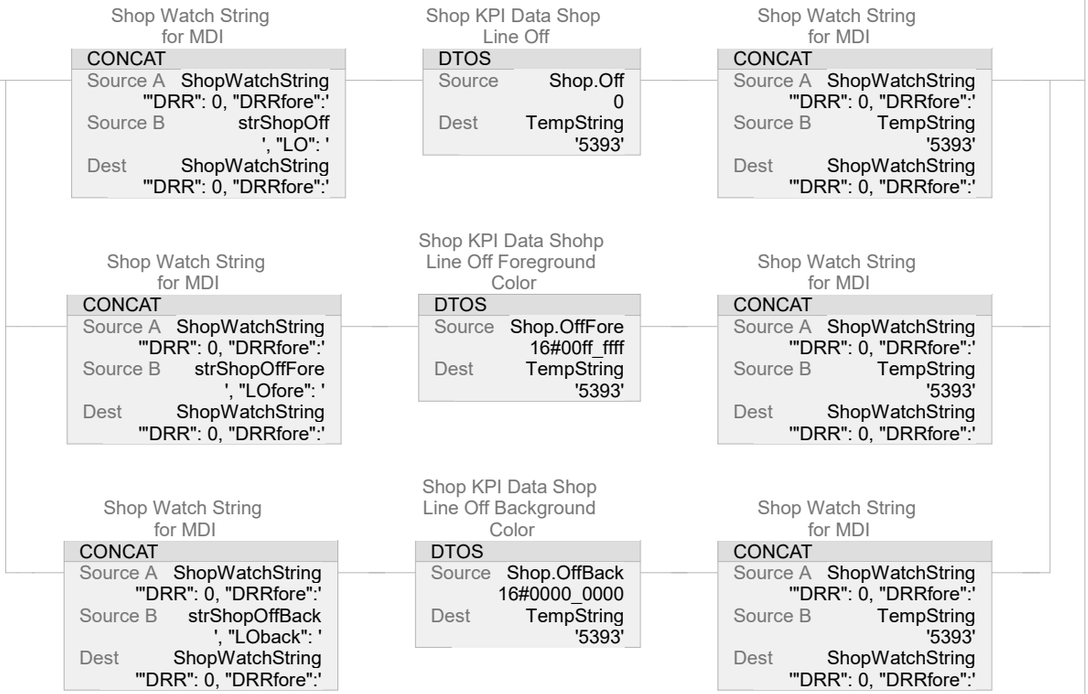
6



Shop CGP String

Shop Line Off

7



Shop CGP String

Shop OT

8

Shop Watch String for MDI
CONCAT
Source A ShopWatchString '"DRR": 0, "DRRfore":'
Source B strShopOT ', "OT": '
Dest ShopWatchString '"DRR": 0, "DRRfore":'

Shop KPI Data Shop Overtime
DTOS
Source Shop.OT 0.0
Dest TempString '5393'

Shop Watch String for MDI
CONCAT
Source A ShopWatchString '"DRR": 0, "DRRfore":'
Source B TempString '5393'
Dest ShopWatchString '"DRR": 0, "DRRfore":'

Shop Watch String for MDI
CONCAT
Source A ShopWatchString '"DRR": 0, "DRRfore":'
Source B strShopOTfore ', "OTfore": '
Dest ShopWatchString '"DRR": 0, "DRRfore":'

Shop KPI Data Shop Overtime Foreground Color
DTOS
Source Shop.OTfore 16#00ff_ffff
Dest TempString '5393'

Shop Watch String for MDI
CONCAT
Source A ShopWatchString '"DRR": 0, "DRRfore":'
Source B TempString '5393'
Dest ShopWatchString '"DRR": 0, "DRRfore":'

Shop Watch String for MDI
CONCAT
Source A ShopWatchString '"DRR": 0, "DRRfore":'
Source B strShopOTback ', "OTback": '
Dest ShopWatchString '"DRR": 0, "DRRfore":'

Shop KPI Data Shop Overtime Background Color
DTOS
Source Shop.OTback 16#0000_0000
Dest TempString '5393'

Shop Watch String for MDI
CONCAT
Source A ShopWatchString '"DRR": 0, "DRRfore":'
Source B TempString '5393'
Dest ShopWatchString '"DRR": 0, "DRRfore":'

Heartbeat Count (1-9999)

9

Shop Watch String for MDI
CONCAT
Source A ShopWatchString '"DRR": 0, "DRRfore":'
Source B strHeartbeat ', "HB": '
Dest ShopWatchString '"DRR": 0, "DRRfore":'

Shop Watch String Update Heartbeat
DTOS
Source ShopStringUpdateHB 4340
Dest TempString '5393'

Shop Watch String for MDI
CONCAT
Source A ShopWatchString '"DRR": 0, "DRRfore":'
Source B TempString '5393'
Dest ShopWatchString '"DRR": 0, "DRRfore":'

Shop CGP String

Copy Shop String to MDI Channel

10

Smartwatch Interface Transmit Channels

COP	
Source	ShopWatchString
Dest	MDI.Tx[12]
Length	1

Store Previous Values

11

Shop KPI Data Previous

COP	
Source	Shop
Dest	ShopPrev
Length	1

Exit from Routine

12

lblExit
LBL

NOP

(End)

Heartbeat CGP String
(Date/Time)

Convert AndonDisplayWatch scalar data to Mobile Data Interface (MDI) string
Example JSON object data

"hour": 15,
"min": 25,
"sec": 32

0

NOP

If Watch Line Data hasn't changed, don't update Watch Line String
(text manipulation is SLOW)

1

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

EQ
Source A CLX.SysTime[3]
<EDC.CLX.SysTime[3]> 15
Source B HeartbeatPrev[3] 15

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

EQ
Source A CLX.SysTime[4]
<EDC.CLX.SysTime[4]> 19
Source B HeartbeatPrev[4] 24

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

EQ
Source A CLX.SysTime[5]
<EDC.CLX.SysTime[5]> 21
Source B HeartbeatPrev[5] 59

lblExit
JMP

Heartbeat Hours

2

Heartbeat Watch
String for MDI

COP
Source strHeartbeatHours
Dest HeartbeatWatchString
Length 1

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[3]
<EDC.CLX.SysTime[3]> 15
Dest TempString '5393'

Heartbeat Watch
String for MDI

CONCAT
Source A HeartbeatWatchString '"hour": 15, "min": 2'
Source B TempString '5393'
Dest HeartbeatWatchString '"hour": 15, "min": 2'

Heartbeat Minutes

3

Heartbeat Watch
String for MDI

CONCAT
Source A HeartbeatWatchString '"hour": 15, "min": 2'
Source B strHeartbeatMinutes ', "min": '
Dest HeartbeatWatchString '"hour": 15, "min": 2'

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[4]
<EDC.CLX.SysTime[4]> 19
Dest TempString '5393'

Heartbeat Watch
String for MDI

CONCAT
Source A HeartbeatWatchString '"hour": 15, "min": 2'
Source B TempString '5393'
Dest HeartbeatWatchString '"hour": 15, "min": 2'

Heartbeat CGP String
(Date/Time)

Heartbeat Seconds

4

Heartbeat Watch
String for MDI

CONCAT	
Source A	HeartbeatWatchString '"hour": 15, "min": 2'
Source B	strHeartbeatSeconds ', "sec": '
Dest	HeartbeatWatchString '"hour": 15, "min": 2'

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DTOS	
Source	CLX.SysTime[5] <EDC.CLX.SysTime[5]> 21
Dest	TempString '5393'

Heartbeat Watch
String for MDI

CONCAT	
Source A	HeartbeatWatchString '"hour": 15, "min": 2'
Source B	TempString '5393'
Dest	HeartbeatWatchString '"hour": 15, "min": 2'

Copy Line String to MDI Channel

5

Smartwatch Interface
Transmit Channels

COP	
Source	HeartbeatWatchString
Dest	MDI.Tx[0]
Length	1

Store Previous Values

6

Heartbeat Data
Previous

COP	
Source	CLX.SysTime[0] <EDC.CLX.SysTime[0]>
Dest	HeartbeatPrev[0]
Length	13

Exit from Routine

7

lblExit
LBL

NOP

(End)

KPI Popup Display

KPI Display
Display KPI Popups when no Event Popups

0

NOP

KPI Rotation Timer

1

Andon Display
KPI Rotation Timer

MOVE	
Source	2500
Dest	KPI.RotateTMR.PRE 2500

Andon Display
KPI Rotation Timer

TON		
Timer	KPI.RotateTMR	EN
Preset	2500	DN
Accum	106	

Exit if KPI Rotation not needed or if KPI Display is not enabled

2

Andon Display
KPI Rotation Timer
KPI.RotateTMR.DN
/

Andon Display
KPI Enable KPI Display
when no Events
KPI.Cfg.DisplayEN
/

lbl_Exit
JMP

Reset KPI Rotation Timer

3

Andon Display
KPI Rotation Timer
KPI.RotateTMR
RES

KPI Display Items

JSR	
Routine Name	_f_KpiDisplayItems

KPI Popup Display

Increment KPI Index
for Display of 1 or 2 KPI Value

4

Andon Display
KPI Enable Display
of 2 KPI Events in
Popups
(if off, displays 1
KPI Event in Popups)
KPI.Cfg.Display2EN

Andon Display
KPI Index
ADD
Source A KPI.i 0
Source B 1
Dest KPI.i 0

Andon Display
KPI Index
LT
Source A KPI.i 0
Source B 0

Andon Display
KPI Index
MOVE
Source 0
Dest KPI.i 0

Andon Display
KPI Index
GE
Source A KPI.i 0
Source B KPI.Cfg.ItemQTY 4

Andon Display
KPI Enable Display
of 2 KPI Events in
Popups
(if off, displays 1
KPI Event in Popups)
KPI.Cfg.Display2EN

Andon Display
KPI Index
ADD
Source A KPI.i 0
Source B 2
Dest KPI.i 0

KPI Popup Display

Increment KPI 2nd Index
for Display of 2 KPI Values

5

Andon Display
KPI Enable Display
of 2 KPI Events in
Popups
(if off, displays 1
KPI Event in Popups)
KPI.Cfg.Display2EN

Andon Display
KPI 2nd Index
LT
Source A KPI.i2 1
Source B 0

Andon Display
KPI 2nd Index
GE
Source A KPI.i2 1
Source B KPI.Cfg.ItemQTY 4

Andon Display
KPI 2nd Index
ADD
Source A KPI.i 0
Source B 1
Dest KPI.i2 1

Andon Display
KPI 2nd Index
MOVE
Source 0
Dest KPI.i2 1

Andon Display
KPI 2nd Index Blank
KPI.i2Blank
L

KPI Popup Display

Load 1 KPI into 2 Popups
- 1st Popup contains Name
- 2nd Popup contains Value (REAL or DINT)

6

Andon Display
KPI Enable Display
of 2 KPI Events in
Popups
(if off, displays 1
KPI Event in Popups)
KPI.Cfg.Display2EN

Andon Display
KPI Popup 1

COP
Source KPI.Item[KPI.i].Name
Dest KPI.Popup1.Text
Length 1

Andon Display
KPI Popup 1

COP
Source KPI.Item[KPI.i].Color
Dest KPI.Popup1.Color
Length 1

Andon Display
KPI Data Value is
REAL (not Integer)
KPI.Item[KPI.i].Real

Real to String in
NN.NN format

JSR
Routine Name _y_RTOS_2Decimal
Input Par KPI.Item[KPI.i].Value
Return Par KPI.Popup2.Text

Andon Display
KPI Popup 2

COP
Source KPI.Item[KPI.i].Color
Dest KPI.Popup2.Color
Length 1

Andon Display
KPI Data Value is
REAL (not Integer)
KPI.Item[KPI.i].Real

MOVE
Source KPI.Item[KPI.i].Value
87.993706
Dest TempDINT
406

DTOS
Source TempDINT
406
Dest KPI.Popup2.Text
'OA: 98.66'

KPI Popup Display

Load 2 KPI into 2 Popups
- 1st Popup contains Name & Value (REAL or DINT)
- 2nd Popup contains Name & Value (REAL or DINT)
(Name must contain separator/space for 'Name: Value')

7

Andon Display
KPI Enable Display
of 2 KPI Events in
Popups
(if off, displays 1
KPI Event in Popups)
KPI.Cfg.Display2EN

Andon Display
KPI Popup 1

COP	
Source	KPI.Item[KPI.i].Name
Dest	KPI.Popup1.Text
Length	1

Andon Display
KPI Data Value is
REAL (not Integer)
KPI.Item[KPI.i].Real

Real to String in
NN.NN format

JSR	
Routine Name	_y_RTOS_2Decimal
Input Par	KPI.Item[KPI.i].Value
Return Par	TempString

Andon Display
KPI Popup 1

CONCAT	
Source A	KPI.Popup1.Text 'OR: 87.99'
Source B	TempString '5393'
Dest	KPI.Popup1.Text 'OR: 87.99'

Andon Display
KPI Data Value is
REAL (not Integer)
KPI.Item[KPI.i].Real

MOVE	
Source	KPI.Item[KPI.i].Value 87.993706
Dest	TempDINT 406

DTOS	
Source	TempDINT 406
Dest	TempString '5393'

Andon Display
KPI Popup 1

COP	
Source	KPI.Item[KPI.i].Color
Dest	KPI.Popup1.Color
Length	1

Andon Display
KPI 2nd Index Blank
KPI.i2Blank

Andon Display
KPI Popup 2

COP	
Source	KPI.Item[KPI.i2].Name
Dest	KPI.Popup2.Text
Length	1

Andon Display
KPI Data Value is
REAL (not Integer)
KPI.Item[KPI.i2].Real

Real to String in
NN.NN format

JSR	
Routine Name	_y_RTOS_2Decimal
Input Par	KPI.Item[KPI.i2].Value
Return Par	TempString

Andon Display
KPI Popup 2

CONCAT	
Source A	KPI.Popup2.Text 'OA: 98.66'
Source B	TempString '5393'
Dest	KPI.Popup2.Text 'OA: 98.66'

Andon Display
KPI Data Value is
REAL (not Integer)
KPI.Item[KPI.i2].Real

MOVE	
Source	KPI.Item[KPI.i2].Value 98.65576
Dest	TempDINT 406

DTOS	
Source	TempDINT 406
Dest	TempString '5393'

Andon Display
KPI Popup 2

COP	
Source	KPI.Item[KPI.i2].Color
Dest	KPI.Popup2.Color
Length	1

KPI Popup Display

Length 1

Andon Display
KPI 2nd Index Blank
KPI.i2Blank

Andon Display
KPI Popup 2

FLL	
Source	0
Dest	KPI.Popup2.Text
Length	1

Andon Display
KPI Popup 2

COP	
Source	conBlack
Dest	KPI.Popup2.Color
Length	1

Convert KPI Display to Mobility String

Convert KPI Display
to Mobility String

JSR	
Routine Name	_f_KpiDisplay_WatchString

8

Exit Routine

lbl_Exit
LBL

9

NOP

(End)

KPI Display Items

KPI Display
Load KPI Items for Popup Displays

0

NOP

Line OR%

1

Andon Display
KPI Data Value is
REAL (not Integer)
KPI.Item[0].Real

L

Andon Display
KPI Data Name

COP
Source strKPI_LineOR
Dest KPI.Item[0].Name
Length 1

Andon Display
KPI Data Value

MOVE
Source Line.OR
87.99378
Dest KPI.Item[0].Value
87.993706

Andon Display
KPI Background Color

MOVE
Source Line.ORback
16#0000_ff00
Dest KPI.Item[0].Color
65280

Line OA%

2

Andon Display
KPI Data Value is
REAL (not Integer)
KPI.Item[1].Real

L

Andon Display
KPI Data Name

COP
Source strKPI_LineOA
Dest KPI.Item[1].Name
Length 1

Andon Display
KPI Data Value

MOVE
Source Line.OA
98.65577
Dest KPI.Item[1].Value
98.65576

Andon Display
KPI Background Color

MOVE
Source Line.OAback
16#0000_ff00
Dest KPI.Item[1].Color
65280

Shop Plan

3

Andon Display
KPI Data Value is
REAL (not Integer)
KPI.Item[2].Real

U

Andon Display
KPI Data Name

COP
Source strKPI_Plan
Dest KPI.Item[2].Name
Length 1

Andon Display
KPI Data Value

MOVE
Source Shop.Plan
999
Dest KPI.Item[2].Value
999.0

Andon Display
KPI Background Color

MOVE
Source Shop.PlanBack
16#0000_ff00
Dest KPI.Item[2].Color
65280

KPI Display Items

Line Off

4

Andon Display
KPI Data Value is
REAL (not Integer)
KPI.Item[3].Real

U

Andon Display
KPI Data Name

COP
Source strKPI_LineOff
Dest KPI.Item[3].Name
Length 1

Andon Display
KPI Data Value

MOVE
Source Line.Off
406
Dest KPI.Item[3].Value
406.0

Andon Display
KPI Background Color

MOVE
Source Line.OffBack
16#0000_ff00
Dest KPI.Item[3].Color
65280

(End)

Convert KPI Display to Mobility String

Convert KPI scalar data to Mobile Data Interface (MDI) string
Example JSON object data

"alarm": 0
, "topColor": 16711680
, "topText": "OR: 98.2"
, "btmColor": 16776960
, "btmText": "OA: 99.1"
, "HB": 59

0

NOP

Alarm Count
(always 0 for KPI Display)

1

KPI Watch String

COP	
Source	strGroupDynAlarm
Dest	strKPI_WatchString
Length	1

DTOS	
Source	0
Dest	TempString '5393'

KPI Watch String

CONCAT	
Source A	strKPI_WatchString '"alarm": 0, "topColo'
Source B	TempString '5393'
Dest	strKPI_WatchString '"alarm": 0, "topColo'

Top Half Color
Popup 0 Color Code

2

KPI Watch String

CONCAT	
Source A	strKPI_WatchString '"alarm": 0, "topColo'
Source B	strGroupDynTopColor ', "topColor": '
Dest	strKPI_WatchString '"alarm": 0, "topColo'

Andon Display
KPI Popup 1

DTOS	
Source	KPI.Popup1.Color 16#0000_ff00
Dest	TempString '5393'

KPI Watch String

CONCAT	
Source A	strKPI_WatchString '"alarm": 0, "topColo'
Source B	TempString '5393'
Dest	strKPI_WatchString '"alarm": 0, "topColo'

Top Half Text
Popup 0 Text

3

KPI Watch String

CONCAT	
Source A	strKPI_WatchString '"alarm": 0, "topColo'
Source B	strGroupDynTopText ', "topText": '
Dest	strKPI_WatchString '"alarm": 0, "topColo'

KPI Watch String

CONCAT	
Source A	strKPI_WatchString '"alarm": 0, "topColo'
Source B	strDblQuote '"'
Dest	strKPI_WatchString '"alarm": 0, "topColo'

KPI Watch String

CONCAT	
Source A	strKPI_WatchString '"alarm": 0, "topColo'
Source B	KPI.Popup1.Text 'OR: 87.99'
Dest	strKPI_WatchString '"alarm": 0, "topColo'

KPI Watch String

CONCAT	
Source A	strKPI_WatchString '"alarm": 0, "topColo'
Source B	strDblQuote '"'
Dest	strKPI_WatchString '"alarm": 0, "topColo'

Convert KPI Display to Mobility String

Bottom Half Color
Popup 1 Color Code

KPI Watch String
CONCAT
Source A strKPI_WatchString '"alarm": 0, "topColo'
Source B strGroupDynBtmColor ', "btmColor": '
Dest strKPI_WatchString '"alarm": 0, "topColo'

Andon Display
KPI Popup 2
DTOS
Source KPI.Popup2.Color 16#0000_ff00
Dest TempString '5393'

KPI Watch String
CONCAT
Source A strKPI_WatchString '"alarm": 0, "topColo'
Source B TempString '5393'
Dest strKPI_WatchString '"alarm": 0, "topColo'

Bottom Half Text
Popup 1 Text

KPI Watch String
CONCAT
Source A strKPI_WatchString '"alarm": 0, "topColo'
Source B strGroupDynBtmText ', "btmText": '
Dest strKPI_WatchString '"alarm": 0, "topColo'

KPI Watch String
CONCAT
Source A strKPI_WatchString '"alarm": 0, "topColo'
Source B strDblQuote '"'
Dest strKPI_WatchString '"alarm": 0, "topColo'

KPI Watch String
CONCAT
Source A strKPI_WatchString '"alarm": 0, "topColo'
Source B KPI.Popup2.Text 'OA: 98.66'
Dest strKPI_WatchString '"alarm": 0, "topColo'

KPI Watch String
CONCAT
Source A strKPI_WatchString '"alarm": 0, "topColo'
Source B strDblQuote '"'
Dest strKPI_WatchString '"alarm": 0, "topColo'

Heartbeat

KPI Watch String
CONCAT
Source A strKPI_WatchString '"alarm": 0, "topColo'
Source B strHeartbeat ', "HB": '
Dest strKPI_WatchString '"alarm": 0, "topColo'

Andon Display
WatchOS Control
Watch Dynamic String
Heartbeat
DTOS
Source GRP_WatchStringCtl[0].UpdateDynHB 9223
Dest TempString '5393'

KPI Watch String
CONCAT
Source A strKPI_WatchString '"alarm": 0, "topColo'
Source B TempString '5393'
Dest strKPI_WatchString '"alarm": 0, "topColo'

Exit from Routine

lblExit
LBL

NOP

(End)

Andon Scrolling Message
Popup Display

Scrolling Message Display
Display Scrolling Message when no Event Popups

0

NOP

Scrolling Mesage Rotation Timer
this controls how fast the message characters scroll across the popup

1

Andon Display
Scrolling Message
Rotation Timer

MOVE
Source 100
Dest ScrollMsg.RotateTMR.PRE
100

Andon Display
Scrolling Message
Rotation Timer

TON
Timer ScrollMsg.RotateTMR
Preset 100
Accum 100
EN
DN

Exit if Scrolling Message Rotation not needed or if Scrolling Message Display is not enabled

2

SAW_AndonDisplayTextMessage.Visible

Andon Display
Scrolling Message
Rotation Timer
ScrollMsg.RotateTMR.DN

Andon Display
Scrolling Message
Enable Scrolling
Message Display
ScrollMsg.Cfg.DisplayEN

lbl_Exit
JMP

Reset Scrolling Message Rotation Timer

3

Andon Display
Scrolling Message
Rotation Timer
ScrollMsg.RotateTMR
RES

Andon Scrolling Message
Popup Display

Create the Scrolling Message
Leading dots + Message + Leading dots
.....Message.....
(this allows the message to scroll completely off display before scrolling back onto display)

4

Scrolling Message
Leading Dots

CONCAT	
Source A	strScrollMsgLeadIn '.....'
Source B	SAW_AndonDisplayTextMessage.Text 'PLAN STOP @ 9:10 FOR'
Dest	ScrollMsg.Text '.....PLAN STOP @'

Andon Display
Scrolling Message
Message Text

CONCAT	
Source A	ScrollMsg.Text '.....PLAN STOP @'
Source B	strScrollMsgLeadIn '.....'
Dest	ScrollMsg.Text '.....PLAN STOP @'

Compute End Index
for Scrolling Display of n characters

5

Andon Display
Scrolling Message
Message End Index

CPT	
Dest	ScrollMsg.ei 57
Expression	ScrollMsg.Text.LEN-ScrollMsg.Cfg.DisplayLEN

Increment Start Index
for Scrolling Display of n characters

6

Andon Display
Scrolling Message
Message Start Index

ADD	
Source A	ScrollMsg.si 25
Source B	1
Dest	ScrollMsg.si 25

Andon Display
Scrolling Message
Message Start Index

LT	
Source A	ScrollMsg.si 25
Source B	0

Andon Display
Scrolling Message
Message Start Index

GT	
Source A	ScrollMsg.si 25
Source B	ScrollMsg.ei 57

Andon Display
Scrolling Message
Message Start Index

MOVE	
Source	0
Dest	ScrollMsg.si 25

Andon Scrolling Message
Popup Display

Load Scrolling Message into 2 Popups
- 1st Popup contains 'ALERT'
- 2nd Popup contains n Scrolling Characters of Message
(2nd branch handles wrap around from end of Message to start)

7

Andon Display Scrolling Message Popup 1

COP
Source strScrollMsgAlert
Dest ScrollMsg.Popup1.Text
Length 1

Andon Display Scrolling Message Popup 1

COP
Source conLightGray
Dest ScrollMsg.Popup1.Color
Length 1

Andon Display Scrolling Message Popup 2

COP
Source ScrollMsg.Text.DATA[ScrollMsg.si]
Dest ScrollMsg.Popup2.Text.DATA[0]
Length ScrollMsg.Cfg.DisplayLEN

Andon Display Scrolling Message Popup 2

MOVE
Source ScrollMsg.Cfg.DisplayLEN
9
Dest ScrollMsg.Popup2.Text.LEN
9

Andon Display Scrolling Message Popup 2

COP
Source conLightGray
Dest ScrollMsg.Popup2.Color
Length 1

Convert Scrolling Message Display to Mobility String

8

Convert Scrolling Message Display to Mobility String

JSR
Routine Name _g_ScrollMsgDisplay_WatchString

Exit Routine

9

lbl_Exit
LBL

NOP

(End)

Convert Scrolling Message Display to Mobility String

Convert Scrolling Message scalar data to Mobile Data Interface (MDI) string
Example JSON object data

"alarm": 999
, "topColor": 16711680
, "topText": "ALERT"
, "btmColor": 16776960
, "btmText": "..Message"
, "HB": 59

0

NOP

Alarm Count
(always 999 to force an Alarm once when Message is visible)

1

Scrolling Message
Watch String

COP
Source strGroupDynAlarm
Dest strScrollMsg_WatchString
Length 1

DTOS
Source 999
Dest TempString
'5393'

Scrolling Message
Watch String

CONCAT
Source A strScrollMsg_WatchString
'"alarm": 999, "topCo'
Source B TempString
'5393'
Dest strScrollMsg_WatchString
'"alarm": 999, "topCo'

Top Half Color
Popup 0 Color Code

2

Scrolling Message
Watch String

CONCAT
Source A strScrollMsg_WatchString
'"alarm": 999, "topCo'
Source B strGroupDynTopColor
', "topColor": '
Dest strScrollMsg_WatchString
'"alarm": 999, "topCo'

Andon Display
Scrolling Message
Popup 1

DTOS
Source ScrollMsg.Popup1.Color
16#00aa_aaaa
Dest TempString
'5393'

Scrolling Message
Watch String

CONCAT
Source A strScrollMsg_WatchString
'"alarm": 999, "topCo'
Source B TempString
'5393'
Dest strScrollMsg_WatchString
'"alarm": 999, "topCo'

Convert Scrolling Message Display to Mobility String

3

Top Half Text
Popup 0 Text

Scrolling Message Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	strGroupDynTopText ', "topText": '
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

Scrolling Message Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	strDblQuote '"'
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

Scrolling Message Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	ScrollMsg.Popup1.Text 'ALERT'
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

Scrolling Message Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	strDblQuote '"'
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

4

Bottom Half Color
Popup 1 Color Code

Scrolling Message Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	strGroupDynBtmColor ', "btmColor": '
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

Andon Display Scrolling Message Popup 2

DTOS	
Source	ScrollMsg.Popup2.Color 16#00aa_aaaa
Dest	TempString '5393'

Scrolling Message Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	TempString '5393'
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

5

Bottom Half Text
Popup 1 Text

Scrolling Message Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	strGroupDynBtmText ', "btmText": '
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

Scrolling Message Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	strDblQuote '"'
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

Scrolling Message Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	ScrollMsg.Popup2.Text ' FOR PROD'
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

Scrolling Message Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	strDblQuote '"'
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

Convert Scrolling Message Display to Mobility String

6

Heartbeat

Scrolling Message
Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	strHeartbeat ', "HB": '
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

Andon Display
WatchOS Control
Watch Dynamic String
Heartbeat

DTOS	
Source	GRP_WatchStringCtl[0].UpdateDynHB 9223
Dest	TempString '5393'

Scrolling Message
Watch String

CONCAT	
Source A	strScrollMsg_WatchString '"alarm": 999, "topCo'
Source B	TempString '5393'
Dest	strScrollMsg_WatchString '"alarm": 999, "topCo'

Exit from Routine

7

lblExit
LBL

NOP

(End)

Real to String in NN.NN format

Real-to-String with NN.NN format

v2.6 - REAL values change constantly - causing continuous update to CGP channel -- convert REAL to DINT values for conversion to STRING
(this also reduces the number of decimal positions generated by RTOS)

This routine provides 2 decimal positions (i.e. 98.00)

0

SBR Input
Real for RTOS

SBR	
Input Par	y_inReal

NOP

1

Convert REAL to Whole/Remainder DINTS

MUL	
Source A	y_inReal 98.65552
Source B	100
Dest	y_inDint 9866

MOD	
Source A	y_inDint 9866
Source B	100
Dest	y_rem 66

CPT	
Dest	y_trn 98
Expression	(y_inDint-y_rem)/100

2

Convert Whole/Remainder DINTS to nn.nn String

DTOS	
Source	y_trn 98
Dest	y_outString '98.66'

SBR Output
STRING from RTOS

CONCAT	
Source A	y_outString '98.66'
Source B	strPeriod '.'
Dest	y_outString '98.66'

LT	
Source A	y_rem 66
Source B	10

DTOS	
Source	0
Dest	TempString '5393'

SBR Output
STRING from RTOS

CONCAT	
Source A	y_outString '98.66'
Source B	TempString '5393'
Dest	y_outString '98.66'

DTOS	
Source	y_rem 66
Dest	TempString '5393'

SBR Output
STRING from RTOS

CONCAT	
Source A	y_outString '98.66'
Source B	TempString '5393'
Dest	y_outString '98.66'

3

SBR Output
STRING from RTOS

RET	
Return Par	y_outString

Real to String in NN.NN format

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
LNi, EQi = Line, Equipment Index
Num = Transaction Number

Transaction Log
Control SBR Input
Parameter --
Transaction Code

0

```
SBR
Input Par   Trans.Code
      <EDC.Trans.Code>
Input Par   Trans.Time
      <EDC.Trans.Time>
Input Par   Trans.LNi
      <EDC.Trans.LNi>
Input Par   Trans.EQi
      <EDC.Trans.EQi>
Input Par   Trans.Num
      <EDC.Trans.Num>
```

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan
Time & Fault Routine
Routine Number
currently Scanned

```
MOVE
Source      901
Dest        Prog.Routine
            130
```

Local Program Scan
Time & Fault Routine
Routine Scan
Counters (Indexed by
Routine Number)

```
ADD
Source A    Prog.RoutineScan[901]
            4492
Source B    1
Dest        Prog.RoutineScan[901]
            4492
```

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT	
Low Limit	Trans.Filter.CodeLow <EDC.Trans.Filter.CodeLow> 0
Test	Trans.Code <EDC.Trans.Code> 2021
High Limit	Trans.Filter.CodeHigh <EDC.Trans.Filter.CodeHigh> 0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT	
Low Limit	Trans.Filter.LNiLow <EDC.Trans.Filter.LNiLow> 0
Test	Trans.LNi <EDC.Trans.LNi> 4
High Limit	Trans.Filter.LNiHigh <EDC.Trans.Filter.LNiHigh> 0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT	
Low Limit	Trans.Filter.EQiLow <EDC.Trans.Filter.EQiLow> 0
Test	Trans.EQi <EDC.Trans.EQi> 21
High Limit	Trans.Filter.EQiHigh <EDC.Trans.Filter.EQiHigh> 0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

ADD
Source A Trans.LOGi
<EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGi
<EDC.Trans.LOGi> 211

Transaction Log Control -- Transaction Log Index

LT
Source A Trans.LOGi
<EDC.Trans.LOGi> 211
Source B 0

Transaction Log Control -- Transaction Log Index

GE
Source A Trans.LOGi
<EDC.Trans.LOGi> 211
Source B CFG.TransLOG.n
<EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index

CLR
Dest Trans.LOGi
<EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

ADD
Source A Trans.LOGi
<EDC.Trans.LOGi> 211
Source B 1
Dest Trans.LOGx
<EDC.Trans.LOGx> 212

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.LOGx
<EDC.Trans.LOGx> 212
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.LOGx
<EDC.Trans.LOGx> 212
Source B CFG.TransLOG.n
<EDC.CFG.TransLOG.n> 250

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.LOGx
<EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGi]
Length 1

Transaction Log
FLL
Source 0
Dest EDC_TransLog[Trans.LOGx]
Length 1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log
MOVE
Source Trans.Code
<EDC.Trans.Code> 2021
Dest EDC_TransLog[Trans.LOGi].TransCode
1011

Transaction Log
MOVE
Source Trans.Time
<EDC.Trans.Time> 0
Dest EDC_TransLog[Trans.LOGi].TransTime
0

Transaction Log
MOVE
Source Trans.LNi
<EDC.Trans.LNi> 4
Dest EDC_TransLog[Trans.LOGi].LNi
15

Transaction Log
MOVE
Source Trans.EQi
<EDC.Trans.EQi> 21
Dest EDC_TransLog[Trans.LOGi].EQi
17

Transaction Log
MOVE
Source Trans.Num
<EDC.Trans.Num> 4971
Dest EDC_TransLog[Trans.LOGi].Num
5631

Transaction Log
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]> 20261030
Dest EDC_TransLog[Trans.LOGi].DateStamp
20261030

Transaction Log
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]> 151921
Dest EDC_TransLog[Trans.LOGi].TimeStamp
152006

Transaction Log
MOVE
Source CLX.SysTime[6]
<EDC.CLX.SysTime[6]> 904687
Dest EDC_TransLog[Trans.LOGi].TimeMicroSec
955653

Transaction Log
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount> 466525176
Dest EDC_TransLog[Trans.LOGi].ScanCount
466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log
MOVE
Source Trans.Opt[1]
<EDC.Trans.Opt[1]> -1
Dest EDC_TransLog[Trans.LOGi].Opt1
1517

Transaction Log
MOVE
Source Trans.Opt[2]
<EDC.Trans.Opt[2]> -1
Dest EDC_TransLog[Trans.LOGi].Opt2
0

Transaction Log
MOVE
Source Trans.Opt[3]
<EDC.Trans.Opt[3]> -1
Dest EDC_TransLog[Trans.LOGi].Opt3
-1

Transaction Log
MOVE
Source Trans.Opt[4]
<EDC.Trans.Opt[4]> -1
Dest EDC_TransLog[Trans.LOGi].Opt4
-1

Transaction Log
MOVE
Source Trans.Opt[5]
<EDC.Trans.Opt[5]> -1
Dest EDC_TransLog[Trans.LOGi].Opt5
-1

Log Transactions

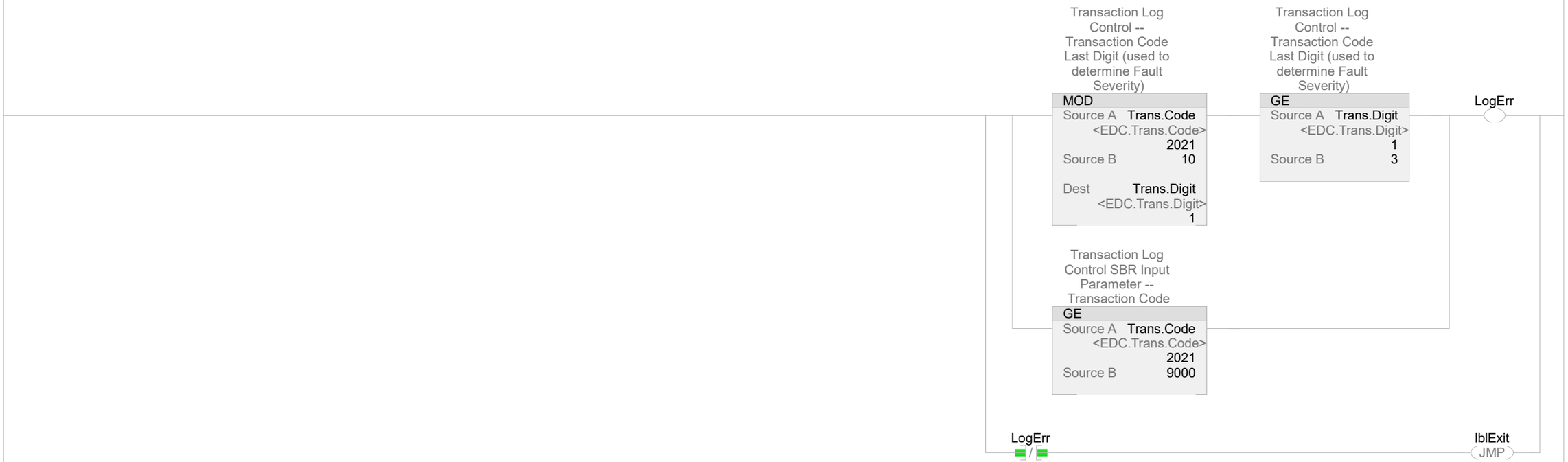
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

EDC2 - v2.00 added PdM with QualityRx, TrendRx and Server Interface

PdM Data Collection
for Tools and Trends

v1.01 Don't clear NetMod data on intermittent loss of StatusOK - delay 30 seconds for StatusFault
v3.02 Replace COP with MOV for ToolData & ToolArray local tags (faster scan)
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion ...
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1001
Dest	Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 704823770
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 704823770

Reset

Reset Event Configuration
(use before Download All to defrag the Event table for faster searches)

2

Gateway Reset CFGResetPdm

ManualReset

Gateway Reset CFGResetPdm (U) ManualReset (U) AutoConfig (L)

Reset

JSR	
Routine Name	_1010_Reset

Configuration

3

AutoConfig

Toggle On to Configure ManualConfig

S:FS

AutoConfig (U)

Toggle On to Configure ManualConfig (U)

JSR
Routine Name _1011_Config

Configure Network Module Types
define the Module layout of each Network Type (i.e. Banner, Balluff)

4

S:FS

Toggle On to Configure Net ManualConfigNet

Toggle On to Configure Net ManualConfigNet (U)

Configure Network Modules

JSR
Routine Name _1012_ConfigNet

Network Modules
Manage Modules and Copy Data from Modules to Network Arrays

*** Modules are serviced by PdM Controller and NetMod data is Messaged to Gateway array ***

5

NOP

Update Sample Rates and Periods
Tools & Trends use Sample Rates and Periods to determine when to sample data.

6

Sample Rates and Periods

JSR
Routine Name _1030_Sample

Tool Loop
service all Tools each Scan for fast update of Samples and Triggers

The Tool Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Tool to be scanned continuously.
(WARNING: Disabling the Tool FOR Loop will cause the unscanned Tools to stop updating)

7

PdM Config Disable Tool Loop for testing
PdM_Cfg.DisableToolLoop
/

Tool Data
FOR
Routine Name _1040_Tool
Index TLi 915
Initial Value PdM_Cfg.TL.si
Terminal Value PdM_Cfg.TL.ei
Step Size 1

PdM Config Disable Tool Loop for testing
PdM_Cfg.DisableToolLoop

Tool Loop Index
MOVE
Source _ViewTLi 0
Dest TLi 915

Tool Data
JSR
Routine Name _1040_Tool

Trend Loop
service all Trends each Scan for fast update of Samples and Triggers

The Trend Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Trend to be scanned continuously.
(WARNING: Disabling the Trend FOR Loop will cause the unscanned Trends to stop updating)

8

PdM Config Disable Trend Loop for testing
PdM_Cfg.DisableTrendLoop
/

Trend Data
FOR
Routine Name _1050_Trend
Index TRNi 4
Initial Value PdM_Cfg.TRN.si
Terminal Value PdM_Cfg.TRN.ei
Step Size 1

PdM Config Disable Trend Loop for testing
PdM_Cfg.DisableTrendLoop

Trend Loop Index
MOVE
Source _ViewTRNi 0
Dest TRNi 4

Trend Data
JSR
Routine Name _1050_Trend

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

9

CLx Module Status
Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan
Time & Fault Routine
Clear Scan History
Prog.Scan.Clear
L

Local Program Scan
Time & Fault Routine
Program Fault
Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

10

Local Program Scan
Time & Fault Routine
Last Scan
GT
Source A Prog.Scan.Last
5609
Source B CLxScanHighLimit
100000

Program
High Scan
Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan
Events
CLxProgScanHigh.10
L

CLx Module Status
Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program
High Scan
Count
CLR
Dest ProgScanHighCount
0

(End)

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV	
Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD	
Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

2

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 20

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 20
Source B	1
Dest	Prog.LOGmi 49

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 1
Source B	1
Dest	Prog.LOGi 1

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 1
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 1
Source B	Prog.LOGmi 49

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 1

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
8293

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
1042
Dest Prog.Log[Prog.LOGi].Routine
1002

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
113947485

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
20220705

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
124500

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
126

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source TLi
915
Dest Prog.Log[Prog.LOGi].Opt1
4

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source TRNi
4
Dest Prog.Log[Prog.LOGi].Opt2
1

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1002
Dest Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 704823770
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 704823770

5

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 1766812086

6

(End)

Reset

PdM Reset

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1010
Dest	Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 704823770
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 704823770

Reset PdM Tool and Trend Arrays

2

Network Modules Network Module

SIZE	
Source	PdM_NetMod[0] ??
Dim. To Vary	0
Size	LEN 20

Network Modules Network Module

FLL	
Source	0
Dest	PdM_NetMod[0]
Length	LEN

PdM Tool Array Pdm Tool

SIZE	
Source	PdM_Tool[0] ??
Dim. To Vary	0
Size	LEN 20

PdM Tool Array Pdm Tool

FLL	
Source	0
Dest	PdM_Tool[0]
Length	LEN

PdM Trend Array Pdm Trend

SIZE	
Source	PdM_Trend[0] ??
Dim. To Vary	0
Size	LEN 20

PdM Trend Array Pdm Trend

FLL	
Source	0
Dest	PdM_Trend[0]
Length	LEN

Reset PdM Config

3

PdM Config

FLL	
Source	0
Dest	PdM_Cfg
Length	1

(End)

PdM Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1011
Dest Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 704823770
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 704823770

Tool Array

-- Use Array to determine maximum values for TLi
-- Set initial values for Tool Loop Start Index and End Index
(End Index will be modified by Configuration)

2

PdM Tool Array Pdm Tool

SIZE
Source PdM_Tool[0] ??
Dim. To Vary 0
Size PdM_Cfg.TL.n 3000

PdM Config Array Max Index

SUB
Source A PdM_Cfg.TL.n 3000
Source B 1
Dest PdM_Cfg.TL.mi 2999

PdM Config Array Start Index

MOVE
Source 0
Dest PdM_Cfg.TL.si 0

AFI

PdM Config Array End Index

MOVE
Source 0
Dest PdM_Cfg.TL.ei 914

Tool Data

-- Use Array to determine maximum values for DATi

3

Pdm Tool Tool Raw Data

SIZE
Source ThisTool.Data[0] ??
Dim. To Vary 0
Size PdM_Cfg.DAT.n 10

PdM Config Array Max Index

SUB
Source A PdM_Cfg.DAT.n 10
Source B 1
Dest PdM_Cfg.DAT.mi 9

Trend Array

-- Use Array to determine maximum values for TRNi
-- Set initial values for Trend Loop Start Index and End Index
(End Index will be modified by Configuration)

4

PdM Trend Array Pdm Trend

SIZE
Source PdM_Trend[0] ??
Dim. To Vary 0
Size PdM_Cfg.TRN.n 1000

PdM Config Array Max Index

SUB
Source A PdM_Cfg.TRN.n 1000
Source B 1
Dest PdM_Cfg.TRN.mi 999

PdM Config Array Start Index

MOVE
Source 0
Dest PdM_Cfg.TRN.si 0

AFI

PdM Config Array End Index

MOVE
Source 0
Dest PdM_Cfg.TRN.ei 3

5

Module Network Array

-- Use Array to determine maximum values for NETi

Network Modules Network Module

SIZE
Source PdM_NetMod[0] ??
Dim. To Vary 0
Size PdM_Cfg.NET.n 300

PdM Config Array Max Index

SUB
Source A PdM_Cfg.NET.n 300
Source B 1
Dest PdM_Cfg.NET.mi 299

PdM Config Array Start Index

MOVE
Source 0
Dest PdM_Cfg.NET.si 0

PdM Config Array End Index

MOVE
Source PdM_Cfg.NET.mi 299
Dest PdM_Cfg.NET.ei 299

6

Module Network Data Array

-- Use Array to determine maximum values for NET DATi

Network Module Sensor Data Array

SIZE
Source PdM_NetMod[0].Data[0] 3935
Dim. To Vary 0
Size PdM_Cfg.NetDAT.n 250

PdM Config Array Max Index

SUB
Source A PdM_Cfg.NetDAT.n 250
Source B 1
Dest PdM_Cfg.NetDAT.mi 249

PdM Config Array Start Index

MOVE
Source 0
Dest PdM_Cfg.NetDAT.si 0

PdM Config Array End Index

MOVE
Source PdM_Cfg.NetDAT.mi 249
Dest PdM_Cfg.NetDAT.ei 249

Sample Period

-- Use Array to determine maximum values for PERi

7

not used

SIZE
Source SamplePeriod[0] 0
Dim. To Vary 0
Size PdM_Cfg.PER.n 50

PdM Config Array Max Index

SUB
Source A PdM_Cfg.PER.n 50
Source B 1
Dest PdM_Cfg.PER.mi 49

Sample Rate

-- Use Array to determine maximum values for RATi

8

not used

SIZE
Source SampleRate[0] 0
Dim. To Vary 0
Size PdM_Cfg.RAT.n 50

PdM Config Array Max Index

SUB
Source A PdM_Cfg.RAT.n 50
Source B 1
Dest PdM_Cfg.RAT.mi 49

(End)

Configure Network Modules

Configure Network Module Types

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1012
Dest	Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 704823770
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 704823770

Banner DSMR90X1 Network

2

Banner DSMR90X1 Network

JSR	
Routine Name	_1013_ConfigNetType01

(End)

Banner DSMR90X1 Network

SQL dbo.CfgPdmNetworkType - Type1
Banner DSMR90X1 Network Configuration

1-40 QM30VT2 Vibration/Temperature Sensor
41-60 S15C-CT-MQ Current Transformer
61-70 S15C-I-MQ 4-20mA
71-75 S15C-B22-MQ Dual Discrete Input Counter/Timer
76-80 M12FTH3Q Relative Humidity and Temperature Sensor
81-85 S15C-BB_MTGQ Infrared Temperature Sensor

********** PdM Gateway Configuration must match SQL DB Configuration **********

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1013
Dest Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 704823770
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 704823770

initialize array and array index

2

Network Type Configuration

FLL
Source 0
Dest ThisCfgNetwork
Length 1

array index

CLR
Dest PTRi 220

initiialize QM30VT2 sensor size and index

3

sensor length

MOVE
Source 4
Dest LEN 20

sensor count

MOVE
Source 40
Dest CNT 0

Banner DSMR90X1 Network

insert QM30VT2

4

lbl_Loop1 — LBL

Network Type Configuration Sensor I/O Map Sensor Data Configuration

COP	
Source	CfgSensorBannerQM30VT2
Dest	ThisCfgNetwork.Data[PTRi]
Length	LEN

array index

ADD		
Source A	PTRi	220
Source B	LEN	20
Dest	PTRi	220

sensor count

SUB		
Source A	CNT	0
Source B		1
Dest	CNT	0

sensor count

GT		
Source A	CNT	0
Source B		0

lbl_Loop1 — JMP

initiialize S15C-CT-MQ sensor size and index

5

sensor length

MOVE		
Source		1
Dest	LEN	20

sensor count

MOVE		
Source		20
Dest	CNT	0

insert S15C-CT-MQ

6

lbl_Loop2 — LBL

Network Type Configuration Sensor I/O Map Sensor Data Configuration

COP	
Source	CfgSensorBannerS15C_CT_MQ
Dest	ThisCfgNetwork.Data[PTRi]
Length	LEN

array index

ADD		
Source A	PTRi	220
Source B	LEN	20
Dest	PTRi	220

sensor count

SUB		
Source A	CNT	0
Source B		1
Dest	CNT	0

sensor count

GT		
Source A	CNT	0
Source B		0

lbl_Loop2 — JMP

initiialize S15C-I-MQ sensor size and index

7

sensor length

MOVE		
Source		1
Dest	LEN	20

sensor count

MOVE		
Source		10
Dest	CNT	0

Banner DSMR90X1 Network

insert S15C-I-MQ

8

lbl_Loop3 — LBL

Network Type Configuration Sensor I/O Map Sensor Data Configuration

COP
Source CfgSensorBannerS15C_I_MQ
Dest ThisCfgNetwork.Data[PTRi]
Length LEN

array index

ADD
Source A PTRi 220
Source B LEN 20
Dest PTRi 220

sensor count

SUB
Source A CNT 0
Source B 1
Dest CNT 0

sensor count

GT
Source A CNT 0
Source B 0

lbl_Loop3 — JMP

initiialize S15C-B22-MQ sensor size and count

9

sensor length

MOVE
Source 2
Dest LEN 20

sensor count

MOVE
Source 5
Dest CNT 0

insert S15C-B22-MQ

10

lbl_Loop4 — LBL

Network Type Configuration Sensor I/O Map Sensor Data Configuration

COP
Source CfgSensorBannerS15C_B22_MQ
Dest ThisCfgNetwork.Data[PTRi]
Length LEN

array index

ADD
Source A PTRi 220
Source B LEN 20
Dest PTRi 220

sensor count

SUB
Source A CNT 0
Source B 1
Dest CNT 0

sensor count

GT
Source A CNT 0
Source B 0

lbl_Loop4 — JMP

initiialize M12FTH3Q sensor size and count

11

sensor length

MOVE
Source 2
Dest LEN 20

sensor count

MOVE
Source 5
Dest CNT 0

Banner DSMR90X1 Network

insert M12FTH3Q

12

Network Type Configuration Sensor I/O Map Sensor Data Configuration

lbl_Loop5 — LBL

COP
Source CfgSensorBannerM21FTH3Q
Dest ThisCfgNetwork.Data[PTRi]
Length LEN

array index

ADD
Source A PTRi 220
Source B LEN 20
Dest PTRi 220

sensor count

SUB
Source A CNT 0
Source B 1
Dest CNT 0

sensor count

GT
Source A CNT 0
Source B 0

lbl_Loop5 — JMP

initiialize S15C-BB-MTGQ sensor size and count

13

sensor length

MOVE
Source 2
Dest LEN 20

sensor count

MOVE
Source 5
Dest CNT 0

insert S15C-BB-MTGQ

14

Network Type Configuration Sensor I/O Map Sensor Data Configuration

lbl_Loop6 — LBL

COP
Source CfgSensorBannerS15C_BB_MTGQ
Dest ThisCfgNetwork.Data[PTRi]
Length LEN

array index

ADD
Source A PTRi 220
Source B LEN 20
Dest PTRi 220

sensor count

SUB
Source A CNT 0
Source B 1
Dest CNT 0

sensor count

GT
Source A CNT 0
Source B 0

lbl_Loop6 — JMP

Save Network Configuration

15

Network Type Configuration Sensor I/O Map configuration

COP
Source ThisCfgNetwork
Dest CfgNetMod[1]
Length 1

Banner DSMR90X1 Network

(End)

Sample Rates and Periods

Generate the Sample Rates and Periods
Sample Rates are used for Trend Data Samples (i.e. sample 120 data values at 1 second interval)
Sample Periods are used for Tool and Trend Record Samples (i.e. send a record every 1 hour)

The Sample Rate/Period are assigned to the array index that matches the server Sample ID configuration
Each Sample value will increment when its Rate/Period occurs.

Samples can range from 1 second to 12 hours -- using an Off-On transition with a BOOL would be tricky with multi-scan servicing of Tools.
With an incremented value, each Tool can compare it's last Sample value to the current Sample value to determine when to Sample.

Custom Sample Periods can be created -- for example, Twice Daily at 6am, 6pm
Custom samples must be added to the server and Gateway with matching PeriodID

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1030
Dest Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 704823770
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 704823770

Custom Rates (sub-second)
SampleRate13 = 100 msec
SampleRate14 = 10 msec

2

Sample Rates (seconds) - matches server configuration options

CPT
Dest SampleRate[13] 3
Expression (CLX.SysTime[6] / 100000) + 1

Sample Rates (seconds) - matches server configuration options

CPT
Dest SampleRate[14] 29
Expression (CLX.SysTime[6] / 10000) + 1

Sample Rates and Periods

Exit if no Seconds Change

CLx Module Status
CLx System Time
Seconds in Current
Hour (mm*60)+ss

3

EQ
Source A CLX.SecondsInHour
<EDC.CLX.SecondsInHour>
1161
Source B LastSeconds
1496

lbl_Return
JMP

Update Last Seconds value
Initialize Rate Loop

4

MOVE
Source CLX.SecondsInHour
<EDC.CLX.SecondsInHour>
1161
Dest LastSeconds
1496

Loop Index
MOVE
Source 1
Dest LOOPi
13

Update Sample Rates 1-12

Sample Rates
(seconds) - matches
server configuration
options

5

lbl_RateLoop
LBL

MOD
Source A CLX.SecondsInHour
<EDC.CLX.SecondsInHour>
1161
Source B CfgSampleRate[LOOPi]
0
Dest Modulo
56

EQ
Source A Modulo
56
Source B 0

CPT
Dest SampleRate[LOOPi]
3
Expression TRUNC(CLX.SecondsInHour/CfgSampleRate[LOOPi]) + 1

Loop Index
ADD
Source A LOOPi
13
Source B 1
Dest LOOPi
13

Loop Index
LE
Source A LOOPi
13
Source B 12

lbl_RateLoop
JMP

Sample Rates and Periods

Update custom Samples

6

NOP

Exit if no Minutes Change

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

7

EQ
Source A CLX.SysTime[4]
<EDC.CLX.SysTime[4]>
19
Source B LastMinutes
24

lbl_Return
JMP

Update Last Minutes value
Initialize Hour/Minute and Period Loop
Initialize Period Loop

(minutes in day doesn't work for a 24-hour sample -- need to use minutes in a month for an increment past midnight)

8

MOVE
Source CLX.SysTime[4]
<EDC.CLX.SysTime[4]>
19
Dest LastMinutes
24

(hh * 60) + mm

CPT
Dest MinutesInDay
924
Expression (CLX.SysTime[3] * 60) + CLX.SysTime[4]

Loop Index

MOVE
Source 1
Dest LOOPi
13

Sample Rates and Periods

Update Sample Periods 1-18

9

lbl_PeriodLoop
LBL

MOD
Source A MinutesInDay 924
Source B CfgSamplePeriod[LOOPi] 120
Dest Modulo 56

EQ
Source A Modulo 56
Source B 0

Sample Periods (minutes)
CPT
Dest SamplePeriod[LOOPi] 8
Expression TRUNC(MinutesInDay/CfgSamplePeriod[LOOPi]) + 1

Loop Index
ADD
Source A LOOPi 13
Source B 1
Dest LOOPi 13

Loop Index
LE
Source A LOOPi 13
Source B 19

lbl_PeriodLoop
JMP

Update custom Periods

10

NOP

24 Hours
(at midnight - toggle between 1 and 2)

11

(hh * 60) + mm
EQ
Source A MinutesInDay 924
Source B 0

24 Hours
NE
Source A SamplePeriod[19] 2
Source B 1

24 Hours
MOVE
Source 1
Dest SamplePeriod[19] 2

24 Hours
EQ
Source A SamplePeriod[19] 2
Source B 1

24 Hours
MOVE
Source 2
Dest SamplePeriod[19] 2

Sample Rates and Periods

Twice Daily
6am, 6pm

12

(hh * 60) + mm

EQ	
Source A	MinutesInDay
	924
Source B	360

6am, 6pm

MOVE	
Source	1
Dest	SamplePeriod[20]
	1

(hh * 60) + mm

EQ	
Source A	MinutesInDay
	924
Source B	1080

6am, 6pm

MOVE	
Source	2
Dest	SamplePeriod[20]
	1

Exit Routine

lbl_Return

13 LBL

NOP

(End)

Tool Data

Tool Loop
service all Tools each Scan for fast update of Samples and Triggers

The Tool Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Tool to be scanned continuously.
(WARNING: Disabling the Tool FOR Loop will cause the unscanned Tools to stop updating)

Index TLi

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1040
Dest Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 704823770
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 704823770

Exit if Tool not Enabled
Clear any existing data from disabled Tools to avoid old data when re-enabled.

2

PdM Tool Array
PdM_Tool[TLi].Cfg.Enable
/

PdM Tool Array
PdM_Tool[TLi].Ctl.Start

PdM Tool Array

FLL
Source 0
Dest PdM_Tool[TLi].Ctl
Length 1

PdM Tool Array

FLL
Source 0
Dest PdM_Tool[TLi].Data[0]
Length PdM_Cfg.DAT.n

PdM Tool Array

FLL
Source 0
Dest PdM_Tool[TLi].Array[0]
Length PdM_Cfg.DAT.n

lbl_Return
JMP

Copy Array Tags to Local Tags

3

Pdm Tool

COP
Source PdM_Tool[TLi]
Dest ThisTool
Length 1

Tool Data

Get LNi and EQi from EQID
Exit if not a Valid EQID

4

MOD
Source A ThisTool.Cfg.EQID
1004
Source B 100
Dest EQi
71

CPT
Dest LNi
9
Expression (ThisTool.Cfg.EQID-EQi)/100

LIMIT
Low Limit 0
Test LNi
9
High Limit CFG.LN.mi
<EDC.CFG.LN.mi>
28

LIMIT
Low Limit 0
Test EQi
71
High Limit CFG.EQ.mi
<EDC.CFG.EQ.mi>
99

ToolValidEQID

PdM Config Tool
Processing Error -
Invalid EQID

ToolValidEQID PdM_Cfg.ToolInvalidEQID

MOVE
Source LNi
9
Dest ToolInvalidEQID_LNi
0

MOVE
Source EQi
71
Dest ToolInvalidEQID_EQi
10

ADD
Source A ToolInvalidEQID_Count
103100
Source B 1
Dest ToolInvalidEQID_Count
103100

lbl_Return
JMP

v3.03 Exit if Equipment is not Enabled
Clear any existing data from disabled Tools to avoid old data when re-enabled.
(this handles the case when the Equipment is Disabled but the PdM Tool is still Configured)

Equipment Data
Collection Lines and
Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.Enable

5

PdM Tool Array
PdM_Tool[TLi].Ctl.Start

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TLi].Ctl
Length 1

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TLi].Data[0]
Length PdM_Cfg.DAT.n

PdM Tool Array
FLL
Source 0
Dest PdM_Tool[TLi].Array[0]
Length PdM_Cfg.DAT.n

lbl_Return
JMP

Tool Data

Exit if Sample Tool and QualityMsg EQID is already in use by another Sample Tool
multple Tools with same EQID can be configured to Sample data on the same scan -- need to load these to RingBuffer one-at-a-time

6

Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.QualityMsg.PdmSampleSend

lbl_Return
JMP

Exit if Tool Data Config not within array limits

7

Pdm Tool Data Start Index

LIMIT	
Low Limit	0
Test	ThisTool.Cfg.DATsi[0]
High Limit	PdM_Cfg.DAT.mi[9]

Pdm Tool Data End Index

LIMIT	
Low Limit	0
Test	ThisTool.Cfg.DATei[3]
High Limit	PdM_Cfg.DAT.mi[9]

ToolDataIndexOK

ToolDataIndexOK

PdM Config Tool Processing Error - Invalid Data Index
PdM_Cfg.ToolInvalidDataIndex
L

MOVE	
Source	TLi[915]
Dest	ToolDataIndexNotOK_TLi[0]

MOVE	
Source	ThisTool.Cfg.DATsi[0]
Dest	ToolDataIndexNotOK_DATsi[-1]

MOVE	
Source	ThisTool.Cfg.DATei[3]
Dest	ToolDataIndexNotOK_DATei[-1]

ADD	
Source A	ToolDataIndexNotOK_Count[0]
Source B	1
Dest	ToolDataIndexNotOK_Count[0]

lbl_Return
JMP

Tool Data

Get Sample Period ID
(clear if unused to avoid illegal indexes on next rung)

8

Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable

Period to send record

CLR
Dest SamplePeriodID
0

Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable

Period to send record

MOVE
Source ThisTool.Cfg.SamplePeriodID
5
Dest SamplePeriodID
0

Exit if Sample PeriodID not within array limits

9

Period to send record

LIMIT
Low Limit 1
Test SamplePeriodID
0
High Limit PdM_Cfg.PER.mi
49

ToolSamplePeriodID_OK

Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable

ToolSamplePeriodID_OK

PdM Config Tool Processing Error - Invalid Sample Period
PdM_Cfg.ToolInvalidSamplePeriod

MOVE
Source TLi
915
Dest ToolPeriodIDNotOK_TLi
191

MOVE
Source SamplePeriodID
0
Dest ToolPeriodIDNotOK_PeriodID
0

ADD
Source A ToolPeriodIDNotOK_Count
1888977
Source B 1
Dest ToolPeriodIDNotOK_Count
1888977

lbl_Return
JMP

Tool Data

Get Tool Trigger/Sample Status
if Triggered, use Start/Send signals and Job Data
if Sampled, use Sample Period to generate Start/Send signals

If Trigger sends Start/Send at same time, one sample with current value will be sent
For Samples, Data Collection is stored until next Sample Period - if QualityMsg doesn't process the PdmSampleSend, the data will be lost
(with minimum 1 minute Period, it's very unlikely)

If new Tool Config (PeriodLast = -1), Initialize to the Sample Period without sampling

10

Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable

Pdm Tool Trigger - Start data collection (from EqDriver)
ThisTool.Ctl.TriggerStart (U)

Pdm Tool Start data collection
ThisTool.Ctl.Start (L)

Pdm Tool Start data collection initiation
ThisTool.Ctl.StartInit (L)

Pdm Tool Trigger - Send data collection (from EqDriver)
ThisTool.Ctl.TriggerSend (U)

Pdm Tool Send data colleection
ThisTool.Ctl.Send (L)

Pdm Tool Start data collection
ThisTool.Ctl.Start (U)

Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable (/)

Pdm Tool Start data collection
ThisTool.Ctl.Start (/)

Pdm Tool Start data collection
ThisTool.Ctl.Start (L)

Pdm Tool Start data collection initiation
ThisTool.Ctl.StartInit (L)

Pdm Tool Last Sample Period

EQ
Source A ThisTool.Ctl.PeriodLast 185
Source B -1

Pdm Tool Last Sample Period

MOVE
Source SamplePeriod[SamplePeriodID] 0
Dest ThisTool.Ctl.PeriodLast 185

Sample Periods (minutes)

NE
Source A SamplePeriod[SamplePeriodID] 0
Source B ThisTool.Ctl.PeriodLast 185

Pdm Tool Last Sample Period

MOVE
Source SamplePeriod[SamplePeriodID] 0
Dest ThisTool.Ctl.PeriodLast 185

Pdm Tool Send data colleection
ThisTool.Ctl.Send (L)

Pdm Tool Start data collection
ThisTool.Ctl.Start (U)

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.QualityMsg.PdmSampleSend (L)

Equipment Data Collection Lines and Equipment Array

MOVE
Source ThisTool.Cfg.ToolID 32
Dest EDC.Line[LNi].Equip[EQi].CTL.QualityMsg.PdmSampleToolID 0

Tool Data

Initialize Data Collection

11

Pdm Tool Start data collection initiation
ThisTool.Ctl.StartInit

Pdm Tool Start data collection initiation
ThisTool.Ctl.StartInit
(U)

Pdm Tool Reset data collection
ThisTool.Ctl.Reset
(L)

Pdm Tool Tool Collection Array

FLL	
Source	0
Dest	ThisTool.Array[0]
Length	PdM_Cfg.DAT.n

Reset Raw Data
also Reset If Triggered and Start/Send are both off -- this keeps the Last data value ready
in case the EqDriver sends Start/Send Triggers in one Message

12

Pdm Tool Reset data collection
ThisTool.Ctl.Reset

Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable

Pdm Tool Start data collection
ThisTool.Ctl.Start

Pdm Tool Start data collection initiation
ThisTool.Ctl.StartInit

Pdm Tool Send data colleection
ThisTool.Ctl.Send

Pdm Tool Reset data collection
ThisTool.Ctl.Reset
(U)

Pdm Tool Tool Raw Data

FLL	
Source	0
Dest	ThisTool.Data[0]
Length	PdM_Cfg.DAT.n

Raw Data Loop
Get each Data for this Tool

continuous update for Triggers - reset on Start/Send

13

Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable

Pdm Tool Start data collection
ThisTool.Ctl.Start

Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable

FOR

Routine Name	_1041_ToolData
Index	DATI
	4
Initial Value	ThisTool.Cfg.DATsi
Terminal Value	ThisTool.Cfg.DATei
Step Size	1

Tool Data

Data Collection Loop
Store Configured Data values for Quality Tool

continuous update for Triggers

14

Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable
/
Pdm Tool Enable EqDriver Trigger Start/Send
ThisTool.Cfg.TriggerEnable

Pdm Tool Send data colleection
ThisTool.Ctl.Send

FOR
Routine Name _1042_ToolArray
Index DATI
4
Initial Value ThisTool.Cfg.DATsi
Terminal Value ThisTool.Cfg.DATei
Step Size 1

PdM Tool Array
COP
Source ThisTool.Array[0]
Dest PdM_Tool[TLi].Array[0]
Length PdM_Cfg.DAT.n

Reset Send

15

Pdm Tool Send data colleection
ThisTool.Ctl.Send

Pdm Tool Send data colleection
ThisTool.Ctl.Send
U

Copy Local Tag to Array Tags

16

PdM Tool Array
COP
Source ThisTool.Ctl
Dest PdM_Tool[TLi].Ctl
Length 1

PdM Tool Array
COP
Source ThisTool.Data[0]
Dest PdM_Tool[TLi].Data[0]
Length PdM_Cfg.DAT.n

Exit Routine

17

lbl_Return
LBL

NOP

(End)

Update Count, Total, Avg, Min and Max for new Data Value
all values are unscaled - scaling will be applied when data is sent to EDC

Index TLi, DATi

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1041
Dest Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 704823770
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 704823770

Exit if Data not Enabled

2

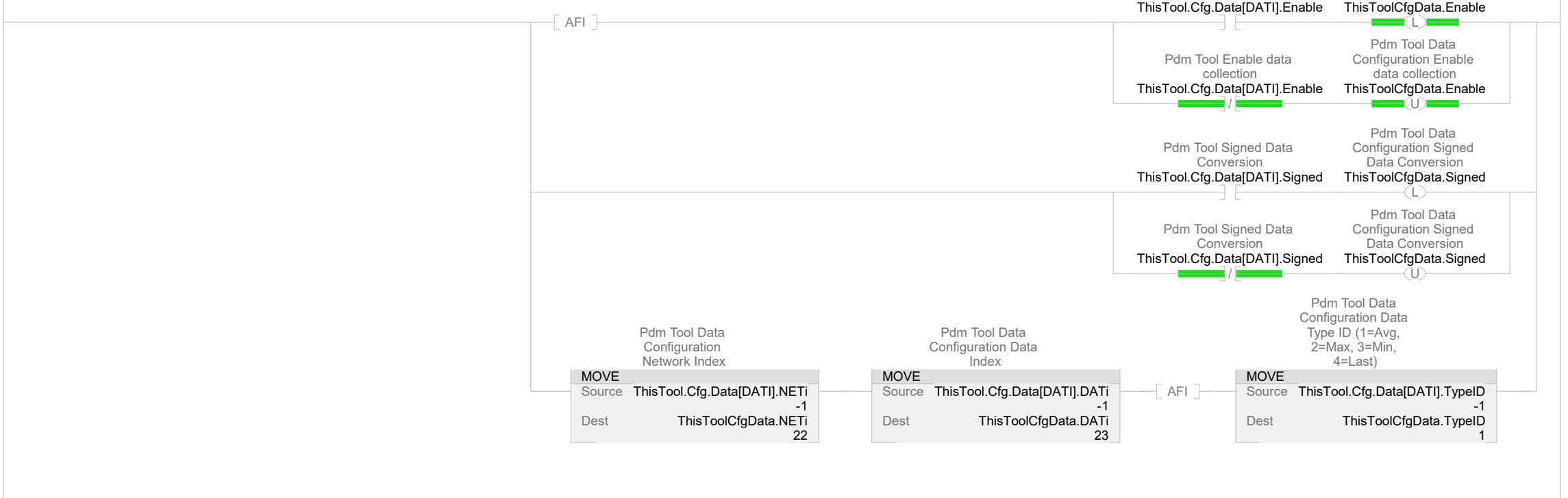
Pdm Tool Enable data collection
ThisTool.Cfg.Data[DATI].Enable
/

lbl_Return
JMP

MOV Config from Array to local tags - faster than COP by 25%
Use Data Arrays - faster than moving to local tags

AFI'd TypeID - not used, so not copied to local tags

3



Exit if Data Config not within array limits

4

Pdm Tool Data Configuration Network Index

LIMIT
Low Limit 0
Test ThisToolCfgData.NETi 22
High Limit PdM_Cfg.NET.mi 299

Pdm Tool Data Configuration Data Index

LIMIT
Low Limit 0
Test ThisToolCfgData.DATi 23
High Limit PdM_Cfg.NetDAT.mi 249

NetDataOK

NetDataOK

PdM Config Tool Processing Error - Invalid Network Data

PdM_Cfg.ToolInvalidNetworkData (L)

MOVE
Source TLi 915
Dest ToolNetDataNotOK_TLi 0

MOVE
Source DATI 4
Dest ToolNetDataNotOK_DATi 0

MOVE
Source ThisToolCfgData.NETi 22
Dest ToolNetDataNotOK_NETi 0

MOVE
Source ThisToolCfgData.DATi 23
Dest ToolNetDataNotOK_NetDATi 0

ADD
Source A ToolNetDataNotOK_Count 0
Source B 1
Dest ToolNetDataNotOK_Count 0

lbl_Return
JMP

TBD -- should Data be ignored if Network Status is not OK
(the data will be 0'd by the PdM Module controller - is there a difference between collecting 0's and not collecting?)

PdM_NetMod.Ctl (including StatusOK) does not update in real time
The PdM Gateway clears the data array when the module is faulted for 30 seconds.
The EDC Gateway stores whatever is sent by the PdM Module without concern for the Module Status.

5

AFI

Network Modules
PdM_NetMod[ThisToolCfgData.NETi].Ctl.StatusOK

lbl_Return
JMP

Get Data Sample from Network

** SPECIAL CASE ** -- if negative value and NOT signed data, convert 16-bit unsigned INT to 32-bit signed DINT by adding 65,536
(Banner DXM uses unsigned 16-bit integer for all sensors EXCEPT temperature which uses signed 16-bit integer)

NOTE: the unsigned INT correction only works if the result is held in a DINT

6

Pdm Tool Sensor Data Last Value

MOVE
Source PdM_NetMod[ThisToolCfgData.NETi].Data[ThisToolCfgData.DATi] 0
Dest ThisTool.Data[DATI].Last 0

Pdm Tool Data Configuration Signed Data Conversion

ThisToolCfgData.Signed

Pdm Tool Sensor Data Last Value

LT
Source A ThisTool.Data[DATI].Last 0
Source B 0

Pdm Tool Sensor Data Last Value

ADD
Source A ThisTool.Data[DATI].Last 0
Source B 65536
Dest ThisTool.Data[DATI].Last 0

Increment Count

DINT allows 2,147,483,647 samples
(6.8 years at 10 samples/second)

7

Pdm Tool Sample Count

ADD
Source A ThisTool.Data[DATI].Count 0
Source B 1
Dest ThisTool.Data[DATI].Count 0

Add Sample to Total

uses REAL to avoid overflow with large Counts
(DINT can only support ~32,000 INT samples -- <1 Hour sampling at 10/second)

8

Pdm Tool Sensor Data Total Value

ADD
Source A ThisTool.Data[DATI].Total 0.0
Source B ThisTool.Data[DATI].Last 0
Dest ThisTool.Data[DATI].Total 0.0

Update Average

9

Pdm Tool Sensor Data
Average Value

DIV	
Source A	ThisTool.Data[DATI].Total 0.0
Source B	ThisTool.Data[DATI].Count 0
Dest	ThisTool.Data[DATI].Avg 0

Update Minimum
(first sample is initial Minimum value)

10

Pdm Tool Sensor Data
Last Value

LT	
Source A	ThisTool.Data[DATI].Last 0
Source B	ThisTool.Data[DATI].Min 0

Pdm Tool Sample
Count

EQ	
Source A	ThisTool.Data[DATI].Count 0
Source B	1

Pdm Tool Sensor Data
Minimum Value

MOVE	
Source	ThisTool.Data[DATI].Last 0
Dest	ThisTool.Data[DATI].Min 0

Update Maximum
(first sample is initial Maximum value)

11

Pdm Tool Sensor Data
Last Value

GT	
Source A	ThisTool.Data[DATI].Last 0
Source B	ThisTool.Data[DATI].Max 0

Pdm Tool Sample
Count

EQ	
Source A	ThisTool.Data[DATI].Count 0
Source B	1

Pdm Tool Sensor Data
Maximum Value

MOVE	
Source	ThisTool.Data[DATI].Last 0
Dest	ThisTool.Data[DATI].Max 0

Copy Local Tag to Array Tag

12 NOP

Exit Routine

lbl_Return
13 LBL NOP

(End)

Store final data array for Data 1-10
for use by the Quality Tool

Index TLi, DATi

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1042
Dest Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 704823770
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 704823770

Exit if Data not Enabled

2

Pdm Tool Enable data collection
ThisTool.Cfg.Data[DATI].Enable

lbl_Return
JMP

MOV Config from Array to local tags - faster than COP by 25%
Use Data Arrays - faster than moving to local tags

AFI'd Enable & Signed - not used, so not copied to local tags

3

AFI

Pdm Tool Enable data collection
ThisTool.Cfg.Data[DATI].Enable

Pdm Tool Data Configuration Enable data collection
ThisToolCfgData.Enable
L

Pdm Tool Enable data collection
ThisTool.Cfg.Data[DATI].Enable
/

Pdm Tool Data Configuration Enable data collection
ThisToolCfgData.Enable
U

AFI

Pdm Tool Signed Data Conversion
ThisTool.Cfg.Data[DATI].Signed

Pdm Tool Data Configuration Signed Data Conversion
ThisToolCfgData.Signed
L

Pdm Tool Signed Data Conversion
ThisTool.Cfg.Data[DATI].Signed
/

Pdm Tool Data Configuration Signed Data Conversion
ThisToolCfgData.Signed
U

Pdm Tool Data Configuration Network Index

MOVE	
Source	ThisTool.Cfg.Data[DATI].NETi -1
Dest	ThisToolCfgData.NETi 22

Pdm Tool Data Configuration Data Index

MOVE	
Source	ThisTool.Cfg.Data[DATI].DATi -1
Dest	ThisToolCfgData.DATi 23

Pdm Tool Data Configuration Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

MOVE	
Source	ThisTool.Cfg.Data[DATI].TypeID -1
Dest	ThisToolCfgData.TypeID 1

Get Network Type (i.e. Banner)
0 is undefined

4

MOVE	
Source	PdM_NetMod[ThisToolCfgData.NETi].Cfg.TypeID 1
Dest	NetTypeID 1

Network Type Configuration Sensor I/O Map configuration

SIZE	
Source	CfgNetMod[0] ??
Dim. To Vary	0
Size	LEN 20

GE	
Source A	NetTypeID 1
Source B	LEN 20

MOVE	
Source	0
Dest	NetTypeID 1

5

Update Scale
Use NetMod TypeID to lookup Scaling for this Data
(default to 1 if configuration is invalid)

Pdm Tool Sensor Data Scaling

MOVE	
Source	1
	0
Dest	ThisTool.Array[DATI].Scale
	0

NE	
Source A	NetTypeID
	1
Source B	0

Pdm Tool Sensor Data Scaling

MOVE	
Source	CfgNetMod[NetTypeID].Data[ThisToolCfgData.DATi].m
	1000
Dest	ThisTool.Array[DATI].Scale
	0

Load Collection Value
(default to Average if illegal Type)

6

Pdm Tool Sensor Data Value

MOVE
Source ThisTool.Data[DATI].Avg 0
Dest ThisTool.Array[DATI].Data 0

Pdm Tool Data Configuration Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

EQ
Source A ThisToolCfgData.TypeID 1
Source B 2

Pdm Tool Sensor Data Value

MOVE
Source ThisTool.Data[DATI].Max 0
Dest ThisTool.Array[DATI].Data 0

Pdm Tool Data Configuration Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

EQ
Source A ThisToolCfgData.TypeID 1
Source B 3

Pdm Tool Sensor Data Value

MOVE
Source ThisTool.Data[DATI].Min 0
Dest ThisTool.Array[DATI].Data 0

Pdm Tool Data Configuration Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

EQ
Source A ThisToolCfgData.TypeID 1
Source B 4

Pdm Tool Sensor Data Value

MOVE
Source ThisTool.Data[DATI].Last 0
Dest ThisTool.Array[DATI].Data 0

Copy Local Tag to Array Tag

7

NOP

Exit Routine

8

lbl_Return
LBL

NOP

(End)

Trend Data

Trend Loop
service all Trends each Scan for fast update of Samples and Triggers

The Trend Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Trend to be scanned continuously.
(WARNING: Disabling the Trend FOR Loop will cause the unscanned Trends to stop updating)

Index TRNi

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1050
Dest Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 704823770
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 704823770

Exit if Trend not Enabled
Clear any existing data from disabled Tools to avoid old data when re-enabled.

2

PdM Trend Array
PdM_Trend[TRNi].Cfg.Enable
/

PdM Trend Array
PdM_Trend[TRNi].Ctl.Start

PdM Trend Array

FLL
Source 0
Dest PdM_Trend[TRNi].Ctl
Length 1

PdM Trend Array

FLL
Source 0
Dest PdM_Trend[TRNi].Data
Length 1

PdM Trend Array

FLL
Source 0
Dest PdM_Trend[TRNi].Array
Length 1

lbl_Return
JMP

Copy Array Tags to Local Tags

3

Pdm Trend

COP
Source PdM_Trend[TRNi]
Dest ThisTrend
Length 1

Trend Data

Get LNi and EQi from EQID
Exit if not a Valid EQID

4

MOD
Source A ThisTrend.Cfg.EQID 967
Source B 100
Dest EQi 71

CPT
Dest LNi 9
Expression (ThisTrend.Cfg.EQID-EQi)/100

LIMIT
Low Limit 0
Test LNi 9
High Limit CFG.LN.mi <EDC.CFG.LN.mi> 28

LIMIT
Low Limit 0
Test EQi 71
High Limit CFG.EQ.mi <EDC.CFG.EQ.mi> 99

TrendValidEQID

PdM Config Trend Processing Error - Invalid EQID

TrendValidEQID /
PdM_Cfg.TrendInvalidEQID (L)

MOVE
Source LNi 9
Dest TrendInvalidEQID_LNi 0

MOVE
Source EQi 71
Dest TrendInvalidEQID_EQi 10

ADD
Source A TrendInvalidEQID_Count 8248
Source B 1
Dest TrendInvalidEQID_Count 8248

lbl_Return (JMP)

v3.03 Exit if Equipment is not Enabled
Clear any existing data from disabled Tools to avoid old data when re-enabled.
(this handles the case when the Equipment is Disabled but the PdM Tool is still Configured)

5

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.Enable /

PdM Trend Array
PdM_Trend[TRNi].Ctl.Start

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNi].Ctl
Length 1

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNi].Data
Length 1

PdM Trend Array
FLL
Source 0
Dest PdM_Trend[TRNi].Array
Length 1

lbl_Return (JMP)

Trend Data

Exit if Sample Trend and QualityMsg EQID is already in use by another Sample Trend
multple Trends with same EQID can be configured to Sample data on the same scan -- need to load these to RingBuffer one-at-a-time

6

Pdm Trend Enable EqDriver Trigger Start/Send: ThisTrend.Cfg.TriggerEnable (/)

Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CTL.TrendMsg.PdmSampleSend

lbl_Return JMP

Get Sample Period ID & RateID
(clear PeriodID if unused to avoid illegal indexes on next rung)

7

Pdm Trend Enable EqDriver Trigger Start/Send: ThisTrend.Cfg.TriggerEnable

Period to send record

CLR
Dest SamplePeriodID
0

Rate to record Trend data

MOVE
Source ThisTrend.Cfg.SampleRateID
14
Dest SampleRateID
14

Pdm Trend Enable EqDriver Trigger Start/Send: ThisTrend.Cfg.TriggerEnable (/)

Period to send record

MOVE
Source ThisTrend.Cfg.SamplePeriodID
0
Dest SamplePeriodID
0

Trend Data

Exit if Sample PeriodID not within array limits

8

Period to send record

LIMIT
Low Limit 1
Test SamplePeriodID 0
High Limit PdM_Cfg.PER.mi 49

Pdm Trend Enable EqDriver Trigger Start/Send
ThisTrend.Cfg.TriggerEnable

TrendSamplePeriodID_OK

PdM Config Trend Processing Error - Invalid Sample Period

TrendSamplePeriodID_OK /
PdM_Cfg.TrendInvalidSamplePeriod L

MOVE
Source SamplePeriodID 0
Dest TrendPeriodIDNotOK_PeriodID 0

ADD
Source A TrendPeriodIDNotOK_Count 0
Source B 1
Dest TrendPeriodIDNotOK_Count 0

lbl_Return JMP

Exit if Sample RateID not within array limits

9

Rate to record Trend data

LIMIT
Low Limit 1
Test SampleRateID 14
High Limit PdM_Cfg.RAT.mi 49

TrendSampleRateID_OK

PdM Config Trend Processing Error - Invalid Sample Rate

TrendSampleRateID_OK /
PdM_Cfg.TrendInvalidSampleRate L

MOVE
Source SampleRateID 14
Dest TrendRateIDNotOK_RateID 0

ADD
Source A TrendRateIDNotOK_Count 0
Source B 1
Dest TrendRateIDNotOK_Count 0

lbl_Return JMP

Trend Data

Get Trend Trigger/Sample Status
if Triggered, use Start/Send signals and Job Data
if Sampled, use Sample Period to generate Start/Send signals

Initialize Sample Rate on Start to clear collection registers for 1st sample

If new Trend Config (PeriodLast = -1), Initialize to the Sample Period without sampling

10

Pdm Trend Enable EqDriver Trigger Start/Send: ThisTrend.Cfg.TriggerEnable

Pdm Trend Trigger - Start data collection (from EqDriver): ThisTrend.Ctl.TriggerStart (U)

Pdm Trend Start data collection: ThisTrend.Ctl.Start (L)

Pdm Trend Start data collection initiation: ThisTrend.Ctl.StartInit (L)

Pdm Trend Trigger - Send data collection (from EqDriver): ThisTrend.Ctl.TriggerSend (U)

Pdm Trend Start data collection: ThisTrend.Ctl.Start (U)

Pdm Trend Enable EqDriver Trigger Start/Send: ThisTrend.Cfg.TriggerEnable (/)

Pdm Trend Start data collection: ThisTrend.Ctl.Start (/)

Pdm Trend Last Sample Period Sent

EQ
- Source A: ThisTrend.Ctl.PeriodLast 0
- Source B: -1

Pdm Trend Last Sample Period Sent

MOVE
- Source: SamplePeriod[SamplePeriodID] 0
- Dest: ThisTrend.Ctl.PeriodLast 0

Sample Periods (minutes)

NE
- Source A: SamplePeriod[SamplePeriodID] 0
- Source B: ThisTrend.Ctl.PeriodLast 0

Pdm Trend Last Sample Period Sent

MOVE
- Source: SamplePeriod[SamplePeriodID] 0
- Dest: ThisTrend.Ctl.PeriodLast 0

Pdm Trend Start data collection: ThisTrend.Ctl.Start (L)

Pdm Trend Start data collection initiation: ThisTrend.Ctl.StartInit (L)

Pdm Trend Start data collection: ThisTrend.Ctl.Start

Pdm Trend Done with data collection: ThisTrend.Ctl.Done

Pdm Trend Start data collection: ThisTrend.Ctl.Start (U)

Pdm Trend Done with data collection: ThisTrend.Ctl.Done (U)

Pdm Trend Save data value: ThisTrend.Ctl.Save (U)

Equipment Data Collection Lines and Equipment Array: EDC.Line[LNi].Equip[EQi].CTL.TrendMsg.PdmSampleSend (L)

Equipment Data Collection Lines and Equipment Array

MOVE
- Source: ThisTrend.Cfg.ToolID 1
- Dest: EDC.Line[LNi].Equip[EQi].CTL.TrendMsg.PdmSampleToolID 0

Trend Data

Initialize Data Collection

11

Pdm Trend Start data collection initiation
ThisTrend.Ctl.StartInit

Pdm Trend Start data collection initiation
ThisTrend.Ctl.StartInit
(U)

Pdm Trend Save data value
ThisTrend.Ctl.Save
(U)

Pdm Trend Done with data collection
ThisTrend.Ctl.Done
(U)

Pdm Trend Reset data collection
ThisTrend.Ctl.Reset
(L)

Pdm Trend Last Sample Rate Saved

MOVE	
Source	SampleRate[SampleRateID] 29
Dest	ThisTrend.Ctl.RateLast 0

Pdm Trend Collection Array Index

CLR	
Dest	ThisTrend.Ctl.ARRi 0

Pdm Trend Collection Array Data Count

CLR	
Dest	ThisTrend.Ctl.DataCount 0

Pdm Trend Trend Data Collection Array

FLL	
Source	0
Dest	ThisTrend.Array
Length	1

Trend Data

Trend Data Collection

collect up to 120 samples at Sample Rate (or until Send Trigger is received)
between each collection, data is accumulated each scan with avg, min and max calculations

12

Pdm Trend Start data collection
ThisTrend.Ctl.Start

Pdm Trend Start data collection initiation
ThisTrend.Ctl.StartInit

Pdm Trend Done with data collection
ThisTrend.Ctl.Done

Sample Rates (seconds) - matches server configuration options

NE	
Source A	SampleRate[SampleRateID] 29
Source B	ThisTrend.Ctl.RateLast 0

Pdm Trend Last Sample Rate Saved

MOVE	
Source	SampleRate[SampleRateID] 29
Dest	ThisTrend.Ctl.RateLast 0

Pdm Trend Save data value
ThisTrend.Ctl.Save
L

Pdm Trend Sensor Data Value

MOVE	
Source	ThisTrend.Data.Avg 0
Dest	ThisTrend.Array.Data[ThisTrend.Ctl.ARRi] 0

Pdm Trend Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

EQ	
Source A	ThisTrend.Cfg.Data.TypeID 3
Source B	2

Pdm Trend Sensor Data Value

MOVE	
Source	ThisTrend.Data.Max 0
Dest	ThisTrend.Array.Data[ThisTrend.Ctl.ARRi] 0

Pdm Trend Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

EQ	
Source A	ThisTrend.Cfg.Data.TypeID 3
Source B	3

Pdm Trend Sensor Data Value

MOVE	
Source	ThisTrend.Data.Min 0
Dest	ThisTrend.Array.Data[ThisTrend.Ctl.ARRi] 0

Pdm Trend Data Type ID (1=Avg, 2=Max, 3=Min, 4=Last)

EQ	
Source A	ThisTrend.Cfg.Data.TypeID 3
Source B	4

Pdm Trend Sensor Data Value

MOVE	
Source	ThisTrend.Data.Last 0
Dest	ThisTrend.Array.Data[ThisTrend.Ctl.ARRi] 0

Trend Data

Trend Collection Done based on Sample Count
EqDriver can also use TriggerSend to Stop/Send Trend before 120 samples are taken

13

Pdm Trend Save data value: ThisTrend.Ctl.Save

ADD
- Source A: ThisTrend.Ctl.ARRi[0]
- Source B: 1
- Dest: ThisTrend.Ctl.ARRi[0]

Pdm Trend Collection Array Data Count

MOVE
- Source: ThisTrend.Ctl.ARRi[0]
- Dest: ThisTrend.Ctl.DataCount[0]

Pdm Trend Collection Array Index

GE
- Source A: ThisTrend.Ctl.ARRi[0]
- Source B: 120

Pdm Trend Done with data collection: ThisTrend.Ctl.Done (L)

Pdm Trend Collection Array Index

MOVE
- Source: 119
- Dest: ThisTrend.Ctl.ARRi[0]

Get Network Type (i.e. Banner)
0 is undefined

14

Pdm Trend Save data value: ThisTrend.Ctl.Save

MOVE
- Source: PdM_NetMod[ThisTrend.Cfg.Data.NETi].Cfg.TypeID[1]
- Dest: NetTypeID[1]

Network Type Configuration Sensor I/O Map configuration

SIZE
- Source: CfgNetMod[0]??
- Dim. To Vary: 0
- Size: LEN[20]

GE
- Source A: NetTypeID[1]
- Source B: LEN[20]

MOVE
- Source: 0
- Dest: NetTypeID[1]

Update Scale
Use NetMod TypeID to lookup Scaling for this Data
(default to 1 if configuration is invalid)

15

Pdm Trend Save data value: ThisTrend.Ctl.Save

Pdm Trend Sensor Data Scaling

MOVE
- Source: 1
- Dest: ThisTrend.Array.Scale[0]

NE
- Source A: NetTypeID[1]
- Source B: 0

Pdm Trend Sensor Data Scaling

MOVE
- Source: CfgNetMod[NetTypeID].Data[ThisTrend.Cfg.Data.DATi].m[100]
- Dest: ThisTrend.Array.Scale[0]

Data Value Save to Array complete

16

Pdm Trend Save data value: ThisTrend.Ctl.Save

Pdm Trend Save data value: ThisTrend.Ctl.Save (U)

Pdm Trend Reset data collection: ThisTrend.Ctl.Reset (L)

Trend Data

Reset Data

17

Pdm Trend Reset data collection
ThisTrend.Ctl.Reset

Pdm Trend Reset data collection
ThisTrend.Ctl.Reset
U

Pdm Trend Trend Raw Data
FLL
Source 0
Dest ThisTrend.Data
Length 1

Data Update
Get Data Sample for this Trend

18

Pdm Trend Start data collection
ThisTrend.Ctl.Start

Pdm Trend Done with data collection
ThisTrend.Ctl.Done
/

JSR
Routine Name _1051_TrendData

Copy Local Tag to Array Tags

19

PdM Trend Array
COP
Source ThisTrend.Ctl
Dest PdM_Trend[TRNi].Ctl
Length 1

PdM Trend Array
COP
Source ThisTrend.Data
Dest PdM_Trend[TRNi].Data
Length 1

PdM Trend Array
COP
Source ThisTrend.Array
Dest PdM_Trend[TRNi].Array
Length 1

Exit Routine

20

lbl_Return
LBL

NOP

(End)

Update Count, Total, Avg, Min and Max for new Data Value
all values are unscaled - scaling will be applied when data is sent to EDC

Index TRNi

0 NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1051
Dest Prog.Routine 1042

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 704823770
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 704823770

Exit if Data not Enabled

Pdm Trend Enable data collection
ThisTrend.Cfg.Data.Enable
2 /

lbl_Return JMP

Copy Array Tags to Local Tags

3 NOP

Exit if Data Config not within array limits

4

Pdm Trend Network Index

LIMIT
Low Limit 0
Test ThisTrend.Cfg.Data.NETi 61
High Limit PdM_Cfg.NET.mi 299

Pdm Trend Data Index

LIMIT
Low Limit 0
Test ThisTrend.Cfg.Data.DATi 168
High Limit PdM_Cfg.NetDAT.mi 249

NetDataOK

NetDataOK /

PdM Config Trend Processing Error - Invalid Network Data

PdM_Cfg.TrendInvalidNetworkData (L)

MOVE
Source TRNi 4
Dest TrendNetDataNotOK_TRNi 0

MOVE
Source ThisTrend.Cfg.Data.NETi 61
Dest TrendNetDataNotOK_NETi 0

MOVE
Source ThisTrend.Cfg.Data.DATi 168
Dest TrendNetDataNotOK_NetDATi 0

ADD
Source A TrendNetDataNotOK_Count 0
Source B 1
Dest TrendNetDataNotOK_Count 0

lbl_Return JMP

TBD -- should Data be ignored if Network Status is not OK
(the data will be 0'd by the PdM Module controller - is there a difference between collecting 0's and not collecting?)

5

AFI

Network Modules

PdM_NetMod[ThisTrend.Cfg.Data.NETi].Ctl.StatusOK /

lbl_Return JMP

Get Data Sample from Network

** SPECIAL CASE ** -- if negative value and NOT signed data, convert 16-bit unsigned INT to 32-bit signed DINT by adding 65,536
(Banner DXM uses unsigned 16-bit integer for all sensors EXCEPT temperature which uses signed 16-bit integer)

NOTE: the unsigned INT correction only works if the result is held in a DINT
The Trend data array is INT[120] - when the DINT values are copied back to the INT array, they return to the original negative values
There is no way to fix this without changing the Trend data packet to TM/SQL.
So instead correcting negative numbers, they must be capped at the maximum positive INT (32,767)

6

Pdm Trend Sensor Data Last Value

MOVE
Source PdM_NetMod[ThisTrend.Cfg.Data.NETi].Data[ThisTrend.Cfg.Data.DATi]
0
Dest ThisTrend.Data.Last
0

Pdm Trend Signed Data Conversion

ThisTrend.Cfg.Data.Signed

Pdm Trend Sensor Data Last Value

LT
Source A ThisTrend.Data.Last
0
Source B 0

AFI

Pdm Trend Sensor Data Last Value

ADD
Source A ThisTrend.Data.Last
0
Source B 65536
Dest ThisTrend.Data.Last
0

Pdm Trend Sensor Data Last Value

MOVE
Source 32767
Dest ThisTrend.Data.Last
0

Increment Count

DINT allows 2,147,483,647 samples
(6.8 years at 10 samples/second)

7

Pdm Trend Sample Count

ADD
Source A ThisTrend.Data.Count
0
Source B 1
Dest ThisTrend.Data.Count
0

Add Sample to Total

uses REAL to avoid overflow with large Counts
(DINT can only support ~32,000 INT samples -- <1 Hour sampling at 10/second)

8

Pdm Trend Sensor
Data Total Value

ADD	
Source A	ThisTrend.Data.Total 0.0
Source B	ThisTrend.Data.Last 0
Dest	ThisTrend.Data.Total 0.0

Update Average

9

Pdm Trend Sensor
Data Average Value

DIV	
Source A	ThisTrend.Data.Total 0.0
Source B	ThisTrend.Data.Count 0
Dest	ThisTrend.Data.Avg 0

Update Minimum
(first sample is initial Minimum value)

10

Pdm Trend Sensor
Data Last Value

LT	
Source A	ThisTrend.Data.Last 0
Source B	ThisTrend.Data.Min 0

Pdm Trend Sample
Count

EQ	
Source A	ThisTrend.Data.Count 0
Source B	1

Pdm Trend Sensor
Data Minimum Value

MOVE	
Source	ThisTrend.Data.Last 0
Dest	ThisTrend.Data.Min 0

Update Maximum
(first sample is initial Maximum value)

11

Pdm Trend Sensor
Data Last Value

GT	
Source A	ThisTrend.Data.Last[0]
Source B	ThisTrend.Data.Max[0]

Pdm Trend Sample
Count

EQ	
Source A	ThisTrend.Data.Count[0]
Source B	1

Pdm Trend Sensor
Data Maximum Value

MOVE	
Source	ThisTrend.Data.Last[0]
Dest	ThisTrend.Data.Max[0]

Copy Local Tag to Array Tag

12

NOP

Exit Routine

lbl_Return

13

LBL

NOP

(End)

Live Data Display
v3.50 - L980 Controller

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion ...
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	1201
Dest	Prog.Routine 1221

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 557231608
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 557231608

2

Line Loop Initial/Terminal values for n Lines/Scan
reduce scan time by updating portion of lines each scan but maintain <500 msec update of HMI data

Loop Increment = updates LineLoopInc+1 Lines/scan

DIV
Source A CFG.LN.n <EDC.CFG.LN.n> 29
Source B 4
Dest LineLoopInc 7

ADD
Source A LineLoopTerm 15
Source B 1
Dest LineLoopInit 0

LT
Source A LineLoopInit 0
Source B 0

GT
Source A LineLoopInit 0
Source B CFG.LN.ei <EDC.CFG.LN.ei> 28

MOVE
Source CFG.LN.si <EDC.CFG.LN.si> 0
Dest LineLoopInit 0

ADD
Source A LineLoopInit 0
Source B LineLoopInc 7
Dest LineLoopTerm 15

LT
Source A LineLoopTerm 15
Source B 0

GT
Source A LineLoopTerm 15
Source B CFG.LN.ei <EDC.CFG.LN.ei> 28

MOVE
Source CFG.LN.si <EDC.CFG.LN.si> 0
Dest LineLoopTerm 15

MOVE
Source CFG.LN.ei <EDC.CFG.LN.ei> 28
Dest LineLoopTerm 15

3

Line Loop
service n Lines each Scan

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop will cause the unscanned Equipment to stop updating)

CFG_DisableLineLoop

FOR
Routine Name _1220_LineLoop
Index LNi 8
Initial Value LineLoopInit
Terminal Value LineLoopTerm
Step Size 1

Line Index

CFG_DisableLineLoop

MOVE
Source _ViewLNi 0
Dest LNi 8

JSR
Routine Name _1220_LineLoop

Line State
Transaction Manager Transaction

4

Line State
Transaction

JSR	
Routine Name	_1221_LineStateTx

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

5

Local Program Scan
Time & Fault Routine
Fault Routine
Executed -- Log in
Transaction Log
Prog.LogFault

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Routine
	702
Dest	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Type
	4
Dest	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Code
	21
Dest	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	Prog.FaultCount
	0
Dest	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	Prog.LOGi
	23
Dest	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1

Log Transactions

JSR	
Routine Name	_901_TransLogSBR
Input Par	9003
Input Par	0
Input Par	LNi
Input Par	EQi
Input Par	0

Local Program Scan
Time & Fault Routine
Fault Routine
Executed -- Log in
Transaction Log
Prog.LogFault
(U)

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

6

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
L

Local Program Scan Time & Fault Routine Program Fault Occurrences
CLR
Dest Prog.FaultCount
0

Program Scan
aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp
<EDC.CLX.ProgScanTemp>

Program Scan Exceeds High Limit (configured for 100 msec)

7

Local Program Scan Time & Fault Routine Last Scan
GT
Source A Prog.Scan.Last
329
Source B CLxScanHighLimit
100000

Program High Scan Count
ADD
Source A ProgScanHighCount
0
Source B
1
Dest ProgScanHighCount
0

Program High Scan Events
CLxProgScanHigh.12
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
<EDC.CLX.ProgScan.Clear>

Program High Scan Count
CLR
Dest ProgScanHighCount
0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

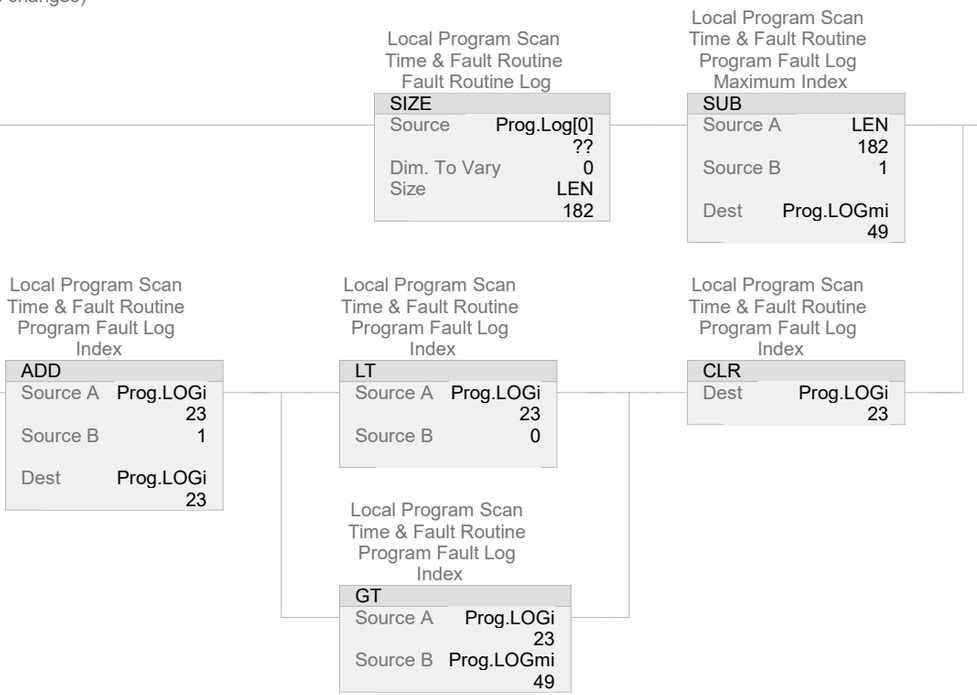
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
4428

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
1221
Dest Prog.Log[Prog.LOGi].Routine
702

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
49925311

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
20220628

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
143141

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
866

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source LNi
8
Dest Prog.Log[Prog.LOGi].Opt1
12

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source EQi
22
Dest Prog.Log[Prog.LOGi].Opt2
2

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt3
4

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
99

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1202
Dest Prog.Routine 1221

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 557231608
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 557231608

5

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 369168272

6

(End)

Line Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Line Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Line to be scanned continuously.
(WARNING: Disabling the Line FOR Loop will cause the unscanned Equipment to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1220
Dest Prog.Routine 1221

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 557231608
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 557231608

Exit if Line not Enabled

2

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].CFG.EN
/

lbl_Exit
JMP

Copy Controller Tags to Local Tags

3

Equipment Data Collection - Line Equipment Data Control

COP
Source EDC.Line[LNi].DataCTL
Dest LineDataCTL
Length 1

4

Initialize Temp Line tags before accumulating worst-case Line, Pitch and Position status
(default Line State is Planned Stop - white; default Pitch and Position status is OK - green)

S:FS

Line-Pitch-Position Status Array

SIZE	
Source	LDD_PositionState ??
Dim. To Vary	1
Size	PITn 60

Line-Pitch End Index

SUB	
Source A	PITn 60
Source B	1
Dest	PITei 59

Line-Pitch-Position Array Size

MUL	
Source A	PITn 60
Source B	PitchPOSn 6
Dest	LinePOSn 360

Line-Pitch-Position Status Array

SIZE	
Source	LDD_PositionState ??
Dim. To Vary	2
Size	PitchPOSn 6

Pitch-Position End Index

SUB	
Source A	PitchPOSn 6
Source B	1
Dest	PitchPOSei 5

MOVE	
Source	0
Dest	TempLineState 4

FLL	
Source	1
Dest	TempLinePitchState[0]
Length	PITn

FLL	
Source	1
Dest	TempLinePosState[0,0]
Length	LinePOSn

Equipment Loop
Only scan the Enabled Equipment on each Line

One set of Interface Routines is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop will cause the unscanned Equipment to stop updating)

5

Enabled Equipment
End Index

MOVE	
Source	EDC.Line[LNi].CFG.EQ.ENei 0
Dest	EquipENei 25

CFG_DisableEqLoop

FOR	
Routine Name	_1230_EquipLoop
Index	EQi 22
Initial Value	0
Terminal Value	EquipENei
Step Size	1

Equipment
Index

CFG_DisableEqLoop

MOVE	
Source	_ViewEQi 0
Dest	EQi 22

JSR	
Routine Name	_1230_EquipLoop

LDD Line KPI State
Line State 3=DT expanded to indicate the KPI item
30=Equip (Default) or 31-35=Option 1-5

6

COP
Source TempLineState
Dest TempKpiLineState
Length 1

EQ
Source A TempKpiLineState 4
Source B 3

MOVE
Source 30
Dest TempKpiLineState 4

Equipment Data Collection - Line Equipment Data Control Config Option 5
LineDataCTL.Option05

MOVE
Source 35
Dest TempKpiLineState 4

Equipment Data Collection - Line Equipment Data Control Config Option 4
LineDataCTL.Option04

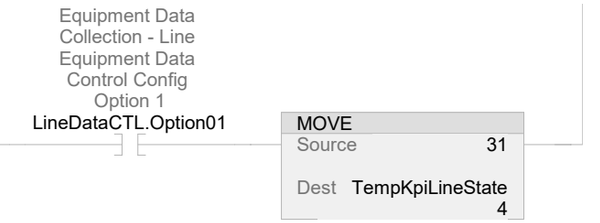
MOVE
Source 34
Dest TempKpiLineState 4

Equipment Data Collection - Line Equipment Data Control Config Option 3
LineDataCTL.Option03

MOVE
Source 33
Dest TempKpiLineState 4

Equipment Data Collection - Line Equipment Data Control Config Option 2
LineDataCTL.Option02

MOVE
Source 32
Dest TempKpiLineState 4


Equipment Data
Collection - Line
Equipment Data
Control Config
Option 1
LineDataCTL.Option01
MOVE
Source
31
Dest
TempKpiLineState
4

7

Update Line Arrays
and set Transaction Request on change of state

NE
Source A TempLineState 4
Source B LDD_LineState[LNi] 0

Line Status Array
MOVE
Source TempLineState 4
Dest LDD_LineState[LNi] 0

NE
Source A TempKpiLineState 4
Source B LDD_LineKpiState[LNi] 0

Line KPI Status Array
MOVE
Source TempKpiLineState 4
Dest LDD_LineKpiState[LNi] 0

Line Index
LIMIT
Low Limit 0
Test LNi 8
High Limit 29

Line State Packet 1 Tx Request
LineStateReq1
L

Line Index
LIMIT
Low Limit 30
Test LNi 8
High Limit 59

Line State Packet 2 Tx Request
LineStateReq2
L

Line Index
LIMIT
Low Limit 60
Test LNi 8
High Limit 89

Line State Packet 3 Tx Request
LineStateReq3
L

Line Index
LIMIT
Low Limit 90
Test LNi 8
High Limit 119

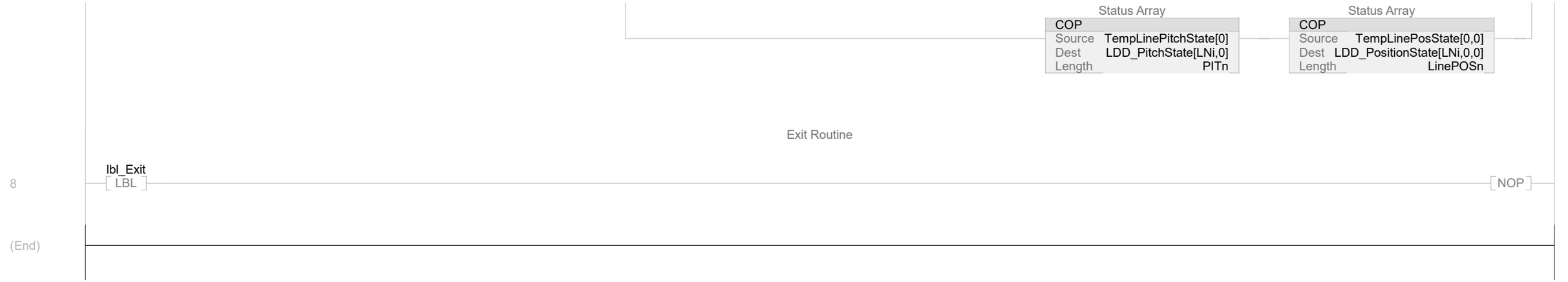
Line State Packet 4 Tx Request
LineStateReq4
L

Line Index
LIMIT
Low Limit 120
Test LNi 8
High Limit 149

Line State Packet 5 Tx Request
LineStateReq5
L

Line-Pitch

Line-Pitch-Position


Status Array
COP
Source TempLinePitchState[0]
Dest LDD_PitchState[LNi,0]
Length PITn
Status Array
COP
Source TempLinePosState[0,0]
Dest LDD_PositionState[LNi,0,0]
Length LinePOSn
Exit Routine
lbl_Exit
LBL
NOP

8

(End)

Line State Transaction

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1221
Dest Prog.Routine 1221

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 557231608
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 557231608

Transaction AntiRepeat Timer
throttles messages for noisy signals - worst case, all 150 Lines update in 1 second
if no Ack from TM/SQL, retry every 5 seconds
(preset is modified by Sent & Sent Ackd below)

2

Line State Transaction Anti-Repeat Timer

TON
Timer LineStateReqTMR
Preset 200
Accum 212
EN
DN

Line State Transaction

Refresh Line State for all Lines
Lines update on Change-of-State; refresh is catch-all to ensure status is active or report Error Event

this could be a bit more clever by detecting if Req has been sent in the past 30 seconds,
but it's not worth the coding effort to avoid sending an extra set of Requests every 30 seconds
(sometimes a hammer is as good as a scalpel)

3

Line Status Refresh Timer
MOVE
Source 30000
Dest LineStateRefreshTMR.PRE 30000

Line Status Refresh Timer
TON
Timer LineStateRefreshTMR
Preset 30000
Accum 11308
EN
DN

Line Status Refresh Timer
LineStateRefreshTMR.DN

Line Status Refresh Timer
LineStateRefreshTMR
RES

Line State Packet 1 Tx Request
LineStateReq1
L

EDC Config Array End Index
GE
Source A CFG.LN.ei <EDC.CFG.LN.ei> 28
Source B 30

Line State Packet 2 Tx Request
LineStateReq2
L

EDC Config Array End Index
GE
Source A CFG.LN.ei <EDC.CFG.LN.ei> 28
Source B 60

Line State Packet 3 Tx Request
LineStateReq3
L

EDC Config Array End Index
GE
Source A CFG.LN.ei <EDC.CFG.LN.ei> 28
Source B 90

Line State Packet 4 Tx Request
LineStateReq4
L

EDC Config Array End Index
GE
Source A CFG.LN.ei <EDC.CFG.LN.ei> 28
Source B 120

Line State Packet 5 Tx Request
LineStateReq5
L

Line State Transaction

Exit if nothing to process

4

LineStateSend | LineStateSent | Line State Packet 1 Tx Request LineStateReq1 | Line State Packet 2 Tx Request LineStateReq2 | Line State Packet 3 Tx Request LineStateReq3 | Line State Packet 4 Tx Request LineStateReq4 | Line State Packet 5 Tx Request LineStateReq5

Line Status Transaction Number
EQ
Source A LDD_LineStateTransNum 5847
Source B LDD_LineStateTransNumAck 5847

Send Packet Index
CLR
Dest LineStateSendPacket 0

lbl_Exit
JMP

Send Packet 1 Request
Lines 0-29

5

Line State Transaction Anti-Repeat Timer LineStateReqTMR.DN | LineStateSend

Send Packet Index
LE
Source A LineStateSendPacket 0
Source B 1

Line State Packet 1 Tx Request LineStateReq1 | Line State Packet 1 Tx Request LineStateReq1 (U) | LineStateSend (L) | Line State Packet 1 Tx Send LineStateSend1 (L)

Line Status Transaction Line Offset
MOVE
Source 0
Dest LDD_LineOffset 0

Line Status Transaction Data
COP
Source LDD_LineKpiState[LDD_LineOffset]
Dest LDD_LineStateSend[0]
Length 30

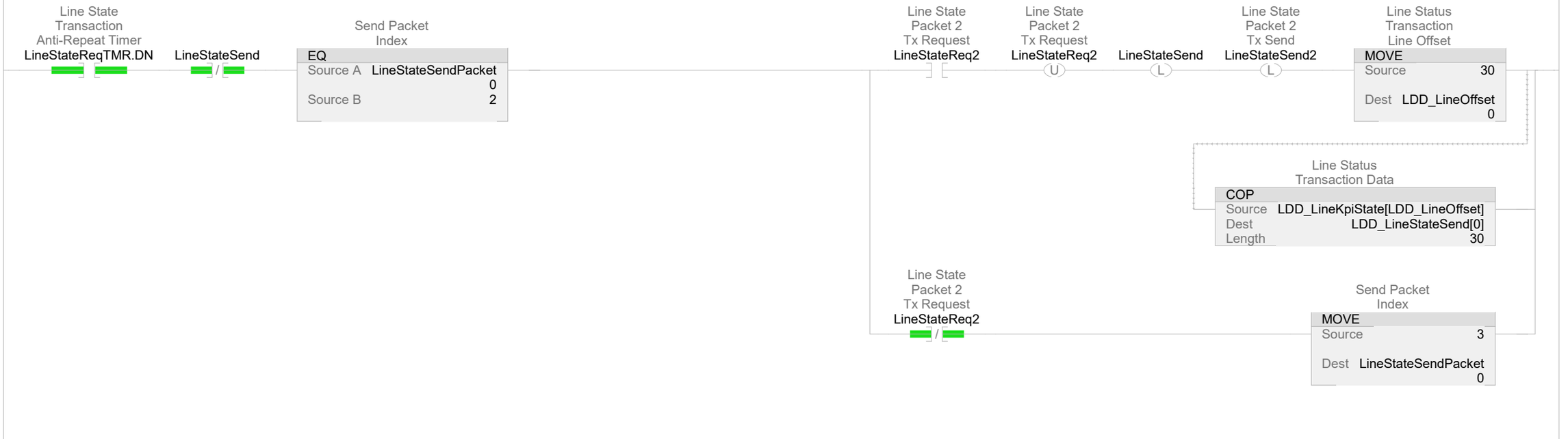
Line State Packet 1 Tx Request LineStateReq1

Send Packet Index
MOVE
Source 2
Dest LineStateSendPacket 0

Line State Transaction

Send Packet 2 Request
Lines 30-59

6



Line State Transaction

Send Packet 3 Request
Lines 60-89

7

Line State Transaction Anti-Repeat Timer
LineStateReqTMR.DN

LineStateSend

Send Packet Index
EQ
Source A: LineStateSendPacket 0
Source B: 3

Line State Packet 3 Tx Request
LineStateReq3

Line State Packet 3 Tx Request
LineStateReq3
U

LineStateSend
L

Line State Packet 3 Tx Send
LineStateSend3
L

Line Status Transaction Line Offset
MOVE
Source: 60
Dest: LDD_LineOffset 0

Line Status Transaction Data
COP
Source: LDD_LineKpiState[LDD_LineOffset]
Dest: LDD_LineStateSend[0]
Length: 30

Line State Packet 3 Tx Request
LineStateReq3

Send Packet Index
MOVE
Source: 4
Dest: LineStateSendPacket 0

Line State Transaction

Send Packet 4 Request
Lines 90-119

8

Line State Transaction Anti-Repeat Timer
LineStateReqTMR.DN

LineStateSend

Send Packet Index
EQ
Source A: LineStateSendPacket 0
Source B: 4

Line State Packet 4 Tx Request
LineStateReq4

Line State Packet 4 Tx Request
LineStateReq4
U

LineStateSend
L

Line State Packet 4 Tx Send
LineStateSend4
L

Line Status Transaction Line Offset
MOVE
Source: 90
Dest: LDD_LineOffset 0

Line Status Transaction Data
COP
Source: LDD_LineKpiState[LDD_LineOffset]
Dest: LDD_LineStateSend[0]
Length: 30

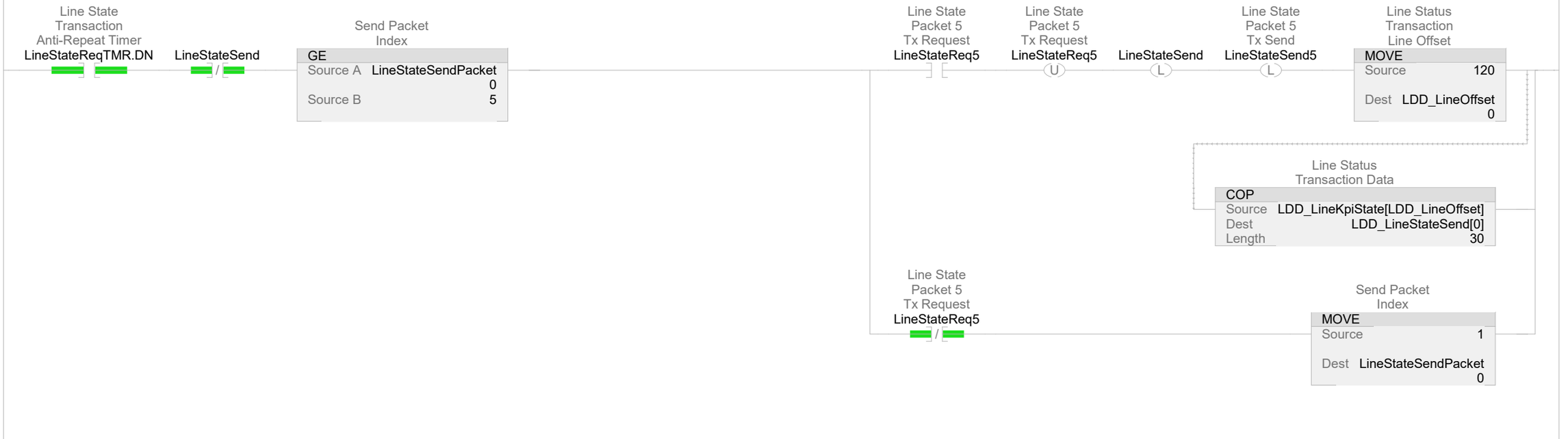
Line State Packet 4 Tx Request
LineStateReq4

Send Packet Index
MOVE
Source: 5
Dest: LineStateSendPacket 0

Line State Transaction

Send Packet 5 Request
Lines 120-149

9



Line State Transaction

Transaction Sent
Increment TransNum to Trigger Transaction
(LineStateReqTMR will resend same message if no ack)

when Msg is Sent, ReqTMR is set to 5 second for retry
when Msg Sent Ack recived, ReqTmr is set to 200 msec to throttle send rate
(150 Lines worst-case update in 1 second - fast enough for LDD)

10

Line State Transaction Anti-Repeat Timer LineStateReqTMR.DN

LineStateSend

LineStateSent (L)

ADD
- Source A: LineStateTransNum 5974
- Source B: 1
- Dest: LineStateTransNum 5974

LT
- Source A: LineStateTransNum 5974
- Source B: 1

GT
- Source A: LineStateTransNum 5974
- Source B: 9999

MOVE
- Source: 1
- Dest: LineStateTransNum 5974

Line Status Transaction Number
MOVE
- Source: LineStateTransNum 5974
- Dest: LDD_LineStateTransNum 5847

Line State Transaction Anti-Repeat Timer LineStateReqTMR (RES)

Line State Transaction Anti-Repeat Timer
MOVE
- Source: 5000
- Dest: LineStateReqTMR.PRE 200

Line State Transaction

Transaction Sent Ack
(SQL/TM returns the Transaction Number if successful)

when Msg is Sent, ReqTMR is set to 5 second for retry
when Msg Sent Ack recived, ReqTmr is set to 200 msec to throttle send rate
(150 Lines worst-case update in 1 second - fast enough for LDD)

LineStateSendPacket is used as a handshake between the packets to share access to the Transaction
the handshake ensures each packet has equal access to the Transaction
after a successfull Send, each Packet increments the Handshake to the next Packet
this prevents one packet from repeatedly Sending while other packet Requests are pending

11

LineStateSent

Line Status
Transaction Number

EQ
Source A LDD_LineStateTransNum 5847
Source B LDD_LineStateTransNumAck 5847

LineStateSent (U)

LineStateSend (U)

Line State
Transaction
Anti-Repeat Timer

MOVE
Source 200
Dest LineStateReqTMR.PRE 200

Line State
Packet 1
Tx Send
LineStateSend1

Line State
Packet 1
Tx Send
LineStateSend1 (U)

Send Packet
Index

MOVE
Source 2
Dest LineStateSendPacket 0

ADD
Source A LineStateSent1 1231391
Source B 1
Dest LineStateSent1 1231391

Line State
Packet 2
Tx Send
LineStateSend2

Line State
Packet 2
Tx Send
LineStateSend2 (U)

Send Packet
Index

MOVE
Source 3
Dest LineStateSendPacket 0

ADD
Source A LineStateSent2 2
Source B 1
Dest LineStateSent2 2

Line State
Packet 3
Tx Send
LineStateSend3

Line State
Packet 3
Tx Send
LineStateSend3 (U)

Send Packet
Index

MOVE
Source 4
Dest LineStateSendPacket 0

ADD
Source A LineStateSent3 2
Source B 1
Dest LineStateSent3 2

Line State
Packet 4
Tx Send

Line State
Packet 4
Tx Send

Send Packet
Index

Line State Transaction

LineStateSend4 LineStateSend4 (U)

MOVE
Source 5
Dest LineStateSendPacket 0

ADD
Source A LineStateSent4 2
Source B 1
Dest LineStateSent4 2

Line State Packet 5 Tx Send LineStateSend5 Line State Packet 5 Tx Send LineStateSend5 (U)

Send Packet Index

MOVE
Source 1
Dest LineStateSendPacket 0

ADD
Source A LineStateSent5 2
Source B 1
Dest LineStateSent5 2

Transaction Sent Error
No Transaction Number Ack received from TM/SQL

12 LineStateSent

MOVE
Source 5000
Dest LineStateSentTMR.PRE 5000

TON
Timer LineStateSentTMR
Preset 5000
Accum 0
EN
DN

LineStateSentTMR.DN

Line Status Transaction Error
LDD_LineStateSendER

Exit Routine

13 lbl_Exit
LBL

NOP

(End)

Equipment Loop

One set of Interfaces is shared by all Equipment
LNi, EQi = Line, Equipment Index

The Equipment Loop can be disabled for Debug & Test Purposes ONLY -- this allows a single Equipment to be scanned continuously.
(WARNING: Disabling the Equipment FOR Loop will cause the unscanned Equipment to stop updating)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1230
Dest Prog.Routine 1221

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 557231608
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 557231608

Exit if Line not Enabled

2

Equipment Data Collection Lines and Equipment Array
EDC.Line[LNi].Equip[EQi].CTL.Config

lbl_Exit
JMP

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

3

Equip Local Tag Equipment Configuration

COP
Source EDC.Line[LNi].Equip[EQi].CFG
Dest Equip.CFG
Length 1

Equip Local Tag Equipment Control

COP
Source EDC.Line[LNi].Equip[EQi].CTL
Dest Equip.CTL
Length 1

Equip Local Tag Equipment Data Control

COP
Source EDC.Line[LNi].Equip[EQi].DataCTL
Dest Equip.DataCTL
Length 1

Equip Callout Local Tag Data - Predefined Status

COP
Source EDC_EqCallout[LNi,EQi]
Dest EqCallout.PreDef
Length 1

Equipment Update

4

LDD State for Equipment

JSR
Routine Name _1231_EquipUpdate

Equipment Position Update
worst-case status of all Equipment in Position

uses EquipState tag from _1231_EquipUpdate

5

LDD State for Equipment Position

JSR
Routine Name _1232_EquipPosUpdate

Equipment Pitch Update
worst-case status of all Equipment in Pitch

uses EquipState tag from _1231_EquipUpdate

6

LDD State for Equipment Pitch

JSR
Routine Name _1233_EquipPitchUpdate

Equipment Line Update
(use Configuration to determine whether to rollup worst-case or use KPI Downtime EQi)

uses EquipState tag from _1231_EquipUpdate

7

Equipment Index

EQ
Source A EQi 22
Source B EDC_CfgGateway.LdtEQi 0

KPI Downtime Equipment LineKpiDT

KPI Downtime Equipment LineKpiDT

Gateway Config Download Line Downtime Rollup Enable
EDC_CfgGateway.LdtRollupEnable

LDD State for Equipment Line

JSR
Routine Name _1234_EquipLineUpdate

Exit Routine

8

lbl_Exit
LBL

NOP

(End)

LDD State for Equipment

Equipment Update

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1231
Dest Prog.Routine 1221

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 557231608
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 557231608

LDD State for Equipment

Load Equipment Display to Temp

0=Not Enabled (Gray)
1=OK (Green)
2=Alarm (Yellow)
3=Fault (Red)
4=InterlockOff (Orange)
5=Alert (Cyan)

2

Equip
Local Tag Equipment
Enabled (Configured
and not Disabled)
Equip.CTL.Enable

MOVE
Source 0
Dest EquipState
1

Equip
Local Tag Equipment
Enabled (Configured
and not Disabled)
Equip.CTL.Enable

Equip
Local Tag Alert
Condition (Lt Blue)
Equip.DataCTL.Alert

Equip
Local Tag Alarm
Condition (Yellow)
Equip.DataCTL.Alarm

Equip
Local Tag Fault
Condition (Red)
Equip.DataCTL.Fault

Equip
Local Tag Interlock
Off Condition
(Orange)
Equip.DataCTL.InterlockOff

MOVE
Source 1
Dest EquipState
1

Equip
Local Tag Alert
Condition (Lt Blue)
Equip.DataCTL.Alert

MOVE
Source 5
Dest EquipState
1

Equip
Local Tag Alarm
Condition (Yellow)
Equip.DataCTL.Alarm

MOVE
Source 2
Dest EquipState
1

Equip
Local Tag Fault
Condition (Red)
Equip.DataCTL.Fault

MOVE
Source 3
Dest EquipState
1

Equip

LDD State for Equipment

Local Tag Interlock
Off Condition
(Orange)
Equip.DataCTL.InterlockOff

MOVE
Source 4
Dest EquipState
1

Update LDD
Equipment State & PreDef

3

Line-Equipment
Status Array

MOVE
Source EquipState
1
Dest LDD_EqState[LNi,EQi]
0

Line-Equipment
PreDefined Event
Array

MOVE
Source EqCallout.PreDef
16#0000_0000
Dest LDD_EqPreDefined[LNi,EQi]
0

(End)

LDD State for Equipment Position

Equipment Position Update
worst-case status of all Equipment in Position

uses EquipState tag from _1231_EquipUpdate

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1231
Dest Prog.Routine 1221

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 557231608
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 557231608

Validate Indexes

2

Equip Local Tag Pitch ID (0-350)

LIMIT
Low Limit 0
Test Equip.CFG.Pitch 15
High Limit PITei 59

Equip Local Tag Position ID (0-5)

LIMIT
Low Limit 0
Test Equip.CFG.Position 2
High Limit PitchPOSei 5

IndexOK

IndexOK
/

lbl_Exit
JMP

LDD State for Equipment Position

Interlock Off

3

EQ
Source A EquipState
1
Source B 4

MOVE
Source 4
Dest TempLinePosState[Equip.CFG.Pitch,Equip.CFG.Position]
1

lbl_Exit
JMP

EQ
Source A TempLinePosState[Equip.CFG.Pitch,Equip.CFG.Position]
1
Source B 4

Fault

4

EQ
Source A EquipState
1
Source B 3

MOVE
Source 3
Dest TempLinePosState[Equip.CFG.Pitch,Equip.CFG.Position]
1

lbl_Exit
JMP

EQ
Source A TempLinePosState[Equip.CFG.Pitch,Equip.CFG.Position]
1
Source B 3

Alarm

5

EQ
Source A EquipState
1
Source B 2

MOVE
Source 2
Dest TempLinePosState[Equip.CFG.Pitch,Equip.CFG.Position]
1

lbl_Exit
JMP

EQ
Source A TempLinePosState[Equip.CFG.Pitch,Equip.CFG.Position]
1
Source B 2

LDD State for Equipment Position

Alert

6

EQ
Source A EquipState
1
Source B 5

MOVE
Source 5
Dest TempLinePosState[Equip.CFG.Pitch,Equip.CFG.Position]
1

lbl_Exit
JMP

EQ
Source A TempLinePosState[Equip.CFG.Pitch,Equip.CFG.Position]
1
Source B 5

OK
(default - initialized at start of line loop)

7

NOP

Exit Routine

lbl_Exit
8 LBL

NOP

(End)

LDD State for Equipment Pitch

Equipment Pitch Update
worst-case status of all Equipment in Pitch

uses EquipState tag from _1231_EquipUpdate

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1232
Dest Prog.Routine 1221

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 557231608
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 557231608

Validate Indexes

2

Equip Local Tag Pitch ID (0-350)

LIMIT
Low Limit 0
Test Equip.CFG.Pitch 15
High Limit PITei 59

IndexOK

IndexOK
/

lbl_Exit
JMP

LDD State for Equipment Pitch

Interlock Off

3

EQ
Source A EquipState
1
Source B 4

MOVE
Source 4
Dest TempLinePitchState[Equip.CFG.Pitch]
1

lbl_Exit
JMP

EQ
Source A TempLinePitchState[Equip.CFG.Pitch]
1
Source B 4

Fault

4

EQ
Source A EquipState
1
Source B 3

MOVE
Source 3
Dest TempLinePitchState[Equip.CFG.Pitch]
1

lbl_Exit
JMP

EQ
Source A TempLinePitchState[Equip.CFG.Pitch]
1
Source B 3

Alarm

5

EQ
Source A EquipState
1
Source B 2

MOVE
Source 2
Dest TempLinePitchState[Equip.CFG.Pitch]
1

lbl_Exit
JMP

EQ
Source A TempLinePitchState[Equip.CFG.Pitch]
1
Source B 2

LDD State for Equipment Pitch

Alert

6

EQ
Source A EquipState
1
Source B 5

MOVE
Source 5
Dest TempLinePitchState[Equip.CFG.Pitch]
1

lbl_Exit
JMP

EQ
Source A TempLinePitchState[Equip.CFG.Pitch]
1
Source B 5

OK
(default - initialized at start of line loop)

7

NOP

Exit Routine

lbl_Exit
8
LBL

NOP

(End)

LDD State for Equipment Line

Equipment Line Update
(use Configuration to determine whether to rollup worst-case or use KPI Downtime EQi)

uses EquipState tag from _1231_EquipUpdate

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1234
Dest Prog.Routine 1221

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 557231608
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 557231608

LDD State for Equipment Line

Determine Equipment Line State

0=No OverAll Start -Pr0
1=Full -Pr26
2=Short -Pr24 or 25
3=Stopped -Pr4 or Pr2 + Pr3
4=Running - Pr1
5=Other - No Pr1 and nothing else
6=Tool Change -Pr28

2

Gateway Config Download Line Downtime Rollup Enable
EDC_CfgGateway.LdtRollupEnable

Equip Local Tag Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

MOVE
Source 0
Dest LineState 4

Equip Local Tag Equipment Enabled (Configured and not Disabled)
Equip.CTL.Enable

Equip Callout Local Tag Data - Predefined Status
EqCallout.PreDef.0

Equip Callout Local Tag Data - Predefined Status
EqCallout.PreDef.1

MOVE
Source 5
Dest LineState 4

Equip Callout Local Tag Data - Predefined Status
EqCallout.PreDef.1

Equip Callout Local Tag Data - Predefined Status
EqCallout.PreDef.2

Equip Callout Local Tag Data - Predefined Status
EqCallout.PreDef.3

MOVE
Source 3
Dest LineState 4

Equip Callout Local Tag Data - Predefined Status
EqCallout.PreDef.4

Equip Callout Local Tag Data - Predefined Status
EqCallout.PreDef.24

MOVE
Source 2
Dest LineState 4

Equip Callout Local Tag Data - Predefined Status
EqCallout.PreDef.25

MOVE
Source 7
Dest LineState 4

LDD State for Equipment Line

Equip Callout Local Tag Data - Predefined Status EqCallout.PreDef.26

MOVE
Source 1
Dest LineState 4

Equip Callout Local Tag Data - Predefined Status EqCallout.PreDef.1

MOVE
Source 4
Dest LineState 4

Equip Callout Local Tag Data - Predefined Status EqCallout.PreDef.0

MOVE
Source 0
Dest LineState 4

Equip Callout Local Tag Data - Predefined Status EqCallout.PreDef.28

MOVE
Source 6
Dest LineState 4

Line State = KPI Downtime Equipment State
(one Equipment - no rollup)

3

Gateway Config Download Line Downtime Rollup Enable EDC_CfgGateway.LdtRollupEnable

MOVE
Source LineState 4
Dest TempLineState 4

lbl_Exit
JMP

LDD State for Equipment Line

Line State = rolllup of all Equipment (rarely or never used)

0=No OverAll Start -Pr0
1=Full -Pr26
2=Short -Pr24 or 25
3=Stopped -Pr4 or Pr2 + Pr3
4=Running - Pr1
5=Other - No Pr1 and nothing else
6=Tool Change -Pr28

4

NOP

Tool Change

5

EQ
Source A LineState 4
Source B 6

MOVE
Source 6
Dest TempLineState 4

lbl_Exit
JMP

EQ
Source A TempLineState 4
Source B 6

No OverAll Start

6

EQ
Source A LineState 4
Source B 0

MOVE
Source 0
Dest TempLineState 4

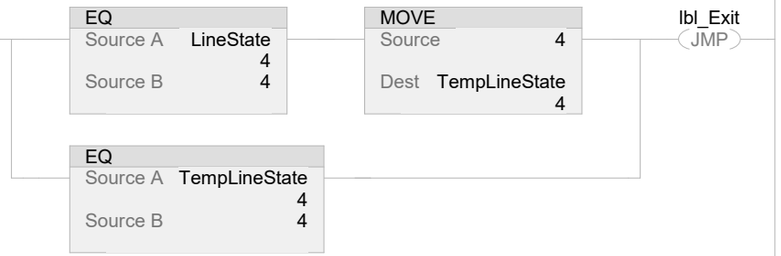
lbl_Exit
JMP

EQ
Source A TempLineState 4
Source B 0

LDD State for Equipment Line

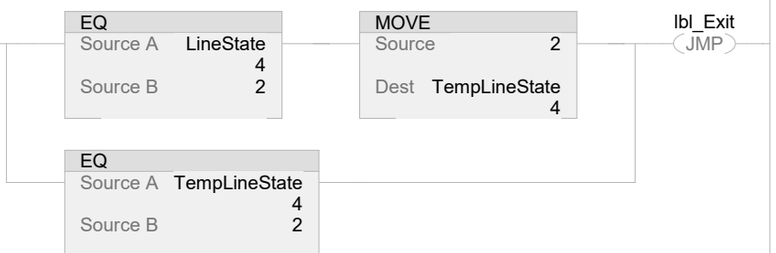
Running

7



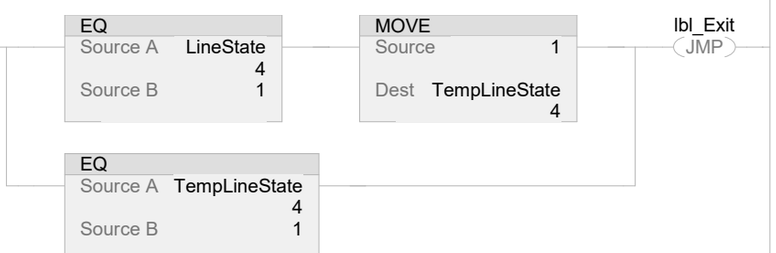
Short

8



Full

9



LDD State for Equipment Line

Stopped

10

EQ
Source A LineState 4
Source B 3

EQ
Source A TempLineState 4
Source B 3

MOVE
Source 3
Dest TempLineState 4

lbl_Exit
JMP

Exit Routine

11

lbl_Exit
LBL

NOP

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
LNi, EQi = Line, Equipment Index
Num = Transaction Number

Transaction Log
Control SBR Input
Parameter --
Transaction Code

0

```
SBR
Input Par   Trans.Code
      <EDC.Trans.Code>
Input Par   Trans.Time
      <EDC.Trans.Time>
Input Par   Trans.LNi
      <EDC.Trans.LNi>
Input Par   Trans.EQi
      <EDC.Trans.EQi>
Input Par   Trans.Num
      <EDC.Trans.Num>
```

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

AFI Prog.Routine to prevent TransLogSBR overwrite of calling Routine - obscures casue of Program Faults

1

AFI

Local Program Scan
Time & Fault Routine
Routine Number
currently Scanned

```
MOVE
Source      901
Dest        Prog.Routine
            1221
```

Local Program Scan
Time & Fault Routine
Routine Scan
Counters (Indexed by
Routine Number)

```
ADD
Source A    Prog.RoutineScan[901]
            5014
Source B    1
Dest        Prog.RoutineScan[901]
            5014
```

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

2

Transaction Log Control Enable Code Filters
Trans.Filter.CodeEN
<EDC.Trans.Filter.CodeEN>

Transaction Log Control Enable LNi Filters
Trans.Filter.LNiEN
<EDC.Trans.Filter.LNiEN>

Transaction Log Control Enable EQi Filters
Trans.Filter.EQiEN
<EDC.Trans.Filter.EQiEN>

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit Trans.Filter.CodeLow
<EDC.Trans.Filter.CodeLow>
0
Test Trans.Code
<EDC.Trans.Code>
2021
High Limit Trans.Filter.CodeHigh
<EDC.Trans.Filter.CodeHigh>
0

Transaction Log Control SBR Input Parameter -- Line Index

LIMIT
Low Limit Trans.Filter.LNiLow
<EDC.Trans.Filter.LNiLow>
0
Test Trans.LNi
<EDC.Trans.LNi>
4
High Limit Trans.Filter.LNiHigh
<EDC.Trans.Filter.LNiHigh>
0

Transaction Log Control SBR Input Parameter -- Equipment Index

LIMIT
Low Limit Trans.Filter.EQiLow
<EDC.Trans.Filter.EQiLow>
0
Test Trans.EQi
<EDC.Trans.EQi>
21
High Limit Trans.Filter.EQiHigh
<EDC.Trans.Filter.EQiHigh>
0

Transaction Log Control Log Record
Trans.Filter.Log
<EDC.Trans.Filter.Log>

lblExit
JMP

Log Transactions

Increment Log Index and check for legal value

3

ADD
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: 1
Dest: Trans.LOGi <EDC.Trans.LOGi> 211

LT
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: 0

GE
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

CLR
Dest: Trans.LOGi <EDC.Trans.LOGi> 211

Generate Log Index for clearing Next Transaction

4

ADD
Source A: Trans.LOGi <EDC.Trans.LOGi> 211
Source B: 1
Dest: Trans.LOGx <EDC.Trans.LOGx> 212

LT
Source A: Trans.LOGx <EDC.Trans.LOGx> 212
Source B: 0

GE
Source A: Trans.LOGx <EDC.Trans.LOGx> 212
Source B: CFG.TransLOG.n <EDC.CFG.TransLOG.n> 250

CLR
Dest: Trans.LOGx <EDC.Trans.LOGx> 212

Log Transactions

Initialize Transaction and Next Transaction

5

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGi]
Length	1

Transaction Log

FLL	
Source	0
Dest	EDC_TransLog[Trans.LOGx]
Length	1

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

6

Transaction Log

MOVE	
Source	Trans.Code
	<EDC.Trans.Code>
	2021
Dest	EDC_TransLog[Trans.LOGi].TransCode
	1011

Transaction Log

MOVE	
Source	Trans.Time
	<EDC.Trans.Time>
	0
Dest	EDC_TransLog[Trans.LOGi].TransTime
	0

Transaction Log

MOVE	
Source	Trans.LNi
	<EDC.Trans.LNi>
	4
Dest	EDC_TransLog[Trans.LOGi].LNi
	15

Transaction Log

MOVE	
Source	Trans.EQi
	<EDC.Trans.EQi>
	21
Dest	EDC_TransLog[Trans.LOGi].EQi
	17

Transaction Log

MOVE	
Source	Trans.Num
	<EDC.Trans.Num>
	4971
Dest	EDC_TransLog[Trans.LOGi].Num
	5631

Transaction Log

MOVE	
Source	CLX.SysTime[8]
	<EDC.CLX.SysTime[8]>
	20261030
Dest	EDC_TransLog[Trans.LOGi].DateStamp
	20261030

Transaction Log

MOVE	
Source	CLX.SysTime[9]
	<EDC.CLX.SysTime[9]>
	151921
Dest	EDC_TransLog[Trans.LOGi].TimeStamp
	152006

Transaction Log

MOVE	
Source	CLX.SysTime[6]
	<EDC.CLX.SysTime[6]>
	904687
Dest	EDC_TransLog[Trans.LOGi].TimeMicroSec
	955653

Transaction Log

MOVE	
Source	CLX.ScanCount
	<EDC.CLX.ScanCount>
	466525176
Dest	EDC_TransLog[Trans.LOGi].ScanCount
	466528353

Store Optional Transaction Details
(loaded before calling Log Routine)

7

Transaction Log

MOVE	
Source	Trans.Opt[1]
	<EDC.Trans.Opt[1]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt1
	1517

Transaction Log

MOVE	
Source	Trans.Opt[2]
	<EDC.Trans.Opt[2]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt2
	0

Transaction Log

MOVE	
Source	Trans.Opt[3]
	<EDC.Trans.Opt[3]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt3
	-1

Transaction Log

MOVE	
Source	Trans.Opt[4]
	<EDC.Trans.Opt[4]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt4
	-1

Transaction Log

MOVE	
Source	Trans.Opt[5]
	<EDC.Trans.Opt[5]>
	-1
Dest	EDC_TransLog[Trans.LOGi].Opt5
	-1

Log Transactions

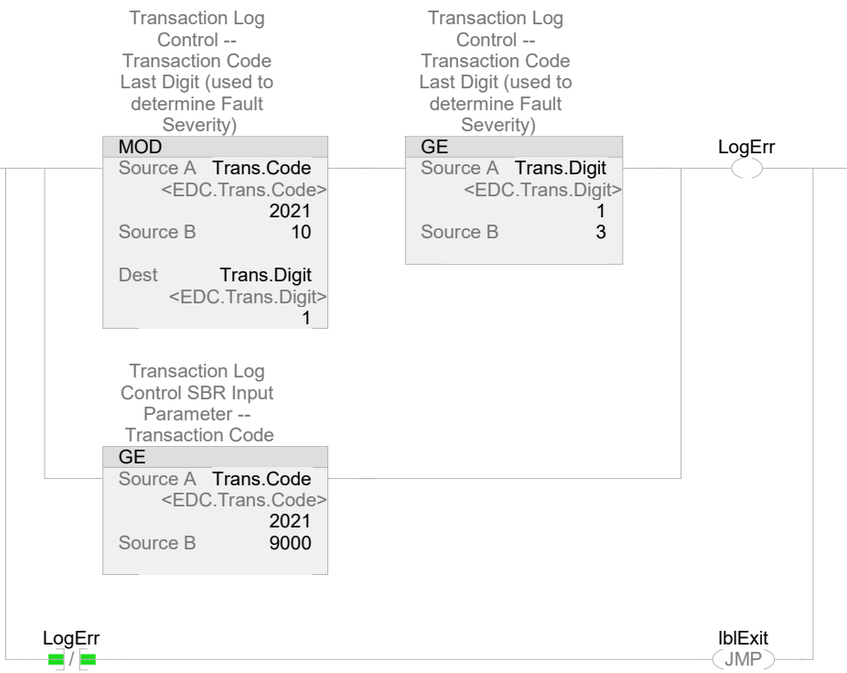
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xxx0 = Enable -- Yellow Status for Normal Operation in Process
xxx1 = Done -- Green Status for Normal Operation Completed
xxx2 = Retry -- Orange Status for Normal Operation in Retry
xxx3 = Error -- Red Status for Normal Operation Failed
xxx4-xxx9 = Other Errors -- Read Status (see TransLog Definitions)

9xxx = Initialization Events

8



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

9

NOP

Log Transactions

Increment Error Log Index and check for legal value

10

Transaction Log Control -- Transaction Error Log Index

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Transaction Log Control -- Transaction Error Log Index

LT
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index

CLR
Dest Trans.ERRi <EDC.Trans.ERRi> 62

Generate Error Log Index for clearing Next Transaction

11

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A Trans.ERRi <EDC.Trans.ERRi> 62
Source B 1
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A Trans.ERRx <EDC.Trans.ERRx> 63
Source B CFG.TransERR.n <EDC.CFG.TransERR.n> 100

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest Trans.ERRx <EDC.Trans.ERRx> 63

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

12

Error Log

COP
Source EDC_TransLog[Trans.LOGi]
Dest EDC_TransErr[Trans.ERRi]
Length 1

Error Log

FLL
Source 0
Dest EDC_TransErr[Trans.ERRx]
Length 1

Exit Routine

13

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

14

Transaction Log
Control SBR External
Load -- Optional
Parameters

FLL
Source -1
Dest Trans.Opt[0]
<EDC.Trans.Opt[0]>
Length Trans.OptN
<EDC.Trans.OptN>

(End)

Main Routine

EDC Equipment Driver
Sends Fault, Quality and Trend to the EDC Gateway

Configure Driver in the following Routines:
_090_USER_Config
_092_USER_LogixRemote
_093_USER_LogixFaults
_094_USER_LogixTools
_095_USER_LogixTrends
_096_USER_LogixCycles

Reference the EDC Driver Configuration document for additional details.

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Main Routine

v1.4 - add Program Version and Time Synch
Manually Toggle UpdateVersion to send Version to Gateway
Manually Toggle UpdateTimeSync to Sync Time from Gateway
v1.11 DHF018 - Refresh EQ Driver Version and Controller Hardware/Firmware Version
v1.16 DHF024 Add Quality Targets
v1.17 DHF029 Add Alert Status to PreDefined Fault Events
v1.18 DHF024 Quality Target Update to support Scaled and Unscaled Target Values
v1.19 DHF031 Duplicate Message EN Inhibit on Startup
v2.00 EDC2 New Features
v2.01 EDC2 PreDefined Changes
v2.02 Trend/Cycle Changes
v2.03 FEC_Share and Toyopuc Share Changes
v2.04 Driver Reset cleared FEC DateTime on EQID change
v2.05 Remove Pre.06 from MaintActive; move Toyopuc TimeSync to FEC
v2.06 TpucEnable - use Toyopuc Module Instance to make Program Uninhibit asynchronous with other Programs
v2.08 - expand Tpuc Tools from 120 to 255
v2.09 - prep for K1A deploy
v2.10 - Tool and Trend Judgement 1-5
v3.02 - Ring Buffer Read Message Delay Time Optimization; Data Refresh
v3.50 Bulk Fault Message (10 Records / Message)

2

S:FS

Equipment Data Collection - User Config Toggle ON to Update Program Version and send to Gateway
UserCfg.UpdateVersion

Message Control Send Data Initiate
VersionMsgCtl.SendInit

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	aoiProgVersion
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Reset / Configuration / Initialization Routines

3

Configuration and Initialization Routine

JSR	
Routine Name	_009_ConfigInit

Main Routine

v2.01 Move Version and TImeSync below ConfigInit
ConfigInit was clearing Version and TimeSync CTL words on initial EQID setup

4

S:FS

Message Control Send Data
VersionMsgCtl.Send
L

Equipment Data Collection - User Config Toggle ON to Update Program Version and send to Gateway
UserCfg.UpdateVersion
U

Equipment Data Collection - User Config Toggle ON to Update Program Version and send to Gateway
UserCfg.UpdateVersion

S:FS

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get
L

Equipment Data Collection - User Config Toggle ON to Update Time Sync from Gateway
UserCfg.UpdateTimeSync
U

Equipment Data Collection - User Config Toggle ON to Update Time Sync from Gateway
UserCfg.UpdateTimeSync

CLx Controller Functions

5

Monitors CLx Status
JSR
Routine Name _030_CLxStatus

Constants
(ASCII Table Initialization, Strings, etc.)

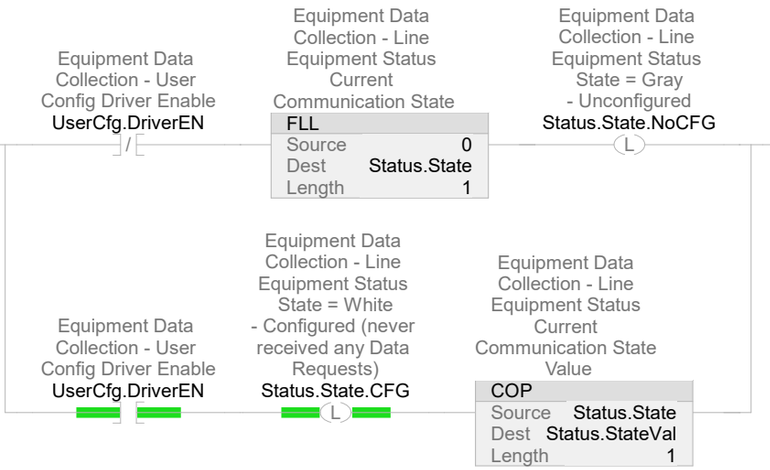
6

Define ASCII Table
JSR
Routine Name _031_AsciiTable

Main Routine

Initialize Status to NoCFG or CFG

7



Main Routine

v1.8 Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)

After power-up, block Driver Messages and monitor the Message control bits for activity
Keep Alive Time (5 minutes) + 30 seconds provides enough time to see activity during non-production windows

8

S:FS

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR
RES

Duplicate Message Detection Control Duplicate Message Delay Timer

ADD	
Source A	CFG.LiveMsgResendTime 31000
Source B	30000
Dest	DuplicateMsg.DelayTMR.PRE 61000

Duplicate Message Detection Control Duplicate Message Delay Timer

TON		
Timer	DuplicateMsg.DelayTMR	EN
Preset	61000	DN
Accum	61000	

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN
/

Listen for Duplicate Message Tag activity before enabling Driver Messages

JSR	
Routine Name	_050_DuplicateMsgDetect

v2.09 Data Refresh signal (daily, pre-shift)
Send all configured Fault Packets to refresh any bits stuck on or off after communication outages, buffer overruns, etc.
(this has mostly been an issue after shutdowns)

EqDriver uses its EQID and the Scan Counter to randomize the sending of the refresh data.
10,000 Scans take 3-4 minutes at ~20msec scan

9

Scheduled Time for Data Refresh 1-shot
DataRefreshTime

Data Refresh Setup (anti-repeat)
DataRefreshSetup
L

Data Refresh Setup (anti-repeat)
DataRefreshSetup

Scan Count 0-9999

MOD	
Source A	CLX.ScanCount 1106690484
Source B	10000
Dest	DataRefreshScanCount 6

Scan Count 0-9999

EQ	
Source A	DataRefreshScanCount 6
Source B	UserCfg.EQID 6

Diagnostic manual reset

ADD	
Source A	DataRefreshCount 97
Source B	1
Dest	DataRefreshCount 97

Data Refresh 1-shot
DataRefresh

Data Refresh Setup (anti-repeat)
DataRefreshSetup
U

Main Routine

User Configuration

Equipment Driver
User Configuration

10

JSR
Routine Name _090_USER_Config

Main Routine

USER Mapping of Data from Driver Source
Toyopuc I/O Module, CIP Remote Messages (i.e. CompactLogix, PLC-5) or embedded driver in Logix

11

Equipment Data Collection - User Config Toyopuc Driver Enable
UserCfg.TpucDriverEN

NOP

Equipment Data Collection - User Config Logix Driver Enable
UserCfg.LogixDriverEN

JSR
Routine Name _092_USER_LogixRemote

Equipment Data Collection - User Config Remote Driver Enable
UserCfg.RemDriverEN

Equipment Data Collection - User Config Logix Driver Enable
UserCfg.LogixDriverEN

Simultation for Fault Data Enable Simulation
SimFault.EN

EqDriver embedded in Logix Controller
JSR
Routine Name _093_USER_LogixFaults

Equipment Data Collection - User Config Quality Message Enable
UserCfg.QualityMsgEN

Equipment Quality/Health Data Mapping
JSR
Routine Name _094_USER_LogixTools

Equipment Data Collection - User Config Trend Message Enable
UserCfg.TrendMsgEN

Equipment Trend Data Mapping
JSR
Routine Name _095_USER_LogixTrends

Equipment Data Collection - User Config Cycle Message Enable
UserCfg.CycleMsgEN

Equipment Cycle Data Mapping
JSR
Routine Name _096_USER_LogixCycles

Equipment Data Collection - User Config Remote Driver Enable
UserCfg.RemDriverEN

NOP

Main Routine

Send Live Callout Data to EDC Gateway
v1.5 Separate EqFault and PreFault routines - scan EqFault first to update before evaluating for Fault/Alarm status
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

Collect Data Packets in reverse order so Packet 0 w/ PreDefined is updated and loaded last to RingBuffer -- after all other packets have been processed
(this should make Fault, Alarm, Alert events Start/Stop times align with the underlying events)

12

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Collect Fault/Status Data
FOR
Routine Name _121_FaultDataCollect
Index PKTi
-1
Initial Value CFG.FaultEq.ei
Terminal Value 0
Step Size -1

Message Control Send Data
CalloutMsgCtl.Send

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend

Fault Ring Buffer Message Control for multiplexed packets Data Change
NE
Source A FaultMsgCtl.Change
0
Source B 0

Callout Data Load for Message
FOR
Routine Name _112_CalloutDataLoad
Index PKTi
-1
Initial Value 0
Terminal Value CFG.FaultEq.ei
Step Size 1

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Callout Data Message
JSR
Routine Name _114_CalloutDataMsg

Main Routine

Send Buffered Fault/Status Data to EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

13

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Fault Message Enable
UserCfg.FaultMsgEN

Load Read Records from RingBuffer to Gateway Message
JSR
Routine Name _123_FaultRingBuffer

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Fault Data Ring Buffer and Message
JSR
Routine Name _124_FaultDataMsg

Send Buffered Quality/Health Data to EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete
v1.16 DHF024 Add Quality Targets
v2.00 Remove Quality Targets - configured on server

14

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Quality Message Enable
UserCfg.QualityMsgEN

Load Read Records from RingBuffer to Gateway Message
JSR
Routine Name _133_QualityRingBuffer

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Quality Data Ring Buffer and Message
JSR
Routine Name _134_QualityDataMsg

Main Routine

Send Buffered Trend Data to EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

15

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Trend Message Enable
UserCfg.TrendMsgEN

Load Read Records from RingBuffer to Gateway Message
JSR
Routine Name _143_TrendRingBuffer

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Trend Data Ring Buffer and Message
JSR
Routine Name _144_TrendDataMsg

Send Version and Sync Time every morning before 1st Shfit
use EQID as a seconds-delay so all Equpment don't sync at same time
v1.10 - Moved Time Sync from 5am to 6am to avoid changing during 2nd Shift
v1.10 Changed EQID multiplier from 1000 to 100 (allow 10 EQ/sec instead of 1 EQ/sec - finishes updates in 3 minutes instead of 30)
v1.11 DHF018 - Refresh EQ Driver Version and Controller Hardware/Firmware Version

16

CLx Module Status CLx System Time (from WALLCLOCKTIME)
EQ
Source A CLX.SysTime[3] 15
Source B 6

CLx Module Status CLx System Time (from WALLCLOCKTIME)
EQ
Source A CLX.SysTime[4] 24
Source B 0

CLx Module Status CLx System Time (from WALLCLOCKTIME)
EQ
Source A CLX.SysTime[5] 52
Source B 0

Time Sync Message Control Get Data Initiate
TimeSyncMsgCtl.GetInit
ONS

Message Control Send Data Initiate
VersionMsgCtl.SendInit

Time Sync Message Control Message Delay Timer
MUL
Source A UserCfg.EQID 6
Source B 100
Dest TimeSyncMsgCtl.DelayTMR.PRE 600

Time Sync Message Control Message Delay Timer
TimeSyncMsgCtl.DelayTMR.EN

Time Sync Message Control Message Delay Timer
TON
Timer TimeSyncMsgCtl.DelayTMR
Preset 600
Accum 0
EN
DN

Main Routine

Send Version and Sync Time every morning before 1st Shfit
use EQID as a seconds-delay so all Equpment don't sync at same time
v1.7 - Time Sync if invalid TransTime to avoid bad Fault Event time stamps
v1.16 DHF024 Add Quality Target Periodic Update
v2.00 Remove Quality Target

17

Time Sync Message Control Message Delay Timer
TimeSyncMsgCtl.DelayTMR.DN

Time Sync Message Control Message Delay Timer
TimeSyncMsgCtl.DelayTMR
RES

Equipment Data Collection - User Config Time Synchronization Message Enable
UserCfg.TimeSyncEN

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get
L

Equipment Data Collection - User Config Version Message Enable
UserCfg.VersionEN

Message Control Send Data
VersionMsgCtl.Send
L

CLx Module Status CLx Transaction Time Invalid
CLX.TransTimeInvalid

Equipment Data Collection - User Config Time Synchronization Message Enable
UserCfg.TimeSyncEN

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get
L

Send Buffered Version Data to EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

18

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Version Message Enable
UserCfg.VersionEN

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Message Control Send Data
VersionMsgCtl.Send

Message Control Write Data
VersionMsgCtl.Write
/

Version Data Load for Message
JSR
Routine Name _152_VersionDataLoad

Logix and Code Version Data Message
JSR
Routine Name _154_VersionDataMsg

Main Routine

Synchronizae Time from EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

19

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Time Synchronization Message Enable
UserCfg.TimeSyncEN

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Time Synchronization Data Load for Message
JSR
Routine Name _162_TimeSyncDataLoad

Time Synchronization Data Message
JSR
Routine Name _164_TimeSyncMsg

Send Buffered Cycle Data to EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

20

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Cycle Message Enable
UserCfg.CycleMsgEN

Load Read Records from RingBuffer to Gateway Message
JSR
Routine Name _173_CycleRingBuffer

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Trend Data Ring Buffer and Message
JSR
Routine Name _174_CycleDataMsg

Fault Event Simulation Mode
blocks EqDriver mapped inputs and generates a bit pattern for PreDefined and EqDefined Faults

21

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Simultation for Fault Data Enable Simulation
SimFault.EN

Simulation of Fault Data
JSR
Routine Name _800_SimFault

Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

22

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Routine
	0
Dest	Trans.Ctl.Opt[1]
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Type
	0
Dest	Trans.Ctl.Opt[2]
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.Log[Prog.LOGi].Fault.Code
	0
Dest	Trans.Ctl.Opt[3]
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.FaultCount
	0
Dest	Trans.Ctl.Opt[4]
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	Prog.LOGi
	0
Dest	Trans.Ctl.Opt[5]
	-1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	903
Input Par	0
Input Par	0

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

23

CLx Module Status Clear Scan History
CLX.ProgScan.Clear

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
(L)

Program Scan

aoiEDC_ProgScan	
aoiEDC_ProgScan	aoiProgScan [...]
Scan	Prog.Scan
ScanTemp	CLX.ProgScanTemp

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

24

Local Program Scan Time & Fault Routine Last Scan

GT
Source A: Prog.Scan.Last 144
Source B: CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A: ProgScanHighCount 0
Source B: 1
Dest: ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.31 (L)

CLx Module Status Clear Scan History

CLX.ProgScan.Clear

Program High Scan Count

CLR
Dest: ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

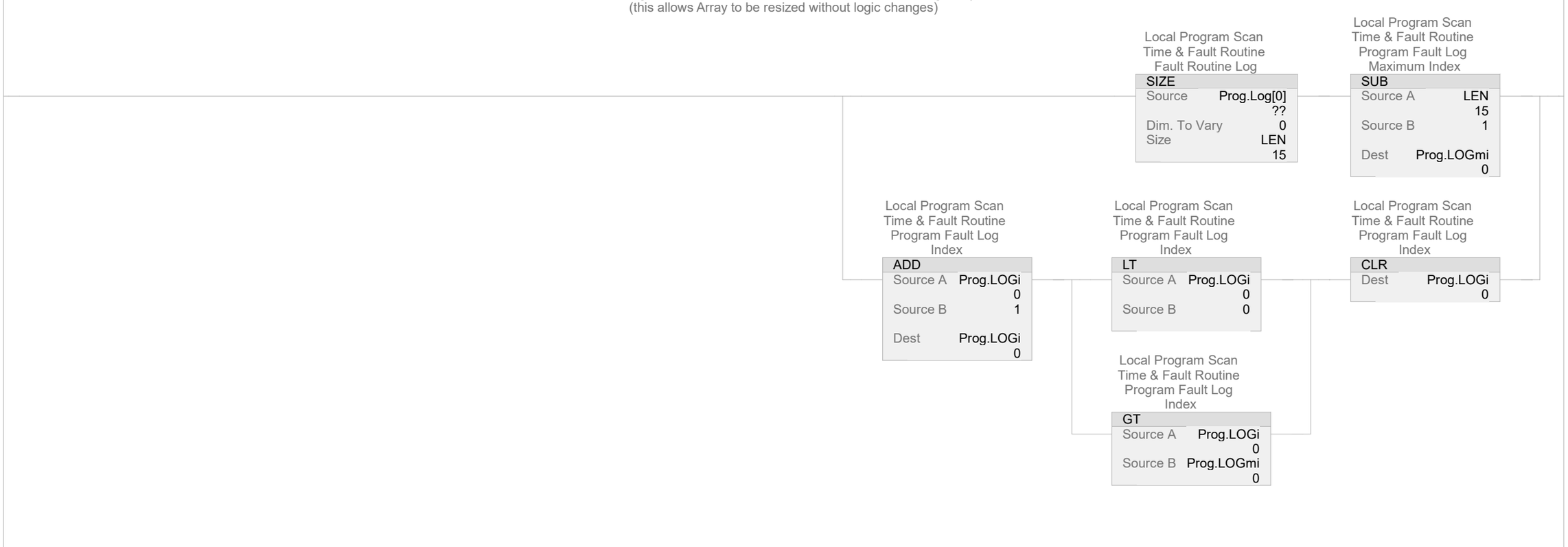
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
154
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
1106690484
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
152452
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
273
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 2
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
/

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration and Initialization Routine

EDC Driver Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 9
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Reset Program Scan Clear after all Programs have been reset

2

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
U

Reset ALL Structures Prior to Configuration & Initialization
v1.4 Move UserCfg.UpdateEQID from Config to Reset to force all Messages to abort and reload updated configuration

3

EDC Config Manual Reset -- Clears all Data Structures (use this to create Clean file before configuration)
CFG.ResetManual

Equipment Data Collection - User Config Toggle ON if EQID Changed
UserCfg.UpdateEQID

Reset all Structures (manually invoked to create clean Master for distribution)

JSR
Routine Name _010_Reset

EDC Config Manual Reset -- Clears all Data Structures (use this to create Clean file before configuration)
CFG.ResetManual
U

Equipment Data Collection - User Config Toggle ON if EQID Changed
UserCfg.UpdateEQID
U

EDC Config Reset ONS (Log Transaction)
CFG.ResetONS

Configuration and Initialization Routine

Configure EDC Parameters
ConfigAuto - set by Download (also performs ConfigInit automatically)
ConfigManual - set manually to force Configuration
v1.4 Move UserCfg.UpdateEQID from Config to Reset to force all Messages to abort and reload updated configuration

4

EDC Config Auto Configuration -- Loads Driver Configuration after Download CFG.ConfigAuto

EDC Config Auto Initialization -- Initializes Driver after Download CFG.InitAuto (L)

EDC Config Manual Configuration -- Loads Driver Configuration CFG.ConfigManual

EDC Config Manual Initialization -- Initializes Driver CFG.InitManual (L)

EDC Config Reset ONS (Log Transaction) CFG.ResetONS

S:FS

EDC Config Auto Configuration -- Loads Driver Configuration after Download CFG.ConfigAuto (U)

EDC Config Manual Configuration -- Loads Driver Configuration CFG.ConfigManual (U)

EDC Config Configuration ONS (Log Transaction) CFG.ConfigONS

Configuration
JSR
Routine Name _011_Config

Initiaize EDC Parameters

5

EDC Config Auto Initialization -- Initializes Driver after Download CFG.InitAuto

EDC Config Manual Initialization -- Initializes Driver CFG.InitManual

EDC Config Reset ONS (Log Transaction) CFG.ResetONS

S:FS

EDC Config Auto Initialization -- Initializes Driver after Download CFG.InitAuto (U)

EDC Config Manual Initialization -- Initializes Driver CFG.InitManual (U)

EDC Config Initialization ONS (Log Transaction) CFG.InitONS

Initialize CLx Module
JSR
Routine Name _012_Initialize

(End)

Reset all Structures
(manually invoked to create clean Master for distribution)

Reset ALL Structures Prior to Configuration & Initialization

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 10
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Reset all Structures
(manually invoked to create clean Master for distribution)

Clear Message Data
v1.3 Replaced CFG value with LEN-1 to set Fault, Quality and Trend ReadPtr - can't rely on CFG values if never set

2

EDC Message Tx Data
FLL
Source 0
Dest EqTxData
<EQ0006_TxData>
Length 1

EDC Message Rx Data
FLL
Source 0
Dest EqRxData
<EQ0006_RxData>
Length 1

Equipment Data Callout/Status Message
FLL
Source 0
Dest CalloutMsg
Length 1

Message Control
FLL
Source 0
Dest CalloutMsgCtl
Length 1

Equipment Data Fault/Status Message
FLL
Source 0
Dest FaultMsg
Length 1

Fault Ring Buffer Message Control for multiplexed packets
FLL
Source 0
Dest FaultMsgCtl
Length 1

EDC Ring Buffer Control
FLL
Source 0
Dest FaultRBCtl
Length 1

Equipment Data Fault/Status Message
SIZE
Source FaultRB[0]
??
Dim. To Vary 0
Size LEN
15

Equipment Data Fault/Status Message
FLL
Source 0
Dest FaultRB[0]
Length LEN

EDC Ring Buffer Control Next Read Record
SUB
Source A LEN
15
Source B 1
Dest FaultRBCtl.ReadPtr
140

Equipment Data Quality/Health Message
FLL
Source 0
Dest QualityMsg
Length 1

Ring Buffer Message Control
FLL
Source 0
Dest QualityMsgCtl
Length 1

EDC Ring Buffer Control
FLL
Source 0
Dest QualityRBCtl
Length 1

Equipment Data Quality/Health Message
SIZE
Source QualityRB[0]
??
Dim. To Vary 0
Size LEN
15

Equipment Data Quality/Health Message
FLL
Source 0
Dest QualityRB[0]
Length LEN

EDC Ring Buffer Control Next Read Record
SUB
Source A LEN
15
Source B 1
Dest QualityRBCtl.ReadPtr
42

Reset all Structures
(manually invoked to create clean Master for distribution)

Equipment Data Trend Message
FLL
Source 0
Dest TrendMsg
Length 1

EDC Gateway Trend Message Control
FLL
Source 0
Dest TrendMsgCtl
Length 1

EDC Ring Buffer Control
FLL
Source 0
Dest TrendRBCtl
Length 1

Equipment Data Trend Message
SIZE
Source TrendRB[0]
??
Dim. To Vary 0
Size LEN
15

Equipment Data Trend Message
FLL
Source 0
Dest TrendRB[0]
Length LEN

EDC Ring Buffer Control Next Read Record
SUB
Source A LEN
15
Source B 1
Dest TrendRBCtl.ReadPtr
49

Equipment Data Cycle Message
FLL
Source 0
Dest CycleMsg
Length 1

Ring Buffer Message Control
FLL
Source 0
Dest CycleMsgCtl
Length 1

EDC Ring Buffer Control
FLL
Source 0
Dest CycleRBCtl
Length 1

Equipment Data Cycle Message
SIZE
Source CycleRB[0]
??
Dim. To Vary 0
Size LEN
15

Equipment Data Cycle Message
FLL
Source 0
Dest CycleRB[0]
Length LEN

EDC Ring Buffer Control Next Read Record
SUB
Source A LEN
15
Source B 1
Dest CycleRBCtl.ReadPtr
19

Equipment Data Version Message Format
FLL
Source 0
Dest VersionMsg
Length 1

Message Control
FLL
Source 0
Dest VersionMsgCtl
Length 1

Time Sync Message Control
FLL
Source 0
Dest TimeSyncMsgCtl
Length 1

Reset all Structures
(manually invoked to create clean Master for distribution)

v1.5 Clear Message Errors
Reset of MsgCtl clears any existing Retry - also need to clear MsgER or Message won't refire.

3



Reset all Structures
(manually invoked to create clean Master for distribution)

Write Message
VersionWriteMsg.EN
<EQ0006_VersionWriteMsg.EN>

Write Message
VersionWriteMsg.ER
<EQ0006_VersionWriteMsg.ER>

Write Message
VersionWriteMsg.ER
<EQ0006_VersionWriteMsg.ER>
(U)

Clear EDC Control Structures
block reset of UserCfg when reset by UpdateEQID -- caused EQID 0000 during Config/Init

4

EDC Config
FLL
Source 0
Dest CFG
Length 1

CLx Module Status
FLL
Source 0
Dest CLX
Length 1

Local Program Scan Time & Fault Routine
FLL
Source 0
Dest Prog
Length 1

Equipment Data Collection - Line Equipment Status
FLL
Source 0
Dest Status
Length 1

Transaction Log Control
FLL
Source 0
Dest Trans
Length 1

Equipment Data Collection - User Config Toggle ON if EQID Changed
UserCfg.UpdateEQID

Equipment Data Collection - User Config
FLL
Source 0
Dest UserCfg
Length 1

Clear Message Data Miscellaneous Structures

5

Fault Ring Buffer Message Control for multiplexed packets Predefined Faults Last
FLL
Source 0
Dest FaultMsgCtl.PreDefLast
Length 1

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined
SIZE
Source FaultMsgCtl.EqDefLast[0]
??
Dim. To Vary 0
Size LEN
15

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined
FLL
Source 0
Dest FaultMsgCtl.EqDef[0]
Length LEN

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined
FLL
Source 0
Dest FaultMsgCtl.EqDefLast[0]
Length LEN

Equipment Data Trend Message Data
FLL
Source 0
Dest TrendCollect
Length 1

Trend Collect Control
FLL
Source 0
Dest TrendCollectCtl
Length 1

Trend Data Collection
SIZE
Source TrendCfg[0]
??
Dim. To Vary 0
Size LEN
15

Trend Data Collection
FLL
Source 0
Dest TrendCfg[0]
Length LEN

Reset all Structures
(manually invoked to create clean Master for distribution)

Clear Transaction Logs

6

Transaction Log Control Transaction Log
SIZE
Source Trans.Err[0] ??
Dim. To Vary 0
Size LEN 15

Transaction Log Control Transaction Log
FLL
Source 0
Dest Trans.Err[0]
Length LEN

Transaction Log Control Transaction Log
FLL
Source 0
Dest Trans.ErrSnap[0]
Length LEN

Transaction Log Control Transaction Log
SIZE
Source Trans.Log[0] ??
Dim. To Vary 0
Size LEN 15

Transaction Log Control Transaction Log
FLL
Source 0
Dest Trans.Log[0]
Length LEN

Transaction Log Control Transaction Log
FLL
Source 0
Dest Trans.LogSnap[0]
Length LEN

v1.8 Duplicate Message Check
v1.10 -- DHF011 UserCfg.UpdateEQID was clearing the DuplicateMsg DnONS and ErONS bits
This caused a nuisance Duplicate Detection if any Msg.DN or Msg.ER bits were on.

7

AFI

Duplicate Message Detection Control
FLL
Source 0
Dest DuplicateMsg
Length 1

Reset all Structures
(manually invoked to create clean Master for distribution)

Initialize Event Masks

8

EqDefined Ignore Mask from Server Equipment Data Fault/Status Equipment Defined

SIZE	
Source	EqDefIgnoreMask[0] ??
Dim. To Vary	0
Size	LEN 15

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

SIZE	
Source	EqDefIgnoreMask[0].Word[0] 16#ffff_ffff
Dim. To Vary	0
Size	LEN2 8

Temp Result for Array Size

MUL	
Source A	LEN 15
Source B	LEN2 8
Dest	LEN 15

EqDefined Ignore Mask from Server Equipment Data Fault/Status Equipment Defined

FLL	
Source	16#ffff_ffff
Dest	EqDefIgnoreMask[0]
Length	LEN

EqDefined Fault Mask from Server Equipment Data Fault/Status Equipment Defined

FLL	
Source	16#ffff_ffff
Dest	EqDefFaultMask[0]
Length	LEN

EqDefined Maint Mask from Server Equipment Data Fault/Status Equipment Defined

FLL	
Source	16#ffff_ffff
Dest	EqDefMaintMask[0]
Length	LEN

EqDefined Alarm Mask from Server Equipment Data Fault/Status Equipment Defined

FLL	
Source	0
Dest	EqDefAlarmMask[0]
Length	LEN

EqDefined Alert Mask from Server Equipment Data Fault/Status Equipment Defined

FLL	
Source	0
Dest	EqDefAlertMask[0]
Length	LEN

(End)

Configuration

EDC Driver Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 11
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Re-enable Fault Handlers

2

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
U

Message Transaction Number Rollover

3

EDC Config Message Transaction Number Rollover

MOVE
Source 9999
Dest CFG.MsgTransNumRollover 9999

Configuration

Message Delay Timer Preset
Message ER Retry Timer Preset

4

EDC Config Message Delay Timer Preset

MOVE	
Source	250
Dest	CFG.MsgDelayTime 250

EDC Config Message Error Retry Timer Preset

MOVE	
Source	5000
Dest	CFG.MsgRetryTime 5000

Ring Buffer Read Message Delay Time Optimization presets

The Limits and Time increase/decrease values should be skewed to quickly increase the delay when Gateway Read notOKs and slowly decrease the delay when Gateway Read OK
The intent is to find the preset that minimizes wasted traffic while maximizing the throughput without too-frequent changes.

5

EDC Config RingBuffer Read OK Count for reducing Delay Time

MOVE	
Source	5
Dest	CfgRB.ReadOkLimit 5

EDC Config RingBuffer Read Not OK Count for increasing Delay Time

MOVE	
Source	2
Dest	CfgRB.ReadNotOkLimit 2

EDC Config RingBuffer Read OK Delay Subtract (msec)

MOVE	
Source	10
Dest	CfgRB.ReadOkSub 10

EDC Config RingBuffer Read Not OK Delay Add (msec)

MOVE	
Source	20
Dest	CfgRB.ReadNotOkAdd 20

EDC Config RingBuffer Read Delay Time Minimum (msec)

MOVE	
Source	100
Dest	CfgRB.ReadDelayMin 100

EDC Config RingBuffer Read Delay Time Maximum (msec)

MOVE	
Source	500
Dest	CfgRB.ReadDelayMax 500

Message Resend Timer Preset
Keep Alive
different times for Live Callout Message and Buffered Fault, Quality, Trend Messages
Callout/Fault are naturally synched by Fault/Status Event changes -- use odd presets to force Keep Alive out of synch
(i.e. 31 seconds and 301 seconds -- not 30 and 300)
v3.02 Reduce Callout Resend from 121 seconds to 31 seconds for faster Mask Download and Data Refresh

6

EDC Config Live Message Keep Alive Resend Timer Preset

MOVE	
Source	31000
Dest	CFG.LiveMsgResendTime 31000

EDC Config Buffer Message Keep Alive Resend Timer Preset

MOVE	
Source	301000
Dest	CFG.BuffMsgResendTime 301000

Configuration

Configure Message Timeout
This forces a Timeout for Unconnected or Connected Timeout failure.
(On occasion, have seen Messages hang indefinitely.)

This should be set higher than Unconnected and Connected Timeouts above.

7

EDC Config CLx Message Timeout (msec)

MOVE
Source 5000
Dest CFG.CLx.MsgTimeout 5000

Configure Unconnected & Connected Timeout for ENET Socket Messages
Unconnected: 4,000,000 = 4.0 seconds (CLx default = 30000000 = 30 seconds)
Connection Rate: 1,000,000 = 4.0 seconds (CLx default = 7500000 = 7.5 seconds)
Connection Rate Timeout Multiplier: 0 = *4 (CLx default = 0 = *4 multiplier)

8

EDC Config CLx Unconnected Message Timeout

MOVE
Source 4000000
Dest CFG.CLx.UC_Timeout 4000000

EDC Config CLx Connected Message Timeout Rate

DIV
Source A CFG.CLx.UC_Timeout 4000000
Source B 4
Dest CFG.CLx.ConnectionRate 1000000

EDC Config CLx Connected Message Timeout Multiplier

MOVE
Source 0
Dest CFG.CLx.ConnectionMult 0

Transaction Log Filters
-- Filters are OFF by default -- manually toggle EN and set filter values for Code, LNi and EQi

9

Transaction Log Control Enable Code Filters
Trans.Ctl.Filter.CodeEN
U

Transaction Log Control Code Filter Low Value

MOVE
Source 0
Dest Trans.Ctl.Filter.CodeLow 0

Transaction Log Control Code Filter High Value

MOVE
Source 0
Dest Trans.Ctl.Filter.CodeHigh 0

Configuration

Fault/Status Equipment Defined Array Indexes
Array for multiplexed data Packets

10

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined

SIZE
Source FaultMsgCtl.EqDef[0] ??
Dim. To Vary 0
Size CFG.FaultEq.n8

EDC Config Array Max Index

SUB
Source A CFG.FaultEq.n8
Source B 1
Dest CFG.FaultEq.mi7

EDC Config Array Start Index

MOVE
Source 0
Dest CFG.FaultEq.si0

EDC Config Array End Index

MOVE
Source 0
Dest CFG.FaultEq.ei2

Fault EqDefined Packet Index

MOVE
Source 0
Dest PKTi-1

Configuration

Fault/Status Equipment Defined Array Indexes
Array for multiplexed data Packets
Use Packet Config BIts to determine End Index

11

lbl_FaultEqLoop
LBL

Equipment Data Collection - User Config Equipment Defined Packet EN bit array
UserCfg.EqDefPacketEN.[PKTi]

EDC Config Array End Index
MOVE
Source PKTi -1
Dest CFG.FaultEq.ei 2

Fault EqDefined Packet Index
ADD
Source A PKTi -1
Source B 1
Dest PKTi -1

Fault EqDefined Packet Index
LE
Source A PKTi -1
Source B CFG.FaultEq.mi 7

lbl_FaultEqLoop
JMP

Fault/Status RB Array Indexes

12

Equipment Data Fault/Status Message
SIZE
Source FaultRB[0] ??
Dim. To Vary 0
Size CFG.FaultRB.n 250

EDC Config Array Max Index
SUB
Source A CFG.FaultRB.n 250
Source B 1
Dest CFG.FaultRB.mi 249

EDC Config Array Start Index
MOVE
Source 0
Dest CFG.FaultRB.si 0

EDC Config Array End Index
MOVE
Source CFG.FaultRB.mi 249
Dest CFG.FaultRB.ei 249

Configuration

Quality/Health RB Array Indexes

13

Equipment Data Quality/Health Message

SIZE
Source QualityRB[0] ??
Dim. To Vary 0
Size CFG.QualityRB.n 100

EDC Config Array Max Index

SUB
Source A CFG.QualityRB.n 100
Source B 1
Dest CFG.QualityRB.mi 99

EDC Config Array Start Index

MOVE
Source 0
Dest CFG.QualityRB.si 0

EDC Config Array End Index

MOVE
Source CFG.QualityRB.mi 99
Dest CFG.QualityRB.ei 99

Trend RB Array Indexes

14

Equipment Data Trend Message

SIZE
Source TrendRB[0] ??
Dim. To Vary 0
Size CFG.TrendRB.n 50

EDC Config Array Max Index

SUB
Source A CFG.TrendRB.n 50
Source B 1
Dest CFG.TrendRB.mi 49

EDC Config Array Start Index

MOVE
Source 0
Dest CFG.TrendRB.si 0

EDC Config Array End Index

MOVE
Source CFG.TrendRB.mi 49
Dest CFG.TrendRB.ei 49

Configuration

Trend Array Indexes

15

Trend Data Collection

SIZE
Source TrendCfg[0] ??
Dim. To Vary 0
Size CFG.Trend.n 21

EDC Config Array Max Index

SUB
Source A CFG.Trend.n 21
Source B 1
Dest CFG.Trend.mi 20

EDC Config Array Start Index

MOVE
Source 1
Dest CFG.Trend.si 1

EDC Config Array End Index

MOVE
Source CFG.Trend.mi 20
Dest CFG.Trend.ei 20

Trend Data Collection Trend Data

SIZE
Source TrendCfg[0].Data.Trend[0] 0
Dim. To Vary 0
Size CFG.TrData.n 120

EDC Config Array Max Index

SUB
Source A CFG.TrData.n 120
Source B 1
Dest CFG.TrData.mi 119

EDC Config Array Start Index

MOVE
Source 1
Dest CFG.TrData.si 1

EDC Config Array End Index

MOVE
Source CFG.TrData.mi 119
Dest CFG.TrData.ei 119

Configuration

Cycle RB Array Indexes

16

Equipment Data Cycle Message

SIZE	
Source	CycleRB[0] ??
Dim. To Vary	0
Size	CFG.CycleRB.n 20

EDC Config Array Max Index

SUB	
Source A	CFG.CycleRB.n 20
Source B	1
Dest	CFG.CycleRB.mi 19

EDC Config Array Start Index

MOVE	
Source	0
Dest	CFG.CycleRB.si 0

EDC Config Array End Index

MOVE	
Source	CFG.CycleRB.mi 19
Dest	CFG.CycleRB.ei 19

Configuration

Cycle Array Indexes

17

Cycle Data Collection

SIZE
Source CycleCfg[0] ??
Dim. To Vary 0
Size CFG.Cycle.n 5

EDC Config Array Max Index

SUB
Source A CFG.Cycle.n 5
Source B 1
Dest CFG.Cycle.mi 4

EDC Config Array Start Index

MOVE
Source 1
Dest CFG.Cycle.si 1

EDC Config Array End Index

MOVE
Source CFG.Cycle.mi 4
Dest CFG.Cycle.ei 4

Cycle Data Collection Step Number (0-65,536)

SIZE
Source CycleCfg[0].Data.Step[0] 0
Dim. To Vary 0
Size CFG.CycData.n 120

EDC Config Array Max Index

SUB
Source A CFG.CycData.n 120
Source B 1
Dest CFG.CycData.mi 119

EDC Config Array Start Index

MOVE
Source 1
Dest CFG.CycData.si 1

EDC Config Array End Index

MOVE
Source CFG.CycData.mi 119
Dest CFG.CycData.ei 119

Configuration

18

Callout Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

EDC Message Tx Data Callout/Status Receive Message

SIZE
Source EqTxData.Callout[0] <EQ0006_TxData.Callout[0]> 9993
Dim. To Vary 0
Size CFG.CalloutMSG.n 14

EDC Config Array Max Index

SUB
Source A CFG.CalloutMSG.n 14
Source B 1
Dest CFG.CalloutMSG.mi 13

EDC Config Array End Count (udt size)

MOVE
Source 14
Dest CFG.CalloutMSG.e 14

EDC Config Array End Count (udt size)

GT
Source A CFG.CalloutMSG.e 14
Source B CFG.CalloutMSG.n 14

EDC Config Array End Count (udt size)

MOVE
Source CFG.CalloutMSG.n 14
Dest CFG.CalloutMSG.e 14

EDC Config Array End Index (used for copies)

SUB
Source A CFG.CalloutMSG.e 14
Source B 1
Dest CFG.CalloutMSG.ei 13

Configuration

Bulk Fault Record & Message Arrays
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

19

EDC Message Tx Data Bulk Fault/Status Receive Message

SIZE
Source EqTxData.Fault[0] <EQ0006_TxData.Fault[0]> 3758
Dim. To Vary 0
Size CFG.FaultMSG.n 143

EDC Config Array Max Index

SUB
Source A CFG.FaultMSG.n 143
Source B 1
Dest CFG.FaultMSG.mi 142

EDC Config Array End Count (udt size)

MOVE
Source 143
Dest CFG.FaultMSG.e 143

EDC Config Array End Count (udt size)

GT
Source A CFG.FaultMSG.e 143
Source B CFG.FaultMSG.n 143

EDC Config Array End Count (udt size)

MOVE
Source CFG.FaultMSG.n 143
Dest CFG.FaultMSG.e 143

EDC Config Array End Index (used for copies)

SUB
Source A CFG.FaultMSG.e 143
Source B 1
Dest CFG.FaultMSG.ei 142

EDC Config Array Size

MOVE
Source 14
Dest CFG.FaultRec.n 14

EDC Config Array Max Index

SUB
Source A CFG.FaultRec.n 14
Source B 1
Dest CFG.FaultRec.mi 13

EDC Config Array End Count (udt size)

MOVE
Source 14
Dest CFG.FaultRec.e 14

EDC Config Array End Count (udt size)

GT
Source A CFG.FaultRec.e 14
Source B CFG.FaultRec.n 14

EDC Config Array End Count (udt size)

MOVE
Source CFG.FaultRec.n 14
Dest CFG.FaultRec.e 14

EDC Config Array End Index (used for copies)

SUB
Source A CFG.FaultRec.e 14

Configuration

Source B	1
Dest	CFG.FaultRec.ei 13

Quality Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array
v1.6 Increased array from 29 to 34 DINTs for 10 Quality Data

20

EDC Message Tx Data Quality/Health Receive Message

SIZE

Source	EqTxData.Quality[0] <EQ0006_TxData.Quality[0]> 4157
Dim. To Vary	0
Size	CFG.QualityMSG.n 39

EDC Config Array Max Index

SUB

Source A	CFG.QualityMSG.n 39
Source B	1
Dest	CFG.QualityMSG.mi 38

EDC Config Array End Count (udt size)

MOVE

Source	39
Dest	CFG.QualityMSG.e 39

EDC Config Array End Count (udt size)

GT

Source A	CFG.QualityMSG.e 39
Source B	CFG.QualityMSG.n 39

EDC Config Array End Count (udt size)

MOVE

Source	CFG.QualityMSG.n 39
Dest	CFG.QualityMSG.e 39

EDC Config Array End Index (used for copies)

SUB

Source A	CFG.QualityMSG.e 39
Source B	1
Dest	CFG.QualityMSG.ei 38

Configuration

Trend Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

21

EDC Message Tx Data Trend Receive Message

SIZE
Source EqTxData.Trend[0] <EQ0006_TxData.Trend[0]>
0
Dim. To Vary 0
Size CFG.TrendMSG.n 70

EDC Config Array Max Index

SUB
Source A CFG.TrendMSG.n 70
Source B 1
Dest CFG.TrendMSG.mi 69

EDC Config Array End Count (udt size)

MOVE
Source 70
Dest CFG.TrendMSG.e 70

EDC Config Array End Count (udt size)

GT
Source A CFG.TrendMSG.e 70
Source B CFG.TrendMSG.n 70

EDC Config Array End Count (udt size)

MOVE
Source CFG.TrendMSG.n 70
Dest CFG.TrendMSG.e 70

EDC Config Array End Index (used for copies)

SUB
Source A CFG.TrendMSG.e 70
Source B 1
Dest CFG.TrendMSG.ei 69

Configuration

22

Version Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

EDC Message Tx Data Version Receive Message

SIZE
Source EqTxData.Version[0] <EQ0006_TxData.Version[0]> 677
Dim. To Vary 0
Size CFG.VersionMSG.n 30

EDC Config Array Max Index

SUB
Source A CFG.VersionMSG.n 30
Source B 1
Dest CFG.VersionMSG.mi 29

EDC Config Array End Count (udt size)

MOVE
Source 30
Dest CFG.VersionMSG.e 30

EDC Config Array End Count (udt size)

GT
Source A CFG.VersionMSG.e 30
Source B CFG.VersionMSG.n 30

EDC Config Array End Count (udt size)

MOVE
Source CFG.VersionMSG.n 30
Dest CFG.VersionMSG.e 30

EDC Config Array End Index (used for copies)

SUB
Source A CFG.VersionMSG.e 30
Source B 1
Dest CFG.VersionMSG.ei 29

Configuration

Cycle Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

23

EDC Message Tx Data Cycle Receive Message

SIZE
Source EqTxData.Cycle[0] <EQ0006_TxData.Cycle[0]>
0
Dim. To Vary 0
Size CFG.CycleMSG.n 134

EDC Config Array Max Index

SUB
Source A CFG.CycleMSG.n 134
Source B 1
Dest CFG.CycleMSG.mi 133

EDC Config Array End Count (udt size)

MOVE
Source 134
Dest CFG.CycleMSG.e 134

EDC Config Array End Count (udt size)

GT
Source A CFG.CycleMSG.e 134
Source B CFG.CycleMSG.n 134

EDC Config Array End Count (udt size)

MOVE
Source CFG.CycleMSG.n 134
Dest CFG.CycleMSG.e 134

EDC Config Array End Index (used for copies)

SUB
Source A CFG.CycleMSG.e 134
Source B 1
Dest CFG.CycleMSG.ei 133

Configuration

Transaction Log
-- TransactionLogEN is OFF by default -- manually toggle ON to log Transactions
-- TransLogAll is OFF by default -- manually toggle ON to log Keep Alive Messages during debug/deployment.
-- Initialize Size
Initialize Transaction Error Log Size

24

EDC Config Enable Transaction Logging
CFG.TransLogEN
U

EDC Config Log all Transactions (includes all Keep Alives)
CFG.TransLogAll
U

Transaction Log Control Transaction Log
SIZE
Source Trans.Log[0] ??
Dim. To Vary 0
Size CFG.TransLOG.n 150

EDC Config Array Max Index
SUB
Source A CFG.TransLOG.n 150
Source B 1
Dest CFG.TransLOG.mi 149

EDC Config Array Start Index
MOVE
Source 0
Dest CFG.TransLOG.si 0

EDC Config Array End Index
MOVE
Source CFG.TransLOG.mi 149
Dest CFG.TransLOG.ei 149

Transaction Log Control Transaction Log
SIZE
Source Trans.Err[0] ??
Dim. To Vary 0
Size CFG.TransERR.n 50

EDC Config Array Max Index
SUB
Source A CFG.TransERR.n 50
Source B 1
Dest CFG.TransERR.mi 49

EDC Config Array Start Index
MOVE
Source 0
Dest CFG.TransERR.si 0

EDC Config Array End Index
MOVE
Source CFG.TransERR.mi 49
Dest CFG.TransERR.ei 49

Transaction Log Control SBR External Load -- Optional Parameters
SIZE
Source Trans.Ctl.Opt[0] -1
Dim. To Vary 0
Size Trans.Ctl.OptN

Configuration

Size Trans.Ctl.OptN
6

Calculate Line and Equipment Indexes from EQID

25

EDC Config Line Index from EQID

CPT
Dest CFG.MsgDest.LNi
0
Expression TRUNC(UserCfg.EQID/100)

EDC Config Equipment Index from EQID

MOD
Source A UserCfg.EQID
6
Source B 100
Dest CFG.MsgDest.EQi
6

Configuration

Gateway Message Destination Tag
EDC_EqRxBuffer.

26

EDC Config Message Remote Destination Tag
MOVE
Source 15
Dest CFG.MsgDest.Tag.LEN 15

EDC Config Message Remote Destination Tag
MOVE
Source ascii.E 'E'
Dest CFG.MsgDest.Tag.DATA[0] 'E'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.D 'D'
Dest CFG.MsgDest.Tag.DATA[1] 'D'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.C 'C'
Dest CFG.MsgDest.Tag.DATA[2] 'C'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.Uscore ' '
Dest CFG.MsgDest.Tag.DATA[3] ' '

EDC Config Message Remote Destination Tag
MOVE
Source ascii.E 'E'
Dest CFG.MsgDest.Tag.DATA[4] 'E'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._q 'q'
Dest CFG.MsgDest.Tag.DATA[5] 'q'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.R 'R'
Dest CFG.MsgDest.Tag.DATA[6] 'R'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._x 'x'
Dest CFG.MsgDest.Tag.DATA[7] 'x'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.B 'B'
Dest CFG.MsgDest.Tag.DATA[8] 'B'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._u 'u'
Dest CFG.MsgDest.Tag.DATA[9] 'u'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._f 'f'
Dest CFG.MsgDest.Tag.DATA[10] 'f'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._f 'f'
Dest CFG.MsgDest.Tag.DATA[11] 'f'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._e 'e'
Dest CFG.MsgDest.Tag.DATA[12] 'e'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._r 'r'
Dest CFG.MsgDest.Tag.DATA[13] 'r'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.Period '.'
Dest CFG.MsgDest.Tag.DATA[14] '.'

Configuration

Gateway Message Destination Tag
EDC_EqRxBufferAux.

27

EDC Config Message Remote Destination Tag
COP
Source CFG.MsgDest.Tag
Dest CFG.MsgDest.TagAux
Length 1

EDC Config Message Remote Destination Tag
MOVE
Source 18
Dest CFG.MsgDest.TagAux.LEN
18

EDC Config Message Remote Destination Tag
MOVE
Source ascii.A
'A'
Dest CFG.MsgDest.TagAux.DATA[14]
'A'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._u
'u'
Dest CFG.MsgDest.TagAux.DATA[15]
'u'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._x
'x'
Dest CFG.MsgDest.TagAux.DATA[16]
'x'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.Period
'.'
Dest CFG.MsgDest.TagAux.DATA[17]
'.'

Gateway Message Destination Tag
EDC_EqRxBufferFault.

28

EDC Config Message Remote Destination Tag
COP
Source CFG.MsgDest.Tag
Dest CFG.MsgDest.TagFault
Length 1

EDC Config Message Remote Destination Tag
MOVE
Source 20
Dest CFG.MsgDest.TagFault.LEN
20

EDC Config Message Remote Destination Tag
MOVE
Source ascii.F
'F'
Dest CFG.MsgDest.TagFault.DATA[14]
'F'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._a
'a'
Dest CFG.MsgDest.TagFault.DATA[15]
'a'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._u
'u'
Dest CFG.MsgDest.TagFault.DATA[16]
'u'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._l
'l'
Dest CFG.MsgDest.TagFault.DATA[17]
'l'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._t
't'
Dest CFG.MsgDest.TagFault.DATA[18]
't'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.Period
'.'
Dest CFG.MsgDest.TagFault.DATA[19]
'.'

Configuration

29

Gateway Message Destination Tag Line[nn].

EDC Config Message Remote Destination Line

MOVE
Source 7
Dest CFG.MsgDest.Line.LEN 8

EDC Config Message Remote Destination Line

MOVE
Source ascii.L 'L'
Dest CFG.MsgDest.Line.DATA[0] 'L'

EDC Config Message Remote Destination Line

MOVE
Source ascii._i 'i'
Dest CFG.MsgDest.Line.DATA[1] 'i'

EDC Config Message Remote Destination Line

MOVE
Source ascii._n 'n'
Dest CFG.MsgDest.Line.DATA[2] 'n'

EDC Config Message Remote Destination Line

MOVE
Source ascii._e 'e'
Dest CFG.MsgDest.Line.DATA[3] 'e'

EDC Config Message Remote Destination Line

MOVE
Source ascii.LtBracket '['
Dest CFG.MsgDest.Line.DATA[4] '['

EDC Config Message Remote Destination Line

MOVE
Source ascii.RtBracket ']'
Dest CFG.MsgDest.Line.DATA[5] '0'

EDC Config Message Remote Destination Line

MOVE
Source ascii.Period '.'
Dest CFG.MsgDest.Line.DATA[6] ']'

EDC Config Line Index from EQID

DTOS
Source CFG.MsgDest.LNi 0
Dest CFG.MsgDest.LineIndex '0'

EDC Config Message Remote Destination Line

INSERT
Source A CFG.MsgDest.Line 'Line[0].'
Source B CFG.MsgDest.LineIndex '0'
Start 6
Dest CFG.MsgDest.Line 'Line[0].'

Configuration

30

Gateway Message Destination Tag
Equip[nn].

EDC Config Message
Remote Destination
Equipment

MOVE
Source 8
Dest CFG.MsgDest.Equip.LEN 9

EDC Config Message
Remote Destination
Equipment

MOVE
Source ascii.E 'E'
Dest CFG.MsgDest.Equip.DATA[0] 'E'

EDC Config Message
Remote Destination
Equipment

MOVE
Source ascii._q 'q'
Dest CFG.MsgDest.Equip.DATA[1] 'q'

EDC Config Message
Remote Destination
Equipment

MOVE
Source ascii._u 'u'
Dest CFG.MsgDest.Equip.DATA[2] 'u'

EDC Config Message
Remote Destination
Equipment

MOVE
Source ascii._i 'i'
Dest CFG.MsgDest.Equip.DATA[3] 'i'

EDC Config Message
Remote Destination
Equipment

MOVE
Source ascii._p 'p'
Dest CFG.MsgDest.Equip.DATA[4] 'p'

EDC Config Message
Remote Destination
Equipment

MOVE
Source ascii.LtBracket '['
Dest CFG.MsgDest.Equip.DATA[5] '['

EDC Config Message
Remote Destination
Equipment

MOVE
Source ascii.RtBracket ']'
Dest CFG.MsgDest.Equip.DATA[6] '6'

EDC Config Message
Remote Destination
Equipment

MOVE
Source ascii.Period '.'
Dest CFG.MsgDest.Equip.DATA[7] ']'

EDC Config Equipment
Index from EQID

DTOS
Source CFG.MsgDest.EQi 6
Dest CFG.MsgDest.EquipIndex '6'

EDC Config Message
Remote Destination
Equipment

INSERT
Source A CFG.MsgDest.Equip 'Equip[6].'
Source B CFG.MsgDest.EquipIndex '6'
Start 7
Dest CFG.MsgDest.Equip 'Equip[6].'

Configuration

31

Gateway Message Destination Tag Callout[0]

EDC Config Message Remote Destination Element

MOVE
Source 10
Dest CFG.MsgDest.Callout.LEN 10

EDC Config Message Remote Destination Element

MOVE
Source ascii.C 'C'
Dest CFG.MsgDest.Callout.DATA[0] 'C'

EDC Config Message Remote Destination Element

MOVE
Source ascii._a 'a'
Dest CFG.MsgDest.Callout.DATA[1] 'a'

EDC Config Message Remote Destination Element

MOVE
Source ascii._l 'l'
Dest CFG.MsgDest.Callout.DATA[2] 'l'

EDC Config Message Remote Destination Element

MOVE
Source ascii._l 'l'
Dest CFG.MsgDest.Callout.DATA[3] 'l'

EDC Config Message Remote Destination Element

MOVE
Source ascii._o 'o'
Dest CFG.MsgDest.Callout.DATA[4] 'o'

EDC Config Message Remote Destination Element

MOVE
Source ascii._u 'u'
Dest CFG.MsgDest.Callout.DATA[5] 'u'

EDC Config Message Remote Destination Element

MOVE
Source ascii._t 't'
Dest CFG.MsgDest.Callout.DATA[6] 't'

EDC Config Message Remote Destination Element

MOVE
Source ascii.LtBracket '['
Dest CFG.MsgDest.Callout.DATA[7] '['

EDC Config Message Remote Destination Element

MOVE
Source ascii._0 '0'
Dest CFG.MsgDest.Callout.DATA[8] '0'

EDC Config Message Remote Destination Element

MOVE
Source ascii.RtBracket ']'
Dest CFG.MsgDest.Callout.DATA[9] ']'

Configuration

32

Gateway Message Destination Tag
Fault[0]

EDC Config Message
Remote Destination
Element

MOVE
Source 8
Dest CFG.MsgDest.Fault.LEN
8

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.F
'F'
Dest CFG.MsgDest.Fault.DATA[0]
'F'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._a
'a'
Dest CFG.MsgDest.Fault.DATA[1]
'a'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._u
'u'
Dest CFG.MsgDest.Fault.DATA[2]
'u'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._l
'l'
Dest CFG.MsgDest.Fault.DATA[3]
'l'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._t
't'
Dest CFG.MsgDest.Fault.DATA[4]
't'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.LtBracket
'['
Dest CFG.MsgDest.Fault.DATA[5]
'['

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._0
'0'
Dest CFG.MsgDest.Fault.DATA[6]
'0'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.RtBracket
']'
Dest CFG.MsgDest.Fault.DATA[7]
']'

Configuration

33

Gateway Message Destination Tag Quality[0]

EDC Config Message Remote Destination Element

MOVE
Source 10
Dest CFG.MsgDest.Quality.LEN 10

EDC Config Message Remote Destination Element

MOVE
Source ascii.Q 'Q'
Dest CFG.MsgDest.Quality.DATA[0] 'Q'

EDC Config Message Remote Destination Element

MOVE
Source ascii._u 'u'
Dest CFG.MsgDest.Quality.DATA[1] 'u'

EDC Config Message Remote Destination Element

MOVE
Source ascii._a 'a'
Dest CFG.MsgDest.Quality.DATA[2] 'a'

EDC Config Message Remote Destination Element

MOVE
Source ascii._l 'l'
Dest CFG.MsgDest.Quality.DATA[3] 'l'

EDC Config Message Remote Destination Element

MOVE
Source ascii._i 'i'
Dest CFG.MsgDest.Quality.DATA[4] 'i'

EDC Config Message Remote Destination Element

MOVE
Source ascii._t 't'
Dest CFG.MsgDest.Quality.DATA[5] 't'

EDC Config Message Remote Destination Element

MOVE
Source ascii._y 'y'
Dest CFG.MsgDest.Quality.DATA[6] 'y'

EDC Config Message Remote Destination Element

MOVE
Source ascii.LtBracket '['
Dest CFG.MsgDest.Quality.DATA[7] '['

EDC Config Message Remote Destination Element

MOVE
Source ascii._0 '0'
Dest CFG.MsgDest.Quality.DATA[8] '0'

EDC Config Message Remote Destination Element

MOVE
Source ascii.RtBracket ']'
Dest CFG.MsgDest.Quality.DATA[9] ']'

Configuration

Gateway Message Destination Tag
Trend[0]

34

EDC Config Message
Remote Destination
Element

MOVE
Source 8
Dest CFG.MsgDest.Trend.LEN
8

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.T
'T'
Dest CFG.MsgDest.Trend.DATA[0]
'T'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._r
'r'
Dest CFG.MsgDest.Trend.DATA[1]
'r'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._e
'e'
Dest CFG.MsgDest.Trend.DATA[2]
'e'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._n
'n'
Dest CFG.MsgDest.Trend.DATA[3]
'n'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._d
'd'
Dest CFG.MsgDest.Trend.DATA[4]
'd'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.LtBracket
'['
Dest CFG.MsgDest.Trend.DATA[5]
'['

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._0
'0'
Dest CFG.MsgDest.Trend.DATA[6]
'0'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.RtBracket
']'
Dest CFG.MsgDest.Trend.DATA[7]
']'

Configuration

35

Gateway Message Destination Tag Version[0]

EDC Config Message Remote Destination Element

MOVE	
Source	10
Dest	CFG.MsgDest.Version.LEN 10

EDC Config Message Remote Destination Element

MOVE	
Source	ascii.V 'V'
Dest	CFG.MsgDest.Version.DATA[0] 'V'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._e 'e'
Dest	CFG.MsgDest.Version.DATA[1] 'e'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._r 'r'
Dest	CFG.MsgDest.Version.DATA[2] 'r'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._s 's'
Dest	CFG.MsgDest.Version.DATA[3] 's'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._i 'i'
Dest	CFG.MsgDest.Version.DATA[4] 'i'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._o 'o'
Dest	CFG.MsgDest.Version.DATA[5] 'o'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._n 'n'
Dest	CFG.MsgDest.Version.DATA[6] 'n'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii.LtBracket '['
Dest	CFG.MsgDest.Version.DATA[7] '['

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._0 '0'
Dest	CFG.MsgDest.Version.DATA[8] '0'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii.RtBracket ']'
Dest	CFG.MsgDest.Version.DATA[9] ']'

Configuration

36

Gateway Message Destination Tag Cycle[0]

EDC Config Message Remote Destination Element

MOVE	
Source	8
Dest	CFG.MsgDest.Cycle.LEN 8

EDC Config Message Remote Destination Element

MOVE	
Source	ascii.C 'C'
Dest	CFG.MsgDest.Cycle.DATA[0] 'C'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._y 'y'
Dest	CFG.MsgDest.Cycle.DATA[1] 'y'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._c 'c'
Dest	CFG.MsgDest.Cycle.DATA[2] 'c'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._l 'l'
Dest	CFG.MsgDest.Cycle.DATA[3] 'l'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._e 'e'
Dest	CFG.MsgDest.Cycle.DATA[4] 'e'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii.LtBracket '['
Dest	CFG.MsgDest.Cycle.DATA[5] '['

EDC Config Message Remote Destination Element

MOVE	
Source	ascii._0 '0'
Dest	CFG.MsgDest.Cycle.DATA[6] '0'

EDC Config Message Remote Destination Element

MOVE	
Source	ascii.RtBracket ']'
Dest	CFG.MsgDest.Cycle.DATA[7] ']'

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Gateway Message Destination Tag Callout

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.Tag 'EDC_EqRxBuffer.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.CalloutDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.CalloutDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.CalloutDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.CalloutDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Callout 'Callout[0]'
Dest	CFG.MsgDest.CalloutDest 'EDC_EqRxBuffer.Line['

Configuration

38

Gateway Message Destination Tag Bulk Fault

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.TagFault 'EDC_EqRxBufferFault.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.FaultDest 'EDC_EqRxBufferFault.'

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.FaultDest 'EDC_EqRxBufferFault.'
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.FaultDest 'EDC_EqRxBufferFault.'

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.FaultDest 'EDC_EqRxBufferFault.'
Source B	CFG.MsgDest.Fault 'Fault[0]'
Dest	CFG.MsgDest.FaultDest 'EDC_EqRxBufferFault.'

39

Gateway Message Destination Tag Quality

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.Tag 'EDC_EqRxBuffer.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.QualityDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.QualityDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.QualityDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.QualityDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Quality 'Quality[0]'
Dest	CFG.MsgDest.QualityDest 'EDC_EqRxBuffer.Line['

40

Gateway Message Destination Tag Trend

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.Tag 'EDC_EqRxBuffer.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.TrendDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.TrendDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.TrendDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.TrendDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Trend 'Trend[0]'
Dest	CFG.MsgDest.TrendDest 'EDC_EqRxBuffer.Line['

Configuration

41

Gateway Message Destination Tag Version

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.Tag 'EDC_EqRxBuffer.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.VersionDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.VersionDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.VersionDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.VersionDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Version 'Version[0]'
Dest	CFG.MsgDest.VersionDest 'EDC_EqRxBuffer.Line['

42

Gateway Message Destination Tag Cycle
** Cycle uses the EDC_EqRxBufferAux tag **

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.TagAux 'EDC_EqRxBufferAux.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.CycleDest 'EDC_EqRxBufferAux.Li'

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.CycleDest 'EDC_EqRxBufferAux.Li'
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.CycleDest 'EDC_EqRxBufferAux.Li'

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.CycleDest 'EDC_EqRxBufferAux.Li'
Source B	CFG.MsgDest.Cycle 'Cycle[0]'
Dest	CFG.MsgDest.CycleDest 'EDC_EqRxBufferAux.Li'

(End)

Initialize CLx Module

Initialize CLx Module

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 12
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Re-enable any Disabled features

2

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
U

Initialize and Start Module & Scan Diagnostics

3

CLx Module Status Clear Scan History
CLX.TaskScan.Clear
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
L

Clear Performance Data

4

Equipment Data Collection - Line Equipment Status

FLL
Source 0
Dest Status
Length 1

Initialize CLx Module

v1.3 Fault Data Change
Initialize Callout/Fault Message Data Packet 0
(ensures good data for Keep Alive when no Faults occur)
v3.02 Init Data Refresh Packets 0-7 (block on first scan when all EqDrv use time-randomized refresh of all packets)

5

S:FS

Data Refresh By Packet DataRefreshPacket.0	Data Refresh By Packet DataRefreshPacket.1	Data Refresh By Packet DataRefreshPacket.2	Data Refresh By Packet DataRefreshPacket.3	Data Refresh By Packet DataRefreshPacket.4	Data Refresh By Packet DataRefreshPacket.5	Data Refresh By Packet DataRefreshPacket.6	Data Refresh By Packet DataRefreshPacket.7
L	L	L	L	L	L	L	L

v1.8 Duplicate Message Check
(only initialize on First Scan -- no need to repeat check for Configuration changes)

v1.10 -- DHF011 Clear entire DuplicateMsg structure to initialize Diagnostic Counts, ONS, etc.
(moved from Reset routine to avoid nuisance DuplicateMsg detects during UserCfg.UpdateEQID)

6

S:FS

Duplicate Message Detection Control

FLL	
Source	0
Dest	DuplicateMsg
Length	1

(End)

Monitors CLx Status

Service CLx Controller Functions

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 30
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

1

Controller Product, Revision, Serial Number, Name
(Note: Revision Major/Minor of 20/11 = v20.011)

2

S:FS

CLx Module Status Controller Type (56=L63; see GSV Help for Lookup Table)

GSV
Class Name ControllerDevice
Instance Name
Attribute Name ProductCode
Dest CLX.Version.ProductCode 309

High Byte = Minor
Low Byte = Major

GSV
Class Name ControllerDevice
Instance Name
Attribute Name ProductRev
Dest ProductRev 16#0b26

CLx Module Status Controller Serial Number (Hex)

GSV
Class Name ControllerDevice
Instance Name
Attribute Name SerialNumber
Dest CLX.Version.SerialNumber 16#016b_5c03

CLx Module Status Controller Name

GSV
Class Name Controller
Instance Name
Attribute Name Name
Dest CLX.Version.Name 'eAssy_EDC_Gway'

CLx Module Status Controller Major Product Revision

CLR
Dest CLX.Version.ProductRevMajor 38

CLx Module Status Controller Minor Product Revision

CLR
Dest CLX.Version.ProductRevMinor 11

CLx Module Status Controller Major Product Revision

BTD
Source ProductRev 16#0b26
Source Bit 0
Dest CLX.Version.ProductRevMajor 38
Dest Bit 0
Length 8

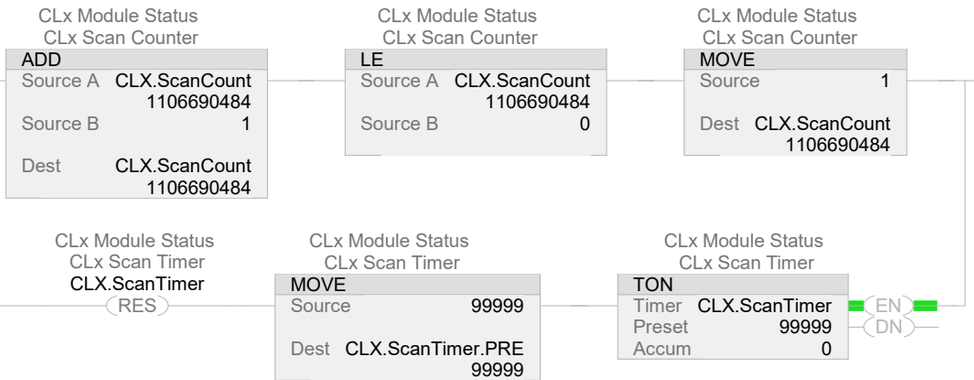
CLx Module Status Controller Minor Product Revision

BTD
Source ProductRev 16#0b26
Source Bit 8
Dest CLX.Version.ProductRevMinor 11
Dest Bit 0
Length 8

Monitors CLx Status

Increment Scan Counter and Initialize Scan Timer
v3.02 Data Refresh only works with positive scan counts

3



Monitors CLx Status

Create DateStamp & TimeStamp DINTs for Transaction Logs
Note: All events on same scan will have same Date/Time Stamp

[0]-[2] = Year YYYY, Month MM, Day DD
[3]-[6] = Hour HH, Minute MM, Second SS, microsec UUUUUU
[7] = YY, [8] = YYYYMMDD, [9] = HHMMSS
[10] = last SS, [11] = msec, [12] = fractional sec .ss

4

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

GSV
Class Name	WallClockTime
Instance Name	
Attribute Name	LocalDateTime
Dest	CLX.SysTime[0] 2026

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DIV
Source A	CLX.SysTime[6] 273675
Source B	1000
Dest	CLX.SysTime[11] 273

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DIV
Source A	CLX.SysTime[11] 273
Source B	10
Dest	CLX.SysTime[12] 27

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

NE
Source A	CLX.SysTime[5] 52
Source B	CLX.SysTime[10] 52

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

MOVE
Source	CLX.SysTime[5] 52
Dest	CLX.SysTime[10] 52

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

MOD
Source A	CLX.SysTime[0] 2026
Source B	100
Dest	CLX.SysTime[7] 26

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

CPT
Dest	CLX.SysTime[8] 20261030
Expression	CLX.SysTime[0]*10000+(CLX.SysTime[1]*100)+CLX.SysTime[2]

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

CPT
Dest	CLX.SysTime[9] 152452
Expression	CLX.SysTime[3]*10000+(CLX.SysTime[4]*100)+CLX.SysTime[5]

Monitors CLx Status

Create DateTime Stamp STRING from SysTime arrays
-- Clear & Initialize to 0's for auto-padding of single-digit Month, Day, etc.
-- Insert each Date/Time field based on string length to use 0-pad

- Date (yyyymmdd)
- Time (hhmmss)
- millisec (sss)

5

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

FLL
Source 0
Dest CLX.DateTime
Length 1

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

MOVE
Source 17
Dest CLX.DateTime.LEN
17

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

FLL
Source ascii._0
Dest CLX.DateTime.DATA[0]
Length CLX.DateTime.LEN

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[8]
20261030
Dest TempString
'390'

Temp Result for
Array Size

SUB
Source A 9
Source B TempString.LEN
3
Dest LEN
15

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

DELETE
Source CLX.DateTime
'20261030152452273'
Qty TempString.LEN
3
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

INSERT
Source A CLX.DateTime
'20261030152452273'
Source B TempString
'390'
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[9]
152452
Dest TempString
'390'

Temp Result for
Array Size

SUB
Source A 15
Source B TempString.LEN
3
Dest LEN
15

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

DELETE
Source CLX.DateTime
'20261030152452273'
Qty TempString.LEN
3
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

INSERT
Source A CLX.DateTime
'20261030152452273'
Source B TempString
'390'
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[11]
273
Dest TempString
'390'

Temp Result for
Array Size

SUB
Source A 18
Source B TempString.LEN
3
Dest LEN
15

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

DELETE
Source CLX.DateTime
'20261030152452273'
Qty TempString.LEN
3
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

CLx Module Status
CLx System DateTime
(YYYYMMDDHHMMSSsss)

INSERT
Source A CLX.DateTime
'20261030152452273'
Source B TempString
'390'
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

Monitors CLx Status

DHF014 -- External Point Status Monitor

Calcluate seconds in hour for use as Heartbeat (0-3600)

6

CLx Module Status
CLx System Time
Seconds in Current
Hour (mm*60)+ss

CPT
Dest CLX.SecondsInHour 1492
Expression CLX.SysTime[4]*60+CLX.SysTime[5]

CLx Transaction Timestamp for Messages
v1.7 - Invalid Transaction Time if Year<2017

7

CLx Module Status
Year (YYYY)

MOVE
Source CLX.SysTime[0] 2026
Dest CLX.TransTime.Year 2026

CLx Module Status
Month (MM)

MOVE
Source CLX.SysTime[1] 10
Dest CLX.TransTime.Month 10

CLx Module Status
Day (DD)

MOVE
Source CLX.SysTime[2] 30
Dest CLX.TransTime.Day 30

CLx Module Status
Hour (HH)

MOVE
Source CLX.SysTime[3] 15
Dest CLX.TransTime.Hour 15

CLx Module Status
Minute (MM)

MOVE
Source CLX.SysTime[4] 24
Dest CLX.TransTime.Min 24

CLx Module Status
Second (SS)

MOVE
Source CLX.SysTime[5] 52
Dest CLX.TransTime.Sec 52

CLx Module Status
Fractional Seconds
(.ss)

MOVE
Source CLX.SysTime[12] 27
Dest CLX.TransTime.FracSec 27

CLx Module Status
Year (YYYY)

LT
Source A CLX.TransTime.Year 2026
Source B 2017

CLx Module Status
CLx Transaction Time
Invalid
CLX.TransTimeInvalid

Monitors CLx Status

CLx Task Scan Time
-- Clear Values
-- Initialize Array Length

8

CLx Module Status
Clear Scan History
CLx.TaskScan.Clear

S:FS

CLx Module Status
CLx Task Scan Time

FLL	
Source	0
Dest	CLX.TaskScan
Length	1

CLx Module Status
Enable Scan Monitor
CLX.TaskScan.EN

L

CLx Module Status
History Array

SIZE	
Source	CLX.TaskScan.History[0]
	15370
Dim. To Vary	0
Size	LEN
	15

CLx Module Status
Avg Scan CONTROL

MOVE	
Source	LEN
	15
Dest	CLX.TaskScan.AvgCTL.LEN
	100

v2.09 Data Refresh signal (daily, pre-shift)
Send all configured Fault Packets to refresh any bits stuck on or off after communication outages, buffer overruns, etc.
(this has mostly been an issue after shutdowns)

DataRefreshTime is located in CLxStatus because it's related to CLxTime and is a common routine for the FEC and Logix EqDriver. (avoid duplicate code in every FEC EqDriver)
each Refresh reloads the RefreshTime configuration

9

CLx Module Status
Hour (HH)

EQ	
Source A	CLX.TransTime.Hour
	15
Source B	DataRefreshHour
	5

CLx Module Status
Minute (MM)

EQ	
Source A	CLX.TransTime.Min
	24
Source B	DataRefreshMin
	30

CLx Module Status
Second (SS)

EQ	
Source A	CLX.TransTime.Sec
	52
Source B	DataRefreshSec
	0

DataRefreshTimeONS

ONS

s:fs

Scheduled Time
for Data Refresh
1-shot
DataRefreshTime

MOVE	
Source	5
Dest	DataRefreshHour
	5

MOVE	
Source	30
Dest	DataRefreshMin
	30

MOVE	
Source	0
Dest	DataRefreshSec
	0

Monitors CLx Status

10

CLx Task Scan Time
-- Store New Value in History Array
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value

CLx Module Status
Enable Scan Monitor
CLX.TaskScan.EN

CLx Module Status
Last Scan

GSV	
Class Name	Task
Instance Name	THIS
Attribute Name	LastScanTime
Dest	CLX.TaskScan.Last 13553

CLx Module Status
History Array

MOVE	
Source	CLX.TaskScan.Last 13553
Dest	CLX.TaskScan.History[CLX.TaskScan.i] 15370

CLx Module Status
Last Scan

GT	
Source A	CLX.TaskScan.Last 13553
Source B	CLX.TaskScan.Max 1883372

CLx Module Status
Max Scan

MOVE	
Source	CLX.TaskScan.Last 13553
Dest	CLX.TaskScan.Max 1883372

CLx Module Status
Last Scan

LT	
Source A	CLX.TaskScan.Last 13553
Source B	CLX.TaskScan.Min 598

CLx Module Status
Last Scan

GT	
Source A	CLX.TaskScan.Last 13553
Source B	0

CLx Module Status
Min Scan

MOVE	
Source	CLX.TaskScan.Last 13553
Dest	CLX.TaskScan.Min 598

CLx Module Status
Min Scan

EQ	
Source A	CLX.TaskScan.Min 598
Source B	0

CLx Module Status
History Array Index

ADD	
Source A	CLX.TaskScan.i 0
Source B	1
Dest	CLX.TaskScan.i 0

CLx Module Status
History Array Index

GE	
Source A	CLX.TaskScan.i 0
Source B	CLX.TaskScan.AvgCTL.LEN 100

CLx Module Status
History Array Index

CLR	
Dest	CLX.TaskScan.i 0

CLx Module Status
History Array Index

LT	
Source A	CLX.TaskScan.i 0
Source B	0

CLx Module Status
Avg Scan CONTROL
CLX.TaskScan.AvgCTL
(RES)

CLx Module Status
Avg Scan CONTROL

AVE		
Array	CLX.TaskScan.History[0]	(EN)
Dim. To Vary	0	
Dest	CLX.TaskScan.Avg 14242	(DN)
Control	CLX.TaskScan.AvgCTL	(ER)
Length	100	
Position	99	

Monitors CLx Status

11

CLx Configuration & Initialization Program Scan Time
-- Clear Values
-- Initialize Array Length

CLx Module Status
Clear Scan History
CLX.ProgScan.Clear

S:FS

CLx Module Status
CLx Total Program
Scan Time (All
Programs)

FLL
Source 0
Dest CLX.ProgScan
Length 1

CLx Module Status
Clear Scan History
CLX.ProgScan.Clear

L

CLx Module Status
Enable Scan Monitor
CLX.ProgScan.EN

L

CLx Module Status
History Array

SIZE
Source CLX.ProgScan.History[0]
148
Dim. To Vary 0
Size LEN
15

CLx Module Status
Avg Scan CONTROL

MOVE
Source LEN
15
Dest CLX.ProgScan.AvgCTL.LEN
100

Monitors CLx Status

CLx Configuration & Initialization Program Scan Time
-- Last Value & History Array are summed by logic in each Program
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value

12

S:FS /

CLx Module Status Enable Scan Monitor
CLX.ProgScan.EN

CLx Module Status Last Scan
MOVE
Source CLX.ProgScanTemp 140
Dest CLX.ProgScan.Last 141

CLx Module Status History Array
MOVE
Source CLX.ProgScan.Last 141
Dest CLX.ProgScan.History[Prog.Scan.i] 144

CLx Module Status Last Scan
GT
Source A CLX.ProgScan.Last 141
Source B CLX.ProgScan.Max 305

CLx Module Status Max Scan
MOVE
Source CLX.ProgScan.Last 141
Dest CLX.ProgScan.Max 305

CLx Module Status Last Scan
LT
Source A CLX.ProgScan.Last 141
Source B CLX.ProgScan.Min 47

CLx Module Status Last Scan
GT
Source A CLX.ProgScan.Last 141
Source B 0

CLx Module Status Min Scan
MOVE
Source CLX.ProgScan.Last 141
Dest CLX.ProgScan.Min 47

CLx Module Status Min Scan
EQ
Source A CLX.ProgScan.Min 47
Source B 0

CLx Module Status History Array Index
ADD
Source A CLX.ProgScan.i 99
Source B 1
Dest CLX.ProgScan.i 99

CLx Module Status History Array Index
GE
Source A CLX.ProgScan.i 99
Source B CLX.ProgScan.AvgCTL.LEN 100

CLx Module Status History Array Index
CLR
Dest CLX.ProgScan.i 99

CLx Module Status History Array Index
LT
Source A CLX.ProgScan.i 99
Source B 0

CLx Module Status Avg Scan CONTROL
CLX.ProgScan.AvgCTL
RES

CLx Module Status Avg Scan CONTROL
AVE
Array CLX.ProgScan.History[0]
Dim. To Vary 0
Dest CLX.ProgScan.Avg 143
Control CLX.ProgScan.AvgCTL
Length 100
Position 0
EN
DN
ER

Monitors CLx Status

13

Log Transactions from Configuration Program

(Clear Trans.LOGi when TransLog is initialized on Reset)

EDC Config Reset ONS
(Log Transaction)
CFG.ResetONS

Transaction Log Control -- Transaction Log Index
CLR
Dest Trans.Ctl.LOGi
0

Transaction Log Control -- Transaction Error Log Index
CLR
Dest Trans.Ctl.ERRi
0

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 9000
Input Par 0
Input Par -1

EDC Config Configuration ONS
(Log Transaction)
CFG.ConfigONS

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 9001
Input Par 0
Input Par -1

EDC Config Initialization ONS
(Log Transaction)
CFG.InitONS

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 9002
Input Par 0
Input Par -1

14

DHF010 -- Log Transaction for Task Scan Exceeds Threshold (hard-coded to 100 msec)

CLx Module Status Last Scan
GT
Source A CLX.TaskScan.Last 13553
Source B 100000

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source CLX.TaskScan.Last 13553
Dest Trans.Ctl.Opt[1] -1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 9004
Input Par 0
Input Par -1

Monitors CLx Status

Clear Total Program Scan & History Array for next Scan values

CLx Module Status
CLx Total Program
Scan Time - Temp
Summation

CLR	
Dest	CLX.ProgScanTemp
	140

15

(End)

Define ASCII Table

ASCII Table Enumeration
2008 Aug 22 -- Initial Release by Dan Guendelsberger (Rockwell Automation)
2009 Jun 11 -- Updated by Dan Guendelsberger
2010 Apr 08 -- Converted to Ladder Logic by Dan Guendelsberger

Creates udt-based ASCII Table for use in enumerations to avoid ASCII table lookups (and mistakes).
ascii._1 = 48
ascii._2 = 49
etc.
ascii.A = 65
ascii.B = 66
etc.
ascii._a = 97
ascii._b = 98
etc.

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 31
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Skip Value Initialize Loop if already initialized
2009Jun11 -- execute loop once to save scan time -- about 1 msec

2

ASCII Table Enumeration Control Structure SINT Array of Values 0-127 for loading ASCII udt

EQ
Source A asciiTable.Values[127] 127
Source B 127

lbl_InitDone
JMP

Define ASCII Table

Reset Index for Initialization Loop

3

ASCII Table Enumeration Control Structure Table Loop Index

CLR
Dest asciiTable.LOOPi 128

Initialize value array (0-127)

4

lbl_Init
LBL

ASCII Table Enumeration Control Structure SINT Array of Values 0-127 for loading ASCII udt

MOVE
Source asciiTable.LOOPi 128
Dest asciiTable.Values[asciiTable.LOOPi] ??

ASCII Table Enumeration Control Structure Table Loop Index

ADD
Source A asciiTable.LOOPi 128
Source B 1
Dest asciiTable.LOOPi 128

ASCII Table Enumeration Control Structure Table Loop Index

LE
Source A asciiTable.LOOPi 128
Source B 127

lbl_Init
JMP

copy value array (0-127) to the ASCII Table udt
2009Jun11 -- continuously execute to ensure ASCII values aren't mistakenly overwritten by users

5

lbl_InitDone
LBL

ASCII Table Enumeration

COP
Source asciiTable.Values[0]
Dest ascii
Length 1

(End)

Listen for Duplicate Message Tag activity before enabling Driver Messages

v1.8 Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 50
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

2

EDC Gateway Callout Read Message CalloutReadMsg.EN <EQ0006_CalloutReadMsg.EN>
Duplicate Message Detection Control Duplicate Callout Read Enable 1-Shot DuplicateMsg.CalloutReadEnONS
ONS

EDC Gateway Callout Read Message CalloutReadMsg.DN <EQ0006_CalloutReadMsg.DN>
Duplicate Message Detection Control Duplicate Callout Read Done 1-Shot DuplicateMsg.CalloutReadDnONS
ONS

EDC Gateway Callout Read Message CalloutReadMsg.ER <EQ0006_CalloutReadMsg.ER>
Duplicate Message Detection Control Duplicate Callout Read Done 1-Shot DuplicateMsg.CalloutReadErONS
ONS

S:FS /

Duplicate Message Detection Control Duplicate Callout Read Count

ADD
Source A DuplicateMsg.CalloutReadCount 0
Source B 1
Dest DuplicateMsg.CalloutReadCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 135
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD
L

EDC Gateway Callout Write Message CalloutWriteMsg.EN <EQ0006_CalloutWriteMsg.EN>
Duplicate Message Detection Control Duplicate Callout Write Enable 1-Shot DuplicateMsg.CalloutWriteEnONS
ONS

EDC Gateway Callout Write Message CalloutWriteMsg.DN <EQ0006_CalloutWriteMsg.DN>
Duplicate Message Detection Control Duplicate Callout Write Done 1-Shot DuplicateMsg.CalloutWriteDnONS
ONS

EDC Gateway Callout Write Message CalloutWriteMsg.ER <EQ0006_CalloutWriteMsg.ER>
Duplicate Message Detection Control Duplicate Callout Write Done 1-Shot DuplicateMsg.CalloutWriteErONS
ONS

S:FS /

Duplicate Message Detection Control Duplicate Callout Write Count

ADD
Source A DuplicateMsg.CalloutWriteCount 0
Source B 1
Dest DuplicateMsg.CalloutWriteCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 145
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

3

EDC Gateway Fault Read Message
FaultReadMsg.EN
<EQ0006_FaultReadMsg.EN>

Duplicate Message Detection Control Duplicate Fault Read Enable 1-Shot
DuplicateMsg.FaultReadEnONS
ONS

EDC Gateway Fault Read Message
FaultReadMsg.DN
<EQ0006_FaultReadMsg.DN>

Duplicate Message Detection Control Duplicate Fault Read Done 1-Shot
DuplicateMsg.FaultReadDnONS
ONS

EDC Gateway Fault Read Message
FaultReadMsg.ER
<EQ0006_FaultReadMsg.ER>

Duplicate Message Detection Control Duplicate Fault Read Done 1-Shot
DuplicateMsg.FaultReadErONS
ONS

S:FS
/

Duplicate Message Detection Control Duplicate Fault Read Count

ADD
Source A DuplicateMsg.FaultReadCount
0
Source B 1
Dest DuplicateMsg.FaultReadCount
0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 235
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found
DuplicateMsg.FD
L

EDC Gateway Fault Write Message
FaultWriteMsg.EN
<EQ0006_FaultWriteMsg.EN>

Duplicate Message Detection Control Duplicate Fault Write Enable 1-Shot
DuplicateMsg.FaultWriteEnONS
ONS

EDC Gateway Fault Write Message
FaultWriteMsg.DN
<EQ0006_FaultWriteMsg.DN>

Duplicate Message Detection Control Duplicate Fault Write Done 1-Shot
DuplicateMsg.FaultWriteDnONS
ONS

EDC Gateway Fault Write Message
FaultWriteMsg.ER
<EQ0006_FaultWriteMsg.ER>

Duplicate Message Detection Control Duplicate Fault Write Done 1-Shot
DuplicateMsg.FaultWriteErONS
ONS

S:FS
/

Duplicate Message Detection Control Duplicate Fault Write Count

ADD
Source A DuplicateMsg.FaultWriteCount
0
Source B 1
Dest DuplicateMsg.FaultWriteCount
0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 245
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

4

EDC Gateway Quality Read Message QualityReadMsg.EN <EQ0006_QualityReadMsg.EN>
Duplicate Message Detection Control Duplicate Quality Read Enable 1-Shot DuplicateMsg.QualityReadEnONS
ONS

EDC Gateway Quality Read Message QualityReadMsg.DN <EQ0006_QualityReadMsg.DN>
Duplicate Message Detection Control Duplicate Quality Read Done 1-Shot DuplicateMsg.QualityReadDnONS
ONS

EDC Gateway Quality Read Message QualityReadMsg.ER <EQ0006_QualityReadMsg.ER>
Duplicate Message Detection Control Duplicate Quality Read Done 1-Shot DuplicateMsg.QualityReadErONS
ONS

EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>
Duplicate Message Detection Control Duplicate Quality Write Enable 1-Shot DuplicateMsg.QualityWriteEnONS
ONS

EDC Gateway Quality Write Message QualityWriteMsg.DN <EQ0006_QualityWriteMsg.DN>
Duplicate Message Detection Control Duplicate Quality Write Done 1-Shot DuplicateMsg.QualityWriteDnONS
ONS

EDC Gateway Quality Write Message QualityWriteMsg.ER <EQ0006_QualityWriteMsg.ER>
Duplicate Message Detection Control Duplicate Quality Write Done 1-Shot DuplicateMsg.QualityWriteErONS
ONS

S:FS
/

Duplicate Message Detection Control Duplicate Quality Read Count

ADD
Source A DuplicateMsg.QualityReadCount 0
Source B 1
Dest DuplicateMsg.QualityReadCount 0

Duplicate Message Detection Control Duplicate Quality Write Count

ADD
Source A DuplicateMsg.QualityWriteCount 0
Source B 1
Dest DuplicateMsg.QualityWriteCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 335
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 345
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD
L

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

5

EDC Gateway Trend Read Message TrendReadMsg.EN <EQ0006_TrendReadMsg.EN>
Duplicate Message Detection Control Duplicate Trend Read Enable 1-Shot DuplicateMsg.TrendReadEnONS
ONS

EDC Gateway Trend Read Message TrendReadMsg.DN <EQ0006_TrendReadMsg.DN>
Duplicate Message Detection Control Duplicate Trend Read Done 1-Shot DuplicateMsg.TrendReadDnONS
ONS

EDC Gateway Trend Read Message TrendReadMsg.ER <EQ0006_TrendReadMsg.ER>
Duplicate Message Detection Control Duplicate Trend Read Done 1-Shot DuplicateMsg.TrendReadErONS
ONS

S:FS
/

Duplicate Message Detection Control Duplicate Trend Read Count

ADD
Source A DuplicateMsg.TrendReadCount 0
Source B 1
Dest DuplicateMsg.TrendReadCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 435
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD
L

EDC Gateway Trend Write Message TrendWriteMsg.EN <EQ0006_TrendWriteMsg.EN>
Duplicate Message Detection Control Duplicate Trend Write Enable 1-Shot DuplicateMsg.TrendWriteEnONS
ONS

EDC Gateway Trend Write Message TrendWriteMsg.DN <EQ0006_TrendWriteMsg.DN>
Duplicate Message Detection Control Duplicate Trend Write Done 1-Shot DuplicateMsg.TrendWriteDnONS
ONS

EDC Gateway Trend Write Message TrendWriteMsg.ER <EQ0006_TrendWriteMsg.ER>
Duplicate Message Detection Control Duplicate Trend Write Done 1-Shot DuplicateMsg.TrendWriteErONS
ONS

S:FS
/

Duplicate Message Detection Control Duplicate Trend Write Count

ADD
Source A DuplicateMsg.TrendWriteCount 0
Source B 1
Dest DuplicateMsg.TrendWriteCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 445
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

6

EDC Gateway Version Write Message VersionWriteMsg.EN <EQ0006_VersionWriteMsg.EN>
Duplicate Message Detection Control Duplicate Version Write Enable 1-Shot DuplicateMsg.VersionWriteEnONS
ONS

EDC Gateway Version Write Message VersionWriteMsg.DN <EQ0006_VersionWriteMsg.DN>
Duplicate Message Detection Control Duplicate Version Write Done 1-Shot DuplicateMsg.VersionWriteDnONS
ONS

EDC Gateway Version Write Message VersionWriteMsg.ER <EQ0006_VersionWriteMsg.ER>
Duplicate Message Detection Control Duplicate Version Write Done 1-Shot DuplicateMsg.VersionWriteErONS
ONS

S:FS /

Duplicate Message Detection Control Duplicate Version Write Count

ADD
Source A DuplicateMsg.VersionWriteCount 0
Source B 1
Dest DuplicateMsg.VersionWriteCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 545
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD
L

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

7

EDC Gateway Time Sync Read Message TimeSyncReadMsg.EN <EQ0006_TimeSyncReadMsg.EN>
Duplicate Message Detection Control Duplicate TimeSync Read Enable 1-Shot DuplicateMsg.TimeSyncReadEnONS
ONS

EDC Gateway Time Sync Read Message TimeSyncReadMsg.DN <EQ0006_TimeSyncReadMsg.DN>
Duplicate Message Detection Control Duplicate TimeSync Read Done 1-Shot DuplicateMsg.TimeSyncReadDnONS
ONS

EDC Gateway Time Sync Read Message TimeSyncReadMsg.ER <EQ0006_TimeSyncReadMsg.ER>
Duplicate Message Detection Control Duplicate TimeSync Read Done 1-Shot DuplicateMsg.TimeSyncReadErONS
ONS

S:FS /

Duplicate Message Detection Control Duplicate TimeSync Read Count

ADD
Source A DuplicateMsg.TimeSyncReadCount 0
Source B 1
Dest DuplicateMsg.TimeSyncReadCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 635
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD
L

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

8

EDC Gateway Cycle Read Message CycleReadMsg.EN <EQ0006_CycleReadMsg.EN>

Duplicate Message Detection Control Duplicate Cycle Read Enable 1-Shot DuplicateMsg.CycleReadEnONS ONS

EDC Gateway Cycle Read Message CycleReadMsg.DN <EQ0006_CycleReadMsg.DN>

Duplicate Message Detection Control Duplicate Cycle Read Done 1-Shot DuplicateMsg.CycleReadDnONS ONS

EDC Gateway Cycle Read Message CycleReadMsg.ER <EQ0006_CycleReadMsg.ER>

Duplicate Message Detection Control Duplicate Cycle Read Done 1-Shot DuplicateMsg.CycleReadErONS ONS

S:FS /

Duplicate Message Detection Control Duplicate Cycle Read Count

ADD
Source A DuplicateMsg.CycleReadCount 0
Source B 1
Dest DuplicateMsg.CycleReadCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 735
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD (L)

EDC Gateway Cycle Write Message CycleWriteMsg.EN <EQ0006_CycleWriteMsg.EN>

Duplicate Message Detection Control Duplicate Cycle Write Enable 1-Shot DuplicateMsg.CycleWriteEnONS ONS

EDC Gateway Cycle Write Message CycleWriteMsg.DN <EQ0006_CycleWriteMsg.DN>

Duplicate Message Detection Control Duplicate Cycle Write Done 1-Shot DuplicateMsg.CycleWriteDnONS ONS

EDC Gateway Cycle Write Message CycleWriteMsg.ER <EQ0006_CycleWriteMsg.ER>

Duplicate Message Detection Control Duplicate Cycle Write Done 1-Shot DuplicateMsg.CycleWriteErONS ONS

S:FS /

Duplicate Message Detection Control Duplicate Cycle Write Count

ADD
Source A DuplicateMsg.CycleWriteCount 0
Source B 1
Dest DuplicateMsg.CycleWriteCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 745
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Listen for Duplicate Message Tag activity before enabling Driver Messages

Disable/Inhibit Program if Duplicate Message Tag detected

9

Duplicate Message Detection Control Duplicate Message Found
DuplicateMsg.FD

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	905
Input Par	DuplicateMsg.DelayTMR.ACC
Input Par	0

Duplicate Message Detection Control Program Disable Flag for SSV Command

MOVE	
Source	-1
Dest	DuplicateMsg.DisableFlag
	0

Duplicate Message Detection Control Program Disable Flag for SSV Command

SSV	
Class Name	Program
Instance Name	THIS
Attribute Name	DisableFlag
Source	DuplicateMsg.DisableFlag
	0

(End)

Equipment Driver User Configuration

Equipment Data Collection
User Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 90
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Driver Enable

Driver Interface - choose Toyopuc, RemMsg or Logix
TpucDriverEN - FEC driver with Toyopuc data via I/O Module
RemMsgEN - Remote Message data from CompactLogix, PLC-5, etc.
LogixDriverEN - driver embedded in Logix Controller or FEC driver with Remote Messages

2

Equipment Data Collection - User Config Toyopuc Driver Enable
UserCfg.TpucDriverEN
U

Equipment Data Collection - User Config Remote Driver Enable
UserCfg.RemDriverEN
U

Equipment Data Collection - User Config Logix Driver Enable
UserCfg.LogixDriverEN
L

TimeSyncEN - enables Time Synchronization of EqDriver from EDC Gateway wallclock

3

Equipment Data Collection - User Config Time Synchronization Message Enable
UserCfg.TimeSyncEN
U

Equipment Driver User Configuration

EDC Gateway Logix Path

Configure Message Path for the EDC Gateway Logix Controller
(This Configuration Message is not used - it serves as a placeholder for the Path used in Driver Messages.)

Use Gateway EN2T in Slot 5 (or future 6), Logix in Slot 3
Path = Local path to Ethernet Network, Gateway Slot 5 IP Address, 0, 3

4

AFI

EDC Gateway Message
Path Configuration

MSG
Message Control UserCfgMsg ...
<EQ0006_UserCfgMsg>
EN
DN
ER

Equipment Driver User Configuration

v1.3 - Dec2016 - Multi-packet support for >8 DINT Equipment-Defined Faults
Packet Enable - Latch bit for each 8-DINT Packet used below
Word Count - Total number of FaultMsg_EqDef Data Words used in each Packet
(unused Words are ignored by Server to reduce processing of raw data)

Toggle the UserCfg.UpdateEQID if Fault Packet Enables are added/removed from the configuration.
v1.5 - moved Packet EN and Counts from FaultMsgCtl to UserCfg to fix Reset, Config, Initialize sequence

5

Equipment Data Collection - User Config Toggle ON if EQID Changed
UserCfg.UpdateEQID

NOP

Equipment Data Collection - User Config Equipment Defined Packet EN bit array
UserCfg.EqDefPacketEN.0 (L)
UserCfg.EqDefPacketEN.1 (L)
UserCfg.EqDefPacketEN.2 (L)
UserCfg.EqDefPacketEN.3 (U)
UserCfg.EqDefPacketEN.4 (U)
UserCfg.EqDefPacketEN.5 (U)
UserCfg.EqDefPacketEN.6 (U)
UserCfg.EqDefPacketEN.7 (U)

Equipment Data Collection - User Config Equipment Defined Packet Data Word Count

MOVE	Source	Dest
UserCfg.EqDefDataCount[0]	8	8
UserCfg.EqDefDataCount[1]	8	8
UserCfg.EqDefDataCount[2]	8	8
UserCfg.EqDefDataCount[3]	8	8
UserCfg.EqDefDataCount[4]	8	8
UserCfg.EqDefDataCount[5]	8	8
UserCfg.EqDefDataCount[6]	8	8
UserCfg.EqDefDataCount[7]	8	8

(End)

Equipment Data Collection
User Configuration for Logix and Remote Drivers
(Toyopuc Drivers configured programatically in Toyopuc routines)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 92
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Driver Options Enable

Latch EN for each Driver function used:
DriverEN - enables Live Callout Msg
FaultMsgEN - enables Buffered Fault/Status Msg
QualityMsgEN - enables Buffered Quality/Health Msg
TrendMsgEN - enables Buffered Trend Msg
CycleMsgEN - enables Buffered Cycle Msg
VersionEN - enables Version Msg
TimeSyncEN - enables Time Synchronization from EDC Gateway wallclock

2

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN
L

Equipment Data Collection - User Config Fault Message Enable
UserCfg.FaultMsgEN
L

Equipment Data Collection - User Config Quality Message Enable
UserCfg.QualityMsgEN
L

Equipment Data Collection - User Config Trend Message Enable
UserCfg.TrendMsgEN
U

Equipment Data Collection - User Config Cycle Message Enable
UserCfg.CycleMsgEN
U

Equipment Data Collection - User Config Version Message Enable
UserCfg.VersionEN
L

Equipment ID

Load unique ID assigned to this Equipment Driver
(This must match the EQID configured on the EDC Server.)
v1.4 Add UpdateVersion and UpdateTimeSync
v2.04 Change default EQID and EQIDprev to 9999

3

Equipment Data Collection - User Config Equipment ID
MOVE
Source 6
Dest UserCfg.EQID 6

Equipment Data Collection - User Config Equipment ID
NE
Source A UserCfg.EQID 6
Source B 0

Equipment Data Collection - User Config Equipment ID
NE
Source A UserCfg.EQID 6
Source B UserCfg.EQIDprev 6

Equipment Data Collection - User Config Toggle ON if EQID Changed
UserCfg.UpdateEQID
L

Equipment Data Collection - User Config Toggle ON to Update Program Version and send to Gateway
UserCfg.UpdateVersion
L

Equipment Data Collection - User Config Toggle ON to Update Time Sync from Gateway
UserCfg.UpdateTimeSync
L

Equipment Data Collection - User Config Previous Equipment ID (to detect changes)
MOVE
Source UserCfg.EQID 6
Dest UserCfg.EQIDprev 6

(End)

EqDriver embedded in Logix Controller

EqDriver embedded in Logix Controller

This routine is only executed when the driver is configured for Logix

Replace with User Logic
xxUserCfgFaultBOOL
U

0

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	93
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

1

EqDriver embedded in Logix Controller

v1.6 Optional Message to get Overall Start from Gateway
Configure Message Source Element Array Index for desired EQID and remove AFI to Enable Rung
EQID PreDefined Fault word is loaded into RemPreDef

i.e. set Message Source to EDC_EqCallout[1,1].PreDef for EQID 101 -- loads RemotePreDef.OverallStartOn, ConvRunning and FIxedStopPos

2 AFI

EDC Gateway Overall Start Message Timer
MOVE
Source 1000
Dest OverallStartMsgTMR.PRE 250

EDC Gateway Overall Start Message Timer
TON
Timer OverallStartMsgTMR
Preset 250
Accum 0
EN
DN

EDC Gateway Overall Start Message Timer
OverallStartMsgTMR.DN

EDC Gateway Overall Start Read Message
OverallStartReadMsg.EN
<EQ0006_OverallStartReadMsg.EN>

EDC Gateway Overall Start Read Message
MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest OverallStartReadMsg.UnconnectedTimeout <EQ0006_OverallStartReadMsg.UnconnectedTimeout> 4000000

EDC Gateway Overall Start Read Message
MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest OverallStartReadMsg.ConnectionRate <EQ0006_OverallStartReadMsg.ConnectionRate> 1000000

EDC Gateway Overall Start Read Message
MOVE
Source CFG.CLx.ConnectionMult 0
Dest OverallStartReadMsg.TimeoutMultiplier <EQ0006_OverallStartReadMsg.TimeoutMultiplier> 0

EDC Gateway Overall Start Read Message
OverallStartReadMsg.EN_CC
<EQ0006_OverallStartReadMsg.EN_CC>
U

EDC Gateway Overall Start Read Message
COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest OverallStartReadMsg.Path <EQ0006_OverallStartReadMsg.Path>
Length 1

EDC Gateway Overall Start Read Message
MSG
Message Control OverallStartReadMsg ... <EQ0006_OverallStartReadMsg>
EN
DN
ER

EDC Gateway Overall Start Message Timer
OverallStartMsgTMR
RES

EDC Gateway Overall Start Read Message
OverallStartReadMsg.ER
<EQ0006_OverallStartReadMsg.ER>

Remote PreDefined Faults from selected EQID
FLL
Source 0
Dest RemotePreDef <EQ0006_RemotePreDef>
Length 1

Gateway Overall Start Message ER
OverallStartReadMsgER
L

EqDriver embedded in Logix Controller

EDC Gateway Overall Start Read Message
OverallStartReadMsg.DN
<EQ0006_OverallStartReadMsg.DN>

Remote PreDefined Faults from selected EQID
COP
Source RemotePreDefDINT
<EQ0006_RemotePreDefDINT>
Dest RemotePreDef
<EQ0006_RemotePreDef>
Length 1

Gateway Overall Start Message ER
OverallStartReadMsgER
U

Overall Start On
used by EDC to determine Equipment Downtime and OA%

Signal comes from Andon Consumed or EDC Gateway Read Message

3

Remote PreDefined Faults from selected EQID Line Production Work Time
RemotePreDef._00
<EQ0006_RemotePreDef._00>

Equipment Data Fault/Status Message Line Production Work Time
FaultMsg.PreDef._00

Conveyor Running
used by EDC to determine Equipment Downtime and OA%

Signal comes from Andon Consumed or EDC Gateway Read Message

4

Remote PreDefined Faults from selected EQID Line Conveyor Running
RemotePreDef._01
<EQ0006_RemotePreDef._01>

AFI

Equipment Data Fault/Status Message Line Conveyor Running
FaultMsg.PreDef._01

Fixed Stop Position
used by EDC to determine Equipment Downtime and OA%

Signal comes from Andon Consumed or EDC Gateway Read Message

5

Remote PreDefined Faults from selected EQID Line Conveyor Stopped at Fixed Stop Position (L10)
RemotePreDef._02
<EQ0006_RemotePreDef._02>

AFI

Equipment Data Fault/Status Message Line Conveyor Stopped at Fixed Stop Position (L10)
FaultMsg.PreDef._02

EqDriver embedded in Logix Controller

Fixed Stop Request
used by EDC to determine Equipment Downtime and OA%

6 AFI

Equipment Data
Fault/Status Message
EQ Fixed Stop
Request
FaultMsg.PreDef._03

Temp Stop Request
used by EDC to determine Equipment Downtime and OA%

7 AFI

Equipment Data
Fault/Status Message
Temp Stop Request
FaultMsg.PreDef._04

v2.0 Productive Stop Request, Fault_On, Alarm_On generated by EqDriver in _100_PreDefFaultData based on Event Mask configuration

8 NOP

Master On is OFF

9 AFI

Equipment Data
Fault/Status Message
Master On is OFF
FaultMsg.PreDef._08

Continuous is OFF

10 AFI

Equipment Data
Fault/Status Message
Continuous is OFF
FaultMsg.PreDef._09

Manual

11 AFI

Equipment Data
Fault/Status Message
Manual is ON
FaultMsg.PreDef._10

EqDriver embedded in Logix Controller

Cycle Time Over

12 AFI

Equipment Data
Fault/Status Message
Cycle Time is Over
FaultMsg.PreDef._11

Overload On

13 AFI

Equipment Data
Fault/Status Message
Overload is ON
FaultMsg.PreDef._12

v1.3 - Maintenance Call

14 AFI

Equipment Data
Fault/Status Message
Maintenance Call
FaultMsg.PreDef._13

v1.4 - Interlock Off - Equipment Bypassed

15 AFI

Equipment Data
Fault/Status Message
Interlock Off -
Equipment Bypassed
FaultMsg.PreDef._14

v2.1 Fault Memory generated by EqDriver in _100_PreDefFaultData

16 NOP

v2.0 - Don't send the HB as an Event (toggles continuously)
HB is monitored and reported as an Equipment Data Error if it stops

17 AFI

Equipment Data
Fault/Status Message
Inch Mode
FaultMsg.PreDef._16

v2.0 Alert_On generated by EqDriver in _100_PreDefFaultData based on Event Mask configuration

18 NOP

EqDriver embedded in Logix Controller

Equipment Data Fault/Status Message Line Down Callout Escalation
FaultMsg.PreDef._18

19 AFI

Health Alarm and Health Fault generated by Gateway from Server status download
Invalid Wall Clock and EqDataER generated by EqDriver

20 NOP

v2.1 Body Weld signal - Auto Run without Transfer

Equipment Data Fault/Status Message Continuous Off2 (BW Auto Run without Transfer)
FaultMsg.PreDef._23

21 AFI

v2.1 Delay (production hold - charge OR% but not OA%)

Equipment Data Fault/Status Message Delay (Prodution Hold)
FaultMsg.PreDef._24

22 AFI

v2.1 Short (charge OR% but not OA%)

Equipment Data Fault/Status Message Short or SubShort
FaultMsg.PreDef._25

23 AFI

v2.1 Full (charge OR% but not OA%)

Equipment Data Fault/Status Message Full
FaultMsg.PreDef._26

24 AFI

EqDriver embedded in Logix Controller

v2.1 In Demand Off (charge OR% but not OA%)
Equipment is not In Demand -- something else is holding line

25 AFI

Equipment Data
Fault/Status Message
In Demand Off (don't charge OA)
FaultMsg.PreDef._27

v2.1 Spare

26 AFI

Equipment Data
Fault/Status Message
Tool Change Mode
FaultMsg.PreDef._28

v2.1 Spare

27 AFI

Equipment Data
Fault/Status Message
Spare
FaultMsg.PreDef._29

RingBuffer FIlling and Message Error generated by EqDriver

28 NOP

Equipment Defined Fault/Status
8 DINTs - 256 signals unique to this Equipment

v1.3 Packet 0 - DINTs 0-7

Shop Status for Communications (Callout, Fault, Quality, Trend)
Bit 18 - Callout Sequence Error Blocked (normal occurrence for short duration events)

29

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM
Source 0
Mask 16#0002_0fff
Dest EqDefHold[0].Word[0] 16#0000_0000

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM
Source EDC.Perf.Sum.Shop.StateVal 16#0000_0a7e
Mask 16#fffd_f000
Dest EqDefHold[0].Word[0] 16#0000_0000

EqDriver embedded in Logix Controller

Fault Ring Buffer

30

Equipment Data Collection Read Buffer Alarm EDC.RB_Fault01.BufferAlarm / Equipment Data Collection Read Buffer Alarm EDC.RB_Fault02.BufferAlarm / Equipment Data Collection Read Buffer Alarm EDC.RB_Fault03.BufferAlarm / Equipment Data Collection Read Buffer Alarm EDC.RB_Fault04.BufferAlarm / Equipment Data Collection Read Buffer Alarm EDC.RB_Fault05.BufferAlarm / Equipment Data Collection Read Buffer Alarm EDC.RB_Fault06.BufferAlarm / NoFaultBufferAlarm

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].0

NoFaultBufferAlarm /

Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault01.WritePtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault02.WritePtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault03.WritePtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault04.WritePtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault05.WritePtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault06.WritePtrErr / NoFaultWritePtrErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].1

NoFaultWritePtrErr /

Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault01.ReadPtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault02.ReadPtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault03.ReadPtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault04.ReadPtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault05.ReadPtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault06.ReadPtrErr / NoFaultReadPtrErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].2

NoFaultReadPtrErr /

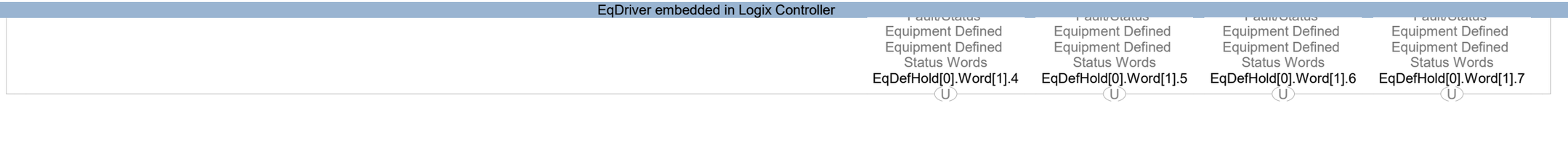
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault01.WriteBlockErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault02.WriteBlockErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault03.WriteBlockErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault04.WriteBlockErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault05.WriteBlockErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault06.WriteBlockErr / NoFaultWriteBlockErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].3

NoFaultWriteBlockErr /

Equipment Data Fault/Status Equipment Data Fault/Status Equipment Data Fault/Status Equipment Data Fault/Status

EqDriver embedded in Logix Controller



EqDriver embedded in Logix Controller

Quality Ring Buffer
DN005 Health Status Alarm Callout

31

Equipment Data Collection Read Buffer Alarm EDC.RB_Quality01.BufferAlarm
Equipment Data Collection Read Buffer Alarm EDC.RB_Quality02.BufferAlarm
Equipment Data Collection Read Buffer Alarm EDC.RB_Quality03.BufferAlarm
Equipment Data Collection Read Buffer Alarm EDC.RB_Quality04.BufferAlarm
NoQualityBufferAlarm

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].8
NoQualityBufferAlarm

Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality01.WritePtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality02.WritePtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality03.WritePtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality04.WritePtrErr
NoQualityWritePtrErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].9
NoQualityWritePtrErr

Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality01.ReadPtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality02.ReadPtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality03.ReadPtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality04.ReadPtrErr
NoQualityReadPtrErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].10
NoQualityReadPtrErr

Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality01.WriteBlockErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality02.WriteBlockErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality03.WriteBlockErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality04.WriteBlockErr
NoQualityWriteBlockErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].11
NoQualityWriteBlockErr

Equipment Data

EqDriver embedded in Logix Controller

Equipment Data Collection Response has CIS Error - Send Callout Bit
EDC.ServerStat.Ctl.CisErrorSendCallout

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[0].Word[1].12

Equipment Data Collection Response has Health Status Error - Send Callout Bit
EDC.ServerStat.Ctl.HealthStatusSendCallout

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[0].Word[1].13

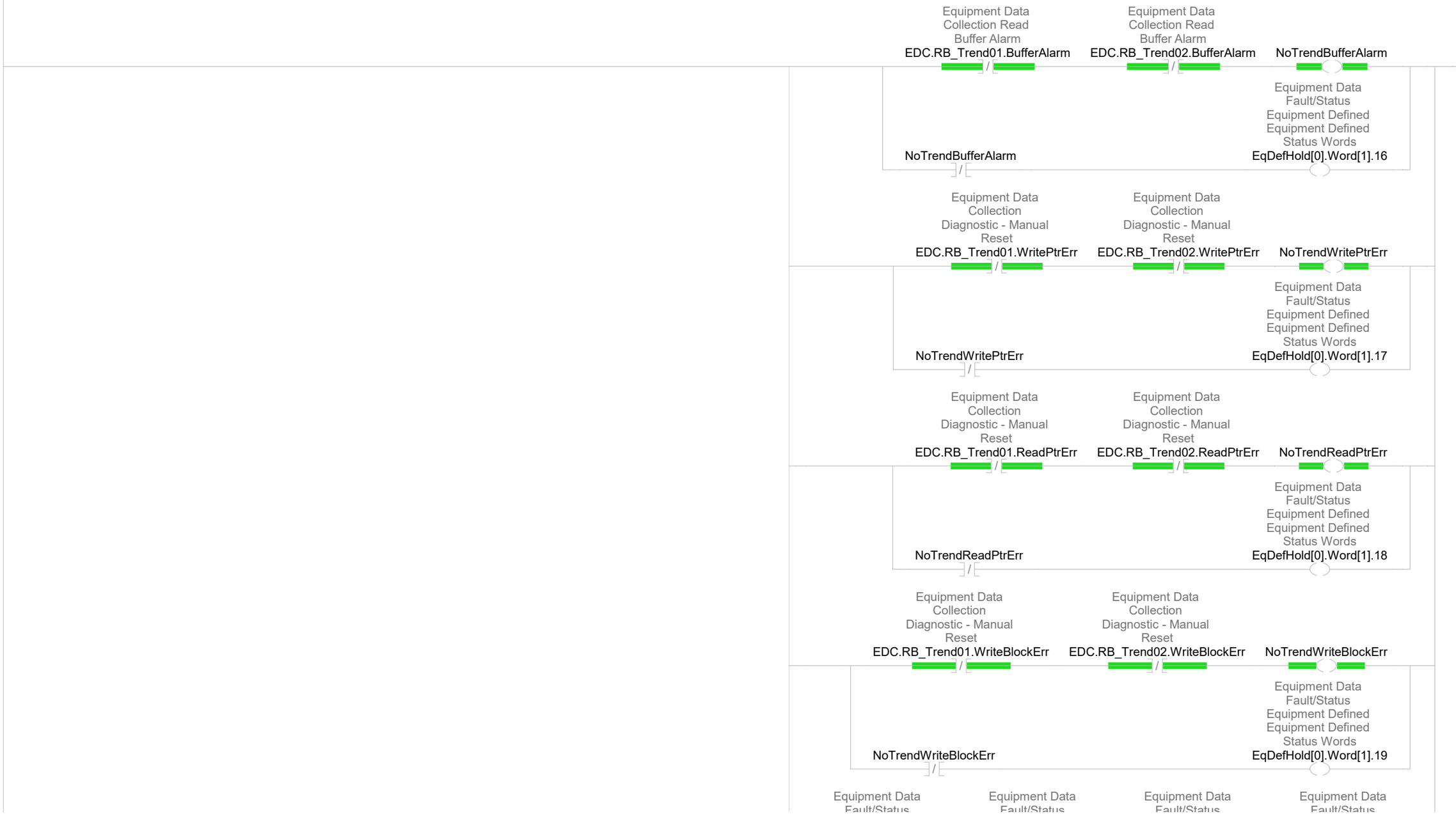
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[0].Word[1].14
U

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[0].Word[1].15
U

EqDriver embedded in Logix Controller

Trend Ring Buffer

32



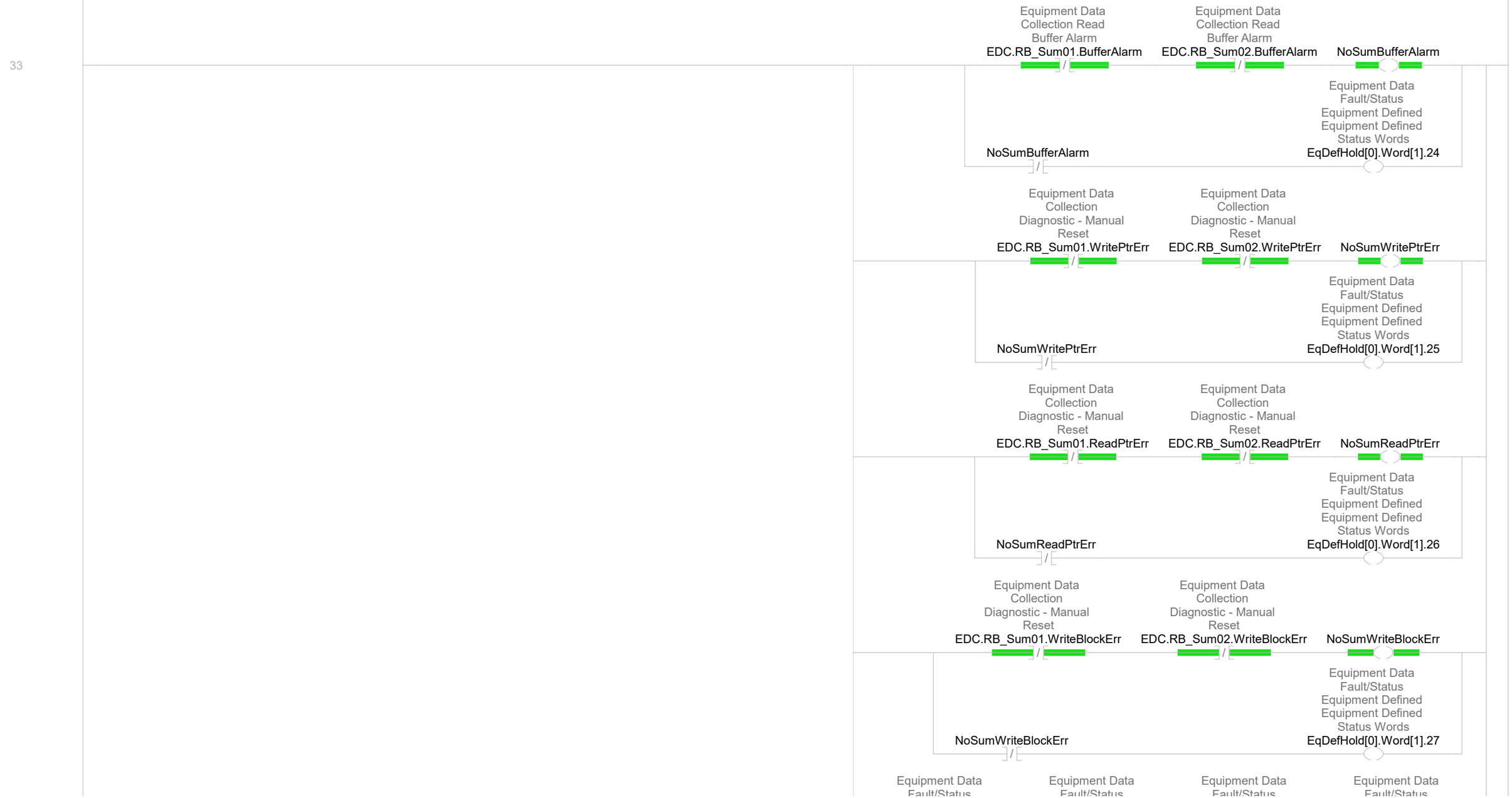
EqDriver embedded in Logix Controller

FaultStatus	FaultStatus	FaultStatus	FaultStatus
Equipment Defined	Equipment Defined	Equipment Defined	Equipment Defined
Equipment Defined	Equipment Defined	Equipment Defined	Equipment Defined
Status Words	Status Words	Status Words	Status Words
EqDefHold[0].Word[1].20	EqDefHold[0].Word[1].21	EqDefHold[0].Word[1].22	EqDefHold[0].Word[1].23
U	U	U	U

EqDriver embedded in Logix Controller

Summary Ring Buffer

33



EqDriver embedded in Logix Controller

Equipment Defined
Equipment Defined
Status Words
EqDefHold[0].Word[1].28
U

Equipment Defined
Equipment Defined
Status Words
EqDefHold[0].Word[1].29
U

Equipment Defined
Equipment Defined
Status Words
EqDefHold[0].Word[1].30
U

Equipment Defined
Equipment Defined
Status Words
EqDefHold[0].Word[1].31
U

Server Config Download

34

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
MVM
Source 0
Mask 16#0000FFFF
Dest EqDefHold[0].Word[2]
16#0000_0000

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
MVM
Source EDC_Status01.ServerCfg.StateVal
16#0000_0004
Mask 16#FFFF0000
Dest EqDefHold[0].Word[2]
16#0000_0000

Server Status Interface

35

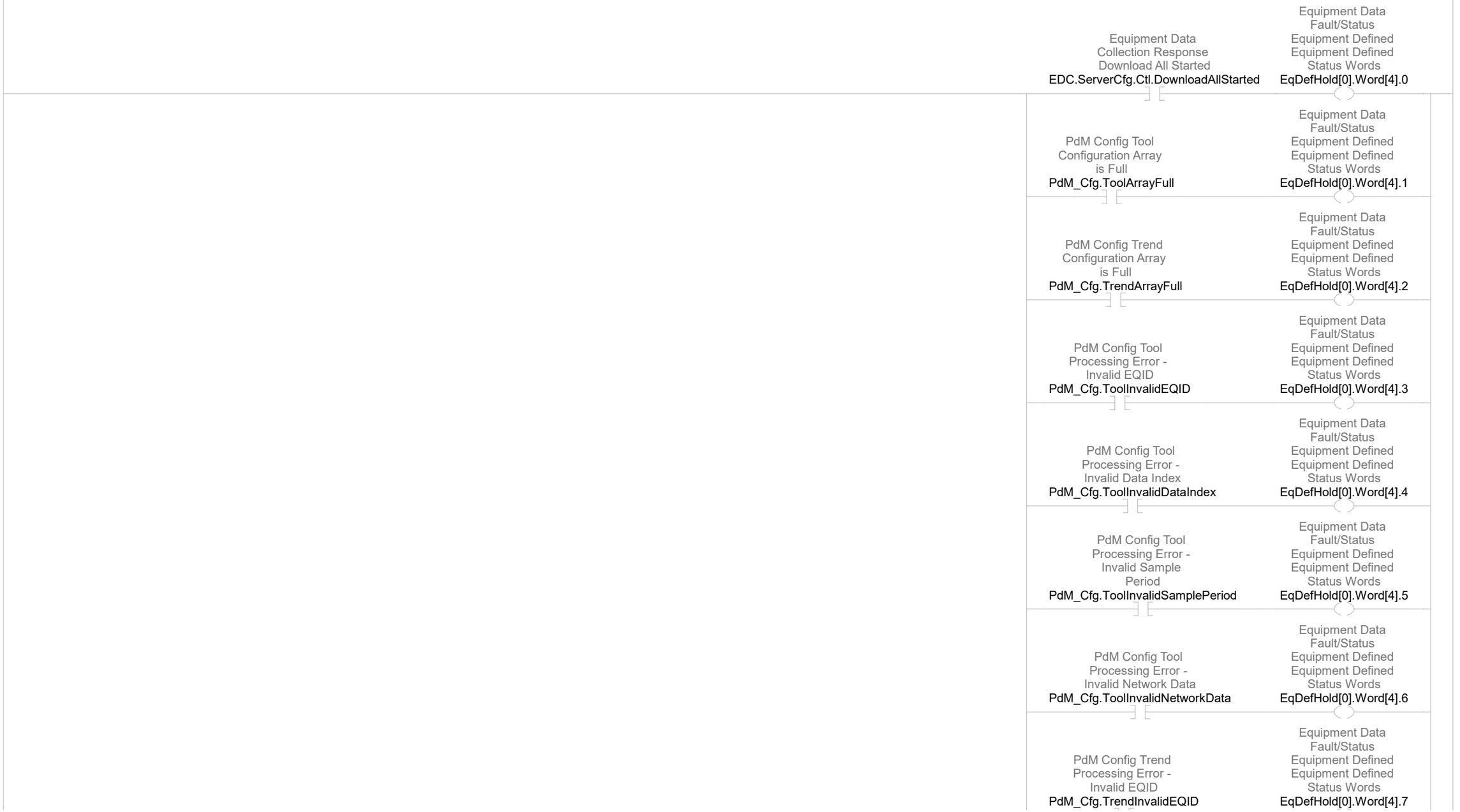
Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
MVM
Source 0
Mask 16#0000FFFF
Dest EqDefHold[0].Word[3]
16#0000_0000

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
MVM
Source EDC_Status01.ServerStat.StateVal
16#0000_0004
Mask 16#FFFF0000
Dest EqDefHold[0].Word[3]
16#0000_0000

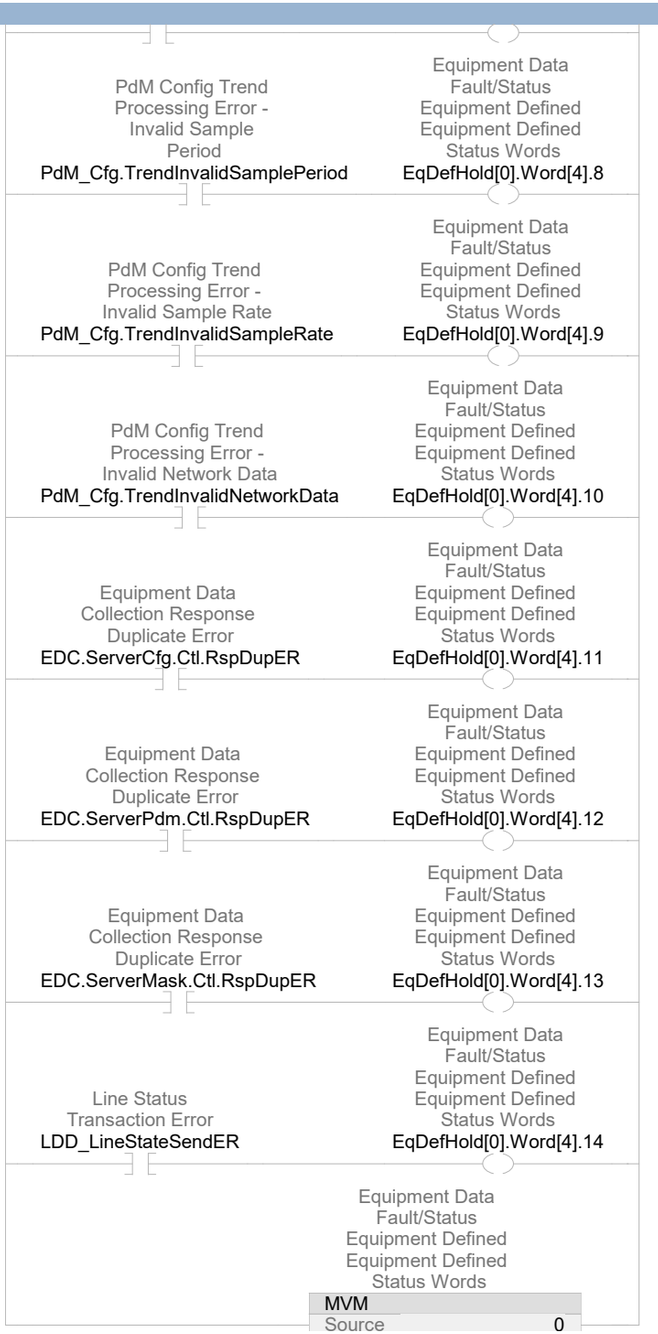
EqDriver embedded in Logix Controller

DHF013 - Nov 27, 2017 -- Add Server Config Download All Event 4.0

36



EqDriver embedded in Logix Controller



EqDriver embedded in Logix Controller

Mask	16#FFFF8000
Dest	EqDefHold[0].Word[4] 16#0000_0000

Server Health Interface

37

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM

Source	0
Mask	16#0000FFFF
Dest	EqDefHold[0].Word[5] 16#0000_0000

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM

Source	EDC_Status01.ServerHealth.StateVal 16#0000_0004
Mask	16#FFFF0000
Dest	EqDefHold[0].Word[5] 16#0000_0000

Server Pdm Interface

38

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM

Source	0
Mask	16#0000FFFF
Dest	EqDefHold[0].Word[6] 16#0000_0000

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM

Source	EDC_Status01.ServerPdm.StateVal 16#0000_0004
Mask	16#FFFF0000
Dest	EqDefHold[0].Word[6] 16#0000_0000

EqDriver embedded in Logix Controller

Server Mask Interface

39

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM	
Source	0
Mask	16#0000FFFF
Dest	EqDefHold[0].Word[7] 16#0000_0000

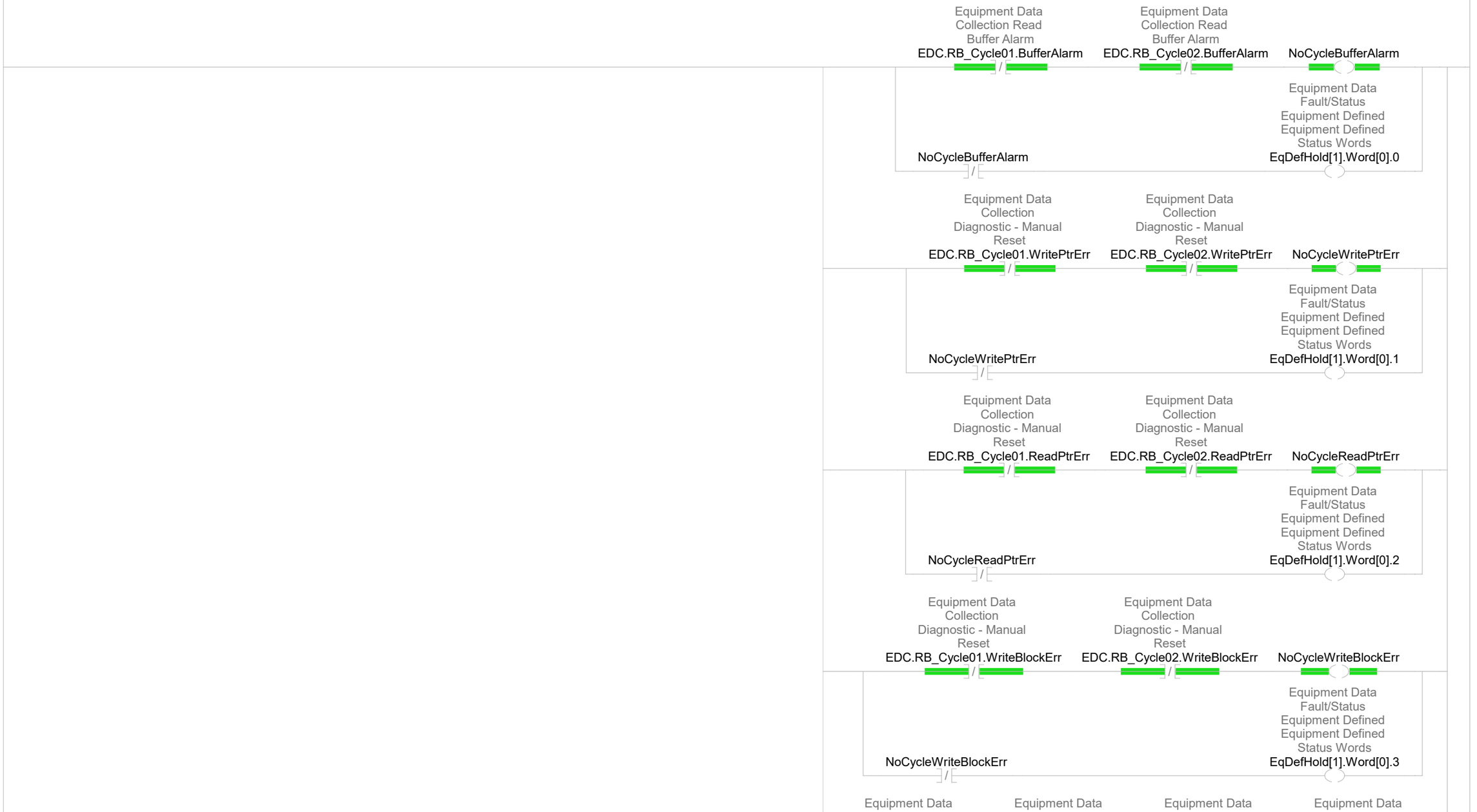
Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM	
Source	EDC_Status01.ServerMask.StateVal 16#0000_0004
Mask	16#FFFF0000
Dest	EqDefHold[0].Word[7] 16#0000_0000

EqDriver embedded in Logix Controller

Cycle Ring Buffer

40



EqDriver embedded in Logix Controller

Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[0].4
U

Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[0].5
U

Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[0].6
U

Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[0].7
U

Packet 1 Spare

41

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM
Source 0
Mask 16#FFFFFF00
Dest EqDefHold[1].Word[0]
16#0000_0000

Event Text Storage

42

CPT
Dest TextQtyLimit
175500
Expression EDC_HMI.CFG.EvTxt01.n *6 *0.9

Event
Controls Event Text
Qty

GT
Source A EDC_Event.CTL.TextQty
45032
Source B TextQtyLimit
175500

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].0

Server Interface
Status State = Red -
Equip Event Array
Full ER
EDC_Status01.ServerCfg.Update.EquipEventArrayFull

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].1

EqDriver embedded in Logix Controller

Ethernet 00 Status

43

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet00.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].2

Ethernet Module Status Last Scan
GE
Source A EDC_Enet00.CPU_Util.Last
580
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].4

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].4

Ethernet Module Status Last Scan
GE
Source A EDC_Enet00.CPU_Util.Last
580
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].4
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet00.CPU_Util.Last
580
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet00.CPU_Util.Last
580
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].3

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].3

Ethernet Module Status Last Scan
GE
Source A EDC_Enet00.CPU_Util.Last
580
Source B 675

EqDriver embedded in Logix Controller

Ethernet 01 Status

44

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet01.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].5

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].7

Ethernet Module Status Last Scan
GE
Source A EDC_Enet01.CPU_Util.Last 440
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].7

Ethernet Module Status Last Scan
GE
Source A EDC_Enet01.CPU_Util.Last 440
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].7
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet01.CPU_Util.Last 440
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet01.CPU_Util.Last 440
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].6

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].6

Ethernet Module Status Last Scan
GE
Source A EDC_Enet01.CPU_Util.Last 440
Source B 675

EqDriver embedded in Logix Controller

Ethernet 02 Status

45

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet02.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].8

Ethernet Module Status Last Scan
GE
Source A EDC_Enet02.CPU_Util.Last 620
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].10

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].10

Ethernet Module Status Last Scan
GE
Source A EDC_Enet02.CPU_Util.Last 620
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].10
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet02.CPU_Util.Last 620
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet02.CPU_Util.Last 620
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].9

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].9

Ethernet Module Status Last Scan
GE
Source A EDC_Enet02.CPU_Util.Last 620
Source B 675

EqDriver embedded in Logix Controller

Ethernet 03 Status

46

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet03.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].11

Ethernet Module Status Last Scan
GE
Source A EDC_Enet03.CPU_Util.Last 350
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].13

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].13

Ethernet Module Status Last Scan
GE
Source A EDC_Enet03.CPU_Util.Last 350
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].13
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet03.CPU_Util.Last 350
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet03.CPU_Util.Last 350
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].12

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].12

Ethernet Module Status Last Scan
GE
Source A EDC_Enet03.CPU_Util.Last 350
Source B 675

EqDriver embedded in Logix Controller

Ethernet 04 Status

47

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet04.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].14

Ethernet Module Status Last Scan
GE
Source A EDC_Enet04.CPU_Util.Last 130
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].16

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].16

Ethernet Module Status Last Scan
GE
Source A EDC_Enet04.CPU_Util.Last 130
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].16
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet04.CPU_Util.Last 130
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet04.CPU_Util.Last 130
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].15

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].15

Ethernet Module Status Last Scan
GE
Source A EDC_Enet04.CPU_Util.Last 130
Source B 675

EqDriver embedded in Logix Controller

Ethernet 05 Status

48

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet05.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].17

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].19

Ethernet Module Status Last Scan
GE
Source A EDC_Enet05.CPU_Util.Last 0
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].19

Ethernet Module Status Last Scan
GE
Source A EDC_Enet05.CPU_Util.Last 0
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].19
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet05.CPU_Util.Last 0
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet05.CPU_Util.Last 0
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].18

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].18

Ethernet Module Status Last Scan
GE
Source A EDC_Enet05.CPU_Util.Last 0
Source B 675

EqDriver embedded in Logix Controller

Ethernet 06 Status

49

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet06.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].20

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].22

Ethernet Module Status Last Scan
GE
Source A EDC_Enet06.CPU_Util.Last 0
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].22

Ethernet Module Status Last Scan
GE
Source A EDC_Enet06.CPU_Util.Last 0
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].22
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet06.CPU_Util.Last 0
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet06.CPU_Util.Last 0
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].21

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].21

Ethernet Module Status Last Scan
GE
Source A EDC_Enet06.CPU_Util.Last 0
Source B 675

EqDriver embedded in Logix Controller

Ethernet 07 Status

50 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].23

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].25

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].24

Ethernet 08 Status

51 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].26

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].28

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].27

EqDriver embedded in Logix Controller

Ethernet 09 Status

52 AFI

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[1].29

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[1].31

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[1].30

Ethernet 10 Status

53 AFI

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[2].0

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[2].2

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[2].1

EqDriver embedded in Logix Controller

Ethernet 11 Status

54 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].3

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].5

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].4

Ethernet 12 Status

55 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].6

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].8

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].7

EqDriver embedded in Logix Controller

Ethernet 13 Status

56 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].9

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].11

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].10

Ethernet 14 Status

57 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].12

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].14

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].13

EqDriver embedded in Logix Controller

Ethernet 15 Status

58 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].15

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].17

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].16

Ethernet 16 Status

59 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].18

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].20

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].19

EqDriver embedded in Logix Controller

Unused
(future)

60

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[2].21	EqDefHold[1].Word[2].22	EqDefHold[1].Word[2].23	EqDefHold[1].Word[2].24	EqDefHold[1].Word[2].25	EqDefHold[1].Word[2].26	EqDefHold[1].Word[2].27	EqDefHold[1].Word[2].28	EqDefHold[1].Word[2].29
U	U	U	U	U	U	U	U	U

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[2].30	EqDefHold[1].Word[2].31
U	U

EqDriver embedded in Logix Controller

CLx Time Sync Errors

61

Logix
Wall Clock
Invalid Year
(force time sync)
SNTP_InvalidYear

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[3].0

SNTP Query and Clock
Sync Instruction
There was an error
retrieving the time.
SNTP_AOI_TAG.ER

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[3].1

SNTP Query and Clock
Sync Instruction
SNTP_AOI_TAG.Enable_Instruction

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[3].2

SNTP Query and Clock
Sync Instruction
SNTP_AOI_TAG.Sync_Controller_Clock

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[3].3

System time master
CST_CurrentStatus.2

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[3].4

Wall Clock Time
Time Zone
matches Config
WCT_TimeZoneOK

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[3].5

Wall Clock Time
DST
matches Config
WCT_DSTok

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[3].6

EqDriver embedded in Logix Controller

62

CLx High Task Scan Time > 0.5 sec
(500,000 microseconds)

CLxTaskScanHigh

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].7

63

AFI

Unused
(future)

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].8

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].9

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].10

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].11

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].12

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].13

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].14

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].15

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].16

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].17

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].18

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].19

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].20

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].21

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].22

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].23

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].24

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].25

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].26

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].27

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].28

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].29

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].30

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].31

EqDriver embedded in Logix Controller

64

Packet 1 Word 3-7
Fault Word from FEC1-FEC4 located in Slots 3-7
(FEC Faults are Messaged from FECs to Gateway using Slot# as Array Index)

If any FEC sends 'first scan' signal, refresh all Masks
(Gateway doesn't know which EqDrv are in which FEC -- so need to use the BIG hammer)

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
MOVE
Source FEC_EqDrvFault[3] 16#0000_0000
Dest EqDefHold[1].Word[4] 16#0000_0000

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
MOVE
Source FEC_EqDrvFault[4] 16#0000_0000
Dest EqDefHold[1].Word[5] 16#0000_0000

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
MOVE
Source FEC_EqDrvFault[5] 16#0000_0000
Dest EqDefHold[1].Word[6] 16#0000_0000

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
MOVE
Source FEC_EqDrvFault[6] 16#0000_0000
Dest EqDefHold[1].Word[7] 16#0000_0000

FEC1 Faults
FEC_EqDrvFault[3].4

FEC2 Faults
FEC_EqDrvFault[4].4

FEC3 Faults
FEC_EqDrvFault[5].4

FEC4 Faults
FEC_EqDrvFault[6].4

FEC_EqDrvFault_FirstScanONS
ONS

Any FEC First Scan
FEC_EqDrvFault_FirstScan

CLx High Program Scan Time

65

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
MOVE
Source CLxProgScanHigh 0
Dest EqDefHold[2].Word[0] 16#0000_0000

EqDriver embedded in Logix Controller

Spare Packets 2-7

66

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[2].Word[1]
Length 7

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[3].Word[0]
Length 8

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[4].Word[0]
Length 8

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[5].Word[0]
Length 8

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[6].Word[0]
Length 8

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[7].Word[0]
Length 8

generate a bunch of data very fast
to test the 10-record batch Fault Event driver

67

AFI

TON
Timer xxTestDataTMR
Preset 10000
Accum 0
EN
DN

xxTestDataTMR.DN

ADD
Source A xxTestCount 0
Source B 1
Dest xxTestCount 0

MOD
Source A xxTestCount 0
Source B 2
Dest xxTestResult 0

EQ
Source A xxTestResult 0
Source B 0

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].8
U

EQ
Source A xxTestResult 0
Source B 1

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].8
L

GE
Source A xxTestCount 0
Source B 12

CLR
Dest xxTestCount 0

xxTestDataTMR
RES

(End)

Equipment Quality/Health Data Mapping

Equipment Quality/Health Data
User Data Mapping

Load Job Data
Load Job Data ONS
Send Tool Data
Send Tool Data ONS
Cfg Fault Code Clear
Cfg Fault Code Trigger

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 94
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

2

Tool02_Enable

MOVE
Source 20000
Dest Tool02_TMR.PRE 20000

TON
Timer Tool02_TMR
Preset 20000
Accum 0
EN
DN

Tool02_TMR.DN

Tool02_Start (L)
Tool02_Send (L)

ADD
Source A Tool02_Count 0
Source B 1
Dest Tool02_Count 0

GE
Source A Tool02_Count 0
Source B 10

Tool02_TMR (RES)

CLR
Dest Tool02_Count 0

Equipment Quality/Health Data Mapping

Load Tool Vehicle Data

3

ShiftReset

ToolVehONS[1]
ONS

Year
NE
Source A WCT_DstTool[0]
0
Source B 0

Tool01_Start

Tool01_Send

Tool02_Start

Tool02_Send

Equipment Data Quality/Health Message Program Number from SLCP
MOVE
Source 0
Dest QualityMsg.Data.AviPrgNum
0

Equipment Data Quality/Health Message Program Number Executed
MOVE
Source 0
Dest QualityMsg.Data.ActPrgNum
0

Equipment Data Quality/Health Message Assembly Body Number
MOVE
Source 0
Dest QualityMsg.Data.BodyNum
0

Equipment Data Quality/Health Message Assembly Sequence Number
MOVE
Source 0
Dest QualityMsg.Data.AssyNum
0

Replace with User Logic
FLL
Source 0
Dest QualityMsg.Data.VinNum[0]
Length 18

Equipment Data Quality/Health Message VIN Number
CLR
Dest QualityMsg.Data.VinNum[18]
'$00'

Equipment Data Quality/Health Message VIN Number
CLR
Dest QualityMsg.Data.VinNum[19]
'$00'

Equipment Data Quality/Health Message Employee Badge Number for Job Release
FLL
Source 0
Dest QualityMsg.Data.BadgeNum[0]
Length 12

Equipment Quality/Health Data Mapping

4

Tool 1
PdM Trigger
(Tool 1 Config has been changed to PdM Sample)

Tool01_Start
Tool01_Start
U
Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.0

Tool01_Send
Tool01_Send
U
Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.1

Equipment Data Quality/Health Message Tool Number
MOVE
Source 1
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 1
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Tool 2
PdM Trigger

tool to demo ability to reset Trigger data by resending Start signal
each Start Signal performs Pdm Reset/Init on Data array

5

Tool02_Start Tool02_Start (U) — Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send: QualityMsg.Data.Spare1.0

Tool02_Send Tool02_Send (U) — Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send: QualityMsg.Data.Spare1.1

Equipment Data Quality/Health Message Tool Number
MOVE
Source 2
Dest QualityMsg.Data.ToolNum 100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 4
Dest QualityMsg.Data.DataCount 10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus 1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode 0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source Tool02_Count 0
Dest QualityMsg.Data.CycleTime 0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source Tool02_Count 0
Dest QualityMsg.Data.CycleCount 0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Wallclock TimeZone and DST
Tool Number 100

6

Year
NE
Source A WCT_DstTool[0] 0
Source B 0

Equipment Data Quality/Health Message Tool Number
MOVE
Source 100
Dest QualityMsg.Data.ToolNum 100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 10
Dest QualityMsg.Data.DataCount 10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus 1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode 0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime 0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount 0

Time Zone Udpated
EQ
Source A WCT_DstTool[7] 0
Source B 1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 1
Dest QualityMsg.Data.FaultCode 0

DST Udpated
EQ
Source A WCT_DstTool[8] 0
Source B 1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 2
Dest QualityMsg.Data.FaultCode 0

Time Zone Udpated
EQ
Source A WCT_DstTool[7] 0
Source B 1

DST Udpated
EQ
Source A WCT_DstTool[8] 0
Source B 1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 3
Dest QualityMsg.Data.FaultCode 0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source WCT_DstTool[0] 0
Dest QualityMsg.Data.Data1 2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source WCT_DstTool[1] 10
Dest QualityMsg.Data.Data2 10

Equipment Data Quality/Health Message Quality Data 3
MOVE
Source WCT_DstTool[2] 30
Dest QualityMsg.Data.Data3 30

Equipment Data Quality/Health Message Quality Data 4
MOVE
Source WCT_DstTool[3] 15
Dest QualityMsg.Data.Data4 15

Equipment Data Quality/Health Message Quality Data 5
MOVE
Source WCT_DstTool[4] 0
Dest QualityMsg.Data.Data5 0

Equipment Quality/Health Data Mapping

Equipment Data Quality/Health Message Quality Data 6

MOVE
Source WCT_DstTool[5] 0
Dest QualityMsg.Data.Data6 0

Equipment Data Quality/Health Message Quality Data 7

MOVE
Source WCT_DstTool[9] 2
Dest QualityMsg.Data.Data7 2

Equipment Data Quality/Health Message Quality Data 8

MOVE
Source WCT_DstTool[11] 1
Dest QualityMsg.Data.Data8 1

Equipment Data Quality/Health Message Quality Data 9

CLR
Dest QualityMsg.Data.Data9 0

Equipment Data Quality/Health Message Quality Data 10

CLR
Dest QualityMsg.Data.Data10 0

Code Rule for DST
0=Disable
1=Disable Dec-Feb
2=Enable Apr-Oct
3=Enable Mar
4=Enable Nov
5=Disable Mar/Nov

GT
Source A WCT_DstTool[9] 2
Source B 2

Equipment Data Quality/Health Message Quality Data 9

MOVE
Source WCT_DstTool[14] 3
Dest QualityMsg.Data.Data9 0

Equipment Data Quality/Health Message Quality Data 10

MOVE
Source WCT_DstTool[15] 29
Dest QualityMsg.Data.Data10 0

Year

CLR
Dest WCT_DstTool[0] 0

Load Quality Data record to Ring Buffer

JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Fault/Event RB Usage by Shift
Tool Number 101

7 ShiftReset ToolSendONS[1] ONS

Equipment Data Quality/Health Message Tool Number
MOVE
Source 101
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 6
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source EDC.RB_Fault01.PrevShiftCnt
85089
Dest QualityMsg.Data.Data1
2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source EDC.RB_Fault02.PrevShiftCnt
25487
Dest QualityMsg.Data.Data2
10

Equipment Data Quality/Health Message Quality Data 3
MOVE
Source EDC.RB_Fault03.PrevShiftCnt
172357
Dest QualityMsg.Data.Data3
30

Equipment Data Quality/Health Message Quality Data 4
MOVE
Source EDC.RB_Fault04.PrevShiftCnt
357752
Dest QualityMsg.Data.Data4
15

Equipment Data Quality/Health Message Quality Data 5
MOVE
Source EDC.RB_Fault05.PrevShiftCnt
161029
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 6
MOVE
Source EDC.RB_Fault06.PrevShiftCnt
401978
Dest QualityMsg.Data.Data6
0

Equipment Data Quality/Health Message Quality Data 7
CLR
Dest QualityMsg.Data.Data7
2

Equipment Data Quality/Health Message Quality Data 8
CLR
Dest QualityMsg.Data.Data8
1

Equipment Data Quality/Health Message Quality Data 9
CLR
Dest QualityMsg.Data.Data9
0

Equipment Data Quality/Health

Equipment Quality/Health Data Mapping

Message Quality Data 10

CLR
Dest QualityMsg.Data.Data10[0]

Load Quality Data record to Ring Buffer

JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Quality/Tool RB Usage by Shift
Tool Number 102

8 ShiftReset ToolSendONS[2] ONS

Equipment Data Quality/Health Message Tool Number
MOVE
Source 102
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 4
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source EDC.RB_Quality01.PrevShiftCnt
101079
Dest QualityMsg.Data.Data1
2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source EDC.RB_Quality02.PrevShiftCnt
102048
Dest QualityMsg.Data.Data2
10

Equipment Data Quality/Health Message Quality Data 3
MOVE
Source EDC.RB_Quality03.PrevShiftCnt
125974
Dest QualityMsg.Data.Data3
30

Equipment Data Quality/Health Message Quality Data 4
MOVE
Source EDC.RB_Quality04.PrevShiftCnt
121316
Dest QualityMsg.Data.Data4
15

Equipment Data Quality/Health Message Quality Data 5
CLR
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 6
CLR
Dest QualityMsg.Data.Data6
0

Equipment Data Quality/Health Message Quality Data 7
CLR
Dest QualityMsg.Data.Data7
2

Equipment Data Quality/Health Message Quality Data 8
CLR
Dest QualityMsg.Data.Data8
1

Equipment Data Quality/Health Message Quality Data 9
CLR
Dest QualityMsg.Data.Data9
0

Equipment Data Quality/Health Message Quality Data 10
CLR
Dest QualityMsg.Data.Data10
0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Equipment Quality/Health Data Mapping

Trend RB Usage by Shift
Tool Number 103

9 ShiftReset ToolSendONS[3] ONS

Equipment Data Quality/Health Message Tool Number
MOVE
Source 103
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 2
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source EDC.RB_Trend01.PrevShiftCnt
2553
Dest QualityMsg.Data.Data1
2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source EDC.RB_Trend02.PrevShiftCnt
3106
Dest QualityMsg.Data.Data2
10

Equipment Data Quality/Health Message Quality Data 3
CLR
Dest QualityMsg.Data.Data3
30

Equipment Data Quality/Health Message Quality Data 4
CLR
Dest QualityMsg.Data.Data4
15

Equipment Data Quality/Health Message Quality Data 5
CLR
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 6
CLR
Dest QualityMsg.Data.Data6
0

Equipment Data Quality/Health Message Quality Data 7
CLR
Dest QualityMsg.Data.Data7
2

Equipment Data Quality/Health Message Quality Data 8
CLR
Dest QualityMsg.Data.Data8
1

Equipment Data Quality/Health Message Quality Data 9
CLR
Dest QualityMsg.Data.Data9
0

Equipment Data Quality/Health Message Quality Data 10
CLR
Dest QualityMsg.Data.Data10
0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Equipment Quality/Health Data Mapping

Summary RB Usage by Shift
Tool Number 104

10

ShiftReset ToolSendONS[4] ONS

Equipment Data Quality/Health Message Tool Number
MOVE
Source 104
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 2
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source EDC.RB_Sum01.PrevShiftCnt
132552
Dest QualityMsg.Data.Data1
2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source EDC.RB_Sum02.PrevShiftCnt
132844
Dest QualityMsg.Data.Data2
10

Equipment Data Quality/Health Message Quality Data 3
CLR
Dest QualityMsg.Data.Data3
30

Equipment Data Quality/Health Message Quality Data 4
CLR
Dest QualityMsg.Data.Data4
15

Equipment Data Quality/Health Message Quality Data 5
CLR
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 6
CLR
Dest QualityMsg.Data.Data6
0

Equipment Data Quality/Health Message Quality Data 7
CLR
Dest QualityMsg.Data.Data7
2

Equipment Data Quality/Health Message Quality Data 8
CLR
Dest QualityMsg.Data.Data8
1

Equipment Data Quality/Health Message Quality Data 9
CLR
Dest QualityMsg.Data.Data9
0

Equipment Data Quality/Health Message Quality Data 10
CLR
Dest QualityMsg.Data.Data10
0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Equipment Quality/Health Data Mapping

Cycle RB Usage by Shift
Tool Number 105

11 ShiftReset ToolSendONS[5] ONS

Equipment Data Quality/Health Message Tool Number
MOVE
Source 105
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 2
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source EDC.RB_Cycle01.PrevShiftCnt
8603
Dest QualityMsg.Data.Data1
2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source EDC.RB_Cycle02.PrevShiftCnt
8544
Dest QualityMsg.Data.Data2
10

Equipment Data Quality/Health Message Quality Data 3
CLR
Dest QualityMsg.Data.Data3
30

Equipment Data Quality/Health Message Quality Data 4
CLR
Dest QualityMsg.Data.Data4
15

Equipment Data Quality/Health Message Quality Data 5
CLR
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 6
CLR
Dest QualityMsg.Data.Data6
0

Equipment Data Quality/Health Message Quality Data 7
CLR
Dest QualityMsg.Data.Data7
2

Equipment Data Quality/Health Message Quality Data 8
CLR
Dest QualityMsg.Data.Data8
1

Equipment Data Quality/Health Message Quality Data 9
CLR
Dest QualityMsg.Data.Data9
0

Equipment Data Quality/Health Message Quality Data 10
CLR
Dest QualityMsg.Data.Data10
0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

(End)

Equipment Trend Data Mapping

Equipment Trend Data
User Data Mapping

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	95
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

(End)

Equipment Cycle Data Mapping

Equipment Cycle Data
User Data Mapping

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 96
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

(End)

Pre-Defined Fault/Status Data Mapping

Predefined Fault/Status Data
v2.0 - Aug2022 - PreDefined Events defined by EqDriver

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 100
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

v1.13 EDC EQ Driver - Wallclock Time Invalid

2

CLx Module Status
CLx Transaction Time Invalid
CLX.TransTimeInvalid

Equipment Data Fault/Status Message Invalid Wallclock Time (from EqDriver)
FaultMsg.PreDef._21

Pre-Defined Fault/Status Data Mapping

v1.5 EDC EQ Driver - Equipment Data Error
FEC EQ Driver not Polling Fault, Quality and/or Trend Data
v1.6 Optional Message to get Overall Start from Gateway

3

Gateway Overall Start Message ER
OverallStartReadMsgER

Equipment Data Collection - User Config Toyopuc Driver Enable
UserCfg.TpucDriverEN

TpucDriverER

Equipment Data Collection - User Config Remote Driver Enable
UserCfg.RemDriverEN

RemoteDriverER

Equipment Data Fault/Status Message Equipment Data Error (from EqDriver)
FaultMsg.PreDef._22

EDC EQ Driver - Ring Buffers Not Empty after 5 minutes
v2.1 - combined all RB Filling to a single Event
v2.09 Change RBCtl.NotEmptyTMR.DN to RBCtl.BufferAlarm to include Not Empty or Full condition

4

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Fault Message Enable
UserCfg.FaultMsgEN

EDC Ring Buffer Control Read Buffer Alarm
FaultRBCtl.BufferAlarm

Diagnostic - manual reset
RingBufferFIlling_Fault

Equipment Data Fault/Status Message RingBuffer Filling (from EqDriver)
FaultMsg.PreDef._30

Equipment Data Collection - User Config Quality Message Enable
UserCfg.QualityMsgEN

EDC Ring Buffer Control Read Buffer Alarm
QualityRBCtl.BufferAlarm

Diagnostic - manual reset
RingBufferFIlling_Quality

Equipment Data Collection - User Config Trend Message Enable
UserCfg.TrendMsgEN

EDC Ring Buffer Control Read Buffer Alarm
TrendRBCtl.BufferAlarm

Diagnostic - manual reset
RingBufferFIlling_Trend

Equipment Data Collection - User Config Cycle Message Enable
UserCfg.CycleMsgEN

EDC Ring Buffer Control Read Buffer Alarm
CycleRBCtl.BufferAlarm

Diagnostic - manual reset
RingBufferFIlling_Cycle

Pre-Defined Fault/Status Data Mapping

EDC EQ Driver - Message Error
v2.1 combined Callout, Fault, Quality, Trend, Time Sync, Version and Cycle to a single Event
(Note: If Callout Error is ON, Callout Msg with this status is not delivered to Gateway -- this Error is just a placeholder -- the EDC Gateway will detect Callout Msg Timeout and annunciate Error)

5

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - Line Equipment Status State = Red - Callout Tx ER
Status.State.CalloutTxER

Diagnostic - manual reset
MsgER_Callout
L

Equipment Data Fault/Status Message Gateway Message Error (from EqDriver)
FaultMsg.PreDef._31

EDC Gateway Callout Read Message
CalloutReadMsg.ER
<EQ0006_CalloutReadMsg.ER>

EDC Gateway Callout Write Message
CalloutWriteMsg.ER
<EQ0006_CalloutWriteMsg.ER>

Equipment Data Collection - User Config Fault Message Enable
UserCfg.FaultMsgEN

Equipment Data Collection - Line Equipment Status State = Red - Fault Tx ER
Status.State.FaultTxER

Diagnostic - manual reset
MsgER_Fault
L

EDC Gateway Fault Read Message
FaultReadMsg.ER
<EQ0006_FaultReadMsg.ER>

EDC Gateway Fault Write Message
FaultWriteMsg.ER
<EQ0006_FaultWriteMsg.ER>

Equipment Data Collection - User Config Quality Message Enable
UserCfg.QualityMsgEN

Equipment Data Collection - Line Equipment Status State = Red - Quality Tx ER
Status.State.QualityTxER

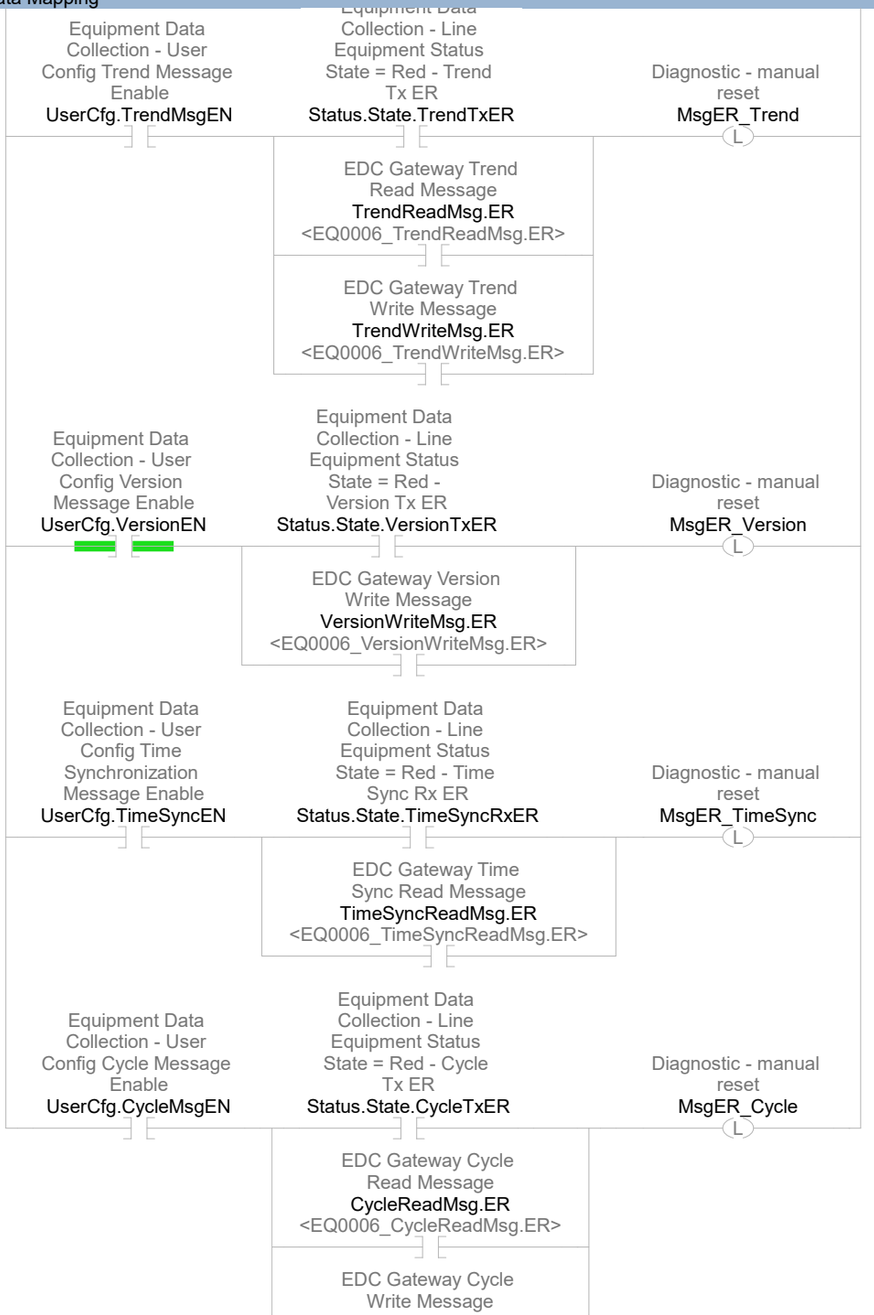
Diagnostic - manual reset
MsgER_Quality
L

EDC Gateway Quality Read Message
QualityReadMsg.ER
<EQ0006_QualityReadMsg.ER>

EDC Gateway Quality Write Message
QualityWriteMsg.ER
<EQ0006_QualityWriteMsg.ER>

Equipment Data

Pre-Defined Fault/Status Data Mapping



Pre-Defined Fault/Status Data Mapping

CycleWriteMsg.ER
<EQ0006_CycleWriteMsg.ER>

Move PreDef bits to DINT for Masking
Load hard-coded Mask for Events used to determine Fault, Alarm, Alert, Maint
(PreDefMaskUsed - ignore PreDefined signals that are generated as a result of Fault, Alarm, Alert)
EDC2 Sep 30, 2022 - remove Pre.15 Fault Memory from PreDefMaskUsed

6

COP	
Source	FaultMsg.PreDef
Dest	PreDefVal
Length	1

Events used to determine Fault, Alarm, Alert, Maint

MOVE	
Source	16#fffd3f00
Dest	PreDefMaskUsed 16#fffd_3f00

Equipment Fault - EqDef and PreDef
used by EDC to determine Equipment Downtime and OA%
(Red Status on Overview)
v2.00 - Fault Mask downloaded from EDC Server

7

Checks EqDef Events against the selected Mask

aoiEDCv2_00_EqDefMaskCheck	
aoiEDCv2_00_EqDefMas...	aoiEqDefFaultCheck
EqDef	FaultMsgCtl.EqDef
EqDefMask	EqDefFaultMask
UserCfg	UserCfg
EventOn	EqDefFaultOn
	0

AND	
Source A	PreDefVal 16#0000_0021
Source B	PreDefMaskUsed 16#fffd_3f00
Dest	PreDefFault 16#0000_0000

AND	
Source A	PreDefFault 16#0000_0000
Source B	PreDefMaskFault 16#2035_cb50
Dest	PreDefFault 16#0000_0000

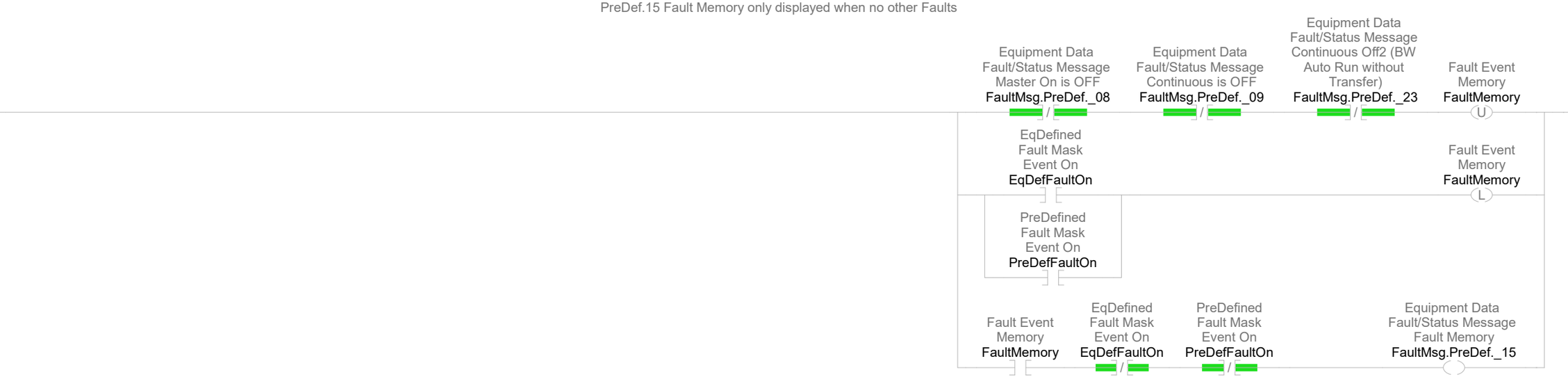
NE	
Source A	PreDefFault 16#0000_0000
Source B	0

PreDefined Fault Mask Event On
PreDefFaultOn

Pre-Defined Fault/Status Data Mapping

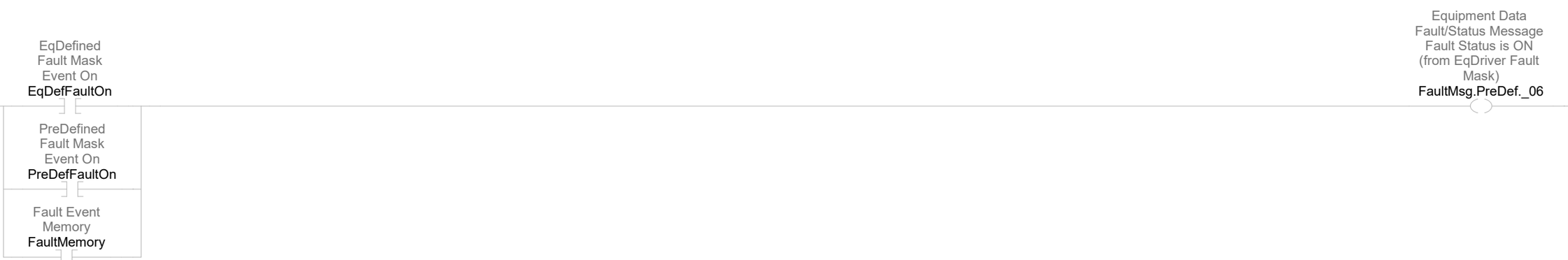
Equipment Fault - Fault Memory
On with any Fault
Off when ready to run
PreDef.15 Fault Memory only displayed when no other Faults

8



Equipment Fault - EqDef, PreDef or PreDefMemory

9



Pre-Defined Fault/Status Data Mapping

Equipment Alarm
(Yellow Status on Overview)
v2.00 - Alarm Mask downloaded from EDC Server

10

Checks EqDef Events against the selected Mask

aoiEDCv2_00_EqDefMaskCheck

aoiEDCv2_00_EqDefMas...	aoiEqDefAlarmCheck
EqDef	FaultMsgCtl.EqDef
EqDefMask	EqDefAlarmMask
UserCfg	UserCfg
EventOn	EqDefAlarmOn
	0

AND

Source A	PreDefVal 16#0000_0021
Source B	PreDefMaskUsed 16#fffd_3f00
Dest	PreDefAlarm 16#0000_0000

AND

Source A	PreDefAlarm 16#0000_0000
Source B	PreDefMaskAlarm 16#c0c8_2088
Dest	PreDefAlarm 16#0000_0000

NE

Source A	PreDefAlarm 16#0000_0000
Source B	0

PreDefined Alarm Mask Event On
PreDefAlarmOn

EqDefined Alarm Mask Event On
EqDefAlarmOn

PreDefined Alarm Mask Event On
PreDefAlarmOn

Equipment Data Fault/Status Message Alarm Status is ON (from EqDriver Alarm Mask)
FaultMsg.PreDef._07

Pre-Defined Fault/Status Data Mapping

v1.17 - Equipment Alert
(Lt Blue Status on Overview)
v2.00 - Alert Mask downloaded from EDC Server

11

Checks EqDef Events against the selected Mask

aoiEDCv2_00_EqDefMaskCheck

aoiEDCv2_00_EqDefMas...	aoiEqDefAlertCheck
EqDef	FaultMsgCtl.EqDef
EqDefMask	EqDefAlertMask
UserCfg	UserCfg
EventOn	EqDefAlertOn
	0

AND

Source A	PreDefVal 16#0000_0021
Source B	PreDefMaskUsed 16#fffd_3f00
Dest	PreDefAlert 16#0000_0000

AND

Source A	PreDefAlert 16#0000_0000
Source B	PreDefMaskAlert 16#1002_0020
Dest	PreDefAlert 16#0000_0000

NE

Source A	PreDefAlert 16#0000_0000
Source B	0

PreDefined Alert Mask Event On
PreDefAlertOn

EqDefined Alert Mask Event On
EqDefAlertOn

PreDefined Alert Mask Event On
PreDefAlertOn

Equipment Data Fault/Status Message Alert Status is ON (from EqDriver Alert Mask)
FaultMsg.PreDef._17

Pre-Defined Fault/Status Data Mapping

Stop Request caused by Production
used by EDC to determine Equipment Downtime and OA%
v2.00 - Maintenance Mask downloaded from EDC Server

12

Checks EqDef Events against the selected Mask

aoiEDCv2_00_EqDefMaskCheck

aoiEDCv2_00_EqDefMas...	aoiEqDefMaintCheck
EqDef	FaultMsgCtl.EqDef
EqDefMask	EqDefMaintMask
UserCfg	UserCfg
EventOn	EqDefMaintOn
	0

AND

Source A	PreDefVal
	16#0000_0021
Source B	PreDefMaskUsed
	16#fffd_3f00
Dest	PreDefMaint
	16#0000_0000

AND

Source A	PreDefMaint
	16#0000_0000
Source B	PreDefMaskMaint
	16#2035_cb40
Dest	PreDefMaint
	16#0000_0000

NE

Source A	PreDefMaint
	16#0000_0000
Source B	0

PreDefined Maint Mask Event On
PreDefMaintOn

EqDefined Maint Mask Event On
EqDefMaintOn

PreDefined Maint Mask Event On
PreDefMaintOn

MaintActive

MaintActive
/

Equipment Data Fault/Status Message Production Stop Request (from EqDriver Mask Prod/Maint Mask)
FaultMsg.PreDef._05

(End)

Callout Data Load for Message

Equipment Fault/Status Data
Callout Data Load
v1.3 - Dec2016 - Multi-packet support for >8 DINT Equipment-Defined Faults

PKTi - Packet Index for multiplexed Fault Data

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 112
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Exit if This Packet not Enabled

2

Fault Ring Buffer Message Control for multiplexed packets Data Change
FaultMsgCtl.Change.[PKTi]
/

lbl_Exit
JMP

Send This Packet

3

Fault Ring Buffer Message Control for multiplexed packets Data Change
FaultMsgCtl.Change.[PKTi]
U

Message Control Send Data
CalloutMsgCtl.Send
L

Message Control Read Buffer Ready
CalloutMsgCtl.ReadOK
U

Message Control Write Command Done
CalloutMsgCtl.WriteOK
U

Callout Data Load for Message

Increment Transaction Number

4

Message Control TM Transaction Trigger

ADD

Source A	CalloutMsgCtl.TransNum 9993
Source B	1
Dest	CalloutMsgCtl.TransNum 9993

Message Control TM Transaction Trigger

LT

Source A	CalloutMsgCtl.TransNum 9993
Source B	1

Message Control TM Transaction Trigger

MOVE

Source	1
Dest	CalloutMsgCtl.TransNum 9993

Message Control TM Transaction Trigger

GT

Source A	CalloutMsgCtl.TransNum 9993
Source B	CFG.MsgTransNumRollover 9999

Callout Data Load for Message

Load Message Header, Timestamp, Data and Footer
v1.3 - Added Header.PacketID for multiplexed data packets
v2.00 Footer.Spare = 2 indicates v2 with Fault Event Mask in Callout Read Packets

5

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

MOVE
Source CalloutMsgCtl.TransNum 9993
Dest CalloutMsg.Header.TransNum 9993

Equipment Data Callout/Status Message Header - Command Number

MOVE
Source 0
Dest CalloutMsg.Header.CmdNum 0

Equipment Data Callout/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source UserCfg.EQID 6
Dest CalloutMsg.Header.EQID 6

Equipment Data Callout/Status Message Packet ID - generates offset for 8-DINT mulitplexed data packet

MOVE
Source PKTi -1
Dest CalloutMsg.Header.PacketID 1

Equipment Data Callout/Status Message Transaction Time Stamp

COP
Source CLX.TransTime
Dest CalloutMsg.TransTime
Length 1

Equipment Data Callout/Status Message Data - Predefined Status

COP
Source FaultMsg.PreDef
Dest CalloutMsg.PreDef
Length 1

Equipment Data Callout/Status Message Data - Equipment Defined Status

COP
Source FaultMsgCtl.EqDef[PKTi]
Dest CalloutMsg.EqDef
Length 1

Equipment Data Callout/Status Message Footer - Spare (fills unused portion of DINT)

MOVE
Source 2
Dest CalloutMsg.Footer.Spare 2

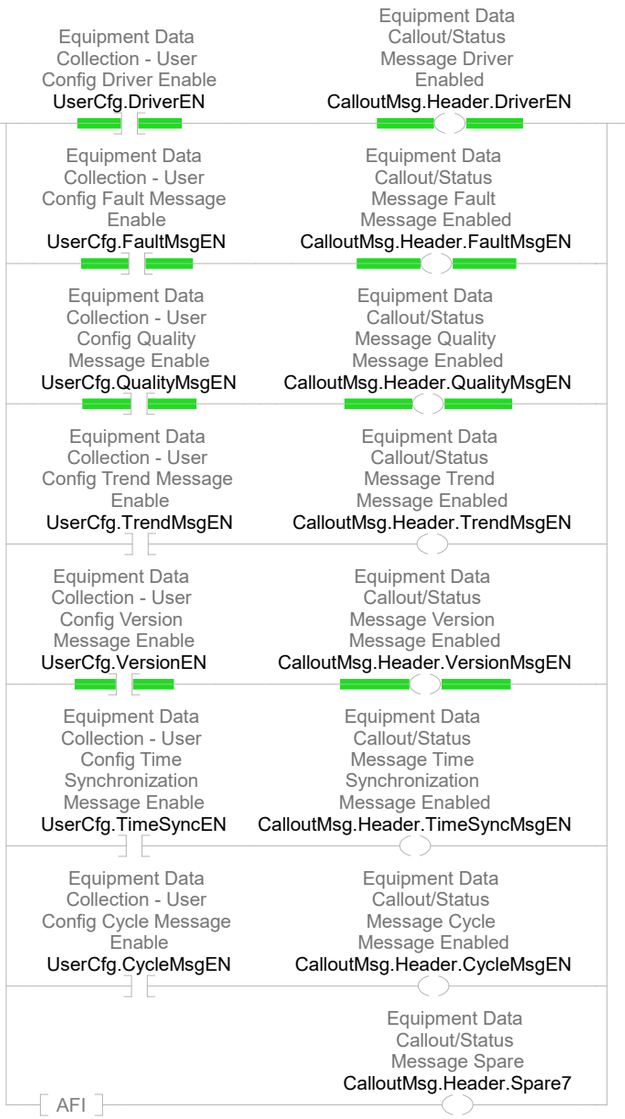
Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

MOVE
Source CalloutMsgCtl.TransNum 9993
Dest CalloutMsg.Footer.TransNum 9993

Callout Data Load for Message

Load Message Header Config Bits

6



Callout Data Load for Message

Break out of the FOR Loop when a Packet is found for Callout Message

7

BRK

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Callout Data Message

Equipment Fault/Status Data
Callout Live Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 114
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

Message Control Message Transaction Timer

MOVE
Source 999999999
Dest CalloutMsgCtl.Trans.TMR.PRE 999999999

Message Control Message Transaction Timer

TON
Timer CalloutMsgCtl.Trans.TMR
Preset 999999999
Accum 24725
EN
DN

Message Delay Timer
prevent Message from sending too quickly for Gateway
Gateway processes one Line per scan -- Equipment Message could take 500-1000 msec to process
any attempt to send faster will generate extra Read Message traffic until previous message has been processed

3

EDC Gateway Callout Write Message
CalloutWriteMsg.EN
<EQ0006_CalloutWriteMsg.EN>

EDC Gateway Callout Read Message
CalloutReadMsg.EN
<EQ0006_CalloutReadMsg.EN>

Message Control Message Delay Timer

MOVE
Source CFG.MsgDelayTime 250
Dest CalloutMsgCtl.DelayTMR.PRE 250

Message Control Message Delay Timer

TON
Timer CalloutMsgCtl.DelayTMR
Preset 250
Accum 253
EN
DN

Callout Data Message

Message Resend Timer
Keep Alive - resends last data without increment of Transaction Number

4

Message Control Send Data
CalloutMsgCtl.Send

EDC Gateway Callout Write Message
CalloutWriteMsg.EN
<EQ0006_CalloutWriteMsg.EN>

EDC Gateway Callout Read Message
CalloutReadMsg.EN
<EQ0006_CalloutReadMsg.EN>

Message Control Message Resend Timer

MOVE	
Source	CFG.LiveMsgResendTime 31000
Dest	CalloutMsgCtl.ResendTMR.PRE 31000

Message Control Message Resend Timer

TON		
Timer	CalloutMsgCtl.ResendTMR	EN
Preset	31000	DN
Accum	24698	

Message Control Message Resend Timer
CalloutMsgCtl.ResendTMR.DN

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend
L

Message Control Message Resend Timer
CalloutMsgCtl.ResendTMR
RES

Message Control Read Command
CalloutMsgCtl.Read
U

Message Control Read Buffer Ready
CalloutMsgCtl.ReadOK
U

Message Control Write Data
CalloutMsgCtl.Write
U

Message Control Write Command Done
CalloutMsgCtl.WriteOK
U

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Header.PacketID 1
Dest	Trans.Ctl.Opt[1] -1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	122
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Send Data Initialize

5

Message Control Send Data
CalloutMsgCtl.Send

Message Control Send Data 1-shot
CalloutMsgCtl.SendONS
ONS

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend

Message Control Message Transaction Timer
CalloutMsgCtl.Trans.TMR
RES

Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR	
Dest	CalloutMsgCtl.Trans.Last 15

Callout Data Message

Read Message
Check Gateway Buffer for Empty Record
(Gateway clears Buffer after processing Record)

v2.00 - Gateway also sends the Fault, Alarm, Alert, Ignore and Maintenance Masks in the 'Empty Record'
v3.50 jumper CalloutMsg Delay TMR - need fast updates for accurate timing and fast PdM Triggers

6

Message Control Send Data
CalloutMsgCtl.Send

Message Control Read Buffer Ready
CalloutMsgCtl.ReadOK

Message Control Message Delay Timer
CalloutMsgCtl.DelayTMR.DN

EDC Gateway Callout Read Message
CalloutReadMsg.EN
<EQ0006_CalloutReadMsg.EN>

EDC Gateway Callout Write Message
CalloutWriteMsg.EN
<EQ0006_CalloutWriteMsg.EN>

EDC Gateway Callout Read Message
CalloutReadMsg.ER
<EQ0006_CalloutReadMsg.ER>

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend

Message Control Message Retry Timer
CalloutMsgCtl.RetryTMR.DN

Message Control Read Command
CalloutMsgCtl.Read
(L)

Message Control Read Command 1-Shot
CalloutMsgCtl.ReadONS

EDC Gateway Callout Read Message

COP
Source UserCfgMsg.Path
<EQ0006_UserCfgMsg.Path>
Dest CalloutReadMsg.Path
<EQ0006_CalloutReadMsg.Path>
Length 1

EDC Gateway Callout Read Message

COP
Source CFG.MsgDest.CalloutDest
Dest CalloutReadMsg.RemoteElement
<EQ0006_CalloutReadMsg.RemoteElement>
Length 1

EDC Gateway Callout Read Message

MOVE
Source 14
Dest CalloutReadMsg.REQ_LEN
<EQ0006_CalloutReadMsg.REQ_LEN>
14

EDC Gateway Callout Read Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest CalloutReadMsg.UnconnectedTimeout
<EQ0006_CalloutReadMsg.UnconnectedTimeout>
4000000

EDC Gateway Callout Read Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest CalloutReadMsg.ConnectionRate
<EQ0006_CalloutReadMsg.ConnectionRate>
1000000

EDC Gateway Callout Read Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest CalloutReadMsg.TimeoutMultiplier
<EQ0006_CalloutReadMsg.TimeoutMultiplier>
0

EDC Gateway Callout Read Message
CalloutReadMsg.EN_CC
<EQ0006_CalloutReadMsg.EN_CC>
(U)

EDC Gateway Callout Read Message

MSG
Message Control CalloutReadMsg ...
<EQ0006_CalloutReadMsg>
EN
DN
ER

Callout Data Message

Read Message EN Performance & Diagnostics

7

Message Control Read
Command 1-Shot
CalloutMsgCtl.ReadONS

Performance
Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	CalloutMsgCtl.Trans
PerfTrans	Status.CalloutTxRead.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	130
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Callout Data Message

Read Message OK
Gateway Buffer has Empty Record
(Gateway clears Buffer after processing Record and loads Mask data)

MEQ used to get first and last INTs holding Header and Footer TransNum
Callout Rx Message contains non-zero Mask data in the adjacent INTs

8

Message Control Read Command
CalloutMsgCtl.Read

EDC Gateway Callout Read Message
CalloutReadMsg.DN
<EQ0006_CalloutReadMsg.DN>

Message Control Read Command
CalloutMsgCtl.Read
(U)

EDC Message Rx Data Callout/Status Receive Message

MEQ
Source EqRxData.Callout[0]
<EQ0006_RxData.Callout[0]> -65536
Mask 16#0000_ffff
Compare 0

EDC Message Rx Data Callout/Status Receive Message

MEQ
Source EqRxData.Callout[CFG.CalloutMSG.ei]
<EQ0006_RxData.Callout[CFG.CalloutMSG.ei]> 248
Mask 16#ffff_0000
Compare 0

Message Control Read Buffer Ready
CalloutMsgCtl.ReadOK
(L)

Store Event Mask in Callout Read Message

JSR
Routine Name _115_CalloutDataMask

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans CalloutMsgCtl.Trans
PerfTrans Status.CalloutTxRead.DN

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans CalloutMsgCtl.Trans
PerfTrans Status.CalloutTxRead.KA

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EqRxData.Callout[0]
<EQ0006_RxData.Callout[0]> -65536
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EqRxData.Callout[CFG.CalloutMSG.ei]
<EQ0006_RxData.Callout[CFG.CalloutMSG.ei]> 248
Dest Trans.Ctl.Opt[2]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 131
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

Callout Data Message

Read Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

9

EDC Gateway Callout Read Message
CalloutReadMsg.EN
<EQ0006_CalloutReadMsg.EN>

Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest CalloutMsgCtl.ReadTimeoutTMR.PRE
5000

Message Control Message Timeout Timer

TON
Timer CalloutMsgCtl.ReadTimeoutTMR
Preset 5000
Accum 0
EN
DN

Message Control Message Timeout Timer
CalloutMsgCtl.ReadTimeoutTMR.DN

EDC Gateway Callout Read Message
CalloutReadMsg.TO
<EQ0006_CalloutReadMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 134
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

Read Message ER Performance & Diagnostics

10

EDC Gateway Callout Read Message
CalloutReadMsg.ER
<EQ0006_CalloutReadMsg.ER>

Message Control Read Error 1-shot
CalloutMsgCtl.ReadErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Callout Tx ER
Status.Update.CalloutTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans CalloutMsgCtl.Trans
PerfTrans Status.CalloutTxRead.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 133
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

Callout Data Message

Write Message

v3.50 remove CalloutMsg Delay TMR - need fast updates for accurate timing and fast PdM Triggers

11

Message Control Send Data
CalloutMsgCtl.Send

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend

Message Control Read Buffer Ready
CalloutMsgCtl.ReadOK

Message Control Write Command Done
CalloutMsgCtl.WriteOK

EDC Gateway Callout Read Message
CalloutReadMsg.EN
<EQ0006_CalloutReadMsg.EN>

EDC Gateway Callout Write Message
CalloutWriteMsg.EN
<EQ0006_CalloutWriteMsg.EN>

EDC Gateway Callout Write Message
CalloutWriteMsg.ER
<EQ0006_CalloutWriteMsg.ER>

Message Control Message Retry Timer
CalloutMsgCtl.RetryTMR.DN

Message Control Write Data
CalloutMsgCtl.Write
L

Message Control Message Delay Timer
CalloutMsgCtl.DelayTMR
RES

EDC Gateway Callout Write Message

COP
Source UserCfgMsg.Path
<EQ0006_UserCfgMsg.Path>
Dest CalloutWriteMsg.Path
<EQ0006_CalloutWriteMsg.Path>
Length 1

EDC Gateway Callout Write Message

COP
Source CFG.MsgDest.CalloutDest
Dest CalloutWriteMsg.RemoteElement
<EQ0006_CalloutWriteMsg.RemoteElement>
Length 1

EDC Gateway Callout Write Message

MOVE
Source CFG.CalloutMSG.e
14
Dest CalloutWriteMsg.REQ_LEN
<EQ0006_CalloutWriteMsg.REQ_LEN>
14

EDC Gateway Callout Write Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest CalloutWriteMsg.UnconnectedTimeout
<EQ0006_CalloutWriteMsg.UnconnectedTimeout>
4000000

EDC Gateway Callout Write Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest CalloutWriteMsg.ConnectionRate
<EQ0006_CalloutWriteMsg.ConnectionRate>
1000000

EDC Gateway Callout Write Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest CalloutWriteMsg.TimeoutMultiplier
<EQ0006_CalloutWriteMsg.TimeoutMultiplier>
0

EDC Gateway Callout Write Message
CalloutWriteMsg.EN_CC
<EQ0006_CalloutWriteMsg.EN_CC>
U

EDC Message Tx Data Callout/Status Receive Message

COP
Source CalloutMsg
Dest EqTxData.Callout[0]
<EQ0006_TxData.Callout[0]>
Length CFG.CalloutMSG.n

EDC Gateway Callout Write Message

MSG
Message Control CalloutWriteMsg ...
<EQ0006_CalloutWriteMsg>
EN
DN
ER

Callout Data Message

Message EN Performance & Diagnostics

12

Message Control Write Data: CalloutMsgCtl.Write

Message Control Write Data 1-shot: CalloutMsgCtl.WriteONS

ONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	CalloutMsgCtl.Trans
PerfTrans	Status.CalloutTxWrite.EN

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Header.PacketID
	1
Dest	Trans.Ctl.Opt[1]
	-1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	140
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Callout Data Message

13

Message Control Write Data CalloutMsgCtl.Write

EDC Gateway Callout Write Message CalloutWriteMsg.DN <EQ0006_CalloutWriteMsg.DN>

Message Done

Message Control Write Data CalloutMsgCtl.Write (U)

Message Control Write Command Done CalloutMsgCtl.WriteOK (L)

Message Control Send Data CalloutMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Callout Tx DN Status.Update.CalloutTxOK (L)

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Message Control Resend Data (Keep Alive) CalloutMsgCtl.Resend

Equipment Data Collection - Line Equipment Status State = Green - Callout Keep Alive Tx DN Status.Update.CalloutKaTxOK (L)

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	CalloutMsgCtl.Trans
PerfTrans	Status.CalloutTxWrite.DN

Message Control Resend Data (Keep Alive) CalloutMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	CalloutMsgCtl.Trans
PerfTrans	Status.CalloutTxWrite.KA

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Header.PacketID 1
Dest	Trans.Ctl.Opt[1] -1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	141
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Message Control Send Data CalloutMsgCtl.Send (U)

Message Control Resend Data (Keep Alive) CalloutMsgCtl.Resend (U)

Callout Data Message

Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

14

EDC Gateway Callout Write Message
CalloutWriteMsg.EN
<EQ0006_CalloutWriteMsg.EN>

Message Control Message Timeout Timer

MOVE	
Source	CFG.CLx.MsgTimeout
	5000
Dest	CalloutMsgCtl.WriteTimeoutTMR.PRE
	5000

Message Control Message Timeout Timer

TON		
Timer	CalloutMsgCtl.WriteTimeoutTMR	EN
Preset	5000	DN
Accum	0	

Message Control Message Timeout Timer
CalloutMsgCtl.WriteTimeoutTMR.DN

EDC Gateway Callout Write Message
CalloutWriteMsg.TO
<EQ0006_CalloutWriteMsg.TO>
L

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Header.PacketID
	1
Dest	Trans.Ctl.Opt[1]
	-1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	144
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Message ER Performance & Diagnostics

15

EDC Gateway Callout Write Message
CalloutWriteMsg.ER
<EQ0006_CalloutWriteMsg.ER>

Message Control Write Error 1-shot
CalloutMsgCtl.WriteErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Callout Tx ER
Status.Update.CalloutTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate		
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate	...
Status	Status	

Performance Transaction

aoiEDCv1_5_EqPerfTrans		
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans	...
PerfMsgTrans	CalloutMsgCtl.Trans	
PerfTrans	Status.CalloutTxWrite.ER	

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Header.PacketID
	1
Dest	Trans.Ctl.Opt[1]
	-1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	143
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Callout Data Message

Message Error Retry

16

EDC Gateway Callout
Read Message
CalloutReadMsg.ER
<EQ0006_CalloutReadMsg.ER>

EDC Gateway Callout
Write Message
CalloutWriteMsg.ER
<EQ0006_CalloutWriteMsg.ER>

Message Control
Message Retry Timer
MOVE
Source CFG.MsgRetryTime
5000
Dest CalloutMsgCtl.RetryTMR.PRE
0

Message Control
Message Retry Timer
TON
Timer CalloutMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

Message Control
Message Retry Timer
CalloutMsgCtl.RetryTMR.DN

Message Control
Retry 1-shot
CalloutMsgCtl.RetryONS
ONS

EDC Gateway Callout
Read Message
CalloutReadMsg.ER
<EQ0006_CalloutReadMsg.ER>

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source CalloutMsg.Header.PacketID
1
Dest Trans.Ctl.Opt[1]
-1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 132
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

EDC Gateway Callout
Write Message
CalloutWriteMsg.ER
<EQ0006_CalloutWriteMsg.ER>

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source CalloutMsg.Header.PacketID
1
Dest Trans.Ctl.Opt[1]
-1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 142
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

(End)

Store Event Mask in Callout Read Message

Equipment Fault/Status Data
Callout Data Mask Store from Read Message

Callout Read Message contains Mask for one Packet (8 words) for Fault, Alarm, Alert, Ignore or Maint
or a Request to Resend Data for one Packet

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 115
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Store Event Mask in Callout Read Message

Check for Valid EQID
Exit if Invalid

2

Equipment Data Callout/Status Message

COP
Source EqRxData.Callout[0]
<EQ0006_RxData.Callout[0]>
Dest CalloutMsgMask
Length 1

Equipment Data Callout/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE
Source A CalloutMsgMask.Header.EQID
6
Source B UserCfg.EQID
6

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsgMask.Header.EQID
6
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsgMask.Header.PacketID
1
Dest Trans.Ctl.Opt[2]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsgMask.Header.CmdNum
-1
Dest Trans.Ctl.Opt[3]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 136
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Check for Valid PacketID
Exit if Invalid

3

Equipment Data
Callout/Status
Message Packet ID -
generates offset for
8-DINT mulitplexed
data packet

LT
Source A CalloutMsgMask.Header.PacketID
1
Source B 0

Equipment Data
Callout/Status
Message Packet ID -
generates offset for
8-DINT mulitplexed
data packet

GT
Source A CalloutMsgMask.Header.PacketID
1
Source B CFG.FaultEq.ei
2

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source CalloutMsgMask.Header.EQID
6
Dest Trans.Ctl.Opt[1]
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source CalloutMsgMask.Header.PacketID
1
Dest Trans.Ctl.Opt[2]
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source CalloutMsgMask.Header.CmdNum
-1
Dest Trans.Ctl.Opt[3]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 137
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

lbl_Exit
JMP

Load Valid PacketID to PKi index

4

Mask Storage
Packet Index

MOVE
Source CalloutMsgMask.Header.PacketID
1
Dest PKi
1

Store Event Mask in Callout Read Message

Refresh Packet if Mask Changed
Save Maintenance Mask

5

Equipment Data
Callout/Status
Message Header -
Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum
-1
Source B -1

Data Refresh
By Packet
DataRefreshPacket.[PKi]
/

COP
Source CalloutMsgMask.PreDef
Dest PreDef
Length 1

NE
Source A PreDef
16#2035_cb40
Source B PreDefMaskMaint
16#2035_cb40

PreDef Mask
Change
RefreshPreDef

PreDef Mask
Change
RefreshPreDef
/

EqDef Mask
Change
MaskFSC
RES

EqDef Mask
Change
MOVE
Source CFG.FaultEq.n
8
Dest MaskFSC.LEN
8

EqDef Mask
Change
CLR
Dest MaskFSC.POS
7

EqDef Mask
Change
MaskFSC.FD

EqDef Mask
Change
FSC
Control MaskFSC
Length 8
Position 7
Mode ALL
Expression CalloutMsgMask.EqDef.Word[MaskFSC.POS]<>EqDefMaintMask[PKi].Word[MaskFSC.POS]
EN
DN
ER

Data Refresh
By Packet
DataRefreshPacket.[PKi]
L

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

COP
Source CalloutMsgMask.PreDef
Dest PreDefMaskMaint
Length 1

COP
Source CalloutMsgMask.EqDef.Word[0]
Dest EqDefMaintMask[PKi].Word[0]
Length CFG.FaultEq.n

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Store Event Mask in Callout Read Message

Refresh Packet if Mask Changed
Save Ignore Mask

6

Equipment Data Callout/Status Message Header - Command Number

EQ
Source A: CalloutMsgMask.Header.CmdNum
-1
Source B: -2

Data Refresh By Packet
DataRefreshPacket.[PKi]

COP
Source: CalloutMsgMask.PreDef
Dest: PreDef
Length: 1

NE
Source A: PreDef
16#2035_cb40
Source B: PreDefMaskIgnore
16#ffff_ffff

PreDef Mask Change
RefreshPreDef

PreDef Mask Change
RefreshPreDef

EqDef Mask Change
MaskFSC
RES

EqDef Mask Change

MOVE
Source: CFG.FaultEq.n
8
Dest: MaskFSC.LEN
8

EqDef Mask Change

CLR
Dest: MaskFSC.POS
7

EqDef Mask Change
MaskFSC.FD

EqDef Mask Change

FSC
Control: MaskFSC
Length: 8
Position: 7
Mode: ALL
Expression: CalloutMsgMask.EqDef.Word[MaskFSC.POS]<>EqDefIgnoreMask[PKi].Word[MaskFSC.POS]
EN
DN
ER

Data Refresh By Packet
DataRefreshPacket.[PKi]
L

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

COP
Source: CalloutMsgMask.PreDef
Dest: PreDefMaskIgnore
Length: 1

COP
Source: CalloutMsgMask.EqDef.Word[0]
Dest: EqDefIgnoreMask[PKi].Word[0]
Length: CFG.FaultEq.n

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Store Event Mask in Callout Read Message

Refresh Packet if Mask Changed
Save Fault Mask

7

Equipment Data
Callout/Status
Message Header -
Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum
-1
Source B -3

Data Refresh
By Packet
DataRefreshPacket.[PKi]

COP
Source CalloutMsgMask.PreDef
Dest PreDef
Length 1

NE
Source A PreDef
16#2035_cb40
Source B PreDefMaskFault
16#2035_cb50

PreDef Mask
Change
RefreshPreDef

PreDef Mask
Change
RefreshPreDef

EqDef Mask
Change
MaskFSC
RES

EqDef Mask
Change
MOVE
Source CFG.FaultEq.n
8
Dest MaskFSC.LEN
8

EqDef Mask
Change
CLR
Dest MaskFSC.POS
7

EqDef Mask
Change
MaskFSC.FD

EqDef Mask
Change
FSC
Control MaskFSC
Length 8
Position 7
Mode ALL
Expression CalloutMsgMask.EqDef.Word[MaskFSC.POS]<>EqDefFaultMask[PKi].Word[MaskFSC.POS]
EN
DN
ER

Data Refresh
By Packet
DataRefreshPacket.[PKi]
L

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

COP
Source CalloutMsgMask.PreDef
Dest PreDefMaskFault
Length 1

COP
Source CalloutMsgMask.EqDef.Word[0]
Dest EqDefFaultMask[PKi].Word[0]
Length CFG.FaultEq.n

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Store Event Mask in Callout Read Message

Refresh Packet if Mask Changed
Save Alarm Mask

8

Equipment Data
Callout/Status
Message Header -
Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum
-1
Source B -4

Data Refresh
By Packet
DataRefreshPacket.[PKi]

COP
Source CalloutMsgMask.PreDef
Dest PreDef
Length 1

NE
Source A PreDef
16#2035_cb40
Source B PreDefMaskAlarm
16#c0c8_2088

PreDef Mask
Change
RefreshPreDef

PreDef Mask
Change
RefreshPreDef

EqDef Mask
Change
MaskFSC
RES

EqDef Mask
Change

MOVE
Source CFG.FaultEq.n
8
Dest MaskFSC.LEN
8

EqDef Mask
Change

CLR
Dest MaskFSC.POS
7

EqDef Mask
Change
MaskFSC.FD

EqDef Mask
Change

FSC
Control MaskFSC
Length 8
Position 7
Mode ALL
Expression CalloutMsgMask.EqDef.Word[MaskFSC.POS]<>EqDefAlarmMask[PKi].Word[MaskFSC.POS]
EN
DN
ER

Data Refresh
By Packet
DataRefreshPacket.[PKi]
L

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

COP
Source CalloutMsgMask.PreDef
Dest PreDefMaskAlarm
Length 1

COP
Source CalloutMsgMask.EqDef.Word[0]
Dest EqDefAlarmMask[PKi].Word[0]
Length CFG.FaultEq.n

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Store Event Mask in Callout Read Message

Refresh Packet if Mask Changed
Save Alert Mask

9

Equipment Data
Callout/Status
Message Header -
Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum
-1
Source B -5

Data Refresh
By Packet
DataRefreshPacket.[PKi]

COP
Source CalloutMsgMask.PreDef
Dest PreDef
Length 1

NE
Source A PreDef
16#2035_cb40
Source B PreDefMaskAlert
16#1002_0020

PreDef Mask
Change
RefreshPreDef

PreDef Mask
Change
RefreshPreDef

EqDef Mask
Change
MaskFSC
RES

EqDef Mask
Change

MOVE
Source CFG.FaultEq.n
8
Dest MaskFSC.LEN
8

EqDef Mask
Change

CLR
Dest MaskFSC.POS
7

EqDef Mask
Change
MaskFSC.FD

EqDef Mask
Change

FSC
Control MaskFSC
Length 8
Position 7
Mode ALL
Expression CalloutMsgMask.EqDef.Word[MaskFSC.POS]<>EqDefAlertMask[PKi].Word[MaskFSC.POS]
EN
DN
ER

Data Refresh
By Packet
DataRefreshPacket.[PKi]
L

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

COP
Source CalloutMsgMask.PreDef
Dest PreDefMaskAlert
Length 1

COP
Source CalloutMsgMask.EqDef.Word[0]
Dest EqDefAlertMask[PKi].Word[0]
Length CFG.FaultEq.n

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Data Refresh Request by Packet
for Gateway first scan or Gateway Status Display Refresh PB

10

Equipment Data
Callout/Status
Message Header -
Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum
-1
Source B -10

Data Refresh
By Packet
DataRefreshPacket.[PKi]
L

lbl_Exit
JMP

Invalid Command Number

11

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source CalloutMsgMask.Header.EQID
6
Dest Trans.Ctl.Opt[1]
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source CalloutMsgMask.Header.PacketID
1
Dest Trans.Ctl.Opt[2]
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source CalloutMsgMask.Header.CmdNum
-1
Dest Trans.Ctl.Opt[3]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 138
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

lbl_Exit
JMP

Exit Routine

12

lbl_Exit
LBL

NOP

(End)

Collect Fault/Status Data

Equipment Fault/Status Data
Collect Data for Callout Live Message and Fault Buffered Message
v1.3 - Dec2016 - Multi-packet support for >8 DINT Equipment-Defined Faults

PKTi - Packet Index for multiplexed Fault Data

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 121
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Exit if this Packet not Enabled

2

Equipment Data Collection - User Config Equipment Defined Packet EN bit array
UserCfg.EqDefPacketEN.[PKTi]

lbl_Exit
JMP

Collect Fault/Status Data

Filter Equipment Defined Fault/Status Packet with Ignore Mask downloaded by Server Config
Sep 22, 2022 - Replaced MVM with AND (MVM doesn't update masked bits)

3

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[0] ??
Source B EqDefIgnoreMask[PKTi].Word[0] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[0] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[1] ??
Source B EqDefIgnoreMask[PKTi].Word[1] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[1] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[2] ??
Source B EqDefIgnoreMask[PKTi].Word[2] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[2] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[3] ??
Source B EqDefIgnoreMask[PKTi].Word[3] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[3] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[4] ??
Source B EqDefIgnoreMask[PKTi].Word[4] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[4] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[5] ??
Source B EqDefIgnoreMask[PKTi].Word[5] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[5] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[6] ??
Source B EqDefIgnoreMask[PKTi].Word[6] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[6] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[7] ??
Source B EqDefIgnoreMask[PKTi].Word[7] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[7] ??

EqDriver Internal PreDefined Faults
(RingBuffer Filling, Message Faults, Maint/Fault/Alarm/Alert Masks etc.)

v2.0 Update PreDefined when everything else is complete to provide correct EQ Status to Gateway

4

Fault EqDefined Packet Index

EQ
Source A PKTi -1
Source B 0

Simultation for Fault Data Enable Simulation

SimFault.EN

Pre-Defined Fault/Status Data Mapping

JSR
Routine Name _100_PreDefFaultData

Collect Fault/Status Data

Check for Data Change in this Packet
Predefined must be copied to DINT for EQU

PreDefined changes are always sent in Packet 0
(This sometimes causes additional data packet, but simplifies logic and server data processing)

5

Fault EqDefined Packet Index

EQ
Source A PKTi -1
Source B 0

Fault Ring Buffer Message Control for multiplexed packets Predefined Faults DINT

COP
Source FaultMsg.PreDef
Dest FaultMsgCtl.PreDefVal
Length 1

Fault Ring Buffer Message Control for multiplexed packets Predefined Faults Last DINT

COP
Source FaultMsgCtl.PreDefLast
Dest FaultMsgCtl.PreDefLastVal
Length 1

Fault Ring Buffer Message Control for multiplexed packets Predefined Faults DINT

MEQ
Source FaultMsgCtl.PreDefVal 16#0000_0021
Mask 16#ffff_ffff
Compare FaultMsgCtl.PreDefLastVal 16#0000_0021

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A FaultMsgCtl.EqDef[PKTi].Word[0] ??
Source B FaultMsgCtl.EqDefLast[PKTi].Word[0] ??

Fault EqDefined Packet Index

NE
Source A PKTi -1
Source B 0

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A FaultMsgCtl.EqDef[PKTi].Word[1] ??
Source B FaultMsgCtl.EqDefLast[PKTi].Word[1] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A FaultMsgCtl.EqDef[PKTi].Word[2] ??
Source B FaultMsgCtl.EqDefLast[PKTi].Word[2] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A FaultMsgCtl.EqDef[PKTi].Word[3] ??
Source B FaultMsgCtl.EqDefLast[PKTi].Word[3] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A FaultMsgCtl.EqDef[PKTi].Word[4] ??
Source B FaultMsgCtl.EqDefLast[PKTi].Word[4] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A FaultMsgCtl.EqDef[PKTi].Word[5] ??
Source B FaultMsgCtl.EqDefLast[PKTi].Word[5] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A FaultMsgCtl.EqDef[PKTi].Word[6] ??
Source B FaultMsgCtl.EqDefLast[PKTi].Word[6] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A FaultMsgCtl.EqDef[PKTi].Word[7] ??
Source B FaultMsgCtl.EqDefLast[PKTi].Word[7] ??

Fault Ring Buffer Message Control for multiplexed packets No Data Change
FaultMsgCtl.NoChange.[PKTi]

Collect Fault/Status Data

Data Change in this Packet
Update Last Values
Load Fault/Status Ring Buffer
Special Case to generate a Callout Message when Gateway Mask Download Required

PreDefined changes are always sent in Packet 0
(This sometimes causes additional data packet, but simplifies logic and server data processing)

v2.09 Data Refresh signal (daily, pre-shift)
v3.02 Data Refresh Packet signal (first scan, mask updates and Gateway request); block Packet refresh until all Mask Downloads are complete

6

Fault Ring Buffer Message Control for multiplexed packets No Data Change
FaultMsgCtl.NoChange.[PKTi]
/

Load Fault RingBuffer
FaultMsgChange

Data Refresh 1-shot
DataRefresh

Equipment Data Callout/Status Message Footer - Spare (fills unused portion of DINT)
CalloutMsgMask.Footer.Spare.[PKTi]
/

Data Refresh By Packet
DataRefreshPacket.[PKTi]

Data Refresh By Packet
DataRefreshPacket.[PKTi]
U

Equipment Data Callout/Status Message Footer - Spare (fills unused portion of DINT)
CalloutMsgMask.Footer.Spare.[PKTi]

Fault Ring Buffer Message Control for multiplexed packets Data Change

EQ	
Source A	FaultMsgCtl.Change
	0
Source B	0

Generate Callout Message for Mask
MaskDownloadRequired

Fault Ring Buffer Message Control for multiplexed packets Data Change
FaultMsgCtl.Change.[PKTi]
L

Load Fault RingBuffer
FaultMsgChange

Fault EqDefined Packet Index

EQ	
Source A	PKTi
	-1
Source B	0

Fault Ring Buffer Message Control for multiplexed packets Predefined Faults Last

COP	
Source	FaultMsg.PreDef
Dest	FaultMsgCtl.PreDefLast
Length	1

Collect Fault/Status Data

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined

COP
Source FaultMsgCtl.EqDef[PKTi]
Dest FaultMsgCtl.EqDefLast[PKTi]
Length 1

Equipment Data Collection - User Config Fault Message Enable
UserCfg.FaultMsgEN

CLx Module Status CLx Transaction Time Invalid
CLX.TransTimeInvalid

Load Fault Data record to Ring Buffer

JSR
Routine Name _122_FaultDataLoadRB

Exit Routine

7

lbl_Exit
LBL

NOP

(End)

Load Fault Data record to Ring Buffer

Equipment Fault/Status Data
Ring Buffer Load
v1.3 - Dec2016 - Multi-packet support for >8 DINT Equipment-Defined Faults

PKTi - Packet Index for multiplexed Fault Data

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 122
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

EDC Ring Buffer Control Transaction Number

ADD
Source A FaultRBCtl.TransactionNumber 3759
Source B 1
Dest FaultRBCtl.TransactionNumber 3759

EDC Ring Buffer Control Transaction Number

LT
Source A FaultRBCtl.TransactionNumber 3759
Source B 1

EDC Ring Buffer Control Transaction Number

GT
Source A FaultRBCtl.TransactionNumber 3759
Source B CFG.MsgTransNumRollover 9999

EDC Ring Buffer Control Transaction Number

MOVE
Source 1
Dest FaultRBCtl.TransactionNumber 3759

Load Fault Data record to Ring Buffer

Load Message Header, Timestamp and Footer Data
v1.3 - Added Header.PacketID for multiplexed data packets
Copy This Packet to FaultMsg.EqDef

3

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

MOVE
Source FaultRBCtl.TransactionNumber 3759
Dest FaultMsg.Header.TransNum 3759

Equipment Data Fault/Status Message Header - Command Number

MOVE
Source 0
Dest FaultMsg.Header.CmdNum 0

Equipment Data Fault/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source UserCfg.EQID 6
Dest FaultMsg.Header.EQID 6

Equipment Data Fault/Status Message Header - Data Count - Total Qty of Words used in all Packets

MOVE
Source UserCfg.EqDefDataCount[PKTi] ??
Dest FaultMsg.Header.DataCount 8

Equipment Data Fault/Status Message Header - Packet ID - generates offset for 8-DINT mulitplexed data packet

MOVE
Source PKTi -1
Dest FaultMsg.Header.PacketID 0

Equipment Data Fault/Status Message Transaction Time Stamp

COP
Source CLX.TransTime
Dest FaultMsg.TransTime
Length 1

Equipment Data Fault/Status Message Data - Predefined Status

COP
Source FaultMsg.PreDef
Dest FaultMsg.PreDef
Length 1

Equipment Data Fault/Status Message Data - Equipment Defined Status

COP
Source FaultMsgCtl.EqDef[PKTi]
Dest FaultMsg.EqDef
Length 1

Equipment Data Fault/Status Message Footer - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest FaultMsg.Footer.Spare 0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

MOVE
Source FaultRBCtl.TransactionNumber 3759
Dest FaultMsg.Footer.TransNum 3759

Load Fault Data record to Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

4

EDC Ring Buffer Control Next Write Record

LT
Source A FaultRBCtl.WritePtr 141
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A FaultRBCtl.WritePtr 141
Source B CFG.FaultRB.mi 249

EDC Ring Buffer Control Diagnostic - Manual Reset

FaultRBCtl.WritePtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source FaultRBCtl.WritePtr 141
Dest FaultRBCtl.WritePtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.WritePtr 141
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.ReadPtr 140
Dest Trans.Ctl.Opt[2] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultMsg.Header.PacketID 0
Dest Trans.Ctl.Opt[3] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 213
Input Par 0
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest FaultRBCtl.WritePtr 141

Load Fault Data record to Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

5

EDC Ring Buffer Control Next Write Record

EQ	
Source A	FaultRBCtl.WritePtr 141
Source B	FaultRBCtl.ReadPtr 140

EDC Ring Buffer Control Buffer Full - Block Writes
FaultRBCtl.FullBlockWrite

EDC Ring Buffer Control Diagnostic - Manual Reset
FaultRBCtl.WriteBlockErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD	
Source A	FaultRBCtl.WriteBlockErrCnt 1557127
Source B	1
Dest	FaultRBCtl.WriteBlockErrCnt 1557127

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	FaultRBCtl.WritePtr 141
Dest	Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	FaultRBCtl.ReadPtr 140
Dest	Trans.Ctl.Opt[2] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	FaultMsg.Header.PacketID 0
Dest	Trans.Ctl.Opt[3] -1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	214
Input Par	0
Input Par	FaultMsg.Header.TransNum

Message Control Transaction Timing

FLL	
Source	0
Dest	TempMsgTrans
Length	1

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans (...)
PerfMsgTrans	TempMsgTrans
PerfTrans	Status.FaultRx.ER

lbl_Exit (JMP)

Load Fault Data record to Ring Buffer

RB Write -- Load Message into Ring Buffer

6

Equipment Data Fault/Status Message

COP
Source FaultMsg
Dest FaultRB[FaultRBCtl.WritePtr]
Length 1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.WritePtr
141
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.ReadPtr
140
Dest Trans.Ctl.Opt[2]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultMsg.Header.PacketID
0
Dest Trans.Ctl.Opt[3]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 211
Input Par 0
Input Par FaultMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans TempMsgTrans
PerfTrans Status.FaultRx.DN

RB Write -- Increment Write Pointer

7

EDC Ring Buffer Control Next Write Record

ADD
Source A FaultRBCtl.WritePtr
141
Source B 1
Dest FaultRBCtl.WritePtr
141

EDC Ring Buffer Control Next Write Record

LT
Source A FaultRBCtl.WritePtr
141
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A FaultRBCtl.WritePtr
141
Source B CFG.FaultRB.mi
249

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest FaultRBCtl.WritePtr
141

Exit Routine

8

lbl_Exit
LBL

NOP

Load Fault Data record to Ring Buffer

(End)

Load Read Records from RingBuffer to Gateway Message

v3.5 Equipment Bulk Fault/Status Data
Ring Buffer Unload to Gateway Message

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 123
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Load Read Records from RingBuffer to Gateway Message

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Next Read Record

LT
Source A FaultRBCtl.ReadPtr 140
Source B 0

EDC Ring Buffer Control Next Read Record

GT
Source A FaultRBCtl.ReadPtr 140
Source B CFG.FaultRB.mi 249

EDC Ring Buffer Control Diagnostic - Manual Reset

FaultRBCtl.ReadPtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source FaultRBCtl.ReadPtr 140
Dest FaultRBCtl.ReadPtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.WritePtr 141
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.ReadPtr 140
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 223
Input Par 0
Input Par FaultRBCtl.TransactionNumber

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest FaultRBCtl.WritePtr 141

EDC Ring Buffer Control Next Read Record

MOVE
Source CFG.FaultRB.mi 249
Dest FaultRBCtl.ReadPtr 140

Load Read Records from RingBuffer to Gateway Message

Next Read Location
ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

ADD
Source A FaultRBCtl.ReadPtr 140
Source B 1
Dest FaultRBCtl.ReadNextPtr 141

EDC Ring Buffer Control Next Read Record

GT
Source A FaultRBCtl.ReadNextPtr 141
Source B CFG.FaultRB.mi 249

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest FaultRBCtl.ReadNextPtr 141

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Next Read Record

EQ
Source A FaultRBCtl.ReadPtr 140
Source B FaultRBCtl.WritePtr 141

EDC Ring Buffer Control Buffer Full
FaultRBCtl.Full

EDC Ring Buffer Control Read Buffer Record in Progress
FaultRBCtl.ReadInProgress

EDC Ring Buffer Control Next Read Record

EQ
Source A FaultRBCtl.ReadNextPtr 141
Source B FaultRBCtl.WritePtr 141

EDC Ring Buffer Control Buffer Empty
FaultRBCtl.Empty

Load Read Records from RingBuffer to Gateway Message Buffer Count

5

EDC Ring Buffer Control Read Buffer Record in Progress: FaultRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete: FaultRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record

GT
Source A: FaultRBCtl.WritePtr 141
Source B: FaultRBCtl.ReadPtr 140

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest: FaultRBCtl.RecordCount 0
Expression: FaultRBCtl.WritePtr-FaultRBCtl.ReadPtr

EDC Ring Buffer Control Next Write Record

LT
Source A: FaultRBCtl.WritePtr 141
Source B: FaultRBCtl.ReadPtr 140

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest: FaultRBCtl.RecordCount 0
Expression: CFG.FaultRB.n-(FaultRBCtl.ReadPtr-FaultRBCtl.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress: FaultRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete: FaultRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record

GE
Source A: FaultRBCtl.WritePtr 141
Source B: FaultRBCtl.ReadNextPtr 141

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest: FaultRBCtl.RecordCount 0
Expression: FaultRBCtl.WritePtr-FaultRBCtl.ReadNextPtr

EDC Ring Buffer Control Next Write Record

LT
Source A: FaultRBCtl.WritePtr 141
Source B: FaultRBCtl.ReadNextPtr 141

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest: FaultRBCtl.RecordCount 0
Expression: CFG.FaultRB.n-(FaultRBCtl.ReadNextPtr-FaultRBCtl.WritePtr)

EDC Ring Buffer Control Ring Buffer Record Count

GT
Source A: FaultRBCtl.RecordCount 0
Source B: FaultRBCtl.RecordCountMax 249

EDC Ring Buffer Control Ring Buffer Record Count Max

MOVE
Source: FaultRBCtl.RecordCount 0
Dest: FaultRBCtl.RecordCountMax 249

Load Read Records from RingBuffer to Gateway Message

Read Next
Ready for next data to Gateway Message

6

EDC Ring Buffer Control Buffer Empty FaultRBCtl.Empty /
EDC Ring Buffer Control Read Buffer Record in Progress FaultRBCtl.ReadInProgress /
Fault Ring Buffer Message Control for multiplexed packets Send Data FaultMsgCtl.Send /
Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive) FaultMsgCtl.Resend /
EDC Ring Buffer Control Read Next Buffer Record 1-shot FaultRBCtl.ReadNextONS [ONS]
EDC Ring Buffer Control Read Next Buffer Record FaultRBCtl.ReadNext (L)
EDC Ring Buffer Control Read Buffer Record in Progress FaultRBCtl.ReadInProgress (L)
EDC Ring Buffer Control Read Buffer Record Complete FaultRBCtl.ReadComplete (U)

Skip ReadNext code if not loading new records
(faster execution)

7

EDC Ring Buffer Control Read Next Buffer Record FaultRBCtl.ReadNext /
lbl_SkipReadNext (JMP)

Next Message
Initialize Bulk Fault Message Loop
use RecordCount to load up to 10 messages

8

EDC Ring Buffer Control Read Next Buffer Record FaultRBCtl.ReadNext

EDC Message Tx Data Bulk Fault/Status Receive Message
FLL
Source 0
Dest EqTxData.Fault[0] <EQ0006_TxData.Fault[0]>
Length CFG.FaultMSG.e

Record Index
CLR
Dest RECi 2

Record Max Index
SUB
Source A FaultRBCtl.RecordCount 0
Source B 1
Dest RECmi 1

Record Max Index
LE
Source A RECmi 1
Source B 0

Record Max Index
MOVE
Source 0
Dest RECmi 1

Record Max Index
GT
Source A RECmi 1
Source B 9

Record Max Index
MOVE
Source 9
Dest RECmi 1

diagnostic record count
ADD
Source A xxRECmi[RECmi] 5794
Source B 1
Dest xxRECmi[RECmi] 5794

Load Read Records from RingBuffer to Gateway Message

Next Message
Bulk Fault Message Loop

9

lbl_MsgLoop
LBL

NOP

Read Next
Increment Read Pointer

10

EDC Ring Buffer Control Read Next Buffer Record
FaultRBCtl.ReadNext

EDC Ring Buffer Control Next Read Record
ADD
Source A FaultRBCtl.ReadPtr 140
Source B 1
Dest FaultRBCtl.ReadPtr 140

EDC Ring Buffer Control Next Read Record
LT
Source A FaultRBCtl.ReadPtr 140
Source B 0

EDC Ring Buffer Control Next Read Record
GT
Source A FaultRBCtl.ReadPtr 140
Source B CFG.FaultRB.mi 249

EDC Ring Buffer Control Next Read Record
MOVE
Source 0
Dest FaultRBCtl.ReadPtr 140

Load Read Records from RingBuffer to Gateway Message

Read Next
Load Read Record

calculate offset for each RECi
14 DINT/record + 2 DINT Header

11

EDC Ring Buffer Control Read Next Buffer Record
FaultRBCtl.ReadNext

Fault Msg Record Index
CPT
Dest FLTi 16
Expression (CFG.FaultRec.e * RECi) + 2

EDC Message Tx Data Bulk Fault/Status Receive Message
COP
Source FaultRB[FaultRBCtl.ReadPtr]
Dest EqTxData.Fault[FLTi] <EQ0006_TxData.Fault[FLTi]>
Length CFG.FaultRec.e

Fault Ring Buffer Message Control for multiplexed packets Transaction Number
MOVE
Source FaultRB[FaultRBCtl.ReadPtr].Header.TransNum 3759
Dest FaultMsgCtl.TransNum 3759

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source FaultRBCtl.WritePtr 141
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source FaultRBCtl.ReadPtr 140
Dest Trans.Ctl.Opt[2] -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source FaultRB[FaultRBCtl.ReadPtr].Header.PacketID 0
Dest Trans.Ctl.Opt[3] -1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 221
Input Par 0
Input Par FaultMsgCtl.TransNum

Load Read Records from RingBuffer to Gateway Message

load Bulk Message Header and Footer Transaction Number from first record
overwrite DataCount/PacketID with Bulk Message RecordCount

the footer location and Write Message length are based on record count
(most messages will have a single record - no point in sending 10x the data when not needed)
14 DINT/record + 2 DINT Header

12

EDC Ring Buffer Control Read Next Buffer Record: FaultRBCtl.ReadNext

Record Index

EQ
- Source A: RECi = 2
- Source B: 0

Equipment Data Message Header

COP
- Source: FaultRB[FaultRBCtl.ReadPtr].Header
- Dest: BulkHeader
- Length: 1

Equipment Data Message Header Header - Data Count - Total Qty of Words used in all Packets

ADD
- Source A: RECmi = 1
- Source B: 1
- Dest: BulkHeader.DataCount = 2

Equipment Data Message Header Header - Packet ID - generates offset for 8-DINT mulitplexed data packet

CLR
- Dest: BulkHeader.PacketID = 0

EDC Message Tx Data Bulk Fault/Status Receive Message

COP
- Source: BulkHeader
- Dest: EqTxData.Fault[0] <EQ0006_TxData.Fault[0]>
- Length: 2

CPT
- Dest: FLTei = 30
- Expression: (CFG.FaultRec.e * (RECmi+1)) + 2

EDC Message Tx Data Bulk Fault/Status Receive Message

COP
- Source: FaultRB[FaultRBCtl.ReadPtr].Footer
- Dest: EqTxData.Fault[FLTei] <EQ0006_TxData.Fault[FLTei]>
- Length: 1

Increment Loop Pointer and Jump until all Records loaded

13

EDC Ring Buffer Control Read Next Buffer Record: FaultRBCtl.ReadNext

Record Index

ADD
- Source A: RECi = 2
- Source B: 1
- Dest: RECi = 2

Record Index

LE
- Source A: RECi = 2
- Source B: RECmi = 1

lbl_MsgLoop
JMP

Load Read Records from RingBuffer to Gateway Message

Read Next
Set Data Ready Handshake

14

EDC Ring Buffer Control Read Next Buffer Record
FaultRBCtl.ReadNext

Fault Ring Buffer Message Control for multiplexed packets Send Data
FaultMsgCtl.Send
(L)

Fault Ring Buffer Message Control for multiplexed packets Read Command
FaultMsgCtl.Read
(U)

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready
FaultMsgCtl.ReadOK
(U)

Fault Ring Buffer Message Control for multiplexed packets Write Command
FaultMsgCtl.Write
(U)

Fault Ring Buffer Message Control for multiplexed packets Write Command Done
FaultMsgCtl.WriteOK
(U)

Skip ReadNext code if not loading new records
(faster execution)

15

lbl_SkipReadNext
LBL

NOP

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

16

Fault Ring Buffer Message Control for multiplexed packets Send Data
FaultMsgCtl.Send

Fault Ring Buffer Message Control for multiplexed packets Write Command Done
FaultMsgCtl.WriteOK

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive)
FaultMsgCtl.Resend

Fault Ring Buffer Message Control for multiplexed packets Send Data
FaultMsgCtl.Send
(U)

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive)
FaultMsgCtl.Resend
(U)

EDC Ring Buffer Control Read Buffer Record Complete
FaultRBCtl.ReadComplete
(L)

Load Read Records from RingBuffer to Gateway Message

Read Complete Delay
Allow Gateway processing time before attempting next record send.

17

EDC Ring Buffer Control Read Buffer Record Complete
FaultRBCtl.ReadComplete

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source 100
Dest FaultRBCtl.ReadCompleteDelayTMR.PRE
100

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer FaultRBCtl.ReadCompleteDelayTMR
Preset 100
Accum 107
EN
DN

EDC Ring Buffer Control Read Complete Delay Timer
FaultRBCtl.ReadCompleteDelayTMR.DN

EDC Ring Buffer Control Read Buffer Record Complete Delay
FaultRBCtl.ReadCompleteDelay

EDC Ring Buffer Control Read Buffer Record in Progress
FaultRBCtl.ReadInProgress
U

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

18

EDC Ring Buffer Control Buffer Empty
FaultRBCtl.Empty
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 180000
Dest FaultRBCtl.NotEmptyTMR.PRE
180000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer FaultRBCtl.NotEmptyTMR
Preset 180000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A FaultRBCtl.NotEmptyTMR.ACC
0
Source B FaultRBCtl.NotEmptyMax
180003

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source FaultRBCtl.NotEmptyTMR.ACC
0
Dest FaultRBCtl.NotEmptyMax
180003

Load Read Records from RingBuffer to Gateway Message

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

19

EDC Ring Buffer Control Buffer Full
FaultRBCtl.Full

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
FaultRBCtl.NotEmptyTMR.DN

EDC Ring Buffer Control Read Buffer Alarm
FaultRBCtl.BufferAlarm

(End)

Fault Data Ring Buffer and Message

Equipment Fault/Status Data
Buffered Fault Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	124
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

Fault Ring Buffer Message Control for multiplexed packets Message Transaction Timer

MOVE	
Source	999999999
Dest	FaultMsgCtl.Trans.TMR.PRE 999999999

Fault Ring Buffer Message Control for multiplexed packets Message Transaction Timer

TON		
Timer	FaultMsgCtl.Trans.TMR	EN
Preset	999999999	DN
Accum	211470	

Fault Data Ring Buffer and Message

Read Message Delay Timer
If driver sends data too fast, it will generate extra Read Message traffic until previous message has been processed
Optimization uses the ReadOk and ReadNotOk counts to tune the Delay Time to the Gateway response

3

EDC Gateway Fault Write Message
FaultWriteMsg.EN
<EQ0006_FaultWriteMsg.EN>

EDC Gateway Fault Read Message
FaultReadMsg.EN
<EQ0006_FaultReadMsg.EN>

Fault Ring Buffer Message Control for multiplexed packets Read Message Ok Count

GE
Source A FaultMsgCtl.ReadOkCount 4
Source B CfgRB_ReadOkLimit <CfgRB.ReadOkLimit> 5

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

SUB
Source A FaultMsgCtl.DelayTMR.PRE 100
Source B CfgRB_ReadOkSub <CfgRB.ReadOkSub> 10
Dest FaultMsgCtl.DelayTMR.PRE 100

Fault Ring Buffer Message Control for multiplexed packets Read Message Not Ok Count

GE
Source A FaultMsgCtl.ReadNotOkCount 0
Source B CfgRB_ReadNotOkLimit <CfgRB.ReadNotOkLimit> 2

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

ADD
Source A FaultMsgCtl.DelayTMR.PRE 100
Source B CfgRB_ReadNotOkAdd <CfgRB.ReadNotOkAdd> 20
Dest FaultMsgCtl.DelayTMR.PRE 100

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

LE
Source A FaultMsgCtl.DelayTMR.PRE 100
Source B CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

MOVE
Source CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100
Dest FaultMsgCtl.DelayTMR.PRE 100

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

GT
Source A FaultMsgCtl.DelayTMR.PRE 100
Source B CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500

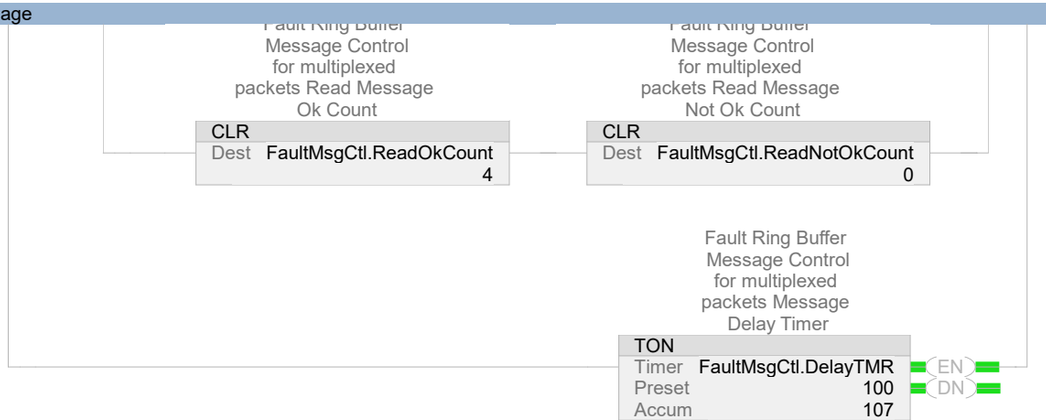
Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

MOVE
Source CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500
Dest FaultMsgCtl.DelayTMR.PRE 100

Fault Ring Buffer

Fault Ring Buffer

Fault Data Ring Buffer and Message



Fault Data Ring Buffer and Message

Message Resend Timer
Keep Alive - resends last data without increment of Transaction Number

4

Fault Ring Buffer Message Control for multiplexed packets Send Data FaultMsgCtl.Send

EDC Gateway Fault Write Message FaultWriteMsg.EN <EQ0006_FaultWriteMsg.EN>

EDC Gateway Fault Read Message FaultReadMsg.EN <EQ0006_FaultReadMsg.EN>

Fault Ring Buffer Message Control for multiplexed packets Message Resend Timer

MOVE
Source CFG.BuffMsgResendTime 301000
Dest FaultMsgCtl.ResendTMR.PRE 301000

Fault Ring Buffer Message Control for multiplexed packets Message Resend Timer

TON
Timer FaultMsgCtl.ResendTMR
Preset 301000
Accum 211442
EN
DN

Fault Ring Buffer Message Control for multiplexed packets Message Resend Timer FaultMsgCtl.ResendTMR.DN

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive) FaultMsgCtl.Resend
L

Fault Ring Buffer Message Control for multiplexed packets Message Resend Timer FaultMsgCtl.ResendTMR
RES

Fault Ring Buffer Message Control for multiplexed packets Read Command FaultMsgCtl.Read
U

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready FaultMsgCtl.ReadOK
U

Fault Ring Buffer Message Control for multiplexed packets Write Command FaultMsgCtl.Write
U

Fault Ring Buffer Message Control for multiplexed packets Write Command Done FaultMsgCtl.WriteOK
U

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.WritePtr 141
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.ReadPtr 140
Dest Trans.Ctl.Opt[2] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRB[FaultRBCtl.ReadPtr].Header.PacketID 0
Dest Trans.Ctl.Opt[3] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 222
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message
Send Data Initialize

5

Fault Ring Buffer Message Control for multiplexed packets Send Data
FaultMsgCtl.Send

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive)
FaultMsgCtl.Resend

Fault Ring Buffer Message Control for multiplexed packets Send Data 1-Shot
FaultMsgCtl.SendONS
ONS

Fault Ring Buffer Message Control for multiplexed packets Message Transaction Timer
FaultMsgCtl.Trans.TMR
RES

Fault Ring Buffer Message Control for multiplexed packets Transaction Elapsed Time for Last Event (used to calculate differential)
CLR
Dest FaultMsgCtl.Trans.Last
14

Fault Data Ring Buffer and Message

Read Message
Check Gateway Buffer for Empty Record
(Gateway clears Buffer after processing Record)

variable Read Msg length for variable record count - read back what was sent with last WriteMsg

6

Fault Ring Buffer Message Control for multiplexed packets Send Data FaultMsgCtl.Send

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready FaultMsgCtl.ReadOK

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer FaultMsgCtl.DelayTMR.DN

EDC Gateway Fault Read Message FaultReadMsg.EN <EQ0006_FaultReadMsg.EN>

EDC Gateway Fault Write Message FaultWriteMsg.EN <EQ0006_FaultWriteMsg.EN>

EDC Gateway Fault Read Message FaultReadMsg.ER <EQ0006_FaultReadMsg.ER>

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive) FaultMsgCtl.Resend

Fault Ring Buffer Message Control for multiplexed packets Message Retry Timer FaultMsgCtl.RetryTMR.DN

Fault Ring Buffer Message Control for multiplexed packets Read Command FaultMsgCtl.Read (L)

Fault Ring Buffer Message Control for multiplexed packets Read Command 1-Shot FaultMsgCtl.ReadONS

EDC Gateway Fault Read Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest FaultReadMsg.Path <EQ0006_FaultReadMsg.Path>
Length 1

EDC Gateway Fault Read Message

COP
Source CFG.MsgDest.FaultDest
Dest FaultReadMsg.RemoteElement <EQ0006_FaultReadMsg.RemoteElement>
Length 1

EDC Gateway Fault Read Message

MOVE
Source FaultWriteMsg.REQ_LEN <EQ0006_FaultWriteMsg.REQ_LEN>
31
Dest FaultReadMsg.REQ_LEN <EQ0006_FaultReadMsg.REQ_LEN>
31

EDC Gateway Fault Read Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest FaultReadMsg.UnconnectedTimeout <EQ0006_FaultReadMsg.UnconnectedTimeout>
4000000

EDC Gateway Fault Read Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest FaultReadMsg.ConnectionRate <EQ0006_FaultReadMsg.ConnectionRate>
1000000

EDC Gateway Fault Read Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest FaultReadMsg.TimeoutMultiplier <EQ0006_FaultReadMsg.TimeoutMultiplier>
0

EDC Gateway Fault Read Message FaultReadMsg.EN_CC <EQ0006_FaultReadMsg.EN_CC> (U)

EDC Gateway Fault Read Message

MSG
Message Control FaultReadMsg <EQ0006_FaultReadMsg> (...)
EN
DN
ER

Fault Data Ring Buffer and Message

Read Message EN Performance & Diagnostics

7

Fault Ring Buffer Message Control for multiplexed packets Read Command 1-Shot
FaultMsgCtl.ReadONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	FaultMsgCtl.Trans
PerfTrans	Status.FaultTxRead.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	230
Input Par	FaultMsgCtl.Trans.TMR.ACC
Input Par	FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

Read Message OK - Gateway Buffer has Empty Record
(Gateway clears Buffer after processing Record)

Increment ReadOk and ReadNotOk counts
only increment ReadOk count when multiple records in Buffer (single records are always ready to send and don't test Gateway servicing speed)

variable Read Msg length for variable record count - check TransNum=0 in first & last word sent with last WriteMsg

8

Fault Ring Buffer Message Control for multiplexed packets Read Command
FaultMsgCtl.Read

EDC Gateway Fault Read Message
FaultReadMsg.DN
<EQ0006_FaultReadMsg.DN>

Fault Ring Buffer Message Control for multiplexed packets Read Command
FaultMsgCtl.Read
U

EDC Message Rx Data Bulk Fault/Status Receive Message
EQ
Source A EqRxData.Fault[0]
<EQ0006_RxData.Fault[0]>
0
Source B 0

EDC Message Rx Data Bulk Fault/Status Receive Message
EQ
Source A EqRxData.Fault[FaultReadMsg.REQ_LEN-1]
<EQ0006_RxData.Fault[FaultReadMsg.REQ_LEN-1]>
0
Source B 0

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready
FaultMsgCtl.ReadOK
L

EDC Ring Buffer Control Ring Buffer Record Count
GE
Source A FaultRBCtl.RecordCount
0
Source B 2

Fault Ring Buffer Message Control for multiplexed packets Read Message Ok Count
ADD
Source A FaultMsgCtl.ReadOkCount
4
Source B 1
Dest FaultMsgCtl.ReadOkCount
4

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready
FaultMsgCtl.ReadOK
/

Fault Ring Buffer Message Control for multiplexed packets Read Message Not Ok Count
ADD
Source A FaultMsgCtl.ReadNotOkCount
0
Source B 1
Dest FaultMsgCtl.ReadNotOkCount
0

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans FaultMsgCtl.Trans
PerfTrans Status.FaultTxRead.DN

Fault Ring Buffer Message Control for multiplexed packets Resend Data

Performance

Fault Data Ring Buffer and Message

(Keep Alive)
FaultMsgCtl.Resend

Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans FaultMsgCtl.Trans
PerfTrans Status.FaultTxRead.KA

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source EqRxData.Fault[0]
<EQ0006_RxData.Fault[0]> 0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source EqRxData.Fault[FaultReadMsg.REQ_LEN-1]
<EQ0006_RxData.Fault[FaultReadMsg.REQ_LEN-1]> 0
Dest Trans.Ctl.Opt[2]
-1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 231
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

Read Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

9

EDC Gateway Fault
Read Message
FaultReadMsg.EN
<EQ0006_FaultReadMsg.EN>

Fault Ring Buffer
Message Control
for multiplexed
packets Message
Timeout Timer
MOVE
Source CFG.CLx.MsgTimeout
5000
Dest FaultMsgCtl.ReadTimeoutTMR.PRE
5000

Fault Ring Buffer
Message Control
for multiplexed
packets Message
Timeout Timer
TON
Timer FaultMsgCtl.ReadTimeoutTMR
Preset 5000
Accum 0
EN
DN

Fault Ring Buffer
Message Control
for multiplexed
packets Message
Timeout Timer
FaultMsgCtl.ReadTimeoutTMR.DN

EDC Gateway Fault
Read Message
FaultReadMsg.TO
<EQ0006_FaultReadMsg.TO>
L

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 234
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

Read Message ER Performance & Diagnostics

10

EDC Gateway Fault Read Message
FaultReadMsg.ER
<EQ0006_FaultReadMsg.ER>

Fault Ring Buffer Message Control for multiplexed packets Read Error 1-shot
FaultMsgCtl.ReadErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Fault Tx ER
Status.Update.FaultTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	FaultMsgCtl.Trans
PerfTrans	Status.FaultTxRead.ER

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	233
Input Par	FaultMsgCtl.Trans.TMR.ACC
Input Par	FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

Write Message
Write Data is loaded by Ring Buffer routine

variable Write Msg length for variable record count - only send what is needed (most messages will only have 1 record)
14 DINT/record + 3 DINT Header/Footer

11

Fault Ring Buffer Message Control for multiplexed packets Send Data FaultMsgCtl.Send

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready FaultMsgCtl.ReadOK

Fault Ring Buffer Message Control for multiplexed packets Write Command Done FaultMsgCtl.WriteOK

EDC Gateway Fault Read Message FaultReadMsg.EN <EQ0006_FaultReadMsg.EN>

EDC Gateway Fault Write Message FaultWriteMsg.EN <EQ0006_FaultWriteMsg.EN>

EDC Gateway Fault Write Message FaultWriteMsg.ER <EQ0006_FaultWriteMsg.ER>

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive) FaultMsgCtl.Resend

Fault Ring Buffer Message Control for multiplexed packets Message Retry Timer FaultMsgCtl.RetryTMR.DN

Fault Ring Buffer Message Control for multiplexed packets Write Command FaultMsgCtl.Write (L)

EDC Gateway Fault Write Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest FaultWriteMsg.Path <EQ0006_FaultWriteMsg.Path>
Length 1

EDC Gateway Fault Write Message

COP
Source CFG.MsgDest.FaultDest
Dest FaultWriteMsg.RemoteElement <EQ0006_FaultWriteMsg.RemoteElement>
Length 1

Equipment Data Message Header

COP
Source EqTxData.Fault[0] <EQ0006_TxData.Fault[0]>
Dest BulkHeader
Length 2

EDC Gateway Fault Write Message

CPT
Dest FaultWriteMsg.REQ_LEN <EQ0006_FaultWriteMsg.REQ_LEN>
31
Expression (BulkHeader.DataCount * CFG.FaultRec.e) + 3

EDC Gateway Fault Write Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest FaultWriteMsg.UnconnectedTimeout <EQ0006_FaultWriteMsg.UnconnectedTimeout>
4000000

EDC Gateway Fault Write Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest FaultWriteMsg.ConnectionRate <EQ0006_FaultWriteMsg.ConnectionRate>
1000000

EDC Gateway Fault Write Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest FaultWriteMsg.TimeoutMultiplier <EQ0006_FaultWriteMsg.TimeoutMultiplier>
0

EDC Gateway Fault Write Message FaultWriteMsg.EN_CC <EQ0006_FaultWriteMsg.EN_CC> (U)

EDC Gateway Fault Write Message

MSG
Message Control FaultWriteMsg <EQ0006_FaultWriteMsg>
EN
DN
ER

Fault Data Ring Buffer and Message

Write Message EN Performance & Diagnostics

12

Fault Ring Buffer Message Control for multiplexed packets Write Command
FaultMsgCtl.Write

Fault Ring Buffer Message Control for multiplexed packets Write Command 1-Shot
FaultMsgCtl.WriteONS
ONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	FaultMsgCtl.Trans
PerfTrans	Status.FaultTxWrite.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	240
Input Par	FaultMsgCtl.Trans.TMR.ACC
Input Par	FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

13

Write Message Done

Fault Ring Buffer Message Control for multiplexed packets Write Command
FaultMsgCtl.Write

EDC Gateway Fault Write Message
FaultWriteMsg.DN
<EQ0006_FaultWriteMsg.DN>

Fault Ring Buffer Message Control for multiplexed packets Write Command
FaultMsgCtl.Write
U

Fault Ring Buffer Message Control for multiplexed packets Write Command Done
FaultMsgCtl.WriteOK
L

Fault Ring Buffer Message Control for multiplexed packets Send Data
FaultMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Fault Tx DN
Status.Update.FaultTxOK
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive)
FaultMsgCtl.Resend

Equipment Data Collection - Line Equipment Status State = Green - Fault Keep Alive Tx DN
Status.Update.FaultKaTxOK
L

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	FaultMsgCtl.Trans
PerfTrans	Status.FaultTxWrite.DN

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive)
FaultMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	FaultMsgCtl.Trans
PerfTrans	Status.FaultTxWrite.KA

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	241
Input Par	FaultMsgCtl.Trans.TMR.ACC
Input Par	FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

Write Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

14

EDC Gateway Fault Write Message
FaultWriteMsg.EN
<EQ0006_FaultWriteMsg.EN>

Fault Ring Buffer Message Control for multiplexed packets Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest FaultMsgCtl.WriteTImeoutTMR.PRE
5000

Fault Ring Buffer Message Control for multiplexed packets Message Timeout Timer

TON
Timer FaultMsgCtl.WriteTImeoutTMR
Preset 5000
Accum 0
EN
DN

Fault Ring Buffer Message Control for multiplexed packets Message Timeout Timer
FaultMsgCtl.WriteTImeoutTMR.DN

EDC Gateway Fault Write Message
FaultWriteMsg.TO
<EQ0006_FaultWriteMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 244
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

Write Message ER Performance & Diagnostics

15

EDC Gateway Fault Write Message
FaultWriteMsg.ER
<EQ0006_FaultWriteMsg.ER>

Fault Ring Buffer Message Control for multiplexed packets Write Error 1-shot
FaultMsgCtl.WriteErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Fault Tx ER
Status.Update.FaultTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans FaultMsgCtl.Trans
PerfTrans Status.FaultTxWrite.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 243
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

Read/Write Message Error Retry

16

EDC Gateway Fault Read Message
FaultReadMsg.ER
<EQ0006_FaultReadMsg.ER>

EDC Gateway Fault Write Message
FaultWriteMsg.ER
<EQ0006_FaultWriteMsg.ER>

Fault Ring Buffer Message Control for multiplexed packets Message Retry Timer
FaultMsgCtl.RetryTMR.DN

Fault Ring Buffer Message Control for multiplexed packets Retry 1-Shot
FaultMsgCtl.RetryONS
ONS

Fault Ring Buffer Message Control for multiplexed packets Message Retry Timer

MOVE
Source CFG.MsgRetryTime
5000
Dest FaultMsgCtl.RetryTMR.PRE
5000

Fault Ring Buffer Message Control for multiplexed packets Message Retry Timer

TON
Timer FaultMsgCtl.RetryTMR
Preset 5000
Accum 0
EN
DN

EDC Gateway Fault Read Message
FaultReadMsg.ER
<EQ0006_FaultReadMsg.ER>

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 232
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

EDC Gateway Fault Write Message
FaultWriteMsg.ER
<EQ0006_FaultWriteMsg.ER>

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 242
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

(End)

Load Quality Data record to Ring Buffer

Equipment Quality/Health Data
Ring Buffer Load

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 132
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

EDC Ring Buffer Control Transaction Number

ADD
Source A QualityRBCtl.TransactionNumber 4157
Source B 1
Dest QualityRBCtl.TransactionNumber 4157

EDC Ring Buffer Control Transaction Number

LT
Source A QualityRBCtl.TransactionNumber 4157
Source B 1

EDC Ring Buffer Control Transaction Number

GT
Source A QualityRBCtl.TransactionNumber 4157
Source B CFG.MsgTransNumRollover 9999

EDC Ring Buffer Control Transaction Number

MOVE
Source 1
Dest QualityRBCtl.TransactionNumber 4157

Load Quality Data record to Ring Buffer

Load Message Header, Timestamp and Footer Data
v1.6 - Packet Type added (0=5 data, 1=10 data)
v1.16 DHF024 Allow Quality Targets to pass -1, -2 Commands to CmdNum using Spare1
v2.00 Remove Targets from CmdNum; Add PdM Triggers to CmdNum

3

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

MOVE
Source QualityRBCtl.TransactionNumber 4157
Dest QualityMsg.Header.TransNum 4157

Equipment Data Quality/Health Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source UserCfg.EQID 6
Dest QualityMsg.Header.EQID 6

Equipment Data Quality/Health Message Header - PacketType (0=5 Data, 1=10 Data)

MOVE
Source 1
Dest QualityMsg.Header.PacketType 1

Equipment Data Quality/Health Message Header - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest QualityMsg.Header.Spare 0

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source 0
Dest QualityMsg.Header.CmdNum 0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.1

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source -11
Dest QualityMsg.Header.CmdNum 0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.1

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source -12
Dest QualityMsg.Header.CmdNum 0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.1

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source -13
Dest QualityMsg.Header.CmdNum 0

Equipment Data Quality/Health Message Transaction Time Stamp

COP
Source CLX.TransTime
Dest QualityMsg.TransTime
Length 1

Load Quality Data record to Ring Buffer

Equipment Data Quality/Health Message Footer - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest QualityMsg.Footer.Spare 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

MOVE
Source QualityRBCtl.TransactionNumber 4157
Dest QualityMsg.Footer.TransNum 4157

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

4

EDC Ring Buffer Control Next Write Record

LT
Source A QualityRBCtl.WritePtr 43
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A QualityRBCtl.WritePtr 43
Source B CFG.QualityRB.mi 99

EDC Ring Buffer Control Diagnostic - Manual Reset
QualityRBCtl.WritePtrErr
(L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source QualityRBCtl.WritePtr 43
Dest QualityRBCtl.WritePtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.WritePtr 43
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.ReadPtr 42
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 313
Input Par 0
Input Par QualityMsg.Header.TransNum

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest QualityRBCtl.WritePtr 43

Load Quality Data record to Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

5

EDC Ring Buffer Control Next Write Record

EQ
Source A QualityRBCtl.WritePtr 43
Source B QualityRBCtl.ReadPtr 42

EDC Ring Buffer Control Buffer Full - Block Writes

QualityRBCtl.FullBlockWrite

EDC Ring Buffer Control Diagnostic - Manual Reset

QualityRBCtl.WriteBlockErr

L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A QualityRBCtl.WriteBlockErrCnt 221
Source B 1
Dest QualityRBCtl.WriteBlockErrCnt 221

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.WritePtr 43
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.ReadPtr 42
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 314
Input Par 0
Input Par QualityMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans TempMsgTrans
PerfTrans Status.QualityRx.ER

lbl_Exit

JMP

Load Quality Data record to Ring Buffer

RB Write -- Load Message into Ring Buffer

6

Equipment Data Quality/Health Message

COP	
Source	QualityMsg
Dest	QualityRB[QualityRBCtl.WritePtr]
Length	1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	QualityRBCtl.WritePtr 43
Dest	Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	QualityRBCtl.ReadPtr 42
Dest	Trans.Ctl.Opt[2] -1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	311
Input Par	0
Input Par	QualityMsg.Header.TransNum

Message Control Transaction Timing

FLL	
Source	0
Dest	TempMsgTrans
Length	1

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	TempMsgTrans
PerfTrans	Status.QualityRx.DN

RB Write -- Increment Write Pointer

7

EDC Ring Buffer Control Next Write Record

ADD	
Source A	QualityRBCtl.WritePtr 43
Source B	1
Dest	QualityRBCtl.WritePtr 43

EDC Ring Buffer Control Next Write Record

LT	
Source A	QualityRBCtl.WritePtr 43
Source B	0

EDC Ring Buffer Control Next Write Record

GT	
Source A	QualityRBCtl.WritePtr 43
Source B	CFG.QualityRB.mi 99

EDC Ring Buffer Control Next Write Record

MOVE	
Source	0
Dest	QualityRBCtl.WritePtr 43

Load Quality Data record to Ring Buffer

Exit Routine

8

lbl_Exit
LBL

NOP

(End)

Load Read Records from RingBuffer to Gateway Message

Equipment Quality/Health Data
Ring Buffer Unload to Gateway Message

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 133
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Load Read Records from RingBuffer to Gateway Message

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Next Read Record

LT
Source A QualityRBCtl.ReadPtr 42
Source B 0

EDC Ring Buffer Control Next Read Record

GT
Source A QualityRBCtl.ReadPtr 42
Source B CFG.QualityRB.mi 99

EDC Ring Buffer Control Diagnostic - Manual Reset

QualityRBCtl.ReadPtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source QualityRBCtl.ReadPtr 42
Dest QualityRBCtl.ReadPtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.WritePtr 43
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.ReadPtr 42
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 323
Input Par 0
Input Par QualityRBCtl.TransactionNumber

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest QualityRBCtl.WritePtr 43

EDC Ring Buffer Control Next Read Record

MOVE
Source CFG.QualityRB.mi 99
Dest QualityRBCtl.ReadPtr 42

Load Read Records from RingBuffer to Gateway Message

Next Read Location
ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

ADD
Source A QualityRBCtl.ReadPtr 42
Source B 1
Dest QualityRBCtl.ReadNextPtr 43

EDC Ring Buffer Control Next Read Record

GT
Source A QualityRBCtl.ReadNextPtr 43
Source B CFG.QualityRB.mi 99

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest QualityRBCtl.ReadNextPtr 43

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Next Read Record

EQ
Source A QualityRBCtl.ReadPtr 42
Source B QualityRBCtl.WritePtr 43

EDC Ring Buffer Control Buffer Full
QualityRBCtl.Full

EDC Ring Buffer Control Read Buffer Record in Progress
QualityRBCtl.ReadInProgress
/

EDC Ring Buffer Control Next Read Record

EQ
Source A QualityRBCtl.ReadNextPtr 43
Source B QualityRBCtl.WritePtr 43

EDC Ring Buffer Control Buffer Empty
QualityRBCtl.Empty

Load Read Records from RingBuffer to Gateway Message

5

EDC Ring Buffer Control Read Buffer Record in Progress
QualityRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
QualityRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record
GT
Source A QualityRBCtl.WritePtr 43
Source B QualityRBCtl.ReadPtr 42

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest QualityRBCtl.RecordCount 0
Expression QualityRBCtl.WritePtr-QualityRBCtl.ReadPtr

EDC Ring Buffer Control Next Write Record
LT
Source A QualityRBCtl.WritePtr 43
Source B QualityRBCtl.ReadPtr 42

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest QualityRBCtl.RecordCount 0
Expression CFG.QualityRB.n-(QualityRBCtl.ReadPtr-QualityRBCtl.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
QualityRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
QualityRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record
GE
Source A QualityRBCtl.WritePtr 43
Source B QualityRBCtl.ReadNextPtr 43

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest QualityRBCtl.RecordCount 0
Expression QualityRBCtl.WritePtr-QualityRBCtl.ReadNextPtr

EDC Ring Buffer Control Next Write Record
LT
Source A QualityRBCtl.WritePtr 43
Source B QualityRBCtl.ReadNextPtr 43

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest QualityRBCtl.RecordCount 0
Expression CFG.QualityRB.n-(QualityRBCtl.ReadNextPtr-QualityRBCtl.WritePtr)

EDC Ring Buffer Control Ring Buffer Record Count
GT
Source A QualityRBCtl.RecordCount 0
Source B QualityRBCtl.RecordCountMax 99

EDC Ring Buffer Control Ring Buffer Record Count Max
MOVE
Source QualityRBCtl.RecordCount 0
Dest QualityRBCtl.RecordCountMax 99

Load Read Records from RingBuffer to Gateway Message

Read Next
Ready for next data to Gateway Message

6

EDC Ring Buffer Control Buffer Empty: QualityRBCtl.Empty (/)
EDC Ring Buffer Control Read Buffer Record in Progress: QualityRBCtl.ReadInProgress (/)
Ring Buffer Message Control Send Data: QualityMsgCtl.Send (/)
Ring Buffer Message Control Resend Data (Keep Alive): QualityMsgCtl.Resend (/)
EDC Ring Buffer Control Read Next Buffer Record 1-shot: QualityRBCtl.ReadNextONS (ONS)
EDC Ring Buffer Control Read Next Buffer Record: QualityRBCtl.ReadNext ( )
EDC Ring Buffer Control Read Buffer Record in Progress: QualityRBCtl.ReadInProgress (L)
EDC Ring Buffer Control Read Buffer Record Complete: QualityRBCtl.ReadComplete (U)

Read Next
Increment Read Pointer

7

EDC Ring Buffer Control Read Next Buffer Record: QualityRBCtl.ReadNext

ADD
Source A: QualityRBCtl.ReadPtr 42
Source B: 1
Dest: QualityRBCtl.ReadPtr 42

EDC Ring Buffer Control Next Read Record

LT
Source A: QualityRBCtl.ReadPtr 42
Source B: 0

EDC Ring Buffer Control Next Read Record

GT
Source A: QualityRBCtl.ReadPtr 42
Source B: CFG.QualityRB.mi 99

EDC Ring Buffer Control Next Read Record

MOVE
Source: 0
Dest: QualityRBCtl.ReadPtr 42

Read Next
Load Read Record

8

EDC Ring Buffer Control Read Next Buffer Record: QualityRBCtl.ReadNext

EDC Message Tx Data Quality/Health Receive Message

COP
Source: QualityRB[QualityRBCtl.ReadPtr]
Dest: EqTxData.Quality[0] <EQ0006_TxData.Quality[0]>
Length: CFG.QualityMSG.e

Ring Buffer Message Control Transaction Number

MOVE
Source: QualityRB[QualityRBCtl.ReadPtr].Header.TransNum 4157
Dest: QualityMsgCtl.TransNum 4157

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: QualityRBCtl.WritePtr 43
Dest: Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: QualityRBCtl.ReadPtr 42
Dest: Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name: _902_TransLogSBR
Input Par: 321
Input Par: 0
Input Par: QualityMsgCtl.TransNum

Load Read Records from RingBuffer to Gateway Message

Read Next
Set Data Ready Handshake

9

EDC Ring Buffer Control Read Next Buffer Record
QualityRBCtl.ReadNext

Ring Buffer Message Control Send Data
QualityMsgCtl.Send (L)

Ring Buffer Message Control Read Command
QualityMsgCtl.Read (U)

Ring Buffer Message Control Read Buffer Ready
QualityMsgCtl.ReadOK (U)

Ring Buffer Message Control Write Command
QualityMsgCtl.Write (U)

Ring Buffer Message Control Write Command Done
QualityMsgCtl.WriteOK (U)

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

10

Ring Buffer Message Control Send Data
QualityMsgCtl.Send

Ring Buffer Message Control Write Command Done
QualityMsgCtl.WriteOK

Ring Buffer Message Control Resend Data (Keep Alive)
QualityMsgCtl.Resend

Ring Buffer Message Control Send Data
QualityMsgCtl.Send (U)

Ring Buffer Message Control Resend Data (Keep Alive)
QualityMsgCtl.Resend (U)

EDC Ring Buffer Control Read Buffer Record Complete
QualityRBCtl.ReadComplete (L)

Read Complete Delay
Allow Gateway processing time before attempting next record send.

11

EDC Ring Buffer Control Read Buffer Record Complete
QualityRBCtl.ReadComplete

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source 100
Dest QualityRBCtl.ReadCompleteDelayTMR.PRE
100

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer QualityRBCtl.ReadCompleteDelayTMR
Preset 100
Accum 113
EN
DN

EDC Ring Buffer Control Read Complete Delay Timer
QualityRBCtl.ReadCompleteDelayTMR.DN

EDC Ring Buffer Control Read Buffer Record Complete Delay
QualityRBCtl.ReadCompleteDelay

EDC Ring Buffer Control Read Buffer Record in Progress
QualityRBCtl.ReadInProgress (U)

Load Read Records from RingBuffer to Gateway Message

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

12

EDC Ring Buffer
Control Buffer Empty
QualityRBCtl.Empty
/

EDC Ring Buffer
Control Buffer Not
Empty Alarm Timer

MOVE
Source 300000
Dest QualityRBCtl.NotEmptyTMR.PRE
300000

EDC Ring Buffer
Control Buffer Not
Empty Alarm Timer

TON
Timer QualityRBCtl.NotEmptyTMR
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer
Control Buffer Not
Empty Alarm Timer

GT
Source A QualityRBCtl.NotEmptyTMR.ACC
0
Source B QualityRBCtl.NotEmptyMax
300001

EDC Ring Buffer
Control Buffer Not
Empty Max Accum

MOVE
Source QualityRBCtl.NotEmptyTMR.ACC
0
Dest QualityRBCtl.NotEmptyMax
300001

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

13

EDC Ring Buffer
Control Buffer Full
QualityRBCtl.Full

EDC Ring Buffer
Control Buffer Not
Empty Alarm Timer
QualityRBCtl.NotEmptyTMR.DN

EDC Ring Buffer
Control Read Buffer
Alarm
QualityRBCtl.BufferAlarm

(End)

Quality Data Ring Buffer and Message

Equipment Quality Data
Buffered Fault Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 134
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

Ring Buffer Message Control Message Transaction Timer

MOVE
Source 999999999
Dest QualityMsgCtl.Trans.TMR.PRE 999999999

Ring Buffer Message Control Message Transaction Timer

TON
Timer QualityMsgCtl.Trans.TMR
Preset 999999999
Accum 287979
EN
DN

Quality Data Ring Buffer and Message

Read Message Delay Timer
If driver sends data too fast, it will generate extra Read Message traffic until previous message has been processed
Optimization uses the ReadOk and ReadNotOk counts to tune the Delay Time to the Gateway response

3

EDC Gateway Quality Write Message
QualityWriteMsg.EN
<EQ0006_QualityWriteMsg.EN>

EDC Gateway Quality Read Message
QualityReadMsg.EN
<EQ0006_QualityReadMsg.EN>

Ring Buffer Message Control Read Message Ok Count

GE		
Source A	QualityMsgCtl.ReadOkCount	3
Source B	CfgRB_ReadOkLimit <CfgRB.ReadOkLimit>	5

Ring Buffer Message Control Message Delay Timer

SUB		
Source A	QualityMsgCtl.DelayTMR.PRE	100
Source B	CfgRB_ReadOkSub <CfgRB.ReadOkSub>	10
Dest	QualityMsgCtl.DelayTMR.PRE	100

Ring Buffer Message Control Message Delay Timer

LE		
Source A	QualityMsgCtl.DelayTMR.PRE	100
Source B	CfgRB_ReadDelayMin <CfgRB.ReadDelayMin>	100

Ring Buffer Message Control Message Delay Timer

MOVE		
Source	CfgRB_ReadDelayMin <CfgRB.ReadDelayMin>	100
Dest	QualityMsgCtl.DelayTMR.PRE	100

Ring Buffer Message Control Read Message Not Ok Count

GE		
Source A	QualityMsgCtl.ReadNotOkCount	0
Source B	CfgRB_ReadNotOkLimit <CfgRB.ReadNotOkLimit>	2

Ring Buffer Message Control Message Delay Timer

ADD		
Source A	QualityMsgCtl.DelayTMR.PRE	100
Source B	CfgRB_ReadNotOkAdd <CfgRB.ReadNotOkAdd>	20
Dest	QualityMsgCtl.DelayTMR.PRE	100

Ring Buffer Message Control Message Delay Timer

GT		
Source A	QualityMsgCtl.DelayTMR.PRE	100
Source B	CfgRB_ReadDelayMax <CfgRB.ReadDelayMax>	500

Ring Buffer Message Control Message Delay Timer

MOVE		
Source	CfgRB_ReadDelayMax <CfgRB.ReadDelayMax>	500
Dest	QualityMsgCtl.DelayTMR.PRE	100

Ring Buffer Message Control Read Message Ok Count

CLR		
Dest	QualityMsgCtl.ReadOkCount	3

Ring Buffer Message Control Read Message Not Ok Count

CLR		
Dest	QualityMsgCtl.ReadNotOkCount	0

Ring Buffer Message Control Message Delay Timer

TON			
Timer	QualityMsgCtl.DelayTMR		EN
Preset		100	DN
Accum		101	

Quality Data Ring Buffer and Message

Message Resend Timer
Keep Alive - resends last data without increment of Transaction Number

4

Ring Buffer Message Control Send Data QualityMsgCtl.Send

EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>

EDC Gateway Quality Read Message QualityReadMsg.EN <EQ0006_QualityReadMsg.EN>

Ring Buffer Message Control Message Resend Timer

MOVE
Source CFG.BuffMsgResendTime 301000
Dest QualityMsgCtl.ResendTMR.PRE 301000

Ring Buffer Message Control Message Resend Timer

TON
Timer QualityMsgCtl.ResendTMR
Preset 301000
Accum 287951
EN
DN

Ring Buffer Message Control Message Resend Timer QualityMsgCtl.ResendTMR.DN

Ring Buffer Message Control Resend Data (Keep Alive) QualityMsgCtl.Resend (L)

Ring Buffer Message Control Message Resend Timer QualityMsgCtl.ResendTMR (RES)

Ring Buffer Message Control Read Command QualityMsgCtl.Read (U)

Ring Buffer Message Control Read Buffer Ready QualityMsgCtl.ReadOK (U)

Ring Buffer Message Control Write Command QualityMsgCtl.Write (U)

Ring Buffer Message Control Write Command Done QualityMsgCtl.WriteOK (U)

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 322
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Send Data Initialize

5

Ring Buffer Message Control Send Data QualityMsgCtl.Send

Ring Buffer Message Control Send Data 1-Shot QualityMsgCtl.SendONS (ONS)

Ring Buffer Message Control Resend Data (Keep Alive) QualityMsgCtl.Resend

Ring Buffer Message Control Message Transaction Timer QualityMsgCtl.Trans.TMR (RES)

Ring Buffer Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR
Dest QualityMsgCtl.Trans.Last 13

Quality Data Ring Buffer and Message

Read Message
Check Gateway Buffer for Empty Record
(Gateway clears Buffer after processing Record)

6

Ring Buffer Message Control Send Data QualityMsgCtl.Send
Ring Buffer Message Control Read Buffer Ready QualityMsgCtl.ReadOK
Ring Buffer Message Control Message Delay Timer QualityMsgCtl.DelayTMR.DN
EDC Gateway Quality Read Message QualityReadMsg.EN <EQ0006_QualityReadMsg.EN>
EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>
EDC Gateway Quality Read Message QualityReadMsg.ER <EQ0006_QualityReadMsg.ER>
Ring Buffer Message Control Resend Data (Keep Alive) QualityMsgCtl.Resend
Ring Buffer Message Control Message Retry Timer QualityMsgCtl.RetryTMR.DN

Ring Buffer Message Control Read Command QualityMsgCtl.Read (L)
Ring Buffer Message Control Read Command 1-Shot QualityMsgCtl.ReadONS

EDC Gateway Quality Read Message
COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest QualityReadMsg.Path <EQ0006_QualityReadMsg.Path>
Length 1

EDC Gateway Quality Read Message
COP
Source CFG.MsgDest.QualityDest
Dest QualityReadMsg.RemoteElement <EQ0006_QualityReadMsg.RemoteElement>
Length 1

EDC Gateway Quality Read Message
MOVE
Source CFG.QualityMSG.e 39
Dest QualityReadMsg.REQ_LEN <EQ0006_QualityReadMsg.REQ_LEN> 39

EDC Gateway Quality Read Message
MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest QualityReadMsg.UnconnectedTimeout <EQ0006_QualityReadMsg.UnconnectedTimeout> 4000000

EDC Gateway Quality Read Message
MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest QualityReadMsg.ConnectionRate <EQ0006_QualityReadMsg.ConnectionRate> 1000000

EDC Gateway Quality Read Message
MOVE
Source CFG.CLx.ConnectionMult 0
Dest QualityReadMsg.TimeoutMultiplier <EQ0006_QualityReadMsg.TimeoutMultiplier> 0

EDC Gateway Quality Read Message QualityReadMsg.EN_CC <EQ0006_QualityReadMsg.EN_CC> (U)

EDC Gateway Quality Read Message
MSG
Message Control QualityReadMsg <EQ0006_QualityReadMsg> ...
EN
DN
ER

Quality Data Ring Buffer and Message

Read Message EN Performance & Diagnostics

7

Ring Buffer Message Control Read Command 1-Shot
QualityMsgCtl.ReadONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans [...]
PerfMsgTrans	QualityMsgCtl.Trans
PerfTrans	Status.QualityTxRead.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	330
Input Par	QualityMsgCtl.Trans.TMR.ACC
Input Par	QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message

Read Message OK - Gateway Buffer has Empty Record
(Gateway clears Buffer after processing Record)

Increment ReadOk and ReadNotOk counts
only increment ReadOk count when multiple records in Buffer (single records are always ready to send and don't test Gateway servicing speed)

8

Ring Buffer Message Control Read Command QualityMsgCtl.Read

EDC Gateway Quality Read Message QualityReadMsg.DN <EQ0006_QualityReadMsg.DN>

Ring Buffer Message Control Read Command QualityMsgCtl.Read (U)

EDC Message Rx Data Quality/Health Receive Message

EQ
Source A EqRxData.Quality[0] <EQ0006_RxData.Quality[0]> 0
Source B 0

EDC Message Rx Data Quality/Health Receive Message

EQ
Source A EqRxData.Quality[CFG.QualityMSG.ei] <EQ0006_RxData.Quality[CFG.QualityMSG.ei]> 0
Source B 0

Ring Buffer Message Control Read Buffer Ready QualityMsgCtl.ReadOK (L)

EDC Ring Buffer Control Ring Buffer Record Count

GE
Source A QualityRBCtl.RecordCount 0
Source B 2

Ring Buffer Message Control Read Message Ok Count

ADD
Source A QualityMsgCtl.ReadOkCount 3
Source B 1
Dest QualityMsgCtl.ReadOkCount 3

Ring Buffer Message Control Read Buffer Ready QualityMsgCtl.ReadOK (/)

Ring Buffer Message Control Read Message Not Ok Count

ADD
Source A QualityMsgCtl.ReadNotOkCount 0
Source B 1
Dest QualityMsgCtl.ReadNotOkCount 0

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans QualityMsgCtl.Trans
PerfTrans Status.QualityTxRead.DN

Ring Buffer Message Control Resend Data (Keep Alive) QualityMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans QualityMsgCtl.Trans
PerfTrans Status.QualityTxRead.KA

Transaction Log Control SBR External Load -- Optional Parameters

Transaction Log Control SBR External Load -- Optional Parameters

Log Transactions

Quality Data Ring Buffer and Message

MOVE
Source EqRxData.Quality[0]
<EQ0006_RxData.Quality[0]>
0
Dest Trans.Ctl.Opt[1]
-1

MOVE
Source EqRxData.Quality[CFG.QualityMSG.ei]
<EQ0006_RxData.Quality[CFG.QualityMSG.ei]>
0
Dest Trans.Ctl.Opt[2]
-1

JSR
Routine Name _902_TransLogSBR
Input Par 331
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Read Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

9

EDC Gateway Quality
Read Message
QualityReadMsg.EN
<EQ0006_QualityReadMsg.EN>

Ring Buffer Message
Control Message
Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest QualityMsgCtl.ReadTimeoutTMR.PRE
5000

Ring Buffer Message
Control Message
Timeout Timer

TON
Timer QualityMsgCtl.ReadTimeoutTMR
Preset 5000
Accum 0
EN
DN

Ring Buffer Message
Control Message
Timeout Timer
QualityMsgCtl.ReadTimeoutTMR.DN

EDC Gateway Quality
Read Message
QualityReadMsg.TO
<EQ0006_QualityReadMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 334
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message
Read Message ER Performance & Diagnostics

10

EDC Gateway Quality Read Message
QualityReadMsg.ER
<EQ0006_QualityReadMsg.ER>

Ring Buffer Message Control Read Error 1-shot
QualityMsgCtl.ReadErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Quality Tx ER
Status.Update.QualityTxER
L

Update Status State for Transactions
aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans QualityMsgCtl.Trans
PerfTrans Status.QualityTxRead.ER

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 333
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message

Write Message
Write Data is loaded by Ring Buffer routine

11

Ring Buffer Message Control Send Data QualityMsgCtl.Send
Ring Buffer Message Control Read Buffer Ready QualityMsgCtl.ReadOK
Ring Buffer Message Control Write Command Done QualityMsgCtl.WriteOK
EDC Gateway Quality Read Message QualityReadMsg.EN <EQ0006_QualityReadMsg.EN>
EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>
EDC Gateway Quality Write Message QualityWriteMsg.ER <EQ0006_QualityWriteMsg.ER>

Ring Buffer Message Control Resend Data (Keep Alive) QualityMsgCtl.Resend

Ring Buffer Message Control Message Retry Timer QualityMsgCtl.RetryTMR.DN

Ring Buffer Message Control Write Command QualityMsgCtl.Write (L)

EDC Gateway Quality Write Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest QualityWriteMsg.Path <EQ0006_QualityWriteMsg.Path>
Length 1

EDC Gateway Quality Write Message

COP
Source CFG.MsgDest.QualityDest
Dest QualityWriteMsg.RemoteElement <EQ0006_QualityWriteMsg.RemoteElement>
Length 1

EDC Gateway Quality Write Message

MOVE
Source CFG.QualityMSG.e 39
Dest QualityWriteMsg.REQ_LEN <EQ0006_QualityWriteMsg.REQ_LEN> 39

EDC Gateway Quality Write Message

MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest QualityWriteMsg.UnconnectedTimeout <EQ0006_QualityWriteMsg.UnconnectedTimeout> 4000000

EDC Gateway Quality Write Message

MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest QualityWriteMsg.ConnectionRate <EQ0006_QualityWriteMsg.ConnectionRate> 1000000

EDC Gateway Quality Write Message

MOVE
Source CFG.CLx.ConnectionMult 0
Dest QualityWriteMsg.TimeoutMultiplier <EQ0006_QualityWriteMsg.TimeoutMultiplier> 0

EDC Gateway Quality Write Message QualityWriteMsg.EN_CC <EQ0006_QualityWriteMsg.EN_CC> (U)

EDC Gateway Quality Write Message

MSG
Message Control QualityWriteMsg ... <EQ0006_QualityWriteMsg>
EN
DN
ER

Quality Data Ring Buffer and Message

Write Message EN Performance & Diagnostics

12

Ring Buffer Message Control Write Command
QualityMsgCtl.Write

Ring Buffer Message Control Write Command 1-Shot
QualityMsgCtl.WriteONS
ONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	QualityMsgCtl.Trans
PerfTrans	Status.QualityTxWrite.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	340
Input Par	QualityMsgCtl.Trans.TMR.ACC
Input Par	QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message

Write Message Done

13

Ring Buffer Message Control Write Command
QualityMsgCtl.Write

EDC Gateway Quality Write Message
QualityWriteMsg.DN
<EQ0006_QualityWriteMsg.DN>

Ring Buffer Message Control Write Command
QualityMsgCtl.Write
U

Ring Buffer Message Control Write Command Done
QualityMsgCtl.WriteOK
L

Ring Buffer Message Control Send Data
QualityMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Quality Tx DN
Status.Update.QualityTxOK
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate ...
Status	Status

Ring Buffer Message Control Resend Data (Keep Alive)
QualityMsgCtl.Resend

Equipment Data Collection - Line Equipment Status State = Green - Quality Keep Alive Tx DN
Status.Update.QualityKaTxOK
L

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	QualityMsgCtl.Trans
PerfTrans	Status.QualityTxWrite.DN

Ring Buffer Message Control Resend Data (Keep Alive)
QualityMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	QualityMsgCtl.Trans
PerfTrans	Status.QualityTxWrite.KA

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	341
Input Par	QualityMsgCtl.Trans.TMR.ACC
Input Par	QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message

Write Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

14

EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>

Ring Buffer Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest QualityMsgCtl.WriteTImeoutTMR.PRE
5000

Ring Buffer Message Control Message Timeout Timer

TON
Timer QualityMsgCtl.WriteTImeoutTMR
Preset 5000
Accum 0
EN
DN

Ring Buffer Message Control Message Timeout Timer QualityMsgCtl.WriteTImeoutTMR.DN

EDC Gateway Quality Write Message QualityWriteMsg.TO <EQ0006_QualityWriteMsg.TO>

L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 344
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Write Message ER Performance & Diagnostics

15

EDC Gateway Quality Write Message QualityWriteMsg.ER <EQ0006_QualityWriteMsg.ER>

Ring Buffer Message Control Write Error 1-shot QualityMsgCtl.WriteErONS

ONS

Equipment Data Collection - Line Equipment Status State = Red - Quality Tx ER Status.Update.QualityTxER

L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans QualityMsgCtl.Trans
PerfTrans Status.QualityTxWrite.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 343
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message
Read/Write Message Error Retry

16

EDC Gateway Quality
Read Message
QualityReadMsg.ER
<EQ0006_QualityReadMsg.ER>

EDC Gateway Quality
Write Message
QualityWriteMsg.ER
<EQ0006_QualityWriteMsg.ER>

Ring Buffer Message
Control Message
Retry Timer
QualityMsgCtl.RetryTMR.DN

Ring Buffer Message
Control Retry 1-Shot
QualityMsgCtl.RetryONS
ONS

Ring Buffer Message
Control Message
Retry Timer
MOVE
Source CFG.MsgRetryTime
5000
Dest QualityMsgCtl.RetryTMR.PRE
0

Ring Buffer Message
Control Message
Retry Timer
TON
Timer QualityMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Quality
Read Message
QualityReadMsg.ER
<EQ0006_QualityReadMsg.ER>

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 332
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

EDC Gateway Quality
Write Message
QualityWriteMsg.ER
<EQ0006_QualityWriteMsg.ER>

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 342
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

(End)

Collect Trend Data

Equipment Trend Data
Collect Data

0

Trend Index
SBR
Input Par TRi

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned
MOVE
Source 141
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)
ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Exit Routine if Index is out of range

2

Trend Index
LT
Source A TRi 4
Source B 0

Trend Index
GT
Source A TRi 4
Source B CFG.Trend.mi 20

lbl_Exit
JMP

Copy Trend Arrays to local tags

3

Trend Collect Control
COP
Source TrendCfg[TRi].CTL
Dest TrendCollectCtl
Length 1

Equipment Data Trend Message Data
COP
Source TrendCfg[TRi].Data
Dest TrendCollect
Length 1

Collect Trend Data

Trend Initialize
Load user Sample Rate if within limits or default to 1 second
(Rate = Samples/Second)

4

Trend Collect Control Trend Start Sampling
TrendCollectCtl.Start

Trend Collect Control Trend Start 1-Shot
TrendCollectCtl.StartONS
ONS

Trend Collect Control Trend Init Sampling
TrendCollectCtl.Init

Trend Collect Control Trend Done Sampling
TrendCollectCtl.Done
U

Trend Collect Control Trend Save
TrendCollectCtl.Loaded
U

Equipment Data Trend Message Data
FLL
Source 0
Dest TrendCollect
Length 1

Trend Collect Control Trend Sample Rate Actual
MOVE
Source 1
Dest TrendCollectCtl.RateAct
0

Trend Collect Control Trend Sample Rate Requested (1-10 tenths of second)
LIMIT
Low Limit 1
Test TrendCollectCtl.Rate
0
High Limit 10

Trend Collect Control Trend Sample Rate Actual
MOVE
Source TrendCollectCtl.Rate
0
Dest TrendCollectCtl.RateAct
0

Trend Collect Control Trend Sample Time Preset
DIV
Source A 1000
Source B TrendCollectCtl.RateAct
0
Dest TrendCollectCtl.SampleTimePre
0

Trend Collect Control Trend Sample Last Time
CLR
Dest TrendCollectCtl.SampleTimeLast
0

Trend Collect Control Trend Array Index
CLR
Dest TrendCollectCtl.ARRi
0

Collect Trend Data

Trend Init
JobStatus and FaultCode are updates on Start and Send

5

Trend Collect
Control Trend Init
Sampling
TrendCollectCtl.Init

Equipment Data
Trend Message Data
Tool Number

MOVE	
Source	TRi 4
Dest	TrendCollect.ToolNum 0

Equipment Data
Trend Message Data
Data Values Used
(1-120)

MOVE	
Source	CFG.TrData.n 120
Dest	TrendCollect.DataCount 0

Equipment Data
Trend Message Data
Trend Rate - samples
per second (1-10)

MOVE	
Source	TrendCollectCtl.RateAct 0
Dest	TrendCollect.Rate 0

Equipment Data
Trend Message Data
Trend Scale

MOVE	
Source	TrendCollectCtl.Scale 0
Dest	TrendCollect.Scale 0

Equipment Data
Trend Message Data
Job Status

MOVE	
Source	TrendMsg.Data.JobStatus 0
Dest	TrendCollect.JobStatus 0

Equipment Data
Trend Message Data
Fault Code

MOVE	
Source	TrendMsg.Data.FaultCode 0
Dest	TrendCollect.FaultCode 0

Collect Trend Data

Trend Vehicle Data

6

Trend Collect
Control Trend Load
Vehicle Data
TrendCollectCtl.LoadVeh

Equipment Data
Trend Message Data
Program Number from
SLCP

MOVE
Source TrendMsg.Data.AviPrgNum 0
Dest TrendCollect.AviPrgNum 0

Equipment Data
Trend Message Data
Program Number
Executed

MOVE
Source TrendMsg.Data.ActPrgNum 0
Dest TrendCollect.ActPrgNum 0

Equipment Data
Trend Message Data
Assembly Body Number

MOVE
Source TrendMsg.Data.BodyNum 0
Dest TrendCollect.BodyNum 0

Equipment Data
Trend Message Data
Assembly Sequence
Number

MOVE
Source TrendMsg.Data.AssyNum 0
Dest TrendCollect.AssyNum 0

Equipment Data
Trend Message Data
Pdm Triggers: Start
(-11) and Send (-12)

MOVE
Source TrendMsg.Data.PdmTrigger 0
Dest TrendCollect.PdmTrigger 0

Collect Trend Data

Trend Sample TImer
Use free-running timer and samle interval to avoid accumulating Timer overruns throughout test.

7

Trend Collect Control Trend Start Sampling
TrendCollectCtl.Start

Trend Collect Control Trend Stop Sampling
TrendCollectCtl.Stop

Trend Collect Control Trend Send
TrendCollectCtl.Send

Trend Collect Control Trend Done Sampling
TrendCollectCtl.Done

Trend Collect Control Trend Sample Timer

MOVE
Source 99999999
Dest TrendCollectCtl.SampleTMR.PRE
0

Trend Collect Control Trend Sample Timer

TON
Timer TrendCollectCtl.SampleTMR
Preset 0
Accum 0
EN
DN

Trend Collect Control Trend Sample Accumulation

SUB
Source A TrendCollectCtl.SampleTMR.ACC
0
Source B TrendCollectCtl.SampleTimeLast
0
Dest TrendCollectCtl.SampleTimeAcc
0

Trend Collect Control Trend Sample Accumulation

GE
Source A TrendCollectCtl.SampleTimeAcc
0
Source B TrendCollectCtl.SampleTimePre
0

Trend Collect Control Trend Sample Last Time

ADD
Source A TrendCollectCtl.SampleTimeLast
0
Source B TrendCollectCtl.SampleTimePre
0
Dest TrendCollectCtl.SampleTimeLast
0

Trend Collect Control Trend Sample
TrendCollectCtl.Sample

Collect Trend Data

Trend Sample

8

Trend Collect
Control Trend Init
Sampling
TrendCollectCtl.Init

Trend Collect
Control Trend Sample
TrendCollectCtl.Sample

Equipment Data
Trend Message Data
Trend Data

MOVE
Source TrendCollectCtl.Data
0
Dest TrendCollect.Trend[TrendCollectCtl.ARRi]
0

Trend Collect
Control Trend Array
Index

ADD
Source A TrendCollectCtl.ARRi
0
Source B 1
Dest TrendCollectCtl.ARRi
0

Trend Collect
Control Trend Array
Index

LT
Source A TrendCollectCtl.ARRi
0
Source B 0

Trend Collect
Control Trend Array
Index

GT
Source A TrendCollectCtl.ARRi
0
Source B CFG.TrData.mi
119

Trend Collect
Control Trend Done
Sampling
TrendCollectCtl.Done
L

Collect Trend Data

Trend Samples Done
Update Status/Fault and Load RIng Buffer

9

Trend Collect Control Trend Send TrendCollectCtl.Send

Trend Collect Control Trend Save TrendCollectCtl.Loaded

Trend Collect Control Trend Send TrendCollectCtl.Send

Equipment Data Trend Message Data Job Status

MOVE
Source TrendMsg.Data.JobStatus 0
Dest TrendCollect.JobStatus 0

Equipment Data Trend Message Data Fault Code

MOVE
Source TrendMsg.Data.FaultCode 0
Dest TrendCollect.FaultCode 0

Equipment Data Trend Message Data Data Values Used (1-120)

MOVE
Source TrendCollectCtl.ARRi 0
Dest TrendCollect.DataCount 0

Load Trend Data record to Ring Buffer

JSR
Routine Name _142_TrendDataLoadRB

Trend Collect Control Trend Save TrendCollectCtl.Loaded (L)

Trend Collect Control Trend Start Sampling TrendCollectCtl.Start (U)

Trend Collect Control Trend Send TrendCollectCtl.Send (U)

Copy local tags to Trend Array

10

Trend Data Collection Trend Control

COP
Source TrendCollectCtl
Dest TrendCfg[TRi].CTL
Length 1

Trend Data Collection Trend Data

COP
Source TrendCollect
Dest TrendCfg[TRi].Data
Length 1

Exit Routine

11

lbl_Exit
LBL

NOP

(End)

Load Trend Data record to Ring Buffer

Equipment Trend Data
Ring Buffer Load

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 142
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

EDC Ring Buffer Control Transaction Number

ADD
Source A TrendRBCtl.TransactionNumber 0
Source B 1
Dest TrendRBCtl.TransactionNumber 0

EDC Ring Buffer Control Transaction Number

LT
Source A TrendRBCtl.TransactionNumber 0
Source B 1

EDC Ring Buffer Control Transaction Number

GT
Source A TrendRBCtl.TransactionNumber 0
Source B CFG.MsgTransNumRollover 9999

EDC Ring Buffer Control Transaction Number

MOVE
Source 1
Dest TrendRBCtl.TransactionNumber 0

Load Trend Data record to Ring Buffer

Load Message Header, Timestamp, Trend Data and Footer Data

3

Equipment Data Trend Message Header - Transaction Number (must match Footer)

MOVE
Source TrendRBCtl.TransactionNumber 0
Dest TrendMsg.Header.TransNum 0

Equipment Data Trend Message Header - Command Number

MOVE
Source 0
Dest TrendMsg.Header.CmdNum 0

Equipment Data Trend Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source UserCfg.EQID 6
Dest TrendMsg.Header.EQID 0

Equipment Data Trend Message Header - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest TrendMsg.Header.Spare 0

Equipment Data Trend Message Transaction Time Stamp

COP
Source CLX.TransTime
Dest TrendMsg.TransTime
Length 1

Equipment Data Trend Message Data - Trend

COP
Source TrendCollect
Dest TrendMsg.Data
Length 1

Equipment Data Trend Message Footer - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest TrendMsg.Footer.Spare 0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

MOVE
Source TrendRBCtl.TransactionNumber 0
Dest TrendMsg.Footer.TransNum 0

Load Trend Data record to Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

4

EDC Ring Buffer Control Next Write Record

LT
Source A: TrendRBCtl.WritePtr 0
Source B: 0

EDC Ring Buffer Control Next Write Record

GT
Source A: TrendRBCtl.WritePtr 0
Source B: CFG.TrendRB.mi 49

EDC Ring Buffer Control Diagnostic - Manual Reset

TrendRBCtl.WritePtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source: TrendRBCtl.WritePtr 0
Dest: TrendRBCtl.WritePtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: TrendRBCtl.WritePtr 0
Dest: Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: TrendRBCtl.ReadPtr 49
Dest: Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name: _902_TransLogSBR
Input Par: 413
Input Par: 0
Input Par: TrendMsg.Header.TransNum

EDC Ring Buffer Control Next Write Record

MOVE
Source: 0
Dest: TrendRBCtl.WritePtr 0

Load Trend Data record to Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

5

EDC Ring Buffer Control Next Write Record

EQ
Source A TrendRBCtl.WritePtr 0
Source B TrendRBCtl.ReadPtr 49

EDC Ring Buffer Control Buffer Full - Block Writes
TrendRBCtl.FullBlockWrite

EDC Ring Buffer Control Diagnostic - Manual Reset
TrendRBCtl.WriteBlockErr
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A TrendRBCtl.WriteBlockErrCnt 0
Source B 1
Dest TrendRBCtl.WriteBlockErrCnt 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendRBCtl.WritePtr 0
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendRBCtl.ReadPtr 49
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 414
Input Par 0
Input Par TrendMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans TempMsgTrans
PerfTrans Status.TrendRx.ER

lbl_Exit
JMP

Load Trend Data record to Ring Buffer

RB Write -- Load Message into Ring Buffer

6

Equipment Data Trend Message

COP
Source TrendMsg
Dest TrendRB[TrendRBCtl.WritePtr]
Length 1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendRBCtl.WritePtr
0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendRBCtl.ReadPtr
49
Dest Trans.Ctl.Opt[2]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 411
Input Par 0
Input Par TrendMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TempMsgTrans
PerfTrans Status.TrendRx.DN

RB Write -- Increment Write Pointer

7

EDC Ring Buffer Control Next Write Record

ADD
Source A TrendRBCtl.WritePtr
0
Source B 1
Dest TrendRBCtl.WritePtr
0

EDC Ring Buffer Control Next Write Record

LT
Source A TrendRBCtl.WritePtr
0
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A TrendRBCtl.WritePtr
0
Source B CFG.TrendRB.mi
49

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest TrendRBCtl.WritePtr
0

Exit Routine

8

lbl_Exit
LBL

NOP

Load Trend Data record to Ring Buffer

(End)

Load Read Records from RingBuffer to Gateway Message

Equipment Trend Data
Ring Buffer Unload to Gateway Message

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 143
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Load Read Records from RingBuffer to Gateway Message

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Next Read Record

LT
Source A: TrendRBCtl.ReadPtr 49
Source B: 0

EDC Ring Buffer Control Next Read Record

GT
Source A: TrendRBCtl.ReadPtr 49
Source B: CFG.TrendRB.mi 49

EDC Ring Buffer Control Diagnostic - Manual Reset

TrendRBCtl.ReadPtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source: TrendRBCtl.ReadPtr 49
Dest: TrendRBCtl.ReadPtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: TrendRBCtl.WritePtr 0
Dest: Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: TrendRBCtl.ReadPtr 49
Dest: Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name: _902_TransLogSBR
Input Par: 423
Input Par: 0
Input Par: TrendRBCtl.TransactionNumber

EDC Ring Buffer Control Next Write Record

MOVE
Source: 0
Dest: TrendRBCtl.WritePtr 0

EDC Ring Buffer Control Next Read Record

MOVE
Source: CFG.TrendRB.mi 49
Dest: TrendRBCtl.ReadPtr 49

Load Read Records from RingBuffer to Gateway Message

Next Read Location
ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

ADD
Source A TrendRBCtl.ReadPtr 49
Source B 1
Dest TrendRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Next Read Record

GT
Source A TrendRBCtl.ReadNextPtr 0
Source B CFG.TrendRB.mi 49

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest TrendRBCtl.ReadNextPtr 0

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Next Read Record

EQ
Source A TrendRBCtl.ReadPtr 49
Source B TrendRBCtl.WritePtr 0

EDC Ring Buffer Control Buffer Full
TrendRBCtl.Full

EDC Ring Buffer Control Read Buffer Record in Progress
TrendRBCtl.ReadInProgress
/

EDC Ring Buffer Control Next Read Record

EQ
Source A TrendRBCtl.ReadNextPtr 0
Source B TrendRBCtl.WritePtr 0

EDC Ring Buffer Control Buffer Empty
TrendRBCtl.Empty

Load Read Records from RingBuffer to Gateway Message

5

Buffer Count

EDC Ring Buffer Control Read Buffer Record in Progress
TrendRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
TrendRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record

GT
Source A TrendRBCtl.WritePtr 0
Source B TrendRBCtl.ReadPtr 49

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest TrendRBCtl.RecordCount 0
Expression TrendRBCtl.WritePtr-TrendRBCtl.ReadPtr

EDC Ring Buffer Control Next Write Record

LT
Source A TrendRBCtl.WritePtr 0
Source B TrendRBCtl.ReadPtr 49

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest TrendRBCtl.RecordCount 0
Expression CFG.TrendRB.n-(TrendRBCtl.ReadPtr-TrendRBCtl.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
TrendRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
TrendRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record

GE
Source A TrendRBCtl.WritePtr 0
Source B TrendRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest TrendRBCtl.RecordCount 0
Expression TrendRBCtl.WritePtr-TrendRBCtl.ReadNextPtr

EDC Ring Buffer Control Next Write Record

LT
Source A TrendRBCtl.WritePtr 0
Source B TrendRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest TrendRBCtl.RecordCount 0
Expression CFG.TrendRB.n-(TrendRBCtl.ReadNextPtr-TrendRBCtl.WritePtr)

EDC Ring Buffer Control Ring Buffer Record Count

GT
Source A TrendRBCtl.RecordCount 0
Source B TrendRBCtl.RecordCountMax 0

EDC Ring Buffer Control Ring Buffer Record Count Max

MOVE
Source TrendRBCtl.RecordCount 0
Dest TrendRBCtl.RecordCountMax 0

Load Read Records from RingBuffer to Gateway Message

Read Next
Ready for next data to Gateway Message

6

EDC Ring Buffer Control Buffer Empty	EDC Ring Buffer Control Read Buffer Record in Progress	EDC Gateway Trend Message Control Send Data	EDC Gateway Trend Message Control Resend Data (Keep Alive)	EDC Ring Buffer Control Read Next Buffer Record 1-shot	EDC Ring Buffer Control Read Next Buffer Record	EDC Ring Buffer Control Read Buffer Record in Progress	EDC Ring Buffer Control Read Buffer Record Complete
TrendRBCtl.Empty	TrendRBCtl.ReadInProgress	TrendMsgCtl.Send	TrendMsgCtl.Resend	TrendRBCtl.ReadNextONS	TrendRBCtl.ReadNext	TrendRBCtl.ReadInProgress	TrendRBCtl.ReadComplete
/	/	/	/	ONS		L	U

Read Next
Increment Read Pointer

7

EDC Ring Buffer Control Read Next Buffer Record	EDC Ring Buffer Control Next Read Record	EDC Ring Buffer Control Next Read Record	EDC Ring Buffer Control Next Read Record
TrendRBCtl.ReadNext	ADD	LT	MOVE
	Source A TrendRBCtl.ReadPtr 49	Source A TrendRBCtl.ReadPtr 49	Source 0
	Source B 1	Source B 0	Dest TrendRBCtl.ReadPtr 49
	Dest TrendRBCtl.ReadPtr 49		

EDC Ring Buffer Control Next Read Record

GT

Source A TrendRBCtl.ReadPtr 49

Source B CFG.TrendRB.mi 49

Load Read Records from RingBuffer to Gateway Message

Read Next
Load Read Record

8

EDC Ring Buffer Control Read Next Buffer Record
TrendRBCtl.ReadNext

EDC Message Tx Data Trend Receive Message

COP	
Source	TrendRB[TrendRBCtl.ReadPtr]
Dest	EqTxData.Trend[0]
	<EQ0006_TxData.Trend[0]>
Length	CFG.TrendMSG.e

EDC Gateway Trend Message Control Transaction Number

MOVE	
Source	TrendRB[TrendRBCtl.ReadPtr].Header.TransNum
	0
Dest	TrendMsgCtl.TransNum
	0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	TrendRBCtl.WritePtr
	0
Dest	Trans.Ctl.Opt[1]
	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	TrendRBCtl.ReadPtr
	49
Dest	Trans.Ctl.Opt[2]
	-1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	421
Input Par	0
Input Par	TrendMsgCtl.TransNum

Read Next
Set Data Ready Handshake

9

EDC Ring Buffer Control Read Next Buffer Record
TrendRBCtl.ReadNext

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send (L)

EDC Gateway Trend Message Control Read Command
TrendMsgCtl.Read (U)

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK (U)

EDC Gateway Trend Message Control Write Command
TrendMsgCtl.Write (U)

EDC Gateway Trend Message Control Write Command Done
TrendMsgCtl.WriteOK (U)

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

10

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send

EDC Gateway Trend Message Control Write Command Done
TrendMsgCtl.WriteOK

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send (U)

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend (U)

EDC Ring Buffer Control Read Buffer Record Complete
TrendRBCtl.ReadComplete (L)

Load Read Records from RingBuffer to Gateway Message

Read Complete Delay
Allow Gateway processing time before attempting next record send.

11

EDC Ring Buffer Control Read Buffer Record Complete
TrendRBCtl.ReadComplete

EDC Ring Buffer Control Read Complete Delay Timer

MOVE	
Source	100
Dest	TrendRBCtl.ReadCompleteDelayTMR.PRE 0

EDC Ring Buffer Control Read Complete Delay Timer

TON		
Timer	TrendRBCtl.ReadCompleteDelayTMR	EN
Preset	0	DN
Accum	0	

EDC Ring Buffer Control Read Complete Delay Timer
TrendRBCtl.ReadCompleteDelayTMR.DN

EDC Ring Buffer Control Read Buffer Record Complete Delay
TrendRBCtl.ReadCompleteDelay

EDC Ring Buffer Control Read Buffer Record in Progress
TrendRBCtl.ReadInProgress (U)

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

12

EDC Ring Buffer Control Buffer Empty
TrendRBCtl.Empty

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE	
Source	300000
Dest	TrendRBCtl.NotEmptyTMR.PRE 0

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON		
Timer	TrendRBCtl.NotEmptyTMR	EN
Preset	0	DN
Accum	0	

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT	
Source A	TrendRBCtl.NotEmptyTMR.ACC 0
Source B	TrendRBCtl.NotEmptyMax 0

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE	
Source	TrendRBCtl.NotEmptyTMR.ACC 0
Dest	TrendRBCtl.NotEmptyMax 0

Load Read Records from RingBuffer to Gateway Message

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

13

EDC Ring Buffer Control Buffer Full
TrendRBCtl.Full

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
TrendRBCtl.NotEmptyTMR.DN

EDC Ring Buffer Control Read Buffer Alarm
TrendRBCtl.BufferAlarm

(End)

Trend Data Ring Buffer and Message

Equipment Trend Data
Buffered Fault Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	144
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

EDC Gateway Trend Message Control Message Transaction Timer

MOVE	
Source	999999999
Dest	TrendMsgCtl.Trans.TMR.PRE 0

EDC Gateway Trend Message Control Message Transaction Timer

TON		
Timer	TrendMsgCtl.Trans.TMR	EN
Preset	0	DN
Accum	0	

Trend Data Ring Buffer and Message

Read Message Delay Timer
If driver sends data too fast, it will generate extra Read Message traffic until previous message has been processed
Optimization uses the ReadOk and ReadNotOk counts to tune the Delay Time to the Gateway response

3

EDC Gateway Trend Write Message
TrendWriteMsg.EN
<EQ0006_TrendWriteMsg.EN>

EDC Gateway Trend Read Message
TrendReadMsg.EN
<EQ0006_TrendReadMsg.EN>

EDC Gateway Trend Message Control Read Message Ok Count

GE
Source A TrendMsgCtl.ReadOkCount 0
Source B CfgRB_ReadOkLimit <CfgRB.ReadOkLimit> 5

EDC Gateway Trend Message Control Message Delay Timer

SUB
Source A TrendMsgCtl.DelayTMR.PRE 0
Source B CfgRB_ReadOkSub <CfgRB.ReadOkSub> 10
Dest TrendMsgCtl.DelayTMR.PRE 0

EDC Gateway Trend Message Control Message Delay Timer

LE
Source A TrendMsgCtl.DelayTMR.PRE 0
Source B CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100

EDC Gateway Trend Message Control Message Delay Timer

MOVE
Source CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100
Dest TrendMsgCtl.DelayTMR.PRE 0

EDC Gateway Trend Message Control Read Message Not Ok Count

GE
Source A TrendMsgCtl.ReadNotOkCount 0
Source B CfgRB_ReadNotOkLimit <CfgRB.ReadNotOkLimit> 2

EDC Gateway Trend Message Control Message Delay Timer

ADD
Source A TrendMsgCtl.DelayTMR.PRE 0
Source B CfgRB_ReadNotOkAdd <CfgRB.ReadNotOkAdd> 20
Dest TrendMsgCtl.DelayTMR.PRE 0

EDC Gateway Trend Message Control Message Delay Timer

GT
Source A TrendMsgCtl.DelayTMR.PRE 0
Source B CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500

EDC Gateway Trend Message Control Message Delay Timer

MOVE
Source CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500
Dest TrendMsgCtl.DelayTMR.PRE 0

EDC Gateway Trend Message Control Read Message Ok Count

CLR
Dest TrendMsgCtl.ReadOkCount 0

EDC Gateway Trend Message Control Read Message Not Ok Count

CLR
Dest TrendMsgCtl.ReadNotOkCount 0

EDC Gateway Trend Message Control Message Delay Timer

TON
Timer TrendMsgCtl.DelayTMR
Preset 0
Accum 0
EN
DN

Trend Data Ring Buffer and Message

Message Resend Timer
Keep Alive - resends last data without increment of Transaction Number

4

EDC Gateway Trend Message Control Send Data TrendMsgCtl.Send

EDC Gateway Trend Write Message TrendWriteMsg.EN <EQ0006_TrendWriteMsg.EN>

EDC Gateway Trend Read Message TrendReadMsg.EN <EQ0006_TrendReadMsg.EN>

EDC Gateway Trend Message Control Message Resend Timer

MOVE
Source CFG.BuffMsgResendTime 301000
Dest TrendMsgCtl.ResendTMR.PRE 0

EDC Gateway Trend Message Control Message Resend Timer

TON
Timer TrendMsgCtl.ResendTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Trend Message Control Message Resend Timer TrendMsgCtl.ResendTMR.DN

EDC Gateway Trend Message Control Resend Data (Keep Alive) TrendMsgCtl.Resend
L

EDC Gateway Trend Message Control Message Resend Timer TrendMsgCtl.ResendTMR
RES

EDC Gateway Trend Message Control Read Command TrendMsgCtl.Read
U

EDC Gateway Trend Message Control Read Buffer Ready TrendMsgCtl.ReadOK
U

EDC Gateway Trend Message Control Write Command TrendMsgCtl.Write
U

EDC Gateway Trend Message Control Write Command Done TrendMsgCtl.WriteOK
U

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 422
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Send Data Initialize

5

EDC Gateway Trend Message Control Send Data TrendMsgCtl.Send

EDC Gateway Trend Message Control Send Data 1-Shot TrendMsgCtl.SendONS
ONS

EDC Gateway Trend Message Control Resend Data (Keep Alive) TrendMsgCtl.Resend

EDC Gateway Trend Message Control Message Transaction Timer TrendMsgCtl.Trans.TMR
RES

EDC Gateway Trend Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR
Dest TrendMsgCtl.Trans.Last 0

Trend Data Ring Buffer and Message

Read Message
Check Gateway Buffer for Empty Record
(Gateway clears Buffer after processing Record)

6

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK

EDC Gateway Trend Message Control Message Delay Timer
TrendMsgCtl.DelayTMR.DN

EDC Gateway Trend Read Message
TrendReadMsg.EN
<EQ0006_TrendReadMsg.EN>

EDC Gateway Trend Write Message
TrendWriteMsg.EN
<EQ0006_TrendWriteMsg.EN>

EDC Gateway Trend Read Message
TrendReadMsg.ER
<EQ0006_TrendReadMsg.ER>

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend

EDC Gateway Trend Message Control Message Retry Timer
TrendMsgCtl.RetryTMR.DN

EDC Gateway Trend Message Control Read Command
TrendMsgCtl.Read
(L)

EDC Gateway Trend Message Control Read Command 1-Shot
TrendMsgCtl.ReadONS
( )

EDC Gateway Trend Read Message
COP
Source UserCfgMsg.Path
<EQ0006_UserCfgMsg.Path>
Dest TrendReadMsg.Path
<EQ0006_TrendReadMsg.Path>
Length 1

EDC Gateway Trend Read Message
COP
Source CFG.MsgDest.TrendDest
Dest TrendReadMsg.RemoteElement
<EQ0006_TrendReadMsg.RemoteElement>
Length 1

EDC Gateway Trend Read Message
MOVE
Source CFG.TrendMSG.e
70
Dest TrendReadMsg.REQ_LEN
<EQ0006_TrendReadMsg.REQ_LEN>
70

EDC Gateway Trend Read Message
MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest TrendReadMsg.UnconnectedTimeout
<EQ0006_TrendReadMsg.UnconnectedTimeout>
30000000

EDC Gateway Trend Read Message
MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest TrendReadMsg.ConnectionRate
<EQ0006_TrendReadMsg.ConnectionRate>
7500000

EDC Gateway Trend Read Message
MOVE
Source CFG.CLx.ConnectionMult
0
Dest TrendReadMsg.TimeoutMultiplier
<EQ0006_TrendReadMsg.TimeoutMultiplier>
0

EDC Gateway Trend Read Message
TrendReadMsg.EN_CC
<EQ0006_TrendReadMsg.EN_CC>
(U)

EDC Gateway Trend Read Message
MSG
Message Control TrendReadMsg ...
<EQ0006_TrendReadMsg>
EN
DN
ER

Trend Data Ring Buffer and Message

Read Message EN Performance & Diagnostics

7

EDC Gateway Trend Message Control Read Command 1-Shot
TrendMsgCtl.ReadONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans (...)
PerfMsgTrans	TrendMsgCtl.Trans
PerfTrans	Status.TrendTxRead.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	430
Input Par	TrendMsgCtl.Trans.TMR.ACC
Input Par	TrendMsgCtl.TransNum

Trend Data Ring Buffer and Message

Read Message OK - Gateway Buffer has Empty Record
(Gateway clears Buffer after processing Record)

Increment ReadOk and ReadNotOk counts
only increment ReadOk count when multiple records in Buffer (single records are always ready to send and don't test Gateway servicing speed)

8

EDC Gateway Trend Message Control Read Command
TrendMsgCtl.Read

EDC Gateway Trend Read Message
TrendReadMsg.DN
<EQ0006_TrendReadMsg.DN>

EDC Gateway Trend Message Control Read Command
TrendMsgCtl.Read
U

EDC Message Rx Data Trend Receive Message

EQ	
Source A	EqRxData.Trend[0]
	<EQ0006_RxData.Trend[0]>
	0
Source B	0

EDC Message Rx Data Trend Receive Message

EQ	
Source A	EqRxData.Trend[CFG.TrendMSG.ei]
	<EQ0006_RxData.Trend[CFG.TrendMSG.ei]>
	0
Source B	0

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK
L

EDC Ring Buffer Control Ring Buffer Record Count

GE	
Source A	TrendRBCtl.RecordCount
	0
Source B	2

EDC Gateway Trend Message Control Read Message Ok Count

ADD	
Source A	TrendMsgCtl.ReadOkCount
	0
Source B	1
Dest	TrendMsgCtl.ReadOkCount
	0

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK
/

EDC Gateway Trend Message Control Read Message Not Ok Count

ADD	
Source A	TrendMsgCtl.ReadNotOkCount
	0
Source B	1
Dest	TrendMsgCtl.ReadNotOkCount
	0

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TrendMsgCtl.Trans
PerfTrans	Status.TrendTxRead.DN

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TrendMsgCtl.Trans

Trend Data Ring Buffer and Message

PerfTrans Status.TrendTxRead.KA

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EqRxData.Trend[0]
<EQ0006_RxData.Trend[0]> 0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EqRxData.Trend[CFG.TrendMSG.ei]
<EQ0006_RxData.Trend[CFG.TrendMSG.ei]> 0
Dest Trans.Ctl.Opt[2]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 431
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Read Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

9

EDC Gateway Trend Read Message
TrendReadMsg.EN
<EQ0006_TrendReadMsg.EN>

EDC Gateway Trend Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest TrendMsgCtl.ReadTimeoutTMR.PRE
0

EDC Gateway Trend Message Control Message Timeout Timer

TON
Timer TrendMsgCtl.ReadTimeoutTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Trend Message Control Message Timeout Timer
TrendMsgCtl.ReadTimeoutTMR.DN

EDC Gateway Trend Read Message
TrendReadMsg.TO
<EQ0006_TrendReadMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 434
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Trend Data Ring Buffer and Message

Read Message ER Performance & Diagnostics

10

EDC Gateway Trend Read Message
TrendReadMsg.ER
<EQ0006_TrendReadMsg.ER>

EDC Gateway Trend Message Control Read Error 1-shot
TrendMsgCtl.ReadErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Trend Tx ER
Status.Update.TrendTxER
L

Update Status State for Transactions
aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate
Status Status

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TrendMsgCtl.Trans
PerfTrans Status.TrendTxRead.ER

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 433
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Trend Data Ring Buffer and Message

Write Message
Write Data is loaded by Ring Buffer routine

11

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK

EDC Gateway Trend Message Control Write Command Done
TrendMsgCtl.WriteOK

EDC Gateway Trend Read Message
TrendReadMsg.EN
<EQ0006_TrendReadMsg.EN>

EDC Gateway Trend Write Message
TrendWriteMsg.EN
<EQ0006_TrendWriteMsg.EN>

EDC Gateway Trend Write Message
TrendWriteMsg.ER
<EQ0006_TrendWriteMsg.ER>

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend

EDC Gateway Trend Message Control Message Retry Timer
TrendMsgCtl.RetryTMR.DN

EDC Gateway Trend Message Control Write Command
TrendMsgCtl.Write
L

EDC Gateway Trend Write Message

COP	
Source	UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest	TrendWriteMsg.Path <EQ0006_TrendWriteMsg.Path>
Length	1

EDC Gateway Trend Write Message

COP	
Source	CFG.MsgDest.TrendDest
Dest	TrendWriteMsg.RemoteElement <EQ0006_TrendWriteMsg.RemoteElement>
Length	1

EDC Gateway Trend Write Message

MOVE	
Source	CFG.TrendMSG.e 70
Dest	TrendWriteMsg.REQ_LEN <EQ0006_TrendWriteMsg.REQ_LEN> 70

EDC Gateway Trend Write Message

MOVE	
Source	CFG.CLx.UC_Timeout 4000000
Dest	TrendWriteMsg.UnconnectedTimeout <EQ0006_TrendWriteMsg.UnconnectedTimeout> 30000000

EDC Gateway Trend Write Message

MOVE	
Source	CFG.CLx.ConnectionRate 1000000
Dest	TrendWriteMsg.ConnectionRate <EQ0006_TrendWriteMsg.ConnectionRate> 7500000

EDC Gateway Trend Write Message

MOVE	
Source	CFG.CLx.ConnectionMult 0
Dest	TrendWriteMsg.TimeoutMultiplier <EQ0006_TrendWriteMsg.TimeoutMultiplier> 0

EDC Gateway Trend Write Message
TrendWriteMsg.EN_CC
<EQ0006_TrendWriteMsg.EN_CC>
U

EDC Gateway Trend Write Message

MSG	
Message Control	TrendWriteMsg <EQ0006_TrendWriteMsg> ...

EN
DN
ER

Trend Data Ring Buffer and Message

Write Message EN Performance & Diagnostics

12

EDC Gateway Trend Message Control Write Command
TrendMsgCtl.Write

EDC Gateway Trend Message Control Write Command 1-Shot
TrendMsgCtl.WriteONS
ONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TrendMsgCtl.Trans
PerfTrans	Status.TrendTxWrite.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	440
Input Par	TrendMsgCtl.Trans.TMR.ACC
Input Par	TrendMsgCtl.TransNum

Trend Data Ring Buffer and Message

Write Message Done

13

EDC Gateway Trend Message Control Write Command TrendMsgCtl.Write

EDC Gateway Trend Write Message TrendWriteMsg.DN <EQ0006_TrendWriteMsg.DN>

EDC Gateway Trend Message Control Write Command TrendMsgCtl.Write (U)

EDC Gateway Trend Message Control Write Command Done TrendMsgCtl.WriteOK (L)

EDC Gateway Trend Message Control Send Data TrendMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Trend Tx DN Status.Update.TrendTxOK (L)

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

EDC Gateway Trend Message Control Resend Data (Keep Alive) TrendMsgCtl.Resend

Equipment Data Collection - Line Equipment Status State = Green - Trend Keep Alive Tx DN Status.Update.TrendKaTxOK (L)

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TrendMsgCtl.Trans
PerfTrans	Status.TrendTxWrite.DN

EDC Gateway Trend Message Control Resend Data (Keep Alive) TrendMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TrendMsgCtl.Trans
PerfTrans	Status.TrendTxWrite.KA

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	441
Input Par	TrendMsgCtl.Trans.TMR.ACC
Input Par	TrendMsgCtl.TransNum

Trend Data Ring Buffer and Message

Write Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

14

EDC Gateway Trend Write Message TrendWriteMsg.EN <EQ0006_TrendWriteMsg.EN>

EDC Gateway Trend Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout 5000
Dest TrendMsgCtl.WriteTImeoutTMR.PRE 0

EDC Gateway Trend Message Control Message Timeout Timer

TON
Timer TrendMsgCtl.WriteTImeoutTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Trend Message Control Message Timeout Timer TrendMsgCtl.WriteTImeoutTMR.DN

EDC Gateway Trend Write Message TrendWriteMsg.TO <EQ0006_TrendWriteMsg.TO>

L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 444
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Write Message ER Performance & Diagnostics

15

EDC Gateway Trend Write Message TrendWriteMsg.ER <EQ0006_TrendWriteMsg.ER>

EDC Gateway Trend Message Control Write Error 1-shot TrendMsgCtl.WriteErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Trend Tx ER Status.Update.TrendTxER

L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans TrendMsgCtl.Trans
PerfTrans Status.TrendTxWrite.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 443
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Trend Data Ring Buffer and Message

Read/Write Message Error Retry

16

EDC Gateway Trend
Read Message
TrendReadMsg.ER
<EQ0006_TrendReadMsg.ER>

EDC Gateway Trend
Write Message
TrendWriteMsg.ER
<EQ0006_TrendWriteMsg.ER>

EDC Gateway Trend
Message Control
Message Retry Timer
TrendMsgCtl.RetryTMR.DN

EDC Gateway Trend
Message Control
Retry 1-Shot
TrendMsgCtl.RetryONS
ONS

EDC Gateway Trend
Message Control
Message Retry Timer
MOVE
Source CFG.MsgRetryTime
5000
Dest TrendMsgCtl.RetryTMR.PRE
0

EDC Gateway Trend
Message Control
Message Retry Timer
TON
Timer TrendMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Trend
Read Message
TrendReadMsg.ER
<EQ0006_TrendReadMsg.ER>

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 432
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

EDC Gateway Trend
Write Message
TrendWriteMsg.ER
<EQ0006_TrendWriteMsg.ER>

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 442
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

(End)

Version Data Load for Message

Equipment Version Data
Data Load

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	152
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

Message Control TM Transaction Trigger

ADD	
Source A	VersionMsgCtl.TransactionNumber 677
Source B	1
Dest	VersionMsgCtl.TransactionNumber 677

Message Control TM Transaction Trigger

LT	
Source A	VersionMsgCtl.TransactionNumber 677
Source B	1

Message Control TM Transaction Trigger

GT	
Source A	VersionMsgCtl.TransactionNumber 677
Source B	CFG.MsgTransNumRollover 9999

Message Control TM Transaction Trigger

MOVE	
Source	1
Dest	VersionMsgCtl.TransactionNumber 677

Version Data Load for Message

Load Message Header, Timestamp and Footer Data

3

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

MOVE
Source VersionMsgCtl.TransactionNumber 677
Dest VersionMsg.Header.TransNum 677

Equipment Data Version Message Format Header - Command Number

MOVE
Source 0
Dest VersionMsg.Header.CmdNum 0

Equipment Data Version Message Format Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source UserCfg.EQID 6
Dest VersionMsg.Header.EQID 6

Equipment Data Version Message Format Header - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest VersionMsg.Header.Spare 0

Equipment Data Version Message Format Transaction Time Stamp

COP
Source CLX.TransTime
Dest VersionMsg.TransTime
Length 1

Equipment Data Version Message Format Footer - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest VersionMsg.Footer.Spare 0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

MOVE
Source VersionMsgCtl.TransactionNumber 677
Dest VersionMsg.Footer.TransNum 677

Load Logix Controller & Driver Code Versions

4

Equipment Data Version Message Format Logix Controller Version

COP
Source CLX.Version
Dest VersionMsg.LogixVersion
Length 1

Equipment Data Version Message Format Equipment Driver Program Version

COP
Source Prog.Version
Dest VersionMsg.DriverVersion
Length 1

(End)

Logix and Code Version Data Message

Equipment Version Data
Version Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 154
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Reset Delay & Transaction Timer

2

Message Control Send Data
VersionMsgCtl.Send

Message Control Send Data 1-shot
VersionMsgCtl.SendONS
ONS

Message Control Message Transaction Timer
VersionMsgCtl.Trans.TMR
RES

Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR
Dest VersionMsgCtl.Trans.Last 14

Message Transaction Timer

3

Message Control Message Transaction Timer

MOVE
Source 999999999
Dest VersionMsgCtl.Trans.TMR.PRE 999999999

Message Control Message Transaction Timer

TON
Timer VersionMsgCtl.Trans.TMR
Preset 999999999
Accum 33891337
EN
DN

Logix and Code Version Data Message

Message Delay Timer
prevent Message from sending too quickly for Gateway
Gateway processes one Line per scan -- Equipment Message could take 500-1000 msec to process
any attempt to send faster will generate extra Read Message traffic until previous message has been processed

4

EDC Gateway Version
Write Message
VersionWriteMsg.EN
<EQ0006_VersionWriteMsg.EN>

Message Control
Message Delay Timer

MOVE	
Source	CFG.MsgDelayTime
	250
Dest	VersionMsgCtl.DelayTMR.PRE
	250

Message Control
Message Delay Timer

TON		
Timer	VersionMsgCtl.DelayTMR	EN
Preset	250	DN
Accum	250	

Logix and Code Version Data Message

Write Message

5

Message Control Send Data VersionMsgCtl.Send

Message Control Message Delay Timer VersionMsgCtl.DelayTMR.DN

EDC Gateway Version Write Message VersionWriteMsg.EN <EQ0006_VersionWriteMsg.EN>

EDC Gateway Version Write Message VersionWriteMsg.ER <EQ0006_VersionWriteMsg.ER>

Message Control Message Retry Timer VersionMsgCtl.RetryTMR.DN

Message Control Write Data VersionMsgCtl.Write (L)

Message Control Message Delay Timer VersionMsgCtl.DelayTMR (RES)

EDC Gateway Version Write Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest VersionWriteMsg.Path <EQ0006_VersionWriteMsg.Path>
Length 1

EDC Gateway Version Write Message

COP
Source CFG.MsgDest.VersionDest
Dest VersionWriteMsg.RemoteElement <EQ0006_VersionWriteMsg.RemoteElement>
Length 1

EDC Gateway Version Write Message

MOVE
Source CFG.VersionMSG.e 30
Dest VersionWriteMsg.REQ_LEN <EQ0006_VersionWriteMsg.REQ_LEN> 30

EDC Gateway Version Write Message

MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest VersionWriteMsg.UnconnectedTimeout <EQ0006_VersionWriteMsg.UnconnectedTimeout> 4000000

EDC Gateway Version Write Message

MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest VersionWriteMsg.ConnectionRate <EQ0006_VersionWriteMsg.ConnectionRate> 1000000

EDC Gateway Version Write Message

MOVE
Source CFG.CLx.ConnectionMult 0
Dest VersionWriteMsg.TimeoutMultiplier <EQ0006_VersionWriteMsg.TimeoutMultiplier> 0

EDC Gateway Version Write Message VersionWriteMsg.EN_CC <EQ0006_VersionWriteMsg.EN_CC> (U)

EDC Message Tx Data Version Receive Message

COP
Source VersionMsg
Dest EqTxData.Version[0] <EQ0006_TxData.Version[0]>
Length CFG.VersionMSG.n

EDC Gateway Version Write Message

MSG
Message Control VersionWriteMsg ... <EQ0006_VersionWriteMsg>
(EN) (DN) (ER)

Logix and Code Version Data Message

Message EN Performance & Diagnostics

6

Message Control Write Data: VersionMsgCtl.Write

Message Control Write Data 1-shot: VersionMsgCtl.WriteONS (ONS)

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	VersionMsgCtl.Trans
PerfTrans	Status.VersionTx.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	540
Input Par	VersionMsgCtl.Trans.TMR.ACC
Input Par	VersionMsgCtl.TransactionNumber

Message Done

7

Message Control Write Data: VersionMsgCtl.Write

EDC Gateway Version Write Message: VersionWriteMsg.DN <EQ0006_VersionWriteMsg.DN>

Message Control Send Data: VersionMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Version Tx DN: Status.Update.VersionTxOK (L)

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Message Control Send Data: VersionMsgCtl.Send (U)

Message Control Write Data: VersionMsgCtl.Write (U)

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	VersionMsgCtl.Trans
PerfTrans	Status.VersionTx.DN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	541
Input Par	VersionMsgCtl.Trans.TMR.ACC
Input Par	VersionMsgCtl.TransactionNumber

Logix and Code Version Data Message

Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

8

EDC Gateway Version Write Message
VersionWriteMsg.EN
<EQ0006_VersionWriteMsg.EN>

Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest VersionMsgCtl.TimeoutTMR.PRE
5000

Message Control Message Timeout Timer

TON
Timer VersionMsgCtl.TimeoutTMR
Preset 5000
Accum 0
EN
DN

Message Control Message Timeout Timer
VersionMsgCtl.TimeoutTMR.DN

EDC Gateway Version Write Message
VersionWriteMsg.TO
<EQ0006_VersionWriteMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 544
Input Par VersionMsgCtl.Trans.TMR.ACC
Input Par VersionMsgCtl.TransactionNumber

Message ER Performance & Diagnostics

9

EDC Gateway Version Write Message
VersionWriteMsg.ER
<EQ0006_VersionWriteMsg.ER>

Message Control Write Error 1-shot
VersionMsgCtl.WriteErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Version Tx ER
Status.Update.VersionTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans VersionMsgCtl.Trans
PerfTrans Status.VersionTx.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 543
Input Par VersionMsgCtl.Trans.TMR.ACC
Input Par VersionMsgCtl.TransactionNumber

Logix and Code Version Data Message

Message Error Retry

10

Message Control
Write Data
VersionMsgCtl.Write

EDC Gateway Version
Write Message
VersionWriteMsg.ER
<EQ0006_VersionWriteMsg.ER>

Message Control
Message Retry Timer

MOVE
Source CFG.MsgRetryTime
5000
Dest VersionMsgCtl.RetryTMR.PRE
0

Message Control
Message Retry Timer

TON
Timer VersionMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

Message Control
Message Retry Timer
VersionMsgCtl.RetryTMR.DN

Message Control
Retry 1-shot
VersionMsgCtl.RetryONS
ONS

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 542
Input Par VersionMsgCtl.Trans.TMR.ACC
Input Par VersionMsgCtl.TransactionNumber

(End)

Time Synchronization Data Load for Message

Equipment Time Sync Data
Time Sync Data Load

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	162
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get

Time Sync Message Control Get Data Increment Transaction Number
TimeSyncMsgCtl.GetInc
ONS

Time Sync Message Control TM Transaction Trigger

ADD	
Source A	TimeSyncMsgCtl.TransactionNumber 0
Source B	1
Dest	TimeSyncMsgCtl.TransactionNumber 0

Time Sync Message Control TM Transaction Trigger

LT	
Source A	TimeSyncMsgCtl.TransactionNumber 0
Source B	1

Time Sync Message Control TM Transaction Trigger

GT	
Source A	TimeSyncMsgCtl.TransactionNumber 0
Source B	CFG.MsgTransNumRollover 9999

Time Sync Message Control TM Transaction Trigger

MOVE	
Source	1
Dest	TimeSyncMsgCtl.TransactionNumber 0

Time Synchronization Data Load for Message

Sync Time with Gateway
if Values are within limits

3

Time Sync Message Control Get Data TimeSyncMsgCtl.Get

Time Sync Message Control Read Data OK TimeSyncMsgCtl.ReadOK

Time Sync Message Control Get Data TimeSyncMsgCtl.Get (U)

Time Sync Message Control Read Data OK TimeSyncMsgCtl.ReadOK (U)

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	2016
Test	EqRxData.TimeSync[0]
<EQ0006_RxData.TimeSync[0]>	0
High Limit	9999

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	1
Test	EqRxData.TimeSync[1]
<EQ0006_RxData.TimeSync[1]>	0
High Limit	12

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	1
Test	EqRxData.TimeSync[2]
<EQ0006_RxData.TimeSync[2]>	0
High Limit	31

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	0
Test	EqRxData.TimeSync[3]
<EQ0006_RxData.TimeSync[3]>	0
High Limit	23

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	0
Test	EqRxData.TimeSync[4]
<EQ0006_RxData.TimeSync[4]>	0
High Limit	59

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	0
Test	EqRxData.TimeSync[5]
<EQ0006_RxData.TimeSync[5]>	0
High Limit	59

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	0
Test	EqRxData.TimeSync[6]
<EQ0006_RxData.TimeSync[6]>	0
High Limit	999999

EDC Message Rx Data Time Sync Receive Message

SSV	
Class Name	WallClockTime
Instance Name	
Attribute Name	LocalDateTime
Source	EqRxData.TimeSync[0]
<EQ0006_RxData.TimeSync[0]>	0

(End)

Time Synchronization Data Message

Equipment Time Sync Data
Time Sync Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 164
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

Time Sync Message Control Message Transaction Timer

MOVE
Source 999999999
Dest TimeSyncMsgCtl.Trans.TMR.PRE 0

Time Sync Message Control Message Transaction Timer

TON
Timer TimeSyncMsgCtl.Trans.TMR
Preset 0
Accum 0
EN
DN

Get Data Initialize

3

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get

Time Sync Message Control Get Data 1-Shot
TimeSyncMsgCtl.GetONS
ONS

Time Sync Message Control Message Transaction Timer
TimeSyncMsgCtl.Trans.TMR
RES

Time Sync Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR
Dest TimeSyncMsgCtl.Trans.Last 0

Time Synchronization Data Message
Read Message

4

Time Sync Message Control Get Data TimeSyncMsgCtl.Get

EDC Gateway Time Sync Read Message TimeSyncReadMsg.EN <EQ0006_TimeSyncReadMsg.EN>

EDC Gateway Time Sync Read Message TimeSyncReadMsg.ER <EQ0006_TimeSyncReadMsg.ER>

Time Sync Message Control Message Retry Timer TimeSyncMsgCtl.RetryTMR.DN

Time Sync Message Control Read Data TimeSyncMsgCtl.Read (L)

EDC Gateway Time Sync Read Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest TimeSyncReadMsg.Path <EQ0006_TimeSyncReadMsg.Path>
Length 1

EDC Gateway Time Sync Read Message

MOVE
Source 7
Dest TimeSyncReadMsg.REQ_LEN <EQ0006_TimeSyncReadMsg.REQ_LEN>
7

EDC Gateway Time Sync Read Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest TimeSyncReadMsg.UnconnectedTimeout <EQ0006_TimeSyncReadMsg.UnconnectedTimeout>
30000000

EDC Gateway Time Sync Read Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest TimeSyncReadMsg.ConnectionRate <EQ0006_TimeSyncReadMsg.ConnectionRate>
7500000

EDC Gateway Time Sync Read Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest TimeSyncReadMsg.TimeoutMultiplier <EQ0006_TimeSyncReadMsg.TimeoutMultiplier>
0

EDC Gateway Time Sync Read Message TimeSyncReadMsg.EN_CC <EQ0006_TimeSyncReadMsg.EN_CC> (U)

EDC Gateway Time Sync Read Message

MSG
Message Control TimeSyncReadMsg ... <EQ0006_TimeSyncReadMsg>
EN
DN
ER

Time Synchronization Data Message

Message EN Performance & Diagnostics

5

Time Sync Message Control Read Data
TimeSyncMsgCtl.Read

Time Sync Message Control Read Data 1-shot
TimeSyncMsgCtl.ReadONS
ONS

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TimeSyncMsgCtl.Trans
PerfTrans Status.TimeSyncRx.EN

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 630
Input Par TimeSyncMsgCtl.Trans.TMR.ACC
Input Par TimeSyncMsgCtl.TransactionNumber

Message Done

6

Time Sync Message Control Read Data
TimeSyncMsgCtl.Read

EDC Gateway Time Sync Read Message
TimeSyncReadMsg.DN
<EQ0006_TimeSyncReadMsg.DN>

Time Sync Message Control Read Data
TimeSyncMsgCtl.Read
U

Time Sync Message Control Read Data OK
TimeSyncMsgCtl.ReadOK
L

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get

Equipment Data Collection - Line Equipment Status State = Green - Time Sync Rx DN
Status.Update.TimeSyncRxOK
L

Update Status State for Transactions
aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate
Status Status

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TimeSyncMsgCtl.Trans
PerfTrans Status.TimeSyncRx.DN

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 631
Input Par TimeSyncMsgCtl.Trans.TMR.ACC
Input Par TimeSyncMsgCtl.TransactionNumber

Time Synchronization Data Message

Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

7

EDC Gateway Time Sync Read Message TimeSyncReadMsg.EN <EQ0006_TimeSyncReadMsg.EN>

Time Sync Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout 5000
Dest TimeSyncMsgCtl.TimeoutTMR.PRE 0

Time Sync Message Control Message Timeout Timer

TON
Timer TimeSyncMsgCtl.TimeoutTMR
Preset 0
Accum 0
EN
DN

Time Sync Message Control Message Timeout Timer TimeSyncMsgCtl.TimeoutTMR.DN

EDC Gateway Time Sync Read Message TimeSyncReadMsg.TO <EQ0006_TimeSyncReadMsg.TO>

L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 634
Input Par TimeSyncMsgCtl.Trans.TMR.ACC
Input Par TimeSyncMsgCtl.TransactionNumber

Message ER Performance & Diagnostics

8

EDC Gateway Time Sync Read Message TimeSyncReadMsg.ER <EQ0006_TimeSyncReadMsg.ER>

Time Sync Message Control Read Error 1-shot TimeSyncMsgCtl.ReadErONS

ONS

Equipment Data Collection - Line Equipment Status State = Red - Time Sync Rx ER Status.Update.TimeSyncRxER

L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans TimeSyncMsgCtl.Trans
PerfTrans Status.TimeSyncRx.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 633
Input Par TimeSyncMsgCtl.Trans.TMR.ACC
Input Par TimeSyncMsgCtl.TransactionNumber

Time Synchronization Data Message

Message Error Retry

9

Time Sync Message Control Read Data
TimeSyncMsgCtl.Read

EDC Gateway Time Sync Read Message
TimeSyncReadMsg.ER
<EQ0006_TimeSyncReadMsg.ER>

Time Sync Message Control Message Retry Timer

MOVE
Source CFG.MsgRetryTime
5000
Dest TimeSyncMsgCtl.RetryTMR.PRE
0

Time Sync Message Control Message Retry Timer

TON
Timer TimeSyncMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

Time Sync Message Control Message Retry Timer
TimeSyncMsgCtl.RetryTMR.DN

Time Sync Message Control Retry 1-shot
TimeSyncMsgCtl.RetryONS
ONS

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 632
Input Par TimeSyncMsgCtl.Trans.TMR.ACC
Input Par TimeSyncMsgCtl.TransactionNumber

(End)

Collect Cycle Data

Equipment Cycle Data
Collect Data

0

Cycle Index
SBR
Input Par CYi

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned
MOVE
Source 171
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)
ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Exit Routine if Index is out of range

2

Cycle Index
LT
Source A CYi 4
Source B 0

Cycle Index
GT
Source A CYi 4
Source B CFG.Cycle.mi 4

lbl_Exit
JMP

Copy Cycle Arrays to local tags

3

Cycle Collect Control
COP
Source CycleCfg[CYi].CTL
Dest CycleCollectCtl
Length 1

Equipment Data Cycle Message Data
COP
Source CycleCfg[CYi].Data
Dest CycleCollect
Length 1

Collect Cycle Data

Cycle Initialize
Load Initial Step Number into StepPrev to begin Cycle Timing

4

Cycle Collect Control Cycle Start Sampling
CycleCollectCtl.Start

Cycle Collect Control Cycle Start 1-Shot
CycleCollectCtl.StartONS
ONS

Cycle Collect Control Cycle Init Sampling
CycleCollectCtl.Init

Cycle Collect Control Cycle Done Sampling
CycleCollectCtl.Done
U

Cycle Collect Control Cycle Save
CycleCollectCtl.Loaded
U

Equipment Data Cycle Message Data
FLL
Source 0
Dest CycleCollect
Length 1

Cycle Collect Control Step Duration Timer
CycleCollectCtl.DurationTMR
RES

Cycle Collect Control Duration Prev Time
CLR
Dest CycleCollectCtl.DurationPrev
36010

Cycle Collect Control Step Previous Value
MOVE
Source CycleCollectCtl.Data
31
Dest CycleCollectCtl.StepPrev
31

Cycle Collect Control Cycle Array Index
CLR
Dest CycleCollectCtl.ARRi
30

Cycle Init
JobStatus and FaultCode are updates on Start and Send

5

Cycle Collect Control Cycle Init Sampling
CycleCollectCtl.Init

Equipment Data Cycle Message Data Sequence Number
MOVE
Source CYi
4
Dest CycleCollect.SeqNum
4

Equipment Data Cycle Message Data Step Numbers Used (1-120)
MOVE
Source CFG.CycData.n
120
Dest CycleCollect.StepCount
120

Equipment Data Cycle Message Data Job Status
MOVE
Source CycleMsg.Data.JobStatus
0
Dest CycleCollect.JobStatus
1

Equipment Data Cycle Message Data Fault Code
MOVE
Source CycleMsg.Data.FaultCode
0
Dest CycleCollect.FaultCode
0

Collect Cycle Data

Cycle Vehicle Data

6

Cycle Collect
Control Cycler Load
Vehicle Data
CycleCollectCtl.LoadVeh

Equipment Data
Cycle Message Data
Program Number from
SLCP

MOVE	
Source	CycleMsg.Data.AviPrgNum 0
Dest	CycleCollect.AviPrgNum 0

Equipment Data
Cycle Message Data
Program Number
Executed

MOVE	
Source	CycleMsg.Data.ActPrgNum 0
Dest	CycleCollect.ActPrgNum 0

Equipment Data
Cycle Message Data
Assembly Body Number

MOVE	
Source	CycleMsg.Data.BodyNum 0
Dest	CycleCollect.BodyNum 0

Equipment Data
Cycle Message Data
Assembly Sequence
Number

MOVE	
Source	CycleMsg.Data.AssyNum 0
Dest	CycleCollect.AssyNum 0

Equipment Data
Cycle Message Data
VIN Number

COP	
Source	CycleMsg.Data.VinNum[0]
Dest	CycleCollect.VinNum[0]
Length	20

Equipment Data
Cycle Message Data
Spare

MOVE	
Source	CycleMsg.Data.Spare1 0
Dest	CycleCollect.Spare1 0

Equipment Data
Cycle Message Data
Spare

MOVE	
Source	CycleMsg.Data.Spare2 0
Dest	CycleCollect.Spare2 0

Collect Cycle Data

Cycle Sample TImer
Use free-running timer and samle interval to avoid accumulating Timer overruns throughout test.
Take a data sample whenever the Step Number changes

Stop suspends timiing but doesn't suspend sampling
Send stores final step and duration if array isn't full

7

Cycle Collect Control Cycle Start Sampling
CycleCollectCtl.Start

Cycle Collect Control Cycle Send
CycleCollectCtl.Send

Cycle Collect Control Cycle Done Sampling
CycleCollectCtl.Done

Cycle Collect Control Cycle Stop Sampling
CycleCollectCtl.Stop

Cycle Collect Control Step Duration Timer

MOVE
Source 99999999
Dest CycleCollectCtl.DurationTMR.PRE 99999999

Cycle Collect Control Step Duration Timer

RTO
Timer CycleCollectCtl.DurationTMR
Preset 99999999
Accum 36484
EN
DN

Cycle Collect Control Cycle Send
CycleCollectCtl.Send

Cycle Collect Control Cycle Send ONS
CycleCollectCtl.SendONS
ONS

Cycle Collect Control Step Data

NE
Source A CycleCollectCtl.Data 31
Source B CycleCollectCtl.StepPrev 31

Cycle Collect Control Duration Accumulation

SUB
Source A CycleCollectCtl.DurationTMR.ACC 36484
Source B CycleCollectCtl.DurationPrev 36010
Dest CycleCollectCtl.DurationAcc 1200

Cycle Collect Control Cycle Sample
CycleCollectCtl.Sample

Cycle Collect Control Cycle Send
CycleCollectCtl.Send

Collect Cycle Data

Cycle Sample
store previous Step Number and Accum after the Step Number changes

8

Cycle Collect
Control Cycle Sample
CycleCollectCtl.Sample

Equipment Data
Cycle Message Data
Step Number
(0-65,536)

MOVE
Source CycleCollectCtl.StepPrev 31
Dest CycleCollect.Step[CycleCollectCtl.ARRi] 0

Equipment Data
Cycle Message Data
Duration (0-655.36
seconds)

DIV
Source A CycleCollectCtl.DurationAcc 1200
Source B 10
Dest CycleCollect.Duration[CycleCollectCtl.ARRi] 0

Cycle Collect
Control Cycle Array
Index

ADD
Source A CycleCollectCtl.ARRi 30
Source B 1
Dest CycleCollectCtl.ARRi 30

Cycle Collect
Control Cycle Array
Index

LT
Source A CycleCollectCtl.ARRi 30
Source B 0

Cycle Collect
Control Cycle Array
Index

GT
Source A CycleCollectCtl.ARRi 30
Source B CFG.CycData.mi 119

Cycle Collect
Control Cycle Done
Sampling
CycleCollectCtl.Done
L

Cycle Collect
Control Step
Previous Value

MOVE
Source CycleCollectCtl.Data 31
Dest CycleCollectCtl.StepPrev 31

Cycle Collect
Control Duration
Prev Time

ADD
Source A CycleCollectCtl.DurationPrev 36010
Source B CycleCollectCtl.DurationAcc 1200
Dest CycleCollectCtl.DurationPrev 36010

Collect Cycle Data

Cycle Samples Done
Update Status/Fault and Job Data
Load RIng Buffer

9

Cycle Collect Control Cycle Send CycleCollectCtl.Send

Cycle Collect Control Cycle Save CycleCollectCtl.Loaded /

Cycle Collect Control Cycle Send CycleCollectCtl.Send

Equipment Data Cycle Message Data Job Status

MOVE
Source CycleMsg.Data.JobStatus 0
Dest CycleCollect.JobStatus 1

Equipment Data Cycle Message Data Fault Code

MOVE
Source CycleMsg.Data.FaultCode 0
Dest CycleCollect.FaultCode 0

Equipment Data Cycle Message Data Step Numbers Used (1-120)

MOVE
Source CycleCollectCtl.ARRi 30
Dest CycleCollect.StepCount 120

Load Cycle Data record to Ring Buffer

JSR
Routine Name _172_CycleDataLoadRB

Cycle Collect Control Cycle Save CycleCollectCtl.Loaded (L)

Cycle Collect Control Cycle Start Sampling CycleCollectCtl.Start (U)

Cycle Collect Control Cycle Send CycleCollectCtl.Send (U)

Copy local tags to Cycle Array

10

Cycle Data Collection Cycle Control

COP
Source CycleCollectCtl
Dest CycleCfg[CYi].CTL
Length 1

Cycle Data Collection Cycle Data

COP
Source CycleCollect
Dest CycleCfg[CYi].Data
Length 1

Exit Routine

11

lbl_Exit
LBL

NOP

(End)

Load Cycle Data record to Ring Buffer

Equipment Cycle Data
Ring Buffer Load

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 172
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

EDC Ring Buffer Control Transaction Number

ADD
Source A CycleRBCtl.TransactionNumber 0
Source B 1
Dest CycleRBCtl.TransactionNumber 0

EDC Ring Buffer Control Transaction Number

LT
Source A CycleRBCtl.TransactionNumber 0
Source B 1

EDC Ring Buffer Control Transaction Number

GT
Source A CycleRBCtl.TransactionNumber 0
Source B CFG.MsgTransNumRollover 9999

EDC Ring Buffer Control Transaction Number

MOVE
Source 1
Dest CycleRBCtl.TransactionNumber 0

Load Cycle Data record to Ring Buffer

Load Message Header, Timestamp, Cycle Data and Footer Data

3

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

MOVE	
Source	CycleRBCtl.TransactionNumber 0
Dest	CycleMsg.Header.TransNum 0

Equipment Data Cycle Message Header - Command Number

MOVE	
Source	0
Dest	CycleMsg.Header.CmdNum 0

Equipment Data Cycle Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE	
Source	UserCfg.EQID 6
Dest	CycleMsg.Header.EQID 0

Equipment Data Cycle Message Header - Spare (fills unused portion of DINT)

MOVE	
Source	0
Dest	CycleMsg.Header.Spare 0

Equipment Data Cycle Message Transaction Time Stamp

COP	
Source	CLX.TransTime
Dest	CycleMsg.TransTime
Length	1

Equipment Data Cycle Message Data - Cycle

COP	
Source	CycleCollect
Dest	CycleMsg.Data
Length	1

Equipment Data Cycle Message Footer - Spare (fills unused portion of DINT)

MOVE	
Source	0
Dest	CycleMsg.Footer.Spare 0

Equipment Data Cycle Message Footer - Transaction Number (must match Footer)

MOVE	
Source	CycleRBCtl.TransactionNumber 0
Dest	CycleMsg.Footer.TransNum 0

Load Cycle Data record to Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

4

EDC Ring Buffer Control Next Write Record

LT
Source A: CycleRBCtl.WritePtr 0
Source B: 0

EDC Ring Buffer Control Next Write Record

GT
Source A: CycleRBCtl.WritePtr 0
Source B: CFG.CycleRB.mi 19

EDC Ring Buffer Control Diagnostic - Manual Reset

CycleRBCtl.WritePtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source: CycleRBCtl.WritePtr 0
Dest: CycleRBCtl.WritePtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: CycleRBCtl.WritePtr 0
Dest: Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: CycleRBCtl.ReadPtr 19
Dest: Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name: _902_TransLogSBR
Input Par: 713
Input Par: 0
Input Par: CycleMsg.Header.TransNum

EDC Ring Buffer Control Next Write Record

MOVE
Source: 0
Dest: CycleRBCtl.WritePtr 0

Load Cycle Data record to Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

5

EDC Ring Buffer Control Next Write Record

EQ
Source A CycleRBCtl.WritePtr 0
Source B CycleRBCtl.ReadPtr 19

EDC Ring Buffer Control Buffer Full - Block Writes
CycleRBCtl.FullBlockWrite

EDC Ring Buffer Control Diagnostic - Manual Reset
CycleRBCtl.WriteBlockErr
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A CycleRBCtl.WriteBlockErrCnt 0
Source B 1
Dest CycleRBCtl.WriteBlockErrCnt 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.WritePtr 0
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.ReadPtr 19
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 714
Input Par 0
Input Par CycleMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans TempMsgTrans
PerfTrans Status.CycleRx.ER

lbl_Exit
JMP

Load Cycle Data record to Ring Buffer

RB Write -- Load Message into Ring Buffer

6

Equipment Data Cycle Message

COP
Source CycleMsg
Dest CycleRB[CycleRBCtl.WritePtr]
Length 1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.WritePtr
0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.ReadPtr
19
Dest Trans.Ctl.Opt[2]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 711
Input Par 0
Input Par CycleMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TempMsgTrans
PerfTrans Status.CycleRx.DN

RB Write -- Increment Write Pointer

7

EDC Ring Buffer Control Next Write Record

ADD
Source A CycleRBCtl.WritePtr
0
Source B 1
Dest CycleRBCtl.WritePtr
0

EDC Ring Buffer Control Next Write Record

LT
Source A CycleRBCtl.WritePtr
0
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A CycleRBCtl.WritePtr
0
Source B CFG.CycleRB.mi
19

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest CycleRBCtl.WritePtr
0

Exit Routine

8

lbl_Exit
LBL

NOP

Load Cycle Data record to Ring Buffer

(End)

Load Read Records from RingBuffer to Gateway Message

Equipment Cycle Data
Ring Buffer Unload to Gateway Message

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 173
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Load Read Records from RingBuffer to Gateway Message

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Next Read Record

LT
Source A: CycleRBCtl.ReadPtr 19
Source B: 0

EDC Ring Buffer Control Next Read Record

GT
Source A: CycleRBCtl.ReadPtr 19
Source B: CFG.CycleRB.mi 19

EDC Ring Buffer Control Diagnostic - Manual Reset

CycleRBCtl.ReadPtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source: CycleRBCtl.ReadPtr 19
Dest: CycleRBCtl.ReadPtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: CycleRBCtl.WritePtr 0
Dest: Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: CycleRBCtl.ReadPtr 19
Dest: Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name: _902_TransLogSBR
Input Par: 723
Input Par: 0
Input Par: CycleRBCtl.TransactionNumber

EDC Ring Buffer Control Next Write Record

MOVE
Source: 0
Dest: CycleRBCtl.WritePtr 0

EDC Ring Buffer Control Next Read Record

MOVE
Source: CFG.CycleRB.mi 19
Dest: CycleRBCtl.ReadPtr 19

Load Read Records from RingBuffer to Gateway Message

Next Read Location
ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

ADD
Source A CycleRBCtl.ReadPtr 19
Source B 1
Dest CycleRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Next Read Record

GT
Source A CycleRBCtl.ReadNextPtr 0
Source B CFG.CycleRB.mi 19

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest CycleRBCtl.ReadNextPtr 0

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Next Read Record

EQ
Source A CycleRBCtl.ReadPtr 19
Source B CycleRBCtl.WritePtr 0

EDC Ring Buffer Control Buffer Full
CycleRBCtl.Full

EDC Ring Buffer Control Read Buffer Record in Progress
CycleRBCtl.ReadInProgress

EDC Ring Buffer Control Next Read Record

EQ
Source A CycleRBCtl.ReadNextPtr 0
Source B CycleRBCtl.WritePtr 0

EDC Ring Buffer Control Buffer Empty
CycleRBCtl.Empty

Load Read Records from RingBuffer to Gateway Message

5

Buffer Count

EDC Ring Buffer Control Read Buffer Record in Progress
CycleRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
CycleRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record

GT
Source A CycleRBCtl.WritePtr 0
Source B CycleRBCtl.ReadPtr 19

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest CycleRBCtl.RecordCount 0
Expression CycleRBCtl.WritePtr-CycleRBCtl.ReadPtr

EDC Ring Buffer Control Next Write Record

LT
Source A CycleRBCtl.WritePtr 0
Source B CycleRBCtl.ReadPtr 19

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest CycleRBCtl.RecordCount 0
Expression CFG.CycleRB.n-(CycleRBCtl.ReadPtr-CycleRBCtl.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
CycleRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
CycleRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record

GE
Source A CycleRBCtl.WritePtr 0
Source B CycleRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest CycleRBCtl.RecordCount 0
Expression CycleRBCtl.WritePtr-CycleRBCtl.ReadNextPtr

EDC Ring Buffer Control Next Write Record

LT
Source A CycleRBCtl.WritePtr 0
Source B CycleRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest CycleRBCtl.RecordCount 0
Expression CFG.CycleRB.n-(CycleRBCtl.ReadNextPtr-CycleRBCtl.WritePtr)

EDC Ring Buffer Control Ring Buffer Record Count

GT
Source A CycleRBCtl.RecordCount 0
Source B CycleRBCtl.RecordCountMax 0

EDC Ring Buffer Control Ring Buffer Record Count Max

MOVE
Source CycleRBCtl.RecordCount 0
Dest CycleRBCtl.RecordCountMax 0

Load Read Records from RingBuffer to Gateway Message

Read Next
Ready for next data to Gateway Message

6

EDC Ring Buffer Control Buffer Empty CycleRBCtl.Empty
EDC Ring Buffer Control Read Buffer Record in Progress CycleRBCtl.ReadInProgress
Ring Buffer Message Control Send Data CycleMsgCtl.Send
Ring Buffer Message Control Resend Data (Keep Alive) CycleMsgCtl.Resend
EDC Ring Buffer Control Read Next Buffer Record 1-shot CycleRBCtl.ReadNextONS
EDC Ring Buffer Control Read Next Buffer Record CycleRBCtl.ReadNext
EDC Ring Buffer Control Read Buffer Record in Progress CycleRBCtl.ReadInProgress
EDC Ring Buffer Control Read Buffer Record Complete CycleRBCtl.ReadComplete

Read Next
Increment Read Pointer

7

EDC Ring Buffer Control Read Next Buffer Record CycleRBCtl.ReadNext

ADD
Source A CycleRBCtl.ReadPtr 19
Source B 1
Dest CycleRBCtl.ReadPtr 19

EDC Ring Buffer Control Next Read Record

LT
Source A CycleRBCtl.ReadPtr 19
Source B 0

EDC Ring Buffer Control Next Read Record

GT
Source A CycleRBCtl.ReadPtr 19
Source B CFG.CycleRB.mi 19

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest CycleRBCtl.ReadPtr 19

Read Next
Load Read Record

8

EDC Ring Buffer Control Read Next Buffer Record CycleRBCtl.ReadNext

EDC Message Tx Data Cycle Receive Message

COP
Source CycleRB[CycleRBCtl.ReadPtr]
Dest EqTxData.Cycle[0] <EQ0006_TxData.Cycle[0]>
Length CFG.CycleMSG.e

Ring Buffer Message Control Transaction Number

MOVE
Source CycleRB[CycleRBCtl.ReadPtr].Header.TransNum 0
Dest CycleMsgCtl.TransNum 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.WritePtr 0
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.ReadPtr 19
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 721
Input Par 0
Input Par CycleMsgCtl.TransNum

Load Read Records from RingBuffer to Gateway Message

Read Next
Set Data Ready Handshake

9

EDC Ring Buffer Control Read Next Buffer Record
CycleRBCtl.ReadNext

Ring Buffer Message Control Send Data
CycleMsgCtl.Send (L)

Ring Buffer Message Control Read Command
CycleMsgCtl.Read (U)

Ring Buffer Message Control Read Buffer Ready
CycleMsgCtl.ReadOK (U)

Ring Buffer Message Control Write Command
CycleMsgCtl.Write (U)

Ring Buffer Message Control Write Command Done
CycleMsgCtl.WriteOK (U)

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

10

Ring Buffer Message Control Send Data
CycleMsgCtl.Send

Ring Buffer Message Control Write Command Done
CycleMsgCtl.WriteOK

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend

Ring Buffer Message Control Send Data
CycleMsgCtl.Send (U)

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend (U)

EDC Ring Buffer Control Read Buffer Record Complete
CycleRBCtl.ReadComplete (L)

Read Complete Delay
Allow Gateway processing time before attempting next record send.

11

EDC Ring Buffer Control Read Buffer Record Complete
CycleRBCtl.ReadComplete

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source 100
Dest CycleRBCtl.ReadCompleteDelayTMR.PRE
0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer CycleRBCtl.ReadCompleteDelayTMR
Preset 0
Accum 0
EN
DN

EDC Ring Buffer Control Read Complete Delay Timer
CycleRBCtl.ReadCompleteDelayTMR.DN

EDC Ring Buffer Control Read Buffer Record Complete Delay
CycleRBCtl.ReadCompleteDelay ( )

EDC Ring Buffer Control Read Buffer Record in Progress
CycleRBCtl.ReadInProgress (U)

Load Read Records from RingBuffer to Gateway Message

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

12

EDC Ring Buffer Control Buffer Empty
CycleRBCtl.Empty

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest CycleRBCtl.NotEmptyTMR.PRE 0

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer CycleRBCtl.NotEmptyTMR
Preset 0
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A CycleRBCtl.NotEmptyTMR.ACC 0
Source B CycleRBCtl.NotEmptyMax 0

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source CycleRBCtl.NotEmptyTMR.ACC 0
Dest CycleRBCtl.NotEmptyMax 0

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

13

EDC Ring Buffer Control Buffer Full
CycleRBCtl.Full

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
CycleRBCtl.NotEmptyTMR.DN

EDC Ring Buffer Control Read Buffer Alarm
CycleRBCtl.BufferAlarm

(End)

Trend Data Ring Buffer and Message

Equipment Cycle Data
Buffered Fault Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	174
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

Ring Buffer Message Control Message Transaction Timer

MOVE	
Source	999999999
Dest	CycleMsgCtl.Trans.TMR.PRE 0

Ring Buffer Message Control Message Transaction Timer

TON		
Timer	CycleMsgCtl.Trans.TMR	EN
Preset	0	DN
Accum	0	

Trend Data Ring Buffer and Message

Read Message Delay Timer
If driver sends data too fast, it will generate extra Read Message traffic until previous message has been processed
Optimization uses the ReadOk and ReadNotOk counts to tune the Delay Time to the Gateway response

3

EDC Gateway Cycle Write Message
CycleWriteMsg.EN
<EQ0006_CycleWriteMsg.EN>

EDC Gateway Cycle Read Message
CycleReadMsg.EN
<EQ0006_CycleReadMsg.EN>

Ring Buffer Message Control Read Message Ok Count

GE	
Source A	CycleMsgCtl.ReadOkCount 0
Source B	CfgRB_ReadOkLimit <CfgRB.ReadOkLimit> 5

Ring Buffer Message Control Message Delay Timer

SUB	
Source A	CycleMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadOkSub <CfgRB.ReadOkSub> 10
Dest	CycleMsgCtl.DelayTMR.PRE 0

Ring Buffer Message Control Message Delay Timer

LE	
Source A	CycleMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100

Ring Buffer Message Control Message Delay Timer

MOVE	
Source	CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100
Dest	CycleMsgCtl.DelayTMR.PRE 0

Ring Buffer Message Control Read Message Not Ok Count

GE	
Source A	CycleMsgCtl.ReadNotOkCount 0
Source B	CfgRB_ReadNotOkLimit <CfgRB.ReadNotOkLimit> 2

Ring Buffer Message Control Message Delay Timer

ADD	
Source A	CycleMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadNotOkAdd <CfgRB.ReadNotOkAdd> 20
Dest	CycleMsgCtl.DelayTMR.PRE 0

Ring Buffer Message Control Message Delay Timer

GT	
Source A	CycleMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500

Ring Buffer Message Control Message Delay Timer

MOVE	
Source	CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500
Dest	CycleMsgCtl.DelayTMR.PRE 0

Ring Buffer Message Control Read Message Ok Count

CLR	
Dest	CycleMsgCtl.ReadOkCount 0

Ring Buffer Message Control Read Message Not Ok Count

CLR	
Dest	CycleMsgCtl.ReadNotOkCount 0

Ring Buffer Message Control Message Delay Timer

TON		
Timer	CycleMsgCtl.DelayTMR	EN
Preset	0	DN
Accum	0	

Trend Data Ring Buffer and Message

Message Resend Timer
Keep Alive - resends last data without increment of Transaction Number

4

Ring Buffer Message Control Send Data
CycleMsgCtl.Send
/

EDC Gateway Cycle Write Message
CycleWriteMsg.EN
<EQ0006_CycleWriteMsg.EN>
/

EDC Gateway Cycle Read Message
CycleReadMsg.EN
<EQ0006_CycleReadMsg.EN>
/

Ring Buffer Message Control Message Resend Timer

MOVE
Source CFG.BuffMsgResendTime
301000
Dest CycleMsgCtl.ResendTMR.PRE
0

Ring Buffer Message Control Message Resend Timer

TON
Timer CycleMsgCtl.ResendTMR
Preset 0
Accum 0
EN
DN

Ring Buffer Message Control Message Resend Timer
CycleMsgCtl.ResendTMR.DN

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend
L

Ring Buffer Message Control Message Resend Timer
CycleMsgCtl.ResendTMR
RES

Ring Buffer Message Control Read Command
CycleMsgCtl.Read
U

Ring Buffer Message Control Read Buffer Ready
CycleMsgCtl.ReadOK
U

Ring Buffer Message Control Write Command
CycleMsgCtl.Write
U

Ring Buffer Message Control Write Command Done
CycleMsgCtl.WriteOK
U

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 722
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Send Data Initialize

5

Ring Buffer Message Control Send Data
CycleMsgCtl.Send

Ring Buffer Message Control Send Data 1-Shot
CycleMsgCtl.SendONS
ONS

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend

Ring Buffer Message Control Message Transaction Timer
CycleMsgCtl.Trans.TMR
RES

Ring Buffer Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR
Dest CycleMsgCtl.Trans.Last
0

Trend Data Ring Buffer and Message

Read Message
Check Gateway Buffer for Empty Record
(Gateway clears Buffer after processing Record)

6

Ring Buffer Message Control Send Data CycleMsgCtl.Send
Ring Buffer Message Control Read Buffer Ready CycleMsgCtl.ReadOK
Ring Buffer Message Control Message Delay Timer CycleMsgCtl.DelayTMR.DN
EDC Gateway Cycle Read Message CycleReadMsg.EN <EQ0006_CycleReadMsg.EN>
EDC Gateway Cycle Write Message CycleWriteMsg.EN <EQ0006_CycleWriteMsg.EN>
EDC Gateway Cycle Read Message CycleReadMsg.ER <EQ0006_CycleReadMsg.ER>
Ring Buffer Message Control Resend Data (Keep Alive) CycleMsgCtl.Resend
Ring Buffer Message Control Message Retry Timer CycleMsgCtl.RetryTMR.DN

Ring Buffer Message Control Read Command CycleMsgCtl.Read (L)
Ring Buffer Message Control Read Command 1-Shot CycleMsgCtl.ReadONS

EDC Gateway Cycle Read Message
COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest CycleReadMsg.Path <EQ0006_CycleReadMsg.Path>
Length 1

EDC Gateway Cycle Read Message
COP
Source CFG.MsgDest.CycleDest
Dest CycleReadMsg.RemoteElement <EQ0006_CycleReadMsg.RemoteElement>
Length 1

EDC Gateway Cycle Read Message
MOVE
Source CFG.CycleMSG.e 134
Dest CycleReadMsg.REQ_LEN <EQ0006_CycleReadMsg.REQ_LEN> 134

EDC Gateway Cycle Read Message
MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest CycleReadMsg.UnconnectedTimeout <EQ0006_CycleReadMsg.UnconnectedTimeout> 30000000

EDC Gateway Cycle Read Message
MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest CycleReadMsg.ConnectionRate <EQ0006_CycleReadMsg.ConnectionRate> 7500000

EDC Gateway Cycle Read Message
MOVE
Source CFG.CLx.ConnectionMult 0
Dest CycleReadMsg.TimeoutMultiplier <EQ0006_CycleReadMsg.TimeoutMultiplier> 0

EDC Gateway Cycle Read Message CycleReadMsg.EN_CC <EQ0006_CycleReadMsg.EN_CC> (U)

EDC Gateway Cycle Read Message
MSG
Message Control CycleReadMsg <EQ0006_CycleReadMsg> (...)
EN
DN
ER

Trend Data Ring Buffer and Message

Read Message EN Performance & Diagnostics

7

Ring Buffer Message Control Read Command 1-Shot

CycleMsgCtl.ReadONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans [...]
PerfMsgTrans	CycleMsgCtl.Trans
PerfTrans	Status.CycleTxRead.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	730
Input Par	CycleMsgCtl.Trans.TMR.ACC
Input Par	CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message

Read Message OK
Gateway Buffer has Empty Record
(Gateway clears Buffer after processing Record)

8

Ring Buffer Message Control Read Command CycleMsgCtl.Read

EDC Gateway Cycle Read Message CycleReadMsg.DN <EQ0006_CycleReadMsg.DN>

Ring Buffer Message Control Read Command CycleMsgCtl.Read (U)

EDC Message Rx Data Cycle Receive Message

EQ
Source A EqRxData.Cycle[0]
<EQ0006_RxData.Cycle[0]>
0
Source B 0

EDC Message Rx Data Cycle Receive Message

EQ
Source A EqRxData.Cycle[CFG.CycleMSG.ei]
<EQ0006_RxData.Cycle[CFG.CycleMSG.ei]>
0
Source B 0

Ring Buffer Message Control Read Buffer Ready CycleMsgCtl.ReadOK (L)

EDC Ring Buffer Control Ring Buffer Record Count

GE
Source A CycleRBCtl.RecordCount
0
Source B 2

Ring Buffer Message Control Read Message Ok Count

ADD
Source A CycleMsgCtl.ReadOkCount
0
Source B 1
Dest CycleMsgCtl.ReadOkCount
0

Ring Buffer Message Control Read Buffer Ready CycleMsgCtl.ReadOK (/)

Ring Buffer Message Control Read Message Not Ok Count

ADD
Source A CycleMsgCtl.ReadNotOkCount
0
Source B 1
Dest CycleMsgCtl.ReadNotOkCount
0

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans (...)
PerfMsgTrans CycleMsgCtl.Trans
PerfTrans Status.CycleTxRead.DN

Ring Buffer Message Control Resend Data (Keep Alive) CycleMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans (...)
PerfMsgTrans CycleMsgCtl.Trans
PerfTrans Status.CycleTxRead.KA

Trend Data Ring Buffer and Message

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source EqRxData.Cycle[0]
<EQ0006_RxData.Cycle[0]>
0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source EqRxData.Cycle[CFG.CycleMSG.ei]
<EQ0006_RxData.Cycle[CFG.CycleMSG.ei]>
0
Dest Trans.Ctl.Opt[2]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 731
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Read Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

9

EDC Gateway Cycle
Read Message
CycleReadMsg.EN
<EQ0006_CycleReadMsg.EN>

Ring Buffer Message
Control Message
Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest CycleMsgCtl.ReadTimeoutTMR.PRE
0

Ring Buffer Message
Control Message
Timeout Timer

TON
Timer CycleMsgCtl.ReadTimeoutTMR
Preset 0
Accum 0
EN
DN

Ring Buffer Message
Control Message
Timeout Timer
CycleMsgCtl.ReadTimeoutTMR.DN

EDC Gateway Cycle
Read Message
CycleReadMsg.TO
<EQ0006_CycleReadMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 734
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message
Read Message ER Performance & Diagnostics

10

EDC Gateway Cycle Read Message
CycleReadMsg.ER
<EQ0006_CycleReadMsg.ER>

Ring Buffer Message Control Read Error 1-shot
CycleMsgCtl.ReadErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Cycle Tx ER
Status.Update.CycleTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	CycleMsgCtl.Trans
PerfTrans	Status.CycleTxRead.ER

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	733
Input Par	CycleMsgCtl.Trans.TMR.ACC
Input Par	CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message

Write Message
Write Data is loaded by Ring Buffer routine

11

Ring Buffer Message Control Send Data CycleMsgCtl.Send
Ring Buffer Message Control Read Buffer Ready CycleMsgCtl.ReadOK
Ring Buffer Message Control Write Command Done CycleMsgCtl.WriteOK
EDC Gateway Cycle Read Message CycleReadMsg.EN <EQ0006_CycleReadMsg.EN>
EDC Gateway Cycle Write Message CycleWriteMsg.EN <EQ0006_CycleWriteMsg.EN>
EDC Gateway Cycle Write Message CycleWriteMsg.ER <EQ0006_CycleWriteMsg.ER>
Ring Buffer Message Control Resend Data (Keep Alive) CycleMsgCtl.Resend
Ring Buffer Message Control Message Retry Timer CycleMsgCtl.RetryTMR.DN

Ring Buffer Message Control Write Command CycleMsgCtl.Write (L)

EDC Gateway Cycle Write Message
COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest CycleWriteMsg.Path <EQ0006_CycleWriteMsg.Path>
Length 1

EDC Gateway Cycle Write Message
COP
Source CFG.MsgDest.CycleDest
Dest CycleWriteMsg.RemoteElement <EQ0006_CycleWriteMsg.RemoteElement>
Length 1

EDC Gateway Cycle Write Message
MOVE
Source CFG.CycleMSG.e 134
Dest CycleWriteMsg.REQ_LEN <EQ0006_CycleWriteMsg.REQ_LEN> 134

EDC Gateway Cycle Write Message
MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest CycleWriteMsg.UnconnectedTimeout <EQ0006_CycleWriteMsg.UnconnectedTimeout> 30000000

EDC Gateway Cycle Write Message
MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest CycleWriteMsg.ConnectionRate <EQ0006_CycleWriteMsg.ConnectionRate> 7500000

EDC Gateway Cycle Write Message
MOVE
Source CFG.CLx.ConnectionMult 0
Dest CycleWriteMsg.TimeoutMultiplier <EQ0006_CycleWriteMsg.TimeoutMultiplier> 0

EDC Gateway Cycle Write Message CycleWriteMsg.EN_CC <EQ0006_CycleWriteMsg.EN_CC> (U)

EDC Gateway Cycle Write Message
MSG
Message Control CycleWriteMsg <EQ0006_CycleWriteMsg> (...)
EN
DN
ER

Trend Data Ring Buffer and Message

Write Message EN Performance & Diagnostics

12

Ring Buffer Message Control Write Command
CycleMsgCtl.Write

Ring Buffer Message Control Write Command 1-Shot
CycleMsgCtl.WriteONS
ONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	CycleMsgCtl.Trans
PerfTrans	Status.CycleTxWrite.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	740
Input Par	CycleMsgCtl.Trans.TMR.ACC
Input Par	CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message

Write Message Done

13

Ring Buffer Message Control Write Command
CycleMsgCtl.Write

EDC Gateway Cycle Write Message
CycleWriteMsg.DN
<EQ0006_CycleWriteMsg.DN>

Ring Buffer Message Control Write Command
CycleMsgCtl.Write
U

Ring Buffer Message Control Write Command Done
CycleMsgCtl.WriteOK
L

Ring Buffer Message Control Send Data
CycleMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Cycle Tx DN
Status.Update.CycleTxOK
L

Update Status State for Transactions
aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend

Equipment Data Collection - Line Equipment Status State = Green - Cycle Keep Alive Tx DN
Status.Update.CycleKaTxOK
L

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans CycleMsgCtl.Trans
PerfTrans Status.CycleTxWrite.DN

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans CycleMsgCtl.Trans
PerfTrans Status.CycleTxWrite.KA

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 741
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message

Write Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

14

EDC Gateway Cycle Write Message CycleWriteMsg.EN <EQ0006_CycleWriteMsg.EN>

Ring Buffer Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest CycleMsgCtl.WriteTImeoutTMR.PRE
0

Ring Buffer Message Control Message Timeout Timer

TON
Timer CycleMsgCtl.WriteTImeoutTMR
Preset 0
Accum 0
EN
DN

Ring Buffer Message Control Message Timeout Timer CycleMsgCtl.WriteTImeoutTMR.DN

EDC Gateway Cycle Write Message CycleWriteMsg.TO <EQ0006_CycleWriteMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 744
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Write Message ER Performance & Diagnostics

15

EDC Gateway Cycle Write Message CycleWriteMsg.ER <EQ0006_CycleWriteMsg.ER>

Ring Buffer Message Control Write Error 1-shot CycleMsgCtl.WriteErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Cycle Tx ER Status.Update.CycleTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans CycleMsgCtl.Trans
PerfTrans Status.CycleTxWrite.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 743
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message
Read/Write Message Error Retry

16

EDC Gateway Cycle Read Message
CycleReadMsg.ER
<EQ0006_CycleReadMsg.ER>

EDC Gateway Cycle Write Message
CycleWriteMsg.ER
<EQ0006_CycleWriteMsg.ER>

Ring Buffer Message Control Message Retry Timer
CycleMsgCtl.RetryTMR.DN

Ring Buffer Message Control Retry 1-Shot
CycleMsgCtl.RetryONS
ONS

Ring Buffer Message Control Message Retry Timer

MOVE
Source CFG.MsgRetryTime
5000
Dest CycleMsgCtl.RetryTMR.PRE
0

Ring Buffer Message Control Message Retry Timer

TON
Timer CycleMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Cycle Read Message
CycleReadMsg.ER
<EQ0006_CycleReadMsg.ER>

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 732
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

EDC Gateway Cycle Write Message
CycleWriteMsg.ER
<EQ0006_CycleWriteMsg.ER>

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 742
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

(End)

Simulation of Fault Data

Simulation
Fault Data

Modified to use Scan Interval for Timing
v1.3 Dec2016 - Modified to send 2 Packets

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 800
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Configuration
Toggle BIts to Enable Simulation and Tests

EN - enables Simulation and blocks User Mapping of Data
Test1EN - enables Test 1 - each bit turned on/off in order
Test2EN - enables Test 2 - each bit turned on in order then off in order

2

Simultation for Fault Data Enable Simulation
SimFault.EN

Simultation for Fault Data Test 1 Enable
SimFault.Test1EN

Simultation for Fault Data Test 2 Enable
SimFault.Test2EN

NOP

Simulation of Fault Data

Test 1 Delay Timer

3

Simultation for Fault Data Enable Simulation
SimFault.EN

Simultation for Fault Data Test 1 Enable
SimFault.Test1EN

Simultation for Fault Data Test 1 Delay Timer

MOVE
Source 60000
Dest SimFault.Test1DelayTMR.PRE
60000

Simultation for Fault Data Test 1 Delay Timer

TON
Timer SimFault.Test1DelayTMR
Preset 60000
Accum 0
EN
DN

Simultation for Fault Data Test 1 Enable 1-shot
SimFault.Test1EN_ONS
ONS

Simultation for Fault Data Test 1 Done
SimFault.Test1DN
U

Simultation for Fault Data Test 2 Bit Set Done
SimFault.Test2SetDN
U

Simultation for Fault Data Test 2 Done
SimFault.Test2DN
U

Simultation for Fault Data Fault Packet Index

CLR
Dest SimFault.PKTi
0

Simultation for Fault Data Fault Word Index

CLR
Dest SimFault.WDi
0

Simultation for Fault Data Fault Bit Index

CLR
Dest SimFault.BITi
0

Equipment Data Fault/Status Message Data - Predefined Status

FLL
Source 0
Dest FaultMsg.PreDef
Length 1

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined

FLL
Source 0
Dest FaultMsgCtl.EqDef[0]
Length CFG.FaultEq.n

Simultation for Fault Data Test 1 Interval Preset Time

MOVE
Source 1000
Dest SimFault.Test1IntervalPre
0

Simultation for Fault Data Test 1 Interval Last Time

CLR
Dest SimFault.Test1IntervalLast
0

Simulation of Fault Data

Test 1 Signal Test Timer
Use free-running timer and 1-second interval to avoid accumulating Timer overruns throughout test.

4

Simultation for Fault Data Test 1 Delay Timer
SimFault.Test1DelayTMR.DN

Simultation for Fault Data Test 1 Interval Timer

MOVE	
Source	999999999
Dest	SimFault.Test1IntervalTMR.PRE 999999999

Simultation for Fault Data Test 1 Interval Timer

TON		
Timer	SimFault.Test1IntervalTMR	EN
Preset	999999999	DN
Accum	0	

Simultation for Fault Data Test 1 Interval Accum Time

SUB	
Source A	SimFault.Test1IntervalTMR.ACC 0
Source B	SimFault.Test1IntervalLast 0
Dest	SimFault.Test1IntervalAcc 0

Simultation for Fault Data Test 1 Interval Accum Time

GE	
Source A	SimFault.Test1IntervalAcc 0
Source B	SimFault.Test1IntervalPre 0

Simultation for Fault Data Test 1 Interval Last Time

ADD	
Source A	SimFault.Test1IntervalLast 0
Source B	SimFault.Test1IntervalPre 0
Dest	SimFault.Test1IntervalLast 0

Simultation for Fault Data Test 1 Interval
SimFault.Test1Interval

Simulation of Fault Data

Test 1 - Set Next Bit

5

Simultation for Fault Data Test 1 Interval: SimFault.Test1Interval

Equipment Data Fault/Status Message Data - Predefined Status

FLL
- Source: 0
- Dest: FaultMsg.PreDef
- Length: 1

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined

FLL
- Source: 0
- Dest: FaultMsgCtl.EqDef[0]
- Length: CFG.FaultEq.n

Simultation for Fault Data Test 1 Done: SimFault.Test1DN (/)

Simultation for Fault Data Temp DINT

CLR
- Dest: SimFault.TempDINT
- 16#0000_0000

Simultation for Fault Data Temp DINT: SimFault.TempDINT.[SimFault.BITi] (L)

Simultation for Fault Data Fault Word Index

EQ
- Source A: SimFault.WDi
- 0
- Source B: 0

Equipment Data Fault/Status Message Data - Predefined Status

COP
- Source: SimFault.TempDINT
- Dest: FaultMsg.PreDef
- Length: 1

Simultation for Fault Data Fault Word Index

NE
- Source A: SimFault.WDi
- 0
- Source B: 0

Simultation for Fault Data Equipment Defined Fault Word Index

SUB
- Source A: SimFault.WDi
- 0
- Source B: 1
- Dest: SimFault.EqWDi
- 7

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

COP
- Source: SimFault.TempDINT
- Dest: FaultMsgCtl.EqDef[SimFault.PKTi].Word[SimFault.EqWDi]
- Length: 1

Test 1 - Increment Indexes
All Bits Set & Last Bit Reset - Test Complete

6

Simultation for Fault Data Test 1 Interval: SimFault.Test1Interval

Simultation for Fault Data Test 1 Done: SimFault.Test1DN

Simultation for Fault Data Test 1 Enable: SimFault.Test1EN (U)

Simulation of Fault Data

Test 1 - Increment Indexes using Packet/DataCount configuration for limits
Check if all Test Bits have been Set
(skip Word 0 when PKTi>0)

7

Simultation for Fault Data Test 1 Interval
SimFault.Test1Interval

Simultation for Fault Data Test 1 Done
SimFault.Test1DN

Simultation for Fault Data Fault Bit Index

ADD	
Source A	SimFault.BITi 0
Source B	1
Dest	SimFault.BITi 0

Simultation for Fault Data Fault Bit Index

GT	
Source A	SimFault.BITi 0
Source B	31

Simultation for Fault Data Fault Bit Index

MOVE	
Source	0
Dest	SimFault.BITi 0

Simultation for Fault Data Fault Word Index

ADD	
Source A	SimFault.WDi 0
Source B	1
Dest	SimFault.WDi 0

Simultation for Fault Data Fault Word Index

GT	
Source A	SimFault.WDi 0
Source B	UserCfg.EqDefDataCount[SimFault.PKTi] 8

Simultation for Fault Data Fault Word Index

MOVE	
Source	1
Dest	SimFault.WDi 0

Simultation for Fault Data Fault Packet Index

ADD	
Source A	SimFault.PKTi 0
Source B	1
Dest	SimFault.PKTi 0

Simultation for Fault Data Fault Packet Index

GT	
Source A	SimFault.PKTi 0
Source B	CFG.FaultEq.ei 2

Simultation for Fault Data Fault Packet Index

CLR	
Dest	SimFault.PKTi 0

Simultation for Fault Data Fault Word Index

CLR	
Dest	SimFault.WDi 0

Simultation for Fault Data Fault Bit Index

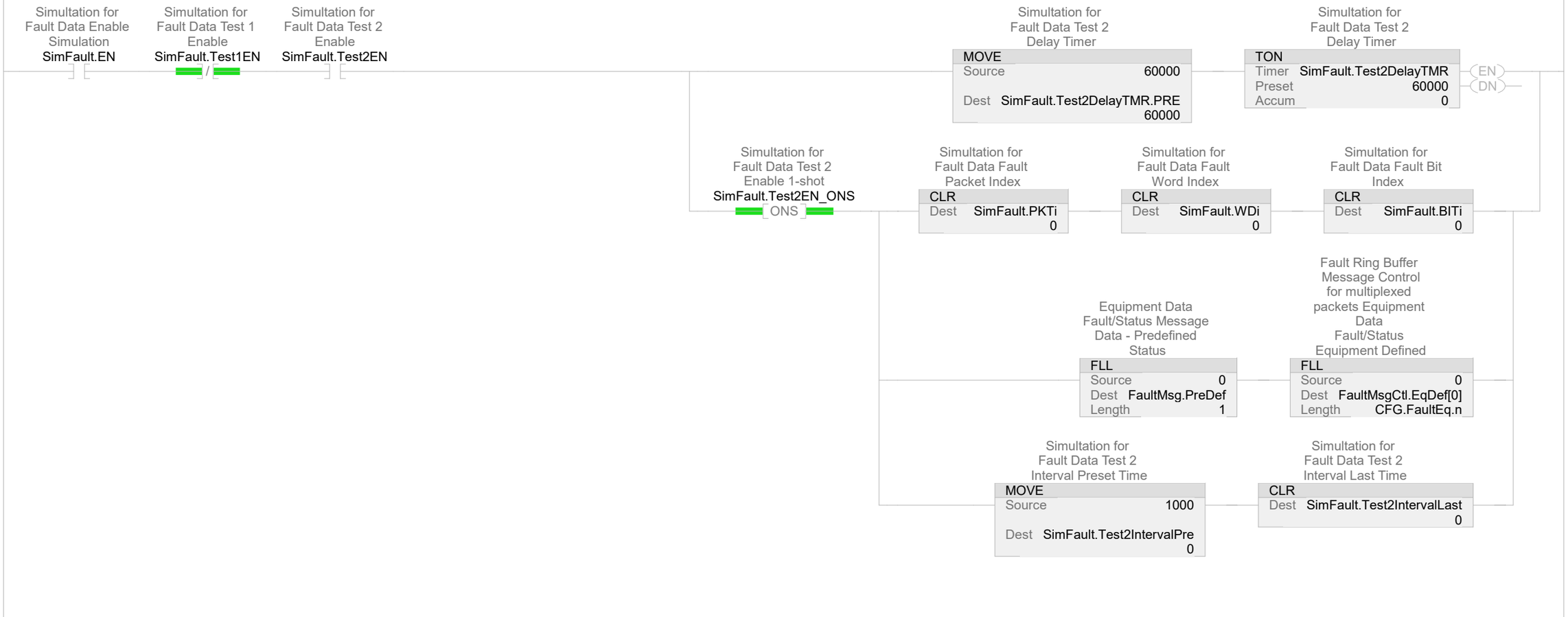
CLR	
Dest	SimFault.BITi 0

Simultation for Fault Data Test 1 Done
SimFault.Test1DN
(L)

Simulation of Fault Data

Test 2 Delay Timer

8



Simulation of Fault Data

Test 2 Signal Interval Timer
Use free-running timer and 1-second interval to avoid accumulating Timer overruns throughout test.

9

Simultation for Fault Data Test 2 Delay Timer
SimFault.Test2DelayTMR.DN

Simultation for Fault Data Test 2 Interval Timer

MOVE	
Source	999999999
Dest	SimFault.Test2IntervalTMR.PRE 999999999

Simultation for Fault Data Test 2 Interval Timer

TON	
Timer	SimFault.Test2IntervalTMR
Preset	999999999
Accum	0

EN
DN

Simultation for Fault Data Test 2 Interval Accum Time

SUB	
Source A	SimFault.Test2IntervalTMR.ACC 0
Source B	SimFault.Test2IntervalLast 0
Dest	SimFault.Test2IntervalAcc 0

Simultation for Fault Data Test 2 Interval Accum Time

GE	
Source A	SimFault.Test2IntervalAcc 0
Source B	SimFault.Test2IntervalPre 0

Simultation for Fault Data Test 2 Interval Last Time

ADD	
Source A	SimFault.Test2IntervalLast 0
Source B	SimFault.Test2IntervalPre 0
Dest	SimFault.Test2IntervalLast 0

Simultation for Fault Data Test 2 Interval
SimFault.Test2Interval

Simulation of Fault Data

Test 2 - Set Next Bit without Reset of Previous Bits

10

Simulation for Fault Data Test 2 Interval: SimFault.Test2Interval

Simulation for Fault Data Test 2 Done: SimFault.Test2DN

Simulation for Fault Data Fault Word Index

EQ
Source A: SimFault.WDi 0
Source B: 0

Simulation for Fault Data Temp DINT

COP
Source: FaultMsg.PreDef
Dest: SimFault.TempDINT
Length: 1

Simulation for Fault Data Test 2 Bit Set Done: SimFault.Test2SetDN

Simulation for Fault Data Temp DINT: SimFault.TempDINT.[SimFault.BITi] (L)

Simulation for Fault Data Test 2 Bit Set Done: SimFault.Test2SetDN

Simulation for Fault Data Temp DINT: SimFault.TempDINT.[SimFault.BITi] (U)

Equipment Data Fault/Status Message Data - Predefined Status

COP
Source: SimFault.TempDINT
Dest: FaultMsg.PreDef
Length: 1

Simulation for Fault Data Fault Word Index

NE
Source A: SimFault.WDi 0
Source B: 0

Simulation for Fault Data Equipment Defined Fault Word Index

SUB
Source A: SimFault.WDi 0
Source B: 1
Dest: SimFault.EqWDi 7

Simulation for Fault Data Temp DINT

COP
Source: FaultMsgCtl.EqDef[SimFault.PKTi].Word[SimFault.EqWDi]
Dest: SimFault.TempDINT
Length: 1

Simulation for Fault Data Test 2 Bit Set Done: SimFault.Test2SetDN

Simulation for Fault Data Temp DINT: SimFault.TempDINT.[SimFault.BITi] (L)

Simulation for Fault Data Test 2 Bit Set Done: SimFault.Test2SetDN

Simulation for Fault Data Temp DINT: SimFault.TempDINT.[SimFault.BITi] (U)

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

COP
Source: SimFault.TempDINT
Dest: FaultMsgCtl.EqDef[SimFault.PKTi].Word[SimFault.EqWDi]
Length: 1

Test 2 - Increment Indexes
All Bits Set & Reset - Test Complete

11

Simulation for Fault Data Test 2 Interval: SimFault.Test2Interval

Simulation for Fault Data Test 2 Done: SimFault.Test2DN

Simulation for Fault Data Test 2 Enable: SimFault.Test2EN (U)

Simulation of Fault Data

Test 2 - Increment Indexes using Packet/DataCount configuration for limits
Check if all Test Bits have been Set
(skip Word 0 when PKTi>0)

12

Simultation for Fault Data Test 2 Interval
SimFault.Test2Interval

Simultation for Fault Data Test 2 Done
SimFault.Test2DN

Simultation for Fault Data Fault Bit Index
ADD
Source A SimFault.BITi 0
Source B 1
Dest SimFault.BITi 0

Simultation for Fault Data Fault Bit Index
GT
Source A SimFault.BITi 0
Source B 31

Simultation for Fault Data Fault Bit Index
MOVE
Source 0
Dest SimFault.BITi 0

Simultation for Fault Data Fault Word Index
ADD
Source A SimFault.WDi 0
Source B 1
Dest SimFault.WDi 0

Simultation for Fault Data Fault Word Index
GT
Source A SimFault.WDi 0
Source B UserCfg.EqDefDataCount[SimFault.PKTi] 8

Simultation for Fault Data Fault Word Index
MOVE
Source 1
Dest SimFault.WDi 0

Simultation for Fault Data Fault Packet Index
ADD
Source A SimFault.PKTi 0
Source B 1
Dest SimFault.PKTi 0

Simultation for Fault Data Fault Packet Index
GT
Source A SimFault.PKTi 0
Source B CFG.FaultEq.ei 2

Simultation for Fault Data Test 2 Bit Set Done
SimFault.Test2SetDN

Simultation for Fault Data Test 2 Done
SimFault.Test2DN
L

Simultation for Fault Data Fault Packet Index
CLR
Dest SimFault.PKTi 0

Simultation for Fault Data Fault Word Index
CLR
Dest SimFault.WDi 0

Simultation for Fault Data Fault Bit Index
CLR
Dest SimFault.BITi 0

Simultation for Fault Data Test 2 Bit Set Done
SimFault.Test2SetDN
L

Simulation of Fault Data

Disable Simulation when Tests Complete

13

Simultation for Fault Data Test 1 Enable
SimFault.Test1EN

Simultation for Fault Data Test 2 Enable
SimFault.Test2EN

Simultation for Fault Data All Tests Done Timer

MOVE
Source 60000
Dest SimFault.TestsDoneTMR.PRE
60000

Simultation for Fault Data All Tests Done Timer

TON
Timer SimFault.TestsDoneTMR
Preset 60000
Accum 0
EN
DN

Simultation for Fault Data All Tests Done Timer
SimFault.TestsDoneTMR.DN

Simultation for Fault Data Enable Simulation
SimFault.EN
U

Simultation for Fault Data All Tests Done Timer
SimFault.TestsDoneTMR
RES

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
Num = Transaction Number

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR	
Input Par	Trans.Ctl.Code
Input Par	Trans.Ctl.Time
Input Par	Trans.Ctl.Num

NOP

Exit if Transaction Log is not Enabled

1

EDC Config Enable Transaction Logging
CFG.TransLogEN

lblExit
JMP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

2

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	902
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

3

Transaction Log Control Enable Code Filters
Trans.Ctl.Filter.CodeEN
/

Transaction Log Control Log Record
Trans.Ctl.Filter.Log

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit: Trans.Ctl.Filter.CodeLow 0
Test: Trans.Ctl.Code 141
High Limit: Trans.Ctl.Filter.CodeHigh 0

Transaction Log Control Log Record
Trans.Ctl.Filter.Log
/

lblExit
JMP

Increment Log Index and check for legal value

4

Transaction Log Control -- Transaction Log Index

ADD
Source A: Trans.Ctl.LOGi 0
Source B: 1
Dest: Trans.Ctl.LOGi 0

Transaction Log Control -- Transaction Log Index

LT
Source A: Trans.Ctl.LOGi 0
Source B: 0

Transaction Log Control -- Transaction Log Index

GE
Source A: Trans.Ctl.LOGi 0
Source B: CFG.TransLOG.n 150

Transaction Log Control -- Transaction Log Index

CLR
Dest: Trans.Ctl.LOGi 0

Log Transactions

Generate Log Index for clearing Next Transaction

5

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A Trans.Ctl.LOGi 0
Source B 1
Dest Trans.Ctl.LOGx 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A Trans.Ctl.LOGx 0
Source B 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A Trans.Ctl.LOGx 0
Source B CFG.TransLOG.n 150

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest Trans.Ctl.LOGx 0

Initialize Transaction and Next Transaction

6

Transaction Log Control Transaction Log

FLL
Source 0
Dest Trans.Log[Trans.Ctl.LOGi]
Length 1

Transaction Log Control Transaction Log

FLL
Source 0
Dest Trans.Log[Trans.Ctl.LOGx]
Length 1

Log Transactions

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

7

Transaction Log Control Transaction Code

MOVE	
Source	Trans.Ctl.Code
	141
Dest	Trans.Log[Trans.Ctl.LOGi].TransCode
	0

Transaction Log Control Transaction Time

MOVE	
Source	Trans.Ctl.Time
	15
Dest	Trans.Log[Trans.Ctl.LOGi].TransTime
	0

Transaction Log Control Transaction Number

MOVE	
Source	Trans.Ctl.Num
	9993
Dest	Trans.Log[Trans.Ctl.LOGi].Num
	0

Transaction Log Control Transaction Date (YYMMDD)

MOVE	
Source	CLX.SysTime[8]
	20261030
Dest	Trans.Log[Trans.Ctl.LOGi].DateStamp
	0

Transaction Log Control Transaction Time (HHMMSS)

MOVE	
Source	CLX.SysTime[9]
	152452
Dest	Trans.Log[Trans.Ctl.LOGi].TimeStamp
	0

Transaction Log Control Transaction Time (microseconds)

MOVE	
Source	CLX.SysTime[6]
	273675
Dest	Trans.Log[Trans.Ctl.LOGi].TimeMicroSec
	0

Transaction Log Control Transaction Scan Couner

MOVE	
Source	CLX.ScanCount
	1106690484
Dest	Trans.Log[Trans.Ctl.LOGi].ScanCount
	0

Store Optional Transaction Details
(loaded before calling Log Routine)

8

Transaction Log Control Optional Parameter 1 (defined by Transaction)

MOVE	
Source	Trans.Ctl.Opt[1]
	-1
Dest	Trans.Log[Trans.Ctl.LOGi].Opt1
	0

Transaction Log Control Optional Parameter 2 (defined by Transaction)

MOVE	
Source	Trans.Ctl.Opt[2]
	-1
Dest	Trans.Log[Trans.Ctl.LOGi].Opt2
	0

Transaction Log Control Optional Parameter 3 (defined by Transaction)

MOVE	
Source	Trans.Ctl.Opt[3]
	-1
Dest	Trans.Log[Trans.Ctl.LOGi].Opt3
	0

Transaction Log Control Optional Parameter 4 (defined by Transaction)

MOVE	
Source	Trans.Ctl.Opt[4]
	-1
Dest	Trans.Log[Trans.Ctl.LOGi].Opt4
	0

Transaction Log Control Optional Parameter 5 (defined by Transaction)

MOVE	
Source	Trans.Ctl.Opt[5]
	-1
Dest	Trans.Log[Trans.Ctl.LOGi].Opt5
	0

Log Transactions

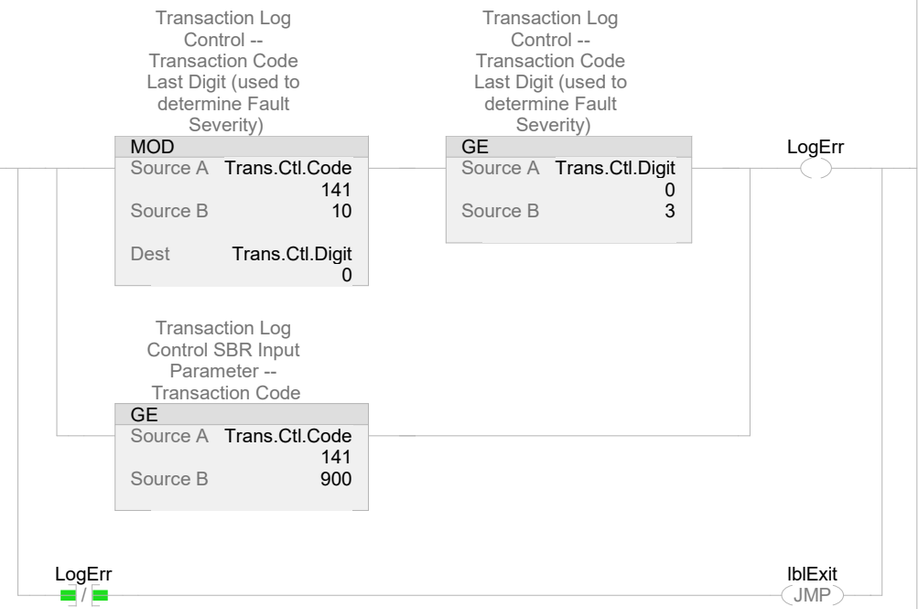
Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xx0 = Enable -- Yellow Status for Normal Operation in Process
xx1 = Done -- Green Status for Normal Operation Completed
xx2 = Retry -- Orange Status for Operation in Retry
xx3 = Error -- Red Status for Operation Failed
xx4-xx9 = Other Errors -- see TransLog Definitions

9xx = Initialization Events
v1.3 Block Retry events from Error Log

9



Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

10

NOP

Log Transactions

Increment Error Log Index and check for legal value

11

Transaction Log Control -- Transaction Error Log Index

ADD
Source A: Trans.Ctl.ERRi 0
Source B: 1
Dest: Trans.Ctl.ERRi 0

Transaction Log Control -- Transaction Error Log Index

LT
Source A: Trans.Ctl.ERRi 0
Source B: 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A: Trans.Ctl.ERRi 0
Source B: CFG.TransERR.n 50

Transaction Log Control -- Transaction Error Log Index

CLR
Dest: Trans.Ctl.ERRi 0

Generate Error Log Index for clearing Next Transaction

12

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A: Trans.Ctl.ERRi 0
Source B: 1
Dest: Trans.Ctl.ERRx 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A: Trans.Ctl.ERRx 0
Source B: 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A: Trans.Ctl.ERRx 0
Source B: CFG.TransERR.n 50

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest: Trans.Ctl.ERRx 0

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

13

Transaction Log Control Transaction Log

COP
Source Trans.Log[Trans.Ctl.LOGi]
Dest Trans.Err[Trans.Ctl.ERRi]
Length 1

Transaction Log Control Transaction Log

FLL
Source 0
Dest Trans.Err[Trans.Ctl.ERRx]
Length 1

Exit Routine

14

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

15

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Ctl.Opt[0]
Length Trans.Ctl.OptN

(End)

Main Routine

Summary Data Event Log

0

S:FS

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	EDC_aoiProgVersion
Major	1
Minor	0
Build	1
Patch	8
Version	Prog.Version

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	601
Dest	Prog.Routine 650

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 26194212
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 26194212

Configure Summary Log Data Collection
Enable - toggle ON to log data
EQID - select Equipment ID for logging

2

AFI

Summary Log Enable
SumLogEN

Equipment ID

EQ	
Source A	EQID 420
Source B	EQID 420

NOP

Main Routine

Initialize Log Size and Max Index

3

s:FS

Summary Log Summary Log Event

SIZE
Source SumLog[0] ??
Dim. To Vary 0
Size LOGn 1000

Log Max Index

SUB
Source A LOGn 1000
Source B 1
Dest LOGmi 999

Equipment Summary Event Log for EQID

4

Summary Log Enable SumLogEN

Equipment Summary Event Log

JSR
Routine Name _650_SumLog

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

5

CLx Module Status Clear Scan History CLX.ProgScan.Clear <EDC.CLX.ProgScan.Clear>

Local Program Scan Time & Fault Routine Clear Scan History Prog.Scan.Clear

L

Local Program Scan Time & Fault Routine Program Fault Occurrences

CLR
Dest Prog.FaultCount 0

Program Scan

aoiEDC_ProgScan
aoiEDC_ProgScan EDC_aoiProgScan ...
Scan Prog.Scan
ScanTemp CLX.ProgScanTemp <EDC.CLX.ProgScanTemp>

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2

Local Program Scan Time & Fault Routine Fault Routine Log

SIZE	
Source	Prog.Log[0] ??
Dim. To Vary	0
Size	LEN 1000

Local Program Scan Time & Fault Routine Program Fault Log Maximum Index

SUB	
Source A	LEN 1000
Source B	1
Dest	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

ADD	
Source A	Prog.LOGi 0
Source B	1
Dest	Prog.LOGi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

LT	
Source A	Prog.LOGi 0
Source B	0

Local Program Scan Time & Fault Routine Program Fault Log Index

GT	
Source A	Prog.LOGi 0
Source B	Prog.LOGmi 0

Local Program Scan Time & Fault Routine Program Fault Log Index

CLR	
Dest	Prog.LOGi 0

Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
650
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
<EDC.CLX.ScanCount>
466525176
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
<EDC.CLX.SysTime[8]>
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
<EDC.CLX.SysTime[9]>
151921
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
<EDC.CLX.SysTime[11]>
904
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 602
Dest Prog.Routine 650

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 26194212
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 26194212

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
<EDC.CFG.DisableFaultHandler>
/

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Equipment Summary Event Log

Equipment Summary Event Log
for EQID

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 650
Dest Prog.Routine 650

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 26194212
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 26194212

Calculate LNi, EQi for Summary Log EQID

2

Equipment Index

MOD
Source A EQID 420
Source B 100
Dest EQi 20

Line Index

CPT
Dest LNi 4
Expression (EQID - EQi) / 100

Line Index

LIMIT
Low Limit 0
Test LNi 4
High Limit CFG.LN.mi <EDC.CFG.LN.mi> 28

Equipment Index

LIMIT
Low Limit 0
Test EQi 20
High Limit CFG.EQ.mi <EDC.CFG.EQ.mi> 99

ValidEQID

Exit if invalid EQID

3

ValidEQID
/

lbl_Exit
JMP

Equipment Summary Event Log

Initialize EquipLast for new EQID

4

ValidEQID ValidEQID_ONS ONS

lbl_Save JMP

Equipment ID

NE	
Source A	EQID 420
Source B	EQIDprev 420

Copy Equipment Arrayed Data to Working Tags
Working Tags are used for speed and logic readabliity (ie. can be xref'd)

5

Local Copy of Equip Array Tag for LNi, EQi

COP	
Source	EDC.Line[LNi].Equip[EQi]
Dest	Equip
Length	1

Equipment Summary Event Log

Detect New Event
- Change in Work Status
- Change in Line Stop Maintenance
- Change in Line Stop Production

6

xxTestEventLog

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start) Equip.DataCTL.Work

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start) EquipPrev.DataCTL.Work

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start) Equip.DataCTL.Work

Local Copy of Equip Array Tag for LNi, EQi Work Time (Overall Start) EquipPrev.DataCTL.Work

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OA Equip.DataCTL.LdtOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OA EquipPrev.DataCTL.LdtOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OA Equip.DataCTL.LdtOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime OA EquipPrev.DataCTL.LdtOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Maintenance OA Equip.DataCTL.LdtMaintOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Maintenance OA EquipPrev.DataCTL.LdtMaintOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Maintenance OA Equip.DataCTL.LdtMaintOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Maintenance OA EquipPrev.DataCTL.LdtMaintOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Production OA Equip.DataCTL.LdtProdOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Production OA EquipPrev.DataCTL.LdtProdOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Production OA Equip.DataCTL.LdtProdOA

Local Copy of Equip Array Tag for LNi, EQi Line Downtime Production OA EquipPrev.DataCTL.LdtProdOA

xxTestEventLog

U

Equipment Summary Event Log

JSR

Routine Name _651_SumLogEvent

Equipment Summary Event Log

Save Equipment for next comparison

7

lbl_Save
LBL

Local Copy of Equip Array Tag for LNi, EQi

COP	
Source	Equip
Dest	EquipPrev
Length	1

EQID Previous Value

MOVE	
Source	EQID 420
Dest	EQIDprev 420

Exit Routine

8

lbl_Exit
LBL

NOP

(End)

Equipment Summary Event Log

Equipment Summary Event Log
for EQID

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 651
Dest Prog.Routine 650

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 26194212
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 26194212

Increment Log Index

2

Log Index

ADD
Source A LOGi 353
Source B 1
Dest LOGi 353

Log Index

LT
Source A LOGi 353
Source B 0

Log Index

GT
Source A LOGi 353
Source B LOGmi 999

Log Index

CLR
Dest LOGi 353

Equipment Summary Event Log

Log Equipment Summary Event

3

Summary Log Event Equipment ID

MOVE
Source EQID 420
Dest SumLog[LOGi].EQID 420

Summary Log Event Transaction Timestamp

COP
Source CLX.TransTime <EDC.CLX.TransTime>
Dest SumLog[LOGi].TransTime
Length 1

Summary Log Event Data Control

COP
Source Equip.DataCTL
Dest SumLog[LOGi].DataCtl
Length 1

Summary Log Event Data Control Previous

COP
Source EquipPrev.DataCTL
Dest SumLog[LOGi].DataCtlPrev
Length 1

Summary Log Event Work Data - Time/Count

COP
Source Equip.Data.Work
Dest SumLog[LOGi].WorkData
Length 1

Summary Log Event Line Data - Time/Count

COP
Source Equip.Data.LdtOA.Maint
Dest SumLog[LOGi].LineData
Length 1

Summary Log Event Equip Data - Time/Count

COP
Source Equip.Data.LdtOA.Prod
Dest SumLog[LOGi].EquipData
Length 1

(End)

eAssy_EDC_Gway
MainTask
Gateway
_001_Main
Ladder Diagram 1
_002_Fault
Ladder Diagram 12
_009_ConfigInit
Ladder Diagram 16
_010_Reset
Ladder Diagram 20
_011_Config
Ladder Diagram 32
_012_ConfigCLx
Ladder Diagram 36
_014_ConfigLine
Ladder Diagram 39
_015_ConfigLineLoop
Ladder Diagram 43
_020_InitializeCLx
Ladder Diagram 46
_021_InitializeEventMasks
Ladder Diagram 50
_029_CLxTimeSyncSNTP
Ladder Diagram 53
_029_CLxTimeZoneDST
Ladder Diagram 58
_030_CLxStatus
Ladder Diagram 69
_031_Enet00_NW_L8x
Ladder Diagram 83
_031_Enet01_GW_L9x_THIS
Ladder Diagram 94
_031_Enet02_PdM_L8x
Ladder Diagram 105
_031_Enet03_FEC1_L8x
Ladder Diagram 116
_031_Enet04_FEC2_L8x
Ladder Diagram 127
_031_Enet05_FEC3_L8x
Ladder Diagram 138
_031_Enet06_FEC4_L8x
Ladder Diagram 149
_038_AsciiTable
Ladder Diagram 160
_039_Constants
Ladder Diagram 162
_080_TransLogSnapshot

Ladder Diagram 163
_090_PerfSummaryLine
Ladder Diagram 166
_091_PerfSummaryEquip
Ladder Diagram 170
_095_PerfSummaryEquipSum
Ladder Diagram 183
_096_PerfSummaryReset
Ladder Diagram 188
_097_PerfSummaryResetLine
Ladder Diagram 192
_098_PerfSummaryResetEquip
Ladder Diagram 193
_901_TransLogSBR
Ladder Diagram 194
EquipRxLiveCallout
_101_Main
Ladder Diagram 201
_102_Fault
Ladder Diagram 208
_111_Config
Ladder Diagram 212
_120_LineLoop
Ladder Diagram 226
_130_EquipLoop
Ladder Diagram 239
_131_EquipCalloutRx
Ladder Diagram 254
_132_EquipCalloutRxLoad
Ladder Diagram 268
_133_EquipCalloutRxMapV1
Ladder Diagram 279
_134_EquipCalloutRxResponse
Ladder Diagram 280
_135_EquipEvents
Ladder Diagram 288
_140_EquipData
Ladder Diagram 303
_141_OfflineGroupDataInit
Ladder Diagram 327
_142_OfflineGroupDataReset
Ladder Diagram 330
_143_OfflineGroupData
Ladder Diagram 332
_150_SumDataMsg
Ladder Diagram 339
_151_SumLoadRB01
Ladder Diagram 341

_152_SumLoadRB02
Ladder Diagram 348
_901_TransLogSBR
Ladder Diagram 355
EquipRxBufferData
_201_Main
Ladder Diagram 362
_202_Fault
Ladder Diagram 366
_211_Config
Ladder Diagram 370
_220_LineLoop
Ladder Diagram 378
_225_EquipLoop
Ladder Diagram 381
_227_EquipFaultRxBulk
Ladder Diagram 389
_230_EquipFaultRx
Ladder Diagram 405
_231_EquipFaultRxLoadRB01
Ladder Diagram 422
_232_EquipFaultRxLoadRB02
Ladder Diagram 428
_233_EquipFaultRxLoadRB03
Ladder Diagram 434
_234_EquipFaultRxLoadRB04
Ladder Diagram 440
_235_EquipFaultRxLoadRB05
Ladder Diagram 446
_236_EquipFaultRxLoadRB06
Ladder Diagram 452
_240_EquipQualityRx
Ladder Diagram 458
_241_EquipQualityRxLoadRB01
Ladder Diagram 477
_242_EquipQualityRxLoadRB02
Ladder Diagram 484
_243_EquipQualityRxLoadRB03
Ladder Diagram 491
_244_EquipQualityRxLoadRB04
Ladder Diagram 498
_245_EquipQualityRxCarrierTrack
Ladder Diagram 505
_245_EquipQualityRxCarrierTrack_CfgAnim
Ladder Diagram 522
_246_EquipQualityRxMapV1_5
Ladder Diagram 524
_247_EquipQualityRxPdmTriggerStart

Ladder Diagram 528
_248_EquipQualityRxPdmTriggerSend
Ladder Diagram 530
_249_EquipQualityRxPdmSampleSend
Ladder Diagram 535
_250_EquipTrendRx
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_251_EquipTrendRxLoadRB01
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_252_EquipTrendRxLoadRB02
Ladder Diagram 564
_257_EquipTrendRxPdmTriggerStart
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_258_EquipTrendRxPdmTriggerSend
Ladder Diagram 573
_259_EquipTrendRxPdmSampleSend
Ladder Diagram 575
_260_EquipVersionRx
Ladder Diagram 579
_270_EquipCycleRx
Ladder Diagram 589
_271_EquipCycleRxLoadRB01
Ladder Diagram 602
_272_EquipCycleRxLoadRB02
Ladder Diagram 609
_901_TransLogSBR
Ladder Diagram 616
RingBufferFault01
_301_Main
Ladder Diagram 623
_302_Fault
Ladder Diagram 626
_311_Config
Ladder Diagram 630
_320_RingBuffer
Ladder Diagram 632
_321_RingBufferKeepAlive
Ladder Diagram 651
_901_TransLogSBR
Ladder Diagram 657
RingBufferFault02
_301_Main
Ladder Diagram 664
_302_Fault
Ladder Diagram 667
_311_Config
Ladder Diagram 671
_320_RingBuffer

Ladder Diagram 673
_321_RingBufferKeepAlive
Ladder Diagram 692
_901_TransLogSBR
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_301_Main
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_302_Fault
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_311_Config
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_320_RingBuffer
Ladder Diagram 714
_321_RingBufferKeepAlive
Ladder Diagram 733
_901_TransLogSBR
Ladder Diagram 739
RingBufferFault04
_301_Main
Ladder Diagram 746
_302_Fault
Ladder Diagram 749
_311_Config
Ladder Diagram 753
_320_RingBuffer
Ladder Diagram 755
_321_RingBufferKeepAlive
Ladder Diagram 774
_901_TransLogSBR
Ladder Diagram 780
RingBufferFault05
_301_Main
Ladder Diagram 787
_302_Fault
Ladder Diagram 790
_311_Config
Ladder Diagram 794
_320_RingBuffer
Ladder Diagram 796
_321_RingBufferKeepAlive
Ladder Diagram 815
_901_TransLogSBR
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RingBufferFault06
_301_Main
Ladder Diagram 828
_302_Fault

Ladder Diagram 831
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Ladder Diagram 835
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_402_Fault
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_420_RingBuffer
Ladder Diagram 878
_421_RingBufferKeepAlive
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_901_TransLogSBR
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_401_Main
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_402_Fault
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_411_Config
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_520_RingBuffer
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_521_RingBufferKeepAlive
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_602_Fault
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Ladder Diagram 1188
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_1101_Main
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_901_TransLogSBR
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_701_Main
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Ladder Diagram 1313
_716_OviewEventStatus
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_717_ZoneEventStatus
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_720_LineLoop
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_730_EquipLoop
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_891_MaskRequestInit
Ladder Diagram 1677
_895_MaskResponse
Ladder Diagram 1680
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Ladder Diagram 1689
_901_TransLogSBR
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_901_Main
Ladder Diagram 1704
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Ladder Diagram 1708
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Ladder Diagram 1713
_920_CarrierRequest
Ladder Diagram 1716
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Ladder Diagram 1719

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Ladder Diagram 1731
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Ladder Diagram 1738
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Ladder Diagram 1938
_1233_EquipPitchUpdate
Ladder Diagram 1941
_1234_EquipLineUpdate

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_901_TransLogSBR
Ladder Diagram 1950
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_001_Main
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_002_Fault
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_009_ConfigInit
Ladder Diagram 1975
_010_Reset
Ladder Diagram 1977
_011_Config
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